THE MIDPOINT RIEMANN–VON MANGOLDT SYNTHETIC ZETA SYSTEM: CANONICAL PRODUCTS, EXPONENTIAL SUMS, PRIME COUNTS, MERTENS TRANSFORMS, AND THE TRANSFORMATION TO ζ_{Z1}

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Abstract. We study the zeta-like system generated by the midpoint inverse of the smooth Riemann–von Mangoldt counting law. The midpoint ordinates τ_k, defined by $\frac{\tau_k}{2\pi}\log\frac{\tau_k}{2\pi e}+\frac{7}{8}=k-\frac12$ and given by the Lambert formula $\tau_k=2\pi(k-\frac{11}{8})/W_0((k-\frac{11}{8})/e)$, are those of França and LeClair; the constructions built on them are, to the author's knowledge, new. The synthetic zeta function ζ_{RVM} has all its nontrivial zeros at $\frac12\pm i\tau_k$. We prove normal convergence of its canonical product, a uniform logarithmic bound for the exponential sums of the τ_k, the explicit-formula prime count π_{RVM} with the unconditional estimate $\pi_{\mathrm{RVM}}(x)-\mathrm{li}(x)=O(\sqrt{x}(\log\log x)^2/\log x)$, the transition theorem $\zeta_{\mathrm{RVM}}=\varpi_{\mathrm{RVM}}(1+w)(1+O(1/|t|))$ with $\varpi_{\mathrm{RVM}}<0.9796$, and the synthetic Mertens bound $M^{\mathrm{crit}}_{\mathrm{RVM}}(x)=L+O_\varepsilon(x^{1/2+\varepsilon})$, $L=1/\varpi_{\mathrm{RVM}}$.

Exponential sums over the ordinates. We compare $\sum_k e^{iw\gamma_k}$, over the ordinates γ_k of the zeros of ζ, with Cramér's function $V_+(w)=\sum_{\Im\rho>0}e^{\rho w}$, for which we give a representation that involves no sum over the zeros, valid on a half-plane containing $(0,\infty)$. This yields equivalent forms of the Riemann hypothesis written with the ordinates alone, which we have not found in the literature consulted: RH holds if and only if $\sum_k e^{-y_0\gamma_k}=e^{-iy_0/2}V_+(iy_0)$ for one $y_0\in(0,4\pi]$; and RH holds if and only if $\sum_k\widehat{\mathsf f}(\gamma_k)=\int\mathsf f(u)e^{-u/2}V_+(u)\,du$ for every test function $\mathsf f$ supported in the window $(\log2,\log3)$, where the explicit formula has no prime atom, that is, if and only if the triangular centroids $\sum_{k\le N}(1-\frac{k-1}{N})e^{iu\gamma_k}$ converge there to $e^{-u/2}V_+(u)$ in the sense of distributions. Without hypothesis the defect is an explicit absolutely convergent sum over the zeros off the line. The rung functional of the endpoint is that same sum: $(\log x)(dJ^{\mathrm M}_{Z1}-dJ)/dx=\frac{2}{\sqrt x}\sum_k(\cosh(d_k\log x)-1)\cos(\gamma_k\log x)$, $d_k=\Re\rho_k-\frac12$, so that the real part of the window identity is the single cell $(2,3)$, and the whole ladder of cells, equivalently RH, is that cell together with a sine identity on the window. If only finitely many zeros were off the line, $J^{\mathrm M}_{Z1}$ would be constant on no cell $(n,n+1)$. A Frullani identity writes $\log L+\log\varpi_{Z1}$ as $\lim_K 2\int_0^\infty u^{-1}e^{-u/2}\sum_{k\le K}(\cos\gamma_k u-\cos\tau_k u)\,du$. Under RH the triangular centroid of $e^{i\gamma_k}$ converges to $e^{-1/2}V_+(1)$, and that of $e^{i\tau_k}$ converges without hypothesis; the two limits are, numerically, $0.77689426079\ldots+0.76785566886\ldots\,i$, whose real part is $\frac{1}{2\sqrt e}(e-\frac{1}{e^2-1})$, and $0.041569+1.757541\,i$.

The family and its endpoint. The ordinates move along $\gamma_k(\lambda)=(1-\lambda)\gamma_k+\lambda\tau_k$; every member ζ_λ has its nontrivial zeros on $\Re s=\frac12$ by construction, $\zeta_1=\zeta_{\mathrm{RVM}}$, and $\zeta_0=\zeta_{Z1}$. The final step is the identity $\zeta_{Z1}=\zeta$, which by Hamburger's theorem is equivalent to the condition (D) that ζ_{Z1} is an absolutely convergent Dirichlet series in $\Re s>1$, and to RH. We prove that $\varpi_{Z1}=\lim_{\sigma\to+\infty}\zeta_{Z1}(\sigma)=\varpi_{\mathrm{RVM}}\lim_K\prod_{k\le K}\frac{1/4+\tau_k^2}{1/4+\gamma_k^2}\ge1$, with equality exactly under (D); that $\log\varpi_\lambda$ is strictly convex in λ, and that if RH fails the family crosses $\varpi_\lambda=1$ at exactly one λ^*, and $\lambda^*<2\cdot10^{-11}$ by the verified zeros; that every member admits an atomization by smooth cells which is an absolutely convergent Dirichlet series; that (D) is equivalent to the vanishing of every cell defect, to a positivity criterion, to Hamburger's theorem for the pair (continuous, atomized) in several forms, and to the existence of a list in a compact space of ordinate lists whose continuous and atomized zeta functions coincide; and, for completeness, that $\zeta_{Z1}=\zeta$ implies $M(x)=O_\varepsilon(x^{1/2+\varepsilon})$. A last section treats route B for finite sections of the rung system: an exact solution whose ordinates stay at bounded distance from the clock, with arbitrary real masses, is the zeta list with the von Mangoldt masses, so that route B is equivalent to RH; since RH is verified up to height $3\cdot10^{12}$, no finite stage decides it; on one window, RH is equivalent to the existence of a list of ordinates reproducing Cramér's function, and to the complete

Date: 15 October 2026 – version 25.

monotonicity of $e^{-iy/2}V_+(iy)$; and fifteen finite sections up to height 487 are solved exactly, with a proof in ball arithmetic.

Tables. We tabulate and draw the first 1000 values of τ_k against γ_k, of the RVM prime levels p_j^{RVM} against p_j, of $\pi_{\mathrm{RVM}}(n)$ against $\pi(n)$ and of $M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ against $M(n)$, and the walks of $e^{i\gamma_k}$, $e^{i\tau_k}$ and $e^{i(\gamma_k+\tau_k)/2}$ in three dimensions, computed from 2 001 052 zeros.

What is proved and what is not. The mathematical statements of the first three paragraphs are theorems of the paper, with the hypotheses (RH or (D)) indicated where they are used; the decimal values of the two centroid limits are computed, the bound on λ^* uses the zeros verified by Platt and Trudgian, and the paragraph on tables describes computations. The identity $\zeta_{Z1} = \zeta$, equivalently (D), equivalently RH, equivalently each of the imaginary-frequency, window and cell criteria above, is not proved. The computations are experiments and are not used in any proof, except for the interval evaluation of a few explicit inequalities in the proof of $\varpi_{\mathrm{RVM}} < 1$ and the ball-arithmetic verification of the finite sections of Theorem 18.78.

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Notation

Every symbol below has a single meaning throughout the paper. The last column links to the defining equation; hovering over the link shows the formula. Symbols introduced inside a single proof (indices j, k_0, m, auxiliary variables v, y, thresholds t_0, t_1, t_2, radii r_1, r_2, r_3) are local to that proof and do not appear elsewhere; so are the following local symbols, which coincide in shape with global ones but never with their meaning: E_k and T_m in the proofs of Theorems 4.1 and 7.1, ϕ in Lemmas 6.4 and 9.3, $K_m^{\pm}$ and r in Section 9, J, S, C_0, E and A in the proofs of Section 14, U as a window length in Subsection 15.15, $\mathsf{X}_{\mathfrak{P}}$ for the prime diagonal operator, $\mathfrak{g}_1$ in Subsection 18.7, and c_1 in Propositions 14.17 and 15.10.

Symbol	Meaning	Defined
Variables and the Riemann zeta function		
$s = \sigma + it$	complex variable, $\sigma = \Re s$, $t = \Im s$	–
$z = s - \frac{1}{2} = \sigma' + it$	centred variable, $\sigma' = \sigma - \frac{1}{2}$	(9.2)
$q(s) = s(s-1)$	symmetric quadratic, $q(1-s) = q(s)$	(3.1)
ζ, ρ	Riemann zeta function; ρ runs over its nontrivial zeros with multiplicity	§2
γ_k	kth positive ordinate of the nontrivial zeros of ζ, with multiplicity, nondecreasing	(2.1)
$N_\zeta(T)$, $N_\zeta^*(T)$	zero count up to height T; its symmetric version	(2.1), (2.10)
$S(T)$	remainder in the Riemann–von Mangoldt formula	(2.9)
$G(s)$	$s(s-1)\Gamma(s/2)\pi^{-s/2}$	(3.3)
$P_\zeta = 2\xi$	completed product $G\zeta$	(10.1)
μ, $M(x)$	Möbius function; Mertens function $\sum_{n\le x}\mu(n)$	(1.1)
$\pi(x)$, li	prime-counting function; $\mathrm{li}(x) := \mathrm{PV}\int_0^x dt/\log t$	–
$\chi(s)$, ϑ	functional-equation factor; Riemann–Siegel theta function	(9.1), (9.29)
The midpoint clock		
$\mathcal{N}_{\mathrm{RVM}}(T)$	midpoint Riemann–von Mangoldt clock	(2.3)
τ_k	midpoint ordinates: the unique solutions in $[2\pi, \infty)$ of $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$	(2.4)
W_0	principal branch of Lambert's W-function	(2.5)
$\mathcal{T}$	inverse function of the clock	(4.4)
ℓ_k	$\log(\tau_k/2\pi)$	text after (4.4)
$n_{\mathrm{RVM}}(H)$	number of $\tau_k \le H$	(4.12)
η_k, C_η	$\eta_k = \tau_k - \gamma_k$; a bound $\lvert\eta_k\rvert \le C_\eta$	(2.11), (2.12)
The synthetic zeta function and its sums		
$P_{\mathrm{RVM}}^{\mathrm{crit}}$, ξ_{RVM}, ζ_{RVM}	canonical product, completed and synthetic zeta	(3.2), (3.4)
ρ_k^{RVM}	synthetic zero $\frac{1}{2} + i\tau_k$	(3.6)
$S_{\mathrm{RVM}}(N;u)$, $C_{\mathrm{RVM}}^{\mathrm{pts}}$, $C_{\mathrm{RVM}}^{\mathrm{path}}$	exponential sum; point and path centroids	(4.1), (4.2)
u_1, u_2	a frequency interval $[u_1, u_2] \subset (0, \infty)$	(4.3)
$M_H(x)$, $M_H^{\mathrm{RVM}}(x)$	finite-height phase averages (not Mertens functions)	(4.14), (4.13)
Ei	exponential integral on $\Im w \neq 0$	(6.1)
J_{RVM}, J_{RVM}^*, π_{RVM}	explicit-formula prime-power count; truncation; synthetic prime count (real-valued, not an integer count)	(6.2), (6.6), (6.7)
$\mathcal{A}_{\mathrm{RVM}}(u)$, $\ell(t)$	weighted clock sum $\sum_k e^{iu\tau_k}/\tau_k$; $\log(t/2\pi)$	(6.10)
$\mathfrak{s}_1$; $\mathfrak{e}$, $\mathfrak{e}_1$	primitive of the midpoint sawtooth; $N_\zeta - \mathcal{N}_{\mathrm{RVM}}$ and its primitive	(9.36), (13.1)

Symbol	Meaning	Defined
ϖ_{Z1}	value of ζ_{Z1} at $+\infty$	(13.4)
J_X^{M}, J_X, C_X	Mellin and explicit-formula prime-power counts of ζ_X, $X \in \{\mathrm{RVM}, Z1\}$; their difference	(13.14), (13.17)
b_n	jumps of J_{Z1}^{M} at the integers	Thm. 13.7
ϖ_λ; $\Upsilon(\lambda)$	value of ζ_λ at $+\infty$; flow at infinity	(14.1), (14.3)
φ_n, $\tilde{c}_n(\lambda)$, $\tilde{a}_N(\lambda)$	smooth cells; atomized masses and coefficients	(14.7), (14.8)
$\tilde{\zeta}_{\mathbb{Z},\lambda}$, $\tilde{h}_\lambda$	atomized family; joining factor	(14.7), (14.15)
$\mathfrak{a}_s$, $\mathfrak{h}_n(\lambda; s)$, $\mathfrak{L}$	atomization kernel; cell defects; $\int_1^x (1-y^{-1})dy/\log y$	(14.10)
V_n, W_n	ordinate kernels of the cell n	(14.21)
$\tilde{\mu}_\lambda$, $\tilde{M}_\lambda$	atomized Möbius coefficients; summatory function	(14.34)
B_{Z1}	Mellin form of the endpoint Mertens function	(14.36)
c_n^{RVM}, $\zeta_{\mathbb{Z},\mathrm{RVM}}$	integer weights; weighted Euler product	(7.1), (7.2)
$J_{\mathbb{Z},\mathrm{RVM}}$, $J_{\mathrm{RVM}}^{*,\mathbb{Z}}$, h_{RVM}	integer-step counts; Mellin correction	(7.3), (7.4), (7.20)
θ	unit-cell step, $0 < \theta \leq 1$	(7.5)
p_j^{RVM}	real RVM prime levels $\pi_{\mathrm{RVM}}^{-1}(j)$ on monotone branches	(1.6)
$\mathcal{M}(n)$, ς_n, μ_h, M_h	target samples of $M_{\mathrm{RVM}}^{\mathrm{crit}}$; signs; ternary walk and its sum (companion object; h is a label)	(1.9), (1.10)
$a_N^{\mathbb{Z}}$	Dirichlet coefficients of $\zeta_{\mathbb{Z},\mathrm{RVM}}$	(7.14)
C_J	Lipschitz constant of J_{RVM}	(7.5)
$\mathfrak{d}_n^{\mathrm{RVM}}$	prime defect $\mathbf{1}_{\mathrm{prime}}(n) - c_n^{\mathrm{RVM}}$	(10.11)
Mertens transforms and the transition theorem		
$F_{\mathrm{RVM}} = \widehat{M}_{\mathrm{RVM}}^{\mathrm{crit}}$	$1/\zeta_{\mathrm{RVM}}$	(8.1)
c, U	Perron abscissa; truncation height	(9.51)
$M_{\mathrm{RVM}}^{\mathrm{crit}}$, $M_{\mathrm{RVM},G}^{\mathrm{crit}}$	symmetric Perron inverse; Gaussian-smoothed inverse	(9.51), (8.5)
τ_*, $w(s)$	transition point $-iz = t - i\sigma'$; transition variable $e^{-2\pi i \mathcal{N}_{\mathrm{RVM}}(\tau_*)}$	(9.2)
B; σ_1', σ_2'	upper bound for σ'; a range $[\sigma_1', \sigma_2'] \subset (0, \frac{1}{2})$	(9.4), (9.5)
$\mathcal{E}_0, \mathcal{E}$	error factors of the transition theorem	(9.4), (9.5)
ϖ_{RVM}, L	transition constant, $\varpi_{\mathrm{RVM}} < 0.9796$ (Theorem 9.11), floating-point value 0.9619415789; $L = 1/\varpi_{\mathrm{RVM}} > 1$	(9.28), (9.33)
$\mathscr{W}$, $\mathscr{I}_0, \mathscr{I}_n$, $\mathfrak{M}$, $\mathbf{B}_2, \mathbf{B}_3$	weight $d\tau/\tau = \mathscr{W}(u)\,du$ of the clock; pieces of ν_4; lower bound in the proof of Theorem 9.11; Bernoulli polynomials	(9.47), (9.48)
$f, \Lambda, \Lambda_{\mathrm{sm}}, \Lambda_{\mathrm{saw}}$	logarithm of the product and its smooth and sawtooth parts (used in the transition and related clock estimates)	(9.6), (9.9)
$\psi, \psi_r, \Phi_r, \nu_1, \ldots, \nu_4$	sawtooth, Abel means, ray remainder, real constants (used in the transition and related clock estimates; ν_3 in closed form (9.43), ν_4 by (9.39))	(9.8), (9.20)
α_c	$\min(c - \frac{1}{2}, 1)$	(9.35)
Comparison with ζ		
$\mathcal{Q}_{\mathrm{RVM}}$, $\mathcal{Q}_{\mathrm{RVM}}^\vee$	quotient $\zeta/\zeta_{\mathrm{RVM}}$; its inverse Mellin transform	(10.6), (10.8)
The λ-family and the endpoint ζ_{Z1}		
λ	family parameter; $\lambda = 1$: ζ_{RVM}, $\lambda = 0$: ζ_{Z1}	(11.1)
$\gamma_k(\lambda)$	moving ordinates $(1-\lambda)\gamma_k + \lambda\tau_k$	(11.1)

Symbol	Meaning	Defined
P_λ, ζ_λ, F_λ	family product, family zeta, $1/\zeta_\lambda$	(11.2), (11.3)
$A_{k,\lambda}$	$\frac{1}{4}+\gamma_k(\lambda)^2$	(11.10)
$\mathcal{V}_\lambda$	infinitesimal generator $\partial_\lambda \log P_\lambda$	(11.10)
Π_{tr}	total multiplier $P_{Z1}/P^{\mathrm{crit}}_{\mathrm{RVM}}$	(11.15)
$\rho_{k,\lambda}$, $\mathfrak{r}_{k,\lambda}$	moving zero $\frac{1}{2}+i\gamma_k(\lambda)$; residue of $1/(s\zeta_\lambda)$	(11.19)
P_{Z1}, ζ_{Z1}, F_{Z1}	endpoint product, endpoint zeta, $1/\zeta_{Z1}$	(12.1), (12.34)
$\mathscr{R}_{Z1}$	ratio ζ_{Z1}/ζ	(12.7)
$\mathcal{K}$	multiplier $\zeta_{Z1}/\zeta_{\mathbb{Z},\mathrm{RVM}}$	(12.12)
(D), a_n	Dirichlet condition for ζ_{Z1} and its coefficients	(12.8)
C_H	constant in Hamburger's theorem	(12.10)
μ_{Z1}, M_{Z1}	coefficients of F_{Z1}; summatory function	(12.37)
$M_{\lambda,U}$, $M_{Z1,U}$	finite-height Perron integrals	(12.35)
δ, ε, $\omega(\delta)$	margins $\sigma \geq \frac{1}{2}+\delta$, exponent $\varepsilon>0$, three-circle exponent	(12.16)
Criteria in the half-plane, the rung system and the models		
$\mathsf{D}_n(\lambda)$, $B^\vee_\lambda$	$\tilde{c}_n(\lambda)-\tilde{c}_n(0)$; dual function of the atomized family	(15.59), (15.60)
d, Y, $\mathsf{e}_1,\mathsf{e}_2$	difference of two counts; auxiliary entire function; exponents	(15.63), (15.65), (15.69)
J_f, J_g, b_n, ϖ_f	counts and coefficients of a general pair (f,g); value of f at $+\infty$	(15.67), Thm. 15.39
(B1), (B2)	the two hypotheses of the bridge theorem	Thm. 15.39
v_λ	phase defect of the family on the critical line	(15.71)
o_n, L_n, j	one-ordinate difference of cell masses; $-\log(1-n^{-s})$; one-ordinate joining factor	(15.74), (15.77)
f, c_f, k, k_f	test function in $u=\log x$; its cosine transform; the part of G and the pole	(15.79)
R_f, S_f, $\mathsf{m}_j(\mathsf{f})$	rung functional; transport; moments sampled on the clock	(15.80), (15.81), (15.84)
$\mathfrak{t}_0$, I_m	zero of the clock, $\mathcal{N}_{\mathrm{RVM}}(\mathfrak{t}_0)=0$; Poisson integrals	(15.85)
a_n, $\dot{\kappa}_k$, $\dot{\mathsf{a}}_n$	free masses of the rung system; a direction of variation	(16.6), (16.10)
J, x_0, H_1, y, H, Z	objects of the proof of Theorem 16.7	(16.11)–(16.13)
$\langle\cdot,\cdot\rangle_\sigma$, $\mathsf{K}_{\boldsymbol{\kappa}}$, r_m, u_m, $\mathsf{Q}_{\boldsymbol{\kappa}}$	pairing of Dirichlet series; $\tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}}/\zeta$; gradient data of V_σ	(16.15), (16.16)
z, $\mathfrak{k}$, $\mathfrak{C}_{\boldsymbol{\kappa}}$, N	variable, half-type and canonical product of the models; $e^{i\mathfrak{k}\mathsf{z}}\mathfrak{C}_{\boldsymbol{\kappa}}$	(17.1)
$\mathscr{F}$, $\mathscr{F}^+$, $\mathfrak{l}$, $\mathfrak{w}_\mathfrak{l}$, $\mathfrak{b}_\mathfrak{l}$, $\mathfrak{c}_\mathfrak{l}$	frequencies of a discrete category; a frequency; masses, exponents, coefficients	(17.3)–(17.5)
$\mathfrak{p},\mathfrak{q},\mathfrak{v}$, U, X, $\mathfrak{D}$, $\mathfrak{S}$	data, unitary matrix, diagonal factor, determinant and sine function of the two-edge model	(17.6)–(17.8)
F, $\mathsf{F}_{\boldsymbol{\kappa}}$, F_λ, $\mathfrak{y}_k$, $\mathfrak{x}_k$	finite Euler product, its list functions and family; its ordinates and its clock	(17.11)–(17.14)
$\mathscr{L}$, $\mathfrak{P}$	ℓ^2 over the primes; $\sum_p p^{-s}$	(17.20), (17.21)
$\theta_{\mathrm{RVM}}(s)$, τ_0	bounded half-plane phase of the clock; fixed transition threshold $\tau_0=4\pi$ (not a clock ordinate)	(9.37), §9.1

Symbol	Meaning	Defined
$\mathfrak{m}_n$, $\mathfrak{u}_\lambda$, N_λ, $\mathfrak{e}^{(\lambda)}$, C_λ	cell baseline; first-cell term; moving count and count defect; Mellin/explicit-count constant	(14.22), (14.15), (14.2)
A_λ, $\mathsf{W}(\lambda)$, T_λ, $k_n(\lambda)$, V_σ	representing function; one-number defect; theta defect; quotient coefficients; vertical defect	(15.23), (15.6), (15.7)
Θ_λ, $\mathfrak{G}_\lambda$, $\mathcal{J}_T$, Δ_T, $K(T)$, L_T	auxiliary criteria and multiscale clock-Jacobian quantities	§15
E_n, Ξ_n, ε_r	cell-error and defect-kernel quantities; local regularization parameter where displayed	§15
$\boldsymbol{\kappa}$, $N_{\boldsymbol{\kappa}}$, τ_0^+	generic ordinate list, its count, and lower list threshold	(16.1)
$\mathcal{G}_\varepsilon$, $\mathsf{P}_{\varepsilon,M}$, $\mathsf{R}_{\varepsilon,M}$	additive interpolation and finite completed Dirichlet polynomials	(F.1), (F.6)
$\nu_5, \dots, \nu_{12}$	real constants, each local to the statement where it occurs	–
Route B for finite sections (Section 18)		
$\mathbb{K}(C,\mathbb{M})$, $\mathbb{M}_*$, a_n^ζ	box of C-admissible lists with masses $\le \mathbb{M}$; $\log 7/(2\sqrt{7})$; prescribed masses $\Lambda(n)/(2\sqrt{n})$	§18.3
$\mathfrak{R}(\boldsymbol{\kappa},\mathsf{a})$, c_φ, $\mathscr{T}_m$, s_m	residual distribution of the rung system; cosine transform of a test; nested test sets; section residual	(18.5), (18.6)
$\mathcal{L}$, $\mathscr{K}$, $\mathscr{S}_{\boldsymbol{\kappa}}$, $\mathscr{A}_\zeta$, $\mathscr{E}_{\boldsymbol{\kappa}}$, $\mathfrak{H}_{\boldsymbol{\kappa}}$, $\mathfrak{g}$, $\mathrm{c}_{\mathscr{K}}$	Laplace transform; Archimedean, spectral, arithmetic and total residual distributions; $\sum_k \kappa_k^{-2}\cos(\kappa_k u)$; auxiliary function and constant	(18.10)–(18.15)
$\mathbb{F}_m$, $\mathbb{J}_m$, $\mathbb{Q}_m$, $\mathbb{L}_m$, $\mathbb{H}_m$, h_m, l_m, k_m^*, t_m, $\mathbb{T}_m$	residual map, Jacobian, right inverse and its norm, Lipschitz constant, start residual and contraction index of stage m; numbers of unknowns and tests; contraction map	(18.21)–(A_m)
q_m	graded contraction factors	Prop. 18.29
P_{X}, ζ_{X}, Λ_{X}, p_{X}, m_{X}, n_{X}, Θ_{X}, $\boldsymbol{\kappa}^{(\mathrm{X})}$	Gonek's truncated Euler product and symmetrised function with parameter X; its prime phase and two constants; the Gonek phase; the Gonek list	(18.27)–(18.30)
$\widehat{\mathsf{R}}$, $\mathbb{E}(T)$, υ, $\mathfrak{B}(C)$, $\mathbb{B}, \mathbb{B}_0, \dots, \mathbb{B}_3$, $\mathrm{c}_1, \dots, \mathrm{c}_6$, c_ℓ	Laplace residual of a section; its L^1 size on $\Re s = 2$ and growth ratio; constants of the a priori bound	(18.40)–(18.52)
$\mathcal{Y}$, $\mathcal{D}_{\mathsf{a}}$, $\mathbb{W}$, $\mathbb{S}$, $\mathbb{J}$, $\mathfrak{f}$, $\mathfrak{F}_T$, $\mathfrak{R}_T$, $\mathbb{G}_T$, n_T, $\mathbb{D}(T)$	Dirichlet factor and polynomial; $\log\lvert\zeta_{\boldsymbol{\kappa}}\rvert$; half-strip; compact set; log-factor; Backlund function; rectangle, path, zero count and bound of the counting argument	§18.5
$\mathscr{X}$, x_k, $\Psi_{\mathscr{X}}$, $\mathbb{P}_{\mathrm{y}}$, $\mathbb{Y}_{\mathrm{y}}$, $\mathfrak{G}_{\mathscr{X}}$, $\mathcal{F}_{\mathscr{X}}$, $\mathcal{F}_0$, $\mathfrak{z} = \mathrm{x} + i\mathrm{y}$	unfolded list and its points; discrepancy; Poisson and conjugate operators; partial-fraction series; generating function; $\cos\pi\mathfrak{z}$; unfolded complex variable	(18.55)–(18.60)
A, B, $\mathrm{d}_\star$, $\mathbb{U}$, $\mathcal{W}$, w, d_k, $\mathfrak{Z}_m$, $\mathfrak{J}_m$, $\mathfrak{n}_m$, D_m, $\mathbb{O}$, $\mathbb{V}$, $\mathbb{U}_{\mathrm{loc}}$	mean and oscillating parts of the discrepancy and their bounds; sector function; weight; displacements, index blocks, intervals, lengths, oscillations and the constants of the variable-length theorem	§18.6
$\hbar$, $\hbar_T$, $\mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}$, $\mathbb{P}_t^\circ$, c_{JN}, $\mathfrak{I}_T$, c_F	line height and its t-scale; the two kernels, the John–Nirenberg constant, the interval and Fujii's constant in the obstruction theorem	§18.7.3
w_m, T_m, u_m^*, $\mathbb{J}'$, $\mathbb{A}^{(3)}$, $\mathsf{l}(\mathrm{r})$	test width, height and window of stage m; second- and third-derivative matrices; Banach index of the computed certificates	§18.8
$\mathfrak{W}_r(\mathsf{f})$, H_0	constant of a test in the bound for the zeta list; height of the verified zeros, $3 \cdot 10^{12}$	(18.18), Prop. 18.20
$\mathscr{Z}_{\boldsymbol{\kappa}}$, $\mathfrak{T}_C$, $\mathfrak{E}_W$, $\mathscr{P}_{\boldsymbol{\kappa}}$, $\mathscr{G}$	ordinate transform of an unknown list; box of lists; window residual, its gradient function and the function of its first-order condition	§18.4

Symbol	Meaning	Defined
N_8, $\varphi^{\mathrm{N}}_{c,h}$, h_m, ρ_m	cardinal B-spline of order 8, B-spline test, its width at stage m, radius of the verified ball	§18.8.5
Exponential sums over the ordinates (Section 5)		
$\rho_k = \frac{1}{2} + d_k + i\gamma_k$	the zero with positive ordinate of index k; d_k its distance to the critical line	Sec. 5
$\widehat{\mathsf{f}}$	$\int \mathsf{f}(u)e^{iuz}du$	(5.1)
V_+	Cramér's function $\sum_{\Im\rho>0} e^{\rho z}$	(5.4)
$\mathscr{Z}_\gamma$, $\mathscr{Z}_\tau$, $\mathscr{Z}_\xi$, $\mathfrak{X}$	ordinate, clock and zero transforms; their defect $\mathscr{Z}_\gamma - \mathscr{Z}_\xi$	(5.3), (5.6)
g; $\mathscr{V}_{\mathrm{b}}, \mathscr{V}_{\mathrm{R}}, \mathscr{V}_{\mathrm{L}}, \mathscr{V}_{\mathrm{p}}$; Ω_θ	archimedean part of ξ'/ξ; pieces of the zero-free representation; its half-plane	(5.10), (5.12)
$\mathscr{Y}$	off-line cosine distribution $\sum_k (\cosh(d_k u) - 1)\cos(\gamma_k u)$	(5.15)
$\mathscr{Q}_K$, $\mathscr{Q}_\infty$	Frullani integrals of the cosine difference	(5.20)
C^{path}_ζ, $C^{\mathrm{path}}_{\mathrm{RVM}}(\infty; u)$	triangular centroid of the ordinates; limit of the clock centroid	(5.22), (5.23)
b_k, d^{b}_k	$(e^{iu(\tau_{k+1}-\tau_k)} - 1)^{-1}$, $b_{k-1} - b_k$	Thm. 5.13
$f(\lambda)$, λ^*	$\log \varpi_\lambda$; crossing point	Prop. 15.13

The three counts. Three counting symbols occur and are never interchanged: $\mathcal{N}_{\mathrm{RVM}}(T) = \mathcal{N}_{\mathrm{RVM}}(T)$ is the *continuous* midpoint clock (2.3), a real-analytic increasing function; $n_{\mathrm{RVM}}(T) = \#\{k : \tau_k \le T\} = \lfloor \mathcal{N}_{\mathrm{RVM}}(T) + \frac{1}{2} \rfloor$ (4.12) is the *integer* count of the clock ordinates; and $N_{\boldsymbol{\kappa}}(T) = \#\{k : \kappa_k \le T\}$ is the integer count of a general list (with N_ζ for the ordinates of ζ and N_λ for the list $(\gamma_k(\lambda))$). Differences such as $N_\zeta - \mathcal{N}_{\mathrm{RVM}}$ compare an integer staircase with the continuous clock; differences such as $N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}}$ compare two staircases. When a mass $\Lambda(n)/(2\sqrt{n})$ is called a "von Mangoldt mass" it is this scaled mass that is meant. A distributional identity is stated with its space: $\mathcal{D}'(0,\infty)$ (the origin excluded), $\mathcal{S}'(\mathbb{R})$, or distributions on $\mathbb{R}$ supported in $[0,\infty)$ (the origin included); the constants b of Appendix F.4 and the polynomials of Lemma 18.14 live exactly in the difference between the first and the third.

Collision check. Global symbols in the table have one meaning throughout the paper. The interpolation parameter of Appendix F.4 is written $\mathsf{t} \in [0,1]$ (θ is the unit-cell step), and the constants of that appendix are $\mathbb{K}_{\mathrm{E}}, \mathbb{K}'_{\mathrm{E}}$ (not $\mathbb{K}(C,\mathbb{M})$, the box of Section 18), B_2 (not $\mathbb{B}_2$ or B_2 of that section) and C^*; the letters K, N, T, L and C with different fonts or indices are different symbols, listed separately in the table. Local proof variables do not collide with global notation: in particular c_0 is the de la Vallée Poussin constant, $\varkappa_j$ and $\mathcal{P}_j$ are local van-der-Corput parameters, ω_0 and ω_{test} are test weights, d_ρ is a horizontal zero displacement, T_{ver} is the verified-zero height, $\mathcal{H}^\pm$ are transition half-planes, and $\Psi^\pm_m$ are local stationary phases; R_0 is reserved for the clock half-plane threshold and R_{end} for the endpoint clock-decomposition threshold. The symbol c is reserved for a Perron abscissa, ε for an exponent/margin, ε for the additive-family parameter, θ for the unit-cell step, and ϑ for the Riemann–Siegel function. The fixed number $\tau_0 = 4\pi$ is a transition threshold and is not a member of the sequence (τ_k). Dotted variables $\dot{\kappa}_k, \dot{\mathsf{a}}_n$ denote a direction of variation.

1. Introduction

1.1. **Background.** The theory of the Riemann zeta function joins four structures that are usually introduced in different languages. On the spectral side there are the nontrivial zeros and their Riemann–von Mangoldt counting law; on the analytic side the completed function ξ, its functional equation and its Hadamard product; on the prime side the prime-power measure and the explicit

formula; and on the inverse side the reciprocal Dirichlet series $1/\zeta$ and the Mertens function

$$M(x) := \sum_{n \le x} \mu(n). \tag{1.1}$$

Riemann's memoir made the first three structures inseparable, and the work of von Mangoldt, Hadamard and Hamburger supplied, respectively, the rigorous zero count and explicit formula, the entire-product factorization, and a converse characterization of ζ from its Dirichlet-series and functional-equation data [56, 62, 41, 42, 61, 37]. This paper asks what remains of that architecture when the zero staircase is replaced, at the outset, by the exact midpoint inverse of its smooth Riemann–von Mangoldt clock, and how the resulting object is carried back to ζ.

1.2. **Known results on the Riemann–von Mangoldt law, the ordinates and the clock.** This subsection collects the known theorems that the paper uses or compares with. They are quoted, not proved here; the theorems of the later sections are new unless a citation is given. Let ϑ be the Riemann–Siegel theta function, $\vartheta(T) = \arg\Gamma(\frac{1}{4} + \frac{iT}{2}) - \frac{T}{2}\log\pi$, and $S(T) = \frac{1}{\pi}\arg\zeta(\frac{1}{2} + iT)$ as in (2.2).

Theorem 1.1 (Riemann–von Mangoldt; [62], [61, Thms. 9.3–9.4])**.** *For $T > 0$ not an ordinate, $N_\zeta(T) = 1 + \frac{1}{\pi}\vartheta(T) + S(T)$, with $1 + \frac{1}{\pi}\vartheta(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8} + O(T^{-1})$ and $S(T) = O(\log T)$.*

The smooth part is the midpoint clock $\mathcal{N}_{\mathrm{RVM}}$ of (2.3). Explicit forms of the bound for $S(T)$ were given by Trudgian [74]; Hasanalizade, Shen and Wong [72] proved $\left|N_\zeta(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e}\right| \le 0.1038\log T + 0.2573\log\log T + 9.3675$ for $T \ge e$, and Bellotti and Wong [73] improved the constants to 0.10076, 0.24460, 8.08344.

Theorem 1.2 (Littlewood; Selberg; [75], [61, Thm. 9.9], [58])**.** $\int_0^T S(t)\,dt = O(\log T)$*, and* $\int_0^T S(t)^2\,dt = \frac{T}{2\pi^2}\log\log T + O\big(T(\log\log T)^{1/2}\big)$.

Gram points. The Gram points g_n are defined by $\vartheta(g_n) = n\pi$; Gram's law is the statement $(-1)^n Z(g_n) > 0$ for Hardy's function Z. The name and the first failure are due to Hutchinson [76]; the law fails infinitely often (Titchmarsh) and, by Trudgian [77], a positive proportion of the time; Brent's computations [78] used Gram blocks and Rosser's rule. By Theorem 1.1, $\mathcal{N}_{\mathrm{RVM}}(g_{k-1}) = k + O(g_{k-1}^{-1})$, while $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$: the midpoint ordinates lie half-way between consecutive Gram points in clock units.

Computations. The Riemann hypothesis has been verified for all zeros up to height $3 \cdot 10^{12}$ (Platt and Trudgian [55]); Platt [79] isolated the zeros up to height about $3.06 \cdot 10^{10}$ in interval arithmetic.

Theorem 1.3 (Landau's formula; [67], uniform version [68])**.** *For fixed $x > 1$, $\sum_{0<\gamma\le T} x^\rho = -\frac{T}{2\pi}\Lambda(x) + O_x(\log T)$, where $\Lambda(x) = 0$ unless x is a prime power.*

Theorem 1.4 (Hlawka; [80], see also [81])**.** *For every real $\alpha \neq 0$ the sequence $(\alpha\gamma_k)_{k\ge1}$ is uniformly distributed modulo 1; in particular $\frac{1}{N}\sum_{k\le N} e^{iu\gamma_k} \to 0$ for every $u \neq 0$.*

Theorem 1.5 (Cramér; [66])**.** *The function $V_+(z) = \sum_{\Im\rho>0} e^{\rho z}$, holomorphic in $\Im z > 0$, has a meromorphic continuation to the plane cut along the non-positive imaginary axis; its poles are simple and lie at the points $\pm\log p^m$.*

The explicit formulas of Riemann and von Mangoldt, in the distributional form of Guinand and Weil [56, 62, 20, 63], are used throughout. For the Mertens function, $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ is equivalent to RH [61, §14.25]; the Mertens conjecture $|M(x)| < \sqrt{x}$ is false [52]; $M(x)$ has been computed to 10^{14} by Kotnik and van de Lune [84] and to 10^{16} by Hurst [83], who proved $\limsup M(x)/\sqrt{x} \ge 1.826054$ and $\liminf M(x)/\sqrt{x} \le -1.837625$; under RH and further hypotheses, Ng [85] determined the limiting distribution of $e^{-y/2}M(e^y)$.

The clock in the literature. The midpoint clock and its Lambert inversion are due to França and LeClair: equation (24) of [38], $\frac{\tilde y_n}{2\pi}\log\frac{\tilde y_n}{2\pi e} = n - \frac{11}{8}$, is the midpoint condition (2.4), and its solution

(25), $\tilde{y}_n = 2\pi(n - \frac{11}{8})/W_0((n - \frac{11}{8})/e)$, is (2.5); thus $\tau_k = \tilde{y}_k$. França and LeClair obtain $\tilde{y}_n$ as the leading-order solution of their transcendental equation for the nth zero, in which the term $\arg\zeta$ is dropped, use it as an approximation to γ_n, study the statistics of the zeros with it, and reconstruct $\pi(x)$ numerically from the explicit formula with these approximations in place of the zeros [38, 71]. In the literature consulted we have not found a zeta function built on the clock ordinates, nor the prime count, the prime levels and the Mertens function constructed from it below, nor the family joining it to ζ.

Beurling systems with a Riemann hypothesis. Zhang [86] constructed Beurling generalized primes whose zeta function has no zeros in $\Re s > \frac{1}{2}$ and whose integer counting function is $x + O(x^{1/2}e^{c(\log x)^{2/3}})$; Broucke, Debruyne and Vindas [26] proved the existence, by a probabilistic construction, of a Beurling system with $\pi_B(x) = \mathrm{Li}(x) + O(\sqrt{x})$ and large oscillation of the integers; see also [34, 88, 87]. These are genuine Beurling systems, with non-decreasing counting functions, obtained by existence arguments. The synthetic count π_{RVM} of this paper is deterministic and is built from the clock, with $\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O(\sqrt{x}(\log\log x)^2/\log x)$ without hypothesis (Theorem 6.5); it is not the counting function of a Beurling system, since its density $\frac{1}{\log y} + 2\cos(2\pi y) + \cdots$ (Theorem 15.18) changes sign. The comparison is therefore one of size of the error term, not of the structure.

1.3. **The midpoint RVM system.** The point of departure is the smooth counting function (2.3),

$$\mathcal{N}_{\mathrm{RVM}}(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}, \tag{1.2}$$

and the real ordinates τ_k determined by the midpoint condition $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$. The half-integer convention is the midpoint convention for a staircase and leads to the exact Lambert inversion (2.5),

$$\tau_k = \frac{2\pi(k - 11/8)}{W_0\big((k - 11/8)/e\big)}. \tag{1.3}$$

The clock, the midpoint convention (the constant $\frac{11}{8}$) and the Lambert inversion are those of França and LeClair [38, Sec. III.A, Eqs. (24)–(25)], who obtained (1.3) as the leading-order solution of their equation for the nth zero and used it as an approximation to γ_n (Subsection 1.2); for the function W see [29, 36]. What is new here is what is built on the clock: the midpoint inverse is treated as the exact spectrum of a deterministic system and is propagated, without the fluctuating term $S(T)$, through canonical products, explicit prime counting and Mertens inversion.

The first object is the canonical product and the synthetic zeta function

$$P^{\mathrm{crit}}_{\mathrm{RVM}}(s) = \prod_{k\ge1}\left(1 + \frac{s(s-1)}{\frac{1}{4} + \tau_k^2}\right), \qquad \zeta_{\mathrm{RVM}}(s) = \frac{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}{G(s)}, \qquad G(s) = s(s-1)\Gamma(s/2)\pi^{-s/2}. \tag{1.4}$$

Because $\tau_k \asymp k/\log k$, the factors form a normally convergent product in the symmetric variable $s(s-1)$; $P^{\mathrm{crit}}_{\mathrm{RVM}}$ is entire and invariant under $s \mapsto 1 - s$, and ζ_{RVM} has a residue-one pole at 1, the trivial zeros, and the simple nontrivial zeros $\rho_k^{\mathrm{RVM}} = \frac{1}{2} \pm i\tau_k$ (Theorem 3.1). The real part $\frac{1}{2}$ is part of the definition.

Generalized prime systems. In Beurling's theory the generalized primes are arbitrary reals $1 < p_1 \le p_2 \le \cdots \to \infty$; their multiplicative semigroup defines generalized integers and a zeta function by a Dirichlet series, an Euler product or a Mellin–Stieltjes transform, separating prime distribution from the additive structure of the integers [23, 13, 33]. Beurling systems with prescribed analytic behaviour, critical zeros and large oscillation show that zeta-like zero data alone do not determine the arithmetic realization [34, 26]. The RVM system runs in the opposite direction: it starts from a prescribed zero clock, constructs an entire completed product first, obtains a continuous prime-power count from its explicit formula, and only afterwards asks for an atomic realization on the integers.

Continuous and discrete prime systems. The passage from a continuous prime system to a discrete one is classical in the theory of Beurling generalized primes [23, 34, 27, 35]: the atoms are placed at suitably chosen real points, and the discretization changes the zeta function by a factor holomorphic and zero-free in $\Re s > \frac{1}{2}$ (compare Theorem 14.13(iii)). Measures with a continuous part and non-integer multiplicities also appear in the theory of complex dimensions [47, Ch. 3]. Grosswald and Schnitzer [39] replaced each prime p_n by a real q_n with $p_n \le q_n \le p_{n+1}$ and proved that the modified Euler product is ζ times a factor holomorphic and zero-free in $\Re s > 0$, so that it has the same zeros as ζ there; Corollary 7.5 and Theorem 14.13(iv) are of the same type for ζ_{RVM}. Here the masses of $d\pi_\lambda$ are placed at the integers, with real exponents $\tilde{c}_n(\lambda)$; this yields an ordinary Dirichlet series for every λ, absolutely convergent in $\Re s > 1$ (Corollary 14.12), and makes Hamburger's theorem [42, 59] available at the endpoint. Hamburger's theorem and its extensions [25, 43] need absolute convergence in $\Re s > 1$, and Knopp [45] showed that a half-plane of absolute convergence further to the right does not suffice. We are not aware of an earlier use of this combination.

Prime coordinate. From the RVM zeros we define the prime-power count J_{RVM} by (6.2), with the branch (6.1) of Ei. Finite Möbius inversion in the exponent variable, which relates Riemann's prime-power count to $\pi(x)$ [56, 61, 37], is used here as the *definition* of the synthetic prime count (6.7),

$$\pi_{\mathrm{RVM}}(x) = \sum_{1 \le m \le \log_2 x} \frac{\mu(m)}{m} J^*_{\mathrm{RVM}}(x^{1/m}), \qquad J^*_{\mathrm{RVM}}(x) = \sum_{1 \le m \le \log_2 x} \frac{1}{m} \pi_{\mathrm{RVM}}(x^{1/m}). \tag{1.5}$$

The function π_{RVM} is continuous away from the points 2^j with $\mu(j) \neq 0$, where it has the jump $\mu(j) J_{\mathrm{RVM}}(2)/j$; in particular it is not a staircase on the integers. It supports two notions of RVM prime: the continuous measure $d\pi_{\mathrm{RVM}}$, and, on every interval where π_{RVM} is continuous and strictly increasing, the real level crossings

$$p_j^{\mathrm{RVM}} := \pi^{-1}_{\mathrm{RVM}}(j), \qquad \pi_{\mathrm{RVM}}(p_j^{\mathrm{RVM}}) = j, \tag{1.6}$$

which need not be integers. In the computations of Section 19 the levels are the first crossings $\inf\{x : \pi_{\mathrm{RVM}}(x) \ge j\}$, which agree with (1.6) when the crossing lies on such an interval, evaluated on a grid. The synthetic count has square-root accuracy without any hypothesis on ζ: $J_{\mathrm{RVM}}(x) - \mathrm{li}(x)$ and $\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x)$ are $O(\sqrt{x}(\log\log x)^2/\log x)$ (Theorem 6.5). The proof reduces the zero sum to $\sum_k e^{iu\tau_k}/\tau_k$, which is $O(\log^2 u)$ by the second-derivative test on dyadic blocks of the clock (Lemma 6.4). Hence $\mathrm{li}(x)/\pi_{\mathrm{RVM}}(x) \to 1$, and with the prime number theorem $\pi(x)/\mathrm{li}(x) \to 1$ and $\pi(x)/\pi_{\mathrm{RVM}}(x) \to 1$ (Corollary 6.6, Figure 14). Sampling on unit cells gives the masses $c_n^{\mathrm{RVM}} = \pi_{\mathrm{RVM}}(n) - \pi_{\mathrm{RVM}}(n-1)$ and the weighted Euler object $\zeta_{\mathbb{Z},\mathrm{RVM}}$ of (7.2). The masses are not 0 or 1 and need not be positive, but they are bounded: J_{RVM} is Lipschitz on $[2,\infty)$ (Theorem 7.1), so $c_n^{\mathrm{RVM}} = O(1)$ and $\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$ (Corollaries 7.2 and 7.3). The unit-periodic first harmonic of the clock integrates to zero on every integer cell (Lemma 7.6); under the cell hypothesis one obtains $c_n^{\mathrm{RVM}} \sim 1/\log n$ (Proposition 7.8).

Mertens coordinate. The Mertens coordinate is the symmetric inverse Mellin object of $F_{\mathrm{RVM}} = 1/\zeta_{\mathrm{RVM}}$. A smooth/sawtooth Stieltjes decomposition of the clock product with Abel regularization of the sawtooth gives on every band to the right of $\frac{1}{2}$

$$\zeta_{\mathrm{RVM}}(s) = \varpi_{\mathrm{RVM}}(1 + w(s))(1 + O(1/|t|)), \qquad w(s) = \exp\{-2\pi i \mathcal{N}_{\mathrm{RVM}}(-i(s - \tfrac{1}{2}))\}, \tag{1.7}$$

and $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+\chi)(1+O(1/|t|))$ on closed substrips of $\frac{1}{2} < \Re s < 1$ (Theorem 9.1). Hence $F_{\mathrm{RVM}}(\sigma + it) \to L = 1/\varpi_{\mathrm{RVM}}$ for every fixed $\sigma > \frac{1}{2}$; neither ζ_{RVM} nor F_{RVM} is an absolutely convergent Dirichlet series in any half-plane (Theorem 9.13); and after subtracting the vertical limit the Perron inverse converges absolutely, with

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon}) \tag{1.8}$$

(Theorem 9.14). The RVM Mertens function therefore exists as a continuous spectral inverse although no coefficient sequence for F_{RVM} exists. Since the primitive of the midpoint sawtooth is bounded, the same decomposition without contours gives $C^{-1} \le |\zeta_{\mathrm{RVM}}| \le C$ on every half-plane $\Re s \ge \sigma_0 > 1$ and $\zeta_{\mathrm{RVM}}(s) \to \varpi_{\mathrm{RVM}}$ as $\Re s \to +\infty$ (Theorem 9.10).

Ternary discretization. An integer-valued representative of $M^{\mathrm{crit}}_{\mathrm{RVM}}$ is obtained by sampling the target $\mathcal{M}(n) := M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ and choosing signs on the squarefree support:

$$\mu_h(n) = \mu(n)^2 \varsigma_n, \qquad \varsigma_n \in \{-1, +1\}, \qquad M_h(n) := \sum_{m \le n} \mu_h(m), \tag{1.9}$$

$$\varsigma_n = \begin{cases} +1, & M_h(n-1) < \mathcal{M}(n), \\ -1, & M_h(n-1) \ge \mathcal{M}(n), \end{cases} \qquad (\mu(n)^2 = 1). \tag{1.10}$$

This is a greedy sigma–delta feedback quantization; feedback quantization and noise shaping are standard themes of sigma–delta theory [30, 40, 14]. Here the target is a zero-generated Mertens inverse and the admissible times are the squarefree integers. The walk μ_h is a comparison object: it is not claimed to be multiplicative or equal to μ. The two discretizations are distinct: the cell masses c_n^{RVM} come from π_{RVM}, the signs ς_n track $M^{\mathrm{crit}}_{\mathrm{RVM}}$.

Comparison with ζ. At the level of transforms the defect is the exact quotient $\mathcal{Q}_{\mathrm{RVM}} = \zeta/\zeta_{\mathrm{RVM}}$ of (10.6), with $F_{\mathrm{RVM}} = \mathcal{Q}_{\mathrm{RVM}} \cdot (1/\zeta)$. Since $|\zeta_{\mathrm{RVM}}|$ is bounded above and below on $\frac{1}{2} + \varepsilon \le \sigma \le c'$, $|t| \ge t_0$ (Corollary 10.3), on each bounded band $\frac{1}{2} + \varepsilon \le \sigma \le c'$, every bound for ζ or $1/\zeta$ is equivalent to the corresponding bound for $\mathcal{Q}_{\mathrm{RVM}}$ by Corollary 10.3; for $\sigma > c'$ one uses Theorem 9.10. For M itself the bound $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ is equivalent to the Riemann hypothesis [61, §14.25], and the disproof of the Mertens conjecture shows that the ε cannot simply be dropped [52].

1.4. **The core of the method.** The transfer from ζ_{RVM} to ζ is a transformation of ζ_{RVM} depending on a parameter $\lambda \in [0, 1]$. The ordinates move along

$$\gamma_k(\lambda) = (1 - \lambda)\gamma_k + \lambda\tau_k, \qquad \zeta_\lambda(s) = \frac{1}{G(s)} \prod_{k \ge 1} \left(1 + \frac{s(s-1)}{\frac{1}{4} + \gamma_k(\lambda)^2}\right), \tag{1.11}$$

where γ_k are the ordinates of the nontrivial zeros of ζ; thus $\zeta_1 = \zeta_{\mathrm{RVM}}$ and $\zeta_0 = \zeta_{Z1}$ (Section 11).

(i) Every member ζ_λ has its nontrivial zeros exactly at $\frac{1}{2} \pm i\gamma_k(\lambda)$. The real part $\frac{1}{2}$ is part of the definition of the family, like the gamma factor, and never changes along it: it is impossible for ζ_{Z1}, or for any ζ_λ, to have a zero off the critical line other than the trivial zeros. For $0 < \lambda \le 1$ all zeros are simple.

(ii) All differences between the moving zetas lie in the numbers $\gamma_k - \tau_k$, which are bounded (Lemma 2.2); the quotient of two members is an explicit product in γ_k and $\lambda(\tau_k - \gamma_k)$ (Remark 11.2).

(iii) The final step is the identity $\zeta_{Z1} = \zeta$. It is the single and complete test: while $\zeta_{Z1} = \zeta$ there is nothing outside $\Re s = \frac{1}{2}$ to speak of, since the identity itself places every nontrivial zero of ζ at $\frac{1}{2} \pm i\gamma_k$, and $\zeta_{Z1} \ne \zeta$ is the only possible obstruction.

The endpoint has the functional equation, the pole, the residue and the order of ζ (Proposition 12.1). Hamburger's theorem belongs to the family of converse theorems that recover Dirichlet series from analytic continuation, functional equation and growth [42, 54]; applied to ζ_{Z1} it gives (Theorem 12.2)

$$\zeta_{Z1} = \zeta \quad \Longleftrightarrow \quad \text{(D): } \zeta_{Z1} \text{ is an absolutely convergent Dirichlet series in } \Re s > 1. \tag{1.12}$$

With the integer-weighted zeta $\zeta_{\mathbb{Z},\mathrm{RVM}}$, which is a Dirichlet series, (D) is equivalent to an absolutely convergent Dirichlet expansion of the explicit multiplier $\mathcal{K} = \zeta_{Z1}/\zeta_{\mathbb{Z},\mathrm{RVM}} = \Pi_{\mathrm{tr}} e^{-h_{\mathrm{RVM}}}$ (Theorem 12.3); since $1/\zeta_{\mathbb{Z},\mathrm{RVM}}$ is also an absolutely convergent Dirichlet series in $\Re s > 1$, the

equivalence holds on the whole half-plane. Section 13 gives two sharper forms of (D). First, the clock decomposition applied to the ordinates γ_k gives the value of the endpoint at infinity,

$$\varpi_{Z1} = \lim_{\sigma\to+\infty} \zeta_{Z1}(\sigma) = \varpi_{\mathrm{RVM}} \lim_{K\to\infty} \prod_{k\le K} \frac{\frac14+\tau_k^2}{\frac14+\gamma_k^2} \ \ge 1, \tag{1.13}$$

and $\zeta_{Z1} = \zeta$ holds if and only if $\varpi_{Z1} = 1$ (Theorems 13.1 and 13.2, Corollary 13.3): the final step is the single ordered-limit identity $\lim_{K\to\infty}\prod_{k\le K}(\frac14+\gamma_k^2)/(\frac14+\tau_k^2) = \varpi_{\mathrm{RVM}}$ between the displacements and the transition constant. Second, the prime-power counts $J^{\mathrm{M}}_{\mathrm{RVM}}$ and J^{M}_{Z1} are the Mellin transforms of $\log\zeta_{\mathrm{RVM}}$ and $\log\zeta_{Z1}$ and satisfy a distributional explicit formula (Theorems 13.5 and 13.6); this proves the Mellin–Stieltjes identification of the integer weights (Theorem 7.4), and (D) holds if and only if J^{M}_{Z1} is constant between consecutive integers and its jumps satisfy the growth hypothesis of Theorem 13.7. Third, along the family $\frac{d}{d\lambda}\log\varpi_\lambda = -\Upsilon(\lambda)$, the generator at infinity, and (D) is the identity $\int_0^1 \Upsilon(\lambda)\,d\lambda = \log L$ (Proposition 14.2). Section 14 atomizes every member: $\tilde\zeta_{\mathbb{Z},\lambda} = \zeta_\lambda e^{\tilde h_\lambda}$ keeps the Euler product and the Dirichlet series while ζ_λ keeps the functional equation; the joining factor $\tilde h_\lambda$ is written cell by cell and is holomorphic in $\Re s > \frac12$ (Theorems 14.7 and 14.13); the cell masses have mean square $X^{1+\varepsilon}$ on $(X,2X]$ uniformly in λ, so every $\tilde\zeta_{\mathbb{Z},\lambda}$, the endpoint included, is an absolutely convergent Dirichlet series in $\Re s > 1$ (Theorem 14.11, Corollary 14.12); and (D) is equivalent to the vanishing of every cell defect at $\lambda = 0$ (Theorem 14.14). At the clock endpoint the joining factor $\tilde h_1$ is not constant, and the endpoint Mertens function has a Mellin form B_{Z1} with $B_{Z1} = M$ exactly when (D) holds (Propositions 14.16 and 14.17). The remaining steps form the programme of Subsection 14.8. In RVM language, the final step asks whether the zero-generated analytic object ζ_{Z1} belongs to the discrete Dirichlet category; if it does, Hamburger uniqueness makes it ζ, and the prime and Mertens coordinates become those of the integers. Step 3 is Theorem 12.8:

$$\zeta_{Z1} = \zeta \quad\Longrightarrow\quad M(x) = O_\varepsilon(x^{1/2+\varepsilon}) \qquad (\varepsilon > 0), \tag{1.14}$$

with a self-contained proof by the Borel–Carathéodory lemma, Hadamard's three-circle theorem and the truncated Perron formula. Theorem 12.9 rewrites the same endpoint consequence through the synthetic factorization. Its estimate for M is a corollary of Theorem 12.8; the synthetic Mertens bound is not an independent input to the classical implication. The identity $\zeta_{Z1} = \zeta$ itself, equivalently (D), is not proved in this paper; it is the open step of the method (Remark 12.12).

The relations among the principal objects are

$$\{\tau_k\} \longrightarrow \zeta_{\mathrm{RVM}} \longrightarrow J_{\mathrm{RVM}} \longleftrightarrow \pi_{\mathrm{RVM}} \longrightarrow (d\pi_{\mathrm{RVM}},\, p_j^{\mathrm{RVM}},\, c_n^{\mathrm{RVM}}), \tag{1.15}$$

$$F_{\mathrm{RVM}} = \zeta_{\mathrm{RVM}}^{-1} \longrightarrow M^{\mathrm{crit}}_{\mathrm{RVM}} \longrightarrow (\mathcal{M}, \mu_h, M_h), \tag{1.16}$$

$$\zeta_{\mathrm{RVM}} = \zeta_1 \xrightarrow{\lambda\downarrow 0} \zeta_0 = \zeta_{Z1} \overset{?}{=} \zeta \implies M(x) = O_\varepsilon(x^{1/2+\varepsilon}). \tag{1.17}$$

The first chain is the prime coordinate, the second the Mertens coordinate, the third the λ-family; they share the midpoint spectrum and solve different realization problems.

1.5. **Exponential sums over the ordinates and the Riemann hypothesis.** The exponential sums of the two spectra behave differently. For the clock, $\sum_{k\le N} e^{iu\tau_k} \ll 1+\log N$ for every $u > 0$ (Theorem 4.1), and the triangular centroid

$$C^{\mathrm{path}}(N;u) = \frac{1}{N}\sum_{n\le N}\sum_{k\le n} e^{iuo_k} = \sum_{k\le N}\Big(1-\frac{k-1}{N}\Big)e^{iuo_k} \tag{1.18}$$

of the list $o_k = \tau_k$ converges, without hypothesis, to an explicit series (Theorem 5.13); at $u = 1$ the limit is $0.041569 + 1.757541\,i$. For the ordinates $o_k = \gamma_k$ the sums resonate at the frequencies $\log p^m$ (Landau), and the natural comparison is with Cramér's function $V_+(z) = \sum_{\Im\rho>0} e^{\rho z}$, which sees the complete zeros. Section 5 proves the following statements, all but the last without any hypothesis.

(a) *Zero-free representation.* V_+ is given on the half-plane Ω_θ, which contains $(0,\infty)$, by an explicit formula in terms of Λ, Γ and ζ on a real segment, with no sum over the zeros (Proposition 5.3).

(b) *One imaginary frequency.* For any $y_0 \in (0, 4\pi]$,

$$\mathrm{RH} \iff \sum_{k\ge1} e^{-y_0\gamma_k} = e^{-iy_0/2}V_+(iy_0) \tag{1.19}$$

(Proposition 5.2); the defect is $\sum_k (1-\cos(y_0 d_k))e^{-y_0\gamma_k} \ge 0$, $d_k = \Re\rho_k - \frac12$.

(c) *The window form.* With $\widehat{\mathsf{f}}(z) = \int \mathsf{f}(u)e^{iuz}du$ and $J = (\log 2, \log 3)$, where the prime side of the explicit formula has no atom,

$$\mathrm{RH} \iff \sum_{k\ge1} \widehat{\mathsf{f}}(\gamma_k) = \int_J \mathsf{f}(u)\, e^{-u/2}V_+(u)\, du \quad \text{for every } \mathsf{f} \in C_c^\infty(J), \tag{1.20}$$

equivalently the triangular centroids $C_\zeta^{\mathrm{path}}(N;u)$ converge to $e^{-u/2}V_+(u)$ in $\mathscr{D}'(J)$ (Theorem 5.5). Without hypothesis the difference of the two sides is $\sum_k (\widehat{\mathsf{f}}(\gamma_k) - \frac12\widehat{\mathsf{f}}(\gamma_k - id_k) - \frac12\widehat{\mathsf{f}}(\gamma_k + id_k))$, an absolutely convergent sum over the zeros off the line. After inserting the representation of (a), both sides are expressed without the abscissas of the zeros. The proof uses a one-variable edge-of-the-wedge lemma and positivity at an imaginary frequency.

(d) *The endpoint defect is a sum over the zeros off the line.* For every test function, $\mathsf{R}_\mathsf{f}(0) - \frac12\sum_n \Lambda(n)n^{-1/2}\mathsf{f}(\log n) = \sum_k \int \mathsf{f}(u)(\cosh(d_k u) - 1)\cos(\gamma_k u)\, du$; equivalently $(\log x)(dJ_{Z1}^{\mathrm{M}} - dJ)/dx = \frac{2}{\sqrt{x}}\sum_k (\cosh(d_k \log x) - 1)\cos(\gamma_k \log x)$; and the cosine difference $\sum_k (\mathsf{c}_\mathsf{f}(\gamma_k) - \mathsf{c}_\mathsf{f}(\tau_k))$ equals the defect of the clock minus that sum (Theorem 5.7). If only finitely many zeros were off the line, J_{Z1}^{M} would be constant on no cell $(n, n+1)$ (Proposition 5.10).

(e) *One window is one cell.* The real part of (1.20) is the constancy of J_{Z1}^{M} on the single cell $(2,3)$, and (D), equivalently RH, is equivalent to that cell together with the sine identity $\sum_k \int \mathsf{f}(u)\sin(\gamma_k u)du = \int \mathsf{f}(u)e^{-u/2}\Im V_+(u)du$ on J (Corollary 5.8). The whole ladder of cells is thereby compressed into one spectral window.

(f) *Frullani form.* $2\int_0^\infty \frac{e^{-u/2}}{u}\sum_{k\le K}(\cos\gamma_k u - \cos\tau_k u)\, du \to \log L + \log \varpi_{Z1} \ge \log L$, with equality exactly under RH (Theorem 5.11).

(g) *The centroid of $e^{i\gamma_k}$ under RH.* If RH holds, $C_\zeta^{\mathrm{path}}(N;u) \to e^{-u/2}V_+(u)$ for every $u > 0$ that is not the logarithm of a prime power; at $u = 1$ the limit is

$$e^{-1/2}V_+(1) = \frac{1}{2\sqrt{e}}\Big(e - \frac{1}{e^2-1}\Big) + i\,0.76785566886\ldots = 0.77689426079\ldots + 0.76785566886\ldots\, i \tag{1.21}$$

(Theorem 5.15). The proof uses Landau's formula, Selberg's mean value of S, the zero-free representation and a local Tauberian lemma; the converse implication is not proved.

The numerical side of these statements (walks of $e^{i\gamma_k}$, $e^{i\tau_k}$ and $e^{i(\gamma_k+\tau_k)/2}$ in three dimensions, the centroids with 2 001 052 zeros, and $V_+(1)$ from the zero-free formula) is in Subsection 19.5. Subsection 19.8 and Appendix A give the first 1000 values of τ_k against γ_k, of the RVM prime levels p_j^{RVM} against p_j, of $\pi_{\mathrm{RVM}}(n)$ against $\pi(n)$ and of $M_{\mathrm{RVM}}^{\mathrm{crit}}(n)$ against $M(n)$, with graphs.

1.6. **Originality, scope and status.** *What this paper does not prove.* Before the list of results: this paper does not prove the Riemann hypothesis; it does not prove (D), the identity $\zeta_{Z1} = \zeta$, or any of the statements shown below to be equivalent to it; it does not prove the existence of a solution of the rung system (16.6) in $\mathscr{O}$, nor of an interior exact-atomic path, nor the convergence of the finite sections of Section 18 to a distributional limit; and, apart from the interval evaluation of a few explicit elementary expressions in the proof of Theorem 9.11 (which the margins allow to be checked by hand with five-decimal values of $\log 2$, $\log\pi$ and $\log\tau_1$), it contains no computer-assisted proof, the other computations being floating-point experiments. The only points left open by the method

are these: the endpoint identity (D) in any of its equivalent forms; the existence of a solution of the infinite rung system; the injectivity of the clock-only linearization (Theorem 16.7 proves injectivity at a solution, not at the clock); and the passage to the limit from finite sections. Every "equivalent form" below is a reduction to (D), not a resolution of it.

The ingredients are standard or belong to neighbouring theories: the Riemann–von Mangoldt law, Lambert inversion, Hadamard products, explicit formulas, Möbius and Perron inversion, Beurling generalized primes, sigma–delta feedback, and Hamburger's converse theorem. The contribution is their assembly into one midpoint RVM space with four coordinated levels: (i) an exact midpoint zero clock and its canonical zeta-like product; (ii) a continuous prime-power and prime coordinate, with real prime levels and non-integral cell masses; (iii) a continuous synthetic Mertens inverse with its ternary feedback discretization; (iv) the λ-family from the midpoint spectrum to the ordinates of ζ, ending in the single identity $\zeta_{Z1} = \zeta$ and the Mertens bound that follows from it. To the author's knowledge the cited literature contains the individual mechanisms but not this chain. Standard material is cited at the point of use; statements without citation are proved here.

Statement	Status
Midpoint clock and Lambert inversion (due to França–LeClair [38]); bounded displacement (from $S(T) = O(\log T)$)	theorem
Canonical product $P^{\mathrm{crit}}_{\mathrm{RVM}}$ and ζ_{RVM}	theorem
$S_{\mathrm{RVM}}(N;u) \ll 1 + \log N$; centroid and M_H^{RVM} decay	theorem
Clock path centroid converges for every $u > 0$, with the limit as an explicit series	theorem
Zero-free representation of Cramér's function V_+ on Ω_θ	theorem (continuation due to Cramér)
RH $\Leftrightarrow \sum_k e^{-y_0\gamma_k} = e^{-iy_0/2}V_+(iy_0)$ for one $y_0 \in (0, 4\pi]$	theorem
RH $\Leftrightarrow$ window identity on $(\log 2, \log 3)$; exact defect as a sum over zeros off the line	theorem
$\mathsf{R}_{\mathsf{f}}(0) = \langle \mathscr{Y}, \mathsf{f}\rangle$; $(\log x)(dJ^{\mathrm{M}}_{Z1} - dJ)/dx = 2x^{-1/2}\mathscr{Y}(\log x)$; cosine difference	theorem, unconditional
(D) $\Leftrightarrow$ the cell $(2,3)$ together with the sine identity on the window	theorem
Finitely many zeros off the line $\Rightarrow$ every cell fails	theorem
Frullani form: RH $\Leftrightarrow \mathscr{Q}_\infty = \log L$	theorem
Local Tauberian lemma	theorem (self-contained case of known results)
Triangular centroid of $e^{iu\gamma_k}$ tends to $e^{-u/2}V_+(u)$	theorem under RH; converse open
Compact-uniform existence of J_{RVM}; finite inversion	theorem
Real prime levels p_j^{RVM}	defined on monotone branches of π_{RVM}
Square-root estimate for $J_{\mathrm{RVM}}, \pi_{\mathrm{RVM}}$	theorem
$\mathrm{li}\,/\pi_{\mathrm{RVM}} \to 1$; $\pi/\pi_{\mathrm{RVM}} \to 1$	theorem (with the prime number theorem)
J_{RVM} Lipschitz, $c_n^{\mathrm{RVM}} = O(1)$	theorem
$\zeta_{\mathbb{Z},\mathrm{RVM}}$ absolutely convergent Dirichlet series in $\Re s > 1$	theorem
Mellin–Stieltjes identification; $\zeta_{\mathbb{Z},\mathrm{RVM}} = \zeta_{\mathrm{RVM}} e^{h_{\mathrm{RVM}}}$	theorem
$\lvert\zeta\rvert \asymp \lvert\mathcal{Q}_{\mathrm{RVM}}\rvert$ for $\sigma \geq \frac{1}{2} + \varepsilon$, $\lvert t\rvert \geq t_0$	theorem
Transition theorem $\varpi_{\mathrm{RVM}}(1+w)$, $\varpi_{\mathrm{RVM}}(1+\chi)$	theorem
$\varpi_{\mathrm{RVM}} < 0.9796$; closed form $\nu_3 = 8 - \frac{33}{4}\log 2 - 2\log\pi$	theorem (explicit inequalities checked in interval arithmetic, program `s14`)
$C^{-1} \leq \lvert\zeta_{\mathrm{RVM}}\rvert \leq C$ on $\Re s \geq \sigma_0 > 1$; $\zeta_{\mathrm{RVM}} \to \varpi_{\mathrm{RVM}}$ at $+\infty$	theorem
Unsmoothed $M^{\mathrm{crit}}_{\mathrm{RVM}} = L + O_\varepsilon(x^{1/2+\varepsilon})$	theorem
Dirichlet expansion of $\zeta^{\pm 1}_{\mathrm{RVM}}$	impossible (theorem)
Ternary walk μ_h, M_h	definition; comparison object

Statement	Status
Exact Hadamard product of 2ξ	theorem
λ-family, generator, multiplier, simplicity for $\lambda > 0$	theorem
Analytic class of ζ_{Z1}	theorem
$\zeta_{Z1} = \zeta \Leftrightarrow$ (D) $\Leftrightarrow$ RH	theorem
(D) $\Leftrightarrow$ Dirichlet expansion of $\mathcal{K} = \zeta_{Z1}/\zeta_{\mathbb{Z},\mathrm{RVM}}$ in $\Re s > 1$	theorem
$\varpi_{Z1} = \varpi_{\mathrm{RVM}} \lim_{K\to\infty} \prod_{k\le K} \frac{1/4+\tau_k^2}{1/4+\gamma_k^2}$ (ordered product); (D) $\Leftrightarrow \varpi_{Z1} = 1$	theorem
Mellin prime-power counts; distributional explicit formula	theorem
(D) $\Leftrightarrow J_{Z1}^{\mathrm{M}}$ constant between integers with the jump-growth bound of Theorem 13.7	theorem
$\frac{d}{d\lambda} \log \varpi_\lambda = -\Upsilon$; (D) $\Leftrightarrow \int_0^1 \Upsilon = \log L$	theorem
Atomized family; joining factor holomorphic in $\Re s > \frac{1}{2}$	theorem
Atomized family: Dirichlet series in $\Re s > 1$ for every λ	theorem
(D) $\Leftrightarrow$ vanishing cell defects at $\lambda = 0$	theorem
ζ_λ not a Dirichlet series in $\Re s > 1$ for $0 < \lambda \le 1$	theorem
$\lambda \mapsto \log \zeta_\lambda(\sigma)$ strictly convex for real $\sigma > 1$	theorem
Hamburger's theorem for the pair $(\zeta_\lambda, \tilde{\zeta}_{\mathbb{Z},\lambda})$ with symmetric joining factor	theorem
(D) $\Leftrightarrow \zeta_{Z1}$ has a positive representing measure; then $\varpi_{Z1} \le 1$	theorem
Density of $d\pi_{\mathrm{RVM}}$; cell masses and defects at $\lambda = 1$	theorem
Height kernel of the cell masses; Gaussian defect of the atomization	theorem
$\zeta_{Z1}(\sigma) \ge \zeta(\sigma)$ for real $\sigma > 1$; (D) $\Leftrightarrow$ equality at one σ	theorem
Hamburger's theorem for the pair in theta form and in vertical mean-square form	theorem
Symmetry of the joining factor fails at $\lambda = 1$; $\tilde{\zeta}_{\mathbb{Z},\lambda} = \zeta$ for finitely many λ at most	theorem
Summation formula for every ζ_λ; $\varpi_\lambda - 1$ as one number $\mathsf{W}(\lambda)$	theorem
(D) $\Leftrightarrow \varpi_\lambda < 1$ on the open path $0 < \lambda \le 1$	theorem (uses $\varpi_{\mathrm{RVM}} < 1$, Theorem 9.11)
Crossing point: unique, $f(0)/\Upsilon(0) \le \lambda^* \le f(0)/(f(0)+\lvert f(1)\rvert)$, $\lambda^* < 2 \cdot 10^{-11}$ if RH fails	theorem (the last bound uses the verified zeros)
A family with all the listed properties and $\varpi_0 > 1$	theorem
Space $\mathscr{O}$ of ordinate lists: rigidity of the coincidence; (D) $\Leftrightarrow$ existence of a coincidence list	theorem
(D) $\Leftrightarrow \tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$ for every n; every point of E is a witness of (D)	theorem
Symmetric joining factors: order one is needed; the symmetry criterion has no slack at $\lambda = 0$	theorem
Ladder: $\tilde{h}_\lambda(\sigma) = O(\sigma N^{-\sigma}) \Leftrightarrow$ no mass of dJ_λ^{M} outside the integers below N	theorem
Bridge theorem in phase form; (D) $\Leftrightarrow$ (B1), (B2) for $\tilde{h}_0$; impossible for $\lambda > 0$	theorem
Phase of ζ_λ on the critical line; cell masses as sums of a one-ordinate function	theorem
Transport $\mathsf{S}_{\mathsf{f}}(\lambda)$: entire of exponential type; moment series on the clock; defect of the clock by Poisson summation	theorem

Statement	Status
Multiscale clock Jacobian: lower frame bound on $U \asymp \log T$, no fixed-window bound, dense range, smooth tails	theorem; quantitative nonlinear continuation open
(D) $\Leftrightarrow$ solvability of the rung system with free masses; the solution is nondegenerate	theorem
Gradient of V_σ on $\mathscr{O}$	theorem
Acceptance criterion (window and frequencies); two-edge model; the finite Euler product over $2, 3$ with the final step proved	theorem
No object of finite rank in the window of ζ; the three abscissas of the diagonal operator over the primes	theorem
Phase defect along the path (rotation of the one-ordinate terms)	heuristic
Endpoint-only interpolation audit; clock-only injectivity problem; a priori rung displacement bound	Appendix F: theorem/proposition/open problem as labelled
Route B: compactness in a box; uniqueness in every tube with free masses (an exact solution is the zeta list with the von Mangoldt masses, and RH holds); B $\Leftrightarrow$ (D)	theorem (Section 18)
Finite stages do not decide route B: the zeta list solves every finite set of tests up to $(3 \cdot 10^{12})^{1-r}$	theorem (uses the zeros verified up to $3{\cdot}10^{12}$)
Route B on one window: RH $\Leftrightarrow$ an unknown list reproduces Cramér's function on $(\log 2, \log 3)$ $\Leftrightarrow$ $e^{-iy/2}V_+(iy)$ completely monotone; minimizers on windows: exact or degenerate; RH $\Leftrightarrow$ solvability on every window	theorem; the exclusion of degenerate minimizers is open
Laplace transform of the residual with prescribed masses; uniqueness; RH $\Leftrightarrow$ B$'(C)$	theorem; classical in substance
Certificate theorem for staged sections; graded extension; Gonek lists in a tube	theorem; the certificate hypotheses (A_m), (B) and (G1), (G2) are open and imply RH
Counting bound $\lvert N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}}\rvert \le (\mathrm{c}_1 + \mathrm{c}_2\upsilon)\log(T+8) + \mathrm{c}_3\mathbb{B} + \mathrm{c}_4$ for quasi-solutions	theorem; void for finite sections with masses, since then $\upsilon = \infty$ (Lemma 18.43)
Block Kadec–Avdonin criterion; variable-length Avdonin theorem	theorem; the block criterion is close to the sufficiency half of the Hruščev–Nikol'skii–Pavlov theorem; the variable-length version was not found in the literature consulted
No (A_2) for the unfolded zeta ordinates on any horizontal line	theorem, unconditional
Candidate Banach certificates of exact square sections up to $T = 487$; basin $\sim T^{-2}$	computation (floating point, Gaussian tests; not verified)
Exact solutions of fifteen square sections up to $T = 487$ with B-spline tests	theorem, computer-assisted (ball arithmetic)
First 1000 values of τ_k, p_j^{RVM}, $\pi_{\mathrm{RVM}}(n)$, $M_{\mathrm{RVM}}^{\mathrm{crit}}(n)$; walks and centroids	computation
$\zeta_{Z1} = \zeta \Rightarrow M(x) = O_\varepsilon(x^{1/2+\varepsilon})$	theorem (Step 3)
$\zeta_{Z1} = \zeta$	open

What is proved and what is not. Four groups of statements are proved in full. *(1) The synthetic system.* The clock and its Lambert inversion, the canonical product and ζ_{RVM}, the logarithmic bound for the exponential sums, the prime coordinate J_{RVM}, π_{RVM} with its square-root estimate and its density (Theorem 15.18), the transition theorem, the impossibility of a Dirichlet expansion

of $\zeta_{\mathrm{RVM}}^{\pm 1}$ and the synthetic Mertens bound. *(2) The family and its endpoint.* Every ζ_λ has the functional equation, order one and all its nontrivial zeros on $\Re s = \frac{1}{2}$; $\zeta_{Z1} = \zeta$ is equivalent to (D), to $\varpi_{Z1} = 1$ and to the constancy of J_{Z1}^{M} between consecutive integers together with the jump-growth bound of Theorem 13.7; $\varpi_{Z1} \geq 1$ and $\zeta_{Z1}(\sigma) \geq \zeta(\sigma)$ for real $\sigma > 1$; $\log \zeta_\lambda(\sigma)$ is strictly convex in λ; and $\zeta_{Z1} = \zeta$ implies $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ (Step 3). *(3) The atomized family and the criteria.* Every $\tilde{\zeta}_{\mathbb{Z},\lambda}$ is an absolutely convergent Dirichlet series in $\Re s > 1$, zero-free in $\Re s > \frac{1}{2}$, joined to ζ_λ by an explicit factor; no ζ_λ with $\lambda > 0$ is a Dirichlet series; Hamburger's theorem holds for the pair in three forms (Theorems 15.3, 15.4); the positivity criterion (Theorem 15.7); and the rigidity of the coincidence in the space of ordinate lists (Theorem 16.2). *(4) Criteria written where the joining factor is defined, and models.* The reduction of (D) to the cell masses (Corollary 15.29); the ladder and the half-line form of Hamburger's theorem for a pair (Theorems 15.36, 15.37); the bridge theorem in phase form (Theorem 15.39, Corollary 15.40); the transport from the clock and the exact defect of the clock (Theorem 15.45, Proposition 15.46); the existence form with free masses and the nondegeneracy of the solution (Theorems 16.6, 16.7); the acceptance criterion and the models in which the final step is a theorem (Theorems 17.1, 17.3 for the primes $2, 3$, Proposition 17.2).

Not proved: the identity $\zeta_{Z1} = \zeta$. Each of the following statements is proved to be equivalent to it, and none of them is proved: (D); $\varpi_{Z1} \leq 1$; $\zeta_{Z1}(\sigma) \leq \zeta(\sigma)$ for one $\sigma > 1$; $\varpi_\lambda < 1$ for all $\lambda \in (0, 1]$ (this equivalence uses Theorem 9.11); J_{Z1}^{M} non-decreasing; $\mathsf{W}(0) \geq 0$ (15.23); $\tilde{h}_0$ constant; the theta relation $T_0 \equiv 0$ (15.6); $\mathsf{V}_\sigma(0) = 0$ (15.7); the existence of a list in $\mathscr{O}$ whose continuous and atomized zeta functions coincide (Corollary 16.4); $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$ for every $n \geq 2$ (Corollary 15.29); $\tilde{h}_0(\sigma) = O_N(\sigma N^{-\sigma})$ for every N (15.68); the conditions (B1) and (B2) for $\tilde{h}_0$ (Corollary 15.40); $\mathsf{S}_{\mathsf{f}}(0) = \mathsf{R}_{\mathsf{f}}(1)$ for every f whose support avoids the points $\log n$, together with the cell jump-growth condition (Proposition 15.44); the solvability of the rung system (16.6) in $\mathscr{O}$ (Theorem 16.6); the identity (5.9) at one imaginary frequency; the window identity of Theorem 5.5(ii); the constancy of J_{Z1}^{M} on $(2, 3)$ together with the sine identity (5.19); $\mathscr{Q}_\infty = \log L$ (Theorem 5.11). The symmetry of the joining factor $e^{\tilde{h}_0}$ with finite order is a sufficient condition (Theorem 15.3), and it is also necessary (Remark 15.33).

1.7. **Organization.** Section 2 treats the clock, its Lambert inversion and the bounded displacement; Section 3 the canonical product; Section 4 the exponential sums of the clock; Section 5 the exponential sums over the ordinates of ζ, Cramér's function, the window form of RH, the endpoint defect as a sum over the zeros off the line, and the centroids; Sections 6 and 7 the prime coordinate and the integer weights; Sections 8 and 9 the Mertens transform and the transition theorem; Section 10 the exact Hadamard product and the comparison with ζ; Section 11 the λ-family; Section 12 the endpoint, Hamburger's theorem and Step 3; Section 13 the value of the endpoint at infinity and the Euler products of ζ_{RVM} and ζ_{Z1}; Section 14 the atomized family, Hamburger's hypotheses at the two endpoints and the programme for the final step; Section 15 the open path, convexity, Hamburger's theorem for the pair, the positivity criterion, the density of the clock, the Gaussian defect, the reduction to the cell masses, the half-line form, the bridge theorem and the transport from the clock; Section 16 the space of ordinate lists, the rigidity of the coincidence, the existence form of the final step, the rung system with free masses and the gradient of the vertical defect; Section 17 the acceptance criterion and the models in which the final step is a theorem; Section 18 route B for finite sections, the prescribed-mass uniqueness, the staged certificates, the Gonek lists, the a priori counting bound, the block Kadec–Avdonin criterion and its failure for the zeta list, with the certificate computations; Section 19 the numerical illustration; Section 20 the conclusions and open problems; Appendix F records the endpoint-only interpolation audit, the autonomous clock-rigidity question, the a priori rung bound, and the documented bounded finite-section diagnostics; Appendix A gives the tables of the first 1000 values.

2. The midpoint Riemann–von Mangoldt clock

Let ρ run over the nontrivial zeros of ζ, counted with multiplicity. No nontrivial zero is real, so the positive ordinates can be listed with multiplicity as $0 < \gamma_1 \le \gamma_2 \le \cdots$. Put

$$N_\zeta(T) := \#\{\rho : \ 0 < \Im\rho \le T\} = \#\{k : \gamma_k \le T\}. \tag{2.1}$$

The Riemann–von Mangoldt formula [61, Thms. 9.3–9.4], [62] states that for $T \ge 2$, $T \ne \gamma_k$,

$$N_\zeta(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8} + S(T) + O\Big(\frac{1}{T}\Big), \qquad S(T) := \frac{1}{\pi}\arg\zeta\big(\tfrac12 + iT\big) = O(\log T), \tag{2.2}$$

where $\arg\zeta(\frac12 + iT)$ is obtained by continuous variation of the argument along the segments from 2 to $2 + iT$ and from $2 + iT$ to $\frac12 + iT$, starting from $\arg\zeta(2) = 0$. The $O(1/T)$ term comes from Stirling's formula for Γ and is explicit; $S(T)$ is the fluctuating part, with a jump $+m$ at an ordinate of multiplicity m. The *midpoint RVM clock* is the smooth part of (2.2):

$$\mathcal{N}_{\mathrm{RVM}}(T) := \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}. \tag{2.3}$$

It is strictly increasing on $[2\pi, \infty)$. The midpoint ordinates τ_k are the unique solutions in $[2\pi, \infty)$ of

$$\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac12, \qquad k \ge 1, \tag{2.4}$$

that is, the abscissae on the increasing branch where the smooth count crosses the midpoints $k - \frac12$ of the steps of a unit staircase. For $k = 1$ the same level has a second positive solution in $(0, 2\pi)$ on the W_{-1} branch; it is not part of the midpoint clock used here. Thus $\mathcal{N}_{\mathrm{RVM}}$ replaces the staircase N_ζ by its smooth part, and τ_k is the point where the kth step would be crossed half-way (Figure 3).

The Lambert W-function is the multivalued inverse of $v \mapsto ve^v$: $W(z)e^{W(z)} = z$. On $[-1/e, \infty)$ the equation $ve^v = z$ has exactly one solution $v \ge -1$; this defines the principal branch $W_0 : [-1/e, \infty) \to [-1, \infty)$, which is real, increasing, with $W_0(0) = 0$ and $W_0(z) \sim \log z - \log\log z$ as $z \to \infty$ [36, §4.13], [29].

Proposition 2.1 (Lambert inversion)**.** *For every $k \ge 1$,*

$$\tau_k = \frac{2\pi(k - 11/8)}{W_0\big((k - 11/8)/e\big)}, \tag{2.5}$$

The quotient is positive also for $k = 1$.

Proof. Put $y = \tau/(2\pi)$. Equation (2.4) reads

$$y(\log y - 1) = k - \frac{11}{8}. \tag{2.6}$$

The function $y \mapsto y(\log y - 1)$ decreases on $(0, 1)$, increases on $(1, \infty)$ and has minimum -1 at $y = 1$; the clock is increasing exactly where $y > 1$, and τ_k is the solution with $y > 1$. With $v = \log y - 1$ one has $y = e^{v+1}$ and $ve^v = (k - 11/8)/e$; the solution with $y > 1$ has $v > -1$, so $v = W_0((k - 11/8)/e)$ and $y = (k - 11/8)/W_0((k - 11/8)/e)$. For $k = 1$, $(k - 11/8)/e = -3/(8e) \in (-1/e, 0)$, where W_0 is real with $W_0(-3/(8e)) \in (-1, 0)$; numerator and denominator are both negative, and $y = e^{1+W_0(-3/(8e))} \in (1, e)$, so $2\pi < \tau_1 < 2\pi e$. (The other solution of (2.6) for $k = 1$, with $y < 1$, is given by the branch W_{-1} and is not a clock point.) □

Equation (2.6) and its solution (2.5) are equations (24) and (25) of França and LeClair [38, Sec. III.A], who derived them as the leading-order approximation to the ordinate of the nth zero; here they define the clock exactly, and nothing is approximated.

The clock derivative and the resulting growth are

$$\mathcal{N}'_{\mathrm{RVM}}(T) = \frac{1}{2\pi}\log\frac{T}{2\pi}, \tag{2.7}$$

$$\tau_k \asymp \frac{k}{\log k} \qquad (k \geq 2). \tag{2.8}$$

In terms of the clock, (2.2) reads

$$N_\zeta(T) = \mathcal{N}_{\mathrm{RVM}}(T) + S(T) + O(T^{-1}), \qquad S(T) = O(\log T) \qquad (T \geq 2). \tag{2.9}$$

Let

$$N_\zeta^*(T) := \tfrac{1}{2}(N_\zeta(T+0) + N_\zeta(T-0)) \tag{2.10}$$

be the symmetric count, and define the displacements

$$\eta_k := \tau_k - \gamma_k. \tag{2.11}$$

Lemma 2.2 (Bounded displacement)**.** *There is a constant C_η such that*

$$|\eta_k| \leq C_\eta \qquad (k \geq 1). \tag{2.12}$$

Proof. Let $\gamma = \gamma_k$ and let $m = m(\gamma)$ be the number of indices j with $\gamma_j = \gamma$; these indices form a block $j_0 + 1, \ldots, j_0 + m$ containing k. Then $N_\zeta(\gamma - 0) = j_0$ and $N_\zeta(\gamma + 0) = j_0 + m$, so $N_\zeta^*(\gamma) = j_0 + m/2$ and

$$|N_\zeta^*(\gamma_k) - (k - \tfrac{1}{2})| \leq \frac{m(\gamma_k)}{2}. \tag{2.13}$$

By (2.9), applied at the one-sided limits $N_\zeta(\gamma+1+0)$ and $N_\zeta(\gamma-1-0)$ if $\gamma \pm 1$ is itself an ordinate, $m(\gamma) \leq N_\zeta(\gamma+1+0) - N_\zeta(\gamma-1-0) = O(\log\gamma)$, and likewise $N_\zeta^*(\gamma) = \mathcal{N}_{\mathrm{RVM}}(\gamma) + O(\log\gamma)$. With (2.4),

$$|\mathcal{N}_{\mathrm{RVM}}(\gamma_k) - \mathcal{N}_{\mathrm{RVM}}(\tau_k)| = O(\log\gamma_k). \tag{2.14}$$

Explicitly, $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$ and $\mathcal{N}_{\mathrm{RVM}}(\gamma_k) = N_\zeta^*(\gamma_k) + O(\log\gamma_k) = k + O(\log\gamma_k)$; since $\mathcal{N}_{\mathrm{RVM}}(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + O(1)$ is increasing with $\mathcal{N}_{\mathrm{RVM}}(T) \asymp T\log T$, both relations give $\gamma_k \asymp \tau_k \asymp k/\log k$ and hence $\log\gamma_k \asymp \log\tau_k \asymp \log k$. Hence there is k_1 such that for $k \geq k_1$ both γ_k and τ_k exceed $2\pi e$, where $\mathcal{N}_{\mathrm{RVM}}$ is increasing with $\mathcal{N}'_{\mathrm{RVM}} \geq \frac{1}{2\pi}$. For $k \geq k_1$, by the mean-value theorem, $\mathcal{N}_{\mathrm{RVM}}(\gamma_k) - \mathcal{N}_{\mathrm{RVM}}(\tau_k) = \mathcal{N}'_{\mathrm{RVM}}(\xi_k)(\gamma_k - \tau_k)$ with ξ_k between γ_k and τ_k, and by (2.7) $\mathcal{N}'_{\mathrm{RVM}}(\xi_k) = \frac{1}{2\pi}\log\frac{\xi_k}{2\pi} \gg \log k$ because $\xi_k \geq \min(\gamma_k, \tau_k) \gg k/\log k$; with (2.14) this gives $|\gamma_k - \tau_k| = O(1)$. The finitely many indices $k < k_1$ are absorbed into C_η. □

3. The pure critical RVM zeta function

Put

$$q(s) := s(s-1), \tag{3.1}$$

$$P_{\mathrm{RVM}}^{\mathrm{crit}}(s) := \prod_{k\geq 1}\left(1 + \frac{q(s)}{\frac{1}{4} + \tau_k^2}\right), \tag{3.2}$$

$$G(s) := s(s-1)\Gamma(s/2)\pi^{-s/2}, \tag{3.3}$$

and

$$\xi_{\mathrm{RVM}}(s) := \frac{1}{2}P_{\mathrm{RVM}}^{\mathrm{crit}}(s), \qquad \zeta_{\mathrm{RVM}}(s) := \frac{P_{\mathrm{RVM}}^{\mathrm{crit}}(s)}{G(s)}. \tag{3.4}$$

Theorem 3.1 (Canonical characterization of ζ_{RVM})**.** *The product* (3.2) *converges absolutely and locally uniformly in $s \in \mathbb{C}$; as a function of the symmetric variable $q = q(s)$ it is a canonical product of genus zero. It defines an entire function with*

$$P_{\mathrm{RVM}}^{\mathrm{crit}}(1-s) = P_{\mathrm{RVM}}^{\mathrm{crit}}(s), \qquad P_{\mathrm{RVM}}^{\mathrm{crit}}(\bar{s}) = \overline{P_{\mathrm{RVM}}^{\mathrm{crit}}(s)}, \qquad P_{\mathrm{RVM}}^{\mathrm{crit}}(0) = P_{\mathrm{RVM}}^{\mathrm{crit}}(1) = 1. \tag{3.5}$$

Its zeros are exactly the simple zeros

$$\rho_k^{\mathrm{RVM}} = \frac{1}{2} + i\tau_k, \qquad \overline{\rho_k^{\mathrm{RVM}}} = \frac{1}{2} - i\tau_k, \qquad k \geq 1. \tag{3.6}$$

Consequently ζ_{RVM} *has a simple pole at* $s = 1$ *with residue* 1*, simple trivial zeros at* $s = -2, -4, -6, \dots$*, and nontrivial zeros exactly at* (3.6)*; moreover* $\xi_{\mathrm{RVM}}(s) = \xi_{\mathrm{RVM}}(1-s)$*.*

Proof. By (2.8), $\sum_k \tau_k^{-2} < \infty$. On a compact set K,

$$\sup_{s\in K}\left|\frac{q(s)}{\frac{1}{4}+\tau_k^2}\right| \ll_K \tau_k^{-2}, \tag{3.7}$$

so the Weierstrass criterion gives normal convergence. Each factor depends on s only through $q(s)$ and equals 1 at $s = 0, 1$, which gives (3.5); real coefficients give the conjugation symmetry. The kth factor vanishes exactly when

$$\Big(s-\frac{1}{2}\Big)^2 + \tau_k^2 = 0. \tag{3.8}$$

The clock is strictly increasing on $[2\pi, \infty)$ and $\tau_1 > 2\pi$, so the τ_k are distinct; the derivative $2s - 1 = \pm 2i\tau_k \neq 0$ of the vanishing factor and the non-vanishing of the other factors make the zeros simple. At $s = 1$, $P^{\mathrm{crit}}_{\mathrm{RVM}}(1) = 1$ and $G(s) = (s-1)(1 + O(s-1))$, so the residue is 1. At $s = -2n$ ($n \geq 1$) the simple pole of $\Gamma(s/2)$ gives a simple zero of $1/G$, while $P^{\mathrm{crit}}_{\mathrm{RVM}}(-2n) > 0$ because $q(-2n) > 0$, so there is no cancellation. At $s = 0$, $s\Gamma(s/2) = 2\Gamma(1 + s/2) \to 2$, so $G(0) = -2 \neq 0$ and $s = 0$ is not a zero. □

4. RVM exponential sums, centroids, and the height transform

For $u \in \mathbb{R}$ and $N \geq 1$ define

$$S_{\mathrm{RVM}}(N; u) := \sum_{k\leq N} e^{iu\tau_k}, \qquad C^{\mathrm{pts}}_{\mathrm{RVM}}(N; u) := \frac{S_{\mathrm{RVM}}(N; u)}{N}, \tag{4.1}$$

and

$$C^{\mathrm{path}}_{\mathrm{RVM}}(N; u) := \frac{1}{N}\sum_{n\leq N} S_{\mathrm{RVM}}(n; u) = \sum_{k\leq N} \frac{N-k+1}{N} e^{iu\tau_k}. \tag{4.2}$$

Theorem 4.1 (Uniform logarithmic exponential-sum bound)**.** *Let* $0 < u_1 \leq u_2 < \infty$*. Then*

$$\sup_{u\in[u_1,u_2]} |S_{\mathrm{RVM}}(N; u)| \ll_{u_1,u_2} 1 + \log N. \tag{4.3}$$

Proof. Let $\mathcal{T} : [-\frac{1}{8}, \infty) \to [2\pi, \infty)$ be the inverse of the clock, so $\tau_k = \mathcal{T}(k - \frac{1}{2})$, and put $\ell_k = \log(\tau_k/2\pi) > 0$. Differentiating $\mathcal{N}_{\mathrm{RVM}}(\mathcal{T}(y)) = y$,

$$\mathcal{T}'(y) = \frac{2\pi}{\log(\mathcal{T}(y)/2\pi)}, \qquad \mathcal{T}''(y) = -\frac{4\pi^2}{\mathcal{T}(y)\log^3(\mathcal{T}(y)/2\pi)}. \tag{4.4}$$

Since $\mathcal{T}'$ decreases and $|\mathcal{T}''|$ decreases,

$$\frac{2\pi}{\ell_{k+1}} \leq \tau_{k+1} - \tau_k \leq \frac{2\pi}{\ell_k}, \tag{4.5}$$

$$|\tau_{k+2} - 2\tau_{k+1} + \tau_k| \leq \frac{4\pi^2}{\tau_k \ell_k^3}. \tag{4.6}$$

From (2.4), $\tau_k\ell_k = 2\pi(k - \frac{11}{8}) + \tau_k$, hence

$$\tau_k\ell_k \asymp k \qquad (k \geq 2). \tag{4.7}$$

Fix $u \in [u_1, u_2]$ and put $\Delta_k = u(\tau_{k+1} - \tau_k)$, $E_k = e^{iu\tau_k}$. Choose $k_0 = k_0(u_2)$ with $\ell_{k_0} \geq 2u_2$; then $0 < \Delta_k \leq \pi$ for $k \geq k_0$. With $\mathfrak{b}_k^{\mathrm{sbp}} := (e^{i\Delta_k} - 1)^{-1}$ one has $E_{k+1} = e^{i\Delta_k}E_k$, hence

$$E_k = \mathfrak{b}_k^{\mathrm{sbp}}(E_{k+1} - E_k). \tag{4.8}$$

Summing (4.8) over $k_0 \le k \le N$ and shifting the index in the first part gives the exact identity

$$\sum_{k=k_0}^{N} E_k = \mathfrak{b}_N^{\mathrm{sbp}} E_{N+1} - \mathfrak{b}_{k_0}^{\mathrm{sbp}} E_{k_0} + \sum_{k=k_0+1}^{N} (\mathfrak{b}_{k-1}^{\mathrm{sbp}} - \mathfrak{b}_k^{\mathrm{sbp}}) E_k. \tag{4.9}$$

For $0 < x \le \pi$, $|e^{ix} - 1| = 2\sin(x/2) \ge 2x/\pi$. Using (4.5) and (4.6),

$$|\mathfrak{b}_k^{\mathrm{sbp}}| \le \frac{\pi}{2\Delta_k} \le \frac{\ell_{k+1}}{4u_1}, \quad |\mathfrak{b}_{k-1}^{\mathrm{sbp}} - \mathfrak{b}_k^{\mathrm{sbp}}| \le |\Delta_k - \Delta_{k-1}|\,|\mathfrak{b}_{k-1}^{\mathrm{sbp}}||\mathfrak{b}_k^{\mathrm{sbp}}| \le \frac{\pi^2 u_2}{4u_1^2} \frac{\ell_{k+1}^2}{\tau_{k-1}\ell_{k-1}^3} \ll_{u_1,u_2} \frac{1}{\tau_k \ell_k} \ll \frac{1}{k}, \tag{4.10}$$

where $\tau_{k\pm1} \asymp \tau_k$ and $\ell_{k\pm1} \asymp \ell_k$ uniformly for $k \ge k_0$ by (4.5), and the last step is (4.7). Inserting (4.10) into (4.9) gives $|\sum_{k_0}^{N} E_k| \ll \ell_{N+1} + \ell_{k_0+1} + \sum_{k \le N} k^{-1} \ll 1 + \log N$. The indices $k < k_0$ contribute at most k_0. All constants depend only on u_1, u_2. □

Corollary 4.2 (Point centroid)**.** *Uniformly for $u \in [u_1, u_2] \subset (0, \infty)$,*

$$C_{\mathrm{RVM}}^{\mathrm{pts}}(N; u) = O_{u_1,u_2}\Big(\frac{1 + \log N}{N}\Big) \longrightarrow 0. \tag{4.11}$$

By direct averaging, Theorem 4.1 gives only $C_{\mathrm{RVM}}^{\mathrm{path}}(N; u) = O(\log N)$ for the path centroid (4.2), not even boundedness; the path centroid converges by a second summation by parts (Theorem 5.13).

Proof. The first assertion is (4.3) divided by N. For the second, the identity $C_{\mathrm{RVM}}^{\mathrm{path}} = N^{-1} \sum_{n \le N} S_{\mathrm{RVM}}(n; u)$ with $S_{\mathrm{RVM}}(n; u) = O(\log n)$ gives $O(\log N)$ and nothing better. □

Define the midpoint count

$$n_{\mathrm{RVM}}(H) := \#\{k : \tau_k \le H\}. \tag{4.12}$$

For $x > 0$ and $n_{\mathrm{RVM}}(H) \ge 1$ define

$$M_H^{\mathrm{RVM}}(x) := \frac{1}{n_{\mathrm{RVM}}(H)} \sum_{\tau_k \le H} x^{i\tau_k}, \tag{4.13}$$

and for the ordinates of ζ

$$M_H(x) := \frac{1}{N_\zeta(H)} \sum_{\gamma_k \le H} x^{i\gamma_k}. \tag{4.14}$$

Corollary 4.3 (Synthetic height-transform decay)**.** *For every fixed $x > 1$, with $N = n_{\mathrm{RVM}}(H)$,*

$$M_H^{\mathrm{RVM}}(x) = O_x\Big(\frac{1 + \log N}{N}\Big) \longrightarrow 0 \qquad (H \to \infty). \tag{4.15}$$

Proof. Take $u = \log x > 0$ in Theorem 4.1; (4.13) is the point centroid at $N = n_{\mathrm{RVM}}(H)$. □

Remark 4.4. Neither (4.13) nor (4.14) is the Mertens function; both are normalized phase averages.

5. Exponential sums over the ordinates of ζ: centroids, Cramér's function and the window form of the Riemann hypothesis

Section 4 bounded the exponential sums of the clock. This section studies the same sums over the ordinates γ_k of ζ and compares them with the sums over the complete zeros. The comparison gives three equivalent forms of the Riemann hypothesis written with the ordinates alone (Proposition 5.2, Theorem 5.5, Corollary 5.8), which we have not found in the literature consulted (Remark 5.6(d)), an exact formula for the defect between the two sums (Proposition 5.1, (5.14)), an unconditional identity between the rung functional of ζ_{Z1} and the zeros off the critical line (Theorem 5.7), and the value of the triangular centroid of $e^{i\gamma_k}$ under RH (Theorem 5.15). The clock centroid converges without any hypothesis (Theorem 5.13).

Throughout, ρ runs over the nontrivial zeros of ζ with multiplicity, and the zeros with positive ordinate are listed as $\rho_k = \frac{1}{2} + d_k + i\gamma_k$, $k \geq 1$, so that (γ_k) is the list of Section 2 and $d_k = \Re\rho_k - \frac{1}{2}$; zeros with equal ordinate are listed in any order. By the functional equation the zeros with $d_k \neq 0$ come in pairs $\{\rho, 1-\bar{\rho}\}$ with the same ordinate, opposite d and equal multiplicity, and $|d_k| < \frac{1}{2}$ because ζ has no zeros on $\Re s = 0$ and $\Re s = 1$ [61]. RH is the statement that $d_k = 0$ for every k. For a function $\mathsf{f} \in C_c^\infty(\mathbb{R})$ we write

$$\widehat{\mathsf{f}}(z) := \int_{\mathbb{R}} \mathsf{f}(u)\, e^{iuz}\, du \qquad (z \in \mathbb{C}); \tag{5.1}$$

for real f and real t, $\Re\widehat{\mathsf{f}}(t) = \mathsf{c}_\mathsf{f}(t)$ is the cosine transform of (15.79). If $\operatorname{supp}\mathsf{f} \subset [u_1, u_2]$, M integrations by parts give

$$|\widehat{\mathsf{f}}(t+iy)| \leq C_{\mathsf{f},M}\, e^{u_2|y|}(1+|t|)^{-M} \qquad (t, y \in \mathbb{R},\ M \geq 0). \tag{5.2}$$

5.1. **The ordinate transform and Cramér's function.** For $\Im w > 0$ define

$$\mathscr{Z}_\gamma(w) := \sum_{k\geq 1} e^{iw\gamma_k}, \qquad \mathscr{Z}_\tau(w) := \sum_{k\geq 1} e^{iw\tau_k}, \qquad \mathscr{Z}_\xi(w) := \sum_{k\geq 1} e^{w(\rho_k - \frac{1}{2})} = e^{-w/2}\, V_+(w), \tag{5.3}$$

where

$$V_+(z) := \sum_{\Im\rho > 0} e^{\rho z} \qquad (\Im z > 0) \tag{5.4}$$

is Cramér's function [66]. Since $|e^{iw\gamma}| = e^{-\gamma\Im w}$, $|e^{w(\rho-\frac{1}{2})}| \leq e^{|\Re w|/2} e^{-\gamma\Im w}$ and $N_\zeta(T) \ll T\log T$, the three series converge absolutely and locally uniformly on $\Im w > 0$ and define holomorphic functions there; summation by parts gives

$$|\mathscr{Z}_\gamma(u+iy)| \leq \sum_k e^{-y\gamma_k} \leq C\, y^{-2} \qquad (u \in \mathbb{R},\ 0 < y \leq 1). \tag{5.5}$$

The defect between the ordinate transform and the zero transform is

$$\mathfrak{X}(w) := \mathscr{Z}_\gamma(w) - \mathscr{Z}_\xi(w) \qquad (\Im w > 0). \tag{5.6}$$

Proposition 5.1 (Pair decomposition of the defect)**.** *For $\Im w > 0$,*

$$\mathfrak{X}(w) = -\sum_{k\geq 1} \big(\cosh(wd_k) - 1\big) e^{iw\gamma_k}, \tag{5.7}$$

the series converging absolutely; here k runs over both members of each pair $\{\frac{1}{2} \pm d + i\gamma\}$, with multiplicity, and only the zeros with $d_k \neq 0$ contribute. In particular $\mathfrak{X} \equiv 0$ under RH, and for $y > 0$

$$\mathfrak{X}(iy) = \sum_{k\geq 1} \big(1 - \cos(yd_k)\big) e^{-y\gamma_k} \ \geq 0. \tag{5.8}$$

Proof. $e^{w(\rho_k - \frac{1}{2})} = e^{wd_k} e^{iw\gamma_k}$. The zeros with $d_k \neq 0$ form pairs with ordinates γ and abscissas $\pm d$, so $\sum_k e^{wd_k} e^{iw\gamma_k} = \sum_k \cosh(wd_k) e^{iw\gamma_k}$, the rearrangement being allowed by absolute convergence. Subtracting from $\mathscr{Z}_\gamma$ gives (5.7), and $\cosh(iyd) = \cos(yd)$ gives (5.8). □

Proposition 5.2 (RH at one imaginary frequency)**.** *For every $y_0 \in (0, 4\pi]$,*

$$\text{RH} \iff \sum_{k\geq 1} e^{-y_0\gamma_k} = e^{-iy_0/2}\, V_+(iy_0). \tag{5.9}$$

Proof. The right side of (5.9) is $\mathscr{Z}_\xi(iy_0)$ and the left side is $\mathscr{Z}_\gamma(iy_0)$. Under RH, $\mathfrak{X} \equiv 0$. Conversely, if $\mathfrak{X}(iy_0) = 0$, every term of (5.8) vanishes, so $y_0 d_k \in 2\pi\mathbb{Z}$ for every k; since $|y_0 d_k| < 4\pi \cdot \frac{1}{2} = 2\pi$, $d_k = 0$. □

The right side of (5.9) has a representation that does not use the zeros (Proposition 5.3; $iy_0 \in \Omega_\theta$). The identity is invisible numerically: if every zero up to height T_{ver} is on the critical line, $0 \le \mathfrak{X}(iy) \le 2\sum_{\gamma_k > T_{\mathrm{ver}}} e^{-y\gamma_k}$, which is of size $T_{\mathrm{ver}} \log T_{\mathrm{ver}}\, e^{-yT_{\mathrm{ver}}}$. At a real frequency the coefficients $1-\cosh(ud_k)$ of (5.7) keep a sign but the phases $e^{iu\gamma_k}$ do not, and the argument of the proof does not apply; Theorem 5.5 replaces one real point by an interval.

Here and below, a *zero-free representation* of V_+ is a representation that involves no sum over the zeros of ζ; the term does not refer to zeros of V_+.

Proposition 5.3 (Zero-free representation of Cramér's function)**.** *Let $a > 0$ and $0 < \theta < \frac{\pi}{2}$, and put*

$$g(s) := \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\,\frac{\Gamma'}{\Gamma}\Big(\frac{s}{2}\Big), \qquad \text{so that} \qquad \frac{\xi'}{\xi}(s) = g(s) + \frac{\zeta'}{\zeta}(s). \tag{5.10}$$

For $\Im z > 0$,

$$2\pi i\, V_+(z) = \mathscr{V}_{\mathrm{b}}(z) + \mathscr{V}_{\mathrm{R}}(z) + \mathscr{V}_{\mathrm{L}}(z) + \mathscr{V}_{\mathrm{p}}(z), \tag{5.11}$$

$$\begin{aligned} &\mathscr{V}_{\mathrm{b}}(z) = \int_{-a}^{1+a} e^{\sigma z}\,\frac{\xi'}{\xi}(\sigma)\,d\sigma, \qquad \mathscr{V}_{\mathrm{R}}(z) = i\,e^{(1+a)z}\int_0^{\infty e^{i\theta}} e^{itz}\,g(1+a+it)\,dt, \\ &\mathscr{V}_{\mathrm{L}}(z) = i\,e^{-az}\int_0^{\infty e^{i\theta}} e^{itz}\,g(1+a-it)\,dt, \qquad \mathscr{V}_{\mathrm{p}}(z) = \sum_{n\ge 2}\Lambda(n)\Big(\frac{e^{(1+a)(z-\log n)}}{z-\log n} + \frac{n^{-1-a}e^{-az}}{z+\log n}\Big), \end{aligned} \tag{5.12}$$

the two t-integrals being taken along the ray $\arg t = \theta$. The right side of (5.11) *does not depend on a and θ, and it continues V_+ meromorphically to the half-plane $\Omega_\theta := \{z = x+iy :\ x\sin\theta + y\cos\theta > 0\}$, which contains the positive real axis; in Ω_θ the only singularities are simple poles at the points $z = \log n$, n a prime power, with residues $\Lambda(n)/2\pi i$. In particular $\mathscr{Z}_\xi(u) = e^{-u/2}V_+(u)$ is real-analytic on $(0,\infty)\setminus\{\log p^m\}$, and*

$$2\Re V_+(u) = e^u - \frac{1}{e^{2u}-1} \qquad (\log 2 < u < \log 3). \tag{5.13}$$

Proof. Contour. ξ is entire, its zeros are the ρ, and $\xi(\sigma) > 0$ for real σ, so ξ'/ξ is meromorphic with simple poles at the ρ (residue the multiplicity) and no pole on $\mathbb{R}$. Fix $z = x+iy$ with $y > 0$ and let R_T be the rectangle with vertices $-a$, $1+a$, $1+a+iT$, $-a+iT$, with T not an ordinate, oriented positively: the bottom side $s = \sigma$, $-a \le \sigma \le 1+a$; the right side $s = 1+a+it$, t from 0 to T; the top side from $1+a+iT$ to $-a+iT$; the left side $s = -a+it$, t from T to 0. The residue theorem gives $\sum_{0<\gamma<T} e^{\rho z} = \frac{1}{2\pi i}\oint_{\partial R_T} e^{sz}\frac{\xi'}{\xi}(s)\,ds$. There are $T_j \in [j, j+1]$ with $\frac{\zeta'}{\zeta}(\sigma + iT_j) \ll \log^2 T_j$ uniformly for $-1 \le \sigma \le 2$ [31, Ch. 17]; with $\xi'/\xi(s) = -\xi'/\xi(1-s)$, the boundedness of ζ'/ζ on $\Re s \ge 2$ and Stirling's formula, $\frac{\xi'}{\xi}(\sigma + iT_j) \ll \log^2 T_j$ for $-a \le \sigma \le 1+a$. Since $|e^{sz}| \le e^{(1+a)|x|}e^{-T_j y}$ on the top side, its contribution tends to 0, and the left side tends to $V_+(z)$. On the right side $s = 1+a+it$, $\frac{\xi'}{\xi} = g + \frac{\zeta'}{\zeta}$ with $\frac{\zeta'}{\zeta}(s) = -\sum_n \Lambda(n)n^{-s}$; on the left side $s = -a+it$, $\frac{\xi'}{\xi}(s) = -g(1+a-it) - \frac{\zeta'}{\zeta}(1+a-it)$. The integrals converge absolutely because $|e^{(c+it)z}| = e^{cx-ty}$ and $g(1+a\pm it) \ll \log(2+t)$, and the Dirichlet series may be integrated termwise; with $\int_0^\infty e^{it(z\mp\log n)}\,dt = i/(z\mp\log n)$ for $\Im z > 0$ this gives (5.11) with the rays on the positive real axis ($\theta = 0$).

Rotation. For $0 \le \varphi \le \theta$ and $t = re^{i\varphi}$, $\Im(tz) = r(x\sin\varphi + y\cos\varphi)$; for $z \in \Omega_\theta$ with $y > 0$ the sinusoid $\varphi \mapsto x\sin\varphi + y\cos\varphi$ is positive at $\varphi = 0$ and $\varphi = \theta$, hence on $[0,\theta]$, since $\theta < \pi$. The singularities of $t \mapsto g(1+a+it)$ lie on the positive imaginary axis and those of $t \mapsto g(1+a-it)$ on the negative imaginary axis, and $g(1+a\pm it) \ll \log(2+|t|)$ in the closed sector $0 \le \arg t \le \theta$. Cauchy's theorem in the sector moves the rays to $\arg t = \theta$. The rotated integrals converge absolutely and locally uniformly on Ω_θ, where they are holomorphic; $\mathscr{V}_{\mathrm{b}}$ is entire; and $\mathscr{V}_{\mathrm{p}}$ converges absolutely away from $\{\pm\log n\}$ and is meromorphic on $\mathbb{C}$ with simple poles at $\pm\log n$, of residue $\Lambda(n)$ at $\log n$;

the points $-\log n$ are not in Ω_θ. For two admissible choices (a,θ) and (a',θ') the two expressions agree with V_+ on $\Omega_\theta\cap\Omega_{\theta'}\cap\{\Im z>0\}$, which is non-empty and open; $\Omega_\theta\cap\Omega_{\theta'}$ is a half-plane, hence connected, and the identity theorem gives agreement on all of it. The continuation is therefore unique and independent of a and θ.

The window. Von Mangoldt's explicit formula $\psi_0(x)=x-\sum_\rho x^\rho/\rho-\log 2\pi-\frac12\log(1-x^{-2})$ [37, Ch. 3], [20, 63], differentiated in $\mathscr{D}'(1,\infty)$ and written in $u=\log x$, is $\sum_\rho e^{\rho u}=e^u-(e^{2u}-1)^{-1}-\sum_n\Lambda(n)\delta_{\log n}$ in $\mathscr{D}'(0,\infty)$, the sum over ρ being taken in symmetric order. Pairing $V_+(\cdot+iy)$ with $\mathsf{f}\in C_c^\infty(0,\infty)$ gives $\sum_{\Im\rho>0}e^{iy\rho}\int\mathsf{f}(u)e^{\rho u}du$, and by (5.2) and dominated convergence its limit as $y\downarrow 0$ is $\sum_{\Im\rho>0}\int\mathsf{f}(u)e^{\rho u}du$. The zeros with negative ordinate are the conjugates, so $2\Re\,\mathrm{bv}\,V_+=e^u-(e^{2u}-1)^{-1}-\sum_n\Lambda(n)\delta_{\log n}$ in $\mathscr{D}'(0,\infty)$, where bv denotes the distributional boundary value from $\Im z>0$. On $(\log 2,\log 3)$ there is no atom and V_+ is analytic, which gives (5.13). □

The full meromorphic continuation of V_+, with a logarithmic singularity at 0, is Cramér's theorem [66]; only the part in Ω_θ is used here. The representation (5.11) is the unilateral form of the explicit formula: on $(\log 2,\log 3)$ there are no prime atoms, but the prime terms $\Lambda(n)e^{(1+a)(u-\log n)}/(u-\log n)$ of $\mathscr{V}_{\mathrm{p}}$ do not vanish there. Numerically (Experiment 19.7) $e^{-1/2}V_+(1)=0.77689426079325\ldots+0.76785566886\ldots\,i$, the same for $a=1$ and $a=2$ up to the truncation of the prime sum.

5.2. **The window form of the Riemann hypothesis.**

Lemma 5.4 (Vanishing boundary values)**.** *Let $J\subset\mathbb{R}$ be an open interval, $h>0$, and F holomorphic on $J\times(0,h)$ with $|F(u+iy)|\le Cy^{-m}$ there for some $m\ge 0$. If $\lim_{y\downarrow0}\int F(u+iy)\mathsf{f}(u)\,du=0$ for every $\mathsf{f}\in C_c^\infty(J)$, then $F\equiv 0$.*

Proof. Fix $[a',b']\subset J$, $w_0\in(a',b')\times(0,h)$, and let $F_1(w)=\int_{w_0}^w F$, $F_{j+1}(w)=\int_{w_0}^w F_j$, the integrals taken along paths in the rectangle $R=(a',b')\times(0,h)$; R is convex, so by Cauchy's theorem they do not depend on the path, and $F_{j+1}'=F_j$. Integrating along vertical segments, $|F_j(u+iy)|\le C_jy^{-m+j}$ for $j<m$, $|F_m|\le C_m(1+|\log y|)$, and F_{m+1} is bounded on $(a',b')\times(0,h/2]$. For z,w in this convex set, $F_{m+2}(z)-F_{m+2}(w)=\int_{[w,z]}F_{m+1}(\zeta)\,d\zeta$, so F_{m+2} is uniformly Lipschitz there and extends continuously to $[a',b']$, with boundary function β. For $\mathsf{f}\in C_c^\infty((a',b'))$, $m+2$ integrations by parts in u (using $\partial_uF_{j+1}=F_j$) give $\int F(u+iy)\mathsf{f}(u)du=(-1)^{m+2}\int F_{m+2}(u+iy)\mathsf{f}^{(m+2)}(u)du$, and letting $y\downarrow0$, by the uniform convergence of $F_{m+2}(\cdot+iy)$ to β, $\int\beta\,\mathsf{f}^{(m+2)}=0$, that is, $\beta^{(m+2)}=0$ in $\mathscr{D}'((a',b'))$. A distribution on an interval with vanishing derivative is a constant, because every $\psi\in C_c^\infty((a',b'))$ with $\int\psi=0$ is the derivative of a test function; applied $m+2$ times, β coincides in $\mathscr{D}'$ with a polynomial P of degree at most $m+1$, and, β being continuous, $\beta=P$ at every point of (a',b'). The function $E=F_{m+2}-P$ is holomorphic in R, continuous up to (a',b'), and vanishes identically there. By the Schwarz reflection principle [57, Thm. 11.14] (whose hypothesis $\Im E(z_n)\to0$ as $z_n\to u\in(a',b')$ holds), the function equal to E on R, to 0 on (a',b') and to $\overline{E(\bar z)}$ on the reflected rectangle is holomorphic on $(a',b')\times(-h,h)$; it vanishes on (a',b'), hence $E\equiv0$, $F_{m+2}=P$ on R and $F=P^{(m+2)}=0$. Since $J\times(0,h)$ is connected, $F\equiv0$. □

Theorem 5.5 (The window form of the Riemann hypothesis)**.** *Let $J\subset(0,\infty)$ be an open interval containing no point $\log p^m$, for instance $J=(\log 2,\log 3)$. Without any hypothesis, for every $\mathsf{f}\in C_c^\infty(J)$,*

$$\sum_{k\ge1}\widehat{\mathsf{f}}(\gamma_k)-\int_J\mathsf{f}(u)\,e^{-u/2}V_+(u)\,du=\sum_{k\ge1}\Big(\widehat{\mathsf{f}}(\gamma_k)-\tfrac12\widehat{\mathsf{f}}(\gamma_k-id_k)-\tfrac12\widehat{\mathsf{f}}(\gamma_k+id_k)\Big),\tag{5.14}$$

both series converging absolutely; only the zeros with $d_k\neq0$ contribute to the right side. The following are equivalent:

(i) *RH;*

(ii) *for every real* $\mathsf{f} \in C_c^\infty(J)$, $\sum_{k\geq 1} \widehat{\mathsf{f}}(\gamma_k) = \int_J \mathsf{f}(u)\, e^{-u/2} V_+(u)\, du$;

(iii) *the triangular centroids* $C_\zeta^{\mathrm{path}}(N;u) := \sum_{k\leq N}(1-\frac{k-1}{N})e^{iu\gamma_k}$ *converge in* $\mathscr{D}'(J)$ *to* $e^{-u/2}V_+(u)$.

Proof. For $y>0$, $\int \mathscr{Z}_\gamma(u+iy)\mathsf{f}(u)du = \sum_k e^{-y\gamma_k}\widehat{\mathsf{f}}(\gamma_k)$ and $\int \mathscr{Z}_\xi(u+iy)\mathsf{f}(u)du = \sum_k e^{iy(d_k+i\gamma_k)}\widehat{\mathsf{f}}(\gamma_k - id_k)$. By (5.2) with $|d_k|<\frac12$ and dominated convergence, the limits as $y\downarrow 0$ are $\sum_k \widehat{\mathsf{f}}(\gamma_k)$ and $\sum_k \widehat{\mathsf{f}}(\gamma_k - id_k)$; the second equals $\int \mathsf{f}(u)e^{-u/2}V_+(u)du$ because $\mathscr{Z}_\xi$ is analytic on a neighbourhood of $\operatorname{supp}\mathsf{f}$ (Proposition 5.3). The multiset $\{(\gamma_k,d_k)\}$ is invariant under $d_k\mapsto -d_k$ (the functional equation pairs ρ with $1-\bar\rho$, with equal multiplicity), so $\sum_k \widehat{\mathsf{f}}(\gamma_k - id_k) = \sum_k \widehat{\mathsf{f}}(\gamma_k + id_k)$, for real or complex f; this gives (5.14). (i)⇒(ii): every term of the right side of (5.14) vanishes. For (ii)⇔(iii): $\int C_\zeta^{\mathrm{path}}(N;u)\mathsf{f}(u)du = \sum_{k\leq N}(1-\frac{k-1}{N})\widehat{\mathsf{f}}(\gamma_k)$ is a Cesàro mean of the absolutely convergent series $\sum_k \widehat{\mathsf{f}}(\gamma_k)$, so for every $\mathsf{f}\in C_c^\infty(J)$ it converges to $\sum_k \widehat{\mathsf{f}}(\gamma_k)$, with no hypothesis. (ii)⇒(iii): by (ii) and linearity this limit is $\int \mathsf{f}(u)e^{-u/2}V_+(u)du$ for every complex f, which is the convergence in $\mathscr{D}'(J)$. (iii)⇒(ii): the limit of the pairings is then both $\int \mathsf{f}e^{-u/2}V_+$ and $\sum_k \widehat{\mathsf{f}}(\gamma_k)$. (ii)⇒(i): $\mathfrak{X} = \mathscr{Z}_\gamma - \mathscr{Z}_\xi$ is holomorphic on the half-plane $\Im w>0$, which is connected. Fix a compact interval $J'\subset J$. By (5.5), and because $\mathscr{Z}_\xi$ is holomorphic, hence bounded, on a neighbourhood of $J'\times[0,1]$ (its only singularities in Ω_θ are the points $\log p^m \notin J$, Proposition 5.3), $|\mathfrak{X}(u+iy)|\leq C_{J'}y^{-2}$ on $J'\times(0,1]$. By (ii), linearity and the first part, $\lim_{y\downarrow 0}\int \mathfrak{X}(u+iy)\mathsf{f}(u)du = 0$ for every $\mathsf{f}\in C_c^\infty(J)$; this is the vanishing of the distributional boundary value. Lemma 5.4, applied on the interior of J' with $m=2$ and $h=1$, gives $\mathfrak{X}=0$ there, and the identity theorem gives $\mathfrak{X}\equiv 0$ on $\Im w>0$. In particular $\mathfrak{X}(iy_0)=0$, and Proposition 5.2, through the positivity of every term of (5.8), gives $d_k=0$ for every k, that is, RH. □

Remark 5.6 (What the window form needs and what it does not). (a) After inserting (5.11), both sides of (ii) are expressed without the abscissas of the zeros: the left side uses the ordinates only, the right side is given in terms of Λ, Γ and the values of ζ on a real segment. Identity (ii) is the explicit formula of Guinand and Weil [20, 63] written for the measure of the ordinates instead of the measure of the zeros, and required only in one gap between prime powers, where its prime side has no atom.

(b) The interval cannot be replaced by the argument at one real point: at $u=1$ the defect $\mathfrak{X}(1+i0)$ is a sum with positive coefficients $\cosh d_k - 1$ and free phases $e^{i\gamma_k}$, and the triangular centroid at $u=1$ (Theorem 5.15) is implied by RH, not shown equivalent to it. Nor can the one-sided transform be replaced by the symmetric one: the cosine part of (ii) controls only $\Re\,\mathrm{bv}\,\mathfrak{X}$ on J (Corollary 5.8), and Lemma 5.4 needs the full boundary value. The window itself is used only to make $\mathscr{Z}_\xi$ holomorphic across J, so that the right side of (ii) is an ordinary integral. Cancellation on J between the terms of different zeros off the line is excluded by the half-plane: $\mathfrak{X}$ is holomorphic in $\Im w>0$ because the ordinates are positive, and the boundary value of such a function of polynomial growth cannot vanish on an interval unless the function vanishes (Lemma 5.4); positivity at an imaginary frequency then gives $d_k=0$ for every k.

(c) The Davenport–Heilbronn function [92, 93] has a functional equation of the same type and a Dirichlet series, and zeros off its critical line; an argument proving (ii) must therefore use the Euler product.

(d) Literature. Cramér [66] continued V_+; Guillera [82] proved height-regularized forms of the real part of such sums at non-prime-power arguments and stated Abel summation in the ordinal index (his Corollary 4.4), leaving a version with ordinal weights depending on N as a conjecture (his Conjecture 4.5); Arias de Reyna [2] proved that RH is equivalent to the temperedness of a measure built from the primes, which then equals the Fourier transform of the symmetric measure of the ordinates. The form (ii) differs in three respects: it is local (one window without prime atoms), one-sided ($\sum_k e^{iu\gamma_k}$ against V_+), and its mechanism is

Lemma 5.4 with the positivity (5.8) at an imaginary frequency. We have not found it in the literature; the search included the Fourier-quasicrystal literature [48, 46] but was not exhaustive.

5.3. **One window is one cell: the cosine and sine identities.** Define the distribution

$$\langle \mathscr{Y}, \mathsf{f} \rangle := \sum_{k \geq 1} \int_0^\infty \mathsf{f}(u)\big(\cosh(d_k u) - 1\big)\cos(\gamma_k u)\, du \qquad (\mathsf{f} \in C_c^\infty(0,\infty)), \tag{5.15}$$

which converges absolutely by (5.2); only zeros off the critical line contribute, and $\mathscr{Y} = 0$ under RH. By (5.7), $\Re \operatorname{bv} \mathfrak{X} = -\mathscr{Y}$ on every interval of $(0,\infty)$ free of the points $\log p^m$. Let $J(x) = \sum_{p^m \leq x} \frac{1}{m}$ be Riemann's prime-power count.

Theorem 5.7 (The rung functional of the endpoint is a sum over the zeros off the line)**.** *For every real* $\mathsf{f} \in C_c^\infty(0,\infty)$*,*

$$\mathsf{R}_\mathsf{f}(0) - \frac{1}{2}\sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}}\, \mathsf{f}(\log n) = \langle \mathscr{Y}, \mathsf{f} \rangle. \tag{5.16}$$

Equivalently, in $\mathscr{D}'(1,\infty)$*,*

$$(\log x)\Big(\frac{dJ_{Z1}^{\mathrm{M}}}{dx} - \frac{dJ}{dx}\Big) = \frac{2}{\sqrt{x}}\, \mathscr{Y}(\log x). \tag{5.17}$$

Here the derivatives are distributional derivatives on $(1,\infty)$ *(*dJ/dx *is the measure* $\sum_{p^m} m^{-1}\delta_{p^m}$*), and* $\mathscr{Y}(\log x)$ *denotes the distribution* $\sum_k (\cosh(d_k \log x) - 1)\cos(\gamma_k \log x)$ *on* $(1,\infty)$*, the series converging in* $\mathscr{D}'(1,\infty)$*. In particular, if the support of* f *contains no point* $\log n$*, then* $\mathsf{R}_\mathsf{f}(0) = \langle \mathscr{Y}, \mathsf{f} \rangle$ *and the cosine difference of Proposition 15.44 is*

$$\mathsf{S}_\mathsf{f}(0) = \sum_{k \geq 1} \big(\mathsf{c}_\mathsf{f}(\gamma_k) - \mathsf{c}_\mathsf{f}(\tau_k)\big) = \mathsf{R}_\mathsf{f}(1) - \langle \mathscr{Y}, \mathsf{f} \rangle. \tag{5.18}$$

Proof. By (15.80), $\mathsf{R}_\mathsf{f}(0) = \mathsf{k}_\mathsf{f} - \sum_k \mathsf{c}_\mathsf{f}(\gamma_k)$. Pair the explicit formula $\sum_n \Lambda(n)\delta_n = 1 - \sum_\rho x^{\rho-1} - x^{-1}(x^2-1)^{-1}$ in $\mathscr{D}'(1,\infty)$ (proof of Proposition 5.3) with $\frac{1}{2}\mathsf{f}(\log x)x^{-1/2}$ and put $u = \log x$: the term 1 and the last term give k_f as in the proof of Proposition 15.44, and the zeros give $\frac{1}{2}\sum_\rho \int \mathsf{f}(u)e^{(\rho-\frac{1}{2})u}du = \sum_k \int \mathsf{f}(u)\cosh(d_k u)\cos(\gamma_k u)du$ (pairs $\rho, \bar\rho$ and the pair symmetry in d). Hence $\frac{1}{2}\sum_n \Lambda(n)n^{-1/2}\mathsf{f}(\log n) = \mathsf{k}_\mathsf{f} - \sum_k \int \mathsf{f}(u)\cosh(d_k u)\cos(\gamma_k u)du$, and subtraction gives (5.16). Theorem 13.6 for $Z1$ and the same explicit formula for J, whose derivative is $(\log x)^{-1}\sum_n \Lambda(n)\delta_n$, give (5.17) by subtraction. The last statement is (5.16) with the prime sum empty, together with (15.81). □

Theorem 5.7 makes the defect of the endpoint explicit: on every rung, the defect $\mathsf{R}_\mathsf{f}(0)$, which (D) requires to vanish (Proposition 15.44), is the pairing of f with the zeros off the critical line, and the transport from the clock is $\mathsf{R}_\mathsf{f}(1) - \langle \mathscr{Y}, \mathsf{f} \rangle$. Any lemma that proves $\mathsf{S}_\mathsf{f}(0) = \mathsf{R}_\mathsf{f}(1)$ on a rung proves $\langle \mathscr{Y}, \mathsf{f} \rangle = 0$ there.

Corollary 5.8 (One window is one cell)**.** *Let* $J = (\log 2, \log 3)$*.*

(a) $\Re \operatorname{bv} \mathfrak{X} = 0$ *on* $J \iff \langle \mathscr{Y}, \mathsf{f} \rangle = 0$ *for every* $\mathsf{f} \in C_c^\infty(J) \iff \mathsf{R}_\mathsf{f}(0) = 0$ *for every* $\mathsf{f} \in C_c^\infty(J) \iff J_{Z1}^{\mathrm{M}}$ *is constant on* $(2,3)$*, the rung* $N = 3$ *of Theorem 15.36.*

(b) *RH holds if and only if* J_{Z1}^{M} *is constant on* $(2,3)$ *and, for every real* $\mathsf{f} \in C_c^\infty(J)$*, the sine identity*

$$\sum_{k \geq 1} \int_J \mathsf{f}(u)\sin(\gamma_k u)\, du = \int_J \mathsf{f}(u)\, e^{-u/2}\, \Im V_+(u)\, du \tag{5.19}$$

holds, with V_+ *given on* J *by* (5.11)*.*

(c) *Consequently (D), which is equivalent to the absence of mass of* dJ_{Z1}^{M} *outside the integers on every rung (Theorem 15.36, Corollary 15.38), is equivalent to the single rung* $(2,3)$ *together with* (5.19) *on* J*.*

Proof. (a) On $(2,3)$, $dJ=0$, so by (5.17) J^{M}_{Z1} is constant there if and only if $\mathscr{Y}=0$ on J; the other equivalences are (5.16) and $\Re\,\mathrm{bv}\,\mathfrak{X}=-\mathscr{Y}$. (b) $\mathrm{bv}\,\mathfrak{X}=0$ on J if and only if its real and imaginary parts vanish; by Theorem 5.5 and (5.13) the real part is (a) and the imaginary part is (5.19). (c) is (b) with the equivalence of (D) and RH (Theorem 12.2). □

Remark 5.9 (The conjugate part is not local). Put $Q(u):=2\mathscr{Y}(u)=u\,e^{u/2}(\frac{dJ^{\mathrm{M}}_{Z1}}{dx}-\frac{dJ}{dx})(e^u)$. For a holomorphic function on the upper half-plane with suitable decay, the boundary values satisfy $\mathrm{bv}=-\frac12(Q+i\mathcal{H}Q)$ with $\mathcal{H}$ the Hilbert transform; $\mathfrak{X}$ grows like $e^{\max|d_k|\,|\Re w|}$ and the relation holds only after a regularization, so the following is a heuristic description, not used in any proof. If $Q=0$ on J, then on J, $\mathcal{H}Q(u)=\frac1\pi\,\mathrm{pv}\!\int_{\mathbb{R}\setminus J}\frac{Q(v)}{u-v}dv$ is produced by the values of Q outside the window: a mass at the end point, $Q=\delta_{\log 2}$, vanishes on J and has $\mathcal{H}Q(u)=1/(\pi(u-\log 2))\neq 0$ there. The terms $\Lambda(n)e^{(1+a)(u-\log n)}/(u-\log n)$ of (5.12) are of this kind. Thus the cell $(2,3)$ controls the cosine part of the window, and the sine identity (5.19) carries the information of all other cells into the window.

Proposition 5.10 (Finitely many zeros off the line)**.** *Suppose that the set of zeros with $d_k\neq 0$ is finite and non-empty. Then for every open interval $I\subset(0,\infty)$ there is a real $\mathsf{f}\in C_c^\infty(I)$ with $\langle\mathscr{Y},\mathsf{f}\rangle>0$. Consequently, if J^{M}_{Z1} is constant on one interval $(n,n+1)$, $n\geq 1$, then either RH holds or infinitely many zeros lie off the critical line.*

Proof. $\mathscr{Y}$ is then the finite exponential polynomial $\mathscr{Y}(u)=\sum_{k\in O}(\cosh(d_ku)-1)\cos(\gamma_ku)$, O the finite set of indices with $d_k\neq 0$. Writing $\cosh(du)\cos(\gamma u)=\frac14\sum_{\epsilon,\epsilon'=\pm1}e^{(\epsilon d+i\epsilon'\gamma)u}$, let $A=\max_{k\in O}|d_k|>0$ and γ_0 the ordinate of an index where it is attained. The exponent $A+i\gamma_0$ occurs only for the $m\geq 1$ indices with $|d_k|=A$, $\gamma_k=\gamma_0$, each with coefficient $\frac14$, so its total coefficient is $m/4>0$, also when several indices share the ordinate γ_0; the terms $-\cos(\gamma_ku)$ have exponents of real part 0. Distinct exponentials are linearly independent on every interval, so $\mathscr{Y}\not\equiv 0$ on I, and $\mathscr{Y}$ is real-analytic. Take $\mathsf{f}=\chi\mathscr{Y}$ with $\chi\geq 0$ a bump in $C_c^\infty(I)$ positive on a subinterval; then $\langle\mathscr{Y},\mathsf{f}\rangle=\int\chi\mathscr{Y}^2>0$. The last statement follows from (5.17), because $dJ=0$ on $(n,n+1)$. □

The proposition is the contrapositive of the easy direction for a finite exceptional set; with infinitely many zeros off the line $\mathscr{Y}$ is only a distribution, and a test function supported in one cell does not separate ordinates closer than about $1/\log 2$, so infinitely many terms might in principle conspire.

Theorem 5.11 (Frullani form of the value at infinity)**.** *For $K\geq 1$ put*

$$\mathscr{Q}_K:=2\int_0^\infty\frac{e^{-u/2}}{u}\sum_{k\leq K}\big(\cos(\gamma_ku)-\cos(\tau_ku)\big)\,du. \tag{5.20}$$

Then $\mathscr{Q}_K=\sum_{k\leq K}\log\frac{\frac14+\tau_k^2}{\frac14+\gamma_k^2}$, the limit $\mathscr{Q}_\infty=\lim_{K\to\infty}\mathscr{Q}_K$ exists, and

$$\mathscr{Q}_\infty=\log L+\log\varpi_{Z1}=\log L+\sum_{pairs}\log c_\gamma\ \geq\ \log L>0.0206, \tag{5.21}$$

with c_γ the pair factor of (13.11)*. Hence RH holds if and only if $\mathscr{Q}_\infty=\log L=-\log\varpi_{\mathrm{RVM}}$.*

Proof. For real t, Frullani's integral gives $\int_0^\infty(1-\cos tu)e^{-u/2}u^{-1}du=\frac12\log(1+4t^2)$, so $\log(\frac14+t^2)=-\log 4+2\int_0^\infty(1-\cos tu)e^{-u/2}u^{-1}du$ and $\log\frac{\frac14+\tau^2}{\frac14+\gamma^2}=2\int_0^\infty(\cos\gamma u-\cos\tau u)e^{-u/2}u^{-1}du$, an absolutely convergent integral. Summing over $k\leq K$ gives the first statement. By Theorem 13.1, $\lim_K\prod_{k\leq K}\frac{1/4+\tau_k^2}{1/4+\gamma_k^2}=\varpi_{Z1}/\varpi_{\mathrm{RVM}}=\varpi_{Z1}L$; Theorem 13.2 gives $\log\varpi_{Z1}=\sum\log c_\gamma\geq 0$ with equality exactly under RH, and Theorem 9.11 gives $\log L>0.020615$. □

Remark 5.12 (The arithmetic of both sides and the size of the defect). (a) Inserting the explicit formula, $\sum_k \cos(\gamma_k u) = \mathsf{k}(u) - \frac{1}{2}\sum_n \Lambda(n) n^{-1/2}\delta_{\log n} - \mathscr{Y}(u)$ (proof of Theorem 5.7), into (5.20) returns the same identity: by Frullani, $\int_0^\infty e^{-u/2}(\cosh(du)-1)\cos(\gamma u)\frac{du}{u} = \frac{1}{4}\log\frac{(\frac{1}{4}+\gamma^2)^2}{|\rho|^2|1-\rho|^2} = -\frac{1}{4}\log c_\gamma$ for each zero $\rho = \frac{1}{2} + d + i\gamma$, so the off-line part of (5.20) is exactly $-\log\varpi_{Z1}$, and what remains is $\zeta(\sigma) \to 1$ and the value ϖ_{RVM} already used in Theorem 13.2. With this kernel the arithmetic of the two sides contracts with ratio exactly 1; a proof of $\log\varpi_{Z1} \le 0$ along these lines needs a kernel and a sign on the prime side giving a ratio < 1, the structure of Weil's positivity and Li's criterion.

(b) Every estimate gives $\log\varpi_{Z1} \le \varepsilon$ with $\varepsilon > 0$: by Proposition 14.3, $0 \le \log\varpi_{Z1} < 4\cdot 10^{-13}$, and zero-density estimates improve the constant only. There is no gap below: one pair with $d \to 0$ or $\gamma \to \infty$ contributes $\log c_\gamma \approx 2d^2/\gamma^2 \to 0$.

(c) The criterion $\log\varpi_{Z1} = \sum \log c_\gamma \ge 0$, with equality exactly under RH, is of the type of the integral criteria of Volchkov [90], Balazard–Saias–Yor [89] and Sekatskii–Beltraminelli–Merlini [91], which also express a sum over the zeros off the line, with weights of first order in d, as an explicit quantity; here the weights are of second order, $\log c_\gamma = \frac{2d^2(\gamma^2-\frac{1}{4})+d^4}{(\frac{1}{4}+\gamma^2)^2} + O(d^4\gamma^{-4})$.

5.4. **Centroids of the two spectra.** The triangular centroid of a list (o_k) at the frequency u is the mean of the partial sums of its exponential walk,

$$C^{\mathrm{path}}(N;u) = \frac{1}{N}\sum_{n\le N}\sum_{k\le n} e^{iuo_k} = \sum_{k\le N}\Big(1 - \frac{k-1}{N}\Big)e^{iuo_k}; \tag{5.22}$$

for $o_k = \tau_k$ it is the path centroid $C^{\mathrm{path}}_{\mathrm{RVM}}(N;u)$ of (4.2), for $o_k = \gamma_k$ it is $C^{\mathrm{path}}_{\zeta}(N;u)$ of Theorem 5.5. In three dimensions, with k on the vertical axis, the walk $(\Re S_k, \Im S_k, k)$ has its centroid at $(\Re C^{\mathrm{path}}, \Im C^{\mathrm{path}}, \frac{N+1}{2})$ (Figures 10, 11). The normalized sums S_N/N tend to 0 for both lists (Corollary 4.2 and Hlawka's theorem [80]); the centroid is the mean of the partial sums, and it does not tend to 0.

Theorem 5.13 (The clock centroid converges)**.** *Let $u > 0$, $E_k = e^{iu\tau_k}$, $S_n = \sum_{k\le n} E_k$, $b_k = (e^{iu(\tau_{k+1}-\tau_k)} - 1)^{-1}$ (this is $\mathfrak{b}^{\mathrm{sbp}}_k$ of the proof of Theorem 4.1), and let $k_0 = k_0(u)$ be as in that proof. The series*

$$C^{\mathrm{path}}_{\mathrm{RVM}}(\infty;u) := \sum_{k<k_0} E_k - b_{k_0}E_{k_0} + \sum_{k>k_0}(b_{k-1}-b_k)E_k \tag{5.23}$$

converges, $S_n - b_nE_{n+1} = C^{\mathrm{path}}_{\mathrm{RVM}}(\infty;u) + O_u(\log n/n)$, and

$$C^{\mathrm{path}}_{\mathrm{RVM}}(N;u) = C^{\mathrm{path}}_{\mathrm{RVM}}(\infty;u) + O_u\Big(\frac{(\log N)^2}{N}\Big). \tag{5.24}$$

Proof. Put $d^{\mathrm{b}}_k := b_{k-1} - b_k$. By (4.9), for $n \ge k_0$, $S_n = \sum_{k<k_0} E_k - b_{k_0}E_{k_0} + \sum_{k_0<k\le n} d^{\mathrm{b}}_k E_k + b_nE_{n+1}$. Write $b_k = \beta(D(k))$ with $\beta(x) = (e^{ix}-1)^{-1}$ and $D(y) = u(\mathcal{T}(y+\frac{1}{2}) - \mathcal{T}(y-\frac{1}{2}))$, $\mathcal{T}$ the inverse clock. On $0 < x \le \pi$, $|\beta^{(j)}(x)| \ll_j x^{-1-j}$; by (4.4), differentiating $\mathcal{T}'' = -4\pi^2/(\mathcal{T}\ell^3)$ with $\ell = \log(\mathcal{T}/2\pi)$ and $\ell' = \mathcal{T}'/\mathcal{T}$ gives $\mathcal{T}''' = 4\pi^2\mathcal{T}'(\ell^3+3\ell^2)/(\mathcal{T}^2\ell^6) = 8\pi^3(\ell+3)/(\mathcal{T}^2\ell^5)$, so that $\mathcal{T}' \asymp \ell^{-1}$, $|\mathcal{T}''| \asymp 1/(\mathcal{T}\ell^3)$ and $\mathcal{T}''' \ll 1/(\mathcal{T}^2\ell^4)$; hence $D \asymp 1/\ell$, $D' \ll 1/(\mathcal{T}\ell^3)$ and $D'' \ll 1/(\mathcal{T}^2\ell^4)$, so $(\beta\circ D)' \ll 1/(\mathcal{T}\ell)$ and $(\beta\circ D)'' \ll 1/(\mathcal{T}\ell)^2$. With $\tau_k\ell_k \asymp k$ (4.7): $|d^{\mathrm{b}}_k| \ll 1/k$ and $|d^{\mathrm{b}}_k - d^{\mathrm{b}}_{k+1}| \ll 1/k^2$. Abel summation with $S_k = O(\log k)$ (Theorem 4.1) gives, for $M > n$, $\sum_{n<k\le M} d^{\mathrm{b}}_k E_k = d^{\mathrm{b}}_M S_M - d^{\mathrm{b}}_{n+1}S_n + \sum_{n<k<M}(d^{\mathrm{b}}_k - d^{\mathrm{b}}_{k+1})S_k = O(\log n/n)$ uniformly in M. Hence the series (5.23) converges and $S_n = C^{\mathrm{path}}_{\mathrm{RVM}}(\infty;u) + b_nE_{n+1} + O(\log n/n)$. Averaging over $n \le N$, the error terms contribute $O((\log N)^2/N)$ and the indices $n < k_0$ contribute $O(k_0\log k_0/N)$. Finally

$E_{n+1} = b_{n+1}(E_{n+2} - E_{n+1})$ by (4.8); with $c_n := b_n b_{n+1}$ the exact identity

$$\sum_{n=k_0}^{N} b_n E_{n+1} = \sum_{n=k_0}^{N} c_n(E_{n+2} - E_{n+1}) = c_N E_{N+2} - c_{k_0} E_{k_0+1} - \sum_{n=k_0+1}^{N} (c_n - c_{n-1}) E_{n+1}$$

holds, and $|c_n| \ll \log^2 n$, $|c_n - c_{n-1}| = |b_n|\,|b_{n+1} - b_{n-1}| \le |b_n|(|d_n^{\mathrm{b}}| + |d_{n+1}^{\mathrm{b}}|) \ll \log n/n$ give $\sum_{k_0 \le n \le N} b_n E_{n+1} = O((\log N)^2)$. This is (5.24). □

This answers the question left open after Corollary 4.2: Theorem 4.1 alone does not give the convergence of the path centroid, and the second summation by parts does. Numerically $C_{\mathrm{RVM}}^{\mathrm{path}}(\infty; 1) = 0.041569 + 1.757541\,i$ (Experiment 19.7).

For the ordinates the same quantity is controlled by Cramér's function. We use the following local Tauberian lemma; Tauberian theorems with local boundary behaviour are developed in [70, 69], and we give a self-contained proof of the case needed.

Lemma 5.14 (A local Tauberian lemma)**.** *Let $F : [0, \infty) \to \mathbb{C}$ be locally absolutely continuous, $F = 0$ on $[0, x_0]$, with $|F(x)| \le C \log(2 + x)$ and $|F'(x)| \le C \log(2 + x)/(1 + x)$ almost everywhere. Suppose that for $\Re s > 0$*

$$\int_0^\infty F(x) e^{-sx}\, dx = \frac{c}{s} + c' \operatorname{Log} s + \Phi(s), \tag{5.25}$$

with constants c, c', Log *the principal branch on $\Re s > 0$, and Φ holomorphic in a disc $|s| < r$. Then $F(x) \to c$ as $x \to \infty$.*

Proof. Let $K(y) = \kappa_0(\sin(\alpha y)/(\alpha y))^6$ with $\alpha = r/12$ and κ_0 chosen so that $\int K = 1$; its Fourier transform $\widehat{K}(\xi) = \int K(y) e^{-i\xi y} dy$ is continuous and supported in $[-r/2, r/2]$, and $K(y) \ll y^{-6}$. Put $f = F - c$ on $(0, \infty)$ and $f = 0$ on $(-\infty, 0]$; the jump at 0 is harmless, and the transform used below is the unilateral Laplace transform of $F - c$, which is $c' \operatorname{Log} s + \Phi(s)$ by (5.25). For $\sigma > 0$, $f_\sigma(x) = f(x) e^{-\sigma x}$ is integrable with Fourier transform $c' \operatorname{Log}(\sigma + i\xi) + \Phi(\sigma + i\xi)$ on $|\xi| < r$, so $(f_\sigma * K)(x) = \frac{1}{2\pi} \int (c' \operatorname{Log}(\sigma + i\xi) + \Phi(\sigma + i\xi)) \widehat{K}(\xi) e^{i\xi x} d\xi$. As $\sigma \downarrow 0$ the left side tends to $(f * K)(x)$ by dominated convergence, and the right side tends to the same integral with $\sigma = 0$, because $|\operatorname{Log}(\sigma + i\xi)| \le |\log|\xi|| + C$ on $|\xi| \le r/2$; the boundary values are $\operatorname{Log}(i\xi) = \log|\xi| + \frac{i\pi}{2} \operatorname{sgn} \xi$. The function $(c' \operatorname{Log}(i\xi) + \Phi(i\xi)) \widehat{K}(\xi)$ is integrable, so $(f * K)(x) \to 0$ by the Riemann–Lebesgue lemma. Finally $f(x) - (f * K)(x) = \int (f(x) - f(x - y)) K(y) dy$ (using $\int K = 1$); let $x \ge 4x_0 + 4$. For $|y| \le \sqrt{x}$, $x - y \ge x - \sqrt{x} > x_0$, and $|f(x) - f(x - y)| \le |y| \sup_{[x - \sqrt{x}, x + \sqrt{x}]} |F'| \ll |y| \log x/x$, which integrates against K to $O(\log x/x)$. For $|y| > \sqrt{x}$ with $x - y > 0$, $|f(x) - f(x - y)| \le |F(x)| + |F(x - y)| + 2|c| \ll \log(2 + x) + \log(2 + |y|)$; for $|y| > \sqrt{x}$ with $x - y \le 0$, $f(x - y) = 0$ and $|f(x) - f(x - y)| = |f(x)| \ll \log(2 + x)$. Both cases integrate against $K(y) \ll y^{-6}$ over $|y| > \sqrt{x}$ to $O(\log(2 + x)\, x^{-5/2})$. Hence $f(x) \to 0$. □

Theorem 5.15 (The ordinate centroid under RH)**.** *Assume RH. For every $u \in (0, \infty) \setminus \{\log p^m\}$,*

$$\lim_{N \to \infty} C_\zeta^{\mathrm{path}}(N; u) = \lim_{N \to \infty} \sum_{k \le N} \Big(1 - \frac{k - 1}{N}\Big) e^{iu\gamma_k} = e^{-u/2}\, V_+(u). \tag{5.26}$$

For $u = 1$ the limit is $e^{-1/2} V_+(1)$, whose real part is $\frac{1}{2\sqrt{e}}(e - \frac{1}{e^2 - 1}) = 0.7768942607932\ldots$ by (5.13)*, and whose numerical value is $0.77689426079325\ldots + 0.76785566886\ldots\, i$ (Experiment 19.7).*

Proof. Let $B_u(t) = \sum_{\gamma_k \le t} e^{iu\gamma_k}$ and $\mathfrak{e} = N_\zeta - \mathcal{N}_{\mathrm{RVM}}$ as in (13.1). Under RH, $x^\rho = \sqrt{x}\, x^{i\gamma}$ and Landau's formula $\sum_{0 < \gamma \le T} x^\rho = -\frac{T}{2\pi} \Lambda(x) + O_x(\log T)$ [67, 68] at $x = e^u$, which is not a prime power, give $B_u(T) = O_u(\log T)$.

Step 1 (index form). It suffices to take N at the end of a block of equal ordinates. A block has $m(\gamma) \ll \log \gamma$ elements (proof of Lemma 2.2), and $|S_n| \le |B_u(\gamma_n)| + m(\gamma_n) \ll \log \gamma_n$; if N' is the end of the block containing N, then $C_\zeta^{\mathrm{path}}(N') - C_\zeta^{\mathrm{path}}(N) = \frac{1}{N'} \sum_{N < n \le N'} S_n + (\frac{1}{N'} - \frac{1}{N}) \sum_{n \le N} S_n$,

and both terms are $O(m(\gamma_N)\log\gamma_N/N) = O(\log^2\gamma_N/N)$. Put $T=\gamma_N$. Summing over the block of each ordinate, $\sum_{k\le N}(N-k+1)e^{iu\gamma_k} = \int_{(0,T]}(N+\frac12 - N^*_\zeta(t))e^{iut}dN_\zeta(t)$ exactly, with the symmetric count (2.10): for a block $k=j+1,\dots,j+m$ at the ordinate γ, $N^*_\zeta(\gamma)=j+\frac{m}{2}$ and $\sum_{k=j+1}^{j+m}(N-k+1) = m(N+1)-mj-\frac{m(m+1)}{2} = m(N+\frac12-N^*_\zeta(\gamma))$, which is the mass of the right side at γ. Write $N^*_\zeta = \mathcal{N}_{\mathrm{RVM}}+\mathfrak{e}^*$, $\mathfrak{e}^*=\frac12(\mathfrak{e}(t+0)+\mathfrak{e}(t-0))$, and $dN_\zeta = \mathcal{N}'_{\mathrm{RVM}}dt + d\mathfrak{e}$. With $G_N = N+\frac12-\mathcal{N}_{\mathrm{RVM}}$, continuous, and $G_N(T)=\mathfrak{e}(T)+\frac12 = O(\log T)$,

$$\int_{(0,T]} G_N e^{iut}dN_\zeta = G_N(T)B_u(T) + \int_0^T B_u(t)\mathcal{N}'_{\mathrm{RVM}}(t)\,dt.$$

The remaining term is $\int \mathfrak{e}^* e^{iut}dN_\zeta = \int_{t_*}^T \mathfrak{e}\,e^{iut}\mathcal{N}'_{\mathrm{RVM}}dt + \frac12\int_{(t_*,T]} e^{iut}d(\mathfrak{e}^2)$, $t_* = \tau_0 = 4\pi < \gamma_1$ (below t_* the measure dN_ζ vanishes), because $\mathfrak{e}^*d\mathfrak{e} = \frac12 d(\mathfrak{e}^2)$: at a jump from a to b, $\frac{a+b}{2}(b-a) = \frac{b^2-a^2}{2}$, and between the jumps $\mathfrak{e}$ is smooth and $\mathfrak{e}^*=\mathfrak{e}$. Integration by parts and Selberg's bound $\int_0^T S(t)^2dt \ll T\log\log T$ [58] give $\frac12\int e^{iut}d(\mathfrak{e}^2) = O(\log^2 T)+O(T\log\log T)$. Let $F_u(t)=\int_{t_*}^t \mathfrak{e}(v)e^{iuv}dv$. Since $B_u(t) = \int_{t_*}^t e^{iuv}\mathcal{N}'_{\mathrm{RVM}}(v)dv + [e^{iuv}\mathfrak{e}(v)]_{t_*}^t - iuF_u(t)$ and the first integral is $O_u(\log t)$, $F_u(t)=O_u(\log t)$, so $\int_{t_*}^T \mathfrak{e}\,e^{iut}\mathcal{N}'_{\mathrm{RVM}}dt = \mathcal{N}'_{\mathrm{RVM}}(T)F_u(T) - \int_{t_*}^T F_u\mathcal{N}''_{\mathrm{RVM}}dt = O(\log^2 T)$. As $N\asymp T\log T$,

$$C^{\mathrm{path}}_\zeta(N;u) = \frac{1}{N}\int_0^T B_u(t)\mathcal{N}'_{\mathrm{RVM}}(t)\,dt + o(1).$$

Step 2. Let $A_u(x)=\int_0^x B_u$ and $\mathcal{M}_u(x)=A_u(x)/x$. If $\mathcal{M}_u(x)\to c$, write $A_u(t)=ct+r(t)$; for every $\epsilon>0$ there is t_ϵ with $|r(t)|\le\epsilon t$ for $t\ge t_\epsilon$. Then $\int_0^T B_u\mathcal{N}'_{\mathrm{RVM}} = \mathcal{N}'_{\mathrm{RVM}}(T)A_u(T) - \int_{t_*}^T A_u\mathcal{N}''_{\mathrm{RVM}}$, with $\mathcal{N}''_{\mathrm{RVM}}(t) = 1/(2\pi t)$, $\mathcal{N}'_{\mathrm{RVM}}(T)A_u(T) = cT\mathcal{N}'_{\mathrm{RVM}}(T)+o(T\log T)$ and $\int_{t_*}^T A_u\mathcal{N}''_{\mathrm{RVM}} = \frac{c(T-t_*)}{2\pi} + O_\epsilon(1)+O(\epsilon T)$; since $T\mathcal{N}'_{\mathrm{RVM}}(T) - \frac{T}{2\pi} = \mathcal{N}_{\mathrm{RVM}}(T) - \frac78$, this is $c\mathcal{N}_{\mathrm{RVM}}(T)+o(T\log T)$. With $N=\mathcal{N}_{\mathrm{RVM}}(T)+O(\log T)$ this gives $C^{\mathrm{path}}_\zeta(N;u)\to c$.

Step 3. For $\Re s>0$, $\int_0^\infty e^{-sx}dB_u(x) = \mathscr{Z}_\gamma(u+is) =: h(s)$. Under RH $h(s)=\mathscr{Z}_\xi(u+is) = e^{-(u+is)/2}V_+(u+is)$, which by Proposition 5.3 is holomorphic in $|s|<r_u = \min(u, \min_n|u-\log n|)$. Since $|B_u(x)|\ll\log(2+x)$ and $|A_u(x)|\ll x\log(2+x)$, two integrations by parts give $\int_0^\infty e^{-sx}A_u(x)dx = h(s)/s^2$ for $\Re s>0$. For real $s>0$, Fubini's theorem (the double integral of $|A_u(x)|e^{-\sigma x}$ over $x>0$, $\sigma>s$ is $\int|A_u|e^{-sx}x^{-1}dx<\infty$) gives $\int_0^\infty e^{-sx}\mathcal{M}_u(x)dx = \int_0^\infty A_u(x)\int_s^\infty e^{-\sigma x}d\sigma\,dx = \int_s^\infty h(\sigma)\sigma^{-2}d\sigma$. Writing $h(\sigma)=h(0)+h'(0)\sigma+(h(\sigma)-h(0)-h'(0)\sigma)$ on $(s,r_u/2)$ gives, for real $0<s<r_u/2$, $\int_s^\infty h(\sigma)\sigma^{-2}d\sigma = \frac{h(0)}{s} - h'(0)\operatorname{Log}s + \Phi(s)$, where $\Phi'(s) = -(h(s)-h(0)-h'(0)s)/s^2$ is holomorphic in $|s|<r_u$. Both sides of this identity are holomorphic in $\{\Re s>0\}\cap\{|s|<r_u\}$ (the left side as a Laplace transform), so it holds there by analytic continuation, which is the form (5.25). Moreover $|\mathcal{M}_u|\ll\log x$ and $\mathcal{M}'_u = (B_u-\mathcal{M}_u)/x \ll \log x/x$. Lemma 5.14 gives $\mathcal{M}_u(x)\to h(0)=e^{-u/2}V_+(u)$, and Step 2 gives (5.26). □

Remark 5.16 (Status of the centroid value). Without RH neither the existence of $\lim_N C^{\mathrm{path}}_\zeta(N;1)$ nor its value is proved: Step 3 uses RH to identify $\mathscr{Z}_\gamma$ with the zero-free function $\mathscr{Z}_\xi$, and Step 1 uses $B_u(T)=O(\log T)$. The equality (5.26) at $u=1$ is implied by RH; the converse does not follow from the argument of Proposition 5.2 (Remark 5.6(b)). The real part of the value is Cramér's and Guillera's [66, 82]; we have not found the imaginary part $0.76785566886\ldots$, or the complex limit of the triangular centroid, in the literature consulted. At $u=\log p^m$ the centroid diverges. Under RH, Landau's formula gives $B_u(T) = -\frac{T}{2\pi}\frac{\log p}{p^{m/2}} + O(\log T)$; with $\gamma_n\sim 2\pi n/\log n$ and $|S_n - B_u(\gamma_n)|\le m(\gamma_n)$ this is $S_n = -\frac{\log p}{p^{m/2}}\frac{n}{\log n}(1+o(1))$, and averaging, $C^{\mathrm{path}}_\zeta(N;u) = -\frac{\log p}{2p^{m/2}}\frac{N}{\log N}(1+o(1)) = -\frac{\log p}{4\pi p^{m/2}}\gamma_N(1+o(1))$. For the clock there is no resonance (Theorem 5.13 holds for every $u>0$).

6. The RVM explicit formula, J_{RVM}, and π_{RVM}

For $w = a + ib$ with $b > 0$ define

$$\mathrm{Ei}(w) := \int_{-\infty}^{a} \frac{e^{v+ib}}{v+ib}\, dv, \qquad \mathrm{Ei}(\bar{w}) := \overline{\mathrm{Ei}(w)}. \tag{6.1}$$

The path is the horizontal line $\Im = b$, which avoids 0; this is the branch of $\mathrm{Ei}(\rho \log x)$ used in Riemann's explicit formula [37]. For $x > 1$ put

$$J_{\mathrm{RVM}}(x) := \mathrm{li}(x) - 2\Re \sum_{k\geq 1} \mathrm{Ei}\Big(\big(\frac{1}{2} + i\tau_k\big)\log x\Big) - \log 2 + \int_x^{\infty} \frac{dy}{y(y^2-1)\log y}, \tag{6.2}$$

the zero sum being taken in the order of k.

Lemma 6.1 (Asymptotics of Ei off the real axis)**.** *For $w = a + ib$, $b > 0$,*

$$\mathrm{Ei}(w) = \frac{e^w}{w} + \varrho(w), \qquad |\varrho(w)| \leq \frac{e^a}{b^2}. \tag{6.3}$$

Proof. Integrate (6.1) by parts: $\int_{-\infty}^{a} e^{v+ib}(v+ib)^{-1}dv = e^w/w + \int_{-\infty}^{a} e^{v+ib}(v+ib)^{-2}dv$, the boundary term at $-\infty$ being 0. The last integral is bounded by $\int_{-\infty}^{a} e^v b^{-2} dv = e^a/b^2$. □

Theorem 6.2 (Compact-uniform existence of the RVM zero sum)**.** *The series $\sum_k \Re\,\mathrm{Ei}\big((\frac{1}{2} + i\tau_k)u\big)$ converges uniformly for u in every compact interval $[u_1, u_2] \subset (0, \infty)$. Hence J_{RVM} is a continuous function on $(1, \infty)$.*

Proof. Apply Lemma 6.1 with $w = (\frac{1}{2} + i\tau_k)u$, $a = u/2$, $b = \tau_k u$. Since $\Re\big(e^{iu\tau_k}/(\frac{1}{2} + i\tau_k)\big) = (\frac{1}{2}\cos u\tau_k + \tau_k \sin u\tau_k)/(\frac{1}{4} + \tau_k^2)$,

$$\Re\,\mathrm{Ei}\big((\tfrac{1}{2} + i\tau_k)u\big) = \frac{e^{u/2}}{u}\,\frac{\sin(u\tau_k)}{\tau_k} + O_{u_1,u_2}(\tau_k^{-2}), \tag{6.4}$$

uniformly for $u \in [u_1, u_2]$. The error series converges absolutely and uniformly. For the main series, Abel summation with $S_{\mathrm{RVM}}(K; u) = O_{u_1,u_2}(1 + \log K)$ (Theorem 4.1) and

$$\frac{1}{\tau_K} - \frac{1}{\tau_{K+1}} \leq \frac{\tau_{K+1} - \tau_K}{\tau_K^2} \leq \frac{2\pi}{\ell_K \tau_K^2} \ll \frac{\log K}{K^2} \tag{6.5}$$

reduces the uniform Cauchy estimate to the convergent majorant $\sum_K (\log K)^2 K^{-2}$ (a majorant for the differences produced by Abel summation, not for the original terms), while the boundary term $S_{\mathrm{RVM}}(K; u)/\tau_K \to 0$ uniformly. Hence $\sum_k e^{iu\tau_k}/\tau_k$ converges uniformly, and so does its imaginary part. □

Define the support-truncated version

$$J^*_{\mathrm{RVM}}(x) := \begin{cases} J_{\mathrm{RVM}}(x), & x \geq 2, \\ 0, & 0 < x < 2, \end{cases} \tag{6.6}$$

set $\pi_{\mathrm{RVM}}(x) = 0$ for $0 < x < 2$, and for $x \geq 2$

$$\pi_{\mathrm{RVM}}(x) := \sum_{1 \leq m \leq \log_2 x} \frac{\mu(m)}{m}\, J^*_{\mathrm{RVM}}(x^{1/m}). \tag{6.7}$$

Proposition 6.3 (Exact finite prime-power inversion)**.** *For every $x \geq 2$,*

$$J^*_{\mathrm{RVM}}(x) = \sum_{1 \leq m \leq \log_2 x} \frac{1}{m}\, \pi_{\mathrm{RVM}}(x^{1/m}). \tag{6.8}$$

Proof. All sums are finite because $x^{1/m} < 2$ for $m > \log_2 x$. Substituting (6.7) into the right side of (6.8), the coefficient of $J^*_{\mathrm{RVM}}(x^{1/r})$ is

$$\frac{1}{r}\sum_{d\mid r}\mu(d), \tag{6.9}$$

which is 1 for $r = 1$ and 0 otherwise. □

6.1. **The square-root estimate.** For $u > 0$ put

$$\mathcal{A}_{\mathrm{RVM}}(u) := \sum_{k\ge 1}\frac{e^{iu\tau_k}}{\tau_k}, \tag{6.10}$$

summed in the order of k, and write $\ell(t) = \log(t/2\pi)$, so that $\mathcal{N}'_{\mathrm{RVM}}(t) = \ell(t)/2\pi$ and $\ell(\tau_k) = \ell_k$.

Lemma 6.4 (The weighted clock sum)**.** *For every $u \ge \log 2$ the series* (6.10) *converges, and*

$$|\mathcal{A}_{\mathrm{RVM}}(u)| \le \frac{\log^2(2+2u)}{4\pi} + \frac{\log(2+2u)}{2\pi} + C_{\mathcal{A}}, \tag{6.11}$$

with an absolute constant $C_{\mathcal{A}}$.

Proof. Put $Y_j = 2^j \cdot 4\pi(1+u)$ for $j \ge 0$.

Head $\tau_k \le Y_0$. Let $n_{\mathrm{RVM}}(t) = \#\{k : \tau_k \le t\}$. By (2.4), $\tau_k \le t$ if and only if $k \le \mathcal{N}_{\mathrm{RVM}}(t) + \frac12$, so $n_{\mathrm{RVM}}(t) \le \mathcal{N}_{\mathrm{RVM}}(t) + \frac12 = \frac{t}{2\pi}(\ell(t) - 1) + \frac{11}{8} \le \frac{t\,\ell(t)}{2\pi}$ for $t \ge \tau_1 > \frac{11\pi}{4}$. Stieltjes integration gives

$$\sum_{\tau_k\le Y_0}\frac{1}{\tau_k} = \frac{n_{\mathrm{RVM}}(Y_0)}{Y_0} + \int_{\tau_1}^{Y_0}\frac{n_{\mathrm{RVM}}(t)}{t^2}\,dt \le \frac{\ell(Y_0)}{2\pi} + \int_{\tau_1}^{Y_0}\frac{\ell(t)}{2\pi t}\,dt \le \frac{\ell(Y_0)}{2\pi} + \frac{\ell(Y_0)^2}{4\pi}, \tag{6.12}$$

and $\ell(Y_0) = \log(2+2u)$.

Blocks. Let $\mathcal{B}_j = \{k : Y_j < \tau_k \le Y_{j+1}\}$, let $\mathcal{T}$ be the inverse clock of (4.4), and put $\phi(y) = u\,\mathcal{T}(y - \frac12)/2\pi$, so that $e^{2\pi i\phi(k)} = e^{iu\tau_k}$. By (4.4),

$$\phi''(y) = -\frac{2\pi u}{\mathcal{T}\,\ell(\mathcal{T})^3}, \qquad \mathcal{T} = \mathcal{T}(y - \tfrac12). \tag{6.13}$$

On $\mathcal{B}_j$ put $\varkappa_j = 2\pi u/(Y_{j+1}\ell(Y_{j+1})^3)$. Since $Y_{j+1} = 2Y_j$ and $\ell(Y_{j+1}) \le 2\ell(Y_j)$ (because $\ell(Y_j) \ge \ell(Y_0) \ge \log 2$), we have $\varkappa_j \le -\phi'' \le 16\varkappa_j$ at the integers of the block, and $\varkappa_j \le -\phi'' \le 256\varkappa_j$ on the whole real interval $[a,b]$ over which a run of $\mathcal{B}_j$ is summed, since that interval may begin inside the previous block, across which $-\phi''$ changes by at most another factor 16; in particular ϕ'' keeps a constant sign on $[a,b]$, as the second-derivative test requires. The number of indices in $\mathcal{B}_j$ is at most $\mathcal{N}_{\mathrm{RVM}}(Y_{j+1}) - \mathcal{N}_{\mathrm{RVM}}(Y_j) + 1 \le Y_j\ell(Y_{j+1})/2\pi + 1$. The second-derivative test for exponential sums [61, Thm. 5.9] gives, for every run $a < k \le b$ of consecutive indices of $\mathcal{B}_j$,

$$\Big|\sum_{a<k\le b} e^{iu\tau_k}\Big| \le A_{\mathrm{vdC}}\Big(256\Big(\frac{Y_j\ell(Y_{j+1})}{2\pi} + 1\Big)\varkappa_j^{1/2} + \varkappa_j^{-1/2}\Big) =: \mathcal{P}_j, \tag{6.14}$$

with an absolute constant A_{vdC}; if the real interval $[a,b]$ contains at most one integer, the sum has at most one term and is bounded trivially by 1, and $1 \le \varkappa_j^{-1/2}$ because $\varkappa_j \le u/(4(1+u)\log^3 2) < 1$. The weights $1/\tau_k$ are positive and decreasing, and $1/\tau_k < 1/Y_j$ on $\mathcal{B}_j$, so Abel's inequality bounds the contribution of any run in $\mathcal{B}_j$ by $\mathcal{P}_j/Y_j$:

$$\frac{\mathcal{P}_j}{Y_j} \le A_{\mathrm{vdC}}\left(\frac{16}{2\pi}\Big(\frac{\pi u}{Y_j\ell(Y_{j+1})}\Big)^{1/2} + \frac{16}{Y_j} + \Big(\frac{\ell(Y_{j+1})^3}{\pi u Y_j}\Big)^{1/2}\right). \tag{6.15}$$

With $Y_j = 2^j \cdot 4\pi(1+u)$ and $\ell(Y_{j+1}) = \log(2^{j+2}(1+u)) \le j + 2 + \log(1+u)$, each of the three terms is at most an absolute constant times $(j+2)^{3/2}2^{-j/2}$, uniformly in $u \ge \log 2$, because $\sup_{u\ge\log 2}(1+\log(1+u))^3/(u(1+u)) < \infty$. Hence $\sum_{j\ge0}\mathcal{P}_j/Y_j \le C_{\mathcal{A}}$ with $C_{\mathcal{A}}$ absolute.

Convergence and bound. Any finite range $K' < k \le K''$ of indices, with $\tau_{K'+1} \in \mathcal{B}_J$, consists of at most two partial runs (the initial segment of $\mathcal{B}_J$ and the terminal segment of the block containing $\tau_{K''}$) and of complete blocks in between; each run, partial or complete, contributes at most $\mathcal{P}_j/Y_j$ by (6.14) and Abel's inequality, so $|\sum_{K'<k\le K''} e^{iu\tau_k}/\tau_k| \le \sum_{j\ge J} \mathcal{P}_j/Y_j \to 0$ as $K' \to \infty$ (then $J \to \infty$), uniformly in K''. The series converges, and (6.12) together with $\sum_j \mathcal{P}_j/Y_j \le C_{\mathcal{A}}$ gives (6.11). □

Theorem 6.5 (Square-root estimate for J_{RVM} and π_{RVM})**.** *For $x \ge 8$,*

$$J_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O\left(\frac{\sqrt{x}(\log\log x)^2}{\log x}\right), \tag{6.16}$$

$$\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O\left(\frac{\sqrt{x}(\log\log x)^2}{\log x}\right), \tag{6.17}$$

with absolute implied constants. No hypothesis on ζ is used.

Proof. Let $u = \log x \ge \log 8$ and $\rho_k^{\mathrm{RVM}} = \frac{1}{2} + i\tau_k$. Lemma 6.1 with $w = \rho_k^{\mathrm{RVM}} u$, $a = u/2$, $b = \tau_k u$ gives

$$\mathrm{Ei}(\rho_k^{\mathrm{RVM}} u) = \frac{e^{\rho_k^{\mathrm{RVM}} u}}{\rho_k^{\mathrm{RVM}} u} + \varrho_k, \qquad |\varrho_k| \le \frac{e^{u/2}}{\tau_k^2 u^2}. \tag{6.18}$$

Moreover $\frac{1}{\rho_k^{\mathrm{RVM}}} = -\frac{i}{\tau_k} + r_k$ with $r_k = \frac{i/2}{\tau_k \rho_k^{\mathrm{RVM}}}$, so $|r_k| \le \frac{1}{2\tau_k^2}$. Since $\sum_k \tau_k^{-2} < \infty$, Lemma 6.4 shows that the series in (6.2) converges in the order of k and

$$\sum_{k\ge 1} \mathrm{Ei}(\rho_k^{\mathrm{RVM}} u) = \frac{e^{u/2}}{u}\Big(-i\,\mathcal{A}_{\mathrm{RVM}}(u) + \sum_k r_k e^{iu\tau_k}\Big) + \sum_k \varrho_k = \frac{e^{u/2}}{u}(-i\,\mathcal{A}_{\mathrm{RVM}}(u) + O(1)). \tag{6.19}$$

For $u \ge 2$ one has $\log(2+2u) \le \log(2u) + \log 2 \le 3\log u$, so (6.11) gives $\mathcal{A}_{\mathrm{RVM}}(u) = O(\log^2 u)$. The remaining terms $-\log 2 + \int_x^\infty dy/(y(y^2-1)\log y)$ of (6.2) are bounded for $x \ge 2$. With $e^{u/2} = \sqrt{x}$ and $\log u = \log\log x$ this proves (6.16).

For (6.17), (6.7) gives

$$\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = \big(J_{\mathrm{RVM}}(x) - \mathrm{li}(x)\big) + \sum_{2\le m\le \log_2 x} \frac{\mu(m)}{m}\, J^*_{\mathrm{RVM}}(x^{1/m}). \tag{6.20}$$

By (6.16), $\mathrm{li}(y) \ll y/\log y$, and the continuity of J_{RVM} on $[2,8]$ (Theorem 6.2), $|J^*_{\mathrm{RVM}}(y)| \le C\,y/\log y$ for all $y \ge 2$. The term $m = 2$ is therefore $O(\sqrt{x}/\log x)$. For $3 \le m \le \log_2 x$, $x^{1/m} \ge 2$ and $\frac{1}{m} J^*_{\mathrm{RVM}}(x^{1/m}) \le \frac{C}{m} \cdot \frac{x^{1/m}}{\log(x^{1/m})} = \frac{C\,x^{1/m}}{\log x} \le \frac{C\,x^{1/3}}{\log x}$, so these terms add up to at most $C\,x^{1/3}\log_2 x/\log x = O(x^{1/3})$. Both are absorbed in (6.17). □

Corollary 6.6 (The three limits)**.** *As $x \to \infty$,*

$$\frac{\mathrm{li}(x)}{\pi_{\mathrm{RVM}}(x)} \to 1, \qquad \frac{\pi(x)}{\mathrm{li}(x)} \to 1, \qquad \frac{\pi(x)}{\pi_{\mathrm{RVM}}(x)} \to 1. \tag{6.21}$$

More precisely, for some absolute $c_0 > 0$,

$$\pi(x) - \pi_{\mathrm{RVM}}(x) = O\Big(x\, e^{-c_0\sqrt{\log x}}\Big). \tag{6.22}$$

Proof. Since $\mathrm{li}(x) \sim x/\log x$, (6.17) gives $\pi_{\mathrm{RVM}}(x) = \mathrm{li}(x)(1 + O(x^{-1/2}(\log\log x)^2))$; in particular $\pi_{\mathrm{RVM}}(x) > 0$ for large x, and the first limit follows. The second is the prime number theorem [41, 32], in the form $\pi(x) = \mathrm{li}(x) + O(x\,e^{-c_0\sqrt{\log x}})$ [31, Ch. 18]. Subtracting (6.17) gives (6.22), and dividing by $\pi_{\mathrm{RVM}}(x) \sim x/\log x$ gives the third limit. □

The first limit and (6.17) concern only the synthetic system; the second and third compare it with the primes and use the prime number theorem. Figures 12 and 14 show them numerically.

7. Integer weights and the continuous-to-atomic problem

Define

$$c_n^{\mathrm{RVM}} := \pi_{\mathrm{RVM}}(n) - \pi_{\mathrm{RVM}}(n-1), \qquad n \geq 2, \tag{7.1}$$

and, initially for $\Re s > 1$,

$$\zeta_{\mathbb{Z},\mathrm{RVM}}(s) := \prod_{n\geq 2} (1-n^{-s})^{-c_n^{\mathrm{RVM}}}. \tag{7.2}$$

No positivity or 0/1 property is built into (7.1). Define the integer-step prime-power count and the right-continuous extension

$$J_{\mathbb{Z},\mathrm{RVM}}(x) := \sum_{1\leq m\leq \log_2 x} \frac{1}{m}\, \pi_{\mathrm{RVM}}(\lfloor x^{1/m}\rfloor), \tag{7.3}$$

$$J^{*,\mathbb{Z}}_{\mathrm{RVM}}(x) := J^{*}_{\mathrm{RVM}}(\lfloor x\rfloor). \tag{7.4}$$

Theorem 7.1 (Lipschitz control of J_{RVM})**.** *There is an absolute constant C_J such that*

$$|J_{\mathrm{RVM}}(x+\theta) - J_{\mathrm{RVM}}(x)| \leq C_J\,\theta \qquad (x\geq 2,\ 0<\theta\leq 1). \tag{7.5}$$

Proof. Put $u_1 = \log x$, $u_2 = \log(x+\theta)$, so $0 < u_2 - u_1 = \log(1+\theta/x) \leq \theta/x$ and $u_2 - u_1 \leq \log 2$. The terms li and $\int_x^\infty dy/(y(y^2-1)\log y)$ of (6.2) have derivatives bounded by $1/\log 2$ and $3/\log 2$ on $[2,\infty)$, so they change by $O(\theta)$. It remains to bound $\sum_k D_k$, where, by (6.1) and $\frac{d}{dv}\,\mathrm{Ei}(\rho v) = e^{\rho v}/v$,

$$D_k := \mathrm{Ei}(\rho_k^{\mathrm{RVM}} u_2) - \mathrm{Ei}(\rho_k^{\mathrm{RVM}} u_1) = \int_{u_1}^{u_2} \frac{e^{v/2}e^{iv\tau_k}}{v}\, dv. \tag{7.6}$$

The series $\sum_k D_k$ converges (Theorem 6.2 and its proof). By Abel's theorem for Dirichlet series with the increasing exponents τ_k (not the index k), $\sum_k D_k = \lim_{\alpha\downarrow 0}\sum_k e^{-\alpha\tau_k} D_k$: with $R_K := \sum_{k>K} D_k$, summation by parts gives $\sum_{k>K} e^{-\alpha\tau_k} D_k = e^{-\alpha\tau_{K+1}} R_K - \sum_{k>K}(e^{-\alpha\tau_k} - e^{-\alpha\tau_{k+1}})R_k$, of modulus at most $2\sup_{k\geq K}|R_k|$ uniformly in $\alpha > 0$ because $e^{-\alpha\tau_k}$ decreases in k, while the finitely many terms $k \leq K$ converge as $\alpha \downarrow 0$. Moreover, for $\alpha > 0$ Fubini gives

$$\sum_k e^{-\alpha\tau_k} D_k = \int_{u_1}^{u_2} \frac{e^{v/2}}{v}\, S_\alpha(v)\, dv, \qquad S_\alpha(v) := \sum_{k\geq 1} g(\tau_k), \quad g(\tau) := e^{(iv-\alpha)\tau}. \tag{7.7}$$

Clock decomposition. With $\tau_0 = 4\pi$ and the sawtooth count (9.8) (proved later in Section 9), Stieltjes integration exactly as in the forward reference (9.9) ($g(\infty) = 0$, $g' \in L^1$) gives

$$S_\alpha(v) = \int_{\tau_0}^\infty g\, d\mathcal{N}_{\mathrm{RVM}} + g(\tau_0)\mathcal{N}_{\mathrm{RVM}}(\tau_0) + \int_{\tau_0}^\infty \psi(\mathcal{N}_{\mathrm{RVM}}(\tau) + \tfrac12) g'(\tau)\, d\tau. \tag{7.8}$$

The middle term is at most $\frac12$. For the first, two integrations by parts with $\mathcal{N}'_{\mathrm{RVM}}(\tau) = \frac{1}{2\pi}\log\frac{\tau}{2\pi}$, $\mathcal{N}''_{\mathrm{RVM}}(\tau) = \frac{1}{2\pi\tau}$ and $v \geq \log 2$ give $|\int_{\tau_0}^\infty g\, d\mathcal{N}_{\mathrm{RVM}}| \leq C/v$, uniformly in $\alpha \in (0,1]$.

Harmonics. The partial sums of $\psi(y+\frac12) = -\sum_{m\geq 1}(-1)^m \sin(2\pi m y)/(\pi m)$ are uniformly bounded, $\sup_{M\geq 1}\sup_{y\in\mathbb{R}}|\sum_{m\leq M}\sin(2\pi m y)/m| < \infty$ (for $0 < y \leq \frac12$ split at $M_0 := \lfloor 1/y\rfloor$: the terms $m \leq M_0$ contribute at most $\sum_{m\leq M_0} 2\pi y \leq 2\pi$, and Abel summation with $|\sum_{M_0<m\leq M'}\sin(2\pi m y)| \leq 1/\sin(\pi y) \leq 1/(2y)$ and $1/m \leq 1/M_0 \leq 2y$ bounds the rest by 2; the sum is odd and 1-periodic in y), and the series converges to $\psi(y+\frac12)$ for every $y \notin \frac12 + \mathbb{Z}$; the exceptional set $\{\tau : \mathcal{N}_{\mathrm{RVM}}(\tau) + \frac12 \in \mathbb{Z}\}$ is discrete, hence Lebesgue-null. Since $g' \in L^1(\tau_0,\infty)$ for $\alpha > 0$, dominated convergence allows termwise integration:

$$\int_{\tau_0}^\infty \psi(\mathcal{N}_{\mathrm{RVM}} + \tfrac12) g'\, d\tau = -\sum_{m\geq 1} \frac{(-1)^m}{2\pi i\, m}(K_m^+ - K_m^-), \qquad K_m^\pm := \int_{\tau_0}^\infty (iv-\alpha) e^{-\alpha\tau} e^{i\Psi_m^\pm(\tau)}\, d\tau, \tag{7.9}$$

with $\Psi_m^\pm(\tau) = v\tau \pm 2\pi m \mathcal{N}_{\mathrm{RVM}}(\tau)$. The weight $e^{-\alpha\tau}$ is positive and decreasing and $|iv-\alpha| \leq v+1$ for $0 < \alpha \leq 1$; the second mean-value theorem (Bonnet's form), applied to the real and to the

imaginary part of $e^{i\Psi_m^{\pm}}$ separately, bounds $|\int_a^b e^{-\alpha\tau}e^{i\Psi_m^{\pm}}d\tau|$ by $2e^{-\alpha a}\sup_{\xi\in[a,b]}|\int_a^{\xi} e^{i\Psi_m^{\pm}}d\tau|$, so each integral is at most $2(v+1)\sup_{\xi}|\int_a^{\xi} e^{i\Psi_m^{\pm}}d\tau|$ over the relevant interval $[a,b]$. We use the first- and second-derivative tests [61, Lemmas 4.2 and 4.4] for a real phase Θ: $|\int_a^b e^{i\Theta}|\le 4/\mu$ if Θ' is monotone with $|\Theta'|\ge\mu$, and $\le 8/\sqrt{\lambda}$ if $|\Theta''|\ge\lambda$.

For K_m^+: $(\Psi_m^+)' = v + m\log\frac{\tau}{2\pi} \ge v + m\log 2$ is increasing, so $|K_m^+|\le 8(v+1)/(v+m\log 2)$.

For K_m^-: $(\Psi_m^-)' = v - m\log\frac{\tau}{2\pi}$ is decreasing and vanishes at $T_m := 2\pi e^{v/m} > 2\pi$, and $(\Psi_m^-)'' = -m/\tau$. On $[\tau_0, T_m/2]$ (if nonempty) $(\Psi_m^-)' \ge m\log 2$; on $[2T_m,\infty)$, $(\Psi_m^-)' \le -m\log 2$; on $[\max(\tau_0, T_m/2), 2T_m]$, $|(\Psi_m^-)''|\ge m/(2T_m)$. Hence

$$|K_m^-| \le 2(v+1)\Big(\frac{8}{m\log 2} + 8\sqrt{\frac{2T_m}{m}}\Big) = 2(v+1)\Big(\frac{8}{m\log 2} + 16\sqrt{\pi}\,\frac{e^{v/(2m)}}{\sqrt{m}}\Big). \tag{7.10}$$

Summing with the weights $1/(2\pi m)$ and using $\sum_m m^{-3/2}e^{v/(2m)} \le e^{v/2} + \zeta(\frac{3}{2})e^{v/4}$,

$$|S_\alpha(v)| \le C\,(v+1)\,e^{v/2} \qquad (v\ge\log 2,\ 0<\alpha\le 1), \tag{7.11}$$

with an absolute constant C.

Conclusion. Inserting (7.11) into (7.7), for $v\in[u_1,u_2]$ we have $e^{v/2}\le\sqrt{2x}$ and $(v+1)/v \le 1 + 1/\log 2$, so

$$\Big|\sum_k e^{-\alpha\tau_k}D_k\Big| \le C'(u_2-u_1)\,x \le C'\theta, \tag{7.12}$$

uniformly in α. Letting $\alpha\downarrow 0$ and taking $2\Re$ proves (7.5). □

The $m=1$ harmonic, with stationary point $T_1 = 2\pi x$, produces the unit-periodic ripple of J_{RVM} (Figure 12); the harmonics $m\ge 2$ have stationary points $2\pi x^{1/m}$ and contribute $O(x^{-1/2+1/(2m)}m^{-3/2})$ to a unit increment.

Corollary 7.2 (Bounded cell masses)**.**

$$\pi_{\mathrm{RVM}}(y) - \pi_{\mathrm{RVM}}(\lfloor y\rfloor) = O(1) \quad (y\ge 2), \qquad c_n^{\mathrm{RVM}} = O(1) \quad (n\ge 2), \tag{7.13}$$

with absolute implied constants, uniform in $y\ge 2$ and $n\ge 2$.

Proof. By (6.7), $c_n^{\mathrm{RVM}} = \sum_{m\le\log_2 n}\frac{\mu(m)}{m}(J^*_{\mathrm{RVM}}(n^{1/m}) - J^*_{\mathrm{RVM}}((n-1)^{1/m}))$. The term $m=1$ is $O(1)$ by (7.5). For $m\ge 2$ and $(n-1)^{1/m}\ge 2$, (7.5) with $\theta = n^{1/m} - (n-1)^{1/m} \le n^{1/m-1}$ bounds the term by $C_J n^{1/m-1}\le C_J n^{-1/2}$, and there are at most $\log_2 n$ such terms. The jump of J^*_{RVM} at 2 enters only for $n=2^m$, once, with size $|J_{\mathrm{RVM}}(2)|/m$. The same argument applies to $\pi_{\mathrm{RVM}}(y) - \pi_{\mathrm{RVM}}(\lfloor y\rfloor)$. □

Corollary 7.3 (The integer-weighted zeta is a Dirichlet series)**.** *The product* (7.2) *converges absolutely for $\Re s > 1$ and*

$$\zeta_{\mathbb{Z},\mathrm{RVM}}(s) = \sum_{N\ge 1}\frac{a_N^{\mathbb{Z}}}{N^s}, \qquad a_1^{\mathbb{Z}} = 1, \qquad \sum_{N\ge 1}\frac{|a_N^{\mathbb{Z}}|}{N^\sigma} < \infty \quad (\sigma>1). \tag{7.14}$$

The inverse is also an absolutely convergent Dirichlet series on the whole half-plane:

$$\frac{1}{\zeta_{\mathbb{Z},\mathrm{RVM}}(s)} = \sum_{N\ge 1}\frac{\bar a_N^{\mathbb{Z}}}{N^s}, \qquad \bar a_1^{\mathbb{Z}} = 1, \qquad \sum_{N\ge 1}\frac{|\bar a_N^{\mathbb{Z}}|}{N^\sigma} < \infty \quad (\sigma>1). \tag{7.15}$$

Proof. By Corollary 7.2,

$$\log\zeta_{\mathbb{Z},\mathrm{RVM}}(s) = \sum_{n\ge 2} c_n^{\mathrm{RVM}}\sum_{m\ge 1}\frac{n^{-ms}}{m} = \sum_{N\ge 2}\frac{b_N}{N^s}, \qquad b_N := \sum_{n^m=N}\frac{c_n^{\mathrm{RVM}}}{m}, \tag{7.16}$$

with $\sum_N |b_N| N^{-\sigma} \le C \sum_{n\ge 2} \sum_m n^{-m\sigma}/m < \infty$ for $\sigma > 1$. The exponential of an absolutely convergent Dirichlet series without constant term is an absolutely convergent Dirichlet series with $a_1^{\mathbb{Z}} = 1$, and $\sum_N |a_N^{\mathbb{Z}}| N^{-\sigma} \le \exp\bigl(\sum_N |b_N| N^{-\sigma}\bigr)$. The same applies to $1/\zeta_{\mathbb{Z},\mathrm{RVM}} = \exp\bigl(-\sum_N b_N N^{-s}\bigr)$, which gives (7.15). □

Numerically the cell masses lie in $[-0.38, 1.04]$ for $n \le 1000$ (Experiment 19.9).

Theorem 7.4 (Mellin–Stieltjes identification)**.** *There is a function $J^{\mathrm{M}}_{\mathrm{RVM}}$ on $(1,\infty)$ with $J^{\mathrm{M}}_{\mathrm{RVM}}(x)x^{-c} \in L^2((1,\infty), dx/x)$ for every $c > 1$ such that*

$$\log \zeta_{\mathrm{RVM}}(s) = s \int_1^\infty J^{\mathrm{M}}_{\mathrm{RVM}}(x)\, x^{-s-1}\, dx \qquad (\Re s > 1), \tag{7.17}$$

and $J^{\mathrm{M}}_{\mathrm{RVM}} - J_{\mathrm{RVM}}$ is constant on $(1,\infty)$. In particular $J^{\mathrm{M}}_{\mathrm{RVM}} = J^{*}_{\mathrm{RVM}} + C_{\mathrm{RVM}}$ on $[2,\infty)$ for a constant C_{RVM}.

The proof is given in Section 13 (Theorems 13.5 and 13.6); it uses the half-plane bound of Theorem 9.10.

Corollary 7.5 (Weighted factorization)**.**

$$J_{\mathbb{Z},\mathrm{RVM}}(x) - J^{*}_{\mathrm{RVM}}(x) = O(1 + \log\log x), \tag{7.18}$$

and

$$\zeta_{\mathbb{Z},\mathrm{RVM}}(s) = \zeta_{\mathrm{RVM}}(s)\, e^{h_{\mathrm{RVM}}(s)}, \tag{7.19}$$

where

$$h_{\mathrm{RVM}}(s) := s \int_1^\infty (J_{\mathbb{Z},\mathrm{RVM}}(x) - J^{\mathrm{M}}_{\mathrm{RVM}}(x)) x^{-s-1}\, dx \tag{7.20}$$

is holomorphic for $\Re s > 0$. The two functions have the same zeros in $\Re s > 0$, and $\zeta_{\mathbb{Z},\mathrm{RVM}}$ inherits through $e^{h_{\mathrm{RVM}}}$ the pole of ζ_{RVM} at $s = 1$; compare the modified Euler products of Grosswald and Schnitzer [39].

Proof. By (7.13) replacing $x^{1/m}$ by its integer part costs $O(1/m)$ in the mth summand; summing m^{-1} up to $\log_2 x$ gives (7.18). The product (7.2) is initially defined for $\Re s > 1$. For non-integral exponents we use the principal logarithmic branch, equivalently $-\log(1 - n^{-s}) = \sum_{m\ge 1} n^{-ms}/m$, which is unambiguous because $|n^{-s}| < 1$ there. It gives $\log \zeta_{\mathbb{Z},\mathrm{RVM}}(s) = s \int_1^\infty J_{\mathbb{Z},\mathrm{RVM}}(x) x^{-s-1} dx$ for $\Re s > 1$, because $\sum_{n\le y} c_n^{\mathrm{RVM}} = \pi_{\mathrm{RVM}}(\lfloor y \rfloor)$. Subtracting (7.17) gives (7.19) on $\Re s > 1$. On $[2,\infty)$, $J_{\mathbb{Z},\mathrm{RVM}} - J^{\mathrm{M}}_{\mathrm{RVM}} = (J_{\mathbb{Z},\mathrm{RVM}} - J^{*}_{\mathrm{RVM}}) - C_{\mathrm{RVM}} = O(1 + \log\log x)$ by (7.18); on $(1,2)$, $J_{\mathbb{Z},\mathrm{RVM}} = 0$ and $J^{\mathrm{M}}_{\mathrm{RVM}}$ is square-integrable. Hence (7.20) converges locally normally for $\Re s > 0$, and (7.19) defines the holomorphic continuation of $\zeta_{\mathbb{Z},\mathrm{RVM}}$ from $\Re s > 1$ to $\Re s > 0$; exponentiation introduces no zeros. □

Lemma 7.6 (Exact first-harmonic cell cancellation)**.** *For every integer n,*

$$\int_{n-1}^{n} 2\cos(2\pi x)\, dx = 0. \tag{7.21}$$

Hypothesis 7.7 (Uniform cell remainder away from the exceptional powers of 2; proved in Corollary 15.19)**.** For $n \ge 3$ with $n \notin \{2^j : \mu(j) \ne 0\}$,

$$c_n^{\mathrm{RVM}} - \int_{n-1}^{n} \left(\frac{1}{\log x} + 2\cos(2\pi x)\right) dx = o\!\left(\frac{1}{\log n}\right). \tag{7.22}$$

Hypothesis 7.7 is proved below, in Corollary 15.19, for exactly the set of n in its statement; it is stated here as a hypothesis only because its proof needs the density theorem of Section 15, and the proposition that follows is therefore unconditional.

Proposition 7.8 (Eventual positivity)**.** *Hypothesis 7.7 implies*

$$c_n^{\mathrm{RVM}} \sim \frac{1}{\log n}, \tag{7.23}$$

hence $0 < c_n^{\mathrm{RVM}} < 1$ *for all sufficiently large* n *in that set.*

Proof. The main-density integral is $1/\log n + O(1/(n\log^2 n))$; the first harmonic vanishes by Lemma 7.6. □

8. The critical RVM Mertens transform

Define

$$F_{\mathrm{RVM}}(s) = \widehat{M}^{\mathrm{crit}}_{\mathrm{RVM}}(s) := \frac{1}{\zeta_{\mathrm{RVM}}(s)} = \frac{G(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}. \tag{8.1}$$

Formally the associated summatory object is the inverse Mellin transform

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = \frac{1}{2\pi i}\int_{(c)} F_{\mathrm{RVM}}(s)\frac{x^s}{s}\,ds; \tag{8.2}$$

its existence as a symmetric limit, for every $c > \frac{1}{2}$, is proved in Theorem 9.14.

Proposition 8.1 (Synthetic pole abscissa)**.** $F_{\mathrm{RVM}} = G/P^{\mathrm{crit}}_{\mathrm{RVM}}$ *is meromorphic on* $\mathbb{C}$ *and holomorphic in* $\Re s > \frac{1}{2}$*. The poles of this meromorphic function are the simple poles*

$$\frac{1}{2} \pm i\tau_k \quad (k \geq 1) \qquad \textit{and} \qquad s = -2, -4, -6, \ldots \tag{8.3}$$

and $s = 1$ *is a simple zero of* F_{RVM}*.*

Proof. This is Theorem 3.1 applied to $F_{\mathrm{RVM}} = G/P^{\mathrm{crit}}_{\mathrm{RVM}}$: the zeros of $P^{\mathrm{crit}}_{\mathrm{RVM}}$ are (3.6), and the poles of G are those of $\Gamma(s/2)$ except $s = 0$, where $s\Gamma(s/2) \to 2$. □

Proposition 8.2 (Gaussian square-root bound under vertical growth)**.** *Fix* $c > 1$ *and* $0 < \varepsilon < c - \frac{1}{2}$*. Assume, uniformly for* $\frac{1}{2} + \varepsilon \leq \sigma \leq c$*,*

$$F_{\mathrm{RVM}}(\sigma + it) = \exp(o(t^2)) \qquad (|t| \to \infty). \tag{8.4}$$

Define

$$M^{\mathrm{crit}}_{\mathrm{RVM},G}(x) := \frac{1}{2\pi i}\int_{(c)} e^{s^2} F_{\mathrm{RVM}}(s)\frac{x^s}{s}\,ds. \tag{8.5}$$

Then

$$M^{\mathrm{crit}}_{\mathrm{RVM},G}(x) = O_\varepsilon(x^{1/2+\varepsilon}). \tag{8.6}$$

Proof. On $s = \sigma + it$, $|e^{s^2}| = e^{\sigma^2 - t^2}$, so (8.4) gives absolute convergence on vertical lines and makes the horizontal integrals tend to 0. The strip $\frac{1}{2} + \varepsilon \leq \Re s \leq c$ contains no pole of F_{RVM} (Proposition 8.1) and does not contain $s = 0$, so Cauchy's theorem moves the line to $\Re s = \frac{1}{2} + \varepsilon$, where $|x^s| = x^{1/2+\varepsilon}$. □

Theorem 9.14 below removes both the assumption (8.4) and the smoothing.

9. The transition theorem and the unsmoothed synthetic Mertens bound

Let

$$\chi(s) := 2^s\pi^{s-1}\sin\Big(\frac{\pi s}{2}\Big)\Gamma(1-s) = \pi^{s-1/2}\frac{\Gamma((1-s)/2)}{\Gamma(s/2)}, \tag{9.1}$$

the factor of the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$ [61, §2.1]. Write

$$s = \tfrac12 + z, \quad z = \sigma' + it, \qquad \tau_* := -iz = t - i\sigma', \qquad w(s) := \exp\{-2\pi i\,\mathcal{N}_{\mathrm{RVM}}(\tau_*)\}. \tag{9.2}$$

Stirling's formula gives, uniformly on compact σ-sets,

$$|\chi(\sigma+it)| = \Big(\frac{|t|}{2\pi}\Big)^{1/2-\sigma}\big(1+O(|t|^{-1})\big). \tag{9.3}$$

Theorem 9.1 (Transition evaluation of the midpoint product)**.** *There is a constant $\varpi_{\mathrm{RVM}} > 0$ with the following properties.*

(a) *For every $B > 0$ there are $t_0(B)$ and C_B such that, for $0 < \sigma' \le B$ and $t \ge t_0(B)$,*

$$\zeta_{\mathrm{RVM}}(s) = \varpi_{\mathrm{RVM}}\big(1+w(s)\big)\big(1+\mathcal{E}_0(s)\big), \qquad |\mathcal{E}_0(s)| \le \frac{C_B}{t}. \tag{9.4}$$

(b) *For every $0 < \sigma_1' < \sigma_2' < \frac12$ there are t_0 and C such that, for $\sigma_1' \le \sigma' \le \sigma_2'$ and $t \ge t_0$,*

$$\zeta_{\mathrm{RVM}}(s) = \varpi_{\mathrm{RVM}}\big(1+\chi(s)\big)\big(1+\mathcal{E}(s)\big), \qquad |\mathcal{E}(s)| \le \frac{C}{t}. \tag{9.5}$$

For $t < 0$ the same statements hold by $\zeta_{\mathrm{RVM}}(\bar s) = \overline{\zeta_{\mathrm{RVM}}(s)}$.

The phase identity $w = \chi(s)(1+O(1/t))$ alone does not give (9.4); the product has to be evaluated across the transition region $\tau_k \asymp t$. Log denotes the principal logarithm and $\operatorname{Log}\Gamma$ the branch of $\log\Gamma$ on $\Re v > 0$ that is real on $(0,\infty)$. Nothing about ζ is used.

9.1. **Proof of Theorem 9.1.** Throughout, $t > 0$, $0 < \sigma' \le B$, $\tau_0 := 4\pi$, $t^2 \ge 2(\tau_0^2 + B^2)$, and implied constants depend only on B (and on σ_1', σ_2' in part (b)).

Step 0: reduction to a sum over the clock. Since $q(s) = z^2 - \frac14$, each factor of (3.2) equals $(\tau_k^2 + z^2)/(\tau_k^2 + \frac14)$, so

$$\frac{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac12)} = \prod_{k\ge1}\Big(1+\frac{z^2}{\tau_k^2}\Big) = \exp\Lambda(z), \qquad \Lambda(z) := \sum_{k\ge1} f(\tau_k), \quad f(\tau) := \operatorname{Log}\Big(1+\frac{z^2}{\tau^2}\Big). \tag{9.6}$$

For $\tau > 0$, $\Im(1 + z^2/\tau^2) = 2\sigma' t/\tau^2 > 0$, so f is smooth on $(0,\infty)$, $f(\tau) = O(|z|^2\tau^{-2})$, the series converges absolutely, and

$$f'(\tau) = -\frac{2z^2}{\tau(\tau^2+z^2)} = -\frac{2}{\tau} + \frac{1}{\tau - iz} + \frac{1}{\tau + iz}. \tag{9.7}$$

Step 1: the midpoint sawtooth. The clock increases on $[2\pi,\infty)$, $\mathcal{N}_{\mathrm{RVM}}(\tau_0) = 2\log\frac{2}{e} + \frac78 \in (0,\frac12)$, and $\tau_1 > \tau_0$. With $\psi(y) := y - \lfloor y\rfloor - \frac12$, the count (4.12) satisfies, for $\tau \ge \tau_0$,

$$n_{\mathrm{RVM}}(\tau) = \lfloor \mathcal{N}_{\mathrm{RVM}}(\tau) + \tfrac12\rfloor = \mathcal{N}_{\mathrm{RVM}}(\tau) - \psi\big(\mathcal{N}_{\mathrm{RVM}}(\tau) + \tfrac12\big), \qquad n_{\mathrm{RVM}}(\tau_0) = 0. \tag{9.8}$$

Stieltjes integration of f against (9.8), followed by integration by parts in the ψ-part ($f(\infty) = 0$, $|\psi| \le \frac12$, $\psi(\mathcal{N}_{\mathrm{RVM}}(\tau_0) + \frac12) = \mathcal{N}_{\mathrm{RVM}}(\tau_0)$), gives

$$\begin{aligned} &\Lambda(z) = \Lambda_{\mathrm{sm}}(z) + \Lambda_{\mathrm{saw}}(z), \qquad \Lambda_{\mathrm{sm}}(z) := \int_{\tau_0}^{\infty} f\,d\mathcal{N}_{\mathrm{RVM}} + f(\tau_0)\mathcal{N}_{\mathrm{RVM}}(\tau_0), \\ &\Lambda_{\mathrm{saw}}(z) := \int_{\tau_0}^{\infty} \psi\big(\mathcal{N}_{\mathrm{RVM}}(\tau) + \tfrac12\big) f'(\tau)\,d\tau. \end{aligned} \tag{9.9}$$

Lemma 9.2 (Smooth part)**.** *With* $\log G(s) := \operatorname{Log}(z^2-\frac14) + \operatorname{Log}\Gamma(\frac{s}{2}) - \frac{s}{2}\log\pi$,

$$\Lambda_{\mathrm{sm}}(z) - \log G(s) = \nu_3 + O(t^{-1}), \qquad \nu_3 \in \mathbb{R} \text{ independent of } z. \tag{9.10}$$

Proof. A second integration by parts ($f(\tau) = O(|z|^2\tau^{-2})$ and $\mathcal{N}_{\mathrm{RVM}}(\tau) = O(\tau\log\tau)$, so $f\mathcal{N}_{\mathrm{RVM}} \to 0$ at ∞) cancels the boundary term:

$$\Lambda_{\mathrm{sm}}(z) = -\int_{\tau_0}^{\infty} \mathcal{N}_{\mathrm{RVM}} f'\, d\tau = \int_{\tau_0}^{\infty} \mathcal{N}_{\mathrm{RVM}}(\tau)\Big(\frac{2}{\tau} - \frac{2\tau}{\tau^2+z^2}\Big) d\tau. \tag{9.11}$$

The constant $\frac78$ of the clock contributes $\frac78\operatorname{Log}(1+z^2/\tau_0^2)$, because $\operatorname{Log}\frac{\tau^2}{\tau^2+z^2}$ is an antiderivative of the bracket on $(0,\infty)$ tending to 0 at ∞. As $0 < \arg z < \frac{\pi}{2}$, $\arg(1+\tau_0^2/z^2) = O(t^{-2})$ and

$$\tfrac78\operatorname{Log}\Big(1+\frac{z^2}{\tau_0^2}\Big) = \tfrac74\operatorname{Log} z - \tfrac74\log\tau_0 + O(t^{-2}). \tag{9.12}$$

The term $\frac{\tau}{2\pi}\log\frac{\tau}{2\pi e}$ contributes $\frac1\pi\int_{\tau_0}^{\infty}\log\frac{\tau}{2\pi e}\frac{z^2}{\tau^2+z^2}\,d\tau$. Put

$$\mathcal{I}(z) := \frac1\pi\int_0^{\infty} \log\frac{\tau}{2\pi e}\,\frac{z^2}{\tau^2+z^2}\,d\tau. \tag{9.13}$$

The integrand is holomorphic in z on $\Re z > 0$ (its singularities are at $z = \pm i\tau$), $O(|z|^2\tau^{-2}\log\tau)$ at ∞ and $O(|\log\tau|)$ at 0, locally uniformly; so $\mathcal{I}$ is holomorphic on $\Re z > 0$. For real $z = x > 0$ the substitution $\tau = xv$ and $\int_0^\infty \frac{dv}{1+v^2} = \frac{\pi}{2}$, $\int_0^\infty \frac{\log v}{1+v^2}\,dv = 0$ give $\mathcal{I}(x) = \frac{x}{2}\log\frac{x}{2\pi e}$; by the identity theorem $\mathcal{I}(z) = \frac{z}{2}\operatorname{Log}\frac{z}{2\pi e}$ on $\Re z > 0$. On $[0,\tau_0]$, $|\tau^2+z^2| \ge t^2/2$, so $\frac{z^2}{\tau^2+z^2} = 1 + O(t^{-2})$ there. Hence

$$\Lambda_{\mathrm{sm}}(z) = \frac{z}{2}\operatorname{Log}\frac{z}{2\pi e} + \frac74\operatorname{Log} z + \nu_1 + O(t^{-2}), \qquad \nu_1 = -\tfrac74\log\tau_0 - \tfrac1\pi\int_0^{\tau_0}\log\tfrac{\tau}{2\pi e}\,d\tau. \tag{9.14}$$

Stirling's formula $\operatorname{Log}\Gamma(v) = (v-\frac12)\operatorname{Log} v - v + \frac12\log 2\pi + O(|v|^{-1})$ [36, §5.11] with $v = \frac{s}{2} = \frac{z+1/2}{2}$ and $\operatorname{Log} v = \operatorname{Log} z - \log 2 + \frac{1}{2z} + O(t^{-2})$ gives

$$\operatorname{Log}\Gamma(\tfrac{s}{2}) = \tfrac{z}{2}\operatorname{Log}\tfrac{z}{2} - \tfrac14\operatorname{Log} z + \tfrac14\log 2 - \tfrac{z}{2} + \tfrac12\log 2\pi + O(t^{-1}), \tag{9.15}$$

and with $\operatorname{Log}(z^2-\frac14) = 2\operatorname{Log} z + O(t^{-2})$, $-\frac{s}{2}\log\pi = -\frac{z}{2}\log\pi - \frac14\log\pi$,

$$\log G(s) = \frac{z}{2}\operatorname{Log}\frac{z}{2\pi e} + \frac74\operatorname{Log} z + \nu_2 + O(t^{-1}), \qquad \nu_2 = \tfrac14\log 2 + \tfrac12\log 2\pi - \tfrac14\log\pi. \tag{9.16}$$

Subtracting gives (9.10) with $\nu_3 = \nu_1 - \nu_2$. □

Lemma 9.3 (The clock as a holomorphic phase)**.** $\mathcal{N}_{\mathrm{RVM}}(\tau) = \frac{\tau}{2\pi}(\operatorname{Log}\frac{\tau}{2\pi} - 1) + \frac78$ *is holomorphic on* $\Re\tau > 0$, $\mathcal{N}_{\mathrm{RVM}}(\bar\tau) = \overline{\mathcal{N}_{\mathrm{RVM}}(\tau)}$, *and for* $\Re\tau \ge \tau_0$,

$$2\pi\,\Im\mathcal{N}_{\mathrm{RVM}}(\tau) \ge (\log 2)\,\Im\tau \quad (\Im\tau \ge 0), \qquad 2\pi\,\Im\mathcal{N}_{\mathrm{RVM}}(\tau) \le (\log 2)\,\Im\tau \quad (\Im\tau \le 0). \tag{9.17}$$

Proof. For $\tau = X + iY$, $X \ge \tau_0$, $Y > 0$, put $v = Y/X$. Then

$$2\pi\Im\mathcal{N}_{\mathrm{RVM}}(\tau) = Y\log\frac{|\tau|}{2\pi e} + X\arg\tau = Y\Big[\log\frac{X}{2\pi e} + \tfrac12\log(1+v^2) + \frac{\arctan v}{v}\Big] \ge Y\big[\log 2 + \varphi(v)\big], \tag{9.18}$$

using $X \ge 4\pi$, where $\varphi(v) = \frac12\log(1+v^2) + \frac{\arctan v}{v} - 1$. Since $\varphi(0^+) = 0$ and $\varphi'(v) = (v - \arctan v)/v^2 \ge 0$, $\varphi \ge 0$. The case $Y < 0$ follows by conjugation. □

In particular $|w(s)| = e^{2\pi\Im\mathcal{N}_{\mathrm{RVM}}(\tau_*)} \le 2^{-\sigma'} < 1$, and on the vertical rays $|e^{\pm 2\pi i m\mathcal{N}_{\mathrm{RVM}}(\tau_0 \pm iy)}| \le 2^{-my}$.

Lemma 9.4 (Abel-regularized harmonic sum)**.**

$$\Lambda_{\mathrm{saw}}(z) = \operatorname{Log}\big(1+w(s)\big) + \nu_4 + O(t^{-2}), \qquad \nu_4 \in \mathbb{R} \text{ independent of } z. \tag{9.19}$$

Proof. Regularization. For $0 < r < 1$ let

$$\psi_r(y) := -\sum_{m\ge 1} r^m \frac{\sin 2\pi m y}{\pi m} = \frac{1}{\pi}\Im \operatorname{Log}(1 - re^{2\pi i y}). \tag{9.20}$$

Since $\Re(1 - re^{2\pi iy}) > 0$, $|\psi_r| \le \frac{1}{2}$, and $\psi_r(y) \to \psi(y)$ as $r \uparrow 1$ for $y \notin \mathbb{Z}$ (Abel summation of the convergent Fourier series of ψ). The set $\{\tau : \mathcal{N}_{\mathrm{RVM}}(\tau) + \frac{1}{2} \in \mathbb{Z}\}$ is discrete and $f' \in L^1(\tau_0, \infty)$, so by dominated convergence

$$\Lambda_{\mathrm{saw}}(z) = \lim_{r\uparrow 1} \Lambda_r(z), \qquad \Lambda_r(z) := \int_{\tau_0}^{\infty} \psi_r(\mathcal{N}_{\mathrm{RVM}} + \tfrac{1}{2}) f' \, d\tau. \tag{9.21}$$

Termwise integration. For fixed $r < 1$ the series for $\psi_r(\mathcal{N}_{\mathrm{RVM}} + \frac{1}{2})$, with $\sin 2\pi m(y + \frac{1}{2}) = (-1)^m \sin 2\pi m y$, is dominated by $\sum_m r^m/m$; by Fubini,

$$\Lambda_r(z) = -\sum_{m\ge 1} \frac{(-r)^m}{2\pi i\, m}(I_m^+ - I_m^-), \qquad I_m^{\pm} := \int_{\tau_0}^{\infty} e^{\pm 2\pi i m \mathcal{N}_{\mathrm{RVM}}(\tau)} f'(\tau)\, d\tau. \tag{9.22}$$

Contours. Let $\mathcal{H}^{\pm} = \{\Re\tau \ge \tau_0,\ \pm\Im\tau \ge 0\}$. On $\mathcal{H}^{\pm}$, $|e^{\pm 2\pi i m \mathcal{N}_{\mathrm{RVM}}}| \le 1$ by Lemma 9.3, and $|f'(\tau)| \le 4|z|^2|\tau|^{-3}$ for $|\tau| \ge 2|z|$, so the quarter-circle arcs of radius R contribute $O(|z|^2 R^{-2}) \to 0$. The poles of f' are 0, $iz = -t + i\sigma' \notin \mathcal{H}^{\pm}$, and $\tau_* = -iz = t - i\sigma' \in \mathcal{H}^-$, the last with residue 1 (from $\frac{1}{\tau + iz}$). Define the ray integrals

$$R_m^+ := i\int_0^{\infty} e^{2\pi i m \mathcal{N}_{\mathrm{RVM}}(\tau_0 + iy)} f'(\tau_0 + iy)\, dy, \qquad R_m^- := -i\int_0^{\infty} e^{-2\pi i m \mathcal{N}_{\mathrm{RVM}}(\tau_0 - iy)} f'(\tau_0 - iy)\, dy. \tag{9.23}$$

On $\mathcal{H}^+$ Cauchy's theorem gives $I_m^+ = R_m^+$. On $\mathcal{H}^-$ the positively oriented boundary of the quarter disc runs down the ray, along the arc, and back along $[\tau_0, \tau_0 + R]$ from right to left; letting $R \to \infty$,

$$R_m^- - I_m^- = 2\pi i\, e^{-2\pi i m \mathcal{N}_{\mathrm{RVM}}(\tau_*)} = 2\pi i\, w^m. \tag{9.24}$$

Therefore, using $|rw| < 1$,

$$\Lambda_r(z) = -\sum_{m\ge 1} \frac{(-rw)^m}{m} + \Phi_r(z) = \operatorname{Log}(1 + rw) + \Phi_r(z), \qquad \Phi_r(z) := -\sum_{m\ge 1} \frac{(-r)^m}{2\pi i\, m}(R_m^+ - R_m^-). \tag{9.25}$$

The rays. Split $f'(\tau) = -\frac{2}{\tau} + g_z(\tau)$, $g_z(\tau) = \frac{2\tau}{\tau^2 + z^2}$, and correspondingly $R_m^{\pm} = R_m^{\pm,(0)} + R_m^{\pm,(z)}$, $\Phi_r = \Phi_r^{(0)} + \Phi_r^{(z)}$. The ray integrands carry the factor 2^{-my}. Since $|2/\tau| \le 2/\tau_0$ on the rays, $|R_m^{\pm,(0)}| \le 2/(\tau_0 m \log 2)$, so the series for $\Phi_r^{(0)}$ is dominated by $\sum_m C/m^2$ uniformly in r, and $\Phi_r^{(0)} \to \nu_4 := \Phi_1^{(0)}$, a constant depending only on τ_0. It is real: $\mathcal{N}_{\mathrm{RVM}}(\bar\tau) = \overline{\mathcal{N}_{\mathrm{RVM}}(\tau)}$ gives $R_m^{-,(0)} = \overline{R_m^{+,(0)}}$, so each term equals $-(-r)^m \Im R_m^{+,(0)}/(\pi m)$. For $\tau = \tau_0 \pm iy$, $\Re(\tau^2 + z^2) = \tau_0^2 + \sigma'^2 - y^2 - t^2 \le -\frac{1}{2}(y^2 + t^2)$ because $t^2 \ge 2(\tau_0^2 + B^2)$; hence $|g_z(\tau)| \le 4(\tau_0 + y)t^{-2}$ and

$$|R_m^{\pm,(z)}| \le \frac{4}{t^2}\int_0^{\infty} 2^{-my}(\tau_0 + y)\, dy = \frac{4}{t^2}\Big(\frac{\tau_0}{m\log 2} + \frac{1}{(m\log 2)^2}\Big). \tag{9.26}$$

So $|\Phi_r^{(z)}(z)| \le C\, t^{-2} \sum_m m^{-2}$, uniformly in $0 < r < 1$ and $0 < \sigma' \le B$, and $\Phi_r^{(z)}$ converges as $r \uparrow 1$ by dominated convergence of the series. Since $\operatorname{Log}(1 + rw) \to \operatorname{Log}(1 + w)$, letting $r \uparrow 1$ in (9.25) and (9.21) gives (9.19). *Uniformity.* The three limits $r \uparrow 1$ taken in this proof are dominated by r-independent majorants: $|\psi_r| \le \frac{1}{2}$ against $|f'| \in L^1$ in (9.21); after multiplication by the exterior factor $r^m/m \le 1/m$, the ray bounds $|R_m^{\pm,(0)}| \le 2/(\tau_0 m \log 2)$ and (9.26) give the summable, r-independent majorants C/m^2 and Ct^{-2}/m^2 for the series $\Phi_r^{(0)}$ and $\Phi_r^{(z)}$, so dominated convergence applies to these series as $r \uparrow 1$; and $|rw| \le |w| < 1$. Hence the remainder $O(t^{-2})$ in (9.19) is uniform in $0 < \sigma' \le B$ and $t \ge t_0(B)$, and does not depend on the regularization. □

Step 2: assembly. By (9.6), (9.9) and Lemmas 9.2 and 9.4,

$$\begin{aligned}\zeta_{\mathrm{RVM}}(s) &= \frac{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}{G(s)} = P^{\mathrm{crit}}_{\mathrm{RVM}}(\tfrac12)\exp\bigl(\Lambda_{\mathrm{sm}}(z) - \log G(s) + \Lambda_{\mathrm{saw}}(z)\bigr)\\ &= P^{\mathrm{crit}}_{\mathrm{RVM}}(\tfrac12)\, e^{\nu_3+\nu_4}(1+w)\bigl(1+O(t^{-1})\bigr),\end{aligned} \tag{9.27}$$

uniformly for $0 < \sigma' \le B$. This is (9.4) with

$$\varpi_{\mathrm{RVM}} := P^{\mathrm{crit}}_{\mathrm{RVM}}(\tfrac12)\, e^{\nu_3+\nu_4} > 0, \tag{9.28}$$

which depends only on the clock; its numerical value $0.9619415789\ldots$ plays no role in the proof. Theorem 9.11 below proves $\varpi_{\mathrm{RVM}} < 1$.

Step 3: from the clock phase to χ. Let

$$\vartheta(v) := \frac{1}{2i}\Bigl(\operatorname{Log}\Gamma\bigl(\tfrac14 + \tfrac{iv}{2}\bigr) - \operatorname{Log}\Gamma\bigl(\tfrac14 - \tfrac{iv}{2}\bigr)\Bigr) - \frac{v}{2}\log\pi, \tag{9.29}$$

holomorphic on $|\Im v| < \frac12$. The identity $\chi(\frac12 + iv) = e^{-2i\vartheta(v)}$ holds for real v, hence on the strip, and $s = \frac12 + i\tau_*$ with $|\Im\tau_*| = \sigma' < \frac12$. Stirling's formula in the strip gives $\vartheta(v) = \frac{v}{2}\operatorname{Log}\frac{v}{2\pi} - \frac{v}{2} - \frac{\pi}{8} + O(|v|^{-1})$, i.e.

$$\mathcal{N}_{\mathrm{RVM}}(v) = \frac{\vartheta(v)}{\pi} + 1 + O(|v|^{-1}). \tag{9.30}$$

Therefore

$$w = e^{-2i\vartheta(\tau_*) - 2\pi i + O(1/t)} = \chi(s)\bigl(1 + O(t^{-1})\bigr). \tag{9.31}$$

For $\sigma' \ge \sigma'_1$, (9.3) gives $|\chi(s)| \le \frac12$ when $t \ge t_0(\sigma'_1)$, so $1 + w = (1+\chi)(1 + (w-\chi)/(1+\chi)) = (1+\chi)(1+O(t^{-1}))$, which with (9.4) gives (9.5). This completes the proof of Theorem 9.1. □

Lemma 9.5 (Extension to the critical line)**.** *The function $\mathcal{E}_0 = \zeta_{\mathrm{RVM}}/(\varpi_{\mathrm{RVM}}(1+w)) - 1$ extends holomorphically to $\{|\sigma'| < \frac12,\ t > t_0\}$, where $t_0 = t_0(\frac14)$ is the threshold of Theorem 9.1(a) with $B = \frac14$, and $|\mathcal{E}_0| \le C/t$ for $0 \le \sigma' \le \frac14$.*

Proof. By Lemma 9.3, $\Im\mathcal{N}_{\mathrm{RVM}}(\tau_*) \neq 0$ for $\sigma' \neq 0$, so $1 + w$ vanishes in the strip only on $\sigma' = 0$, exactly at $t = \tau_k$; these zeros are simple because $\frac{d}{dt}w = -i\log(\tau_*/2\pi)\, w \neq 0$. The zeros of ζ_{RVM} in the strip are the same simple points. Hence $\mathcal{E}_0$ has only removable singularities, and the bound extends to $\sigma' = 0$ by continuity. □

Corollary 9.6 (Derivative at the zeros)**.** *For k with $\tau_k > t_0$,*

$$\zeta'_{\mathrm{RVM}}(\rho^{\mathrm{RVM}}_k) = \varpi_{\mathrm{RVM}}\log\frac{\tau_k}{2\pi}\Bigl(1 + O\Bigl(\frac{1}{\tau_k}\Bigr)\Bigr). \tag{9.32}$$

Proof. By Lemma 9.5, $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+\mathcal{E}_0)$ holds holomorphically near $\sigma' = 0$. At $s = \rho^{\mathrm{RVM}}_k$, $1 + w = 0$, so $\zeta'_{\mathrm{RVM}}(\rho^{\mathrm{RVM}}_k) = \varpi_{\mathrm{RVM}} w'(\rho^{\mathrm{RVM}}_k)(1 + \mathcal{E}_0(\rho^{\mathrm{RVM}}_k))$. From (9.2), $w'(s) = -2\pi i\,\mathcal{N}'_{\mathrm{RVM}}(\tau_*)\cdot(-i)\, w(s) = -\log(\tau_*/2\pi)\, w(s)$, and at the zero $w = -1$, $\tau_* = \tau_k$. Finally $|\mathcal{E}_0| \le C/\tau_k$. □

Remark 9.7 (Structure of the proof). The delicate points are handled as follows: the smooth part is computed with principal branches by the identity theorem (Lemma 9.2); the clock estimate is quantified (Lemma 9.3); the sawtooth series is replaced by its Abel means, termwise integration is justified for $r < 1$, and $r \uparrow 1$ is taken after the contour argument uniformly in r and z (Lemma 9.4); the orientation of the lower contour and the sign of the residue are written out in (9.24); the z-dependence of the rays is isolated in g_z and is $O(t^{-2})$, the rest is the explicit real constant ν_4; Stirling is used uniformly for $0 < \sigma' \le B$, and the only place where $\sigma' < \frac12$ enters is Step 3.

Corollary 9.8 (Vertical constant limit)**.** *For every fixed $\sigma > \frac{1}{2}$,*

$$\zeta_{\mathrm{RVM}}(\sigma + it) \longrightarrow \varpi_{\mathrm{RVM}}, \qquad F_{\mathrm{RVM}}(\sigma + it) \longrightarrow L := \frac{1}{\varpi_{\mathrm{RVM}}} \qquad (|t| \to \infty). \tag{9.33}$$

More precisely, with $\sigma' = \sigma - \frac{1}{2}$,

$$|w(\sigma + it)| \le (|t|/2\pi)^{-\sigma'}(1 + O_B(|t|^{-1})) \qquad (0 < \sigma' \le B), \tag{9.34}$$

and, with $\alpha_c := \min(c - \frac{1}{2}, 1)$ for $c > \frac{1}{2}$,

$$F_{\mathrm{RVM}}(c + it) - L = O_c(|t|^{-\alpha_c}). \tag{9.35}$$

Proof. Expanding the clock at $\tau_* = t - i\sigma'$ gives $\Im\mathcal{N}_{\mathrm{RVM}}(\tau_*) = -\frac{\sigma'}{2\pi}\log\frac{t}{2\pi} + O_\sigma(t^{-1})$, hence $|w| = e^{2\pi\Im\mathcal{N}_{\mathrm{RVM}}(\tau_*)} = O_\sigma(t^{-\sigma'})$. Inverting (9.4) for large t gives $F_{\mathrm{RVM}} = L(1 + O(|w|) + O(t^{-1}))$. □

9.2. **Bounds on the whole half-plane.** The constants of Theorem 9.1 depend on the width B of the band. The following estimate holds on the whole half-plane and does not use contours. Put

$$\mathfrak{s}_1(\tau) := \int_{\tau_0}^{\tau} \psi(\mathcal{N}_{\mathrm{RVM}}(v) + \tfrac{1}{2})\, dv \qquad (\tau \ge \tau_0). \tag{9.36}$$

Lemma 9.9 (Bounded primitive of the midpoint sawtooth)**.** $|\mathfrak{s}_1(\tau)| \le \dfrac{\pi}{4\log 2}$ *for all $\tau \ge \tau_0$.*

Proof. Substitute $y = \mathcal{N}_{\mathrm{RVM}}(v)$, $dv = \mathcal{T}'(y)\,dy$ with $\mathcal{T}'(y) = 2\pi/\log(\mathcal{T}(y)/2\pi)$ from (4.4). The weight $\mathcal{T}'$ is positive and decreasing, with $\mathcal{T}' \le 2\pi/\log 2$ on $y \ge \mathcal{N}_{\mathrm{RVM}}(\tau_0)$ because $\log(\tau_0/2\pi) = \log 2$. The function $\Phi(y) := \frac{1}{2}(\{y + \frac{1}{2}\} - \frac{1}{2})^2 - \frac{1}{8}$ is continuous, 1-periodic, with values in $[-\frac{1}{8}, 0]$, and $\Phi'(y) = \{y + \frac{1}{2}\} - \frac{1}{2} = \psi(y + \frac{1}{2})$ for $y \notin \frac{1}{2} + \mathbb{Z}$; it is therefore a primitive of $y \mapsto \psi(y + \frac{1}{2})$, so every integral of $\psi(y + \frac{1}{2})$ over an interval has modulus at most $\frac{1}{8}$. The second mean-value theorem gives $|\mathfrak{s}_1(\tau)| \le \frac{2\pi}{\log 2} \cdot \frac{1}{8}$. □

Theorem 9.10 (Two-sided bound and limit at $+\infty$)**.** *There is an absolute constant R_0 such that, for $s = \frac{1}{2} + z$, $z = \sigma' + it$, $\sigma' > \frac{1}{2}$ and $|z| \ge R_0$,*

$$\zeta_{\mathrm{RVM}}(s) = \varpi_{\mathrm{RVM}}\, e^{\theta_{\mathrm{RVM}}(s)}, \qquad |\theta_{\mathrm{RVM}}(s)| \le \frac{\pi^2}{2\log 2} \cdot \frac{1}{\sigma'} + \frac{C}{|z|}. \tag{9.37}$$

Consequently, for every $\sigma_0 > 1$ there is $C(\sigma_0)$ with

$$C(\sigma_0)^{-1} \le |\zeta_{\mathrm{RVM}}(s)| \le C(\sigma_0) \qquad (\Re s \ge \sigma_0), \tag{9.38}$$

and $\zeta_{\mathrm{RVM}}(s) \to \varpi_{\mathrm{RVM}}$ uniformly in t as $\Re s \to +\infty$. Moreover

$$\nu_4 = -2\int_{\tau_0}^{\infty} \frac{\mathfrak{s}_1(\tau)}{\tau^2}\, d\tau. \tag{9.39}$$

Proof. Smooth part. For $\Re z > 0$ the computation in the proof of Lemma 9.2 is exact:

$$\Lambda_{\mathrm{sm}}(z) = \frac{z}{2}\operatorname{Log}\frac{z}{2\pi e} + \frac{7}{8}\operatorname{Log}\Big(1 + \frac{z^2}{\tau_0^2}\Big) - \frac{1}{\pi}\int_0^{\tau_0} \log\frac{\tau}{2\pi e}\,\frac{z^2}{\tau^2 + z^2}\, d\tau. \tag{9.40}$$

For $|z| \ge 2\tau_0$ one has $|\tau^2 + z^2| \ge \frac{3}{4}|z|^2$ on $[0, \tau_0]$, and Stirling's formula is uniform in $|\arg(s/2)| < \frac{\pi}{2}$; the expansions (9.14) and (9.16) were derived for $t > 0$, but every step (the branches of $\operatorname{Log}(z^2)$, of $\operatorname{Log}(1 + z^2/\tau_0^2)$, of $\operatorname{Log}(z^2 - \frac{1}{4})$ and of $f(\tau) = \operatorname{Log}(1 + z^2/\tau^2)$ on $[\tau_0, \infty)$) only uses $|\arg z| < \frac{\pi}{2}$, which holds on $\Re z > 0$ for either sign of t; they therefore hold with $O(|z|^{-1})$ in place of $O(t^{-1})$, uniformly on $\Re z > \frac{1}{2}$, and $\Lambda_{\mathrm{sm}}(z) - \log G(s) = \nu_3 + O(|z|^{-1})$.

Sawtooth part. Integrating (9.9) by parts with (9.36) ($\mathfrak{s}_1(\tau_0) = 0$, $f'(\tau) = O(|z|^2\tau^{-3})$),

$$\Lambda_{\mathrm{saw}}(z) = -\int_{\tau_0}^{\infty} \mathfrak{s}_1 f''\, d\tau, \qquad f''(\tau) = \frac{2}{\tau^2} - \frac{1}{(\tau - iz)^2} - \frac{1}{(\tau + iz)^2}. \tag{9.41}$$

Since $|\tau \mp iz|^2 = (\tau \pm t)^2 + \sigma'^2$ and $\int_{\mathbb{R}} \frac{d\tau}{(\tau-a)^2+\sigma'^2} = \frac{\pi}{\sigma'}$, Lemma 9.9 gives

$$\Big|\Lambda_{\mathrm{saw}}(z) + 2\int_{\tau_0}^{\infty} \frac{\mathfrak{s}_1(\tau)}{\tau^2}\, d\tau\Big| \le \frac{\pi}{4\log 2}\cdot\frac{2\pi}{\sigma'}. \tag{9.42}$$

Assembly. By (9.27), $\log\zeta_{\mathrm{RVM}}(s) = \log\varpi_\infty + \theta_{\mathrm{RVM}}(s)$ with $\varpi_\infty := P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac12)\exp(\nu_3 - 2\int_{\tau_0}^{\infty}\mathfrak{s}_1\tau^{-2}d\tau) > 0$ and θ_{RVM} bounded as in (9.37). Both ϖ_∞ and ϖ_{RVM} are positive real numbers, and we use their real logarithms. Taking moduli, $|\zeta_{\mathrm{RVM}}(s)| = \varpi_\infty e^{\Re\theta_{\mathrm{RVM}}(s)}$; for fixed σ' and $t\to\infty$, Corollary 9.8 gives $|\zeta_{\mathrm{RVM}}(s)|\to\varpi_{\mathrm{RVM}}$, hence $|\log\varpi_{\mathrm{RVM}} - \log\varpi_\infty| \le \pi^2/(2\sigma'\log 2)$, with no $2\pi i$-ambiguity; letting $\sigma'\to\infty$ gives $\varpi_\infty = \varpi_{\mathrm{RVM}}$, which is (9.39) by (9.28). On the compact set $\{\Re s\ge\sigma_0,\ |z|\le R_0\}$ the function ζ_{RVM} is continuous and has no zeros or poles, which gives (9.38). □

Theorem 9.11 (The transition constant is less than one)**.** *The smooth constant of Lemma 9.2 is*

$$\nu_3 = 8 - \frac{33}{4}\log 2 - 2\log\pi = -0.0079240113\ldots, \tag{9.43}$$

the sawtooth constant of Lemma 9.4 satisfies $\nu_4 < -0.012691$, and

$$\varpi_{\mathrm{RVM}} = P^{\mathrm{crit}}_{\mathrm{RVM}}(\tfrac12)\, e^{\nu_3+\nu_4} < P^{\mathrm{crit}}_{\mathrm{RVM}}(\tfrac12)\, e^{-0.020615} < 0.97960. \tag{9.44}$$

In particular $\varpi_{\mathrm{RVM}} < 1$ and $L = 1/\varpi_{\mathrm{RVM}} > 1$.

Proof. (a) The canonical factor. $q(\frac12) = -\frac14$, so $P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac12) = \prod_k(1 - \frac{1/4}{1/4+\tau_k^2}) = \prod_k \frac{\tau_k^2}{\frac14+\tau_k^2}$; every factor lies in $(0,1)$ and $\sum_k \tau_k^{-2} < \infty$, so $0 < P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac12) < 1$. No numerical value of this product is used.

(b) The smooth constant. In (9.14), with $\tau_0 = 4\pi$, $\int_0^{\tau_0}\log\frac{\tau}{2\pi e}\,d\tau = \tau_0\log\frac{\tau_0}{2\pi e} - \tau_0 = 4\pi\log 2 - 8\pi$, so $\nu_1 = 8 - 4\log 2 - \frac74\log(4\pi) = 8 - \frac{15}{2}\log 2 - \frac74\log\pi$; in (9.16), $\nu_2 = \frac14\log 2 + \frac12\log 2\pi - \frac14\log\pi = \frac34\log 2 + \frac14\log\pi$. Hence $\nu_3 = \nu_1 - \nu_2$ is (9.43). Its sign is the inequality $e^8 < 2^{33/4}\pi^2$: with $\log 2 > 0.69314$ and $\log\pi > 1.14472$, $\frac{33}{4}\log 2 + 2\log\pi > 5.718405 + 2.28944 > 8.0078$, so $\nu_3 < -0.0078$.

(c) The sawtooth constant as a sum over the periods of the clock. By (9.39) and one integration by parts ($\mathfrak{s}_1' = \psi(\mathcal{N}_{\mathrm{RVM}} + \frac12)$, $\mathfrak{s}_1(\tau_0) = 0$, $\mathfrak{s}_1$ bounded by Lemma 9.9),

$$\nu_4 = -2\lim_{X\to\infty}\int_{\tau_0}^{X} \psi(\mathcal{N}_{\mathrm{RVM}}(\tau) + \tfrac12)\frac{d\tau}{\tau}, \tag{9.45}$$

and the limit may be taken along $X = \tau_{M+1}$, $M\to\infty$, because the left side of the integration by parts converges absolutely. Since $\mathcal{N}_{\mathrm{RVM}}$ increases on $[2\pi,\infty)$ and $0 < \mathcal{N}_{\mathrm{RVM}}(\tau_0) = 2\log\frac{2}{e} + \frac78 < \frac12$, on $[\tau_0,\tau_1)$ one has $\mathcal{N}_{\mathrm{RVM}} + \frac12 \in (\frac12, 1)$ and $\psi(\mathcal{N}_{\mathrm{RVM}} + \frac12) = \mathcal{N}_{\mathrm{RVM}}$; so the first piece is

$$\mathscr{I}_0(\tau_1) := \int_{\tau_0}^{\tau_1}\mathcal{N}_{\mathrm{RVM}}(\tau)\frac{d\tau}{\tau} = \frac{\tau_1\ell_1 - 2\tau_1 - \tau_0\log 2 + 2\tau_0}{2\pi} + \frac78\log\frac{\tau_1}{\tau_0}, \qquad \ell_1 := \log\frac{\tau_1}{2\pi}. \tag{9.46}$$

For $n\ge 1$, on $[\tau_n,\tau_{n+1})$ one has $\mathcal{N}_{\mathrm{RVM}} + \frac12 \in [n, n+1)$ and $\psi(\mathcal{N}_{\mathrm{RVM}} + \frac12) = \mathcal{N}_{\mathrm{RVM}} + \frac12 - n - \frac12$. Substitute $u = \mathcal{N}_{\mathrm{RVM}}(\tau) + \frac12$, i.e. $\tau = \mathcal{T}(u - \frac12)$ with the inverse clock of (4.4); then

$$\frac{d\tau}{\tau} = \mathscr{W}(u)\,du, \qquad \mathscr{W}(u) := \frac{2\pi}{\mathcal{T}\,\log(\mathcal{T}/2\pi)}, \qquad \mathcal{T} = \mathcal{T}(u-\tfrac12), \tag{9.47}$$

and $\mathscr{I}_n := \int_{\tau_n}^{\tau_{n+1}}\psi(\mathcal{N}_{\mathrm{RVM}} + \frac12)\frac{d\tau}{\tau} = \int_n^{n+1}(u - n - \frac12)\mathscr{W}(u)\,du$. With $\ell = \log(\mathcal{T}/2\pi) > 0$ for $u\ge 1$,

$$\mathscr{W}'(u) = -\frac{4\pi^2(\ell+1)}{\mathcal{T}^2\ell^3} < 0, \qquad \frac{d}{d\mathcal{T}}\log\frac{\ell+1}{\mathcal{T}^2\ell^3} = -\frac{2\ell^2+4\ell+3}{\mathcal{T}\ell(\ell+1)} < 0,$$

so $-\mathscr{W}'$ is decreasing in u: $\mathscr{W}$ is decreasing and convex on $[1,\infty)$, and $\mathscr{W}(u), \mathscr{W}'(u)\to 0$ as $u\to\infty$. Let $\mathbf{B}_2(x) = x^2 - x + \frac16$ and $\mathbf{B}_3(x) = x^3 - \frac32x^2 + \frac12x$ be the Bernoulli polynomials, $\bar{\mathbf{B}}_j(u) := \mathbf{B}_j(u - \lfloor u\rfloor)$;

since $\frac{1}{2}\mathbf{B}_2'(x) = x - \frac{1}{2}$, $\frac{1}{6}\mathbf{B}_3' = \frac{1}{2}\mathbf{B}_2$, $\mathbf{B}_2(0) = \mathbf{B}_2(1) = \frac{1}{6}$ and $\mathbf{B}_3(0) = \mathbf{B}_3(1) = 0$, two integrations by parts (the Euler–Maclaurin formula on one period) give

$$\mathscr{I}_n = \frac{\mathscr{W}(n+1) - \mathscr{W}(n)}{12} + \frac{1}{6}\int_n^{n+1} \bar{\mathbf{B}}_3(u)\,\mathscr{W}''(u)\,du.$$

Summing over $1 \le n \le M$ and letting $M \to \infty$,

$$\sum_{n\ge1} \mathscr{I}_n = -\frac{\mathscr{W}(1)}{12} + \frac{1}{6}\int_1^{\infty} \bar{\mathbf{B}}_3\,\mathscr{W}''\,du, \qquad \Big|\frac{1}{6}\int_1^{\infty} \bar{\mathbf{B}}_3\,\mathscr{W}''\,du\Big| \le \frac{\sqrt{3}}{216}\int_1^{\infty} \mathscr{W}''\,du = \frac{\sqrt{3}}{216}\,|\mathscr{W}'(1)|,$$

because $\max_{[0,1]}|\mathbf{B}_3| = \mathbf{B}_3\big(\frac{3-\sqrt{3}}{6}\big) = \frac{\sqrt{3}}{36}$ and $\mathscr{W}'' \ge 0$. Since $\mathcal{T}(\frac{1}{2}) = \tau_1$, $\mathscr{W}(1) = 2\pi/(\tau_1\ell_1)$ and $|\mathscr{W}'(1)| = 4\pi^2(\ell_1+1)/(\tau_1^2\ell_1^3)$. With (9.45),

$$\nu_4 = -2\Big(\mathscr{I}_0(\tau_1) + \sum_{n\ge1}\mathscr{I}_n\Big) \le -2\,\mathfrak{M}(\tau_1), \qquad \mathfrak{M}(t) := \mathscr{I}_0(t) - \frac{\pi}{6t\ell} - \frac{\sqrt{3}}{216}\cdot\frac{4\pi^2(\ell+1)}{t^2\ell^3}, \quad \ell = \log\frac{t}{2\pi}. \tag{9.48}$$

(d) The numerical inequalities. Since

$$\mathcal{N}_{\mathrm{RVM}}(14.52134) = 0.4999991\ldots < \tfrac{1}{2} < 0.5000004\ldots = \mathcal{N}_{\mathrm{RVM}}(14.52135),$$

$\tau_1 \in [14.52134, 14.52135]$ (the Lambert formula (2.5) gives $\tau_1 = 14.5213469530\ldots$). On this interval $\mathscr{I}_0 \in [0.054079, 0.054086]$, $\pi/(6t\ell) \in [0.0430408, 0.0430410]$, the last term of $\mathfrak{M}$ lies in $[0.0046925, 0.0046926]$, and therefore $\mathfrak{M}(\tau_1) \in [0.0063457, 0.0063526]$; hence $\nu_4 \le -0.0126914$. These enclosures, and those of (b), are computed in interval arithmetic with outward rounding by the program `s14_varpi_bound.py` (Appendix B.15, output in Appendix C.14); the margins ($\nu_3 < -0.0079$, $\mathfrak{M} > 0.0063$ against terms of size 0.05) are large enough for a hand computation with five decimals.

(e) Conclusion. By (9.28), $\log\varpi_{\mathrm{RVM}} = \log P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac{1}{2}) + \nu_3 + \nu_4 < 0 - 0.0079240 - 0.0126914 < -0.020615$, and $e^{-0.020615} < 0.97960$. □

Remark 9.12 (High-precision values; what the proof uses). The proof uses only $P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac{1}{2}) < 1$ and the two upper bounds $\nu_3 < 0$, $\nu_4 \le -2\mathfrak{M}(\tau_1)$. The floating-point values computed by the same program (Part B of its output, not part of the proof) are $\nu_4 = -0.0251569083$ (period-by-period quadrature of $\mathscr{I}_n$ for $n \le 2000$ and the Euler–Maclaurin tail), $e^{\nu_3+\nu_4} = 0.9674602699$, $P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac{1}{2}) = 0.9942956924$ (Lambert clock for $k \le 2\cdot10^6$ and the tail integral), and $\varpi_{\mathrm{RVM}} = 0.9619415789$. The Richardson value 0.9619415789 of Experiment 19.4 is obtained from $\zeta_{\mathrm{RVM}} = P^{\mathrm{crit}}_{\mathrm{RVM}}/G$ on $\Re s = 2$ without using ν_3 or ν_4; the agreement to $2\cdot10^{-11}$ is an independent check of (9.28) and (9.39). The Euler–Maclaurin remainder $\frac{1}{6}\int\bar{\mathbf{B}}_3\mathscr{W}''$ is $+0.00154$, about a third of the bound used, which is why the true ν_4 is about twice the bound.

Theorem 9.10 extends the two-sided bound (10.9) of Corollary 10.3 from bands to the whole half-plane $\Re s \ge \sigma_0 > 1$, and identifies ϖ_{RVM} also as the limit of $\zeta_{\mathrm{RVM}}(\sigma)$ as $\sigma \to +\infty$.

Theorem 9.13 (No absolutely convergent Dirichlet expansion)**.** *Neither ζ_{RVM} nor $F_{\mathrm{RVM}} = 1/\zeta_{\mathrm{RVM}}$ is represented by a Dirichlet series $\sum_n a_n n^{-s}$ converging absolutely in some half-plane $\Re s > \sigma_0$.*

Proof. Suppose $F_{\mathrm{RVM}}(s) = \sum_{n\ge1} a_n n^{-s}$ absolutely for $\Re s > \sigma_0$. Fix $\sigma > \max(\sigma_0, \frac{1}{2})$ and put $\tilde a_1 = a_1 - L$, $\tilde a_n = a_n$ $(n \ge 2)$, so $F_{\mathrm{RVM}}(\sigma+it) - L = \sum_n \tilde a_n n^{-\sigma-it}$ with $\sum_n |\tilde a_n| n^{-\sigma} < \infty$. Then

$$\frac{1}{2T}\int_{-T}^{T}\big|F_{\mathrm{RVM}}(\sigma+it) - L\big|^2 dt = \sum_{m,n}\tilde a_m\overline{\tilde a_n}(mn)^{-\sigma}\,\frac{1}{2T}\int_{-T}^{T}\Big(\frac{n}{m}\Big)^{it} dt, \tag{9.49}$$

the interchange being justified by absolute convergence of the double series, uniformly in t. The inner average is 1 for $m = n$ and is bounded by $\min\big(1, 1/(T|\log(n/m)|)\big) \to 0$ for $m \ne n$; each term

is bounded by $|\tilde{a}_m\tilde{a}_n|(mn)^{-\sigma}$, which is summable, so dominated convergence gives

$$\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}\bigl|F_{\mathrm{RVM}}(\sigma+it)-L\bigr|^2dt=\sum_{n\geq 1}|\tilde{a}_n|^2n^{-2\sigma}. \tag{9.50}$$

By (9.33) the integrand tends to 0 and is bounded on the line (continuous, with a finite limit), so the left side of (9.50) is 0. Hence every $\tilde{a}_n=0$, $F_{\mathrm{RVM}}\equiv L$ in the half-plane and, by analytic continuation, everywhere, which contradicts $F_{\mathrm{RVM}}(1)=0$. The same argument applied to $\zeta_{\mathrm{RVM}}-\varpi_{\mathrm{RVM}}$ forces an absolutely convergent Dirichlet series for ζ_{RVM} to be the constant ϖ_{RVM}, which contradicts its pole at $s=1$. □

Theorem 9.14 (Unsmoothed synthetic Mertens bound)**.** *For every $x>1$ and every $c>\frac{1}{2}$ the symmetric Perron integral*

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x):=\lim_{U\to\infty}\frac{1}{2\pi i}\int_{c-iU}^{c+iU}F_{\mathrm{RVM}}(s)\frac{x^s}{s}\,ds \tag{9.51}$$

exists, is independent of $c>\frac{1}{2}$, and equals the absolutely convergent expression

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x)=L+\frac{1}{2\pi i}\int_{(c)}(F_{\mathrm{RVM}}(s)-L)\frac{x^s}{s}\,ds. \tag{9.52}$$

For every $\varepsilon\in(0,\frac{1}{2})$,

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x)=L+O_\varepsilon(x^{1/2+\varepsilon}). \tag{9.53}$$

Proof. Fix $c>\frac{1}{2}$. By (9.35) and continuity of F_{RVM} on $\Re s=c$ (no poles there by Proposition 8.1), the integral in (9.52) converges absolutely. The Perron kernel gives, for $x>1$ and $c>0$ [31, Ch. 17] (for $0<x<1$ the limit is 0),

$$\lim_{U\to\infty}\frac{1}{2\pi i}\int_{c-iU}^{c+iU}\frac{x^s}{s}\,ds=1. \tag{9.54}$$

Splitting $F_{\mathrm{RVM}}=L+(F_{\mathrm{RVM}}-L)$ proves existence of (9.51) and (9.52). If $\frac{1}{2}<c<c'$, the integrand of (9.51) is holomorphic on the closed strip $c\leq\Re s\leq c'$ (Proposition 8.1); by Theorem 9.1(a) with $B=c'-\frac{1}{2}$, F_{RVM} is bounded on the horizontal segments $\Im s=\pm U$, $U\geq t_0(B)$, so these contribute $O(x^{c'}U^{-1})\to 0$, and Cauchy's theorem gives independence of c. Take $c=\frac{1}{2}+\varepsilon$ in (9.52):

$$|M^{\mathrm{crit}}_{\mathrm{RVM}}(x)-L|\leq\frac{x^{1/2+\varepsilon}}{2\pi}\int_{-\infty}^{\infty}\frac{|F_{\mathrm{RVM}}(\frac{1}{2}+\varepsilon+it)-L|}{|\frac{1}{2}+\varepsilon+it|}\,dt, \tag{9.55}$$

and the integral is finite by (9.35), the integrand being $O(|t|^{-1-\alpha_c})$. □

Remark 9.15 (Scope). Theorem 9.14 concerns $1/\zeta_{\mathrm{RVM}}$. The transfer to M goes through the endpoint of the λ-family and is Theorem 12.8.

10. The exact Hadamard product and the comparison with ζ

Put

$$P_\zeta(s):=G(s)\zeta(s)=2\xi(s),\qquad \xi(s)=\tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \tag{10.1}$$

Theorem 10.1 (Exact Hadamard product)**.** *Let ρ run over the nontrivial zeros of ζ with $\Im\rho>0$, with multiplicity, ordered by $\Im\rho$. The product*

$$P_\zeta(s)=\prod_{\Im\rho>0}\left(1+\frac{q(s)}{\rho(1-\rho)}\right) \tag{10.2}$$

converges absolutely and locally uniformly in $s\in\mathbb{C}$, and the identity holds with no exponential or constant factor.

Proof. ξ is entire of order one with $\xi(0) = \frac{1}{2}$, so Hadamard's factorization [41], [61, §2.12], [31, Ch. 12] gives

$$2\xi(s) = e^{Bs} \prod_{\rho} \Big(1 - \frac{s}{\rho}\Big) e^{s/\rho}, \tag{10.3}$$

the product running over all nontrivial zeros and converging absolutely and locally uniformly: for $|s| \le K$ and $|\rho| \ge 2K$ the logarithm of the factor $(1 - s/\rho)e^{s/\rho}$ is $O_K(|\rho|^{-2})$, and $\sum_\rho |\rho|^{-2} < \infty$. The functional equation $\xi(s) = \xi(1 - s)$ makes $\rho \mapsto 1 - \rho$ an involution of the zero multiset; no nontrivial zero is real, since $\eta(\sigma) := (1 - 2^{1-\sigma})\zeta(\sigma) = \sum_{n \ge 1} (-1)^{n-1} n^{-\sigma} > 0$ for $0 < \sigma < 1$ (an alternating series with decreasing terms) while $1 - 2^{1-\sigma} < 0$ there, so $\zeta(\sigma) < 0$; so this involution exchanges $\Im\rho > 0$ and $\Im\rho < 0$. Grouping ρ with $1 - \rho$, which absolute convergence permits,

$$\Big(1 - \frac{s}{\rho}\Big)\Big(1 - \frac{s}{1-\rho}\Big) e^{s/\rho + s/(1-\rho)} = \Big(1 + \frac{q(s)}{\rho(1-\rho)}\Big) e^{s/(\rho(1-\rho))}. \tag{10.4}$$

Since $|\rho(1-\rho)| \ge (\Im\rho)^2$ and $\sum_{\Im\rho > 0} (\Im\rho)^{-2} < \infty$, the numbers $1/(\rho(1-\rho))$ are absolutely summable, so the exponentials collect into a single factor:

$$2\xi(s) = e^{B's}\, \Pi_\zeta(s), \qquad \Pi_\zeta(s) := \prod_{\Im\rho > 0} \Big(1 + \frac{q(s)}{\rho(1-\rho)}\Big), \qquad B' = B + \sum_{\Im\rho > 0} \frac{1}{\rho(1-\rho)}. \tag{10.5}$$

Π_ζ depends on s only through $q(s)$, hence $\Pi_\zeta(1 - s) = \Pi_\zeta(s)$. From $\xi(1 - s) = \xi(s)$ we get $e^{B's} = e^{B'(1-s)}$ on the open set where $\Pi_\zeta \ne 0$, so $B' = 0$. No constant factor is left either: the general factorization (10.3) carries the factor e^{A+Bs} with $e^A = 2\xi(0) = 1$, which is why it was written without A; equivalently, at $s = 0$ one has $q(0) = 0$, $\Pi_\zeta(0) = 1$ and $2\xi(0) = 1$, so (10.5) with $B' = 0$ is (10.2). □

Define the exact quotient

$$\mathcal{Q}_{\mathrm{RVM}}(s) := \frac{\zeta(s)}{\zeta_{\mathrm{RVM}}(s)} = \frac{P_\zeta(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}. \tag{10.6}$$

Proposition 10.2 (Exact transform relation)**.** *For $\Re s > 1$, and by meromorphic continuation,*

$$F_{\mathrm{RVM}}(s) = \mathcal{Q}_{\mathrm{RVM}}(s)\, \frac{1}{\zeta(s)}. \tag{10.7}$$

If the inverse Mellin transform $\mathcal{Q}^\vee_{\mathrm{RVM}}$ of $\mathcal{Q}_{\mathrm{RVM}}$ and the inverse defining $M^{\mathrm{crit}}_{\mathrm{RVM}}$ exist in a common distribution space with Mellin convolution, then

$$M^{\mathrm{crit}}_{\mathrm{RVM}} = \mathcal{Q}^\vee_{\mathrm{RVM}} *_{\mathrm{Mel}} M. \tag{10.8}$$

(10.7) is the identity $1/\zeta_{\mathrm{RVM}} = (\zeta/\zeta_{\mathrm{RVM}})(1/\zeta)$. Equation (10.8) is only notation for the Mellin-convolution theorem when both inverse transforms belong to a specified common convolution space; no such space is constructed here.

Corollary 10.3 (All growth of ζ to the right of $\frac{1}{2}$ lies in $\mathcal{Q}_{\mathrm{RVM}}$)**.** *For every $c' > \frac{1}{2}$ and $0 < \varepsilon < c' - \frac{1}{2}$ there are $t_0 = t_0(\varepsilon, c')$ and $0 < c_-(\varepsilon, c') \le c_+(\varepsilon, c')$ such that*

$$c_- \le |\zeta_{\mathrm{RVM}}(\sigma + it)| \le c_+ \qquad (\tfrac{1}{2} + \varepsilon \le \sigma \le c',\ |t| \ge t_0), \tag{10.9}$$

and therefore, in the same region,

$$c_- |\mathcal{Q}_{\mathrm{RVM}}(s)| \le |\zeta(s)| \le c_+ |\mathcal{Q}_{\mathrm{RVM}}(s)|, \qquad \frac{1}{c_+ |\mathcal{Q}_{\mathrm{RVM}}(s)|} \le \Big|\frac{1}{\zeta(s)}\Big| \le \frac{1}{c_- |\mathcal{Q}_{\mathrm{RVM}}(s)|}. \tag{10.10}$$

Consequently:

(i) *the Lindelöf hypothesis holds if and only if $\mathcal{Q}_{\mathrm{RVM}}(\sigma + it) = O_{\sigma,\varepsilon}(|t|^\varepsilon)$ for every $\sigma > \frac{1}{2}$;*

(ii) *every bound for ζ or $1/\zeta$ on $\frac{1}{2} + \varepsilon \le \sigma \le c'$, $|t| \ge t_0$, is equivalent, up to the constants $c_\pm$, to the same bound for $\mathcal{Q}_{\mathrm{RVM}}$ or $1/\mathcal{Q}_{\mathrm{RVM}}$.*

Proof. By Theorem 9.1(a) with $B = c' - \frac{1}{2}$, $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+\mathcal{E}_0)$ with $|\mathcal{E}_0| \le C_B/|t|$ and, by the uniform form (9.34), $|w| \le C_B(|t|/2\pi)^{-\varepsilon}$; for $|t| \ge t_0$ this gives $\frac{1}{2} \le |1+w| \le 2$ and $\frac{1}{2} \le |1+\mathcal{E}_0| \le 2$, hence (10.9) with $c_- = \varpi_{\mathrm{RVM}}/4$, $c_+ = 4\varpi_{\mathrm{RVM}}$. Then (10.10) follows from $\zeta = \mathcal{Q}_{\mathrm{RVM}}\zeta_{\mathrm{RVM}}$. For (i): the Lindelöf hypothesis is equivalent to $\zeta(\sigma+it) = O(|t|^{\varepsilon})$ for every $\sigma > \frac{1}{2}$ [61, Ch. XIII], and (10.10) transfers this to $\mathcal{Q}_{\mathrm{RVM}}$. (ii) is (10.10). □

On the prime side define

$$\mathfrak{d}_n^{\mathrm{RVM}} := \mathbf{1}_{\{n \text{ prime}\}} - c_n^{\mathrm{RVM}}. \tag{10.11}$$

Since $\pi_{\mathrm{RVM}}(1) = 0$, for every integer $N \ge 2$

$$\sum_{2 \le n \le N} \mathfrak{d}_n^{\mathrm{RVM}} = \pi(N) - \pi_{\mathrm{RVM}}(N). \tag{10.12}$$

11. The λ-family from ζ_{RVM} to ζ_{Z1}

For $\lambda \in [0,1]$ define the moving ordinates, the family product and the family zeta function

$$\gamma_k(\lambda) := (1-\lambda)\gamma_k + \lambda\tau_k = \gamma_k + \lambda\eta_k, \tag{11.1}$$

$$P_\lambda(s) := \prod_{k \ge 1} \left(1 + \frac{q(s)}{\frac{1}{4} + \gamma_k(\lambda)^2}\right), \tag{11.2}$$

$$\zeta_\lambda(s) := \frac{P_\lambda(s)}{G(s)}, \qquad F_\lambda(s) := \frac{1}{\zeta_\lambda(s)}, \tag{11.3}$$

and the endpoint

$$P_{Z1} := P_0, \qquad \zeta_{Z1} := \zeta_0, \qquad \zeta_1 = \zeta_{\mathrm{RVM}}. \tag{11.4}$$

Theorem 11.1 (The λ-family)**.** *The product* (11.2) *converges absolutely and locally uniformly jointly on* $[0,1] \times \mathbb{C}$. *For every* λ, P_λ *is entire of order exactly one,* $P_\lambda(1-s) = P_\lambda(s)$, $P_\lambda(\bar{s}) = \overline{P_\lambda(s)}$, $P_\lambda(0) = P_\lambda(1) = 1$, *and its zeros, with multiplicity, are exactly*

$$\frac{1}{2} \pm i\gamma_k(\lambda), \qquad k \ge 1. \tag{11.5}$$

For $0 < \lambda \le 1$ *the ordinates* $\gamma_k(\lambda)$ *are strictly increasing in* k *and all zeros of* P_λ *are simple; at* $\lambda = 0$ *the multiplicities are those of the ordinate list* (γ_k). *For every* $j \ge 0$, $P_\lambda^{(j)} \to P_{\lambda_0}^{(j)}$ *locally uniformly as* $\lambda \to \lambda_0$. *In particular no member* ζ_λ *of the family,* $\zeta_{Z1} = \zeta_0$ *included, can have a zero off the critical line other than* $-2, -4, \ldots$.

Proof. By Lemma 2.2, $|\gamma_k(\lambda) - \gamma_k| \le C_\eta$, and $\gamma_k \asymp k/\log k$; so $\gamma_k(\lambda) \ge \frac{1}{2}\gamma_k$ for $k \ge k_0$, uniformly in λ. On a compact $K \subset \mathbb{C}$,

$$\sum_{k \ge 1} \sup_{s \in K,\ 0 \le \lambda \le 1} \left| \frac{q(s)}{\frac{1}{4} + \gamma_k(\lambda)^2} \right| < \infty, \tag{11.6}$$

which gives joint normal convergence and, by Cauchy's formula, convergence of all s-derivatives. For $|s| \le R_{\mathrm{fam}}$, uniformly in λ,

$$\log \max_{|s| \le R_{\mathrm{fam}}} |P_\lambda(s)| \le \sum_{k \ge 1} \log\Big(1 + \frac{R_{\mathrm{fam}}^2 + R_{\mathrm{fam}}}{\frac{1}{4} + \gamma_k(\lambda)^2}\Big) = O(R_{\mathrm{fam}} \log R_{\mathrm{fam}}), \tag{11.7}$$

so the order is at most one; the zero count is $\asymp T\log T$, so the exponent of convergence is one and the order is exactly one. Symmetries, normalization and divisor follow factor by factor as in Theorem 3.1. For $0 < \lambda \le 1$,

$$\gamma_{k+1}(\lambda) - \gamma_k(\lambda) = (1-\lambda)(\gamma_{k+1} - \gamma_k) + \lambda(\tau_{k+1} - \tau_k) \ge \lambda(\tau_{k+1} - \tau_k) > 0, \tag{11.8}$$

so distinct factors vanish at distinct points and each zero is simple; at $\lambda = 0$ equal ordinates give equal factors. □

Remark 11.2 (All differences lie in $\gamma_k - \tau_k$). For $\lambda, \lambda' \in [0,1]$ and s off the divisors,

$$\frac{\zeta_\lambda(s)}{\zeta_{\lambda'}(s)} = \prod_{k\geq 1} \frac{\big(\frac{1}{4} + (\gamma_k + \lambda'\eta_k)^2\big)\big(\frac{1}{4} + (\gamma_k + \lambda\eta_k)^2 + q(s)\big)}{\big(\frac{1}{4} + (\gamma_k + \lambda\eta_k)^2\big)\big(\frac{1}{4} + (\gamma_k + \lambda'\eta_k)^2 + q(s)\big)}. \tag{11.9}$$

The gamma factor cancels; the members differ only through $\eta_k = \tau_k - \gamma_k$, and all of them coincide if every η_k vanishes.

Theorem 11.3 (Infinitesimal generator)**.** *Put $A_{k,\lambda} := \frac{1}{4} + \gamma_k(\lambda)^2$. Off the divisor of P_λ,*

$$\mathcal{V}_\lambda(s) := \partial_\lambda \log P_\lambda(s) = -2q(s) \sum_{k\geq 1} \frac{\gamma_k(\lambda)\,\eta_k}{A_{k,\lambda}\big(A_{k,\lambda} + q(s)\big)}, \tag{11.10}$$

the series converging locally uniformly on compact sets separated from the divisor. Since G does not depend on λ,

$$\partial_\lambda \log \zeta_\lambda = \mathcal{V}_\lambda, \qquad \partial_\lambda \log F_\lambda = -\mathcal{V}_\lambda. \tag{11.11}$$

Proof. Since $\partial_\lambda A_{k,\lambda} = 2\gamma_k(\lambda)\eta_k$,

$$\partial_\lambda \log\Big(1 + \frac{q}{A_{k,\lambda}}\Big) = -\frac{2q\,\gamma_k(\lambda)\eta_k}{A_{k,\lambda}(A_{k,\lambda} + q)}. \tag{11.12}$$

On compact sets separated from the divisor the summand is $O_K(|\eta_k|\gamma_k^{-3})$, and $\sum_k \gamma_k^{-3} < \infty$, so termwise differentiation is justified. □

Proposition 11.4 (Integrated multiplier)**.** *On every component of $\mathbb{C} \setminus \bigcup_{\lambda'\in[0,1]} \{\frac{1}{2} \pm i\gamma_k(\lambda') : k \geq 1\}$,*

$$\zeta_\lambda(s) = \zeta_{Z1}(s) \exp\Big(\int_0^\lambda \mathcal{V}_{\lambda'}(s)\, d\lambda'\Big), \tag{11.13}$$

$$F_\lambda(s) = F_{Z1}(s) \exp\Big(-\int_0^\lambda \mathcal{V}_{\lambda'}(s)\, d\lambda'\Big), \tag{11.14}$$

and the total multiplier is

$$\Pi_{\mathrm{tr}}(s) := \exp\Big(-\int_0^1 \mathcal{V}_{\lambda'}(s)\, d\lambda'\Big) = \frac{P_{Z1}(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}, \qquad \zeta_{Z1} = \zeta_{\mathrm{RVM}}\,\Pi_{\mathrm{tr}}. \tag{11.15}$$

The identities extend meromorphically to $\mathbb{C}$ through the product form.

Proof. (11.13) and (11.14) are the fundamental theorem of calculus applied to (11.11). For (11.15), integrate each mode with $r = \gamma_k(\lambda')$, $dr = \eta_k\, d\lambda'$:

$$\int_0^1 \partial_{\lambda'} \log\Big(1 + \frac{q}{A_{k,\lambda'}}\Big) d\lambda' \equiv \log \frac{1 + q/(\frac{1}{4} + \tau_k^2)}{1 + q/(\frac{1}{4} + \gamma_k^2)} \pmod{2\pi i}. \tag{11.16}$$

Normal convergence permits summation, and exponentiation removes the branch ambiguity. □

Corollary 11.5 (Expansion at the endpoint)**.** *Let $K \subset \mathbb{C}$ be compact, disjoint from the zeros of P_{Z1} and from $s = 1$ (equivalently one may state the expansion for the product ratio P_λ/P_{Z1} without this exclusion). Uniformly for $s \in K$, as $\lambda \downarrow 0$,*

$$\zeta_\lambda(s) = \zeta_{Z1}(s)\big(1 + \lambda\mathcal{V}_0(s) + O_K(\lambda^2)\big), \tag{11.17}$$

$$F_\lambda(s) = F_{Z1}(s)\big(1 - \lambda\mathcal{V}_0(s) + O_K(\lambda^2)\big). \tag{11.18}$$

Proof. By Theorem 11.1 the zeros of P_λ converge to those of P_{Z1}, so K avoids the moving divisor for small λ. The series for $\mathcal{V}_\lambda$ and its λ-derivative converge normally on K, uniformly for λ near 0: the summands of (11.10) are $O_K(|\eta_k|\gamma_k^{-3})$, and, since $\partial_\lambda\gamma_k(\lambda) = \eta_k$ and $\partial_\lambda A_{k,\lambda} = 2\gamma_k(\lambda)\eta_k$,

$$\partial_\lambda \mathcal{V}_\lambda(s) = -2q(s) \sum_{k\geq 1} \eta_k^2 \Big(\frac{1}{A_{k,\lambda}(A_{k,\lambda} + q(s))} - \frac{2\gamma_k(\lambda)^2\,(2A_{k,\lambda} + q(s))}{A_{k,\lambda}^2(A_{k,\lambda} + q(s))^2}\Big),$$

whose summands are $O_K(\eta_k^2\gamma_k^{-4})$. Hence $\mathcal{V}_\lambda = \mathcal{V}_0 + O_K(\lambda)$, and (11.13) gives $\log(\zeta_\lambda/\zeta_{Z1}) = \lambda\mathcal{V}_0 + O_K(\lambda^2)$. □

Proposition 11.6 (Residues along the family)**.** *Suppose γ_k is a simple ordinate (it occurs once in the list). Put $\rho_{k,\lambda} = \frac{1}{2} + i\gamma_k(\lambda)$ and, for $0 < \lambda \le 1$,*

$$\mathfrak{r}_{k,\lambda} := \operatorname*{Res}_{s=\rho_{k,\lambda}} \frac{1}{s\,\zeta_\lambda(s)} = \frac{G(\rho_{k,\lambda})}{\rho_{k,\lambda}P'_\lambda(\rho_{k,\lambda})}. \tag{11.19}$$

Then

$$\mathfrak{r}_{k,\lambda} \longrightarrow \frac{G(\frac{1}{2}+i\gamma_k)}{(\frac{1}{2}+i\gamma_k)P'_{Z1}(\frac{1}{2}+i\gamma_k)} \qquad (\lambda \downarrow 0). \tag{11.20}$$

Proof. Theorem 11.1 gives convergence of the zero locations and local uniform convergence of P'_λ; the limit denominator is nonzero because the ordinate γ_k occurs once in the list, so the corresponding endpoint zero $\frac{1}{2} + i\gamma_k$ of P_{Z1} is simple; and G is continuous. □

Remark 11.7 (Multiplicity). For $0 < \lambda \le 1$ every zero of ζ_λ is simple. At $\lambda = 0$ a repeated ordinate of ζ appears in ζ_{Z1} with the same multiplicity, and the residues of Proposition 11.6 are replaced by the full principal part. Multiplicity plays no role in Step 3: Theorem 12.8 uses only that ζ does not vanish in $\Re s > \frac{1}{2}$.

12. The endpoint ζ_{Z1}, Hamburger's theorem, and Step 3

12.1. The analytic class of the endpoint.

Proposition 12.1 (Analytic class of ζ_{Z1})**.** *The endpoint*

$$P_{Z1}(s) = \prod_{k\ge1}\left(1 + \frac{q(s)}{\frac{1}{4}+\gamma_k^2}\right), \qquad \zeta_{Z1}(s) = \frac{P_{Z1}(s)}{G(s)}, \tag{12.1}$$

has the following properties.

(i) *P_{Z1} is entire of order exactly one, $P_{Z1}(1-s) = P_{Z1}(s)$, $P_{Z1}(0) = P_{Z1}(1) = 1$;*

(ii) *the nontrivial zeros of ζ_{Z1} are exactly*

$$\frac{1}{2} \pm i\gamma_k, \qquad k \ge 1, \tag{12.2}$$

with the multiplicities of the ordinate list; its other zeros are $s = -2, -4, \dots$. It is impossible for ζ_{Z1} to vanish at any point off the critical line other than $-2, -4, \dots$: every factor of P_{Z1} vanishes only at $\frac{1}{2} \pm i\gamma_k$;

(iii) *ζ_{Z1} has a simple pole at $s = 1$ with residue 1;*

(iv) *ζ_{Z1} satisfies the functional equation of ζ:*

$$\pi^{-s/2}\Gamma(s/2)\zeta_{Z1}(s) = \pi^{-(1-s)/2}\Gamma((1-s)/2)\zeta_{Z1}(1-s); \tag{12.3}$$

(v) *$(s-1)\zeta_{Z1}(s)$ is entire of order exactly one.*

Proof. (i) and the divisor in (ii) are the case $\lambda = 0$ of Theorem 11.1; the trivial zeros and the absence of other zeros come from $1/G$ as in Theorem 3.1. For (iii), $P_{Z1}(1) = 1$ and

$$G(s) = (s-1) + O\big((s-1)^2\big) \qquad (s \to 1). \tag{12.4}$$

For (iv),

$$\pi^{-s/2}\Gamma(s/2)\zeta_{Z1}(s) = \frac{P_{Z1}(s)}{q(s)}, \tag{12.5}$$

and both P_{Z1} and q are invariant under $s \mapsto 1-s$. For (v),

$$(s-1)\zeta_{Z1}(s) = P_{Z1}(s)\,\frac{\pi^{s/2}}{s\,\Gamma(s/2)}, \tag{12.6}$$

where $s\Gamma(s/2) \to 2$ at $s = 0$ and $1/\Gamma$ is entire of order one; the product has order at most one, and the zero count $\asymp T \log T$ gives order at least one. □

Define the ratio

$$\mathscr{R}_{Z1}(s) := \frac{\zeta_{Z1}(s)}{\zeta(s)} = \frac{P_{Z1}(s)}{P_\zeta(s)}, \tag{12.7}$$

a meromorphic function; $\mathscr{R}_{Z1} \equiv 1$ is the identity $\zeta_{Z1} = \zeta$.

12.2. **Hamburger's theorem at the endpoint.**

Theorem 12.2 (Hamburger closure)**.** *Consider the condition*

$$\textit{(D)} \qquad \zeta_{Z1}(s) = \sum_{n\geq 1} \frac{a_n}{n^s} \qquad \textit{absolutely for } \Re s > 1. \tag{12.8}$$

Then

$$\text{(D)} \iff \mathscr{R}_{Z1} \equiv 1 \iff \zeta_{Z1} = \zeta \iff \text{RH}. \tag{12.9}$$

Proof. Assume (D). Then $\zeta_{Z1}(1-s) = \sum_n a_n n^{-(1-s)}$ converges absolutely for $\Re s < 0$, $(s-1)\zeta_{Z1}$ is entire of finite order (Proposition 12.1(v)), and (12.3) holds. These are the hypotheses of Hamburger's theorem [42], [61, §2.13], which gives

$$\zeta_{Z1} = C_H\, \zeta \tag{12.10}$$

for a constant C_H. Both functions have residue 1 at $s = 1$, so $C_H = 1$. If $\zeta_{Z1} = \zeta$, the nontrivial zeros of ζ are those of ζ_{Z1}, namely (12.2), which is RH. If RH holds, every ρ with $\Im\rho > 0$ is $\frac12 + i\gamma_k$, so $\rho(1-\rho) = \frac14 + \gamma_k^2$, and Theorem 10.1 gives

$$P_\zeta(s) = \prod_{k\geq 1} \Big(1 + \frac{q(s)}{\frac14 + \gamma_k^2}\Big) = P_{Z1}(s) \tag{12.11}$$

exactly, hence $\zeta = \zeta_{Z1}$. Finally $\zeta = \sum_n n^{-s}$ converges absolutely for $\Re s > 1$, so $\zeta_{Z1} = \zeta$ gives (D). □

12.3. **Condition (D) and the integer-weighted zeta.** By Corollary 7.3, $\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$ with first coefficient 1, unconditionally. Define the multiplier

$$\mathcal{K}(s) := \frac{\zeta_{Z1}(s)}{\zeta_{\mathbb{Z},\mathrm{RVM}}(s)}. \tag{12.12}$$

Theorem 12.3 (Reduction of (D) to the multiplier)**.** (i) *If $\mathcal{K}$ has an expansion $\mathcal{K}(s) = \sum_{n\geq 1} k_n n^{-s}$ that converges absolutely in $\Re s > 1$, then* (12.8) *holds.*

(ii) *If* (12.8) *holds, then $\mathcal{K}(s) = \sum_{n\geq 1} k_n n^{-s}$ absolutely on the whole half-plane $\Re s > 1$.*

Thus (D) holds if and only if $\mathcal{K}$ is an absolutely convergent Dirichlet series in $\Re s > 1$. Hence, by Theorem 12.2, an absolutely convergent Dirichlet expansion of $\mathcal{K}$ in $\Re s > 1$ gives $\zeta_{Z1} = \zeta$, RH, and, by Theorem 12.8, $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$.

Proof. (i) $\zeta_{Z1} = \zeta_{\mathbb{Z},\mathrm{RVM}}\,\mathcal{K}$ is the Dirichlet convolution of two absolutely convergent series in $\Re s > 1$, and $\sum_N |(a^{\mathbb{Z}} * k)_N| N^{-\sigma} \leq \sum_N |a^{\mathbb{Z}}_N| N^{-\sigma} \sum_n |k_n| n^{-\sigma} < \infty$. (ii) By (7.15), $1/\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$, and so is its product $\mathcal{K}$ with (12.8). □

By (11.15) and Corollary 7.5,

$$\mathcal{K}(s) = \Pi_{\mathrm{tr}}(s)\, e^{-h_{\mathrm{RVM}}(s)} = \frac{P_{Z1}(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}\, e^{-h_{\mathrm{RVM}}(s)}, \tag{12.13}$$

an explicit function of the displacements $\gamma_k - \tau_k$ and of the cell masses c_n^{RVM}. Condition (D) is thereby located in a single object built from the two constructions of this paper: the λ-family and the integer weights.

Remark 12.4 (Continuous-to-atomic meaning of (D)). The prime-power count J_{Z1}^{M} of ζ_{Z1} (Theorem 13.5) is an L^2_{loc} count built from the zeros, and (D) makes its distributional derivative the atomic measure $\sum_{p,m} m^{-1}\delta_{p^m}$. The final step of the method is therefore the passage of ζ_{Z1} into the discrete Dirichlet category. Theorem 13.7 states it exactly: (D) holds if and only if J_{Z1}^{M} is constant between consecutive integers and its jumps obey $\sum_{n\le X}|\mathsf{b}_n| = O_\varepsilon(X^{1+\varepsilon})$, and Corollary 13.3 reduces it to the single identity (13.12) between the displacements $\gamma_k - \tau_k$ and ϖ_{RVM}.

12.4. **Step 3: the Mertens bound from $\zeta_{Z1} = \zeta$.**

Lemma 12.5 (Vertical growth of ζ). *For $\frac14 \le \sigma \le 4$ and $|t| \ge 2$,*

$$|\zeta(\sigma + it)| \le 15|t|. \tag{12.14}$$

Proof. For $\sigma > 0$ [61, (2.1.5)],

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \{y\} y^{-s-1}\,dy. \tag{12.15}$$

For $|t| \ge 2$ and $\sigma \le 4$ one has $|s| \le 3|t|$, $|s/(s-1)| \le |s|/|t| \le 3$, and the integral is at most $1/\sigma \le 4$; hence $|\zeta(s)| \le 3 + 12|t| \le 15|t|$. □

Lemma 12.6 (Logarithm of ζ off the critical line). *Assume $\zeta(s) \ne 0$ for $\Re s > \frac12$. Let $0 < \delta \le \frac14$ and*

$$\omega(\delta) := \frac{\log(3-2\delta)}{\log(3-\delta)} < 1. \tag{12.16}$$

Then, with the branch of $\log\zeta$ equal to $\sum_{n\ge2}\Lambda(n)n^{-s}/\log n$ on $\Re s > 1$,

$$|\log\zeta(\sigma+it)| \le \Big(\frac{48}{\delta}\log|t|\Big)^{\omega(\delta)} \qquad (\sigma \ge \tfrac12 + \delta,\ |t| \ge 4). \tag{12.17}$$

Consequently, for every $\varepsilon > 0$ there is $t_2 = t_2(\delta,\varepsilon)$ with

$$\Big|\frac{1}{\zeta(\sigma+it)}\Big| \le |t|^{\varepsilon} \qquad (\sigma \ge \tfrac12 + \delta,\ |t| \ge t_2). \tag{12.18}$$

Proof. Fix t with $|t| \ge 4$ and put $s_0 = 2 + it$, $R = \frac32 - \frac{\delta}{4}$, $r_3 = \frac32 - \frac{\delta}{2}$. The closed disc $|s - s_0| \le R$ lies in $\frac12 + \frac{\delta}{4} \le \Re s \le \frac72$, $|\Im s - t| \le \frac32$; it avoids $s = 1$ and, by hypothesis, every zero of ζ. Hence $\log\zeta$ has a holomorphic branch on a neighbourhood of this disc, chosen to agree with the Dirichlet-series branch on the connected overlap with $\Re s > 1$. Such branches are defined component by component on zero-free regions; no global branch on $\Re s > \frac12$ is asserted. On the disc $|\Im s| \ge |t|/2 \ge 2$ and $|\Im s| \le 2|t|$, so Lemma 12.5 gives $\Re\log\zeta = \log|\zeta| \le \log(30|t|)$ there, while $|\log\zeta(s_0)| \le \log\zeta(2)$. The Borel–Carathéodory lemma [60, §5.5], applied with radii $r_3 < R$,

$$\max_{|s-s_0|\le r_3} |\log\zeta(s)| \le \frac{2r_3}{R-r_3}\max_{|s-s_0|=R}\Re\log\zeta(s) + \frac{R+r_3}{R-r_3}|\log\zeta(s_0)|, \tag{12.19}$$

together with $R - r_3 = \delta/4$, $2r_3 \le 3$, $R + r_3 \le 3$ and $\log 30 + \log\zeta(2) < 4 \le 3\log|t|$, gives

$$\max_{|s-s_0|\le r_3} |\log\zeta(s)| \le \frac{12}{\delta}\big(\log(30|t|) + \log\zeta(2)\big) \le \frac{48}{\delta}\log|t|. \tag{12.20}$$

Now let $\frac12 + \delta \le \sigma < \frac32$ and set $r_1 = \frac12$, $r_2 = 2 - \sigma \in (\frac12, \frac32 - \delta]$. On $|s - s_0| = r_1$ one has $\Re s \ge \frac32$, so $|\log\zeta(s)| \le \log\zeta(\frac32) < 1$. Hadamard's three-circle theorem [60, §5.3] gives, with $M_j = \max_{|s-s_0|=r_j}|\log\zeta|$,

$$M_2 \le M_1^{1-\omega} M_3^{\omega}, \qquad \omega = \frac{\log(r_2/r_1)}{\log(r_3/r_1)} \le \frac{\log(3-2\delta)}{\log(3-\delta)} = \omega(\delta). \tag{12.21}$$

Since $M_1 < 1$ and $M_3 \le \frac{48}{\delta}\log|t|$ with $\frac{48}{\delta}\log|t| > 1$, (12.21) gives (12.17) at $s = \sigma + it$, which lies on the circle $|s - s_0| = r_2$ because $|(\sigma+it) - (2+it)| = 2 - \sigma = r_2$. For $\sigma \ge \frac32$, $|\log\zeta| \le \log\zeta(\frac32) < 1$.

Finally $|1/\zeta| = \exp(-\Re \log \zeta) \le \exp|\log\zeta|$, and $(\frac{48}{\delta}\log|t|)^{\omega(\delta)} \le \varepsilon \log|t|$ for $|t| \ge t_2(\delta,\varepsilon)$ because $\omega(\delta) < 1$. □

Lemma 12.7 (Perron kernel)**.** *For $y > 0$, $y \ne 1$, $c > 0$, $T > 0$ [31, Ch. 17],*

$$\left| \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{y^s}{s}\, ds - \mathbf{1}_{\{y>1\}} \right| \le y^c \min\Big(1, \frac{1}{T|\log y|}\Big). \tag{12.22}$$

Theorem 12.8 (Step 3)**.** *If $\zeta_{Z1} = \zeta$, then for every $\varepsilon > 0$*

$$M(x) = O_\varepsilon(x^{1/2+\varepsilon}) \qquad (x \ge 1). \tag{12.23}$$

Proof. Non-vanishing. By Proposition 12.1(ii), ζ_{Z1} has no zeros in $\Re s > \frac12$; since $\zeta = \zeta_{Z1}$, neither has ζ, and Lemma 12.6 applies.

Truncation. Let $x = N + \frac12$ with $N \ge 3$ an integer, $c = 1 + 1/\log x$ and $T = x$. Applying (12.22) with $y = x/n$ to each term of $\sum_n \mu(n) n^{-s} = 1/\zeta(s)$, which converges absolutely on $\Re s = c$, gives

$$M(x) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{x^s}{s\,\zeta(s)}\, ds + O\Big(\sum_{n\ge1} \Big(\frac{x}{n}\Big)^c \min\Big(1, \frac{1}{T|\log(x/n)|}\Big)\Big). \tag{12.24}$$

For $n \le x/2$ or $n \ge 2x$, $|\log(x/n)| \ge \log 2$, and these terms contribute at most $x^c\zeta(c)/(T\log 2) \le e\,x(1+\log x)/(T\log 2)$. For $x/2 < n < 2x$, $(x/n)^c \le 4$ and $|\log(x/n)| \ge |n-x|/(2x)$; as $|n-x| \in \{j - \frac12 : j \ge 1\}$, each value taken at most twice, these terms contribute at most $8\sum_{1\le j\le 2x} \min(1, 4x/(T(2j-1))) \ll 1 + (x/T)\log x$. Hence

$$M(x) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{x^s}{s\,\zeta(s)}\, ds + O(\log x). \tag{12.25}$$

Contour shift. Fix $0 < \delta \le \frac14$. The integrand is holomorphic on the closed rectangle with vertices $c \pm iT$, $\frac12 + \delta \pm iT$: ζ has no zeros there, $s = 1$ is a zero of $1/\zeta$, and $s = 0$ lies outside. By Cauchy's theorem

$$\int_{c-iT}^{c+iT} = \int_{\frac12+\delta-iT}^{\frac12+\delta+iT} + \int_{\frac12+\delta+iT}^{c+iT} - \int_{\frac12+\delta-iT}^{c-iT}. \tag{12.26}$$

For $x \ge t_2(\delta,\varepsilon)$, (12.18) on $|t| = T = x$ bounds the horizontal integrals by

$$\frac{1}{2\pi} \int_{\frac12+\delta}^{c} \frac{x^\sigma T^\varepsilon}{T}\, d\sigma \le x^c T^{\varepsilon-1} = e\,x^\varepsilon. \tag{12.27}$$

On the left side, $1/\zeta$ is continuous on the compact segment $\{\frac12 + \delta + it : |t| \le t_2\}$, bounded by a constant $C(\delta,\varepsilon)$, and (12.18) holds for $|t| \ge t_2$; hence

$$\frac{1}{2\pi} \int_{-T}^{T} \frac{x^{1/2+\delta}}{|\frac12+\delta+it|\,|\zeta(\frac12+\delta+it)|}\, dt \le x^{1/2+\delta}\Big(C'(\delta,\varepsilon) + \int_{t_2}^{T} t^{\varepsilon-1} dt\Big) \ll_{\delta,\varepsilon} x^{1/2+\delta+\varepsilon}. \tag{12.28}$$

Conclusion. (12.25)–(12.28) give $M(x) \ll_{\delta,\varepsilon} x^{1/2+\delta+\varepsilon}$ for $x = N + \frac12$; it suffices to prove the assertion for $0 < \varepsilon_0 \le \frac12$, since larger exponents follow from smaller ones, and for such ε_0 the choice $\delta = \varepsilon = \varepsilon_0/2$ satisfies $\delta \le \frac14$ and gives $O_{\varepsilon_0}(x^{1/2+\varepsilon_0})$. For arbitrary $x \ge 3$, $M(x) = M(\lfloor x\rfloor + \frac12)$ and $\lfloor x\rfloor + \frac12 \le 2x$; for $1 \le x < 3$, $|M(x)| \le 2$. □

Theorem 12.9 (Transfer of the synthetic Mertens bound to M)**.** *Assume $\zeta_{Z1} = \zeta$ and let $0 < \delta \le \frac14$, $\varepsilon > 0$. Then:*

(i) *on $\Re s > \frac12$,*

$$\frac{1}{\zeta(s)} = F_{\mathrm{RVM}}(s)\,\Pi_{\mathrm{tr}}(s)^{-1}, \qquad \Pi_{\mathrm{tr}} = \frac{P_{Z1}}{P^{\mathrm{crit}}_{\mathrm{RVM}}} = \frac{\zeta}{\zeta_{\mathrm{RVM}}}, \tag{12.29}$$

where Π_{tr} is holomorphic and zero-free in $\Re s > \frac12$ and $\Pi_{\mathrm{tr}}(1) = 1$;

(ii) *there is* $t_3 = t_3(\delta,\varepsilon)$ *such that, for* $\frac{1}{2}+\delta \le \sigma \le 2$ *and* $|t| \ge t_3$,

$$|F_{\mathrm{RVM}}(\sigma+it)| \le 4L, \qquad |\Pi_{\mathrm{tr}}(\sigma+it)^{\pm 1}| \le |t|^{\varepsilon}; \tag{12.30}$$

(iii) *the following classical consequence of RH is recovered at the endpoint:*

$$M(x) = O_\varepsilon(x^{1/2+\varepsilon}). \tag{12.31}$$

Proof. (i) By (11.15), $\zeta_{Z1} = \zeta_{\mathrm{RVM}}\Pi_{\mathrm{tr}}$; with $\zeta_{Z1} = \zeta$ this is (12.29). In $\Re s > \frac{1}{2}$, $\zeta = \zeta_{Z1}$ has no zeros (Proposition 12.1(ii)) and ζ_{RVM} has none (Theorem 3.1); both have a simple pole of residue 1 at $s=1$, so Π_{tr} is holomorphic there with $\Pi_{\mathrm{tr}}(1)=1$, and it is zero-free in $\Re s > \frac{1}{2}$.

(ii) By Theorem 9.1(a) with $B = \frac{3}{2}$, for $\delta \le \sigma' \le \frac{3}{2}$ and $|t| \ge t_0$ we have $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+\mathcal{E}_0)$ with $|w| \le C|t|^{-\delta} \le \frac{1}{2}$ and $|\mathcal{E}_0| \le \frac{1}{2}$ for $|t|$ large; hence $|F_{\mathrm{RVM}}| \le 4L$ and

$$\log \zeta_{\mathrm{RVM}} = \log \varpi_{\mathrm{RVM}} + \mathrm{Log}(1+w) + \mathrm{Log}(1+\mathcal{E}_0) = O(1). \tag{12.32}$$

By Lemma 12.6, which applies because $\zeta = \zeta_{Z1}$ has no zeros in $\Re s > \frac{1}{2}$, $|\log \zeta(\sigma+it)| \le (\frac{48}{\delta}\log|t|)^{\omega(\delta)}$ with $\omega(\delta) < 1$. Therefore

$$|\log \Pi_{\mathrm{tr}}(\sigma+it)| = |\log\zeta - \log\zeta_{\mathrm{RVM}}| \le \Big(\frac{48}{\delta}\log|t|\Big)^{\omega(\delta)} + O(1) \le \varepsilon \log|t| \tag{12.33}$$

for $|t| \ge t_3(\delta,\varepsilon)$, which gives (12.30).

(iii) The required estimate for $1/\zeta$ already follows from the zero-free hypothesis $\zeta = \zeta_{Z1}$ through Lemma 12.6 and Theorem 12.8. The factorization (12.29) is therefore a reformulation of the same estimate, not an independent transfer from the synthetic Mertens bound. By (i)–(ii), Π_{tr}^{-1} is holomorphic in $\Re s > \frac{1}{2}$ and grows at most like $|t|^{\varepsilon}$. Hence $|1/\zeta(\sigma+it)| \le 4L|t|^{\varepsilon}$ for $\frac{1}{2}+\delta \le \sigma \le 2$, $|t| \ge t_3$, and the truncated Perron formula, contour shift and estimates (12.25)–(12.28) of Theorem 12.8, applied with this bound, give $M(x) \ll_{\delta,\varepsilon} x^{1/2+\delta+\varepsilon}$. Choosing $\delta = \varepsilon = \varepsilon_0/2$ gives (12.31). □

Remark 12.10 (The transfer in Mellin form). Formally, (12.29) reads $M = (\Pi_{\mathrm{tr}}^{-1})^{\vee} *_{\mathrm{Mel}} M_{\mathrm{RVM}}^{\mathrm{crit}}$: the Mertens function is the synthetic Mertens function convolved with the inverse Mellin transform of the inverse transport multiplier. The synthetic factor F_{RVM} contributes the exponent $\frac{1}{2}$ and is bounded (Theorem 9.1); the multiplier contributes at most $|t|^{\varepsilon}$. All the information of the λ-family enters through $\Pi_{\mathrm{tr}} = P_{Z1}/P_{\mathrm{RVM}}^{\mathrm{crit}}$, that is, through the displacements $\gamma_k - \tau_k$.

Corollary 12.11 (Step 3 through (D))**.** *Condition* (12.8) *implies* $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ *for every* $\varepsilon > 0$.

Proof. Theorem 12.2 and Theorem 12.8. □

Remark 12.12 (Status of Step 3). The implication $\zeta_{Z1} = \zeta \Rightarrow M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ is the classical consequence RH $\Rightarrow M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ (see [61, §14.25]) rewritten and reproved in the endpoint notation. It requires nothing beyond the identity: no separate statement about the zeros of ζ enters, because $\zeta = \zeta_{Z1}$ already carries the divisor (12.2). The identity $\zeta_{Z1} = \zeta$ itself, equivalently (D), equivalently an absolutely convergent Dirichlet expansion of the multiplier $\mathcal{K}$ of (12.12) in $\Re s > 1$ (Theorem 12.3), is not proved in this paper. It is the open step of the method.

12.5. **Reciprocal endpoint and finite-height Perron integrals.** Put

$$F_{Z1}(s) := \frac{1}{\zeta_{Z1}(s)} = F_{\mathrm{RVM}}(s)\,\Pi_{\mathrm{tr}}(s)^{-1}, \tag{12.34}$$

the second identity by (11.15). For $c > 1$ and finite U define

$$M_{\lambda,U}(x) := \frac{1}{2\pi i}\int_{c-iU}^{c+iU} F_\lambda(s)\frac{x^s}{s}\,ds, \qquad M_{Z1,U} := M_{0,U}. \tag{12.35}$$

Proposition 12.13 (Finite-height Perron integrals along the family)**.** *For fixed $c > 1$, $U < \infty$ and $x > 0$,*

$$M_{\lambda,U}(x) \longrightarrow M_{Z1,U}(x) \qquad (\lambda \downarrow 0). \tag{12.36}$$

Proof. The segment $\{c + it : |t| \le U\}$ is compact and disjoint from every divisor (11.5), and $P_\lambda \to P_{Z1}$ uniformly on it, so $F_\lambda \to F_{Z1}$ uniformly there. □

Theorem 12.14 (Reciprocal-coefficient criterion)**.** *Assume that in $\Re s > 1$*

$$F_{Z1}(s) = \sum_{n\ge1} \frac{\mu_{Z1}(n)}{n^s} \tag{12.37}$$

absolutely, and let $M_{Z1}(x) = \sum_{n\le x} \mu_{Z1}(n)$, so that

$$\mu_{Z1}(n) = M_{Z1}(n) - M_{Z1}(n-1). \tag{12.38}$$

Then the following are equivalent: (i) $M_{Z1}(n) = M(n)$ for every $n \ge 1$; (ii) $\mu_{Z1} = \mu$; (iii) $F_{Z1} = 1/\zeta$ in $\Re s > 1$; (iv) $\zeta_{Z1} = \zeta$; (v) RH.

Proof. (i)⇔(ii) by (12.38); (ii)⇔(iii) by uniqueness of absolutely convergent Dirichlet coefficients; (iii)⇔(iv) by analytic continuation, both functions being nonzero in $\Re s > 1$; (iv)⇔(v) by Theorem 12.2. □

Remark 12.15 (A route that does not close Step 3). The following approach does not work. One may try to expand both factors of $F_{Z1} = F_{\mathrm{RVM}}\,\Pi_{\mathrm{tr}}^{-1}$ in (12.34) as absolutely convergent Dirichlet series and identify their Dirichlet convolution with μ. By Theorem 9.13 the expansion of F_{RVM} does not exist in any half-plane, so there are no coefficients to convolve. The endpoint criterion concerns ζ_{Z1} itself, as in Theorem 12.2.

13. The endpoint at infinity and the Euler products

This section applies the clock decomposition of Theorem 9.10 to the ordinates γ_k of the endpoint, and constructs the prime-power counts of ζ_{RVM} and ζ_{Z1} as Mellin transforms of their logarithms. No hypothesis is used.

13.1. **The value of the endpoint at infinity.** With the zero count N_ζ of (2.2) put

$$\mathfrak{e}(\tau) := N_\zeta(\tau) - \mathcal{N}_{\mathrm{RVM}}(\tau), \qquad \mathfrak{e}_1(\tau) := \int_{\tau_0}^{\tau} \mathfrak{e}(v)\,dv \qquad (\tau \ge \tau_0 = 4\pi). \tag{13.1}$$

By (2.2), $\mathfrak{e}(\tau) = S(\tau) + O(1/\tau)$, and Littlewood's bound $\int_0^T S(t)\,dt = O(\log T)$ [61, Thm. 9.9] gives an absolute constant $C_{\mathfrak{e}}$ with

$$|\mathfrak{e}_1(\tau)| \le C_{\mathfrak{e}} \log\tau \qquad (\tau \ge \tau_0). \tag{13.2}$$

For the clock the corresponding primitive is $-\mathfrak{s}_1$, bounded by Lemma 9.9.

Theorem 13.1 (Clock decomposition of the endpoint)**.** *Let $s = \frac12 + z$, $z = \sigma' + it$, $\sigma' > 0$. There is an absolute constant $R_{\mathrm{end}} > 0$ such that*

$$\log\zeta_{Z1}(s) = \log\varpi_{Z1} - \int_{\tau_0}^{\infty} \mathfrak{e}_1(\tau)\Big(\frac{1}{(\tau - iz)^2} + \frac{1}{(\tau + iz)^2}\Big)d\tau + O\Big(\frac{1}{|z|}\Big) \qquad (|z| \ge R_{\mathrm{end}}), \tag{13.3}$$

where

$$\varpi_{Z1} := P_{Z1}(\tfrac12)\exp\Big(\nu_3 + 2\int_{\tau_0}^{\infty} \frac{\mathfrak{e}_1(\tau)}{\tau^2}\,d\tau\Big) > 0. \tag{13.4}$$

Consequently:

(i) *for $\sigma' > 0$ and $|z| \ge R_{\mathrm{end}}$,*

$$\big|\log\zeta_{Z1}(s) - \log\varpi_{Z1}\big| \le \frac{2(\pi+2)C_{\mathfrak{e}}}{\sigma'}\Big(1 + \log(2|t| + 2\sigma' + \tau_0)\Big) + \frac{C}{|z|}; \tag{13.5}$$

(ii) $\zeta_{Z1}(\sigma) \to \varpi_{Z1}$ *as* $\sigma \to +\infty$*;*
(iii) *the value at infinity is the ordered product*

$$\varpi_{Z1} = \varpi_{\mathrm{RVM}} \prod_{k\ge 1} \frac{\frac14+\tau_k^2}{\frac14+\gamma_k^2} := \varpi_{\mathrm{RVM}} \lim_{K\to\infty} \prod_{k\le K} \frac{\frac14+\tau_k^2}{\frac14+\gamma_k^2}. \tag{13.6}$$

Proof. As in (9.6), $\log P_{Z1}(s) - \log P_{Z1}(\frac12) = \sum_k f(\gamma_k)$ with $f(\tau) = \mathrm{Log}(1 + z^2/\tau^2)$. Since $\gamma_1 > \tau_0$, $N_\zeta(\tau_0) = 0$ and $\mathfrak{e}(\tau_0) = -\mathcal{N}_{\mathrm{RVM}}(\tau_0)$. Stieltjes integration against $N_\zeta = \mathcal{N}_{\mathrm{RVM}} + \mathfrak{e}$ and two integrations by parts ($f\mathfrak{e} \to 0$ and $\mathfrak{e}_1 f' \to 0$ at ∞, because $f(\tau) = O(|z|^2\tau^{-2})$, $f'(\tau) = O(|z|^2\tau^{-3})$, $\mathfrak{e}(\tau) = O(\log\tau)$ and $\mathfrak{e}_1(\tau) = O(\log\tau)$ by (13.2); and $\mathfrak{e}_1(\tau_0) = 0$) give

$$\sum_k f(\gamma_k) = \Lambda_{\mathrm{sm}}(z) + \int_{\tau_0}^{\infty} \mathfrak{e}_1 f''\, d\tau, \tag{13.7}$$

with the same Λ_{sm} as in (9.9). For $\sigma' > \frac12$ the proof of Theorem 9.10 gives $\Lambda_{\mathrm{sm}} - \log G = \nu_3 + O(|z|^{-1})$; for $0 < \sigma' \le \frac12$ the same estimate holds because (9.40) is an exact formula valid on all of $\Re z > 0$ (the singularities of f' at $\tau = \pm iz$ lie off the positive axis), $|\tau^2 + z^2| \ge \frac34|z|^2$ on $[0,\tau_0]$ for $|z| \ge 2\tau_0$ whatever $\sigma' > 0$, Stirling's formula with remainder $O(|v|^{-1})$ is uniform in $|\arg v| \le \frac{\pi}{2}$, and $\mathrm{Log}(z^2 - \frac14) = 2\,\mathrm{Log}\, z + O(|z|^{-2})$ since $\Im(z^2 - \frac14) = 2\sigma' t$ has the sign of t, so no cut is crossed; an O-estimate is not transported by analytic continuation, and none is needed. Thus, inserting f'' from (9.41) gives (13.3).

For (i), $|\tau \mp iz|^2 = (\tau \pm t)^2 + \sigma'^2$. For real a, splitting at $X = 2|a| + 2\sigma' + \tau_0$ and using $(\tau - a)^2 \ge \tau^2/4$ for $\tau \ge X$,

$$\int_{\tau_0}^{\infty} \frac{\log\tau}{(\tau-a)^2+\sigma'^2}\, d\tau \le \frac{\pi}{\sigma'}\log X + 4\,\frac{1+\log X}{X} \le \frac{\pi+2}{\sigma'}(1+\log X). \tag{13.8}$$

With (13.2) this gives (13.5). For (ii) take $t = 0$: the right side of (13.5) tends to 0 as $\sigma' \to \infty$.

For (iii), Theorem 9.10 gives $\varpi_{\mathrm{RVM}} = P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac12)\exp(\nu_3 - 2\int \mathfrak{s}_1\tau^{-2}d\tau)$, so

$$\frac{\varpi_{Z1}}{\varpi_{\mathrm{RVM}}} = \frac{P_{Z1}(\frac12)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac12)} \exp\Big(2\int_{\tau_0}^{\infty} \frac{\mathfrak{e}_1(\tau) + \mathfrak{s}_1(\tau)}{\tau^2}\, d\tau\Big). \tag{13.9}$$

Since $\mathfrak{e} + \psi(\mathcal{N}_{\mathrm{RVM}} + \frac12) = N_\zeta - n_{\mathrm{RVM}}$ and $(\mathfrak{e}_1 + \mathfrak{s}_1)(X)/X \to 0$,

$$\int_{\tau_0}^{\infty} \frac{\mathfrak{e}_1 + \mathfrak{s}_1}{\tau^2}\, d\tau = \lim_{X\to\infty} \int_{\tau_0}^{X} \frac{N_\zeta(\tau) - n_{\mathrm{RVM}}(\tau)}{\tau}\, d\tau = \lim_{K\to\infty} \sum_{k\le K} \log\frac{\tau_k}{\gamma_k}. \tag{13.10}$$

In the last step, $\int_{\tau_0}^{X}(N_\zeta - n_{\mathrm{RVM}})\tau^{-1}d\tau = \sum_{\gamma_k\le X}\log\frac{X}{\gamma_k} - \sum_{\tau_k\le X}\log\frac{X}{\tau_k}$; the indices counted by only one of the two sums are $O(\log X)$ in number and, by Lemma 2.2, have ordinates in $[X - C_\eta, X]$, where $0 \le \log(X/\gamma_k) \le -\log(1 - C_\eta/X) = O(1/X)$, so they contribute $O(\log X/X)$. Finally $P_{Z1}(\frac12)/P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac12) = \prod_k \frac{\gamma_k^2}{\frac14+\gamma_k^2}\cdot\frac{\frac14+\tau_k^2}{\tau_k^2}$, an absolutely convergent product, and multiplying by $\prod_{k\le K}\tau_k^2/\gamma_k^2$ gives (13.6). □

The ordinates enter ϖ_{Z1} only through the displacements $\gamma_k - \tau_k$, in agreement with Remark 11.2. The product in (13.6) is an *ordered* product: its factors are $1 + O(|\eta_k|/\tau_k)$, and $\sum_k |\eta_k|/\tau_k$ is not known to converge ($\sum_k \tau_k^{-1}$ diverges), so absolute convergence is not asserted and no rearrangement is permitted. The limit is that of the partial products over $k \le K$ in increasing height, which is the order in which (13.10) is evaluated, and this order is part of the definition of ϖ_{Z1}; the equality of the ordered limit with ϖ_{Z1} is what (13.10) proves, and the value is not claimed to be independent of other groupings.

Theorem 13.2 (The value at infinity decides the endpoint identity)**.** $\varpi_{Z1} \ge 1$*, and* $\varpi_{Z1} = 1$ *if and only if* $\zeta_{Z1} = \zeta$*.*

Proof. Write $\zeta_{Z1} = \zeta\,\mathscr{R}_{Z1}$ with $\mathscr{R}_{Z1} = P_{Z1}/P_\zeta$ as in (12.7), and compare the products factor by factor, with the zeros $\rho_k = \beta_k + i\gamma_k$ listed with the ordinates (Theorem 10.1). Factors with $\rho_k(1-\rho_k) = \frac14 + \gamma_k^2$ cancel. The remaining factors of P_ζ come in pairs $\{\rho, 1-\bar\rho\}$ with the same ordinate γ; let $\mathcal{P}$ be a set of representatives, one zero ρ with $\Im\rho > 0$ and $\Re\rho \neq \frac12$ from each pair, repeated according to multiplicity. With $d_\rho = \Re\rho - \frac12$ and $W_\rho = (s-\frac12)^2 + \gamma^2$, the corresponding factor of $\mathscr{R}_{Z1}$ is

$$\frac{W_\rho^2}{(W_\rho - d_\rho^2)^2 + 4d_\rho^2\gamma^2}\cdot c_\gamma, \qquad c_\gamma = \frac{|\rho(1-\rho)|^2}{(\frac14+\gamma^2)^2} = 1 + \frac{2d_\rho^2(\gamma^2 - \frac14) + d_\rho^4}{(\frac14+\gamma^2)^2} > 1. \tag{13.11}$$

For real $\sigma > 1$, $W_\rho \geq \gamma^2 \geq \gamma_1^2$, and the logarithm $\ell_\rho(\sigma)$ of the first factor satisfies $|\ell_\rho(\sigma)| \leq 14d_\rho^2/W_\rho \leq 14d_\rho^2/\gamma^2$, while $0 < \log c_\gamma \leq c_\gamma - 1 \leq 3d_\rho^2/\gamma^2$. Hence $\log\mathscr{R}_{Z1}(\sigma) = \sum_{\rho\in\mathcal{P}}(\ell_\rho(\sigma) + \log c_\gamma)$ is a series dominated, uniformly in $\sigma > 1$, by $\sum_{\rho\in\mathcal{P}} 17d_\rho^2/\gamma^2 \leq \frac{17}{4}\sum_{\rho\in\mathcal{P}}\gamma^{-2} < \infty$; each term tends to $\log c_\gamma$ as $\sigma\to\infty$ because $W_\rho\to\infty$; so, by dominated convergence for series, $\log\mathscr{R}_{Z1}(\sigma) \to \sum_{\rho\in\mathcal{P}}\log c_\gamma$. With $\zeta(\sigma)\to 1$ this gives $\varpi_{Z1} = \lim_{\sigma\to\infty}\zeta_{Z1}(\sigma) = \prod_{\rho\in\mathcal{P}} c_\gamma \geq 1$ (an absolutely convergent product of factors > 1), with equality exactly when there are no such pairs, that is, when $\mathscr{R}_{Z1}\equiv 1$. □

Corollary 13.3 (The endpoint identity as an identity of the displacements)**.** *The following are equivalent: (D); $\zeta_{Z1} = \zeta$; $\varpi_{Z1} = 1$;*

$$\lim_{K\to\infty}\prod_{k\leq K}\frac{\frac14+\gamma_k^2}{\frac14+\tau_k^2} = \varpi_{\mathrm{RVM}} \tag{13.12}$$

(ordered product). Unconditionally $\lim_{\sigma\to+\infty}\Pi_{\mathrm{tr}}(\sigma) = \varpi_{Z1}/\varpi_{\mathrm{RVM}} = \varpi_{Z1}L \geq L$, and the equality $\lim_{\sigma\to+\infty}\Pi_{\mathrm{tr}}(\sigma) = L = 1/\varpi_{\mathrm{RVM}}$ is a fourth condition equivalent to the three above.

Proof. Theorems 12.2, 13.1 and 13.2, and $\Pi_{\mathrm{tr}} = \zeta_{Z1}/\zeta_{\mathrm{RVM}} \to \varpi_{Z1}/\varpi_{\mathrm{RVM}}$ by Theorems 9.10 and 13.1(ii). □

Proposition 13.4 (Boundedness of the endpoint)**.** *Let $\sigma_0 > 1$. Then $|\zeta_{Z1}|$ is bounded above and below on $\Re s \geq \sigma_0$ if and only if*

$$\sup_{\sigma'\geq\sigma_0-\frac12,\ t\geq0}\Big|\Re\int_{\tau_0}^\infty\frac{\mathfrak{e}_1(\tau)+\mathfrak{s}_1(\tau)}{(\tau - t + i\sigma')^2}\,d\tau\Big| < \infty. \tag{13.13}$$

Proof. $\tau + iz = \tau - t + i\sigma'$. For $t \geq 0$ the term with $(\tau - iz)^{-2}$ in (13.3) is bounded by (13.2), since $|\tau - iz| \geq \tau + t$; the separate integral $\int_{\tau_0}^\infty \mathfrak{s}_1(\tau)(\tau+iz)^{-2}d\tau$ is bounded directly by Lemma 9.9 and $\int_{\tau_0}^\infty|\tau+iz|^{-2}d\tau \ll 1/\sigma'$. By (13.3), $\log|\zeta_{Z1}|$ is bounded on $\Re s\geq\sigma_0$, $t\geq 0$, exactly when (13.13) holds, and $\zeta_{Z1}(\bar s) = \overline{\zeta_{Z1}(s)}$ covers $t < 0$. □

The function in (13.13) is, after one integration by parts, the conjugate Poisson integral at distance σ' of the count difference $N_\zeta - n_{\mathrm{RVM}}$; it depends only on the displacements $\gamma_k - \tau_k$.

13.2. The Euler products of ζ_{RVM} and ζ_{Z1}.

Theorem 13.5 (Prime-power counts as Mellin transforms)**.** *Let $X \in \{\mathrm{RVM}, Z1\}$. There is a unique function J_X^{M} on $(1,\infty)$ with $J_X^{\mathrm{M}}(x)x^{-c} \in L^2((1,\infty), dx/x)$ for every $c > 1$ and*

$$\log\zeta_X(s) = s\int_1^\infty J_X^{\mathrm{M}}(x)\,x^{-s-1}\,dx \qquad (\Re s > 1), \tag{13.14}$$

the branch of $\log\zeta_X$ being real on $(1,\infty)$. We use the notation $\int x^{-s}\,dJ_X^{\mathrm{M}} := s\int_1^\infty J_X^{\mathrm{M}}(x)x^{-s-1}dx$ distributionally; no bounded-variation representative is asserted. With this convention $\zeta_X(s) = \exp(\int x^{-s}\,dJ_X^{\mathrm{M}}(x))$.

Proof. ζ_X has no zeros or poles in $\Re s > 1$, so $\log \zeta_X$ is holomorphic there. For $c > 1$, Theorem 9.10 ($X = \mathrm{RVM}$) and Theorem 13.1 ($X = Z1$), together with continuity on the compact part, give $|\log \zeta_X(s)| \le C_c(1 + \log(|t| + 2))$ on $\Re s \ge c$. Hence $\mathsf{F}(s) := \log \zeta_X(s)/s$ is holomorphic on $\Re s > c$ and satisfies $\sup_{\sigma > c} \int_{\mathbb{R}} |\mathsf{F}(\sigma + it)|^2 dt \le B_c^2 < \infty$: $\mathsf{F}(\,\cdot + c)$ belongs to the Hardy space H^2 of the right half-plane. By the Paley–Wiener theorem for H^2 of a half-plane [57, Thm. 19.2] there is $g \in L^2(0, \infty)$ with $\mathsf{F}(s) = \int_0^\infty g(u) e^{-(s-c)u} du$ for $\Re s > c$, the integral converging absolutely. Put $j(u) := g(u) e^{cu}$ for $u > 0$; then

$$\frac{\log \zeta_X(s)}{s} = \int_0^\infty j(u)\, e^{-su}\, du \qquad (\Re s > c), \qquad e^{-cu} j \in L^2(0, \infty). \tag{13.15}$$

The function j does not depend on c: if $c' > c$, the representation obtained from c' agrees with (13.15) on $\Re s > c'$, and the Laplace transform is injective on functions with $e^{-c'u} j \in L^2$ (the Fourier transform of $e^{-\sigma u} j(u) \mathbf{1}_{u>0}$ is $\mathsf{F}(\sigma + i\cdot)$, and the Fourier–Plancherel transform is injective). Define $J_X^{\mathrm{M}}(e^u) := j(u)$ for $u > 0$, i.e. $J_X^{\mathrm{M}}(x) = j(\log x)$ on $(1, \infty)$; the substitution $x = e^u$ turns (13.15) into (13.14), and $e^{-cu} j \in L^2(0, \infty)$ is the stated condition $J_X^{\mathrm{M}}(x) x^{-c} \in L^2((1, \infty), dx/x)$ for every $c > 1$. Uniqueness is the injectivity just used. □

Theorem 13.6 (Distributional explicit formula)**.** *Let $X \in \{\mathrm{RVM}, Z1\}$, with ordinates $\kappa_k = \tau_k$ for $X = \mathrm{RVM}$ and $\kappa_k = \gamma_k$ for $X = Z1$. In $\mathcal{D}'(1, \infty)$,*

$$(\log x)\, \frac{dJ_X^{\mathrm{M}}}{dx} = 1 - \frac{2}{\sqrt{x}}\, \Re \sum_{k \ge 1} x^{i\kappa_k} - \frac{1}{x(x^2 - 1)}, \tag{13.16}$$

the series converging in $\mathcal{D}'(1, \infty)$. The explicit-formula count

$$J_X(x) := \mathrm{li}(x) - 2\Re \sum_{k \ge 1} \mathrm{Ei}\Big(\big(\tfrac{1}{2} + i\kappa_k\big) \log x \Big) - \log 2 + \int_x^\infty \frac{dy}{y(y^2 - 1) \log y} \tag{13.17}$$

converges in $\mathcal{D}'(1, \infty)$ (J_{RVM} is the function (6.2)), and $J_X^{\mathrm{M}} - J_X$ is a constant C_X on $(1, \infty)$.

Proof. With $\rho_k = \frac{1}{2} + i\kappa_k$, the factorization $\zeta_X = P_X/G$, $P_X'/P_X = \sum_k \big(\frac{1}{s - \rho_k} + \frac{1}{s - \bar\rho_k}\big)$, and the digamma expansion $\frac{1}{2}\frac{\Gamma'}{\Gamma}\big(\frac{s}{2}\big) + \frac{1}{s} = -\frac{\gamma_E}{2} + \sum_{n \ge 1} \big(\frac{1}{2n} - \frac{1}{s + 2n}\big)$ give, for $\Re s > 1$,

$$-\frac{\zeta_X'}{\zeta_X}(s) = \frac{1}{s - 1} - \sum_{k \ge 1} \Big(\frac{1}{s - \rho_k} + \frac{1}{s - \bar\rho_k} \Big) + \sum_{n \ge 1} \Big(\frac{1}{2n} - \frac{1}{s + 2n} \Big) - \frac{\gamma_E}{2} - \frac{1}{2} \log \pi. \tag{13.18}$$

Each term is a Mellin–Stieltjes transform $\int_{[1,\infty)} x^{-s} d\mu$: $dx \mapsto \frac{1}{s-1}$, $x^{\rho - 1} dx \mapsto \frac{1}{s - \rho}$, and the sum over the trivial zeros is kept as a whole, $\sum_n \big(\frac{1}{2n} - \frac{1}{s+2n}\big) = \int_1^\infty \frac{(1 - x^{-s})\, dx}{x(x^2 - 1)}$, a tempered distribution in $u = \log x$ whose restriction to $(1, \infty)$ is the density $-1/(x(x^2 - 1))$ (the two pieces $\sum \frac{1}{2n}$ and $x^{-2n-1} dx$ are not separately summable, nor is $1/(x(x^2 - 1))$ integrable at $x = 1$, so the split is made only on $(1, \infty)$); $\delta_1 \mapsto 1$ accounts for the constant term. After multiplication by x^{-c} and the change $u = \log x$, the Fourier transforms of the partial sums over k and over n converge locally uniformly with polynomial bounds, so the series converge in $\mathcal{S}'$, and on $(1, \infty)$ the limit measure is the right side of (13.16) times dx. Differentiating (13.14) in s shows that $-\zeta_X'/\zeta_X$ is the distributional Mellin transform of $(\log x)\, dJ_X^{\mathrm{M}}$; injectivity gives (13.16). The series over k converges in $\mathcal{D}'(1, \infty)$ because $\int \varphi(x) x^{-1/2 + i\kappa} dx$ is rapidly decreasing in κ for $\varphi \in C_c^\infty(1, \infty)$ and $\kappa_k \asymp k/\log k$. By Lemma 6.1, $\mathrm{Ei}(\rho_k u) = e^{\rho_k u}/(\rho_k u) + \varrho_k$ with $\rho_k = \frac{1}{2} + i\kappa_k$ and $\sum_k |\varrho_k|$ locally uniformly convergent, so (13.17) converges in $\mathcal{D}'(1, \infty)$ for the same reason; differentiating termwise with $\frac{d}{dx} \mathrm{Ei}(\rho \log x) = x^{\rho - 1}/\log x$, $(\log x) J_X'$ equals the right side of (13.16). A distribution on an interval with zero derivative is constant. □

Proof of Theorem 7.4. Take $X = \mathrm{RVM}$ in Theorems 13.5 and 13.6. By Theorem 6.2, J_{RVM} is continuous on $(1, \infty)$, so $J_{\mathrm{RVM}}^{\mathrm{M}} = J_{\mathrm{RVM}} + C_{\mathrm{RVM}}$ pointwise, and $J_{\mathrm{RVM}}^{*} = J_{\mathrm{RVM}}$ on $[2, \infty)$. □

Theorem 13.7 (Cell form of the endpoint identity)**.** *The following are equivalent.*

(i) *(D), that is,* $\zeta_{Z1} = \zeta$.
(ii) J^{M}_{Z1} *is constant on every interval* $(n, n+1)$, $n \geq 1$, *and its jumps* $\mathsf{b}_n := v_n - v_{n-1}$, *where* v_n *is its value on* $(n, n+1)$ *and* $v_0 := 0$, *satisfy* $\sum_{n \leq X} |\mathsf{b}_n| = O_\varepsilon(X^{1+\varepsilon})$ *for every* $\varepsilon > 0$.
(iii) *For every* $n \geq 1$, *in* $\mathcal{D}'(n, n+1)$,

$$2\,\Re \sum_{k \geq 1} \big(x^{i\gamma_k} - x^{i\tau_k}\big) = \sqrt{x}\,\log x\; J'_{\mathrm{RVM}}(x), \tag{13.19}$$

and the jumps of J^{M}_{Z1} *satisfy the bound in (ii).*

In this case J^{M}_{Z1} *is Riemann's prime-power count* $J(x) = \sum_{p^m \leq x} 1/m$.

Proof. (ii)⇒(i): $|v_n| \leq \sum_{m \leq n} |\mathsf{b}_m| \ll n^{1+\varepsilon}$, so for $\Re s > 1$ summation by parts in (13.14) gives $\log \zeta_{Z1}(s) = \sum_n v_n(n^{-s} - (n+1)^{-s}) = \sum_{n \geq 1} \mathsf{b}_n n^{-s}$, absolutely convergent by partial summation. The exponential of an absolutely convergent Dirichlet series is again an absolutely convergent Dirichlet series, so ζ_{Z1} satisfies (12.8).

(i)⇒(ii): by Theorem 12.2, $\zeta_{Z1} = \zeta$, and $\log \zeta(s) = s \int_1^\infty J(x) x^{-s-1} dx$ for $\Re s > 1$; by uniqueness in Theorem 13.5, $J^{\mathrm{M}}_{Z1} = J$, which is constant between consecutive integers, with jumps $1/m$ at p^m and $\sum_{n \leq X} |\mathsf{b}_n| = J(X) \ll X$.

(ii)⇔(iii): Theorem 13.6 for $X = Z1$ and $X = \mathrm{RVM}$, with $dJ^{\mathrm{M}}_{\mathrm{RVM}}/dx = J'_{\mathrm{RVM}}$, gives in $\mathcal{D}'(1, \infty)$

$$(\log x)\Big(\frac{dJ^{\mathrm{M}}_{Z1}}{dx} - J'_{\mathrm{RVM}}(x)\Big) = -\frac{2}{\sqrt{x}}\,\Re \sum_{k \geq 1} \big(x^{i\gamma_k} - x^{i\tau_k}\big), \tag{13.20}$$

so $dJ^{\mathrm{M}}_{Z1}/dx = 0$ on $(n, n+1)$ is (13.19). □

Remark 13.8 (The final step in cell form). Theorem 13.7 is the exact form of the continuous-to-atomic passage of Remark 12.4. Both sides of (13.19) are built from the midpoint construction: the right side from the clock alone (J'_{RVM} is bounded on $[2, \infty)$ by Theorem 7.1, equal to $1/\log x$ plus the unit-periodic first harmonic and the harmonics with stationary points $2\pi x^{1/m}$, Section 7), the left side from the displacements $\gamma_k - \tau_k$. The final step of the method is the identity (13.19) on every cell, equivalently (13.12). Numerically, the explicit formula (13.17) with the γ_k is constant between consecutive integers up to the truncation of the list (Experiment 19.9, Figure 12), and the partial products of (13.12) approach ϖ_{RVM} (Experiment 19.3).

14. The atomized family and the programme for the final step

The continuous family ζ_λ keeps, for every λ, the functional equation and the zeros on the critical line. Its atomization $\tilde{\zeta}_{\mathbb{Z},\lambda}$, built from the cell masses of the prime coordinate, keeps the Euler product and the Dirichlet series. The two are joined by a factor $e^{\tilde{h}_\lambda}$, and both families have a limit at $\lambda = 0$. Hamburger's theorem needs the functional equation and the Dirichlet series on one and the same function; at $\lambda = 0$ this is the vanishing of the joining factor. This section constructs the atomized family for every λ, proves its analytic properties, writes the joining factor cell by cell, and lists in Subsection 14.8 the steps that remain, in the order in which they will be followed. The passage from a continuous prime system to a discrete one is classical for Beurling generalized primes [23, 34, 27, 35]; the atomization used here places the masses at the integers, which produces ordinary Dirichlet series.

Throughout, $\eta_k = \tau_k - \gamma_k$, $\gamma_k(\lambda) = (1-\lambda)\gamma_k + \lambda\tau_k$, $\rho_{k,\lambda} = \frac{1}{2} + i\gamma_k(\lambda)$, and λ' denotes an integration variable in $[0, 1]$.

14.1. **The family at infinity.** Put $N_\lambda(\tau)=\#\{k:\gamma_k(\lambda)\le\tau\}$, $\mathfrak{e}^{(\lambda)}=N_\lambda-\mathcal{N}_{\mathrm{RVM}}$ and $\mathfrak{e}_1^{(\lambda)}(\tau)=\int_{\tau_0}^{\tau}\mathfrak{e}^{(\lambda)}$, so that $\mathfrak{e}_1^{(0)}=\mathfrak{e}_1$ and $\mathfrak{e}_1^{(1)}=-\mathfrak{s}_1$.

Proposition 14.1 (Clock decomposition along the family)**.** *There is an absolute constant C with $|\mathfrak{e}_1^{(\lambda)}(\tau)|\le C\log\tau$ for $\tau\ge\tau_0$ and all $\lambda\in[0,1]$. Consequently Theorem 13.1 holds for every ζ_λ with $\mathfrak{e}_1$ replaced by $\mathfrak{e}_1^{(\lambda)}$, uniformly in λ; in particular $\varpi_\lambda:=\lim_{\sigma\to+\infty}\zeta_\lambda(\sigma)$ exists and*

$$\varpi_\lambda=\varpi_{\mathrm{RVM}}\lim_{K\to\infty}\prod_{k\le K}\frac{\frac14+\tau_k^2}{\frac14+\gamma_k(\lambda)^2},\qquad \varpi_1=\varpi_{\mathrm{RVM}},\quad \varpi_0=\varpi_{Z1}. \tag{14.1}$$

Proof. $\gamma_k(\lambda)$ lies between γ_k and τ_k, so $\gamma_1(\lambda)>\tau_0$. For $X\ge\tau_0$,

$$\int_{\tau_0}^{X}(N_\lambda-n_{\mathrm{RVM}})=\sum_k\big((X-\gamma_k(\lambda))^+-(X-\tau_k)^+\big)=(1-\lambda)\sum_{\gamma_k(\lambda),\tau_k\le X}\eta_k+O(\log X), \tag{14.2}$$

the indices counted by only one of the two sums being $O(\log X)$ in number with ordinates in $[X-C_\eta,X]$ (Lemma 2.2). The case $\lambda=0$ of (14.2) identifies the last sum with $\mathfrak{e}_1+\mathfrak{s}_1+O(\log X)=O(\log X)$, by (13.2) and Lemma 9.9. Since $\mathfrak{e}_1^{(\lambda)}=\int(N_\lambda-n_{\mathrm{RVM}})-\mathfrak{s}_1$, the bound follows, and the proof of Theorem 13.1 applies verbatim. □

Define the *flow at infinity*

$$\Upsilon(\lambda):=\sum_{k\ge1}\frac{2\gamma_k(\lambda)\,\eta_k}{\frac14+\gamma_k(\lambda)^2}\qquad\text{(summed in the order of } k). \tag{14.3}$$

Proposition 14.2 (The value at infinity along the family)**.** *The series* (14.3) *converges uniformly on $[0,1]$, $\lambda\mapsto\log\varpi_\lambda$ is C^1 on $[0,1]$, and*

$$\frac{d}{d\lambda}\log\varpi_\lambda=-\Upsilon(\lambda)=\lim_{\sigma\to+\infty}\mathcal{V}_\lambda(\sigma),\qquad \Upsilon'(\lambda)=\sum_{k\ge1}\frac{2\eta_k^2\big(\frac14-\gamma_k(\lambda)^2\big)}{\big(\frac14+\gamma_k(\lambda)^2\big)^2}<0, \tag{14.4}$$

with $\mathcal{V}_\lambda$ the generator of Theorem 11.3. Moreover

$$\int_0^1\Upsilon(\lambda)\,d\lambda=\sum_{k\ge1}\log\frac{\frac14+\tau_k^2}{\frac14+\gamma_k^2}=\log L+\log\varpi_{Z1}, \tag{14.5}$$

so that (D) holds if and only if $\int_0^1\Upsilon(\lambda)\,d\lambda=\log L$, and $\Upsilon(1)<\log L+\log\varpi_{Z1}<\Upsilon(0)$.

Proof. Put $g(u)=2u/(\frac14+u^2)$. The series $\Upsilon(1)=\sum_k g(\tau_k)\eta_k$ converges by summation by parts against $N_\zeta-n_{\mathrm{RVM}}$, exactly as in (13.10). Since $|g(\gamma_k(\lambda))-g(\tau_k)|\le C|\eta_k|\gamma_k^{-2}$ and $\sum_k\eta_k^2\gamma_k^{-2}<\infty$, the series $\Upsilon(\lambda)-\Upsilon(1)$ converges absolutely and uniformly; the same holds for the termwise derivative, which is the series in (14.4). Each term of that series is ≤0 because $\gamma_k(\lambda)>\frac12$, and the term $k=1$ is <0 because $\eta_1\ne0$: the first ordinate satisfies $\gamma_1<14.14$ (its rigorous location $\gamma_1=14.1347\ldots$ is classical, [61, Ch. XV]), whereas $\mathcal{N}_{\mathrm{RVM}}(14.14)=0.4499\ldots<\frac12=\mathcal{N}_{\mathrm{RVM}}(\tau_1)$ by (2.3), so $\tau_1>14.14>\gamma_1$ by the monotonicity of $\mathcal{N}_{\mathrm{RVM}}$ on $[2\pi,\infty)$, and $\eta_1=\tau_1-\gamma_1>0$. Hence $\Upsilon'<0$; the formula alone gives only $\Upsilon'\le0$, with strict inequality exactly when $\boldsymbol{\gamma}\ne\boldsymbol{\tau}$. Differentiating (14.1) termwise gives $\frac{d}{d\lambda}\log\varpi_\lambda=-\Upsilon(\lambda)$, and the limit $\sigma\to+\infty$ in the generator (11.10) is justified by Abel summation of the ordered ordinate sum: its partial sums are controlled by the count-difference integral, while the multiplier $q/(A+q)$ is bounded and monotone in A. This gives the same ordered series. Integrating from 0 to 1 and using $\varpi_1=\varpi_{\mathrm{RVM}}$, $\varpi_0=\varpi_{Z1}$ gives (14.5); the last statement follows from Theorem 13.2 and the monotonicity of Υ. □

Proposition 14.3 (The value at infinity and the verified zeros)**.** *Let $T_{\mathrm{ver}} \ge e^4$. If every zero $\beta + i\gamma$ with $0 < \gamma \le T_{\mathrm{ver}}$ has $\beta = \frac{1}{2}$, then*

$$0 \le \log \varpi_{Z1} \le 0.255 \sum_{\gamma > T_{\mathrm{ver}}} \frac{1}{\gamma^2} \le 0.255\Big(\frac{\log(T_{\mathrm{ver}}/2\pi) + 1}{2\pi T_{\mathrm{ver}}} + \frac{2\log T_{\mathrm{ver}} + 1}{T_{\mathrm{ver}}^2} \Big), \tag{14.6}$$

the sum running over the zeros with $\gamma > T_{\mathrm{ver}}$, with multiplicity. With $T_{\mathrm{ver}} = 3 \cdot 10^{12}$ [55] the right side is less than $3.78 \cdot 10^{-13}$; in particular $0 \le \log \varpi_{Z1} < 4 \cdot 10^{-13}$.

Proof. By the proof of Theorem 13.2, $\log \varpi_{Z1} = \sum_{\rho \in \mathcal{P}} \log c_\gamma$, and every $\rho \in \mathcal{P}$ has $\gamma > T_{\mathrm{ver}}$. For $|d_\rho| \le \frac{1}{2}$ and $\gamma \ge T_{\mathrm{ver}}$, (13.11) gives $\log c_\gamma \le c_\gamma - 1 \le (2d_\rho^2 \gamma^2 + d_\rho^4)\gamma^{-4} \le 0.51\,\gamma^{-2}$. Each $\rho \in \mathcal{P}$ stands for the two distinct zeros ρ and $1 - \bar\rho$ with ordinate γ, so $\sum_{\rho \in \mathcal{P}} 0.51\gamma^{-2} \le 0.255 \sum_{\gamma > T_{\mathrm{ver}}} \gamma^{-2}$, the sum over all zeros; this is the first inequality. For the second, Trudgian's explicit bound [74, Cor. 1] gives $N_\zeta(t) = \mathcal{M}(t) + \mathcal{R}(t)$ with $\mathcal{M}(t) := \frac{t}{2\pi}\log\frac{t}{2\pi e} + \frac{7}{8}$ and $|\mathcal{R}(t)| \le 0.112\log t + 0.278\log\log t + 2.51 + 0.2/t \le \log t$ for $t \ge e^4$. Stieltjes integration and one integration by parts (the boundary term at ∞ vanishes because $N_\zeta(t) = O(t\log t)$) give

$$\sum_{\gamma > T} \frac{1}{\gamma^2} = \int_T^\infty \frac{\mathcal{M}'(t)}{t^2}\,dt - \frac{\mathcal{R}(T)}{T^2} + 2\int_T^\infty \frac{\mathcal{R}(t)}{t^3}\,dt \le \frac{\log(T/2\pi)+1}{2\pi T} + \frac{\log T}{T^2} + \frac{\log T + \frac{1}{2}}{T^2}$$

for $T \ge e^4$, since $\mathcal{M}'(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$ and $2\int_T^\infty t^{-3}\log t\,dt = (\log T + \frac{1}{2})/T^2$; this is the second inequality. For $T_{\mathrm{ver}} = 3 \cdot 10^{12}$, $\log(T_{\mathrm{ver}}/2\pi) + 1 = 27.8917\ldots$, the first term is $1.4797\ldots \cdot 10^{-12}$, the second is below 10^{-23}, and $0.255 \cdot 1.4798 \cdot 10^{-12} < 3.78 \cdot 10^{-13}$. □

14.2. **Atomization by smooth cells.** Fix $\varphi_n \in C_c^\infty(n-1, n+1)$, $n \ge 2$, with $\sum_{n \ge 2} \varphi_n = 1$ on $[2, \infty)$ and $\varphi_n(m) = \delta_{nm}$ for integers m. For $\lambda \in [0,1]$ let J_λ be the explicit-formula count (13.17) with the ordinates $\gamma_k(\lambda)$ (a distribution on $(1,\infty)$ by Theorem 13.6, which applies to every ζ_λ by Theorem 13.1), $J_\lambda^* = J_\lambda \mathbf{1}_{[2,\infty)}$, and $\pi_\lambda = \sum_{j \ge 1} \frac{\mu(j)}{j} J_\lambda^*(x^{1/j})$.

Definition 14.4 (Atomized family)**.** The cell masses, the atomized zeta function and its coefficients are

$$\tilde c_n(\lambda) := \langle d\pi_\lambda, \varphi_n \rangle, \qquad \tilde\zeta_{\mathbb{Z},\lambda}(s) := \prod_{n \ge 2} (1 - n^{-s})^{-\tilde c_n(\lambda)} = \sum_{N \ge 1} \frac{\tilde a_N(\lambda)}{N^s}, \tag{14.7}$$

$$\tilde a_N(\lambda) = \sum_{\prod_n n^{r_n} = N} \prod_{n:\, r_n \ge 1} \binom{\tilde c_n(\lambda) + r_n - 1}{r_n}, \qquad \tilde a_1(\lambda) = 1. \tag{14.8}$$

Unlike the discretizations of Beurling systems, where the atoms are placed at chosen real points [34, 27, 35], the masses of $d\pi_\lambda$ are placed at the integers with real exponents; measures with continuous part and non-integer multiplicities of this kind appear also in [47, Ch. 3]. In (14.8), $\binom{c+r-1}{r} := c(c+1)\cdots(c+r-1)/r!$ for real c and integers $r \ge 1$. Thus $\tilde a_p(\lambda) = \tilde c_p(\lambda)$ for primes p, $\tilde a_4 = \tilde c_4 + \frac{1}{2}\tilde c_2(\tilde c_2 + 1)$, $\tilde a_6 = \tilde c_6 + \tilde c_2 \tilde c_3$. The pairing with a fixed smooth partition defines $\tilde c_n(\lambda)$ for every λ without point values of π_λ, and $\lambda \mapsto \tilde c_n(\lambda)$ is continuous. The individual masses depend on the partition (φ_n), which is fixed throughout; the closure statements below are made for this fixed partition. If $d\pi_0 = \sum_p \delta_p$, then $\tilde c_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$ and $\tilde a_N(0) = 1$ for all N. Along the family, in $\mathcal{D}'(1,\infty)$,

$$\frac{\partial J_\lambda}{\partial \lambda}(x) = 2\sum_{k \ge 1} \eta_k\, \Im \frac{x^{\rho_{k,\lambda}}}{\rho_{k,\lambda}}, \qquad \frac{\partial}{\partial\lambda}\Big((\log x)\frac{dJ_\lambda}{dx}\Big) = \frac{2\log x}{\sqrt{x}} \sum_{k \ge 1} \eta_k \sin\big(\gamma_k(\lambda)\log x\big). \tag{14.9}$$

14.3. **The joining factor cell by cell.** Define the *atomization kernel* and the *cell defects*

$$\mathfrak{a}_s(x) := \sum_{n\ge 2} \varphi_n(x)\, n^{-s} - x^{-s}, \qquad \mathfrak{h}_n(\lambda; s) := \langle d\pi_\lambda,\ \varphi_n(x)\big(n^{-s} - x^{-s}\big)\rangle. \tag{14.10}$$

$\mathfrak{a}_s$ vanishes at every integer ≥ 2, and $\mathfrak{h}_n(\lambda; s)$ measures how far the mass of the cell n is from being concentrated at n.

The explicit-formula count J_λ need not vanish on $(1,2)$, whereas π_λ is built from $J_\lambda^* = J_\lambda \mathbf{1}_{[2,\infty)}$. The first cell $(1,2)$ enters the joining factor through

$$\mathfrak{u}_\lambda(s) := C_\lambda\, 2^{-s} + s\int_1^2 J_\lambda^{\mathrm{M}}(x)\, x^{-s-1}\, dx, \qquad C_\lambda := J_\lambda^{\mathrm{M}} - J_\lambda, \tag{14.11}$$

where J_λ^{M} is the Mellin count of ζ_λ (Theorem 13.5, which applies to every ζ_λ by Proposition 14.1) and C_λ is the constant of Theorem 13.6.

Proposition 14.5 (The first cell)**.** *Let $\lambda \in [0,1]$.*

(i) *$\mathfrak{u}_\lambda$ is entire, and for $\Re s > 1$*

$$\log \zeta_\lambda(s) = \mathfrak{u}_\lambda(s) + \langle dJ_\lambda^*, x^{-s}\rangle, \qquad \langle dJ_\lambda^*, x^{-s}\rangle := s\int_2^\infty J_\lambda(x)\, x^{-s-1}\, dx. \tag{14.12}$$

(ii) *$\mathfrak{u}_\lambda(\sigma + it) = o(|t|)$ as $|t| \to \infty$, uniformly for σ in compact sets.*

(iii) *$\mathfrak{u}_\lambda$ is constant if and only if $J_\lambda = 0$ on $(1,2)$; in that case $\mathfrak{u}_\lambda \equiv C_\lambda$.*

For Riemann's count $J(x) = \sum_{p^m \le x} 1/m$, which vanishes on $(1,2)$, the corresponding term is 0.

Proof. (i) $J_\lambda^{\mathrm{M}}(x)x^{-c} \in L^2((1,\infty), dx/x)$ for $c > 1$, so $J_\lambda^{\mathrm{M}} \in L^1(1,2)$ and the integral in (14.11) is entire in s. By (13.14), $\log\zeta_\lambda(s) = s\int_1^2 J_\lambda^{\mathrm{M}} x^{-s-1}dx + s\int_2^\infty (J_\lambda + C_\lambda)x^{-s-1}dx$, and $s\int_2^\infty C_\lambda x^{-s-1}dx = C_\lambda 2^{-s}$. The second integral is the pairing of x^{-s} with the distributional derivative of J_λ^*, which vanishes on $(0,2)$. (ii) $C_\lambda 2^{-s}$ is bounded on vertical strips, and $\int_1^2 J_\lambda^{\mathrm{M}}(x)x^{-s-1}dx = \int_0^{\log 2} J_\lambda^{\mathrm{M}}(e^u)e^{-\sigma u}e^{-itu}du \to 0$ as $|t|\to\infty$ by the Riemann–Lebesgue lemma, uniformly for σ in compact sets. (iii) For a constant c one has $c = c\,2^{-s} + s\int_1^2 c\, x^{-s-1}dx$, so $\mathfrak{u}_\lambda(s) - c = s\int_1^\infty H(x)x^{-s-1}dx$ with $H = (J_\lambda^{\mathrm{M}} - c)\mathbf{1}_{(1,2)} + (C_\lambda - c)\mathbf{1}_{[2,\infty)}$. By the injectivity used in Theorem 13.5, $\mathfrak{u}_\lambda \equiv c$ if and only if $H = 0$ almost everywhere, that is, $C_\lambda = c$ and $J_\lambda^{\mathrm{M}} = C_\lambda$ on $(1,2)$, which is $J_\lambda = 0$ on $(1,2)$. □

Lemma 14.6 (Mean-square convergence of the ordinate series)**.** *Let $\boldsymbol{\kappa}$ be a list with $|\kappa_k - \tau_k| \le C_\eta$ (in particular $\boldsymbol{\kappa} = (\gamma_k(\lambda))$ for any $\lambda \in [0,1]$), $\rho_k := \frac12 + i\kappa_k$, and let $I \subset (1,\infty)$ be compact. The partial sums $S_K(x) := \sum_{k\le K}(x^{\rho_k}/\rho_k + x^{\bar\rho_k}/\bar\rho_k)$ and $E_K(x) := \sum_{k\le K}(\mathrm{Ei}(\rho_k\log x) + \mathrm{Ei}(\bar\rho_k\log x))$ converge in $L^2(I)$ as $K\to\infty$, with*

$$\|S_{K'} - S_K\|_{L^2(I)}^2 + \|E_{K'} - E_K\|_{L^2(I)}^2 \ \le\ C_I \sum_{K<k\le K'} \frac{\log(\kappa_k + 2)}{\kappa_k^2} \qquad (K < K'), \tag{14.13}$$

where C_I depends only on I and C_η. Consequently: (a) *for every Lipschitz ψ with compact support in $(1,\infty)$, the pairing $\langle dJ_X^{\mathrm{M}}, \psi\rangle := -\int J_X^{\mathrm{M}}\psi'\,dx$ equals the series obtained by integrating* (13.16) *term by term, which converges; if moreover ψ' is of bounded variation, the series converges absolutely;* (b) *$\lambda \mapsto J_\lambda$ and $\lambda \mapsto \partial_\lambda J_\lambda$ are continuous from $[0,1]$ to $L^2(I)$;* (c) *the constant $C_\lambda := J_\lambda^{\mathrm{M}} - J_\lambda$ is continuous on $[0,1]$, hence $\sup_\lambda |C_\lambda| < \infty$.*

Proof. In $u = \log x$, $u \in \log I =: [u_1, u_2]$, the difference $S_{K'} - S_K$ is $e^{u/2}\sum_{K<k\le K'}(e^{i\kappa_k u}/\rho_k + e^{-i\kappa_k u}/\bar\rho_k)$. Let $\omega \in C_c^\infty(\mathbb{R})$ be ≥ 1 on $[u_1,u_2]$. Expanding the square and using $|\hat\omega(\xi)| \ll_N$

$(1+|\xi|)^{-N}$,

$$\int \omega(u)\Big|\sum_{K<k\le K'}\frac{e^{i\kappa_k u}}{\rho_k}\Big|^2 du \ll \sum_{K<k,k'\le K'}\frac{(1+|\kappa_k-\kappa_{k'}|)^{-2}}{\kappa_k\kappa_{k'}}$$
$$\ll \sum_{K<k\le K'}\frac{1}{\kappa_k^2}\sum_{j\ge 0}\frac{\#\{k' : j\le|\kappa_{k'}-\kappa_k|<j+1\}}{(1+j)^2} \ll \sum_{K<k\le K'}\frac{\log(\kappa_k+2)}{\kappa_k^2}, \tag{14.14}$$

because the tube gives $\#\{k' : |\kappa_{k'}-t|\le 1\}\ll\log(t+2)$ uniformly (the $\kappa_{k'}$ in $[t-1,t+1]$ have their clock points in $[t-1-C_\eta,t+1+C_\eta]$, an interval of length $2+2C_\eta$, which contains $O(\log(t+2))$ clock points), $\kappa_{k'}\asymp\kappa_k$ on the pairs that matter, and the pairs with $|\kappa_k-\kappa_{k'}|>1$ are summed with the weight $(1+|\kappa_k-\kappa_{k'}|)^{-2}$ against the same local count. The same bound holds for the Ei sums by Lemma 6.1, since $\mathrm{Ei}(\rho u)=e^{\rho u}/(\rho u)+\varrho_k$ with $\sum_k|\varrho_k|$ locally uniformly convergent and $u\ge u_1>0$. This is (14.13), and the right side tends to 0 as $K\to\infty$ because $\kappa_k\asymp k/\log k$.

(a) For $\psi\in C_c^\infty(1,\infty)$ the identity is the convergence in $\mathcal{D}'(1,\infty)$ established in Theorem 13.6. For Lipschitz ψ with compact support in $(1,\infty)$, $\psi'\in L^\infty\subset L^2(I)$, and both sides are continuous in the $L^2(I)$ topology of the partial sums: the left side is the pairing of an $L^2(I)$ limit with ψ', and the terms of the right side are the pairings of the summands with ψ', whose partial sums converge by (14.13) and the Cauchy–Schwarz inequality. Lipschitz continuity alone gives convergence, not absolute convergence: the k-th term is $\int\psi'(x)\,\Re(x^{\rho_k}/\rho_k)\,dx$, of size $O(1/\kappa_k)$ in general. When ψ' is of bounded variation, one integration by parts in $u=\log x$ (against $e^{i\kappa_k u}$, with ψ' absorbing the boundary terms) gains a factor $1/\kappa_k$ and the terms become $O(\kappa_k^{-2}\,\mathrm{TV}(\psi'))$, so the series converges absolutely; this is the case for the test functions $\mathfrak{a}_{ms}(y^j)\mathbf{1}_{y\ge2}$ used below, whose derivatives are piecewise smooth.

(b) With $\boldsymbol{\kappa}=(\gamma_k(\lambda))$ and $\boldsymbol{\kappa}'=(\gamma_k(\lambda'))$, $|\gamma_k(\lambda)-\gamma_k(\lambda')|=|\lambda-\lambda'||\eta_k|\le C_\eta|\lambda-\lambda'|$; the difference of the k-th summands is $O(|\lambda-\lambda'|\log x\,x^{1/2}/\kappa_k)$ termwise, and the tail beyond K is controlled uniformly in λ by (14.13). Hence $J_\lambda\to J_{\lambda'}$ in $L^2(I)$ as $\lambda\to\lambda'$. For $\partial_\lambda J_\lambda$ the summands are $\eta_k\,\partial_\kappa(x^\rho/\rho)=i\eta_k(\log x-1/\rho_k)x^{\rho_k}/\rho_k$, of modulus at most C_I/κ_k on I because $|\eta_k|\le C_\eta$; the part $-i\eta_k x^{\rho_k}/\rho_k^2$ converges absolutely, and for the part $i\eta_k(\log x)x^{\rho_k}/\rho_k$ the proof of (14.13) applies verbatim with $\omega(u)u^2$ in place of $\omega(u)$ and the coefficients bounded by C_η, which gives the same uniform tail bound and hence the continuity of $\partial_\lambda J_\lambda$.

(c) By Plancherel on a fixed line $\Re s=c>1$ (proof of Proposition 14.9), $\lambda\mapsto J_\lambda^{\mathrm M}$ is continuous into the weighted L^2, hence into $L^2(I)$. Pairing $J_\lambda^{\mathrm M}-J_\lambda$ with a fixed $\phi\in C_c^\infty(2,3)$, $\int\phi=1$, gives C_λ as a difference of two continuous functions of λ. □

Theorem 14.7 (The joining factor)**.** *Let $\tilde h_\lambda:=\log\tilde\zeta_{\mathbb{Z},\lambda}-\log\zeta_\lambda$ on the half-plane where $\tilde\zeta_{\mathbb{Z},\lambda}$ converges absolutely (Theorem 14.13). Then*

$$\tilde\zeta_{\mathbb{Z},\lambda}=\zeta_\lambda\,e^{\tilde h_\lambda},\qquad \tilde h_\lambda(s)=-\mathfrak{u}_\lambda(s)+\sum_{m\ge1}\frac1m\sum_{n\ge2}\mathfrak{h}_n(\lambda;ms), \tag{14.15}$$

and, explicitly,

$$\tilde h_\lambda(s)=-\mathfrak{u}_\lambda(s)+\sum_{m\ge1}\frac1m\sum_{j\ge1}\frac{\mu(j)}{j}\int_2^\infty\mathfrak{a}_{ms}(y^j)\,\frac{1}{\log y}\Big(1-\frac{2}{\sqrt y}\Re\sum_{k\ge1}y^{i\gamma_k(\lambda)}-\frac{1}{y(y^2-1)}\Big)dy, \tag{14.16}$$

the series of the integrated terms over k converging absolutely. Its derivative along the family is

$$\partial_\lambda\tilde h_\lambda(s)=-\partial_\lambda\mathfrak{u}_\lambda(s)+\sum_{m\ge1}\frac1m\sum_{j\ge1}\frac{\mu(j)}{j}\int_2^\infty\mathfrak{a}_{ms}(y^j)\,\frac{2}{\sqrt y}\sum_{k\ge1}\eta_k\sin\big(\gamma_k(\lambda)\log y\big)\,dy. \tag{14.17}$$

Here $\partial_\lambda\mathfrak{u}_\lambda(s)=\mathcal{V}_\lambda(s)-\partial_\lambda\langle dJ_\lambda^*,x^{-s}\rangle$ by (14.12) *and* (11.11)*. Moreover $\mathfrak{u}_\lambda(\sigma)\to\log\varpi_\lambda$ as $\sigma\to+\infty$, and $\mathfrak{u}_\lambda\equiv\log\varpi_\lambda$ if J_λ vanishes on $(1,2)$.*

Proof. By the Euler product, $\log\tilde{\zeta}_{\mathbb{Z},\lambda}(s)=\sum_{m\geq1}\frac{1}{m}\sum_n\tilde{c}_n(\lambda)n^{-ms}$. By Proposition 14.5, $\log\zeta_\lambda(s)=\mathfrak{u}_\lambda(s)+\langle dJ_\lambda^*,x^{-s}\rangle$; finite inversion (Proposition 6.3, whose proof applies to J_λ^*) gives $dJ_\lambda^*=\sum_m\frac{1}{m}\,d[\pi_\lambda(x^{1/m})]$, and $\langle d[\pi_\lambda(x^{1/m})],x^{-s}\rangle=\langle d\pi_\lambda,y^{-ms}\rangle$. Inserting $1=\sum_n\varphi_n$ on $[2,\infty)$ gives $\langle d\pi_\lambda,y^{-ms}\rangle=\sum_n(\tilde{c}_n(\lambda)n^{-ms}-\mathfrak{h}_n(\lambda;ms))$, hence (14.15). Letting $\sigma\to+\infty$, $\log\tilde{\zeta}_{\mathbb{Z},\lambda}\to0$ (because $\tilde{a}_1=1$), $\log\zeta_\lambda\to\log\varpi_\lambda$ (Proposition 14.1) and the double series tends to 0 by the bounds of Theorem 14.13; so $\mathfrak{u}_\lambda(\sigma)\to\log\varpi_\lambda$, and a constant $\mathfrak{u}_\lambda$ (Proposition 14.5(iii)) equals $\log\varpi_\lambda$. Atoms of $d\pi_\lambda$ at integers contribute nothing because the test functions vanish there. Formula (14.16) is (14.15) with $d\pi_\lambda$ written by (13.16) and the substitution $x=y^j$; the integrated terms in k are $O_{m,j,s}(\gamma_k(\lambda)^{-2})$ because $\mathfrak{a}_{ms}(2)=0$, and the termwise integration against the Lipschitz test $\mathfrak{a}_{ms}(y^j)\mathbf{1}_{y\geq2}$ is justified by Lemma 14.6(a). The remaining double summation is absolutely controlled by $\sum_{m\geq1}m^{-1}\sum_{j\geq1}j^{-1}(mj|s|)^2 2^{-jm\sigma}<\infty$, so the exchanges of the m- and j-sums with the integrated ordinate sum are justified. Differentiating in λ with (14.9) gives (14.17). The differentiability of $\mathfrak{u}_\lambda(s)$ in λ rests on three facts, for a fixed $\sigma>1$. (1) $|\mathcal{V}_\lambda(\sigma+it)|\ll_\sigma\log^2(|t|+2)$ uniformly in $\lambda\in[0,1]$: in (11.10), $|A_{k,\lambda}+q(s)|=|\gamma_k(\lambda)-t+i(\sigma-\frac12)|\,|\gamma_k(\lambda)+t-i(\sigma-\frac12)|$, so each of the $O(\log(|t|+2))$ ordinates with $|\gamma_k(\lambda)-t|\leq1$ contributes $O_\sigma(1)$, through $|\eta_k|\leq C_\eta$, and the remaining ordinates contribute $O(\log^2(|t|+2))$ in total by summation against the count. (2) $\lambda\mapsto\log\zeta_\lambda(\sigma+it)/(\sigma+it)$ is C^1 into $L^2(\mathbb{R},dt)$ with derivative $\mathcal{V}_\lambda(\sigma+it)/(\sigma+it)$: by (11.11) and the mean-value theorem the difference quotients are dominated by $C_\sigma\log^2(|t|+2)/|\sigma+it|\in L^2(\mathbb{R})$, and $\lambda\mapsto\mathcal{V}_\lambda(\sigma+it)$ is continuous for each t by dominated convergence on the ordinate sum. (3) By Plancherel, J_λ^{M} corresponds to $\log\zeta_\lambda(\sigma+it)/(\sigma+it)$ isometrically (Theorem 13.5), so $\lambda\mapsto J_\lambda^{\mathrm{M}}$ is C^1 in the corresponding weighted L^2 space; the constant $C_\lambda=J_\lambda^{\mathrm{M}}-J_\lambda$ is obtained by pairing with one fixed test function and is C^1 as well, by Lemma 14.6(b). In particular $\|\partial_\lambda J_\lambda\|_{L^2(2,3)}$ is locally uniformly bounded. □

Proposition 14.8 (Explicit cell defects)**.** *For every $n\geq2$, $\lambda\in[0,1]$ and $\Re s>1$,*

$$\mathfrak{h}_n(\lambda;s)=\sum_{j\geq1}\frac{\mu(j)}{j}\int_2^\infty\varphi_n(y^j)(n^{-s}-y^{-js})\,\frac{1}{\log y}\Big(1-\frac{2}{\sqrt{y}}\Re\sum_{k\geq1}y^{i\gamma_k(\lambda)}-\frac{1}{y(y^2-1)}\Big)dy, \tag{14.18}$$

$$\partial_\lambda\mathfrak{h}_n(\lambda;s)=\sum_{j\geq1}\frac{\mu(j)}{j}\int_2^\infty\varphi_n(y^j)(n^{-s}-y^{-js})\,\frac{2}{\sqrt{y}}\sum_{k\geq1}\eta_k\sin\big(\gamma_k(\lambda)\log y\big)\,dy, \tag{14.19}$$

the sums over j being finite and the series of the integrated terms over k converging absolutely, because the test functions are smooth with compact support. At $\lambda=1$ the cell defect $\mathfrak{h}_n(1;\cdot)$ is the dispersion of the cell n for ζ_{RVM}, computed from the clock alone; the absolutely continuous part of $d\pi_{\mathrm{RVM}}$ has bounded density there by Theorem 7.1, while the atoms at 2^j are treated separately, so $\mathfrak{h}_n(1;\cdot)\equiv0$ holds only if this density vanishes on the support of φ_n.

Proof. Insert (13.16) for $d\pi_\lambda=\sum_j\frac{\mu(j)}{j}\,d[J_\lambda^*(x^{1/j})]$ into the definition (14.10) and substitute $x=y^j$; the atoms of $d\pi_\lambda$ at integers, for instance the jump of J_λ^* at 2, contribute nothing because $n^{-s}-x^{-s}$ vanishes at $x=n$ and φ_n vanishes at the other integers. Differentiation in λ uses (14.9); the factor $\log y$ cancels. For the last statement, $\mathfrak{h}_n(1;\cdot)\equiv0$ forces $\varphi_n\,d\pi_{\mathrm{RVM}}=\tilde{c}_n(1)\delta_n$ by the injectivity used in Proposition 14.9, and, away from the explicit atoms at 2^j, an absolutely continuous measure equals a multiple of δ_n only if both vanish; at the powers of 2 the boundary atom is handled by (14.23). □

In the form of the path, the final step requires that in every cell the flow of the displacements cancel exactly the dispersion of the clock, (14.32).

Proposition 14.9 (The limit $\lambda\to0$ and the meaning of a vanishing cell)**.** *For every $n\geq2$ and $\Re s>1$, $\mathfrak{h}_n(\lambda;s)\to\mathfrak{h}_n(0;s)$ and $\tilde{c}_n(\lambda)\to\tilde{c}_n(0)$ as $\lambda\to0^+$, and*

$$\mathfrak{h}_n(0;s)=\mathfrak{h}_n(1;s)-\int_0^1\partial_\lambda\mathfrak{h}_n(\lambda;s)\,d\lambda. \tag{14.20}$$

Moreover $\mathfrak{h}_n(0;s)=0$ for all $\Re s>1$ if and only if $\varphi_n\,d\pi_{Z1}=\tilde{c}_n(0)\,\delta_n$.

Proof. The test function $\varphi_n(x)(n^{-s}-x^{-s})$ is smooth with compact support, and the integrated terms over k are dominated uniformly in λ because $|\gamma_k(\lambda)-\gamma_k|\le C_\eta$; dominated convergence gives the limits for cells away from the powers of 2. For $n=2^j$, the boundary term (14.23) also converges: by Plancherel, $\log\zeta_\lambda(\sigma+it)/(\sigma+it)\to\log\zeta_{Z1}(\sigma+it)/(\sigma+it)$ in $L^2(dt)$ on a fixed line $\sigma>1$, hence $J_\lambda^{\mathrm{M}}\to J_{Z1}^{\mathrm{M}}$ in the corresponding weighted L^2 space, and therefore $J_\lambda\to J_{Z1}$ in $L^2(2,3)$ by Lemma 14.6(b),(c). This proves all stated limits and (14.20). Since $\mathfrak{h}_n(0;s)=\tilde{c}_n(0)n^{-s}-\langle\varphi_n\,d\pi_{Z1},x^{-s}\rangle$, its vanishing for all s says that the compactly supported distribution $\varphi_n\,d\pi_{Z1}$ has the Mellin transform of $\tilde{c}_n(0)\delta_n$; the Mellin transform is injective on such distributions. □

14.4. **Mean square of the cell masses.** From now on the cells are translates of one function: $\varphi_n(x)=\varphi(x-n)$, where $\varphi=\beta_0/\sum_{m\in\mathbb{Z}}\beta_0(\cdot-m)$ for a fixed bump $\beta_0\in C_c^\infty(-1,1)$ positive on $[-\frac12,\frac12]$. Then $\sum_{n\ge2}\varphi_n=1$ on $[2,\infty)$, $\varphi_n(m)=\delta_{nm}$, and $\|\varphi_n^{(A)}\|_\infty=\|\varphi^{(A)}\|_\infty=:C_A$ for every $A\ge0$, uniformly in n. For $n\ge2$ put

$$V_n(\gamma):=\sum_{j\ge1}\frac{\mu(j)}{j}\int_2^\infty\varphi_n(y^j)\,y^{-1/2}e^{i\gamma\log y}\,dy,\qquad W_n(\gamma):=\sum_{j\ge1}\frac{\mu(j)}{j}\int_2^\infty\varphi_n(y^j)\,\frac{y^{-1/2}e^{i\gamma\log y}}{\log y}\,dy, \tag{14.21}$$

so that $W_n'=iV_n$; both are rapidly decreasing in γ when n is not a power of 2 (for $n=2^j$ the term j carries the cut of J_λ^* at $y=2$ and is treated in the proof of Theorem 14.11). For n not a power of 2, the density of $d\pi_\lambda$ used in (14.18) and the derivative (14.9) give, with $\eta_k=\tau_k-\gamma_k$:

$$\tilde{c}_n(\lambda)=\mathfrak{m}_n-2\,\Re\sum_{k\ge1}W_n\big(\gamma_k(\lambda)\big),\qquad\partial_\lambda\tilde{c}_n(\lambda)=2\sum_{k\ge1}\eta_k\,\Im V_n\big(\gamma_k(\lambda)\big), \tag{14.22}$$

where $\mathfrak{m}_n$ collects the terms that do not involve the ordinates and is $O(1)$. If $n=2^j$, the jth Möbius-inversion term crosses the cut at $y=2$ and contributes instead the additional boundary term

$$-\frac{\mu(j)}{j}\int_2^3 J_\lambda(y)\,\frac{d}{dy}\varphi_n(y^j)\,dy, \tag{14.23}$$

whose λ-derivative is obtained by replacing J_λ by $\partial_\lambda J_\lambda$.

Lemma 14.10 (Mean value over the moving ordinates)**.** *There is C such that for all $\lambda\in[0,1]$, $T\ge2$, $u_0\in\mathbb{R}$ and complex a_k with $|a_k|\le1$,*

$$\int_{u_0-1}^{u_0+1}\Big|\sum_{\gamma_k(\lambda)\le T}a_k\,e^{i\gamma_k(\lambda)u}\Big|^2du\le C\,T\log^2T. \tag{14.24}$$

Proof. Let $\omega_0(u)=C_0\big(\sin(\frac{u-u_0}{2})/\frac{u-u_0}{2}\big)^2$ with C_0 such that $\omega_0\ge1$ on $|u-u_0|\le1$; the Fourier transform of ω_0 is bounded and supported in $[-1,1]$. Expanding the square,

$$\begin{aligned}\int\omega_0(u)\Big|\sum_k a_ke^{i\gamma_k(\lambda)u}\Big|^2du&=\sum_{k,k'}a_k\bar{a}_{k'}\,e^{i(\gamma_k(\lambda)-\gamma_{k'}(\lambda))u_0}\,\widehat{\omega_0}\big(\gamma_k(\lambda)-\gamma_{k'}(\lambda)\big)\\&\ll\#\{(k,k'):\gamma_k(\lambda)\le T,\ |\gamma_k(\lambda)-\gamma_{k'}(\lambda)|\le1\}.\end{aligned} \tag{14.25}$$

Since $|\gamma_k(\lambda)-\tau_k|\le C_\eta$, every interval $[t,t+1]$ contains $O(\log(t+2))$ of the $\gamma_k(\lambda)$, uniformly in λ, because the corresponding τ_k lie in $[t-C_\eta,t+1+C_\eta]$, an interval of length $1+2C_\eta$ containing $O(\log(t+2))$ clock points. The number of pairs is therefore $O(T\log^2T)$. □

Theorem 14.11 (Mean square of the cell masses along the family)**.** *For every $\varepsilon > 0$ there is C_ε such that for all $X \geq 2$ and all $\lambda \in [0,1]$*

$$\sum_{X<n\leq 2X} |\tilde{c}_n(\lambda)|^2 \leq C_\varepsilon X^{1+\varepsilon}, \qquad \sum_{X<n\leq 2X} |\partial_\lambda \tilde{c}_n(\lambda)|^2 \leq C_\varepsilon X^{1+\varepsilon}. \tag{14.26}$$

In particular the same bound already includes $\lambda = 0$: $\sum_{X<n\leq 2X} |\tilde{c}_n(0)|^2 \leq C_\varepsilon X^{1+\varepsilon}$.

Proof. Fix ε, X, λ and put $T = X^{1+\varepsilon}$. For n not a power of 2, (14.22) gives $\tilde{c}_n(\lambda) - \mathfrak{m}_n = -2\Re \sum_j \frac{\mu(j)}{j} A_{n,j}$, with

$$A_{n,j} = \sum_k \int_2^\infty \varphi_n(y^j)\, \frac{y^{-1/2+i\gamma_k(\lambda)}}{\log y}\, dy, \qquad 1 \leq j \leq J := \Big\lfloor \frac{\log(2X+1)}{\log 2} \Big\rfloor. \tag{14.27}$$

Large ordinates. In $u = \log y$ the function $\varphi_n(e^{ju})e^{u/2}/u$ has support of length $O((jn)^{-1})$, size $O(n^{1/(2j)})$ and r-th derivative $O_r((jn)^r n^{1/(2j)})$: by Faà di Bruno's formula the r-th derivative of $\varphi_n(e^{ju})$ is a sum of terms $\varphi_n^{(m)}(e^{ju})(je^{ju})^m j^{r-m}$, $m \leq r$, and $\|\varphi_n^{(m)}\|_\infty = C_m$ does not depend on n; the derivatives of $e^{u/2}/u$ are $O_r(e^{u/2})$ because $u \geq \log 2$. Integrating by parts A times, each term with $\gamma_k(\lambda) > T$ is $O_A\big((jn/\gamma_k(\lambda))^A n^{1/(2j)-1}\big)$, and their sum over k is $O(X^{-1})$ once $A > (3+\varepsilon)/\varepsilon$. *The cell $n = 2^j$.* Each block contains exactly one $n = 2^j$; for it the term j carries the cut of J_λ^* at $y = 2$ and is treated directly. It equals $-\frac{\mu(j)}{j}\int_2^3 J_\lambda(y)\, \frac{d}{dy}\varphi_n(y^j)\, dy$; the weight has L^2 norm $O((jn)^{1/2})$, and $\|J_\lambda\|_{L^2(2,3)}$ is bounded uniformly in λ (Theorem 13.5 with the uniform bound of Proposition 14.1 for J_λ^{M}, and Lemma 14.6(c) for $\sup_\lambda |C_\lambda|$). This term is therefore $O((n/j)^{1/2})$ and contributes $O(X)$ to the sum of squares; the same holds for $\partial_\lambda \tilde{c}_n$, with $\partial_\lambda J_\lambda$ in place of J_λ. *Small ordinates.* Let $\mathcal{S}_{n,j,\lambda}(y) = y^{-1/2}(\log y)^{-1} \sum_{\gamma_k(\lambda)\leq T} y^{i\gamma_k(\lambda)}$ and let $I_{n,j}$ be the support of $y \mapsto \varphi_n(y^j)$, of length $O(X^{1/j-1})$. By Cauchy–Schwarz in the cell,

$$\Big| \int \varphi_n(y^j) \mathcal{S}_{n,j,\lambda}(y)\, dy \Big|^2 \leq \|\varphi\|_\infty^2\, |I_{n,j}| \int_{I_{n,j}} |\mathcal{S}_{n,j,\lambda}|^2 dy. \tag{14.28}$$

For fixed j the intervals $I_{n,j}$, $X < n \leq 2X$, cover each point at most twice and lie in $[(X-1)^{1/j}, (2X+1)^{1/j}]$, an interval of length at most 2 in $u = \log y$. With $dy/y = du$ and Lemma 14.10 ($a_k = 1$),

$$\sum_{X<n\leq 2X} \Big| \int \varphi_n(y^j) S\, dy \Big|^2 \ll X^{1/j-1} \Big(\frac{j}{\log X}\Big)^2 \int_{u_0-1}^{u_0+1} \Big| \sum_{\gamma_k(\lambda)\leq T} e^{i\gamma_k(\lambda)u} \Big|^2 du \ll j^2 X^{1/j+2\varepsilon}. \tag{14.29}$$

Assembly. By Minkowski's inequality over j, the ℓ^2 norm over $X < n \leq 2X$ of $\tilde{c}_n(\lambda) - \mathfrak{m}_n$ is $\ll \sum_{j\leq J} (X^{1/(2j)+\varepsilon} + X^{-1/2}) \ll X^{1/2+\varepsilon}$, and $\sum_{X<n\leq 2X} \mathfrak{m}_n^2 \ll X$. This is the first bound with 3ε in place of ε. Away from powers of 2, the argument for $\partial_\lambda \tilde{c}_n$ is the same, by (14.22), with coefficients $a_k = \eta_k/C_\eta$ in Lemma 14.10 and without the factor $(\log y)^{-1}$. □

Corollary 14.12 (Abscissa 1 for every member)**.** *For every $\lambda \in [0,1]$ and $\sigma > 1$,*

$$\sum_{n\geq 2} |\tilde{c}_n(\lambda)|\, n^{-\sigma} < \infty; \tag{14.30}$$

hence $\tilde{\zeta}_{\mathbb{Z},\lambda}$ and $1/\tilde{\zeta}_{\mathbb{Z},\lambda}$ are absolutely convergent Dirichlet series in $\Re s > 1$, for every λ, including the endpoint $\lambda = 0$. Moreover, uniformly in λ, $\#\{n \leq X : |\tilde{c}_n(\lambda)| > n^\delta\} \ll_\varepsilon X^{1-2\delta+\varepsilon}$ for every $\delta > 0$.

Proof. By Cauchy–Schwarz on dyadic blocks, $\sum_{X<n\leq 2X} |\tilde{c}_n(\lambda)| \leq (X \cdot C_\varepsilon X^{1+\varepsilon})^{1/2}$, and summing over $X = 2^i$ gives (14.30) for $\sigma > 1 + \varepsilon/2$, with ε arbitrary. Then $\log \tilde{\zeta}_{\mathbb{Z},\lambda} = \sum_m \frac{1}{m} \sum_n \tilde{c}_n(\lambda) n^{-ms}$ converges absolutely in $\Re s > 1$, and the exponentials of $\pm$ this series are absolutely convergent Dirichlet series there. The last statement is Chebyshev's inequality on dyadic blocks. □

The abscissa $\frac{3}{2}$ obtained below from the primitive $\pi_\lambda - \mathfrak{L}$ (Theorem 14.13(ii)) is not the true one: a bound on the size of $\pi_\lambda - \mathfrak{L}$ does not see the cancellation of the density of $d\pi_\lambda$ in mean square, which Theorem 14.11 captures. Absolute convergence in $\Re s > 1$ is exactly what Hamburger's theorem requires, and Knopp's obstruction (Subsection 14.7) does not arise.

14.5. **Analytic continuation of the atomized family.** Let $\mathfrak{L}(x) = \int_1^x (1 - y^{-1})\, dy/\log y$; then $\log \frac{s}{s-1} = \int_1^\infty x^{-s}\, d\mathfrak{L}(x)$ for $\Re s > 1$ (both sides have derivative $\frac{1}{s} - \frac{1}{s-1}$ and tend to 0 as $s \to +\infty$), and $\mathfrak{L}(x) = \mathrm{li}(x) - \log\log x - \gamma_E$.

Theorem 14.13 (Continuation of the atomized family)**.** *Let $\lambda \in [0,1]$.*

(i) *For every $c > \frac{1}{2}$, $(\pi_\lambda - \mathfrak{L})\, x^{-c} \in L^2((2,\infty), dx/x)$.*

(ii) *$|\tilde{c}_n(\lambda)| \le C(1+I_n^{1/2})$ with I_n as below, and $\sum_n |\tilde{c}_n(\lambda)| n^{-\sigma} < \infty$ for $\sigma > \frac{3}{2}$; by Corollary 14.12 this holds for $\sigma > 1$, and $\tilde{\zeta}_{\mathbb{Z},\lambda}$ is an absolutely convergent Dirichlet series in $\Re s > 1$.*

(iii) *$\tilde{h}_\lambda$ extends holomorphically to $\Re s > \frac{1}{2}$. Hence $\tilde{\zeta}_{\mathbb{Z},\lambda} = \zeta_\lambda e^{\tilde{h}_\lambda}$ is meromorphic in $\Re s > \frac{1}{2}$, with a single pole at $s = 1$ and no zeros.*

(iv) *For $\lambda = 1$, $\tilde{h}_1$ extends holomorphically to $\Re s > 0$, and the zeros of $\tilde{\zeta}_{\mathbb{Z},1}$ in $\Re s > 0$ are exactly $\frac{1}{2} \pm i\tau_k$.*

Proof. (i) By Proposition 14.1, for $c > \frac{1}{2}$ there is C_c with $|\log \zeta_\lambda(s) - \log \frac{s}{s-1}| \le C_c(1+\log(|t|+2))$ on $\Re s \ge c$; the function $\log \zeta_\lambda - \log \frac{s}{s-1}$ is holomorphic on $\Re s > \frac{1}{2}$, where ζ_λ has no zeros, and its pole at 1 is removed. It equals $s\int_1^\infty (J_\lambda^{\mathrm{M}} - \mathfrak{L}) x^{-s-1} dx$ for $\Re s > 1$. The argument of Theorem 13.5, applied on the lines $\Re s = c > \frac{1}{2}$, gives $(J_\lambda^{\mathrm{M}} - \mathfrak{L}) x^{-c} \in L^2$. The constant $C_\lambda = J_\lambda^{\mathrm{M}} - J_\lambda$, the truncation at 2, the terms $j \ge 2$ of the Möbius inversion and $\mathfrak{L}(x) - \mathfrak{L}(x^{1/j})$ preserve the property for $c > \frac{1}{2}$: with the substitution $y = x^{1/j}$, $\int_1^\infty |(J_\lambda^{\mathrm{M}} - \mathfrak{L})(x^{1/j})|^2 x^{-2c} \frac{dx}{x} = j\int_1^\infty |(J_\lambda^{\mathrm{M}} - \mathfrak{L})(y)|^2 y^{-2cj} \frac{dy}{y} < \infty$ because $cj > \frac{1}{2}$, and $\mathfrak{L}(x^{1/j}) \ll x^{1/j}/\log x$ lies in the weighted space for $2c > 1$.

(ii) Put $E = \pi_\lambda - \mathfrak{L}$ and $I_n = \int_{n-1}^{n+1} |E|^2 dx$. Since $\int \varphi_n' = 0$, $\tilde{c}_n(\lambda) = \int \varphi_n\, d\mathfrak{L} - \int E\varphi_n'\, dx$, so $|\tilde{c}_n(\lambda)| \le C(1 + I_n^{1/2})$. By (i), $\sum_n I_n n^{-2c-1} < \infty$, and Cauchy–Schwarz gives $\sum_n I_n^{1/2} n^{-\sigma} < \infty$ for $\sigma > c + 1$.

(iii) With $\psi_n = \varphi_n(x)(n^{-s} - x^{-s})$ one has $|\psi_n'| \le C|s| n^{-\sigma-1}$ and ψ_n vanishes at the ends of the cell, so

$$|\mathfrak{h}_n(\lambda; s)| \le C|s|\, n^{-\sigma-1}(1 + I_n^{1/2}), \qquad \sum_n I_n^{1/2} n^{-\sigma-1} < \infty \quad (\sigma > c). \tag{14.31}$$

Choosing $c \in (\frac{1}{2}, \sigma)$, the series (14.15) converges locally uniformly on $\Re s > \frac{1}{2}$; the terms $m \ge 2$ are evaluated at $\Re(ms) \ge 2\sigma$. Since $\mathfrak{u}_\lambda$ is entire (Proposition 14.5), $\tilde{h}_\lambda$ is holomorphic on $\Re s > \frac{1}{2}$, and ζ_λ has no zeros there.

(iv) By Corollary 7.2, π_{RVM} oscillates by $O(1)$ on every unit cell; integrating by parts, $|\mathfrak{h}_n(1; s)| \le C|s| n^{-\sigma-1}$, and the series converges for $\sigma > 0$. The factor $e^{\tilde{h}_1}$ has no zeros, so the zeros of $\tilde{\zeta}_{\mathbb{Z},1}$ in $\Re s > 0$ are those of ζ_{RVM}. □

Theorem 14.14 (Closing criterion; via Hamburger's theorem [42])**.** *The following are equivalent.*

(i) *(D), that is, $\zeta_{Z1} = \zeta$.*

(ii) *$\tilde{h}_0 \equiv 0$.*

(iii) *$J_{Z1} = 0$ on $(1,2)$, and for every $n \ge 2$ and every $\Re s > 1$,*

$$\int_0^1 \partial_\lambda \mathfrak{h}_n(\lambda; s)\, d\lambda = \mathfrak{h}_n(1; s). \tag{14.32}$$

In this case $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$, $\tilde{a}_N(0) = 1$ for all N, and $\varpi_{Z1} = 1$.

Proof. (ii)⇒(i): $\tilde{h}_0 \equiv 0$ gives $\zeta_{Z1} = \tilde{\zeta}_{\mathbb{Z},0} = \exp\big(\sum_m \frac{1}{m} \sum_n \tilde{c}_n(0) n^{-ms}\big)$, an absolutely convergent Dirichlet series in $\Re s > 1$ by Corollary 14.12. (iii)⇒(ii): by (14.20) every $\mathfrak{h}_n(0;\cdot)$ vanishes, and

$\mathfrak{u}_0 \equiv \log \varpi_{Z1}$ (Theorem 14.7), so $\tilde{h}_0 \equiv -\log \varpi_{Z1}$ is constant; then $\zeta_{Z1} = \varpi_{Z1}\tilde{\zeta}_{\mathbb{Z},0}$ is an absolutely convergent Dirichlet series in $\Re s > 1$ (Corollary 14.12), Theorem 12.2 gives $\zeta_{Z1} = \zeta$, and comparing the first coefficients, $\varpi_{Z1} = 1$. (i)⇒(iii): if $\zeta_{Z1} = \zeta$, Theorem 13.7 gives $J_{Z1}^{\mathrm{M}} = J$; and the explicit-formula count J_{Z1} built from the ordinates (Theorem 13.6) also equals J, because every zero of ζ is then $\frac{1}{2} \pm i\gamma_k$ and the series defining J_{Z1} is von Mangoldt's explicit formula for J [37]; so J_{Z1} vanishes on $(1, 2)$, $d\pi_{Z1} = \sum_p \delta_p$, $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$, every $\mathfrak{h}_n(0;\cdot)$ vanishes, and (14.20) gives (14.32). □

Remark 14.15 (The pointwise size of the cell masses). Theorem 14.14 needs no pointwise bound on $\tilde{c}_n(0)$: its proof uses the cell masses only through (14.30). The bound $\tilde{c}_n(0) = O(n^{\varepsilon})$ for every n is not proved directly; Corollary 14.12 restricts the exceptions to a set of $n \le X$ of size $O(X^{1-2\delta+\varepsilon})$ for the threshold n^{δ}, uniformly in λ, and under the closure every cell mass is bounded, $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$. For n not a power of 2, (14.22) gives the trivial pointwise bound $\tilde{c}_n(\lambda) = O(n^{1/2+\varepsilon})$; the mean square (14.26) shows that the typical cell mass is $O(n^{\varepsilon})$.

14.6. **The two endpoints of the atomized family.**

Proposition 14.16 (Atomization at the clock endpoint)**.** *At $\lambda = 1$:*

(i) *$\tilde{c}_n(1) = O(1)$, and $\tilde{\zeta}_{\mathbb{Z},1}$ is an absolutely convergent Dirichlet series in $\Re s > 1$;*
(ii) *$\tilde{h}_1(\sigma) \to -\log \varpi_{\mathrm{RVM}} = \log L$ as $\sigma \to +\infty$; this limit is $\neq 0$ exactly when $\varpi_{\mathrm{RVM}} \neq 1$, and it is > 0 by Theorem 9.11;*
(iii) *$\tilde{h}_1$ is not constant: for every $\sigma > 1$, $\zeta_{\mathrm{RVM}}(\sigma + it) \to \varpi_{\mathrm{RVM}}$ as $|t| \to \infty$, whereas $\tilde{\zeta}_{\mathbb{Z},1}(\sigma + it)$ has no limit as $|t| \to \infty$.*

Proof. (i) By Corollary 7.2, π_{RVM} oscillates by $O(1)$ on every unit cell, and $\tilde{c}_n(1) = -\int \pi_{\mathrm{RVM}}\varphi_n' \, dx$ with $\int \varphi_n' = 0$. (ii) is (14.15) with $\sigma \to +\infty$; the limit is $\neq 0$ if and only if $\varpi_{\mathrm{RVM}} \neq 1$. (iii) The first limit is (9.33). If $\tilde{\zeta}_{\mathbb{Z},1}(\sigma + it)$ had a limit ℓ as $|t| \to \infty$, the mean-square argument (9.49)–(9.50) applied to $\tilde{\zeta}_{\mathbb{Z},1} - \ell$ on the line $\Re s = \sigma$ would give $\tilde{a}_N(1) = 0$ for $N \ge 2$, so $\tilde{\zeta}_{\mathbb{Z},1} \equiv 1$, which contradicts its pole at $s = 1$ (Theorem 14.13(iii)). If $\tilde{h}_1$ were constant, ζ_{RVM} would be a constant multiple of an absolutely convergent Dirichlet series, which Theorem 9.13 excludes. □

The two families thus keep complementary structures and are joined by a zero-free factor:

$$\underbrace{\zeta_\lambda}_{\substack{\text{functional equation}\\ \text{zeros } \frac{1}{2}\pm i\gamma_k(\lambda)}} \cdot e^{\tilde{h}_\lambda} \;=\; \underbrace{\tilde{\zeta}_{\mathbb{Z},\lambda}}_{\substack{\text{Dirichlet series}\\ \text{Euler product}}}, \qquad \tilde{h}_1 \text{ not constant}, \qquad (D) \iff \tilde{h}_0 \equiv 0. \tag{14.33}$$

At $\lambda = 1$ the joining factor carries the whole difference between the clock and an Euler product; Theorem 14.14 states that this difference disappears at $\lambda = 0$.

The reciprocal of the atomized family is again an Euler product:

$$\frac{1}{\tilde{\zeta}_{\mathbb{Z},\lambda}(s)} = \prod_{n\ge 2} \left(1 - n^{-s}\right)^{\tilde{c}_n(\lambda)} = \sum_{N\ge 1} \frac{\tilde{\mu}_\lambda(N)}{N^s}, \qquad \tilde{\mu}_\lambda(N) = \sum_{\prod_n n^{r_n} = N} \prod_{n:\, r_n \ge 1} (-1)^{r_n} \binom{\tilde{c}_n(\lambda)}{r_n}, \tag{14.34}$$

absolutely convergent in $\Re s > 1$ for every λ (Corollary 14.12), because it is the exponential of the absolutely convergent series $-\sum_m \frac{1}{m} \sum_n \tilde{c}_n(\lambda) n^{-ms}$. Thus $\tilde{\mu}_\lambda(p) = -\tilde{c}_p(\lambda)$, and the atomized Mertens functions

$$\tilde{M}_\lambda(x) := \sum_{N \le x} \tilde{\mu}_\lambda(N), \qquad \frac{1}{\tilde{\zeta}_{\mathbb{Z},\lambda}} = e^{-\tilde{h}_\lambda} F_\lambda, \tag{14.35}$$

exist for every $\lambda \in [0, 1]$. Under the conditions of Theorem 14.14, $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$, so $\tilde{\mu}_0 = \mu$ and $\tilde{M}_0 = M$.

Proposition 14.17 (Mellin form of the endpoint Mertens function)**.** *There are $c_1 \geq 2$ and a unique measurable B_{Z1} on $(0,\infty)$, vanishing on $(0,1)$, with $x^{-c}B_{Z1} \in L^2((1,\infty), dx/x)$ and*

$$F_{Z1}(s) = s\int_1^\infty B_{Z1}(x)\, x^{-s-1}\, dx = \exp\Big(-\int_{[1,\infty)} x^{-s}\, dJ^{\mathrm{M}}_{Z1}(x)\Big) \qquad (\Re s > c_1), \tag{14.36}$$

for every $c > c_1$. Moreover (D) holds if and only if $B_{Z1} = M$ almost everywhere; and if F_{Z1} has an absolutely convergent Dirichlet expansion (12.37)*, then $B_{Z1} = M_{Z1}$.*

Proof. Put $A = 2(\pi+2)C_{\mathfrak{e}}$. By (13.5), for $\sigma' \geq \max(4A, R_{\mathrm{end}})$, $|F_{Z1}(s)| \leq C(|t|+\sigma)^{1/4}$; take $c_1 = \max(2, \frac{1}{2}+4A, \frac{1}{2}+R_{\mathrm{end}})$. Then $\int_{\mathbb{R}} |F_{Z1}(\sigma+it)|^2|\sigma+it|^{-2}dt \leq C'\sigma^{-1/2}$ for $\sigma > c_1$, and the argument of Theorem 13.5 gives B_{Z1}, its vanishing on $(0,1)$, its independence of c and its uniqueness; the second equality in (14.36) is Theorem 13.5. If $\zeta_{Z1} = \zeta$, then $F_{Z1}(s) = 1/\zeta(s) = s\int_1^\infty M(x)x^{-s-1}dx$ and uniqueness gives $B_{Z1} = M$. Conversely, $B_{Z1} = M$ gives $F_{Z1} = 1/\zeta$ on $\Re s > c_1$, hence $\zeta_{Z1} = \zeta$ by analytic continuation. Under (12.37), partial summation gives $F_{Z1}(s) = s\int_1^\infty M_{Z1}(x)x^{-s-1}dx$, and uniqueness gives $B_{Z1} = M_{Z1}$. □

14.7. **Hamburger's hypotheses at the two endpoints.** Hamburger's theorem [42], in the form of [61, §2.13] (see also [59, 25]), has four hypotheses: (H0) $f = G/P$ with G entire of finite order and P a polynomial; (H1) $f(s) = \sum_n a_n n^{-s}$ absolutely for $\sigma > 1$, with arbitrary complex a_n; (H2) $f(s)\Gamma(\frac{s}{2})\pi^{-s/2} = g(1-s)\Gamma(\frac{1-s}{2})\pi^{-(1-s)/2}$; (H3) $g(1-s) = \sum_n b_n n^{s-1}$ absolutely for $\sigma < -\alpha < 0$. It contains no hypothesis on the location of zeros, and its conclusion is $f = C\zeta$. Absolute convergence for $\sigma > 1$ in (H1) cannot be weakened to some half-plane: Knopp constructed infinitely many linearly independent solutions of (H0), (H2), (H3) that are Dirichlet series absolutely convergent only in some half-plane [45, 53]. Within the Selberg class, the functions of degree 1 are classified in [43]. The table records the status at the two endpoints and is updated as the programme advances.

	ζ_{Z1} (continuous limit)	$\tilde{\zeta}_{\mathbb{Z},0}$ (atomized limit)
H0	proved: $(s-1)\zeta_{Z1}$ entire of order 1 (Proposition 12.1)	open: meromorphic in $\Re s > \frac{1}{2}$ only (Theorem 14.13)
H1	open: this is (D)	proved in $\Re s > 1$ (Corollary 14.12)
H2	proved, with $g = f$ (Proposition 12.1)	open
H3	follows from H1	open
zeros	no hypothesis	no hypothesis

The two columns meet when $\tilde{h}_0 \equiv 0$: then $\zeta_{Z1} = \tilde{\zeta}_{\mathbb{Z},0}$ has H0–H3, and Hamburger's theorem gives $\zeta_{Z1} = \zeta$ (Theorem 14.14). It suffices that the joining factor be symmetric (Theorem 15.3), or that ζ_{Z1} have a positive representing measure (Theorem 15.7).

The parallel between the two families, topic by topic of the paper, is the following.

Topic of the paper	continuous family ζ_λ	atomized family $\tilde{\zeta}_{\mathbb{Z},\lambda}$
clock, τ_k, bounded displacement	same	same
definition	Hadamard product over $\frac{1}{2} \pm i\gamma_k(\lambda)$	Euler product over $n \geq 2$ with masses $\tilde{c}_n(\lambda) = \langle d\pi_\lambda, \varphi_n\rangle$
zeros	exactly $\frac{1}{2} \pm i\gamma_k(\lambda)$; impossible elsewhere (Theorem 11.1)	none in $\Re s > \frac{1}{2}$ for every λ, since $\tilde{h}_\lambda$ is holomorphic there (Theorem 14.13(iii)); for $\lambda = 1$, exactly $\frac{1}{2} \pm i\tau_k$ in $\Re s > 0$ (Theorem 14.13(iv))
functional equation	proved for every λ	open; the factor $e^{\tilde{h}_\lambda}$ carries no functional equation

Topic of the paper	continuous family ζ_λ	atomized family $\tilde\zeta_{\mathbb{Z},\lambda}$
Dirichlet series	not for $\lambda=1$ in any half-plane (Theorem 9.13); not in $\Re s>1$ for $0<\lambda<1$ (Proposition 15.1); at $\lambda=0$ this is (D)	proved in $\Re s>1$ for every λ, including $\lambda=0$ (Corollary 14.12)
first coefficient ($\sigma\to+\infty$)	$\varpi_\lambda=\varpi_{\mathrm{RVM}}\prod_k \frac{1/4+\tau_k^2}{1/4+\gamma_k(\lambda)^2}$ (14.1)	$\tilde a_1(\lambda)=1$ for every λ
vertical behaviour	$\lambda=1$: $\zeta_{\mathrm{RVM}}(\sigma+it)\to\varpi_{\mathrm{RVM}}$ as $\lvert t\rvert\to\infty$ (9.33)	almost periodic on every line of absolute convergence; $\lambda=1$: no limit as $\lvert t\rvert\to\infty$ (Proposition 14.16)
half-plane bounds	$\lambda=1$: $C^{-1}\le\lvert\zeta_{\mathrm{RVM}}\rvert\le C$ on $\Re s\ge\sigma_0>1$ (Theorem 9.10); every λ: $\lvert\log\zeta_\lambda-\log\frac{s}{s-1}\rvert\ll_c 1+\log(\lvert t\rvert+2)$ on $\Re s\ge c>\frac12$	$\lvert\tilde\zeta_{\mathbb{Z},\lambda}\rvert\le\sum_N\lvert\tilde a_N(\lambda)\rvert N^{-\sigma_0}$ and $\lvert\tilde\zeta_{\mathbb{Z},\lambda}^{-1}\rvert\le\sum_N\lvert\tilde\mu_\lambda(N)\rvert N^{-\sigma_0}$ on $\Re s\ge\sigma_0>1$, for every λ
prime coordinate	counts J_λ, π_λ (distributions, (13.17))	masses $\tilde c_n(\lambda)$ at the integers; $\tilde c_n(1)=O(1)$; mean square $X^{1+\varepsilon}$ on $(X,2X]$ for every λ (Theorem 14.11)
reciprocal	$F_\lambda=1/\zeta_\lambda$; F_{Z1} is the Mellin transform of B_{Z1} (Proposition 14.17)	Dirichlet series with coefficients $\tilde\mu_\lambda(N)$ (14.34)
Mertens	$M_{\mathrm{RVM}}^{\mathrm{crit}}=L+O_\varepsilon(x^{1/2+\varepsilon})$ (Theorem 9.14); $B_{Z1}=M\iff(D)$	$\tilde M_\lambda=\sum_{N\le x}\tilde\mu_\lambda(N)$ exists for every λ (14.35); the bound $x^{1/2+\varepsilon}$ is open (it needs control of $e^{-\tilde h_\lambda}$)
flow in λ	generator $\mathcal{V}_\lambda$ on the Hadamard product; $\Upsilon(\lambda)$ at $+\infty$ (Proposition 14.2)	$\log\tilde\zeta_{\mathbb{Z},\lambda}=\sum_n\tilde c_n(\lambda)\big(-\log(1-n^{-s})\big)$, each mass differentiable in λ by (14.9); joining factor (14.17)
limit $\lambda\to0^+$	$\zeta_\lambda\to\zeta_{Z1}$	$\tilde c_n(\lambda)\to\tilde c_n(0)$, $\mathfrak{h}_n(\lambda;\cdot)\to\mathfrak{h}_n(0;\cdot)$ (Proposition 14.9)
endpoint: Hamburger's hypotheses	H0 and H2 proved; H1 is (D), open	H1 proved in $\Re s>1$ (Corollary 14.12); H0 and H2 open
relation between the endpoints	$\tilde\zeta_{\mathbb{Z},0}=\zeta_{Z1}\,e^{\tilde h_0}$	same
what closes the route	ζ_{Z1} satisfies H1	$\tilde\zeta_{\mathbb{Z},0}$ satisfies H0 and H2
common form of the closure	$\tilde h_0\equiv0$	$\tilde h_0\equiv0$
equivalent forms of the closure	ζ_{Z1} has a positive representing measure (Theorem 15.7)	$e^{\tilde h_0}$ of finite order and symmetric under $s\mapsto1-s$ (Theorem 15.3)
consequence of the closure	$\zeta_{Z1}=\zeta$; Step 3 gives $M(x)=O_\varepsilon(x^{1/2+\varepsilon})$ (Theorem 12.8)	$\tilde a_N(0)=1$, $\tilde c_n(0)=\mathbf{1}_{\mathrm{prime}}(n)$, $\tilde\mu_0=\mu$, $\tilde M_0=M$; $\tilde\zeta_{\mathbb{Z},0}=\zeta=\zeta_{Z1}$

How the two routes complement each other. Along λ each family keeps exactly the half of Hamburger's hypotheses that the other one lacks:

$$\underbrace{\zeta_\lambda}_{\substack{\text{keeps the functional equation}\\ \text{and the zeros on } \Re s=\frac12}} \quad\longleftrightarrow\quad \underbrace{\tilde\zeta_{\mathbb{Z},\lambda}}_{\substack{\text{keeps the Dirichlet series}\\ \text{and the Euler product}}}, \qquad \tilde\zeta_{\mathbb{Z},\lambda}=\zeta_\lambda\,e^{\tilde h_\lambda}. \tag{14.37}$$

The closure is the same in both columns, the vanishing of the joining factor at the endpoint: by (14.15) and Theorem 14.14,

$$\tilde h_0\equiv0 \iff J_{Z1}=0 \text{ on } (1,2),\quad \mathfrak{h}_n(0;s)=0 \text{ for all } n\ge2,\ \Re s>1,\ \text{and } \varpi_{Z1}=1. \tag{14.38}$$

No further condition is needed in the atomized column: the cell masses have mean square $X^{1+\varepsilon}$ uniformly in λ (Theorem 14.11), so $\tilde{\zeta}_{\mathbb{Z},0}$ satisfies H1 (Corollary 14.12).

14.8. **Programme.** The steps are numbered in the order in which they are followed. The status column records what is proved in this paper.

Step	Statement	Status
A1	Atomization by smooth cells, coefficients $\tilde{a}_N(\lambda)$	proved (Definition 14.4)
A2	Joining factor and explicit cell defects, (14.15)–(14.19)	proved (Thm. 14.7, Prop. 14.8)
A3	L^2 square-root estimate for every λ; Dirichlet series in $\Re s > \frac{3}{2}$	proved (Theorem 14.13); superseded by A6
A4	$\tilde{h}_\lambda$ holomorphic in $\Re s > \frac{1}{2}$; in $\Re s > 0$ for $\lambda = 1$	proved (Theorem 14.13)
A5	Limits $\lambda \to 0^+$ of $\tilde{c}_n$, $\mathfrak{h}_n$, ϖ_λ; flow at infinity Υ	proved (Propositions 14.2, 14.9)
A5′	Endpoints: $\tilde{h}_1$ not constant; atomized Möbius coefficients $\tilde{\mu}_\lambda$; Mellin form B_{Z1}, with $B_{Z1} = M_{Z1}$ only under (12.37)	proved (Propositions 14.16, 14.17)
A6	Mean square of the cell masses uniformly in λ; Dirichlet series in $\Re s > 1$ for every λ, including $\lambda = 0$	proved (Theorem 14.11, Corollary 14.12)
A7	Cell identity (14.32) for every $n \geq 2$, and $J_{Z1} = 0$ on the first cell $(1, 2)$	open; reduced to one number for each cell in A20
A8	Equivalent scalar form: $\int_0^1 \Upsilon \, d\lambda = \log L$, i.e. (13.12)	open; follows from A7
A9	$\zeta_{Z1} = \zeta$ by Hamburger; $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ by Step 3	proved as implication from A7 (Thms. 14.14, 12.8)
A10	Continuation of $\tilde{h}_\lambda$ to $0 < \Re s \leq \frac{1}{2}$ for $\lambda < 1$	open; not needed for A9; at $\lambda = 0$ it follows from (B1), (B2) (A23)
A11	ζ_λ is not a Dirichlet series and $\tilde{h}_\lambda$ is not constant for $0 < \lambda \leq 1$; convexity of $\log \zeta_\lambda(\sigma)$ in λ	proved (Propositions 15.1, 15.2)
A12	Hamburger's theorem for the pair: A9 follows if $e^{\tilde{h}_0}$ is of finite order and symmetric under $s \mapsto 1 - s$	proved as implication (Theorem 15.3); at $\lambda = 0$ the hypothesis is equivalent to (D) (Remark 15.33)
A13	Positivity: A9 follows if ζ_{Z1} has a positive representing measure; then $\varpi_{Z1} \leq 1$	proved as implication (Theorem 15.7, Corollary 15.8)
A14	Clock end: density plus explicit atoms of $d\pi_{\mathrm{RVM}}$, cell masses $\frac{1}{\log n} + \mathbf{1}_{\mathrm{Pow}_2}(n) \frac{\mu(\log_2 n)}{\log_2 n} J_{\mathrm{RVM}}(2) + O(n^{-1/4})$, defects $O(\lvert s\rvert n^{-\sigma-4/3})$	proved (Theorem 15.18, Corollary 15.19)
A15	Height kernel of the cell masses; flow as count difference; Gaussian defect and the density condition $\tilde{h}_0(1) = 0$	proved (Propositions 15.20, 15.21, 15.25)
A16	Hamburger's theorem for the pair without continuation: A9 follows if $T_0 \equiv 0$, or if $\mathsf{V}_\sigma(0) = 0$; symmetry fails at $\lambda = 1$; exceptional set finite	proved (Theorem 15.4, Propositions 15.5, 15.28)
A17	$\zeta_{Z1}(\sigma) \geq \zeta(\sigma)$ for $\sigma > 1$; A9 follows from the reverse inequality at one σ, or from $\varpi_\lambda \leq 1$ along $\lambda \to 0^+$, or from $\mathsf{W}(0) \geq 0$	proved (Theorem 15.11, Propositions 15.10, 15.12)
A18	Space $\mathscr{O}$ of ordinate lists: at most one coincidence list; A9 is equivalent to its existence	proved (Theorem 16.2, Corollary 16.4)
A19	Existence of a list in $\mathscr{O}$ with $\tilde{h}_{\boldsymbol{\kappa}}$ constant, equivalently $\min_{\mathscr{O}} \mathsf{V}_\sigma = 0$	open; equivalent to A7
A20	A9 holds if and only if $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$ for every n; every $\lambda \in \mathsf{E}$ is a witness; the identity (15.61) cell by cell	proved (Corollary 15.29, Propositions 15.30, 15.32); the identity (15.61) itself is open
A21	Symmetric joining factors: order one is needed, and A12 has no slack at $\lambda = 0$	proved (Proposition 15.34, Remarks 15.33, 15.35)

Step	Statement	Status
A22	Half-line form: A9 follows if $\tilde h_0(\sigma) = O_N(\sigma N^{-\sigma})$ for every N; the Nth rung is the constancy of J^{M}_{Z1} on $(N-1, N)$	proved (Theorems 15.36, 15.37, Corollary 15.38)
A23	Bridge theorem: A9 follows if $\Im\tilde h_0$ vanishes on $\Re s = \frac12$ and $\tilde h_0$ is temperate; the two conditions hold for no $\lambda > 0$	proved (Theorem 15.39, Corollary 15.40); (B1), (B2) at $\lambda = 0$ open
A24	Transport from the clock: $\mathsf{S_f}$ entire of type $C_\eta \log N$, moment series on the clock, defect of the clock by Poisson summation	proved (Proposition 15.44, Theorem 15.45, Proposition 15.46)
A24b	Multiscale Jacobian at the clock: stable finite ordinate block on windows $U \asymp \log T$, fixed-window obstruction, dense range, and rapidly summable smooth tails	proved (Theorem 15.47); quantitative multiscale continuation to a solution open
A25	Rung system with free masses: A9 holds if and only if it has a solution in $\mathscr{O}$; the solution is unique and nondegenerate	proved (Theorems 16.6, 16.7); existence open
A26	Coordinate gradient of V_σ on $\mathscr{O}$; A9 would follow from a boundary-aware coordinate/KKT criterion forcing every minimizer to have zero defect	proved (Proposition 16.9); the forcing statement is open
A27	Acceptance criterion and models: the final step is a theorem for the finite Euler product over the primes $2, 3$ (not for more primes, see after Theorem 17.3); no object of finite rank in the window of ζ	proved (Theorems 17.1, 17.3, Propositions 17.2, 17.4)

Remark 14.18 (Input for A7). Steps A1–A6 use only two properties of an ordinate list: bounded displacement from the clock and a primitive of the count difference of size $O(\log\tau)$. The list (τ_k) has both properties, and for it the corresponding function ζ_{RVM} is not a Dirichlet series (Theorem 9.13). Step A7 must therefore use a property of the ordinates γ_k that the clock ordinates do not have; the natural candidate is the arithmetic of ζ carried by the γ_k, for instance Landau's formula $\sum_{0<\gamma\le T} x^\rho = -\frac{T}{2\pi}\Lambda(x) + O(\log T)$, in which the prime powers appear as resonances of the ordinates. The multiscale analysis below separates a different issue: the linearized clock has enough directions, but only on windows whose length grows like $\log T$; thus the obstruction is not fixed-height injectivity but quantitative nonlinear continuation across scales.

15. Criteria for the final step and the clock end of the family

This section collects the consequences of Corollary 14.12 for the final step. Along the open path $0 < \lambda \le 1$ no member ζ_λ is a Dirichlet series (Proposition 15.1); the family is convex in λ on the real axis (Proposition 15.2); Hamburger's theorem applies to the pair $(\zeta_\lambda, \tilde\zeta_{\mathbb{Z},\lambda})$ as soon as the joining factor is symmetric (Theorem 15.3); positivity of the measure representing ζ_{Z1} gives $\varpi_{Z1} \le 1$ and closes the route (Theorem 15.7). At the clock end the density of $d\pi_{\mathrm{RVM}}$ is computed explicitly (Theorem 15.18), and the flow of the cell masses is written in the heights of the ordinates (Propositions 15.20 and 15.21). Subsections 15.10–15.14 reduce the final step to the cell masses (Corollary 15.29), state two criteria whose hypotheses are written where the joining factor is defined, the half-line form (Theorem 15.37) and the bridge theorem (Theorem 15.39), and compute the transport from the clock on each cell (Theorem 15.45).

Throughout this section the function χ of Subsection 14.4 is even, so that φ is even, and

$$\hat\varphi(\xi) := \int_{\mathbb{R}} \varphi(t) e^{-2\pi i\xi t}\, dt, \qquad \mathsf{M}_r := \int_{\mathbb{R}} t^r \varphi(t)\, dt, \qquad \mathsf{C}_r := \int_{\mathbb{R}} t^r \varphi(t)\cos(2\pi t)\, dt. \tag{15.1}$$

$\hat\varphi$ is real and even, and $\mathsf{M}_1 = \mathsf{C}_1 = 0$. Since $\sum_{m\in\mathbb{Z}} \varphi(t-m) = 1$,

$$\hat\varphi(q) = \int_0^1 \sum_{m\in\mathbb{Z}} \varphi(t-m)\, e^{-2\pi i q t}\, dt = \delta_{q0} \qquad (q \in \mathbb{Z}), \tag{15.2}$$

so $\mathsf{M}_0 = 1$ and $\mathsf{C}_0 = \hat{\varphi}(1) = 0$.

15.1. **The open path.**

Proposition 15.1 (No member of the open path is a Dirichlet series)**.** *Let $0 < \lambda \le 1$. Then ζ_λ is not a constant multiple of a Dirichlet series $\sum_n a_n n^{-s}$ converging absolutely in $\Re s > 1$. In particular the joining factor $\tilde{h}_\lambda$ is not constant.*

Proof. Suppose $\zeta_\lambda(s) = \sum_n a_n n^{-s}$ absolutely for $\Re s > 1$. By Theorem 11.1, exactly as in Proposition 12.1, $(s-1)\zeta_\lambda(s)$ is entire of order one and ζ_λ satisfies (12.3); these are the hypotheses of Hamburger's theorem, so $\zeta_\lambda = C\zeta$ with $C \neq 0$. The nontrivial zeros of ζ with positive ordinate are then $\frac{1}{2} + i\gamma_k(\lambda)$, $k \ge 1$, with multiplicity. The ordinates of ζ, listed in non-decreasing order, are the γ_k, and the $\gamma_k(\lambda)$ are increasing; hence $\gamma_k(\lambda) = \gamma_k$, that is $\lambda\eta_k = 0$, for every k. Since $\lambda > 0$, every η_k vanishes, all members of the family coincide (Remark 11.2), and $\zeta_{\mathrm{RVM}} = \zeta_\lambda$ is an absolutely convergent Dirichlet series, against Theorem 9.13. If $\tilde{h}_\lambda$ were a constant c, then $\zeta_\lambda = e^{-c}\tilde{\zeta}_{\mathbb{Z},\lambda}$ would be such a series by Corollary 14.12. □

The two columns of (14.37) exist for every λ, and for every $\lambda \in (0,1]$ they are different functions. The joining factor converges, $\tilde{h}_\lambda \to \tilde{h}_0$ as $\lambda \to 0^+$, through non-constant functions; (D) is the statement that the limit is constant. Any property of the displacement sequence that would imply (D) must distinguish the full sequence (η_k) from every scaled sequence $(c\,\eta_k)$ with $0 \le c < 1$: the latter correspond to members $\lambda > 0$ of the family, for which Proposition 15.1 excludes closure. Thus this observation is a necessary discrimination test for a proposed criterion, not an assertion that the actual displacement sequence already satisfies such a criterion.

15.2. **Convexity along the family.**

Proposition 15.2 (Convexity on the real axis)**.** *For every real $\sigma > 1$ the function $\lambda \mapsto \log\zeta_\lambda(\sigma)$ is strictly convex on $[0,1]$:*

$$\partial_\lambda^2 \log\zeta_\lambda(\sigma) = \sum_{k\ge1} \eta_k^2 \Big(\mathcal{C}\big(\sigma - \tfrac{1}{2}, \gamma_k(\lambda)\big) - \mathcal{C}\big(\tfrac{1}{2}, \gamma_k(\lambda)\big) \Big) > 0, \qquad \mathcal{C}(a,\gamma) := \frac{2(a^2-\gamma^2)}{(a^2+\gamma^2)^2}. \tag{15.3}$$

Consequently, with the generator $\mathcal{V}_\lambda$ of Theorem 11.3,

$$\log\zeta_{\mathrm{RVM}}(\sigma) - \mathcal{V}_1(\sigma) < \log\zeta_{Z1}(\sigma) < \log\zeta_{\mathrm{RVM}}(\sigma) - \mathcal{V}_0(\sigma) \qquad (\sigma > 1). \tag{15.4}$$

Proof. With $a = \sigma - \frac{1}{2} > \frac{1}{2}$ one has $q(\sigma) = a^2 - \frac{1}{4}$, so by (11.2) $\log\zeta_\lambda(\sigma) = -\log G(\sigma) + \sum_k \big(\log(a^2 + \gamma_k(\lambda)^2) - \log(\frac{1}{4} + \gamma_k(\lambda)^2)\big)$. Since $\partial_\gamma^2 \log(a^2+\gamma^2) = \mathcal{C}(a,\gamma)$ and $\partial_\lambda \gamma_k(\lambda) = \eta_k$, termwise differentiation gives the series in (15.3); its terms are $O(\eta_k^2 \gamma_k^{-2})$, so it converges uniformly on $[0,1]$. Fix $\gamma > \frac{1}{2}$. Then $\mathcal{C}(\frac{1}{2}, \gamma) < 0$; $\partial_a \mathcal{C}(a,\gamma) = 4a(3\gamma^2 - a^2)(a^2+\gamma^2)^{-3} > 0$ for $0 < a < \sqrt{3}\,\gamma$; and $\mathcal{C}(a,\gamma) > 0$ for $a > \gamma$. Hence $\mathcal{C}(a,\gamma) > \mathcal{C}(\frac{1}{2},\gamma)$ for every $a > \frac{1}{2}$. All ordinates satisfy $\gamma_k(\lambda) > \tau_0 > \frac{1}{2}$, so every term of the series is ≥ 0; the sum is > 0 exactly when some $\eta_k \neq 0$, that is, when $\boldsymbol{\gamma} \neq \boldsymbol{\tau}$, and this holds with $k = 1$: $\eta_1 = \tau_1 - \gamma_1 > 0$ by the comparison $\gamma_1 < 14.14 < \tau_1$ made in the proof of Proposition 14.2. So the sum is positive. A strictly convex function F on $[0,1]$ satisfies $F(1) - F'(1) < F(0) < F(1) - F'(0)$; with $F(\lambda) = \log\zeta_\lambda(\sigma)$ and $F' = \mathcal{V}_\lambda(\sigma)$ (11.11) this is (15.4). □

As $\sigma \to +\infty$, (15.3) becomes $-\Upsilon'(\lambda) > 0$ and (15.4) becomes the last statement of Proposition 14.2. The sign in (15.3) comes from the test function $x^{-\sigma}$, whose transform is not oscillatory in the ordinate; for the cell functions φ_n the corresponding second derivative $\partial_\lambda^2 \tilde{c}_n(\lambda) = -2\Re\sum_k \eta_k^2 W_n''(\gamma_k(\lambda))$ has terms of both signs.

15.3. **Hamburger's theorem for the pair.**

Theorem 15.3 (Hamburger's theorem for the pair $(\zeta_\lambda, \tilde\zeta_{\mathbb{Z},\lambda})$)**.** *Let $\lambda \in [0,1]$. Suppose that the joining factor $e^{\tilde h_\lambda}$ extends to a meromorphic function on $\mathbb{C}$ of finite order with finitely many poles, and that*

$$e^{\tilde h_\lambda(s)} = e^{\tilde h_\lambda(1-s)}. \tag{15.5}$$

Then $\tilde\zeta_{\mathbb{Z},\lambda} = \zeta$. For $\lambda = 0$, moreover, $\zeta_{Z1} = \zeta$ and $\tilde h_0 \equiv 0$.

Proof. Put $f = \tilde\zeta_{\mathbb{Z},\lambda} = \zeta_\lambda e^{\tilde h_\lambda}$. By Theorem 11.1, $(s-1)\zeta_\lambda$ is entire of order one, so $f = \mathsf{G}/\mathsf{P}$ with G entire of finite order and P a polynomial: this is (H0). By (12.3) for ζ_λ and (15.5), $f(s)\Gamma(\frac{s}{2})\pi^{-s/2} = f(1-s)\Gamma(\frac{1-s}{2})\pi^{-(1-s)/2}$: this is (H2) with $g = f$. (H1) is Corollary 14.12, and it gives (H3) for $g = f$. Hamburger's theorem gives $f = C\zeta$, and $C = 1$ because $\tilde a_1(\lambda) = 1$. By Theorem 14.13(iii), $f = \zeta$ has no zeros in $\Re s > \frac12$, and by the functional equation none in $\Re s < \frac12$ apart from the trivial ones: the Riemann hypothesis holds. For $\lambda = 0$, Theorem 12.2 (RH $\Leftrightarrow \zeta_{Z1} = \zeta$) then gives $\zeta_{Z1} = \zeta = \tilde\zeta_{\mathbb{Z},0}$, and $e^{\tilde h_0} = 1$. □

Theorem 15.3 uses the functional equation of ζ_λ and the Dirichlet series of $\tilde\zeta_{\mathbb{Z},\lambda}$ as they stand; the only hypothesis is on the factor that joins them, and it asks for symmetry instead of constancy. At $\lambda = 1$ the two hypotheses do not hold together: the conclusion $\tilde\zeta_{\mathbb{Z},1} = \zeta$ would place the zeros of ζ at $\frac12 \pm i\tau_k$ (Theorem 14.13(iv)), so $\zeta = \zeta_{\mathrm{RVM}}$ by Theorem 10.1, against Theorem 9.13.

Theorem 15.4 (Real-variable and vertical forms of Hamburger's theorem for the pair)**.** *For $\lambda \in [0,1]$, $y > 0$ and $\sigma > 1$ put*

$$T_\lambda(y) := \sum_{N\ge1} \tilde a_N(\lambda)\, e^{-\pi N^2 y} - y^{-1/2} \sum_{N\ge1} \tilde a_N(\lambda)\, e^{-\pi N^2/y} - \tfrac12(y^{-1/2} - 1), \tag{15.6}$$

$$k_n(\lambda) := \sum_{d\mid n} \tilde a_d(\lambda)\, \mu(n/d), \qquad \mathsf{V}_\sigma(\lambda) := \sum_{n\ge2} |k_n(\lambda)|^2\, n^{-2\sigma}. \tag{15.7}$$

(i) *$\tilde\zeta_{\mathbb{Z},\lambda}/\zeta = \sum_{n\ge1} k_n(\lambda) n^{-s}$ converges absolutely in $\Re s > 1$, $k_1(\lambda) = 1$, and*

$$\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \left| \frac{\tilde\zeta_{\mathbb{Z},\lambda}(\sigma+it)}{\zeta(\sigma+it)} \right|^2 dt = 1 + \mathsf{V}_\sigma(\lambda) \ \ge 1. \tag{15.8}$$

(ii) *For each λ the following are equivalent:* (a) $\tilde\zeta_{\mathbb{Z},\lambda} = \zeta$*;* (b) $T_\lambda(y) = 0$ *for every $y > 0$;* (c) $\mathsf{V}_\sigma(\lambda) = 0$ *for some, equivalently every, $\sigma > 1$. Each of them implies (D).*

(iii) *$\lambda \mapsto T_\lambda(y)$ and $\lambda \mapsto \mathsf{V}_\sigma(\lambda)$ are continuous on $[0,1]$, and*

$$\text{(D)} \iff T_0 \equiv 0 \iff \mathsf{V}_\sigma(0) = 0 \iff \lim_{\lambda\to0^+} T_\lambda(y) = 0 \ \ (y>0) \iff \lim_{\lambda\to0^+} \mathsf{V}_\sigma(\lambda) = 0. \tag{15.9}$$

At the clock end $T_1 \not\equiv 0$ and $\mathsf{V}_\sigma(1) > 0$.

Proof. (i) $1/\zeta = \sum_n \mu(n) n^{-s}$ and $\tilde\zeta_{\mathbb{Z},\lambda}$ (Corollary 14.12) converge absolutely in $\Re s > 1$, and so does their product, with $k_1 = \tilde a_1 = 1$. The mean square of an absolutely convergent Dirichlet series on a vertical line is $\sum_n |k_n|^2 n^{-2\sigma}$, as in (9.49)–(9.50). (ii) (a)⇔(c) because $\mathsf{V}_\sigma(\lambda) = 0$ means $k_n(\lambda) = 0$ for $n \ge 2$. (a)⇒(b) is the theta relation $\sum_{n\ge1} e^{-\pi n^2 y} = y^{-1/2} \sum_{n\ge1} e^{-\pi n^2/y} + \frac12(y^{-1/2} - 1)$. (b)⇒(a): the relation $T_\lambda \equiv 0$ is the identity reached in the first half of the proof of Hamburger's theorem in [61, §2.13], with $b_n = a_n = \tilde a_n(\lambda)$; the second half of that proof uses only this identity and the absolute convergence of $\sum_n a_n n^{-\sigma'}$ for $\sigma' > 1$, and gives $a_k = a_{k+1}$ for every k, hence $\tilde a_N(\lambda) = \tilde a_1(\lambda) = 1$. If (a) holds, then by Theorem 14.13(iii) and the functional equation the nontrivial zeros of ζ lie on $\Re s = \frac12$; they are the points $\frac12 \pm i\gamma_k$, and Theorem 12.2 gives (D). (iii) By Theorem 14.11, for every $\sigma' > 1$ the sums $\sum_n |\tilde c_n(\lambda)| n^{-\sigma'}$ are bounded uniformly in λ; hence so are $\sum_N |\tilde a_N(\lambda)| N^{-\sigma'} \le \exp(\sum_m \frac1m \sum_n |\tilde c_n(\lambda)| n^{-m\sigma'})$ and $\sum_n |k_n(\lambda)| n^{-\sigma'}$. Each $\tilde a_N(\lambda)$ and each

$k_n(\lambda)$ is a polynomial in finitely many cell masses, which are continuous in λ. The tails $\sum_{N>N_0}$ of (15.6) and of (15.7) (with $1<\sigma'<\sigma$) are small uniformly in λ, which gives the continuity. Under (D), $\tilde a_N(0)=1$ (Theorem 14.14), so $T_0\equiv 0$ and $\mathsf{V}_\sigma(0)=0$; the converse is (ii) at $\lambda=0$, and the two limits are the values at $\lambda=0$ by continuity. Finally $\tilde\zeta_{\mathbb{Z},1}\neq\zeta$ by the remark after Theorem 15.3. □

Theorem 15.4 needs no continuation of the joining factor: (15.6) is a real function of y built from the atomized coefficients alone, and (15.8) lives in $\Re s>1$, where

$$\frac{\tilde\zeta_{\mathbb{Z},\lambda}}{\zeta}=e^{\tilde h_\lambda}\,\frac{\zeta_\lambda}{\zeta} \tag{15.10}$$

is a quotient of three explicit functions. Its vertical mean is 1 and its vertical mean square is at least 1 for every λ; (D) is the statement that at $\lambda=0$ the mean square does not exceed 1.

Proposition 15.5 (At the clock end the symmetry fails)**.** *The joining factor $e^{\tilde h_1}$ has no meromorphic continuation to $\mathbb{C}$ that satisfies* (15.5)*.*

Proof. $e^{\tilde h_1}$ is holomorphic and zero-free in $\Re s>0$ (Theorem 14.13(iv)). A meromorphic continuation with (15.5) is holomorphic and zero-free also in $\Re s<1$, hence entire and zero-free, so it equals e^g with g entire, $g=\tilde h_1$ on $\Re s>0$, and $g(s)-g(1-s)$ constant. For $\Re s\ge\frac14$, the bound $|\mathfrak{h}_n(1;ms)|\le C\,m|s|\,n^{-m\sigma-1}$ of Theorem 14.13(iv) gives $\sum_m\frac1m\sum_n|\mathfrak{h}_n(1;ms)|\le C|s|\sum_{n\ge2}n^{-1}(n^{1/4}-1)^{-1}$, and $|\mathfrak{u}_1(s)|\le C(1+|s|)$ by (14.11); so $|g(s)|\le C(1+|s|)$ on $\Re s\ge\frac14$, and by the symmetry also on $\Re s\le\frac34$. An entire function of linear growth is a polynomial of degree at most 1, and $g(s)-g(1-s)$ constant then forces g to be constant. This contradicts Proposition 14.16(iii). □

At $\lambda=1$ it is therefore the symmetry (15.5) itself that fails in Theorem 15.3, whatever the order of the continuation.

15.4. **Positivity and the value at infinity.** For an even Schwartz function g on $\mathbb{R}$ put $\hat{\mathsf{g}}(\xi)=\int_{\mathbb{R}}\mathsf{g}(u)e^{-2\pi i\xi u}du$ and $\mathsf{g}^\sharp(s)=\int_0^\infty\mathsf{g}(u)u^{s-1}du$.

Lemma 15.6 (Summation formula for a representing measure)**.** *Let $\lambda\in[0,1]$ and let A be a positive measure on $[1,\infty)$ with*

$$\zeta_\lambda(s)=\int_{[1,\infty)}u^{-s}\,d\mathsf{A}(u)\qquad(\Re s>1). \tag{15.11}$$

Then for every even Schwartz function g

$$\mathsf{g}(0)+2\int_{[1,\infty)}\mathsf{g}\,d\mathsf{A}=\hat{\mathsf{g}}(0)+2\int_{[1,\infty)}\hat{\mathsf{g}}\,d\mathsf{A}. \tag{15.12}$$

Proof. $\mathsf{g}^\sharp$ is meromorphic with simple poles at $s=0,-2,-4,\dots$, residue $\mathsf{g}(0)$ at $s=0$, and decays faster than any power of $|t|$ on vertical strips; $\mathsf{g}^\sharp(1)=\frac12\hat{\mathsf{g}}(0)$; and

$$\pi^{-s/2}\Gamma(\tfrac s2)\,\hat{\mathsf{g}}^\sharp(1-s)=\pi^{-(1-s)/2}\Gamma(\tfrac{1-s}{2})\,\mathsf{g}^\sharp(s) \tag{15.13}$$

for $0<\Re s<1$, and then by meromorphic continuation: the Fourier transform of the even homogeneous distribution $|\xi|^{-s}$ is a multiple of $|u|^{s-1}$, and the multiple is fixed by $\mathsf{g}(u)=e^{-\pi u^2}$, for which $\hat{\mathsf{g}}=\mathsf{g}$ and $\mathsf{g}^\sharp(s)=\frac12\pi^{-s/2}\Gamma(\frac s2)$. Fix $1<c<2$. By Mellin inversion and Fubini, $\int\mathsf{g}\,d\mathsf{A}=\frac{1}{2\pi i}\int_{(c)}\zeta_\lambda(s)\mathsf{g}^\sharp(s)\,ds$, since $\int u^{-c}d\mathsf{A}<\infty$. On $\Re s\ge c$, $|\zeta_\lambda(s)|\le\zeta_\lambda(c)$; on $\Re s=1-c$ the functional equation (12.3), valid for every ζ_λ by Theorem 11.1, gives $\zeta_\lambda(s)=\zeta_\lambda(1-s)G(1-s)/G(s)$, a power of $|t|$ times $\zeta_\lambda(c)$. The Phragmén–Lindelöf principle, applied to the entire function $(s-1)\zeta_\lambda(s)$ of finite order on the strip $1-c\le\Re s\le c$, bounds it by a power of $|t|$ there, and dividing by $s-1$, with $|s-1|\ge1$ for $|t|\ge1$, gives the same for ζ_λ on $1-c\le\Re s\le c$, $|t|\ge1$. Moving the line to $\Re s=1-c$ crosses $s=1$, where ζ_λ has residue 1, and $s=0$, where $\zeta_\lambda(0)=P_\lambda(0)/G(0)=-\frac12$:

$$\int\mathsf{g}\,d\mathsf{A}=\tfrac12\hat{\mathsf{g}}(0)-\tfrac12\mathsf{g}(0)+\frac{1}{2\pi i}\int_{(1-c)}\zeta_\lambda(s)\mathsf{g}^\sharp(s)\,ds. \tag{15.14}$$

By (12.3) and (15.13), $\zeta_\lambda(s)\mathsf{g}^\sharp(s) = \zeta_\lambda(1-s)\hat{\mathsf{g}}^\sharp(1-s)$, and the last integral is $\frac{1}{2\pi i}\int_{(c)} \zeta_\lambda(s)\hat{\mathsf{g}}^\sharp(s)\,ds = \int \hat{\mathsf{g}}\,d\mathsf{A}$. □

Theorem 15.7 (Positivity criterion)**.** *Let $\lambda \in [0,1]$. If ζ_λ has a representation* (15.11) *with a positive measure A, then*

$$\varpi_\lambda = \mathsf{A}(\{1\}) \le 1. \tag{15.15}$$

In particular, if ζ_{Z1} has such a representation, then $\varpi_{Z1} = 1$ and $\zeta_{Z1} = \zeta$.

Proof. By monotone convergence $\zeta_\lambda(\sigma) = \int u^{-\sigma} d\mathsf{A} \to \mathsf{A}(\{1\})$ as $\sigma \to +\infty$, and the limit is ϖ_λ (Proposition 14.1). Let $\mathsf{T}(\xi) = (1-|\xi|)_+$ and $\mathsf{B} = \mathsf{T} * \mathsf{T}$, so that $\mathsf{B}(\xi) = \frac{2}{3} - \xi^2 + \frac{1}{2}|\xi|^3$ for $|\xi| \le 1$, $\mathsf{B}(\xi) = \frac{1}{6}(2-|\xi|)^3$ for $1 \le |\xi| \le 2$, $\mathsf{B} = 0$ elsewhere, and $\mathsf{B}''(\xi) = \mathsf{T}(\xi+1) - 2\mathsf{T}(\xi) + \mathsf{T}(\xi-1)$. Put

$$\mathsf{g}(u) := \Big(\frac{\sin \pi u}{\pi u}\Big)^4 (1-u^2), \qquad \hat{\mathsf{g}}(\xi) = \mathsf{B}(\xi) + \frac{1}{4\pi^2}\mathsf{B}''(\xi). \tag{15.16}$$

Then $\mathsf{g}(0) = 1$; $\mathsf{g}(u) \le 0$ for $|u| \ge 1$; $\hat{\mathsf{g}}(0) = \frac{2}{3} - \frac{1}{2\pi^2}$ and $\hat{\mathsf{g}}(1) = \frac{1}{6} + \frac{1}{4\pi^2}$; and $\hat{\mathsf{g}} \ge 0$ on $\mathbb{R}$: for $0 \le \xi \le 1$, $\hat{\mathsf{g}}(\xi) = \frac{2}{3} - \xi^2 + \frac{1}{2}\xi^3 + \frac{3\xi - 2}{4\pi^2}$ is positive at $\xi = 0$ and $\xi = 1$, and its derivative $-2\xi + \frac{3}{2}\xi^2 + \frac{3}{4\pi^2}$ has the roots $\xi_\pm = \frac{1}{3}(2 \pm \sqrt{4 - 9/(2\pi^2)})$, $\xi_- = 0.0391\ldots$, $\xi_+ = 1.294\ldots$, so on $[0,1]$ it is positive on $[0, \xi_-)$ and negative on $(\xi_-, 1]$, and the minimum of $\hat{\mathsf{g}}$ on $[0,1]$ is $\min(\hat{\mathsf{g}}(0), \hat{\mathsf{g}}(1)) = \frac{1}{6} + \frac{1}{4\pi^2} > 0$; and for $1 \le \xi \le 2$, $\hat{\mathsf{g}}(\xi) = \frac{1}{6}(2-\xi)^3 + \frac{2-\xi}{4\pi^2} \ge 0$. For $\epsilon > 0$ let $\mathsf{g}_\epsilon(u) = \mathsf{g}(u)e^{-\epsilon\pi u^2}$, an even Schwartz function with $\mathsf{g}_\epsilon(0) = 1$, $\mathsf{g}_\epsilon \le 0$ on $|u| \ge 1$, and $\hat{\mathsf{g}}_\epsilon = \hat{\mathsf{g}} * (\epsilon^{-1/2} e^{-\pi\xi^2/\epsilon}) \ge 0$. By Lemma 15.6, since $\mathsf{A} \ge 0$ is carried by $[1,\infty)$ and has mass ϖ_λ at $u = 1$,

$$1 \;\ge\; \mathsf{g}_\epsilon(0) + 2\int \mathsf{g}_\epsilon\, d\mathsf{A} = \hat{\mathsf{g}}_\epsilon(0) + 2\int \hat{\mathsf{g}}_\epsilon\, d\mathsf{A} \;\ge\; \hat{\mathsf{g}}_\epsilon(0) + 2\varpi_\lambda\, \hat{\mathsf{g}}_\epsilon(1). \tag{15.17}$$

Letting $\epsilon \to 0$, $\hat{\mathsf{g}}_\epsilon \to \hat{\mathsf{g}}$ pointwise because $\hat{\mathsf{g}}$ is continuous, and

$$\varpi_\lambda \le \frac{1 - \hat{\mathsf{g}}(0)}{2\hat{\mathsf{g}}(1)} = \frac{\frac{1}{3} + \frac{1}{2\pi^2}}{\frac{1}{3} + \frac{1}{2\pi^2}} = 1. \tag{15.18}$$

The last statement follows from Theorem 13.2. □

Corollary 15.8 (Positive form of the final step)**.** *The following are equivalent:* (i) *(D);* (ii) *ζ_{Z1} is the Mellin–Stieltjes transform* (15.11) *of a positive measure on $[1,\infty)$;* (iii) *J^{M}_{Z1}, extended by 0 to $(0,1]$, is non-decreasing on $(0,\infty)$.*

Proof. (i)⇒(iii): then $J^{\mathrm{M}}_{Z1} = J$ (Theorem 13.7). (iii)⇒(ii): $J^{\mathrm{M}}_{Z1}(x)x^{-c} \in L^2((1,\infty), dx/x)$ for every $c > 1$, and a non-decreasing J^{M}_{Z1}, which is ≥ 0, satisfies $J^{\mathrm{M}}_{Z1}(x)^2\,(2x)^{-2c}\log 2 \le \int_x^{2x} J^{\mathrm{M}}_{Z1}(y)^2 y^{-2c}\,dy/y$ for $x \ge 1$; so J^{M}_{Z1} is $O_c(x^c)$ and dJ^{M}_{Z1} is a positive measure on $[1,\infty)$ with $\int x^{-\sigma} dJ^{\mathrm{M}}_{Z1} < \infty$ for $\sigma > 1$. By Theorem 13.5, $\zeta_{Z1}(s) = \exp\int x^{-s} dJ^{\mathrm{M}}_{Z1} = \int u^{-s} d\mathsf{A}(u)$, where $\mathsf{A} = \sum_{r\ge 0} \frac{1}{r!}(dJ^{\mathrm{M}}_{Z1})^{*r}$ (multiplicative convolution powers) is a positive measure on $[1,\infty)$. (ii)⇒(i) is Theorem 15.7. □

In Theorem 15.7 the Dirichlet series is not used: (H1) is replaced by the positivity of the representing measure, the functional equation gives $\varpi_{Z1} \le 1$ through (15.17), and the Hadamard product gives $\varpi_{Z1} \ge 1$ (Theorem 13.2). The function (15.16) is extremal for the integers: for $\mathsf{A} = \sum_{n\ge 1} \delta_n$ both inequalities in (15.17) are equalities in the limit, since $\mathsf{g}(n) = 0$ for $n \ge 1$ and $\hat{\mathsf{g}}(n) = 0$ for $n \ge 2$. At the clock end $\varpi_{\mathrm{RVM}} < 1$ (Theorem 9.11); unconditionally, the density of dJ_{RVM} takes negative values in every cell $(n, n+1)$ with n large (Theorem 15.18).

Lemma 15.9 (Mellin form and summation formula for every member)**.** *There is $c_1 \ge 2$ such that for every $\lambda \in [0,1]$ there is a unique measurable function A_λ on $(0,\infty)$, vanishing on $(0,1)$, with $x^{-c}\mathsf{A}_\lambda \in L^2((1,\infty), dx/x)$ for every $c > c_1$ and*

$$\zeta_\lambda(s) = s\int_1^\infty \mathsf{A}_\lambda(x)\, x^{-s-1}\,dx \qquad (\Re s > c_1). \tag{15.19}$$

If ζ_λ has a representation (15.11)*, then $\mathsf{A}_\lambda(x) = \mathsf{A}([1,x])$ almost everywhere. With the pairing $\langle \mathsf{A}_\lambda, \mathsf{g}\rangle := -\int_1^\infty \mathsf{A}_\lambda(x)\,\mathsf{g}'(x)\,dx$, every even Schwartz function g satisfies*

$$\mathsf{g}(0) + 2\langle \mathsf{A}_\lambda, \mathsf{g}\rangle = \hat{\mathsf{g}}(0) + 2\langle \mathsf{A}_\lambda, \hat{\mathsf{g}}\rangle. \tag{15.20}$$

Equivalently, the even tempered distribution $\delta_0 + d\mathsf{A}_\lambda(|u|)$ on $\mathbb{R}$ is its own Fourier transform.

Proof. By Proposition 14.1 the bound (13.5) holds for every ζ_λ with constants independent of λ, so, as in the proof of Proposition 14.17, $|\zeta_\lambda(s)| \le C(|t|+\sigma)^{1/4}$ for $\Re s > c_1$, and the argument of Theorem 13.5 applied to $\zeta_\lambda(s)/s$ gives A_λ, its vanishing on $(0,1)$ and its uniqueness. For a positive measure A as in (15.11), partial integration gives (15.19) with $\mathsf{A}([1,x])$, and uniqueness applies.

Let $c > c_1$. In the variable $u = \log x$, the function $\mathsf{A}_\lambda(e^u)e^{-cu}$ has Fourier transform $\zeta_\lambda(c+it)/(c+it)$, and $-\mathsf{g}'(e^u)e^{(c+1)u}$ has Mellin transform $\int_0^\infty(-\mathsf{g}'(x))x^s dx = s\,\mathsf{g}^\sharp(s)$ on $\Re s = c$; Parseval's formula gives

$$\langle \mathsf{A}_\lambda, \mathsf{g}\rangle = \frac{1}{2\pi i}\int_{(c)} \zeta_\lambda(s)\,\mathsf{g}^\sharp(s)\,ds. \tag{15.21}$$

ζ_λ is bounded by a power of $|t|$ on every vertical strip (finite order, the functional equation and the bound on $\Re s \ge c$), and $\mathsf{g}^\sharp$ decays faster than any power. Move the line to $\Re s = 1-c$. The poles of $\mathsf{g}^\sharp$ at $s = -2, -4, \dots$ are cancelled by the trivial zeros of ζ_λ; the remaining poles are $s = 1$ and $s = 0$, exactly as in (15.14). On the new line (12.3) and (15.13) give $\zeta_\lambda(s)\mathsf{g}^\sharp(s) = \zeta_\lambda(1-s)\hat{\mathsf{g}}^\sharp(1-s)$, and (15.21) for $\hat{\mathsf{g}}$ gives (15.20). □

Proposition 15.10 (The value at infinity as one number)**.** *Let g_ϵ be as in the proof of Theorem 15.7, $\mathsf{w}_\epsilon := \hat{\mathsf{g}}_\epsilon - \mathsf{g}_\epsilon$, which is non-negative on $|u| \ge 1$, and $\mathsf{A}'_\lambda(x) := \mathsf{A}_\lambda(x) - \varpi_\lambda$ for $x \ge 1$. For every $\lambda \in [0,1]$ and $\epsilon > 0$,*

$$\varpi_\lambda\,\mathsf{w}_\epsilon(1) = \tfrac{1}{2}(1 - \hat{\mathsf{g}}_\epsilon(0)) - \langle \mathsf{A}'_\lambda, \mathsf{w}_\epsilon\rangle, \qquad \langle \mathsf{A}'_\lambda, \mathsf{w}_\epsilon\rangle := -\int_1^\infty \mathsf{A}'_\lambda(x)\,\mathsf{w}'_\epsilon(x)\,dx, \tag{15.22}$$

the limit $\mathsf{W}(\lambda) := \lim_{\epsilon\to 0}\langle \mathsf{A}'_\lambda, \mathsf{w}_\epsilon\rangle$ exists, and

$$\varpi_\lambda - 1 = -\frac{\mathsf{W}(\lambda)}{\hat{\mathsf{g}}(1)}, \qquad \hat{\mathsf{g}}(1) = \frac{1}{6} + \frac{1}{4\pi^2}. \tag{15.23}$$

Consequently $\mathsf{W}(0) \le 0$; $\mathsf{W}(1) > 0$, since this is equivalent to $\varpi_{\mathrm{RVM}} < 1$ (Theorem 9.11); and (D) holds if and only if $\mathsf{W}(0) \ge 0$. If A_λ is non-decreasing, then $\langle \mathsf{A}'_\lambda, \mathsf{w}_\epsilon\rangle = \int_{(1,\infty)} \mathsf{w}_\epsilon\,d\mathsf{A}_\lambda \ge 0$, which is Theorem 15.7.

Proof. By (15.20) for g_ϵ, with $\mathsf{g}_\epsilon(0) = 1$, $2\langle \mathsf{A}_\lambda, \mathsf{w}_\epsilon\rangle = 1 - \hat{\mathsf{g}}_\epsilon(0)$. Since $\mathsf{A}_\lambda = \varpi_\lambda + \mathsf{A}'_\lambda$ on $[1,\infty)$ and $-\int_1^\infty \mathsf{w}'_\epsilon = \mathsf{w}_\epsilon(1)$, $\langle \mathsf{A}_\lambda, \mathsf{w}_\epsilon\rangle = \varpi_\lambda \mathsf{w}_\epsilon(1) + \langle \mathsf{A}'_\lambda, \mathsf{w}_\epsilon\rangle$; this is (15.22). As $\epsilon \to 0$, $\mathsf{w}_\epsilon(1) = \hat{\mathsf{g}}_\epsilon(1) \to \hat{\mathsf{g}}(1)$, because $\mathsf{g}_\epsilon(1) = 0$, and $\frac{1}{2}(1 - \hat{\mathsf{g}}_\epsilon(0)) \to \frac{1}{2}(1 - \hat{\mathsf{g}}(0)) = \hat{\mathsf{g}}(1)$ by (15.18); hence the limit exists and (15.23) holds. The sign statements follow from (15.23): at $\lambda = 0$ from $\varpi_{Z1} \ge 1$ (Theorem 13.2), and at $\lambda = 1$ from $\varpi_1 = \varpi_{\mathrm{RVM}}$, so that $\mathsf{W}(1) > 0$ is the same statement as $\varpi_{\mathrm{RVM}} < 1$. □

Theorem 15.11 (The endpoint dominates ζ on the real axis)**.** *For every real $\sigma > 1$,*

$$\zeta_{Z1}(\sigma) \;\ge\; \zeta(\sigma), \tag{15.24}$$

and equality for one $\sigma > 1$ holds if and only if $\zeta_{Z1} = \zeta$. In particular

$$\prod_{k\ge 1}\Big(1 + \frac{2}{\frac{1}{4} + \gamma_k^2}\Big) \;\ge\; \frac{\pi}{3}, \qquad \textit{with equality if and only if } (\mathrm{D}). \tag{15.25}$$

Proof. By the proof of Theorem 13.2, $\zeta_{Z1}(\sigma)/\zeta(\sigma) = \mathscr{R}_{Z1}(\sigma)$ is the convergent product of the factors (13.11), with $W = (\sigma - \frac{1}{2})^2 + \gamma^2$, which is strictly increasing in $\sigma > \frac{1}{2}$, hence on $\sigma \ge 1$. At $\sigma = 1$,

$W = \frac{1}{4} + \gamma^2$ and the factor equals 1 by the definition of c_γ. The logarithmic derivative in W of $W_\rho^2/((W_\rho - d_\rho^2)^2 + 4d_\rho^2\gamma^2)$ is

$$\frac{2}{W} - \frac{2(W - d_\rho^2)}{(W - d_\rho^2)^2 + 4d_\rho^2\gamma^2} = \frac{2d_\rho^2\,(4\gamma^2 + d_\rho^2 - W)}{W\big((W - d_\rho^2)^2 + 4d_\rho^2\gamma^2\big)}, \tag{15.26}$$

positive for $W < 4\gamma^2 + d_\rho^2$ and negative afterwards, and the factor tends to $c_\gamma > 1$ as $W \to \infty$. A function that equals 1 at $\sigma = 1$, increases and then decreases to a limit greater than 1 is greater than 1 for every $\sigma > 1$. Hence every factor of the convergent product $\mathscr{R}_{Z1}(\sigma)$ is ≥ 1, and every factor attached to a pair with $d_\rho \neq 0$ is > 1; so $\mathscr{R}_{Z1}(\sigma) \geq 1$, with equality if and only if no such factor exists, that is, if and only if the product is empty and $\mathscr{R}_{Z1} \equiv 1$. For $\sigma = 2$: $q(2) = 2$, $G(2) = 2/\pi$ and $P_\zeta(2) = G(2)\zeta(2) = \pi/3$, so (15.24) reads $P_{Z1}(2) \geq \pi/3$. □

Proposition 15.12 (The final step on the open path)**.** *The following are equivalent:* (i) *(D);* (ii) $\varpi_\lambda < 1$ *for every* $\lambda \in (0, 1]$*;* (iii) $\varpi_{\lambda_m} \leq 1$ *for some sequence* $\lambda_m \to 0^+$*;* (iv) $\mathsf{W}(\lambda) > 0$ *for every* $\lambda \in (0, 1]$*. The implication (i)⇒(ii) uses* $\varpi_{\mathrm{RVM}} < 1$ *(Theorem 9.11), which is the case* $\lambda = 1$ *of (ii); the implications (ii)⇒(iii)⇒(i) and (ii)⇔(iv) do not use it.*

Proof. $f(\lambda) = \log \varpi_\lambda$ is continuous and strictly convex on $[0, 1]$ (Proposition 14.2: $f' = -\Upsilon$, $\Upsilon' < 0$), $f(1) = \log \varpi_{\mathrm{RVM}} < 0$ by Theorem 9.11, and $f(0) \geq 0$ (Theorem 13.2). If $f(0) = 0$, convexity gives $f(\lambda) < (1 - \lambda)f(0) + \lambda f(1) < 0$ for $0 < \lambda < 1$. If $f(\lambda_m) \leq 0$ along $\lambda_m \to 0^+$, then $f(0) \leq 0$ by continuity, so $f(0) = 0$, which is (D) by Theorem 13.2. (ii)⇔(iv) is (15.23). □

15.5. **The crossing point of the family and what convexity can give.** Write $f(\lambda) := \log \varpi_\lambda$ for $0 \leq \lambda \leq 1$. By Proposition 14.2, f is C^1 with $f' = -\Upsilon$ and $\Upsilon' < 0$, so f is strictly convex; $f(1) = \log \varpi_{\mathrm{RVM}} < -0.020615$ by Theorem 9.11, and $f(0) = \log \varpi_{Z1} \geq 0$, with equality exactly under RH, by Theorem 13.2.

Proposition 15.13 (The crossing point)**.** (i) $\Upsilon(0) > -\log \varpi_{\mathrm{RVM}} > 0.020615$*, without hypothesis.*

(ii) *If RH holds, then for every* $\lambda \in (0, 1]$

$$f(\lambda) \leq \lambda f(1) \qquad \textit{and} \qquad f(\lambda) + \lambda\Upsilon(\lambda) \leq 0. \tag{15.27}$$

(iii) *If RH fails, there is exactly one* $\lambda^* \in (0, 1)$ *with* $\varpi_{\lambda^*} = 1$*;* $f > 0$ *on* $[0, \lambda^*)$*,* $f < 0$ *on* $(\lambda^*, 1]$*, and*

$$\frac{f(0)}{\Upsilon(0)} \leq \lambda^* \leq \frac{f(0)}{f(0) + |f(1)|}. \tag{15.28}$$

With the verified zeros (Proposition 14.3), $\lambda^* < 4 \cdot 10^{-13}/0.020615 < 1.95 \cdot 10^{-11}$.

Proof. (i) Proposition 14.2 gives $\Upsilon(0) > \log L + \log \varpi_{Z1} \geq \log L = -\log \varpi_{\mathrm{RVM}}$. (ii) Under RH $f(0) = 0$; the chord of the convex function f gives $f(\lambda) \leq (1 - \lambda)f(0) + \lambda f(1) = \lambda f(1)$, and the tangent at λ gives $0 = f(0) \geq f(\lambda) - \lambda f'(\lambda) = f(\lambda) + \lambda\Upsilon(\lambda)$. (iii) If $f(0) > 0 > f(1)$, f has a zero; if it had two, $\lambda_1 < \lambda_2$, convexity on $[\lambda_1, 1]$ would give $f(\lambda_2) \leq \frac{1-\lambda_2}{1-\lambda_1} f(\lambda_1) + \frac{\lambda_2 - \lambda_1}{1-\lambda_1} f(1) < 0$. The chord at λ^* gives $0 \leq (1 - \lambda^*)f(0) + \lambda^* f(1)$, which is the upper bound, and the tangent at 0 gives $0 = f(\lambda^*) \geq f(0) - \lambda^*\Upsilon(0)$, which is the lower bound. □

For a fixed λ a violation of either inequality (15.27) would refute RH, and their validity is necessary, not sufficient. Along a sequence $\lambda_m \to 0^+$ the first one is sufficient: it gives $f(0) = \lim f(\lambda_m) \leq 0$, which is Proposition 15.12(iii). If RH fails, the set on which $f > 0$ is the single interval $[0, \lambda^*)$, of length $\lambda^* < 2 \cdot 10^{-11}$ (Proposition 15.13(iii)); the information that decides RH is therefore asymptotic in $\lambda \to 0^+$ and of the same nature as the information at $\lambda = 0$.

Remark 15.14 (What convexity cannot give). For a convex function, values in the interior bound the interior from the end points (chord), and the end points only from below (tangents): $f(0) \geq f(\lambda)+\lambda\Upsilon(\lambda)$. The only upper bound for $f(0)$ obtained from convexity on $[0,1]$ is $f(0) < f(1)+\Upsilon(0)$, which cannot give $f(0) \leq 0$ because $\Upsilon(0) > |f(1)|$ by (i). If f were extended as a strictly convex function to $[-c,1]$, the chord through $-c$ and 1 would give $f(0) < U_c := \frac{cf(1)+f(-c)}{1+c}$, but $U_c > f(0) \geq 0$ for every $c > 0$, whether $f(0) = 0$ or $f(0) > 0$; and $U_c \to f(0)$ as $c \to 0^+$. The same holds for the other convexity principles that bound interior values by boundary values (three lines, Phragmén–Lindelöf, Hardy's convexity, Lyapunov's inequality). The properties used in Proposition 15.13 are also satisfied by a family whose endpoint has $f(0) > 0$; Proposition 15.15 gives one.

Proposition 15.15 (A family with the same properties and the opposite endpoint)**.** *Let* $\kappa_1 = 12.6$ *and, for* $0 \leq \lambda \leq 1$, *let* $\boldsymbol{\kappa}(\lambda)$ *be the list* $((1-\lambda)\kappa_1 + \lambda\tau_1, \tau_2, \tau_3, \dots)$, $\zeta_\lambda^{\boldsymbol{\kappa}} = P_{\boldsymbol{\kappa}(\lambda)}/G$ *with* $P_{\boldsymbol{\kappa}}(s) = \prod_k (1 + q(s)/(\frac{1}{4}+\kappa_k^2))$, *and* $\varpi_\lambda^{\boldsymbol{\kappa}} = \lim_{\sigma\to+\infty} \zeta_\lambda^{\boldsymbol{\kappa}}(\sigma)$. *Then every* $\zeta_\lambda^{\boldsymbol{\kappa}}$ *has order one, the functional equation* (12.3) *and its nontrivial zeros exactly at* $\frac{1}{2} \pm i\kappa_k(\lambda)$; $\zeta_1^{\boldsymbol{\kappa}} = \zeta_{\mathrm{RVM}}$ *and* $\varpi_1^{\boldsymbol{\kappa}} = \varpi_{\mathrm{RVM}} < 1$; $\lambda \mapsto \log \varpi_\lambda^{\boldsymbol{\kappa}}$ *is strictly convex; and* $\varpi_0^{\boldsymbol{\kappa}} > 1$.

Proof. Only the first factor differs from $P_{\mathrm{RVM}}^{\mathrm{crit}}$, so the first three properties are those of ζ_{RVM} (Theorem 3.1), and $\zeta_\lambda^{\boldsymbol{\kappa}}/\zeta_{\mathrm{RVM}} = \frac{1+q(s)/(\frac{1}{4}+\kappa_1(\lambda)^2)}{1+q(s)/(\frac{1}{4}+\tau_1^2)} \to \frac{\frac{1}{4}+\tau_1^2}{\frac{1}{4}+\kappa_1(\lambda)^2}$ as $\sigma \to +\infty$, with $\kappa_1(\lambda) = (1-\lambda)\kappa_1 + \lambda\tau_1$. With $\zeta_{\mathrm{RVM}}(\sigma) \to \varpi_{\mathrm{RVM}}$ (Theorem 9.10), $\log \varpi_\lambda^{\boldsymbol{\kappa}} = \log \varpi_{\mathrm{RVM}} + \log(\frac{1}{4}+\tau_1^2) - \log(\frac{1}{4}+\kappa_1(\lambda)^2)$, whose second derivative in λ is $2(\tau_1-\kappa_1)^2(\kappa_1(\lambda)^2 - \frac{1}{4})/(\frac{1}{4}+\kappa_1(\lambda)^2)^2 > 0$. For the endpoint we bound ϖ_{RVM} from below. By (9.28), $\log \varpi_{\mathrm{RVM}} = \log P_{\mathrm{RVM}}^{\mathrm{crit}}(\frac{1}{2}) + \nu_3 + \nu_4$. First, $\log P_{\mathrm{RVM}}^{\mathrm{crit}}(\frac{1}{2}) = -\sum_k \log(1 + \frac{1}{4\tau_k^2}) \geq -\frac{1}{4}\sum_k \tau_k^{-2}$; from $k - \frac{1}{2} = \mathcal{N}_{\mathrm{RVM}}(\tau_k) \leq \frac{\tau_k}{2\pi}\log\frac{\tau_k}{2\pi} \leq (\frac{\tau_k}{2\pi})^{3/2}$ (because $\tau_k \geq \tau_1 > \frac{7\pi}{4}$ and $\log y \leq \sqrt{y}$), $\tau_k \geq 2\pi(k-\frac{1}{2})^{2/3}$ and $\sum_k \tau_k^{-2} \leq \frac{1}{4\pi^2}(2^{4/3} + 3\cdot 2^{1/3}) < 0.1596$. Second, $\nu_3 > -0.007925$ by (9.43). Third, by (9.39) and Lemma 9.9, $|\nu_4| \leq \frac{2\pi}{4\log 2}\int_{\tau_0}^\infty \tau^{-2} d\tau = \frac{1}{8\log 2} < 0.18034$. Hence $\log \varpi_{\mathrm{RVM}} > -0.0399 - 0.007925 - 0.18034 > -0.2283$ and $\varpi_{\mathrm{RVM}} > 0.7958$. Since $\tau_1 > 14.52134$, $\frac{\frac{1}{4}+\tau_1^2}{\frac{1}{4}+\kappa_1^2} > \frac{211.119}{159.01} > 1.3277$, and $\varpi_0^{\boldsymbol{\kappa}} > 0.7958 \cdot 1.3277 > 1.056$. □

With the floating-point value $\varpi_{\mathrm{RVM}} = 0.96194$, already $\kappa_1 = 14$ gives $\varpi_0^{\boldsymbol{\kappa}} \approx 1.035$. The list $\boldsymbol{\kappa}(0)$ differs from the clock in one entry, by 1.93, and lies at bounded distance from it. Proposition 15.15 does not concern ζ: it shows that the properties "zeros on the critical line, functional equation, order one, bounded displacement, C^1 strictly convex $\log \varpi_\lambda$, the endpoint $\lambda = 1$ equal to the clock with $\varpi < 1$" do not imply $\varpi_0 \leq 1$, equivalently $\mathsf{W}(0) \geq 0$ in Proposition 15.10. A transport lemma from the clock end to $\lambda = 0$ must therefore use that the list at $\lambda = 0$ is the list of the ordinates of ζ; in the form of Theorem 5.7 it must show that the sum over the zeros off the line vanishes. The positivity transported along the family is the number $\mathsf{W}(1) > 0$; the monotonicity of the measure A_λ is a hypothesis of Proposition 15.10 and, at $\lambda = 0$, is equivalent to (D) (Corollary 15.8).

A statement that holds as an identity only at the endpoint cannot be reached as a limit of the same identity along the path, where it is false (Proposition 15.1); an inequality can. Proposition 15.12 is of this form: the inequality $\varpi_\lambda \leq 1$ on the open path passes to the limit, and the opposite inequality $\varpi_{Z1} \geq 1$ holds at the endpoint. By Theorem 15.11 the same scheme is available at every real $\sigma > 1$: (D) holds if and only if $\zeta_{Z1}(\sigma) \leq \zeta(\sigma)$ for one $\sigma > 1$.

Remark 15.16 (The hypothesis added to the functional equation). The functional equation, the order and the zeros on the critical line do not characterize ζ: every ζ_λ has them. A converse theorem for ζ_{Z1} needs one further hypothesis on the function A_{Z1} of Lemma 15.9, and each of the following suffices and is equivalent to (D): A_{Z1} is a step function with jumps b_n at the integers satisfying $\sum_{n\leq X} |\mathsf{b}_n| = O_\varepsilon(X^{1+\varepsilon})$, so that ζ_{Z1} is a Dirichlet series absolutely convergent in $\Re s > 1$ (Hamburger's theorem, in the form of Theorem 13.7; the growth condition on the jumps is part of

the hypothesis, since Hamburger's theorem requires the absolute convergence); A_{Z1} is non-decreasing (Theorem 15.7); $\mathsf{W}(0) \geq 0$ (Proposition 15.10), which is the weakest of the three, being equivalent to $\varpi_{Z1} \leq 1$. The inequality (15.17) is the one-dimensional case of the linear programming bound of Cohn and Elkies [28]. By Lemma 15.9 the measure $\delta_0 + d\mathsf{A}_\lambda(|u|)$ is self-dual for every λ; measures on $\mathbb{R}$ whose support and spectrum are both uniformly discrete are classified in [48].

15.6. **The density of the clock.** Let $\Xi(t) := t\log t - t$ for $t > 0$, and for $M \geq 1$, $0 < t_0 \leq \frac{1}{2}$ let

$$\mathcal{U}(M, t_0) := \int_{t_0}^{\infty} e^{-iM\Xi(t)}\, dt := \lim_{T\to\infty} \int_{t_0}^{T} e^{-iM\Xi(t)}\, dt. \tag{15.29}$$

Lemma 15.17 (Stationary phase for Ξ)**.** *The limit* (15.29) *exists, and there is an absolute constant $C_{\mathcal{U}}$ with*

$$\Big|\mathcal{U}(M, t_0) - \sqrt{\frac{2\pi}{M}}\, e^{i(M-\pi/4)}\Big| \leq \frac{C_{\mathcal{U}}}{M} \qquad (M \geq 1,\ 0 < t_0 \leq \tfrac{1}{2}). \tag{15.30}$$

Proof. $\Xi'(t) = \log t$ is increasing and unbounded, so the limit exists by the first-derivative test. Ξ is convex with minimum $\Xi(1) = -1$. Put $\mathsf{w} := \operatorname{sgn}(t-1)\sqrt{2(\Xi(t)+1)}$. Near $t = 1$, $\Xi(t) + 1 = \frac{1}{2}(t-1)^2(1 + O(t-1))$ with an analytic factor, so $t \mapsto \mathsf{w}$ is analytic with derivative 1 at $t = 1$; elsewhere $\mathsf{w}\, d\mathsf{w} = \log t\, dt$. Hence $t \mapsto \mathsf{w}$ is an increasing analytic bijection of $(0, \infty)$ onto $(-\sqrt{2}, \infty)$, with

$$\mathsf{p}(\mathsf{w}) := \frac{dt}{d\mathsf{w}} = \frac{\mathsf{w}}{\log t}, \qquad \mathsf{p}(0) = 1, \qquad \mathsf{q}(\mathsf{w}) := \frac{\mathsf{p}(\mathsf{w}) - 1}{\mathsf{w}} = \frac{1}{\log t} - \frac{1}{\mathsf{w}}. \tag{15.31}$$

q is analytic on $(-\sqrt{2}, \infty)$, bounded, tends to 0 as $\mathsf{w} \to \infty$, and $\mathsf{q}' = -\mathsf{w}/(t\log^3 t) + \mathsf{w}^{-2}$. Away from $\mathsf{w} = 0$ one has $\int |\mathsf{w}|\, |t\log^3 t|^{-1} d\mathsf{w} = \int dt/(t\log^2 t)$, which converges at $t = 0$ and at $t = \infty$, and $\int \mathsf{w}^{-2} d\mathsf{w}$ converges at ∞; so $\int_{-\sqrt{2}}^{\infty} |\mathsf{q}'|\, d\mathsf{w} < \infty$. With $\mathsf{w}_0 = \mathsf{w}(t_0) \leq \mathsf{w}(\frac{1}{2}) = -\sqrt{1 - \log 2} < -\frac{1}{2}$ and $e^{-iM\Xi} = e^{iM} e^{-iM\mathsf{w}^2/2}$,

$$e^{-iM}\, \mathcal{U}(M, t_0) = \int_{\mathsf{w}_0}^{\infty} e^{-iM\mathsf{w}^2/2}\, d\mathsf{w} + \int_{\mathsf{w}_0}^{\infty} e^{-iM\mathsf{w}^2/2}\, \mathsf{w}\, \mathsf{q}(\mathsf{w})\, d\mathsf{w}. \tag{15.32}$$

The first integral is $\int_{\mathbb{R}} e^{-iM\mathsf{w}^2/2} d\mathsf{w} = \sqrt{2\pi/M}\, e^{-i\pi/4}$ minus $\int_{-\infty}^{\mathsf{w}_0}$, and the latter is at most $4/(M|\mathsf{w}_0|) \leq 8/M$ by the first-derivative test [61, Lemma 4.2]. Since $\mathsf{w}\, e^{-iM\mathsf{w}^2/2} = \frac{i}{M}\frac{d}{d\mathsf{w}} e^{-iM\mathsf{w}^2/2}$, integration by parts bounds the second integral by $M^{-1}(\sup|\mathsf{q}| + \int|\mathsf{q}'|)$. □

For $m \geq 1$ and $y > 1$ put

$$\mathcal{R}_m(y) := -2(-1)^m\, m^{-3/2}\, y^{\frac{1}{2m} - \frac{1}{2}} \cos\Big(2\pi m\, y^{1/m} - \frac{(7m+1)\pi}{4}\Big), \tag{15.33}$$

so that $\mathcal{R}_1(y) = 2\cos(2\pi y)$ and $\mathcal{R}_2(y) = -2^{-1/2} y^{-1/4} \cos(4\pi\sqrt{y} + \frac{\pi}{4})$.

Theorem 15.18 (The density of the clock)**.** *J_{RVM} is continuously differentiable on $(2, \infty)$, and for $y \geq 16$*

$$J'_{\mathrm{RVM}}(y) = \frac{1}{\log y} + 2\cos(2\pi y) - \frac{y^{-1/4}}{\sqrt{2}} \cos\Big(4\pi\sqrt{y} + \frac{\pi}{4}\Big) + O(y^{-1/3}). \tag{15.34}$$

On $[2, \infty)$ the measure $d\pi_{\mathrm{RVM}}$ is the sum of the atoms $\frac{\mu(j)}{j} J_{\mathrm{RVM}}(2)\, \delta_{2^j}$, $j \geq 1$, and of a density that is continuous off the points 2^j and equals $J'_{\mathrm{RVM}}(y) + O(y^{-1/2})$.

Proof. An exact series. We use the notation of the proof of Theorem 7.1: $v = \log y$, $S_\alpha(v) = \sum_k e^{(iv-\alpha)\tau_k}$, the decomposition (7.8) and the harmonics $K_m^{\pm}$ of (7.9), with phases $\Psi_m^{\pm}(\tau) = v\tau \pm 2\pi m \mathcal{N}_{\mathrm{RVM}}(\tau)$. Put

$$\mathcal{H}_m^{\pm}(v) := \int_{\tau_0}^{\infty} e^{i\Psi_m^{\pm}(\tau)}\, d\tau, \tag{15.35}$$

improper integrals that converge, locally uniformly in $v \ge \log 2$, by the first-derivative test, because $(\Psi_m^{\pm})'$ is monotone and unbounded. As $\alpha \downarrow 0$, $K_m^{\pm} \to iv\,\mathcal{H}_m^{\pm}(v)$, and the first two terms of (7.8) tend to a continuous function $S_{\mathrm{sm}}(v)$ with $|S_{\mathrm{sm}}(v)| \le C/v + \frac{1}{2}$. The bounds for $K_m^{\pm}$ in that proof are uniform in the Abel parameter and, for K_m^{-}, include the factor displayed in (7.10); only the finite range $m \le v/\log 2$ can contain the stationary point. After this finite range is separated, the remaining bounds are summable against $1/(2\pi m)$, so dominated convergence in (7.7) and (7.9)

$$\sum_k D_k = \int_{u_1}^{u_2} \frac{e^{v/2}}{v}\, S_0(v)\, dv, \qquad S_0(v) := S_{\mathrm{sm}}(v) + \sum_{m\ge 1} \frac{(-1)^m\, v}{2\pi m}\big(\mathcal{H}_m^{-}(v) - \mathcal{H}_m^{+}(v)\big), \tag{15.36}$$

the series converging uniformly on compact sets. Thus S_0 is continuous, J_{RVM} is continuously differentiable on $(2,\infty)$, and by (6.2)

$$J'_{\mathrm{RVM}}(y) = \frac{1}{\log y}\Big(1 - \frac{1}{y(y^2-1)} - \frac{2}{\sqrt{y}}\,\Re S_0(\log y)\Big). \tag{15.37}$$

The terms without a stationary point. $(\Psi_m^{+})' \ge v + m\log 2$, so $|\mathcal{H}_m^{+}| \le 4/(v + m\log 2)$ and $\sum_m \frac{v}{2\pi m}|\mathcal{H}_m^{+}| \ll 1 + \log v$. For $m > v/\log 2$, $(\Psi_m^{-})' \le v - m\log 2 < 0$ on $[\tau_0,\infty)$; the first-derivative test, and the second-derivative test on $[\tau_0, 2\tau_0]$ with $|(\Psi_m^{-})''| = m/\tau$, give $|\mathcal{H}_m^{-}| \ll \min\big((m\log 2 - v)^{-1}, m^{-1/2}\big)$, and $\sum_{m>v/\log 2} \frac{v}{2\pi m}|\mathcal{H}_m^{-}| \ll 1 + \log v$.

The terms $3 \le m \le v/\log 2$. By the argument for (7.10), $|\mathcal{H}_m^{-}| \le \frac{8}{m\log 2} + 16\sqrt{\pi}\, e^{v/(2m)} m^{-1/2}$, so these terms contribute to S_0 at most $C\,v\sum_{m\ge 3}\big(m^{-2} + m^{-3/2}e^{v/(2m)}\big) \ll v\, e^{v/6}$.

The terms $m = 1, 2$. The stationary point of Ψ_m^{-} is $\mathtt{t}_m := 2\pi e^{v/m} = 2\pi y^{1/m}$. With $\tau = \mathtt{t}_m t$ and (2.3), $\Psi_m^{-}(\mathtt{t}_m t) = -m\mathtt{t}_m \Xi(t) - \frac{7\pi m}{4}$, hence

$$\mathcal{H}_m^{-}(v) = \mathtt{t}_m\, e^{-7\pi i m/4}\, \mathcal{U}\big(m\mathtt{t}_m, \tau_0/\mathtt{t}_m\big), \qquad \tau_0/\mathtt{t}_m = 2e^{-v/m} \le \tfrac{1}{2} \quad (y \ge 4^m). \tag{15.38}$$

By Lemma 15.17 with $M = m\mathtt{t}_m$,

$$\frac{(-1)^m v}{2\pi m}\,\mathcal{H}_m^{-}(v) = (-1)^m\,\frac{v}{m^{3/2}}\, y^{\frac{1}{2m}} \exp\Big(i\Big(2\pi m\, y^{1/m} - \frac{(7m+1)\pi}{4}\Big)\Big) + O\Big(\frac{v}{m^2}\Big). \tag{15.39}$$

Inserting the four estimates into (15.37) gives $J'_{\mathrm{RVM}}(y) = \frac{1}{\log y} + \mathcal{R}_1(y) + \mathcal{R}_2(y) + O(y^{-1/3})$, which is (15.34).

The prime coordinate. By (6.7), on each interval $(2^r, 2^{r+1})$ the function π_{RVM} is the finite sum of the continuously differentiable functions $\frac{\mu(j)}{j} J_{\mathrm{RVM}}(x^{1/j})$, $j \le r$; the terms $j \ge 2$ have derivative $\frac{\mu(j)}{j^2} x^{1/j-1} J'_{\mathrm{RVM}}(x^{1/j})$, and $|J'_{\mathrm{RVM}}| \le C_J$ by (7.5), so their sum is $O(x^{-1/2})$. At $x = 2^j$ the term j jumps by $\frac{\mu(j)}{j} J_{\mathrm{RVM}}(2)$. □

Corollary 15.19 (Cell masses and cell defects at the clock end)**.** *For $n \ge 17$, with $\mathbf{1}_{\mathrm{Pow}_2}(n) = 1$ if $n = 2^j$ for an integer $j \ge 1$ and 0 otherwise (the product below is 0 when n is not a power of 2, and $\log_2 n = j$ when it is),*

$$c_n^{\mathrm{RVM}} = \int_{n-1}^{n} \Big(\frac{1}{\log x} + 2\cos(2\pi x)\Big)dx + \mathbf{1}_{\mathrm{Pow}_2}(n)\,\frac{\mu(\log_2 n)}{\log_2 n}\, J_{\mathrm{RVM}}(2) + O\big(n^{-1/4}\big), \tag{15.40}$$

$$\tilde{c}_n(1) = \frac{1}{\log n} + \mathbf{1}_{\mathrm{Pow}_2}(n)\,\frac{\mu(\log_2 n)}{\log_2 n}\, J_{\mathrm{RVM}}(2) + O\big(n^{-1/4}\big), \tag{15.41}$$

and for $0 < \Re s \le \sigma_1$

$$|\mathfrak{h}_n(1;s)| \le C_{\sigma_1}\big(|s|\, n^{-\sigma-4/3} + |s|(1+|s|)\, n^{-\sigma-2}\big). \tag{15.42}$$

In particular Hypothesis 7.7 is proved (it holds along the integers n that are not of the form 2^j with $\mu(j) \ne 0$, which is its whole range), so Proposition 7.8 is unconditional, and $c_n^{\mathrm{RVM}} \sim 1/\log n$, $0 < c_n^{\mathrm{RVM}} < 1$ for all large such n; at $n = 2^j$, $c_n^{\mathrm{RVM}} = \big(1 + \mu(j) J_{\mathrm{RVM}}(2)\log 2\big)/\log n + O(n^{-1/4})$.

Proof. (15.40): $c_n^{\mathrm{RVM}} = \pi_{\mathrm{RVM}}(n) - \pi_{\mathrm{RVM}}(n-1)$ is the integral of the density over $(n-1,n)$ plus the atom at n, and $\int_{n-1}^{n}(|\mathcal{R}_2| + O(x^{-1/3}))dx = O(n^{-1/4})$. (15.41): $\varphi_n(2^j) = 1$ if $n = 2^j$ and 0 otherwise; $\int \varphi(t)\,dt/\log(n+t) = \frac{1}{\log n} + O(\frac{1}{n\log^2 n})$ because $\mathsf{M}_0 = 1$; and $\int \varphi(t)\,2\cos(2\pi(n+t))\,dt = 2\mathsf{C}_0 = 0$ by (15.2). (15.42): the test function $\varphi_n(x)(n^{-s} - x^{-s})$ vanishes at the atoms, and $n^{-s} - (n+t)^{-s} = s\,n^{-s-1}t + O_{\sigma_1}(|s|(1+|s|)n^{-\sigma-2})$ for $|t| \le 1$. The first moment of the density is $\int t\varphi(t)\pi'_{\mathrm{RVM}}(n+t)\,dt = \frac{\mathsf{M}_1}{\log n} + 2\mathsf{C}_1 + O(n^{-1/3}) = O(n^{-1/3})$, because $\mathsf{M}_1 = \mathsf{C}_1 = 0$, $\mathcal{R}_2(n+t) = \mathcal{R}_2(n) + O(n^{-3/4})$ and the density is bounded. The last statements follow from (15.40) and Lemma 7.6. □

By (15.34), $\frac{d\pi_{\mathrm{RVM}}}{dy} = (\frac{1}{\log y} - 1) + (1 + 2\cos(2\pi y)) + O(y^{-1/4})$: the clock carries the first harmonic of the comb $\sum_n \delta_n = 1 + 2\sum_{q\ge1}\cos(2\pi qy)$ of all integers, with weight 1, on a smooth background. This harmonic comes from the ordinates τ_k near the height $2\pi y$, the stationary point of Φ_1^-; by (15.2) it does not enter the cell masses (15.41), which are $1/\log n$ to leading order, and it enters the cell defects only through the moments C_r, $r \ge 2$.

15.7. **The cell masses in the heights of the ordinates.**

Proposition 15.20 (Height kernel of the cell masses)**.** *For every $\varepsilon > 0$, $\lambda \in [0,1]$ and $n \ge 3$ that is not a power of 2,*

$$\tilde{c}_n(\lambda) = \mathfrak{m}_n - \frac{2}{\sqrt{n}\,\log n}\sum_{k\ge1}\hat{\varphi}\Big(\frac{\gamma_k(\lambda)}{2\pi n}\Big)\cos\big(\gamma_k(\lambda)\log n\big) + E_n(\lambda), \tag{15.43}$$

where $\sum_{X<n\le 2X}|E_n(\lambda)|^2 \le C_\varepsilon X^{1/2+\varepsilon}$ uniformly in λ (sum over the n that are not powers of 2). The kernel $\hat{\varphi}(\gamma/2\pi n)$ vanishes at the heights $\gamma = 2\pi qn$, $q = 1,2,\dots$

Proof. The term $j = 1$ of W_n in (14.21) is $\int \varphi(t)(n+t)^{-1/2+i\gamma}\,dt/\log(n+t)$. Uniformly for $|t| \le 1$ and $0 < \gamma \le n^{1+\varepsilon}$, $(1+t/n)^{-1/2+i\gamma} = e^{i\gamma t/n}(1 + O((1+\gamma/n)/n))$ and $\frac{1}{\log(n+t)} = \frac{1}{\log n}(1 + O(\frac{1}{n}))$ on $|t| \le 1$, so this term equals $\frac{n^{-1/2+i\gamma}}{\log n}\hat{\varphi}(\frac{\gamma}{2\pi n}) + O\big(n^{-3/2}(1+\gamma/n)\big)$, because $\int \varphi(t)e^{i\gamma t/n}dt = \hat{\varphi}(\gamma/2\pi n)$ for even φ. Summed over the $O(n^{1+\varepsilon}\log n)$ ordinates $\gamma_k(\lambda) \le n^{1+\varepsilon}$ the error is at most $O(n^{1+\varepsilon}\log n)\cdot n^{-3/2}(1+n^{\varepsilon}) = O(n^{-1/2+2\varepsilon}\log n) = O(n^{-1/2+3\varepsilon})$. For $\gamma > n^{1+\varepsilon}$ both the term and $\hat{\varphi}(\gamma/2\pi n)$ are $O_A((n/\gamma)^A)$, as in the proof of Theorem 14.11. The terms $j \ge 2$ have sum of squares $O(X^{1/2+3\varepsilon})$ over $X < n \le 2X$ by (14.29). The kernel vanishes at $\gamma = 2\pi qn$, $q \in \mathbb{Z}\setminus\{0\}$, because the partition of unity gives $\hat{\varphi}(q) = 0$ for $q \in \mathbb{Z}\setminus\{0\}$, (15.2). □

Since $\gamma_k(\lambda)\log n = \tau_k \log n - (1-\lambda)\eta_k \log n$, the passage along the family rotates each clock phase $n^{i\tau_k}$ by the angle $-(1-\lambda)\eta_k\log n$: up to the change of $\hat{\varphi}$ over a distance $O(1/n)$, the sum in (15.43) is $\Re\,\Xi_n\big((1-\lambda)\log n\big)$ with

$$\Xi_n(\Omega) := \sum_{k\ge1}\hat{\varphi}\Big(\frac{\tau_k}{2\pi n}\Big)n^{i\tau_k}e^{-i\Omega\eta_k} \qquad (0 \le \Omega \le \log n). \tag{15.44}$$

At $\Omega = 0$ this is the clock sum, whose real part is $O(n^{1/4}\log n)$: the kernel of the term $j = 1$ is, up to the $O(1/n)$ change of kernel, the ordinate part of $\int \varphi(t)J'_{\mathrm{RVM}}(n+t)\,dt$ in (15.37), and Theorem 15.18 gives $\int \varphi(t)\big[2\cos(2\pi(n+t)) + \mathcal{R}_2(n+t) + O((n+t)^{-1/3})\big]dt = O(n^{-1/4})$ pointwise because $C_0 = 0$ (the bound is not obtained from $\tilde{c}_n(1)$, which carries the terms $j \ge 2$ whose remainder is controlled only in mean square), because the resonant heights $\tau_k \approx 2\pi n$ are zeros of the kernel; the endpoint of the family is $\Omega = \log n$.

Proposition 15.21 (The flow of a cell mass and the count difference)**.** *For every $n \ge 3$ that is not a power of 2,*

$$\tilde{c}_n(0) - \tilde{c}_n(1) = -2\int_0^1 \sum_{k\ge1}\eta_k\,\Im V_n\big(\gamma_k(\lambda)\big)\,d\lambda = -2\,\Im\int_{\tau_0}^{\infty} V_n(t)\big(N_\zeta(t) - n_{\mathrm{RVM}}(t)\big)\,dt. \tag{15.45}$$

Proof. The first equality is (14.22) and the continuity of $\tilde{c}_n$ at $\lambda = 0$. By (14.22), $\tilde{c}_n(0) - \tilde{c}_n(1) = 2\Re \sum_k (W_n(\tau_k) - W_n(\gamma_k)) = 2\Re \int_{\tau_0}^{\infty} W_n \, d(n_{\mathrm{RVM}} - N_\zeta)$. Both counts vanish at τ_0, their difference is $O(\log t)$ by Lemma 2.2, and W_n is rapidly decreasing; integration by parts with $W_n' = iV_n$ gives $2\Re(i \int V_n (N_\zeta - n_{\mathrm{RVM}})\, dt)$. □

Proposition 15.22 (Height kernel of the cell defects)**.** *For every $\varepsilon > 0$, $\lambda \in [0,1]$, $\Re s > 0$ and $n \geq 3$ that is not a power of 2, with an implied constant uniform for s in compact subsets of $\Re s > 0$ and for $\lambda \in [0,1]$,*

$$\mathfrak{h}_n(\lambda; s) = -\frac{s\, n^{-s-3/2}}{\pi \log n} \sum_{k \geq 1} \hat{\varphi}'\Big(\frac{\gamma_k(\lambda)}{2\pi n}\Big) \sin\big(\gamma_k(\lambda) \log n\big) + O_{s,\varepsilon}\big(n^{-\sigma-3/4+\varepsilon}\big). \tag{15.46}$$

Proof. For $|t| \leq 1$, $n^{-s} - (n+t)^{-s} = s\, n^{-s-1} t + \psi_2(t)$, where ψ_2 and all its derivatives in t are $O_s(n^{-\sigma-2})$. *The term $j = 1$.* The part of the density that does not involve the ordinates contributes $O_s(n^{-\sigma-2})$, because $\int t\varphi(t)\, dt / \log(n+t) = O(1/(n \log^2 n))$ ($\mathsf{M}_1 = 0$). The pairing of $\varphi\psi_2$ with the ordinate sum is $O_s(n^{-\sigma-2})$ times $O(n^{1/2+\varepsilon})$, as in Remark 14.15. For the real function $t\varphi(t)$, the computation in the proof of Proposition 15.20, with $\int t\varphi(t) e^{2\pi i \xi t} dt = -\frac{i}{2\pi}\hat{\varphi}'(\xi)$ for even φ and $-2\Re(-\frac{i}{2\pi} n^{i\gamma}) = -\frac{1}{\pi}\sin(\gamma \log n)$, gives the sum in (15.46) with an error $O_s(n^{-\sigma-3/2+3\varepsilon})$. *The terms $j \geq 2$.* The test function $\varphi_n(y^j)(n^{-s} - y^{-js})$ is $O(|s| n^{-\sigma-1})$ on a support of length $O(n^{1/j-1})$; using the uniform upper bound $O(n^{1+\varepsilon} \log n)$ for the number of relevant ordinates, which is more than sufficient for every $j \geq 2$, and $y^{-1/2} = O(n^{-1/(2j)})$, these terms are $O_s(n^{-\sigma-1+1/(2j)+2\varepsilon})$, at most $O_s(n^{-\sigma-3/4+2\varepsilon})$. The bounds used depend on s only through $|s|$ and σ, hence are uniform on compact subsets of $\Re s > 0$; replacing ε by $\varepsilon/3$ gives the stated error. □

The kernel $\hat{\varphi}$ of the masses vanishes at the heights $2\pi q n$ and the kernel $\hat{\varphi}'$ of the defects in general does not: the masses and the defects of a cell read complementary heights of the ordinates. At $\lambda = 1$ the resonant heights $\tau_k \approx 2\pi n$ contribute to $\sum_k n^{i\tau_k}$ the real quantity $-\sqrt{n} \log n\, e^{2\pi i n}$ (proof of Theorem 15.18), which the sine in (15.46) does not see: the first harmonic of the clock is centred at the integers.

Remark 15.23 (The scale of the displacements)*.* By Proposition 15.1, a property of the sequence (η_k) that yields (D) is false for every sequence $(c\,\eta_k)$ with $0 \leq c < 1$. Properties that are invariant under $\eta_k \mapsto c\,\eta_k$ with $c > 0$ (the signs of the η_k, the non-vanishing of the η_k, their boundedness) hold along the whole open path and do not suffice; the property fixes the size of the displacements. In the form (15.44) it is a condition on the angles $\eta_k \log n$ modulo 2π, jointly in k and n. The construction contains one exact relation of this kind: for a simple ordinate γ_k, the midpoint value of the count is $N_\zeta^*(\gamma_k) = \frac{1}{2}(N_\zeta(\gamma_k + 0) + N_\zeta(\gamma_k - 0)) = k - \frac{1}{2}$, and (2.4) and (2.13) give $\mathcal{N}_{\mathrm{RVM}}(\tau_k) - \mathcal{N}_{\mathrm{RVM}}(\gamma_k) = N_\zeta^*(\gamma_k) - \mathcal{N}_{\mathrm{RVM}}(\gamma_k)$, that is,

$$\eta_k\, \mathcal{N}_{\mathrm{RVM}}'(\xi_k) = N_\zeta^*(\gamma_k) - \mathcal{N}_{\mathrm{RVM}}(\gamma_k) \qquad \text{for some } \xi_k \text{ between } \gamma_k \text{ and } \tau_k. \tag{15.47}$$

15.8. **The Gaussian defect of the atomization.**

Lemma 15.24 (Vertical growth of the joining factor)**.** *For every $\lambda \in [0,1]$ and $\frac{1}{2} < \sigma_0 < \sigma_1$, $\tilde{h}_\lambda(\sigma + it) = o(|t|)$ as $|t| \to \infty$, uniformly for $\sigma_0 \leq \sigma \leq \sigma_1$.*

Proof. By (14.15) and Proposition 14.5(ii) it suffices to treat the double series. Fix $c \in (\frac{1}{2}, \sigma_0)$. With $I_n := \int_{n-1}^{n+1} |\pi_\lambda - \mathfrak{L}|^2 dx$ as in the proof of Theorem 14.13, (14.31) gives $\sum_{m \geq 1} \frac{1}{m} \sum_{n > N} |\mathfrak{h}_n(\lambda; ms)| \leq C|s| \sum_{n>N} n^{-\sigma_0-1}(1 + I_n^{1/2})$ and $\sum_{m>N} \frac{1}{m} \sum_{n \geq 2} |\mathfrak{h}_n(\lambda; ms)| \leq C|s| \sum_{n \geq 2} n^{-N\sigma_0-1}(1 + I_n^{1/2})$; both are at most $\epsilon_N |s|$ with $\epsilon_N \to 0$ as $N \to \infty$. For the finitely many pairs $m, n \leq N$, with $E = \pi_\lambda - \mathfrak{L}$

as in the proof of Theorem 14.13,

$$\begin{aligned}\mathfrak{h}_n(\lambda;ms) = &\int \varphi_n(x)(n^{-ms}-x^{-ms})d\mathfrak{L}(x) - \int E(x)\varphi_n'(x)(n^{-ms}-x^{-ms})dx \\ &- ms\int E(x)\varphi_n(x)\,x^{-ms-1}dx.\end{aligned} \tag{15.48}$$

The first two terms are bounded in t, uniformly for $\sigma_0 \le \sigma \le \sigma_1$. In $u = \log x$ the last integral is the Fourier transform at mt of the amplitude $E(e^u)\varphi_n(e^u)e^{-m\sigma u}$; as σ ranges over $[\sigma_0,\sigma_1]$ these amplitudes form a compact subset of L^1 (a continuous image of a compact interval, because $E \in L^1(n-1,n+1)$), so the Riemann–Lebesgue convergence to 0 as $|t|\to\infty$ is uniform in σ. Hence $\limsup_{|t|\to\infty}|t|^{-1}\sup_\sigma|\tilde h_\lambda(\sigma+it)| \le 2\epsilon_N$ for every N. □

For $y>0$ and $c>1$ put

$$\Theta_\lambda(y) := \frac{1}{4\pi i}\int_{(c)} \zeta_\lambda(s)\,\Gamma(\tfrac{s}{2})\,\pi^{-s/2}\,y^{-s/2}\,ds, \qquad \mathfrak{G}_\lambda(y) := \sum_{N\ge1}\tilde a_N(\lambda)\,e^{-\pi N^2 y} - \Theta_\lambda(y). \tag{15.49}$$

For ζ itself $\Theta(y)=\sum_{n\ge1}e^{-\pi n^2y}$, and if ζ_λ has a representation (15.11), then $\Theta_\lambda(y)=\int e^{-\pi u^2y}\,d\mathsf{A}(u)$. $\mathfrak{G}_\lambda$ is the change produced by the atomization in the Gaussian count of the integers of the system.

Proposition 15.25 (Gaussian defect)**.** *Let $\lambda\in[0,1]$.*

(i) $\Theta_\lambda(y) = y^{-1/2}\,\Theta_\lambda(1/y) + \frac12(y^{-1/2}-1)$ *for $y>0$, and $\Theta_\lambda(y)=O_A(y^{-A})$ as $y\to\infty$ for every A.*

(ii) *The coefficients $\tilde a_N(\lambda)$ satisfy the theta relation of ζ,*

$$\sum_{N\ge1}\tilde a_N(\lambda)\,e^{-\pi N^2y} = y^{-1/2}\sum_{N\ge1}\tilde a_N(\lambda)\,e^{-\pi N^2/y} + \tfrac12(y^{-1/2}-1) \qquad (y>0), \tag{15.50}$$

if and only if

$$\mathfrak{G}_\lambda(y) = y^{-1/2}\,\mathfrak{G}_\lambda(1/y) \qquad (y>0). \tag{15.51}$$

Under the hypotheses of Theorem 15.3, (15.51) *holds.*

(iii) *For every $\delta>0$, as $y\to0^+$,*

$$\mathfrak{G}_\lambda(y) = \tfrac12(e^{\tilde h_\lambda(1)}-1)\,y^{-1/2} + O_\delta(y^{-1/4-\delta}), \tag{15.52}$$

and $e^{\tilde h_\lambda(1)} = \operatorname{Res}_{s=1}\tilde\zeta_{\mathbb{Z},\lambda}(s)$.

(iv) *If* (15.51) *holds, then $\tilde h_\lambda(1)=0$.*

Proof. (i) ζ_λ is bounded by a power of $|t|$ on vertical strips (proof of Lemma 15.6; on $\Re s\ge c$ use Proposition 14.1), and $\Gamma(\frac{s}{2})$ decays exponentially. Moving the line to $\Re s = 1-c$ crosses $s=1$, where the integrand has residue $\Gamma(\frac12)\pi^{-1/2}y^{-1/2} = y^{-1/2}$, and $s=0$, with residue $2\zeta_\lambda(0)=-1$; the prefactor $\frac{1}{4\pi i} = \frac12\cdot\frac{1}{2\pi i}$ turns these residues into the term $\frac12(y^{-1/2}-1)$. On the new line (12.3) and $y^{-s/2} = y^{-1/2}(1/y)^{-(1-s)/2}$ give $y^{-1/2}\Theta_\lambda(1/y)$. The decay follows by moving the line to the right, to $\Re s = 2A$ with $2A>c$: no pole is crossed, since $\zeta_\lambda(s)\Gamma(\frac{s}{2})$ is holomorphic on $\Re s>1$, the integrals converge uniformly by the same bounds, and $|y^{-s/2}| = y^{-A}$ on the new line. (ii) By $e^{-x} = \frac{1}{2\pi i}\int_{(c/2)}\Gamma(w)x^{-w}dw$ and Corollary 14.12, $\sum_N \tilde a_N(\lambda)e^{-\pi N^2y} = \frac{1}{4\pi i}\int_{(c)}\tilde\zeta_{\mathbb{Z},\lambda}(s)\Gamma(\frac{s}{2})\pi^{-s/2}y^{-s/2}ds$; subtracting (i) from the theta relation gives the equivalence. Under the hypotheses of Theorem 15.3, $\tilde\zeta_{\mathbb{Z},\lambda} = \zeta$. (iii) By (ii), $2\mathfrak{G}_\lambda(y) = \frac{1}{2\pi i}\int_{(c)}\zeta_\lambda(s)(e^{\tilde h_\lambda(s)}-1)\Gamma(\frac{s}{2})\pi^{-s/2}y^{-s/2}ds$. By Lemma 15.24, $|e^{\tilde h_\lambda(s)}| \le e^{o(|t|)}$ on $\frac12+2\delta\le\Re s\le c$, so the line can be moved to $\Re s = \frac12+2\delta$. The only pole crossed is $s=1$, where ζ_λ has residue 1 and $\tilde h_\lambda$ is holomorphic; the residue is $(e^{\tilde h_\lambda(1)}-1)y^{-1/2}$, and the remaining integral is $O_\delta(y^{-1/4-\delta})$. (iv) $\sum_N\tilde a_N(\lambda)e^{-\pi N^2y} = O(e^{-\pi y})$ as $y\to\infty$, so by (i) $\mathfrak{G}_\lambda(y)=O_A(y^{-A})$ there; then (15.51) gives $\mathfrak{G}_\lambda(y)=O_A(y^A)$ as $y\to0^+$ for every A, and (15.52)

forces $e^{\tilde{h}_\lambda(1)} = 1$; since $\tilde{h}_\lambda$ is real on the real axis, $\tilde{h}_\lambda(1)$ is real, so $\tilde{h}_\lambda(1) = 0$ and not $2\pi i m$ with $m \neq 0$. □

Corollary 15.26 (The density of the atomized integers)**.** *$\sum_{N\le x} \tilde{a}_N(\lambda)$ has Gaussian mean density $e^{\tilde{h}_\lambda(1)}$ in the sense of* (15.52) *(a statement about the Gaussian weights $e^{-\pi N^2 y}$ as $y \to 0^+$, not a Tauberian statement about the partial sums), and (D) implies*

$$\tilde{h}_0(1) = 0, \qquad \textit{that is,} \qquad \mathfrak{u}_0(1) = \sum_{m\ge1} \frac{1}{m} \sum_{n\ge2} \mathfrak{h}_n(0; m). \tag{15.53}$$

Proof. Under (D), $\tilde{h}_0 \equiv 0$ (Theorem 14.14); the second form is (14.15) at $s = 1$. □

The identity $\tilde{h}_0(\sigma) = 0$ for real σ has two distinguished members: $\sigma = +\infty$ is $\varpi_{Z1} = 1$, the first coefficient (Corollary 13.3), and $\sigma = 1$ is (15.53), the density of the integers.

By Proposition 15.25(i), the function (15.6) is $T_\lambda(y) = \mathfrak{G}_\lambda(y) - y^{-1/2}\mathfrak{G}_\lambda(1/y)$, so (15.51) is $T_\lambda \equiv 0$. At the clock end the expansion (15.52) holds with a smaller remainder: for every $\delta > 0$,

$$\mathfrak{G}_1(y) = \tfrac{1}{2}(e^{\tilde{h}_1(1)} - 1)\, y^{-1/2} + O_\delta(y^{-\delta}) \qquad (y \to 0^+), \tag{15.54}$$

because $\tilde{h}_1$ is holomorphic in $\Re s > 0$ with $\tilde{h}_1(\sigma + it) = o(|t|)$ uniformly on $\sigma_0 \le \sigma \le \sigma_1$ for every $\sigma_0 > 0$ (the proof of Lemma 15.24 with the bound $|\mathfrak{h}_n(1; s)| \le C|s|n^{-\sigma-1}$ of Theorem 14.13(iv)), and the line of integration can be moved to $\Re s = 2\delta$.

15.9. **Entire dependence on λ and the exceptional set.**

Proposition 15.27 (The cell masses as entire functions of λ)**.** *Let $n \ge 3$ not be a power of* 2.

(i) *The real-axis function $\lambda \mapsto \tilde{c}_n(\lambda)$ has the entire extension*

$$\widetilde{C}_n(\lambda) := \mathfrak{m}_n - \sum_k \Big(W_n(\gamma_k + \lambda\eta_k) + \overline{W_n(\gamma_k + \bar{\lambda}\eta_k)} \Big),$$

which is of exponential type at most $C_\eta \log(n+1)$ and agrees with $\tilde{c}_n(\lambda)$ for real λ.

(ii) *For every $r \ge 0$, $\varepsilon > 0$ and $\lambda \in [0, 1]$,*

$$\partial_\lambda^r \tilde{c}_n(\lambda) = -2\,\Re \sum_{k\ge1} \eta_k^r\, W_n^{(r)}(\gamma_k(\lambda)), \qquad |\partial_\lambda^r \tilde{c}_n(\lambda)| \le C_{r,\varepsilon} (C_\eta \log(n+1))^r\, n^{1/2+\varepsilon}. \tag{15.55}$$

(iii) *$|\tilde{c}_n(\lambda) - \tilde{c}_n(\lambda')| \le C_\varepsilon |\lambda - \lambda'|\, n^{1/2+\varepsilon}$ on $[0, 1]$. Hence, for sets of integers n that are not powers of* 2*, $\tilde{c}_n(0) = O_\varepsilon(n^\varepsilon)$ if and only if $\tilde{c}_n(\lambda) = O_\varepsilon(n^\varepsilon)$ uniformly for $0 \le \lambda \le n^{-1/2}$, and $\tilde{c}_n(\lambda) = O_\varepsilon(n^\varepsilon)$ uniformly for $1 - n^{-1/2} \le \lambda \le 1$.*

Proof. For complex γ, $|y^{i\gamma}| \le e^{|\Im\gamma| \log y}$ with $\log y \le \log(n+1)$ on the support of $\varphi_n(y^j)$, and repeated integration by parts in (14.21) gives $|W_n^{(r)}(\gamma)| \le C_{A,r,n}\, e^{|\Im\gamma| \log(n+1)} (1 + |\Re\gamma|)^{-A}$ for every A. With $\gamma = \gamma_k + \lambda\eta_k$, $|\Im\gamma| \le C_\eta |\Im\lambda|$, the two reflected series defining $\widetilde{C}_n$ converge locally uniformly in $\lambda \in \mathbb{C}$, so $\widetilde{C}_n$ is entire; on the real axis it equals $\tilde{c}_n$. Termwise differentiation on the real axis gives the first formula in (15.55). For real λ, $W_n^{(r)}$ carries the factor $(i \log y)^r$ with $\log y \le \log(n+1)$, the ordinates $\gamma_k(\lambda) > n^{1+\varepsilon}$ contribute $O(n^{-1})$, and each of the $O(n^{1+\varepsilon} \log n)$ remaining ordinates contributes $O\big((\log(n+1))^r n^{-1+1/(2j)}\big)$ to the term j, as in Remark 14.15; this is the bound in (15.55). (iii) is the case $r = 1$ and Corollary 15.19. □

The bounds (15.55) do not by themselves control $\tilde{c}_n(0)$ from $\tilde{c}_n(1)$: the function $n^{1/2} \sin\big((1 - \lambda) C_\eta \log n\big)$ is entire of the same type, satisfies the same bounds, vanishes at $\lambda = 1$ and has size $n^{1/2}$ at $\lambda = 0$ for infinitely many n. The Taylor expansion at the clock end, $\tilde{c}_n(0) = \sum_{r\ge0} \frac{(-1)^r}{r!} \partial_\lambda^r \tilde{c}_n(1)$, has the terms $-2\Re \sum_k \eta_k^r W_n^{(r)}(\tau_k)$, clock sums weighted by the powers of the displacements.

Proposition 15.28 (The exceptional set)**.** *The set*

$$\mathsf{E} := \{\lambda \in [0,1] :\ \tilde{\zeta}_{\mathbb{Z},\lambda} = \zeta\} \tag{15.56}$$

is finite, $1 \notin \mathsf{E}$*, and (D) holds if and only if* $0 \in \mathsf{E}$*. For every* $\lambda \notin \mathsf{E}$ *the hypotheses of Theorem 15.3 do not hold together,* $T_\lambda \not\equiv 0$ *and* $\mathsf{V}_\sigma(\lambda) > 0$*.*

Proof. If $\lambda \in \mathsf{E}$, then $\sum_n \tilde{c}_n(\lambda) \sum_m \frac{n^{-ms}}{m} = \log \zeta(s) = \sum_p \sum_m \frac{p^{-ms}}{m}$ for $\Re s > 1$. Comparing the coefficients of N^{-s} in increasing order of N determines the exponents uniquely, so $\tilde{c}_n(\lambda) = \mathbf{1}_{\mathrm{prime}}(n)$ for every n. Choose a composite n, not a power of 2, so large that $\tilde{c}_n(1) > 0$ by (15.41). Then $\tilde{c}_n$ is an entire function of λ (Proposition 15.27) that does not vanish identically; the zeros of a nonzero entire function have no finite accumulation point, so only finitely many of them lie in the compact interval $[0,1]$; they contain E, and 1 is not one of them. $0 \in \mathsf{E}$ implies (D) by Theorem 15.4(ii), and (D) implies $\tilde{\zeta}_{\mathbb{Z},0} = \zeta_{Z1} = \zeta$ by Theorem 14.14. The last statement is Theorems 15.3 and 15.4(ii). □

Proposition 15.1 excludes that ζ_λ and $\tilde{\zeta}_{\mathbb{Z},\lambda}$ coincide for $\lambda > 0$; Proposition 15.28 excludes, outside a finite set, that $\tilde{\zeta}_{\mathbb{Z},\lambda}$ coincides with ζ. For $0 < \lambda < 1$ outside E it is not decided which of the two hypotheses of Theorem 15.3 fails; at $\lambda = 1$ it is the symmetry (Proposition 15.5).

15.10. **The final step in the cell masses.**

Corollary 15.29 (The final step is one number per cell)**.** *The following are equivalent:*

(i) *(D);*
(ii) $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$ *for every* $n \geq 2$*;*
(iii) *there is a list* $\boldsymbol{\kappa} \in \mathscr{O}$ *(Section 16) with* $\tilde{c}_n(\boldsymbol{\kappa}) = \mathbf{1}_{\mathrm{prime}}(n)$ *for every* $n \geq 2$*.*

Proof. (i)⇒(ii) is Theorem 14.14, and (ii)⇒(iii) with the list (γ_k). If (iii) holds, then

$$\tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}}(s) = \prod_{n \geq 2} (1 - n^{-s})^{-\tilde{c}_n(\boldsymbol{\kappa})} = \prod_p (1 - p^{-s})^{-1} = \zeta(s) \qquad (\Re s > 1), \tag{15.57}$$

and Corollary 16.4, (iii)⇒(i), gives (D). □

Step A7 of the programme asks for the vanishing of the functions $\mathfrak{h}_n(0;\cdot)$, one function of s for each cell, and for $J_{Z1} = 0$ on the first cell. Corollary 15.29 asks for one real number for each cell; the position of the mass inside the cell and the first cell then follow from Hamburger's theorem.

Proposition 15.30 (Every point of the exceptional set is a witness)**.** *For the set* E *of* (15.56)*,*

$$\mathsf{E} \neq \varnothing \iff \text{(D)}, \qquad \text{so that either } \mathsf{E} = \varnothing \text{ or } 0 \in \mathsf{E}. \tag{15.58}$$

If (D) holds, then

$$\mathsf{E} \setminus \{0\} = \{\lambda \in (0,1] :\ \mathsf{D}_n(\lambda) = 0 \text{ for every } n \geq 2\}, \qquad \mathsf{D}_n(\lambda) := \tilde{c}_n(\lambda) - \tilde{c}_n(0), \tag{15.59}$$

and for $n \geq 3$ *not a power of* 2*, the real-axis function* $\mathsf{D}_n(\lambda) = -2\Re \sum_{k \geq 1} \big(W_n(\gamma_k + \lambda\eta_k) - W_n(\gamma_k)\big)$ *has the entire extension obtained by replacing each real part by the corresponding holomorphic plus reflected-conjugate pair; this extension is of exponential type at most* $C_\eta \log(n+1)$ *that vanishes at* $\lambda = 0$*.*

Proof. If $\lambda_0 \in \mathsf{E}$, the list $(\gamma_k(\lambda_0))$ belongs to $\mathscr{O}$ and its atomized zeta function is ζ; Corollary 16.4, (iii)⇒(i), gives (D), and then $0 \in \mathsf{E}$ by Proposition 15.28. Under (D), $\tilde{c}_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$, and $\lambda \in \mathsf{E}$ means $\tilde{c}_n(\lambda) = \mathbf{1}_{\mathrm{prime}}(n)$ for every n (proof of Proposition 15.28). The last statement is (14.22) and Proposition 15.27. □

Thus a proof that $\mathsf{E} \cap (0,1] = \varnothing$ does not approach the final step: if (D) fails, E is empty, and if (D) holds, $\mathsf{E} \cap (0,1] = \varnothing$ is the uniqueness of the witness on the path, which is the statement that the entire functions D_n have no common zero in $(0,1]$. This uniqueness is not proved; Proposition 15.28 gives that E is finite and $1 \notin \mathsf{E}$.

Remark 15.31 (The dual function of the atomized family). For $\Re s < \frac{1}{2}$ put $B_\lambda^\vee(s) := \zeta_\lambda(s)\, e^{\tilde h_\lambda(1-s)}$. By the functional equation of ζ_λ,

$$\pi^{-s/2}\Gamma\big(\tfrac{s}{2}\big)B_\lambda^\vee(s) = \pi^{-(1-s)/2}\Gamma\big(\tfrac{1-s}{2}\big)\,\tilde\zeta_{\mathbb{Z},\lambda}(1-s) \qquad (\Re s < \tfrac{1}{2}), \tag{15.60}$$

so $B_\lambda^\vee$ is the function attached to $\tilde\zeta_{\mathbb{Z},\lambda}$ by the functional equation. To ask that $B_\lambda^\vee$ continue to $\Re s > 1$ as an absolutely convergent Dirichlet series is to ask for the hypotheses of Theorem 15.3; no monotonicity in λ is known, and a parameter such as $\inf\{\lambda > 0 : B_\lambda^\vee$ is not a Dirichlet series$\}$ is not shown to be a transition point.

Proposition 15.32 (The identity that remains, cell by cell)**.** *Let $n \geq 17$ not be a power of 2. Then $\tilde c_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$ if and only if*

$$\mathbf{1}_{\mathrm{prime}}(n) - \tilde c_n(1) = -2\,\Im \int_{\tau_0}^{\infty} V_n(t)\big(N_\zeta(t) - n_{\mathrm{RVM}}(t)\big)\,dt, \qquad \tilde c_n(1) = \frac{1}{\log n} + O\big(n^{-1/4}\big). \tag{15.61}$$

Proof. This is (15.45) and (15.41). □

The left side of (15.61) is explicit. The right side is the pairing of the count difference with the kernel V_n, whose term $j = 1$ is $n^{-1/2+it}\hat\varphi(t/2\pi n)$ up to $O\big(n^{-3/2}(1+t/n)\big)$ (proof of Proposition 15.22). For $n = 2^j$ the same identity holds with the cut term of Theorem 14.11 added. No evaluation of the right side of (15.61) is given in this paper.

15.11. **Symmetric joining factors.**

Remark 15.33 (The symmetry criterion has no slack at $\lambda = 0$). At $\lambda = 0$ the hypothesis of Theorem 15.3 is equivalent to its conclusion. If (D) holds, then $\tilde h_0 \equiv 0$ (Theorem 14.14), and $e^{\tilde h_0} = 1$ is symmetric and of finite order; hence

$$e^{\tilde h_0} \text{ symmetric of finite order } \iff \tilde h_0 \equiv 0 \iff \text{(D)}, \tag{15.62}$$

and in the same way $T_0 \equiv 0$ and $\mathsf{V}_\sigma(0) = 0$ are equivalent forms of (D), not weaker conditions. Moreover (15.5) presupposes a continuation of $\tilde h_0$ across the line $\Re s = \frac{1}{2}$ (step A10 of the programme): $\tilde h_0$ is holomorphic in $\Re s > \frac{1}{2}$, and $s \mapsto 1 - s$ maps this half-plane onto $\Re s < \frac{1}{2}$. Theorems 15.37 and 15.39 below are criteria whose hypotheses are written on the half-line $\sigma \to +\infty$ and on the closed half-plane $\Re s \geq \frac{1}{2}$.

Proposition 15.34 (Symmetric joining factors in general)**.** *Let f be meromorphic in $\mathbb{C}$ with $(s-1)f(s)$ entire of order at most one, and let f satisfy the functional equation* (12.3)*. Let $g = f\,e^h$, where e^h is entire, without zeros, of finite order, and $e^{h(s)} = e^{h(1-s)}$. If $g \not\equiv 0$ and $g(s) = \sum_{n\geq 1} a_n n^{-s}$ converges absolutely in $\Re s > 1$, then $g = C_H\zeta$ with $C_H \neq 0$ and e^h is constant. If moreover $\lim_{\sigma\to+\infty} g(\sigma) = \lim_{\sigma\to+\infty} f(\sigma) \neq 0$, then $e^h \equiv 1$ and $f = g = C_H\zeta$.*

Proof. g satisfies (H0)–(H3) of Subsection 14.7 with the second function equal to g, so $g = C_H\zeta$ by Hamburger's theorem. If $C_H = 0$ then $f\,e^h = 0$, which is excluded when $g \not\equiv 0$; assume $C_H \neq 0$. Then $e^h(s-1)f(s) = C_H(s-1)\zeta(s)$ identically, and we compare the Hadamard factorizations of the two entire functions of order at most one, $(s-1)f(s) = e^{A_1+B_1 s}\Pi_1(s)$ and $(s-1)\zeta(s) = e^{A_2+B_2 s}\Pi_2(s)$, with Π_i the canonical products of genus at most one over their zeros. Since e^h has no zeros, the two functions have the same zeros with the same multiplicities, so $\Pi_1 = \Pi_2$ and $e^h = C_H e^{(A_2-A_1)+(B_2-B_1)s}$: h is a polynomial of degree at most one. The symmetry gives $h(s) - h(1-s) \in 2\pi i\mathbb{Z}$, which forces the degree to be zero. The last statement compares the limits at $+\infty$. □

Remark 15.35 (Order one cannot be dropped). For real $\nu_5 \neq 0$ put $f(s) = \zeta(s)\exp\big(-\nu_5(s-\frac{1}{2})^2\big)$ and $e^{h(s)} = \exp\big(\nu_5(s-\frac{1}{2})^2\big)$. Then f is meromorphic of order two with the functional equation, the pole and the zeros of ζ; e^h is entire, without zeros, of finite order and symmetric; $g = f\,e^h = \zeta$ is

an absolutely convergent Dirichlet series in $\Re s > 1$; and e^h is not constant. In Theorem 15.3 the conclusion $\tilde{h}_0 \equiv 0$ uses that $(s-1)\zeta_{Z1}$ has order one (Proposition 12.1). The location of the zeros of f plays no role in Proposition 15.34.

15.12. **The half-line form of Hamburger's theorem for a pair.** The next two theorems are stated for a general pair: a function with the functional equation and a Dirichlet series, joined on the real half-line only.

Theorem 15.36 (The ladder)**.** *Let* $\mathsf{d}\colon [1,\infty) \to \mathbb{R}$ *be measurable with* $\int_1^\infty |\mathsf{d}(x)|\, x^{-\mathsf{e}_1-1}\, dx < \infty$ *for some* $\mathsf{e}_1 \geq 0$*, and put*

$$h(\sigma) := \sigma \int_1^\infty \mathsf{d}(x)\, x^{-\sigma-1}\, dx \qquad (\sigma > \mathsf{e}_1). \tag{15.63}$$

For every integer $N \geq 2$*,*

$$h(\sigma) = O(\sigma N^{-\sigma}) \quad (\sigma \to +\infty) \iff \mathsf{d}(x) = 0 \text{ for almost every } x \in [1, N). \tag{15.64}$$

Proof. If $\mathsf{d} = 0$ almost everywhere on $[1, N)$, then $|h(\sigma)| \leq \sigma N^{-(\sigma-\mathsf{e}_1)} \int_N^\infty |\mathsf{d}| x^{-\mathsf{e}_1-1} dx$ for $\sigma > \mathsf{e}_1$. Conversely, put $d(u) := \mathsf{d}(e^u)$, so that $d\, e^{-\mathsf{e}_1 u} \in L^1(0,\infty)$, and

$$\mathsf{Y}(s) := \int_0^{\log N} d(u)\, e^{s(\log N - u)}\, du. \tag{15.65}$$

Y is entire of exponential type at most $\log N$, bounded on $\Re s \leq 0$ by $\int_0^{\log N} |d|$, and $\mathsf{Y}(\sigma) \to 0$ as $\sigma \to -\infty$. For real $\sigma > \mathsf{e}_1$,

$$\mathsf{Y}(\sigma) = \frac{N^\sigma h(\sigma)}{\sigma} - \int_{\log N}^\infty d(u)\, e^{-\sigma(u - \log N)}\, du, \tag{15.66}$$

where the first term is bounded by hypothesis and the second by $N^{\mathsf{e}_1} \int |d| e^{-\mathsf{e}_1 u} du$; so Y is bounded on the real axis. The Phragmén–Lindelöf theorem in the two quadrants $\{\Re s > 0,\ \pm \Im s > 0\}$ (angle $\pi/2$, order one, bounded on both sides) shows that Y is bounded in $\mathbb{C}$, hence constant, hence 0. By the uniqueness of the Laplace transform on $(0, \log N)$, $d = 0$ almost everywhere there. □

No functional equation, no continuation of h and no uniformity in t enter Theorem 15.36: the hypothesis lives on $\sigma \to +\infty$.

Theorem 15.37 (Hamburger's theorem for a pair, half-line form)**.** *Let* f *satisfy (H0) and the functional equation* (12.3)*, and suppose that for* $\Re s > 1$

$$\log f(s) = s \int_1^\infty J_f(x)\, x^{-s-1}\, dx, \qquad \int_1^\infty |J_f(x)|\, x^{-\sigma-1}\, dx < \infty \quad (\sigma > 1), \tag{15.67}$$

with J_f *real and measurable. Let* $g(s) = \sum_{n\geq 1} a_n n^{-s}$ *converge absolutely in* $\Re s > 1$*, with* $a_1 = 1$*, and let* $J_g(x) := \sum_{2 \leq n \leq x} b_n$*, where* $\log g(s) = \sum_{n \geq 2} b_n n^{-s}$ *for large* σ*. Put* $h(\sigma) := \log g(\sigma) - \log f(\sigma)$ *for large real* σ*. The following are equivalent:*

(i) $h(\sigma) = O_N(\sigma N^{-\sigma})$ *as* $\sigma \to +\infty$*, for every integer* $N \geq 2$*;*
(ii) $f = g$*;*
(iii) $f = g = \zeta$*.*

In this case J_f *is almost everywhere the step function* $\sum_{p^m \leq x} \frac{1}{m}$*.*

Proof. For large σ, $\log g(\sigma) = \sigma \int_1^\infty J_g(x) x^{-\sigma-1} dx$ by partial summation, so h has the form (15.63) with $\mathsf{d} = J_g - J_f$. (i)⇒(ii): Theorem 15.36 gives $\mathsf{d} = 0$ almost everywhere, so $\log f = \log g$ for large σ and $f = g$ by analytic continuation. (ii)⇒(iii): g then satisfies (H0)–(H3), so $g = C_H \zeta$, and $a_1 = 1$ gives $C_H = 1$. (iii)⇒(i) because $h \equiv 0$. □

Corollary 15.38 (The ladder for the pair $(\zeta_\lambda, \tilde\zeta_{\mathbb{Z},\lambda})$)**.** *Let $\lambda \in [0,1]$ and $N \geq 2$. Then $\tilde h_\lambda(\sigma) = O(\sigma N^{-\sigma})$ as $\sigma \to +\infty$ if and only if the Mellin count J_λ^{M} of ζ_λ coincides almost everywhere on $[1,N)$ with the step function $\sum_{n^m \leq x} \tilde c_n(\lambda)/m$; in particular J_λ^{M} then vanishes on $(1,2)$ and is constant between consecutive integers below N. Consequently*

$$\text{(D)} \iff \tilde h_0(\sigma) = O_N\big(\sigma N^{-\sigma}\big) \text{ for every } N \geq 2, \tag{15.68}$$

and at the clock end the first rung already fails, by Theorem 9.11: $\tilde h_1(\sigma) \to \log L = -\log \varpi_{\mathrm{RVM}}$, which is $\neq 0$ exactly when $\varpi_{\mathrm{RVM}} \neq 1$ (Proposition 14.16(ii)).

Proof. Theorem 15.36 with $\mathsf{d} = J_g - J_\lambda^{\mathrm{M}}$, where J_g is the count of $\log \tilde\zeta_{\mathbb{Z},\lambda} = \sum_m \frac{1}{m} \sum_n \tilde c_n(\lambda) n^{-ms}$; the integrability is Theorem 13.5, whose proof applies to every ζ_λ, and Corollary 14.12. (15.68) is Theorem 15.37 with Corollary 14.12 and Theorem 13.7, and the last statement is Proposition 14.16. □

The joining factor therefore measures the mass of dJ_λ^{M} outside the integers, and its rate of decay on the real axis is the first integer below which there is such mass. The Nth rung at $\lambda = 0$ is the statement that J_{Z1}^{M} is constant on $(N-1, N)$; it is (D) restricted to one cell.

15.13. **The bridge theorem: the phase of the joining factor on the critical line.**

Theorem 15.39 (Bridge theorem, phase form)**.** *Let f be meromorphic in $\mathbb{C}$, real on the real axis, with (H0), the functional equation* (12.3)*, and $\lim_{\sigma\to+\infty} f(\sigma) = \varpi_f > 0$. Let $g(s) = \sum_{n\geq 1} a_n n^{-s}$ converge absolutely in $\Re s > 1$, with real a_n and $a_1 = 1$. Suppose that $g = f\,e^h$ in $\Re s > \frac12$ with h holomorphic there and real on $(1,\infty)$, and that*

(B1) *$\Im h$ extends continuously to $\Re s \geq \frac12$ and vanishes on the line $\Re s = \frac12$;*
(B2) *there are $\mathsf{e}_1, \mathsf{e}_2 \geq 0$ with*

$$|h(s)| \leq C\,(1+|s|)^{\mathsf{e}_1}\Big(1 + \big(\sigma - \tfrac12\big)^{-\mathsf{e}_2}\Big) \qquad (\Re s > \tfrac12). \tag{15.69}$$

Then $h \equiv -\log \varpi_f$, $g = \zeta$ and $f = \varpi_f\,\zeta$.

Proof. Reflection. $v := \Im h$ is harmonic in $\Re s > \frac12$, continuous on the closed half-plane and 0 on its boundary. By the reflection principle [1, Ch. 4, §6.5] the function equal to $v(s)$ for $\Re s \geq \frac12$ and to $-v(1-\bar s)$ for $\Re s < \frac12$ is harmonic in $\mathbb{C}$. It is the imaginary part of an entire function, which can be chosen equal to h in $\Re s > \frac12$; in $\Re s < \frac12$ it differs from $\overline{h(1-\bar s)}$ by a real constant, and this constant is 0 because the real part of an entire function is continuous across the line. Thus h is entire and $h(1-\bar s) = \overline{h(s)}$. Since h is real on $(1,\infty)$, $h(\bar s) = \overline{h(s)}$, and therefore $h(1-s) = h(s)$.

Growth. By (15.69) and the symmetry, $|h(s)| \leq C(1+|s|)^{\mathsf{e}_1}(1+|\sigma-\frac12|^{-\mathsf{e}_2})$ for $\Re s \neq \frac12$. The function $\log|h|$ is subharmonic, so for every s

$$\log|h(s)| \leq \frac{1}{2\pi}\int_0^{2\pi} \log\big|h\big(s+e^{i\phi}\big)\big|\,d\phi \leq \log C + \mathsf{e}_1 \log(2+|s|) + \log 2 + \frac{\mathsf{e}_2}{2\pi}\int_0^{2\pi} \log^+ \frac{1}{|\sigma - \frac12 + \cos\phi|}\,d\phi. \tag{15.70}$$

For real y, $\frac{1}{2\pi}\int_0^{2\pi} \log|y+\cos\phi|\,d\phi \geq -\log 2$, so the last integral is at most $2\pi\big(\log 2 + \log(2+|s|)\big)$. Hence $|h(s)| \leq C'(2+|s|)^{\mathsf{e}_1+\mathsf{e}_2}$ in $\mathbb{C}$, and h is a polynomial. On the real axis $h(\sigma) = \log g(\sigma) - \log f(\sigma) \to -\log\varpi_f$ as $\sigma \to +\infty$, so the polynomial is the constant $-\log\varpi_f$.

Hamburger. $g = f/\varpi_f$ satisfies (H0), (H1), and (H2)–(H3) with the second function equal to g; so $g = C_H\zeta$, and $a_1 = 1$ gives $C_H = 1$. □

The symmetry of the joining factor, its continuation to $\Re s \leq \frac12$ and its finite order are not hypotheses of Theorem 15.39: they follow from (B1) and (B2), which are written on the closed half-plane $\Re s \geq \frac12$ where the joining factor of the family is defined. Theorem 15.3 assumes the continuation and the symmetry instead, and nothing like (B2); Theorem 15.39 derives both from (B1)–(B2). Neither set of hypotheses contains the other.

Corollary 15.40 (The bridge theorem for the family). (i) *For $0 < \lambda \le 1$ the joining factor $\tilde h_\lambda$ does not satisfy both* (B1) *and* (B2).
(ii) *(D) holds if and only if $\tilde h_0$ satisfies* (B1) *and* (B2).

Proof. Theorem 15.39 applies with $f = \zeta_\lambda$ (Theorem 11.1, Proposition 14.1), $g = \tilde\zeta_{\mathbb{Z},\lambda}$ (Corollary 14.12) and $h = \tilde h_\lambda$ (Theorem 14.13(iii)). If (B1) and (B2) hold, then $\zeta_\lambda = \varpi_\lambda\zeta$ is an absolutely convergent Dirichlet series in $\Re s > 1$, which is excluded for $0 < \lambda \le 1$ by Proposition 15.1; for $\lambda = 0$ it is (D). Conversely (D) gives $\tilde h_0 \equiv 0$ (Theorem 14.14). □

Define the *phase defect* of the family, when the limit exists,

$$\mathsf{v}_\lambda(t) := \lim_{\sigma\to\frac12^+} \Im\tilde h_\lambda(\sigma+it) = \lim_{\sigma\to\frac12^+}\Big(\Im\log\tilde\zeta_{\mathbb{Z},\lambda}(\sigma+it) - \Im\log\zeta_\lambda(\sigma+it)\Big). \tag{15.71}$$

Proposition 15.41 (The phase of the continuous side). *Let $\lambda \in [0,1]$ and $t > 0$ not an ordinate $\gamma_k(\lambda)$. With the branch of $\log\zeta_\lambda$ that is real on $(1,\infty)$, continued along any path in the open quadrant $\{\Re s > \frac12,\ \Im s > 0\}$ (the continuation is well defined there, since ζ_λ has no zeros or poles in the open quadrant: its zeros lie on the line $\Re s = \frac12$ and its pole $s = 1$ on the real axis, both on the boundary of the open quadrant; the limit below is taken at points $\frac12 + it$ that are not zeros),*

$$\lim_{\sigma\to\frac12^+} \Im\log\zeta_\lambda(\sigma+it) = -\vartheta(t) + \pi\big(N_\lambda(t) - 1\big), \qquad N_\lambda(t) = \#\{k : \gamma_k(\lambda) \le t\}, \tag{15.72}$$

and in the sense of distributions in $t > 0$

$$\lim_{\sigma\to\frac12^+} \Im\mathcal{V}_\lambda(\sigma+it) = -\pi\sum_{k\ge1} \eta_k\,\delta\big(t - \gamma_k(\lambda)\big). \tag{15.73}$$

Consequently $\mathsf{v}_\lambda(t) = \lim_{\sigma\to\frac12^+} \Im\log\tilde\zeta_{\mathbb{Z},\lambda}(\sigma+it) + \vartheta(t) - \pi\big(N_\lambda(t) - 1\big)$.

Proof. $\zeta_\lambda = P_\lambda/G$. On $s = \frac12 + it$, $q(s) = -(\frac14 + t^2)$ and $1 + q(s)/A_{k,\lambda} = (\gamma_k(\lambda)^2 - t^2)/A_{k,\lambda}$ are real, so $P_\lambda(\frac12 + it)$ is real and changes sign at each $\gamma_k(\lambda)$; approaching from $\sigma > \frac12$ the argument of the factor k increases by π as t passes $\gamma_k(\lambda)$, since $\Im(\gamma_k(\lambda)^2 + (\sigma - \frac12 + it)^2) = 2t(\sigma - \frac12) > 0$. Modulo 2π, the argument of $G(\frac12 + it) = -(\frac14 + t^2)\Gamma(\frac14 + \frac{it}{2})\pi^{-1/4-it/2}$ is $\pi + \vartheta(t)$ by (9.29). The multiple of 2π is fixed by continuity from $t = 0^+$: for $\frac12 < \sigma < 1$, $P_\lambda(\sigma) > 0$, because $q(\sigma) \ge -\frac14$ makes every factor $\ge 1 - \frac14/A_{k,\lambda} > 0$, and $G(\sigma) = \sigma(\sigma-1)\pi^{-\sigma/2}\Gamma(\frac\sigma2) < 0$, so $\zeta_\lambda(\sigma) < 0$; the branch continued from $(1,\infty)$ through the upper half-plane around the simple pole $s = 1$ has $\Im\log\zeta_\lambda(\sigma) = -\pi$ there. This gives (15.72). For (15.73) it is preferable not to interchange a boundary limit and ∂_λ. Put $z = \sigma' + it$ and $\gamma = \gamma_k(\lambda)$. The kth term of (11.11) is

$$2\gamma\eta_k\left(\frac{1}{\gamma^2 + z^2} - \frac{1}{\frac14 + \gamma^2}\right).$$

The second summand is real, while

$$\frac{2\gamma}{\gamma^2 + z^2} = -i\left(\frac{1}{z - i\gamma} - \frac{1}{z + i\gamma}\right).$$

The Poisson-kernel boundary limit on $t > 0$ is therefore $-\pi\eta_k\delta(t-\gamma)$; the pole at $-\gamma$ lies outside the positive half-line. Summing against a compactly supported test function is justified by the bounded displacement and the same mean-value/summation-by-parts estimates used for the generator (equivalently by Cauchy–Schwarz with $\sum_k \eta_k^2\gamma_k^{-2} < \infty$ after one integration by parts). This proves (15.73) directly in distributions. □

By Proposition 15.41, $\mathsf{v}_\lambda \equiv 0$ asks two things of the discrete function on the critical line: jumps of π of its argument exactly at the ordinates $\gamma_k(\lambda)$, and the phase $-\vartheta(t)$ between them; equivalently, $e^{i\vartheta(t)}\tilde\zeta_{\mathbb{Z},\lambda}(\frac12 + it)$ is real. At the clock end $\tilde h_1$ is holomorphic in $\Re s > 0$ (Theorem 14.13(iv)), so

$\mathsf{v}_1(t) = \Im\tilde h_1(\frac12 + it)$ is a real-analytic function; it does not vanish identically, by Corollary 15.40(i) and the bound $|\tilde h_1(s)| \le C(1+|s|)$ on $\Re s \ge \frac14$ of the proof of Proposition 15.5.

Proposition 15.42 (The cell masses are sums of a one-ordinate function)**.** *For $n \ge 2$ there is a real function $\mathsf{o}_n(\kappa,\kappa')$ of two real variables, depending only on n and the cell φ_n, such that for all lists $\boldsymbol\kappa, \boldsymbol\kappa' \in \mathscr O$*

$$\tilde c_n(\boldsymbol\kappa) - \tilde c_n(\boldsymbol\kappa') = \sum_{k\ge1} \mathsf{o}_n(\kappa_k,\kappa_k'), \qquad \mathsf{o}_n(\kappa,\kappa') = -2\,\Re\big(W_n(\kappa) - W_n(\kappa')\big) \ \text{ if } n \text{ is not a power of } 2, \tag{15.74}$$

and, with $\mathsf{L}_n(s) := -\log(1-n^{-s})$, for $\Re s > 1$

$$\log\tilde\zeta_{\mathbb Z,\boldsymbol\kappa}(s) - \log\tilde\zeta_{\mathbb Z,\boldsymbol\kappa'}(s) = \sum_{n\ge2}\Big(\sum_{k\ge1}\mathsf{o}_n(\kappa_k,\kappa_k')\Big)\mathsf{L}_n(s), \tag{15.75}$$

$$\log\zeta_{\boldsymbol\kappa}(s) - \log\zeta_{\boldsymbol\kappa'}(s) = \sum_{k\ge1}\log\frac{1+q(s)/(\frac14+\kappa_k^2)}{1+q(s)/(\frac14+{\kappa_k'}^2)}. \tag{15.76}$$

Proof. $\tilde c_n(\boldsymbol\kappa) = \langle d\pi_{\boldsymbol\kappa}, \varphi_n\rangle$, where $d\pi_{\boldsymbol\kappa}$ is obtained from $J_{\boldsymbol\kappa}$ by truncation at 2 and finite Möbius inversion, both linear, and by (13.17) the difference $J_{\boldsymbol\kappa} - J_{\boldsymbol\kappa'} = -2\Re\sum_k\big(\mathrm{Ei}((\frac12+i\kappa_k)\log x) - \mathrm{Ei}((\frac12+i\kappa_k')\log x)\big)$ is a sum over k converging in $\mathcal D'(1,\infty)$. This gives (15.74), with the value (14.22) off the powers of 2. (15.75) is the Euler product with Corollary 14.12, and (15.76) is (16.2). □

By Proposition 15.42, the difference of two joining factors is, term by term in n and in k, a sum over the ordinates of one universal function of one ordinate, the joining factor of a single zero factor:

$$\mathsf{j}(\kappa,\kappa';s) := \sum_{n\ge2}\mathsf{o}_n(\kappa,\kappa')\,\mathsf{L}_n(s) - \log\frac{1+q(s)/(\frac14+\kappa^2)}{1+q(s)/(\frac14+\kappa'^2)}, \qquad \tilde h_{\boldsymbol\kappa} - \tilde h_{\boldsymbol\kappa'} = \sum_{k\ge1}\mathsf{j}(\kappa_k,\kappa_k';s) \ \text{ formally.} \tag{15.77}$$

With $\boldsymbol\kappa' = (\tau_k)$ this writes the phase defect as $\mathsf{v}_{\boldsymbol\kappa} = \mathsf{v}_1 + \sum_k \Im\mathsf{j}(\kappa_k,\tau_k;\frac12+it)$, the defect of the clock plus a transport term.

Heuristic 15.43 (What happens to the phase defect along the path)**.** The following is a computation by stationary phase without remainder terms; it is not proved. For n much larger than κ the cell resolves the oscillation, $\Re W_n(\kappa) \approx n^{-1/2}\cos(\kappa\log n)/\log n$, and for n much smaller than $\kappa/2\pi$ the kernel $W_n(\kappa)$ is negligible because φ is smooth (Proposition 15.20). The transition is at $n \approx \kappa/2\pi$, where the phase is $\kappa\log\frac{\kappa}{2\pi} - \kappa = 2\pi\big(\mathcal N_{\mathrm{RVM}}(\kappa) - \frac78\big)$. This suggests

$$\Im\mathsf{j}\big(\kappa,\cdot\,;\tfrac12+it\big) \approx \text{amplitude}\times\cos\big(2\pi\mathcal N_{\mathrm{RVM}}(\kappa) + \text{phase}(t)\big), \qquad \big|\mathsf{j}(\kappa,\cdot\,;\sigma+it)\big| \approx \kappa^{-1/2-\sigma}, \tag{15.78}$$

which suggests absolute convergence of the sum over the ordinates for $\sigma > \frac12$ and critical behaviour at $\sigma = \frac12$, in agreement with Theorem 14.13(iii), and is at the edge of convergence on the critical line. On the clock, $\mathcal N_{\mathrm{RVM}}(\tau_k) = k - \frac12$ and the phases are aligned, which is why $\tilde h_1$ crosses the line. Along the path each term turns by the angle $(1-\lambda)\eta_k\ell_k$, the rotation of (15.44). At $\lambda = 0$, (B1) is the statement that the rotated sum, with the phases $e^{2\pi i\mathcal N_{\mathrm{RVM}}(\gamma_k)}$, converges on the critical line to minus the defect v_1 of the clock.

15.14. **Transport from the clock on the rungs.** For a real function $\mathsf{f} \in C_c^\infty(0,\infty)$ put

$$\mathsf{c}_{\mathsf f}(t) := \int_0^\infty \mathsf f(u)\cos(tu)\,du, \qquad \mathsf k(u) := \frac12\Big(e^{u/2} - \frac{e^{-u/2}}{e^{2u}-1}\Big), \qquad \mathsf k_{\mathsf f} := \int_0^\infty \mathsf f(u)\,\mathsf k(u)\,du. \tag{15.79}$$

Proposition 15.44 (The rung functional)**.** *For every list $\boldsymbol\kappa \in \mathscr O$ (Section 16),*

$$\mathsf R_{\mathsf f}(\boldsymbol\kappa) := \frac12\int_1^\infty \mathsf f(\log x)\,x^{-1/2}\log x\,dJ_{\boldsymbol\kappa}^{\mathrm M}(x) = \mathsf k_{\mathsf f} - \sum_{k\ge1}\mathsf c_{\mathsf f}(\kappa_k), \tag{15.80}$$

the series converging absolutely. Along the family, with $\mathsf{R}_\mathsf{f}(\lambda) := \mathsf{R}_\mathsf{f}\big((\gamma_k(\lambda))\big)$,

$$\mathsf{R}_\mathsf{f}(\lambda) = \mathsf{R}_\mathsf{f}(1) - \mathsf{S}_\mathsf{f}(\lambda), \qquad \mathsf{S}_\mathsf{f}(\lambda) := \sum_{k\ge1}\Big(\mathsf{c}_\mathsf{f}(\gamma_k(\lambda)) - \mathsf{c}_\mathsf{f}(\tau_k)\Big), \qquad \mathsf{S}_\mathsf{f}(1) = 0. \tag{15.81}$$

If the support of f *contains no point* $\log n$, $n \in \mathbb{N}$, *then*

$$\text{(D)} \implies \mathsf{S}_\mathsf{f}(0) = \mathsf{R}_\mathsf{f}(1), \tag{15.82}$$

and (D) holds if and only if $\mathsf{S}_\mathsf{f}(0) = \mathsf{R}_\mathsf{f}(1)$ *for every such* f *and the resulting integer jumps satisfy the growth hypothesis of Theorem 13.7.*

Proof. Pair (13.16), which holds for every list of $\mathscr{O}$ with the same proof, with $\frac12\mathsf{f}(\log x)x^{-1/2}$ and put $u = \log x$: the term 1 gives $\frac12\int \mathsf{f}(u)e^{u/2}du$, the term $x^{-1}(x^2-1)^{-1}$ gives $\frac12\int \mathsf{f}(u)e^{-u/2}(e^{2u}-1)^{-1}du$, and the sum gives $\sum_k \mathsf{c}_\mathsf{f}(\kappa_k)$; c_f is rapidly decreasing. (15.81) is the difference of two cases of (15.80). By Theorem 13.7, the vanishing $\mathsf{R}_\mathsf{f}(0) = 0$ for every f whose support avoids the points $\log n$ is exactly the constancy part of the cell criterion; (D) follows when the resulting integer jumps also satisfy $\sum_{n\le X}|\mathsf{b}_n| = O_\varepsilon(X^{1+\varepsilon})$. □

The number $\mathsf{R}_\mathsf{f}(1)$ belongs to the clock; $\mathsf{S}_\mathsf{f}(0)$ is the transport from the clock to the ordinates of ζ. (D) is the statement that the transport cancels the defect of the clock on every rung.

Theorem 15.45 (The transport is band-limited in λ)**.** *Let* $\mathsf{f} \in C_c^\infty(0,\log N)$. *The function* $\lambda \mapsto \mathsf{S}_\mathsf{f}(\lambda)$ *extends to an entire function with*

$$|\mathsf{S}_\mathsf{f}(\lambda)| \le C_\mathsf{f}(1 + |\lambda|\log(2+|\lambda|))\, e^{C_\eta \log N\,|\Im\lambda|} \qquad (\lambda \in \mathbb{C}), \tag{15.83}$$

and for every $j \ge 1$

$$\partial_\lambda^j \mathsf{S}_\mathsf{f}(1) = \mathsf{m}_j(\mathsf{f}) := \sum_{k\ge1} \eta_k^j\, \mathsf{c}_\mathsf{f}^{(j)}(\tau_k), \qquad \mathsf{S}_\mathsf{f}(0) = \sum_{j\ge1} \frac{(-1)^j}{j!}\, \mathsf{m}_j(\mathsf{f}). \tag{15.84}$$

Moreover $|\mathsf{m}_j(\mathsf{f})| \le C_\mathsf{f}(1+j)^2(C_\eta \log N)^j$ *(the support of* f *lies in some* $[u_1, \log N]$ *with* $u_1 > 0$, *and* C_f *depends on it).*

Proof. By the Paley–Wiener theorem, $|\mathsf{c}_\mathsf{f}^{(j)}(t)| \le C_{A,j}\, e^{\log N|\Im t|}(1+|t|)^{-A}$ for complex t and every A. With $\gamma_k(\lambda) = \gamma_k + \lambda\eta_k$, $|\Im\gamma_k(\lambda)| \le C_\eta|\Im\lambda|$ and $|\gamma_k(\lambda)| \ge \gamma_k - C_\eta|\lambda|$. The ordinates with $\gamma_k \le 2C_\eta|\lambda|$ are $O(|\lambda|\log(2+|\lambda|))$ in number and each contributes at most $C\,e^{C_\eta\log N|\Im\lambda|}$; for the others $|\mathsf{c}_\mathsf{f}(\gamma_k(\lambda))| \le C\,e^{C_\eta\log N|\Im\lambda|}(\gamma_k/2)^{-2}$. This gives (15.83), the termwise differentiation and the convergence of the Taylor series at $\lambda = 1$ evaluated at $\lambda = 0$. Finally $\mathsf{c}_\mathsf{f}^{(j)}(t) = \int \mathsf{f}(u)u^j\cos(tu + \frac{j\pi}{2})du$ and two integrations by parts give $(1+t^2)|\mathsf{c}_\mathsf{f}^{(j)}(t)| \le C_\mathsf{f}(1+j)^2(\log N)^j$. □

The scaled lists $(\tau_k - c'\eta_k)$ are the members $\lambda = 1 - c'$ of the family, so the scale test of Remark 15.23 is the behaviour of the entire function S_f on the real axis. In (15.84) the nodes τ_k and the weights $\mathsf{c}_\mathsf{f}^{(j)}(\tau_k)$ belong to the clock, and the powers η_k^j do not; on the first rung, $N = 2$, the term j is at most $C_\mathsf{f}(1+j)^2(C_\eta\log 2)^j/j!$.

Proposition 15.46 (The defect of the clock on a rung)**.** *Let* $\mathsf{t}_0 \in (2\pi, \tau_0)$ *be the point with* $\mathcal{N}_{\mathrm{RVM}}(\mathsf{t}_0) = 0$, *and let* $\mathsf{f} \in C_c^\infty(0,\infty)$. *Then*

$$\sum_{k\ge1}\mathsf{c}_\mathsf{f}(\tau_k) = \mathsf{I}_0 + 2\sum_{m\ge1}(-1)^m\mathsf{I}_m, \qquad \mathsf{I}_m := \int_{\mathsf{t}_0}^\infty \mathsf{c}_\mathsf{f}(t)\cos(2\pi m\mathcal{N}_{\mathrm{RVM}}(t))\mathcal{N}'_{\mathrm{RVM}}(t)\,dt = O_\mathsf{f}(m^{-2}), \tag{15.85}$$

$$\mathsf{I}_0 = -\int_0^\infty \frac{\mathsf{f}(u)}{2\pi u}\Big(\sin(u\mathsf{t}_0)\log\frac{\mathsf{t}_0}{2\pi} + \frac{\pi}{2} - \int_0^{u\mathsf{t}_0}\frac{\sin y}{y}\,dy\Big)du, \tag{15.86}$$

and therefore $\mathsf{R}_\mathsf{f}(1) = (\mathsf{k}_\mathsf{f} - \mathsf{I}_0) - 2\sum_{m\ge1}(-1)^m\mathsf{I}_m$.

Proof. The function $g(v) := \mathsf{c}_\mathsf{f}(\mathcal{T}(v))$, $v \geq 0$, with $\mathcal{T}$ the inverse of the clock on $[\mathfrak{t}_0, \infty)$, is continuous on $[0, \infty)$, C^∞ on $(0, \infty)$ with $g'(0^+) = \mathsf{c}'_\mathsf{f}(\mathfrak{t}_0)\mathcal{T}'(0)$ finite, and rapidly decreasing with its derivatives, and $\tau_k = \mathcal{T}(k - \frac{1}{2})$. Poisson's summation formula for the even extension of g (continuous, in L^1, with a corner at 0 in general, since $g'(0^+) \neq 0$ unless $\mathsf{c}'_\mathsf{f}(\mathfrak{t}_0) = 0$) at the half-integers gives $2\sum_{k\geq 1} g(k - \frac{1}{2}) = \sum_{m\in\mathbb{Z}} (-1)^m 2\int_0^\infty g(v)\cos(2\pi m v)\, dv$, the Fourier coefficients being $O(m^{-2})$ by two integrations by parts, $\int_0^\infty g(v)\cos(2\pi m v)\, dv = -\frac{g'(0^+)}{(2\pi m)^2} - \frac{1}{(2\pi m)^2}\int_0^\infty g''(v)\cos(2\pi m v)\, dv$, the boundary term coming from the corner (so the decay cannot be improved to $O(m^{-3})$ in general); the substitution $v = \mathcal{N}_{\mathrm{RVM}}(t)$ gives (15.85). For $m = 0$, $\mathcal{N}'_{\mathrm{RVM}}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$; integrating by parts with the primitive $\int \mathsf{f}(u)\sin(tu)u^{-1}du$ of c_f and exchanging the integrals gives (15.86). □

In I_m the phases $2\pi m\mathcal{N}_{\mathrm{RVM}}(t) \mp tu$ have the derivative $m\log\frac{t}{2\pi} \mp u$, which vanishes at $t = 2\pi e^{u/m}$: the defect of the clock at the point $x = e^u$ is produced at the heights $2\pi x^{1/m}$, which is the origin of the terms $2\cos(2\pi y)$ and $y^{-1/4}\cos(4\pi\sqrt{y} + \frac{\pi}{4})$ of (15.34). On the rung $(n, n+1)$ the height for $m = 1$ lies in $(2\pi n, 2\pi(n+1))$. On the first rung it lies in $(2\pi, 4\pi)$, below τ_1 and on both sides of $\mathfrak{t}_0$: the defect of the clock on $(1, 2)$ is an effect of the start of the clock. Experiments 19.14 and 19.15 evaluate (15.81)–(15.85) for one function f.

15.15. **The multiscale Jacobian of the midpoint clock.** The rung system of Subsection 16.2 asks for a list and free masses simultaneously. Before attempting nonlinear continuation from the clock, one must understand the linearization with respect to the ordinates. The point is genuinely multiscale: the gaps of the midpoint clock tend to zero, so no fixed u-window can carry a uniform inverse, while a window of length comparable with $\log T$ is exactly large enough to resolve the ordinates up to height T.

For $T > \tau_2$ put

$$K(T) := \max\{k : \tau_k \leq T\}, \qquad \Delta_T := \min_{1\leq k<K(T)} (\tau_{k+1} - \tau_k),$$

and define the finite ordinate block of the linearized rung map by

$$(\mathcal{J}_T\mathbf{v})(u) := -u\sum_{\tau_k\leq T} v_k \sin(\tau_k u). \tag{15.87}$$

No genuine zero ordinate and no individual displacement $\eta_k = \tau_k - \gamma_k$ enters this definition.

Theorem 15.47 (Multiscale Jacobian of the midpoint clock)**.** *The following statements hold.*

(i) *For every $T > \tau_2$,*

$$\Delta_T \geq \frac{2\pi}{\log(T/2\pi)}. \tag{15.88}$$

(ii) *Fix $a > 0$ and $\varkappa_{\mathrm{win}} > 0$, and put*

$$L_T := (1 + \varkappa_{\mathrm{win}})\log(T/2\pi), \qquad I_T = [a, a + L_T].$$

There is $c_{a,\varkappa_{\mathrm{win}}} > 0$, independent of T, such that for all finitely supported $\mathbf{v} = (v_k)$ on $\{k : \tau_k \leq T\}$,

$$\|\mathcal{J}_T\mathbf{v}\|^2_{L^2(I_T)} \geq c_{a,\varkappa_{\mathrm{win}}} L_T \sum_{\tau_k\leq T} |v_k|^2. \tag{15.89}$$

(iii) *No analogue of* (15.89) *with a positive T-independent lower constant can hold on a fixed interval $(0, U)$. Indeed, for the unit vector $\mathbf{v}^{(k)} = (e_k - e_{k+1})/\sqrt{2}$,*

$$\|\mathcal{J}\mathbf{v}^{(k)}\|_{L^2(0,U)} \leq \frac{U^{5/2}}{\sqrt{10}}(\tau_{k+1} - \tau_k) \longrightarrow 0. \tag{15.90}$$

(iv) *For every finite $U > 0$,*

$$\overline{\operatorname{span}}\{u\sin(\tau_k u) : k \geq 1\} = L^2(0, U).$$

Thus the infinite linearized clock has dense range on every finite window, despite the absence of a bounded inverse on a fixed window.

(v) *Let $\mathsf{f} \in C_c^\infty(0, U)$ and $\mathsf{c}_\mathsf{f}(t) = \int_0^\infty \mathsf{f}(u)\cos(tu)\,du$. For every list $\boldsymbol{\kappa}$ satisfying $|\kappa_k - \tau_k| \leq C_\eta$ and every $A > 1$,*

$$\sum_{\tau_k > T} |\mathsf{c}_\mathsf{f}(\kappa_k) - \mathsf{c}_\mathsf{f}(\tau_k)| \ll_{A,\mathsf{f},C_\eta} T^{1-A}\log T. \tag{15.91}$$

Proof. The midpoint equation is $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$ with

$$\mathcal{N}'_{\mathrm{RVM}}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}.$$

For $\tau_{k+1} \leq T$, the mean-value theorem gives

$$1 = \mathcal{N}'_{\mathrm{RVM}}(\xi_k)(\tau_{k+1} - \tau_k), \qquad \tau_k < \xi_k < \tau_{k+1} \leq T,$$

which proves (15.88).

For (ii), apply Ingham's inequality for non-harmonic exponentials [22] to the separated frequency set $\{\pm\tau_k : \tau_k \leq T\}$. By (15.88),

$$L_T\Delta_T \geq 2\pi(1 + \varkappa_{\mathrm{win}}),$$

so after scaling the frequencies by Δ_T the interval lies a fixed positive distance beyond Ingham's critical length. The lower Ingham constant is therefore bounded below by a positive constant depending only on $\varkappa_{\mathrm{win}}$; rescaling gives a bound of size L_T for the sine sum. Since $u \geq a$ on I_T, multiplication by u gives (15.89).

For (iii), $|\sin(\tau_{k+1}u) - \sin(\tau_k u)| \leq (\tau_{k+1} - \tau_k)u$, whence

$$\|\mathcal{J}\mathbf{v}^{(k)}\|^2_{L^2(0,U)} \leq \frac{(\tau_{k+1} - \tau_k)^2}{2}\int_0^U u^4\,du = \frac{(\tau_{k+1} - \tau_k)^2 U^5}{10}.$$

The same mean-value formula used above shows $\tau_{k+1} - \tau_k \to 0$.

For (iv), suppose $F \in L^2(0, U)$ is orthogonal to every $u\sin(\tau_k u)$. Put $H(u) = uF(u)$ and extend H oddly to $(-U, U)$. Its Fourier transform

$$\widehat{H}(z) = \int_{-U}^{U} H(u)e^{-izu}\,du$$

is entire of exponential type at most U and vanishes at every $\pm\tau_k$. A nonzero entire function of finite exponential type has $O(r)$ zeros in $|z| \leq r$ by Jensen's formula, whereas the midpoint clock has $2\mathcal{N}_{\mathrm{RVM}}(r) + O(1) \asymp r\log r$ points $\pm\tau_k$ in that disk. Hence $\widehat{H} \equiv 0$, so $H = 0$ and therefore $F = 0$ almost everywhere.

For (v), the Paley–Wiener decay of the cosine transform gives $|\mathsf{c}'_\mathsf{f}(t)| \ll_{A,\mathsf{f}} (1 + |t|)^{-A}$ for every A. The mean-value theorem and $|\kappa_k - \tau_k| \leq C_\eta$ give

$$|\mathsf{c}_\mathsf{f}(\kappa_k) - \mathsf{c}_\mathsf{f}(\tau_k)| \ll_{A,\mathsf{f},C_\eta} \tau_k^{-A}$$

for large k. Partial summation with $\mathcal{N}_{\mathrm{RVM}}(t) = O(t\log t)$ yields $\sum_{\tau_k > T}\tau_k^{-A} = O(T^{1-A}\log T)$. □

The theorem identifies the correct scale but does not prove existence of a solution of the rung system. In particular, a fixed-window Banach-space implicit-function argument cannot work because of (15.90). A quantitative continuation must grow the resolved window together with the spectral height.

One precise residual formulation is the following. Fix $r > \frac{1}{2}$, so point masses belong to $H^{-r}(0,U)$, and put

$$\mathsf{D}_{\mathrm{RVM}}(u) := \mathsf{k}(u) - \sum_{k\ge 1} \cos(\tau_k u)$$

in $\mathcal{D}'(0,\infty)$. For $T, U > 0$ define

$$\varepsilon_r(T,U) := \inf_{h,b} \Big\| \mathsf{D}_{\mathrm{RVM}} + u \sum_{\tau_k \le T} h_k \sin(\tau_k u) - \sum_{\log n \le U} b_n \delta(u - \log n) \Big\|_{H^{-r}(0,U)} . \tag{15.92}$$

Theorem 15.47 supplies the finite-height separation, dense-range and tail ingredients. What is not proved is a bound for (15.92), together with control of the minimizing h strong enough to dominate the quadratic remainder in

$$\cos((\tau_k + h_k)u) = \cos(\tau_k u) - h_k u \sin(\tau_k u) + O(h_k^2 u^2),$$

uniformly along a nested sequence with $U \asymp \log T$. Such an estimate would turn the finite-section continuation observed numerically into an existence proof for the infinite rung system. By Theorem 16.6, any resulting solution would then be the zeta ordinates with the von Mangoldt masses. The individual displacements η_k are not data in this formulation.

16. The space of ordinate lists and the existence form of the final step

The constructions of Sections 11–15 use two properties of the ordinates (Remark 14.18). This section applies them to every list with these two properties. The final step then becomes an existence statement in a compact space, and the list of ζ is not part of its formulation.

16.1. **Lists, their zeta functions and their atomizations.** Let $\mathscr{O}$ be the set of sequences $\boldsymbol{\kappa} = (\kappa_k)_{k\ge 1}$ of real numbers with

$$\tau_0^+ \le \kappa_1 \le \kappa_2 \le \cdots, \qquad |\kappa_k - \tau_k| \le C_\eta, \qquad \Big| \int_{\tau_0}^{X} \big(N_{\boldsymbol{\kappa}}(\tau) - \mathcal{N}_{\mathrm{RVM}}(\tau)\big) d\tau \Big| \le C \log X \quad (X \ge \tau_0), \tag{16.1}$$

where τ_0^+ is a fixed number with $\tau_0 < \tau_0^+ < \min(\gamma_1, \tau_1)$, $N_{\boldsymbol{\kappa}}(\tau) = \#\{k : \kappa_k \le \tau\}$, C_η is the constant of Lemma 2.2 and C the constant of Proposition 14.1. The clock (τ_k), the ordinates (γ_k) of ζ and every list $(\gamma_k(\lambda))$, $0 \le \lambda \le 1$, belong to $\mathscr{O}$. For $\boldsymbol{\kappa} \in \mathscr{O}$ (Section 16) define, exactly as for the family,

$$\zeta_{\boldsymbol{\kappa}}(s) := \frac{1}{G(s)} \prod_{k\ge 1} \Big(1 + \frac{q(s)}{\frac{1}{4} + \kappa_k^2}\Big), \qquad \tilde{c}_n(\boldsymbol{\kappa}), \quad \tilde{a}_N(\boldsymbol{\kappa}), \quad k_N(\boldsymbol{\kappa}), \qquad \tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}} = \zeta_{\boldsymbol{\kappa}}\, e^{\tilde{h}_{\boldsymbol{\kappa}}}, \qquad \mathsf{V}_\sigma(\boldsymbol{\kappa}), \tag{16.2}$$

with the cell masses of Definition 14.4, the joining factor of Theorem 14.7 and the function (15.7). The results of the preceding sections that are used below for lists are transferred to $\mathscr{O}$ by the following theorem, which records, result by result, which part of (16.1) each proof uses and which constants are uniform.

Theorem 16.1 (Transfer to $\mathscr{O}$)**.** *Let $\boldsymbol{\kappa} \in \mathscr{O}$. The following statements, proved above for the lists $(\gamma_k(\lambda))_{k\ge 1}$, hold for $\boldsymbol{\kappa}$ with the same proofs, and their constants depend only on τ_0^+, C_η and C in* (16.1) *and on the parameters of the statement (σ, ε, n, N), never on the list: $C = C(\tau_0^+, C_\eta, C; \sigma, \varepsilon, \dots)$, hence they are uniform on $\mathscr{O}$. This uniformity is what Proposition 16.3 and the compactness arguments of Section 18 use.*

(i) *Theorem 11.1: $\zeta_{\boldsymbol{\kappa}}$ has order one, satisfies the functional equation* (12.3)*, and its nontrivial zeros are exactly $\frac{1}{2} \pm i\kappa_k$, with multiplicity. Input: the tube (convergence of the product and $\kappa_k \asymp \tau_k$), the ordering, and $\kappa_1 \ge \tau_0^+$.*

(ii) *Theorem 13.1 in its list form, with $\boldsymbol{\kappa}$ in place of $\boldsymbol{\gamma}$: the clock decomposition of $\zeta_{\boldsymbol{\kappa}}$, the value $\varpi_{\boldsymbol{\kappa}}$ at $+\infty$ as an ordered product, and the bounds of Proposition 14.1. Input: the tube and the third condition of* (16.1)*, which replaces $\int_{\tau_0}^{X}(N_\zeta - n_{\mathrm{RVM}}) = O(\log X)$ in every summation by parts against the count difference.*

(iii) *Theorems 13.5 and 13.6: the prime-power count $J^{\mathrm{M}}_{\boldsymbol{\kappa}}$ as a Mellin transform, its L^2_{loc} bound, and the distributional explicit formula. Input: (i), (ii) and the Perron argument, in which the ordinates enter only through $\log\zeta_{\boldsymbol{\kappa}}$ on vertical lines, controlled by (ii).*

(iv) *Proposition 15.42: the cell masses $\tilde{c}_n(\boldsymbol{\kappa})$ as sums of a one-ordinate function. Input: (iii) and dominated convergence, with the domination supplied by the tube.*

(v) *Lemma 14.10 with κ_k in place of $\gamma_k(\lambda)$: the tube gives $N_{\boldsymbol{\kappa}}(t+1) - N_{\boldsymbol{\kappa}}(t) \le n_{\mathrm{RVM}}(t+1+C_\eta) - n_{\mathrm{RVM}}(t-C_\eta) \ll \log(t+2)$, with an implied constant depending on C_η only.*

(vi) *Theorem 14.11 and Corollary 14.12: $\sum_{X<n\le 2X}|\tilde{c}_n(\boldsymbol{\kappa})|^2 \le C_\varepsilon X^{1+\varepsilon}$, the absolute convergence of $\tilde{\zeta}^{\pm1}_{\mathbb{Z},\boldsymbol{\kappa}}$ in $\Re s > 1$, and the exceptional-set bound. Input: (v) for the cells away from the powers of 2; for the cells $n = 2^j$ the term* (14.23)*, whose bound needs $\|J_{\boldsymbol{\kappa}}\|_{L^2(2,3)}$ bounded uniformly on $\mathscr{O}$, which is (iii) together with the uniform bound of (ii); and the uniformity in n of the bumps φ_n. The constant C_ε depends on ε, τ_0^+, C_η and C only. The statement on $\partial_\lambda\tilde{c}_n$ has no analogue for a single list and is not transferred.*

(vii) *Theorem 14.13(i)–(iii) for $\tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}}$. Input: (vi) and (iii).*

Proof. Each item names the only places where the cited proofs use the ordinates. They are four: the tube $|\kappa_k - \tau_k| \le C_\eta$, which gives $\kappa_k \asymp \tau_k$, $\sum_k \kappa_k^{-2} < \infty$, the local count of (v), and the domination of (iv); the ordering, used in the monotone rearrangements and in the identification of the zeros of $\zeta_{\boldsymbol{\kappa}}$ with the ordinates; the threshold $\kappa_1 \ge \tau_0^+ > \tau_0$, used wherever a branch of $\log(1+q(s)/(\frac14+\kappa^2))$ is fixed on $|s-\frac12| < \tau_0$; and the third condition, used in every summation by parts over the ordered ordinates in place of Proposition 14.1. None of the cited proofs uses that the list is a zero set of ζ or a convex combination with the clock, except in the statements about λ-derivatives, which are excluded in (vi). Reading the proofs with these four inputs in place of their sources gives the items, and the constants are those of the inputs. □

Theorem 16.2 (Rigidity of the coincidence)**.** *The set*

$$\mathscr{C} := \{\boldsymbol{\kappa} \in \mathscr{O} :\ \tilde{h}_{\boldsymbol{\kappa}} \text{ is constant}\} \tag{16.3}$$

of the lists whose continuous and atomized zeta functions coincide up to a constant factor has at most one element. If $\boldsymbol{\kappa} \in \mathscr{C}$, then $\zeta_{\boldsymbol{\kappa}} = \tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}} = \zeta$, $\boldsymbol{\kappa}$ is the list of the ordinates of ζ, and (D) holds. Conversely, (D) implies $\mathscr{C} = \{(\gamma_k)\}$.

Proof. If $\tilde{h}_{\boldsymbol{\kappa}} \equiv c$, then $\zeta_{\boldsymbol{\kappa}} = e^{-c}\tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$ (Corollary 14.12), $(s-1)\zeta_{\boldsymbol{\kappa}}$ is entire of order one and $\zeta_{\boldsymbol{\kappa}}$ satisfies (12.3) (Theorem 11.1). Hamburger's theorem gives $\zeta_{\boldsymbol{\kappa}} = C_H\zeta$, and $C_H = 1$ because both functions have residue 1 at $s = 1$. Hence the nontrivial zeros of ζ with positive ordinate are the points $\frac12 + i\kappa_k$, with multiplicity; this determines $\boldsymbol{\kappa}$ as the list (γ_k), and $\zeta_{Z1} = \zeta_{\boldsymbol{\kappa}} = \zeta$. Then $e^{-c}\tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}} = \zeta$ and the first coefficients give $e^{-c} = 1$; c is real, because $\tilde{h}_{\boldsymbol{\kappa}}$ is real on the real axis, so $c = 0$. Conversely, under (D), $\tilde{h}_0 \equiv 0$ (Theorem 14.14), so $(\gamma_k) \in \mathscr{C}$. □

Proposition 16.3 (Compactness and continuity)**.** *The space $\mathscr{O}$ and the functions* (16.2) *have the following properties.*

(i) *$\mathscr{O}$ is compact in the product topology of $\prod_k[\tau_k - C_\eta, \tau_k + C_\eta]$.*

(ii) *For every $n \ge 2$, $N \ge 1$ and $\sigma > 1$, the functions $\boldsymbol{\kappa} \mapsto \tilde{c}_n(\boldsymbol{\kappa})$, $\tilde{a}_N(\boldsymbol{\kappa})$, $k_N(\boldsymbol{\kappa})$ and $\mathsf{V}_\sigma(\boldsymbol{\kappa})$ are continuous on $\mathscr{O}$.*

(iii) *For every real $\sigma > 1$, $\boldsymbol{\kappa} \mapsto \log\zeta_{\boldsymbol{\kappa}}(\sigma) = -\log G(\sigma) + \sum_k \log\frac{(\sigma-\frac12)^2+\kappa_k^2}{\frac14+\kappa_k^2}$ is a sum of strictly convex functions of the single coordinates κ_k.*

Proof. (i) The product $\mathcal{K} := \prod_k [\tau_k - C_\eta, \tau_k + C_\eta]$ is compact by Tychonoff's theorem and metrizable, being a countable product of compact intervals. Each condition in (16.1) defines a closed subset of $\mathcal{K}$: $\{\kappa_1 \geq \tau_0^+\}$ is closed; the ordering is $\bigcap_k \{\kappa_k \leq \kappa_{k+1}\}$, an intersection of closed sets; and for fixed X the function $F_X(\boldsymbol{\kappa}) := \sum_k (X - \kappa_k)^+ - \int_{\tau_0}^X \mathcal{N}_{\mathrm{RVM}}$ equals $\int_{\tau_0}^X (N_{\boldsymbol{\kappa}} - \mathcal{N}_{\mathrm{RVM}})$, depends continuously on the finitely many coordinates with $\tau_k \leq X + C_\eta$, and the third condition is $\bigcap_{X \geq \tau_0} \{|F_X| \leq C \log X\}$, again an intersection of closed sets. So $\mathscr{O}$ is closed in $\mathcal{K}$, hence compact. (ii) For n not a power of 2, $\tilde{c}_n(\boldsymbol{\kappa}) = \mathfrak{m}_n - 2\Re \sum_k W_n(\kappa_k)$ by (14.22), and the series converges uniformly on $\mathscr{O}$ because W_n is rapidly decreasing and $|\kappa_k - \tau_k| \leq C_\eta$. For $n = 2^j$ the term j is $-\frac{\mu(j)}{j} \int_2^3 J_{\boldsymbol{\kappa}}(y) \frac{d}{dy} \varphi_n(y^j)\, dy$ (proof of Theorem 14.11), and $\boldsymbol{\kappa} \mapsto J_{\boldsymbol{\kappa}} \in L^2(2,3)$ is continuous, by the following three steps. (1) Fix $\sigma > 1$ and put $\mathsf{F}_{\boldsymbol{\kappa}}(t) := \log \zeta_{\boldsymbol{\kappa}}(\sigma + it)/(\sigma + it)$. For fixed t, $\log \zeta_{\boldsymbol{\kappa}}(\sigma + it) = -\log G(\sigma + it) + \sum_k \mathrm{Log}(1 + q(\sigma + it)/(\frac{1}{4} + \kappa_k^2))$ is a series converging uniformly on $\mathscr{O}$ (the k-th term is $O(|q|/\tau_k^2)$ on the tube), hence continuous in the product topology; so $\mathsf{F}_{\boldsymbol{\kappa}}(t)$ is continuous in $\boldsymbol{\kappa}$ for every t. (2) By Theorem 16.1(ii) (the bound of Proposition 14.1 on $\mathscr{O}$), $|\log \zeta_{\boldsymbol{\kappa}}(\sigma + it) - \log \frac{s}{s-1}| \leq C_\sigma (1 + \log(2 + |t|))$ for all $\boldsymbol{\kappa} \in \mathscr{O}$, so that

$$|\mathsf{F}_{\boldsymbol{\kappa}}(t)| \leq \mathsf{H}(t) := \frac{C'_\sigma\,(1 + \log(2 + |t|))}{1 + |t|}, \qquad \mathsf{H} \in L^2(\mathbb{R}), \tag{16.4}$$

a pointwise majorant independent of $\boldsymbol{\kappa}$; if $\boldsymbol{\kappa}^{(m)} \to \boldsymbol{\kappa}$ in $\mathscr{O}$, then $|\mathsf{F}_{\boldsymbol{\kappa}^{(m)}} - \mathsf{F}_{\boldsymbol{\kappa}}|^2 \leq 4\mathsf{H}^2$ and dominated convergence gives $\mathsf{F}_{\boldsymbol{\kappa}^{(m)}} \to \mathsf{F}_{\boldsymbol{\kappa}}$ in $L^2(\mathbb{R})$; by Plancherel and Theorem 13.5, $J^{\mathrm{M}}_{\boldsymbol{\kappa}^{(m)}}(e^u) e^{-\sigma u} \to J^{\mathrm{M}}_{\boldsymbol{\kappa}}(e^u) e^{-\sigma u}$ in $L^2(\mathbb{R}, du)$, hence $J^{\mathrm{M}}_{\boldsymbol{\kappa}^{(m)}} \to J^{\mathrm{M}}_{\boldsymbol{\kappa}}$ in $L^2(2,3)$. (3) The constant $C_{\boldsymbol{\kappa}} := J^{\mathrm{M}}_{\boldsymbol{\kappa}} - J_{\boldsymbol{\kappa}}$ equals $\langle J^{\mathrm{M}}_{\boldsymbol{\kappa}}, \phi \rangle - \langle J_{\boldsymbol{\kappa}}, \phi \rangle$ for a fixed $\phi \in C_c^\infty(2,3)$ with $\int \phi = 1$: the first pairing is continuous by (2); the second is the sum over k of $\int \big(\mathrm{Ei}(\rho_k \log x) + \mathrm{Ei}(\bar{\rho}_k \log x)\big) \phi(x)\, dx$ plus terms independent of the list, a series whose tails are bounded uniformly on $\mathscr{O}$ by (14.13) (whose constant depends on C_η only) and whose terms are continuous in κ_k, hence a continuous function on $\mathscr{O}$. Therefore $J_{\boldsymbol{\kappa}} = J^{\mathrm{M}}_{\boldsymbol{\kappa}} - C_{\boldsymbol{\kappa}}$ is continuous into $L^2(2,3)$, and the cells $n = 2^j$ are continuous. $\tilde{a}_N$ and k_N are polynomials in finitely many cell masses. The sums $\sum_n |k_n(\boldsymbol{\kappa})| n^{-\sigma'}$, $\sigma' > 1$, are bounded uniformly on $\mathscr{O}$ (Theorem 14.11), so the tails of V_σ are uniformly small, as in the proof of Theorem 15.4(iii). (iii) is the computation in the proof of Proposition 15.2: $\partial_\kappa^2 \log \frac{a^2 + \kappa^2}{1/4 + \kappa^2} = \mathcal{C}(a, \kappa) - \mathcal{C}(\frac{1}{2}, \kappa) > 0$ for $a > \frac{1}{2}$, $\kappa > \frac{1}{2}$. □

Corollary 16.4 (Existence form of the final step)**.** *For every $\sigma > 1$ the minimum $\min_{\mathscr{O}} \mathsf{V}_\sigma$ is attained, and the following are equivalent:* (i) *(D);* (ii) *$\mathscr{C} \neq \varnothing$;* (iii) *there is $\boldsymbol{\kappa} \in \mathscr{O}$ (Section 16) with $\tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}} = \zeta$;* (iv) *$\min_{\mathscr{O}} \mathsf{V}_\sigma = 0$.*

Proof. A continuous function on a compact space attains its minimum (Proposition 16.3). (i)⇔(ii) is Theorem 16.2, and (ii)⇒(iii) is part of it. (iii)⇔(iv) because $\mathsf{V}_\sigma(\boldsymbol{\kappa}) = \sum_{n \geq 2} |k_n(\boldsymbol{\kappa})|^2 n^{-2\sigma} \geq 0$, where $k_n(\boldsymbol{\kappa})$ are the coefficients of $\tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}}/\zeta$ (16.15) and $k_1 = 1$, vanishes exactly when $k_n(\boldsymbol{\kappa}) = 0$ for all $n \geq 2$, that is, when $\tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}} = \zeta$. (iii)⇒(i): $\tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}} = \zeta$ has no zeros in $\Re s > \frac{1}{2}$ (Theorem 14.13(iii)), so, with the functional equation, the nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$: the Riemann hypothesis holds, and Theorem 12.2 gives (D). This is not circular: Theorem 12.2 rests on the Hadamard product and Hamburger's theorem, not on the present corollary. □

In this form the final step does not ask to identify ζ_{Z1} with ζ: it asks for the existence of one list in $\mathscr{O}$ whose continuous and atomized zeta functions coincide, or whose atomized zeta function is ζ; Theorem 16.2 then identifies the list. The path $\lambda \mapsto (\gamma_k(\lambda))$ is one segment in $\mathscr{O}$, from the clock to the list of ζ, and along it V_σ is the function of Theorem 15.4.

Remark 16.5 (What an existence proof uses). Compactness, continuity and convexity do not decide whether $\min_{\mathscr{O}} \mathsf{V}_\sigma$ is 0: they do not involve the form of the functional equation. The existence statement uses the functional equation with the factor $\Gamma(\frac{s}{2}) \pi^{-s/2}$, for which a function with both an absolutely convergent Dirichlet series and the functional equation exists, namely ζ. The first

variation of the defect is explicit: by (14.22), for n not a power of 2,

$$\frac{\partial \tilde{c}_n(\boldsymbol{\kappa})}{\partial \kappa_k} = 2\,\Im V_n(\kappa_k), \tag{16.5}$$

so the vanishing of all coordinate derivatives of V_σ at a list is described by the functions V_n of (14.21) alone. Since $\mathscr{O}$ has empty interior in the product topology and the displayed derivatives are coordinate derivatives rather than a proved Fréchet gradient on a fixed Banach space, boundary minimizers must be treated by one-sided/KKT conditions.

16.2. **The rung system with free masses.** With the function k of (15.79), consider, for a list $\boldsymbol{\kappa} \in \mathscr{O}$ (Section 16) and real numbers a_n, $n \geq 2$, on which nothing is assumed, the system

$$\sum_{k\geq 1} \cos(\kappa_k u) + \sum_{n\geq 2} \mathsf{a}_n\, \delta(u - \log n) = \mathsf{k}(u) \qquad \text{in } \mathcal{D}'(0,\infty). \tag{16.6}$$

In the variable $x = e^u$ it states that the Mellin count of $\zeta_{\boldsymbol{\kappa}}$ has no mass outside the integers, the masses at the integers being unknown.

Theorem 16.6 (Existence form with free masses)**.** *(D) holds if and only if* (16.6) *has a solution* $(\boldsymbol{\kappa}, (\mathsf{a}_n))$ *with* $\boldsymbol{\kappa} \in \mathscr{O}$ *(Section 16). The solution is then unique:* $\boldsymbol{\kappa} = (\gamma_k)$*, and*

$$\mathsf{a}_n = \frac{\log p}{2\sqrt{n}} \quad \textit{if } n = p^m \textit{ is a prime power}, \qquad \mathsf{a}_n = 0 \quad \textit{otherwise.} \tag{16.7}$$

Proof. By (13.16) for the list $\boldsymbol{\kappa}$, with $u = \log x$, (16.6) is the statement

$$(\log x)\, dJ^{\mathrm{M}}_{\boldsymbol{\kappa}}(x) = \sum_{n\geq 2} 2\sqrt{n}\, \mathsf{a}_n\, \delta_n \qquad \text{in } \mathcal{D}'(1,\infty). \tag{16.8}$$

If (D) holds, $J^{\mathrm{M}}_{Z1} = \sum_{p^m \leq x} \frac{1}{m}$ (Theorem 13.7), which is (16.8) for the list (γ_k) with the masses (16.7). Conversely let (16.8) hold. Then $J^{\mathrm{M}}_{\boldsymbol{\kappa}}$ is constant on $(1,2)$ and between consecutive integers; the truncation at 2 and the finite Möbius inversion keep the support in the integers, so every cell defect vanishes and, by (14.11) and the proof of Theorem 14.7,

$$\zeta_{\boldsymbol{\kappa}}(s) = \exp\bigl(\nu_6 + \nu_7\, 2^{-s}\bigr)\, \tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}}(s) \qquad (\Re s > 1) \tag{16.9}$$

with real constants ν_6, ν_7. The two factors on the right are the exponential $\exp(\nu_6 + \nu_7 2^{-s}) = e^{\nu_6} \sum_{r\geq 0} \frac{\nu_7^r}{r!} 2^{-rs}$, an absolutely convergent Dirichlet series on all of $\mathbb{C}$ supported on the powers of 2, and $\tilde{\zeta}_{\mathbb{Z},\boldsymbol{\kappa}}$, an absolutely convergent Dirichlet series in $\Re s > 1$ by Corollary 14.12, which Theorem 16.1(vi) extends to every list of $\mathscr{O}$; the product of two absolutely convergent Dirichlet series is one. (Why the support stays in the integers: $J^{\mathrm{M}}_{\boldsymbol{\kappa}}$ constant between consecutive integers means $dJ^{\mathrm{M}}_{\boldsymbol{\kappa}}$ is a combination of δ_n, $n \geq 2$; the truncation J^* at 2 removes the cell $(1,2)$, on which $J^{\mathrm{M}}_{\boldsymbol{\kappa}}$ is a constant, and the finite Möbius inversion (6.7) maps $\sum_n \mathsf{w}_n \delta_n$ to $\sum_m \frac{\mu(m)}{m} \sum_n \mathsf{w}_n \delta_{n^{1/m}}$, whose support is in the integers exactly when the cell masses vanish off the integers, which is the vanishing of every cell defect $\mathfrak{h}_n$; the constant on $(1,2)$ is what produces the factor $e^{\nu_6 + \nu_7 2^{-s}}$ through (14.11).) Hence $\zeta_{\boldsymbol{\kappa}}$ satisfies (H1), and it satisfies (H0), (H2) and (H3) by Theorem 16.1(i) (order one, functional equation (12.3), pole at 1 with residue 1) for every list of $\mathscr{O}$; so Hamburger's theorem in the form of Subsection 14.7 gives $\zeta_{\boldsymbol{\kappa}} = C_H \zeta$, and $C_H = 1$ by the residues at $s = 1$. The zeros of $\zeta_{\boldsymbol{\kappa}}$ are $\frac{1}{2} \pm i\kappa_k$, so $\boldsymbol{\kappa} = (\gamma_k)$ and $\zeta_{Z1} = \zeta$. Then $(\log x)\, dJ^{\mathrm{M}} = \sum_{p^m} \log p\, \delta_{p^m}$, which gives (16.7). □

In Theorem 16.6 the primes are not part of the data: the data are the function k, which comes from the factor G and the pole, the space $\mathscr{O}$, which is defined by the clock, and the condition that the support lie in the points $\log n$. The masses of von Mangoldt are an output. What the theorem proves is *uniqueness*: a solution, if it exists, is the zeta list with the von Mangoldt masses. *Existence* of a solution is exactly (D) and is not proved; the theorem is a reformulation of the final step, not a step towards it. Experiment 19.16 solves a finite section of (16.6) numerically, starting from the

clock; nothing proved here says that finite sections converge to a solution of the infinite distributional system. The uniqueness part of Theorem 16.6 extends to every list at bounded distance from the clock, without the third condition of (16.1) and with no assumption on the masses (Theorem 18.7).

Theorem 16.7 (Nondegeneracy of the solution)**.** *Let $(\boldsymbol{\kappa}, (\mathsf{a}_n))$ be a solution of* (16.6) *with $\boldsymbol{\kappa} \in \mathscr{O}$ (Section 16). Let $(\dot\kappa_k)$ be a bounded real sequence and $(\dot{\mathsf{a}}_n)_{n\ge 2}$ real numbers with $\dot{\mathsf{a}}_n = O_\varepsilon(n^{-1/2+\varepsilon})$ for every $\varepsilon > 0$, such that*

$$\sum_{k\ge 1} \dot\kappa_k\, u \sin(\kappa_k u) = \sum_{n\ge 2} \dot{\mathsf{a}}_n\, \delta(u - \log n) \qquad \text{in } \mathcal{D}'(0,\infty), \tag{16.10}$$

the derivative of (16.6) *in the direction $((\dot\kappa_k), (\dot{\mathsf{a}}_n))$. Then $\dot{\mathsf{a}}_n = 0$ for every n, and $\sum_{k:\,\kappa_k = t} \dot\kappa_k = 0$ for every value t of the list. If the ordinates are simple, $\dot\kappa_k = 0$ for every k.*

Proof. Step 1. The series $\mathsf{J} := \sum_k \dot\kappa_k\, u \sin(\kappa_k u)$ converges in $\mathcal{S}'(\mathbb{R})$: for $\phi \in \mathcal{S}$, three integrations by parts give $\int \phi(u) u \sin(\kappa u)\, du = \kappa^{-3} \int (u\phi)'''(u) \cos(\kappa u)\, du$ (the boundary terms vanish since $u\phi$ and its derivatives are Schwartz), so $|\int \phi\, u \sin(\kappa u)| \le \|(u\phi)'''\|_{L^1} \kappa^{-3}$, and $\sum_k |\dot\kappa_k| \kappa_k^{-3} < \infty$ since $\kappa_k \asymp k/\log k$. Each term $u\sin(\kappa_k u)$ is even, so J is even. By (16.10), J vanishes on $(0, \log 2)$, hence by evenness on $(-\log 2, 0)$, so the restriction of J to $(-\log 2, \log 2)$ is supported at $\{0\}$. We use the following standard fact.

Lemma (point support). A distribution $\mathsf{T} \in \mathcal{D}'(-a, a)$ with $\operatorname{supp} \mathsf{T} \subset \{0\}$ is a finite combination $\mathsf{T} = \sum_{j\le m} c_j \delta_0^{(j)}$, with m the order of T [21, Thm. 2.3.4]; consequently $\langle \mathsf{T}, \chi\rangle = \sum_{j\le m} (-1)^j c_j \chi^{(j)}(0) = 0$ for every $\chi \in C_c^\infty(-a,a)$ vanishing to order m at 0, and the same formula defines $\langle \mathsf{T}, \chi\rangle$ for $\chi \in C_c^m(-a,a)$. Applied to $\mathsf{J}|_{(-\log 2, \log 2)}$ and using evenness, $\mathsf{J} = \sum_{j\le m,\ j \text{ even}} c_j \delta_0^{(j)}$ on $(-\log 2, \log 2)$ for some finite m. Fix an even $\mathsf{x}_0 \in C_c^\infty(-\log 2, \log 2)$ equal to 1 near 0, and for $\Re z > 0$ put

$$\mathsf{H}_1(z) := \sum_{k\ge 1} \dot\kappa_k \int_0^\infty (1 - \mathsf{x}_0(u))\, u \sin(\kappa_k u)\, e^{-zu}\, du = \tfrac12 \big\langle \mathsf{J}, (1-\mathsf{x}_0)\, e^{-z|u|} \big\rangle. \tag{16.11}$$

Approximating by test functions with compact support in $(0,\infty)$ and using (16.10), $\mathsf{H}_1(z) = \sum_{n\ge 2} \dot{\mathsf{a}}_n n^{-z}$ for $\Re z > \frac12$, where the series converges absolutely.

Step 2. For $\Re z > 0$, $\int_0^\infty u \sin(\kappa u) e^{-zu} du = 2\kappa z/(z^2+\kappa^2)^2$, so

$$\mathsf{H}_1(z) = \sum_{k\ge 1} \dot\kappa_k \frac{2\kappa_k z}{(z^2+\kappa_k^2)^2} - \mathsf{y}(z), \qquad \mathsf{y}(z) := \sum_{k\ge 1} \dot\kappa_k \int_0^{\log 2} \mathsf{x}_0(u)\, u \sin(\kappa_k u)\, e^{-zu}\, du. \tag{16.12}$$

y *is a polynomial.* Let m be the order of J at 0 from Step 1 and fix an integer $A \ge \max(m, 3)$. Write $e^{-zu} = \sum_{j\le A} \frac{(-z)^j}{j!} u^j + \mathsf{R}_A(z,u)$ with $\mathsf{R}_A(z,u) = \frac{(-z)^{A+1}}{A!} \int_0^1 (1-\vartheta)^A u^{A+1} e^{-\vartheta z u}\, d\vartheta$, so that $\mathsf{R}_A(z,\cdot)$ is smooth on $[0, \log 2]$ and vanishes to order $A+1$ at $u = 0$. Accordingly $\mathsf{y} = \mathsf{y}_{\mathrm{pol}} + \mathsf{y}_{\mathrm{rem}}$. *Polynomial part.* $\mathsf{y}_{\mathrm{pol}}(z) = \sum_{j\le A} \frac{(-z)^j}{j!} \sum_k \dot\kappa_k\, \mathsf{i}_{j,k}$ with $\mathsf{i}_{j,k} := \int_0^{\log 2} \mathsf{x}_0(u) u^{j+1} \sin(\kappa_k u)\, du$. For $g \in C_c^\infty[0,\infty)$ two integrations by parts give $\int_0^\infty g(u)\sin(\kappa u)\, du = g(0)/\kappa - \kappa^{-2}\int_0^\infty g''(u)\sin(\kappa u)\, du = g(0)/\kappa - g''(0)/\kappa^3 + O_g(\kappa^{-5})$; with $g = \mathsf{x}_0 u^{j+1}$ one has $g(0) = 0$ for all $j \ge 0$, so $\mathsf{i}_{j,k} = O_j(\kappa_k^{-3})$ and each inner series converges absolutely since $\sum_k |\dot\kappa_k| \kappa_k^{-3} < \infty$. Thus $\mathsf{y}_{\mathrm{pol}}$ is a polynomial of degree at most A. *Remainder part.* Put $\chi_z(u) := \mathsf{x}_0(u) \mathsf{R}_A(z, |u|)$, an even function with compact support in $(-\log 2, \log 2)$, of class C^A (because $|u|^{A+1}$ is C^A) and vanishing to order $A+1$ at 0. Since $u\sin(\kappa u)$ is even, $\mathsf{y}_{\mathrm{rem}}(z) = \frac12 \sum_k \dot\kappa_k \int_{\mathbb{R}} \chi_z(u)\, u \sin(\kappa_k u)\, du$. Let $\chi_{z,\epsilon} := \chi_z * \varrho_\epsilon$ with a standard mollifier; for small ϵ these are in $C_c^\infty(-\log 2, \log 2)$, $\|(u\chi_{z,\epsilon})'''\|_{L^1}$ is bounded uniformly in ϵ because $A \ge 3$, and $\chi_{z,\epsilon}^{(j)}(0) \to \chi_z^{(j)}(0) = 0$ for $j \le A$. By Step 1 and the point-support lemma, $\sum_k \dot\kappa_k \int \chi_{z,\epsilon}\, u \sin(\kappa_k u)\, du = \langle \mathsf{J}, \chi_{z,\epsilon}\rangle = \sum_{j\le m} (-1)^j c_j \chi_{z,\epsilon}^{(j)}(0) \to 0$ as $\epsilon \to 0$; on the other side, each term converges to $\dot\kappa_k \int \chi_z\, u \sin(\kappa_k u)\, du$ and the series is dominated by

$\sum_k |\dot\kappa_k|\kappa_k^{-3}\sup_\epsilon \|(u\chi_{z,\epsilon})'''\|_{L^1} < \infty$, so the limit may be taken termwise. Hence $\mathsf{y}_{\rm rem} \equiv 0$ and $\mathsf{y} = \mathsf{y}_{\rm pol}$ is a polynomial.

$\mathsf{y} \equiv 0$. As $z = \sigma \to +\infty$: (a) $\mathsf{H}_1(\sigma) = \sum_{n\ge 2} \dot{\mathsf{a}}_n n^{-\sigma} \to 0$, since for $\sigma \ge 1$ the terms are dominated by $|\dot{\mathsf{a}}_n| n^{-1}$, $\sum_n |\dot{\mathsf{a}}_n| n^{-1} < \infty$ because $\dot{\mathsf{a}}_n = O_\varepsilon(n^{-1/2+\varepsilon})$, and each term tends to 0; (b) the first sum in (16.12) is $O(\sigma^{-1}\log\sigma)$: the terms with $\kappa_k \le \sigma$ are at most $2\kappa_k\sigma/\sigma^4$ and add up to at most $2\sigma^{-3}\sum_{\kappa_k\le\sigma}\kappa_k \le 2\sigma^{-2}N_{\boldsymbol\kappa}(\sigma) = O(\sigma^{-1}\log\sigma)$, while the terms with $\kappa_k > \sigma$ are at most $2\sigma\kappa_k^{-3}$ and, by partial summation with $N_{\boldsymbol\kappa}(t) = O(t\log t)$, $\sum_{\kappa_k>\sigma}\kappa_k^{-3} = \int_\sigma^\infty t^{-3} dN_{\boldsymbol\kappa}(t) = O(\sigma^{-2}\log\sigma)$ (the boundary term $N_{\boldsymbol\kappa}(\sigma)\sigma^{-3}$ included), which gives $O(\sigma^{-1}\log\sigma)$ again; here $(\dot\kappa_k)$ is bounded. Hence the polynomial $\mathsf{y}(\sigma) = \sum_k \dot\kappa_k 2\kappa_k\sigma/(\sigma^2+\kappa_k^2)^2 - \mathsf{H}_1(\sigma) \to 0$ as $\sigma\to+\infty$, so $\mathsf{y}\equiv 0$.

Step 3. Put $\mathsf{H}(z) := \sum_{n\ge2} \frac{\dot{\mathsf{a}}_n}{\log n} n^{-z}$ for $\Re z > \frac12$, so that $\mathsf{H}' = -\mathsf{H}_1$. By Step 2 there is a real constant ν_8 with

$$\mathsf{H}(z) = \nu_8 + \sum_{k\ge1} \dot\kappa_k\Big(\frac{\kappa_k}{z^2+\kappa_k^2} - \frac{1}{\kappa_k}\Big), \tag{16.13}$$

which continues H to a meromorphic function in $\mathbb{C}$, even in z, with at most simple poles at $z = \pm i\kappa_k$, of residue $\pm\frac{1}{2i}\sum_{k':\kappa_{k'}=\kappa_k}\dot\kappa_{k'}$.

Step 4: finite order. By Theorem 16.6, $\zeta_{\boldsymbol\kappa} = \zeta$, with zeros $\frac12 \pm i\kappa_k$ of the multiplicity of the list. Put $\mathsf{Z}(s) := \mathsf{H}(s-\frac12)\,\zeta(s)$. At $s = \frac12 \pm i\kappa$ the function $\mathsf{H}(s-\frac12)$ has at most a simple pole (the residues of the entries of the list with the same value κ add up, (16.13)) and ζ has a zero of order equal to the multiplicity of κ in the list, which is at least 1; so Z is holomorphic there and $(s-1)\mathsf{Z}(s)$ is entire. *Admissible radii.* Call $r \ge 3$ admissible if $\operatorname{dist}(r, \{\kappa_k\}) \ge 1/\log r$. Since $N_{\boldsymbol\kappa}(2R+1) - N_{\boldsymbol\kappa}(R-1) = (\frac{R}{2\pi} + o(R))\log R$ (bounded displacement from the clock), the union of the intervals of radius $1/\log R$ about the $\kappa_k \in [R-1, 2R+1]$ has total length at most $(\frac{1}{\pi}+o(1))R < R$, so every interval $[R, 2R]$ with $R \ge R_0$ contains an admissible radius; choose admissible $r_j \in [2^j R_0, 2^{j+1}R_0]$, so that $r_j \to \infty$ and $r_{j+1} \le 4r_j$. *Bound on admissible circles.* Write (16.13) as $\mathsf{H}(z) = \nu_8 - \sum_k \dot\kappa_k \frac{z^2}{\kappa_k(z^2+\kappa_k^2)}$ and let $|z| = r$ be admissible. Then $|z \mp i\kappa_k| \ge |r - \kappa_k|$ and $|z - i\kappa_k| + |z + i\kappa_k| \ge 2\kappa_k$. Shell $\kappa_k \ge 2r$: $|z^2+\kappa_k^2| \ge \kappa_k^2 - r^2 \ge \frac34\kappa_k^2$, terms $\le \frac43 r^2\kappa_k^{-3}\sup|\dot\kappa|$, sum $O(r^2\cdot r^{-2}\log r) = O(\log r)$ by partial summation. Shell $\kappa_k \le r/2$: $|z^2+\kappa_k^2| \ge \frac34 r^2$, terms $\le \frac43\kappa_k^{-1}\sup|\dot\kappa|$, sum $O(\sum_{\kappa_k\le r}\kappa_k^{-1}) = O(\log^2 r)$. Shell $r/2 < \kappa_k < 2r$: one of $|z \pm i\kappa_k|$ is at least $\kappa_k \ge r/2$ and the other is at least $\max(|r-\kappa_k|, 1/\log r)$, so the term is at most $\frac{4r}{\kappa_k}\cdot\frac{\sup|\dot\kappa|}{\max(|r-\kappa_k|,1/\log r)} \le \frac{8\sup|\dot\kappa|}{\max(|r-\kappa_k|,1/\log r)}$; the $O(\log r)$ ordinates with $|\kappa_k - r| \le 1$ contribute $O(\log^2 r)$, and for $i \ge 0$ the $O(2^i\log r)$ ordinates with $2^i < |\kappa_k - r| \le 2^{i+1}$ contribute $O(\log r)$ each shell, $i \le \log_2 r$, so $O(\log^2 r)$ in all (multiplicities are counted by $N_{\boldsymbol\kappa}$ throughout). Hence $|\mathsf{H}(z)| \le C_{\mathsf{H}}\log^2 r$ on every admissible circle $|z| = r \ge R_0$. *From circles to the plane.* $(s-1)\zeta(s)$ is entire of order one, so $|(s-1)\zeta(s)| \le \exp(|s|^{1+\varepsilon})$ for $|s|$ large. By the maximum principle for the entire function $(s-1)\mathsf{Z}(s)$ on the disc $|s-\frac12| \le r_j$,

$$\max_{|s-\frac12|\le r_j} |(s-1)\mathsf{Z}(s)| \le C_{\mathsf{H}}\log^2 r_j \cdot \max_{|s-\frac12|=r_j} |(s-1)\zeta(s)| \le \exp\big(2r_j^{1+\varepsilon}\big) \tag{16.14}$$

for $j \ge j_0(\varepsilon)$. Given $R \ge r_{j_0}$ choose the least j with $r_j \ge R$; then $r_j \le 4R$ and (16.14) gives $\max_{|s-\frac12|\le R}|(s-1)\mathsf{Z}| \le \exp(2(4R)^{1+\varepsilon})$. So $(s-1)\mathsf{Z}$ is entire of order at most one: this is (H0). Z is the product of $\sum_n \dot{\mathsf{a}}_n\sqrt{n}(\log n)^{-1}n^{-s}$ and $\zeta(s)$, both absolutely convergent in $\Re s > 1$: this is (H1). Since H is even, $\mathsf{Z}(1-s) = \mathsf{H}(s-\frac12)\zeta(1-s)$, so Z satisfies the functional equation of ζ: this is (H2) and (H3) with the second function equal to Z. Hamburger's theorem gives $\mathsf{Z} = C_H\zeta$, that is, $\mathsf{H} \equiv C_H$. Since $\mathsf{H}(\sigma) \to 0$ as $\sigma\to+\infty$, $\mathsf{H}\equiv 0$: every $\dot{\mathsf{a}}_n$ and every residue vanishes. □

Remark 16.8 (Scope of Theorem 16.7). The theorem is conditional on the existence of a solution; it shows that a solution is isolated to first order, the Jacobian of (16.6) being injective there on the directions considered, except for the splitting of multiple ordinates. This is a necessary endpoint

nondegeneracy condition for any continuation scheme; it is not, by itself, an implicit-function theorem, because no isomorphism between fixed Banach spaces is proved. Steps 1–3 do not use that $\boldsymbol{\kappa}$ is a solution: at any list of $\mathscr{O}$ the kernel of the linearization consists of the Dirichlet series H that are even meromorphic functions with simple poles at $\pm i\kappa_k$ of the form (16.13). Such series exist for lists in arithmetic progression, for instance $(2^z+2^{-z}-2\cos\nu_9)^{-1}$, which is even, a power series in 2^{-z} for $\Re z>0$, with simple poles at $z=\pm i(\nu_9+2\pi m)/\log 2$, $m\in\mathbb{Z}$; these lists do not belong to $\mathscr{O}$. For perturbations $\dot{\mathsf{a}}_n$ of polynomial growth the same proof applies with Hamburger's theorem for a finite abscissa of absolute convergence [25]. The injectivity of the Jacobian at the clock and along a path, and with it the existence of a solution, are not proved.

16.3. **The gradient of the vertical defect.** For Dirichlet series $A=\sum_n a_n n^{-s}$, $B=\sum_n b_n n^{-s}$ and $\sigma>1$ put $\langle A,B\rangle_\sigma:=\sum_{n\ge1}a_n\bar b_n\,n^{-2\sigma}$, and for $\boldsymbol{\kappa}\in\mathscr{O}$ (Section 16)

$$\mathsf{K}_{\boldsymbol{\kappa}}:=\frac{\tilde\zeta_{\mathbb{Z},\boldsymbol{\kappa}}}{\zeta}=\sum_{n\ge1}k_n(\boldsymbol{\kappa})\,n^{-s},\qquad \mathsf{V}_\sigma(\boldsymbol{\kappa})=\langle\mathsf{K}_{\boldsymbol{\kappa}},\mathsf{K}_{\boldsymbol{\kappa}}\rangle_\sigma-1,\qquad \mathsf{r}_m(\boldsymbol{\kappa}):=\Re\langle\mathsf{K}_{\boldsymbol{\kappa}}\mathsf{L}_m,\mathsf{K}_{\boldsymbol{\kappa}}\rangle_\sigma, \tag{16.15}$$

with $\mathsf{L}_m(s)=-\log(1-m^{-s})$ as in Proposition 15.42. By Proposition 15.42 the derivative $\partial\tilde c_m(\boldsymbol{\kappa})/\partial\kappa_k$ is a function of m and of the value κ_k alone; denote it by $\mathsf{u}_m(\kappa_k)$, so that $\mathsf{u}_m=2\,\Im V_m$ when m is not a power of 2 (16.5).

Proposition 16.9 (Critical points of the vertical defect)**.** *Let $\sigma>1$ and $\boldsymbol{\kappa}\in\mathscr{O}$ (Section 16). Then $|\mathsf{r}_m(\boldsymbol{\kappa})|\le 2m^{-\sigma}\big(1+\mathsf{V}_\sigma(\boldsymbol{\kappa})\big)$, $|\mathsf{u}_m(t)|\le Cm^{-1/2}$ off the powers of 2 and $|\mathsf{u}_m(t)|\le C$ on them, and*

$$\frac{\partial\mathsf{V}_\sigma}{\partial\kappa_k}(\boldsymbol{\kappa})=2\,\mathsf{Q}_{\boldsymbol{\kappa}}(\kappa_k),\qquad \mathsf{Q}_{\boldsymbol{\kappa}}(t):=\sum_{m\ge2}\mathsf{r}_m(\boldsymbol{\kappa})\,\mathsf{u}_m(t), \tag{16.16}$$

the series converging absolutely and uniformly in t. If $\mathsf{V}_\sigma(\boldsymbol{\kappa})=0$, then $\mathsf{r}_m(\boldsymbol{\kappa})=0$ for every m and $\mathsf{Q}_{\boldsymbol{\kappa}}\equiv0$.

Proof. Since $\log\mathsf{K}_{\boldsymbol{\kappa}}=\sum_m(\tilde c_m(\boldsymbol{\kappa})-\mathbf{1}_{\mathrm{prime}}(m))\mathsf{L}_m$, $\partial\mathsf{K}_{\boldsymbol{\kappa}}/\partial\tilde c_m=\mathsf{K}_{\boldsymbol{\kappa}}\mathsf{L}_m$ and $\partial\mathsf{V}_\sigma/\partial\tilde c_m=2\mathsf{r}_m$. Multiplication by m^{-rs} multiplies the norm $\langle\cdot,\cdot\rangle_\sigma^{1/2}$ by $m^{-r\sigma}$, so $\langle\mathsf{K}\mathsf{L}_m,\mathsf{K}\mathsf{L}_m\rangle_\sigma^{1/2}\le\langle\mathsf{K},\mathsf{K}\rangle_\sigma^{1/2}\sum_{r\ge1}m^{-r\sigma}/r\le 2m^{-\sigma}\langle\mathsf{K},\mathsf{K}\rangle_\sigma^{1/2}$, and the bound for r_m follows by the Cauchy–Schwarz inequality. In (14.21) the support of $\varphi_m(y^j)$ has length at most $\frac{2}{j}m^{1/j-1}$ and the integrand is at most $m^{-1/(2j)}$ in absolute value, so $|V_m(t)|\le\sum_j\frac{2}{j^2}m^{-1+1/(2j)}\le Cm^{-1/2}$; for $m=2^j$ the cut term is $-\frac{\mu(j)}{j}\int_2^3 J_{\boldsymbol{\kappa}}(y)\frac{d}{dy}\varphi_m(y^j)\,dy$ (proof of Theorem 14.11), and its coordinate derivative needs $\partial_{\kappa_k}J_{\boldsymbol{\kappa}}$ on $(2,3)$: in the series (13.17) only the k-th term depends on κ_k, so the derivative exists and equals $\partial_{\kappa_k}J_{\boldsymbol{\kappa}}(y)=-2\,\Re\big(i\,y^{1/2+i\kappa_k}/(\tfrac12+i\kappa_k)\big)$, a bounded continuous function of $y\in(2,3)$ with $|\partial_{\kappa_k}J_{\boldsymbol{\kappa}}(y)|\le 2\sqrt3/\kappa_k\le 2\sqrt3/\tau_0^+$ uniformly on $\mathscr{O}$ (Lemma 14.6(b) is the same computation for the direction η_k); hence $|\mathsf{u}_m(t)|\le\frac{2}{j}\cdot2\sqrt3/t\cdot\mathrm{TV}(\varphi_m(y^j))\le C$ for $m=2^j$, with $\mathrm{TV}(\varphi_m(y^j))=2\|\varphi\|_\infty$. This gives the two bounds for u_m, and they are uniform in $t\ge\tau_0^+$. The chain rule is justified first for finite Euler products $\prod_{m\le M}$ and then by passage to the limit $M\to\infty$ in the weighted space $\ell^1(n^{-\sigma'})$, $1<\sigma'<\sigma$, of cell masses, in which the coordinate derivatives $\partial_{\kappa_k}\tilde c_m=\mathsf{u}_m(\kappa_k)$ are bounded uniformly in m and M by the bounds just proved, so that $\sum_n n^{-2\sigma}\partial_{\kappa_k}|k_n|^2$ converges uniformly in the truncation. It gives (16.16), with $\sum_m|\mathsf{r}_m||\mathsf{u}_m(t)|\le C\sum_m m^{-\sigma-1/2}+C\sum_j 2^{-j\sigma}<\infty$ uniformly in t, which is the asserted absolute and uniform convergence on all of $t\ge\tau_0^+$, not only on compact sets. If $\mathsf{V}_\sigma=0$, then $\mathsf{K}=1$ and $\mathsf{r}_m=\Re\langle\mathsf{L}_m,1\rangle_\sigma=0$, because L_m has no constant term. □

A list $\boldsymbol{\kappa}\in\mathscr{O}$ (Section 16) at which all coordinate derivatives of V_σ vanish is therefore a list all of whose ordinates are zeros of the single real-analytic function $\mathsf{Q}_{\boldsymbol{\kappa}}$. The minimum of V_σ on $\mathscr{O}$ is attained (Corollary 16.4); if every minimizer satisfied the appropriate coordinate/KKT stationarity conditions and those conditions forced $\mathsf{V}_\sigma=0$, the minimum would be 0 and (D) would follow. This is not proved. A count of zeros does not exclude critical points with $\mathsf{V}_\sigma>0$: $\mathsf{Q}_{\boldsymbol{\kappa}}(t)$ is a sum of

terms of frequency $\log m$ with $m \gtrsim t/2\pi$, so it oscillates at the scale $1/\log t$, which is the scale of the gaps between the ordinates.

17. Acceptance criteria and models in which the final step is a theorem

In the construction of this paper the continuous function ζ_λ is built from a list of ordinates and the discrete function $\tilde\zeta_{\mathbb{Z},\lambda}$ from the cells, and the two are joined by $\tilde h_\lambda$. This section states, without ζ, the general criterion under which a list is accepted by both sides, and gives three models: a non-periodic list accepted by a discrete category with two generators, where the acceptance is proved by one unitary matrix; a finite Euler product over the primes, to which the constructions of Sections 2–16 apply verbatim and for which the final step is a theorem; and the diagonal operator over all primes, which shows where the argument of the finite models stops. The complex variable of the first two subsections is denoted by z, and the half-plane $\Im\mathsf{z} > 0$ plays the role of $\Re s > 1$.

17.1. **The acceptance criterion: window and frequencies.** Let $\mathfrak{k} > 0$ and let $\boldsymbol{\kappa} = (\kappa_k)_{k\ge1}$ be a nondecreasing list of positive numbers with $|\kappa_k - k\pi/\mathfrak{k}|$ bounded. The *continuous side* is the canonical product

$$\mathfrak{C}_{\boldsymbol{\kappa}}(\mathsf{z}) := \mathsf{z}\prod_{k\ge1}\Big(1-\frac{\mathsf{z}^2}{\kappa_k^2}\Big), \tag{17.1}$$

an odd entire function of order one with real zeros; the oddness plays the role of the functional equation. We assume that $\mathfrak{C}_{\boldsymbol{\kappa}}$ has exponential type $\mathfrak{k}$ and polynomial growth on the real axis, as $\sin(\mathfrak{k}\mathsf{z})$ has for the list $\kappa_k = k\pi/\mathfrak{k}$. The *discrete side* is given by a closed, locally finite set $\mathscr{F} \subset (0,\infty)$ of frequencies ($\mathscr{F}\cap[0,R]$ is finite for every R; the set need not be uniformly discrete, and in the two-edge model below it is not) that is closed under addition; put $\mathscr{F}^+ := \mathscr{F}\cup\{0\}$. The discrete category consists of the generalized Dirichlet series $\sum_{\mathfrak{l}\in\mathscr{F}^+}\mathfrak{c}_\mathfrak{l}\,e^{i\mathfrak{l}\mathsf{z}}$, absolutely convergent in some half-plane $\Im\mathsf{z} > \nu_{10}$ (the number ν_{10} is not prescribed and may be enlarged), and of their Euler forms $\exp\big(\sum_{\mathfrak{l}\in\mathscr{F}}\mathfrak{b}_\mathfrak{l}\,e^{i\mathfrak{l}\mathsf{z}}\big)$. For $\Im\mathsf{z} > 0$,

$$\frac{\mathfrak{C}'_{\boldsymbol{\kappa}}}{\mathfrak{C}_{\boldsymbol{\kappa}}}(\mathsf{z}) = \frac{1}{\mathsf{z}} + \sum_{k\ge1}\frac{2\mathsf{z}}{\mathsf{z}^2-\kappa_k^2} = -i\int_0^\infty e^{i\mathsf{z}u}\Big(1+2\sum_{k\ge1}\cos(\kappa_k u)\Big)\,du, \tag{17.2}$$

the bracket being a tempered distribution in u: it is the spectral density of the list.

Theorem 17.1 (Acceptance criterion)**.** *For a list $\boldsymbol{\kappa}$ as above the following are equivalent:*

(i) *the spectral density is atomic on the frequencies: there are real $\mathfrak{w}_\mathfrak{l}$ with $\sum_{\mathfrak{l}\in\mathscr{F}}|\mathfrak{w}_\mathfrak{l}|e^{-\nu_{10}\mathfrak{l}} < \infty$ for some admissible ν_{10} and*

$$1 + 2\sum_{k\ge1}\cos(\kappa_k u) = \sum_{\mathfrak{l}\in\mathscr{F}}\mathfrak{w}_\mathfrak{l}\,\delta(u-\mathfrak{l}) \qquad \text{in } \mathcal{D}'(0,\infty); \tag{17.3}$$

(ii) *the continuous function has an Euler form: for $\Im\mathsf{z} > \nu_{10}$, with a constant $\mathfrak{c}_0 \neq 0$,*

$$e^{i\mathfrak{k}\mathsf{z}}\,\mathfrak{C}_{\boldsymbol{\kappa}}(\mathsf{z}) = \mathfrak{c}_0\exp\Big(\sum_{\mathfrak{l}\in\mathscr{F}}\mathfrak{b}_\mathfrak{l}\,e^{i\mathfrak{l}\mathsf{z}}\Big), \qquad \mathfrak{b}_\mathfrak{l} = -\frac{\mathfrak{w}_\mathfrak{l}}{\mathfrak{l}}; \tag{17.4}$$

(iii) *the continuous function is a finite sum in the discrete category:*

$$e^{i\mathfrak{k}\mathsf{z}}\,\mathfrak{C}_{\boldsymbol{\kappa}}(\mathsf{z}) = \sum_{\mathfrak{l}\in\mathscr{F}^+\cap[0,2\mathfrak{k}]}\mathfrak{c}_\mathfrak{l}\,e^{i\mathfrak{l}\mathsf{z}}, \qquad \mathfrak{c}_0 \neq 0. \tag{17.5}$$

Proof. (i)⇔(ii): the two sides of (17.3) extend to distributions on $\mathbb{R}$ supported in $[0,\infty)$ (the left side as in (17.2), the right side as the measure $\sum_\mathfrak{l}\mathfrak{w}_\mathfrak{l}\delta_\mathfrak{l}$), and (i) says that their difference vanishes on $(0,\infty)$; being supported at $\{0\}$ and of finite order, the difference is a finite combination of derivatives of δ_0 [21, Thm. 2.3.4], whose transform $u\mapsto e^{i\mathsf{z}u}$ is a polynomial $\mathsf{p}(\mathsf{z})$. Hence, by (17.2), (i) is the identity $\mathfrak{C}'_{\boldsymbol{\kappa}}/\mathfrak{C}_{\boldsymbol{\kappa}} = -i\sum_\mathfrak{l}\mathfrak{w}_\mathfrak{l}e^{i\mathfrak{l}\mathsf{z}} + \mathsf{p}(\mathsf{z})$ in $\Im\mathsf{z} > \nu_{10}$. Along $\mathsf{z} = iy$, $y\to+\infty$, the exponential sum

tends to 0 and $\mathfrak{C}'_{\boldsymbol{\kappa}}/\mathfrak{C}_{\boldsymbol{\kappa}}(iy) = -i(\frac{1}{y} + \sum_k \frac{2y}{y^2+\kappa_k^2}) \to -i\mathfrak{k}$ (type $\mathfrak{k}$); so the polynomial p has a finite limit along a ray, hence is constant, and the constant is $-i\mathfrak{k}$. Integrating, $\log \mathfrak{C}_{\boldsymbol{\kappa}} = \text{const} - i\mathfrak{k}\mathsf{z} - \sum_{\mathfrak{l}} \frac{\mathfrak{w}_{\mathfrak{l}}}{\mathfrak{l}} e^{i\mathfrak{l}\mathsf{z}}$, and conversely. (ii)⇒(iii): expanding the exponential, $\mathsf{N}(\mathsf{z}) := e^{i\mathfrak{k}\mathsf{z}}\mathfrak{C}_{\boldsymbol{\kappa}}(\mathsf{z}) = \sum_{\mathfrak{l}\in\mathscr{F}^+} \mathfrak{c}_{\mathfrak{l}} e^{i\mathfrak{l}\mathsf{z}}$, absolutely convergent in $\Im\mathsf{z} > \nu_{10}$. On the other hand N satisfies $|\mathsf{N}(\mathsf{z})| \le C(1+|\mathsf{z}|)^N e^{2\mathfrak{k}\max(0,-\Im\mathsf{z})}$: for $\Im\mathsf{z} \ge 0$ it is bounded by a polynomial, since $\mathfrak{C}_{\boldsymbol{\kappa}}$ has type $\mathfrak{k}$ and polynomial growth on the real axis and $|e^{i\mathfrak{k}\mathsf{z}}| = e^{-\mathfrak{k}\Im\mathsf{z}}$ absorbs the exponential growth in the upper half-plane, and for $\Im\mathsf{z} < 0$ the factor $e^{i\mathfrak{k}\mathsf{z}}$ contributes $e^{\mathfrak{k}|\Im\mathsf{z}|}$ on top of the type $\mathfrak{k}$ of $\mathfrak{C}_{\boldsymbol{\kappa}}$. By the Paley–Wiener–Schwartz theorem [21, Thm. 7.3.1], used with the transform $\mathsf{T} \mapsto \langle \mathsf{T}, e^{iz u}\rangle$ of (17.2) (for which a distribution supported in $[0, 2\mathfrak{k}]$ has a transform bounded by $C(1+|\mathsf{z}|)^N e^{2\mathfrak{k}\max(0,-\Im\mathsf{z})}$, and conversely), an entire function with this bound is the transform of a distribution supported in $[0, 2\mathfrak{k}]$, so the Fourier transform of the restriction of N to a horizontal line is supported in $[0, 2\mathfrak{k}]$. On the line $\Im\mathsf{z} = \nu_{10} + 1$ this transform is $\sum_{\mathfrak{l}} \mathfrak{c}_{\mathfrak{l}} e^{-\mathfrak{l}(\nu_{10}+1)}\delta_{\mathfrak{l}}$, so $\mathfrak{c}_{\mathfrak{l}} = 0$ for $\mathfrak{l} > 2\mathfrak{k}$, and the sum is finite because $\mathscr{F}^+ \cap [0, 2\mathfrak{k}]$ is finite. (iii)⇒(ii): write $\mathsf{N} = \mathfrak{c}_0(1+\mathsf{w})$ with $\mathsf{w}(\mathsf{z}) = \sum_{0<\mathfrak{l}\le 2\mathfrak{k}} (\mathfrak{c}_{\mathfrak{l}}/\mathfrak{c}_0) e^{i\mathfrak{l}\mathsf{z}}$; for $\Im\mathsf{z} \ge \nu'$ with $\sum_{0<\mathfrak{l}\le 2\mathfrak{k}} |\mathfrak{c}_{\mathfrak{l}}/\mathfrak{c}_0| e^{-\mathfrak{l}\nu'} \le \frac{1}{2}$ one has $|\mathsf{w}| \le \frac{1}{2}$, so $\log(1+\mathsf{w}) = \sum_{r\ge 1} (-1)^{r+1}\mathsf{w}^r/r$ converges absolutely, and expanding the powers gives an absolutely convergent series of exponentials whose frequencies are finite sums of elements of $\mathscr{F}$, hence elements of $\mathscr{F}$ (closure under addition); this proves (ii) in the half-plane $\Im\mathsf{z} > \nu'$, after increasing ν_{10} to ν' if necessary. □

In model identities of the form $2 + 2\sum_k \cos(\kappa_k u)$, the leading 2 records a double zero at the origin; the normalized criterion (17.3) starts with 1 because $\mathfrak{C}_{\boldsymbol{\kappa}}$ has a simple zero there.

The continuous side contributes a *window*: by its type and its symmetry, the frequencies of $e^{i\mathfrak{k}\mathsf{z}}\mathfrak{C}_{\boldsymbol{\kappa}}$ lie in $[0, 2\mathfrak{k}]$. The discrete side contributes the *admissible frequencies* $\mathscr{F}^+$. A list is accepted by both sides if and only if its function lives in the intersection, and it is then determined by the finitely many coefficients $\mathfrak{c}_{\mathfrak{l}}$, $\mathfrak{l} \in \mathscr{F}^+ \cap [0, \mathfrak{k})$, the others being fixed by the oddness. If $\mathscr{F} = 2\mathfrak{k}\mathbb{N}$, the intersection is $\{0, 2\mathfrak{k}\}$, the only function is a multiple of $\sin(\mathfrak{k}\mathsf{z})$, and the only accepted list is the clock $\kappa_k = k\pi/\mathfrak{k}$. Hamburger's theorem is the statement that for the window fixed by the factor $\Gamma(\frac{s}{2})\pi^{-s/2}$ and the frequencies $\log n$ the space of accepted functions has dimension one. The criterion says nothing about whether a given list, the zeta list in particular, belongs to the intersection; for ζ that membership is (D).

17.2. **A non-periodic accepted list: one unitary matrix.** Let $\mathfrak{p}, \mathfrak{q} > 0$ with $\mathfrak{p}/\mathfrak{q} \notin \mathbb{Q}$, $0 < \mathfrak{v} < 1$, $\mathfrak{k} = \mathfrak{p} + \mathfrak{q}$, and

$$\mathsf{U} := \begin{pmatrix} \mathfrak{v} & \sqrt{1-\mathfrak{v}^2} \\ \sqrt{1-\mathfrak{v}^2} & -\mathfrak{v} \end{pmatrix}, \qquad \mathsf{X}(\mathsf{z}) := \begin{pmatrix} e^{2i\mathfrak{p}\mathsf{z}} & 0 \\ 0 & e^{2i\mathfrak{q}\mathsf{z}} \end{pmatrix}, \tag{17.6}$$

$$\mathfrak{D}(\mathsf{z}) := \det\big(I - \mathsf{U}\mathsf{X}(\mathsf{z})\big) = 1 - \mathfrak{v}\, e^{2i\mathfrak{p}\mathsf{z}} + \mathfrak{v}\, e^{2i\mathfrak{q}\mathsf{z}} - e^{2i(\mathfrak{p}+\mathfrak{q})\mathsf{z}} = -2i\, e^{i(\mathfrak{p}+\mathfrak{q})\mathsf{z}}\, \mathfrak{S}(\mathsf{z}), \tag{17.7}$$

$$\mathfrak{S}(\mathsf{z}) := \sin\big((\mathfrak{p}+\mathfrak{q})\mathsf{z}\big) + \mathfrak{v}\sin\big((\mathfrak{p}-\mathfrak{q})\mathsf{z}\big). \tag{17.8}$$

Proposition 17.2 (The two-edge model)**.** (i) *All zeros of $\mathfrak{S}$ are real. Each interval between consecutive points $(m+\frac{1}{2})\pi/\mathfrak{k}$, $m \in \mathbb{Z}$, contains a zero, and the number of zeros in $[0, T]$, with multiplicity, is $\mathfrak{k}T/\pi + O(1)$; hence the positive zeros form a list $\boldsymbol{\kappa}$ with $|\kappa_k - k\pi/\mathfrak{k}|$ bounded. The set of zeros is not periodic.*

(ii) $\mathfrak{S}(-\mathsf{z}) = -\mathfrak{S}(\mathsf{z})$, *and* $\mathfrak{D}(\mathsf{z}) = -e^{2i\mathfrak{k}\mathsf{z}}\,\overline{\mathfrak{D}(\bar{\mathsf{z}})}$.

(iii) *For $\Im\mathsf{z} > 0$,*

$$\log \mathfrak{D}(\mathsf{z}) = -\sum_{r\ge 1} \frac{1}{r} \operatorname{tr}\big(\mathsf{U}\mathsf{X}(\mathsf{z})\big)^r = \sum_{\mathfrak{l}\in\mathscr{F}} \mathfrak{b}_{\mathfrak{l}}\, e^{i\mathfrak{l}\mathsf{z}}, \qquad \mathscr{F} = \{2(m\mathfrak{p} + m'\mathfrak{q}) : m, m' \ge 0\} \setminus \{0\}, \tag{17.9}$$

absolutely convergent, and the spectral density of the zeros of $\mathfrak{S}$, counted with multiplicity, is atomic on $\mathscr{F}$ in the sense of (17.3).

(iv) $\mathfrak{D}(\mathsf{z}) = -2i\,\mathfrak{S}'(0)\, e^{i\mathfrak{k}\mathsf{z}}\, \mathfrak{C}_{\boldsymbol{\kappa}}(\mathsf{z})$ *for the list $\boldsymbol{\kappa}$ of all positive zeros of $\mathfrak{S}$: the continuous and the discrete functions coincide up to the factor $e^{i\mathfrak{k}\mathsf{z}}$, and the list is accepted.*

Proof. (i) For $\Im \mathsf{z} > 0$ the matrix $\mathsf{UX}(\mathsf{z})$ has norm at most $\max(e^{-2\mathfrak{p}\Im \mathsf{z}}, e^{-2\mathfrak{q}\Im \mathsf{z}}) < 1$, because U is unitary; so $I - \mathsf{UX}(\mathsf{z})$ is invertible and $\mathfrak{D}(\mathsf{z}) \neq 0$. $\mathfrak{S}$ is odd, so it has no zeros in $\Im \mathsf{z} < 0$ either. At $\mathsf{z}_m = (m + \frac{1}{2})\pi/\mathfrak{k}$, $\sin(\mathfrak{k}\mathsf{z}_m) = (-1)^m$ and $|\mathfrak{v} \sin((\mathfrak{p} - \mathfrak{q})\mathsf{z}_m)| < 1$, so $\mathfrak{S}(\mathsf{z}_m)$ has the sign $(-1)^m$ and there is a zero in $(\mathsf{z}_{m-1}, \mathsf{z}_m)$. For the count, let $\mathsf{M}(t) := \mathsf{UX}(t)$, unitary and real-analytic in $t \in \mathbb{R}$, with $\mathsf{M}'(t) = 2i\,\mathsf{M}(t)\mathsf{D}$, $\mathsf{D} := \operatorname{diag}(\mathfrak{p}, \mathfrak{q})$ (diagonal matrices commute). $\mathsf{M}(t)$ is never a scalar matrix, since U has nonzero off-diagonal entries; so its two eigenvalues $e^{i\theta_1(t)}, e^{i\theta_2(t)}$, which can be labelled as continuous functions of t (roots of a quadratic with continuous coefficients), coincide only on a discrete set (the discriminant $\operatorname{tr}^2 - 4\det = \mathfrak{v}^2(e^{2i\mathfrak{p}t} - e^{2i\mathfrak{q}t})^2 + 4e^{2i\mathfrak{k}t}$ is real-analytic and equals 4 at $t = 0$). Where an eigenvalue $\mu = e^{i\theta_j}$ is simple it is analytic with an analytic unit eigenvector v_j, and differentiating $\mathsf{M}v_j = \mu v_j$ and pairing with v_j gives $\mu' = \langle \mathsf{M}'v_j, v_j\rangle$ (the term $\langle \mathsf{M}v_j', v_j\rangle = \langle v_j', \mathsf{M}^*v_j\rangle = \mu\langle v_j', v_j\rangle$ cancels against $\mu\langle v_j', v_j\rangle$), i.e.

$$\theta_j'(t) = 2\langle \mathsf{D}v_j, v_j\rangle = 2\big(\mathfrak{p}|v_{j,1}|^2 + \mathfrak{q}|v_{j,2}|^2\big) \;\geq\; 2\min(\mathfrak{p}, \mathfrak{q}) > 0. \tag{17.10}$$

Hence each continuous eigenphase is strictly increasing with $\theta_j(b) - \theta_j(a) \geq 2\min(\mathfrak{p}, \mathfrak{q})(b - a)$, and $\theta_1 + \theta_2 = 2\mathfrak{k}t + \text{const}$ because $\det \mathsf{M}(t) = -e^{2i\mathfrak{k}t}$. For real t, $\mathfrak{D}(t) = (1 - e^{i\theta_1(t)})(1 - e^{i\theta_2(t)})$, and 1 is never a double eigenvalue ($\mathsf{M}(t) = I$ is impossible), so every real zero of $\mathfrak{D}$ is a simple zero of exactly one factor, at which that eigenphase crosses $2\pi\mathbb{Z}$ with positive speed: the real zeros of $\mathfrak{S}$ are simple, and their number in $[0, T]$ is $\sum_j \#\{t \in [0, T] : \theta_j(t) \in 2\pi\mathbb{Z}\} = \sum_j \big(\frac{\theta_j(T) - \theta_j(0)}{2\pi} + O(1)\big) = \frac{\mathfrak{k}T}{\pi} + O(1)$. If the zero set had a period $y \neq 0$, then $\mathfrak{S}(\mathsf{z} + y)$ and $\mathfrak{S}(\mathsf{z})$ would be entire functions of order one with the same zero multiset, so by Hadamard's factorization $\mathfrak{S}(\mathsf{z} + y) = e^{a\mathsf{z} + b}\mathfrak{S}(\mathsf{z})$ for constants a, b. As $\Im \mathsf{z} \to +\infty$, (17.8) and $\mathfrak{k} > |\mathfrak{p} - \mathfrak{q}|$ give $\mathfrak{S}(\mathsf{z}) = -\frac{1}{2i}e^{-i\mathfrak{k}\mathsf{z}}(1 + o(1))$ uniformly in $\Re \mathsf{z}$, so $e^{a\mathsf{z}+b} = e^{-i\mathfrak{k}y}(1 + o(1))$ as $\Im \mathsf{z} \to +\infty$, which forces $a = 0$. With $a = 0$, comparing the coefficients of $e^{\pm i(\mathfrak{p}+\mathfrak{q})\mathsf{z}}$ and of $e^{\pm i(\mathfrak{p}-\mathfrak{q})\mathsf{z}}$ in $\mathfrak{S}(\mathsf{z} + y) = e^b\mathfrak{S}(\mathsf{z})$ gives $e^{\pm i(\mathfrak{p}+\mathfrak{q})y} = e^b = e^{\pm i(\mathfrak{p}-\mathfrak{q})y}$, hence $(\mathfrak{p} + \mathfrak{q})y \in \pi\mathbb{Z}$ and $(\mathfrak{p} - \mathfrak{q})y \in \pi\mathbb{Z}$, so $2\mathfrak{p}y \in \pi\mathbb{Z}$ and $2\mathfrak{q}y \in \pi\mathbb{Z}$, against $\mathfrak{p}/\mathfrak{q} \notin \mathbb{Q}$. (ii) is immediate; the second identity follows from $\det \mathsf{U} = -1$. (iii) The series is the logarithm of the determinant, convergent because the norm is less than 1; each trace is a combination of exponentials with frequencies in $\mathscr{F}$. The statement on the density is Theorem 17.1, (ii)⇒(i), applied to (iv). (iv) $\mathfrak{S}$ is an odd entire function of exponential type $\mathfrak{k}$, bounded on the real axis, with real zeros; by Hadamard's factorization and the oddness, $\mathfrak{S}(\mathsf{z}) = \mathfrak{S}'(0)\,\mathfrak{C}_{\boldsymbol{\kappa}}(\mathsf{z})$. □

In this model nothing has to be joined: the continuous function and the discrete function are the same determinant. The Euler form comes from the expansion of $\det(I - \mathsf{UX})$ in closed paths, and the real zeros and the symmetry come from the unitarity of U, through the contraction in $\Im \mathsf{z} > 0$. The acceptance criterion at this level is the existence of the common object U. In finite rank this is general: the zeros of $\mathsf{P}(e^{i\mathfrak{l}_1\mathsf{z}}, \dots, e^{i\mathfrak{l}_r\mathsf{z}})$, for a polynomial P without zeros in the product of the open unit discs and in the product of their exteriors, are real and their spectral density is atomic [46], and conversely the lists with unit masses whose spectral density is atomic with locally finite support are zero sets of exponential polynomials with real zeros [51].

17.3. **A finite Euler product: the constructions of this paper with the final step proved.** Let

$$\mathsf{F}(s) := \big(1 - 2^{-s}\big)\big(1 - 3^{-s}\big) = 1 - 2^{-s} - 3^{-s} + 6^{-s} = \sum_{n|6} \mu(n)\, n^{-s}. \tag{17.11}$$

It is a Dirichlet series over the integers with an Euler product over the primes 2 and 3, and $\mathsf{F}(-s) = 6^s\,\mathsf{F}(s)$, with axis of symmetry $\Re s = 0$. Its zeros are $s = 0$, double, and $s = \pm i\mathfrak{y}_k$, where

$$\{\mathfrak{y}_k\}_{k\geq 1} = \Big\{\frac{2\pi m}{\log 2}\Big\}_{m\geq 1} \cup \Big\{\frac{2\pi m}{\log 3}\Big\}_{m\geq 1} \quad \text{in increasing order,} \tag{17.12}$$

all simple, a non-periodic list, and, with $2\sinh(\frac{s\log p}{2}) = s\log p\prod_{m\ge1}(1+\frac{s^2\log^2 p}{4\pi^2m^2})$,

$$\mathsf{F}(s) = 6^{-s/2}\,(\log 2\log 3)\,s^2\prod_{k\ge1}\Big(1+\frac{s^2}{\mathfrak{y}_k^2}\Big). \tag{17.13}$$

The constructions of this paper apply as follows. The count of the ordinates is $\lfloor\frac{T\log 2}{2\pi}\rfloor+\lfloor\frac{T\log 3}{2\pi}\rfloor = \frac{T\log 6}{2\pi}-1+O(1)$ with a remainder of absolute value at most 1; the midpoint clock is $\frac{\mathfrak{x}_k\log 6}{2\pi}-1 = k-\frac12$, that is, $\mathfrak{x}_k = (2k+1)\pi/\log 6$, and $\mathfrak{x}_k-\mathfrak{y}_k$ is bounded. For a list $\boldsymbol{\kappa}$ at bounded distance from the clock,

$$\mathsf{F}_{\boldsymbol{\kappa}}(s) := 6^{-s/2}\,(\log 2\log 3)\,s^2\prod_{k\ge1}\Big(1+\frac{s^2}{\kappa_k^2}\Big), \qquad \mathsf{F}_{\boldsymbol{\kappa}}(-s) = 6^s\,\mathsf{F}_{\boldsymbol{\kappa}}(s), \tag{17.14}$$

and the family is $\mathsf{F}_\lambda := \mathsf{F}_{\boldsymbol{\kappa}(\lambda)}$ with $\kappa_k(\lambda) = (1-\lambda)\mathfrak{y}_k+\lambda\mathfrak{x}_k$, so that $\mathsf{F}_0 = \mathsf{F}$ and, since $\prod_{k\ge0}(1+s^2/\mathfrak{x}_k^2) = \cosh(\frac{s\log 6}{2})$,

$$\mathsf{F}_1(s) = \frac{\log 2\log 3}{2}\cdot\frac{s^2\,(1+6^{-s})}{1+s^2\log^2 6/\pi^2}. \tag{17.15}$$

Theorem 17.3 (The final step in the model)**.** (i) (Poisson, prime by prime.) *In $\mathcal{D}'(\mathbb{R})$,*

$$2+2\sum_{k\ge1}\cos(\mathfrak{y}_k u) = \sum_{p\in\{2,3\}}\sum_{m\in\mathbb{Z}}\cos\frac{2\pi m u}{\log p} = \sum_{p\in\{2,3\}}\log p\sum_{m\in\mathbb{Z}}\delta(u-m\log p). \tag{17.16}$$

The spectral density of the list $(\mathfrak{y}_k)$ is atomic on the points $m\log p$: the continuous function built from the list is the Dirichlet series with Euler product, $\mathsf{F}_0 = \mathsf{F}$.

(ii) (Hamburger's theorem of the model.) *Let f be entire of finite order with $f(-s) = 6^s f(s)$ and $f(0) = 0$, and let $f(s) = \sum_{n\ge1}a_n n^{-s}$ converge absolutely in some half-plane $\Re s > \nu_{12}$. Then $f = a_1\mathsf{F}$.*

(iii) (The family.) *F_λ is an absolutely convergent Dirichlet series in a half-plane if and only if $\lambda = 0$. In particular the clock function F_1 is not a Dirichlet series.*

Proof. (i) is the Poisson summation formula $\sum_{m\in\mathbb{Z}}e^{2\pi i m u/\log p} = \log p\sum_{m\in\mathbb{Z}}\delta(u-m\log p)$ for $p=2$ and $p=3$; the second statement is (17.13). (ii) Put $\sigma_1 := \nu_{12}+1$ and $\sigma_2 := -\sigma_1$. On $\Re s\ge\sigma_1$, $f(s) = \sum_n a_n n^{-s}$ converges absolutely and $|f|\le C := \sum_n|a_n|n^{-\sigma_1}$; on $\Re s\le\sigma_2$, the functional equation gives $f(s) = 6^{-s}f(-s) = \sum_n a_n(n/6)^s$, again absolutely convergent, with $|f(s)|\le C\,6^{-\sigma}$. By the Phragmén–Lindelöf theorem f is bounded in the closed strip $\mathbb{S} := \{\sigma_2\le\Re s\le\sigma_1\}$: f has finite order ρ, so $|f(s)|\le\exp(|s|^{\rho+1})\le\exp(e^{c|\Im s|})$ for every $c>0$ and $|s|$ large, which is the growth condition of the theorem in a strip of any finite width; let $\mathrm{M} := \sup_{\mathbb{S}}|f|$. *Comparison of the two Dirichlet expansions.* For real λ, $\sigma\in[\sigma_2,\sigma_1]$ and $T>0$ put $\mathsf{A}_{\lambda,T}(\sigma) := \frac{1}{2T}\int_{-T}^{T}f(\sigma+it)e^{i\lambda t}\,dt$. Since $f'(\sigma+it) = -i\,\partial_t f(\sigma+it)$, an integration by parts gives

$$\frac{d}{d\sigma}\mathsf{A}_{\lambda,T}(\sigma) = \frac{-i}{2T}\Big[f(\sigma+it)e^{i\lambda t}\Big]_{t=-T}^{t=T} - \lambda\,\mathsf{A}_{\lambda,T}(\sigma), \qquad \text{so} \qquad \Big|\frac{d}{d\sigma}\big(e^{\lambda\sigma}\mathsf{A}_{\lambda,T}(\sigma)\big)\Big| \le \frac{\mathrm{M}}{T}e^{\lambda\sigma}, \tag{17.17}$$

and integrating over $[\sigma_2,\sigma_1]$, $|e^{\lambda\sigma_1}\mathsf{A}_{\lambda,T}(\sigma_1) - e^{\lambda\sigma_2}\mathsf{A}_{\lambda,T}(\sigma_2)|\le\frac{\mathrm{M}}{T}\int_{\sigma_2}^{\sigma_1}e^{\lambda\sigma}d\sigma\to0$ as $T\to\infty$. On the two boundary lines the limits $\mathsf{A}_\lambda(\sigma_i) := \lim_{T\to\infty}\mathsf{A}_{\lambda,T}(\sigma_i)$ exist and are computed termwise from the absolutely convergent expansions (dominated convergence; the mean of $e^{i\mu t}$ is 1 for $\mu=0$ and 0 otherwise): $\mathsf{A}_\lambda(\sigma_1) = a_n n^{-\sigma_1}$ if $\lambda=\log n$ and 0 otherwise; $\mathsf{A}_\lambda(\sigma_2) = a_m(m/6)^{\sigma_2}$ if $\lambda=\log(6/m)$ and 0 otherwise. Hence

$$e^{\lambda\sigma_1}\mathsf{A}_\lambda(\sigma_1) = e^{\lambda\sigma_2}\mathsf{A}_\lambda(\sigma_2) \qquad (\lambda\in\mathbb{R}). \tag{17.18}$$

Take $\lambda=\log n$ with $a_n\ne0$: the left side is $a_n\ne0$, so the right side is nonzero, which forces $\log n = \log(6/m)$ for some m with $a_m\ne0$, i.e. $n\mid 6$ and $m = 6/n$, and then $e^{\lambda\sigma_2}\mathsf{A}_\lambda(\sigma_2) = a_m(m/6)^{\sigma_2}n^{\sigma_2} = a_m$, so $a_n = a_{6/n}$. Thus $a_n = 0$ unless $n\mid 6$, $a_6 = a_1$ and $a_3 = a_2$, and $f(0) = 2a_1+2a_2 = 0$ gives $a_2 = -a_1$ and $f = a_1(1-2^{-s}-3^{-s}+6^{-s}) = a_1\mathsf{F}$. (This is the elementary

case of Bohr's theorem that the Dirichlet exponents of a function almost periodic in a strip are the same on every line of the strip [24]; (17.17) is what allows the coefficient formula to be used on the lines $\Re s < 0$.) (iii) F_λ is entire of order one, satisfies the functional equation and vanishes at 0. If it is a Dirichlet series, (ii) gives $\mathsf{F}_\lambda = a_1 \mathsf{F}$, the lists of zeros coincide, and $\lambda(\mathfrak{x}_k - \mathfrak{y}_k) = 0$ for every k, so $\lambda = 0$. □

In the model every object of the programme exists and the analogue of (D) is proved. What makes $\lambda = 0$ close is a property of the list that the clock and the lists with $\lambda \neq 0$ do not have,

$$\{\mathfrak{y}_k\} = \frac{2\pi}{\log 2}\,\mathbb{N} \;\cup\; \frac{2\pi}{\log 3}\,\mathbb{N}, \tag{17.19}$$

a superposition of one clock for each prime; it passes the scale test of Remark 15.23 and it is the hypothesis of the Poisson formula (17.16). The constructions extend to $\prod_{p\le X}(1-p^{-s}) = \sum_{n\mid X\#}\mu(n)n^{-s}$, with $X\#$ the product of the primes up to X, the functional equation with the factor $(-1)^{\pi(X)}(X\#)^s$, a zero of order $\pi(X)$ at the origin and the list $\bigcup_{p\le X}\frac{2\pi}{\log p}\mathbb{N}$: part (i) of Theorem 17.3 holds prime by prime, and the clock and the family are defined in the same way. The uniqueness statement (ii) does not extend verbatim: for $\pi(X) = 3$ the coefficient conditions $a_{X\#/n} = -a_n$ leave four free coefficients, $f(0) = 0$ is then automatic, and the two remaining conditions $f'(0) = f''(0) = 0$ leave a two-dimensional space, so the analogue of (ii) is false as stated, and for larger X the defect $2^{\pi(X)-1} - \pi(X)$ only grows; since (iii) was proved through (ii), it is proved here only for the two primes $2, 3$. As $X \to \infty$ these functions tend to $1/\zeta$ in $\Re s > 1$ only; their axis of symmetry is $\Re s = 0$, not $\Re s = \frac12$, and their lists do not converge to the ordinates of ζ, which are not a superposition of one clock for each prime.

17.4. **The diagonal operator over all primes and the three abscissas.** On $\mathscr{L} := \ell^2(\{p \text{ prime}\})$ let $\mathsf{X}_{\mathfrak{P}}(s) := \operatorname{diag}(p^{-s})$. The closed paths of this model are the primes themselves, without scattering between them, and

$$\det\bigl(I - \mathsf{X}_{\mathfrak{P}}(s)\bigr) = \prod_p (1-p^{-s}) = \frac{1}{\zeta(s)} \qquad (\Re s > 1). \tag{17.20}$$

$\mathsf{X}_{\mathfrak{P}}(s)$ has norm $2^{-\sigma} < 1$ for $\sigma > 0$, it is of trace class if and only if $\sigma > 1$, and Hilbert–Schmidt if and only if $\sigma > \frac12$. With $\mathfrak{P}(s) := \sum_p p^{-s}$ for $\Re s > 1$, the regularized determinant is the function

$$D_2(s) := \det{}_2\bigl(I - \mathsf{X}_{\mathfrak{P}}(s)\bigr) = \prod_p (1-p^{-s})e^{p^{-s}} = \exp\Bigl(-\sum_{m\ge 2}\frac{1}{m}\,\mathfrak{P}(ms)\Bigr) \qquad (\Re s > \tfrac12), \tag{17.21}$$

holomorphic and without zeros in $\Re s > \frac12$ because $I - \mathsf{X}_{\mathfrak{P}}(s)$ is invertible there; the last expression converges for $\Re s > \frac12$ since $m\sigma > 1$ for $m \ge 2$. For $\Re s > 1$, where $\mathfrak{P}$ and the Euler product converge, $D_2(s) = e^{\mathfrak{P}(s)}/\zeta(s)$; in the strip $\frac12 < \Re s \le 1$ the primary object is D_2, and $e^{\mathfrak{P}}/\zeta$ is defined there only through it, since $\mathfrak{P}$ itself is not given by its series.

Proposition 17.4 (No common object of finite rank in the window of ζ)**.** *Let $X \ge 2$. There is no function $f \not\equiv 0$ that satisfies (H0) and the functional equation* (12.3) *and is an absolutely convergent Dirichlet series in $\Re s > 1$ supported on the integers all of whose prime factors are at most X.*

Proof. By Hamburger's theorem $f = C_H\zeta$, and the coefficients of ζ do not vanish at the integers with a prime factor larger than X; so $C_H = 0$. □

In the models of Subsections 17.2 and 17.3 three boundaries coincide: the boundary of the domain where the matrix is a strict contraction, the boundary of the domain of convergence of the Euler

product, and the axis of symmetry. The real zeros come from this coincidence. For (17.20) they are different:

$$\text{contraction: } \sigma > 0; \qquad \text{convergence of the determinant: } \sigma > 1; \qquad \text{axis of symmetry: } \sigma = \tfrac{1}{2}. \tag{17.22}$$

The contraction, together with the trace-class determinant identity (17.20) valid for $\sigma > 1$, proves that ζ has no zeros in $\sigma > 1$; the contraction together with the Hilbert–Schmidt property proves that $D_2 = \det_2(I - \mathsf{X}_{\mathfrak{P}})$ is holomorphic without zeros in $\sigma > \frac{1}{2}$, and $D_2 = e^{\mathfrak{P}}/\zeta$ there only in the sense that D_2 is the continuation of that quotient from $\sigma > 1$. Neither statement reaches the axis, and in the strip $\frac{1}{2} < \sigma \le 1$ the regularized determinant moves the question to the continuation of $\mathfrak{P}$ (equivalently of ζ). In the language of this paper, $\log \tilde{\zeta}_{\mathbb{Z},\lambda} = \sum_n \tilde{c}_n(\lambda)\mathsf{L}_n$ is the determinant (17.20) with real multiplicities $\tilde{c}_n(\lambda)$ at all integers, and the strip $\frac{1}{2} < \sigma \le 1$ is where the mean square of the cell masses (Theorem 14.11) and the continuation of the joining factor (Theorem 14.13) are used. By Proposition 17.4 the common object of the window of ζ has no exact finite-rank representative in the discrete category described there (approximation in a topology is not excluded); a unitary object of infinite rank from which both ζ_{Z1} and $\tilde{\zeta}_{\mathbb{Z},0}$ descend, as U does in Proposition 17.2, is not constructed in this paper. A model without ζ with the same separation of the three boundaries is a graph with infinitely many edges of lengths $\mathfrak{l}_j$ such that $\sum_j e^{-\mathfrak{l}_j y}$ diverges for $y \le y_0$, with $y_0 > 0$.

18. Route B for finite sections: compactness, uniqueness, certificates, and a Kadec–Avdonin criterion with blocks of variable length

This section treats route B for the midpoint RvM system in the notation of this paper. Its subsections are self-contained and use the preceding sections only through the results quoted by label. The numerical part (Section 18.8) is a computation and is not part of any proof.

18.1. **Introduction to the section.** This paper constructs a synthetic zeta system on the midpoint clock τ_k, defined by $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$, and reduces the remaining step to condition (D). Route B approaches (D) without constructing the zeta ordinates. One solves finite sections of the rung system (18.4), which imposes simultaneously a real critical spectrum and atomic arithmetic on $\{\log n\}$. If the residuals of these sections tend to zero in a fixed box, compactness yields an exact solution. Theorem 18.7 identifies every exact solution whose ordinates lie at bounded distance from the clock, with arbitrary real masses, as the list of zeta ordinates with the von Mangoldt masses, and shows that its existence implies the Riemann hypothesis. In this scheme the ordinates γ_k are the output of an inverse problem, not an input. Section 18.3 makes this precise: route B is equivalent to (D) (Corollary 18.11), and its only open piece is the existence of sections with vanishing residual. Section 18.3.1 gives the same equivalence with the von Mangoldt masses prescribed: the Laplace transform of the residual is identified (Lemma 18.14), and uniqueness then follows from Hadamard's theorem and the functional equation alone, so that route B′ is equivalent to the Riemann hypothesis. Section 18.3.2 shows that no finite stage decides route B: because the Riemann hypothesis is verified up to height $3 \cdot 10^{12}$, the zeta list with the von Mangoldt masses solves every finite set of tests up to an explicit error of order $(3 \cdot 10^{12})^{1-r}$, whether or not the hypothesis holds. Section 18.3.3 gives the existence part a certificate form: two computable quantities per stage, a right inverse of the Jacobian and the residual of the new tests, imply an exact solution of the stage. Section 18.3.4 shows that Gonek's symmetrised Euler products give explicit lists in a tube around the clock. Section 18.8 reports the computations that measure these quantities, and proves in ball arithmetic, with compactly supported tests, that the square sections of fifteen stages, up to height 487, have exact solutions (Theorem 18.78).

Section 18.5 proves an a priori bound for quasi-solutions. Section 18.6 proves a Kadec–Avdonin criterion with blocks of variable length, which is the tool needed for a Fredholm/stability formulation

of the linearised problem. Section 18.7 applies that criterion to lists near the clock and proves that, for the zeta list, the uniform version fails.

References of the form Theorem 16.6 and the other results of the preceding sections are used as stated there. Except in Proposition 18.20 and Section 18.7.3, where the zeta list is the object of study, and in the diagnostic columns of Section 18.8, nothing below uses the ordinates of ζ.

18.2. **Setting.** Throughout, $q(s) := s(s-1)$ and $G(s) := s(s-1)\pi^{-s/2}\Gamma(s/2)$. The smooth clock count is

$$\mathcal{N}_{\mathrm{RVM}}(t) := \frac{t}{2\pi}\log\frac{t}{2\pi e} + \frac{7}{8}, \qquad \mathcal{N}'_{\mathrm{RVM}}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi} \qquad (t>0). \tag{18.1}$$

It is strictly increasing on $[2\pi,\infty)$, with $\mathcal{N}_{\mathrm{RVM}}(2\pi) = -\frac{1}{8}$, and the clock points are the solutions $\tau_k > 2\pi$ of $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$, $k \ge 1$ ($\tau_1 = 14.5213\ldots$). We write $n_{\mathrm{RVM}}(t) := \#\{k \ge 1 : \tau_k \le t\}$. Since $\mathcal{N}_{\mathrm{RVM}}$ is increasing,

$$n_{\mathrm{RVM}}(t) = \big\lfloor \mathcal{N}_{\mathrm{RVM}}(t) + \tfrac{1}{2} \big\rfloor, \qquad |n_{\mathrm{RVM}}(t) - \mathcal{N}_{\mathrm{RVM}}(t)| \le \tfrac{1}{2} \qquad (t \ge 2\pi). \tag{18.2}$$

Definition 18.1 (Admissible lists)**.** For $C>0$, a list $\boldsymbol{\kappa} = (\kappa_k)_{k\ge1}$ is *C-admissible* if it is real and nondecreasing, $\kappa_1 \ge 2\pi$, and $|\kappa_k - \tau_k| \le C$ for all k. Put $N_{\boldsymbol{\kappa}}(t) := \#\{k : \kappa_k \le t\}$ (with multiplicity) and

$$P_{\boldsymbol{\kappa}}(s) := \prod_{k\ge1}\Big(1 + \frac{q(s)}{\frac14 + \kappa_k^2}\Big), \qquad \zeta_{\boldsymbol{\kappa}} := \frac{P_{\boldsymbol{\kappa}}}{G}. \tag{18.3}$$

The clock list $\boldsymbol{\tau} = (\tau_k)$ gives $\zeta_{\boldsymbol{\tau}} = \zeta_{\mathrm{RVM}}$.

$P_{\boldsymbol{\kappa}}$ is entire and real on $\mathbb{R}$, and $P_{\boldsymbol{\kappa}}(1-s) = P_{\boldsymbol{\kappa}}(s)$. Its zeros are exactly $\frac12 \pm i\kappa_k$, with multiplicity, and it is positive on $[-1,2]$, because $q([-1,2]) = [-\frac14, 2]$, so every factor satisfies $1 + q(s)/(\frac14+\kappa_k^2) \ge 1 - \frac{1/4}{\frac14+\kappa_k^2} > 0$ there, and the product converges. Writing $s = \frac12 + w$, one has $\frac14 + \kappa^2 + q(s) = \kappa^2 + w^2$.

The *rung system* 16.6 for a list $\boldsymbol{\kappa}$ and real masses $\mathsf{a} = (\mathsf{a}_n)_{n\ge2}$ is

$$\sum_{k\ge1}\cos(\kappa_k u) + \sum_{n\ge2}\mathsf{a}_n\,\delta(u - \log n) = \mathsf{k}(u) \qquad \text{in } \mathcal{D}'(0,\infty), \tag{18.4}$$

with the fixed distribution k of 15.79. Its residual is the distribution $\mathfrak{R}(\boldsymbol{\kappa},\mathsf{a})$ defined, for $\varphi \in C_c^\infty(0,\infty)$, by

$$\langle \mathfrak{R}(\boldsymbol{\kappa},\mathsf{a}), \varphi\rangle := \sum_{k\ge1}\mathsf{c}_\varphi(\kappa_k) + \sum_{n\ge2}\mathsf{a}_n\varphi(\log n) - \langle \mathsf{k}, \varphi\rangle, \qquad \mathsf{c}_\varphi(t) := \int_0^\infty \varphi(u)\cos(tu)\,du. \tag{18.5}$$

Two integrations by parts, without boundary terms because φ and φ' vanish at the ends of its support, give $|\mathsf{c}_\varphi(t)| \le \|\varphi''\|_{L^1}\,t^{-2}$. Since $\kappa_k \ge \tau_k - C$ and $\tau_k \gg k/\log(k+1)$, the series in (18.5) converges absolutely. The mass sum is finite, because $\operatorname{supp}\varphi \subset [u_1,u^*]$ with $0 < u_1 < u^*$ leaves only $n \le e^{u^*}$.

18.3. **Route B: compactness, selection and the canonical box.** For $C, \mathbb{M} > 0$ let $\mathbb{K}(C,\mathbb{M})$ be the set of pairs $(\boldsymbol{\kappa},\mathsf{a})$ with $\boldsymbol{\kappa}$ C-admissible and $|\mathsf{a}_n| \le \mathbb{M}$ for all n. A *finite section* is an element of $\mathbb{K}(C,\mathbb{M})$ with $\kappa_k = \tau_k$ for $k > k^*$ and $\mathsf{a}_n = 0$ for $n > n^*$, for some k^*, n^*.

Fix finite sets $\mathscr{T}_1 \subset \mathscr{T}_2 \subset \cdots \subset C_c^\infty(0,\infty)$ whose union $\mathscr{T}_\infty$ is dense in $C_c^\infty(0,\infty)$. Such sets exist because $C_c^\infty(0,\infty)$ is separable for its usual LF topology: it is the countable strict inductive limit of the separable Fréchet spaces $C^\infty_{[1/j,j]}$, and a countable union of countable dense subsets of these is dense. For $(\boldsymbol{\kappa},\mathsf{a}) \in \mathbb{K}(C,\mathbb{M})$ put

$$\mathsf{s}_m(\boldsymbol{\kappa},\mathsf{a}) := \max_{\varphi\in\mathscr{T}_m} \big|\langle \mathfrak{R}(\boldsymbol{\kappa},\mathsf{a}), \varphi\rangle\big|. \tag{18.6}$$

Definition 18.2 (Statement B)**.** *Statement $B(C,\mathbb{M})$* says that there are finite sections $(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}) \in \mathbb{K}(C,\mathbb{M})$ with $\mathsf{s}_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}) \to 0$.

Lemma 18.3 (Canonical box of the exact solution)**.** *Let* $\mathbb{M}_* := \max_{n\ge 2} \Lambda(n)/(2\sqrt{n})$*. Then* $\mathbb{M}_* = \log 7/(2\sqrt{7}) = 0.367742\ldots < 1/e$*. If condition (D) holds, the pair* $(\boldsymbol{\gamma}, \mathsf{a}^{\zeta})$*, with* $\mathsf{a}^{\zeta}_n := \Lambda(n)/(2\sqrt{n})$*, solves* (18.4) *and lies in* $\mathbb{K}(C_\eta, \mathbb{M}_*)$*, where* C_η *is the constant of 2.2.*

Proof. $x \mapsto \log x/(2\sqrt{x})$ increases on $[1, e^2]$ and decreases on $[e^2, \infty)$, with $e^2 = 7.389\ldots$. Hence among primes its maximum is at $p = 7$ or $p = 11$; the values are $0.367742\ldots$ and $0.361500\ldots$. A prime power p^j with $j \ge 2$ gives $\log p/(2p^{j/2}) < \log p/(2\sqrt{p}) \le \mathbb{M}_*$. The bound by $1/e$ is the continuous maximum, attained at $x = e^2$. Under (D), $\zeta_{Z1} = \zeta$, so all zeros of ζ are $\frac{1}{2} \pm i\gamma_k$. The rung system with list $\boldsymbol{\gamma}$ and masses a^{ζ} is then the distributional explicit formula 13.6. Finally, $|\gamma_k - \tau_k| \le C_\eta$ by 2.2, and $\gamma_1 = 14.13\ldots \ge 2\pi$. □

Corollary 18.4 ((D) implies B)**.** *If (D) holds, then* $B(C, \mathbb{M})$ *holds for every* $C \ge C_\eta$ *and* $\mathbb{M} \ge \mathbb{M}_*$*.*

Proof. Let $(\boldsymbol{\kappa}^{(k^*,n^*)}, \mathsf{a}^{(k^*,n^*)})$ be the truncation of $(\boldsymbol{\gamma}, \mathsf{a}^{\zeta})$. Its list is the nondecreasing rearrangement of the multiset $\{\gamma_1, \ldots, \gamma_{k^*}\} \cup \{\tau_{k^*+1}, \tau_{k^*+2}, \ldots\}$, and its masses are a^{ζ}_n for $n \le n^*$ and 0 beyond. By Lemma 18.23 (Section 18.3.3) the rearranged list satisfies $|\kappa_k - \tau_k| \le C_\eta$, coincides with (τ_k) beyond some index, and $\mathfrak{R}$ depends only on the multiset. So this is a finite section in $\mathbb{K}(C_\eta, \mathbb{M}_*)$. For fixed φ, $\langle \mathfrak{R}(\boldsymbol{\kappa}^{(k^*,n^*)}, \mathsf{a}^{(k^*,n^*)}), \varphi\rangle - \langle \mathfrak{R}(\boldsymbol{\gamma}, \mathsf{a}^{\zeta}), \varphi\rangle = \sum_{k>k^*} (\mathsf{c}_\varphi(\tau_k) - \mathsf{c}_\varphi(\gamma_k))$ as soon as $e^{u^*} \le n^*$. This tends to 0 as $k^* \to \infty$. Since $\mathscr{T}_m$ is finite and $\mathfrak{R}(\boldsymbol{\gamma}, \mathsf{a}^{\zeta}) = 0$ by Lemma 18.3, k^*_m, n^*_m can be chosen with $\mathsf{s}_m \le 1/m$. □

Proposition 18.5 (Compactness and passage to the limit)**.** *Let* $(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}) \in \mathbb{K}(C, \mathbb{M})$ *with* $\mathsf{s}_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}) \to 0$*. Then a subsequence converges coordinatewise to some* $(\boldsymbol{\kappa}, \mathsf{a}) \in \mathbb{K}(C, \mathbb{M})$*, and every such limit satisfies* $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}) = 0$ *in* $\mathcal{D}'(0, \infty)$*.*

Proof. Each $\kappa^{(m)}_k$ lies in the compact interval $[\tau_k - C, \tau_k + C]$ and each $\mathsf{a}^{(m)}_n$ in $[-\mathbb{M}, \mathbb{M}]$. A diagonal extraction over the countably many coordinates gives a coordinatewise convergent subsequence. The limit is nondecreasing, satisfies $\kappa_1 \ge 2\pi$, $|\kappa_k - \tau_k| \le C$ and $|\mathsf{a}_n| \le \mathbb{M}$, and therefore lies in $\mathbb{K}(C, \mathbb{M})$.

Continuity of each pairing. Fix φ with support in $[u_1, u^*]$. The mass part of (18.5) involves only $n \le e^{u^*}$ and converges. For the spectral part, $|\mathsf{c}_\varphi(\kappa^{(m)}_k)| \le \|\varphi''\|_1 (\tau_k - C)^{-2}$ when $\tau_k > 2C$, and $|\mathsf{c}_\varphi| \le \|\varphi\|_1$ for the finitely many other k. The majorant is summable, so dominated convergence for series gives $\sum_k \mathsf{c}_\varphi(\kappa^{(m)}_k) \to \sum_k \mathsf{c}_\varphi(\kappa_k)$. Hence $\langle \mathfrak{R}(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}), \varphi\rangle \to \langle \mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}), \varphi\rangle$ along the subsequence.

Vanishing. If $\varphi \in \mathscr{T}_\infty$, then $\varphi \in \mathscr{T}_m$ for all large m, so $|\langle \mathfrak{R}(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}), \varphi\rangle| \le \mathsf{s}_m \to 0$ and $\langle \mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}), \varphi\rangle = 0$. The limit $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a})$ is a single distribution, hence continuous on $C^\infty_c(0, \infty)$. It vanishes on the dense set $\mathscr{T}_\infty$, so it vanishes identically. No uniform bound on the residuals is needed for this step. □

Theorem 16.6 proves uniqueness for lists in $\mathscr{O}$. Besides the tube and the ordering, $\mathscr{O}$ (16.1) carries a third condition, $|\int (N_{\boldsymbol{\kappa}} - \mathcal{N}_{\mathrm{RVM}})| \le C \log \mathrm{X}$, and a coordinatewise limit of sections does not inherit it. The following lemma and theorem prove uniqueness for exact solutions in every tube, without the third condition and with no bound on the masses.

Lemma 18.6 (Mean square of the masses of an exact solution)**.** *Let* $\boldsymbol{\kappa}$ *be* C*-admissible for some* $C > 0$*, and let* $\mathsf{a} = (\mathsf{a}_n)_{n\ge 2}$ *be real numbers, with no bound assumed, such that* $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}) = 0$ *in* $\mathcal{D}'(0, \infty)$*. Then for every* $\epsilon > 0$ *there is* C_ϵ*, depending only on* ϵ *and* C*, such that*

$$\sum_{X < n \le 2X} \mathsf{a}_n^2 \le C_\epsilon X^\epsilon \qquad (X \ge 2). \tag{18.7}$$

Proof. Fix an even real $\psi \in C^\infty_c(-\frac{1}{5}, \frac{1}{5})$ with $\psi(0) = 1$, and put $\widehat{\psi}(\xi) := \int \psi(v) \cos(\xi v)\, dv$; then $|\widehat{\psi}(\xi)| \le \|\psi\|_1$, and A integrations by parts give $|\widehat{\psi}(\xi)| \le \|\psi^{(A)}\|_1 |\xi|^{-A}$ for every integer $A \ge 0$. Fix $X \ge 2$ and $\delta \in (0, 1]$, and put $T := X^{1+\delta}$. All implied constants below depend only on ψ, δ and C.

One test for each mass. For $X < n \le 2X$ let $\varphi_n(u) := \psi(X(u - \log n))$, supported in $|u - \log n| < 1/(5X)$. For an integer $m \ge 2$, $m \ne n$, one has $|\log m - \log n| \ge \min\big(\log \frac{n+1}{n}, \log \frac{n}{n-1}\big) \ge \frac{1}{n+1} \ge \frac{1}{2X+1} \ge \frac{1}{5X}$, so $\varphi_n(\log m) = 0$, while $\varphi_n(\log n) = 1$. Since ψ is even, the substitution $u = \log n + v/X$ gives $\mathsf{c}_{\varphi_n}(t) = X^{-1}\widehat{\psi}(t/X)\cos(t\log n)$. Pairing $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}) = 0$ with φ_n,

$$\begin{aligned} \mathsf{a}_n &= \langle \mathsf{k}, \varphi_n \rangle - \Re\mathcal{S}(\log n) - \mathcal{E}_n, \qquad \mathrm{y}_k := \frac{\widehat{\psi}(\kappa_k/X)}{X}, \\ \mathcal{S}(u) &:= \sum_{\kappa_k \le T} \mathrm{y}_k e^{i\kappa_k u}, \qquad \mathcal{E}_n := \sum_{\kappa_k > T} \mathrm{y}_k \cos(\kappa_k \log n). \end{aligned} \tag{18.8}$$

The term k. On the support of φ_n, $u \ge \log 2 - \frac{1}{10} > \frac{1}{2}$, where $|\mathsf{k}(u)| \le \cosh\frac{u}{2} + \frac{e^{-u/2}}{2(1-e^{-2u})} \le 2e^{u/2}$; hence $|\langle \mathsf{k}, \varphi_n \rangle| \le 2e^{1/10}\sqrt{n}\,\|\psi\|_1 X^{-1} \ll X^{-1/2}$.

Large ordinates. By the tube, $N_{\boldsymbol{\kappa}}(t) \le n_{\mathrm{RVM}}(t + C) \ll t\log t$ for $t \ge 2$. Let $A := \lceil 2/\delta \rceil + 1$, so that $\delta(A-1) \ge 2$. Partial summation gives $\sum_{\kappa_k > T} \kappa_k^{-A} \le A\int_T^\infty N_{\boldsymbol{\kappa}}(t) t^{-A-1} dt \ll T^{1-A}\log T$, hence

$$|\mathcal{E}_n| \le \|\psi^{(A)}\|_1 X^{A-1} \sum_{\kappa_k > T} \kappa_k^{-A} \ll X^{A-1}T^{1-A}\log T = X^{-\delta(A-1)}\log T \ll X^{-2}\log X.$$

Small ordinates. The intervals $I_n := [\log n - \frac{1}{6X}, \log n + \frac{1}{6X}]$, $X < n \le 2X$, are pairwise disjoint (consecutive points $\log n$ differ by at least $\frac{1}{2X+1} > \frac{1}{3X}$) and contained in $\mathcal{W} := [\log X - 1, \log X + 2]$. For $g \in C^1(I)$ on an interval I of length ℓ and $t_0 \in I$, $|g(t_0)|^2 \le |g(t)|^2 + 2\int_I |g||g'|$ for every $t \in I$; averaging over $t \in I$,

$$|g(t_0)|^2 \le \ell^{-1}\int_I |g|^2 + 2\int_I |g||g'|.$$

With $g = \mathcal{S}$, $I = I_n$, summing over n and using the Cauchy–Schwarz inequality,

$$\sum_{X < n \le 2X} |\mathcal{S}(\log n)|^2 \le 3X\int_{\mathcal{W}} |\mathcal{S}|^2 + 2\Big(\int_{\mathcal{W}} |\mathcal{S}|^2\Big)^{1/2}\Big(\int_{\mathcal{W}} |\mathcal{S}'|^2\Big)^{1/2}.$$

The proof of Lemma 14.10 uses only the tube (Theorem 16.1(v)); with C in place of C_η it gives $\int_{u_0-1}^{u_0+1} |\sum_{\kappa_k \le T} \mathrm{b}_k e^{i\kappa_k u}|^2 du \ll T\log^2 T$ for complex $|\mathrm{b}_k| \le 1$. The interval $\mathcal{W}$ is covered by two intervals of length 2. Applied with $\mathrm{b}_k = X\mathrm{y}_k/\|\psi\|_1$, and with $\mathrm{b}_k = X\kappa_k\mathrm{y}_k/(T\|\psi\|_1)$ for $\mathcal{S}'(u) = \sum_{\kappa_k \le T} i\kappa_k \mathrm{y}_k e^{i\kappa_k u}$, this gives $\int_{\mathcal{W}} |\mathcal{S}|^2 \ll X^{-2}T\log^2 T$ and $\int_{\mathcal{W}} |\mathcal{S}'|^2 \ll X^{-2}T^3\log^2 T$. Hence

$$\sum_{X < n \le 2X} |\mathcal{S}(\log n)|^2 \ll X^{-1}T\log^2 T + X^{-2}T^2\log^2 T \ll X^{2\delta}\log^2 X.$$

Conclusion. By (18.8), $\mathsf{a}_n^2 \le 3\big(\langle \mathsf{k}, \varphi_n \rangle^2 + |\mathcal{S}(\log n)|^2 + \mathcal{E}_n^2\big)$, and summing over $X < n \le 2X$ gives $\sum_{X < n \le 2X} \mathsf{a}_n^2 \ll 1 + X^{2\delta}\log^2 X + X^{-3}\log^2 X \ll X^{3\delta}$. Take $\delta = \epsilon/3$. □

Theorem 18.7 (Uniqueness in the tube, with free masses)**.** *Let $\boldsymbol{\kappa}$ be C-admissible for some $C > 0$, and let $\mathsf{a} = (\mathsf{a}_n)_{n \ge 2}$ be real numbers, with no bound assumed. If $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}) = 0$ in $\mathcal{D}'(0, \infty)$, then all nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$, $\boldsymbol{\kappa} = \boldsymbol{\gamma}$, $\mathsf{a} = \mathsf{a}^\zeta$, and (D) holds.*

Proof. Step 1: no drift, and an Euler form. We apply Proposition F.2 with $\mathsf{t} = 1$, for which (F.10) is the equation $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}) = 0$. Its hypotheses hold: $0 < 2\pi \le \kappa_1 \le \kappa_2 \le \cdots$ and $\kappa_k \ge \tau_k - C \to \infty$; the tube gives $N_{\boldsymbol{\kappa}}(T+1) - N_{\boldsymbol{\kappa}}(T) \le n_{\mathrm{RVM}}(T+1+C) - n_{\mathrm{RVM}}(T-C) \ll_C \log(2+T)$, which is (F.11); and Lemma 18.6 gives (F.12). Hence the constant b of (F.13) exists and $\kappa_k - \tau_k = -(4\mathrm{b} + o(1))\kappa_k/\log\kappa_k$. Since $|\kappa_k - \tau_k| \le C$, $\mathrm{b} = 0$, and (F.20) with $\mathsf{t} = 1$ and $\mathrm{b} = 0$ reads

$$\zeta_{\boldsymbol{\kappa}}(s) = \frac{P_{\boldsymbol{\kappa}}(s)}{G(s)} = C_1 \exp\Big(\sum_{n \ge 2} \mathsf{c}_n n^{-s}\Big), \qquad \mathsf{c}_n := \frac{2\sqrt{n}\,\mathsf{a}_n}{\log n} \qquad (\Re s > \tfrac{3}{2}), \tag{18.9}$$

with a constant $C_1 > 0$.

Step 2: abscissa 1. Let $\sigma > 1$ and $\epsilon := \sigma - 1$. By the Cauchy–Schwarz inequality and Lemma 18.6 with $\epsilon/2$ in place of ϵ,

$$\sum_{X<n\le 2X} |\mathsf{c}_n| n^{-\sigma} \le \Big(\sum_{X<n\le 2X} \mathsf{a}_n^2\Big)^{1/2} \Big(\sum_{X<n\le 2X} \frac{4n^{1-2\sigma}}{\log^2 n}\Big)^{1/2} \ll_{\sigma,C} X^{\epsilon/4} X^{1-\sigma} = X^{-3\epsilon/4},$$

which is summable over $X = 2^j$. So $\mathsf{D}(s) := \sum_{n\ge 2} \mathsf{c}_n n^{-s}$ converges absolutely on $\Re s > 1$, and so does the Dirichlet series $\exp(\mathsf{D}(s)) = \sum_{j\ge 0} \mathsf{D}(s)^j/j! = \sum_{n\ge 1} \mathsf{e}_n n^{-s}$, since $\sum_n |\mathsf{e}_n| n^{-\sigma} \le \exp(\sum_n |\mathsf{c}_n| n^{-\sigma})$. The function $\zeta_{\boldsymbol{\kappa}} = P_{\boldsymbol{\kappa}}/G$ is holomorphic on $\Re s > 1$, where G has no zeros; by analytic continuation (18.9) holds on $\Re s > 1$, and $\zeta_{\boldsymbol{\kappa}}$ satisfies (H1) of Subsection 14.7.

Step 3: Hamburger's theorem. (H0): $(s-1)\zeta_{\boldsymbol{\kappa}}(s) = P_{\boldsymbol{\kappa}}(s)\,\pi^{s/2}/(2\Gamma(1+\frac{s}{2}))$ is entire of finite order, because $P_{\boldsymbol{\kappa}}$ has order at most one (first step of the proof of Proposition F.2) and $\pi^{s/2}$ and $1/\Gamma$ have order one. (H2), with $g = f = \zeta_{\boldsymbol{\kappa}}$: $\zeta_{\boldsymbol{\kappa}}(s)\Gamma(\frac{s}{2})\pi^{-s/2} = P_{\boldsymbol{\kappa}}(s)/(s(s-1))$ is invariant under $s \mapsto 1-s$. (H3) follows from (H1) for $g = f$. Hamburger's theorem gives $\zeta_{\boldsymbol{\kappa}} = C_H \zeta$. Since $P_{\boldsymbol{\kappa}}(1) = 1$ and $G(s) = (s-1)(1+O(s-1))$ (because $\pi^{-1/2}\Gamma(\frac{1}{2}) = 1$), $\zeta_{\boldsymbol{\kappa}}$ has residue 1 at $s = 1$, as ζ has, and $C_H = 1$.

Step 4: identification. The nontrivial zeros of $\zeta = P_{\boldsymbol{\kappa}}/G$ are the zeros of $P_{\boldsymbol{\kappa}}$, namely $\frac{1}{2} \pm i\kappa_k$ with multiplicity. So they all lie on $\Re s = \frac{1}{2}$, and the nondecreasing list of their positive ordinates, with multiplicity, is $\boldsymbol{\kappa}$: $\boldsymbol{\kappa} = \boldsymbol{\gamma}$, and $\zeta_{Z1} = \zeta_{\boldsymbol{\gamma}} = \zeta$, which is (D). Letting $\sigma \to +\infty$ in (18.9) gives $C_1 = 1$. Then D and $\log\zeta = \sum_n \frac{\Lambda(n)}{\log n} n^{-s}$ are continuous real logarithms of ζ on $(1,\infty)$ that tend to 0 at $+\infty$, so they coincide there, and the uniqueness of the coefficients of an absolutely convergent Dirichlet series gives $\mathsf{c}_n = \Lambda(n)/\log n$, that is, $\mathsf{a}_n = \Lambda(n)/(2\sqrt{n}) = \mathsf{a}_n^{\zeta}$. □

Remark 18.8 (What Theorem 18.7 uses). Theorem 18.7 contains the uniqueness half of Theorem 16.6, for every list at bounded distance from the clock and with no assumption on the masses: neither the third condition of (16.1) nor the threshold $\kappa_1 \ge \tau_0^+$ is used. In the proof of Theorem 16.6 the third condition enters through Theorem 16.1(ii), (iii) and (vi): the bounds of $\log\zeta_{\boldsymbol{\kappa}}$ on vertical lines, the Perron argument and the mean square of the cell masses. For an exact solution in a tube these inputs are replaced by the Laplace identity (F.18), which such a solution satisfies (Proposition F.2, whose mass hypothesis is supplied by Lemma 18.6), and by Lemma 18.6 itself, in which each mass is isolated by one test of width $1/X$ and the remaining exponential sum over the ordinates is controlled by the local mean value of Lemma 14.10. The tube enters twice: it gives the local density used in Lemma 18.6, and it forces $\mathrm{b} = 0$. Proposition F.2 alone does not exclude an exact solution with a linear drift, $\kappa_k - \tau_k \sim -4\mathrm{b}\kappa_k/\log\kappa_k$ (whether such solutions exist is not known); the tube excludes it. The last two steps are the classical route through Hamburger's theorem; what is specific to the rung system is that exactness in $\mathcal{D}'(0,\infty)$ supplies both the Euler form (18.9) and its abscissa.

Corollary 18.9 (Uniqueness in the box)**.** *For all $C, \mathbb{M} > 0$, the only possible zero of $\mathfrak{R}$ in $\mathbb{K}(C,\mathbb{M})$ is $(\boldsymbol{\gamma}, \mathsf{a}^{\zeta})$, and $\mathfrak{R}$ has a zero in $\mathbb{K}(C,\mathbb{M})$ only if the Riemann hypothesis holds.*

Proof. Theorem 18.7, since $|\mathsf{a}_n| \le \mathbb{M}$ is not used there. □

Corollary 18.10 (Selection of the zeta list)**.** *Assume $B(C,\mathbb{M})$, and let $(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)})$ be any sequence in $\mathbb{K}(C,\mathbb{M})$ with $\mathsf{s}_m \to 0$. Then the whole sequence converges coordinatewise to $(\boldsymbol{\gamma}, \mathsf{a}^{\zeta})$, and (D) and the Riemann hypothesis hold.*

Proof. By Proposition 18.5, every subsequence has a further subsequence converging to a zero of $\mathfrak{R}$ in $\mathbb{K}(C,\mathbb{M})$. By Theorem 18.7 that zero is $(\boldsymbol{\gamma}, \mathsf{a}^{\zeta})$, and (D) and the Riemann hypothesis hold. The set $\mathbb{K}(C,\mathbb{M})$ is compact and metrizable in the product topology, and in a compact metric space a sequence all of whose convergent subsequences have the same limit converges to it; so the whole sequence converges to $(\boldsymbol{\gamma}, \mathsf{a}^{\zeta})$. The last step can also be read directly from the limit: $\mathfrak{R}(\boldsymbol{\gamma}, \mathsf{a}^{\zeta}) = 0$ implies (D), as follows. By Lemma 18.14 (Section 18.3.1), $\widehat{\mathsf{R}}$ of (18.40) is the Laplace

transform of a distribution of finite order on $[0,\infty)$ whose restriction to $(0,\infty)$ is $\mathfrak{R}$. If $\mathfrak{R}=0$, that distribution is supported at $\{0\}$, so $\widehat{\mathsf{R}}$ is a polynomial $p(s)$. Since $\sum_n \mathsf{a}_n^{\zeta} n^{\frac12-s} = -\frac12\zeta'/\zeta(s)$, this reads $P_\gamma'/P_\gamma - (G\zeta)'/(G\zeta) = 2p$, so $P_\gamma = c\,G\zeta\,e^{Q}$ with $Q'=2p$. Here $G\zeta = 2\xi$ and P_γ are entire of order 1, so $e^{Q} = P_\gamma/(2c\,\xi)$ is entire, zero-free and of order at most 1, and $\deg Q \le 1$. Both P_γ and ξ are invariant under $s\mapsto 1-s$, so $e^{Q(1-s)} = e^{Q(s)}$, which forces Q to be constant. Hence $\zeta_\gamma = c'\zeta$. Both functions have residue 1 at $s=1$ ($P_\gamma(1)=1$ and $G(s)\sim s-1$), so $c'=1$. This is (D), because $\zeta_\gamma = \zeta_{Z1}$. □

Corollary 18.11 (Route B is equivalent to (D))**.** *(D) holds if and only if $B(C,\mathbb{M})$ holds for some $C,\mathbb{M}>0$; and then $B(C,\mathbb{M})$ holds for all $C\ge C_\eta$ and $\mathbb{M}\ge\mathbb{M}_*$.*

Proof. Corollaries 18.10 and 18.4. □

Remark 18.12 (Logical status of route B). By Corollary 18.11, route B with free masses is an equivalence, $\mathrm{B}(C,\mathbb{M}) \Leftrightarrow$(D) for $C\ge C_\eta$ and $\mathbb{M}\ge\mathbb{M}_*$, with a single open piece, the existence statement B, which is therefore not weaker than (D). Compactness is not an issue in the box: Proposition 18.5 uses only the imposed bounds; and uniqueness holds in every tube, with no bound on the masses (Theorem 18.7). Proposition 18.20 shows that no finite stage of B decides it. In this scheme the ordinates γ_k never enter the construction; they appear only as the unique admissible zero of $\mathfrak{R}$. Lemma 18.3 does *not* show that near-solutions of sections lie in $\mathbb{K}(C_\eta,\mathbb{M}_*)$: it locates only the exact solution. Restricting the sections to that box is a choice, justified a posteriori by Corollary 18.4.

18.3.1. *Route B with prescribed von Mangoldt masses.* The masses $\mathsf{a}_n^{\zeta} = \Lambda(n)/(2\sqrt{n})$ are known and do not depend on (D) or on the ordinates of ζ. Prescribing them leaves the ordinates as the only unknowns, and uniqueness then follows from Hadamard's factorisation and the functional equation alone, without Lemma 18.6 and Hamburger's theorem. The tool is the identification of the Laplace transform of the residual.

Laplace transforms of distributions. Let $\mathscr{T}\in\mathcal{D}'(\mathbb{R})$ have support in $[0,\infty)$ and suppose that $e^{-au}\mathscr{T}$ is a tempered distribution for some $a\in\mathbb{R}$. For $\Re s > a+\frac12$ put $z := s-\frac12$ and

$$\mathcal{L}\mathscr{T}(s) := \big\langle \mathscr{T},\ \mathsf{j}(u)\,e^{-zu}\big\rangle, \tag{18.10}$$

where $\mathsf{j}\in C^\infty(\mathbb{R})$ equals 1 on $[-1,\infty)$ and 0 on $(-\infty,-2]$; the value does not depend on j, and $\mathcal{L}\mathscr{T}$ is holomorphic there [65, Ch. 8]. The shift by $\frac12$ in z is the normalisation of (18.40). Since j' vanishes on the support of $\mathscr{T}$, $\mathcal{L}(\partial_u\mathscr{T})(s) = z\,\mathcal{L}\mathscr{T}(s)$; also $\mathcal{L}\delta_0 = 1$, $\mathcal{L}\delta_0' = z$, and for a locally integrable g on $[0,\infty)$ with $|g(u)|\le ce^{au}$, $\mathcal{L}(\mathfrak{g}\mathbf{1}_+)(s) = \int_0^\infty e^{-zu}g(u)\,du$. We write $\mathbf{1}_+ := \mathbf{1}_{[0,\infty)}$ and $\operatorname{fp}\frac{\mathbf{1}_+}{u}$ for the distribution $\varphi\mapsto\int_0^1\frac{\varphi(u)-\varphi(0)}{u}\,du + \int_1^\infty\frac{\varphi(u)}{u}\,du$, of order 1, supported in $[0,\infty)$, with restriction $1/u$ to $(0,\infty)$.

Lemma 18.13 (The Archimedean distribution)**.** *Let $\mathfrak{g}(u) := \frac{e^{-u/2}}{e^{2u}-1} - \frac{1}{2u}$ for $u>0$; $\mathfrak{g}$ is bounded, $\mathfrak{g}(0^+) = -\frac34$, and $\mathfrak{g}(u) = -\frac{1}{2u} + O(e^{-5u/2})$ as $u\to\infty$, in particular bounded. Put $\mathrm{c}_{\mathscr{K}} := -\frac14\log(2\pi e^{\gamma_E})$, γ_E Euler's constant, and*

$$\mathscr{K} := \tfrac12 e^{u/2}\mathbf{1}_+ - \tfrac12\,\mathfrak{g}\,\mathbf{1}_+ - \tfrac14\operatorname{fp}\frac{\mathbf{1}_+}{u} + \mathrm{c}_{\mathscr{K}}\,\delta_0. \tag{18.11}$$

Then $\mathscr{K}$ is a distribution of order 1 supported in $[0,\infty)$, its restriction to $(0,\infty)$ is the function k of 15.79, and

$$\mathcal{L}\mathscr{K}(s) = \tfrac12\,\frac{G'}{G}(s) \qquad (\Re s>1). \tag{18.12}$$

Proof. On $(0,\infty)$, $\frac12 e^{u/2} - \frac12\mathfrak{g}(u) - \frac{1}{4u} = \frac12\big(e^{u/2} - \frac{e^{-u/2}}{e^{2u}-1}\big) = \mathsf{k}(u)$. The expansion $\frac{e^{-u/2}}{e^{2u}-1} = \frac{1}{2u} - \frac34 + O(u)$ gives the behaviour of $\mathfrak{g}$ at 0. With $z = s-\frac12$ and $\Re s>1$: $\mathcal{L}(\frac12 e^{u/2}\mathbf{1}_+) = \frac{1}{2(z-\frac12)} = \frac{1}{2(s-1)}$. Gauss's

integral $\frac{\Gamma'}{\Gamma}(x)=\int_0^\infty(\frac{e^{-t}}{t}-\frac{e^{-xt}}{1-e^{-t}})dt$ ($\Re x>0$) for the logarithmic derivative of Γ [64, Ch. XII], with $t=2u$ and $x=\frac{s}{2}$, reads

$$\tfrac12\frac{\Gamma'}{\Gamma}(\tfrac{s}{2})=\int_0^\infty\Big(\frac{e^{-2u}}{2u}-\frac{e^{-su}}{1-e^{-2u}}\Big)du=\int_0^\infty\Big(\frac{e^{-2u}}{2u}-e^{-zu}\frac{e^{-u/2}}{e^{2u}-1}\Big)du-\frac{1}{s},\tag{18.13}$$

because $\frac{e^{-su}}{1-e^{-2u}}=e^{-zu}(\frac{e^{-u/2}}{e^{2u}-1}+e^{-u/2})$ and $\int_0^\infty e^{-zu}e^{-u/2}du=\frac1s$. Writing $\frac{e^{-u/2}}{e^{2u}-1}=\mathfrak{g}(u)+\frac{1}{2u}$ and using Frullani's integral $\int_0^\infty\frac{e^{-2u}-e^{-zu}}{u}\,du=\log\frac{z}{2}$ (both sides vanish at $z=2$ and have derivative $\int_0^\infty e^{-zu}du=1/z$ in z, for $\Re z>0$), the integral in (18.13) equals $-\mathcal{L}(\mathfrak{g}\mathbf{1}_+)(s)+\frac12\log\frac{z}{2}$. Hence

$$\tfrac12\frac{G'}{G}(s)=\frac{1}{2s}+\frac{1}{2(s-1)}-\frac14\log\pi+\frac14\frac{\Gamma'}{\Gamma}(\tfrac{s}{2})=\mathcal{L}(\tfrac12e^{u/2}\mathbf{1}_+)(s)-\tfrac12\mathcal{L}(\mathfrak{g}\mathbf{1}_+)(s)+\tfrac14\log\frac{z}{2}-\tfrac14\log\pi.\tag{18.14}$$

Finally $\mathcal{L}(\mathrm{fp}\,\frac{\mathbf{1}_+}{u})(s)=\int_0^1\frac{e^{-zu}-1}{u}\,du+\int_1^\infty\frac{e^{-zu}}{u}\,du=-\gamma_E-\log z$: the left side has derivative $-\int_0^\infty e^{-zu}du=-1/z$ in z, and its value at $z=1$ is $\int_0^1\frac{e^{-u}-1}{u}\,du+\int_1^\infty\frac{e^{-u}}{u}\,du=-\gamma_E$, the classical integral representation of Euler's constant [64, Ch. XII]. Substituting, $\frac14\log\frac{z}{2}=-\frac14\mathcal{L}(\mathrm{fp}\,\frac{\mathbf{1}_+}{u})-\frac14(\gamma_E+\log2)$, and (18.14) becomes (18.12) with the constant $\mathrm{c}_{\mathscr{K}}$. (The identity was also checked numerically to 25 digits at four points s; file `laplace_check.py`.) □

Lemma 18.14 (Laplace transform of the residual)**.** *Let $\boldsymbol{\kappa}$ be C-admissible and put $\mathfrak{H}_{\boldsymbol{\kappa}}(u):=\sum_k\kappa_k^{-2}\cos(\kappa_ku)$, which converges absolutely and uniformly on $\mathbb{R}$ since $\mathfrak{H}_{\boldsymbol{\kappa}}(0)=\sum_k\kappa_k^{-2}<\infty$, and*

$$\mathscr{S}_{\boldsymbol{\kappa}}:=-\partial_u^2(\mathfrak{H}_{\boldsymbol{\kappa}}\mathbf{1}_+)+\mathfrak{H}_{\boldsymbol{\kappa}}(0)\,\delta_0',\qquad\mathscr{A}_\zeta:=\sum_{n\ge2}\mathsf{a}_n^\zeta\,\delta_{\log n},\qquad\mathscr{E}_{\boldsymbol{\kappa}}:=\mathscr{S}_{\boldsymbol{\kappa}}+\mathscr{A}_\zeta-\mathscr{K}.\tag{18.15}$$

Then $\mathscr{E}_{\boldsymbol{\kappa}}$ is a distribution of local order at most 2 (its restriction to every bounded interval is of order ≤2; the support is not compact, since $\mathscr{A}_\zeta$ is a locally finite measure on $[\log2,\infty)$), supported in $[0,\infty)$, Laplace-transformable on $\Re s>1$, whose restriction to $(0,\infty)$ is $\mathfrak{R}(\boldsymbol{\kappa},\mathsf{a}^\zeta)$, and for $\Re s>1$

$$\mathcal{L}\mathscr{E}_{\boldsymbol{\kappa}}(s)=\widehat{\mathsf{R}}(s)=\tfrac12\Big(\frac{P_{\boldsymbol{\kappa}}'}{P_{\boldsymbol{\kappa}}}-\frac{\xi'}{\xi}\Big)(s),\qquad\xi:=\tfrac12G\zeta,\tag{18.16}$$

where $\widehat{\mathsf{R}}$ is (18.40) *with the masses a^ζ for all n. The same holds for a finite section $(\boldsymbol{\kappa},\mathsf{a})$ with $\mathscr{A}_\zeta$ replaced by $\sum_{n\le n^*}\mathsf{a}_n\delta_{\log n}$ and $-\frac12\xi'/\xi$ by $-\frac12G'/G+\sum_{n\le n^*}\mathsf{a}_nn^{\frac12-s}$.*

Proof. Spectral part. $\kappa_k\ge\tau_k-C\gg k/\log(k+1)$, so $\mathfrak{H}_{\boldsymbol{\kappa}}(0)<\infty$ and $\mathfrak{H}_{\boldsymbol{\kappa}}$ is continuous and bounded by $\mathfrak{H}_{\boldsymbol{\kappa}}(0)$. In $\mathcal{D}'(0,\infty)$ a uniformly convergent series may be differentiated termwise, so $-\mathfrak{H}_{\boldsymbol{\kappa}}''=\sum_k\cos(\kappa_ku)$ there; the term $\mathfrak{H}_{\boldsymbol{\kappa}}(0)\delta_0'$ is supported at 0. Hence $\mathscr{S}_{\boldsymbol{\kappa}}|_{(0,\infty)}=\sum_k\cos(\kappa_ku)$, and $\mathscr{S}_{\boldsymbol{\kappa}}$ has order at most 2. The Laplace transform of a distribution $\mathscr{T}$ supported in $[0,\infty)$ is $\mathcal{L}\mathscr{T}(s):=\langle\mathscr{T},\chi e^{-zu}\rangle$, $z=s-\frac12$, with $\chi\in C^\infty(\mathbb{R})$ equal to 1 on a neighbourhood of the support and to 0 on $(-\infty,-1]$; thus $\mathcal{L}(\delta_0')(s)=-\frac{d}{du}e^{-zu}\big|_{u=0}=z$ and $\mathcal{L}(\mathbf{1}_+f)(s)=\int_0^\infty f(u)e^{-zu}du$. For $\Re z>0$, Fubini (the double integral $\sum_k\kappa_k^{-2}\int_0^\infty e^{-\Re z\,u}du$ is finite) gives $\mathcal{L}(\mathfrak{H}_{\boldsymbol{\kappa}}\mathbf{1}_+)(s)=\sum_k\kappa_k^{-2}\frac{z}{z^2+\kappa_k^2}$, so

$$\mathcal{L}\mathscr{S}_{\boldsymbol{\kappa}}(s)=-z^2\sum_k\frac{z}{\kappa_k^2(z^2+\kappa_k^2)}+z\sum_k\frac{1}{\kappa_k^2}=\sum_k\frac{z}{z^2+\kappa_k^2}=\tfrac12\frac{P_{\boldsymbol{\kappa}}'}{P_{\boldsymbol{\kappa}}}(s),\tag{18.17}$$

because $\frac{z}{\kappa^2}-\frac{z^3}{\kappa^2(z^2+\kappa^2)}=\frac{z}{z^2+\kappa^2}$, and $P_{\boldsymbol{\kappa}}=\prod_k\frac{z^2+\kappa_k^2}{\frac14+\kappa_k^2}$ converges locally uniformly, so its logarithmic derivative is the termwise sum $\sum_k\frac{2z}{z^2+\kappa_k^2}$.

Arithmetic part. On $[0,u^*]$ only $n\le e^{u^*}$ contribute, so $\mathscr{A}_\zeta$ is a locally finite measure, of order 0, and $e^{-au}\mathscr{A}_\zeta$ is a finite measure for $a>\frac12$. For $\Re s>1$, $\mathcal{L}\mathscr{A}_\zeta(s)=\sum_n\frac{\Lambda(n)}{2\sqrt{n}}n^{-z}=\frac12\sum_n\Lambda(n)n^{-s}=-\frac12\frac{\zeta'}{\zeta}(s)$.

Assembly. By Lemma 18.13, $\mathcal{L}\mathscr{E}_{\boldsymbol{\kappa}} = \frac{1}{2}P'_{\boldsymbol{\kappa}}/P_{\boldsymbol{\kappa}} - \frac{1}{2}\zeta'/\zeta - \frac{1}{2}G'/G$, and $(G\zeta)'/(G\zeta) = \xi'/\xi$ since $\xi = \frac{1}{2}G\zeta$. The restriction to $(0,\infty)$ is $\sum_k \cos(\kappa_k u) + \sum_n \mathsf{a}_n^{\zeta}\delta(u-\log n) - \mathsf{k}(u)$, which is (18.5). The finite-section case is identical with a finite sum. □

Remark 18.15 (The boundary term). Without the correction $\mathfrak{H}_{\boldsymbol{\kappa}}(0)\delta'_0$ in (18.15), the transform of $-\partial_u^2(\mathfrak{H}_{\boldsymbol{\kappa}}\mathbf{1}_+)$ is $\frac{1}{2}P'_{\boldsymbol{\kappa}}/P_{\boldsymbol{\kappa}} - (s-\frac{1}{2})\mathfrak{H}_{\boldsymbol{\kappa}}(0)$: by (18.17), $\mathcal{L}(-\partial_u^2(\mathfrak{H}_{\boldsymbol{\kappa}}\mathbf{1}_+)) = -z^2\mathcal{L}(\mathfrak{H}_{\boldsymbol{\kappa}}\mathbf{1}_+)$, which differs from $\frac{1}{2}P'_{\boldsymbol{\kappa}}/P_{\boldsymbol{\kappa}}$ by $-z\mathfrak{H}_{\boldsymbol{\kappa}}(0) = -\mathcal{L}(\mathfrak{H}_{\boldsymbol{\kappa}}(0)\delta'_0)$, the transform of the boundary term that the second derivative of the cut-off function produces at $u=0$. Either normalisation gives the same restriction to $(0,\infty)$ and differs by a polynomial in s, which is harmless in the argument below. Both $\mathscr{K}$ and $\mathscr{S}_{\boldsymbol{\kappa}}$ carry singular terms at $u=0$; this is why the rung system is posed in $\mathcal{D}'(0,\infty)$ and not on $[0,\infty)$.

Proposition 18.16 (Uniqueness with prescribed masses)**.** *Let $\boldsymbol{\kappa}$ be C-admissible for some C, and suppose $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}^{\zeta}) = 0$ in $\mathcal{D}'(0,\infty)$. Then all nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$, and $\boldsymbol{\kappa} = \boldsymbol{\gamma}$.*

Proof. By Lemma 18.14, $\mathscr{E}_{\boldsymbol{\kappa}}$ is a distribution of finite order supported in $[0,\infty)$ whose restriction to $(0,\infty)$ vanishes, so it is supported at $\{0\}$. A distribution of finite order supported at a point is a finite linear combination of δ_0 and its derivatives [21, Thm. 2.3.4], so $\widehat{\mathsf{R}} = \mathcal{L}\mathscr{E}_{\boldsymbol{\kappa}}$ is a polynomial p in $z = s - \frac{1}{2}$. By (18.16), $(P_{\boldsymbol{\kappa}}e^{-Q}/\xi)' = 0$ on $\Re s > 1$ with $Q' = 2p$, Q a polynomial. Hence $P_{\boldsymbol{\kappa}} = c\,\xi\,e^{Q}$ on $\Re s > 1$ with $c \neq 0$, and, both sides being entire, the identity holds on all of $\mathbb{C}$. Only now is the quotient discussed: the identity shows that $P_{\boldsymbol{\kappa}}$ and ξ have the same zeros with the same multiplicities, since e^{Q} has none. In the variable $w = s - \frac{1}{2}$, $P_{\boldsymbol{\kappa}}$ is a constant multiple of $\prod_k(1 + w^2/\kappa_k^2)$, and $N_{\boldsymbol{\kappa}}(t) \ll t\log t$, so $P_{\boldsymbol{\kappa}}$ has order 1; ξ has order 1. Comparing the Hadamard factorizations $P_{\boldsymbol{\kappa}} = e^{A_1+B_1 s}\Pi(s)$ and $\xi = e^{A_2+B_2 s}\Pi(s)$, with the same canonical product Π over the common zeros, gives $e^{Q} = c^{-1}e^{(A_1-A_2)+(B_1-B_2)s}$, so $\deg Q \leq 1$. Since $P_{\boldsymbol{\kappa}}(1-s) = P_{\boldsymbol{\kappa}}(s)$ and $\xi(1-s) = \xi(s)$, $e^{Q(1-s)} = e^{Q(s)}$, that is, $Q(1-s) - Q(s) \in 2\pi i\mathbb{Z}$ for every s; the left side is continuous, hence constant, and for $Q(s) = a + bs$ it equals $b(1-2s)$, so $b = 0$. Therefore $P_{\boldsymbol{\kappa}} = c'\xi$. The zeros of $P_{\boldsymbol{\kappa}}$ are $\frac{1}{2} \pm i\kappa_k$, so every zero of ξ, that is, every nontrivial zero of ζ, has real part $\frac{1}{2}$, and the ordinates counted with multiplicity are the κ_k. Both lists are nondecreasing, so $\boldsymbol{\kappa} = \boldsymbol{\gamma}$. □

Definition 18.17 (Statement B′)**.** *Statement $B'(C)$* says that there are C-admissible lists $\boldsymbol{\kappa}^{(m)}$, each equal to the clock beyond some finite index, with $\mathsf{s}_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{\zeta}) \to 0$.

Corollary 18.18 (Route B′)**.** (a) *If $B'(C)$ holds for some C, then every sequence as in Definition 18.17 converges coordinatewise to $\boldsymbol{\gamma}$, and all nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$.*

(b) *If all nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$, then $B'(C)$ holds for every $C \geq C_\eta$.*

Hence the Riemann hypothesis is equivalent to $B'(C)$ for some $C < \infty$.

Proof. (a) $|\mathsf{a}_n^{\zeta}| \leq \mathbb{M}_*$ by the elementary computation in Lemma 18.3, so $(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{\zeta}) \in \mathbb{K}(C, \mathbb{M}_*)$. By Proposition 18.5, every subsequence has a further subsequence converging coordinatewise to some $(\boldsymbol{\kappa}, \mathsf{a}^{\zeta})$ with $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}^{\zeta}) = 0$ and $\boldsymbol{\kappa}$ C-admissible; the masses do not move. By Proposition 18.16, $\boldsymbol{\kappa} = \boldsymbol{\gamma}$ and the zeros lie on the line. The product $\prod_k[\tau_k - C, \tau_k + C]$ is compact and metrizable, for instance by $\mathrm{d}(\boldsymbol{\kappa}, \boldsymbol{\kappa}') = \sum_k 2^{-k}\min(1, |\kappa_k - \kappa'_k|)$, which induces the product topology; in a compact metric space a sequence all of whose convergent subsequences have the same limit converges to it. Hence the whole sequence converges. (b) Under this hypothesis the rung system with $(\boldsymbol{\gamma}, \mathsf{a}^{\zeta})$ is the distributional explicit formula 13.6, and $|\gamma_k - \tau_k| \leq C_\eta$ by 2.2. The truncations in the proof of Corollary 18.4, with the masses kept equal to a^{ζ}, give the sections. □

Remark 18.19 (What prescribing the masses changes, and what it does not). With prescribed masses, uniqueness for every list at bounded distance from the clock follows from Hadamard's theorem and the functional equation; with free masses it needs in addition Lemma 18.6 and Hamburger's theorem (Theorem 18.7). Routes B and B′ have the same single open piece, existence, and each is equivalent to the Riemann hypothesis (Corollaries 18.11 and 18.18). With prescribed masses the

whole window $(0,u^*)$ constrains the ordinates; the part above $\log\Omega$ can no longer be absorbed by the masses, as it could in Heuristic 18.55.

The equivalence itself is classical in substance. The prime side of the rung system is the prime side of the explicit formula of Riemann and von Mangoldt in the form of Guinand and Weil [20, 63], and the uniqueness is Hadamard's factorisation. Arias de Reyna [2] proves that the Riemann hypothesis is equivalent to the temperedness of the measure $-\sum_n \frac{\Lambda(n)}{\sqrt{n}}(\delta_{\log n}+\delta_{-\log n})+2\cosh(x/2)\,dx$, the same prime side up to the factor 2 and the symmetrisation, and identifies its Fourier transform with $\sum_\gamma \delta_{\gamma/2\pi}-2\vartheta'(2\pi t)\,dt$ under that hypothesis; by partial summation his criterion is von Koch's $\sum_{n\le x}\Lambda(n)=x+O(x^{1/2}\log^2 x)$. Statement B′ is the finite-section form of the same equivalence: it asks for a real spectrum in a tube around the clock that reproduces the prime side against a dense family of tests. It is not a weaker intermediate statement, and the finite sections are not more constrained than the global statement; what they offer is a formulation in which each stage is a finite-dimensional problem with computable certificates (Section 18.3.3) and explicit candidates (Section 18.3.4). Dyson's suggestion that the zeros form a one-dimensional quasicrystal, and the classification of crystalline measures by Kurasov and Sarnak [46] and Olevskii and Ulanovskii [51], concern the same object; the zeta list is not in those classes because of its density, as Arias de Reyna points out.

18.3.2. *Finite stages do not decide route B.* Let $r\ge 2$. For f with compact support in $(0,u^*]$ such that $\mathsf{f},\dots,\mathsf{f}^{(r-1)}$ are absolutely continuous and $\mathsf{f}^{(r)}\in L^1$, the residual is defined by (18.5), since $|\mathsf{c}_\mathsf{f}(t)|\le\|\mathsf{f}''\|_1 t^{-2}$, and we put

$$\mathfrak{W}_r(\mathsf{f}):=(1+u^{*2})\cosh\tfrac{u^*}{2}\sum_{j=0}^{r}\binom{r}{j}\|\mathsf{f}^{(r-j)}\|_{L^1}. \tag{18.18}$$

Here $\boldsymbol{\gamma}$ is the nondecreasing list of the ordinates of all nontrivial zeros of ζ with positive imaginary part, with multiplicity, and $d_k=\Re\rho_k-\frac12$ as in (5.15); no hypothesis on the zeros is made.

Proposition 18.20 (The zeta list solves every finite stage up to the verified height)**.** *Let $H_0:=3\cdot10^{12}$. For every $r\ge2$ and every real f as above,*

$$\big|\langle\Re(\boldsymbol{\gamma},\mathsf{a}^\zeta),\mathsf{f}\rangle\big|\le\frac14\,\mathfrak{W}_r(\mathsf{f})\sum_{k:\,d_k\neq0}\gamma_k^{-r}\le\frac{r}{4(r-1)}\,\mathfrak{W}_r(\mathsf{f})\,H_0^{1-r}\log H_0. \tag{18.19}$$

The list $\boldsymbol{\gamma}$ is C_η-admissible. Consequently, for every finite set $\mathscr{T}$ of such test functions (with the same r) and every $\delta>0$, there is a finite section $(\boldsymbol{\kappa},\mathsf{a})\in\mathbb{K}(C_\eta,\mathbb{M}_*)$ whose masses are a_n^ζ for $n\le n^*$ and 0 beyond, such that

$$\max_{\mathsf{f}\in\mathscr{T}}\big|\langle\Re(\boldsymbol{\kappa},\mathsf{a}),\mathsf{f}\rangle\big|\le\max_{\mathsf{f}\in\mathscr{T}}\frac{r}{4(r-1)}\,\mathfrak{W}_r(\mathsf{f})\,H_0^{1-r}\log H_0+\delta, \tag{18.20}$$

whether or not the Riemann hypothesis holds.

Proof. The residual is the sum over the zeros off the line. For $\mathsf{f}\in C_c^\infty(0,\infty)$, Theorem 5.7 and (15.80) give $\langle\Re(\boldsymbol{\gamma},\mathsf{a}^\zeta),\mathsf{f}\rangle=\sum_k\mathsf{c}_\mathsf{f}(\gamma_k)+\frac12\sum_n\Lambda(n)n^{-1/2}\mathsf{f}(\log n)-\mathsf{k}_\mathsf{f}=-\langle\mathscr{Y},\mathsf{f}\rangle$. Both sides are continuous under convergence of $\mathsf{f},\mathsf{f}',\mathsf{f}''$ in L^1 and of f uniformly, with supports in a fixed compact subset of $(0,\infty)$: on the left by $|\mathsf{c}_\mathsf{g}(t)|\le\|\mathsf{g}''\|_1t^{-2}$ and the finiteness of the mass sum, on the right by the bound of the next paragraph with $r=2$. Mollification therefore extends the identity to every f as in the statement.

One zero. Let $d=d_k$, $|d|<\frac12$, and $0\le u\le u^*$. Every derivative of $\mathsf{e}_d(u):=\cosh(du)-1$ is bounded by $d^2(1+u^2)\cosh\frac{u}{2}$: $0\le\mathsf{e}_d(u)\le\frac{d^2u^2}{2}\cosh(du)$; $|\mathsf{e}_d'(u)|=|d\sinh(du)|\le d^2u\cosh(du)$; and $|\mathsf{e}_d^{(j)}(u)|\le|d|^j\cosh(du)\le d^2\,2^{2-j}\cosh(du)$ for $j\ge2$. Hence $\mathsf{g}_k:=\mathsf{f}\,\mathsf{e}_{d_k}$ has $\|\mathsf{g}_k^{(r)}\|_1\le d_k^2\,\mathfrak{W}_r(\mathsf{f})$ by

the Leibniz rule, and r integrations by parts, without boundary terms, give

$$\Big|\int_0^\infty \mathsf{f}(u)\big(\cosh(d_k u)-1\big)\cos(\gamma_k u)\,du\Big| \le \gamma_k^{-r}\|\mathsf{g}_k^{(r)}\|_1 \le \tfrac{1}{4}\,\mathfrak{W}_r(\mathsf{f})\,\gamma_k^{-r},$$

and the term vanishes when $d_k = 0$. This is the first inequality in (18.19).

The verified zeros. By [55] every zero with $0 < \gamma \le H_0$ has $d = 0$, so the sum runs over $\gamma_k > H_0$. By [72], $N_\zeta(t) \le \frac{t}{2\pi}\log\frac{t}{2\pi e} + 0.1038\log t + 0.2573\log\log t + 9.3675 \le \frac{t}{2\pi}\log t$ for $t \ge 100$, and partial summation gives

$$\begin{aligned}\sum_{\gamma_k > H_0} \gamma_k^{-r} &\le r\int_{H_0}^\infty N_\zeta(t)\,t^{-r-1}dt \le \frac{r}{2\pi}\int_{H_0}^\infty t^{-r}\log t\,dt\\ &= \frac{r\,H_0^{1-r}}{2\pi(r-1)}\Big(\log H_0 + \frac{1}{r-1}\Big) \le \frac{r}{r-1}H_0^{1-r}\log H_0.\end{aligned}$$

Sections. $|\gamma_k - \tau_k| \le C_\eta$ by Lemma 2.2, which is unconditional, and $\gamma_1 > 2\pi$, so $\boldsymbol{\gamma}$ is C_η-admissible. The truncations of $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$ in the proof of Corollary 18.4 are finite sections in $\mathbb{K}(C_\eta, \mathbb{M}_*)$, and for $n^* \ge e^{u^*}$ their residual on f differs from $\langle \mathfrak{R}(\boldsymbol{\gamma}, \mathsf{a}^\zeta), \mathsf{f}\rangle$ by $\sum_{k>k^*}(\mathsf{c}_\mathsf{f}(\tau_k) - \mathsf{c}_\mathsf{f}(\gamma_k)) \to 0$ as $k^* \to \infty$; that part of the proof uses no hypothesis. Take k^* large for the finitely many $\mathsf{f} \in \mathscr{T}$. □

Remark 18.21 (Reading of Proposition 18.20). If the Riemann hypothesis holds, the residual in (18.19) vanishes. If it fails, B and B′ fail (Corollaries 18.11 and 18.18), and still every finite stage has sections in the box with residual at most $\frac{r}{4(r-1)}\max_{\mathscr{T}}\mathfrak{W}_r\,H_0^{1-r}\log H_0 + \delta$. No computation on finitely many tests at a precision coarser than this bound distinguishes the two cases: route B is not finitary, and finite sections, computed or certified at such a precision, carry no information on the Riemann hypothesis. A test sees a residual of size η only if $\mathfrak{W}_r(\mathsf{f}) \ge \frac{4(r-1)}{r}\,\eta\,H_0^{r-1}/\log H_0$ for every admissible r. For the B-spline tests $\varphi^{\mathrm{N}}_{c,h}$ of Section 18.8.5, with $h \le 1$ and $r \le 7$, $\mathfrak{W}_r(\varphi^{\mathrm{N}}_{c,h}) \le \mathsf{K}\,3^r h^{1-r}$ with $\mathsf{K} := (1+u^{*2})\cosh\frac{u^*}{2}$ (by (18.82)), so such a test sees η only if $h \le 3\,(3\mathsf{K}\log H_0/\eta)^{1/(r-1)}H_0^{-1}$ for $r = 2,\dots,7$: the test must resolve frequencies of the order of H_0. For the tests used there the bound (18.19) with $r = 7$, evaluated in ball arithmetic by program `certV`, is below 10^{-57} at every stage computed (column ϵ_ζ of Table 7).

18.3.3. *Staged construction: a certificate theorem and a graded extension.*

Definition 18.22 (Staged sections)**.** Fix $u_1^* < u_2^* < \cdots \to \infty$, finite test sets $\mathscr{T}_1 \subset \mathscr{T}_2 \subset \cdots$ supported in $(0, u_m^*)$ with dense union as in Section 18.3, $\mathrm{t}_m := |\mathscr{T}_m|$, and $k_m^* \to \infty$. Stage m starts from a list $\boldsymbol{\kappa}^{(m,0)}$ that equals the clock beyond some finite index, moves the coordinates $k \le k_m^*$ by $x \in \mathbb{R}^{k_m^*}$, and has masses a^ζ. Its residual map is

$$\mathbb{F}_m : \mathbb{R}^{k_m^*} \to \mathbb{R}^{\mathrm{t}_m}, \qquad \mathbb{F}_m(x)_\varphi := \big\langle \mathfrak{R}(\boldsymbol{\kappa}^{(m,0)} + x, \mathsf{a}^\zeta), \varphi\big\rangle \quad (\varphi \in \mathscr{T}_m), \tag{18.21}$$

where $(\boldsymbol{\kappa} + x)_k = \kappa_k + x_k$ for $k \le k_m^*$ and κ_k otherwise, and its Jacobian at 0 is

$$(\mathbb{J}_m)_{\varphi,k} = \partial_{x_k}\mathbb{F}_m(0)_\varphi = -\int_0^\infty \varphi(u)\,u\sin(\kappa_k^{(m,0)}u)\,du \qquad (\varphi \in \mathscr{T}_m,\ k \le k_m^*). \tag{18.22}$$

We use the norms $\|x\|_\infty$ on $\mathbb{R}^{k_m^*}$ and $\mathbb{R}^{\mathrm{t}_m}$ and the induced operator norms $\|\cdot\|_{\infty\to\infty}$ (maximal absolute row sum). Put

$$\mathsf{h}_m := \|\mathbb{F}_m(0)\|_\infty, \qquad \mathbb{H}_m := k_m^* \max_{\varphi\in\mathscr{T}_m}\int_0^\infty |\varphi(u)|\,u^2\,du. \tag{18.23}$$

Lemma 18.23 (Monotone matching)**.** *Let $(\mathrm{a}_k)_{k\ge1}$ be real with $|\mathrm{a}_k - \tau_k| \le C$ for all k, and let $(\mathrm{a}_{(k)})$ be its nondecreasing rearrangement. Then $|\mathrm{a}_{(k)} - \tau_k| \le C$ for all k. If moreover $\mathrm{a}_k = \tau_k$ for $k > k^*$, then $\mathrm{a}_{(k)} = \tau_k$ for all $k > k^* + j$, where $j := \#\{k > k^* : \tau_k \le \tau_{k^*} + C\} < \infty$.*

Proof. At least k of the a_i (those with $i \le k$) are $\le \tau_k + C$, so $\mathrm{a}_{(k)} \le \tau_k + C$; at least k of them (those with $i \ge k$) are $\ge \tau_k - C$, hence at most $k-1$ are $< \tau_k - C$, so $\mathrm{a}_{(k)} \ge \tau_k - C$. For the second claim, the elements of the multiset not exceeding $\tau_{k^*} + C$ are the k^* moved ones (each is at most $\tau_{k^*} + C$) and the j clock points τ_k, $k^* < k \le k^* + j$; the remaining elements are exactly τ_k, $k > k^* + j$, in increasing order. □

Lemma 18.24 (Lipschitz bound for the Jacobian)**.** *For all $x, x' \in \mathbb{R}^{k^*_m}$, $\|D\mathbb{F}_m(x) - D\mathbb{F}_m(x')\|_{\infty\to\infty} \le \mathbb{H}_m \|x - x'\|_\infty$, and*

$$\|\mathbb{F}_m(x) - \mathbb{F}_m(0) - \mathbb{J}_m x\|_\infty \le \tfrac{1}{2}\mathbb{H}_m \|x\|_\infty^2. \tag{18.24}$$

Proof. $\mathbb{F}_m(x)_\varphi = \sum_{k \le k^*_m} \mathsf{c}_\varphi(\kappa_k^{(m,0)} + x_k) + \text{const}$, so $D^2\mathbb{F}_m(x)_\varphi$ is diagonal with entries $\partial^2_{x_k}\mathbb{F}_m(x)_\varphi = -\int \varphi(u) u^2 \cos((\kappa_k + x_k)u)\, du$, of modulus at most $\int |\varphi| u^2$. The row φ of $D\mathbb{F}_m(x) - D\mathbb{F}_m(x')$ has absolute sum at most $\sum_{k \le k^*_m} \int |\varphi| u^2 \, |x_k - x'_k| \le \mathbb{H}_m \|x - x'\|_\infty$. The Taylor bound follows by integrating $D\mathbb{F}_m$ along the segment. □

Theorem 18.25 (Certificate for one stage)**.** *Suppose $\mathbb{J}_m$ has a right inverse $\mathbb{Q}_m : \mathbb{R}^{\mathrm{t}_m} \to \mathbb{R}^{k^*_m}$, $\mathbb{J}_m\mathbb{Q}_m = I_{\mathrm{t}_m}$, with $\|\mathbb{Q}_m\|_{\infty\to\infty} \le \mathbb{L}_m$, and*

$$\mathsf{l}_m := 2\mathbb{L}_m^2 \mathbb{H}_m \mathsf{h}_m < 1. \tag{A_m}$$

Then the map $\mathbb{T}_m(y) := y - \mathbb{F}_m(\mathbb{Q}_m y)$ is a contraction of the ball $\{\|y\|_\infty \le 2\mathsf{h}_m\} \subset \mathbb{R}^{\mathrm{t}_m}$ with constant l_m, and $x_* := \mathbb{Q}_m y_*$, y_* its unique fixed point in that ball, satisfies

$$\mathbb{F}_m(x_*) = 0, \qquad \|x_*\|_\infty \le 2\mathbb{L}_m \mathsf{h}_m. \tag{18.25}$$

The uniqueness is that of x_* among the zeros of $\mathbb{F}_m$ in $\mathbb{Q}_m\{\|y\|_\infty \le 2\mathsf{h}_m\} \subset \operatorname{Ran}\mathbb{Q}_m$; nothing is asserted about zeros of $\mathbb{F}_m$ outside $\operatorname{Ran}\mathbb{Q}_m$.

Proof. $D\mathbb{T}_m(y) = I - D\mathbb{F}_m(\mathbb{Q}_m y)\mathbb{Q}_m = (\mathbb{J}_m - D\mathbb{F}_m(\mathbb{Q}_m y))\mathbb{Q}_m$ because $\mathbb{J}_m\mathbb{Q}_m = I$, so by Lemma 18.24, for $\|y\|_\infty \le 2\mathsf{h}_m$, $\|D\mathbb{T}_m(y)\|_{\infty\to\infty} \le \mathbb{H}_m \|\mathbb{Q}_m y\|_\infty \mathbb{L}_m \le 2\mathbb{H}_m\mathbb{L}_m^2\mathsf{h}_m = \mathsf{l}_m < 1$. Moreover $\mathbb{T}_m(y) = -\mathbb{F}_m(0) - \big(\mathbb{F}_m(\mathbb{Q}_m y) - \mathbb{F}_m(0) - \mathbb{J}_m\mathbb{Q}_m y\big)$, and by (18.24) the bracket has norm at most $\frac{1}{2}\mathbb{H}_m\mathbb{L}_m^2\|y\|_\infty^2 \le 2\mathbb{H}_m\mathbb{L}_m^2\mathsf{h}_m^2$. Hence $\|\mathbb{T}_m(y)\|_\infty \le \mathsf{h}_m(1 + \mathsf{l}_m) < 2\mathsf{h}_m$: the ball is mapped into itself. Banach's fixed point theorem gives a unique fixed point y_* in the ball, $y_* = y_* - \mathbb{F}_m(\mathbb{Q}_m y_*)$, that is, $\mathbb{F}_m(x_*) = 0$ with $\|x_*\|_\infty \le \mathbb{L}_m\|y_*\|_\infty \le 2\mathbb{L}_m\mathsf{h}_m$. The uniqueness is that of y_* in the ball of $\mathbb{R}^{\mathrm{t}_m}$, hence of x_* in $\operatorname{Ran}\mathbb{Q}_m$; when $k^*_m > \mathrm{t}_m$ the Jacobian has a kernel and $\mathbb{F}_m$ may have other zeros in the ball of $\mathbb{R}^{k^*_m}$, about which the theorem says nothing. The iteration $y_{q+1} = \mathbb{T}_m(y_q)$, $y_0 = 0$, converges to y_* geometrically with ratio l_m; in the variable x it is $x_{q+1} = x_q - \mathbb{Q}_m\mathbb{F}_m(x_q)$, a *modified* Newton iteration, the right inverse being frozen at $\mathbb{Q}_m$ instead of being recomputed at each x_q. □

Remark 18.26 (Evaluation errors and order of the ordinates). Theorem 18.25 is stated for exact values of h_m, $\mathbb{L}_m$ and $\mathbb{H}_m$. If the residual is only known up to an error, $\|\tilde{\mathbb{F}}_m(0) - \mathbb{F}_m(0)\|_\infty \le \epsilon_m$, the theorem applies with h_m replaced by $\|\tilde{\mathbb{F}}_m(0)\|_\infty + \epsilon_m$; if the computed matrix $\tilde{\mathbb{Q}}_m$ satisfies only $\|I - \mathbb{J}_m\tilde{\mathbb{Q}}_m\|_{\infty\to\infty} \le \delta_m < 1$, then $\mathbb{Q}_m := \tilde{\mathbb{Q}}_m(\mathbb{J}_m\tilde{\mathbb{Q}}_m)^{-1}$ is an exact right inverse with $\|\mathbb{Q}_m\| \le \|\tilde{\mathbb{Q}}_m\|/(1 - \delta_m)$, and this is the $\mathbb{L}_m$ to use. A rigorous application therefore needs enclosures of $\mathbb{F}_m(0)$ (quadrature and truncation of the ordinate sums), of $\mathbb{J}_m\tilde{\mathbb{Q}}_m$ (rounding) and of $\mathbb{H}_m$; for the Gaussian tests of Section 18.8 none of these enclosures is produced, and the label `FLOAT_PASS` in the console output means only that the floating-point values satisfy the inequalities; for the B-spline tests of Section 18.8.5 all of them are produced in ball arithmetic. The theorem also does not guarantee that the corrected list $\boldsymbol{\kappa}^{(m,0)} + x_*$ is nondecreasing; this holds when $2\mathbb{L}_m\mathsf{h}_m$ is smaller than half the least gap of the moved ordinates, and otherwise the list is rearranged by Lemma 18.23, which preserves the tube but changes the section. Finally, certifying each stage separately gives no compatibility between stages: the solutions $x_*^{(m)}$ need not be restrictions of one another, and the uniform estimates that would be needed to pass to an infinite sequence (a stage-independent bound on $\mathbb{L}_m\mathbb{H}_m\mathsf{h}_m$ together with the compactness of Proposition 18.5) are not available.

Theorem 18.27 (Staged certificate implies route B′)**.** *Let $C < \tau_1 - 2\pi$. Start at $\boldsymbol{\kappa}^{(1,0)} := \boldsymbol{\tau}$ and, for $m \geq 1$, let $\boldsymbol{\kappa}^{(m)}$ be the nondecreasing rearrangement of $\boldsymbol{\kappa}^{(m,0)} + x_*^{(m)}$ given by Theorem 18.25, and $\boldsymbol{\kappa}^{(m+1,0)} := \boldsymbol{\kappa}^{(m)}$. Suppose* ($\mathrm{A}_m$) *holds for every m and*

$$\sum_{j\geq 1} 2\mathbb{L}_j \mathsf{h}_j = \sup_{m\geq 1} \sum_{j\leq m} 2\mathbb{L}_j \mathsf{h}_j \leq C. \tag{B}$$

Then every $\boldsymbol{\kappa}^{(m)}$ is C-admissible and equals the clock beyond a finite index $\widetilde{k}_m^*$, $\mathsf{s}_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^\zeta) = 0$ for every m, $\mathrm{B}'(C)$ holds, and therefore (Corollary 18.18) $\boldsymbol{\kappa}^{(m)} \to \boldsymbol{\gamma}$ coordinatewise and the Riemann hypothesis holds. In (A_{m+1}) *and* (B) *the quantity h_{m+1} is $\max_{\varphi\in\mathscr{T}_{m+1}\setminus\mathscr{T}_m} |\langle \mathfrak{R}(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^\zeta), \varphi\rangle|$, the residual of the new tests only.*

Proof. Induction on m. Suppose $\boldsymbol{\kappa}^{(m,0)}$ is C_{m-1}-admissible with $C_{m-1} := \sum_{j<m} 2\mathbb{L}_j \mathsf{h}_j$ ($C_0 = 0$) and equals the clock beyond a finite index. By Theorem 18.25, $\boldsymbol{\kappa}^{(m,0)} + x_*^{(m)}$ satisfies $|(\cdot)_k - \tau_k| \leq C_{m-1} + 2\mathbb{L}_m \mathsf{h}_m = C_m$ for all k, and only finitely many coordinates differ from the clock. Lemma 18.23 shows that its rearrangement $\boldsymbol{\kappa}^{(m)}$ satisfies $|\kappa_k^{(m)} - \tau_k| \leq C_m \leq C$ and equals the clock beyond some $\widetilde{k}_m^*$; it is nondecreasing, and $\kappa_1^{(m)} \geq \tau_1 - C > 2\pi$. So $\boldsymbol{\kappa}^{(m)}$ is C-admissible. The residual $\mathfrak{R}$ depends only on the multiset of ordinates, so $\langle \mathfrak{R}(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^\zeta), \varphi\rangle = \mathbb{F}_m(x_*^{(m)})_\varphi = 0$ for all $\varphi \in \mathscr{T}_m$, that is, $\mathsf{s}_m = 0$. Since $\mathscr{T}_m \subset \mathscr{T}_{m+1}$, the start of stage $m+1$ has zero residual on $\mathscr{T}_m$, which gives the stated form of h_{m+1}. $\mathrm{B}'(C)$ holds with $\mathsf{s}_m = 0$, and Corollary 18.18(a) gives the rest. □

Remark 18.28 (What the certificate theorem does and does not do)*.* Theorem 18.27 reduces the extension from one stage to the next to two computable quantities per stage, $\mathsf{l}_m = 2\mathbb{L}_m^2 \mathbb{H}_m \mathsf{h}_m$ and $2\mathbb{L}_m \mathsf{h}_m$. It requires no contraction factor q, no separate control of the perturbation of old tests, and no prior "entry into the basin": old tests are kept in $\mathscr{T}_{m+1}$ and are solved again exactly. The two conditions are not independent of the height: $\mathbb{H}_m \geq k_m^* \int |\varphi| u^2$ grows with the number of unknowns, and $\mathbb{L}_m$ is the ℓ^∞ norm of a right inverse of the Jacobian, whose growth is measured in Section 18.8. The conditions (A_m) for all m and (B) together imply the Riemann hypothesis, so they are not a weaker intermediate step; what changes is that each stage is a statement about one finite matrix and one finite residual vector, verifiable by interval arithmetic; Section 18.8.5 does this for square sections. The restriction $C < \tau_1 - 2\pi = 8.24\ldots$ only ensures $\kappa_1 \geq 2\pi$ and can be removed by treating the first coordinate separately.

Proposition 18.29 (Graded extension)**.** *Let $\mathsf{s}_m \geq 0$ satisfy $\mathsf{s}_{m+1} \leq \mathrm{q}_m \mathsf{s}_m + \mathsf{h}_m$ with $0 \leq \mathrm{q}_m \leq 1$ and $\sum_m \mathsf{h}_m < \infty$, and assume*

$$\prod_{j=L}^{m} \mathrm{q}_j \longrightarrow 0 \quad (m \to \infty) \qquad \text{for every } L \geq 1. \tag{18.26}$$

Then $\mathsf{s}_m \to 0$. Condition (18.26) *holds if $\mathrm{q}_m \leq \mathrm{q} < 1$ for all large m, and more generally if $\sum_m (1 - \mathrm{q}_m) = \infty$. If the lists $\boldsymbol{\kappa}^{(m)}$ producing $\mathsf{s}_m = \mathsf{s}_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^\zeta)$ are C-admissible sections for a fixed C, then $B'(C)$ holds.*

Proof. Iterating, $\mathsf{s}_m \leq \mathsf{s}_1 \prod_{j<m} \mathrm{q}_j + \sum_{\ell<m} \mathsf{h}_\ell \prod_{\ell<j<m} \mathrm{q}_j$. Given $\epsilon > 0$ choose L with $\sum_{\ell\geq L} \mathsf{h}_\ell < \epsilon$. For $m > L$ and $\ell < L$, every q_j is at most 1, so $\prod_{\ell<j<m} \mathrm{q}_j \leq \prod_{j=L}^{m-1} \mathrm{q}_j$, and the same bound holds for $\prod_{j<m} \mathrm{q}_j$; hence the first term and the terms with $\ell < L$ are together at most $\big(\mathsf{s}_1 + \sum_{\ell<L} \mathsf{h}_\ell\big) \prod_{j=L}^{m-1} \mathrm{q}_j$, which tends to 0 by (18.26). The terms with $\ell \geq L$ are at most $\sum_{\ell\geq L} \mathsf{h}_\ell < \epsilon$. Hence $\limsup_m \mathsf{s}_m \leq \epsilon$, and $\mathsf{s}_m \to 0$. The weaker hypothesis $\prod_{m\geq 1} \mathrm{q}_m = 0$ would not suffice: with $\mathrm{q}_1 = 0$ and $\mathrm{q}_m = 1$ for $m \geq 2$ the product vanishes, while $\mathsf{s}_{m+1} \leq \mathsf{s}_m + \mathsf{h}_m$ allows $\mathsf{s}_m \to \mathsf{s}_2 > 0$. For the two sufficient conditions, $\prod_{j=L}^{m} \mathrm{q}_j \leq \mathrm{q}^{\,m - \max(L, m_0)}$ when $\mathrm{q}_j \leq \mathrm{q}$ for $j \geq m_0$, and $\prod_{j=L}^{m} \mathrm{q}_j \leq \exp\big(-\sum_{j=L}^{m} (1 - \mathrm{q}_j)\big)$ in general, since $\mathrm{q} \leq e^{-(1-\mathrm{q})}$ on $[0, 1]$. The second claim is Definition 18.17. □

Remark 18.30 (Tube budget, and the two geometries). In Proposition 18.29 the tube $|\kappa_k^{(m)} - \tau_k| \le C$, uniform in m, is a separate hypothesis: summability of the coordinate increments $\sum_m |\kappa_k^{(m+1)} - \kappa_k^{(m)}|$ gives convergence of each coordinate but not that the limit stays in a prescribed tube, so a construction must carry a displacement budget $\sum_{m \ge m(k)} |\kappa_k^{(m+1)} - \kappa_k^{(m)}| \le C - |\kappa_k^{(m(k))} - \tau_k|$. For $\mathrm{B}'(C)$ itself no convergence of the whole sequence is needed: compactness in the product of the intervals $[\tau_k - C, \tau_k + C]$ (Proposition 18.5) extracts a convergent subsequence, and uniqueness identifies the limit. A second point concerns norms. Theorem 18.25 is stated in ℓ^∞, where $\mathbb{H}_m$ has the simple form (18.23) because the Hessian of each c_φ is diagonal. A bound $\|\mathbb{J}_m^\dagger\|_{2\to 2} \le 1/\sigma_{\min}(\mathbb{J}_m)$ is a Euclidean statement; passing to ℓ^∞ costs factors $\sqrt{\mathrm{t}_m}$ and $\sqrt{K_m}$ unless the right inverse is controlled directly in $\ell^\infty_{\mathrm{t}_m} \to \ell^\infty_{K_m}$. The numerical constants of Section 18.8 are computed directly in ℓ^∞.

Heuristic 18.31 (Frame bound from exponential sums of the clock)**.** In the opposite regime $k_m^* \gg \mathrm{t}_m$ one may take $\mathbb{Q}_m := \mathbb{J}_m^T(\mathbb{J}_m\mathbb{J}_m^T)^{-1}$. If $\|\frac{1}{k_m^*}\mathbb{J}_m\mathbb{J}_m^T - D\|_{\infty\to\infty} \le \epsilon < d_-$ for a diagonal $D \ge d_- I$, and $|(\mathbb{J}_m)_{\varphi,k}| \le \mathrm{a}_J$, then $\mathbb{L}_m \le \mathrm{t}_m \mathrm{a}_J/(k_m^*(d_- - \epsilon)) = O(\mathrm{t}_m/k_m^*)$ and $\mathbb{L}_m^2\mathbb{H}_m = O(\mathrm{t}_m^2\mathrm{B}_0/k_m^*)$ with $\mathrm{B}_0 := \max_\varphi \int |\varphi| u^2$, so (A_m) becomes $k_m^* \gg \mathrm{t}_m^2 \mathrm{B}_0 \mathsf{h}_m$: a wide block compensates a moderate residual. In the point model $\varphi \approx \delta_{u_j}$ the entries of $\frac{1}{k_m^*}\mathbb{J}_m\mathbb{J}_m^T$ are $u_j u_\ell \cdot \frac{1}{k_m^*}\sum_{k\in\mathfrak{B}} \sin(\tau_k u_j)\sin(\tau_k u_\ell)$, where $\mathfrak{B}$ is the block of the k_m^* interior indices, so the frame property reduces to cancellation in the exponential sums $\frac{1}{k_m^*}\sum_{k\in\mathfrak{B}} e^{i\mathrm{v}\tau_k}$ at the frequencies $\mathrm{v} = u_j \pm u_\ell \ne 0$. On a block of k_m^* consecutive clock points at height T the spacing is $\Delta_T \approx 2\pi/\log T$ with curvature $O((k_m^*)^2/(T\log^3 T))$, which suggests $|\frac{1}{k_m^*}\sum_{k\in\mathfrak{B}} e^{i\mathrm{v}\tau_k}| \lesssim \frac{\log T}{k_m^*|\mathrm{v}|} + \frac{|\mathrm{v}|(k_m^*)^2}{T\log^3 T}$ and, for tests on a grid of step Δu in $[0, u^*]$ (so that $\Delta u \le |\mathrm{v}| \le 2u^*$), the window $\frac{\log T}{\Delta u} \ll k_m^* \ll (T\log^3 T/u^*)^{1/2}$. None of this is proved here, and the point model omits a decisive feature of smooth tests: a test of width w in u has $|\partial_\kappa \mathsf{c}_\varphi(\kappa)| \lesssim \mathrm{w}^{-1} e^{-(\mathrm{w}\kappa)^2/2}$ for Gaussian φ, so ordinates with $\mathrm{w}\kappa \gg 1$ are invisible to it and the corresponding columns of $\mathbb{J}_m$ are exponentially small. The underdetermined right inverse then has $\mathbb{L}_m$ dominated by the invisible columns; in the floating-point computations of Section 18.8 this choice produced $\mathbb{L}_m$ of size 10^{11} within four stages, and the stable regime was the opposite one, $\mathrm{t}_m \ge k_m^*$ with the unknowns restricted to the visible ordinates (a least-squares regime; the certificate theorem applies to its square subsystems only). The quasi-orthogonality heuristic is therefore a statement about tests that are not in $C_c^\infty(0,\infty)$, and it is not pursued further here.

18.3.4. *Explicit lists in the tube: Gonek's symmetrised Euler products.* Gonek [19] introduced, for $\mathrm{X} \ge 2$, the truncated Euler product $P_{\mathrm{X}}(s) := \exp\big(\sum_{n\le \mathrm{X}^2} \frac{\Lambda_{\mathrm{X}}(n)}{n^s \log n}\big)$ with $\Lambda_{\mathrm{X}}(n) := \Lambda(n)$ for $n \le \mathrm{X}$ and $\Lambda_{\mathrm{X}}(n) := \Lambda(n)\log(\mathrm{X}^2/n)/\log \mathrm{X}$ for $\mathrm{X} < n \le \mathrm{X}^2$, and the symmetrised function $\zeta_{\mathrm{X}}(s) := P_{\mathrm{X}}(s) + \chi(s) P_{\mathrm{X}}(1-s)$, where χ is the factor of the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$. On the critical line $\chi(\frac12 + it) = e^{-2i\vartheta(t)}$ and $P_{\mathrm{X}}(\frac12 - it) = \overline{P_{\mathrm{X}}(\frac12 + it)}$, so with

$$\mathrm{p}_{\mathrm{X}}(t) := \arg P_{\mathrm{X}}(\tfrac12 + it) = -\sum_{n\le \mathrm{X}^2} \frac{\Lambda_{\mathrm{X}}(n)\sin(t\log n)}{\sqrt{n}\log n}, \tag{18.27}$$

and the two constants

$$\mathrm{m}_{\mathrm{X}} := \sum_{n\le \mathrm{X}^2} \frac{\Lambda_{\mathrm{X}}(n)}{\sqrt{n}\log n}, \qquad \mathrm{n}_{\mathrm{X}} := \sum_{n\le \mathrm{X}^2} \frac{\Lambda_{\mathrm{X}}(n)}{\sqrt{n}}, \tag{18.28}$$

one has $\zeta_{\mathrm{X}}(\frac12 + it) = 2|P_{\mathrm{X}}(\frac12 + it)|\, e^{-i\vartheta(t)} \cos\big(\vartheta(t) + \mathrm{p}_{\mathrm{X}}(t)\big)$. Since P_{X} has no zeros, the zeros of ζ_{X} on the line are exactly the solutions of

$$\Theta_{\mathrm{X}}(t) := \vartheta(t) + \mathrm{p}_{\mathrm{X}}(t) \in \tfrac{\pi}{2} + \pi\mathbb{Z}. \tag{18.29}$$

Gonek proves [19, Thm. 6.2 of the arXiv version] that every zero of ζ_{X} in $0 \le \Re s \le 1$ with imaginary part at least an absolute constant C_0 lies on $\Re s = \frac12$, unconditionally, and that under the Riemann

hypothesis the zeros of ζ_{X} approach those of ζ as $\mathrm{X}\to\infty$. The phase (18.29) is the clock phase plus a truncated prime sum: $\mathcal{N}_{\mathrm{RVM}}(t)=1+\vartheta(t)/\pi+\mathrm{r}_\vartheta(t)$ with $|\mathrm{r}_\vartheta(t)|\le \mathrm{c}_r/t$ for $t\ge 1$, by Stirling's formula [64, §12.33], so (18.29) reads $\mathcal{N}_{\mathrm{RVM}}(t)+\mathrm{p}_{\mathrm{X}}(t)/\pi\in\frac12+\mathbb{Z}+\mathrm{r}_\vartheta(t)$, to be compared with $\mathcal{N}_{\mathrm{RVM}}(\tau_k)=k-\frac12$.

Lemma 18.32 (Gonek lists lie in a tube around the clock)**.** *Fix* $\mathrm{X}\ge 2$ *and put* $\mathrm{t}_{\mathrm{X}}:=\max(2\pi e^{2\mathrm{n}_{\mathrm{X}}+2},\,t_1)$*, where* t_1 *is such that* $|\vartheta'(t)-\frac12\log\frac{t}{2\pi}|\le\frac12$ *for* $t\ge t_1$*. Then* Θ_{X} *is strictly increasing on* $[\mathrm{t}_{\mathrm{X}},\infty)$ *with* $\Theta'_{\mathrm{X}}\ge\frac12$*, the solutions of* (18.29) *in* $[\mathrm{t}_{\mathrm{X}},\infty)$ *form a strictly increasing sequence* $\mathrm{t}_1^{(\mathrm{X})}<\mathrm{t}_2^{(\mathrm{X})}<\cdots$*, each a simple zero of* ζ_{X}*, and there is an integer* j_{X} *such that the list*

$$\kappa_k^{(\mathrm{X})}:=\begin{cases}\tau_k, & k\le \mathrm{j}_{\mathrm{X}},\\ \mathrm{t}_{k-\mathrm{j}_{\mathrm{X}}}^{(\mathrm{X})}, & k>\mathrm{j}_{\mathrm{X}},\end{cases}\tag{18.30}$$

is c_{X}*-admissible for some* $\mathrm{c}_{\mathrm{X}}<\infty$ *and, possibly after a finite permutation, nondecreasing: if the junction at* $k=\mathrm{j}_{\mathrm{X}}$ *breaks monotonicity, the list is replaced by its nondecreasing rearrangement, which keeps the multiset of values, reassigns the indices of finitely many entries (those not exceeding* $\tau_{\mathrm{j}_{\mathrm{X}}}+\mathrm{c}_{\mathrm{X}}$*) and preserves the tube bound by Lemma 18.23; and*

$$|\kappa_k^{(\mathrm{X})}-\tau_k|\le\frac{2\mathrm{m}_{\mathrm{X}}+o(1)}{\log(\tau_k/2\pi)}\qquad(k\to\infty).\tag{18.31}$$

Proof. $|\mathrm{p}'_{\mathrm{X}}(t)|=|\sum_{n\le \mathrm{X}^2}\Lambda_{\mathrm{X}}(n)n^{-1/2}\cos(t\log n)|\le \mathrm{n}_{\mathrm{X}}$ and $|\mathrm{p}_{\mathrm{X}}|\le\mathrm{m}_{\mathrm{X}}$. For $t\ge\mathrm{t}_{\mathrm{X}}$, $\vartheta'(t)\ge\frac12\log\frac{t}{2\pi}-\frac12\ge\mathrm{n}_{\mathrm{X}}+\frac12$, so $\Theta'_{\mathrm{X}}\ge\frac12$. Hence Θ_{X} is a strictly increasing bijection of $[\mathrm{t}_{\mathrm{X}},\infty)$ onto $[\Theta_{\mathrm{X}}(\mathrm{t}_{\mathrm{X}}),\infty)$, and the solutions of (18.29) are the preimages of the points $\frac{\pi}{2}+\pi j$, $j\ge j_0$, where j_0 is the least integer with $\frac{\pi}{2}+\pi j_0\ge\Theta_{\mathrm{X}}(\mathrm{t}_{\mathrm{X}})$; they are simple zeros because $\frac{d}{dt}\cos\Theta_{\mathrm{X}}=-\Theta'_{\mathrm{X}}\sin\Theta_{\mathrm{X}}\neq0$ there. The i-th solution $\mathrm{t}_i^{(\mathrm{X})}$ satisfies $\Theta_{\mathrm{X}}(\mathrm{t}_i^{(\mathrm{X})})=\frac{\pi}{2}+\pi(j_0+i-1)=\pi(j_0+i-\frac12)$; we set $\mathrm{j}_{\mathrm{X}}:=j_0+1$, so that $k=\mathrm{j}_{\mathrm{X}}+i$ gives $\Theta_{\mathrm{X}}(\kappa_k^{(\mathrm{X})})=\pi(k-\frac32)$ for $k>\mathrm{j}_{\mathrm{X}}$. Since $\mathcal{N}_{\mathrm{RVM}}=1+\vartheta/\pi+\mathrm{r}_\vartheta$, the constant 1 raises the count by one unit and

$$\mathcal{N}_{\mathrm{RVM}}(\kappa_k^{(\mathrm{X})})=k-\tfrac12-\frac{\mathrm{p}_{\mathrm{X}}(\kappa_k^{(\mathrm{X})})}{\pi}+\mathrm{r}_\vartheta(\kappa_k^{(\mathrm{X})})\qquad(k>\mathrm{j}_{\mathrm{X}}),\tag{18.32}$$

so, since $\mathcal{N}_{\mathrm{RVM}}(\tau_k)=k-\frac12$,

$$|\mathcal{N}_{\mathrm{RVM}}(\kappa_k^{(\mathrm{X})})-\mathcal{N}_{\mathrm{RVM}}(\tau_k)|\le\frac{\mathrm{m}_{\mathrm{X}}}{\pi}+\frac{\mathrm{c}_r}{\kappa_k^{(\mathrm{X})}}\tag{18.33}$$

(the choice $\mathrm{j}_{\mathrm{X}}=j_0+1$, and not j_0, is what makes the two counts agree; in the computations of Section 18.8 the offset was determined by matching). The right-hand side is bounded, and $\mathcal{N}_{\mathrm{RVM}}(2t)-\mathcal{N}_{\mathrm{RVM}}(t)\to\infty$, so $\kappa_k^{(\mathrm{X})}/\tau_k\to1$. By the mean value theorem, $|\kappa_k^{(\mathrm{X})}-\tau_k|\le(\mathrm{m}_{\mathrm{X}}/\pi+o(1))/\mathcal{N}'_{\mathrm{RVM}}(\mathrm{s}_k)$ with s_k between $\kappa_k^{(\mathrm{X})}$ and τ_k, and $\mathcal{N}'_{\mathrm{RVM}}(\mathrm{s}_k)=\frac{1}{2\pi}\log\frac{\tau_k}{2\pi}(1+o(1))$. This is (18.31); it gives $|\kappa_k^{(\mathrm{X})}-\tau_k|\to0$, so the displacements are bounded, the list is eventually nondecreasing, and finitely many initial indices (where the list is the clock) are harmless. If monotonicity of the concatenated list fails at the junction $k=\mathrm{j}_{\mathrm{X}}$, one replaces the prefix τ_k, $k\le\mathrm{j}_{\mathrm{X}}$, by the nondecreasing rearrangement of Lemma 18.23, which keeps the tube. □

Remark 18.33 (What the Gonek lists give and what remains). The lists $\boldsymbol{\kappa}^{(\mathrm{X})}$ use the primes up to X^2 and the clock, and nothing about the zeros of ζ. Their prime side is built in: formally, $\sum_k\delta(t-\kappa_k^{(\mathrm{X})})=\frac{\Theta'_{\mathrm{X}}(t)}{\pi}\big[1+2\sum_{j\ge1}(-1)^j\cos(2j\Theta_{\mathrm{X}}(t))\big]$ where $\Theta'_{\mathrm{X}}>0$, and the $j=0$ term carries $\mathrm{p}'_{\mathrm{X}}/\pi=-\frac{1}{\pi}\sum_{n\le\mathrm{X}^2}\Lambda_{\mathrm{X}}(n)n^{-1/2}\cos(t\log n)$, whose cosine transform is a sum of masses $\Lambda_{\mathrm{X}}(n)/(2\sqrt{n})$ at $u=\log n$, equal to a_n^{ζ} for $n\le\mathrm{X}$. Two statements would make them a solution of B′: (G1) a tube uniform in X, $\sup_{\mathrm{X}\ge\mathrm{X}_0}\sup_k|\kappa_k^{(\mathrm{X})}-\tau_k|\le C$, which by (18.31) concerns the size of p_{X} at its

own crossing points and not its supremum ($\mathrm{m_X} \asymp \sqrt{\mathrm{X}}/\log^2 \mathrm{X}$ by the prime number theorem, and $\mathrm{n_X} \asymp \sqrt{\mathrm{X}}$; the supremum is useless for this); and (G2) $\langle \Re(\boldsymbol{\kappa}^{(\mathrm{X})}, \mathsf{a}^{\zeta}), \varphi\rangle \to 0$ as $\mathrm{X} \to \infty$ for each test φ, whose content is the vanishing of the harmonics $j \geq 1$ against φ. Neither is proved here. The monotonicity threshold $\mathfrak{t}_{\mathrm{X}}$ of Lemma 18.32 grows like $e^{2\mathrm{n_X}}$, so the global construction cannot rely on monotonicity of Θ_{X} at fixed height as $\mathrm{X} \to \infty$; Gonek's theorem on the location of the zeros does not need it. For fixed X the tube constant in (18.31) is $2\mathrm{m_X}$, and the proof only uses $\sup_t |\mathrm{p_X}(t)| \leq \mathrm{m_X}$; by hand $\mathrm{m_3} = 0.707 + 0.577 + 0.185 + 0.239 + 0.087 + 0.013 = 1.81$, while the observed suprema of $|\mathrm{p_X}|$ on the computed ranges are 1.42 ($\mathrm{X} = 3$), 3.02 ($\mathrm{X} = 8$) and 5.98 ($\mathrm{X} = 20$), which are the constants that matter numerically. Numerically (Section 18.8) the lists with $3 \leq \mathrm{X} \leq 20$ lie within 0.19 to 0.035 mean spacings of the zeta ordinates for $k \leq 400$, whereas the clock is within 0.66 spacings.

18.4. **Route B on a window: unknown lists, a Bernstein form, and minimizers.** This subsection combines route B with the exponential sums of Section 5. The ordinates are unknowns, as in route B; the prime side is Cramér's function on one window, as in Theorem 5.5. Uniqueness then needs neither the masses nor Hamburger's theorem: it comes from Lemma 5.4 and from the following lemma, in which the positivity sits on the side of the unknowns.

Lemma 18.34 (Laplace transforms against the zeros)**.** *Let μ be a positive Borel measure on $\mathbb{R}$ with $\int e^{-yt}\, d\mu(t) < \infty$ for $y > y_1$, and suppose that*

$$\int_{\mathbb{R}} e^{-yt}\, d\mu(t) = \sum_{k\geq 1} e^{iyd_k} e^{-y\gamma_k} \qquad (y > y_1). \tag{18.34}$$

Then all nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$, and $\mu = \sum_k \delta_{\gamma_k}$.

Proof. The right side converges absolutely for $y > 0$ and is real, because the zeros with $d_k \neq 0$ come in pairs with opposite d_k. Let $t_1 < t_2 < \cdots$ be the distinct values of the γ_k, m_j the number of k with $\gamma_k = t_j$, and $P_j(y) := \sum_{k:\gamma_k = t_j} e^{iyd_k} = \sum_\omega \mathrm{a}_{j,\omega} e^{iy\omega}$, a trigonometric polynomial whose coefficients $\mathrm{a}_{j,\omega}$, the number of k with $\gamma_k = t_j$ and $d_k = \omega$, are nonnegative integers with $\sum_\omega \mathrm{a}_{j,\omega} = m_j$. For $y \geq y_2 > \max(y_1, 0)$, $\sum_{\gamma_k > t_j} e^{-y(\gamma_k - t_j)} \leq e^{-(y-y_2)(t_{j+1}-t_j)} \sum_k e^{-y_2(\gamma_k - t_j)}$, which tends to 0 as $y \to \infty$.

We show by induction on j that $\mu_j := \mu - \sum_{i<j} m_i \delta_{t_i}$ is a positive measure supported in $[t_j, \infty)$ and that $d_k = 0$ whenever $\gamma_k < t_j$; for $j = 1$ the first claim is about the support of μ. By (18.34) and the induction hypothesis, $\int e^{-yt} d\mu_j(t) = \sum_{\gamma_k \geq t_j} e^{iyd_k} e^{-y\gamma_k}$ for $y > y_1$, and the modulus of the right side is at most Ce^{-yt_j} for $y \geq y_2$. *Support.* Let $s := \inf \operatorname{supp} \mu_j$ and suppose $s < t_j$; for $s < s' < t_j$, $\mu_j((-\infty, s']) > 0$, and $\int e^{-yt} d\mu_j \geq e^{-ys'} \mu_j((-\infty, s'])$, which contradicts the bound Ce^{-yt_j} for large y. So $\operatorname{supp} \mu_j \subset [t_j, \infty)$. *The lowest group.* Then $e^{yt_j} \int e^{-yt} d\mu_j(t) = \int e^{-y(t-t_j)} d\mu_j(t)$ decreases to $\mu_j(\{t_j\})$ as $y \to \infty$, while $e^{yt_j} \sum_{\gamma_k \geq t_j} e^{iyd_k} e^{-y\gamma_k} = P_j(y) + o(1)$. A trigonometric polynomial with a limit c at infinity has all its coefficients at nonzero frequencies equal to 0 and constant term c, because the mean of $|P_j - c|^2$ over $[Y, 2Y]$ tends to $\sum_{\omega \neq 0} \mathrm{a}_{j,\omega}^2 + (\mathrm{a}_{j,0} - c)^2$. Hence $\mathrm{a}_{j,\omega} = 0$ for $\omega \neq 0$, that is, $d_k = 0$ when $\gamma_k = t_j$, and $\mu_j(\{t_j\}) = \mathrm{a}_{j,0} = m_j$. So $\mu_{j+1} = \mu_j - m_j \delta_{t_j}$ is positive, supported in $[t_j, \infty)$, without atom at t_j, and the support argument applied to it with t_{j+1} in place of t_j completes the induction. Since $t_j \to \infty$, every d_k vanishes and $\mu = \sum_j m_j \delta_{t_j} = \sum_k \delta_{\gamma_k}$. □

Theorem 18.35 (Route B on one window, with an unknown list)**.** *Let $J \subset (0, \infty)$ be an open interval containing no point $\log p^m$, for instance $J = (\log 2, \log 3)$. Let $\boldsymbol{\kappa} = (\kappa_k)_{k\geq 1}$ be a nondecreasing sequence of real numbers with $\kappa_1 > 0$ and $N_{\boldsymbol{\kappa}}(T) \ll (1+T)^A$ for some A. If*

$$\sum_{k\geq 1} \widehat{\mathsf{f}}(\kappa_k) = \int_J \mathsf{f}(u)\, e^{-u/2} V_+(u)\, du \qquad \text{for every real } \mathsf{f} \in C_c^\infty(J), \tag{18.35}$$

then all nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$ and $\boldsymbol{\kappa} = \boldsymbol{\gamma}$. Conversely, under RH the list $\boldsymbol{\gamma}$ satisfies (18.35) *(Theorem 5.5).*

Proof. Put $\mathscr{Z}_{\boldsymbol{\kappa}}(w) := \sum_k e^{iw\kappa_k}$ for $\Im w > 0$. By partial summation $|\mathscr{Z}_{\boldsymbol{\kappa}}(u+iy)| \le \sum_k e^{-y\kappa_k} \ll y^{-A} \le y^{-\lceil A\rceil}$ for $0 < y \le 1$, and $\int \mathscr{Z}_{\boldsymbol{\kappa}}(u+iy)\mathsf{f}(u)\,du = \sum_k e^{-y\kappa_k}\widehat{\mathsf{f}}(\kappa_k) \to \sum_k \widehat{\mathsf{f}}(\kappa_k)$ as $y \downarrow 0$, by (5.2) and dominated convergence. The function $\mathscr{Z}_\xi = e^{-w/2}V_+$ is holomorphic, hence bounded, on a neighbourhood of $J' \times [0,1]$ for every compact interval $J' \subset J$ (Proposition 5.3). By (18.35) and linearity, $F := \mathscr{Z}_{\boldsymbol{\kappa}} - \mathscr{Z}_\xi$ satisfies the hypotheses of Lemma 5.4 on the interior of J', so $F \equiv 0$ there, and, the half-plane being connected, $\mathscr{Z}_{\boldsymbol{\kappa}} = \mathscr{Z}_\xi$ on $\Im w > 0$. At $w = iy$ this is (18.34) with $\mu = \sum_k \delta_{\kappa_k}$, because $\mathscr{Z}_\xi(iy) = \sum_k e^{iy(\rho_k - \frac12)} = \sum_k e^{iyd_k}e^{-y\gamma_k}$. Lemma 18.34 gives RH and $\sum_k \delta_{\kappa_k} = \sum_k \delta_{\gamma_k}$; both lists are nondecreasing, so $\boldsymbol{\kappa} = \boldsymbol{\gamma}$. □

Theorem 18.36 (A Bernstein form of the Riemann hypothesis)**.** *The function $y \mapsto \mathscr{Z}_\xi(iy) = e^{-iy/2}V_+(iy) = \sum_k e^{iyd_k}e^{-y\gamma_k}$ is real on $(0,\infty)$ and has the zero-free representation* (5.11) *($iy \in \Omega_\theta$). The following are equivalent:*

(i) *RH;*
(ii) *$y \mapsto \mathscr{Z}_\xi(iy)$ is completely monotone on $(0,\infty)$: $(-1)^n \frac{d^n}{dy^n}\mathscr{Z}_\xi(iy) \ge 0$ for all $n \ge 0$ and $y > 0$;*
(iii) *$y \mapsto \mathscr{Z}_\xi(iy)$ is completely monotone on (a,∞) for some $a \ge 0$.*

Proof. (i)⇒(ii): under RH, $\mathscr{Z}_\xi(iy) = \sum_k e^{-y\gamma_k}$ and $(-1)^n \frac{d^n}{dy^n}\mathscr{Z}_\xi(iy) = \sum_k \gamma_k^n e^{-y\gamma_k} > 0$. (ii)⇒(iii) is trivial. (iii)⇒(i): by Bernstein's theorem [8], [9, Ch. IV, §12], $x \mapsto \mathscr{Z}_\xi(i(a+x))$ is the Laplace transform of a positive measure ν on $[0,\infty)$, so $\mathscr{Z}_\xi(iy) = \int e^{-yt}e^{at}d\nu(t)$ for $y > a$, and Lemma 18.34 with $\mu := e^{at}\nu$ gives RH. □

Remark 18.37 (Where the positivity sits). In Proposition 5.2 one imaginary frequency suffices, because every term of the defect $\mathfrak{X}(iy_0)$ is nonnegative. For an unknown list there is no such sign, and Theorems 18.35 and 18.36 use all frequencies of a half-line; the positivity is that of the unknown measure μ, which Lemma 18.34 turns, group by group from the lowest ordinate up, into the vanishing of the abscissas. Theorem 18.35 is route B on a single window: no masses, no tube (only polynomial counting), no Hamburger theorem, and existence is again equivalent to RH. By the argument of Proposition 18.20, the zeta list satisfies (18.35) up to the contribution of the zeros above H_0, so no finite computation at a coarser precision decides it either. Theorem 18.36 is a positivity criterion in one real variable whose function has a representation with no sum over the zeros (the primes, ξ'/ξ on a real segment and Γ, Proposition 5.3), in the family of the criteria of Weil and Li; we have not searched the literature for it, and novelty is not claimed.

Minimizers of the residual on a window. The second combination uses compactness, as route B does, and the completeness of exponential systems on bounded intervals in place of stability estimates. Fix $C \ge C_\eta$ and the box

$$\mathfrak{T}_C := \prod_{k\ge1} \big[\max(2\pi, \tau_k - C),\ \tau_k + C\big], \tag{18.36}$$

compact and metrizable in the product topology; the residual $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}^\zeta)$ is defined by (18.5) for every $\boldsymbol{\kappa} \in \mathfrak{T}_C$ (it depends only on the multiset of the coordinates), and $\boldsymbol{\gamma} \in \mathfrak{T}_C$. Fix a window $W = (u_1, u_2)$, $0 < u_1 < u_2$, real test functions $\varphi_j \in C_c^\infty(W)$ whose span is dense in the real space $C_c^\infty(W)$, the bounds $\mathsf{B}_j := \sup_{\mathfrak{T}_C} |\langle \mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}^\zeta), \varphi_j\rangle| < \infty$, and weights $\mathrm{w}_j > 0$ with $\sum_j \mathrm{w}_j(1+\mathsf{B}_j)^2(1+\|\varphi_j\|_{C^2}) < \infty$. Put

$$\mathfrak{E}_W(\boldsymbol{\kappa}) := \sum_{j\ge1} \mathrm{w}_j \,\langle \mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}^\zeta), \varphi_j\rangle^2, \qquad \mathscr{P}_{\boldsymbol{\kappa}} := \sum_{j\ge1} \mathrm{w}_j \,\langle \mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}^\zeta), \varphi_j\rangle\, \varphi_j. \tag{18.37}$$

Lemma 18.38 (Minimizers on a window)**.** (a) *$\mathfrak{E}_W$ is continuous on $\mathfrak{T}_C$ and attains its minimum.*

(b) *Let $\boldsymbol{\kappa}^*$ be a minimizer, and put*

$$\mathscr{G}(z) := \int_{u_1}^{u_2} u\, \mathscr{P}_{\boldsymbol{\kappa}^*}(u) \sin(zu)\, du \qquad (z \in \mathbb{C}). \tag{18.38}$$

Then $\mathfrak{E}_W(\boldsymbol{\kappa}^*) = \langle \mathfrak{R}(\boldsymbol{\kappa}^*, \mathsf{a}^\zeta), \mathscr{P}_{\boldsymbol{\kappa}^*} \rangle$, *and* $\mathscr{G}(\kappa_k^*) = 0$ *for every* k *for which* κ_k^* *is an interior point of its interval in* (18.36).

(c) Dichotomy. *Either* $\mathfrak{E}_W(\boldsymbol{\kappa}^*) = 0$, *that is,* $\mathfrak{R}(\boldsymbol{\kappa}^*, \mathsf{a}^\zeta) = 0$ *in* $\mathcal{D}'(W)$; *or there is* c *such that for every* $r \geq 1$ *the interior coordinates of* $\boldsymbol{\kappa}^*$ *take at most* $(u_2/\log 2)\, r + c$ *distinct values in* $(0, r]$.

Proof. (a) Each pairing is continuous on $\mathfrak{T}_C$ (proof of Proposition 18.5) and bounded by B_j, so the series converges uniformly; a continuous function on a compact space attains its minimum. (b) The series for $\mathscr{P}_{\boldsymbol{\kappa}^*}$ converges in C^2, so $\mathscr{P}_{\boldsymbol{\kappa}^*} \in C^2(\mathbb{R})$ is supported in $[u_1, u_2]$ and vanishes with its first two derivatives at the ends. The residual is a distribution of order at most 2 on $(0, \infty)$, because $\sum_k \cos(\kappa_k u)$ is the second derivative of the uniformly convergent series $-\sum_k \kappa_k^{-2} \cos(\kappa_k u)$, the masses form a measure and k is smooth; it is therefore continuous for this convergence, which gives the first claim. For fixed other coordinates, $t \mapsto \langle \mathfrak{R}, \varphi_j \rangle$ is C^1 in the coordinate $\kappa_k = t$ with derivative $\mathsf{c}'_{\varphi_j}(t) = -\int \varphi_j(u) u \sin(tu)\, du$, bounded by $u_2 \|\varphi_j\|_1$; the weights make the differentiated series converge uniformly, so $\partial_{\kappa_k} \mathfrak{E}_W = -2 \int u \mathscr{P}_{\boldsymbol{\kappa}}(u) \sin(\kappa_k u)\, du$. At an interior coordinate of a minimizer this derivative vanishes. (c) If $\mathfrak{E}_W(\boldsymbol{\kappa}^*) > 0$, then $\mathscr{P}_{\boldsymbol{\kappa}^*} \neq 0$ by (b), so the odd extension of $u \mathscr{P}_{\boldsymbol{\kappa}^*}$ is a nonzero function and $\mathscr{G} \not\equiv 0$. $\mathscr{G}$ is odd, entire, and $|\mathscr{G}(z)| \leq \|u \mathscr{P}_{\boldsymbol{\kappa}^*}\|_1 e^{u_2 |\Im z|}$. Write $\mathscr{G}(z) = z^m \mathscr{H}(z)$ with $\mathscr{H}(0) \neq 0$. Jensen's formula gives, for $r \geq 1$, $n_{\mathscr{H}}(r) \log 2 \leq \int_r^{2r} n_{\mathscr{H}}(t) \frac{dt}{t} \leq \log \max_{|z|=2r} |\mathscr{H}| - \log |\mathscr{H}(0)| \leq 2u_2 r + \log \frac{\|u \mathscr{P}_{\boldsymbol{\kappa}^*}\|_1}{|\mathscr{H}(0)|}$. The zeros of $\mathscr{G}$ are symmetric about 0, so $\mathscr{G}$ has at most $\frac{1}{2} n_{\mathscr{H}}(r) \leq (u_2/\log 2) r + c$ zeros in $(0, r]$, and by (b) every interior coordinate of $\boldsymbol{\kappa}^*$ is one of them. If $\mathfrak{E}_W(\boldsymbol{\kappa}^*) = 0$, every pairing $\langle \mathfrak{R}, \varphi_j \rangle$ vanishes, and so does $\mathfrak{R}$ on W, by density. □

Theorem 18.39 (Window solvability)**.** *Let* $C \geq C_\eta$. *The following are equivalent:*

(i) *RH;*

(ii) *for every* $U > 1$ *there is* $\boldsymbol{\kappa} \in \mathfrak{T}_C$ *with* $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}^\zeta) = 0$ *in* $\mathcal{D}'(U^{-1}, U)$;

(iii) *for every* $U > 1$, $\min_{\mathfrak{T}_C} \mathfrak{E}_{(U^{-1},U)} = 0$, *for one (equivalently every) admissible choice of tests and weights.*

In particular RH holds if, for every $U > 1$, *some minimizer of* $\mathfrak{E}_{(U^{-1},U)}$ *has interior coordinates taking more than* $(U/\log 2 + \delta)\, r$ *distinct values in* $(0, r]$ *for some* $\delta > 0$ *and arbitrarily large* r.

Proof. (i)⇒(ii): take $\boldsymbol{\kappa} = \boldsymbol{\gamma}$, which lies in $\mathfrak{T}_C$ by Lemma 2.2 and has $\mathfrak{R}(\boldsymbol{\gamma}, \mathsf{a}^\zeta) = 0$ by Theorem 13.6. (ii)⇔(iii): by Lemma 18.38(a) and density. (ii)⇒(i): let $\boldsymbol{\kappa}^{(m)} \in \mathfrak{T}_C$ solve (ii) with $U = m$. A subsequence converges coordinatewise to some $\boldsymbol{\kappa}^\infty \in \mathfrak{T}_C$. For $\varphi \in C_c^\infty(0, \infty)$, $\operatorname{supp} \varphi \subset (m^{-1}, m)$ for all large m, so $\langle \mathfrak{R}(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^\zeta), \varphi \rangle = 0$ along the subsequence, and continuity of the pairing gives $\mathfrak{R}(\boldsymbol{\kappa}^\infty, \mathsf{a}^\zeta) = 0$ in $\mathcal{D}'(0, \infty)$. The nondecreasing rearrangement of $\boldsymbol{\kappa}^\infty$ is C-admissible (Lemma 18.23; all coordinates are at least 2π), and Proposition 18.16 gives RH. The last statement follows from Lemma 18.38(c) with $u_2 = U$. □

Remark 18.40 (What the dichotomy leaves open). Lemma 18.38 uses no stability estimate: the first-order condition at a minimizer is a completeness statement for $\{u \sin(\kappa_k u)\}$ on a bounded interval, and the clock density $(\log r)/2\pi$ per unit length is far above the density $u_2/\log 2$ that a nonzero function of exponential type u_2 can absorb (Jensen's formula; the sharp constant is u_2/π, by the Cartwright–Levinson theorem). Theorem 18.39 therefore replaces the existence question of route B by a degeneracy question: whether minimizers can saturate the box, or collapse their interior coordinates onto a set of linear density. If RH fails, then for large U every minimizer is degenerate in this sense, while $\min \mathfrak{E}_{(U^{-1},U)} \leq \mathfrak{E}_{(U^{-1},U)}(\boldsymbol{\gamma})$, which is of the size of Proposition 18.20; at $\boldsymbol{\gamma}$ all coordinates are interior when $C > \sup_k |\gamma_k - \tau_k|$, and the gradient does not vanish: the identity $\mathfrak{E}_W = \langle \mathfrak{R}, \mathscr{P}_{\boldsymbol{\kappa}} \rangle$ and the formula for $\partial_{\kappa_k} \mathfrak{E}_W$ in the proof of Lemma 18.38 hold at every $\boldsymbol{\kappa} \in \mathfrak{T}_C$, so if $\mathfrak{E}_W(\boldsymbol{\gamma}) > 0$ the function (18.38) built at $\boldsymbol{\gamma}$ vanishes at $O(r)$ points of $(0, r]$, while the ordinates take $\gg r \log r$ distinct values, because a positive proportion of the zeros is simple

[10]. Under RH and for such C, $\boldsymbol{\gamma}$ is a non-degenerate minimizer, for the same reason. Whether degenerate minimizers can be excluded directly is open; it is not a statement of the form of (D).

Heuristic 18.41 (Window size and zero-free regions)**.** Theorem 5.7 gives, without hypothesis, $\mathsf{k}-\sum_n \mathsf{a}_n^{\zeta}\delta_{\log n}=\sum_k \cosh(d_k u)\cos(\gamma_k u)$ in $\mathcal{D}'(0,\infty)$. So a solution $\boldsymbol{\kappa}\in\mathfrak{T}_C$ on the window (U^{-1},U) satisfies $\sum_k\cos(\kappa_k u)=\sum_k\cosh(d_k u)\cos(\gamma_k u)$ there. Test it with $\psi(u)\cos(\gamma_0 u)$, $\psi\ge 0$ supported in $[U-\mathrm{w},U)$, where $\frac12+d_0+i\gamma_0$ is a zero off the line. The pair of zeros at γ_0 contributes $\int\psi(u)\cosh(d_0u)\,du$; the left side is at most $\int\psi$ times the number of κ_k within about $1/\mathrm{w}$ of γ_0, which the box bounds by about $(C+\frac{1}{\mathrm{w}})\frac{\log\gamma_0}{\pi}$. This suggests

$$\cosh\bigl(d_0(U-\mathrm{w})\bigr)\lesssim\Bigl(C+\frac{1}{\mathrm{w}}\Bigr)\frac{\log\gamma_0}{2\pi},\qquad\text{that is,}\qquad |d_0|\lesssim\frac{\log(C\log\gamma_0)}{U}\,: \tag{18.39}$$

solvability on the window (U^{-1},U) would exclude zeros at distance more than about $\log(C\log\gamma)/U$ from the critical line; for fixed U this is informative only for $\log(C\log\gamma)<U/2$, and graded by U it would give statements between the verified zeros and RH. Three points are not proved. First, the on-line zeros near γ_0 and the κ_k near γ_0 contribute terms $\frac12\int\psi(u)\cos((\gamma_k-\gamma_0)u)\,du$ of indefinite sign, since a bump ψ supported near U has no positive Fourier transform. Second, other zeros off the line with ordinates within $1/\mathrm{w}$ of γ_0 could cancel the main term; a lower bound of Turán type for exponential sums would be needed, and with it the conclusion weakens to $|d_0|\ll\log\gamma_0/U$. Third, the converse, that such a zero-free region allows a solution on the window, is not constructed. With $H_0=3\cdot10^{12}$, $C=1$ and $\mathrm{w}=1$, the right side of (18.39) is about 9.1 at $\gamma_0=H_0$, so the bound says nothing for $|d_0|<\frac12$ unless $U\gtrsim 6.8$: window solvability up to $U\approx 7$ is consistent with the verified zeros, and beyond it would carry information on the zeros above H_0. The stages computed in Section 18.8 use tests supported in $(0,6.4]$.

18.5. **An a priori bound for quasi-solutions.** Throughout this section $(\boldsymbol{\kappa},\mathsf{a})$ is a finite section in $\mathbb{K}(C,\mathbb{M})$ with parameters k^*,n^*.

Definition 18.42 (Laplace residual)**.** For $\Re s>1$ put

$$\widehat{\mathsf{R}}(s):=\tfrac12\frac{P_{\boldsymbol{\kappa}}'}{P_{\boldsymbol{\kappa}}}(s)-\tfrac12\frac{G'}{G}(s)+\sum_{n=2}^{n^*}\mathsf{a}_n\,n^{\frac12-s}. \tag{18.40}$$

For $T\ge0$ put

$$\mathbb{E}(T):=\int_{-T}^{T}\bigl|\widehat{\mathsf{R}}(2+iv)\bigr|\,dv,\qquad \upsilon:=\sup_{T\ge2}\frac{\mathbb{E}(T)}{\log T}\in[0,\infty]. \tag{18.41}$$

Since $\int_0^\infty e^{-(s-\frac12)u}\cos(\kappa u)\,du=\frac{s-\frac12}{(s-\frac12)^2+\kappa^2}=\frac12\frac{d}{ds}\log(1+\frac{q(s)}{\frac14+\kappa^2})$ and $\int_0^\infty e^{-(s-\frac12)u}\delta(u-\log n)\,du=n^{\frac12-s}$, $\widehat{\mathsf{R}}$ is formally the Laplace transform of the residual distribution of (18.4), with the normalisation of k fixed in 15.79; this is made precise in Lemma 18.14. We take (18.40) as the definition. Because the coefficients are real, $\widehat{\mathsf{R}}(\bar s)=\overline{\widehat{\mathsf{R}}(s)}$, so $\mathbb{E}(T)=2\int_0^T|\widehat{\mathsf{R}}(2+iv)|\,dv$. We assume $\upsilon<\infty$ throughout this section; if $\upsilon=\infty$ every bound below is void. By Lemma 18.14, $\widehat{\mathsf{R}}$ is the Laplace transform of the residual distribution of the section.

Lemma 18.43 (When the Laplace residual grows logarithmically)**.** *For a finite section, $\upsilon<\infty$ if and only if $\mathsf{a}_n=0$ for every n. More precisely, with $\mathcal{D}(v):=\sum_{n=2}^{n^*}\mathsf{a}_n n^{-3/2-iv}$,*

$$\mathbb{E}(T)=\int_{-T}^{T}|\mathcal{D}(v)|\,dv+O_{C,k^*}(\log T),\quad \liminf_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}|\mathcal{D}(v)|\,dv\ge\frac{\sum_{n\le n^*}|\mathsf{a}_n|^2n^{-3}}{\sum_{n\le n^*}|\mathsf{a}_n|n^{-3/2}}>0\ \ (\mathsf{a}\neq0). \tag{18.42}$$

Proof. With $z=\frac32+iv$, the spectral part of $\widehat{\mathsf{R}}(2+iv)$ is $\frac12\zeta'_{\mathrm{RVM}}/\zeta_{\mathrm{RVM}}(2+iv)+\sum_{k\le k^*}\bigl[\frac{z}{z^2+\kappa_k^2}-\frac{z}{z^2+\tau_k^2}\bigr]$, because the list equals the clock beyond k^*. Each term of the finite sum is $z(\tau_k^2-\kappa_k^2)/((z^2+\kappa_k^2)(z^2+$

$\tau_k^2)) = O_{C,k}(|v|^{-3})$, integrable over $\mathbb{R}$. By Theorem 9.1(a) with $B = \frac{3}{2}$, $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+\mathcal{E}_0)$ on the band $1 \le \sigma' \le 2$, $t \ge t_0$, with $|w| \ll t^{-3/2}$, $w' = -\log(\tau_*/2\pi)\, w$ (Corollary 9.6) and $|\mathcal{E}_0| \le C/t$; Cauchy's estimate on discs of radius $\frac{1}{4}$ inside the band gives $|\mathcal{E}_0'| \le C'/t$, so $|\zeta'_{\mathrm{RVM}}/\zeta_{\mathrm{RVM}}(2+iv)| \ll (1+|v|)^{-1}$ and its integral over $[-T,T]$ is $O(\log T)$. This is the first statement of (18.42). For the second, $\mathcal{D}$ is a Dirichlet polynomial with distinct frequencies $\log n$, so $\frac{1}{2T}\int_{-T}^{T}|\mathcal{D}|^2 dv \to \sum_n |\mathsf{a}_n|^2 n^{-3}$ (Montgomery–Vaughan, or directly: the off-diagonal terms are $O(1/T)$), while $|\mathcal{D}| \le \sum_n |\mathsf{a}_n| n^{-3/2}$; hence $\int_{-T}^{T}|\mathcal{D}| \ge \int_{-T}^{T}|\mathcal{D}|^2/\sup|\mathcal{D}|$ gives the stated lower bound. If $\mathsf{a} \ne 0$, then $\mathbb{E}(T) \asymp T$ and $\upsilon = \infty$; if $\mathsf{a} = 0$, then $\mathbb{E}(T) = O(\log T)$ and $\upsilon < \infty$. □

Remark 18.44 (Scope of the counting bound). Lemma 18.43 restricts Theorem 18.51 and Corollary 18.52, as stated for finite sections, to the sections without masses, for which $|N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}}| \le k^* + 1$ is immediate. The reason is that on a vertical line the Laplace transform of the prime side is almost periodic and does not decay, so an L^1 residual of logarithmic growth on $\Re s = 2$ demands that the spectral side reproduce the prime side at all heights, which no finite section does; the natural examples with $\upsilon < \infty$ are infinite lists with the full mass sequence, such as $(\boldsymbol{\gamma}, \mathsf{a}^{\zeta})$, for which $\widehat{\mathsf{R}} \equiv 0$ and $\upsilon = 0$. The theorem is kept as the template of the count-to-displacement mechanism (Lemmas 18.45–18.50), which is reused in Appendix F; its extension to infinite lists needs a polynomial growth bound for $P_{\boldsymbol{\kappa}}/P_{\boldsymbol{\tau}}$ on $\Re s \ge 2$, which is not given here (Remark 18.53), and Heuristic 18.54 explains why the Euclidean residual of a numerical section is a different quantity.

Constants. Put $\mathrm{c}_\ell := 2\sum_{n\ge 2} n^{-3/2}/\log n < \infty$, and, for $s \in \mathbb{C}$ and real $x \ge 2\pi$, $\mathfrak{f}(x;s) := \log(1 + \frac{q(s)}{\frac{1}{4}+x^2})$, with the principal branch, which is defined whenever $|s - \frac{1}{2}| < x$. Let $\mathbb{J} := \{2\} \cup \{\sigma + i : \ -1 \le \sigma \le 2\}$, a compact set on which $|s - \frac{1}{2}| \le \frac{5}{2} < 2\pi$, and

$$\mathfrak{B}(C) := \sum_{k\ge 1} \sup_{x \in [\max(\tau_k - C, 2\pi),\, \tau_k + C]} \sup_{s\in\mathbb{J}} |\mathfrak{f}(x;s) - \mathfrak{f}(\tau_k;s)|. \tag{18.43}$$

This is finite. On $\mathbb{J}$, $|q(s)| \le 5$ and $|\partial_x f(x;s)| = |\frac{2xq(s)}{(\frac{1}{4}+x^2)(x^2+(s-\frac{1}{2})^2)}| \le \frac{20}{x^3}$ for $x \ge 4$. The k-th term is therefore at most $160C\tau_k^{-3}$ when $\tau_k \ge 2C$, and $\sum_k \tau_k^{-3} < \infty$. Further, $\mathrm{c}_\tau := \max_{-1\le\sigma\le 2} |\log|\zeta_{\mathrm{RVM}}(\sigma+i)||$ and $\mathrm{c}_{\mathrm{RVM}} := \sup_{\Re s \ge 2} |\log \zeta_{\mathrm{RVM}}(s)|$. Both are finite: the first because ζ_{RVM} has no zeros or poles on that segment, the second by 9.10.

Lemma 18.45 (Phragmén–Lindelöf, two forms)**.** (a) *Let* $\mathbb{S} := \{\mathrm{l}_1 \le \Re s \le \mathrm{l}_2,\ \Im s \ge t_1\}$ *have width* $a := \mathrm{l}_2 - \mathrm{l}_1$. *Let* u *be upper semicontinuous on* $\mathbb{S}$ *and subharmonic in its interior, with* $u \le \mathrm{M}_0$ *on* $\partial\mathbb{S}$ *and* $u(s) \le \mathrm{A}_0 e^{c\Im s}$ *for some* $c < \pi/a$. *Then* $u \le \mathrm{M}_0$ *on* $\mathbb{S}$.
(b) *Let* $\mathfrak{g}$ *be holomorphic in* $\Re s > 1$, *continuous on* $\Re s \ge 2$, *with* $|g| \le 1$ *on* $\Re s = 2$ *and* $|\mathfrak{g}(s)| \le \mathrm{A}_0(2+|s|)^{\mathrm{L}}$ *on* $\Re s \ge 2$. *Then* $|g| \le 1$ *on* $\Re s \ge 2$.

Proof. (a) Choose $c < c' < \pi/a$ and $m := \frac{\mathrm{l}_1+\mathrm{l}_2}{2}$, and put $\omega(s) := e^{c'\Im s}\cos(c'(\Re s - m))$. This is harmonic, being the real part of $e^{-ic'(s-m)}$. On $\mathbb{S}$, $\omega \ge \cos(c'a/2)\, e^{c'\Im s}$, and $\cos(c'a/2) > 0$. For $\mathrm{p} > 0$, $u - \mathrm{M}_0 - \mathrm{p}\omega$ is subharmonic, ≤ 0 on $\partial\mathbb{S}$, and tends to $-\infty$ as $\Im s \to \infty$. The maximum principle on $\mathbb{S} \cap \{\Im s \le t_2\}$, with t_2 large, gives $u \le \mathrm{M}_0 + \mathrm{p}\omega$. Let $\mathrm{p} \to 0$.
(b) For $\mathrm{p} > 0$, $\mathfrak{g}_\eta(s) := \mathfrak{g}(s)\exp(-\mathrm{p}(s-1)^{1/2})$, with the principal branch, satisfies $|\mathfrak{g}_\eta| \le |g|$ on $\Re s \ge 2$, because $\Re(s-1)^{1/2} \ge \cos(\pi/4)|s-1|^{1/2} > 0$ there. Moreover $\mathfrak{g}_\eta(s) \to 0$ as $|s| \to \infty$ in $\Re s \ge 2$. The maximum principle on large half-discs gives $|\mathfrak{g}_\eta| \le 1$. Let $\mathrm{p} \to 0$. □

Lemma 18.46 (The Dirichlet factor)**.** *Put* $\mathcal{D}_{\mathsf{a}}(s) := \sum_{n=2}^{n^*} \frac{2\mathsf{a}_n}{\log n} n^{\frac{1}{2}-s}$ *and* $\mathcal{Y} := \zeta_{\boldsymbol{\kappa}} e^{-\mathcal{D}_{\mathsf{a}}}$. *Then* $\mathcal{Y}$ *is holomorphic and zero-free in* $\Re s > 1$, *and* $|\mathcal{D}_{\mathsf{a}}| \le \mathbb{M}\mathrm{c}_\ell$ *on* $\Re s \ge 2$. *For the branch of* $\log\mathcal{Y}$ *that is real on* $(1,\infty)$,

$$(\log\mathcal{Y})' = 2\,\widehat{\mathsf{R}}, \qquad |\log\mathcal{Y}(2)| \le \mathbb{B}_0 := |\log\zeta_{\mathrm{RVM}}(2)| + \mathfrak{B}(C) + \mathbb{M}\mathrm{c}_\ell, \tag{18.44}$$

and, for all real t,

$$|\log \mathcal{Y}(2+it)| \le \mathbb{B}_0 + \mathbb{E}(|t|) \le \mathbb{B}_0 + \upsilon \log(2+|t|). \tag{18.45}$$

Proof. $P_{\boldsymbol{\kappa}}$ vanishes only on $\Re s = \frac12$, and G is holomorphic and zero-free in $\Re s > 1$. Hence $\zeta_{\boldsymbol{\kappa}}$ and $\mathcal{Y}$ are holomorphic and zero-free there. $|\mathcal{D}_{\mathsf{a}}(s)| \le \sum_n 2M n^{\frac12 - \Re s}/\log n \le \mathbb{M}\mathrm{c}_\ell$ for $\Re s \ge 2$. Differentiating, $(\log \mathcal{Y})' = \frac{P'_{\boldsymbol{\kappa}}}{P_{\boldsymbol{\kappa}}} - \frac{G'}{G} + 2\sum_n \mathsf{a}_n n^{\frac12 - s} = 2\widehat{\mathsf{R}}$. At $s = 2$, $\log \zeta_{\boldsymbol{\kappa}}(2) = \log \zeta_{\mathrm{RVM}}(2) + \sum_{k \le k^*} (\mathfrak{f}(\kappa_k; 2) - \mathfrak{f}(\tau_k; 2))$, a finite sum of real numbers bounded by $\mathfrak{B}(C)$, and $|\mathcal{D}_{\mathsf{a}}(2)| \le \mathbb{M}\mathrm{c}_\ell$. Finally, $\log \mathcal{Y}(2+it) - \log \mathcal{Y}(2) = 2i \int_0^t \widehat{\mathsf{R}}(2+iv)\, dv$ has modulus at most $\mathbb{E}(|t|)$. For $|t| \ge 2$, $\mathbb{E}(|t|) \le \upsilon \log|t|$; for $|t| < 2$, $\mathbb{E}(|t|) \le \mathbb{E}(2) \le \upsilon \log 2$. □

Lemma 18.47 (Bound on the half-plane)**.** *With $\mathbb{W} := \log|\zeta_{\boldsymbol{\kappa}}| = \log|P_{\boldsymbol{\kappa}}| - \log|G|$ and $\mathbb{B}_1 := \mathbb{B}_0 + \upsilon \log(2\sqrt{2})$, $\mathbb{B}_2 := \mathbb{B}_1 + \mathbb{M}\mathrm{c}_\ell$, for $\Re s \ge 2$,*

$$\big|\log|\mathcal{Y}(s)|\big| \le \mathbb{B}_1 + \upsilon \log|s-1|, \qquad |\mathbb{W}(s)| \le \mathbb{B}_2 + \upsilon \log|s-1|. \tag{18.46}$$

Proof. On $\Re s = 2$, $|s-1| = \sqrt{1+t^2} \ge (2+|t|)/(2\sqrt{2})$, so (18.45) gives $|\mathcal{Y}(s)|^{\pm 1} \le e^{B_1}|s-1|^{\upsilon}$. A priori growth: on $\Re s \ge 2$, $|\log|\zeta_{\mathrm{RVM}}|| \le \mathrm{c}_{\mathrm{RVM}}$. The quotient $\zeta_{\boldsymbol{\kappa}}/\zeta_{\mathrm{RVM}}$ is a finite product of k^* factors $\frac{(\kappa_k^2+w^2)(\frac14+\tau_k^2)}{(\frac14+\kappa_k^2)(\tau_k^2+w^2)}$ with $w = s - \frac12$. Since $|\kappa^2 + w^2| = |\kappa - \Im s + i\Re w|\,|\kappa + \Im s - i\Re w| \ge (\Re w)^2 \ge \frac94$, each factor and its inverse are $O((2+|s|)^4)$. Hence $|\mathcal{Y}|^{\pm 1} \le \mathrm{A}_0(2+|s|)^{4k^*}$. Apply Lemma 18.45(b) to $\mathfrak{g}_+ := \mathcal{Y}\,(s-1)^{-\upsilon} e^{-\mathbb{B}_1}$ and to $\mathfrak{g}_- := \mathcal{Y}^{-1}(s-1)^{-\upsilon} e^{-\mathbb{B}_1}$, with the principal branch of $(s-1)^{-\upsilon}$, holomorphic in $\Re s > 1$ with $|(s-1)^{-\upsilon}| = |s-1|^{-\upsilon}$; both are holomorphic in $\Re s > 1$ and of polynomial growth on $\Re s \ge 2$. Finally $\mathbb{W} = \log|\mathcal{Y}| + \Re \mathcal{D}_{\mathsf{a}}$. □

Lemma 18.48 (Bound on the half-strip)**.** *Let $\mathbb{S} := \{-1 \le \Re s \le 2,\ \Im s \ge 1\}$ and $\mathbb{B}_3 := \max\big(\mathbb{B}_2 + \frac34 \log 2,\ \mathrm{c}_\tau + \mathfrak{B}(C)\big)$. Then*

$$\mathbb{W}(s) \le \mathbb{B}_3 + \big(\tfrac32 + \upsilon\big) \log|s+2| \qquad (s \in \mathbb{S}). \tag{18.47}$$

Proof. $\mathbb{W}$ is subharmonic in the interior of $\mathbb{S}$, since $\log|G|$ is harmonic there (G has no zeros or poles with $\Im s \ge 1$), and it is upper semicontinuous on $\mathbb{S}$. The majorant $\mathrm{h}_0(s) := (\frac32 + \upsilon)\log|s+2|$ is harmonic and nonnegative on $\mathbb{S}$. We check the three sides.

Right side. By (18.46), $\mathbb{W}(2+it) \le \mathbb{B}_2 + \upsilon \log|1+it| \le \mathbb{B}_2 + \mathrm{h}_0(2+it)$.

Left side. $P_{\boldsymbol{\kappa}}$ and G are real, so $\mathbb{W}(\bar s) = \mathbb{W}(s)$. Combined with $P_{\boldsymbol{\kappa}}(1-s) = P_{\boldsymbol{\kappa}}(s)$, this gives $\mathbb{W}(-1+it) = \mathbb{W}(2+it) + \log|G(2+it)/G(-1+it)|$. By $\Gamma(z+1) = z\Gamma(z)$ and the reflection formula, $|\Gamma(\frac12 + iy)|^2 = \pi/\cosh \pi y$ and $|\Gamma(1+iy)|^2 = \pi y/\sinh \pi y$, so

$$\Big|\frac{G(2+it)}{G(-1+it)}\Big| = \frac{|1+it|}{2}\,\pi^{-3/2}\Big(\frac{t}{2}\coth\frac{\pi t}{2}\Big)^{1/2} \le (1+t)^{3/2} \qquad (t \ge 1). \tag{18.48}$$

For the inequality: $|1+it| \le 1+t$, $\frac{t}{2}\coth\frac{\pi t}{2} \le \frac{1+t}{2}\coth\frac{\pi}{2}$, and $\frac12 \pi^{-3/2}(\frac12 \coth\frac{\pi}{2})^{1/2} < 1$. Since $(1+t) \le \sqrt{2}|1+it|$ and $|s+2| = |1+it|$ at $s = -1+it$, we get $\mathbb{W}(-1+it) \le \mathbb{B}_2 + \frac34 \log 2 + \mathrm{h}_0(-1+it)$.

Bottom side. For $s = \sigma + i$, $\mathbb{W}(s) = \log|\zeta_{\mathrm{RVM}}(s)| + \sum_{k \le k^*} \Re(\mathfrak{f}(\kappa_k; s) - \mathfrak{f}(\tau_k; s)) \le \mathrm{c}_\tau + \mathfrak{B}(C) \le \mathrm{c}_\tau + \mathfrak{B}(C) + \mathrm{h}_0(s)$.

Growth. $\log|P_{\boldsymbol{\kappa}}(s)| \le \sum_k \log(1 + |q(s)|/\kappa_k^2)$. Split at $\kappa_k \le 2|s|$ and $\kappa_k > 2|s|$: the first part has $N_{\boldsymbol{\kappa}}(2|s|) = O(|s|\log|s|)$ terms, each at most $\log(1 + |q(s)|/4\pi^2) = O(\log|s|)$; in the second, $\log(1 + |q(s)|/\kappa_k^2) \le |q(s)|\kappa_k^{-2}$, and $\sum_{\kappa_k > 2|s|} \kappa_k^{-2} = O(\log|s|/|s|)$ by partial summation with $N_{\boldsymbol{\kappa}}(t) = O(t\log t)$. Hence $\log|P_{\boldsymbol{\kappa}}(s)| = O(|s|\log^2|s|)$, and $-\log|G(s)| \le \frac{\pi}{4}|\Im s| + O(\log|s|)$ on $\mathbb{S}$ by Stirling's formula [64, §12.33]. Hence $\mathbb{W} - \mathrm{h}_0 \le \mathrm{A}_0 e^{c\Im s}$ for any $c > 0$. Lemma 18.45(a), with width 3 and $c < \pi/3$, applied to $\mathbb{W} - \mathrm{h}_0$ gives the claim. □

Lemma 18.49 (Backlund–Jensen on a horizontal segment)**.** *There are absolute constants $T_0 \ge 3$ and c_5 such that, for $T \ge T_0$ with $T \notin \{\kappa_k\}$, the variation of $\arg P_{\boldsymbol{\kappa}}$ along the segment from $2+iT$*

to $\frac{1}{2}+iT$ *is at most* $\pi(n_T+1)$*, where*

$$n_T \le \frac{(\frac{5}{2}+\upsilon)\log(T+8)+\mathbb{E}(T)+\mathbb{B}_0+\mathbb{B}_3+\mathbb{M}\mathrm{c}_\ell+2\mathrm{c}_5}{\log(4/3)}. \tag{18.49}$$

Proof. Choose $\mathrm{a}_T\in\mathbb{R}$ with $e^{i\mathrm{a}_T}P_{\boldsymbol\kappa}(2+iT)>0$ and put $\mathfrak{F}_T(z):=\frac{1}{2}\big(e^{i\mathrm{a}_T}P_{\boldsymbol\kappa}(z+iT)+e^{-i\mathrm{a}_T}P_{\boldsymbol\kappa}(z-iT)\big)$, which is entire. Since $P_{\boldsymbol\kappa}$ is real, $P_{\boldsymbol\kappa}(x-iT)=\overline{P_{\boldsymbol\kappa}(x+iT)}$ for real x, so $\mathfrak{F}_T(x)=\Re(e^{i\mathrm{a}_T}P_{\boldsymbol\kappa}(x+iT))$ and $\mathfrak{F}_T(2)=|P_{\boldsymbol\kappa}(2+iT)|$. If $\mathfrak{F}_T$ has n_T zeros on $[\frac{1}{2},2]$, the segment splits into at most n_T+1 pieces on which $\Re(e^{i\mathrm{a}_T}P_{\boldsymbol\kappa})$ has constant sign. On each piece arg varies by less than π, which gives the first claim [61, §9.4].

The segment $[\frac{1}{2},2]$ lies in the closed disc $|z-2|\le\frac{3}{2}$. For $\frac{3}{2}<r<2$, Jensen's formula on $|z-2|\le R=2$ (valid also if $\mathfrak{F}_T$ has zeros on the circle $|z-2|=2$, which contribute $\log 1=0$, since $\mathfrak{F}_T$ is entire [57, Thm. 15.18]) gives $n_T\log\frac{R}{r}\le\max_{|z-2|=2}\log|\mathfrak{F}_T(z)|-\log|\mathfrak{F}_T(2)|$, because the zeros in the closed disc of radius r, which include the n_T zeros on the segment, contribute at least $n_T\log(R/r)$; letting $r\downarrow\frac{3}{2}$ gives the inequality with $\log\frac{4}{3}$. Moreover $|\mathfrak{F}_T(z)|\le\max_{|\zeta-2|\le2}|P_{\boldsymbol\kappa}(\zeta+iT)|$, because $|P_{\boldsymbol\kappa}(z-iT)|=|P_{\boldsymbol\kappa}(\bar z+iT)|$. Write $\log|P_{\boldsymbol\kappa}|=\mathbb{W}+\log|G|$. On the disc $\{|s-(2+iT)|\le2\}$, the part with $\Re s\le2$ lies in $\mathbb{S}$ and $|s+2|\le T+6$, so Lemma 18.48 gives $\mathbb{W}\le\mathbb{B}_3+(\frac{3}{2}+\upsilon)\log(T+6)$. The part with $\Re s\ge2$ satisfies $\mathbb{W}\le\mathbb{B}_2+\upsilon\log(T+5)$ by Lemma 18.47, a smaller bound. At the centre, (18.45) gives $\mathbb{W}(2+iT)\ge-(\mathbb{B}_0+\mathbb{E}(T)+\mathbb{M}\mathrm{c}_\ell)$. For the oscillation of $\log|G|$ on the disc: by Stirling's formula for Γ'/Γ in $|\arg z|\le\frac{3}{4}\pi$ [64, §12.33], $|G'/G(s)|=|\frac{1}{s}+\frac{1}{s-1}-\frac{1}{2}\log\pi+\frac{1}{2}\frac{\Gamma'}{\Gamma}(\frac{s}{2})|\le\frac{1}{2}\log(T+8)+\mathrm{c}_5$ on the disc for $T\ge T_0$. So the oscillation is at most $\log(T+8)+2\mathrm{c}_5$. Collecting terms gives (18.49). □

Lemma 18.50 (Argument principle)**.** *For* $T>0$*,* $T\notin\{\kappa_k\}$*, let* $\mathbb{G}_T$ *be the path* $2\to2+iT\to\frac{1}{2}+iT$*. Then*

$$\pi N_{\boldsymbol\kappa}(T)=\Delta_{\mathbb{G}_T}\arg P_{\boldsymbol\kappa}=\pi+\vartheta(T)+\Delta_{\mathbb{G}_T}\arg\zeta_{\boldsymbol\kappa}, \tag{18.50}$$

where $\vartheta(T)=\arg\big(\pi^{-iT/2}\Gamma(\frac{1}{4}+\frac{iT}{2})\big)$ *is the Riemann–Siegel theta function, defined by continuous variation from* $\vartheta(0)=0$*.*

Proof. Take the rectangle $\mathfrak{R}_T$ with vertices $2\pm iT$, $-1\pm iT$. $P_{\boldsymbol\kappa}$ has no zeros on $\partial\mathfrak{R}_T$: its zeros lie on $\Re s=\frac{1}{2}$, and $T\notin\{\kappa_k\}$, so $\frac{1}{2}\pm iT$ are not zeros. Its zeros inside $\mathfrak{R}_T$ are $\frac{1}{2}\pm i\kappa_k$ with $\kappa_k<T$, counted with multiplicity (coincident $\kappa_j=\kappa_k$ give a multiple zero), $2N_{\boldsymbol\kappa}(T)$ in total. Reflection $s\mapsto1-\bar s$ maps $\mathbb{G}_T$ to the path $-1\to-1+iT\to\frac{1}{2}+iT$. Since $P_{\boldsymbol\kappa}(1-\bar s)=\overline{P_{\boldsymbol\kappa}(s)}$, the variation of $\arg P_{\boldsymbol\kappa}$ along the reversed image path $\frac{1}{2}+iT\to-1+iT\to-1$ equals $\Delta_{\mathbb{G}_T}\arg P_{\boldsymbol\kappa}$. The two lower quarters are handled in the same way with $s\mapsto\bar s$. The four quarter paths make up $\partial\mathfrak{R}_T$, so $4\Delta_{\mathbb{G}_T}\arg P_{\boldsymbol\kappa}=2\pi\cdot2N_{\boldsymbol\kappa}(T)$.

For the second equality, write $P_{\boldsymbol\kappa}=G\zeta_{\boldsymbol\kappa}$. Along $\mathbb{G}_T$, $s(s-1)$ moves from 2 to $-(\frac{1}{4}+T^2)$ inside the closed upper half-plane, so its argument varies by π. $\pi^{-s/2}\Gamma(s/2)$ is holomorphic and zero-free in $\Re s>0$, so its argument variation along $\mathbb{G}_T$ equals that along $[2,\frac{1}{2}]\cup[\frac{1}{2},\frac{1}{2}+iT]$, which is $0+\vartheta(T)$. □

Theorem 18.51 (Counting bound for quasi-solutions)**.** *There are absolute constants* T_0 *and* c_5 *such that, for every finite section in* $\mathbb{K}(C,\mathbb{M})$ *and every* $T\ge T_0$*,*

$$\big|N_{\boldsymbol\kappa}(T)-n_{\mathrm{RVM}}(T)\big|\le(\mathrm{c}_1+\mathrm{c}_2\upsilon)\log(T+8)+\mathrm{c}_3\mathbb{B}+\mathrm{c}_4, \tag{18.51}$$

with

$$\mathrm{c}_1=\frac{5}{2\log(4/3)}=8.690\ldots,\quad \mathrm{c}_2=\frac{2}{\log(4/3)}+\frac{1}{\pi}=7.270\ldots,\quad \mathrm{c}_3=\frac{1}{\log(4/3)}+\frac{1}{\pi}=3.794\ldots, \tag{18.52}$$

$\mathrm{c}_4:=\frac{3}{2}+\frac{\mathrm{c}_6}{\pi}+\frac{1}{T_0}+\frac{2\mathrm{c}_5}{\log(4/3)}$ *with* c_5 *the constant of Lemma 18.49 and* $\mathrm{c}_6:=\frac{3}{2}$*, and* $\mathbb{B}:=\mathbb{B}_0+\mathbb{B}_3+\mathbb{M}\mathrm{c}_\ell$*. The constants* $\mathrm{c}_1,\ldots,\mathrm{c}_6$ *do not depend on* k^**,* n^**,* C*,* $\mathbb{M}$ *or* υ*;* $\mathbb{B}$ *depends on* C *(through* $\mathfrak{B}(C)$*), on* $\mathbb{M}$*, and linearly on* υ *(through* $\mathbb{B}_2$*), so the total coefficient of* υ *in* (18.51) *is* $\mathrm{c}_2\log(T+8)+\mathrm{c}_3\log(2\sqrt{2})$*. In Corollary 18.52 the parameters* C*,* $\mathbb{M}$*,* υ *are fixed before* $k\to\infty$*.*

Proof. By continuity from the right it suffices to take $T \notin \{\kappa_k\}$. By Lemma 18.50, $N_{\boldsymbol{\kappa}}(T) - 1 - \vartheta(T)/\pi = \pi^{-1}\Delta_{\mathbb{G}_T} \arg \zeta_{\boldsymbol{\kappa}}$.

Vertical piece. $\arg \zeta_{\boldsymbol{\kappa}} = \Im \log \mathcal{Y} + \Im \mathcal{D}_{\mathsf{a}}$, with $\Im \log \mathcal{Y}(2) = \Im \mathcal{D}_{\mathsf{a}}(2) = 0$. Its variation is at most $\mathbb{E}(T) + \mathbb{M}\mathrm{c}_\ell$ by Lemma 18.46.

Horizontal piece. The variation is $\Delta \arg P_{\boldsymbol{\kappa}} - \Delta \arg G$. The first is at most $\pi(n_T + 1)$ by Lemma 18.49. The second is $-\Im \int_{1/2}^{2} (G'/G)(\sigma + iT)\, d\sigma$. Here $|\Im(G'/G)| \le |\frac{1}{s}| + |\frac{1}{s-1}| + \frac{1}{2}|\arg \frac{s}{2}| + O(1/T) \le 1$ for $T \ge T_0$, by Stirling's formula for Γ'/Γ [64, §12.33]. So the second variation is at most $\mathrm{c}_6 := \frac{3}{2}$.

Main term. By the Stirling expansion of ϑ [64, §12.33], $1 + \vartheta(T)/\pi = \mathcal{N}_{\mathrm{RVM}}(T) + O(1/T)$, and this error is at most $1/T$ for $T \ge T_0$. With (18.2):

$$|N_{\boldsymbol{\kappa}}(T) - n_{\mathrm{RVM}}(T)| \le n_T + 1 + \frac{\mathbb{E}(T) + \mathbb{M}\mathrm{c}_\ell + \mathrm{c}_6}{\pi} + \frac{1}{T} + \frac{1}{2}. \tag{18.53}$$

Insert (18.49) and $\mathbb{E}(T) \le \upsilon \log(T + 8)$: the coefficient of $\log(T + 8)$ is $\frac{5/2 + 2\upsilon}{\log(4/3)} + \frac{\upsilon}{\pi} = \mathrm{c}_1 + \mathrm{c}_2\upsilon$, the terms in $\mathbb{B}_0 + \mathbb{B}_3 + \mathbb{M}\mathrm{c}_\ell$ are at most $(\frac{1}{\log(4/3)} + \frac{1}{\pi})\mathbb{B} = \mathrm{c}_3\mathbb{B}$, and the remaining constant is $\frac{2\mathrm{c}_5}{\log(4/3)} + 1 + \frac{\mathrm{c}_6}{\pi} + \frac{1}{T_0} + \frac{1}{2} = \mathrm{c}_4$. □

Corollary 18.52 (Displacement bound)**.** *Assume $T_0 \ge 2\pi e$. For every k with $\tau_k - C \ge T_0$,*

$$|\kappa_k - \tau_k| \le \frac{2\pi\big[(\mathrm{c}_1 + \mathrm{c}_2\upsilon)\log(\tau_k + C + 8) + \mathrm{c}_3\mathbb{B} + \mathrm{c}_4\big]}{\log((\tau_k - C)/2\pi)}. \tag{18.54}$$

In particular $\limsup_{k\to\infty} |\kappa_k - \tau_k| \le 2\pi(\mathrm{c}_1 + \mathrm{c}_2\upsilon)$ *uniformly over sections with given $C, \mathbb{M}, \upsilon$. For every* $\mathrm{p} > 0$ *there is* $k_0(C, \mathbb{M}, \upsilon, \mathrm{p})$ *such that* $|\kappa_k - \tau_k| \le 2\pi(\mathrm{c}_1 + \mathrm{c}_2\upsilon) + \mathrm{p}$ *for* $k \ge k_0$.

Proof. Let $\mathbb{D}(T)$ be the right-hand side of (18.51); it is nondecreasing. Suppose $\kappa_k > \tau_k + \mathrm{q}$ with $0 < \mathrm{q} \le C$, and let $T := \tau_k + \mathrm{q} - \mathrm{h}'$ with small $\mathrm{h}' > 0$. The list is nondecreasing, so $N_{\boldsymbol{\kappa}}(T) \le k - 1$. By (18.2) and because $\mathcal{N}'_{\mathrm{RVM}}$ is increasing, $n_{\mathrm{RVM}}(T) \ge \mathcal{N}_{\mathrm{RVM}}(T) - \frac{1}{2} \ge k - 1 + (\mathrm{q} - \mathrm{h}')\mathcal{N}'_{\mathrm{RVM}}(\tau_k)$. Hence $(\mathrm{q} - \mathrm{h}')\frac{1}{2\pi}\log\frac{\tau_k}{2\pi} \le \mathbb{D}(\tau_k + C)$. Now suppose $\kappa_k < \tau_k - \mathrm{q}$, and let $T := \tau_k - \mathrm{q} + \mathrm{h}'$. Then $N_{\boldsymbol{\kappa}}(T) \ge k$ and $n_{\mathrm{RVM}}(T) \le \mathcal{N}_{\mathrm{RVM}}(T) + \frac{1}{2} \le k - (\mathrm{q} - \mathrm{h}')\mathcal{N}'_{\mathrm{RVM}}(\tau_k - C)$. Let $\mathrm{h}' \to 0$ in both cases. □

Remark 18.53 (What Corollary 18.52 does not give). The proof uses the imposed tube $|\kappa_k - \tau_k| \le C$ twice: through $\mathfrak{B}(C)$ in $\mathbb{B}$, and to replace $\mathbb{D}(\tau_k + \mathrm{q})$ by $\mathbb{D}(\tau_k + C)$. The conclusion is that, inside any finite tube, the coordinates of sufficiently high index lie in the smaller tube of radius $2\pi(\mathrm{c}_1 + \mathrm{c}_2\upsilon) + \mathrm{p}$. So the constraint is inactive there once $C > 2\pi(\mathrm{c}_1 + \mathrm{c}_2\upsilon) + \mathrm{p}$. For the indices $k < k_0(C, \mathbb{M}, \upsilon, \mathrm{p})$ nothing is proved, and the corollary does not show that unconstrained near-minimisers lie in a universal tube. The proof uses finite sections: Lemma 18.47 needs the polynomial growth $\mathrm{A}_0(2 + |s|)^{4k^*}$. Extending it to infinite lists requires a separate growth bound for $P_{\boldsymbol{\kappa}}/P_{\boldsymbol{\tau}}$ on $\Re s \ge 2$, which is not given here. The constant $2\pi\mathrm{c}_1 = 54.60\ldots$ is far from optimal.

Heuristic 18.54 (What a finite section controls)**.** The quantity in Theorem 18.51 is $\mathbb{E}(T)$, the L^1 norm on $\Re s = 2$ of the Laplace transform of the residual. It is not the Euclidean residual that a numerical section minimises over finitely many tests. Let the tests be $\mathrm{j}_0(u)\varphi(u)$ with a fixed window $\mathrm{j}_0 \in C_c^\infty(0, \infty)$, $\mathrm{j}_0 \equiv 1$ on $[u_0, u^*]$, and φ band-limited to $|\omega| \le \mathrm{w}^\sharp$. Then the kernel $\mathrm{j}_0(u)e^{-\frac{3}{2}u}e^{-ivu}$ is, approximately, in the span of the tests only for $|v| \lesssim \mathrm{w}^\sharp$. The residual on $(0, u_0)$ is then not controlled, and for a list at distance C from the clock it is expected to be of size $C\log(1/u)$ as $u \to 0$. Its contribution to $\mathbb{E}(T)$ is then $O(CTu_0\log(1/u_0))$. Since $u_0 \gtrsim 1/\mathrm{w}^\sharp$ in a band-limited section, Theorem 18.51 would apply with a given υ_0 only at heights $T \lesssim \upsilon_0\mathrm{w}^\sharp/(C\log \mathrm{w}^\sharp)$. None of these size estimates is proved here. They indicate which norm a section must make small. Lemma 18.46 shows that this norm is the right one.

Heuristic 18.55 (Informative part of a section)**.** Where the atoms are denser than the resolution of the tests, $e^{-u} \ll \pi/\mathrm{w}^\sharp$, the masses alone can approximate a smooth target by a Riemann sum

$\mathsf{a}_n \approx \mathfrak{g}(\log n)/n$ without moving any ordinate. That part of the window then carries no information on the ordinates. Counting unknowns against equations in the remaining window, $u^* \le \log \mathrm{w}^\sharp + O(1)$, gives an overdetermined system. This proves nothing about compatibility. It only explains why B is an existence statement and not a dimension count.

18.6. **A Kadec–Avdonin criterion with blocks of variable length.**

18.6.1. *Discrepancy and generating function.*

Definition 18.56 (Discrepancy, Poisson and conjugate operators)**.** Let $\mathscr{X} = (\mathrm{x}_k)_{k\in\mathbb{Z}}$ be a real nondecreasing sequence with

$$|\mathrm{x}_k - (k - \tfrac{1}{2})| \le \mathrm{c}_{\mathscr{X}} \log(2 + |k|) \qquad (k \in \mathbb{Z}). \tag{18.55}$$

Its *discrepancy* with respect to the half-integers is

$$\Psi_{\mathscr{X}}(\mathrm{x}) := \sum_{k\in\mathbb{Z}} \big(\mathbf{1}_{[\mathrm{x}_k,\infty)}(\mathrm{x}) - \mathbf{1}_{[k-\frac{1}{2},\infty)}(\mathrm{x})\big). \tag{18.56}$$

The k-th term is ± 1 on the interval between x_k and $k - \frac{1}{2}$ and 0 elsewhere. By (18.55), at most $O(\log(2+|\mathrm{x}|))$ terms are nonzero at x, so $|\Psi_{\mathscr{X}}(\mathrm{x})| \le \mathrm{c}'_{\mathscr{X}} \log(2+|\mathrm{x}|)$ and $\Psi_{\mathscr{X}} \in L^1(d\mathrm{x}/(1+\mathrm{x}^2))$. For $\mathrm{g} \in L^1(d\mathrm{x}/(1+\mathrm{x}^2))$ and $\mathrm{y} > 0$ put

$$\mathbb{P}_{\mathrm{y}}\mathrm{g}(\mathrm{x}) := \frac{1}{\pi}\int \frac{\mathrm{y}\,\mathrm{g}(\mathrm{s})\,d\mathrm{s}}{(\mathrm{x}-\mathrm{s})^2+\mathrm{y}^2}, \qquad \mathbb{Y}_{\mathrm{y}}\mathrm{g}(\mathrm{x}) := \frac{1}{\pi}\int \mathrm{g}(\mathrm{s})\Big[\frac{\mathrm{x}-\mathrm{s}}{(\mathrm{x}-\mathrm{s})^2+\mathrm{y}^2} + \frac{\mathrm{s}}{1+\mathrm{s}^2}\Big]d\mathrm{s}. \tag{18.57}$$

The last term in $\mathbb{Y}_{\mathrm{y}}$ only makes the integral converge. Changing it changes $\mathbb{Y}_{\mathrm{y}}\mathrm{g}$ by a constant, which is irrelevant for all statements below.

Lemma 18.57 (Generating function)**.** *Let $\mathscr{X} = (\mathrm{x}_k)_{k\in\mathbb{Z}}$ satisfy* (18.55)*, understood as a sequence indexed by $\mathbb{Z}$ in which a point of multiplicity m occurs m times. Let $\mathcal{F}_0(\mathfrak{z}) := \cos \pi\mathfrak{z}$, and fix $\mathfrak{z}_0 \in \mathbb{R} \setminus (\mathscr{X} \cup (\mathbb{Z} + \frac{1}{2}))$. The series*

$$\mathfrak{G}_{\mathscr{X}}(\mathfrak{z}) := \sum_{k\in\mathbb{Z}} \Big(\frac{1}{\mathfrak{z} - \mathrm{x}_k} - \frac{1}{\mathfrak{z} - k + \frac{1}{2}}\Big) \tag{18.58}$$

converges absolutely and locally uniformly off $\mathscr{X} \cup (\mathbb{Z} + \frac{1}{2})$, and

$$\mathcal{F}_{\mathscr{X}}(\mathfrak{z}) := \mathcal{F}_0(\mathfrak{z}) \exp \int_{\mathfrak{z}_0}^{\mathfrak{z}} \mathfrak{G}_{\mathscr{X}}(\mathrm{z})\,d\mathrm{z} \tag{18.59}$$

is a real entire function whose zero set is $\mathscr{X}$, with multiplicities. Moreover:

(i) *for $\mathrm{y} > 0$ there is a constant $\mathrm{c}''_{\mathscr{X}}$ such that*

$$\log|\mathcal{F}_{\mathscr{X}}(\mathrm{x}+i\mathrm{y})| = \log|\mathcal{F}_0(\mathrm{x}+i\mathrm{y})| + \pi\,\mathbb{Y}_{\mathrm{y}}\Psi_{\mathscr{X}}(\mathrm{x}) + \mathrm{c}''_{\mathscr{X}}; \tag{18.60}$$

(ii) *$\mathcal{F}_{\mathscr{X}}$ is of exponential type π, and its indicator diagram is the segment $[-i\pi, i\pi]$;*

(iii) *if $\mathscr{X}$ is symmetric, $\mathrm{x}_{1-k} = -\mathrm{x}_k$, and $\mathfrak{z}_0 = 0 \notin \mathscr{X}$, then $\mathcal{F}_{\mathscr{X}}$ is even.*

Proof. Each term of $\mathfrak{G}_{\mathscr{X}}$ equals $\frac{x_k - k + \frac{1}{2}}{(\mathfrak{z}-\mathrm{x}_k)(\mathfrak{z}-k+\frac{1}{2})} = O(\log(2+|k|)/k^2)$ locally uniformly, so the series converges. $\mathfrak{G}_{\mathscr{X}}$ is meromorphic with simple poles, with residue $+m$ at a point of $\mathscr{X}$ of multiplicity m and -1 at each half-integer; coincident poles combine their residues. The residues are integers, so $\exp\int_{\mathfrak{z}_0}^{\mathfrak{z}} \mathfrak{G}_{\mathscr{X}}$ is single-valued. It has a pole of order one at each half-integer not in $\mathscr{X}$, cancelled by the zero of $\mathcal{F}_0$. Hence $\mathcal{F}_{\mathscr{X}}$ is entire with zero set $\mathscr{X}$. $\mathfrak{G}_{\mathscr{X}}(\bar{\mathfrak{z}}) = \overline{\mathfrak{G}_{\mathscr{X}}(\mathfrak{z})}$ and $\mathfrak{z}_0$ is real, so $\mathcal{F}_{\mathscr{X}}$ is real on $\mathbb{R}$. Under the symmetry in (iii), $\mathfrak{G}_{\mathscr{X}}$ is odd, so $\int_0^{\mathfrak{z}} \mathfrak{G}_{\mathscr{X}}$ is even, and $\mathcal{F}_{\mathscr{X}}$ is even.

(i) For $\Im\mathfrak{z} \neq 0$ put $\mathbb{A}_{\mathscr{X}}(\mathfrak{z}) := \int \Psi_{\mathscr{X}}(\mathrm{s})\big[\frac{1}{\mathfrak{z}-\mathrm{s}} + \frac{\mathrm{s}}{1+\mathrm{s}^2}\big]d\mathrm{s}$. For $f(\mathrm{s}) := \frac{1}{\mathfrak{z}-\mathrm{s}}$, the identity $f(\mathrm{x}_k) - f(k-\frac{1}{2}) = -\int f'(\mathrm{s})\big(\mathbf{1}_{[\mathrm{x}_k,\infty)} - \mathbf{1}_{[k-\frac{1}{2},\infty)}\big)(\mathrm{s})\,d\mathrm{s}$ summed over k (Fubini applies because $\sum_k \int |f'||\cdots| < \infty$ by

(18.55)) gives $\mathfrak{G}_{\mathscr{X}}(\mathfrak{z}) = -\int \Psi_{\mathscr{X}}(\mathrm{s})(\mathfrak{z}-\mathrm{s})^{-2}d\mathrm{s} = \mathbb{A}'_{\mathscr{X}}(\mathfrak{z})$. So $\mathcal{F}_{\mathscr{X}} = \mathrm{c}_+\mathcal{F}_0 e^{\mathbb{A}_{\mathscr{X}}}$ on $\Im\mathfrak{z}>0$, with $\mathrm{c}_+\neq 0$. Since $\Re\frac{1}{\mathfrak{z}-\mathrm{s}} = \frac{\mathrm{x}-\mathrm{s}}{(\mathrm{x}-\mathrm{s})^2+\mathrm{y}^2}$, $\Re\mathbb{A}_{\mathscr{X}}(\mathrm{x}+i\mathrm{y}) = \pi\mathbb{Y}_{\mathrm{y}}\Psi_{\mathscr{X}}(\mathrm{x})$ and $\Im\mathbb{A}_{\mathscr{X}}(\mathrm{x}+i\mathrm{y}) = -\pi\mathbb{P}_{\mathrm{y}}\Psi_{\mathscr{X}}(\mathrm{x})$.

(ii) Split $\mathbb{Y}_{\mathrm{y}}\Psi_{\mathscr{X}}(\mathrm{x})$ into $|\mathrm{s}-\mathrm{x}|\le 1$, $1<|\mathrm{s}-\mathrm{x}|$ with $|\mathrm{s}|\le 2|\mathfrak{z}|+2$, and $|\mathrm{s}|>2|\mathfrak{z}|+2$. Using $|\Psi_{\mathscr{X}}(\mathrm{s})|\le \mathrm{c}'_{\mathscr{X}}\log(2+|\mathrm{s}|)$, $\int_{|u|\le 1}\frac{|u|\,du}{u^2+\mathrm{y}^2} = \log(1+\mathrm{y}^{-2})$, and, for $|\mathrm{s}|>2|\mathfrak{z}|+2$, $|\frac{\mathrm{x}-\mathrm{s}}{(\mathrm{x}-\mathrm{s})^2+\mathrm{y}^2}+\frac{\mathrm{s}}{1+\mathrm{s}^2}|\le 8(1+|\mathfrak{z}|)/\mathrm{s}^2$, one gets

$$|\mathbb{Y}_{\mathrm{y}}\Psi_{\mathscr{X}}(\mathrm{x})| \le \mathrm{c}'''_{\mathscr{X}}(1+\log(1+\mathrm{y}^{-2}))\log^2(4+|\mathfrak{z}|) \qquad (\mathfrak{z}=\mathrm{x}+i\mathrm{y},\ \mathrm{y}>0). \tag{18.61}$$

For $|\mathrm{y}|\ge 1$, $\log|\mathcal{F}_{\mathscr{X}}(\mathfrak{z})| \le \pi|\mathrm{y}| + O(\log^2(4+|\mathfrak{z}|))$ by (18.60), (18.61) and $|\mathcal{F}_{\mathscr{X}}(\bar{\mathfrak{z}})| = |\mathcal{F}_{\mathscr{X}}(\mathfrak{z})|$. For $|\mathrm{y}_1|<1$, apply the sub-mean-value property of $\log|\mathcal{F}_{\mathscr{X}}|$ on the circle $|\mathfrak{z}-\mathfrak{z}_1| = \frac12$. The factor $\log(1+\mathrm{y}^{-2})$ is integrable along the circle uniformly in y_1, since it has at most two logarithmic singularities. This gives $\log|\mathcal{F}_{\mathscr{X}}(\mathfrak{z}_1)| \le \frac{3\pi}{2} + O(\log^2(4+|\mathfrak{z}_1|))$ for $|\mathrm{y}_1|<1$. Hence the type is at most π and the indicator satisfies $\mathrm{h}(\mathrm{u}) \le \pi|\sin\mathrm{u}|$. Since (18.60) is an equality and (18.61) is two-sided, $\log|\mathcal{F}_{\mathscr{X}}(i\mathrm{y})| = \log|\cos(i\pi\mathrm{y})| + \pi\mathbb{Y}_{\mathrm{y}}\Psi_{\mathscr{X}}(0) + \mathrm{c}''_{\mathscr{X}} = \pi|\mathrm{y}| + O(\log^2(4+|\mathrm{y}|))$ for $|\mathrm{y}|\ge 1$, so $\mathrm{h}(\pm\frac{\pi}{2}) = \pi$. The indicator diagram lies in $[-i\pi, i\pi]$, whose support function is $\pi|\sin\mathrm{u}|$, and it contains $\pm i\pi$, so it equals that segment. □

18.6.2. *The criterion.* A weight $\mathrm{w}>0$ on $\mathbb{R}$ is in (A_2) if

$$[\mathrm{w}]_{A_2} := \sup_I \Big(\frac{1}{|I|}\int_I w\Big)\Big(\frac{1}{|I|}\int_I w^{-1}\Big) < \infty, \tag{18.62}$$

the supremum being over bounded intervals.

Theorem 18.58 (Block $\frac14$-criterion)**.** *Let $\mathscr{X}$ satisfy* (18.55) *and suppose $\Psi_{\mathscr{X}} = \mathrm{A}+\mathrm{B}$ with measurable A,B such that:*

(i) $\|\mathrm{A}\|_\infty \le \mathrm{d}_\star < \frac14$;

(ii) $\sup_x|\mathbb{Y}_1\mathrm{B}(\mathrm{x}) - \mathrm{c}_{\mathrm{B}}| \le \mathbb{U}$ *for some constant* c_{B}.

Then $\mathrm{w}(\mathrm{x}) := |\mathcal{F}_{\mathscr{X}}(\mathrm{x}+i)|^2$ *is in* (A_2) *and*

$$[\mathrm{w}]_{A_2} \le \frac{\pi^2\coth^2(\pi)\,e^{4\pi\mathbb{U}}}{\cos^2(2\pi\mathrm{d}_\star)}. \tag{18.63}$$

Proof. $\mathrm{B} = \Psi_{\mathscr{X}} - \mathrm{A} \in L^1(d\mathrm{x}/(1+\mathrm{x}^2))$, so $\mathbb{Y}_1\Psi_{\mathscr{X}} = \mathbb{Y}_1\mathrm{A} + \mathbb{Y}_1\mathrm{B}$. By (18.60), $\mathrm{w} = e^{2\mathrm{c}''_{\mathscr{X}}}|\mathcal{F}_0(\mathrm{x}+i)|^2 e^{2\pi\mathbb{Y}_1\mathrm{A}}e^{2\pi\mathbb{Y}_1\mathrm{B}}$.

Put $\mathbb{A}_{\mathrm{A}}(\mathfrak{z}) := \int \mathrm{A}(\mathrm{s})[\frac{1}{\mathfrak{z}-\mathrm{s}} + \frac{\mathrm{s}}{1+\mathrm{s}^2}]d\mathrm{s}$. It is holomorphic on $\Im\mathfrak{z}>0$, and $\mathbb{A}_{\mathrm{A}}(\mathfrak{o}+i)$ is holomorphic on $\Im\mathfrak{o}>-1$. Let $\mathcal{W}(\mathfrak{o}) := \exp(2\mathbb{A}_{\mathrm{A}}(\mathfrak{o}+i))$. For $\mathfrak{o} = \mathrm{x}_1 + i\mathrm{y}_1$ with $\mathrm{y}_1\ge 0$,

$$|\mathcal{W}(\mathfrak{o})| = e^{2\pi\mathbb{Y}_{1+\mathrm{y}_1}\mathrm{A}(\mathrm{x}_1)}, \qquad \arg\mathcal{W}(\mathfrak{o}) = -2\pi\mathbb{P}_{1+\mathrm{y}_1}\mathrm{A}(\mathrm{x}_1) \in [-2\pi\mathrm{d}_\star, 2\pi\mathrm{d}_\star]. \tag{18.64}$$

So $\mathcal{W}$ maps the closed upper half-plane into the sector $|\arg| \le \mathrm{b}_\star := 2\pi\mathrm{d}_\star < \frac{\pi}{2}$, and $|\mathcal{W}(\mathrm{x})| = e^{2\pi\mathbb{Y}_1\mathrm{A}(\mathrm{x})}$ on $\mathbb{R}$.

$\Re\mathcal{W}$ is a positive harmonic function on $\Im\mathfrak{o}>0$, continuous up to $\mathbb{R}$. By the Herglotz representation [18, Ch. I] it equals $\mathrm{c}\,\mathrm{y}_1 + \int \mathbb{P}_{\mathfrak{o}}(t)\,d\mathrm{n}(t)$ with $\mathrm{c}\ge 0$ and $\mathrm{n}\ge 0$, where n is the vague limit of the measures $\Re\mathcal{W}(t+i\mathrm{y})\,dt$ as $\mathrm{y}\downarrow 0$. Since $\Re\mathcal{W}$ is continuous on the closed half-plane, $\Re\mathcal{W}(t+i\mathrm{y}) \to \Re\mathcal{W}(t)$ uniformly on compact sets of t, so $\mathrm{n} = \Re\mathcal{W}(t)\,dt$. Here $\mathbb{P}_{\mathfrak{o}}(t) := \frac{1}{\pi}\frac{\mathrm{y}_1}{(t-\mathrm{x}_1)^2+\mathrm{y}_1^2}$. Hence $\Re\mathcal{W}(\mathfrak{o}) \ge \int \mathbb{P}_{\mathfrak{o}}(t)\Re\mathcal{W}(t)\,dt$. For an interval I with centre x_I, put $\mathfrak{o}_I := \mathrm{x}_I + i|I|/2$. Then $\mathbb{P}_{\mathfrak{o}_I}(t) \ge (\pi|I|)^{-1}$ for $t\in I$, and

$$\frac{1}{|I|}\int_I|\mathcal{W}| \le \frac{1}{\cos\mathrm{b}_\star}\frac{1}{|I|}\int_I\Re\mathcal{W} \le \frac{\pi}{\cos\mathrm{b}_\star}\Re\mathcal{W}(\mathfrak{o}_I) \le \frac{\pi}{\cos\mathrm{b}_\star}|\mathcal{W}(\mathfrak{o}_I)|. \tag{18.65}$$

$1/\mathcal{W}$ maps into the same sector, so the same holds for it. Multiplying, $[|\mathcal{W}|]_{A_2} \le \pi^2/\cos^2\mathrm{b}_\star$.

Finally $\mathrm{w} = \mathrm{g}\,|\mathcal{W}|$ with $\mathrm{g} := e^{2\mathrm{c}''_{\mathscr{X}}}|\mathcal{F}_0(\mathrm{x}+i)|^2 e^{2\pi\mathbb{Y}_1\mathrm{B}}$. Since $|\cos\pi(\mathrm{x}+i)|^2 = \cos^2(\pi\mathrm{x})\cosh^2\pi + \sin^2(\pi\mathrm{x})\sinh^2\pi \in [\sinh^2\pi, \cosh^2\pi]$ and $e^{2\pi\mathbb{Y}_1\mathrm{B}} \in [e^{2\pi(\mathrm{c}_{\mathrm{B}}-\mathbb{U})}, e^{2\pi(\mathrm{c}_{\mathrm{B}}+\mathbb{U})}]$, we have $\sup\mathrm{g}/\inf\mathrm{g} \le \coth^2(\pi)e^{4\pi\mathbb{U}}$. For any weight w_1, $[\mathrm{g}\mathrm{w}_1]_{A_2} \le (\sup\mathrm{g}/\inf\mathrm{g})[\mathrm{w}_1]_{A_2}$. □

Corollary 18.59 (Riesz bases)**.** *Under the hypotheses of Theorem 18.58:*

(a) *if $\mathscr{X}$ is uniformly discrete (in particular without repeated points), $\{e^{i\mathrm{x}_k\mathrm{v}}\}_{k\in\mathbb{Z}}$ is a Riesz basis of $L^2(-\pi,\pi)$;*

(b) *if $\mathscr{X}$ is a finite union of uniformly discrete sets, the generalised divided differences of $\{e^{i\mathrm{x}_k\mathrm{v}}\}$ in the sense of* [12] *form a Riesz basis of $L^2(-\pi,\pi)$. Consequently, for every partition of $\mathbb{Z}$ into consecutive blocks $\mathfrak{B}_m$ that are unions of the groups of* [12]*, the subspaces spanned by $\{\mathrm{v}^j e^{i\mathrm{x}\mathrm{v}}\}$, with $\mathrm{x}=\mathrm{x}_k$ for $k\in\mathfrak{B}_m$ and j below the multiplicity of* x*, form a Riesz basis of subspaces.*

Proof. By Lemma 18.57, $\mathcal{F}_{\mathscr{X}}$ is entire of exponential type with indicator diagram of width 2π and zero set $\mathscr{X}$. By Theorem 18.58, $|\mathcal{F}_{\mathscr{X}}(\mathrm{x}+i)|^2\in(A_2)$, which is equivalent to the Helson–Szegő condition [18, Ch. VI]. Then (a) is Pavlov's criterion [12, Thm. 1], and the first statement of (b) is [12, Thm. 3]. Both are stated for $L^2(0,2\pi)$, and translation by π multiplies each exponential by a unimodular constant. Grouping the vectors of a Riesz basis into disjoint groups gives a Riesz basis of subspaces. □

18.6.3. *Kadec, Avdonin, and variable block length.* Write $\mathrm{x}_k=k-\frac12+\mathrm{d}_k$, $I_k:=[k-1,k)$, $\mathsf{i}_k:=\mathbf{1}_{[\mathrm{x}_k,\infty)}-\mathbf{1}_{[k-\frac12,\infty)}$, so that $\Psi_{\mathscr{X}}=\sum_k\mathsf{i}_k$ and $\int\mathsf{i}_k=-\mathrm{d}_k$.

Proposition 18.60 (Bounded conjugate for block-supported oscillation)**.** *Let $\{\mathfrak{J}_m\}_{m\in\mathbb{Z}}$ be a partition of $\mathbb{R}$ into consecutive intervals of lengths $\mathfrak{n}_m\ge1$. Let $\mathrm{B}=\sum_m\mathrm{B}_m$ with $\operatorname{supp}\mathrm{B}_m\subset\mathfrak{J}_m$, $\int\mathrm{B}_m=0$ and $\|\mathrm{B}_m\|_{L^\infty}\le\mathrm{D}'_m$, $\sup_m\mathrm{D}'_m<\infty$. Put*

$$\mathbb{U}_{\mathrm{loc}}:=\sup_m\sup_{\mathrm{x}\in\mathfrak{J}_{m-1}\cup\mathfrak{J}_m\cup\mathfrak{J}_{m+1}}\Big|\int_{\mathfrak{J}_m}\frac{(\mathrm{x}-\mathrm{s})\,\mathrm{B}_m(\mathrm{s})}{\pi((\mathrm{x}-\mathrm{s})^2+1)}\,d\mathrm{s}\Big|,\qquad\mathbb{O}:=\sup_{m_0}\sum_{|m-m_0|\ge2}\frac{\mathfrak{n}_m^2}{\operatorname{dist}(\mathfrak{J}_{m_0},\mathfrak{J}_m)^2}.\tag{18.66}$$

If $\mathbb{U}_{\mathrm{loc}}<\infty$ and $\mathbb{O}<\infty$, then $\sup_x|\mathbb{Y}_1\mathrm{B}(\mathrm{x})-\mathrm{c}_{\mathrm{B}}|\le3\mathbb{U}_{\mathrm{loc}}+\frac{\mathbb{O}}{2\pi}\sup_m\mathrm{D}'_m$ for a constant c_{B}.

Proof. The term $\frac1\pi\int\frac{\mathrm{s}}{1+\mathrm{s}^2}\mathrm{B}_m(\mathrm{s})\,d\mathrm{s}$ equals $\frac1\pi\int(\frac{\mathrm{s}}{1+\mathrm{s}^2}-\frac{\mathrm{m}_m}{1+\mathrm{m}_m^2})\mathrm{B}_m(\mathrm{s})\,d\mathrm{s}$, where m_m is the centre of $\mathfrak{J}_m$. Since $|\frac{d}{d\mathrm{s}}\frac{\mathrm{s}}{1+\mathrm{s}^2}|\le\frac{1}{1+\mathrm{s}^2}$, its modulus is at most $\mathrm{D}'_m\mathfrak{n}_m^2/(2\pi(1+\operatorname{dist}(0,\mathfrak{J}_m)^2))$. If m_0 is the index with $0\in\mathfrak{J}_{m_0}$, then $\operatorname{dist}(0,\mathfrak{J}_m)\ge\operatorname{dist}(\mathfrak{J}_{m_0},\mathfrak{J}_m)$, so these terms are summable by $\mathbb{O}<\infty$. Their sum is the constant c_{B}. Let $\mathrm{x}\in\mathfrak{J}_{m_0}$. The three blocks with $|m-m_0|\le1$ contribute at most $3\mathbb{U}_{\mathrm{loc}}$. For $|m-m_0|\ge2$, $\int\mathrm{B}_m=0$ gives $\big|\int_{\mathfrak{J}_m}\mathrm{k}(\mathrm{x}-\mathrm{s})\mathrm{B}_m(\mathrm{s})\,d\mathrm{s}\big|\le\mathrm{D}'_m\mathfrak{n}_m\cdot\frac{\mathfrak{n}_m}{2}\sup_{\mathrm{s}\in\mathfrak{J}_m}|\mathrm{k}'(\mathrm{x}-\mathrm{s})|$ with $\mathrm{k}(\mathrm{v}):=\frac{\mathrm{v}}{\pi(\mathrm{v}^2+1)}$. Moreover $|\mathrm{k}'(\mathrm{v})|\le\frac{1}{\pi\mathrm{v}^2}$ and $|\mathrm{x}-\mathrm{s}|\ge\operatorname{dist}(\mathfrak{J}_{m_0},\mathfrak{J}_m)$. Sum over m. □

Corollary 18.61 (Kadec and Avdonin, with the constant $\frac14$)**.** (a) [44] *If $|\mathrm{d}_k|\le\mathrm{d}_\star<\frac14$ for all k, then $\{e^{i\mathrm{x}_k\mathrm{v}}\}$ is a Riesz basis of $L^2(-\pi,\pi)$.*
(b) [11] *If $\mathscr{X}$ is uniformly discrete, $\sup_k|\mathrm{d}_k|\le\mathrm{D}$, and for a fixed $\mathfrak{n}$ the means of d_k over consecutive blocks of $\mathfrak{n}$ indices are at most $\mathrm{d}_\star<\frac14$ in modulus, then $\{e^{i\mathrm{x}_k\mathrm{v}}\}$ is a Riesz basis of $L^2(-\pi,\pi)$.*

Proof. Both statements follow from Theorem 18.62 below: in (a) take $\mathfrak{n}_m\equiv1$, and in (b) take $\mathfrak{n}_m\equiv\mathfrak{n}$. In both cases the blocks at index distance $j\ge2$ from $\mathfrak{J}_{m_0}$ are at distance $(j-1)\mathfrak{n}$, so $\mathbb{O}\le2\sum_{j\ge2}\mathfrak{n}^2/((j-1)\mathfrak{n})^2=\pi^2/3$ and $\mathbb{V}\le2\mathrm{D}\log(2+3\mathfrak{n})$; for $\mathfrak{n}=1$, $\mathrm{D}_m=0$ and $\mathbb{V}=0$. For (a), $\mathrm{d}_\star<\frac14$ gives gaps $\ge1-2\mathrm{d}_\star>\frac12$, so $\mathscr{X}$ is uniformly discrete. □

Theorem 18.62 (Avdonin's theorem with variable block length)**.** *Let $\mathscr{X}$ be uniformly discrete with $\sup_k|\mathrm{d}_k|\le\mathrm{D}$. Let $\{\mathfrak{B}_m\}$ be a partition of $\mathbb{Z}$ into consecutive blocks of $\mathfrak{n}_m$ indices, and $\mathfrak{J}_m:=\bigcup_{k\in\mathfrak{B}_m}I_k$. Let $\bar{\mathrm{d}}_m$ be the mean of d_k over $\mathfrak{B}_m$, $\mathrm{D}_m:=\max_{k\in\mathfrak{B}_m}|\mathrm{d}_k-\bar{\mathrm{d}}_m|$, and $\mathfrak{n}_m^*:=\mathfrak{n}_{m-1}+\mathfrak{n}_m+\mathfrak{n}_{m+1}$. Assume*

(a) $\sup_m|\bar{\mathrm{d}}_m|\le\mathrm{d}_\star<\frac14$;
(b) $\mathbb{O}<\infty$, *with $\mathbb{O}$ as in* (18.66)*;*
(c) $\mathbb{V}:=\sup_m\mathrm{D}_m\log(2+\mathfrak{n}_m^*)<\infty$.

Then $\{e^{i\mathrm{x}_k \mathrm{v}}\}$ *is a Riesz basis of* $L^2(-\pi,\pi)$*, and* (18.63) *holds with*

$$\mathbb{U} = 16(1+\mathrm{D})^3 + \frac{6\mathbb{V}}{\pi} + \frac{\mathbb{V}\mathbb{O}}{2\pi\log 2}. \tag{18.67}$$

Proof. Write $\Psi_{\mathscr{X}} = \mathrm{A} + \mathrm{B}_1 + \mathrm{B}_2$ with $\mathrm{A} := -\sum_m \bar{\mathrm{d}}_m \mathbf{1}_{\mathfrak{J}_m}$, $\mathrm{B}_1 := \sum_k \mathrm{b}_k$ where $\mathrm{b}_k := \mathsf{i}_k + \mathrm{d}_k \mathbf{1}_{I_k}$, and $\mathrm{B}_2 := -\sum_m \sum_{k\in\mathfrak{B}_m} (\mathrm{d}_k - \bar{\mathrm{d}}_m)\mathbf{1}_{I_k}$. Then $\|\mathrm{A}\|_\infty \le \mathrm{d}_\star$ by (a).

B_1. b_k has mean zero, $|\mathrm{b}_k| \le 1+\mathrm{D}$, and support in an interval of length at most $2\mathrm{D}+1$ containing $\mathrm{m}_k := k - \frac12$. With k as above, $\sup|\mathrm{k}| = \frac{1}{2\pi}$ and $|\mathrm{k}'(\mathrm{v})| \le \frac{1}{\pi(1+\mathrm{v}^2)}$, so $|\int \mathrm{k}(\mathrm{x}-\mathrm{s})\mathrm{b}_k(\mathrm{s})\,d\mathrm{s}| = |\int(\mathrm{k}(\mathrm{x}-\mathrm{s}) - \mathrm{k}(\mathrm{x}-\mathrm{m}_k))\mathrm{b}_k(\mathrm{s})\,d\mathrm{s}| \le (1+\mathrm{D})(2\mathrm{D}+1)\min\big(\frac{1}{\pi}, (2\mathrm{D}+1)\sup_{\mathrm{s}\in\operatorname{supp}\mathrm{b}_k}|\mathrm{k}'(\mathrm{x}-\mathrm{s})|\big)$. At most $8\mathrm{D}+5 \le 8(1+\mathrm{D})$ indices have $|\mathrm{x}-\mathrm{m}_k| < 2(2\mathrm{D}+1)$, and they contribute at most $16(1+\mathrm{D})^3/\pi$. For the others $|\mathrm{x}-\mathrm{s}| \ge |\mathrm{x}-\mathrm{m}_k|/2$ on the support, so the term is at most $4(1+\mathrm{D})^3 \cdot \frac{4}{\pi}(4+(\mathrm{x}-\mathrm{m}_k)^2)^{-1}$. Since $\sum_k (4+(\mathrm{x}-\mathrm{m}_k)^2)^{-1} \le \frac12 + \frac{\pi}{2}$, they contribute at most $\frac{16}{\pi}(\frac12+\frac{\pi}{2})(1+\mathrm{D})^3$. The total is below $16(1+\mathrm{D})^3$. The modifying term of $\mathbb{Y}_1$ contributes a convergent sum of constants, as in Proposition 18.60.

B_2. On $\mathfrak{J}_m$ it has mean zero and modulus at most D_m. For x in $\mathfrak{J}_{m-1}\cup\mathfrak{J}_m\cup\mathfrak{J}_{m+1}$, $|\int_{\mathfrak{J}_m} \mathrm{k}(\mathrm{x}-\mathrm{s})\mathrm{B}_2(\mathrm{s})\,d\mathrm{s}| \le \frac{\mathrm{D}_m}{\pi}\int_{|u|\le \mathfrak{n}_m^*} \frac{|u|\,du}{1+u^2} \le \frac{2}{\pi}\mathrm{D}_m \log(2+\mathfrak{n}_m^*) \le \frac{2\mathbb{V}}{\pi}$. Moreover $\sup_m \mathrm{D}_m \le \mathbb{V}/\log 2$. Proposition 18.60, applied with the blocks $\mathfrak{J}_m$, $\mathrm{D}'_m = \mathrm{D}_m$ and $\mathbb{U}_{\mathrm{loc}} \le \frac{2\mathbb{V}}{\pi}$, gives $|\mathbb{Y}_1\mathrm{B}_2 - \mathrm{c}| \le \frac{6\mathbb{V}}{\pi} + \frac{\mathbb{V}\mathbb{O}}{2\pi\log 2}$.

Apply Theorem 18.58 and Corollary 18.59(a). □

Remark 18.63 (The price of variable length)*.* For fixed $\mathfrak{n}$, conditions (b) and (c) hold automatically, and Theorem 18.62 is Avdonin's theorem. If $\mathfrak{n}_m \to \infty$, the averaging condition (a) is still only $|\bar{\mathrm{d}}_m| \le \mathrm{d}_\star < \frac14$. However, (c) then requires the oscillation inside the block to decay like $1/\log \mathfrak{n}_m$. In Theorem 18.58 the oscillation enters only through the bounded-conjugate condition (ii), which allows cancellation inside the blocks and is weaker than (c). Without some condition of this kind the conclusion fails for growing blocks: a block of length $\mathfrak{n}$ with oscillation of size D_m can make $|\mathbb{Y}_1\mathrm{B}|$ of size $\mathrm{D}_m \log \mathfrak{n}$. Lyubarskii and Seip derive Kadec's theorem from their Muckenhoupt characterisation ([49, Thm. 2] for $p=2$), and indicate that the same technique gives results of Avdonin type. The proof of Theorem 18.58 is the sufficiency half of the Helson–Szegő argument, applied on the line $\Im\mathfrak{z} = 1$.

Literature. The theorem of Hruščev, Nikol'skii and Pavlov [3, p. 240], in the form stated by Kozma and Lev [4, Thm. 5.1], characterises the Riesz bases of exponentials $E(\Lambda)$ of $L^2(0,a)$ by three conditions: Λ is separated, $n_\Lambda(x) - ax$ belongs to $\mathrm{BMO}(\mathbb{R})$, and a decomposition condition of Helson–Szegő type on the generating function. Theorem 18.58 with Corollary 18.59(a) is close to the sufficiency half of that theorem, written for the discrepancy $\Psi_{\mathscr{X}}$, with the bounded-conjugate condition (ii) and the explicit constant (18.63). Avdonin's theorem appears with a fixed block length in [11] and in [4, Thm. 3.3]. In the sources consulted ([4, 49, 12], and [3] through its statement in [4]) we did not find a version with block lengths that vary from block to block, as in Theorem 18.62. The search was not exhaustive, so this is weak evidence, and novelty is not claimed.

18.7. **Application to lists near the RvM clock.**

18.7.1. *Unfolding.*

Definition 18.64 (Unfolded list)**.** Let $\mathfrak{t}_0 := 9.6769\ldots$ be the zero of $\mathcal{N}_{\mathrm{RVM}}$ in $(2\pi,\infty)$. For a C-admissible list $\boldsymbol{\kappa}$ with $\kappa_1 > \mathfrak{t}_0$ (true for $\boldsymbol{\tau}$ and $\boldsymbol{\gamma}$), put $\mathrm{x}_k := \mathcal{N}_{\mathrm{RVM}}(\kappa_k) > 0$ for $k \ge 1$ and $\mathrm{x}_{1-k} := -\mathrm{x}_k$. The resulting symmetric sequence $\mathscr{X}_{\boldsymbol{\kappa}}$ has discrepancy $\Psi_{\boldsymbol{\kappa}} := \Psi_{\mathscr{X}_{\boldsymbol{\kappa}}}$, and $\mathcal{F}_{\boldsymbol{\kappa}} := F_{\mathscr{X}_{\boldsymbol{\kappa}}}$ with $\mathfrak{z}_0 = 0$. For the clock, $\mathscr{X}_{\boldsymbol{\tau}} = \mathbb{Z} + \frac12$ and $\mathcal{F}_{\boldsymbol{\tau}} = \mathcal{F}_0$.

Lemma 18.65 (Unfolded discrepancy)**.** *For a* C*-admissible list with* $\kappa_1 > \mathfrak{t}_0$*,* $|\mathrm{x}_k - (k-\frac12)| \le \frac{C}{2\pi}\log\frac{\tau_k + C}{2\pi}$ *for* $k \ge 1$*, so* (18.55) *holds. Moreover, for* $\mathrm{x} = \mathcal{N}_{\mathrm{RVM}}(t)$ *with* $\mathrm{x} > 0$*,*

$$\Psi_{\boldsymbol{\kappa}}(\mathrm{x}) = N_{\boldsymbol{\kappa}}(t) - n_{\mathrm{RVM}}(t), \qquad |N_{\boldsymbol{\kappa}}(t) - n_{\mathrm{RVM}}(t)| \le \frac{C}{\pi}\log\frac{t+C}{2\pi} + 2, \tag{18.68}$$

and $\Psi_{\boldsymbol{\kappa}}(-\mathrm{x}) = -\Psi_{\boldsymbol{\kappa}}(\mathrm{x})$ *for almost every* x.

Proof. By the mean value theorem and the monotonicity of $\mathcal{N}'_{\mathrm{RVM}}$, $|\mathcal{N}_{\mathrm{RVM}}(\kappa_k) - \mathcal{N}_{\mathrm{RVM}}(\tau_k)| \le C\mathcal{N}'_{\mathrm{RVM}}(\tau_k + C)$. For $\mathrm{x} > 0$ the terms of (18.56) with $k \le 0$ vanish, because both x_k and $k - \frac{1}{2}$ are negative. For $k \ge 1$, $\mathrm{x}_k \le \mathrm{x}$ if and only if $\kappa_k \le t$, by monotonicity of $\mathcal{N}_{\mathrm{RVM}}$. If $N_{\boldsymbol{\kappa}}(t) = j$, then $\tau_j \le t + C$ and $\tau_{j+1} > t - C$, so $n_{\mathrm{RVM}}(t - C) \le j \le n_{\mathrm{RVM}}(t + C)$. By (18.2), $n_{\mathrm{RVM}}(t+C) - n_{\mathrm{RVM}}(\max(t-C, 2\pi)) \le \mathcal{N}_{\mathrm{RVM}}(t+C) - \mathcal{N}_{\mathrm{RVM}}(\max(t-C,2\pi)) + 1$. This is at most $\frac{2C}{2\pi}\log\frac{t+C}{2\pi} + 2$, the extra 1 covering $t - C < 2\pi$, where $\mathcal{N}_{\mathrm{RVM}} \ge -\frac{1}{8}$ and $n_{\mathrm{RVM}} = 0$. Oddness follows from the symmetry of $\mathscr{X}_{\boldsymbol{\kappa}}$ and of $\mathbb{Z} + \frac{1}{2}$. □

Proposition 18.66 (The condition of $\mathscr{O}$ gives the averaging hypothesis with blocks $\asymp \log^2 T$)**.** *Let* $\boldsymbol{\kappa}$ *be* C*-admissible with* $\kappa_1 > \mathfrak{t}_0$*, and assume the third condition of* $\mathscr{O}$ *(16.1) in the form*

$$\Big|\int_{2\pi}^{R} (N_{\boldsymbol{\kappa}}(t) - \mathcal{N}_{\mathrm{RVM}}(t))\,dt\Big| \le C_2 \log R \qquad (R \ge 3), \tag{18.69}$$

with a constant $C_2 > 0$*; this is no restriction, since* (18.69) *with one constant holds with every larger one, and it is needed because the block length* $\mathrm{H}(T) := 16C_2 \log T$ *must be positive. Put* $\mathrm{H}(T) := 16C_2 \log T$*. There is* $T_1 = T_1(C, C_2)$ *such that the following holds for the points* $T_{m+1} := T_m + \mathrm{H}(T_m)$ *(*$m \ge 1$*) and the blocks* $\mathfrak{J}_m := [\mathcal{N}_{\mathrm{RVM}}(T_m), \mathcal{N}_{\mathrm{RVM}}(T_{m+1}))$ *with their mirror images* $-\mathfrak{J}_m$*. The step function* $\mathrm{A}_{\boldsymbol{\kappa}}$*, equal to the mean of* $\Psi_{\boldsymbol{\kappa}}$ *on each* $\pm\mathfrak{J}_m$ *and to* 0 *on* $(-\mathcal{N}_{\mathrm{RVM}}(T_1), \mathcal{N}_{\mathrm{RVM}}(T_1))$*, satisfies*

$$\|\mathrm{A}_{\boldsymbol{\kappa}}\|_\infty \le \tfrac{3}{16}, \qquad |\mathfrak{J}_m| = \frac{8C_2}{\pi}\log^2 T_m\,(1 + o(1)) \quad (m \to \infty). \tag{18.70}$$

The remainder $\mathrm{B}_{\boldsymbol{\kappa}} := \Psi_{\boldsymbol{\kappa}} - \mathrm{A}_{\boldsymbol{\kappa}}$ *has mean zero on every block. For* $\boldsymbol{\kappa} = \boldsymbol{\gamma}$*,* (18.69) *holds unconditionally for some* C_2*.*

Proof. Write $\mathfrak{J} = [\mathcal{N}_{\mathrm{RVM}}(T), \mathcal{N}_{\mathrm{RVM}}(T + \mathrm{H}))$. Then

$$\begin{aligned}\int_{\mathfrak{J}} \Psi_{\boldsymbol{\kappa}}\,d\mathrm{x} &= \int_T^{T+\mathrm{H}} (N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}})(t)\,\mathcal{N}'_{\mathrm{RVM}}(t)\,dt \\ &= \mathcal{N}'_{\mathrm{RVM}}(T)\int_T^{T+\mathrm{H}} (N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}}) + \int_T^{T+\mathrm{H}} (N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}})(\mathcal{N}'_{\mathrm{RVM}}(t) - \mathcal{N}'_{\mathrm{RVM}}(T))\,dt.\end{aligned} \tag{18.71}$$

The last integral is at most $\mathrm{H}\cdot\frac{\mathrm{H}}{2\pi T}\big(\frac{C}{\pi}\log\frac{T+\mathrm{H}+C}{2\pi} + 1\big)$, by (18.68) and $\mathcal{N}''_{\mathrm{RVM}}(t) = \frac{1}{2\pi t}$.

Next, $\int_T^{T+\mathrm{H}}(N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}}) = \int_T^{T+\mathrm{H}}(N_{\boldsymbol{\kappa}} - \mathcal{N}_{\mathrm{RVM}}) + \int_T^{T+\mathrm{H}}(\mathcal{N}_{\mathrm{RVM}} - n_{\mathrm{RVM}})$. The first piece is at most $2C_2\log(T + \mathrm{H})$ by (18.69). For the second, substitute $\mathrm{x} = \mathcal{N}_{\mathrm{RVM}}(t)$. By (18.2), $\mathcal{N}_{\mathrm{RVM}} - n_{\mathrm{RVM}} = \mathrm{saw}(\mathrm{x}) := \mathrm{x} - \lfloor \mathrm{x} + \frac{1}{2}\rfloor$, which is 1-periodic with mean zero and $|\mathrm{saw}| \le \frac{1}{2}$, and $dt = \mathfrak{g}_1(\mathrm{x})\,d\mathrm{x}$ with $\mathfrak{g}_1 := 1/\mathcal{N}'_{\mathrm{RVM}}(t(\mathrm{x}))$ positive and decreasing. On a full period $[j - \frac{1}{2}, j + \frac{1}{2})$, $|\int \mathrm{saw}\,\mathfrak{g}_1| = |\int \mathrm{saw}\,(\mathfrak{g}_1 - \mathfrak{g}_1(j))| \le \frac{1}{4}(\mathfrak{g}_1(j - \frac{1}{2}) - \mathfrak{g}_1(j + \frac{1}{2}))$, which telescopes. The two partial periods contribute at most $\mathfrak{g}_1(\mathcal{N}_{\mathrm{RVM}}(T))$. Hence $|\int_T^{T+\mathrm{H}}(\mathcal{N}_{\mathrm{RVM}} - n_{\mathrm{RVM}})| \le \frac{5}{4\mathcal{N}'_{\mathrm{RVM}}(T)}$.

Since $|\mathfrak{J}| \ge \mathcal{N}'_{\mathrm{RVM}}(T)\mathrm{H}$,

$$\Big|\frac{1}{|\mathfrak{J}|}\int_{\mathfrak{J}} \Psi_{\boldsymbol{\kappa}}\Big| \le \frac{2C_2\log(T+\mathrm{H})}{\mathrm{H}} + \frac{5}{4\mathcal{N}'_{\mathrm{RVM}}(T)\mathrm{H}} + \frac{\mathrm{H}}{2\pi T\mathcal{N}'_{\mathrm{RVM}}(T)}\Big(\frac{C}{\pi}\log\frac{T+\mathrm{H}+C}{2\pi} + 2\Big). \tag{18.72}$$

With $\mathrm{H} = 16C_2\log T$, the first term tends to $\frac{1}{8}$ and the others tend to 0, so the bound is at most $\frac{3}{16}$ for $T \ge T_1(C, C_2)$. The length is $|\mathfrak{J}_m| = \mathcal{N}'_{\mathrm{RVM}}(\mathrm{s})\mathrm{H}(T_m)$ with $\mathrm{s} \in [T_m, T_{m+1}]$. The mirror blocks follow by oddness. For $\boldsymbol{\gamma}$: $N_{\boldsymbol{\gamma}}(t) = 1 + \vartheta(t)/\pi + S(t)$ [61, §9.3], $\int_0^R S = O(\log R)$ [61, Thm. 9.9], and $1 + \vartheta/\pi - \mathcal{N}_{\mathrm{RVM}} = O(1/t)$. Finally, $\boldsymbol{\gamma}$ is C_η-admissible, unconditionally, by 2.2. In particular (18.55) holds for $\mathscr{X}_{\boldsymbol{\gamma}}$, by Lemma 18.65. □

Remark 18.67 (Proposition 18.66 is not an equivalence)*.* Proposition 18.66 is an implication. The averaging condition on blocks of length $\asymp \log^2 T$ does not imply (18.69): summing $|\int_{\mathfrak{J}_m}\Psi| \le \frac{3}{16}|\mathfrak{J}_m|$

over the blocks up to height R gives only $O(R\log R)$, not $O(\log R)$. The two conditions do not coincide.

Corollary 18.68 (Reduction to one condition)**.** *For every $\boldsymbol{\kappa}$ as in Proposition 18.66: if $\mathbb{Y}_1\mathrm{B}_{\boldsymbol{\kappa}}$ is bounded up to a constant, then $|\mathcal{F}_{\boldsymbol{\kappa}}(\mathrm{x}+i)|^2\in(A_2)$, with constant* (18.63) *for $\mathrm{d}_\star=\frac{3}{16}$. Corollary 18.59(a) then holds for $\mathscr{X}_{\boldsymbol{\kappa}}$ if it is uniformly discrete, and Corollary 18.59(b) if it is a finite union of uniformly discrete sets.*

Proof. Lemma 18.65, Proposition 18.66 and Theorem 18.58. □

18.7.2. *The weight is a quotient of synthetic zetas.*

Proposition 18.69 (Weight identity)**.** *Let $\boldsymbol{\kappa}$ be C-admissible and satisfy* (18.69)*. For $0<\hbar\le 1$ and $t\ge 2\pi$,*

$$\log\Big|\frac{\zeta_{\boldsymbol{\kappa}}}{\zeta_{\mathrm{RVM}}}\Big(\tfrac12+\hbar+it\Big)\Big|=\int_{2\pi}^{\infty}\frac{(t-\mathrm{s})\,(N_{\boldsymbol{\kappa}}-n_{\mathrm{RVM}})(\mathrm{s})}{(t-\mathrm{s})^2+\hbar^2}\,d\mathrm{s}+\mathrm{c}_{\boldsymbol{\kappa}}+\mathrm{m}_{\boldsymbol{\kappa},\hbar}(t),\tag{18.73}$$

where the integral converges as an improper integral and $\mathrm{c}_{\boldsymbol{\kappa}}$ is a constant. The remainder satisfies $|\mathrm{m}_{\boldsymbol{\kappa},\hbar}(t)|\le \mathrm{c}(C,C_2)\log(2+t)/(1+t)$.

Proof. For $s=\frac12+\hbar+it$, $|\frac14+\kappa^2+q(s)|=|\kappa^2+(\hbar+it)^2|=|\kappa-t+i\hbar|\,|\kappa+t-i\hbar|$. Hence, with $\mathrm{l}_\hbar(u):=\frac12\log(u^2+\hbar^2)$,

$$\log\Big|\frac{P_{\boldsymbol{\kappa}}}{P_{\boldsymbol{\tau}}}(s)\Big|=\sum_k\big[\mathrm{l}_\hbar(t-\kappa_k)-\mathrm{l}_\hbar(t-\tau_k)\big]+\sum_k\big[\mathrm{l}_\hbar(t+\kappa_k)-\mathrm{l}_\hbar(t+\tau_k)\big]-\sum_k\log\frac{\frac14+\kappa_k^2}{\frac14+\tau_k^2}.\tag{18.74}$$

Put $\mathfrak{g}_1:=N_{\boldsymbol{\kappa}}-n_{\mathrm{RVM}}$ and $\mathfrak{G}_1(\mathrm{s}):=\int_{2\pi}^{\mathrm{s}}\mathfrak{g}_1$. By (18.69), by the sawtooth estimate in the proof of Proposition 18.66 on $[\tau_1,\infty)$, and by $|\mathcal{N}_{\mathrm{RVM}}-n_{\mathrm{RVM}}|\le\frac12$ on $[2\pi,\tau_1]$, $|\mathfrak{G}_1(\mathrm{s})|\le \mathrm{c}\log(2+\mathrm{s})$. For f smooth with $\mathrm{f}'(\mathrm{s})=O(1/\mathrm{s})$ and $\mathrm{f}''(\mathrm{s})=O(1/\mathrm{s}^2)$, the sums $\sum_{k\le k^*}[\mathrm{f}(\kappa_k)-\mathrm{f}(\tau_k)]=-\int\mathrm{f}'\mathfrak{g}_{1,k^*}$ converge as $k^*\to\infty$ to $-\int^\infty\mathrm{f}'\mathfrak{g}_1$, an improper integral. Integration by parts against $\mathfrak{G}_1$ shows both the convergence and that the tail contributions vanish. With $\mathrm{f}=\mathrm{l}_\hbar(t-\cdot)$ this gives the integral in (18.73). With $\mathrm{f}=\log(\frac14+(\cdot)^2)$ it gives the constant $\mathrm{c}_{\boldsymbol{\kappa}}$. With $\mathrm{f}=\mathrm{l}_\hbar(t+\cdot)$ it gives $\mathrm{m}_{\boldsymbol{\kappa},\hbar}(t)=\int_{2\pi}^\infty\mathrm{k}_1'(\mathrm{s})\mathfrak{G}_1(\mathrm{s})\,d\mathrm{s}$, where $\mathrm{k}_1(\mathrm{s}):=\frac{t+\mathrm{s}}{(t+\mathrm{s})^2+\hbar^2}$ and $|\mathrm{k}_1'(\mathrm{s})|\le(t+\mathrm{s})^{-2}$. This yields the bound. □

Remark 18.70 (Local scale). With $\hbar=1$ and $\hbar_T:=\hbar/\mathcal{N}'_{\mathrm{RVM}}(T)=2\pi/\log(T/2\pi)$ (the notation of the proof of Theorem 18.71), the substitution $\mathrm{x}=\mathcal{N}_{\mathrm{RVM}}(t)$ maps the kernel $\frac{(t-\mathrm{s})\,ds}{(t-\mathrm{s})^2+\hbar_T^2}$ to $\frac{(\mathrm{x}-\mathrm{x}')\,d\mathrm{x}'}{(\mathrm{x}-\mathrm{x}')^2+1}$ to first order near $t=T$. Comparing (18.73) with (18.60) at $\mathrm{y}=1$, the unfolded weight $|\mathcal{F}_{\boldsymbol{\kappa}}(\mathrm{x}+i)|^2$ near $\mathrm{x}=\mathcal{N}_{\mathrm{RVM}}(T)$ is therefore, to first order, $|\zeta_{\boldsymbol{\kappa}}/\zeta_{\mathrm{RVM}}|^2$ on the line $\Re s=\frac12+\hbar_T$, up to the bounded factor $|\mathcal{F}_0(\mathrm{x}+i)|^2$. The identity (18.73) is exact. The identification with the unfolded weight is local, and its quantitative form for the Poisson part is Step 3 of the proof of Theorem 18.71.

18.7.3. *Obstruction for the zeta list.*

Theorem 18.71 (No (A_2) for the unfolded zeta ordinates)**.** *Let $\mathscr{X}_{\boldsymbol{\gamma}}$ be the unfolded list of the ordinates $\gamma_k>0$ of the nontrivial zeros of ζ, counted with multiplicity, and let $\mathcal{F}$ be any entire function of exponential type whose zero set is $\mathscr{X}_{\boldsymbol{\gamma}}$, with multiplicities. Then $|\mathcal{F}(\mathrm{x}+i\hbar)|^2\notin(A_2)$ for every real $\hbar\neq0$. The proof does not assume the Riemann hypothesis.*

Proof. Write $\Psi:=\Psi_{\boldsymbol{\gamma}}$ and $\mathcal{F}_{\boldsymbol{\gamma}}$ for the function of Definition 18.64. It is even and real by Lemma 18.57(iii), because $\mathrm{x}_1=\mathcal{N}_{\mathrm{RVM}}(\gamma_1)=0.45\ldots>0$, so $0\notin\mathscr{X}_{\boldsymbol{\gamma}}$.

Step 1: reduction to $\mathcal{F}_{\boldsymbol{\gamma}}$ and $\hbar>0$. $\mathcal{F}$ and $\mathcal{F}_{\boldsymbol{\gamma}}$ are entire of order at most 1 (exponential type) with the same zeros, with multiplicity, so by Hadamard's factorisation theorem both are e^{linear} times the same canonical product, and $\mathcal{F}/\mathcal{F}_{\boldsymbol{\gamma}}=e^{\mathrm{a}_1+\mathrm{b}_1\mathfrak{z}}$. Hence $|\mathcal{F}(\mathrm{x}+i\hbar)|^2=\mathrm{c}\,e^{2\mathrm{b}_1'\mathrm{x}}\mathrm{w}_0(\mathrm{x})$ with $\mathrm{b}_1'=\Re\mathrm{b}_1$, a constant $\mathrm{c}>0$, and $\mathrm{w}_0(\mathrm{x}):=|\mathcal{F}_{\boldsymbol{\gamma}}(\mathrm{x}+i\hbar)|^2$, which is even in x. Moreover $|\mathcal{F}_{\boldsymbol{\gamma}}(\mathrm{x}-i\hbar)|=|\mathcal{F}_{\boldsymbol{\gamma}}(\mathrm{x}+i\hbar)|$.

Suppose $e^{2\mathrm{b}_1'\mathrm{x}}\mathrm{w}_0 \in (A_2)$. Reflection preserves (A_2), so $e^{-2\mathrm{b}_1'\mathrm{x}}\mathrm{w}_0 \in (A_2)$ as well. By Step 2 below both logarithms are in BMO, so $4\mathrm{b}_1'\mathrm{x} \in \mathrm{BMO}$, which forces $\mathrm{b}_1' = 0$. So we may assume $\mathcal{F} = \mathcal{F}_\gamma$ and $\hbar > 0$, and that $\mathrm{w}_0 \in (A_2)$.

Step 2: BMO. Let $\mathrm{f} := \log \mathrm{w}_0$ and f_I its mean on I. By Jensen's inequality, $\frac{1}{|I|}\int_I e^{\pm(f-\mathrm{f}_I)} \ge 1$. The product of the two averages is at most $[\mathrm{w}_0]_{A_2}$, so each is at most $[\mathrm{w}_0]_{A_2}$. Since $u^2 \le 2e^{|u|} \le 2(e^u + e^{-u})$,

$$\frac{1}{|I|}\int_I |f - \mathrm{f}_I|^2 \le 4[\mathrm{w}_0]_{A_2} \qquad \text{for every interval } I. \tag{18.75}$$

By (18.60) and the boundedness of $\log|\mathcal{F}_0(\mathrm{x}+i\hbar)|$, $\mathrm{g}_0 := \pi\mathbb{Y}_\hbar\Psi \in \mathrm{BMO}$. Now take $\mathbb{A}_\hbar(\mathfrak{z}) := \mathbb{A}_{\mathscr{X}}(\mathfrak{z}+i\hbar)$, in the notation of the proof of Lemma 18.57, which is holomorphic on $\Im\mathfrak{z} > -\hbar$. Its real part on $\mathbb{R}$ is g_0, and its imaginary part on $\mathbb{R}$ is $-\pi\mathbb{P}_\hbar\Psi$. By (18.61), $\Re\mathbb{A}_\hbar(\mathfrak{z}) = O(\log^2(4+|\mathfrak{z}|))$ on $\Im\mathfrak{z} \ge 0$, and so is the Poisson integral $\mathbb{P}[\mathrm{g}_0]$. Hence $\Re\mathbb{A}_\hbar - \mathbb{P}[\mathrm{g}_0]$ is harmonic on $\Im\mathfrak{z} > 0$, continuous with zero boundary values, and $O(\log^2|\mathfrak{z}|)$. After odd reflection it is harmonic on $\mathbb{C}$ and $o(|\mathfrak{z}|)$, hence constant, hence zero. Therefore $\mathbb{A}_\hbar = \mathbb{P}[\mathrm{g}_0] + i\mathbb{Y}[\mathrm{g}_0] + ic$ on $\Im\mathfrak{z} > 0$, where $\mathbb{P}[\mathrm{g}_0] + i\mathbb{Y}[\mathrm{g}_0]$ is the holomorphic function with real part $\mathbb{P}[\mathrm{g}_0]$ and $\mathbb{Y}[\mathrm{g}_0](\mathrm{x}+iy) = \mathbb{Y}_\mathrm{y}\mathrm{g}_0(\mathrm{x})$. Letting $\mathrm{y} \to 0$, $-\pi\mathbb{P}_\hbar\Psi = \tilde{\mathrm{g}}_0 + c$ almost everywhere on $\mathbb{R}$, where $\tilde{\mathrm{g}}_0$ is the conjugate function of g_0, the a.e. boundary value of $\mathbb{Y}_\mathrm{y}u$ [18, Ch. III]. Conjugation is bounded on BMO [18, Ch. VI], so $\mathbb{P}_\hbar\Psi \in \mathrm{BMO}$. By the John–Nirenberg inequality [18, Ch. VI], there is $\mathrm{c}_{\mathrm{JN}} < \infty$ with

$$\frac{1}{|I|}\int_I \bigl|\mathbb{P}_\hbar\Psi - (\mathbb{P}_\hbar\Psi)_I\bigr|^2 \le \mathrm{c}_{\mathrm{JN}} \qquad \text{for every interval } I. \tag{18.76}$$

Step 3: back to the ordinates. By [61, §9.3], $N_\gamma(t) = 1 + \vartheta(t)/\pi + S(t)$, with $S(t) = O(\log t)$ [61, Thm. 9.4]. Lemma 18.65 and (18.2) then give, for $\mathrm{x} = \mathcal{N}_{\mathrm{RVM}}(t) > 0$, $\Psi(\mathrm{x}) = S(t) + \mathrm{r}(\mathrm{x})$ with $|\mathrm{r}| \le \mathrm{c}_e$. Here c_e bounds $|1 + \vartheta/\pi - \mathcal{N}_{\mathrm{RVM}}| + \frac12$ on $t \ge 2\pi$. Let $\mathfrak{t}_0 := \mathcal{N}_{\mathrm{RVM}}^{-1}(0)$ and $T \to \infty$, with $t \in [T, 2T]$ and $\hbar_t := \hbar/\mathcal{N}_{\mathrm{RVM}}'(t)$, $\hbar_T := \hbar/\mathcal{N}_{\mathrm{RVM}}'(T)$.

(a) The part of $\mathbb{P}_\hbar\Psi(\mathcal{N}_{\mathrm{RVM}}(t))$ coming from $\mathrm{x}' < 0$ is $O(\log T/T)$, since $|\Psi(\mathrm{x}')| \le \mathrm{c}\log(2 + |\mathrm{x}'|)$. The part coming from r is at most c_e. Substituting $\mathrm{x}' = \mathcal{N}_{\mathrm{RVM}}(\mathrm{s})$, the rest is $\int_{\mathfrak{t}_0}^\infty \mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}(\mathrm{s})S(\mathrm{s})\,d\mathrm{s}$ with $\mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}(\mathrm{s}) := \mathbb{P}_\hbar(\mathcal{N}_{\mathrm{RVM}}(t) - \mathcal{N}_{\mathrm{RVM}}(\mathrm{s}))\mathcal{N}_{\mathrm{RVM}}'(\mathrm{s})$, where $\mathbb{P}_\hbar(\mathrm{v}) := \frac{\hbar}{\pi(\mathrm{v}^2+\hbar^2)}$.

(b) Put $\mathbb{P}_t^\circ(\mathrm{s}) := \mathbb{P}_{\hbar_t}(t-\mathrm{s}) = \mathbb{P}_\hbar(\mathcal{N}_{\mathrm{RVM}}'(t)(t-\mathrm{s}))\mathcal{N}_{\mathrm{RVM}}'(t)$. For $|\mathrm{s}-t| \le T^{1/3}$, Taylor's formula with $\mathcal{N}_{\mathrm{RVM}}'' = \frac{1}{2\pi\mathrm{s}}$ gives $|\mathcal{N}_{\mathrm{RVM}}(t) - \mathcal{N}_{\mathrm{RVM}}(\mathrm{s}) - \mathcal{N}_{\mathrm{RVM}}'(t)(t-\mathrm{s})| \le \mathrm{d}_T := T^{-1/3}/\pi$. Hence $|\mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}(\mathrm{s}) - \mathbb{P}_t^\circ(\mathrm{s})| \le (5\mathrm{d}_T/\hbar)\mathbb{P}_t^\circ(\mathrm{s})$ for T large. With $|S| \le \mathrm{c}\log(3T)$ on that range, the core difference is $O(T^{-1/3}\log T)$. For $|\mathrm{s}-t| > T^{1/3}$ we use $\mathbb{P}_\hbar(\mathrm{v}) \le \frac{\hbar}{\pi\mathrm{v}^2}$, the convexity of $\mathcal{N}_{\mathrm{RVM}}$ and $|S(\mathrm{s})| \le \mathrm{c}\log(2+\mathrm{s})$. If $\mathrm{s} > t + T^{1/3}$, then $\mathcal{N}_{\mathrm{RVM}}(\mathrm{s}) - \mathcal{N}_{\mathrm{RVM}}(t) \ge \mathcal{N}_{\mathrm{RVM}}'(t)(\mathrm{s}-t) \ge \mathrm{s}-t$ for T large, so $\int_{t+T^{1/3}}^\infty \mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}(\mathrm{s})|S(\mathrm{s})|\,d\mathrm{s} \le \frac{\mathrm{c}\hbar}{\pi}\int_{T^{1/3}}^\infty \frac{\mathcal{N}_{\mathrm{RVM}}'(t+\mathrm{v})\log(2+t+\mathrm{v})}{\mathrm{v}^2}\,d\mathrm{v} = O(T^{-1/3}\log^2 T)$, splitting the v-integral at $\mathrm{v} = T$. If $\mathfrak{t}_0 \le \mathrm{s} < t - T^{1/3}$, then $\mathcal{N}_{\mathrm{RVM}}(t) - \mathcal{N}_{\mathrm{RVM}}(\mathrm{s}) \ge \mathcal{N}_{\mathrm{RVM}}'(\mathrm{s})(t-\mathrm{s})$ and $\mathcal{N}_{\mathrm{RVM}}'(\mathrm{s}) \ge \mathcal{N}_{\mathrm{RVM}}'(\mathfrak{t}_0) > 0$, so $\mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}(\mathrm{s}) \le \frac{\hbar}{\pi\mathcal{N}_{\mathrm{RVM}}'(\mathrm{s})(t-\mathrm{s})^2} \le \frac{\hbar}{\pi\mathcal{N}_{\mathrm{RVM}}'(\mathfrak{t}_0)(t-\mathrm{s})^2}$ and $\int_{\mathfrak{t}_0}^{t-T^{1/3}} \mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}|S| \le \frac{\mathrm{c}\hbar\log(3T)}{\pi\mathcal{N}_{\mathrm{RVM}}'(\mathfrak{t}_0)}\int_{T^{1/3}}^\infty \mathrm{v}^{-2}\,d\mathrm{v} = O(T^{-1/3}\log T)$. The same bounds hold for $\mathbb{P}_t^\circ$ with $\hbar_t$ in place of $\hbar$ and $\mathcal{N}_{\mathrm{RVM}}'(t)$ in place of $\mathcal{N}_{\mathrm{RVM}}'(\mathrm{s})$, the region $\mathrm{s} \gg T$ being handled by the split at $\mathrm{v} = T$. So that part tends to 0 as well.

(c) $\|\mathbb{P}_a - \mathbb{P}_b\|_{L^1} \le \frac{2}{\pi}|\log(a/b)|$, because $\int|\partial_c\mathbb{P}_c| = \frac{2}{\pi c}$. Since $\hbar_t/\hbar_T \in [1 - \frac{2\log 2}{\log(T/2\pi)}, 1]$, $|\int(\mathbb{P}_{\hbar_t} - \mathbb{P}_{\hbar_T})(t-\mathrm{s})S(\mathrm{s})\,d\mathrm{s}| = O(\log T/\log T) = O(1)$, again with the tails controlled by $|S(\mathrm{s})| \le \mathrm{c}\log(2+\mathrm{s})$.

(d) $\|\mathbb{P}_{\hbar_T} * S - S\|_{L^2[T,2T]} = O(\sqrt{T})$, with $S := 0$ on $(-\infty, \mathfrak{t}_0)$; this is Lemma 18.72 below.

Together, (a)–(d) give $\|\mathbb{P}_\hbar\Psi \circ \mathcal{N}_{\mathrm{RVM}} - S\|_{L^2[T,2T]} = O(\sqrt{T})$.

Step 4: contradiction. Let $\mathfrak{I}_T := [\mathcal{N}_{\mathrm{RVM}}(T), \mathcal{N}_{\mathrm{RVM}}(2T)]$ and $\bar{\mathrm{p}}_T := (\mathbb{P}_\hbar\Psi)_{\mathfrak{I}_T}$. By (18.76), $\int_T^{2T} |\mathbb{P}_\hbar\Psi(\mathcal{N}_{\mathrm{RVM}}(t)) - \bar{\mathrm{p}}_T|^2\mathcal{N}_{\mathrm{RVM}}'(t)\,dt \le \mathrm{c}_{\mathrm{JN}}|\mathfrak{I}_T| \le \mathrm{c}_{\mathrm{JN}}T\mathcal{N}_{\mathrm{RVM}}'(2T)$, so $\int_T^{2T} |\mathbb{P}_\hbar\Psi(\mathcal{N}_{\mathrm{RVM}}(t)) -$

$\bar{\mathrm{p}}_T|^2\,dt \le 2\mathrm{c}_{\mathrm{JN}}T$. With Step 3, $\int_T^{2T}|S-\bar{\mathrm{p}}_T|^2 = O(T)$. But

$$\int_T^{2T}|S-\mathrm{c}|^2 \ge \int_T^{2T} S^2 - \frac{1}{T}\Big(\int_T^{2T} S\Big)^2 = \frac{1}{2\pi^2}T\log\log T\,(1+o(1)) \tag{18.77}$$

for every constant c, by Selberg's theorem $\int_0^T S^2 = \frac{T}{2\pi^2}\log\log T + O(T(\log\log T)^{1/2})$ [58], with the normalisation $S(t)=\pi^{-1}\arg\zeta(\frac12+it)$ used here and in [15, p. 225], and $\int_T^{2T} S = O(\log T)$ [61, Thm. 9.9]. This is a contradiction. □

Lemma 18.72 (Smoothing S at scale $1/\log T$)**.** *Let $\hbar>0$ be fixed, $\hbar_T := 2\pi\hbar/\log(T/2\pi)$, and $S:=0$ on $(-\infty,\mathfrak{t}_0)$. Then $\|\mathbb{P}_{\hbar_T} * S - S\|_{L^2[T,2T]} = O_\hbar(\sqrt{T})$, unconditionally.*

Proof. Fujii's Main Theorem [15, Main Theorem], proved unconditionally by modifying Selberg's method, states for $q=1$, $k=1$, $T^{1/2+a}\le H\le T$ and $0<u\le H-(H/\sqrt{T})^{1/8}$ that $\int_T^{T+H}(S(t+u)-S(t))^2dt = c_1H\cdot 2\log(2+u\log T)\,(1+O((\log(2+u\log T))^{-1/2}))$ with an absolute c_1. The relative error is $O(1)$ when $u\log T$ is bounded, so the statement carries no asymptotic information there, and we use it only as the upper bound it contains: for $0<u\le 1$, $T\ge T_0$ and $H=T$,

$$\int_T^{2T}\big(S(t+u)-S(t)\big)^2dt \le \mathrm{c}_F T\log(2+u\log T), \tag{18.78}$$

with an absolute constant c_F. The range $0<u\le H-(H/\sqrt{T})^{1/8} = T-T^{1/16}$ of [15] contains $0<u\le 1$, and the same bound holds for $-1\le u<0$: the substitution $t\mapsto t-u$ moves the integral to $[T-|u|,2T-|u|]\subset[T/2,2T]$, covered by (18.78) at the base points $T/2$ (with $H=T/2$) and T (with $H=T$). By Minkowski's inequality, $\|\mathbb{P}_{\hbar_T}*S-S\|_{L^2[T,2T]} \le \int \mathbb{P}_{\hbar_T}(u)\|S(\cdot-u)-S\|_{L^2[T,2T]}\,du$. For $0<|u|\le 1$ the norm is at most $(2\mathrm{c}_F T\log(2+|u|\log T))^{1/2}$. For $1<|u|\le T/2$, Selberg's theorem $\int_0^T S^2 = \frac{1}{2\pi^2}T\log\log T + O(T(\log\log T)^{1/2})$ [58] bounds it by $\mathrm{c}(T\log\log T)^{1/2}$. For $|u|>T/2$, $|S(t)|\le \mathrm{c}\log(2+t)$ [61, Thm. 9.4] bounds it by $\mathrm{c}\sqrt{T}\log(3T+|u|)$. Substituting $u=\hbar_T v$ in the first range, where $\hbar_T\log T\to 2\pi\hbar$, and using $\int_{|u|>1}\mathbb{P}_{\hbar_T} = O(\hbar_T)$ and $\int_{|u|>T/2}\mathbb{P}_{\hbar_T}(u)\log(3T+|u|)\,du = O(\hbar_T\log T/T)$, all three contributions are $O(\sqrt{T})$. □

Remark 18.73 (A second proof)*.* Lemma 18.72 also follows from Selberg's mean-square approximation of $S(t)$ by $S_{\mathrm{x}_0}(t) := -\frac{1}{\pi}\sum_{p\le \mathrm{x}} p^{-1/2}\sin(t\log p)$, $\mathrm{x}_0 = T^\delta$ [58], and the Montgomery–Vaughan mean value theorem [50, Cor. 3]. Indeed $\mathbb{P}_{\hbar_T}*S_{\mathrm{x}_0} - S_{\mathrm{x}_0} = -\frac{1}{\pi}\sum_{p\le \mathrm{x}} p^{-1/2}(e^{-\hbar_T\log p}-1)\sin(t\log p)$ has mean square $\ll T\,\hbar_T^2\log^2\mathrm{x} \ll T$ on $[T,2T]$, and the remainder $S-S_{\mathrm{x}_0}$ is handled by Minkowski's inequality. Fujii's announcement [16] states the moment lemma of [15] in the form used in [17, Lemma 1].

Corollary 18.74 (Consequences for the zeta list)**.** (a) *$\{e^{ix_k\xi}\}_{x_k\in\mathscr{X}_\gamma}$ is not a Riesz basis of $L^2(-\pi,\pi)$.*
(b) *If $\mathscr{X}_\gamma$ is a finite union of uniformly discrete sets, its generalised divided differences are not a Riesz basis of $L^2(-\pi,\pi)$.*
(c) *For every decomposition $\Psi_\gamma = A+B$ with $\|A\|_\infty<\frac14$, $\mathbb{Y}_1B$ is unbounded. In particular this holds for the blocks of Proposition 18.66, where the averaging hypothesis is satisfied.*

Proof. (a) and (b): the criteria [12, Thm. 1, Thm. 3] are necessary and sufficient, and require a generating function satisfying Helson–Szegő, equivalently (A_2), on some line $\Im\mathfrak{z}=\hbar$. The line $\hbar=0$ is excluded, since $|\mathcal{F}(\mathrm{x})|^{-2}$ is not locally integrable at a real zero. All lines with $\hbar\neq 0$ are excluded by Theorem 18.71. (c): otherwise Theorem 18.58 would give $|\mathcal{F}_\gamma(\mathrm{x}+i)|^2\in(A_2)$. □

Heuristic 18.75 (Graduated version)**.** The proof of Theorem 18.71 uses intervals $\mathfrak{I}_T$ of length $\asymp T\log T$. On a window of L unfolded points at height T, the mean square oscillation of $\mathbb{P}_\hbar\Psi$ is governed by the variance of increments of S over t-intervals of length $L/\mathcal{N}'_{\mathrm{RVM}}(T)$. By Fujii's asymptotics this is expected to be of size $\frac{1}{\pi^2}\log L$. This suggests that local (A_2) constants on windows of L points grow with L but stay bounded for L bounded. That is the graduated stability

that a Fredholm formulation would need. None of this is proved here. The computations of Section 18.8 measure the ℓ^∞ norm of the inverse of square finite sections directly and find growth like $T^{1.8}$ up to $T = 487$.

Remark 18.76 (Bearing on route B). Theorem 18.71 rules out a uniform Riesz-basis (Fredholm) structure for the unfolded zeta list on the whole line, and hence a linear stability argument with constants independent of the height. It says nothing about statement B or B′. The compactness step of route B (Proposition 18.5) uses no stability, and uniqueness holds in every tube (Theorem 18.7); the open part of route B and of route B′ is existence alone (Corollaries 18.11 and 18.18).

Remark 18.77 (Literature on the zeta ordinates as an exponential system). We have not found any study of Riesz bases of exponentials, or of the condition (A_2), for the ordinates of ζ; the search was not exhaustive. The closest work found is the Fourier interpolation theorem of Bondarenko, Radchenko and Seip [5, Thm. 1.1]: an even function f analytic in a strip $|\Im z| < \frac{1}{2} + \epsilon$, with an integrability condition, is recovered by an explicit series from the values $f^{(j)}((\rho - \frac{1}{2})/i)$, $0 \le j < m(\rho)$, at the nontrivial zeros ρ with positive imaginary part, and the values of $\widehat{f}$ at the points $\log n/(4\pi)$. The theorem is unconditional (zeros off the line enter as complex nodes), it rests on a strengthened form of Knopp's abundance principle for Dirichlet series with a functional equation [45], and removing a single node breaks it. There the zeros and the logarithms of the integers are dual families of nodes, as the ordinates and the points $\log n$ are in the rung system (18.4); Theorem 18.71 concerns instead the Riesz-basis property of the exponentials of the unfolded ordinates, which is not used there.

18.8. **Numerical certificates for finite sections.** This section reports computations. Except for Theorem 18.78, which is proved by the ball-arithmetic computation of Section 18.8.5, they are not part of any proof; their role is to measure the quantities of Theorems 18.25 and 18.27 on actual sections and to locate where the staged construction fails. The programs and their outputs are listed in Appendix D and in `code/routeB/` of the repository. Except in Section 18.8.5, floating-point arithmetic was used and no interval arithmetic was run.

18.8.1. *Set-up.* The tests are Gaussians $\varphi_{c,\mathrm{w}}(u) := e^{-(u-c)^2/2\mathrm{w}^2}$ with centre $c \ge 9\mathrm{w}$, so that their restriction to $u > 0$ is, to working precision, supported in $(0, \infty)$; their cosine transform is $\mathsf{c}_{\varphi_{c,\mathrm{w}}}(t) = \mathrm{w}\sqrt{2\pi}\, e^{-(\mathrm{w}t)^2/2}\cos(ct)$. A Gaussian is not compactly supported, so these tests are not elements of $C_c^\infty(0,\infty)$ and the computed quantities are not literally those of Theorem 18.25, whose tests are in that space. The pairings against a compactly supported bump $\varphi_{c,\mathrm{w}}\chi$, with $\chi \in C_c^\infty(0,\infty)$ equal to 1 on $[c/2, 3c/2]$, differ from the computed ones by the pairings against the tail $\varphi_{c,\mathrm{w}}(1-\chi)$, which were not computed; the alternative, an extension of Theorem 18.25 to Schwartz tests, is not carried out here. For this reason, and because of the floating-point arithmetic, the certificates reported below are *candidate certificates*: floating-point evidence for the inequalities of the theorem evaluated on tests outside its hypothesis, not verified statements. Stage m uses the width $\mathrm{w}_m := 0.08 \cdot 0.85^{m-1}$, the height $T_m := 4/\mathrm{w}_m$, and the new tests φ_{c,w_m} with centres $c \in [9\mathrm{w}_m,\ \log(\max(T_m, 7)/2\pi) + 2]$ in steps of w_m; the test set $\mathscr{T}_m$ is the union of the levels $\le m$. The unknowns of stage m are the ordinates with $\tau_k \le 2.4/\mathrm{w}_m$ ("interior"); the factor $e^{-(\mathrm{w}_m\kappa)^2/2}$ makes the others invisible to the tests of level m. The masses are a_n^ζ for all $n \le e^{c+9\mathrm{w}}$, and k is 15.79. The pairings $\langle \mathsf{k}, \varphi\rangle$ were computed by adaptive quadrature. The zeta ordinates γ_k, $k \le 400$ ($\gamma_{400} = 679.74$), computed with `mpmath`, enter only in the diagnostic columns; no construction uses them.

Two checks fix the normalisation: $\langle \Re(\boldsymbol{\gamma}, \mathsf{a}^\zeta), \varphi\rangle \approx 10^{-16}$ for all tests tried, and $\langle \Re(\boldsymbol{\tau}, \mathsf{a}^\zeta), \varphi\rangle$ between 10^{-3} and 10^{-1}.

18.8.2. *Initialisation: the clock is outside the basin, the Gonek lists are inside. Receding horizon from the clock.* Starting at $\boldsymbol{\kappa} = \boldsymbol{\tau}$, each stage solved the overdetermined least-squares problem for the interior ordinates with all tests of levels $\le m$, with bounds $|\kappa_k - \tau_k| \le 3$ and the non-interior

ordinates frozen at their current values; every residual reported for these runs is therefore the residual of a frozen-tail finite model, not of the infinite system. This is the overdetermined geometry $\mathrm{t}_m \geq k_m^*$: the Jacobian has a left inverse (the pseudo-inverse) and Gauss–Newton is a least-squares iteration. Theorem 18.25 requires a right inverse, $\mathbb{J}_m\mathbb{Q}_m = I_{\mathrm{t}_m}$, which forces $\mathrm{t}_m \leq k_m^*$; the two geometries meet only in the square case $\mathrm{t}_m = k_m^*$, which is the one used for the certificates of Section 18.8.4. The receding-horizon runs are therefore diagnostics of the basin, not evaluations of the theorem. The iteration stalls at the first Gram-type failure of the clock: the fit returns $\kappa_{34} = 109.75$ where $\gamma_{34} = 111.03$ ($\tau_{34} = 110.04$), and $\kappa_{41} = 125.50$ where $\gamma_{41} = 124.26$; from $m = 9$ on the residual on the lagged levels $\leq m-4$ grows from $2 \cdot 10^{-5}$ to $4 \cdot 10^{-2}$ and the front of ordinates within 10^{-2} of $\boldsymbol{\gamma}$ stays at $k \approx 31$.

Receding horizon from a Gonek list. The same iteration started at $\boldsymbol{\kappa}^{(\mathrm{X})}$ with $\mathrm{X} = 8$ (983 roots up to $t = 1400$, that is, every solution of (18.29) from $t \approx 14$ on; this is not the list of Lemma 18.32, whose prefix is the clock below $\mathfrak{t}_8 \approx 2\pi e^{2\mathrm{n}_8+2}$ with $\mathrm{n}_8 \approx 7$, so that the lemma's tube does not cover the list used here, whose monotonicity was checked directly; the roots were computed from the Riemann–Siegel expansion of ϑ and the sum (18.27)) converged through $m = 16$, $T_{16} = 572$, 165 interior ordinates and 4985 tests: all interior ordinates within 10^{-2} of γ_k, 90 of them within 10^{-4}, the first twenty within $5 \cdot 10^{-6}$, residual on the lagged levels between 10^{-6} and $9 \cdot 10^{-6}$ at every stage, and $\max_k |\kappa_k - \tau_k| = 0.99$ throughout. The residual over all tests of level $\leq m$ stays at $2 \cdot 10^{-3}$, which is the contribution of the frozen edge.

Why. The basin of the exact solution, measured by random perturbations of $\boldsymbol{\gamma}$ in the interior block, has radius 0.1 to 0.25 mean spacings. The clock lies at up to 0.66 spacings from $\boldsymbol{\gamma}$ for $k \leq 400$, with 50 indices beyond 0.25; the lists $\boldsymbol{\kappa}^{(\mathrm{X})}$ lie within 0.184 ($\mathrm{X} = 3$), 0.081 ($\mathrm{X} = 5$), 0.053 ($\mathrm{X} = 8$), 0.047 ($\mathrm{X} = 12$) and 0.035 ($\mathrm{X} = 20$) spacings for $k \leq 160$, and within 0.103 spacings for $k \leq 400$ at $\mathrm{X} = 8$ (Figure 1). The ℓ^2 conditioning of the interior block at $\boldsymbol{\gamma}$ is moderate and roughly uniform up to $T = 299$: condition numbers 17 to 75, smallest singular value 0.08 to 0.15. The coupling of the interior to the frozen edge, $\|\mathbb{J}_I^+ \mathbb{J}_E\|_{\infty\to\infty}$, is between 0.25 and 0.47, and the shift it induces on the low ordinates decays from $2 \cdot 10^{-2}$ at $m = 2$ to 10^{-5} for $m \geq 10$.

18.8.3. *Certificates at the Gonek list: the direct bounds fail.* At $\boldsymbol{\kappa}^{(8)}$, with the interior unknowns of stage m and three choices of the right inverse, the quantities of Theorem 18.25 were computed for $m = 1, \ldots, 14$ (T from 50 to 414).

Right inverse and norm	Certificate index	Range over $m = 1, \ldots, 14$
square subsystem (tests chosen by pivoted QR), ℓ^∞	$2\mathbb{L}_m^2 \mathbb{H}_m \mathsf{h}_m$	$56 \to 1.2 \cdot 10^7$
pseudo-inverse of the full system, ℓ^2, $\beta_2 = 1/\sigma_{\min}$	$\beta_2 \mathrm{K}_2 \delta_2$ (Kantorovich)	$0.47 \to 233$
pseudo-inverse of the full system, ℓ^∞, diagonal Hessian	$2\mathrm{c}_{\mathrm{diag}} \mathsf{h}$	$1.3 \to 1.6 \cdot 10^3$

None of the certificates is satisfied, not even at $T = 50$. Yet one Gauss–Newton step from $\boldsymbol{\kappa}^{(8)}$ reduces $\max_k |\kappa_k - \gamma_k|$ by a factor between 4 and 15 at every height ($0.088 \to 0.016$ at $T = 572$), and $\sigma_{\min}$ in ℓ^2 stays between 1/17 and 1/5. The worst-case bounds lose a factor 10^3 to 10^4 through cancellation: $\mathbb{H}_m$ counts every unknown with the full weight $\int |\varphi| u^2$, while the actual second-order term of $\mathbb{F}_m$ along the Newton step is a sum of k_m^* oscillating terms.

18.8.4. *Floating-point candidate certificates for exact square sections.* The iterate of Section 18.8.2 was polished, stage by stage, by Newton's method on the square system (n tests selected by pivoted QR on $\mathbb{J}^T$, n interior unknowns) to residual 10^{-13}. At the polished point x_0 the certificate of Theorem 18.25 was evaluated in the sharper form that uses the separable structure of F_m: with $\mathbb{Q}$ an approximate inverse of the square Jacobian $\mathbb{J}$ at x_0, computed by solving $\mathbb{J}\mathbb{Q} = I$ with an LU

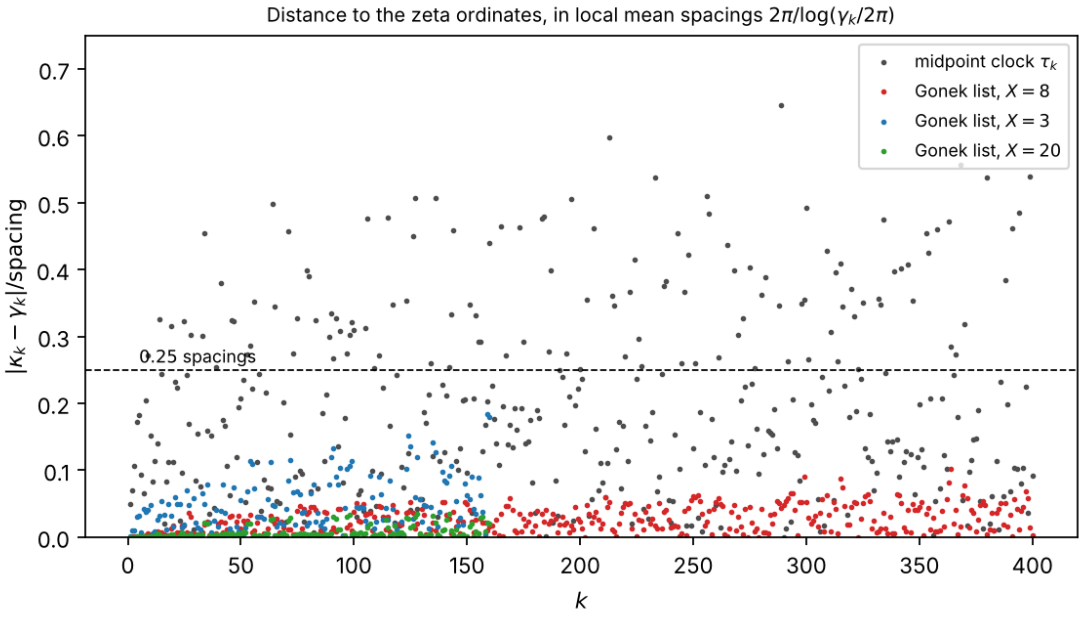


FIGURE 1. Distance to the zeta ordinates, in units of the local mean spacing $2\pi/\log(\gamma_k/2\pi)$, of the midpoint clock and of the Gonek lists $\boldsymbol{\kappa}^{(\mathrm{X})}$ for $\mathrm{X} = 3, 8, 20$ ($k \le 160$ for $\mathrm{X} = 3, 20$; $k \le 400$ for $\mathrm{X} = 8$). The dashed line is the observed radius of the Newton basin. The ordinates γ_k are used here only to measure the distance.

factorization, $\mathsf{d}_0 := \|I - \mathbb{Q}\mathbb{J}\|_{\infty\to\infty}$ its defect, $\mathbb{J}'$ the matrix of second derivatives $\partial^2_{\kappa_k}\mathsf{c}_\varphi$ at x_0, and $\mathbb{A}^{(3)}(\mathrm{r})$ an entrywise bound for the third derivatives on $|x - x_0| \le \mathrm{r}$, recomputed for each r, the map $\mathbb{T}(x) = x - \mathbb{Q}\mathbb{F}(x)$ contracts the ball of radius r in ℓ^∞ if

$$\mathsf{l}(\mathrm{r}) := \mathsf{d}_0 + \|\mathbb{Q}\mathbb{J}'\|_{\infty\to\infty}\,\mathrm{r} + \tfrac{1}{2}\,\big\||\mathbb{Q}|\mathbb{A}^{(3)}(\mathrm{r})\big\|_{\infty\to\infty}\,\mathrm{r}^2 < 1, \qquad \mathsf{h} \le (1 - \mathsf{l}(\mathrm{r}))\,\mathrm{r}, \tag{18.79}$$

in which case the ball contains an exact solution of the square system; here $\mathrm{t}_m = k^*_m$, so the right inverse of Theorem 18.25 is an inverse. The defect enters l because $D\mathbb{T}(x) = (I - \mathbb{Q}\mathbb{J}) + \mathbb{Q}(\mathbb{J} - D\mathbb{F}(x))$, and a fixed point of $\mathbb{T}$ is a zero of $\mathbb{F}$ because $\mathsf{d}_0 < 1$ makes $\mathbb{Q}\mathbb{J}$, hence $\mathbb{Q}$, invertible. The sampled margin of a stage is the largest value of $(1-\mathsf{l}(\mathrm{r}))\mathrm{r}$ over 60 radii r spaced logarithmically in $[10^{-12}, 10^{-1}]$: a sampled estimate, not a maximum over all r, of the largest residual that (18.79) could accept at that point.

m	T_m	n	$\mathbb{L}_m = \|Q\|_{\infty\to\infty}$	sampled margin	s_m on all tests $\le m$	$\max_k \|\kappa_k - \gamma_k\|$
1	50	4	41	$7.0 \cdot 10^{-3}$	$7.7 \cdot 10^{-8}$	$7 \cdot 10^{-7}$
4	81	9	50	$3.1 \cdot 10^{-3}$	$2.6 \cdot 10^{-7}$	$1 \cdot 10^{-6}$
8	156	26	150	$7.5 \cdot 10^{-4}$	$9.6 \cdot 10^{-7}$	$1 \cdot 10^{-5}$
12	299	68	460	$2.2 \cdot 10^{-4}$	$5.5 \cdot 10^{-6}$	$4 \cdot 10^{-5}$
15	487	133	1600	$6.7 \cdot 10^{-5}$	$3.3 \cdot 10^{-3}$	$4 \cdot 10^{-2}$
16	572	165	–	square Jacobian singular	–	–

For stages 1 to 15 the floating-point evaluation of (18.79) holds with r equal to the polishing residual (10^{-15} to $5 \cdot 10^{-13}$), with defect $\mathsf{d}_0 \le 6.3 \cdot 10^{-13}$ and $\mathsf{l}(\mathrm{r}) \le 1.9 \cdot 10^{-9}$. Radii of this size are at the level of rounding and are numerically fragile as individual numbers; what measures the stability of the test is the sampled margin, which shows that at the same point every residual up to $7 \cdot 10^{-3}$ ($m = 1$) to $7 \cdot 10^{-5}$ ($m = 15$) would pass, eight to twelve orders of magnitude above

the polishing residual. If the inequalities held in exact arithmetic for compactly supported tests, Theorem 18.25 would give an exact solution of the square section within r of x_0, inside the tube with $C = 1$; what has been established is the floating-point inequality, without rounding bounds and with Gaussian tests, so these stages carry candidate certificates and not verified ones. For compactly supported tests this is settled in Section 18.8.5: with B-spline tests the same fifteen stages pass in ball arithmetic. The candidate solutions are not unique: at $m = 15$ the polished point is $4 \cdot 10^{-2}$ from $\boldsymbol{\gamma}$. At $m = 16$ the square Jacobian is singular to working precision and the stage has no candidate certificate. Least-squares log-log fits over the computed stages $m \geq 4$ (Figure 2) give the empirical exponents

$$\mathbb{L}_m \sim T^{1.81\pm0.10}, \qquad \|\mathbb{Q}\mathbb{J}'\|_{\infty\to\infty} \sim T^{2.10\pm0.06}, \qquad \text{sampled margin} \sim T^{-2.08\pm0.06} \qquad (T \leq 487); \tag{18.80}$$

these are regressions over the 12 computed stages $4 \leq m \leq 15$, with one standard error of the fitted slope after $\pm$, not asymptotic laws.

18.8.5. *Verified certificates with compactly supported tests. Tests.* Let N_8 be the cardinal B-spline of order 8 on $[0, 8]$, the eightfold convolution of $\mathbf{1}_{[0,1]}$: a piecewise polynomial of degree 7 and class C^6 with $\int \mathrm{N}_8 = 1$. For $c, h > 0$ put $\varphi^{\mathrm{N}}_{c,h}(u) := \mathrm{N}_8\big((u - c)/h + 4\big)$, supported in $[c - 4h, c + 4h]$. Then

$$\mathsf{c}_{\varphi^{\mathrm{N}}_{c,h}}(t) = h\cos(ct)\Big(\frac{\sin(ht/2)}{ht/2}\Big)^8, \qquad |\mathsf{c}_{\varphi^{\mathrm{N}}_{c,h}}(t)| \leq h\Big(\frac{2}{ht}\Big)^8. \tag{18.81}$$

Since $\mathrm{N}_8^{(i)} = \nabla^i \mathrm{N}_{8-i}$ with $\nabla g(x) = g(x) - g(x-1)$ and $\|\mathrm{N}_q\|_1 = 1$, $\|(\varphi^{\mathrm{N}}_{c,h})^{(i)}\|_1 \leq 2^i h^{1-i}$ for $i \leq 7$, and the constant (18.18) satisfies, for $r \leq 7$ and $u^* = c + 4h$,

$$\mathfrak{W}_r(\varphi^{\mathrm{N}}_{c,h}) \leq (1 + u^{*2})\cosh\tfrac{u^*}{2}\; h\Big(1 + \frac{2}{h}\Big)^r. \tag{18.82}$$

The residual $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a})$ is a distribution of order at most 2 on $(0, \infty)$, because $\sum_k \cos(\kappa_k u)$ is the second derivative of the uniformly convergent series $-\sum_k \kappa_k^{-2}\cos(\kappa_k u)$, the masses form a measure and k is smooth. So (18.5) defines $\langle \mathfrak{R}, \varphi\rangle$ for every $\varphi \in C^2_c(0, \infty)$, in particular for $\varphi^{\mathrm{N}}_{c,h}$, and Proposition 18.5, Theorems 18.25 and 18.27 and Corollaries 18.11 and 18.18 hold verbatim for test sets in $C^2_c(0, \infty)$ whose union has a span that approximates every $\varphi \in C^\infty_c(0, \infty)$ in C^2 with supports in a fixed compact set: the proofs use the tests only through $|\mathsf{c}_\varphi(t)| \leq \|\varphi''\|_1 t^{-2}$, the finiteness of the mass sum, and density. B-splines with dyadic widths provide such sets: for $f \in C^\infty_c(0, \infty)$ and $h = 2^{-i}$ the splines $\sum_j f(jh)\,\varphi^{\mathrm{N}}_{jh,h}$ converge to f in C^2, because $\frac{d}{du}\sum_j \mathsf{c}_j \mathrm{N}_q(u/h - j) = h^{-1}\sum_j(\mathsf{c}_j - \mathsf{c}_{j-1})\mathrm{N}_{q-1}(u/h - j)$, $\sum_j \mathrm{N}_q(x - j) = 1$, and the divided differences of f converge uniformly to its derivatives. The computations below use other widths and centres; they concern finite square sections, for which density plays no role.

Sections. Stage m, $1 \leq m \leq 16$, uses $\mathrm{w}_m = 0.08 \cdot 0.85^{m-1}$ and $T_m = 4/\mathrm{w}_m$ as in Section 18.8.1, the width $h_m := \mathrm{w}_m\sqrt{12/8}$ (so that $\varphi^{\mathrm{N}}_{c,h_m}$ has the variance of the Gaussian of width w_m), and the tests $\varphi^{\mathrm{N}}_{c,h_j}$ with centres $c \in [9\mathrm{w}_j, \log(\max(T_j, 7)/2\pi) + 2]$ in steps of w_j, for all levels $j \leq m$; for $m \leq 15$ their supports lie in $[4.1\,\mathrm{w}_j, 6.4]$. The unknowns are the n_m ordinates with $\tau_k \leq 2.4/\mathrm{w}_m$. The other ordinates are frozen: at the receding-horizon iterate of Section 18.8.2 for $n_m < k \leq 60\,000$ (double-precision numbers, taken as exact; they coincide with the clock to 10^{-12} beyond $k = 1000$), and at the clock for $k > 60\,000$. The masses are a^ζ. The square system $\mathbb{F}_m$ consists of n_m tests chosen by pivoted QR on $\mathbb{J}^T$ at the iterate.

Computation. Program `certV` (Appendix D.8) polishes $\mathbb{F}_m$ by Newton's method in double precision and then works in ball arithmetic (Arb [6] through python-flint, 128 bits). It computes enclosures of: the cosine transforms by (18.81); the frozen ordinates $k \leq 60\,000$, summed explicitly;

the clock tail $k > 60\,000$, bounded by

$$\sum_{k>K} |\mathsf{c}_{\varphi^{\mathrm{N}}_{c,h}}(\tau_k)| \le h\Big(\pi h\Big(1-\frac{11}{8K}\Big)\Big)^{-8} \frac{(\log K)^8 K^{-7}}{7(1-\frac{8}{7\log K})}, \tag{18.83}$$

which follows from (18.81), from $\tau_k \ge 2\pi(1-\frac{11}{8K})k/\log k$ for $k > K$ ($W_0(x) \le \log x$ for $x \ge e$) and from the integral test; the masses, exactly; and $\langle \mathsf{k}, \varphi^{\mathrm{N}}_{c,h}\rangle$, by rigorous numerical integration over the eight polynomial pieces of $\varphi^{\mathrm{N}}_{c,h}$ [7]. Two Newton steps in ball arithmetic give the centre x_1. With $\mathbb{Q}$ the double-precision inverse of the Jacobian at the polished point, taken as an exact matrix, the map $\mathbb{T}(x) = x - \mathbb{Q}\mathbb{F}_m(x)$ maps the ball $B(x_1,\rho)$ of ℓ^∞ into itself and contracts if

$$\mathsf{I}(\rho) := \sup_{x\in B(x_1,\rho)} \big\|I - \mathbb{Q}\,D\mathbb{F}_m(x)\big\|_{\infty\to\infty} < 1, \qquad \|\mathbb{Q}\mathbb{F}_m(x_1)\|_\infty \le (1-\mathsf{I}(\rho))\,\rho; \tag{18.84}$$

an upper bound for $\mathsf{I}(\rho)$ is obtained by evaluating the Jacobian on the ball, entry by entry, and multiplying interval matrices. As in Section 18.8.4, $\mathsf{I} < 1$ makes $\mathbb{Q}$ invertible, so the fixed point is the only zero of $\mathbb{F}_m$ in the ball. The program also checks, in ball arithmetic, that the frozen list is increasing and lies within 0.9938 of the clock (with τ_k enclosed through Arb's Lambert function), it bounds the distance of the solution to the zeta ordinates using Arb's rigorous enclosures of the zeros (`acb.zeta_zero`), and it evaluates the bound (18.19) with $r = 7$ and (18.82) on every test of the stage.

Theorem 18.78 (Exact square sections up to height 487; computer-assisted)**.** *For each $m = 1,\dots,15$ the square system $\mathbb{F}_m$ described above has exactly one zero $x_*^{(m)}$ in the ball $\|x - x_1^{(m)}\|_\infty \le \rho_m$; the centres $x_1^{(m)}$ (exact binary numbers) and the radii ρ_m (double-precision numbers) are recorded in the file `certV_centres.json` of the repository, and Table 7 gives ρ_m rounded up. The list formed by $x_*^{(m)}$ and the frozen ordinates is strictly increasing and* 0.995*-admissible, and with the von Mangoldt masses it satisfies exactly the n_m selected equations of the rung system. Its first n_m ordinates are within the distance given in Table 7 of $\gamma_1,\dots,\gamma_{n_m}$.*

Proof. For every stage the ball-arithmetic evaluation of program `certV` gives enclosures that satisfy (18.84) with the values of Table 7, and Banach's fixed point theorem gives the zero and its uniqueness in the ball. The gaps between consecutive ordinates of $x_1^{(m)}$, and between the last of them and the first frozen ordinate, exceed $2\rho_m$ (the smallest gap is 0.71); $\max_k |x_{1,k}^{(m)} - \tau_k| + \rho_m \le 0.9937$, and the frozen ordinates are within 0.9938 of the clock. The distances to γ_k are bounded with the enclosures of the zeros computed by Arb. □

Table 7: Verified square sections with B-spline tests (program `certV`). ρ_m: radius of the ball of Theorem 18.78; upper bounds are rounded up and lower bounds down; $\mathsf{I}(\rho_m)$: upper bound of the contraction constant (18.84); $\mathbb{L}_m = \|\mathbb{Q}\|_{\infty\to\infty}$; margin: largest $(1-\mathsf{I}(\rho))\rho$ over 36 radii in $[10^{-8}, 10^{-1}]$, each value a rigorous lower bound for the residual that (18.84) would accept; $\max_{k\le n}|x_* - \gamma|$: rigorous bound of the distance of the solution to $\gamma_1,\dots,\gamma_{n_m}$; ϵ_ζ: the bound of Proposition 18.20 ($r = 7$) for the residual of the zeta list on every test of level $\le m$.

m	T_m	n_m	ρ_m	$\mathsf{I}(\rho_m)$	$\mathbb{L}_m$	margin	$\max_{k\le n}\lvert x_* - \gamma\rvert$	ϵ_ζ
1	50	4	$6.5\cdot10^{-21}$	$1.3\cdot10^{-11}$	$1.1\cdot10^{2}$	$2.6\cdot10^{-3}$	$7.7\cdot10^{-7}$	$2.2\cdot10^{-64}$
2	59	5	$1.3\cdot10^{-20}$	$2.4\cdot10^{-12}$	$6.9\cdot10^{1}$	$4.3\cdot10^{-3}$	$9.0\cdot10^{-8}$	$6.3\cdot10^{-64}$
3	69	7	$6.1\cdot10^{-20}$	$9.7\cdot10^{-12}$	$1.2\cdot10^{2}$	$1.2\cdot10^{-3}$	$1.1\cdot10^{-6}$	$1.8\cdot10^{-63}$
4	81	9	$2.9\cdot10^{-19}$	$1.3\cdot10^{-11}$	$1.7\cdot10^{2}$	$8.5\cdot10^{-4}$	$1.2\cdot10^{-6}$	$5.0\cdot10^{-63}$
5	96	12	$9.8\cdot10^{-19}$	$1.5\cdot10^{-11}$	$1.7\cdot10^{2}$	$5.2\cdot10^{-4}$	$4.8\cdot10^{-7}$	$1.5\cdot10^{-62}$

m	T_m	n_m	ρ_m	$\mathsf{I}(\rho_m)$	$\mathbb{L}_m$	margin	$\max_{k\le n}\lvert x_*-\gamma\rvert$	ϵ_ζ
6	113	16	$3.9\cdot10^{-18}$	$2.5\cdot10^{-11}$	$2.2\cdot10^{2}$	$3.8\cdot10^{-4}$	$2.6\cdot10^{-6}$	$4.2\cdot10^{-62}$
7	133	20	$2.5\cdot10^{-17}$	$4.9\cdot10^{-11}$	$4.6\cdot10^{2}$	$1.3\cdot10^{-4}$	$1.8\cdot10^{-5}$	$1.3\cdot10^{-61}$
8	156	26	$6.0\cdot10^{-17}$	$3.0\cdot10^{-11}$	$4.2\cdot10^{2}$	$1.7\cdot10^{-4}$	$3.9\cdot10^{-6}$	$3.7\cdot10^{-61}$
9	183	34	$3.1\cdot10^{-16}$	$1.7\cdot10^{-10}$	$6.6\cdot10^{2}$	$8.3\cdot10^{-5}$	$1.7\cdot10^{-5}$	$1.1\cdot10^{-60}$
10	216	43	$1.3\cdot10^{-15}$	$1.3\cdot10^{-10}$	$9.1\cdot10^{2}$	$5.1\cdot10^{-5}$	$2.1\cdot10^{-5}$	$3.2\cdot10^{-60}$
11	254	54	$5.5\cdot10^{-15}$	$1.3\cdot10^{-10}$	$1.2\cdot10^{3}$	$3.6\cdot10^{-5}$	$3.2\cdot10^{-5}$	$9.4\cdot10^{-60}$
12	299	68	$1.7\cdot10^{-14}$	$3.4\cdot10^{-10}$	$1.4\cdot10^{3}$	$2.7\cdot10^{-5}$	$5.3\cdot10^{-5}$	$2.8\cdot10^{-59}$
13	352	85	$9.4\cdot10^{-14}$	$1.8\cdot10^{-9}$	$2.1\cdot10^{3}$	$1.5\cdot10^{-5}$	$3.1\cdot10^{-4}$	$8.4\cdot10^{-59}$
14	414	107	$3.3\cdot10^{-13}$	$7.6\cdot10^{-9}$	$2.5\cdot10^{3}$	$1.1\cdot10^{-5}$	$3.9\cdot10^{-3}$	$2.5\cdot10^{-58}$
15	487	133	$1.1\cdot10^{-12}$	$3.0\cdot10^{-8}$	$2.8\cdot10^{3}$	$8.9\cdot10^{-6}$	$3.7\cdot10^{-2}$	$7.5\cdot10^{-58}$
16	572	165	Newton polishing fails: the square Jacobian is singular to working precision					

Remark 18.79 (What Theorem 18.78 says and does not say). For these fifteen stages the candidate certificates of Section 18.8.4 are replaced by verified ones: the inequalities hold in ball arithmetic, for tests in $C_c^6(0,\infty)$, with every evaluation error included. The radii ρ_m, from $6.5\cdot10^{-21}$ to $1.1\cdot10^{-12}$, are set by the bound (18.83) for the clock tail, which more frozen ordinates would reduce; $\mathsf{I}(\rho_m)\le 3.0\cdot10^{-8}$. The verified margins are smaller than the sampled margins of the Gaussian candidates, because the B-spline transform changes sign and decays only like t^{-8}. At $m=16$, $T=572$, the square Jacobian is singular to working precision, as with Gaussian tests. Three things are not given. The selected equations are n_m of the tests of level $\le m$; on all of them the floating-point residual of the solution is between $2.4\cdot10^{-8}$ and $7.2\cdot10^{-4}$. The solutions of different stages are not restrictions of one another. And by Proposition 18.20 no statement of this kind bears on the Riemann hypothesis: whether or not RH holds, the zeta list with the von Mangoldt masses has residual at most $7.5\cdot10^{-58}$ on every test used here (column ϵ_ζ). The distances to γ_k (at most $7.6\cdot10^{-4}$ for $k\le20$ at every stage) are diagnostics.

18.8.6. *The entry step.* The open point of the staged construction, in the form that Theorem 18.27 makes precise, is whether the residual h_{m+1} of the new tests at the start of stage $m+1$ is below the margin of that stage. This was tested as follows: start from the polished point of stage $m-1$ (Section 18.8.4), put the new band of unknowns (those with $\tau_k\in(2.4/\mathrm{w}_{m-1},2.4/\mathrm{w}_m]$) and everything beyond at their Gonek values, and evaluate (18.79) (i) for the band alone, with the new tests of level m, and (ii) for the full square system of stage m, without polishing.

The band-only test passes (in the same floating-point sense) for $T\le96$ and fails from $T=113$ on; its margin falls from $9\cdot10^{-2}$ to $3\cdot10^{-3}$ while the Gonek residual of the band stays between 10^{-2} and $3\cdot10^{-1}$ (the Gonek error in the band is 0.02 to 0.09). The full-system test passes only at $T=59$. The margin decreases as a power of T by (18.80), whereas the error of the Gonek lists with fixed X does not decrease with T, and with $\mathrm{X}=T^{\mathrm{x}}$ it would decrease at most like $1/\log T$. Within the range computed, no worst-case certificate in ℓ^∞ closes the entry step, and the trend of the constants says that none will at larger T. This is a numerical statement, not a theorem.

The same test in ℓ^2. Program `entry2` (Appendix D.7) evaluates the entry test with the norms $\|\cdot\|_{2\to2}$ and $\|\cdot\|_2$ in place of $\|\cdot\|_{\infty\to\infty}$ and $\|\cdot\|_\infty$ in (18.79). This is legitimate: the second-order term is $\mathbb{Q}\mathbb{J}'\operatorname{diag}(\mathrm{e})\mathrm{e}$, and $\|\operatorname{diag}(\mathrm{e})\|_{2\to2}=\|\mathrm{e}\|_\infty\le\|\mathrm{e}\|_2$; a zero in the ℓ^2 ball lies in the ℓ^∞ ball of the same radius. In ℓ^2 the inverse is smaller ($\|\mathbb{Q}\|_{2\to2}$ is 1.0 to 2.3 times below $\|\mathbb{Q}\|_{\infty\to\infty}$ for the band and 1.7 to 3.9 times for the full system), the ℓ^2 norm of the preconditioned residual of the new tests is up to 1.9 times larger than its ℓ^∞ norm, and the margins grow by factors 1.0 to 2.0 (band) and 1.2 to 2.8 (full system). The pattern does not change: the band test passes for $T\le96$ and fails from $T=113$ on, and the full-system test passes only at $T=59$, in both norms. The moderate ℓ^2 conditioning at the zeta list (Section 18.8.2) does not close the entry step; what fails is the size of the error of the Gonek lists relative to the margin, not the choice of norm.

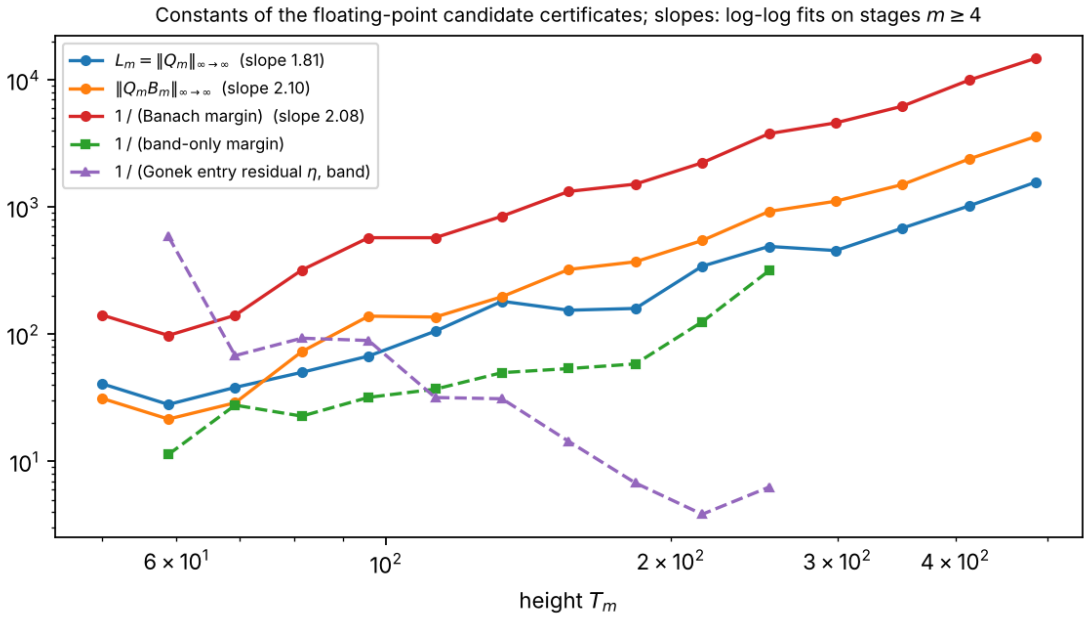


FIGURE 2. Constants of the Banach certificate (18.79) for the square sections of Section 18.8.4 (solid lines, with fitted slopes in $\log T$), and the band-only entry test of Section 18.8.6 (dashed): the reciprocal of the band margin and the reciprocal of the residual of the new band at its Gonek values. The entry test passes, in floating point, where the triangles lie below the squares.

18.8.7. *Reading of the computations.*

(i) The constructions that use only the clock and the primes (the Gonek lists, the square sections, the certificates) behave as predicted by the tube lemma and the certificate theorem: the candidate certificates hold in floating point up to $T = 487$, and fifteen square sections have verified exact solutions (Theorem 18.78). By Proposition 18.20 none of this bears on the Riemann hypothesis: if it were false, the zeta list with the von Mangoldt masses would still solve every test used here up to 10^{-57}, and sections in the box with residuals of that size would exist at every stage. A zero $\frac{1}{2} + d_0 + i\gamma_0$ off the line contributes $2\cosh(d_0 u)\cos(\gamma_0 u)$ to the prime side; a stage detects it only through tests whose constant $\mathfrak{W}_r$ is of order H_0^{r-1}, that is, tests that resolve frequencies of the order of H_0 (Remark 18.21); Heuristic 18.41 describes how the size of the window enters.

(ii) The quantities $\mathbb{L}_m$ and $\|QB\|$ grow polynomially in T. This is the quantitative face of the obstruction theorem of Section 18.7.3: no height-uniform linear stability, hence no height-uniform Newton basin. A certificate for the infinite construction needs a basin that shrinks no faster than the initial error of a γ-free list decreases, and the computed trend is T^{-2} against $1/\log T$.

(iii) The observed contraction of Gauss–Newton (factor 4 to 15 per step at every height) is a mean-square phenomenon that the worst-case constants do not see. A proof of the entry step would have to bound $\|\mathbb{Q}\mathbb{J}'\operatorname{diag}(\mathrm{e})\,\mathrm{e}\|_\infty$ for the actual error vector e of the Gonek list, not for an arbitrary vector of the same norm; that error vector is γ-free, and its residual $\mathbb{F}_m(\boldsymbol{\kappa}^{(\mathrm{X})})$ is controlled through the explicit formula by the moments of $S - S_{\mathrm{X}}$, which are unconditional (Selberg, Fujii). Whether this closes is not known; any proof along this line would contain a proof of the Riemann hypothesis, by Corollary 18.18.

18.9. **Status of the statements.**

Statement	Status
Lemma 18.3, Corollary 18.4	proved; conditional on (D) as stated
Proposition 18.5, Lemma 18.23	proved, unconditional
Lemma 18.6, Theorem 18.7, Corollaries 18.9, 18.10, 18.11	proved, unconditional: an exact solution in a tube, with arbitrary real masses, is $(\boldsymbol{\gamma}, \mathsf{a}^{\zeta})$ and implies RH; $\mathrm{B}(C, \mathbb{M}) \Leftrightarrow$(D); uses Proposition F.2, Lemma 14.10 and Hamburger's theorem
Proposition 18.20	proved, unconditional; uses [55, 72]; no finite stage decides B or B′
Lemmas 18.13, 18.14	proved, unconditional
Proposition 18.16, Corollary 18.18	proved, unconditional: RH $\Leftrightarrow$ B′(C); classical in substance
Lemma 18.24, Theorems 18.25, 18.27, Proposition 18.29	proved; their hypotheses (A_m) for all m and (B) are open and imply RH
Heuristic 18.31	not proved; its point model is not a test of $C_c^{\infty}(0,\infty)$
Lemma 18.32	proved, unconditional; uses [19] only for the interpretation of the list
(G1), (G2) of Remark 18.33	open; together they imply B′ and RH
Lemmas 18.45–18.50, Theorem 18.51, Corollary 18.52	proved, unconditional; uses 9.10; by Lemma 18.43 the hypothesis $\upsilon < \infty$ holds for a finite section only when all masses vanish, so the bound is a template, not a bound for sections with primes (Remark 18.44)
Lemma 18.34, Theorems 18.35, 18.36	proved, unconditional: RH $\Leftrightarrow$ an unknown list solves the window identity $\Leftrightarrow$ $e^{-iy/2}V_{+}(iy)$ is completely monotone; uses [8]; literature not searched for Theorem 18.36
Lemma 18.38, Theorem 18.39	proved, unconditional: dichotomy for minimizers; RH $\Leftrightarrow$ solvability on every window in a fixed box; the exclusion of degenerate minimizers is open
Heuristics 18.54, 18.55, 18.75, 18.41	not proved
Lemma 18.57, Theorem 18.58, Corollary 18.59	proved; Corollary 18.59 uses [12]
Corollary 18.61, Theorem 18.62, Proposition 18.60	proved; the variable-length version was not found in the literature consulted (Remark 18.63)
Lemma 18.65, Proposition 18.66, Proposition 18.69, Corollary 18.68	proved, unconditional
Lemma 18.72, Theorem 18.71, Corollary 18.74	proved, unconditional; uses the upper bound contained in [15, Main Theorem], [58, 61, 18]; no study of (A_2) for the zeta ordinates was found (Remark 18.77)
Theorem 18.78	proved with computer assistance (ball arithmetic, B-spline tests): exact solutions of fifteen square sections up to $T = 487$; no bearing on RH (Proposition 18.20)
Other computations of Section 18.8	floating point; Gaussian candidate certificates and the entry step in ℓ^{∞} and ℓ^{2}; not part of any proof

19. Numerical illustration with 2 001 052 zeros

This section shows what is being constructed and checks that the theorems are consistent with computation. It uses the ordinates $\gamma_1, \dots, \gamma_N$ of the first $N = 2\,001\,052$ zeros of ζ (file `zeross.bin`, $\gamma_N = 1\,132\,490.6587$) and the midpoint ordinates $\tau_1, \dots, \tau_N$ computed from (2.5) (file `zLamb.bin`, $\tau_N = 1\,132\,490.7226$). The programs are listed in Appendix B and their complete console output in

Appendix C; all figures have linear axes. Each item is an *experiment*: it reports a computation and does not replace a proof.

Products over the ordinates are evaluated as follows, in double precision throughout: the first 10^5 factors one by one, the factors $10^5 < k \le N$ through power sums of $(\frac{1}{4}+o_k^2)^{-1}$, and the factors $k > N$ by the midpoint rule for the clock, which gives the tail $\frac{1}{2\pi}(q\frac{L_U+1}{U} - \frac{q^2}{2}\frac{3L_U+1}{9U^3} + \frac{q^3}{3}\frac{5L_U+1}{25U^5})$ with $\mathcal{N}_{\mathrm{RVM}}(U) = N$, $L_U = \log(U/2\pi)$. For the products over the γ_k (P_{Z1}, ζ_{Z1}) this means that the unknown tail of the zeta ordinates beyond the file is *replaced by the clock tail*; every agreement reported for ζ_{Z1} against ζ includes that substitution and is an agreement of the construction with itself beyond height γ_N. "All N ordinates" or "all zeros" below always means all zeros in the supplied file, $\gamma_k \le \gamma_N$. No quantity in this section is certified: the words "exact" and "certified" are not used for floating-point values, and the label `FLOAT_PASS` of the console outputs (Appendix E) marks a floating-point test, as explained in Remark 18.26.

19.1. **The clock and the zeros.**

Experiment 19.1 (Clock, midpoints and displacements)**.** *Object:* $\mathcal{N}_{\mathrm{RVM}}$, N_ζ, τ_k, $\eta_k = \tau_k - \gamma_k$. *Range:* $k \le N$. *Expected:* (2.4) to rounding, and $|\eta_k| \le C_\eta$ (Lemma 2.2). *Observed:* $\max_k |\mathcal{N}_{\mathrm{RVM}}(\tau_k) - (k-\frac{1}{2})| = 6.985 \cdot 10^{-10}$; $\max_k |\eta_k| = 0.9937$ (at $k = 34$), mean $|\eta_k| = 0.1512$, root mean square 0.1881, mean $\eta_k = 1.7 \cdot 10^{-7}$. By blocks of k, $\max|\eta_k|$ is $0.840, 0.994, 0.915, 0.834, 0.786, 0.763, 0.717$ on $[1,10]$, $[11,10^2]$, $[10^2,10^3]$, …, $[10^6, N]$.

Figure 3 shows how the clock places the midpoint ordinates. The staircase N_ζ jumps by one at each γ_k; the smooth clock $\mathcal{N}_{\mathrm{RVM}}$ crosses the half-integer level $k-\frac{1}{2}$ at τ_k. Since $N_\zeta - \mathcal{N}_{\mathrm{RVM}} = S(T) + O(1/T)$ (Figure 4), the horizontal distance between the open square $(\gamma_k, k-\frac{1}{2})$ and the red point $(\tau_k, k-\frac{1}{2})$ is the remainder S converted into ordinate units by the slope $\mathcal{N}'_{\mathrm{RVM}} = \frac{1}{2\pi}\log\frac{T}{2\pi}$. The right panel repeats the picture at height $1.13 \cdot 10^6$, where the mean spacing is 0.519: every τ_k still lies next to its γ_k.

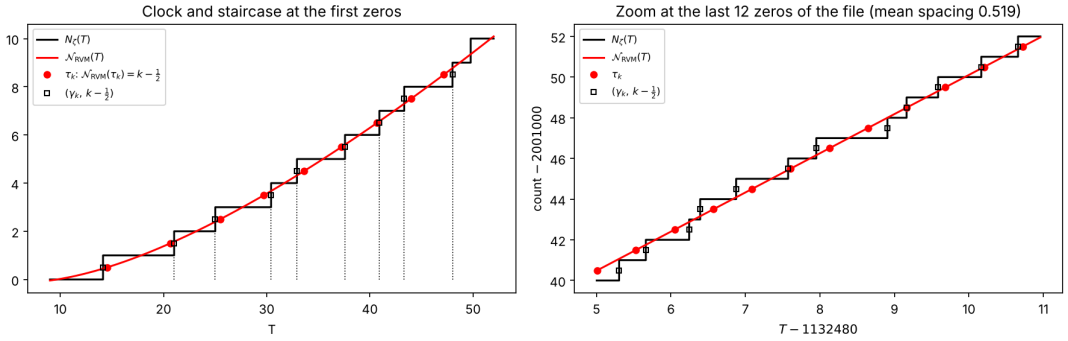


FIGURE 3. The midpoint clock $\mathcal{N}_{\mathrm{RVM}}$ (red), the zero staircase N_ζ (black), the midpoint ordinates τ_k (red points) and the points $(\gamma_k, k-\frac{1}{2})$ (squares). Left: the first nine zeros. Right: the last twelve zeros of the file.

19.2. **The λ-family and the endpoint.** Figure 5 shows the core of the method. The lower left panel draws the moving ordinates $\gamma_k(\lambda)$ in clock units, $\mathcal{N}_{\mathrm{RVM}}(\gamma_k(\lambda)) - (k-\frac{1}{2})$: at $\lambda = 1$ all lines meet at 0 (the τ_k), at $\lambda = 0$ they sit at the ordinates of ζ; for $\lambda > 0$ the ordinates $\gamma_k(\lambda)$ remain strictly increasing in k (Theorem 11.1). The lower right panel draws the real functions $Z_\lambda(t) = e^{i\vartheta(t)}\zeta_\lambda(\frac{1}{2}+it)$ for $\lambda = 0, 0.25, 0.5, 0.75, 1$ together with Hardy's function $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2}+it)$: the zeros of every Z_λ are real, they move continuously with λ, and the curve for $\lambda = 0$ lies on top of Z at the scale of the figure, because it is computed from the supplied ordinates with the clock tail substituted beyond them; this is a property of the computed curves, not evidence for $\zeta_{Z1} = \zeta$.

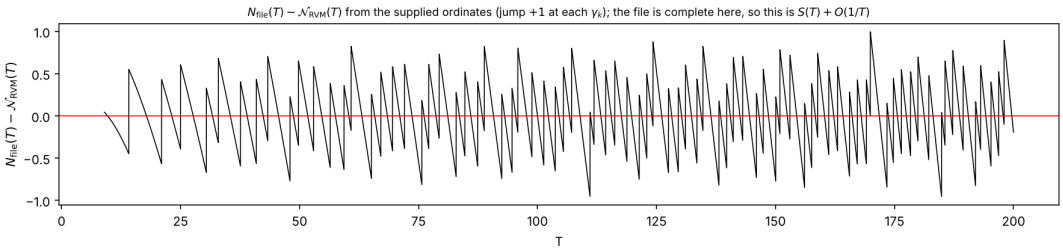


FIGURE 4. The computed difference $N_{\text{file}}(T) - \mathcal{N}_{\text{RVM}}(T)$ for $T \le 200$, where N_{file} counts the supplied ordinates. The file is complete on this range, so the curve is the remainder $N_\zeta(T) - \mathcal{N}_{\text{RVM}}(T) = S(T) + O(1/T)$ of (2.2).

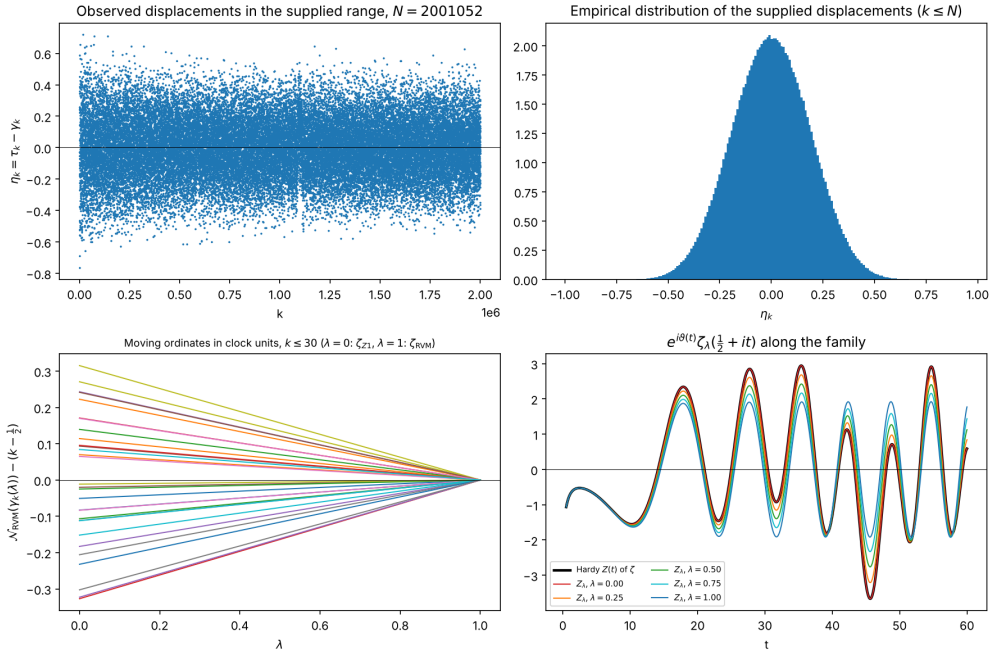


FIGURE 5. Top: the observed displacements η_k for every 50th $k \le N$ of the supplied range and their empirical distribution. Bottom: moving ordinates in clock units, $k \le 30$, and the functions $Z_\lambda(t)$ on the critical line.

Experiment 19.2 (The endpoint ζ_{Z1} against ζ)**.** *Object:* $\zeta_{Z1} = P_{Z1}/G$ computed from the N ordinates γ_k, and ζ computed by mpmath. *Range:* $\sigma \in \{2, 1.25, 0.75\}$, $1 \le t \le 2000$ (499 points per line), and the critical line $0.5 \le t \le 60$. *Expected:* if $\zeta_{Z1} = \zeta$, the ratio is 1 up to the truncation of the product. *Observed:* $\max|\zeta_{Z1}/\zeta - 1| = 8.7 \cdot 10^{-10}$, $1.1 \cdot 10^{-9}$, $2.6 \cdot 10^{-9}$ on the three lines, and $\max|Z_{Z1} - Z|/\max(1, |Z|) = 8.4 \cdot 10^{-10}$ on the critical line; the balanced error $|\zeta_{Z1} - \zeta|/\max(1, |\zeta_{Z1}|, |\zeta|)$, which is not amplified near the zeros, is at most $7.5 \cdot 10^{-10}$, $1.1 \cdot 10^{-9}$, $1.9 \cdot 10^{-9}$ on the three lines, and the imaginary parts discarded in Z_λ are below $3 \cdot 10^{-14}$ (Figure 6); the tail beyond γ_N is the clock tail, so the 10^{-9} figure is consistent with, and is expected to be dominated by, the file truncation and the clock-tail substitution; it does not test (D).

Experiment 19.2 shows numerical agreement at the displayed truncation precision. This does not test (D): Proposition 14.3 already forces $|\log \varpi_{Z1}| < 4 \cdot 10^{-13}$ from verified zeros, while the observed 10^{-9} scale is dominated by truncating the ordered clock tail.

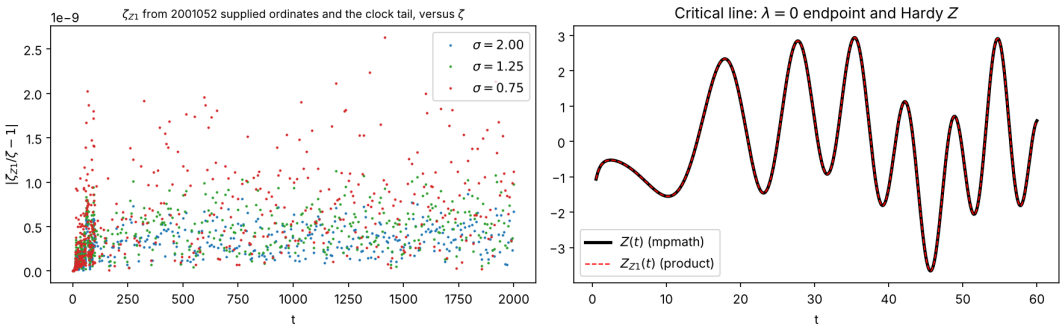


FIGURE 6. Left: $|\zeta_{Z1}/\zeta - 1|$ on three vertical lines, with ζ_{Z1} computed from the supplied ordinates and the clock tail beyond them. Right: Z_{Z1} and Hardy's Z on the critical line.

Experiment 19.3 (The value at infinity: the product identity)**.** *Object:* the ordered partial products $P_K = \prod_{k\le K}(\frac{1}{4}+\gamma_k^2)/(\frac{1}{4}+\tau_k^2)$ of (13.12). *Range:* $K \le N$. *Expected:* by Corollary 13.3, $\zeta_{Z1} = \zeta$ is the statement $P_K \to \varpi_{\mathrm{RVM}}$. *Observed:* $P_K = 0.96256,\ 0.96248,\ 0.96229,\ 0.961951,\ 0.961930,\ 0.9619418189,\ 0.9619415390$ for $K = 10, 10^2, \dots, 10^6, N$, against $\varpi_{\mathrm{RVM}} = 0.9619415789$ from Experiment 19.4; for $K \ge 10^6$, P_K stays in $[0.9619389, 0.9619440]$ with mean 0.9619416633 (Figure 7). Proposition 14.3 already bounds the endpoint discrepancy by $4\cdot 10^{-13}$ under the verified-zero input, so the 10^{-9}–10^{-6} discrepancies in these finite products are truncation/tail errors and are not numerical evidence for (D). The flow at infinity (14.3), truncated to the supplied ordinates $k \le N$, is $\Upsilon(\lambda) = 0.0436950,\ 0.0412445,\ 0.0387987,\ 0.0363572,\ 0.0339198$ for $\lambda = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$, decreasing as in (14.4) with slope close to the auxiliary approximation $-\sum_{k\le N} 2\eta_k^2/\gamma_k^2 = -0.0098211$ (which drops the $\frac{1}{4}$ of the denominators and the terms of higher order in η_k/γ_k), and the truncated flow integral is $\int_0^1 \Upsilon\, d\lambda = \sum_{k\le N} \log\frac{1/4+\tau_k^2}{1/4+\gamma_k^2} = 0.0388016004$, against $\log L = 0.0388015590$; the agreement of P_N with ϖ_{RVM} is an internal check of two computations built on the same clock, not a certification of the infinite product.

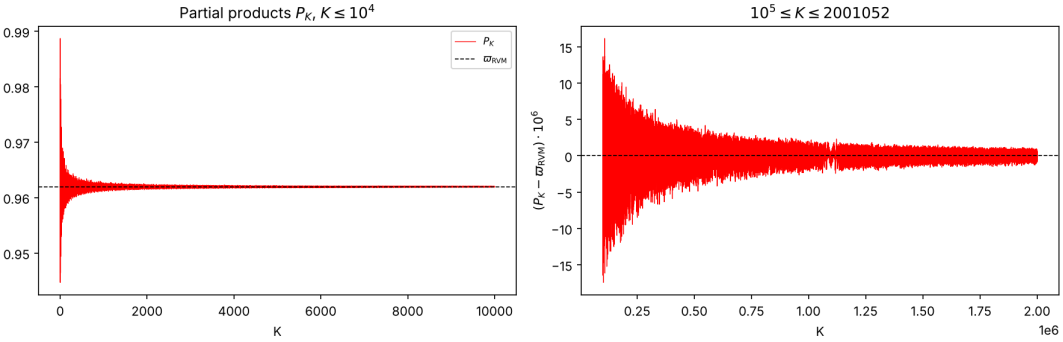


FIGURE 7. The partial products P_K of (13.12) against ϖ_{RVM} (dashed). Left: $K \le 10^4$. Right: $(P_K - \varpi_{\mathrm{RVM}}) \cdot 10^6$ for $10^5 \le K \le N$. The value of the endpoint at $+\infty$ is $\varpi_{Z1} = \varpi_{\mathrm{RVM}}/\lim_K P_K$ (Theorem 13.1).

19.3. The transition theorem and ζ_{RVM} against ζ.

Experiment 19.4 (Transition theorem)**.** *Object:* ζ_{RVM}, $w(s)$, ϖ_{RVM}, L. *Range:* $\sigma' \in \{0.25, 0.5, 1.5\}$, $10 \le t \le 10^4$. *Expected:* Theorem 9.1 and Corollary 9.8: $t\,|\zeta_{\mathrm{RVM}}/(\varpi_{\mathrm{RVM}}(1+w)) - 1|$ bounded,

$|w| = (t/2\pi)^{-\sigma'}(1 + O(1/t))$. *Observed:* Richardson extrapolation on $\Re s = 2$ (which presupposes an expansion in $1/T$ and $1/T^2$ finer than the $O(1/T)$ proved) gives $\varpi_{\mathrm{RVM}} = 0.9619415789$ and $L = 1.0395641710$, values without error bound; for this extrapolation the product is evaluated with 10^6 exact factors and eight power sums, because the default evaluation of the programs (10^5 exact factors, six power sums) has relative error $7 \cdot 10^{-10}$ at $T = 10^4$ and $1.2 \cdot 10^{-5}$ at $T = 2 \cdot 10^4$; the heights $4000, 7000, 10000$ change the extrapolated value by $9 \cdot 10^{-11}$, doubling all heights by $1.8 \cdot 10^{-9}$, and the linear model on $5000, 10000$ by $1.1 \cdot 10^{-7}$; for the computed residual, which contains the $O(1/t)$ term of the theorem together with the truncation of the product, the clock tail, the extrapolated ϖ_{RVM} and rounding, $\max_{t>100} t\,|\zeta_{\mathrm{RVM}}/(\varpi_{\mathrm{RVM}}(1+w)) - 1| = 0.0617, 0.0618, 0.0620$ for the three values of σ', and 0.0790 for $\varpi_{\mathrm{RVM}}(1 + \chi)$ at $\sigma' = 0.25$; $|w|\,t^{\sigma'} = 1.5832$, 2.5066, 15.66–15.75, i.e. $(2\pi)^{\sigma'}$. The vertical limit of Corollary 9.8 is approached at the rate of $|w|$: at $t = 10^4$, $|1/\zeta_{\mathrm{RVM}} - L| = 0.143$, 0.025, $1.6 \cdot 10^{-5}$ for $\sigma' = 0.25, 0.5, 1.5$, so on $\sigma' = 0.25$ the convergence is slow and the limit is not reached numerically in the computed range.

Experiment 19.5 (The transition constant: the inequalities of Theorem 9.11)**.** *Object:* ν_3, the enclosure of τ_1, the lower bound $\mathfrak{M}(\tau_1)$ of (9.48), and, as a cross-check, ν_4, $P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac{1}{2})$ and ϖ_{RVM}. *Method:* Part A in interval arithmetic (`mpmath.iv`, 40 digits, outward rounding); Part B in floating point (25 digits for ν_4, double precision for the product over $k \le 2 \cdot 10^6$). The program generates the clock from the Lambert formula and reads no data file (it reads `output/s02_varpi.txt` only for the comparison in Part B). *Observed (Part A, the proof):* $\nu_3 \in [-0.00792401131834915, -0.00792401131834915]$; $\mathcal{N}_{\mathrm{RVM}}(\tau_0) = 0.261294\ldots \in (0, \frac{1}{2})$; $\mathcal{N}_{\mathrm{RVM}}(14.52134) = 0.49999907 < \frac{1}{2} < 0.50000041 = \mathcal{N}_{\mathrm{RVM}}(14.52135)$; $\mathfrak{M}(\tau_1) \in [0.00634570, 0.00635251]$; hence $\nu_4 \le -0.0126914$, $\nu_3 + \nu_4 \le -0.0206154$ and $\varpi_{\mathrm{RVM}} < 0.979596$. *Observed (Part B, not part of the proof):* $\nu_4 = -0.0251569083$, $e^{\nu_3+\nu_4} = 0.9674602699$, $P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac{1}{2}) = 0.9942956924$, $\varpi_{\mathrm{RVM}} = 0.9619415789$, which agrees with the Richardson value of Experiment 19.4 to $2.1 \cdot 10^{-11}$; the Euler–Maclaurin remainder $\sum_n \mathscr{I}_n + \mathscr{W}(1)/12 = 0.00154$ lies within its bound 0.00469. Console output in Appendix C.14.

The bottom panels of Figure 8 compare $|\zeta|$ and $|\zeta_{\mathrm{RVM}}|$ on two vertical lines: ζ carries the Euler product and oscillates with the primes, whereas ζ_{RVM} approaches the constant ϖ_{RVM} on $\sigma = 2$ and $\varpi_{\mathrm{RVM}}|1 + \chi|$ on $\sigma = 0.75$. This is the analytic content of Theorem 9.13.

19.4. **Exponential sums.**

Experiment 19.6 (Exponential sums over the two spectra)**.** *Object:* $S(N; \log x) = \sum_{k \le N} x^{io_k}$ for $o_k = \tau_k$ and $o_k = \gamma_k$. *Range:* $N \le 2\,001\,052$, $x \in \{2, 3, 4, 6, 10\}$. *Expected:* $|S_{\mathrm{RVM}}| \ll 1 + \log N$ (Theorem 4.1); for γ_k, the main term $-\frac{T}{2\pi}\frac{\Lambda(x)}{\sqrt{x}}$ of Landau's formula (Theorem 1.3), whose error term $O_x(\log T)$ makes the comparison at a finite height illustrative. *Observed:* the maximum over all $N \le 2\,001\,052$ of $|S_{\mathrm{RVM}}(N; \log x)|$ is 3.58, 3.98, 3.96, 4.53, 10.34 (the figure samples every 2000th N); at $N = 2\,001\,052$, $|\sum x^{i\gamma_k}| = 88\,301$, $114\,268$, $62\,435$, 0.26, 2.42, against the main terms $88\,342$, $114\,324$, $62\,467$, 0, 0.

19.5. **Centroids, walks and Cramér's function.**

Experiment 19.7 (Cramér's function and the centroids of the two spectra)**.** *Object:* $e^{-1/2}V_+(1)$ from the zero-free representation (5.11); the triangular centroids $C^{\mathrm{path}}_{\zeta}(N; 1)$ and $C^{\mathrm{path}}_{\mathrm{RVM}}(N; 1)$; the limit $C^{\mathrm{path}}_{\mathrm{RVM}}(\infty; 1)$ of Theorem 5.13. *Method:* program `s17` (Appendix B.18), mpmath with 30 digits, rays at $\theta = \pi/3$, prime powers up to $2 \cdot 10^6$, $a = 1$ and $a = 2$. *Expected:* independence of a; $2\Re V_+(u) = e^u - (e^{2u} - 1)^{-1}$ on $(\log 2, \log 3)$ (5.13); under RH, $C^{\mathrm{path}}_{\zeta}(N; 1) \to e^{-1/2}V_+(1)$ (Theorem 5.15). *Observed:*

- for $a = 2$ the value of $e^{-1/2}V_+(1)$ is $0.7768942607932536 + 0.7678556688644353\,i$, and for $a = 1$ it is $0.7768942607932536 + 0.7678556452589\,i$; the difference $2.4 \cdot 10^{-8}$ is the truncation

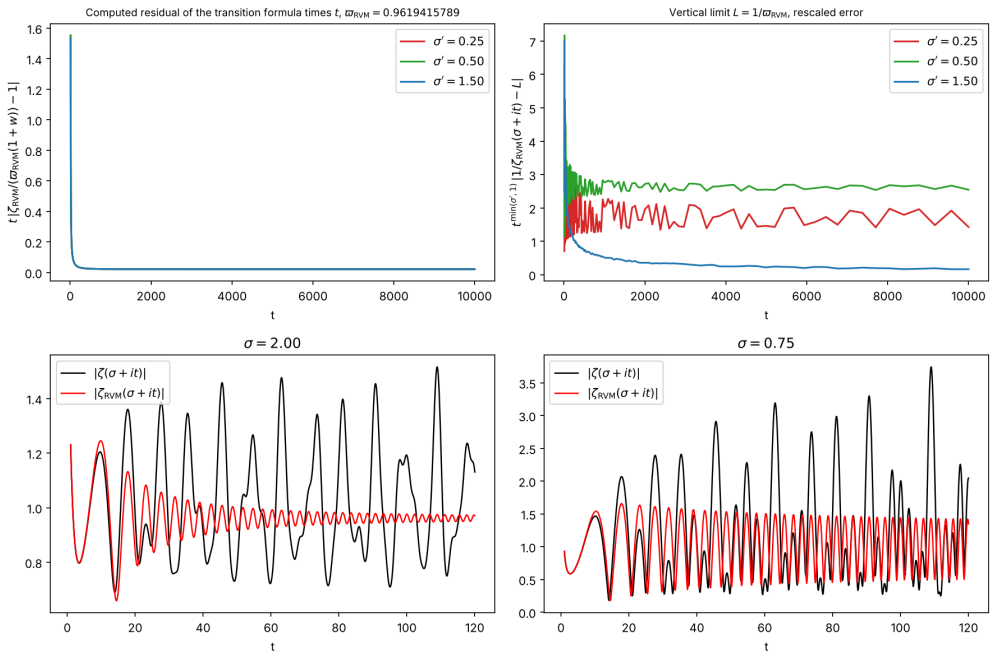


FIGURE 8. Top: the computed residual of the transition formula multiplied by t (it contains the truncation of the product, the tail approximation, the extrapolated value of ϖ_{RVM} and rounding), and the rescaled vertical limit $t^{\min(\sigma',1)}|1/\zeta_{\mathrm{RVM}} - L|$. Bottom: $|\zeta|$ and $|\zeta_{\mathrm{RVM}}|$ on $\sigma = 2$ and $\sigma = 0.75$.

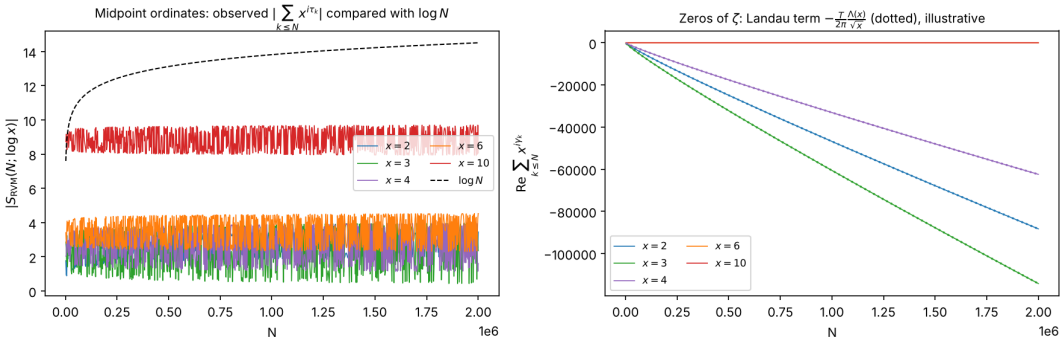


FIGURE 9. Observed size of the exponential sums on the computed range $N \le 2\,001\,052$, compared with $\log N$: the midpoint spectrum shows no prime frequencies (left); the spectrum of ζ resonates at prime powers (right).

of the prime sum, of order $\sum_{n>2\cdot10^6} \Lambda(n)n^{-2}e^2/\log n$ for $a = 1$ and below 10^{-12} for $a = 2$; the real part equals $\frac{1}{2\sqrt{e}}(e - \frac{1}{e^2-1}) = 0.7768942607932536$;

- $2\Re V_+(u)$ agrees with $e^u - (e^{2u} - 1)^{-1}$ to all printed digits at $u = 0.75,\ 0.90,\ 1.05$;
- $C_\zeta^{\mathrm{path}}(N;1) = 0.77776961{+}0.76786173\,i,\ 0.77702246{+}0.76802715\,i,\ 0.77689988{+}0.76784866\,i,\ 0.7769071196 + 0.7678445908\,i$ for $N = 10^4, 10^5, 10^6, 2\,001\,052$: the distance to $e^{-1/2}V_+(1)$ is $1.7 \cdot 10^{-5}$ at $N = 2\,001\,052$;
- $C_{\mathrm{RVM}}^{\mathrm{path}}(N;1) = 0.0415702193 + 1.7575417496\,i$ at $N = 2\,001\,052$, and $S_n - b_nE_{n+1} = 0.0415691734 + 1.7575409824\,i,\ 0.0415688005 + 1.7575410722\,i$ at $n = 10^6$ and $n = 2\,001\,051$, so $C_{\mathrm{RVM}}^{\mathrm{path}}(\infty;1) = 0.041569 + 1.757541\,i$.

Experiment 19.8 (Walks of $e^{i\gamma_k}$, $e^{i\tau_k}$ and $e^{i(\gamma_k+\tau_k)/2}$)**.** *Object:* the walks $S_k = \sum_{j\le k} e^{ix_j}$ for $x = \gamma$, τ and $\theta = (\gamma+\tau)/2$, $k \le 1000$, and their triangular centroids. *Method:* program `s16` (Appendix B.17). *Observed:* $C_\gamma = 0.7720742579 + 0.7626754110\,i$, $C_\tau = 0.0401443298 + 1.7565288414\,i$, $C_\theta = 0.6133423097 + 1.3939939459\,i$; the midpoint of the chord is $\frac{1}{2}(C_\gamma + C_\tau) = 0.4061092939 + 1.2596021262\,i$, and C_θ lies 0.2470 away from it (difference $0.2072 + 0.1344\,i$); $\max_{k\le1000}|S_k| = 3.418$, 2.668, 3.310; $\gamma_k - \tau_k$ has mean $-2.2\cdot10^{-5}$ and standard deviation 0.3587 on $k \le 1000$.

The clock walk is a regular spiral around its centroid, the walk of the zeros an irregular cloud around C_γ, and the walk of the midpoints θ_k is intermediate in shape and position (Figures 10 and 11). Since $e^{i\theta_k} = e^{i\gamma_k}e^{i\eta_k/2}$, the centroid of the midpoints is not the midpoint of the centroids: the passage from the clock to the zeros acts on the walk through the phases $e^{i\eta_k/2}$, not linearly.

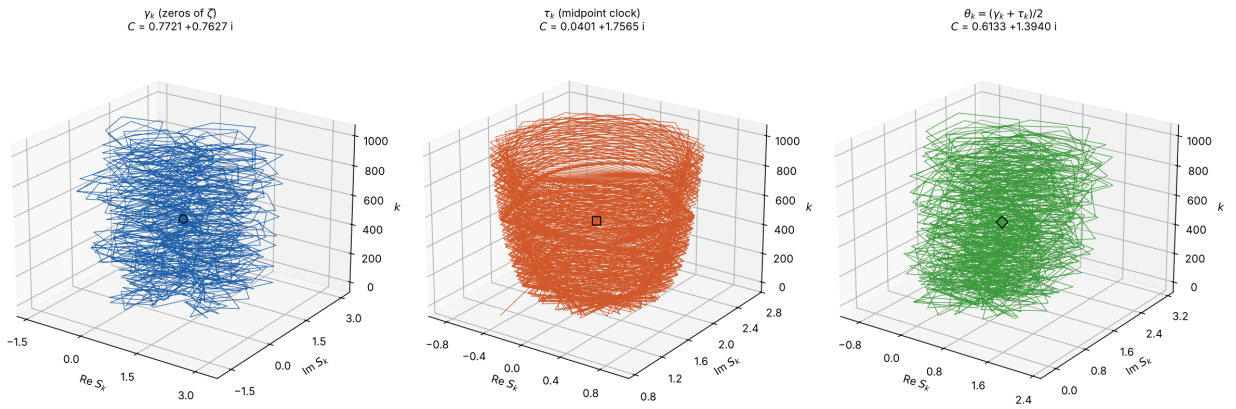


FIGURE 10. The walks $(\Re S_k, \Im S_k, k)$, $k \le 1000$, with k on the vertical axis, for $e^{i\gamma_k}$ (left), $e^{i\tau_k}$ (centre) and $e^{i(\gamma_k+\tau_k)/2}$ (right); the marker is the centroid at height $(N+1)/2$.

19.6. Prime counts: π and π_{RVM}.

Experiment 19.9 (Prime counts and RVM primes)**.** *Object:* J_{RVM}, π_{RVM}, c_n^{RVM}, p_j^{RVM} against J, π, the primes. *Range:* $2 \le x \le 100$ with all N ordinates of each list (3000 points); $2 \le x \le 1000$ with the first $K = 17\,061$ ordinates and a smooth cut (40 000 points). *Expected:* the explicit formula with the γ_k reproduces $\pi(x)$; π_{RVM} is continuous and $\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O(\sqrt{x}(\log\log x)^2/\log x)$ (Theorem 6.5). *Observed:* away from the primes, $|\pi_\gamma - \pi| \le 0.40$ on $[2, 100]$; $\pi_{\mathrm{RVM}}(100) = 26.08$, $\pi(100) = 25$, $\mathrm{li}(100) = 30.13$; $\pi_{\mathrm{RVM}}(1000) = 169.98$, $\pi(1000) = 168$, $\mathrm{li}(1000) = 177.61$. π_{RVM} is not monotone: it carries a ripple of unit scale, with minimum grid increment -0.25 on $[2, 100]$. The linear interpolation of $J_{\mathrm{RVM}}(x^{1/m})$ in the Möbius inversion changes π_{RVM} by at most $5.2\cdot10^{-5}$ when the step in u is doubled, an interpolation error of about $2\cdot10^{-5}$ at the step used. The first grid crossings of the levels $j = 1, \dots, 8$, refined by linear interpolation, are $p_j^{\mathrm{RVM}} = 2.21, 3.15, 5.19, 7.16, 11.14, 14.02, 17.26, 19.18$ against $p_j = 2, 3, 5, 7, 11, 13, 17, 19$.

Figure 12 shows π and π_{RVM} in the usual format. The RVM count follows the prime staircase much more closely than li: replacing each γ_k by τ_k moves every zero by less than one unit (Experiment 19.1), and the low ordinates, which shape $\pi(x)$ for small x, are moved least in relative terms. The steps are replaced by a continuous ripple, so the cell masses c_n^{RVM} of Figure 13 are real numbers, occasionally negative, and are not a prime indicator.

Experiment 19.10 (The three ratios and the square-root scale)**.** *Object:* $\mathrm{li}/\pi_{\mathrm{RVM}}$, π/li, π/π_{RVM}, and $(\pi_{\mathrm{RVM}} - \mathrm{li})/B$, $(\pi - \mathrm{li})/B$ with $B(x) = \sqrt{x}(\log\log x)^2/\log x$. *Range:* the grid of Experiment 19.9

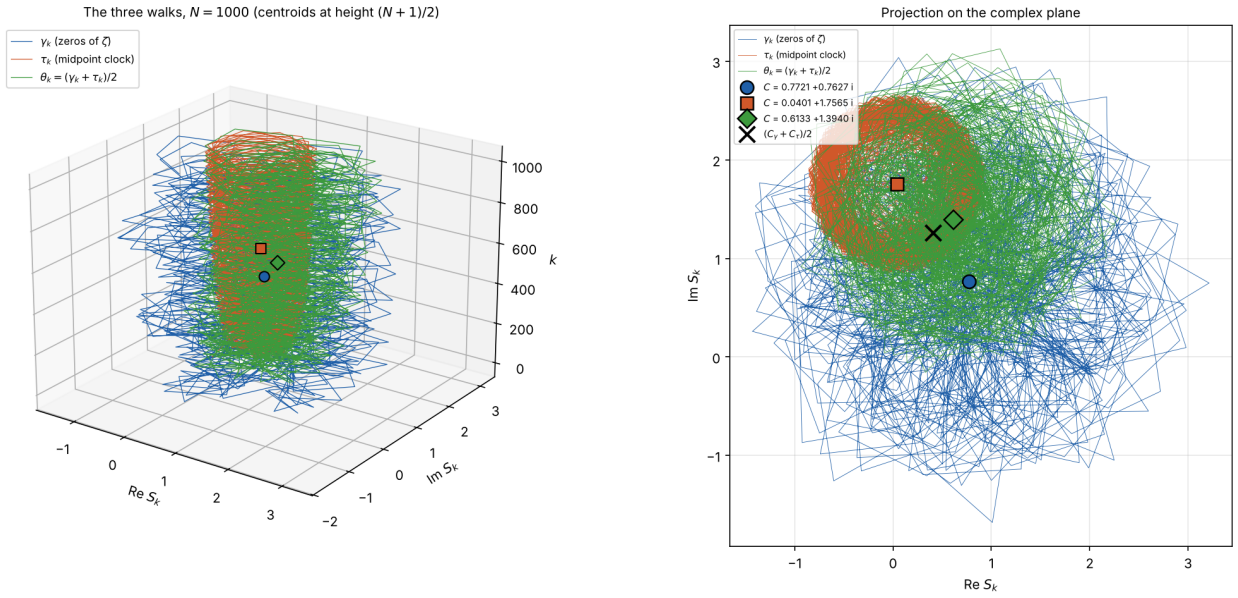


FIGURE 11. Left: the three walks of Figure 10 together. Right: their projection on the complex plane with the three centroids and the midpoint of the chord $\frac{1}{2}(C_\gamma + C_\tau)$ (cross).

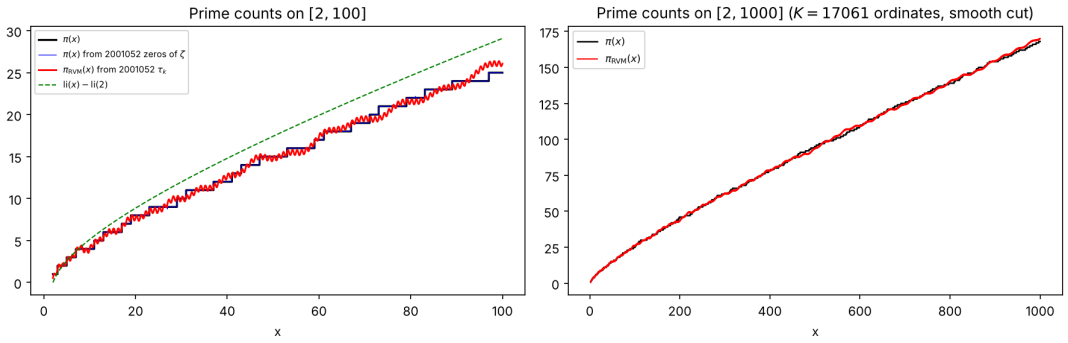


FIGURE 12. $\pi(x)$ (black staircase), the explicit formula with the zeros of ζ (blue), $\pi_{\mathrm{RVM}}(x)$ (red) and $\mathrm{li}(x) - \mathrm{li}(2)$ (green). The distance between the red and green curves is the quantity bounded in Theorem 6.5; the ratios are in Figure 14.

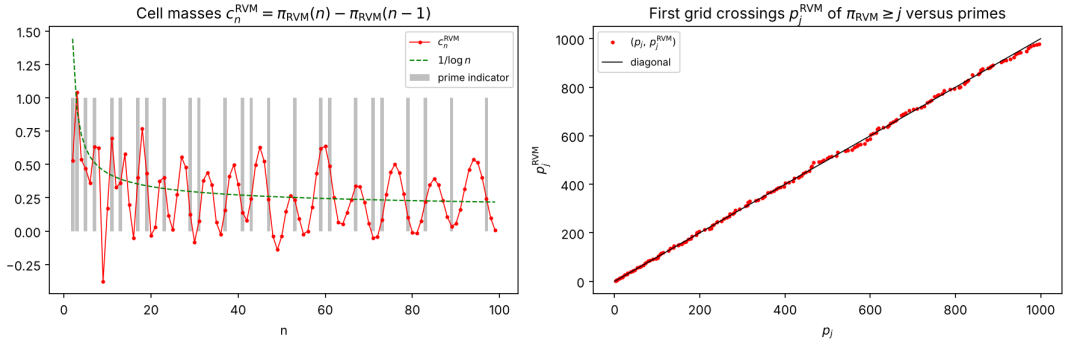


FIGURE 13. Left: cell masses c_n^{RVM}, the prime indicator and $1/\log n$. Right: real RVM prime levels against the primes up to 1000.

on $[8, 1000]$ ($K = 17\,061$ ordinates). *Expected:* the three ratios tend to 1 (Corollary 6.6) and $(\pi_{\mathrm{RVM}} - \mathrm{li})/B$ stays bounded (Theorem 6.5). *Observed:* at $x = 100, 500, 1000$, $\mathrm{li}/\pi_{\mathrm{RVM}} = 1.155, 1.086, 1.045$, $\pi/\mathrm{li} = 0.830, 0.933, 0.946$, $\pi/\pi_{\mathrm{RVM}} = 0.959, 1.013, 0.988$; $\max|\pi_{\mathrm{RVM}} - \mathrm{li}|/B = 2.72$ on $[8, 1000]$, 1.35 on $[50, 1000]$ and 0.72 on $[500, 1000]$, while $\max|\pi - \mathrm{li}|/B = 2.45$ on $[8, 1000]$ (Figure 14).

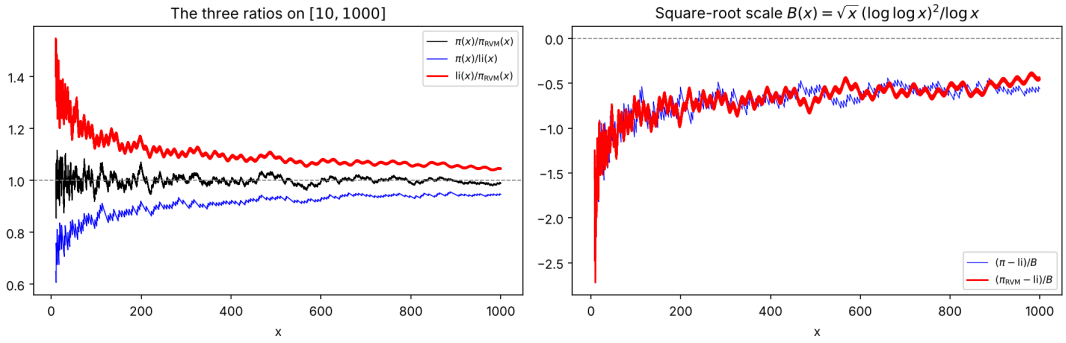


FIGURE 14. Left: the three ratios of Corollary 6.6. Right: $\pi_{\mathrm{RVM}} - \mathrm{li}$ and $\pi - \mathrm{li}$ in units of the scale $B(x)$ of Theorem 6.5. The ratio π/π_{RVM} (black) is the closest of the three to 1; the ratios with li approach 1 at the rate $\frac{1}{2}\,\mathrm{li}(\sqrt{x})/\,\mathrm{li}(x)$ of the bias $J - \pi$.

On $[2, 1000]$ the two normalized errors in Figure 14 have the same size and sign: both counts lie below li by about $\frac{1}{2}\,\mathrm{li}(\sqrt{x})$, the $m = 2$ term of the Möbius inversion (6.20). This is why π/π_{RVM} is much closer to 1 than either ratio with li.

19.7. **The Mertens functions and the ternary division.** The Mertens function is the sum of the ternary sequence $\mu(n) \in \{-1, 0, +1\}$. Its values are specified by two pieces of data:

(a) *the zero set* $\{n : \mu(n) = 0\}$, the non-squarefree integers, of density $1 - 6/\pi^2 = 0.392073$;
(b) *the signs* on the squarefree integers, $\mu(n) = (-1)^{\omega(n)}$, each sign of density $3/\pi^2 = 0.303964$.

The ternary walk μ_h of (1.9)–(1.10) keeps (a) exactly, $\mu_h(n) = 0 \Leftrightarrow \mu(n) = 0$, and replaces (b): at a squarefree n it steps up when $M_h(n-1)$ lies below the target $\mathcal{M}(n) = M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ and down otherwise. By construction the walk follows its target within one unit, so its closeness to $M^{\mathrm{crit}}_{\mathrm{RVM}}$ is a property of the quantizer and carries no information about M; any coincidence of signs between μ_h and μ is reported as an observation without a statistical analysis and is not claimed to have theoretical content. Thus

$$M_h(n) = M_h(n-1) + \mu(n)^2\,\varsigma_n, \qquad |M_h(n) - \mathcal{M}(n)| \text{ small}, \tag{19.1}$$

so that M_h is an integer walk with the zero set of μ that follows $M^{\mathrm{crit}}_{\mathrm{RVM}}$, in the same way as M is the integer walk with the zero set of μ whose signs are $(-1)^{\omega(n)}$.

Experiment 19.11 (Mertens functions and the ternary walk)**.** *Object:* M, $M^{\mathrm{crit}}_{\mathrm{RVM}}$, M_h; the explicit formulas. *Range:* $n \le 10^6$; explicit formulas on $[2, 2000]$ with all N ordinates of each list. *Expected:* the explicit formula with the zeros of ζ reproduces M; $M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = O(x^{1/2+\varepsilon})$ (Theorem 9.14); M_h tracks $M^{\mathrm{crit}}_{\mathrm{RVM}}$ with the densities of μ. *Observed:*

- explicit formula for M with the γ_k: root mean square error 0.0008, 0.0005, 0.0004 on $[2, 2000]$ at distance more than 0.02, 0.05, 0.10 from the integers; the derivatives $\zeta'(\rho_k)$ for $k > 1000$ come from the Riemann–Siegel formula with its first correction term, whose relative difference from mpmath at spot checks is at most $2.3 \cdot 10^{-5}$, and the trivial terms are truncated at six (seven terms change them by $1.5 \cdot 10^{-5}$ at $x = 2$);
- $\zeta'_{\mathrm{RVM}}(\rho_k^{\mathrm{RVM}})/\log(\tau_k/2\pi) = 0.96194$ at $k = 1000$, against $\varpi_{\mathrm{RVM}} = 0.96194$ (Corollary 9.6); the values $|\zeta'(\rho_k)|/\log(\gamma_k/2\pi)$ are spread over $(0, 3)$ (Figure 17);

n	$\mu(n)$	$\mu(n)^2$	$\mathcal{M}(n)$	$M_h(n-1)$	ς_n	$\mu_h(n)$	$M_h(n)$	$M(n)$
1	1	1	0.2925	0	+1	1	1	1
2	−1	1	0.4563	1	−1	−1	0	0
3	−1	1	−0.6523	0	−1	−1	−1	−1
4	0	0	−0.8720	−1	–	0	−1	−1
5	−1	1	−1.7255	−1	−1	−1	−2	−2
6	1	1	−1.3063	−2	+1	1	−1	−1
7	−1	1	−1.5931	−1	−1	−1	−2	−2
8	0	0	−2.4029	−2	–	0	−2	−2
9	0	0	−1.3292	−2	–	0	−2	−2
10	1	1	−1.3280	−2	+1	1	−1	−1
11	−1	1	−2.3272	−1	−1	−1	−2	−2
12	0	0	−2.2067	−2	–	0	−2	−2
13	−1	1	−1.8468	−2	+1	1	−1	−3
14	1	1	−1.9324	−1	−1	−1	−2	−2
15	1	1	−1.5435	−2	+1	1	−1	−1

TABLE 9. The ternary division for $n \le 15$ (the first 40 values are in Appendix C.6). The zero columns of μ and μ_h coincide; the signs of μ_h are chosen by comparing $M_h(n-1)$ with $\mathcal{M}(n)$. For $n \le 12$ the two walks agree; at $n = 13$ the target -1.85 lies above $M_h(12) = -2$, so $\mu_h(13) = +1$ while $\mu(13) = -1$.

- truncated spectral sums $2\sum_{k\le N} |a_k|^2$ over the supplied ordinates: 0.029034 for ζ, 0.027203 for the midpoint system;
- the integer target agrees with the full explicit formula for $M^{\mathrm{crit}}_{\mathrm{RVM}}$ to 0.019 at 100 integers $\le 10^5$ and to 0.018 at 34 integers in $[9\cdot 10^5, 10^6]$; the residues are computed from the product for $k \le 1000$ and modelled beyond, so $\mathcal{M}$ is a tapered-and-interpolated target, not the exact function $M^{\mathrm{crit}}_{\mathrm{RVM}}$; the product formula for $\zeta'_{\mathrm{RVM}}(\rho^{\mathrm{RVM}}_k)$, with its sign $(-1)^{k-1}$ (the factors $j < k$ of $P^{\mathrm{crit}}_{\mathrm{RVM}}{}'(\rho^{\mathrm{RVM}}_k)$ are negative), agrees with a central difference of $P^{\mathrm{crit}}_{\mathrm{RVM}}/G$ to $7\cdot 10^{-11}$ for $k \le 6$, and refitting c on $250 < k \le 1000$ or adding a term d/τ_k^2 to the model changes the target by at most $5.2\cdot 10^{-6}$ at the 134 check integers;
- $\max|M_h - \mathcal{M}| = 2.62$ on $[1, 10^6]$ and 1.65 on $[10^4, 10^6]$;
- densities of $(-1, 0, +1)$ for μ: 0.303857, 0.392074, 0.304069;
- densities of $(-1, 0, +1)$ for μ_h: 0.304003, 0.392074, 0.303923;
- $\mu_h(n) = \mu(n)$ at 50.03% of the squarefree $n \le 10^6$, and $\sum_{n\le 10^6} \mu(n)\mu_h(n)/\#\{n \le 10^6 : \mu(n) \ne 0\} = 0.0007$: the empirical normalized correlation of the sign sequences is 0.0007 on this range (no asymptotic statement is made), although both walks have the same zero set and comparable size;
- $\max_{10\le n\le 10^6} |W(n)|/\sqrt{n} = 0.832, 0.754, 0.688$ for $W = M, \mathcal{M}, M_h$;
- energies $\frac{1}{N}\sum_{n\le N} W(n)^2$ at $N = 10^6$: 14 851, 12 538, 12 538 for $W = M, \mathcal{M}, M_h$; the values for $\mathcal{M}$ carry the truncation, residue-model, taper and interpolation errors of the target.

Figures 15–17 summarize the Mertens side. $M^{\mathrm{crit}}_{\mathrm{RVM}}$ has the size and the visual texture of M, it stays inside $\pm\sqrt{n}$ in agreement with the square-root-scale behaviour of Theorem 9.14, and its residues are deterministic; M_h is its ternary integer version with the zero set of μ. The passage from these synthetic objects to M itself is not numerical: it is the identity $\zeta_{Z1} = \zeta$ of Step 3, which Experiment 19.2 is consistent with.

Remark 19.12 (The opposite phase of M and $M^{\mathrm{crit}}_{\mathrm{RVM}}$ between 10^3 and $2\cdot 10^3$). In the left panel of Figure 15 the curves of M and $M^{\mathrm{crit}}_{\mathrm{RVM}}$ are in opposite phase between $n \approx 10^3$ and $n \approx 2\cdot 10^3$. Heuristically, the visible anticorrelation is largely explained by the first mode, $2\sqrt{x}\,\Re(x^{i\gamma_1}/(\rho_1\zeta'(\rho_1)))$

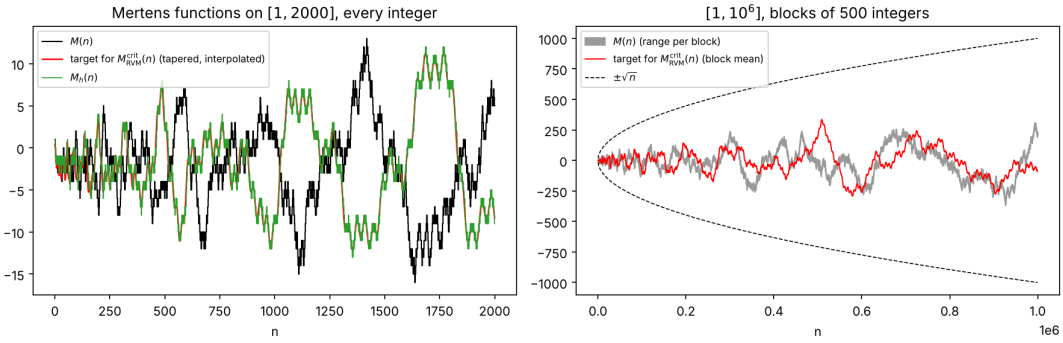


FIGURE 15. Left: M (black), the tapered-and-interpolated target $\mathcal{M}$ for $M^{\mathrm{crit}}_{\mathrm{RVM}}$ (red) and M_h (green) at every integer $n \le 2000$; the residues are computed from the product for $k \le 1000$ and modelled beyond. Right: range of M and mean of $\mathcal{M}$ in blocks of 500 integers up to 10^6, with $\pm\sqrt{n}$.



FIGURE 16. Left: $\mu(n)$ and $\mu_h(n)$ for $n \le 60$: same zeros, different signs. Right: M, the tapered-and-interpolated target $\mathcal{M}$ for $M^{\mathrm{crit}}_{\mathrm{RVM}}$ and M_h for $n \le 300$.

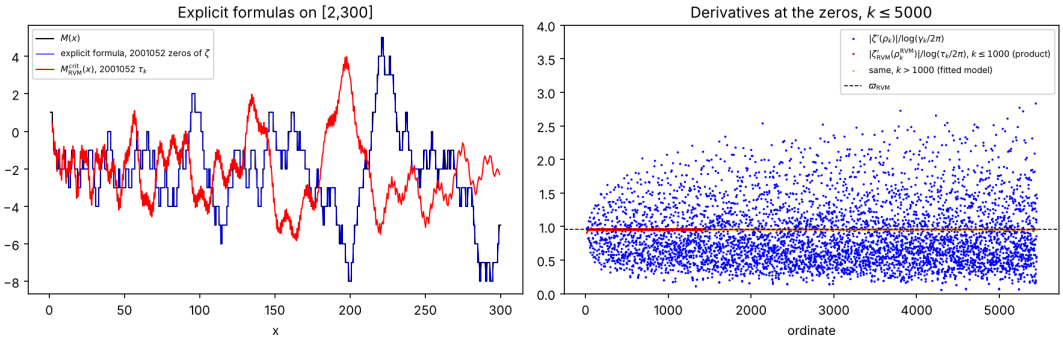


FIGURE 17. Left: $M(x)$, its explicit formula from the zeros of ζ and $M^{\mathrm{crit}}_{\mathrm{RVM}}(x)$ from the τ_k on $[2, 300]$. Right: normalized derivatives at the zeros: erratic for ζ, constant ϖ_{RVM} for ζ_{RVM}.

and $2\sqrt{x}\,\Re(x^{i\tau_1}/(\rho_1^{\mathrm{RVM}}\zeta'_{\mathrm{RVM}}(\rho_1^{\mathrm{RVM}})))$, two oscillations in $\log x$ with the frequencies γ_1 and τ_1. Their phase difference is $\eta_1 \log x$ with $\eta_1 = \tau_1 - \gamma_1 = 14.5213 - 14.1347 = 0.3866$, up to the constant difference of the arguments of the two residues, which is not computed here; $\eta_1 \log x$ equals π at $x = e^{\pi/\eta_1} \approx 3.4 \cdot 10^3$ and lies between 2.67 and 2.94 on $[10^3, 2 \cdot 10^3]$. The beat has period $2\pi/\eta_1 \approx 16.3$ in $\log x$. This is the rotation (15.44) seen on the Mertens side: the passage from the clock to ζ

turns the mode k by the angle $\eta_k \log x$. The higher modes turn at different rates, and at large scale neither a correlation nor an anticorrelation remains, in agreement with the value 0.0007 of Experiment 19.11. The effect depends on η_1 alone and changes sign with x; it does not distinguish $\lambda = 0$.

19.8. **The first** 1000 **values.** The four comparisons of the paper are tabulated for their first 1000 values in Appendix A (Tables 13–16) and drawn in Figures 18–21. The functions are those of the preceding subsections: τ_k from (2.5) (file `zLamb.bin`), γ_k from `zeross.bin`; π_{RVM} from (6.7); p_j^{RVM} the first crossing of the level j by π_{RVM}, as in Experiment 19.9; $M_{\mathrm{RVM}}^{\mathrm{crit}}(n)$ the integer target of Experiment 19.11 and Table 9.

Experiment 19.13 (The first 1000 values)**.** *Object:* τ_k against γ_k, p_j^{RVM} against p_j, $\pi_{\mathrm{RVM}}(n)$ against $\pi(n)$, and $M_{\mathrm{RVM}}^{\mathrm{crit}}(n)$ against $M(n)$, for $k, j, n \le 1000$. *Method:* program `s15` (Appendix B.16). For π_{RVM} on $[2, 8200]$ the zero sum of (6.2) uses the first $K = 183\,029$ ordinates ($\tau_K = 128\,805.3 = 2.5 \cdot 2\pi \cdot 8200$) with the smooth cut of Experiment 19.9, on $1.4 \cdot 10^6$ points in $u = \log x$; the oscillatory sums are evaluated by a nonuniform fast Fourier transform. For $M_{\mathrm{RVM}}^{\mathrm{crit}}$ the tapered explicit formula of Experiment 19.11 is used; it differs from the ordered sum over all $2\,001\,052$ ordinates by at most 0.028 on $n \le 1000$. *Observed:*

- clock: $\max_{k \le 1000} |\eta_k| = 0.9937$ (at $k = 34$), mean $\eta_k = 2.2 \cdot 10^{-5}$, root mean square 0.3587; $\gamma_{1000} = 1419.422481$, $\tau_{1000} = 1419.517765$;
- prime counts: $\pi_{\mathrm{RVM}}(100) = 26.0782$, $\pi_{\mathrm{RVM}}(1000) = 169.9750$, $\pi_{\mathrm{RVM}}(8200) = 1029.688$ against 25, 168, 1028; on $n \le 1000$, $\pi_{\mathrm{RVM}}(n) - \pi(n)$ lies in $[-2.957, 3.779]$, with mean 0.055 and root mean square 1.187; the means of $\pi_{\mathrm{RVM}}(n) - \mathrm{li}(n)$ and $\pi(n) - \mathrm{li}(n)$ are -6.98 and -7.04;
- RVM prime levels: $p_1^{\mathrm{RVM}}, \dots, p_{12}^{\mathrm{RVM}} = 2.2138$, 3.1546, 5.1880, 7.1588, 11.1391, 14.0201, 17.2551, 19.1767, 26.0524, 28.0419, 33.0665, 38.0878; $p_{1000}^{\mathrm{RVM}} = 7928.195$ against $p_{1000} = 7919$; $p_j^{\mathrm{RVM}} - p_j \in [-52.80, 65.25]$, with root mean square 19.26; $\max_{j \le 1000} |p_j^{\mathrm{RVM}} - p_j| / \sqrt{p_j} = 1.069$;
- Mertens: $M_{\mathrm{RVM}}^{\mathrm{crit}}(1000) = -8.646$ against $M(1000) = 2$; $M_{\mathrm{RVM}}^{\mathrm{crit}}(n) - M(n) \in [-17.08, 13.88]$, with root mean square 6.44; the maxima of $|M(n)|/\sqrt{n}$ and of $|M_{\mathrm{RVM}}^{\mathrm{crit}}(n)|/\sqrt{n}$ over $10 \le n \le 1000$ are 0.832 and 0.754; the ternary walk μ_h agrees in sign with μ on 51.5% of the squarefree $n \le 1000$.

The proximity of p_j^{RVM} to p_j is that of π_{RVM} to π read horizontally: the error $p_j^{\mathrm{RVM}} - p_j$ is numerically of the scale $\sqrt{p_j}$ on this range, as $\pi_{\mathrm{RVM}}(x) - \pi(x)$ is numerically of the order of $\sqrt{x}/\log x$ there (Figure 20); it is not small in units of the prime gap. The Mertens functions have the same size and the same $\sqrt{n}$ envelope, and show little sign agreement on the computed range (Remark 19.12). Odlyzko and te Riele [52] used the first 2000 zeros of ζ to disprove the Mertens conjecture; here the explicit formula of $M_{\mathrm{RVM}}^{\mathrm{crit}}$ runs over the clock ordinates with the residues $1/(\rho_k^{\mathrm{RVM}} \zeta'_{\mathrm{RVM}}(\rho_k^{\mathrm{RVM}}))$, which are deterministic.

19.9. **The rung equations: transport, the defect of the clock, and the solution from the clock.** The four experiments of this subsection evaluate the objects of Subsections 15.14 and 16.2. In the first two, f is the function $\exp(-1/(1-y^2))$, $y = (u - 0.36)/0.26$, on $[0.10, 0.62] \subset (0, \log 2)$, normalized to $\int \mathsf{f} = 1$; it tests the first rung. In the programs f, $\mathsf{k_f}$ and $\mathsf{c_f}$ are called `psi`, `W_psi` and `c_psi`.

Experiment 19.14 (Transport on the first rung)**.** *Object:* $\mathsf{R_f}(1)$, $\mathsf{S_f}(\lambda)$ and the moments $\mathsf{m}_j(\mathsf{f})$ of (15.80)–(15.84). *Range:* the first 20 000 ordinates of each list; $\max_{k \ge 1000} |\mathsf{c_f}(\tau_k)| = 2.5 \cdot 10^{-10}$ and $\max_{k \ge 3000} |\mathsf{c_f}(\tau_k)| = 2.1 \cdot 10^{-13}$. *Expected:* if $\zeta_{Z1} = \zeta$, then $\mathsf{S_f}(0) = \mathsf{R_f}(1)$ (Proposition 15.44); the series (15.84) converges to $\mathsf{S_f}(0)$ with terms bounded by $C(1+j)^2 (C_\eta \log 2)^j / j!$ (Theorem 15.45). *Observed:*

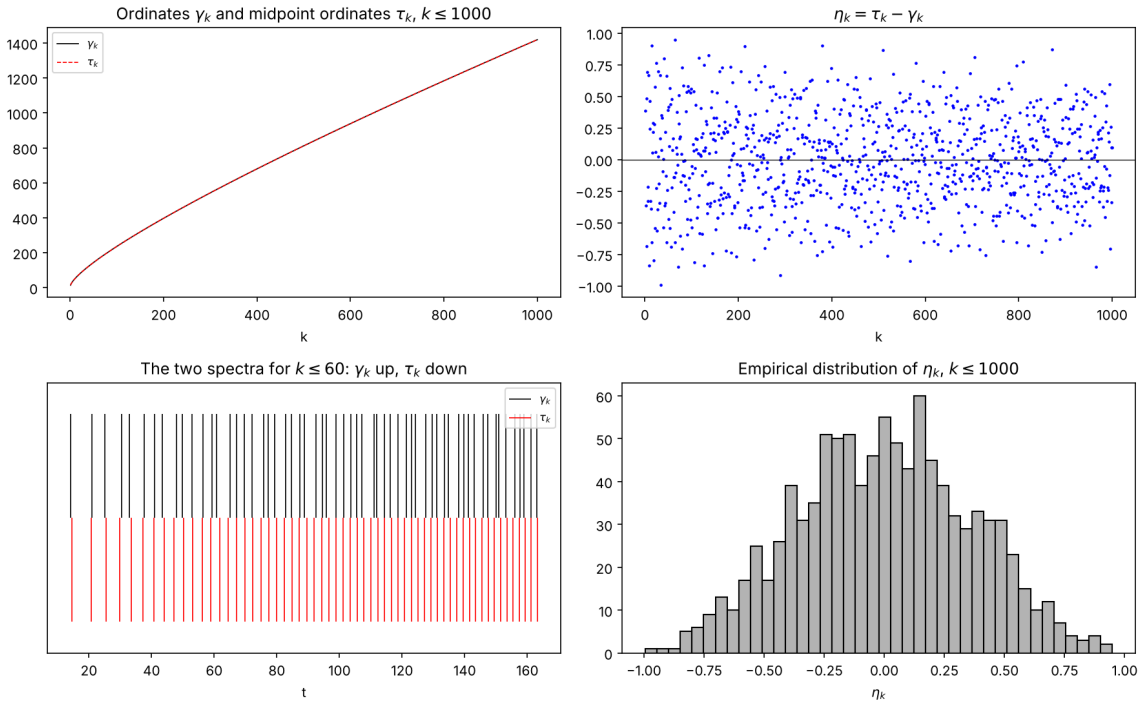


FIGURE 18. The clock against the zeros, $k \le 1000$: γ_k and τ_k; the displacements $\eta_k = \tau_k - \gamma_k$; the two spectra for $k \le 60$; the distribution of η_k (Table 13).

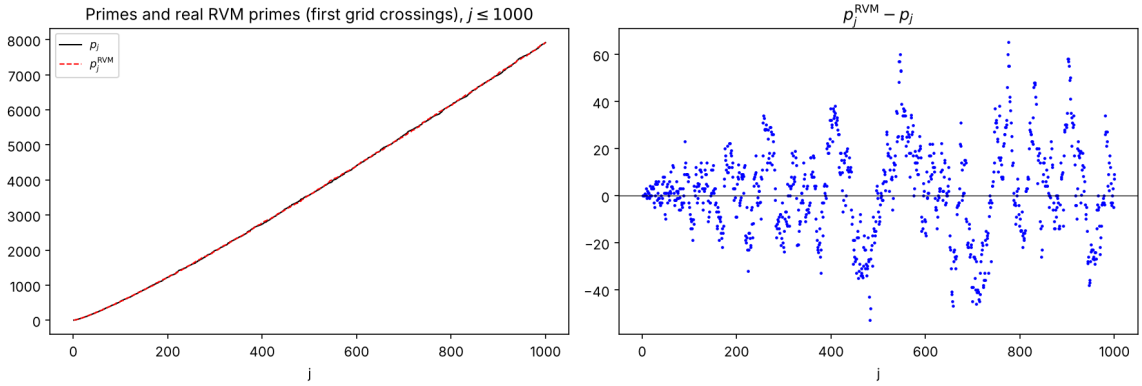


FIGURE 19. The RVM prime levels p_j^{RVM} against the primes p_j, $j \le 1000$, and the difference $p_j^{\mathrm{RVM}} - p_j$ (Table 14).

- $\mathsf{k_f} = 0.133643530916$, $\sum_k \mathsf{c_f}(\tau_k) = 0.176106813076$, $\sum_k \mathsf{c_f}(\gamma_k) = 0.133643530929$;
- defect of the clock $\mathsf{R_f}(1) = -0.042463282161$; transport $\mathsf{S_f}(0) = -0.042463282147$; $\mathsf{R_f}(0) = \mathsf{R_f}(1) - \mathsf{S_f}(0) = -1.4 \cdot 10^{-11}$, and halving the number of quadrature nodes changes $\mathsf{R_f}(1)$ and $\mathsf{S_f}(0)$ by $4 \cdot 10^{-13}$ and $6 \cdot 10^{-13}$; by Theorem 5.7, $\mathsf{R_f}(0) = \langle \mathscr{Y}, \mathsf{f} \rangle$ is a sum over the zeros off the critical line, all of which have $\gamma > 3 \cdot 10^{12}$ [55], where $|\widehat{\mathsf{f}}|$ is far below 10^{-11} by (5.2); the value $-1.4 \cdot 10^{-11}$ is the difference of two quantities computed from the same 20 000 ordinates with the same quadrature, consistent with the truncation and the quadrature error; it is not a measurement of $\mathscr{Y}$;

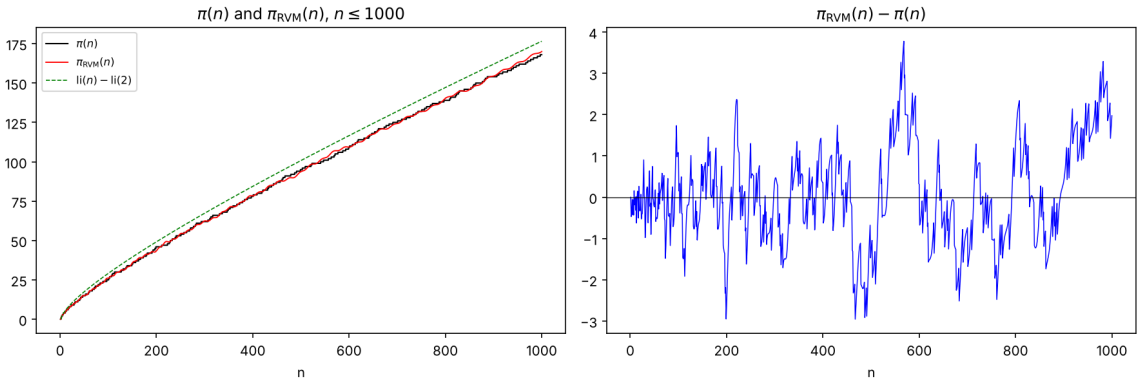


FIGURE 20. $\pi(n)$, $\pi_{\mathrm{RVM}}(n)$ and $\mathrm{li}(n) - \mathrm{li}(2)$ at the integers $n \le 1000$, and the difference $\pi_{\mathrm{RVM}}(n) - \pi(n)$ (Table 15).

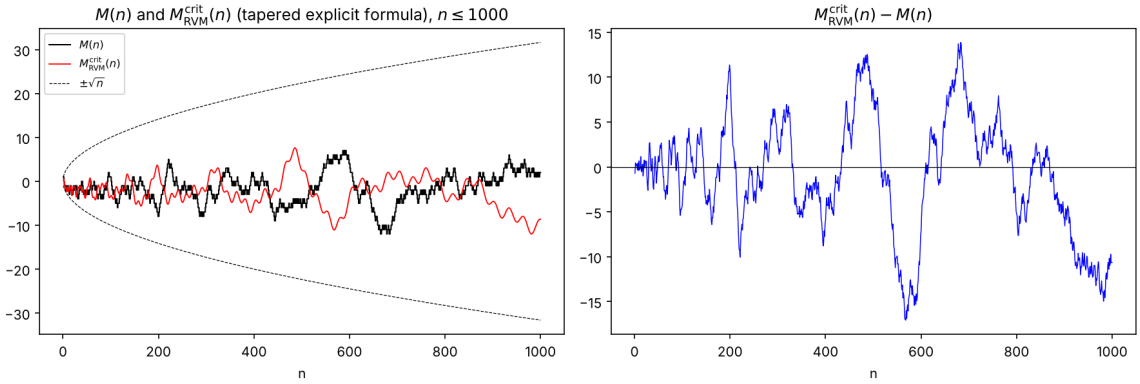


FIGURE 21. $M(n)$ and $M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ at the integers $n \le 1000$ with $\pm\sqrt{n}$, and the difference $M^{\mathrm{crit}}_{\mathrm{RVM}}(n) - M(n)$ (Table 16).

- moment series (Table 10): the term $j = 1$ is 95.1% of $\mathsf{R_f}(1)$, the partial sum with $j \le 2$ is 100.8%, and the partial sums with $j \le 3$, $j \le 5$, $j \le 8$ differ from $\mathsf{R_f}(1)$ by $3.7 \cdot 10^{-6}$, $6.5 \cdot 10^{-9}$, $1.0 \cdot 10^{-11}$;
- along the family (Table 10), extended numerically to $\lambda \in [-1, 2]$ by the same formula $\gamma_k(\lambda) = \gamma_k + \lambda\eta_k$, $\mathsf{R_f}(\lambda)$ is decreasing on the sampled values and almost linear, with slope about -0.044 and a single sign change, at $\lambda = 0$; outside $[0, 1]$ this is only a numerical prolongation of the formula: the products remain defined, since $|\gamma_k(\lambda) - \tau_k| \le (1 + |\lambda|)C_\eta$, but the list is not increasing at $\lambda = -1$, -0.5 and 2, and the results of Section 11 are not claimed there;
- the partial sums $\sum_{k \le K_0} (\mathsf{c_f}(\gamma_k) - \mathsf{c_f}(\tau_k))$ are -0.02086, -0.04381, -0.04649, -0.04221, -0.04249, -0.0424634 for $K_0 = 1, 5, 10, 50, 100, 400$: the first ordinate gives half of the transport and, for this test function, the first-rung sum is numerically stabilised by about fifty ordinates.

Experiment 19.15 (The defect of the clock by Poisson summation)**.** *Object:* the two parts of $\mathsf{R_f}(1)$ in Proposition 15.46. *Expected:* $\mathsf{R_f}(1) = (\mathsf{k_f} - \mathsf{l}_0) - 2\sum_{m \ge 1} (-1)^m \mathsf{l}_m$. *Observed:* $\mathsf{t}_0 = 9.676906787$; $\mathsf{l}_0 = 0.063533498702$ from (15.86); the mismatch of the smooth densities is $\mathsf{k_f} - \mathsf{l}_0 = +0.070110032214$, and the oscillating part, obtained as the difference $\mathsf{R_f}(1) - (\mathsf{k_f} - \mathsf{l}_0)$ and not by summing the l_m, is -0.112573314375; the program does not verify independently that the oscillatory series

j	$\mathsf{m}_j(\mathsf{f})$	$(-1)^j\mathsf{m}_j(\mathsf{f})/j!$	partial sum $-\ \mathsf{R}_\mathsf{f}(1)$	λ	$\mathsf{R}_\mathsf{f}(\lambda)$
1	$+4.0395\cdot10^{-2}$	$-4.0395\cdot10^{-2}$	$+2.07\cdot10^{-3}$	-1.00	$+0.045303554710$
2	$-4.8098\cdot10^{-3}$	$-2.4049\cdot10^{-3}$	$-3.37\cdot10^{-4}$	-0.50	$+0.022413444752$
3	$-2.0432\cdot10^{-3}$	$+3.4053\cdot10^{-4}$	$+3.69\cdot10^{-6}$	-0.25	$+0.011133149816$
4	$-5.5184\cdot10^{-5}$	$-2.2993\cdot10^{-6}$	$+1.39\cdot10^{-6}$	0.00	-0.000000000014
5	$+1.6734\cdot10^{-4}$	$-1.3945\cdot10^{-6}$	$-6.47\cdot10^{-9}$	$+0.25$	-0.010956709511
6	$+5.5721\cdot10^{-6}$	$+7.7390\cdot10^{-9}$	$+1.27\cdot10^{-9}$	$+0.50$	-0.021706821780
7	$+7.2608\cdot10^{-6}$	$-1.4406\cdot10^{-9}$	$-1.69\cdot10^{-10}$	$+1.00$	-0.042463282161
8	$+6.4039\cdot10^{-6}$	$+1.5883\cdot10^{-10}$	$-9.77\cdot10^{-12}$	$+2.00$	-0.080112190380

TABLE 10. Experiment 19.14. Left: the moments $\mathsf{m}_j(\mathsf{f})=\sum_k \eta_k^j \mathsf{c}_\mathsf{f}^{(j)}(\tau_k)$ and the partial sums of the series (15.84) for $\mathsf{S}_\mathsf{f}(0)$, compared with the defect $\mathsf{R}_\mathsf{f}(1) = -0.042463282161$ of the clock. Right: the rung functional along the family.

$-2\sum_{m\ge1}(-1)^m\mathsf{l}_m$ converges to this value. The stationary heights $2\pi e^u$, u in the support of f, fill $[6.944, 11.680]$; only those with $u > 0.4319$ lie above t_0.

Experiment 19.16 (From the clock to the ordinates and to the masses of the primes)**.** *Object:* a finite section of the system (16.6): the unknowns are the ordinates κ_k below a height and the masses a_n, $2\le n\le 11$; the equations are the pairings of (16.6) with $\omega_{\mathrm{test}}(u)\cos(\omega u)$ and $\omega_{\mathrm{test}}(u)\sin(\omega u)$, where ω_{test} is a fixed smooth function supported in $(0,\log 13.5)$ and ω runs over a grid of step 0.1 in $[0,\Omega]$. The ordinates above $\Omega+12$ are kept on the clock; only the first $KT = 6000$ clock ordinates are summed numerically. Although the support window $(0,\log 13.5)$ contains $\log 12$ and $\log 13$, the implementation drops mass columns whose bump weight is below 0.05, so those two masses are fixed to zero in this run. These are explicit truncation sources in the residual. *Method:* Levenberg–Marquardt iteration started at $\kappa_k=\tau_k$; the masses enter linearly and are solved for at each step by ridge regression (the least-squares problem for the masses carries the ridge term $10^{-10}|\mathsf{a}|^2$, so they are not eliminated exactly; this problem and the damped steps are solved by QR/SVD on stacked matrices, not by normal equations, and a step is accepted only if it decreases the residual), with continuation in height $\Omega = 18, 22, \dots, 50$. The proximity to γ_k is never used inside the solver, only as a diagnostic afterwards. The file `zeross.bin` is read only after the iteration, for the comparison. *Expected:* by Theorem 16.6, if $\zeta_{Z1}=\zeta$ the system has the unique solution $\kappa_k=\gamma_k$, $\mathsf{a}_n = \log p/(2\sqrt{n})$ at prime powers and 0 elsewhere. *Observed:* the iteration reaches small residuals at every stage (at two stages it stops because no damping up to 10^{12} decreases the residual further) (residual norms between $4\cdot10^{-4}$ and $5\cdot10^{-3}$, of the size of the contribution of the ordinates kept on the clock); the solution is in Table 11: the first ten ordinates agree with $\gamma_1,\dots,\gamma_{10}$ within $4.4\cdot10^{-4}$, the eleventh within $2.2\cdot10^{-3}$, and the masses agree with (16.7) within $8\cdot10^{-4}$, including the values 0 at $n=6,10$ and the prime-power values at $n=4,8,9$. The ordinates $k=12,13,14$, which lie above the band $\omega\le50$, are not determined (errors 0.10, 0.77, 1.73). With masses up to $n\le34$ and the band $\omega\le60$ the section is not resolved and the iteration does not converge: resolving the ordinates up to a height Ω requires a strict margin $\log N > (1+\varkappa)\log(\Omega/2\pi)$ for some fixed $\varkappa>0$ in a stable finite solve and resolving the masses up to N requires Ω of the order of πN, in agreement with the heights $2\pi x^{1/m}$ of Proposition 15.46. The experiment illustrates the existence form of the final step; it is a computation and not a proof of existence.

Experiment 19.17 (Multiscale Jacobian of the clock)**.** *Object:* the minimum midpoint-clock gap, its bound (15.88), the Ingham critical window $2\pi/\Delta_T$, and the adjacent-pair bound (15.90). *Input:* none of the genuine zeta ordinates; the program `s12_multiscale_jacobian.py` generates the τ_k directly from the Lambert formula up to $K = 2\,001\,052$. *Observed:*

k	τ_k (start)	solution	γ_k	n	a_n (solution)	(16.7)
1	14.521346953	14.134660056	14.134725142	2	+0.245076213	0.245064536
2	20.655740356	21.022091631	21.022039639	3	+0.317145240	0.317142050
3	25.492675432	25.010827129	25.010857580	4	+0.173268937	0.173286795
4	29.739411632	30.424917668	30.424876126	5	+0.359884977	0.359881258
5	33.624531889	32.935072953	32.935061588	6	−0.000004422	0
6	37.257370087	37.586146548	37.586178159	7	+0.367785301	0.367742452
7	40.700551604	40.918919600	40.918719012	8	+0.122615622	0.122532268
8	43.994013953	43.327322117	43.327073281	9	+0.183143344	0.183102048
9	47.165128427	48.004716753	48.005150881	10	−0.000286111	0
10	50.233653391	49.773657726	49.773832478	11	+0.362276961	0.361496314
11	53.214422874	52.972535790	52.970321478			

TABLE 11. Experiment 19.16. Solution of a finite section of the rung system (16.6), started at the clock. Left: the ordinates. Right: the masses at the points $\log n$, which are not assumed, against $\log p/(2\sqrt{n})$ at prime powers $n = p^m$.

K	$T = \tau_K$	Δ_T	$2\pi/\log(T/2\pi)$
10^2	235.987267	1.73462445	1.73286190
10^3	1419.517765	1.15930485	1.15921748
10^4	9877.629616	0.85368118	0.85367617
10^5	74920.891033	0.66939912	0.66939880
$5 \cdot 10^5$	319387.322632	0.57982852	0.57982848
2 001 052	1132490.722588	0.51918344	0.51918343

At the last height, $2\pi/\Delta_T = 12.102053$ while $\log(T/2\pi) = 12.10205288$. Thus the rigorous scale $U \asymp \log T$ is numerically almost exact for the midpoint clock. On every fixed window the adjacent-pair upper bound decreases with the clock gap, illustrating the obstruction in Theorem 15.47(iii). These calculations use no γ_k and no η_k.

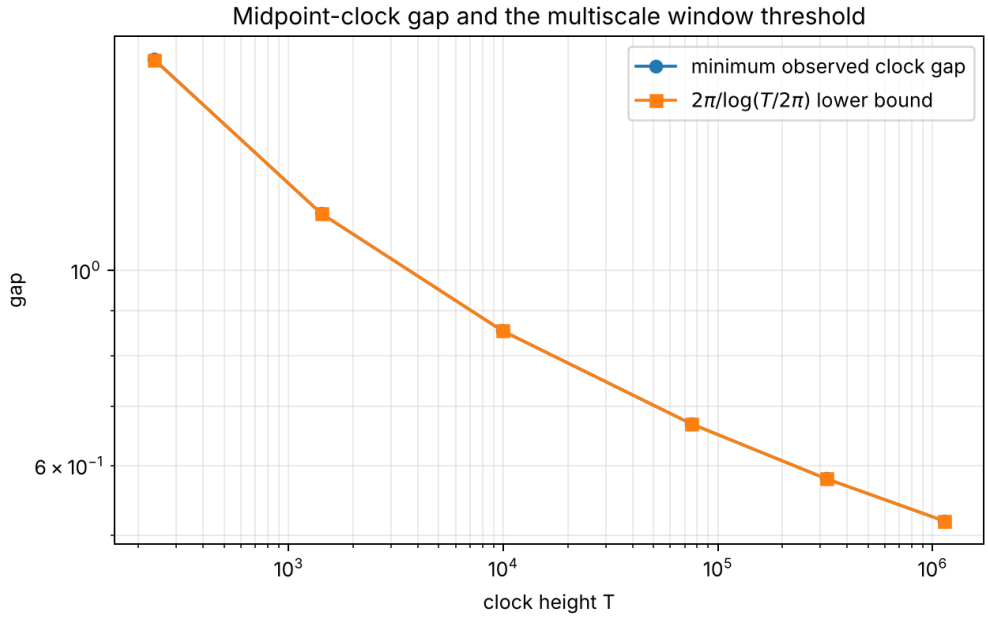


FIGURE 22. Minimum gap of the midpoint clock up to height T and the explicit lower bound $2\pi/\log(T/2\pi)$. Their ratio is already $1 + 2 \cdot 10^{-8}$ at $K = 2\,001\,052$.

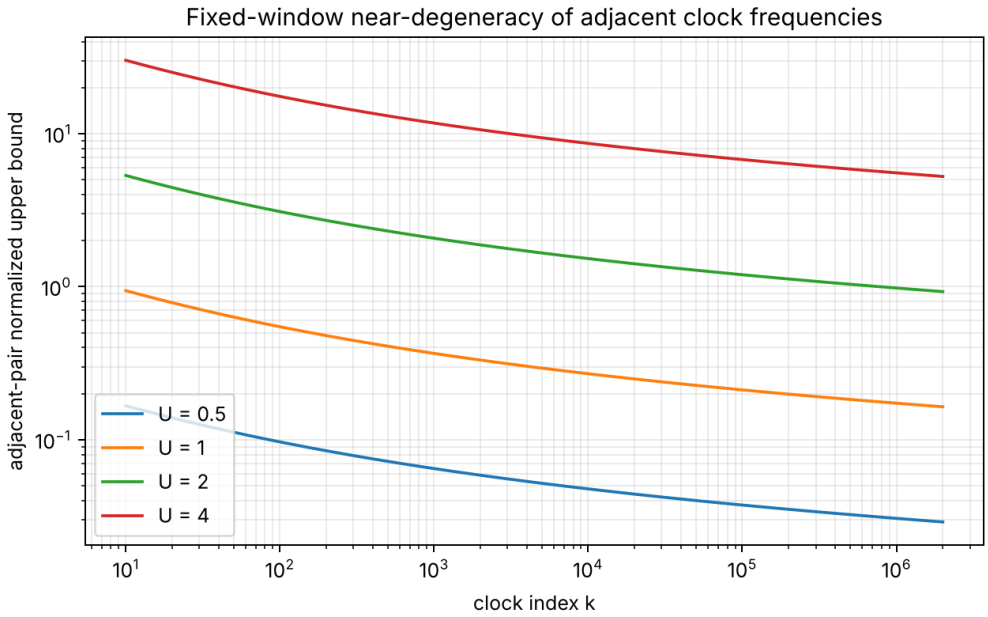


FIGURE 23. Upper bound in (15.90) for adjacent clock frequencies on four fixed u-windows. The decay is slow, of order $1/\log \tau_k$, but it is enough to exclude a height-uniform lower frame bound on any fixed window.

20. Conclusions and open problems

What is new, what is proved, what is assumed, what is open. *New in this paper.* The objects and results that, to the author's knowledge, do not appear elsewhere are: the synthetic zeta function ζ_{RVM} built on the midpoint clock (Sections 2–3; the clock and its Lambert inversion are due to França and LeClair [38], Subsection 1.2); the study of the exponential sums over the ordinates of ζ against Cramér's function, with the zero-free representation of V_+, the forms of RH at one imaginary frequency and on the window $(\log 2, \log 3)$ with the exact defect, the identification of the endpoint rung functional with the sum over the zeros off the line and the reduction of the ladder to the cell $(2, 3)$ with the sine identity, the proposition on finitely many zeros off the line, the Frullani form, the convergence of the clock centroid and the value of the ordinate centroid under RH (Section 5); the crossing point of the family and the family with the opposite endpoint (Subsection 15.5); the tables of the first 1000 values (Appendix A); the unconditional square-root estimate $\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O(\sqrt{x}(\log\log x)^2/\log x)$ for the synthetic prime count, with its continuous prime levels and cell masses (Sections 6–7); the transition theorem $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+O(1/|t|))$ with its constant ϖ_{RVM}, the closed form $\nu_3 = 8 - \frac{33}{4}\log 2 - 2\log\pi$ of its smooth part and the inequality $\varpi_{\mathrm{RVM}} < 1$ (Theorem 9.11) and the synthetic Mertens bound $M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon})$ (Section 9); the λ-family from the clock to the zeta ordinates, with the endpoint ζ_{Z1}, the value ϖ_{Z1} at infinity as an ordered product over the displacements, and the equivalences (D)$\Leftrightarrow \zeta_{Z1} = \zeta \Leftrightarrow \varpi_{Z1} = 1 \Leftrightarrow$RH (Sections 11–13); the atomization of every member of the family by smooth cells, the mean-square bound $X^{1+\varepsilon}$ for the cell masses uniformly in λ, and Hamburger's theorem for the pair (continuous, atomized) in its symmetric, theta and vertical mean-square forms (Sections 14–15); the compact space $\mathscr{O}$ of lists with the rigidity, existence-form and free-mass theorems, and the nondegeneracy theorem for the rung system (Section 16); the acceptance criterion and the two finite models in which the final step is a theorem (Section 17); in Route B for finite sections, the uniqueness of exact solutions in every tube with free masses and the mean-square bound for their masses, the proposition that finite stages do not decide route B, the window forms with an unknown list and the Bernstein form, the dichotomy for minimizers on windows, the identification of the Laplace transform of the residual as $\frac{1}{2}(P'_{\boldsymbol{\kappa}}/P_{\boldsymbol{\kappa}} - \xi'/\xi)$ with the boundary term $\mathfrak{H}(0)\delta'_0$, the certificate theorem for staged sections, the verified exact solutions of fifteen finite sections, Gonek's lists in a tube around the clock, and the Kadec–Avdonin criterion with blocks of variable length (Section 18); and the a priori displacement proposition with its drift constant b (Appendix F.4). The equivalence of B′ with the Riemann hypothesis is classical in substance (Remark 18.19), and no novelty is claimed for the variable-block variant beyond its formulation.

Proved without hypotheses. Every numbered statement (1)–(41) and (43)–(51) below is a theorem of this paper, proved with no hypothesis on ζ; statement (42) assumes RH explicitly; including the secondary lemmas and corollaries that carry them (the bounded displacement lemma, the weighted clock sum, the Lipschitz control of J_{RVM}, the mean-square convergence of the ordinate series, the transfer theorem to $\mathscr{O}$, the compactness and continuity on $\mathscr{O}$, the point-support and finite-order steps of the nondegeneracy theorem, the Laplace-transform lemma of Route B, the monotone matching lemma); Section 18 has its own status table (Section 18.9).

No unproved hypothesis. The inequality $\varpi_{\mathrm{RVM}} < 1$ is Theorem 9.11, with $\varpi_{\mathrm{RVM}} < 0.9796$; the five statements that use it, Proposition 14.16(ii), the sign $\mathsf{W}(1) > 0$ in Proposition 15.10, Proposition 15.12, Corollary 15.38 at the clock end, and the remark after Corollary 15.8, are therefore unconditional. Hypothesis 7.7 is proved (Corollary 15.19) and is a hypothesis only in name. The jump-growth condition in Theorem 13.7(ii) is part of a statement equivalent to (D), not an assumption of the paper. The statements that carry a hypothesis are those that assume (D) explicitly (Theorems 12.8 and 12.9) and the a priori Proposition F.2, which assumes an exact

solution; they are statements about (D) and about hypothetical solutions, not uses of an unproved lemma.

Equivalent to (D) and not proved. The identity $\zeta_{Z1} = \zeta$, (D), RH, and each of the statements listed in Subsection 1.6 and after item (50) below: the identity (5.9); the window identity of Theorem 5.5(ii) and the distributional convergence of the ordinate centroids on the window; the cell $(2, 3)$ together with the sine identity (5.19); $\mathscr{Q}_\infty = \log L$; $\varpi_{Z1} = 1$; the product identity (13.12); the cell identity (13.19); $\mathsf{W}(0) \geq 0$; the vanishing of the cell defects at $\lambda = 0$; the symmetry of $e^{\tilde{h}_0}$; $T_0 \equiv 0$; $\mathsf{V}_\sigma(0) = 0$; the existence of a coincidence list in $\mathscr{O}$; the solvability of the rung system in $\mathscr{O}$ or in any tube; statements B and B′; the window identity for an unknown list (Theorem 18.35); the complete monotonicity of $e^{-iy/2}V_+(iy)$ (Theorem 18.36); the solvability of the rung system on every window in a fixed box (Theorem 18.39). These are reductions, each proved as an equivalence; none of them is resolved.

Open and not claimed. The existence of a solution of the rung system; the injectivity of the clock-only linearization and any continuation from the clock; the convergence of finite sections to a distributional limit; a certified enclosure of ϖ_{RVM} beyond the bounds $0 < \varpi_{\mathrm{RVM}} < 0.9796$ of Theorem 9.11 (the ten digits 0.9619415789 are floating-point values); and any computer-assisted statement beyond Theorem 18.78, which concerns fifteen finite square sections and, by Proposition 18.20, has no bearing on RH (the Gaussian certificates of Section 18.8 are floating-point candidates); the exclusion of degenerate minimizers in Theorem 18.39; and the relation between window solvability and zero-free regions (Heuristic 18.41).

Dependence table. The following table records what each group of results depends on.

Results	Depend on
Clock, ζ_{RVM}, exponential sums, J_{RVM}, π_{RVM}, unit cell, transition theorem, synthetic Mertens bound (Sections 2–9)	the clock only; no property of ζ beyond the Riemann–von Mangoldt formula (for Lemma 2.2)
λ-family, ζ_{Z1}, ϖ_{Z1}, Mellin counts, explicit formula, cell form (Sections 11–13)	the ordinates γ_k as a list in the tube, i.e. membership in $\mathscr{O}$; no hypothesis
Atomized family, mean square of cell masses, joining factor, pair criteria (Sections 14–15)	membership in $\mathscr{O}$; no hypothesis
$\varpi_{\mathrm{RVM}} < 1$ (Theorem 9.11) and the five statements that use it	the clock only; the explicit inequalities of the proof are verified in interval arithmetic (program `s14`, Appendix B.15)
Step 3 and the transfer of the Mertens bound to M (Theorems 12.8, 12.9)	(D), as an explicit hypothesis of the statement
Rigidity, existence form, free masses, nondegeneracy (Section 16)	membership in $\mathscr{O}$; nondegeneracy assumes the existence of a solution
Route B: compactness, uniqueness with free and with prescribed masses, window forms, minimizers, certificate, Gonek lists, (A_2) criterion (Section 18)	C-admissibility (tube) only; Theorem 18.7 also uses Proposition F.2 and Hamburger's theorem, the window forms Lemma 5.4 and Proposition 5.3; B, B′ and the window forms are equivalent to RH and not proved
Proposition 18.20	the zeros verified up to $3 \cdot 10^{12}$ [55] and the bound of [72] for N_ζ
Theorem 18.78	ball-arithmetic computation (program `certV`; Arb through python-flint)
A priori displacement (Appendix F.4)	the existence of an exact solution with the local-density and mean-square hypotheses stated there
$\lvert \log \varpi_{Z1} \rvert < 4 \cdot 10^{-13}$ (Proposition 14.3) and $\lambda^* < 2 \cdot 10^{-11}$ (Proposition 15.13)	the verified zeros up to T_{ver} (input data)

Results	Depend on
Section 5	Landau's formula, Selberg's mean value of S, Cramér's function, the explicit formula; Theorem 5.15 assumes RH explicitly; all other statements are unconditional
Section 19, Subsection 18.8 except Theorem 18.78	floating-point computations with the supplied files; not part of any proof

The proved statements. The following statements are proved (the results of Section 18 are summarised in its status table, Section 18.9):

(1) the midpoint ordinates τ_k are given by the Lambert formula (2.5) and stay within a bounded distance of the ordinates γ_k of ζ (Lemma 2.2);
(2) $P^{\mathrm{crit}}_{\mathrm{RVM}}$ is entire and symmetric, and ζ_{RVM} has exactly the simple zeros $\frac{1}{2} \pm i\tau_k$, the trivial zeros and a residue-one pole at 1 (Theorem 3.1);
(3) $S_{\mathrm{RVM}}(N; u) \ll 1 + \log N$ uniformly on compact positive frequency intervals, so the point centroid and $M_H^{\mathrm{RVM}}(x)$ tend to 0 (Section 4);
(4) J_{RVM} exists compact-uniformly and is related to π_{RVM} by exact finite Möbius inversion (Section 6);
(5) $J_{\mathrm{RVM}}(x) - \mathrm{li}(x)$ and $\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x)$ are $O(\sqrt{x}(\log\log x)^2/\log x)$, so $\mathrm{li}\,/\pi_{\mathrm{RVM}} \to 1$, and with the prime number theorem $\pi/\mathrm{li} \to 1$ and $\pi/\pi_{\mathrm{RVM}} \to 1$ (Theorem 6.5, Corollary 6.6);
(6) J_{RVM} is Lipschitz on $[2, \infty)$, $c_n^{\mathrm{RVM}} = O(1)$, and $\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$ (Section 7);
(7) $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+O(1/|t|))$ on every band to the right of $\frac{1}{2}$ and $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+\chi)(1+O(1/|t|))$ on closed substrips of $\frac{1}{2} < \Re s < 1$ (Theorem 9.1); the transition constant is $\varpi_{\mathrm{RVM}} = P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac{1}{2})e^{\nu_3+\nu_4} < 0.9796$, with $\nu_3 = 8 - \frac{33}{4}\log 2 - 2\log\pi$ and $\nu_4 < -0.0126$ (Theorem 9.11);
(8) $1/\zeta_{\mathrm{RVM}} \to L$ on every vertical line to the right of $\frac{1}{2}$, $\zeta_{\mathrm{RVM}}^{\pm 1}$ have no absolutely convergent Dirichlet expansion, and $M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon})$ (Section 9);
(9) 2ξ equals its Hadamard product over $\rho(1-\rho)$ exactly (Theorem 10.1), and $|\zeta| \asymp |\mathcal{Q}_{\mathrm{RVM}}|$ on $\frac{1}{2} + \varepsilon \le \sigma \le c'$, $|t| \ge t_0$ (Corollary 10.3);
(10) the λ-family from ζ_{RVM} to ζ_{Z1} converges jointly, has simple zeros for $\lambda > 0$, an explicit generator and total multiplier, and its members differ only through $\gamma_k - \tau_k$ (Section 11);
(11) ζ_{Z1} has the functional equation, the pole, the residue and the order of ζ, and $\zeta_{Z1} = \zeta \Leftrightarrow$(D)$\Leftrightarrow$RH (Theorem 12.2);
(12) ζ_{RVM} is bounded above and below on every half-plane $\Re s \ge \sigma_0 > 1$ and tends to ϖ_{RVM} as $\Re s \to +\infty$ (Theorem 9.10); $1/\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$, so (D) is equivalent to an absolutely convergent Dirichlet expansion of $\mathcal{K}$ in $\Re s > 1$ (Theorem 12.3);
(13) no member of the λ-family, ζ_{Z1} included, can have a zero off the critical line other than the trivial zeros (Theorem 11.1);
(14) the endpoint has the value $\varpi_{Z1} = \varpi_{\mathrm{RVM}} \lim_{K\to\infty} \prod_{k\le K}(\frac{1}{4}+\tau_k^2)/(\frac{1}{4}+\gamma_k^2) \ge 1$ at $+\infty$, and $\zeta_{Z1} = \zeta$ if and only if $\varpi_{Z1} = 1$ (Theorems 13.1, 13.2, Corollary 13.3);
(15) the prime-power counts of ζ_{RVM} and ζ_{Z1} are the Mellin transforms of their logarithms and satisfy a distributional explicit formula; this proves the Mellin–Stieltjes identification and the weighted factorization $\zeta_{\mathbb{Z},\mathrm{RVM}} = \zeta_{\mathrm{RVM}} e^{h_{\mathrm{RVM}}}$, and, with the jump-growth hypothesis of Theorem 13.7(ii), (D) holds if and only if the prime-power count of ζ_{Z1} is constant between consecutive integers (Theorems 13.5, 13.6, 7.4, 13.7);

(16) along the family $\frac{d}{d\lambda}\log\varpi_\lambda = -\Upsilon(\lambda)$, and (D) is the identity $\int_0^1 \Upsilon\, d\lambda = \log L$; with the verified zeros, $0 \le \log\varpi_{Z1} < 4\cdot 10^{-13}$ (Propositions 14.2, 14.3);

(17) the atomized family $\tilde\zeta_{\mathbb{Z},\lambda} = \zeta_\lambda e^{\tilde h_\lambda}$ consists of absolutely convergent Dirichlet series in $\Re s > 1$ for every λ, the endpoint included, because the cell masses have mean square $X^{1+\varepsilon}$ on dyadic blocks uniformly in λ; the joining factor is explicit cell by cell and holomorphic in $\Re s > \frac12$ (in $\Re s > 0$ for $\lambda = 1$), and (D) is equivalent to the vanishing of every cell defect at $\lambda = 0$ and of J_{Z1} on the first cell $(1,2)$ (Proposition 14.5, Theorems 14.7, 14.11, 14.13, 14.14);

(18) at the clock endpoint the joining factor is not constant, $\tilde h_1(\sigma) \to \log L$; the atomized reciprocals have Möbius coefficients $\tilde\mu_\lambda$, and the endpoint Mertens function has a Mellin form B_{Z1} with $B_{Z1} = M$ if and only if (D) (Propositions 14.16, 14.17);

(19) Step 3: $\zeta_{Z1} = \zeta$ implies $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ (Theorem 12.8); equivalently, the synthetic bound $M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon})$ transfers to M through $1/\zeta = F_{\mathrm{RVM}}\Pi_{\mathrm{tr}}^{-1}$ (Theorem 12.9);

(20) for $0 < \lambda \le 1$ no ζ_λ is an absolutely convergent Dirichlet series in $\Re s > 1$ and no joining factor $\tilde h_\lambda$ is constant, and $\lambda \mapsto \log\zeta_\lambda(\sigma)$ is strictly convex for every real $\sigma > 1$ (Propositions 15.1, 15.2);

(21) if the joining factor $e^{\tilde h_\lambda}$ is meromorphic of finite order with finitely many poles and symmetric under $s \mapsto 1-s$, Hamburger's theorem gives $\tilde\zeta_{\mathbb{Z},\lambda} = \zeta$, and for $\lambda = 0$ also $\zeta_{Z1} = \zeta$ (Theorem 15.3);

(22) if ζ_λ is the Mellin–Stieltjes transform of a positive measure, then $\varpi_\lambda \le 1$; (D) holds if and only if ζ_{Z1} has such a representation, if and only if J^{M}_{Z1} is non-decreasing (Theorem 15.7, Corollary 15.8);

(23) J_{RVM} is continuously differentiable on $(2,\infty)$ with $J'_{\mathrm{RVM}}(y) = \frac{1}{\log y} + 2\cos(2\pi y) - 2^{-1/2}y^{-1/4}\cos(4\pi\sqrt{y} + \frac{\pi}{4}) + O(y^{-1/3})$; consequently $\tilde c_n(1) = \frac{1}{\log n} + O(n^{-1/4})$ off the powers of 2, and Hypothesis 7.7 holds off the integers 2^j with $\mu(j) \ne 0$ (Theorem 15.18, Corollary 15.19);

(24) the cell masses are sums over the ordinates with the height kernel $\hat\varphi(\gamma/2\pi n)$, which vanishes at the heights $2\pi qn$; their flow is the pairing of V_n with the count difference $N_\zeta - n_{\mathrm{RVM}}$; and the Gaussian defect of the atomization is $\frac12(e^{\tilde h_\lambda(1)} - 1)y^{-1/2} + O(y^{-1/4-\delta})$ (Propositions 15.20, 15.21, 15.25);

(25) $\zeta_{Z1}(\sigma) \ge \zeta(\sigma)$ for every real $\sigma > 1$, with equality at one point if and only if $\zeta_{Z1} = \zeta$; in particular $\prod_k (1 + 2/(\frac14 + \gamma_k^2)) \ge \pi/3$ (Theorem 15.11);

(26) every ζ_λ is the Mellin transform of a function A_λ that satisfies the summation formula of ζ, and $\varpi_\lambda - 1 = -\mathsf{W}(\lambda)/\hat{\mathsf{g}}(1)$ for one explicit number $\mathsf{W}(\lambda)$; (D) is equivalent to $\mathsf{W}(0) \ge 0$ and, by Theorem 9.11, to $\varpi_\lambda < 1$ for all $\lambda \in (0,1]$ (Lemma 15.9, Propositions 15.10, 15.12);

(27) Hamburger's theorem for the pair holds without continuation of the joining factor: $\tilde\zeta_{\mathbb{Z},\lambda} = \zeta$ if and only if the atomized coefficients satisfy the theta relation, if and only if the vertical mean square of $\tilde\zeta_{\mathbb{Z},\lambda}/\zeta$ on a line $\Re s = \sigma > 1$ is 1; both defects are continuous in λ (Theorem 15.4);

(28) at the clock end the joining factor has no symmetric meromorphic continuation, and $\tilde\zeta_{\mathbb{Z},\lambda} = \zeta$ for at most finitely many $\lambda \in [0,1]$, not for $\lambda = 1$ (Propositions 15.5, 15.28);

(29) the cell masses are entire functions of λ of exponential type $O(\log n)$, and the cell defects have the height kernel $\hat\varphi'(\gamma/2\pi n)$ (Propositions 15.27, 15.22);

(30) in the compact space $\mathscr{O}$ of ordinate lists at bounded distance from the clock, at most one list has coinciding continuous and atomized zeta functions; such a list exists if and only if $\zeta_{Z1} = \zeta$, and it is then the list of the ordinates of ζ (Theorem 16.2, Proposition 16.3, Corollary 16.4);

(31) (D) holds if and only if $\tilde c_n(0) = \mathbf{1}_{\mathrm{prime}}(n)$ for every $n \ge 2$, one real number for each cell; every λ with $\tilde\zeta_{\mathbb{Z},\lambda} = \zeta$ is a witness of (D), so the exceptional set is empty or contains 0; cell by cell the remaining identity is (15.61) (Corollary 15.29, Propositions 15.30, 15.32);

(32) for a pair joined by a symmetric factor Hamburger's theorem gives a constant factor when the continuous function has order one, and not in general; at $\lambda = 0$ the symmetry of the joining factor is equivalent to (D) (Proposition 15.34, Remarks 15.33, 15.35);
(33) the ladder: for every $N \geq 2$, $\tilde{h}_\lambda(\sigma) = O(\sigma N^{-\sigma})$ as $\sigma \to +\infty$ if and only if the Mellin count of ζ_λ has no mass outside the integers below N; (D) holds if and only if this holds at $\lambda = 0$ for every N, and at the clock end the first rung fails (Theorems 15.36, 15.37, Corollary 15.38);
(34) the bridge theorem: if the imaginary part of the joining factor of a pair extends continuously to $\Re s \geq \frac{1}{2}$, vanishes on the critical line, and the factor satisfies the temperate bound (15.69), then the factor is constant, the discrete function is ζ and the continuous one is a multiple of ζ; for the family the two conditions hold for no $\lambda \in (0, 1]$, and at $\lambda = 0$ they are equivalent to (D) (Theorem 15.39, Corollary 15.40);
(35) on the critical line $\Im \log \zeta_\lambda = -\vartheta(t) + \pi(N_\lambda(t) - 1)$, and the cell masses of two lists differ by a sum over the ordinates of one function of one ordinate (Propositions 15.41, 15.42);
(36) on each rung the passage from the clock to the ordinates of ζ is the transport $\mathsf{S}_\mathsf{f}(\lambda)$, an entire function of λ of exponential type at most $C_\eta \log N$ whose value at $\lambda = 0$ is the sum of the series of the moments $\sum_k \eta_k^j \mathsf{c}_\mathsf{f}^{(j)}(\tau_k)$; (D) holds if and only if the transport equals the defect of the clock on every rung, and this defect is given exactly by Poisson summation, with stationary heights $2\pi x^{1/m}$ (Proposition 15.44, Theorem 15.45, Proposition 15.46);
(37) (D) holds if and only if the rung system (16.6), with free masses at the points $\log n$, has a solution in $\mathscr{O}$; the solution is then the list of ζ with the masses of von Mangoldt, and its linearization is injective up to the splitting of multiple ordinates (Theorems 16.6, 16.7);
(38) at the midpoint clock, the finite ordinate Jacobian up to height T has a uniform lower frame bound on windows of length $(1 + \varkappa_{\mathrm{win}}) \log(T/2\pi)$, no uniform lower bound is possible on a fixed window, the infinite ordinate block has dense range on every finite window, and smooth bounded-displacement transport tails are $O_{A,\mathsf{f}}(T^{1-A} \log T)$ (Theorem 15.47);
(39) the gradient of the vertical defect on $\mathscr{O}$ is $\partial \mathsf{V}_\sigma / \partial \kappa_k = 2\mathsf{Q}_{\boldsymbol{\kappa}}(\kappa_k)$ with one real-analytic function $\mathsf{Q}_{\boldsymbol{\kappa}}$ (Proposition 16.9);
(40) a list is accepted by a continuous side and a discrete side if and only if its function lies in the intersection of the window of the first with the frequencies of the second; a non-periodic list is accepted through one unitary matrix; for the finite Euler product $(1 - 2^{-s})(1 - 3^{-s})$ the constructions of the paper apply and the endpoint $\lambda = 0$ is the only member of the family that is a Dirichlet series; for $\prod_{p \leq X}(1 - p^{-s})$ with more primes the Poisson part extends and the uniqueness part does not (Theorem 17.1, Proposition 17.2, Theorem 17.3 and the remark after it);
(41) there is no function with the functional equation of ζ and an absolutely convergent Dirichlet series supported on the integers with bounded prime factors, and for the diagonal operator over the primes the contraction, the convergence of the determinant and the symmetry hold in $\sigma > 0$, in $\sigma > 1$ and on $\sigma = \frac{1}{2}$ respectively (Proposition 17.4, Subsection 17.4);
(42) under RH, the triangular centroid $\sum_{k \leq N}(1 - \frac{k-1}{N})e^{iu\gamma_k}$ tends to $e^{-u/2}V_+(u)$ for every $u > 0$ that is not the logarithm of a prime power; at $u = 1$ the limit is $0.77689426079\ldots + 0.76785566886\ldots\, i$, with real part $\frac{1}{2\sqrt{e}}(e - \frac{1}{e^2-1})$ (Theorem 5.15, Lemma 5.14);
(43) the triangular centroid of $e^{iu\tau_k}$ converges for every $u > 0$, with the limit given by the series (5.23) and the rate $O((\log N)^2/N)$ (Theorem 5.13);
(44) Cramér's function has the zero-free representation (5.11) on the half-plane Ω_θ, with simple poles at the points $\log p^m$, and $2\Re V_+(u) = e^u - (e^{2u} - 1)^{-1}$ on $(\log 2, \log 3)$ (Proposition 5.3);
(45) the defect between the ordinate transform $\mathscr{Z}_\gamma$ and the zero transform $\mathscr{Z}_\xi$ is $-\sum_k(\cosh(wd_k) - 1)e^{iw\gamma_k}$, and RH holds if and only if $\sum_k e^{-y_0\gamma_k} = e^{-iy_0/2}V_+(iy_0)$ for one $y_0 \in (0, 4\pi]$ (Propositions 5.1, 5.2);

(46) on every open interval free of the points $\log p^m$, in particular on $(\log 2, \log 3)$, RH holds if and only if $\sum_k \widehat{\mathsf{f}}(\gamma_k) = \int \mathsf{f}(u) e^{-u/2} V_+(u) du$ for every test function supported there, equivalently if and only if the ordinate centroids converge there to $e^{-u/2}V_+(u)$ in the sense of distributions; without hypothesis the difference is the absolutely convergent sum (5.14) over the zeros off the line (Lemma 5.4, Theorem 5.5);

(47) the rung functional of the endpoint is the sum over the zeros off the line, $\mathsf{R}_\mathsf{f}(0) - \frac{1}{2}\sum_n \Lambda(n) n^{-1/2} \mathsf{f}(\log n) = \langle \mathscr{Y}, \mathsf{f} \rangle$; equivalently $(\log x)(dJ_{Z1}^{\mathrm{M}} - dJ)/dx = 2x^{-1/2}\mathscr{Y}(\log x)$; and the cosine difference is $\mathsf{R}_\mathsf{f}(1) - \langle \mathscr{Y}, \mathsf{f} \rangle$ on every rung (Theorem 5.7);

(48) the real part of the window identity on $(\log 2, \log 3)$ is the constancy of J_{Z1}^{M} on the cell $(2, 3)$, and (D) is equivalent to that cell together with the sine identity (5.19) (Corollary 5.8);

(49) if the set of zeros off the critical line is finite and non-empty, then $\langle \mathscr{Y}, \mathsf{f} \rangle > 0$ for some test function supported in any given interval, and J_{Z1}^{M} is constant on no interval $(n, n+1)$ (Proposition 5.10);

(50) $2\int_0^\infty u^{-1} e^{-u/2} \sum_{k \le K} (\cos \gamma_k u - \cos \tau_k u) du \to \log L + \log \varpi_{Z1} \ge \log L$, with equality exactly under RH (Theorem 5.11);

(51) $\log \varpi_\lambda$ satisfies the chord and tangent inequalities (15.27) under RH, and if RH fails it vanishes at exactly one λ^* with $f(0)/\Upsilon(0) \le \lambda^* \le f(0)/(f(0)+|f(1)|)$; the family of Proposition 15.15 has all the listed properties of the λ-family and $\varpi_0 > 1$ (Propositions 15.13, 15.15).

Not proved. The identity $\zeta_{Z1} = \zeta$ is not proved; neither is the Riemann hypothesis, to which it is equivalent. The statements proved to be equivalent to it are listed in Subsection 1.6 and above; in the form of Corollary 16.4 it is the existence of one list in $\mathscr{O}$ whose continuous and atomized zeta functions coincide, and in the form of Corollary 15.40 it is the vanishing on the critical line of the imaginary part of the joining factor $\tilde{h}_0$, with the bound (15.69). The theorems of Subsections 15.10–15.14 and of Sections 16 and 17 are formulations of this step and tools for it; none of them evaluates the pairing (15.61) of the count difference $N_\zeta - n_{\mathrm{RVM}}$, on which every one of them depends at $\lambda = 0$.

The computations of Section 19 with 2 001 052 zeros illustrate only the specific numerical statements explicitly tested there; they do not provide evidence for the unproved equivalent forms of (D).

Remark 20.1 (Statements derived in part or conjectural, not proved here). The following statements arose in the work on the final step; none of them is used in this paper.

(a) *Rates in λ.* Off the powers of 2 the formulas of Proposition 15.27 suggest

$$\tilde{h}_\lambda(s) - \tilde{h}_0(s) = O_s(\lambda), \qquad \big|\mathfrak{h}_n(\lambda; s) - \mathfrak{h}_n(0; s)\big| \ll_{s,\varepsilon} \lambda\, n^{-\sigma - \frac{1}{2} + \varepsilon}; \tag{20.1}$$

the cells $n = 2^j$ and the uniformity in n of the first-cell term are not written.

(b) *Flow form of the density condition.* $\tilde{h}_0(1) = 0$ if and only if $\int_0^1 \partial_\lambda \tilde{h}_\lambda(1)\, d\lambda = \tilde{h}_1(1)$; the differentiability at $s = 1$ uniformly in λ is not written.

(c) *Pointwise bound.* $|\tilde{c}_n(\lambda)| \le C_\varepsilon n^\varepsilon$ uniformly for λ in a neighbourhood of 0 is open; by Corollary 15.29 it is not needed for the final step.

(d) *Leading term of the clock defect.* The form $\mathfrak{h}_n(1; s) \approx \sqrt{2}\,\pi\, \mathsf{M}_2\, s\, n^{-s-7/4} \sin(4\pi\sqrt{n} + \frac{\pi}{4})$ is not proved; the bound (15.42) is.

(e) *A heuristic model of the atomized clock.* By (15.41), $\tilde{\zeta}_{\mathbb{Z},1}(s)$ is close to $\prod_{n \ge 2}(1 - n^{-s})^{-1/\log n}$, whose logarithm has the leading part $\sum_n n^{-s}/\log n = \int_s^\infty (\zeta(y) - 1)\, dy$; no error term is claimed.

(f) The comparison between the two columns of Subsection 14.7 and the use of the classification of [48] remain as discussed in the open-problem paragraph following Proposition 16.9 and are not proved.

Remark 20.2 (Open problems). (i) Prove the identity $\zeta_{Z1} = \zeta$. Each of the following is proved to be equivalent to it, unconditionally and within the class of objects stated: (a) condition (D), an absolutely convergent Dirichlet series for ζ_{Z1} in $\Re s > 1$ (Theorem 12.2); (b) an absolutely convergent Dirichlet expansion of $\mathcal{K} = \zeta_{Z1}/\zeta_{\mathbb{Z},\mathrm{RVM}}$ in $\Re s > 1$; (c) the ordered product identity (13.12) (Corollary 13.3); (d) the cell identity (13.19) together with the jump bound of Theorem 13.7; (e) step A7 of the programme in Subsection 14.8; (f) a representation of ζ_{Z1} as the Mellin–Stieltjes transform (15.11) of a positive measure on $[1, \infty)$ (Corollary 15.8); (g) the existence of a list in $\mathscr{O}$ whose joining factor is constant, equivalently $\min_{\mathscr{O}} \mathsf{V}_\sigma = 0$ (Corollary 16.4, step A19). Conditions that are sufficient but not shown to be equivalent, such as the hypotheses of Theorem 15.3 for $0 < \lambda < 1$, are those of item (iii). This is the final step of the method.

(ii) Hypothesis 7.7 is proved for the integers n that are not of the form 2^j with $\mu(j) \neq 0$ (Corollary 15.19); at $n = 2^j$ the cell mass is $(1 + \mu(j) J_{\mathrm{RVM}}(2) \log 2)/\log n + O(n^{-1/4})$. Determine the sign of $1 \pm J_{\mathrm{RVM}}(2) \log 2$.

(iii) Decide, for $0 < \lambda < 1$ outside the finite set E, which hypothesis of Theorem 15.3 fails, and which of (B1), (B2) fails in Theorem 15.39; decide whether the vanishing of all admissible coordinate derivatives (with the corresponding one-sided/KKT conditions at active constraints) forces $\mathsf{V}_\sigma = 0$ (Proposition 16.9); and decide whether the atomized coefficients $\tilde{a}_N(0)$ control the sign of $\mathsf{W}(0)$ in Proposition 15.10.

(iv) Determine the monotone branches of π_{RVM}, on which the real prime levels p_j^{RVM} are defined, and explain the observed proximity $p_j^{\mathrm{RVM}} \approx p_j$ for small j (Experiment 19.9).

(v) Derive the bound (B2) for $\tilde{h}_\lambda$ from the bounds of the cell defects, and turn Heuristic 15.43 into a lemma with remainder; decide (B1) at $\lambda = 0$.

(vi) Evaluate from the construction the right side of (15.61), or the moments $\mathsf{m}_j(\mathsf{f})$ of (15.84); on the first rung the moments $j = 1, 2$ carry the transport up to $4 \cdot 10^{-6}$ (Experiment 19.14).

(vii) Theorem 15.47 gives the finite-height lower frame bound on the natural scale $U \asymp \log T$, the failure of fixed-window uniform invertibility, dense range on every fixed window, and rapidly summable smooth tails. The exact-atomic interpolation tested later has no established interior path. A remaining numerical existence programme is instead to control bounded endpoint finite sections on an exhausting nested family of tests and pass to a distributional limit; proving the required residual-to-zero and compactness statement would already be an existence theorem for the final rung system.

(viii) Under (D), decide whether $\mathsf{E} = \{0\}$, that is, whether the functions D_n of (15.59) have a common zero in $(0, 1]$.

(ix) Construct an object of infinite rank from which ζ_{Z1} and $\tilde{\zeta}_{\mathbb{Z},0}$ both descend, as the matrix U of Proposition 17.2 does for the two-edge model (Subsection 17.4).

(x) Prove the window identity of Theorem 5.5(ii), or the cell $(2, 3)$ together with the sine identity (5.19), directly from $dN_\zeta = d\mathcal{N}_{\mathrm{RVM}} + d\mathfrak{e}$ and the zero-free representation (5.11), without a contour shift that brings back the terms of the zeros off the line (Remark 5.6, Remark 5.9).

(xi) Decide whether the triangular centroid of $e^{i\gamma_k}$ at $u = 1$ converges without RH, and whether its value $e^{-1/2} V_+(1)$, which RH implies (Theorem 5.15), implies anything about the zeros.

(xii) Find a kernel for which the Frullani form of Theorem 5.11 contracts with ratio < 1 (Remark 5.12).

Appendix A. Tables of the first 1000 values

The tables below list the first 1000 values of the four comparisons of Experiment 19.13; they are produced by program `s15` (Appendix B.16), which also writes them as CSV files in the folder `output` (`s15_clock_1000.csv`, `s15_rvm_primes_1000.csv`, `s15_pi_1000.csv`, `s15_mertens_1000.csv`). The ordinates γ_k are those of the file `zeross.bin` and the midpoint ordinates τ_k those of `zLamb.bin`,

which agree with the Lambert formula (2.5) to $5 \cdot 10^{-13}$ on $k \le 1000$. Every difference column is computed from the unrounded values and rounded on its own. After writing each table, `s15` reads it back and checks that every index $1, \dots, 1000$ occurs exactly once (no missing or repeated rows) and that each displayed difference agrees with the difference of the displayed values up to the rounding of the displayed digits (console output in Appendix C.15).

Table 13: The ordinates γ_k of the zeros of ζ, the midpoint ordinates τ_k of the clock and the displacements $\eta_k = \tau_k - \gamma_k$, $k \le 1000$. The displacements are computed from the full-precision ordinates and rounded independently, so they may differ in the last digit from the difference of the displayed columns.

k	γ_k	τ_k	η_k	k	γ_k	τ_k	η_k
1	14.134725	14.521347	+0.3866	501	812.771108	812.740087	−0.0310
2	21.022040	20.655740	−0.3663	502	814.045914	814.032038	−0.0139
3	25.010858	25.492675	+0.4818	503	814.870540	815.323569	+0.4530
4	30.424876	29.739412	−0.6855	504	816.727738	816.614679	−0.1131
5	32.935062	33.624532	+0.6895	505	818.380669	817.905369	−0.4753
6	37.586178	37.257370	−0.3288	506	819.204642	819.195642	−0.0090
7	40.918719	40.700552	−0.2182	507	820.721898	820.485497	−0.2364
8	43.327073	43.994014	+0.6669	508	821.713454	821.774936	+0.0615
9	48.005151	47.165128	−0.8400	509	822.197757	823.063960	+0.8662
10	49.773832	50.233653	+0.4598	510	824.526294	824.352571	−0.1737
11	52.970321	53.214423	+0.2441	511	826.039287	825.640768	−0.3985
12	56.446248	56.118922	−0.3273	512	826.905811	826.928553	+0.0227
13	59.347044	58.956266	−0.3908	513	828.340174	828.215928	−0.1242
14	60.831779	61.733839	+0.9021	514	829.437011	829.502893	+0.0659
15	65.112544	64.457725	−0.6548	515	830.895884	830.789449	−0.1064
16	67.079811	67.133010	+0.0532	516	831.799778	832.075597	+0.2758
17	69.546402	69.764002	+0.2176	517	833.003641	833.361339	+0.3577
18	72.067158	72.354386	+0.2872	518	834.651915	834.646674	−0.0052
19	75.704691	74.907346	−0.7973	519	836.693576	835.931605	−0.7620
20	77.144840	77.425658	+0.2808	520	837.347335	837.216133	−0.1312
21	79.337375	79.911757	+0.5744	521	838.249022	838.500257	+0.2512
22	82.910381	82.367798	−0.5426	522	839.465395	839.783980	+0.3186
23	84.735493	84.795695	+0.0602	523	841.036390	841.067302	+0.0309
24	87.425275	87.197160	−0.2281	524	842.041354	842.350225	+0.3089
25	88.809111	89.573734	+0.7646	525	844.166197	843.632748	−0.5334
26	92.491899	91.926805	−0.5651	526	844.805994	844.914874	+0.1089
27	94.651344	94.257633	−0.3937	527	846.194770	846.196603	+0.0018
28	95.870634	96.567366	+0.6967	528	847.971718	847.477936	−0.4938
29	98.831194	98.857051	+0.0259	529	848.489281	848.758874	+0.2696
30	101.317851	101.127648	−0.1902	530	849.862274	850.039419	+0.1771
31	103.725538	103.380040	−0.3455	531	850.645448	851.319570	+0.6741
32	105.446623	105.615041	+0.1684	532	853.163113	852.599329	−0.5638
33	107.168611	107.833404	+0.6648	533	854.095512	853.878697	−0.2168
34	111.029536	110.035825	−0.9937	534	855.286710	855.157676	−0.1290
35	111.874659	112.222953	+0.3483	535	856.484117	856.436264	−0.0479
36	114.320221	114.395391	+0.0752	536	857.310741	857.714465	+0.4037
37	116.226680	116.553701	+0.3270	537	858.904026	858.992278	+0.0883
38	118.790783	118.698412	−0.0924	538	860.410671	860.269705	−0.1410
39	121.370125	120.830014	−0.5401	539	861.171098	861.546747	+0.3756
40	122.946829	122.948971	+0.0021	540	863.189720	862.823404	−0.3663
41	124.256819	125.055718	+0.7989	541	864.340824	864.099677	−0.2411
42	127.516684	127.150664	−0.3660	542	865.594664	865.375567	−0.2191
43	129.578704	129.234193	−0.3445	543	866.423740	866.651076	+0.2273
44	131.087689	131.306672	+0.2190	544	867.693123	867.926204	+0.2331

Table 13 (continued)

k	γ_k	τ_k	η_k	k	γ_k	τ_k	η_k
45	133.497737	133.368444	−0.1293	545	868.670494	869.200952	+0.5305
46	134.756510	135.419835	+0.6633	546	870.846902	870.475320	−0.3716
47	138.116042	137.461154	−0.6549	547	872.188751	871.749311	−0.4394
48	139.736209	139.492695	−0.2435	548	873.098979	873.022924	−0.0761
49	141.123707	141.514736	+0.3910	549	873.908389	874.296160	+0.3878
50	143.111846	143.527542	+0.4157	550	875.985285	875.569021	−0.4163
51	146.000982	145.531366	−0.4696	551	876.600826	876.841508	+0.2407
52	147.422765	147.526447	+0.1037	552	877.654698	878.113620	+0.4589
53	150.053520	149.513016	−0.5405	553	879.380952	879.385360	+0.0044
54	150.925258	151.491290	+0.5660	554	880.834649	880.656727	−0.1779
55	153.024694	153.461481	+0.4368	555	882.386697	881.927723	−0.4590
56	156.112909	155.423787	−0.6891	556	883.430332	883.198349	−0.2320
57	157.597592	157.378401	−0.2192	557	884.198743	884.468606	+0.2699
58	158.849988	159.325507	+0.4755	558	885.272304	885.738493	+0.4662
59	161.188964	161.265279	+0.0763	559	886.852802	887.008013	+0.1552
60	163.030710	163.197888	+0.1672	560	888.475567	888.277166	−0.1984
61	165.537069	165.123495	−0.4136	561	889.735294	889.545953	−0.1893
62	167.184440	167.042257	−0.1422	562	890.813132	890.814375	+0.0012
63	169.094515	168.954323	−0.1402	563	892.386433	892.082432	−0.3040
64	169.911976	170.859837	+0.9479	564	893.119118	893.350125	+0.2310
65	173.411537	172.758939	−0.6526	565	894.886292	894.617456	−0.2688
66	174.754192	174.651762	−0.1024	566	895.397920	895.884424	+0.4865
67	176.441434	176.538435	+0.0970	567	896.632252	897.151032	+0.5188
68	178.377408	178.419083	+0.0417	568	899.221523	898.417279	−0.8042
69	179.916484	180.293825	+0.3773	569	899.858885	899.683167	−0.1757
70	182.207078	182.162777	−0.0443	570	900.849740	900.948696	+0.0990
71	184.874468	184.026051	−0.8484	571	902.243208	902.213867	−0.0293
72	185.598784	185.883756	+0.2850	572	903.099674	903.478681	+0.3790
73	187.228923	187.735995	+0.5071	573	904.702903	904.743139	+0.0402
74	189.416159	189.582870	+0.1667	574	905.829941	906.007241	+0.1773
75	192.026656	191.424480	−0.6022	575	907.656729	907.270989	−0.3857
76	193.079727	193.260918	+0.1812	576	908.333544	908.534382	+0.2008
77	195.265397	195.092276	−0.1731	577	910.186334	909.797423	−0.3889
78	196.876482	196.918645	+0.0422	578	911.234951	911.060111	−0.1748
79	198.015310	198.740109	+0.7248	579	912.331046	912.322448	−0.0086
80	201.264752	200.556753	−0.7080	580	912.823999	913.584434	+0.7604
81	202.493595	202.368658	−0.1249	581	914.730097	914.846070	+0.1160
82	204.189672	204.175902	−0.0138	582	916.355001	916.107357	−0.2476
83	205.394697	205.978563	+0.5839	583	917.825378	917.368295	−0.4571
84	207.906259	207.776714	−0.1295	584	918.836535	918.628885	−0.2076
85	209.576510	209.570428	−0.0061	585	919.448344	919.889129	+0.4408
86	211.690863	211.359776	−0.3311	586	921.156396	921.149027	−0.0074
87	213.347919	213.144824	−0.2031	587	922.500629	922.408579	−0.0921
88	214.547045	214.925641	+0.3786	588	923.285720	923.667786	+0.3821
89	216.169539	216.702290	+0.5328	589	924.773484	924.926650	+0.1532
90	219.067596	218.474835	−0.5928	590	926.551553	926.185170	−0.3664
91	220.714919	220.243336	−0.4716	591	927.850859	927.443348	−0.4075
92	221.430706	222.007854	+0.5771	592	928.663659	928.701185	+0.0375
93	224.007000	223.768447	−0.2386	593	929.874093	929.958680	+0.0846
94	224.983325	225.525170	+0.5418	594	931.009211	931.215836	+0.2066
95	227.421444	227.278080	−0.1434	595	931.852741	932.472652	+0.6199
96	229.337413	229.027231	−0.3102	596	934.385307	933.729129	−0.6562
97	231.250189	230.772674	−0.4775	597	934.995425	934.985268	−0.0102
98	231.987235	232.514461	+0.5272	598	936.228649	936.241070	+0.0124
99	233.693404	234.252643	+0.5592	599	937.532926	937.496535	−0.0364
100	236.524230	235.987267	−0.5370	600	939.024301	938.751665	−0.2726

Table 13 (continued)

k	γ_k	τ_k	η_k	k	γ_k	τ_k	η_k
101	237.769820	237.718382	−0.0514	601	939.660941	940.006460	+0.3455
102	239.555478	239.446035	−0.1094	602	941.157000	941.260920	+0.1039
103	241.049158	241.170269	+0.1211	603	942.052342	942.515046	+0.4627
104	242.823272	242.891131	+0.0679	604	944.188036	943.768840	−0.4192
105	244.070898	244.608663	+0.5378	605	945.333563	945.022301	−0.3113
106	247.136990	246.322908	−0.8141	606	946.765842	946.275431	−0.4904
107	248.101990	248.033908	−0.0681	607	947.079183	947.528230	+0.4490
108	249.573690	249.741702	+0.1680	608	948.346646	948.780699	+0.4341
109	251.014948	251.446331	+0.4314	609	950.151613	950.032838	−0.1188
110	253.069987	253.147833	+0.0778	610	951.033249	951.284648	+0.2514
111	255.306256	254.846246	−0.4600	611	952.727989	952.536131	−0.1919
112	256.380714	256.541608	+0.1609	612	954.129719	953.787286	−0.3424
113	258.610439	258.233956	−0.3765	613	954.829309	955.038115	+0.2088
114	259.874407	259.923323	+0.0489	614	956.675479	956.288617	−0.3869
115	260.805085	261.609746	+0.8047	615	957.510053	957.538794	+0.0287
116	263.573894	263.293259	−0.2806	616	958.414593	958.788647	+0.3741
117	265.557852	264.973895	−0.5840	617	959.459169	960.038175	+0.5790
118	266.614974	266.651687	+0.0367	618	961.669572	961.287380	−0.3822
119	267.921915	268.326667	+0.4048	619	963.182087	962.536263	−0.6458
120	269.970449	269.998866	+0.0284	620	963.567040	963.784824	+0.2178
121	271.494056	271.668316	+0.1743	621	965.055580	965.033063	−0.0225
122	273.459609	273.335047	−0.1246	622	966.110755	966.280982	+0.1702
123	275.587493	274.999088	−0.5884	623	967.371154	967.528580	+0.1574
124	276.452050	276.660469	+0.2084	624	968.636302	968.775860	+0.1396
125	278.250744	278.319218	+0.0685	625	970.125611	970.022820	−0.1028
126	279.229251	279.975364	+0.7461	626	971.071491	971.269463	+0.1980
127	282.465115	281.628933	−0.8362	627	973.185361	972.515789	−0.6696
128	283.211186	283.279953	+0.0688	628	973.873079	973.761797	−0.1113
129	284.835964	284.928450	+0.0925	629	974.774635	975.007490	+0.2329
130	286.667445	286.574450	−0.0930	630	976.178502	976.252867	+0.0744
131	287.911921	288.217980	+0.3061	631	976.917202	977.497930	+0.5807
132	289.579855	289.859063	+0.2792	632	978.766672	978.742678	−0.0240
133	291.846291	291.497725	−0.3486	633	980.578001	979.987113	−0.5909
134	293.558434	293.133989	−0.4244	634	981.288615	981.231235	−0.0574
135	294.965370	294.767881	−0.1975	635	982.396485	982.475045	+0.0786
136	295.573255	296.399422	+0.8262	636	983.575076	983.718543	+0.1435
137	297.979277	298.028636	+0.0494	637	985.186929	984.961730	−0.2252
138	299.840326	299.655546	−0.1848	638	986.130515	986.204607	+0.0741
139	301.649325	301.280173	−0.3692	639	986.756008	987.447174	+0.6912
140	302.696750	302.902540	+0.2058	640	988.992622	988.689432	−0.3032
141	304.864371	304.522668	−0.3417	641	990.223918	989.931381	−0.2925
142	305.728913	306.140577	+0.4117	642	991.374294	991.173023	−0.2013
143	307.219496	307.756290	+0.5368	643	992.728696	992.414357	−0.3143
144	310.109463	309.369826	−0.7396	644	993.214581	993.655385	+0.4408
145	311.165142	310.981204	−0.1839	645	994.404591	994.896107	+0.4915
146	312.427801	312.590446	+0.1626	646	996.205336	996.136523	−0.0688
147	313.985286	314.197570	+0.2123	647	997.511935	997.376635	−0.1353
148	315.475616	315.802596	+0.3270	648	998.827547	998.616442	−0.2111
149	317.734806	317.405542	−0.3293	649	999.791572	999.855946	+0.0644
150	318.853104	319.006426	+0.1533	650	1001.349483	1001.095146	−0.2543
151	321.160134	320.605268	−0.5549	651	1002.404305	1002.334044	−0.0703
152	322.144559	322.202085	+0.0575	652	1003.267808	1003.572641	+0.3048
153	323.466970	323.796894	+0.3299	653	1004.675044	1004.810936	+0.1359
154	324.862866	325.389713	+0.5268	654	1005.543420	1006.048930	+0.5055
155	327.443901	326.980559	−0.4633	655	1008.006704	1007.286625	−0.7201
156	329.033072	328.569449	−0.4636	656	1008.795710	1008.524020	−0.2717

Table 13 (continued)

k	γ_k	τ_k	η_k	k	γ_k	τ_k	η_k
157	329.953240	330.156400	+0.2032	657	1009.806591	1009.761116	−0.0455
158	331.474468	331.741427	+0.2670	658	1010.569757	1010.997914	+0.4282
159	333.645379	333.324548	−0.3208	659	1012.410043	1012.234414	−0.1756
160	334.211355	334.905777	+0.6944	660	1013.058638	1013.470617	+0.4120
161	336.841850	336.485131	−0.3567	661	1014.689633	1014.706523	+0.0169
162	338.339993	338.062625	−0.2774	662	1016.060179	1015.942133	−0.1180
163	339.858217	339.638274	−0.2199	663	1017.266402	1017.177448	−0.0890
164	341.042261	341.212093	+0.1698	664	1018.605573	1018.412469	−0.1931
165	342.054878	342.784097	+0.7292	665	1019.912440	1019.647194	−0.2652
166	344.661703	344.354300	−0.3074	666	1020.917475	1020.881626	−0.0358
167	346.347871	345.922717	−0.4252	667	1021.544344	1022.115765	+0.5714
168	347.272678	347.489362	+0.2167	668	1022.885271	1023.349612	+0.4643
169	349.316261	349.054249	−0.2620	669	1025.265724	1024.583166	−0.6826
170	350.408419	350.617391	+0.2090	670	1025.707944	1025.816429	+0.1085
171	351.878649	352.178803	+0.3002	671	1027.467694	1027.049401	−0.4183
172	353.488900	353.738497	+0.2496	672	1028.128964	1028.282083	+0.1531
173	356.017575	355.296486	−0.7211	673	1029.227297	1029.514474	+0.2872
174	357.151302	356.852785	−0.2985	674	1030.897369	1030.746577	−0.1508
175	357.952685	358.407405	+0.4547	675	1031.833180	1031.978391	+0.1452
176	359.743755	359.960359	+0.2166	676	1032.812883	1033.209916	+0.3970
177	361.289362	361.511660	+0.2223	677	1034.612916	1034.441154	−0.1718
178	363.331331	363.061320	−0.2700	678	1036.195917	1035.672105	−0.5238
179	364.736024	364.609351	−0.1267	679	1037.024708	1036.902770	−0.1219
180	366.212710	366.155765	−0.0569	680	1038.087752	1038.133148	+0.0454
181	367.993575	367.700575	−0.2930	681	1039.077401	1039.363241	+0.2858
182	368.968438	369.243791	+0.2754	682	1040.264038	1040.593049	+0.3290
183	370.050919	370.785426	+0.7345	683	1041.621528	1041.822573	+0.2010
184	373.061928	372.325490	−0.7364	684	1043.623954	1043.051813	−0.5721
185	373.864874	373.863995	−0.0009	685	1044.514976	1044.280769	−0.2342
186	375.825913	375.400952	−0.4250	686	1045.107042	1045.509443	+0.4024
187	376.324092	376.936373	+0.6123	687	1047.089817	1046.737835	−0.3520
188	378.436680	378.470267	+0.0336	688	1047.987147	1047.965945	−0.0212
189	379.872975	380.002645	+0.1297	689	1048.953785	1049.193773	+0.2400
190	381.484469	381.533519	+0.0491	690	1049.996284	1050.421321	+0.4250
191	383.443529	383.062899	−0.3806	691	1051.576572	1051.648589	+0.0720
192	384.956117	384.590794	−0.3653	692	1053.245785	1052.875577	−0.3702
193	385.861301	386.117216	+0.2559	693	1054.781039	1054.102286	−0.6788
194	387.222890	387.642173	+0.4193	694	1055.002146	1055.328716	+0.3266
195	388.846128	389.165677	+0.3195	695	1056.688847	1056.554868	−0.1340
196	391.456084	390.687737	−0.7683	696	1057.100044	1057.780743	+0.6807
197	392.245083	392.208362	−0.0367	697	1059.133769	1059.006340	−0.1274
198	393.427744	393.727562	+0.2998	698	1060.139519	1060.231661	+0.0921
199	395.582870	395.245347	−0.3375	699	1061.501304	1061.456706	−0.0446
200	396.381854	396.761726	+0.3799	700	1062.915382	1062.681475	−0.2339
201	397.918736	398.276708	+0.3580	701	1064.071551	1063.905969	−0.1656
202	399.985120	399.790303	−0.1948	702	1065.121855	1065.130189	+0.0083
203	401.839229	401.302519	−0.5367	703	1066.463223	1066.354134	−0.1091
204	402.861918	402.813366	−0.0486	704	1067.418860	1067.577806	+0.1589
205	404.236442	404.322851	+0.0864	705	1067.990000	1068.801205	+0.8112
206	405.134387	405.830985	+0.6966	706	1070.535042	1070.024332	−0.5107
207	407.581460	407.337775	−0.2437	707	1071.618623	1071.247186	−0.3714
208	408.947246	408.843230	−0.1040	708	1072.543998	1072.469769	−0.0742
209	410.513869	410.347359	−0.1665	709	1073.570353	1073.692080	+0.1217
210	411.972268	411.850170	−0.1221	710	1074.747771	1074.914121	+0.1664
211	413.262736	413.351671	+0.0889	711	1076.266626	1076.135892	−0.1307
212	415.018810	414.851870	−0.1669	712	1076.924056	1077.357394	+0.4333

Table 13 (continued)

k	γ_k	τ_k	η_k	k	γ_k	τ_k	η_k
213	415.455215	416.350775	+0.8956	713	1078.647198	1078.578626	−0.0686
214	418.387706	417.848395	−0.5393	714	1079.809965	1079.799589	−0.0104
215	419.861365	419.344737	−0.5166	715	1081.171002	1081.020284	−0.1507
216	420.643828	420.839809	+0.1960	716	1082.952750	1082.240712	−0.7120
217	422.076710	422.333619	+0.2569	717	1083.295467	1083.460872	+0.1654
218	423.716580	423.826175	+0.1096	718	1084.183264	1084.680766	+0.4975
219	425.069882	425.317483	+0.2476	719	1085.647831	1085.900393	+0.2526
220	427.208825	426.807552	−0.4013	720	1086.911999	1087.119754	+0.2078
221	428.127914	428.296388	+0.1685	721	1088.755725	1088.338850	−0.4169
222	430.328745	429.784000	−0.5447	722	1089.795338	1089.557682	−0.2377
223	431.301307	431.270394	−0.0309	723	1090.863191	1090.776248	−0.0869
224	432.138642	432.755577	+0.6169	724	1091.728473	1091.994551	+0.2661
225	433.889218	434.239558	+0.3503	725	1093.440873	1093.212591	−0.2283
226	436.161006	435.722342	−0.4387	726	1094.284458	1094.430367	+0.1459
227	437.581698	437.203936	−0.3778	727	1095.433085	1095.647881	+0.2148
228	438.621739	438.684348	+0.0626	728	1096.401918	1096.865133	+0.4632
229	439.918442	440.163585	+0.2451	729	1098.841015	1098.082123	−0.7589
230	441.683199	441.641652	−0.0415	730	1099.360667	1099.298852	−0.0618
231	442.904546	443.118557	+0.2140	731	1100.574461	1100.515320	−0.0591
232	444.319336	444.594306	+0.2750	732	1101.839111	1101.731529	−0.1076
233	446.860623	446.068907	−0.7917	733	1102.551780	1102.947477	+0.3957
234	447.441704	447.542364	+0.1007	734	1103.732297	1104.163166	+0.4309
235	449.148546	449.014685	−0.1339	735	1105.617189	1105.378596	−0.2386
236	450.126946	450.485876	+0.3589	736	1106.774372	1106.593768	−0.1806
237	451.403308	451.955944	+0.5526	737	1107.774532	1107.808681	+0.0341
238	453.986738	453.424894	−0.5618	738	1109.158919	1109.023337	−0.1356
239	454.974684	454.892733	−0.0820	739	1110.444143	1110.237736	−0.2064
240	456.328427	456.359466	+0.0310	740	1111.443505	1111.451879	+0.0084
241	457.903893	457.825100	−0.0788	741	1112.432995	1112.665765	+0.2328
242	459.513415	459.289641	−0.2238	742	1113.397595	1113.879395	+0.4818
243	460.087944	460.753095	+0.6652	743	1115.065359	1115.092770	+0.0274
244	462.065367	462.215467	+0.1501	744	1116.787254	1116.305890	−0.4814
245	464.057287	463.676764	−0.3805	745	1117.965920	1117.518756	−0.4472
246	465.671539	465.136990	−0.5345	746	1118.684135	1118.731367	+0.0472
247	466.570287	466.596153	+0.0259	747	1119.473247	1119.943726	+0.4705
248	467.439046	468.054257	+0.6152	748	1121.155938	1121.155830	−0.0001
249	469.536005	469.511307	−0.0247	749	1122.458621	1122.367683	−0.0909
250	470.773655	470.967311	+0.1937	750	1123.101117	1123.579282	+0.4782
251	472.799175	472.422272	−0.3769	751	1125.314729	1124.790631	−0.5241
252	473.835232	473.876197	+0.0410	752	1125.763442	1126.001727	+0.2383
253	475.600339	475.329091	−0.2712	753	1127.658024	1127.212573	−0.4455
254	476.769015	476.780959	+0.0119	754	1128.430225	1128.423168	−0.0071
255	478.075264	478.231807	+0.1565	755	1129.728997	1129.633513	−0.0955
256	478.942182	479.681639	+0.7395	756	1130.391598	1130.843608	+0.4520
257	481.830339	481.130461	−0.6999	757	1131.495086	1132.053454	+0.5584
258	482.834783	482.578278	−0.2565	758	1133.708626	1133.263051	−0.4456
259	483.851427	484.025095	+0.1737	759	1134.885655	1134.472399	−0.4133
260	485.539148	485.470918	−0.0682	760	1135.562214	1135.681499	+0.1193
261	486.528718	486.915750	+0.3870	761	1136.929293	1136.890352	−0.0389
262	488.380567	488.359598	−0.0210	762	1138.151590	1138.098958	−0.0526
263	489.661762	489.802466	+0.1407	763	1138.992342	1139.307316	+0.3150
264	491.398822	491.244359	−0.1545	764	1140.721848	1140.515429	−0.2064
265	493.314442	492.685281	−0.6292	765	1141.261023	1141.723295	+0.4623
266	493.957998	494.125238	+0.1672	766	1142.858660	1142.930916	+0.0723
267	495.358829	495.564234	+0.2054	767	1144.782300	1144.138292	−0.6440
268	496.429696	497.002275	+0.5726	768	1145.485328	1145.345422	−0.1399

Table 13 (continued)

k	γ_k	τ_k	η_k	k	γ_k	τ_k	η_k
269	498.580782	498.439363	−0.1414	769	1146.576815	1146.552309	−0.0245
270	500.309085	499.875506	−0.4336	770	1147.501777	1147.758951	+0.2572
271	501.604447	501.310706	−0.2937	771	1148.615277	1148.965350	+0.3501
272	502.276270	502.744968	+0.4687	772	1149.982601	1150.171506	+0.1889
273	504.499773	504.178297	−0.3215	773	1151.562815	1151.377419	−0.1854
274	505.415232	505.610698	+0.1955	774	1152.943129	1152.583090	−0.3600
275	506.464153	507.042174	+0.5780	775	1153.890304	1153.788519	−0.1018
276	508.800700	508.472731	−0.3280	776	1154.697520	1154.993706	+0.2962
277	510.264228	509.902372	−0.3619	777	1156.621568	1156.198653	−0.4229
278	511.562290	511.331102	−0.2312	778	1157.432315	1157.403358	−0.0290
279	512.623145	512.758925	+0.1358	779	1158.001609	1158.607823	+0.6062
280	513.668986	514.185845	+0.5169	780	1159.480657	1159.812048	+0.3314
281	515.435057	515.611867	+0.1768	781	1161.396645	1161.016034	−0.3806
282	517.589669	517.036995	−0.5527	782	1162.487529	1162.219780	−0.2677
283	518.234223	518.461232	+0.2270	783	1163.701032	1163.423288	−0.2777
284	520.106310	519.884584	−0.2217	784	1164.737586	1164.626557	−0.1110
285	521.525193	521.307053	−0.2181	785	1165.271228	1165.829588	+0.5584
286	522.456696	522.728645	+0.2719	786	1166.943613	1167.032382	+0.0888
287	523.960531	524.149363	+0.1888	787	1168.086272	1168.234938	+0.1487
288	525.077386	525.569210	+0.4918	788	1169.698357	1169.437258	−0.2611
289	527.903642	526.988192	−0.9154	789	1170.463639	1170.639341	+0.1757
290	528.406214	528.406312	+0.0001	790	1172.120682	1171.841188	−0.2795
291	529.806226	529.823573	+0.0173	791	1173.305688	1173.042799	−0.2629
292	530.866918	531.239980	+0.3731	792	1174.232767	1174.244175	+0.0114
293	532.688183	532.655537	−0.0326	793	1175.215452	1175.445316	+0.2299
294	533.779631	534.070246	+0.2906	794	1176.632876	1176.646222	+0.0133
295	535.664314	535.484113	−0.1802	795	1177.106304	1177.846894	+0.7406
296	537.069759	536.897141	−0.1726	796	1179.701224	1179.047333	−0.6539
297	538.428526	538.309332	−0.1192	797	1180.653544	1180.247537	−0.4060
298	540.213166	539.720692	−0.4925	798	1181.267318	1181.447509	+0.1802
299	540.631390	541.131224	+0.4998	799	1182.582270	1182.647248	+0.0650
300	541.847437	542.540931	+0.6935	800	1183.712775	1183.846755	+0.1340
301	544.323890	543.949817	−0.3741	801	1185.155843	1185.046030	−0.1098
302	545.636833	545.357885	−0.2789	802	1185.875359	1186.245073	+0.3697
303	547.010912	546.765139	−0.2458	803	1187.345161	1187.443885	+0.0987
304	547.931613	548.171583	+0.2400	804	1188.856444	1188.642467	−0.2140
305	549.497568	549.577220	+0.0797	805	1189.963636	1189.840817	−0.1228
306	550.970010	550.982053	+0.0120	806	1191.482606	1191.038938	−0.4437
307	552.049572	552.386086	+0.3365	807	1192.218611	1192.236829	+0.0182
308	553.764972	553.789322	+0.0243	808	1193.324021	1193.434490	+0.1105
309	555.792021	555.191765	−0.6003	809	1193.857427	1194.631923	+0.7745
310	556.899476	556.593418	−0.3061	810	1196.034672	1195.829126	−0.2055
311	557.564659	557.994283	+0.4296	811	1197.071787	1197.026102	−0.0457
312	559.316237	559.394366	+0.0781	812	1198.686569	1198.222849	−0.4637
313	560.240807	560.793668	+0.5529	813	1199.356514	1199.419369	+0.0629
314	562.559208	562.192194	−0.3670	814	1200.532692	1200.615662	+0.0830
315	564.160879	563.589945	−0.5709	815	1201.810335	1201.811728	+0.0014
316	564.506056	564.986926	+0.4809	816	1203.137351	1203.007567	−0.1298
317	566.698788	566.383140	−0.3156	817	1203.855248	1204.203180	+0.3479
318	567.731758	567.778589	+0.0468	818	1204.985492	1205.398567	+0.4131
319	568.923955	569.173278	+0.2493	819	1206.870500	1206.593729	−0.2768
320	570.051115	570.567208	+0.5161	820	1208.471460	1207.788665	−0.6828
321	572.419984	571.960384	−0.4596	821	1208.989484	1208.983377	−0.0061
322	573.614611	573.352808	−0.2618	822	1209.898030	1210.177865	+0.2798
323	575.093886	574.744483	−0.3494	823	1211.416116	1211.372128	−0.0440
324	575.807247	576.135412	+0.3282	824	1212.113153	1212.566168	+0.4530

Table 13 (continued)

k	γ_k	τ_k	η_k	k	γ_k	τ_k	η_k
325	577.039003	577.525598	+0.4866	825	1213.598373	1213.759984	+0.1616
326	579.098835	578.915045	−0.1838	826	1215.389975	1214.953577	−0.4364
327	580.136959	580.303754	+0.1668	827	1216.183722	1216.146948	−0.0368
328	581.946576	581.691730	−0.2548	828	1217.174482	1217.340096	+0.1656
329	583.236088	583.078975	−0.1571	829	1219.050028	1218.533022	−0.5170
330	584.561706	584.465492	−0.0962	830	1219.614471	1219.725727	+0.1113
331	585.984563	585.851283	−0.1333	831	1220.816348	1220.918210	+0.1019
332	586.742772	587.236352	+0.4936	832	1221.692242	1222.110472	+0.4182
333	588.139663	588.620701	+0.4810	833	1222.952484	1223.302513	+0.3500
334	590.660398	590.004334	−0.6561	834	1225.018330	1224.494335	−0.5240
335	591.725858	591.387252	−0.3386	835	1225.855021	1225.685936	−0.1691
336	592.571358	592.769460	+0.1981	836	1227.231828	1226.877317	−0.3545
337	593.974715	594.150959	+0.1762	837	1227.917142	1228.068480	+0.1513
338	595.728154	595.531752	−0.1964	838	1228.793154	1229.259423	+0.4663
339	596.362768	596.911842	+0.5491	839	1230.584603	1230.450148	−0.1345
340	598.493077	598.291232	−0.2018	840	1231.562274	1231.640654	+0.0784
341	599.545640	599.669924	+0.1243	841	1232.529587	1232.830942	+0.3014
342	601.602137	601.047921	−0.5542	842	1234.277817	1234.021013	−0.2568
343	602.579168	602.425226	−0.1539	843	1235.502549	1235.210867	−0.2917
344	603.625619	603.801841	+0.1762	844	1236.399017	1236.400503	+0.0015
345	604.616218	605.177769	+0.5616	845	1237.977299	1237.589923	−0.3874
346	606.383460	606.553012	+0.1696	846	1238.457233	1238.779126	+0.3219
347	608.413217	607.927574	−0.4856	847	1239.490807	1239.968114	+0.4773
348	609.389575	609.301456	−0.0881	848	1240.813472	1241.156885	+0.3434
349	610.839163	610.674661	−0.1645	849	1243.078076	1242.345442	−0.7326
350	611.774210	612.047192	+0.2730	850	1243.538147	1243.533783	−0.0044
351	613.599779	613.419051	−0.1807	851	1244.851434	1244.721910	−0.1295
352	614.646238	614.790240	+0.1440	852	1245.655866	1245.909822	+0.2540
353	615.538563	616.160763	+0.6222	853	1247.372562	1247.097520	−0.2750
354	618.112831	617.530621	−0.5822	854	1248.063061	1248.285004	+0.2219
355	619.184483	618.899817	−0.2847	855	1249.159888	1249.472275	+0.3124
356	620.272894	620.268353	−0.0045	856	1250.672397	1250.659333	−0.0131
357	621.709295	621.636233	−0.0731	857	1251.659832	1251.846178	+0.1863
358	622.375003	623.003457	+0.6285	858	1253.673578	1253.032811	−0.6408
359	624.269900	624.370029	+0.1001	859	1254.431328	1254.219231	−0.2121
360	626.019283	625.735951	−0.2833	860	1255.408231	1255.405439	−0.0028
361	627.268397	627.101225	−0.1672	861	1256.181214	1256.591436	+0.4102
362	628.325862	628.465854	+0.1400	862	1257.541219	1257.777222	+0.2360
363	630.473887	629.829840	−0.6440	863	1258.779233	1258.962797	+0.1836
364	630.805781	631.193185	+0.3874	864	1260.344548	1260.148161	−0.1964
365	632.225141	632.555891	+0.3308	865	1261.611717	1261.333315	−0.2784
366	633.546858	633.917962	+0.3711	866	1262.556614	1262.518259	−0.0384
367	635.523800	635.279398	−0.2444	867	1263.676733	1263.702993	+0.0263
368	637.397193	636.640202	−0.7570	868	1264.957223	1264.887518	−0.0697
369	637.925514	638.000377	+0.0749	869	1266.179038	1266.071834	−0.1072
370	638.927938	639.359925	+0.4320	870	1267.200346	1267.255941	+0.0556
371	640.694795	640.718847	+0.0241	871	1267.570572	1268.439839	+0.8693
372	641.945500	642.077147	+0.1316	872	1270.118922	1269.623530	−0.4954
373	643.278884	643.434826	+0.1559	873	1271.134300	1270.807012	−0.3273
374	644.990578	644.791886	−0.1987	874	1272.083960	1271.990287	−0.0937
375	646.348192	646.148329	−0.1999	875	1273.261145	1273.173355	−0.0878
376	647.761753	647.504159	−0.2576	876	1274.196221	1274.356216	+0.1600
377	648.786401	648.859376	+0.0730	877	1275.092030	1275.538870	+0.4468
378	650.197519	650.213983	+0.0165	878	1276.842172	1276.721318	−0.1209
379	650.668684	651.567982	+0.8993	879	1277.763092	1277.903560	+0.1405
380	653.649572	652.921375	−0.7282	880	1279.332843	1279.085596	−0.2472

Table 13 (continued)

k	γ_k	τ_k	η_k	k	γ_k	τ_k	η_k
381	654.301921	654.274165	−0.0278	881	1280.155794	1280.267426	+0.1116
382	655.709463	655.626352	−0.0831	882	1281.828727	1281.449052	−0.3797
383	656.964085	656.977940	+0.0139	883	1283.000491	1282.630472	−0.3700
384	658.175614	658.328930	+0.1533	884	1283.335032	1283.811688	+0.4767
385	659.663846	659.679324	+0.0155	885	1284.854795	1284.992700	+0.1379
386	660.716733	661.029125	+0.3124	886	1285.695023	1286.173508	+0.4785
387	662.296586	662.378333	+0.0817	887	1287.410027	1287.354112	−0.0559
388	664.244605	663.726953	−0.5177	888	1289.165352	1288.534513	−0.6308
389	665.342763	665.074984	−0.2678	889	1290.104772	1289.714711	−0.3901
390	666.515148	666.422429	−0.0927	890	1290.417708	1290.894705	+0.4770
391	667.148495	667.769291	+0.6208	891	1291.945871	1292.074498	+0.1286
392	668.975849	669.115570	+0.1397	892	1293.493982	1293.254088	−0.2399
393	670.323585	670.461270	+0.1377	893	1294.118474	1294.433476	+0.3150
394	672.458184	671.806391	−0.6518	894	1295.365364	1295.612663	+0.2473
395	673.043578	673.150937	+0.1074	895	1296.801111	1296.791648	−0.0095
396	674.355898	674.494908	+0.1390	896	1298.256527	1297.970432	−0.2861
397	676.139674	675.838306	−0.3014	897	1299.405171	1299.149015	−0.2562
398	677.230181	677.181134	−0.0490	898	1300.490019	1300.327398	−0.1626
399	677.800445	678.523393	+0.7229	899	1301.495517	1301.505580	+0.0101
400	679.742198	679.865086	+0.1229	900	1302.346742	1302.683563	+0.3368
401	681.894992	681.206213	−0.6888	901	1303.273200	1303.861346	+0.5881
402	682.602735	682.546777	−0.0560	902	1305.401672	1305.038930	−0.3627
403	684.013550	683.886780	−0.1268	903	1306.508393	1306.216314	−0.2921
404	684.972630	685.226223	+0.2536	904	1307.267242	1307.393500	+0.1263
405	686.163224	686.565108	+0.4019	905	1308.988197	1308.570487	−0.4177
406	687.961543	687.903437	−0.0581	906	1309.421532	1309.747276	+0.3257
407	689.368941	689.241212	−0.1277	907	1311.056571	1310.923867	−0.1327
408	690.474735	690.578435	+0.1037	908	1311.966941	1312.100260	+0.1333
409	692.451684	691.915106	−0.5366	909	1313.031599	1313.276455	+0.2449
410	693.176970	693.251229	+0.0743	910	1314.052566	1314.452454	+0.3999
411	694.533909	694.586804	+0.0529	911	1316.212113	1315.628256	−0.5839
412	695.726336	695.921834	+0.1955	912	1317.072986	1316.803861	−0.2691
413	696.626070	697.256320	+0.6303	913	1318.171279	1317.979270	−0.1920
414	699.132095	698.590264	−0.5418	914	1318.947881	1319.154483	+0.2066
415	700.296739	699.923668	−0.3731	915	1319.931083	1320.329500	+0.3984
416	701.301743	701.256532	−0.0452	916	1321.628139	1321.504321	−0.1238
417	702.227343	702.588860	+0.3615	917	1322.258067	1322.678947	+0.4209
418	704.033839	703.920652	−0.1132	918	1324.224979	1323.853379	−0.3716
419	705.125814	705.251911	+0.1261	919	1325.237624	1325.027615	−0.2100
420	706.184655	706.582637	+0.3980	920	1325.981970	1326.201657	+0.2197
421	708.269071	707.912833	−0.3562	921	1327.635281	1327.375505	−0.2598
422	709.229589	709.242500	+0.0129	922	1329.043518	1328.549159	−0.4944
423	711.130274	710.571640	−0.5586	923	1329.205019	1329.722620	+0.5176
424	711.900290	711.900254	−0.0000	924	1330.429937	1330.895887	+0.4659
425	712.749383	713.228344	+0.4790	925	1331.827591	1332.068961	+0.2414
426	714.082772	714.555912	+0.4731	926	1333.673523	1333.241842	−0.4317
427	716.112396	715.882959	−0.2294	927	1334.747329	1334.414531	−0.3328
428	717.482570	717.209487	−0.2731	928	1335.694975	1335.587027	−0.1079
429	718.742787	718.535497	−0.2073	929	1336.690185	1336.759332	+0.0691
430	719.697101	719.860991	+0.1639	930	1337.688792	1337.931444	+0.2427
431	721.351162	721.185970	−0.1652	931	1338.923165	1339.103365	+0.1802
432	722.277505	722.510436	+0.2329	932	1340.426400	1340.275095	−0.1513
433	723.845821	723.834391	−0.0114	933	1341.166272	1341.446634	+0.2804
434	724.562614	725.157836	+0.5952	934	1342.608508	1342.617982	+0.0095
435	727.056403	726.480772	−0.5756	935	1344.156044	1343.789140	−0.3669
436	728.405482	727.803201	−0.6023	936	1345.477106	1344.960107	−0.5170

Table 13 (continued)

k	γ_k	τ_k	η_k	k	γ_k	τ_k	η_k
437	728.758750	729.125125	+0.3664	937	1345.731413	1346.130885	+0.3995
438	730.416482	730.446545	+0.0301	938	1347.519472	1347.301472	−0.2180
439	731.417355	731.767463	+0.3501	939	1348.017238	1348.471871	+0.4546
440	732.818053	733.087879	+0.2698	940	1349.085194	1349.642080	+0.5569
441	734.789643	734.407796	−0.3818	941	1351.296206	1350.812100	−0.4841
442	735.765459	735.727215	−0.0382	942	1352.210465	1351.981932	−0.2285
443	737.052929	737.046137	−0.0068	943	1353.483338	1353.151575	−0.3318
444	738.580421	738.364565	−0.2159	944	1353.886782	1354.321030	+0.4342
445	739.909524	739.682498	−0.2270	945	1355.680595	1355.490297	−0.1903
446	740.573807	740.999939	+0.4261	946	1356.605656	1356.659376	+0.0537
447	741.757336	742.316890	+0.5596	947	1357.771743	1357.828268	+0.0565
448	743.895013	743.633350	−0.2617	948	1358.460160	1358.996973	+0.5368
449	745.344990	744.949323	−0.3957	949	1360.393145	1360.165491	−0.2277
450	746.499306	746.264810	−0.2345	950	1361.393075	1361.333822	−0.0593
451	747.674564	747.579811	−0.0948	951	1363.022329	1362.501967	−0.5204
452	748.242754	748.894328	+0.6516	952	1363.879191	1363.669926	−0.2093
453	750.655950	750.208362	−0.4476	953	1364.576585	1364.837699	+0.2611
454	750.966381	751.521916	+0.5555	954	1365.493734	1366.005286	+0.5116
455	752.887622	752.834990	−0.0526	955	1367.104091	1367.172688	+0.0686
456	754.322370	754.147586	−0.1748	956	1368.330193	1368.339905	+0.0097
457	755.839309	755.459704	−0.3796	957	1369.686949	1369.506936	−0.1800
458	756.768248	756.771347	+0.0031	958	1370.973523	1370.673783	−0.2997
459	758.101729	758.082516	−0.0192	959	1371.686554	1371.840446	+0.1539
460	758.900238	759.393211	+0.4930	960	1373.202915	1373.006924	−0.1960
461	760.282367	760.703435	+0.4211	961	1374.154799	1374.173219	+0.0184
462	762.700033	762.013189	−0.6868	962	1375.302392	1375.339330	+0.0369
463	763.593066	763.322473	−0.2706	963	1376.161780	1376.505257	+0.3435
464	764.307523	764.631290	+0.3238	964	1377.177634	1377.671001	+0.4934
465	766.087540	765.939641	−0.1479	965	1379.683283	1378.836562	−0.8467
466	767.218472	767.247526	+0.0291	966	1380.148578	1380.001941	−0.1466
467	768.281462	768.554948	+0.2735	967	1381.073977	1381.167137	+0.0932
468	769.693407	769.861907	+0.1685	968	1382.345663	1382.332150	−0.0135
469	771.070839	771.168404	+0.0976	969	1383.297591	1383.496982	+0.1994
470	772.961618	772.474442	−0.4872	970	1384.444416	1384.661632	+0.2172
471	774.117745	773.780021	−0.3377	971	1385.663777	1385.826100	+0.1623
472	775.047847	775.085142	+0.0373	972	1387.326648	1386.990387	−0.3363
473	775.999712	776.389808	+0.3901	973	1387.921454	1388.154493	+0.2330
474	777.299749	777.694018	+0.3943	974	1389.565832	1389.318419	−0.2474
475	779.157077	778.997774	−0.1593	975	1390.705490	1390.482163	−0.2233
476	780.348925	780.301078	−0.0478	976	1391.853200	1391.645727	−0.2075
477	782.137664	781.603931	−0.5337	977	1392.644028	1392.809112	+0.1651
478	782.597944	782.906333	+0.3084	978	1393.433402	1393.972316	+0.5389
479	784.288823	784.208287	−0.0805	979	1394.884185	1395.135340	+0.2512
480	785.739090	785.509793	−0.2293	980	1396.544163	1396.298186	−0.2460
481	786.461147	786.810852	+0.3497	981	1397.834623	1397.460852	−0.3738
482	787.468464	788.111467	+0.6430	982	1398.837675	1398.623339	−0.2143
483	790.059092	789.411637	−0.6475	983	1399.839473	1399.785647	−0.0538
484	790.831620	790.711364	−0.1203	984	1400.426946	1400.947777	+0.5208
485	792.427708	792.010650	−0.4171	985	1402.564347	1402.109728	−0.4546
486	792.888653	793.309495	+0.4208	986	1402.973748	1403.271502	+0.2978
487	794.483792	794.607900	+0.1241	987	1404.006292	1404.433097	+0.4268
488	795.606596	795.905868	+0.2993	988	1405.666975	1405.594516	−0.0725
489	797.263470	797.203398	−0.0601	989	1407.085143	1406.755756	−0.3294
490	798.707570	798.500493	−0.2071	990	1408.136307	1407.916820	−0.2195
491	799.654336	799.797152	+0.1428	991	1409.320681	1409.077707	−0.2430
492	801.604246	801.093379	−0.5109	992	1410.024811	1410.238417	+0.2136

Table 13 (continued)

k	γ_k	τ_k	η_k	k	γ_k	τ_k	η_k
493	802.541985	802.389172	−0.1528	993	1411.257057	1411.398950	+0.1419
494	803.243096	803.684534	+0.4414	994	1411.965653	1412.559308	+0.5937
495	804.762239	804.979467	+0.2172	995	1413.843149	1413.719489	−0.1237
496	805.861636	806.273970	+0.4123	996	1415.585785	1414.879495	−0.7063
497	808.151815	807.568045	−0.5838	997	1415.781581	1416.039325	+0.2577
498	809.197783	808.861693	−0.3361	998	1417.102823	1417.198980	+0.0962
499	810.081805	810.154915	+0.0731	999	1418.696964	1418.358460	−0.3385
500	811.184359	811.447713	+0.2634	1000	1419.422481	1419.517765	+0.0953

Table 14: The primes p_j and the RVM prime levels p_j^{RVM}, $j \le 1000$. Here p_j^{RVM} is a discrete version of $\inf\{x : \pi_{\mathrm{RVM}}(x) \ge j\}$: the first point of the computational grid in $u = \log x$ (step $5.9 \cdot 10^{-6}$) at which $\pi_{\mathrm{RVM}} \ge j$, refined by linear interpolation from the previous grid point. It is not the inverse of a monotone function, and its accuracy is limited by the grid, the truncation of the ordinate sum and the smooth cut. The same levels computed by program `s04` with $K = 17\,061$ ordinates on $[2, 1000]$ (40 000 grid points), instead of $K = 183\,029$ ordinates on $[2, 8200]$, differ from these by a median of $1.7 \cdot 10^{-3}$ over the 168 levels $j \le 168$, by more than 0.01 at 29 of them, and by 0.85, 0.85, 0.79 at $j = 148$, 118, 106: at these three levels π_{RVM} has, before the crossing, a local maximum within 0.006 of j, which one truncation lifts above j and the other does not.

j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}} - p_j$	j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}} - p_j$	j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}} - p_j$
1	2	2.2138	+0.2138	335	2251	2240.2009	−10.7991	669	4999	4995.1090	−3.8910
2	3	3.1546	+0.1546	336	2267	2264.1226	−2.8774	670	5003	5000.1908	−2.8092
3	5	5.1880	+0.1880	337	2269	2270.0868	+1.0868	671	5009	5006.2128	−2.7872
4	7	7.1588	+0.1588	338	2273	2274.1702	+1.1702	672	5011	5014.2390	+3.2390
5	11	11.1391	+0.1391	339	2281	2279.1152	−1.8848	673	5021	5043.1625	+22.1625
6	13	14.0201	+1.0201	340	2287	2284.2038	−2.7962	674	5023	5054.1243	+31.1243
7	17	17.2551	+0.2551	341	2293	2297.2222	+4.2222	675	5039	5060.1530	+21.1530
8	19	19.1767	+0.1767	342	2297	2311.1364	+14.1364	676	5051	5065.2030	+14.2030
9	23	26.0524	+3.0524	343	2309	2316.1611	+7.1611	677	5059	5070.1949	+11.1949
10	29	28.0419	−0.9581	344	2311	2320.1971	+9.1971	678	5077	5075.2101	−1.7899
11	31	33.0665	+2.0665	345	2333	2324.1726	−8.8274	679	5081	5081.1521	+0.1521
12	37	38.0878	+1.0878	346	2339	2328.1884	−10.8116	680	5087	5088.1848	+1.1848
13	41	40.2252	−0.7748	347	2341	2333.1621	−7.8379	681	5099	5100.1882	+1.1882
14	43	44.1857	+1.1857	348	2347	2341.1412	−5.8588	682	5101	5114.1940	+13.1940
15	47	46.1156	−0.8844	349	2351	2354.1374	+3.1374	683	5107	5122.1794	+15.1794
16	53	57.1721	+4.1721	350	2357	2361.1040	+4.1040	684	5113	5128.1328	+15.1328
17	59	59.1154	+0.1154	351	2371	2365.2375	−5.7625	685	5119	5133.1056	+14.1056
18	61	61.0717	+0.0717	352	2377	2370.0802	−6.9198	686	5147	5137.1713	−9.8287
19	67	67.0344	+0.0344	353	2381	2374.1044	−6.8956	687	5153	5142.0815	−10.9185
20	71	74.0901	+3.0901	354	2383	2379.0944	−3.9056	688	5167	5146.1533	−20.8467
21	73	76.1129	+3.1129	355	2389	2386.1853	−2.8147	689	5171	5151.1596	−19.8404
22	79	83.0777	+4.0777	356	2393	2409.1394	+16.1394	690	5179	5157.1915	−21.8085
23	83	86.0388	+3.0388	357	2399	2415.1474	+16.1474	691	5189	5167.1708	−21.8292
24	89	92.0775	+3.0775	358	2411	2420.1302	+9.1302	692	5197	5185.1602	−11.8398
25	97	94.0745	−2.9255	359	2417	2425.1210	+8.1210	693	5209	5195.1193	−13.8807
26	101	96.1261	−4.8739	360	2423	2431.1450	+8.1450	694	5227	5201.1508	−25.8492
27	103	103.2182	+0.2182	361	2437	2447.1760	+10.1760	695	5231	5206.2067	−24.7933
28	107	112.1163	+5.1163	362	2441	2458.1209	+17.1209	696	5233	5211.1935	−21.8065
29	109	115.0244	+6.0244	363	2447	2463.1388	+16.1388	697	5237	5216.1890	−20.8110
30	113	118.1253	+5.1253	364	2459	2467.1877	+8.1877	698	5261	5222.1181	−38.8819
31	127	125.1594	−1.8406	365	2467	2471.2079	+4.2079	699	5273	5228.2500	−44.7500
32	131	136.0371	+5.0371	366	2473	2476.1039	+3.1039	700	5279	5240.2593	−38.7407
33	137	138.1768	+1.1768	367	2477	2481.1801	+4.1801	701	5281	5262.1653	−18.8347
34	139	145.1787	+6.1787	368	2503	2493.1792	−9.8208	702	5297	5271.1112	−25.8888
35	149	148.0858	−0.9142	369	2521	2506.1738	−14.8262	703	5303	5277.1190	−25.8810

Table 14 (continued)

j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}} - p_j$	j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}} - p_j$	j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}} - p_j$
36	151	150.1093	−0.8907	370	2531	2512.1287	−18.8713	704	5309	5282.1773	−26.8227
37	157	154.1579	−2.8421	371	2539	2516.2206	−22.7794	705	5323	5288.0995	−34.9005
38	163	160.1227	−2.8773	372	2543	2521.0787	−21.9213	706	5333	5293.2319	−39.7681
39	167	163.1171	−3.8829	373	2549	2525.1356	−23.8644	707	5347	5301.1624	−45.8376
40	173	173.1152	+0.1152	374	2551	2530.2229	−20.7771	708	5351	5323.2036	−27.7964
41	179	176.0959	−2.9041	375	2557	2543.1879	−13.8121	709	5381	5340.1379	−40.8621
42	181	186.1175	+5.1175	376	2579	2559.1228	−19.8772	710	5387	5347.1295	−39.8705
43	191	190.1285	−0.8715	377	2591	2565.1248	−25.8752	711	5393	5352.2287	−40.7713
44	193	201.0969	+8.0969	378	2593	2570.1536	−22.8464	712	5399	5358.0905	−40.9095
45	197	203.1883	+6.1883	379	2609	2576.1280	−32.8720	713	5407	5363.1069	−43.8931
46	199	207.1015	+8.1015	380	2617	2610.1950	−6.8050	714	5413	5368.1864	−44.8136
47	211	212.1460	+1.1460	381	2621	2618.1134	−2.8866	715	5417	5375.1711	−41.8289
48	223	215.0763	−7.9237	382	2633	2624.0998	−8.9002	716	5419	5386.1770	−32.8230
49	227	217.2057	−9.7943	383	2647	2630.1946	−16.8054	717	5431	5403.1571	−27.8429
50	229	230.1447	+1.1447	384	2657	2654.1582	−2.8418	718	5437	5412.1531	−24.8469
51	233	234.1155	+1.1155	385	2659	2663.0956	+4.0956	719	5441	5418.1734	−22.8266
52	239	243.0719	+4.0719	386	2663	2668.0972	+5.0972	720	5443	5423.2029	−19.7971
53	241	246.0130	+5.0130	387	2671	2672.1596	+1.1596	721	5449	5428.1506	−20.8494
54	251	248.1162	−2.8838	388	2677	2677.0942	+0.0942	722	5471	5433.1031	−37.8969
55	257	258.2462	+1.2462	389	2683	2682.1362	−0.8638	723	5477	5438.0936	−38.9064
56	263	263.0536	+0.0536	390	2687	2691.1547	+4.1547	724	5479	5443.1613	−35.8387
57	269	267.1805	−1.8195	391	2689	2709.1416	+20.1416	725	5483	5450.2001	−32.7999
58	271	277.0384	+6.0384	392	2693	2715.2375	+22.2375	726	5501	5465.2375	−35.7625
59	277	279.1633	+2.1633	393	2699	2720.1824	+21.1824	727	5503	5484.1660	−18.8340
60	281	282.2079	+1.2079	394	2707	2725.0835	+18.0835	728	5507	5492.1544	−14.8456
61	283	293.1444	+10.1444	395	2711	2729.1455	+18.1455	729	5519	5498.1613	−20.8387
62	293	297.0717	+4.0717	396	2713	2734.1897	+21.1897	730	5521	5504.1003	−16.8997
63	307	311.0820	+4.0820	397	2719	2743.1740	+24.1740	731	5527	5509.1649	−17.8351
64	311	314.2060	+3.2060	398	2729	2761.1677	+32.1677	732	5531	5515.1948	−15.8052
65	313	326.1105	+13.1105	399	2731	2768.1501	+37.1501	733	5557	5524.1509	−32.8491
66	317	329.2472	+12.2472	400	2741	2773.1571	+32.1571	734	5563	5554.2170	−8.7830
67	331	332.1141	+1.1141	401	2749	2778.1149	+29.1149	735	5569	5566.1453	−2.8547
68	337	335.1199	−1.8801	402	2753	2783.1305	+30.1305	736	5573	5573.1534	+0.1534
69	347	343.1291	−3.8709	403	2767	2790.1587	+23.1587	737	5581	5579.2072	−1.7928
70	349	347.2034	−1.7966	404	2777	2814.2147	+37.2147	738	5591	5586.1546	−4.8454
71	353	351.0568	−1.9432	405	2789	2822.0996	+33.0996	739	5623	5595.1431	−27.8569
72	359	365.1411	+6.1411	406	2791	2827.0857	+36.0857	740	5639	5636.1566	−2.8434
73	367	369.0667	+2.0667	407	2797	2831.1482	+34.1482	741	5641	5645.1130	+4.1130
74	373	372.1471	−0.8529	408	2801	2836.1031	+35.1031	742	5647	5651.1535	+4.1535
75	379	380.2049	+1.2049	409	2803	2841.1735	+38.1735	743	5651	5657.1496	+6.1496
76	383	385.0915	+2.0915	410	2819	2850.1575	+31.1575	744	5653	5663.1924	+10.1924
77	389	388.0514	−0.9486	411	2833	2863.1914	+30.1914	745	5657	5671.1662	+14.1662
78	397	391.0724	−5.9276	412	2837	2870.1900	+33.1900	746	5659	5686.1916	+27.1916
79	401	404.1579	+3.1579	413	2843	2875.1506	+32.1506	747	5669	5706.1647	+37.1647
80	409	408.1667	−0.8333	414	2851	2879.1584	+28.1584	748	5683	5714.1603	+31.1603
81	419	413.1228	−5.8772	415	2857	2883.1256	+26.1256	749	5689	5720.1408	+31.1408
82	421	424.1253	+3.1253	416	2861	2887.1097	+26.1097	750	5693	5725.1545	+32.1545
83	431	427.1525	−3.8475	417	2879	2891.1585	+12.1585	751	5701	5730.1327	+29.1327
84	433	430.1187	−2.8813	418	2887	2897.1097	+10.1097	752	5711	5735.1272	+24.1272
85	439	434.1375	−4.8625	419	2897	2906.2458	+9.2458	753	5717	5740.1864	+23.1864
86	443	446.0793	+3.0793	420	2903	2921.1375	+18.1375	754	5737	5747.1411	+10.1411
87	449	449.2083	+0.2083	421	2909	2927.2070	+18.2070	755	5741	5757.1855	+16.1855
88	457	454.1441	−2.8559	422	2917	2932.1846	+15.1846	756	5743	5777.1652	+34.1652
89	461	470.1710	+9.1710	423	2927	2937.0967	+10.0967	757	5749	5787.1601	+38.1601
90	463	475.1873	+12.1873	424	2939	2941.1732	+2.1732	758	5779	5794.1041	+15.1041
91	467	490.0909	+23.0909	425	2953	2946.2388	−6.7612	759	5783	5799.1482	+16.1482
92	479	493.0870	+14.0870	426	2957	2955.1707	−1.8293	760	5791	5804.1472	+13.1472
93	487	496.0928	+9.0928	427	2963	2977.1315	+14.1315	761	5801	5809.1428	+8.1428
94	491	500.1660	+9.1660	428	2969	2984.1305	+15.1305	762	5807	5814.1876	+7.1876
95	499	508.1736	+9.1736	429	2971	2989.1881	+18.1881	763	5813	5820.2221	+7.2221
96	503	512.1244	+9.1244	430	2999	2995.0915	−3.9085	764	5821	5829.1735	+8.1735
97	509	515.1067	+6.1067	431	3001	3001.1309	+0.1309	765	5827	5848.1921	+21.1921

Table 14 (continued)

j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}} - p_j$	j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}} - p_j$	j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}} - p_j$
98	521	518.1442	−2.8558	432	3011	3028.2015	+17.2015	766	5839	5861.2576	+22.2576
99	523	531.1694	+8.1694	433	3019	3038.0972	+19.0972	767	5843	5869.1412	+26.1412
100	541	536.1441	−4.8559	434	3023	3043.1355	+20.1355	768	5849	5875.1237	+26.1237
101	547	539.2398	−7.7602	435	3037	3048.1091	+11.1091	769	5851	5880.1722	+29.1722
102	557	543.1977	−13.8023	436	3041	3053.1064	+12.1064	770	5857	5886.0965	+29.0965
103	563	553.1345	−9.8655	437	3049	3059.1164	+10.1164	771	5861	5891.2185	+30.2185
104	569	558.0439	−10.9561	438	3061	3069.1605	+8.1605	772	5867	5899.1423	+32.1423
105	571	561.0337	−9.9663	439	3067	3083.1748	+16.1748	773	5869	5915.1969	+46.1969
106	577	563.2463	−13.7537	440	3079	3090.1288	+11.1288	774	5879	5939.1321	+60.1321
107	587	568.1124	−18.8876	441	3083	3095.0947	+12.0947	775	5881	5946.2450	+65.2450
108	593	583.0774	−9.9226	442	3089	3099.1119	+10.1119	776	5897	5952.2356	+55.2356
109	599	587.1276	−11.8724	443	3109	3103.1107	−5.8893	777	5903	5958.1194	+55.1194
110	601	605.1479	+4.1479	444	3119	3107.1361	−11.8639	778	5923	5963.1517	+40.1517
111	607	609.2115	+2.2115	445	3121	3112.1162	−8.8838	779	5927	5969.1067	+42.1067
112	613	613.2031	+0.2031	446	3137	3118.2117	−18.7883	780	5939	5975.1703	+36.1703
113	617	626.1870	+9.1870	447	3163	3134.1748	−28.8252	781	5953	5984.1943	+31.1943
114	619	631.1994	+12.1994	448	3167	3144.1963	−22.8037	782	5981	6000.2382	+19.2382
115	631	635.0364	+4.0364	449	3169	3150.1564	−18.8436	783	5987	6012.2388	+25.2388
116	641	638.0574	−2.9426	450	3181	3155.1151	−25.8849	784	6007	6020.1066	+13.1066
117	643	642.1105	−0.8895	451	3187	3159.1966	−27.8034	785	6011	6025.1631	+14.1631
118	647	652.2519	+5.2519	452	3191	3164.1538	−26.8462	786	6029	6030.1357	+1.1357
119	653	657.1967	+4.1967	453	3203	3170.1508	−32.8492	787	6037	6035.0769	−1.9231
120	659	661.1242	+2.1242	454	3209	3182.1918	−26.8082	788	6043	6039.1408	−3.8592
121	661	666.1305	+5.1305	455	3217	3199.2409	−17.7591	789	6047	6044.1026	−2.8974
122	673	684.1446	+11.1446	456	3221	3206.1529	−14.8471	790	6053	6049.1283	−3.8717
123	677	688.1253	+11.1253	457	3229	3211.1657	−17.8343	791	6067	6055.1625	−11.8375
124	683	693.1262	+10.1262	458	3251	3216.1375	−34.8625	792	6073	6064.1640	−8.8360
125	691	705.1071	+14.1071	459	3253	3221.1711	−31.8289	793	6079	6079.1805	+0.1805
126	701	709.1331	+8.1331	460	3257	3228.1795	−28.8205	794	6089	6091.1295	+2.1295
127	709	712.1804	+3.1804	461	3259	3251.1554	−7.8446	795	6091	6098.1337	+7.1337
128	719	716.0725	−2.9275	462	3271	3260.1233	−10.8767	796	6101	6104.0863	+3.0863
129	727	721.1769	−5.8231	463	3299	3265.1671	−33.8329	797	6113	6109.0764	−3.9236
130	733	736.2032	+3.2032	464	3301	3270.1020	−30.8980	798	6121	6113.1661	−7.8339
131	739	741.0824	+2.0824	465	3307	3274.1585	−32.8415	799	6131	6118.1213	−12.8787
132	743	746.1513	+3.1513	466	3313	3279.1163	−33.8837	800	6133	6123.1215	−9.8785
133	751	762.1431	+11.1431	467	3319	3284.2196	−34.7804	801	6143	6128.2314	−14.7686
134	757	766.1179	+9.1179	468	3323	3293.1799	−29.8201	802	6151	6136.1334	−14.8666
135	761	770.0602	+9.0602	469	3329	3307.1402	−21.8598	803	6163	6148.1902	−14.8098
136	769	774.1790	+5.1790	470	3331	3314.2068	−16.7932	804	6173	6166.1645	−6.8355
137	773	787.1370	+14.1370	471	3343	3320.0863	−22.9137	805	6197	6175.1769	−21.8231
138	787	791.2035	+4.2035	472	3347	3324.1300	−22.8700	806	6199	6182.1224	−16.8776
139	797	795.0554	−1.9446	473	3359	3328.1464	−30.8536	807	6203	6187.2377	−15.7623
140	809	798.0927	−10.9073	474	3361	3332.1809	−28.8191	808	6211	6193.1275	−17.8725
141	811	802.1355	−8.8645	475	3371	3337.1305	−33.8695	809	6217	6198.2448	−18.7552
142	821	814.1596	−6.8404	476	3373	3343.1542	−29.8458	810	6221	6205.1467	−15.8533
143	823	820.1013	−2.8987	477	3389	3358.2128	−30.7872	811	6229	6214.2019	−14.7981
144	827	824.1010	−2.8990	478	3391	3373.1885	−17.8115	812	6247	6249.1596	+2.1596
145	829	829.1200	+0.1200	479	3407	3380.1137	−26.8863	813	6257	6259.1317	+2.1317
146	839	850.0981	+11.0981	480	3413	3385.1452	−27.8548	814	6263	6266.1082	+3.1082
147	853	854.1110	+1.1110	481	3433	3390.1794	−42.8206	815	6269	6272.1128	+3.1128
148	857	858.2501	+1.2501	482	3449	3396.1996	−52.8004	816	6271	6278.1248	+7.1248
149	859	870.1575	+11.1575	483	3457	3409.2101	−47.7899	817	6277	6285.1249	+8.1249
150	863	876.1066	+13.1066	484	3461	3434.1310	−26.8690	818	6287	6294.2094	+7.2094
151	877	879.1848	+2.1848	485	3463	3440.1841	−22.8159	819	6299	6319.2203	+20.2203
152	881	882.1983	+1.1983	486	3467	3446.1293	−20.8707	820	6301	6332.1178	+31.1178
153	883	886.0679	+3.0679	487	3469	3452.1334	−16.8666	821	6311	6339.1017	+28.1017
154	887	890.1740	+3.1740	488	3491	3461.1543	−29.8457	822	6317	6344.1800	+27.1800
155	907	903.1825	−3.8175	489	3499	3490.1483	−8.8517	823	6323	6349.2281	+26.2281
156	911	908.1821	−2.8179	490	3511	3497.0978	−13.9022	824	6329	6355.0914	+26.0914
157	919	912.1256	−6.8744	491	3517	3502.1047	−14.8953	825	6337	6360.1410	+23.1410
158	929	916.1235	−12.8765	492	3527	3507.0822	−19.9178	826	6343	6366.1786	+23.1786
159	937	923.2156	−13.7844	493	3529	3512.0963	−16.9037	827	6353	6375.1452	+22.1452

Table 14 (continued)

j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}} - p_j$	j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}} - p_j$	j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}} - p_j$
160	941	936.1740	−4.8260	494	3533	3518.1307	−14.8693	828	6359	6392.1884	+33.1884
161	947	940.1852	−6.8148	495	3539	3528.2055	−10.7945	829	6361	6407.1625	+46.1625
162	953	944.1096	−8.8904	496	3541	3543.2003	+2.2003	830	6367	6415.1249	+48.1249
163	967	948.1594	−18.8406	497	3547	3551.0938	+4.0938	831	6373	6421.1049	+48.1049
164	971	961.1896	−9.8104	498	3557	3556.0958	−0.9042	832	6379	6426.1171	+47.1171
165	977	968.1073	−8.8927	499	3559	3560.1556	+1.1556	833	6389	6431.1010	+42.1010
166	983	972.0546	−10.9454	500	3571	3564.2246	−6.7754	834	6397	6436.0893	+39.0893
167	991	975.1122	−15.8878	501	3581	3569.1259	−11.8741	835	6421	6441.1147	+20.1147
168	997	979.1111	−17.8889	502	3583	3574.1666	−8.8334	836	6427	6447.1179	+20.1179
169	1009	987.1885	−21.8115	503	3593	3582.1662	−10.8338	837	6449	6454.2179	+5.2179
170	1013	999.1110	−13.8890	504	3607	3603.2141	−3.7859	838	6451	6468.1750	+17.1750
171	1019	1003.2085	−15.7915	505	3613	3612.1946	−0.8054	839	6469	6484.1702	+15.1702
172	1021	1008.1539	−12.8461	506	3617	3618.1614	+1.1614	840	6473	6493.1410	+20.1410
173	1031	1035.1474	+4.1474	507	3623	3623.1826	+0.1826	841	6481	6499.1961	+18.1961
174	1033	1040.2508	+7.2508	508	3631	3629.0905	−1.9095	842	6491	6505.1265	+14.1265
175	1039	1059.1355	+20.1355	509	3637	3635.1365	−1.8635	843	6521	6510.1525	−10.8475
176	1049	1064.2205	+15.2205	510	3643	3648.2322	+5.2322	844	6529	6515.1953	−13.8047
177	1051	1068.1539	+17.1539	511	3659	3670.1553	+11.1553	845	6547	6521.1514	−25.8486
178	1061	1072.0944	+11.0944	512	3671	3677.1069	+6.1069	846	6551	6528.1567	−22.8433
179	1063	1076.1849	+13.1849	513	3673	3682.1500	+9.1500	847	6553	6540.1754	−12.8246
180	1069	1090.1959	+21.1959	514	3677	3687.1971	+10.1971	848	6563	6569.1748	+6.1748
181	1087	1097.1027	+10.1027	515	3691	3693.2007	+2.2007	849	6569	6578.2150	+9.2150
182	1091	1101.0906	+10.0906	516	3697	3703.1970	+6.1970	850	6571	6585.1629	+14.1629
183	1093	1105.0703	+12.0703	517	3701	3728.2204	+27.2204	851	6577	6591.1497	+14.1497
184	1097	1109.2010	+12.2010	518	3709	3735.2415	+26.2415	852	6581	6597.1282	+16.1282
185	1103	1125.2160	+22.2160	519	3719	3741.0998	+22.0998	853	6599	6603.1671	+4.1671
186	1109	1131.1785	+22.1785	520	3727	3745.2187	+18.2187	854	6607	6611.1599	+4.1599
187	1117	1135.2052	+18.2052	521	3733	3750.1669	+17.1669	855	6619	6625.2230	+6.2230
188	1123	1140.0986	+17.0986	522	3739	3756.1017	+17.1017	856	6637	6648.1572	+11.1572
189	1129	1148.2078	+19.2078	523	3761	3763.1526	+2.1526	857	6653	6657.1283	+4.1283
190	1151	1163.1249	+12.1249	524	3767	3778.1695	+11.1695	858	6659	6663.1601	+4.1601
191	1153	1167.1264	+14.1264	525	3769	3789.1701	+20.1701	859	6661	6669.0930	+8.0930
192	1163	1171.0550	+8.0550	526	3779	3795.2208	+16.2208	860	6673	6674.1076	+1.1076
193	1171	1174.1700	+3.1700	527	3793	3800.2202	+7.2202	861	6679	6679.1290	+0.1290
194	1181	1179.1805	−1.8195	528	3797	3805.1339	+8.1339	862	6689	6684.2380	−4.7620
195	1187	1190.1627	+3.1627	529	3803	3810.0886	+7.0886	863	6691	6691.1594	+0.1594
196	1193	1196.2211	+3.2211	530	3821	3815.1156	−5.8844	864	6701	6700.2104	−0.7896
197	1201	1200.2294	−0.7706	531	3823	3822.1383	−0.8617	865	6703	6719.1737	+16.1737
198	1213	1204.1230	−8.8770	532	3833	3847.2074	+14.2074	866	6709	6732.1496	+23.1496
199	1217	1208.1130	−8.8870	533	3847	3858.1242	+11.1242	867	6719	6739.1971	+20.1971
200	1223	1214.1813	−8.8187	534	3851	3864.1525	+13.1525	868	6733	6745.1602	+12.1602
201	1229	1234.1617	+5.1617	535	3853	3870.1187	+17.1187	869	6737	6750.1738	+13.1738
202	1231	1239.1422	+8.1422	536	3863	3876.1484	+13.1484	870	6761	6755.1420	−5.8580
203	1237	1244.1696	+7.1696	537	3877	3885.2086	+8.2086	871	6763	6760.1145	−2.8855
204	1249	1262.1869	+13.1869	538	3881	3917.1363	+36.1363	872	6779	6765.1198	−13.8802
205	1259	1269.0881	+10.0881	539	3889	3924.1419	+35.1419	873	6781	6770.2180	−10.7820
206	1277	1273.0736	−3.9264	540	3907	3930.1158	+23.1158	874	6791	6777.1893	−13.8107
207	1279	1276.1902	−2.8098	541	3911	3936.0980	+25.0980	875	6793	6787.2075	−5.7925
208	1283	1281.0824	−1.9176	542	3917	3942.2406	+25.2406	876	6803	6803.1936	+0.1936
209	1289	1288.1618	−0.8382	543	3919	3967.1976	+48.1976	877	6823	6814.1984	−8.8016
210	1291	1300.1525	+9.1525	544	3923	3980.1423	+57.1423	878	6827	6822.1094	−4.8906
211	1297	1305.0959	+8.0959	545	3929	3986.1390	+57.1390	879	6829	6828.0874	−0.9126
212	1301	1308.2039	+7.2039	546	3931	3991.1290	+60.1290	880	6833	6833.1122	+0.1122
213	1303	1312.0823	+9.0823	547	3943	3996.0869	+53.0869	881	6841	6838.1202	−2.8798
214	1307	1315.2176	+8.2176	548	3947	4000.2131	+53.2131	882	6857	6843.1460	−13.8540
215	1319	1320.2297	+1.2297	549	3967	4006.1341	+39.1341	883	6863	6849.1169	−13.8831
216	1321	1333.1642	+12.1642	550	3989	4013.1954	+24.1954	884	6869	6856.1298	−12.8702
217	1327	1340.1193	+13.1193	551	4001	4026.2110	+25.2110	885	6871	6867.2183	−3.7817
218	1361	1344.1685	−16.8315	552	4003	4037.1856	+34.1856	886	6883	6897.1894	+14.1894
219	1367	1348.1871	−18.8129	553	4007	4044.0995	+37.0995	887	6899	6907.1603	+8.1603
220	1373	1353.2045	−19.7955	554	4013	4049.0810	+36.0810	888	6907	6914.1568	+7.1568
221	1381	1376.1714	−4.8286	555	4019	4053.1223	+34.1223	889	6911	6920.1919	+9.1919

Table 14 (continued)

j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}}-p_j$	j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}}-p_j$	j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}}-p_j$
222	1399	1381.2304	−17.7696	556	4021	4057.1494	+36.1494	890	6917	6927.1058	+10.1058
223	1409	1386.1129	−22.8871	557	4027	4061.2107	+34.2107	891	6947	6934.1205	−12.8795
224	1423	1391.1422	−31.8578	558	4049	4066.1554	+17.1554	892	6949	6943.2164	−5.7836
225	1427	1404.2010	−22.7990	559	4051	4072.1499	+21.1499	893	6959	6976.1840	+17.1840
226	1429	1413.1140	−15.8860	560	4057	4082.1779	+25.1779	894	6961	6988.1276	+27.1276
227	1433	1417.1484	−15.8516	561	4073	4100.1739	+27.1739	895	6967	6995.1400	+28.1400
228	1439	1421.0968	−17.9032	562	4079	4108.1728	+29.1728	896	6971	7001.1551	+30.1551
229	1447	1425.0792	−21.9208	563	4091	4114.1227	+23.1227	897	6977	7007.1397	+30.1397
230	1451	1430.1040	−20.8960	564	4093	4119.1115	+26.1115	898	6983	7013.1620	+30.1620
231	1453	1447.1742	−5.8258	565	4099	4124.0913	+25.0913	899	6991	7020.2063	+29.2063
232	1459	1454.1326	−4.8674	566	4111	4129.1171	+18.1171	900	6997	7032.1631	+35.1631
233	1471	1459.0779	−11.9221	567	4127	4135.1730	+8.1730	901	7001	7059.2235	+58.2235
234	1481	1463.1780	−17.8220	568	4129	4147.1853	+18.1853	902	7013	7070.1505	+57.1505
235	1483	1471.2437	−11.7563	569	4133	4166.1596	+33.1596	903	7019	7077.1204	+58.1204
236	1487	1491.2253	+4.2253	570	4139	4173.2379	+34.2379	904	7027	7082.2226	+55.2226
237	1489	1496.2035	+7.2035	571	4153	4179.1365	+26.1365	905	7039	7088.1016	+49.1016
238	1493	1501.1561	+8.1561	572	4157	4184.1179	+27.1179	906	7043	7093.1208	+50.1208
239	1499	1508.2039	+9.2039	573	4159	4189.0987	+30.0987	907	7057	7098.1800	+41.1800
240	1511	1528.1286	+17.1286	574	4177	4194.1336	+17.1336	908	7069	7104.2009	+35.2009
241	1523	1533.0746	+10.0746	575	4201	4201.1176	+0.1176	909	7079	7112.1650	+33.1650
242	1531	1536.2254	+5.2254	576	4211	4214.1793	+3.1793	910	7103	7124.1912	+21.1912
243	1543	1540.1696	−2.8304	577	4217	4231.1522	+14.1522	911	7109	7140.2028	+31.2028
244	1549	1545.1106	−3.8894	578	4219	4238.1392	+19.1392	912	7121	7150.1656	+29.1656
245	1553	1552.2090	−0.7910	579	4229	4243.1610	+14.1610	913	7127	7157.1338	+30.1338
246	1559	1563.2131	+4.2131	580	4231	4248.0954	+17.0954	914	7129	7163.0859	+34.0859
247	1567	1569.0950	+2.0950	581	4241	4252.1437	+11.1437	915	7151	7168.0901	+17.0901
248	1571	1573.0829	+2.0829	582	4243	4257.0808	+14.0808	916	7159	7172.2505	+13.2505
249	1579	1576.2189	−2.7811	583	4253	4262.0881	+9.0881	917	7177	7177.2239	+0.2239
250	1583	1580.2451	−2.7549	584	4259	4268.1213	+9.1213	918	7187	7183.1121	−3.8879
251	1597	1587.1309	−9.8691	585	4261	4277.1804	+16.1804	919	7193	7189.1335	−3.8665
252	1601	1609.1269	+8.1269	586	4271	4290.1888	+19.1888	920	7207	7197.1626	−9.8374
253	1607	1614.2199	+7.2199	587	4273	4298.2185	+25.2185	921	7211	7214.1783	+3.1783
254	1609	1620.1133	+11.1133	588	4283	4304.1596	+21.1596	922	7213	7233.1440	+20.1440
255	1613	1630.2090	+17.2090	589	4289	4309.1209	+20.1209	923	7219	7242.1073	+23.1073
256	1619	1650.1672	+31.1672	590	4297	4313.1833	+16.1833	924	7229	7248.1581	+19.1581
257	1621	1655.1315	+34.1315	591	4327	4318.0858	−8.9142	925	7237	7254.1270	+17.1270
258	1627	1660.0847	+33.0847	592	4337	4322.1738	−14.8262	926	7243	7259.2123	+16.2123
259	1637	1665.2272	+28.2272	593	4339	4328.1063	−10.8937	927	7247	7265.1601	+18.1601
260	1657	1685.1699	+28.1699	594	4349	4335.2012	−13.7988	928	7253	7272.1091	+19.1091
261	1663	1691.1681	+28.1681	595	4357	4357.2365	+0.2365	929	7283	7280.1560	−2.8440
262	1667	1695.2491	+28.2491	596	4363	4368.2150	+5.2150	930	7297	7297.2521	+0.2521
263	1669	1699.2422	+30.2422	597	4373	4375.1351	+2.1351	931	7307	7318.1900	+11.1900
264	1693	1704.1334	+11.1334	598	4391	4381.0877	−9.9123	932	7309	7327.1551	+18.1551
265	1697	1711.1798	+14.1798	599	4397	4386.1536	−10.8464	933	7321	7334.1023	+13.1023
266	1699	1727.1269	+28.1269	600	4409	4393.1121	−15.8879	934	7331	7339.2000	+8.2000
267	1709	1733.0868	+24.0868	601	4421	4404.1742	−16.8258	935	7333	7345.1297	+12.1297
268	1721	1737.1364	+16.1364	602	4423	4433.1312	+10.1312	936	7349	7351.1041	+2.1041
269	1723	1741.2102	+18.2102	603	4441	4440.1625	−0.8375	937	7351	7357.1568	+6.1568
270	1733	1747.1495	+14.1495	604	4447	4446.1503	−0.8497	938	7369	7365.2112	−3.7888
271	1741	1770.1563	+29.1563	605	4451	4452.1218	+1.1218	939	7393	7383.1842	−9.8158
272	1747	1776.1354	+29.1354	606	4457	4458.1915	+1.1915	940	7411	7403.1632	−7.8368
273	1753	1780.2224	+27.2224	607	4463	4469.1808	+6.1808	941	7417	7412.1108	−4.8892
274	1759	1785.1063	+26.1063	608	4481	4498.1269	+17.1269	942	7433	7418.1327	−14.8673
275	1777	1790.1557	+13.1557	609	4483	4505.1394	+22.1394	943	7451	7423.1983	−27.8017
276	1783	1800.2051	+17.2051	610	4493	4510.2300	+17.2300	944	7457	7428.2188	−28.7812
277	1787	1810.2125	+23.2125	611	4507	4515.2186	+8.2186	945	7459	7433.2203	−25.7797
278	1789	1815.1773	+26.1773	612	4513	4520.2076	+7.2076	946	7477	7439.0995	−37.9005
279	1801	1819.1221	+18.1221	613	4517	4526.1275	+9.1275	947	7481	7444.2001	−36.7999
280	1811	1822.2042	+11.2042	614	4519	4533.1364	+14.1364	948	7487	7451.1815	−35.8185
281	1823	1826.0840	+3.0840	615	4523	4545.1641	+22.1641	949	7489	7461.1463	−27.8537
282	1831	1829.2458	−1.7542	616	4547	4559.2047	+12.2047	950	7499	7475.1602	−23.8398
283	1847	1834.1824	−12.8176	617	4549	4567.1488	+18.1488	951	7507	7486.2426	−20.7574

Table 14 (continued)

j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}} - p_j$	j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}} - p_j$	j	p_j	p_j^{RVM}	$p_j^{\mathrm{RVM}} - p_j$
284	1861	1843.1978	−17.8022	618	4561	4573.0854	+12.0854	952	7517	7494.1905	−22.8095
285	1867	1853.2027	−13.7973	619	4567	4577.1858	+10.1858	953	7523	7500.1934	−22.8066
286	1871	1859.0767	−11.9233	620	4583	4582.1034	−0.8966	954	7529	7506.0829	−22.9171
287	1873	1863.0681	−9.9319	621	4591	4586.1839	−4.8161	955	7537	7510.1940	−26.8060
288	1877	1866.1832	−10.8168	622	4597	4591.1700	−5.8300	956	7541	7515.1291	−25.8709
289	1879	1870.1600	−8.8400	623	4603	4597.1749	−5.8251	957	7547	7520.0902	−26.9098
290	1889	1875.1388	−13.8612	624	4621	4607.1728	−13.8272	958	7549	7525.0855	−23.9145
291	1901	1885.2000	−15.8000	625	4637	4628.1755	−8.8245	959	7559	7530.1406	−28.8594
292	1907	1899.2482	−7.7518	626	4639	4637.1525	−1.8475	960	7561	7537.1119	−23.8881
293	1913	1905.1448	−7.8552	627	4643	4643.1561	+0.1561	961	7573	7546.1598	−26.8402
294	1931	1910.1046	−20.8954	628	4649	4648.2062	−0.7938	962	7577	7563.1873	−13.8127
295	1933	1915.1364	−17.8636	629	4651	4654.0889	+3.0889	963	7583	7577.1558	−5.8442
296	1949	1940.1808	−8.8192	630	4657	4659.1633	+2.1633	964	7589	7585.1594	−3.8406
297	1951	1948.1175	−2.8825	631	4663	4666.1874	+3.1874	965	7591	7591.2224	+0.2224
298	1973	1953.0953	−19.9047	632	4673	4683.2213	+10.2213	966	7603	7597.1469	−5.8531
299	1979	1957.2030	−21.7970	633	4679	4701.2121	+22.2121	967	7607	7602.2052	−4.7948
300	1987	1963.1685	−23.8315	634	4691	4709.1040	+18.1040	968	7621	7608.1175	−12.8825
301	1993	1978.1719	−14.8281	635	4703	4714.1763	+11.1763	969	7639	7614.1082	−24.8918
302	1997	1987.2099	−9.7901	636	4721	4719.2292	−1.7708	970	7643	7621.1189	−21.8811
303	1999	1992.1883	−6.8117	637	4723	4725.1228	+2.1228	971	7649	7631.1663	−17.8337
304	2003	1996.2189	−6.7811	638	4729	4731.1617	+2.1617	972	7669	7656.1998	−12.8002
305	2011	2000.2526	−10.7474	639	4733	4741.1632	+8.1632	973	7673	7669.2208	−3.7792
306	2017	2005.1736	−11.8264	640	4751	4765.1530	+14.1530	974	7681	7677.1972	−3.8028
307	2027	2014.1942	−12.8058	641	4759	4774.1077	+15.1077	975	7687	7684.1274	−2.8726
308	2029	2034.1186	+5.1186	642	4783	4779.2209	−3.7791	976	7691	7690.1320	−0.8680
309	2039	2039.1959	+0.1959	643	4787	4784.2094	−2.7906	977	7699	7696.1559	−2.8441
310	2053	2044.1700	−8.8300	644	4789	4789.1435	+0.1435	978	7703	7703.1624	+0.1624
311	2063	2050.1539	−12.8461	645	4793	4794.1115	+1.1115	979	7717	7713.1613	−3.8387
312	2069	2073.1594	+4.1594	646	4799	4799.1402	+0.1402	980	7723	7749.1757	+26.1757
313	2081	2081.1765	+0.1765	647	4801	4805.1949	+4.1949	981	7727	7761.1384	+34.1384
314	2083	2086.1956	+3.1956	648	4813	4815.1612	+2.1612	982	7741	7768.1766	+27.1766
315	2087	2091.1594	+4.1594	649	4817	4829.1706	+12.1706	983	7753	7774.2311	+21.2311
316	2089	2097.1512	+8.1512	650	4831	4838.1537	+7.1537	984	7757	7780.2035	+23.2035
317	2099	2117.1977	+18.1977	651	4861	4844.1525	−16.8475	985	7759	7786.2199	+27.2199
318	2111	2125.2130	+14.2130	652	4871	4849.1438	−21.8562	986	7789	7793.1874	+4.1874
319	2113	2130.1450	+17.1450	653	4877	4854.0774	−22.9226	987	7793	7802.1813	+9.1813
320	2129	2134.1472	+5.1472	654	4889	4858.1238	−30.8762	988	7817	7819.1849	+2.1849
321	2131	2138.1507	+7.1507	655	4903	4862.2267	−40.7733	989	7823	7837.1870	+14.1870
322	2137	2143.1033	+6.1033	656	4909	4867.2231	−41.7769	990	7829	7846.1774	+17.1774
323	2141	2149.1816	+8.1816	657	4919	4874.1219	−44.8781	991	7841	7853.1183	+12.1183
324	2143	2162.1343	+19.1343	658	4931	4884.1951	−46.8049	992	7853	7858.1992	+5.1992
325	2153	2169.1611	+16.1611	659	4933	4905.1407	−27.8593	993	7867	7864.0833	−2.9167
326	2161	2174.1224	+13.1224	660	4937	4913.2263	−23.7737	994	7873	7869.0898	−3.9102
327	2179	2178.1252	−0.8748	661	4943	4920.1034	−22.8966	995	7877	7874.1153	−2.8847
328	2203	2182.1158	−20.8842	662	4951	4925.1450	−25.8550	996	7879	7879.2346	+0.2346
329	2207	2186.1720	−20.8280	663	4957	4930.1899	−26.8101	997	7883	7886.1803	+3.1803
330	2213	2192.1796	−20.8204	664	4967	4936.1552	−30.8448	998	7901	7896.1478	−4.8522
331	2221	2215.1472	−5.8528	665	4969	4943.2236	−25.7764	999	7907	7914.1644	+7.1644
332	2237	2222.1383	−14.8617	666	4973	4965.2293	−7.7707	1000	7919	7928.1950	+9.1950
333	2239	2227.1649	−11.8351	667	4987	4982.1295	−4.8705				
334	2243	2232.2362	−10.7638	668	4993	4989.1289	−3.8711				

Table 15: The prime count $\pi(n)$, exact, and the synthetic prime count $\pi_{\mathrm{RVM}}(n)$, a real-valued function computed from truncated sums (not an integer count), at the integers $n \leq 1000$, with the difference $\pi_{\mathrm{RVM}}(n) - \pi(n)$.

n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n) - \pi(n)$	n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n) - \pi(n)$	n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n) - \pi(n)$
1	0	0.0000	+0.0000	335	67	67.7510	+0.7510	669	121	120.9426	−0.0574
2	1	0.5294	−0.4706	336	67	67.9877	+0.9877	670	121	120.9419	−0.0581

Table 15 (continued)

n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$	n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$	n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$
3	2	1.5713	−0.4287	337	68	68.1700	+0.1700	671	121	120.9184	−0.0816
4	2	1.9484	−0.0516	338	68	68.3041	+0.3041	672	121	120.8793	−0.1207
5	3	2.5746	−0.4254	339	68	68.4015	+0.4015	673	122	120.8330	−1.1670
6	3	2.9355	−0.0645	340	68	68.4778	+0.4778	674	122	120.7882	−1.2118
7	4	3.5681	−0.4319	341	68	68.5504	+0.5504	675	122	120.7532	−1.2468
8	4	4.1567	+0.1567	342	68	68.6367	+0.6367	676	122	120.7360	−1.2640
9	4	3.8153	−0.1847	343	68	68.7516	+0.7516	677	123	120.7434	−2.2566
10	4	3.9868	−0.0132	344	68	68.9061	+0.9061	678	123	120.7808	−2.2192
11	5	4.6846	−0.3154	345	68	69.1059	+1.1059	679	123	120.8520	−2.1480
12	5	5.0116	+0.0116	346	68	69.3506	+1.3506	680	123	120.9589	−2.0411
13	6	5.3739	−0.6261	347	69	69.6339	+0.6339	681	123	121.1014	−1.8986
14	6	5.9505	−0.0495	348	69	69.9441	+0.9441	682	123	121.2775	−1.7225
15	6	6.1472	+0.1472	349	70	70.2659	+0.2659	683	124	121.4835	−2.5165
16	6	6.0944	+0.0944	350	70	70.5815	+0.5815	684	124	121.7141	−2.2859
17	7	6.4940	−0.5060	351	70	70.8732	+0.8732	685	124	121.9627	−2.0373
18	7	7.2626	+0.2626	352	70	71.1252	+1.1252	686	124	122.2222	−1.7778
19	8	7.6976	−0.3024	353	71	71.3251	+0.3251	687	124	122.4848	−1.5152
20	8	7.6642	−0.3358	354	71	71.4659	+0.4659	688	124	122.7428	−1.2572
21	8	7.6943	−0.3057	355	71	71.5463	+0.5463	689	124	122.9893	−1.0107
22	8	8.0689	+0.0689	356	71	71.5711	+0.5711	690	124	123.2180	−0.7820
23	9	8.4717	−0.5283	357	71	71.5508	+0.5508	691	125	123.4240	−1.5760
24	9	8.5894	−0.4106	358	71	71.5004	+0.5004	692	125	123.6039	−1.3961
25	9	8.5995	−0.4005	359	72	71.4378	−0.5622	693	125	123.7561	−1.2439
26	9	8.8730	−0.1270	360	72	71.3820	−0.6180	694	125	123.8807	−1.1193
27	9	9.4280	+0.4280	361	72	71.3515	−0.6485	695	125	123.9798	−1.0202
28	9	9.9043	+0.9043	362	72	71.3617	−0.6383	696	125	124.0569	−0.9431
29	10	10.0301	+0.0301	363	72	71.4240	−0.5760	697	125	124.1170	−0.8830
30	10	9.9464	−0.0536	364	72	71.5445	−0.4555	698	125	124.1663	−0.8337
31	11	10.0197	−0.9803	365	72	71.7236	−0.2764	699	125	124.2118	−0.7882
32	11	10.3577	−0.6423	366	72	71.9556	−0.0444	700	125	124.2606	−0.7394
33	11	10.8394	−0.1606	367	73	72.2304	−0.7696	701	126	124.3201	−1.6799
34	11	11.1855	+0.1855	368	73	72.5338	−0.4662	702	126	124.3968	−1.6032
35	11	11.2497	+0.2497	369	73	72.8493	−0.1507	703	126	124.4967	−1.5033
36	11	11.2250	+0.2250	370	73	73.1605	+0.1605	704	126	124.6243	−1.3757
37	12	11.3795	−0.6205	371	73	73.4518	+0.4518	705	126	124.7827	−1.2173
38	12	11.7894	−0.2106	372	73	73.7111	+0.7111	706	126	124.9732	−1.0268
39	12	12.2857	+0.2857	373	74	73.9302	−0.0698	707	126	125.1956	−0.8044
40	12	12.6379	+0.6379	374	74	74.1063	+0.1063	708	126	125.4475	−0.5525
41	13	12.7782	−0.2218	375	74	74.2415	+0.2415	709	127	125.7252	−1.2748
42	13	12.8571	−0.1429	376	74	74.3430	+0.3430	710	127	126.0233	−0.9767
43	14	13.0980	−0.9020	377	74	74.4222	+0.4222	711	127	126.3354	−0.6646
44	14	13.5939	−0.4061	378	74	74.4930	+0.4930	712	127	126.6541	−0.3459
45	14	14.2234	+0.2234	379	75	74.5707	−0.4293	713	127	126.9717	−0.0283
46	14	14.7450	+0.7450	380	75	74.6702	−0.3298	714	127	127.2803	+0.2803
47	15	14.9810	−0.0190	381	75	74.8040	−0.1960	715	127	127.5724	+0.5724
48	15	14.9416	−0.0584	382	75	74.9814	−0.0186	716	127	127.8415	+0.8415
49	15	14.8016	−0.1984	383	76	75.2070	−0.7930	717	127	128.0819	+1.0819
50	15	14.7623	−0.2377	384	76	75.4805	−0.5195	718	127	128.2896	+1.2896
51	15	14.9108	−0.0892	385	76	75.7963	−0.2037	719	128	128.4621	+0.4621
52	15	15.1776	+0.1776	386	76	76.1444	+0.1444	720	128	128.5986	+0.5986
53	16	15.4113	−0.5887	387	76	76.5110	+0.5110	721	128	128.7002	+0.7002
54	16	15.5054	−0.4946	388	76	76.8802	+0.8802	722	128	128.7696	+0.7696
55	16	15.4828	−0.5172	389	77	77.2352	+0.2352	723	128	128.8111	+0.8111
56	16	15.4822	−0.5178	390	77	77.5602	+0.5602	724	128	128.8304	+0.8304
57	16	15.6626	−0.3374	391	77	77.8417	+0.8417	725	128	128.8341	+0.8341
58	16	16.0949	+0.0949	392	77	78.0702	+1.0702	726	128	128.8296	+0.8296
59	17	16.7128	−0.2872	393	77	78.2405	+1.2405	727	129	128.8243	−0.1757
60	17	17.3497	+0.3497	394	77	78.3527	+1.3527	728	129	128.8256	−0.1744
61	18	17.8352	−0.1648	395	77	78.4121	+1.4121	729	129	128.8406	−0.1594
62	18	18.0886	+0.0886	396	77	78.4282	+1.4282	730	129	128.8750	−0.1250
63	18	18.1544	+0.1544	397	78	78.4144	+0.4144	731	129	128.9338	−0.0662
64	18	18.2061	+0.2061	398	78	78.3862	+0.3862	732	129	129.0202	+0.0202

Table 15 (continued)

n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$	n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$	n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$
65	18	18.3456	+0.3456	399	78	78.3599	+0.3599	733	130	129.1361	−0.8639
66	18	18.5811	+0.5811	400	78	78.3513	+0.3513	734	130	129.2814	−0.7186
67	19	18.9195	−0.0805	401	79	78.3736	−0.6264	735	130	129.4546	−0.5454
68	19	19.2516	+0.2516	402	79	78.4369	−0.5631	736	130	129.6525	−0.3475
69	19	19.4681	+0.4681	403	79	78.5468	−0.4532	737	130	129.8707	−0.1293
70	19	19.5266	+0.5266	404	79	78.7044	−0.2956	738	130	130.1034	+0.1034
71	20	19.4766	−0.5234	405	79	78.9059	−0.0941	739	131	130.3444	−0.6556
72	20	19.4340	−0.5660	406	79	79.1434	+0.1434	740	131	130.5867	−0.4133
73	21	19.5193	−1.4807	407	79	79.4054	+0.4054	741	131	130.8234	−0.1766
74	21	19.7953	−1.2047	408	79	79.6786	+0.6786	742	131	131.0481	+0.0481
75	21	20.2355	−0.7645	409	80	79.9487	−0.0513	743	132	131.2547	−0.7453
76	21	20.7379	−0.2621	410	80	80.2022	+0.2022	744	132	131.4384	−0.5616
77	21	21.1750	+0.1750	411	80	80.4273	+0.4273	745	132	131.5956	−0.4044
78	21	21.4539	+0.4539	412	80	80.6157	+0.6157	746	132	131.7241	−0.2759
79	22	21.5573	−0.4427	413	80	80.7629	+0.7629	747	132	131.8234	−0.1766
80	22	21.5464	−0.4536	414	80	80.8687	+0.8687	748	132	131.8944	−0.1056
81	22	21.5287	−0.4713	415	80	80.9376	+0.9376	749	132	131.9398	−0.0602
82	22	21.6057	−0.3943	416	80	80.9776	+0.9776	750	132	131.9636	−0.0364
83	23	21.8275	−1.1725	417	80	81.0004	+1.0004	751	133	131.9709	−1.0291
84	23	22.1734	−0.8266	418	80	81.0193	+1.0193	752	133	131.9679	−1.0321
85	23	22.5664	−0.4336	419	81	81.0486	+0.0486	753	133	131.9614	−1.0386
86	23	22.9118	−0.0882	420	81	81.1019	+0.1019	754	133	131.9584	−1.0416
87	23	23.1427	+0.1427	421	82	81.1911	−0.8089	755	133	131.9658	−1.0342
88	23	23.2500	+0.2500	422	82	81.3250	−0.6750	756	133	131.9901	−1.0099
89	24	23.2860	−0.7140	423	82	81.5083	−0.4917	757	134	132.0369	−1.9631
90	24	23.3408	−0.6592	424	82	81.7418	−0.2582	758	134	132.1108	−1.8892
91	24	23.5027	−0.4973	425	82	82.0216	+0.0216	759	134	132.2151	−1.7849
92	24	23.8199	−0.1801	426	82	82.3399	+0.3399	760	134	132.3515	−1.6485
93	24	24.2803	+0.2803	427	82	82.6854	+0.6854	761	135	132.5203	−2.4797
94	24	24.8160	+0.8160	428	82	83.0446	+1.0446	762	135	132.7199	−2.2801
95	24	25.3300	+1.3300	429	82	83.4027	+1.4027	763	135	132.9477	−2.0523
96	24	25.7333	+1.7333	430	82	83.7451	+1.7451	764	135	133.1992	−1.8008
97	25	25.9782	+0.9782	431	83	84.0586	+1.0586	765	135	133.4692	−1.5308
98	25	26.0736	+1.0736	432	83	84.3326	+1.3326	766	135	133.7514	−1.2486
99	25	26.0787	+1.0787	433	84	84.5599	+0.5599	767	135	134.0391	−0.9609
100	25	26.0782	+1.0782	434	84	84.7374	+0.7374	768	135	134.3254	−0.6746
101	26	26.1488	+0.1488	435	84	84.8664	+0.8664	769	136	134.6035	−1.3965
102	26	26.3312	+0.3312	436	84	84.9519	+0.9519	770	136	134.8672	−1.1328
103	27	26.6166	−0.3834	437	84	85.0029	+1.0029	771	136	135.1109	−0.8891
104	27	26.9530	−0.0470	438	84	85.0310	+1.0310	772	136	135.3305	−0.6695
105	27	27.2672	+0.2672	439	85	85.0493	+0.0493	773	137	135.5229	−1.4771
106	27	27.4935	+0.4935	440	85	85.0715	+0.0715	774	137	135.6866	−1.3134
107	28	27.5995	−0.4005	441	85	85.1106	+0.1106	775	137	135.8215	−1.1785
108	28	27.5976	−0.4024	442	85	85.1774	+0.1774	776	137	135.9292	−1.0708
109	29	27.5415	−1.4585	443	86	85.2801	−0.7199	777	137	136.0126	−0.9874
110	29	27.5066	−1.4934	444	86	85.4233	−0.5767	778	137	136.0760	−0.9240
111	29	27.5640	−1.4360	445	86	85.6073	−0.3927	779	137	136.1246	−0.8754
112	29	27.7561	−1.2439	446	86	85.8289	−0.1711	780	137	136.1647	−0.8353
113	30	28.0824	−1.9176	447	86	86.0810	+0.0810	781	137	136.2027	−0.7973
114	30	28.5006	−1.4994	448	86	86.3535	+0.3535	782	137	136.2455	−0.7545
115	30	28.9409	−1.0591	449	87	86.6343	−0.3657	783	137	136.2996	−0.7004
116	30	29.3290	−0.6710	450	87	86.9103	−0.0897	784	137	136.3710	−0.6290
117	30	29.6096	−0.3904	451	87	87.1685	+0.1685	785	137	136.4649	−0.5351
118	30	29.7635	−0.2365	452	87	87.3973	+0.3973	786	137	136.5855	−0.4145
119	30	29.8115	−0.1885	453	87	87.5874	+0.5874	787	138	136.7357	−1.2643
120	30	29.8060	−0.1940	454	87	87.7326	+0.7326	788	138	136.9168	−1.0832
121	30	29.8131	−0.1869	455	87	87.8302	+0.8302	789	138	137.1288	−0.8712
122	30	29.8904	−0.1096	456	87	87.8817	+0.8817	790	138	137.3701	−0.6299
123	30	30.0688	+0.0688	457	88	87.8919	−0.1081	791	138	137.6378	−0.3622
124	30	30.3437	+0.3437	458	88	87.8693	−0.1307	792	138	137.9277	−0.0723
125	30	30.6772	+0.6772	459	88	87.8247	−0.1753	793	138	138.2344	+0.2344
126	30	31.0106	+1.0106	460	88	87.7706	−0.2294	794	138	138.5519	+0.5519

Table 15 (continued)

n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$	n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$	n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$
127	31	31.2839	+0.2839	461	89	87.7202	−1.2798	795	138	138.8737	+0.8737
128	31	31.4262	+0.4262	462	89	87.6859	−1.3141	796	138	139.1931	+1.1931
129	31	31.4309	+0.4309	463	90	87.6789	−2.3211	797	139	139.5033	+0.5033
130	31	31.3888	+0.3888	464	90	87.7075	−2.2925	798	139	139.7985	+0.7985
131	32	31.3009	−0.6991	465	90	87.7772	−2.2228	799	139	140.0731	+1.0731
132	32	31.2297	−0.7703	466	90	87.8896	−2.1104	800	139	140.3230	+1.3230
133	32	31.2332	−0.7668	467	91	88.0428	−2.9572	801	139	140.5448	+1.5448
134	32	31.3489	−0.6511	468	91	88.2316	−2.7684	802	139	140.7369	+1.7369
135	32	31.5837	−0.4163	469	91	88.4474	−2.5526	803	139	140.8989	+1.8989
136	32	31.9130	−0.0870	470	91	88.6797	−2.3203	804	139	141.0319	+2.0319
137	33	32.2892	−0.7108	471	91	88.9165	−2.0835	805	139	141.1383	+2.1383
138	33	32.6552	−0.3448	472	91	89.1457	−1.8543	806	139	141.2217	+2.2217
139	34	32.9609	−1.0391	473	91	89.3559	−1.6441	807	139	141.2868	+2.2868
140	34	33.1772	−0.8228	474	91	89.5374	−1.4626	808	139	141.3390	+2.3390
141	34	33.3035	−0.6965	475	91	89.6832	−1.3168	809	140	141.3844	+1.3844
142	34	33.3672	−0.6328	476	91	89.7892	−1.2108	810	140	141.4291	+1.4291
143	34	33.4154	−0.5846	477	91	89.8551	−1.1449	811	141	141.4791	+0.4791
144	34	33.5018	−0.4982	478	91	89.8840	−1.1160	812	141	141.5402	+0.5402
145	34	33.6717	−0.3283	479	92	89.8825	−2.1175	813	141	141.6172	+0.6172
146	34	33.9498	−0.0502	480	92	89.8598	−2.1402	814	141	141.7142	+0.7142
147	34	34.3339	+0.3339	481	92	89.8272	−2.1728	815	141	141.8341	+0.8341
148	34	34.7950	+0.7950	482	92	89.7972	−2.2028	816	141	141.9785	+0.9785
149	35	35.2855	+0.2855	483	92	89.7826	−2.2174	817	141	142.1478	+1.1478
150	35	35.7509	+0.7509	484	92	89.7951	−2.2049	818	141	142.3406	+1.3406
151	36	36.1439	+0.1439	485	92	89.8448	−2.1552	819	141	142.5547	+1.5547
152	36	36.4354	+0.4354	486	92	89.9392	−2.0608	820	141	142.7865	+1.7865
153	36	36.6214	+0.6214	487	93	90.0827	−2.9173	821	142	143.0312	+1.0312
154	36	36.7228	+0.7228	488	93	90.2763	−2.7237	822	142	143.2835	+1.2835
155	36	36.7794	+0.7794	489	93	90.5175	−2.4825	823	143	143.5374	+0.5374
156	36	36.8394	+0.8394	490	93	90.8004	−2.1996	824	143	143.7866	+0.7866
157	37	36.9470	−0.0530	491	94	91.1164	−2.8836	825	143	144.0251	+1.0251
158	37	37.1310	+0.1310	492	94	91.4550	−2.5450	826	143	144.2471	+1.2471
159	37	37.3987	+0.3987	493	94	91.8042	−2.1958	827	144	144.4474	+0.4474
160	37	37.7339	+0.7339	494	94	92.1517	−1.8483	828	144	144.6216	+0.6216
161	37	38.1016	+1.1016	495	94	92.4862	−1.5138	829	145	144.7668	−0.2332
162	37	38.4570	+1.4570	496	94	92.7974	−1.2026	830	145	144.8808	−0.1192
163	38	38.7565	+0.7565	497	94	93.0777	−0.9223	831	145	144.9633	−0.0367
164	38	38.9685	+0.9685	498	94	93.3222	−0.6778	832	145	145.0148	+0.0148
165	38	39.0808	+1.0808	499	95	93.5290	−1.4710	833	145	145.0377	+0.0377
166	38	39.1038	+1.1038	500	95	93.6998	−1.3002	834	145	145.0353	+0.0353
167	39	39.0679	+0.0679	501	95	93.8391	−1.1609	835	145	145.0122	+0.0122
168	39	39.0165	+0.0165	502	95	93.9540	−1.0460	836	145	144.9737	−0.0263
169	39	38.9962	−0.0038	503	96	94.0538	−1.9462	837	145	144.9260	−0.0740
170	39	39.0462	+0.0462	504	96	94.1488	−1.8512	838	145	144.8755	−0.1245
171	39	39.1901	+0.1901	505	96	94.2499	−1.7501	839	146	144.8288	−1.1712
172	39	39.4304	+0.4304	506	96	94.3672	−1.6328	840	146	144.7922	−1.2078
173	40	39.7489	−0.2511	507	96	94.5096	−1.4904	841	146	144.7718	−1.2282
174	40	40.1109	+0.1109	508	96	94.6838	−1.3162	842	146	144.7726	−1.2274
175	40	40.4729	+0.4729	509	97	94.8939	−2.1061	843	146	144.7989	−1.2011
176	40	40.7926	+0.7926	510	97	95.1411	−1.8589	844	146	144.8537	−1.1463
177	40	41.0375	+1.0375	511	97	95.4231	−1.5769	845	146	144.9389	−1.0611
178	40	41.1922	+1.1922	512	97	95.7351	−1.2649	846	146	145.0550	−0.9450
179	41	41.2604	+0.2604	513	97	96.0695	−0.9305	847	146	145.2012	−0.7988
180	41	41.2640	+0.2640	514	97	96.4166	−0.5834	848	146	145.3753	−0.6247
181	42	41.2376	−0.7624	515	97	96.7654	−0.2346	849	146	145.5742	−0.4258
182	42	41.2205	−0.7795	516	97	97.1045	+0.1045	850	146	145.7935	−0.2065
183	42	41.2481	−0.7519	517	97	97.4230	+0.4230	851	146	146.0283	+0.0283
184	42	41.3444	−0.6556	518	97	97.7110	+0.7110	852	146	146.2729	+0.2729
185	42	41.5167	−0.4833	519	97	97.9605	+0.9605	853	147	146.5216	−0.4784
186	42	41.7547	−0.2453	520	97	98.1664	+1.1664	854	147	146.7686	−0.2314
187	42	42.0325	+0.0325	521	98	98.3262	+0.3262	855	147	147.0082	+0.0082
188	42	42.3150	+0.3150	522	98	98.4406	+0.4406	856	147	147.2356	+0.2356

Table 15 (continued)

n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$	n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$	n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$
189	42	42.5651	+0.5651	523	99	98.5135	−0.4865	857	148	147.4463	−0.5537
190	42	42.7517	+0.7517	524	99	98.5515	−0.4485	858	148	147.6373	−0.3627
191	43	42.8568	−0.1432	525	99	98.5636	−0.4364	859	149	147.8063	−1.1937
192	43	42.8788	−0.1212	526	99	98.5603	−0.4397	860	149	147.9524	−1.0476
193	44	42.8337	−1.1663	527	99	98.5533	−0.4467	861	149	148.0759	−0.9241
194	44	42.7518	−1.2482	528	99	98.5540	−0.4460	862	149	148.1783	−0.8217
195	44	42.6720	−1.3280	529	99	98.5734	−0.4266	863	150	148.2622	−1.7378
196	44	42.6344	−1.3656	530	99	98.6209	−0.3791	864	150	148.3314	−1.6686
197	45	42.6731	−2.3269	531	99	98.7038	−0.2962	865	150	148.3903	−1.6097
198	45	42.8094	−2.1906	532	99	98.8266	−0.1734	866	150	148.4441	−1.5559
199	46	43.0489	−2.9511	533	99	98.9911	−0.0089	867	150	148.4983	−1.5017
200	46	43.3800	−2.6200	534	99	99.1960	+0.1960	868	150	148.5584	−1.4416
201	46	43.7772	−2.2228	535	99	99.4373	+0.4373	869	150	148.6301	−1.3699
202	46	44.2058	−1.7942	536	99	99.7085	+0.7085	870	150	148.7183	−1.2817
203	46	44.6290	−1.3710	537	99	100.0010	+1.0010	871	150	148.8274	−1.1726
204	46	45.0143	−0.9857	538	99	100.3050	+1.3050	872	150	148.9612	−1.0388
205	46	45.3398	−0.6602	539	99	100.6101	+1.6101	873	150	149.1220	−0.8780
206	46	45.5972	−0.4028	540	99	100.9064	+1.9064	874	150	149.3112	−0.6888
207	46	45.7933	−0.2067	541	100	101.1845	+1.1845	875	150	149.5292	−0.4708
208	46	45.9475	−0.0525	542	100	101.4371	+1.4371	876	150	149.7748	−0.2252
209	46	46.0878	+0.0878	543	100	101.6588	+1.6588	877	151	150.0456	−0.9544
210	46	46.2445	+0.2445	544	100	101.8469	+1.8469	878	151	150.3385	−0.6615
211	47	46.4441	−0.5559	545	100	102.0013	+2.0013	879	151	150.6491	−0.3509
212	47	46.7039	−0.2961	546	100	102.1247	+2.1247	880	151	150.9722	−0.0278
213	47	47.0284	+0.0284	547	101	102.2224	+1.2224	881	152	151.3022	−0.6978
214	47	47.4081	+0.4081	548	101	102.3015	+1.3015	882	152	151.6331	−0.3669
215	47	47.8212	+0.8212	549	101	102.3709	+1.3709	883	153	151.9590	−1.0410
216	47	48.2370	+1.2370	550	101	102.4403	+1.4403	884	153	152.2739	−0.7261
217	47	48.6217	+1.6217	551	101	102.5195	+1.5195	885	153	152.5726	−0.4274
218	47	48.9439	+1.9439	552	101	102.6176	+1.6176	886	153	152.8503	−0.1497
219	47	49.1808	+2.1808	553	101	102.7427	+1.7427	887	154	153.1032	−0.8968
220	47	49.3215	+2.3215	554	101	102.9009	+1.9009	888	154	153.3286	−0.6714
221	47	49.3692	+2.3692	555	101	103.0958	+2.0958	889	154	153.5247	−0.4753
222	47	49.3409	+2.3409	556	101	103.3287	+2.3287	890	154	153.6913	−0.3087
223	48	49.2643	+1.2643	557	102	103.5980	+1.5980	891	154	153.8292	−0.1708
224	48	49.1732	+1.1732	558	102	103.8993	+1.8993	892	154	153.9403	−0.0597
225	48	49.1020	+1.1020	559	102	104.2263	+2.2263	893	154	154.0277	+0.0277
226	48	49.0805	+1.0805	560	102	104.5702	+2.5702	894	154	154.0955	+0.0955
227	49	49.1289	+0.1289	561	102	104.9213	+2.9213	895	154	154.1485	+0.1485
228	49	49.2550	+0.2550	562	102	105.2691	+3.2691	896	154	154.1920	+0.1920
229	50	49.4539	−0.5461	563	103	105.6029	+2.6029	897	154	154.2316	+0.2316
230	50	49.7087	−0.2913	564	103	105.9130	+2.9130	898	154	154.2732	+0.2732
231	50	49.9947	−0.0053	565	103	106.1909	+3.1909	899	154	154.3223	+0.3223
232	50	50.2833	+0.2833	566	103	106.4302	+3.4302	900	154	154.3839	+0.3839
233	51	50.5477	−0.4523	567	103	106.6267	+3.6267	901	154	154.4626	+0.4626
234	51	50.7670	−0.2330	568	103	106.7790	+3.7790	902	154	154.5620	+0.5620
235	51	50.9301	−0.0699	569	104	106.8882	+2.8882	903	154	154.6848	+0.6848
236	51	51.0374	+0.0374	570	104	106.9580	+2.9580	904	154	154.8326	+0.8326
237	51	51.1004	+0.1004	571	105	106.9947	+1.9947	905	154	155.0058	+1.0058
238	51	51.1399	+0.1399	572	105	107.0061	+2.0061	906	154	155.2036	+1.2036
239	52	51.1820	−0.8180	573	105	107.0016	+2.0016	907	155	155.4241	+0.4241
240	52	51.2538	−0.7462	574	105	106.9912	+1.9912	908	155	155.6646	+0.6646
241	53	51.3786	−1.6214	575	105	106.9849	+1.9849	909	155	155.9211	+0.9211
242	53	51.5719	−1.4281	576	105	106.9921	+1.9921	910	155	156.1892	+1.1892
243	53	51.8390	−1.1610	577	106	107.0208	+1.0208	911	156	156.4638	+0.4638
244	53	52.1735	−0.8265	578	106	107.0774	+1.0774	912	156	156.7396	+0.7396
245	53	52.5584	−0.4416	579	106	107.1659	+1.1659	913	156	157.0111	+1.0111
246	53	52.9686	−0.0314	580	106	107.2879	+1.2879	914	156	157.2732	+1.2732
247	53	53.3748	+0.3748	581	106	107.4425	+1.4425	915	156	157.5210	+1.5210
248	53	53.7477	+0.7477	582	106	107.6265	+1.6265	916	156	157.7502	+1.7502
249	53	54.0624	+1.0624	583	106	107.8343	+1.8343	917	156	157.9576	+1.9576
250	53	54.3022	+1.3022	584	106	108.0587	+2.0587	918	156	158.1405	+2.1405

Table 15 (continued)

n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$	n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$	n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$
251	54	54.4611	+0.4611	585	106	108.2913	+2.2913	919	157	158.2976	+1.2976
252	54	54.5441	+0.5441	586	106	108.5228	+2.5228	920	157	158.4286	+1.4286
253	54	54.5667	+0.5667	587	107	108.7442	+1.7442	921	157	158.5342	+1.5342
254	54	54.5521	+0.5521	588	107	108.9470	+1.9470	922	157	158.6165	+1.6165
255	54	54.5279	+0.5279	589	107	109.1239	+2.1239	923	157	158.6782	+1.6782
256	54	54.5215	+0.5215	590	107	109.2693	+2.2693	924	157	158.7231	+1.7231
257	55	54.5565	−0.4435	591	107	109.3797	+2.3797	925	157	158.7558	+1.7558
258	55	54.6491	−0.3509	592	107	109.4541	+2.4541	926	157	158.7812	+1.7812
259	55	54.8061	−0.1939	593	108	109.4938	+1.4938	927	157	158.8048	+1.8048
260	55	55.0238	+0.0238	594	108	109.5023	+1.5023	928	157	158.8318	+1.8318
261	55	55.2892	+0.2892	595	108	109.4855	+1.4855	929	158	158.8676	+0.8676
262	55	55.5823	+0.5823	596	108	109.4508	+1.4508	930	158	158.9170	+0.9170
263	56	55.8792	−0.1208	597	108	109.4068	+1.4068	931	158	158.9844	+0.9844
264	56	56.1558	+0.1558	598	108	109.3630	+1.3630	932	158	159.0733	+1.0733
265	56	56.3917	+0.3917	599	109	109.3287	+0.3287	933	158	159.1864	+1.1864
266	56	56.5732	+0.5732	600	109	109.3130	+0.3130	934	158	159.3253	+1.3253
267	56	56.6956	+0.6956	601	110	109.3234	−0.6766	935	158	159.4905	+1.4905
268	56	56.7634	+0.7634	602	110	109.3662	−0.6338	936	158	159.6815	+1.6815
269	57	56.7900	−0.2100	603	110	109.4455	−0.5545	937	159	159.8968	+0.8968
270	57	56.7958	−0.2042	604	110	109.5632	−0.4368	938	159	160.1335	+1.1335
271	58	56.8046	−1.1954	605	110	109.7189	−0.2811	939	159	160.3883	+1.3883
272	58	56.8408	−1.1592	606	110	109.9096	−0.0904	940	159	160.6569	+1.6569
273	58	56.9253	−1.0747	607	111	110.1306	−0.8694	941	160	160.9344	+0.9344
274	58	57.0729	−0.9271	608	111	110.3750	−0.6250	942	160	161.2157	+1.2157
275	58	57.2897	−0.7103	609	111	110.6348	−0.3652	943	160	161.4955	+1.4955
276	58	57.5728	−0.4272	610	111	110.9012	−0.0988	944	160	161.7684	+1.7684
277	59	57.9101	−1.0899	611	111	111.1651	+0.1651	945	160	162.0296	+2.0296
278	59	58.2824	−0.7176	612	111	111.4176	+0.4176	946	160	162.2746	+2.2746
279	59	58.6658	−0.3342	613	112	111.6510	−0.3490	947	161	162.4997	+1.4997
280	59	59.0351	+0.0351	614	112	111.8587	−0.1413	948	161	162.7019	+1.7019
281	60	59.3670	−0.6330	615	112	112.0362	+0.0362	949	161	162.8794	+1.8794
282	60	59.6434	−0.3566	616	112	112.1810	+0.1810	950	161	163.0313	+2.0313
283	61	59.8537	−1.1463	617	113	112.2929	−0.7071	951	161	163.1576	+2.1576
284	61	59.9959	−1.0041	618	113	112.3739	−0.6261	952	161	163.2597	+2.2597
285	61	60.0769	−0.9231	619	114	112.4281	−1.5719	953	162	163.3398	+1.3398
286	61	60.1114	−0.8886	620	114	112.4616	−1.5384	954	162	163.4011	+1.4011
287	61	60.1194	−0.8806	621	114	112.4819	−1.5181	955	162	163.4476	+1.4476
288	61	60.1240	−0.8760	622	114	112.4975	−1.5025	956	162	163.4838	+1.4838
289	61	60.1476	−0.8524	623	114	112.5174	−1.4826	957	162	163.5149	+1.5149
290	61	60.2092	−0.7908	624	114	112.5504	−1.4496	958	162	163.5459	+1.5459
291	61	60.3219	−0.6781	625	114	112.6047	−1.3953	959	162	163.5822	+1.5822
292	61	60.4908	−0.5092	626	114	112.6872	−1.3128	960	162	163.6289	+1.6289
293	62	60.7132	−1.2868	627	114	112.8035	−1.1965	961	162	163.6906	+1.6906
294	62	60.9782	−1.0218	628	114	112.9568	−1.0432	962	162	163.7714	+1.7714
295	62	61.2689	−0.7311	629	114	113.1486	−0.8514	963	162	163.8746	+1.8746
296	62	61.5644	−0.4356	630	114	113.3778	−0.6222	964	162	164.0025	+2.0025
297	62	61.8426	−0.1574	631	115	113.6414	−1.3586	965	162	164.1567	+2.1567
298	62	62.0837	+0.0837	632	115	113.9343	−1.0657	966	162	164.3375	+2.3375
299	62	62.2721	+0.2721	633	115	114.2497	−0.7503	967	163	164.5442	+1.5442
300	62	62.3988	+0.3988	634	115	114.5798	−0.4202	968	163	164.7752	+1.7752
301	62	62.4627	+0.4627	635	115	114.9157	−0.0843	969	163	165.0277	+2.0277
302	62	62.4706	+0.4706	636	115	115.2485	+0.2485	970	163	165.2984	+2.2984
303	62	62.4359	+0.4359	637	115	115.5696	+0.5696	971	164	165.5829	+1.5829
304	62	62.3775	+0.3775	638	115	115.8712	+0.8712	972	164	165.8767	+1.8767
305	62	62.3171	+0.3171	639	115	116.1468	+1.1468	973	164	166.1745	+2.1745
306	62	62.2761	+0.2761	640	115	116.3915	+1.3915	974	164	166.4713	+2.4713
307	63	62.2737	−0.7263	641	116	116.6025	+0.6025	975	164	166.7618	+2.7618
308	63	62.3237	−0.6763	642	116	116.7789	+0.7789	976	164	167.0413	+3.0413
309	63	62.4333	−0.5667	643	117	116.9220	−0.0780	977	165	167.3052	+2.3052
310	63	62.6023	−0.3977	644	117	117.0351	+0.0351	978	165	167.5498	+2.5498
311	64	62.8231	−1.1769	645	117	117.1231	+0.1231	979	165	167.7721	+2.7721
312	64	63.0819	−0.9181	646	117	117.1927	+0.1927	980	165	167.9700	+2.9700

Table 15 (continued)

n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$	n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$	n	$\pi(n)$	$\pi_{\mathrm{RVM}}(n)$	$\pi_{\mathrm{RVM}}(n)-\pi(n)$
313	65	63.3604	−1.6396	647	118	117.2512	−0.7488	981	165	168.1423	+3.1423
314	65	63.6384	−1.3616	648	118	117.3067	−0.6933	982	165	168.2889	+3.2889
315	65	63.8963	−1.1037	649	118	117.3671	−0.6329	983	166	168.4107	+2.4107
316	65	64.1173	−0.8827	650	118	117.4400	−0.5600	984	166	168.5093	+2.5093
317	66	64.2901	−1.7099	651	118	117.5319	−0.4681	985	166	168.5876	+2.5876
318	66	64.4097	−1.5903	652	118	117.6482	−0.3518	986	166	168.6489	+2.6489
319	66	64.4786	−1.5214	653	119	117.7922	−1.2078	987	166	168.6972	+2.6972
320	66	64.5058	−1.4942	654	119	117.9655	−1.0345	988	166	168.7371	+2.7371
321	66	64.5063	−1.4937	655	119	118.1678	−0.8322	989	166	168.7732	+2.7732
322	66	64.4990	−1.5010	656	119	118.3967	−0.6033	990	166	168.8103	+2.8103
323	66	64.5045	−1.4955	657	119	118.6479	−0.3521	991	167	168.8530	+1.8530
324	66	64.5423	−1.4577	658	119	118.9155	−0.0845	992	167	168.9057	+1.9057
325	66	64.6291	−1.3709	659	120	119.1926	−0.8074	993	167	168.9722	+1.9722
326	66	64.7761	−1.2239	660	120	119.4712	−0.5288	994	167	169.0555	+2.0555
327	66	64.9883	−1.0117	661	121	119.7432	−1.2568	995	167	169.1579	+2.1579
328	66	65.2637	−0.7363	662	121	120.0004	−0.9996	996	167	169.2810	+2.2810
329	66	65.5934	−0.4066	663	121	120.2355	−0.7645	997	168	169.4252	+1.4252
330	66	65.9631	−0.0369	664	121	120.4422	−0.5578	998	168	169.5901	+1.5901
331	67	66.3543	−0.6457	665	121	120.6154	−0.3846	999	168	169.7741	+1.7741
332	67	66.7469	−0.2531	666	121	120.7523	−0.2477	1000	168	169.9750	+1.9750
333	67	67.1211	+0.1211	667	121	120.8515	−0.1485				
334	67	67.4600	+0.4600	668	121	120.9140	−0.0860				

Table 16: The Möbius function $\mu(n)$ and the Mertens function $M(n)$, exact (sieve), and the critical RVM Mertens function $M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$, approximate (the tapered explicit formula of program `s05` with the residues computed from the product for $k \le 1000$ and modelled beyond), at the integers $n \le 1000$, with the difference $M^{\mathrm{crit}}_{\mathrm{RVM}}(n) - M(n)$.

n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$	n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$	n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$
1	1	1	0.2925	−0.7075	335	1	2	−1.9107	−3.9107	669	1	−10	−1.6301	+8.3699
2	−1	0	0.4563	+0.4563	336	0	2	−2.2151	−4.2151	670	−1	−11	−1.2832	+9.7168
3	−1	−1	−0.6523	+0.3477	337	−1	1	−2.3953	−3.3953	671	1	−10	−0.8918	+9.1082
4	0	−1	−0.8720	+0.1280	338	0	1	−2.4640	−3.4640	672	0	−10	−0.4714	+9.5286
5	−1	−2	−1.7255	+0.2745	339	1	2	−2.4456	−4.4456	673	−1	−11	−0.0388	+10.9612
6	1	−1	−1.3063	−0.3063	340	0	2	−2.3727	−4.3727	674	1	−10	0.3882	+10.3882
7	−1	−2	−1.5931	+0.4069	341	1	3	−2.2821	−5.2821	675	0	−10	0.7923	+10.7923
8	0	−2	−2.4029	−0.4029	342	0	3	−2.2108	−5.2108	676	0	−10	1.1575	+11.1575
9	0	−2	−1.3292	+0.6708	343	0	3	−2.1911	−5.1911	677	−1	−11	1.4699	+12.4699
10	1	−1	−1.3280	−0.3280	344	0	3	−2.2470	−5.2470	678	−1	−12	1.7185	+13.7185
11	−1	−2	−2.3272	−0.3272	345	−1	2	−2.3914	−4.3914	679	1	−11	1.8958	+12.8958
12	0	−2	−2.2067	−0.2067	346	1	3	−2.6249	−5.6249	680	0	−11	1.9982	+12.9982
13	−1	−3	−1.8468	+1.1532	347	−1	2	−2.9356	−4.9356	681	1	−10	2.0263	+12.0263
14	1	−2	−1.9324	+0.0676	348	0	2	−3.3007	−5.3007	682	−1	−11	1.9844	+12.9844
15	1	−1	−1.5435	−0.5435	349	−1	1	−3.6892	−4.6892	683	−1	−12	1.8806	+13.8806
16	0	−1	−1.0414	−0.0414	350	0	1	−4.0654	−5.0654	684	0	−12	1.7261	+13.7261
17	−1	−2	−1.6970	+0.3030	351	0	1	−4.3933	−5.3933	685	1	−11	1.5347	+12.5347
18	0	−2	−3.0095	−1.0095	352	0	1	−4.6406	−5.6406	686	0	−11	1.3217	+12.3217
19	−1	−3	−3.2869	−0.2869	353	−1	0	−4.7824	−4.7824	687	1	−10	1.1032	+11.1032
20	0	−3	−2.2604	+0.7396	354	−1	−1	−4.8043	−3.8043	688	0	−10	0.8952	+10.8952
21	1	−2	−1.2515	+0.7485	355	1	0	−4.7038	−4.7038	689	1	−9	0.7125	+9.7125
22	1	−1	−1.1120	−0.1120	356	0	0	−4.4909	−4.4909	690	1	−8	0.5678	+8.5678
23	−1	−2	−1.3607	+0.6393	357	−1	−1	−4.1872	−3.1872	691	−1	−9	0.4714	+9.4714
24	0	−2	−1.3669	+0.6331	358	1	0	−3.8234	−3.8234	692	0	−9	0.4302	+9.4302
25	0	−2	−1.3810	+0.6190	359	−1	−1	−3.4364	−2.4364	693	0	−9	0.4474	+9.4474
26	1	−1	−1.9805	−0.9805	360	0	−1	−3.0652	−2.0652	694	1	−8	0.5225	+8.5225
27	0	−1	−3.0141	−2.0141	361	0	−1	−2.7470	−1.7470	695	1	−7	0.6512	+7.6512
28	0	−1	−3.6155	−2.6155	362	1	0	−2.5134	−2.5134	696	0	−7	0.8259	+7.8259

Table 16 (continued)

n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$	n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$	n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$
29	−1	−2	−3.2066	−1.2066	363	0	0	−2.3872	−2.3872	697	1	−6	1.0357	+7.0357
30	−1	−3	−2.1599	+0.8401	364	0	0	−2.3801	−2.3801	698	1	−5	1.2675	+6.2675
31	−1	−4	−1.3649	+2.6351	365	1	1	−2.4917	−3.4917	699	1	−4	1.5065	+5.5065
32	0	−4	−1.2973	+2.7027	366	−1	0	−2.7101	−2.7101	700	0	−4	1.7374	+5.7374
33	1	−3	−1.6435	+1.3565	367	−1	−1	−3.0129	−2.0129	701	−1	−5	1.9448	+6.9448
34	1	−2	−1.7783	+0.2217	368	0	−1	−3.3700	−2.3700	702	0	−5	2.1143	+7.1143
35	1	−1	−1.4711	−0.4711	369	0	−1	−3.7470	−2.7470	703	1	−4	2.2335	+6.2335
36	0	−1	−1.0723	−0.0723	370	−1	−2	−4.1086	−2.1086	704	0	−4	2.2926	+6.2926
37	−1	−2	−1.0774	+0.9226	371	1	−1	−4.4228	−3.4228	705	−1	−5	2.2849	+7.2849
38	1	−1	−1.5897	−0.5897	372	0	−1	−4.6640	−3.6640	706	1	−4	2.2071	+6.2071
39	1	0	−2.2227	−2.2227	373	−1	−2	−4.8153	−2.8153	707	1	−3	2.0598	+5.0598
40	0	0	−2.4881	−2.4881	374	−1	−3	−4.8708	−1.8708	708	0	−3	1.8473	+4.8473
41	−1	−1	−2.2590	−1.2590	375	0	−3	−4.8354	−1.8354	709	−1	−4	1.5774	+5.5774
42	−1	−2	−1.8775	+0.1225	376	0	−3	−4.7247	−1.7247	710	−1	−5	1.2609	+6.2609
43	−1	−3	−1.8384	+1.1616	377	1	−2	−4.5625	−2.5625	711	0	−5	0.9113	+5.9113
44	0	−3	−2.3559	+0.6441	378	0	−2	−4.3787	−2.3787	712	0	−5	0.5438	+5.5438
45	0	−3	−3.1822	−0.1822	379	−1	−3	−4.2057	−1.2057	713	1	−4	0.1745	+4.1745
46	1	−2	−3.8050	−1.8050	380	0	−3	−4.0751	−1.0751	714	1	−3	−0.1803	+2.8197
47	−1	−3	−3.8344	−0.8344	381	1	−2	−4.0139	−2.0139	715	−1	−4	−0.5051	+3.4949
48	0	−3	−3.2670	−0.2670	382	1	−1	−4.0420	−3.0420	716	0	−4	−0.7856	+3.2144
49	0	−3	−2.4450	+0.5550	383	−1	−2	−4.1698	−2.1698	717	1	−3	−1.0104	+1.9896
50	0	−3	−1.7747	+1.2253	384	0	−2	−4.3972	−2.3972	718	1	−2	−1.1706	+0.8294
51	1	−2	−1.4374	+0.5626	385	−1	−3	−4.7131	−1.7131	719	−1	−3	−1.2606	+1.7394
52	0	−2	−1.3016	+0.6984	386	1	−2	−5.0972	−3.0972	720	0	−3	−1.2785	+1.7215
53	−1	−3	−1.0780	+1.9220	387	0	−2	−5.5213	−3.5213	721	1	−2	−1.2263	+0.7737
54	0	−3	−0.5793	+2.4207	388	0	−2	−5.9522	−3.9522	722	0	−2	−1.1091	+0.8909
55	1	−2	0.1096	+2.1096	389	−1	−3	−6.3553	−3.3553	723	1	−1	−0.9355	+0.0645
56	0	−2	0.6666	+2.6666	390	1	−2	−6.6975	−4.6975	724	0	−1	−0.7168	+0.2832
57	1	−1	0.7366	+1.7366	391	1	−1	−6.9507	−5.9507	725	0	−1	−0.4661	+0.5339
58	1	0	0.1610	+0.1610	392	0	−1	−7.0945	−6.0945	726	0	−1	−0.1982	+0.8018
59	−1	−1	−0.9166	+0.0834	393	1	0	−7.1179	−7.1179	727	−1	−2	0.0717	+2.0717
60	0	−1	−2.1293	−1.1293	394	1	1	−7.0205	−8.0205	728	0	−2	0.3288	+2.3288
61	−1	−2	−3.0912	−1.0912	395	1	2	−6.8123	−8.8123	729	0	−2	0.5592	+2.5592
62	1	−1	−3.5946	−2.5946	396	0	2	−6.5126	−8.5126	730	−1	−3	0.7512	+3.7512
63	0	−1	−3.6905	−2.6905	397	−1	1	−6.1482	−7.1482	731	1	−2	0.8951	+2.8951
64	0	−1	−3.6144	−2.6144	398	1	2	−5.7506	−7.7506	732	0	−2	0.9845	+2.9845
65	1	0	−3.6166	−3.6166	399	−1	1	−5.3528	−6.3528	733	−1	−3	1.0163	+4.0163
66	−1	−1	−3.8046	−2.8046	400	0	1	−4.9865	−5.9865	734	1	−2	0.9906	+2.9906
67	−1	−2	−4.0854	−2.0854	401	−1	0	−4.6786	−4.6786	735	0	−2	0.9111	+2.9111
68	0	−2	−4.2334	−2.2334	402	−1	−1	−4.4491	−3.4491	736	0	−2	0.7846	+2.7846
69	1	−1	−4.0355	−3.0355	403	1	0	−4.3091	−4.3091	737	1	−1	0.6209	+1.6209
70	−1	−2	−3.4264	−1.4264	404	0	0	−4.2600	−4.2600	738	0	−1	0.4316	+1.4316
71	−1	−3	−2.5386	+0.4614	405	0	0	−4.2936	−4.2936	739	−1	−2	0.2304	+2.2304
72	0	−3	−1.6446	+1.3554	406	−1	−1	−4.3929	−3.3929	740	0	−2	0.0314	+2.0314
73	−1	−4	−1.0258	+2.9742	407	1	0	−4.5343	−4.5343	741	−1	−3	−0.1510	+2.8490
74	1	−3	−0.8384	+2.1616	408	0	0	−4.6898	−4.6898	742	−1	−4	−0.3032	+3.6968
75	0	−3	−1.0461	+1.9539	409	−1	−1	−4.8299	−3.8299	743	−1	−5	−0.4131	+4.5869
76	0	−3	−1.4474	+1.5526	410	−1	−2	−4.9265	−2.9265	744	0	−5	−0.4709	+4.5291
77	1	−2	−1.7796	+0.2204	411	1	−1	−4.9563	−3.9563	745	1	−4	−0.4691	+3.5309
78	−1	−3	−1.8446	+1.1554	412	0	−1	−4.9022	−3.9022	746	1	−3	−0.4038	+2.5962
79	−1	−4	−1.5955	+2.4045	413	1	0	−4.7557	−4.7557	747	0	−3	−0.2740	+2.7260
80	0	−4	−1.1456	+2.8544	414	0	0	−4.5177	−4.5177	748	0	−3	−0.0824	+2.9176
81	0	−4	−0.7025	+3.2975	415	1	1	−4.1981	−5.1981	749	1	−2	0.1652	+2.1652
82	1	−3	−0.4619	+2.5381	416	0	1	−3.8154	−4.8154	750	0	−2	0.4599	+2.4599
83	−1	−4	−0.5149	+3.4851	417	1	2	−3.3945	−5.3945	751	−1	−3	0.7902	+3.7902
84	0	−4	−0.8092	+3.1908	418	−1	1	−2.9648	−3.9648	752	0	−3	1.1429	+4.1429
85	1	−3	−1.1801	+1.8199	419	−1	0	−2.5574	−2.5574	753	1	−2	1.5031	+3.5031
86	1	−2	−1.4316	+0.5684	420	0	0	−2.2018	−2.2018	754	−1	−3	1.8555	+4.8555
87	1	−1	−1.4295	−0.4295	421	−1	−1	−1.9238	−0.9238	755	1	−2	2.1852	+4.1852
88	0	−1	−1.1643	−0.1643	422	1	0	−1.7429	−1.7429	756	0	−2	2.4781	+4.4781

Table 16 (continued)

n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$	n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$	n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$
89	−1	−2	−0.7568	+1.2432	423	0	0	−1.6704	−1.6704	757	−1	−3	2.7216	+5.7216
90	0	−2	−0.4069	+1.5931	424	0	0	−1.7084	−1.7084	758	1	−2	2.9057	+4.9057
91	1	−1	−0.3096	+0.6904	425	0	0	−1.8499	−1.8499	759	−1	−3	3.0229	+6.0229
92	0	−1	−0.5747	+0.4253	426	−1	−1	−2.0792	−1.0792	760	0	−3	3.0689	+6.0689
93	1	0	−1.1827	−1.1827	427	1	0	−2.3734	−2.3734	761	−1	−4	3.0429	+7.0429
94	1	1	−1.9934	−2.9934	428	0	0	−2.7045	−2.7045	762	−1	−5	2.9470	+7.9470
95	1	2	−2.8017	−4.8017	429	−1	−1	−3.0420	−2.0420	763	1	−4	2.7868	+6.7868
96	0	2	−3.4154	−5.4154	430	−1	−2	−3.3553	−1.3553	764	0	−4	2.5708	+6.5708
97	−1	1	−3.7240	−4.7240	431	−1	−3	−3.6166	−0.6166	765	0	−4	2.3096	+6.3096
98	0	1	−3.7325	−4.7325	432	0	−3	−3.8034	−0.8034	766	1	−3	2.0159	+5.0159
99	0	1	−3.5497	−4.5497	433	−1	−4	−3.9000	+0.1000	767	1	−2	1.7034	+3.7034
100	0	1	−3.3365	−4.3365	434	−1	−5	−3.8991	+1.1009	768	0	−2	1.3864	+3.3864
101	−1	0	−3.2390	−3.2390	435	−1	−6	−3.8019	+2.1981	769	−1	−3	1.0788	+4.0788
102	−1	−1	−3.3304	−2.3304	436	0	−6	−3.6182	+2.3818	770	1	−2	0.7935	+2.7935
103	−1	−2	−3.5847	−1.5847	437	1	−5	−3.3650	+1.6350	771	1	−1	0.5419	+1.5419
104	0	−2	−3.8897	−1.8897	438	−1	−6	−3.0650	+2.9350	772	0	−1	0.3331	+1.3331
105	−1	−3	−4.0934	−1.0934	439	−1	−7	−2.7442	+4.2558	773	−1	−2	0.1733	+2.1733
106	1	−2	−4.0637	−2.0637	440	0	−7	−2.4294	+4.5706	774	0	−2	0.0662	+2.0662
107	−1	−3	−3.7411	−0.7411	441	0	−7	−2.1460	+4.8540	775	0	−2	0.0121	+2.0121
108	0	−3	−3.1633	−0.1633	442	−1	−8	−1.9153	+6.0847	776	0	−2	0.0081	+2.0081
109	−1	−4	−2.4549	+1.5451	443	−1	−9	−1.7527	+7.2473	777	−1	−3	0.0486	+3.0486
110	−1	−5	−1.7869	+3.2131	444	0	−9	−1.6662	+7.3338	778	1	−2	0.1253	+2.1253
111	1	−4	−1.3206	+2.6794	445	1	−8	−1.6557	+6.3443	779	1	−1	0.2275	+1.2275
112	0	−4	−1.1564	+2.8436	446	1	−7	−1.7130	+5.2870	780	0	−1	0.3432	+1.3432
113	−1	−5	−1.3030	+3.6970	447	1	−6	−1.8225	+4.1775	781	1	0	0.4593	+0.4593
114	−1	−6	−1.6789	+4.3211	448	0	−6	−1.9625	+4.0375	782	−1	−1	0.5624	+1.5624
115	1	−5	−2.1415	+2.8585	449	−1	−7	−2.1074	+4.8926	783	0	−1	0.6396	+1.6396
116	0	−5	−2.5354	+2.4646	450	0	−7	−2.2295	+4.7705	784	0	−1	0.6791	+1.6791
117	0	−5	−2.7417	+2.2583	451	1	−6	−2.3017	+3.6983	785	1	0	0.6706	+0.6706
118	1	−4	−2.7129	+1.2871	452	0	−6	−2.3001	+3.6999	786	−1	−1	0.6064	+1.6064
119	1	−3	−2.4825	+0.5175	453	1	−5	−2.2054	+2.7946	787	−1	−2	0.4812	+2.4812
120	0	−3	−2.1486	+0.8514	454	1	−4	−2.0055	+1.9945	788	0	−2	0.2928	+2.2928
121	0	−3	−1.8366	+1.1634	455	−1	−5	−1.6958	+3.3042	789	1	−1	0.0420	+1.0420
122	1	−2	−1.6547	+0.3453	456	0	−5	−1.2799	+3.7201	790	−1	−2	−0.2672	+1.7328
123	1	−1	−1.6571	−0.6571	457	−1	−6	−0.7695	+5.2305	791	1	−1	−0.6282	+0.3718
124	0	−1	−1.8256	−0.8256	458	1	−5	−0.1834	+4.8166	792	0	−1	−1.0314	−0.0314
125	0	−1	−2.0757	−1.0757	459	0	−5	0.4544	+5.4544	793	1	0	−1.4653	−1.4653
126	0	−1	−2.2829	−1.2829	460	0	−5	1.1161	+6.1161	794	1	1	−1.9169	−2.9169
127	−1	−2	−2.3227	−0.3227	461	−1	−6	1.7729	+7.7729	795	−1	0	−2.3720	−2.3720
128	0	−2	−2.1094	−0.1094	462	1	−5	2.3966	+7.3966	796	0	0	−2.8161	−2.8161
129	1	−1	−1.6233	−0.6233	463	−1	−6	2.9627	+8.9627	797	−1	−1	−3.2354	−2.2354
130	−1	−2	−0.9186	+1.0814	464	0	−6	3.4515	+9.4515	798	1	0	−3.6166	−3.6166
131	−1	−3	−0.1091	+2.8909	465	−1	−7	3.8501	+10.8501	799	1	1	−3.9486	−4.9486
132	0	−3	0.6624	+3.6624	466	1	−6	4.1531	+10.1531	800	0	1	−4.2220	−5.2220
133	1	−2	1.2621	+3.2621	467	−1	−7	4.3627	+11.3627	801	0	1	−4.4302	−5.4302
134	1	−1	1.5996	+2.5996	468	0	−7	4.4884	+11.4884	802	1	2	−4.5695	−6.5695
135	0	−1	1.6493	+2.6493	469	1	−6	4.5463	+10.5463	803	1	3	−4.6392	−7.6392
136	0	−1	1.4524	+2.4524	470	−1	−7	4.5568	+11.5568	804	0	3	−4.6414	−7.6414
137	−1	−2	1.1017	+3.1017	471	1	−6	4.5436	+10.5436	805	−1	2	−4.5815	−6.5815
138	−1	−3	0.7119	+3.7119	472	0	−6	4.5309	+10.5309	806	−1	1	−4.4670	−5.4670
139	−1	−4	0.3857	+4.3857	473	1	−5	4.5416	+9.5416	807	1	2	−4.3080	−6.3080
140	0	−4	0.1858	+4.1858	474	−1	−6	4.5951	+10.5951	808	0	2	−4.1160	−6.1160
141	1	−3	0.1181	+3.1181	475	0	−6	4.7056	+10.7056	809	−1	1	−3.9038	−4.9038
142	1	−2	0.1329	+2.1329	476	0	−6	4.8810	+10.8810	810	0	1	−3.6845	−4.6845
143	1	−1	0.1413	+1.1413	477	0	−6	5.1221	+11.1221	811	−1	0	−3.4712	−3.4712
144	0	−1	0.0424	+1.0424	478	1	−5	5.4224	+10.4224	812	0	0	−3.2760	−3.2760
145	1	0	−0.2466	−0.2466	479	−1	−6	5.7686	+11.7686	813	1	1	−3.1096	−4.1096
146	1	1	−0.7654	−1.7654	480	0	−6	6.1420	+12.1420	814	−1	0	−2.9809	−2.9809
147	0	1	−1.4974	−2.4974	481	1	−5	6.5195	+11.5195	815	1	1	−2.8965	−3.8965
148	0	1	−2.3724	−3.3724	482	1	−4	6.8760	+10.8760	816	0	1	−2.8600	−3.8600

Table 16 (continued)

n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$	n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$	n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$
149	−1	0	−3.2821	−3.2821	483	−1	−5	7.1859	+12.1859	817	1	2	−2.8726	−4.8726
150	0	0	−4.1063	−4.1063	484	0	−5	7.4257	+12.4257	818	1	3	−2.9323	−5.9323
151	−1	−1	−4.7412	−3.7412	485	1	−4	7.5754	+11.5754	819	0	3	−3.0347	−6.0347
152	0	−1	−5.1236	−4.1236	486	0	−4	7.6205	+11.6205	820	0	3	−3.1727	−6.1727
153	0	−1	−5.2446	−4.2446	487	−1	−5	7.5531	+12.5531	821	−1	2	−3.3370	−5.3370
154	−1	−2	−5.1491	−3.1491	488	0	−5	7.3724	+12.3724	822	−1	1	−3.5168	−4.5168
155	1	−1	−4.9234	−3.9234	489	1	−4	7.0847	+11.0847	823	−1	0	−3.7000	−3.7000
156	0	−1	−4.6723	−3.6723	490	0	−4	6.7033	+10.7033	824	0	0	−3.8742	−3.8742
157	−1	−2	−4.4931	−2.4931	491	−1	−5	6.2472	+11.2472	825	0	0	−4.0268	−4.0268
158	1	−1	−4.4525	−3.4525	492	0	−5	5.7396	+10.7396	826	−1	−1	−4.1461	−3.1461
159	1	0	−4.5718	−4.5718	493	1	−4	5.2061	+9.2061	827	−1	−2	−4.2217	−2.2217
160	0	0	−4.8235	−4.8235	494	−1	−5	4.6731	+9.6731	828	0	−2	−4.2449	−2.2449
161	1	1	−5.1402	−6.1402	495	0	−5	4.1654	+9.1654	829	−1	−3	−4.2092	−1.2092
162	0	1	−5.4332	−6.4332	496	0	−5	3.7045	+8.7045	830	−1	−4	−4.1110	−0.1110
163	−1	0	−5.6150	−5.6150	497	1	−4	3.3071	+7.3071	831	1	−3	−3.9489	−0.9489
164	0	0	−5.6219	−5.6219	498	−1	−5	2.9838	+7.9838	832	0	−3	−3.7249	−0.7249
165	−1	−1	−5.4291	−4.4291	499	−1	−6	2.7387	+8.7387	833	0	−3	−3.4434	−0.4434
166	1	0	−5.0574	−5.0574	500	0	−6	2.5687	+8.5687	834	−1	−4	−3.1115	+0.8885
167	−1	−1	−4.5679	−3.5679	501	1	−5	2.4646	+7.4646	835	1	−3	−2.7386	+0.2614
168	0	−1	−4.0477	−3.0477	502	1	−4	2.4114	+6.4114	836	0	−3	−2.3359	+0.6641
169	0	−1	−3.5897	−2.5897	503	−1	−5	2.3899	+7.3899	837	0	−3	−1.9160	+1.0840
170	−1	−2	−3.2712	−1.2712	504	0	−5	2.3783	+7.3783	838	1	−2	−1.4922	+0.5078
171	0	−2	−3.1362	−1.1362	505	1	−4	2.3539	+6.3539	839	−1	−3	−1.0781	+1.9219
172	0	−2	−3.1856	−1.1856	506	−1	−5	2.2953	+7.2953	840	0	−3	−0.6868	+2.3132
173	−1	−3	−3.3772	−0.3772	507	0	−5	2.1836	+7.1836	841	0	−3	−0.3304	+2.6696
174	−1	−4	−3.6351	+0.3649	508	0	−5	2.0044	+7.0044	842	1	−2	−0.0195	+1.9805
175	0	−4	−3.8666	+0.1334	509	−1	−6	1.7490	+7.7490	843	1	−1	0.2373	+1.2373
176	0	−4	−3.9822	+0.0178	510	1	−5	1.4147	+6.4147	844	0	−1	0.4338	+1.4338
177	1	−3	−3.9143	−0.9143	511	1	−4	1.0052	+5.0052	845	0	−1	0.5662	+1.5662
178	1	−2	−3.6318	−1.6318	512	0	−4	0.5308	+4.5308	846	0	−1	0.6339	+1.6339
179	−1	−3	−3.1450	−0.1450	513	0	−4	0.0069	+4.0069	847	0	−1	0.6385	+1.6385
180	0	−3	−2.5031	+0.4969	514	1	−3	−0.5468	+2.4532	848	0	−1	0.5846	+1.5846
181	−1	−4	−1.7827	+2.2173	515	1	−2	−1.1076	+0.8924	849	1	0	0.4792	+0.4792
182	−1	−5	−1.0713	+3.9287	516	0	−2	−1.6520	+0.3480	850	0	0	0.3310	+0.3310
183	1	−4	−0.4487	+3.5513	517	1	−1	−2.1571	−1.1571	851	1	1	0.1508	−0.8492
184	0	−4	0.0289	+4.0289	518	−1	−2	−2.6025	−0.6025	852	0	1	−0.0500	−1.0500
185	1	−3	0.3394	+3.3394	519	1	−1	−2.9716	−1.9716	853	−1	0	−0.2591	−0.2591
186	−1	−4	0.4983	+4.4983	520	0	−1	−3.2532	−2.2532	854	−1	−1	−0.4645	+0.5355
187	1	−3	0.5529	+3.5529	521	−1	−2	−3.4422	−1.4422	855	0	−1	−0.6546	+0.3454
188	0	−3	0.5711	+3.5711	522	0	−2	−3.5397	−1.5397	856	0	−1	−0.8191	+0.1809
189	0	−3	0.6257	+3.6257	523	−1	−3	−3.5533	−0.5533	857	−1	−2	−0.9492	+1.0508
190	−1	−4	0.7773	+4.7773	524	0	−3	−3.4962	−0.4962	858	1	−1	−1.0384	−0.0384
191	−1	−5	1.0612	+6.0612	525	0	−3	−3.3866	−0.3866	859	−1	−2	−1.0821	+0.9179
192	0	−5	1.4790	+6.4790	526	1	−2	−3.2459	−1.2459	860	0	−2	−1.0787	+0.9213
193	−1	−6	1.9971	+7.9971	527	1	−1	−3.0974	−2.0974	861	−1	−3	−1.0289	+1.9711
194	1	−5	2.5533	+7.5533	528	0	−1	−2.9647	−1.9647	862	1	−2	−0.9362	+1.0638
195	−1	−6	3.0682	+9.0682	529	0	−1	−2.8696	−1.8696	863	−1	−3	−0.8061	+2.1939
196	0	−6	3.4608	+9.4608	530	−1	−2	−2.8311	−0.8311	864	0	−3	−0.6467	+2.3533
197	−1	−7	3.6636	+10.6636	531	0	−2	−2.8634	−0.8634	865	1	−2	−0.4675	+1.5325
198	0	−7	3.6358	+10.6358	532	0	−2	−2.9756	−0.9756	866	1	−1	−0.2796	+0.7204
199	−1	−8	3.3703	+11.3703	533	1	−1	−3.1705	−2.1705	867	0	−1	−0.0946	+0.9054
200	0	−8	2.8952	+10.8952	534	−1	−2	−3.4451	−1.4451	868	0	−1	0.0755	+1.0755
201	1	−7	2.2680	+9.2680	535	1	−1	−3.7903	−2.7903	869	1	0	0.2189	+0.2189
202	1	−6	1.5648	+7.5648	536	0	−1	−4.1920	−3.1920	870	1	1	0.3250	−0.6750
203	1	−5	0.8665	+5.8665	537	1	0	−4.6320	−4.6320	871	1	2	0.3841	−1.6159
204	0	−5	0.2437	+5.2437	538	1	1	−5.0897	−6.0897	872	0	2	0.3886	−1.6114
205	1	−4	−0.2547	+3.7453	539	0	1	−5.5432	−6.5432	873	0	2	0.3331	−1.6669
206	1	−3	−0.6094	+2.3906	540	0	1	−5.9713	−6.9713	874	−1	1	0.2142	−0.7858
207	0	−3	−0.8324	+2.1676	541	−1	0	−6.3550	−6.3550	875	0	1	0.0316	−0.9684
208	0	−3	−0.9629	+2.0371	542	1	1	−6.6785	−7.6785	876	0	1	−0.2129	−1.2129

Table 16 (continued)

n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$	n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$	n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$
209	1	−2	−1.0581	+0.9419	543	1	2	−6.9309	−8.9309	877	−1	0	−0.5146	−0.5146
210	1	−1	−1.1806	−0.1806	544	0	2	−7.1067	−9.1067	878	1	1	−0.8668	−1.8668
211	−1	−2	−1.3855	+0.6145	545	1	3	−7.2060	−10.2060	879	1	2	−1.2604	−3.2604
212	0	−2	−1.7091	+0.2909	546	1	4	−7.2346	−11.2346	880	0	2	−1.6850	−3.6850
213	1	−1	−2.1610	−1.1610	547	−1	3	−7.2037	−10.2037	881	−1	1	−2.1289	−3.1289
214	1	0	−2.7222	−2.7222	548	0	3	−7.1288	−10.1288	882	0	1	−2.5795	−3.5795
215	1	1	−3.3474	−4.3474	549	0	3	−7.0287	−10.0287	883	−1	0	−3.0244	−3.0244
216	0	1	−3.9734	−4.9734	550	0	3	−6.9238	−9.9238	884	0	0	−3.4512	−3.4512
217	1	2	−4.5299	−6.5299	551	1	4	−6.8352	−10.8352	885	−1	−1	−3.8488	−2.8488
218	1	3	−4.9520	−7.9520	552	0	4	−6.7824	−10.7824	886	1	0	−4.2071	−4.2071
219	1	4	−5.1918	−9.1918	553	1	5	−6.7824	−11.7824	887	−1	−1	−4.5180	−3.5180
220	0	4	−5.2263	−9.2263	554	1	6	−6.8485	−12.8485	888	0	−1	−4.7756	−3.7756
221	1	5	−5.0620	−10.0620	555	−1	5	−6.9888	−11.9888	889	1	0	−4.9760	−4.9760
222	−1	4	−4.7331	−8.7331	556	0	5	−7.2061	−12.2061	890	−1	−1	−5.1183	−4.1183
223	−1	3	−4.2963	−7.2963	557	−1	4	−7.4975	−11.4975	891	0	−1	−5.2038	−4.2038
224	0	3	−3.8211	−6.8211	558	0	4	−7.8544	−11.8544	892	0	−1	−5.2361	−4.2361
225	0	3	−3.3782	−6.3782	559	1	5	−8.2633	−13.2633	893	1	0	−5.2214	−5.2214
226	1	4	−3.0283	−7.0283	560	0	5	−8.7067	−13.7067	894	−1	−1	−5.1677	−4.1677
227	−1	3	−2.8125	−5.8125	561	−1	4	−9.1642	−13.1642	895	1	0	−5.0843	−5.0843
228	0	3	−2.7467	−5.7467	562	1	5	−9.6138	−14.6138	896	0	0	−4.9821	−4.9821
229	−1	2	−2.8197	−4.8197	563	−1	4	−10.0335	−14.0335	897	−1	−1	−4.8723	−3.8723
230	−1	1	−2.9968	−3.9968	564	0	4	−10.4027	−14.4027	898	1	0	−4.7664	−4.7664
231	−1	0	−3.2264	−3.2264	565	1	5	−10.7036	−15.7036	899	1	1	−4.6753	−5.6753
232	0	0	−3.4496	−3.4496	566	1	6	−10.9224	−16.9224	900	0	1	−4.6094	−5.6094
233	−1	−1	−3.6115	−2.6115	567	0	6	−11.0501	−17.0501	901	1	2	−4.5773	−6.5773
234	0	−1	−3.6698	−2.6698	568	0	6	−11.0831	−17.0831	902	−1	1	−4.5864	−5.5864
235	1	0	−3.6032	−3.6032	569	−1	5	−11.0232	−16.0232	903	−1	0	−4.6416	−4.6416
236	0	0	−3.4141	−3.4141	570	1	6	−10.8778	−16.8778	904	0	0	−4.7459	−4.7459
237	1	1	−3.1286	−4.1286	571	−1	5	−10.6589	−15.6589	905	1	1	−4.8997	−5.8997
238	−1	0	−2.7921	−2.7921	572	0	5	−10.3824	−15.3824	906	−1	0	−5.1011	−5.1011
239	−1	−1	−2.4610	−1.4610	573	1	6	−10.0669	−16.0669	907	−1	−1	−5.3458	−4.3458
240	0	−1	−2.1939	−1.1939	574	−1	5	−9.7326	−14.7326	908	0	−1	−5.6276	−4.6276
241	−1	−2	−2.0413	−0.0413	575	0	5	−9.3996	−14.3996	909	0	−1	−5.9383	−4.9383
242	0	−2	−2.0376	−0.0376	576	0	5	−9.0866	−14.0866	910	1	0	−6.2682	−6.2682
243	0	−2	−2.1949	−0.1949	577	−1	4	−8.8099	−12.8099	911	−1	−1	−6.6067	−5.6067
244	0	−2	−2.5015	−0.5015	578	0	4	−8.5818	−12.5818	912	0	−1	−6.9426	−5.9426
245	0	−2	−2.9228	−0.9228	579	1	5	−8.4103	−13.4103	913	1	0	−7.2644	−7.2644
246	−1	−3	−3.4071	−0.4071	580	0	5	−8.2983	−13.2983	914	1	1	−7.5612	−8.5612
247	1	−2	−3.8933	−1.8933	581	1	6	−8.2435	−14.2435	915	−1	0	−7.8231	−7.8231
248	0	−2	−4.3198	−2.3198	582	−1	5	−8.2388	−13.2388	916	0	0	−8.0412	−8.0412
249	1	−1	−4.6343	−3.6343	583	1	6	−8.2723	−14.2723	917	1	1	−8.2087	−9.2087
250	0	−1	−4.8007	−3.8007	584	0	6	−8.3291	−14.3291	918	0	1	−8.3204	−9.3204
251	−1	−2	−4.8046	−2.8046	585	0	6	−8.3913	−14.3913	919	−1	0	−8.3735	−8.3735
252	0	−2	−4.6547	−2.6547	586	1	7	−8.4402	−15.4402	920	0	0	−8.3677	−8.3677
253	1	−1	−4.3808	−3.3808	587	−1	6	−8.4570	−14.4570	921	1	1	−8.3050	−9.3050
254	1	0	−4.0290	−4.0290	588	0	6	−8.4243	−14.4243	922	1	2	−8.1894	−10.1894
255	−1	−1	−3.6541	−2.6541	589	1	7	−8.3275	−15.3275	923	1	3	−8.0274	−11.0274
256	0	−1	−3.3109	−2.3109	590	−1	6	−8.1555	−14.1555	924	0	3	−7.8273	−10.8273
257	−1	−2	−3.0462	−1.0462	591	1	7	−7.9018	−14.9018	925	0	3	−7.5986	−10.5986
258	−1	−3	−2.8917	+0.1083	592	0	7	−7.5650	−14.5650	926	1	4	−7.3524	−11.3524
259	1	−2	−2.8595	−0.8595	593	−1	6	−7.1485	−13.1485	927	0	4	−7.1000	−11.1000
260	0	−2	−2.9407	−0.9407	594	0	6	−6.6609	−12.6609	928	0	4	−6.8532	−10.8532
261	0	−2	−3.1075	−1.1075	595	−1	5	−6.1151	−11.1151	929	−1	3	−6.6234	−9.6234
262	1	−1	−3.3174	−2.3174	596	0	5	−5.5277	−10.5277	930	1	4	−6.4211	−10.4211
263	−1	−2	−3.5207	−1.5207	597	1	6	−4.9181	−10.9181	931	0	4	−6.2558	−10.2558
264	0	−2	−3.6678	−1.6678	598	−1	5	−4.3068	−9.3068	932	0	4	−6.1354	−10.1354
265	1	−1	−3.7172	−2.7172	599	−1	4	−3.7148	−7.7148	933	1	5	−6.0658	−11.0658
266	−1	−2	−3.6418	−1.6418	600	0	4	−3.1615	−7.1615	934	1	6	−6.0509	−12.0509
267	1	−1	−3.4334	−2.4334	601	−1	3	−2.6645	−5.6645	935	−1	5	−6.0921	−11.0921
268	0	−1	−3.1037	−2.1037	602	−1	2	−2.2377	−4.2377	936	0	5	−6.1889	−11.1889

Table 16 (continued)

n	μ	M	$M_{\mathrm{RVM}}^{\mathrm{crit}}$	$M_{\mathrm{RVM}}^{\mathrm{crit}}(n)$ $-M(n)$	n	μ	M	$M_{\mathrm{RVM}}^{\mathrm{crit}}$	$M_{\mathrm{RVM}}^{\mathrm{crit}}(n)$ $-M(n)$	n	μ	M	$M_{\mathrm{RVM}}^{\mathrm{crit}}$	$M_{\mathrm{RVM}}^{\mathrm{crit}}(n)$ $-M(n)$
269	−1	−2	−2.6832	−0.6832	603	0	2	−1.8909	−3.8909	937	−1	4	−6.3382	−10.3382
270	0	−2	−2.2163	−0.2163	604	0	2	−1.6291	−3.6291	938	−1	3	−6.5348	−9.5348
271	−1	−3	−1.7558	+1.2442	605	0	2	−1.4523	−3.4523	939	1	4	−6.7718	−10.7718
272	0	−3	−1.3547	+1.6453	606	−1	1	−1.3558	−2.3558	940	0	4	−7.0406	−11.0406
273	−1	−4	−1.0594	+2.9406	607	−1	0	−1.3302	−1.3302	941	−1	3	−7.3314	−10.3314
274	1	−3	−0.9028	+2.0972	608	0	0	−1.3623	−1.3623	942	−1	2	−7.6336	−9.6336
275	0	−3	−0.9002	+2.0998	609	−1	−1	−1.4360	−0.4360	943	1	3	−7.9363	−10.9363
276	0	−3	−1.0470	+1.9530	610	−1	−2	−1.5337	+0.4663	944	0	3	−8.2288	−11.2288
277	−1	−4	−1.3198	+2.6802	611	1	−1	−1.6366	−0.6366	945	0	3	−8.5009	−11.5009
278	1	−3	−1.6794	+1.3206	612	0	−1	−1.7271	−0.7271	946	−1	2	−8.7433	−10.7433
279	0	−3	−2.0769	+0.9231	613	−1	−2	−1.7890	+0.2110	947	−1	1	−8.9483	−9.9483
280	0	−3	−2.4595	+0.5405	614	1	−1	−1.8090	−0.8090	948	0	1	−9.1098	−10.1098
281	−1	−4	−2.7786	+1.2214	615	−1	−2	−1.7776	+0.2224	949	1	2	−9.2237	−11.2237
282	−1	−5	−2.9955	+2.0045	616	0	−2	−1.6895	+0.3105	950	0	2	−9.2879	−11.2879
283	−1	−6	−3.0866	+2.9134	617	−1	−3	−1.5438	+1.4562	951	1	3	−9.3026	−12.3026
284	0	−6	−3.0459	+2.9541	618	−1	−4	−1.3445	+2.6555	952	0	3	−9.2702	−12.2702
285	−1	−7	−2.8857	+4.1143	619	−1	−5	−1.0998	+3.9002	953	−1	2	−9.1951	−11.1951
286	−1	−8	−2.6338	+5.3662	620	0	−5	−0.8217	+4.1783	954	0	2	−9.0837	−11.0837
287	1	−7	−2.3295	+4.6705	621	0	−5	−0.5252	+4.4748	955	1	3	−8.9438	−11.9438
288	0	−7	−2.0175	+4.9825	622	1	−4	−0.2271	+3.7729	956	0	3	−8.7846	−11.7846
289	0	−7	−1.7419	+5.2581	623	1	−3	0.0547	+3.0547	957	−1	2	−8.6163	−10.6163
290	−1	−8	−1.5394	+6.4606	624	0	−3	0.3026	+3.3026	958	1	3	−8.4492	−11.4492
291	1	−7	−1.4342	+5.5658	625	0	−3	0.5005	+3.5005	959	1	4	−8.2939	−12.2939
292	0	−7	−1.4351	+5.5649	626	1	−2	0.6348	+2.6348	960	0	4	−8.1604	−12.1604
293	−1	−8	−1.5338	+6.4662	627	−1	−3	0.6954	+3.6954	961	0	4	−8.0580	−12.0580
294	0	−8	−1.7061	+6.2939	628	0	−3	0.6758	+3.6758	962	−1	3	−7.9947	−10.9947
295	1	−7	−1.9154	+5.0846	629	1	−2	0.5745	+2.5745	963	0	3	−7.9768	−10.9768
296	0	−7	−2.1174	+4.8826	630	0	−2	0.3943	+2.3943	964	0	3	−8.0088	−11.0088
297	0	−7	−2.2663	+4.7337	631	−1	−3	0.1425	+3.1425	965	1	4	−8.0933	−12.0933
298	1	−6	−2.3206	+3.6794	632	0	−3	−0.1693	+2.8307	966	1	5	−8.2306	−13.2306
299	1	−5	−2.2488	+2.7512	633	1	−2	−0.5265	+1.4735	967	−1	4	−8.4188	−12.4188
300	0	−5	−2.0336	+2.9664	634	1	−1	−0.9116	+0.0884	968	0	4	−8.6539	−12.6539
301	1	−4	−1.6743	+2.3257	635	1	0	−1.3059	−1.3059	969	−1	3	−8.9300	−11.9300
302	1	−3	−1.1871	+1.8129	636	0	0	−1.6902	−1.6902	970	−1	2	−9.2395	−11.2395
303	1	−2	−0.6029	+1.3971	637	0	0	−2.0459	−2.0459	971	−1	1	−9.5732	−10.5732
304	0	−2	0.0364	+2.0364	638	−1	−1	−2.3564	−1.3564	972	0	1	−9.9210	−10.9210
305	1	−1	0.6826	+1.6826	639	0	−1	−2.6079	−1.6079	973	1	2	−10.2719	−12.2719
306	0	−1	1.2874	+2.2874	640	0	−1	−2.7901	−1.7901	974	1	3	−10.6150	−13.6150
307	−1	−2	1.8082	+3.8082	641	−1	−2	−2.8966	−0.8966	975	0	3	−10.9391	−13.9391
308	0	−2	2.2125	+4.2125	642	−1	−3	−2.9258	+0.0742	976	0	3	−11.2340	−14.2340
309	1	−1	2.4821	+3.4821	643	−1	−4	−2.8803	+1.1197	977	−1	2	−11.4902	−13.4902
310	−1	−2	2.6144	+4.6144	644	0	−4	−2.7669	+1.2331	978	−1	1	−11.6999	−12.6999
311	−1	−3	2.6223	+5.6223	645	−1	−5	−2.5965	+2.4035	979	1	2	−11.8565	−13.8565
312	0	−3	2.5320	+5.5320	646	−1	−6	−2.3826	+3.6174	980	0	2	−11.9557	−13.9557
313	−1	−4	2.3794	+6.3794	647	−1	−7	−2.1412	+4.8588	981	0	2	−11.9950	−13.9950
314	1	−3	2.2049	+5.2049	648	0	−7	−1.8891	+5.1109	982	1	3	−11.9741	−14.9741
315	0	−3	2.0480	+5.0480	649	1	−6	−1.6436	+4.3564	983	−1	2	−11.8948	−13.8948
316	0	−3	1.9426	+4.9426	650	0	−6	−1.4208	+4.5792	984	0	2	−11.7609	−13.7609
317	−1	−4	1.9122	+5.9122	651	−1	−7	−1.2348	+5.7652	985	1	3	−11.5781	−14.5781
318	−1	−5	1.9671	+6.9671	652	0	−7	−1.0970	+5.9030	986	−1	2	−11.3536	−13.3536
319	1	−4	2.1031	+6.1031	653	−1	−8	−1.0152	+6.9848	987	−1	1	−11.0959	−12.0959
320	0	−4	2.3022	+6.3022	654	−1	−9	−0.9933	+8.0067	988	0	1	−10.8147	−11.8147
321	1	−3	2.5349	+5.5349	655	1	−8	−1.0311	+6.9689	989	1	2	−10.5198	−12.5198
322	−1	−4	2.7637	+6.7637	656	0	−8	−1.1241	+6.8759	990	0	2	−10.2214	−12.2214
323	1	−3	2.9482	+5.9482	657	0	−8	−1.2639	+6.7361	991	−1	1	−9.9296	−10.9296
324	0	−3	3.0499	+6.0499	658	−1	−9	−1.4391	+7.5609	992	0	1	−9.6534	−10.6534
325	0	−3	3.0370	+6.0370	659	−1	−10	−1.6354	+8.3646	993	1	2	−9.4010	−11.4010
326	1	−2	2.8887	+4.8887	660	0	−10	−1.8368	+8.1632	994	−1	1	−9.1792	−10.1792
327	1	−1	2.5974	+3.5974	661	−1	−11	−2.0268	+8.9732	995	1	2	−8.9932	−10.9932
328	0	−1	2.1702	+3.1702	662	1	−10	−2.1887	+7.8113	996	0	2	−8.8463	−10.8463

Table 16 (continued)

n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$	n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$	n	μ	M	$M^{\mathrm{crit}}_{\mathrm{RVM}}$	$M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ $-M(n)$
329	1	0	1.6280	+1.6280	663	−1	−11	−2.3077	+8.6923	997	−1	1	−8.7399	−9.7399
330	1	1	1.0035	+0.0035	664	0	−11	−2.3706	+8.6294	998	1	2	−8.6731	−10.6731
331	−1	0	0.3371	+0.3371	665	−1	−12	−2.3676	+9.6324	999	0	2	−8.6435	−10.6435
332	0	0	−0.3269	−0.3269	666	0	−12	−2.2923	+9.7077	1000	0	2	−8.6463	−10.6463
333	0	0	−0.9458	−0.9458	667	1	−11	−2.1424	+8.8576					
334	1	1	−1.4828	−2.4828	668	0	−11	−1.9197	+9.0803					

Appendix B. Python programs

The programs below produce every number, table and figure of Section 19. They read the two data files `data/zeross.bin` (ordinates γ_k of the zeros of ζ) and `data/zLamb.bin` (midpoint ordinates τ_k), both little-endian float64 with 2 001 052 entries, and write their console output, CSV, MAT and binary files to `output/` and the figures to `figures/`. They require Python 3 with NumPy, SciPy, mpmath and Matplotlib (`s15` also needs `finufft`). The outputs distributed with the paper were produced with Python 3.13.16, NumPy 2.5.3, SciPy 1.18.1, mpmath 1.4.1, Matplotlib 3.11.2, finufft 2.5.1 and python-flint 0.9.0 (Arb ball arithmetic, used only by `code/routeB/certV.py`) on Linux x86_64, in IEEE double precision except where mpmath or python-flint is called; `code/requirements.txt` pins these versions. A double-precision ordinate is resolved to one unit in the last place, $2.3\cdot10^{-10}$ near the top of the files ($1.13\cdot10^{6}$) and $1.4\cdot10^{-14}$ near $t=100$; differences between computed quantities at or below the level of this representation of the data are not attributed to the algorithms. The programs are run from the directory `code/` in the order `s01`, `s02` (which writes `output/s02_varpi.txt`, read by `s05`, `s07` and `s15`), `s03`–`s10`, and then `s11`–`s17`; `s06` reads the CSV file written by `s04`, and the others depend only on the two data files. The loader of `rvm_common.py` refuses files whose size is not a multiple of 8 bytes, files of different lengths, non-finite or non-positive values, a list τ_k that is not strictly increasing or starts below 2π, and a list γ_k that decreases; the programs `s08`–`s10` and `s13` use NumPy and SciPy only and check the lengths and the monotonicity themselves.

B.1. **rvm_common.py.**

```
"""Common routines for the midpoint Riemann-von Mangoldt (RVM) zeta system.

Data files (little-endian float64, one ordinate per entry):
    data/zeross.bin  ordinates gamma_k of the nontrivial zeros of zeta (2,001,052 values)
    data/zLamb.bin   midpoint RVM ordinates tau_k, N_RVM(tau_k) = k - 1/2

Notation follows the paper: N_RVM is the clock, tau_k the midpoint ordinates,
gamma_k the ordinates of zeta, eta_k = tau_k - gamma_k, q(s) = s(s-1),
G(s) = s(s-1) Gamma(s/2) pi^(-s/2), zeta_lambda = P_lambda / G with ordinates
gamma_k(lambda) = (1-lambda) gamma_k + lambda tau_k.
"""
import os
import sys
import time
import numpy as np
from scipy import special

ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
DATA = os.path.join(ROOT, "data")
OUT = os.path.join(ROOT, "output")
FIG = os.path.join(ROOT, "figures")
os.makedirs(OUT, exist_ok=True)
os.makedirs(FIG, exist_ok=True)

TWO_PI = 2.0 * np.pi


class Tee:
```

```
    """Write console output both to the screen and to output/<name>_console.txt."""

    def __init__(self, name):
        self.f = open(os.path.join(OUT, name + "_console.txt"), "w")
        self.t0 = time.time()

    def __call__(self, fmt, *args):
        s = fmt % args if args else fmt
        sys.stdout.write(s + "\n")
        sys.stdout.flush()
        self.f.write(s + "\n")
        self.f.flush()

    def close(self):
        self("total run time %.1f s" % (time.time() - self.t0))
        self.f.close()


def read_bin(name):
    """Read a data file of little-endian IEEE float64 values (8 bytes each, no header)."""
    path = os.path.join(DATA, name)
    size = os.path.getsize(path)
    if size == 0 or size % 8 != 0:
        raise ValueError("%s has %d bytes, which is not a positive multiple of 8: it is not a "
                         "little-endian float64 file" % (name, size))
    return np.fromfile(path, dtype="<f8")


def load_ordinates(n_required=None):
    """Read gamma_k (zeross.bin) and tau_k (zLamb.bin) and validate them.

    The two files must be little-endian float64 (size a multiple of 8 bytes) and have the same
    length; a different length is an error, never a silent truncation.  Both lists must be finite; tau_k must
     be strictly increasing (the
    clock is strictly increasing on [2 pi, oo)); gamma_k must be nondecreasing (a repeated
    value would be a multiple zero; the supplied file has none, which is reported).
    If n_required is given, at least that many ordinates must be present."""
    gam = read_bin("zeross.bin")
    tau = read_bin("zLamb.bin")
    if gam.size != tau.size:
        raise ValueError("zeross.bin has %d entries and zLamb.bin has %d: the ordinate files have "
                         "different lengths" % (gam.size, tau.size))
    if n_required is not None and gam.size < n_required:
        raise ValueError("%d ordinates required, %d supplied" % (n_required, gam.size))
    if not (np.all(np.isfinite(gam)) and np.all(np.isfinite(tau))):
        raise ValueError("non-finite ordinate in the data files")
    if not (gam[0] > 0 and tau[0] > TWO_PI):
        raise ValueError("the ordinates must be positive and tau_1 > 2 pi (increasing branch of the clock)")
    if not np.all(np.diff(tau) > 0):
        raise ValueError("zLamb.bin is not strictly increasing")
    dg = np.diff(gam)
    if not np.all(dg >= 0):
        raise ValueError("zeross.bin is not nondecreasing")
    if not np.all(dg > 0):
        sys.stdout.write("warning: zeross.bin contains repeated ordinates (multiple zeros)\n")
    return gam, tau


def clock(T):
    """N_RVM(T) = T/(2 pi) log(T/(2 pi e)) + 7/8 (real or complex T, principal log)."""
    return T / TWO_PI * (np.log(T / TWO_PI) - 1.0) + 0.875


def tail_coefficients(U):
    """Midpoint-rule tail of sum_{k>N} log(1 + q/tau_k^2), N = N_RVM(U):
```

```
    returns (c1, c2, c3) with tail(q) = c1 q - c2 q^2 + c3 q^3.

    Approximations, all used for the factors beyond the supplied file only:
    (i) the unknown ordinates beyond the file are replaced by the clock (the 'clock tail');
    (ii) the sum over the clock points is replaced by the integral against dN_RVM, which is
        the midpoint rule because N_RVM(tau_k) = k - 1/2; its error is O(|q| log U / U^3);
    (iii) 1/4 + tau_k^2 is replaced by tau_k^2, a relative change O(U^-2) of each term,
        hence O(|q| log U / U^3) in total;
    (iv) the expansion of log(1 + q/t^2) is cut after q^3; the omitted terms are
        O(|q|^4 log U / U^7).
    For the file used here U = tau_N ~ 1.13e6 and |q| <= 4.1e6 (|s| <= 2000), so (ii)-(iv)
    are below 1e-10; (i) is not bounded here and is part of what the experiments measure."""
    LU = np.log(U / TWO_PI)
    c1 = (LU + 1.0) / U / TWO_PI
    c2 = (3.0 * LU + 1.0) / (9.0 * U ** 3) / 2.0 / TWO_PI
    c3 = (5.0 * LU + 1.0) / (25.0 * U ** 5) / 3.0 / TWO_PI
    return c1, c2, c3


def clock_inverse(y, tol=1e-15, max_iter=60):
    """Solve N_RVM(U) = y on the increasing branch U > 2 pi (y > -1/8).

    Safeguarded Newton iteration: the iterates are kept in a bracket [lo, hi] with
    N_RVM(lo) < y < N_RVM(hi) inside (2 pi, oo), where N_RVM' > 0; a Newton step that
    leaves the bracket is replaced by bisection.  Stops when the relative step is below
    tol and raises RuntimeError if max_iter iterations do not suffice."""
    y = float(y)
    if y <= -0.125:
        raise ValueError("N_RVM(U) = y has no solution with U > 2 pi for y <= -1/8")
    lo = TWO_PI * (1.0 + 1e-12)
    hi = TWO_PI * np.e * 2.0
    while clock(hi) <= y:
        hi *= 2.0
    U = min(max(TWO_PI * max(y, 2.0) / np.log(max(y, 3.0)), lo), hi)
    for _ in range(max_iter):
        f = clock(U) - y
        if f > 0:
            hi = U
        else:
            lo = U
        Unew = U - f / (np.log(U / TWO_PI) / TWO_PI)
        if not (lo < Unew < hi):
            Unew = 0.5 * (lo + hi)
        if abs(Unew - U) <= tol * max(1.0, abs(U)):
            return Unew
        U = Unew
    raise RuntimeError("clock_inverse did not converge for y = %r" % y)


class Product:
    """log P(s) = sum_k log(1 + q(s)/(1/4 + o_k^2)) for an ordinate list o_k.

    Exact summation for k <= K1, power sums (nterms powers of q) for K1 < k <= N, and
    the clock tail for k > N (tail_coefficients).  The power sums are accurate for
    |q(s)| << o_{K1}^2; the omitted power terms are O(|q|^(nterms+1) o_{K1}^(-2 nterms))
    times the number of factors.  The logarithm is the principal one, factor by factor;
    use the result through exp(logP) unless branch continuity is checked separately.  In the
    programs of this repository logP and logG enter only through exp(.) or through their real
    parts (logarithms of moduli); no program reads an argument or a phase from them."""

    def __init__(self, ords, K1=100000, nterms=6):
        self.o = np.asarray(ords, dtype=float)
        self.N = self.o.size
        self.K1 = min(K1, self.N)
        A = 0.25 + self.o[self.K1:] ** 2
```

```
        self.pw = [np.sum(A ** (-j)) for j in range(1, nterms + 1)]
        self.A1 = 0.25 + self.o[: self.K1] ** 2
        self.tail = tail_coefficients(clock_inverse(float(self.N)))

    def logP(self, s):
        s = np.atleast_1d(np.asarray(s, dtype=complex))
        q = s * (s - 1.0)
        out = np.empty(s.shape, dtype=complex)
        for i, qi in enumerate(q):
            v = np.sum(np.log1p(qi / self.A1))
            for j, p in enumerate(self.pw, start=1):
                v += (-1) ** (j + 1) * qi ** j / j * p
            c1, c2, c3 = self.tail
            v += c1 * qi - c2 * qi ** 2 + c3 * qi ** 3
            out[i] = v
        return out


def logG(s):
    """Sum of principal-branch logarithms of s, s - 1, Gamma(s/2) and pi^(-s/2).

    This is a logarithm of G(s) = s(s-1) Gamma(s/2) pi^(-s/2), not necessarily the branch
    continuous along a given path: use it only through exp(logG) unless branch continuity
    has been checked separately (arguments, phases and their derivatives)."""
    s = np.asarray(s, dtype=complex)
    return np.log(s) + np.log(s - 1.0) + special.loggamma(s / 2.0) - s / 2.0 * np.log(np.pi)


def zeta_from_product(prod, s):
    s = np.atleast_1d(np.asarray(s, dtype=complex))
    return np.exp(prod.logP(s) - logG(s))


def chi(s):
    s = np.asarray(s, dtype=complex)
    return np.exp((s - 0.5) * np.log(np.pi) + special.loggamma((1 - s) / 2) - special.loggamma(s / 2))


def w_transition(s):
    """w(s) = exp(-2 pi i N_RVM(tau_*)), tau_* = -i (s - 1/2)."""
    s = np.asarray(s, dtype=complex)
    tstar = -1j * (s - 0.5)
    return np.exp(-2j * np.pi * clock(tstar))


def theta_rs(t):
    t = np.asarray(t, dtype=float)
    return t / 2 * np.log(t / TWO_PI) - t / 2 - np.pi / 8 + 1 / (48 * t) + 7 / (5760 * t ** 3)


def gamma_lambda(gam, tau, lam):
    return (1.0 - lam) * gam + lam * tau


def ei_upper(w):
    """Ei on Im w != 0 along the horizontal line (paper, eq. Ei-def): Ei(w) = -E1(-w)."""
    return -special.exp1(-np.asarray(w, dtype=complex))


def mobius_sieve(X):
    """mu(n) for 0 <= n <= X (mu(0) = 0), by a sieve over the primes p <= X."""
    mu = np.ones(X + 1, dtype=np.int8)
    mu[0] = 0
    is_comp = np.zeros(X + 1, dtype=bool)
    for p in range(2, X + 1):
```

```
        if not is_comp[p]:
            is_comp[2 * p::p] = True        # p itself stays marked prime
            mu[p::p] *= -1
            if p * p <= X:
                mu[p * p::p * p] = 0
    if X >= 12:                              # self-test on the first values
        ref = [0, 1, -1, -1, 0, -1, 1, -1, 0, 0, 1, -1, 0]
        if list(mu[:13]) != ref:
            raise AssertionError("mobius_sieve self-test failed")
    return mu


def primes_upto(X):
    s = np.ones(X + 1, dtype=bool)
    s[:2] = False
    for p in range(2, int(X ** 0.5) + 1):
        if s[p]:
            s[p * p::p] = False
    return np.nonzero(s)[0]


def write_csv(name, header, cols):
    A = np.column_stack(cols)
    np.savetxt(os.path.join(OUT, name), A, delimiter=",", header=",".join(header), comments="", fmt="%.12g")


def savefig(fig, name):
    fig.savefig(os.path.join(FIG, name), dpi=150, bbox_inches="tight")
```

B.2. **s01_clock_and_family.py.**

```
"""Experiment 1: the midpoint clock, the displacements eta_k = tau_k - gamma_k,
the lambda-family zeta_lambda, and the numerical comparison zeta_Z1 versus zeta."""
import numpy as np
import mpmath as mp
import scipy.io as sio
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import (Tee, load_ordinates, clock, Product, zeta_from_product, gamma_lambda,
                        write_csv, savefig, OUT)
import os

say = Tee("s01_clock_and_family")
gam, tau = load_ordinates()
N = gam.size
k = np.arange(1, N + 1)
say("zeros loaded: N = %d, gamma_1 = %.9f, gamma_N = %.6f, tau_1 = %.9f, tau_N = %.6f",
    N, gam[0], gam[-1], tau[0], tau[-1])

# ---- clock check and displacements
clock_err = np.max(np.abs(clock(tau) - (k - 0.5)))
eta = tau - gam
say("max |N_RVM(tau_k) - (k - 1/2)| = %.3e", clock_err)
say("eta_k = tau_k - gamma_k:  max |eta| = %.6f (k = %d),  mean |eta| = %.6f,  rms = %.6f,  mean eta = %.3e",
    np.max(np.abs(eta)), int(np.argmax(np.abs(eta))) + 1, np.mean(np.abs(eta)),
    np.sqrt(np.mean(eta ** 2)), np.mean(eta))
blocks = [(1, 10), (11, 100), (101, 1000), (1001, 10 ** 4), (10 ** 4 + 1, 10 ** 5), (10 ** 5 + 1, 10 ** 6),
     (10 ** 6 + 1, N)]
blocks = [(a, min(b, N)) for a, b in blocks if a <= N]   # no empty block for a shorter file
rows = []
say("%12s %12s %12s %12s", "k from", "k to", "max|eta|", "rms eta")
for a, b in blocks:
    e = eta[a - 1:b]
    rows.append([a, b, np.max(np.abs(e)), np.sqrt(np.mean(e ** 2))])
```

```
    say("%12d %12d %12.6f %12.6f", a, b, rows[-1][2], rows[-1][3])
rows = np.array(rows)
write_csv("s01_eta_blocks.csv", ["k_from", "k_to", "max_abs_eta", "rms_eta"], [rows[:, i] for i in range(4)])
say("first ordinates:")
say("%4s %16s %16s %12s", "k", "gamma_k", "tau_k", "eta_k")
for i in range(12):
    say("%4d %16.9f %16.9f %12.6f", i + 1, gam[i], tau[i], eta[i])

# ---- zeta_Z1 versus zeta on vertical lines
PZ = Product(gam)
PR = Product(tau)
tline = np.concatenate([np.linspace(1.0, 100.0, 199), np.linspace(101.0, 2000.0, 300)])
mp.mp.dps = 20
cmp_rows = []
for sigma in (2.0, 1.25, 0.75):
    s = sigma + 1j * tline
    zz1 = zeta_from_product(PZ, s)
    ze = np.array([complex(mp.zeta(complex(x))) for x in s])
    rel = np.abs(zz1 / ze - 1.0)
    # the relative error can be amplified near a zero of zeta; report also a balanced metric
    bal = np.abs(zz1 - ze) / np.maximum(1.0, np.maximum(np.abs(zz1), np.abs(ze)))
    cmp_rows.append([sigma, np.max(rel), np.median(rel), np.max(bal)])
    say("sigma = %.2f, t in [1,2000] (%d points): max |zeta_Z1/zeta - 1| = %.3e, median = %.3e; "
        "max |zeta_Z1 - zeta|/max(1,|zeta_Z1|,|zeta|) = %.3e",
        sigma, tline.size, np.max(rel), np.median(rel), np.max(bal))
    write_csv("s01_zetaZ1_vs_zeta_sigma%.2f.csv" % sigma, ["t", "abs_zeta", "abs_zetaZ1", "rel_err", "
     balanced_err"],
              [tline, np.abs(ze), np.abs(zz1), rel, bal])
cmp_rows = np.array(cmp_rows)

# ---- Hardy-type functions Z_lambda(t) = exp(i theta(t)) zeta_lambda(1/2+it) on the critical line
def theta_exact(t):
    return np.array([float(mp.siegeltheta(x)) for x in t])

tc = np.linspace(0.5, 60.0, 1200)
th = theta_exact(tc)
lams = [0.0, 0.25, 0.5, 0.75, 1.0]
Zl = {}
for lam in lams:
    P = PZ if lam == 0.0 else (PR if lam == 1.0 else Product(gamma_lambda(gam, tau, lam)))
    Zc = np.exp(1j * th) * zeta_from_product(P, 0.5 + 1j * tc)
    # Z_lambda is real in exact arithmetic; measure the imaginary part before discarding it
    say("lambda = %.2f: max |Im Z_lambda| / max(1,|Z_lambda|) on [0.5,60] = %.3e",
        lam, np.max(np.abs(Zc.imag) / np.maximum(1.0, np.abs(Zc))))
    Zl[lam] = np.real(Zc)
Zhardy = np.array([float(mp.siegelz(x)) for x in tc])
dZ = np.max(np.abs(Zl[0.0] - Zhardy) / np.maximum(1.0, np.abs(Zhardy)))
say("critical line, t in [0.5,60]: max |Z_Z1(t) - Z(t)| / max(1,|Z(t)|) = %.3e", dZ)
write_csv("s01_Z_lambda_critical_line.csv", ["t", "Z_hardy"] + ["Z_lambda_%.2f" % l for l in lams],
          [tc, Zhardy] + [Zl[l] for l in lams])

# ---- figures
dec = np.arange(0, N, 50)
fig, ax = plt.subplots(2, 2, figsize=(12, 8))
ax[0, 0].plot(k[dec], eta[dec], ".", ms=1.5, color="tab:blue")
ax[0, 0].axhline(0, color="k", lw=0.5)
ax[0, 0].set_xlabel("k")
ax[0, 0].set_ylabel(r"$\eta_k=\tau_k-\gamma_k$")
ax[0, 0].set_title(r"Observed displacements in the supplied range, $N=%d$" % N)
ax[0, 1].hist(eta, bins=200, density=True, color="tab:blue")
ax[0, 1].set_xlabel(r"$\eta_k$")
ax[0, 1].set_title(r"Empirical distribution of the supplied displacements ($k\leq N$)")
lam = np.linspace(0, 1, 101)
for kk in range(30):
    gl = (1 - lam) * gam[kk] + lam * tau[kk]
```

```
    ax[1, 0].plot(lam, clock(gl) - (kk + 0.5), "-", lw=1)
ax[1, 0].axhline(0, color="k", lw=0.5)
ax[1, 0].set_xlabel(r"$\lambda$")
ax[1, 0].set_ylabel(r"$\mathcal{N}_{\rm RVM}(\gamma_k(\lambda))-(k-\frac{1}{2})$")
ax[1, 0].set_title(r"Moving ordinates in clock units, $k\leq30$ ($\lambda=0$: $\zeta_{Z1}$, $\lambda=1$: $\
    zeta_{\rm RVM}$)", fontsize=9)
ax[1, 1].plot(tc, Zhardy, "k-", lw=2.2, label=r"Hardy $Z(t)$ of $\zeta$")
cols = ["tab:red", "tab:orange", "tab:green", "tab:cyan", "tab:blue"]
for lam, c in zip(lams, cols):
    ax[1, 1].plot(tc, Zl[lam], "-", color=c, lw=1.0, label=r"$Z_\lambda$, $\lambda=%.2f$" % lam)
ax[1, 1].axhline(0, color="k", lw=0.5)
ax[1, 1].set_xlabel("t")
ax[1, 1].set_title(r"$e^{i\vartheta(t)}\zeta_\lambda(\frac{1}{2}+it)$ along the family")
ax[1, 1].legend(fontsize=7, ncol=2)
fig.tight_layout()
savefig(fig, "fig01_clock_family.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
for sigma, c in zip((2.0, 1.25, 0.75), ("tab:blue", "tab:green", "tab:red")):
    d = np.loadtxt(os.path.join(OUT, "s01_zetaZ1_vs_zeta_sigma%.2f.csv" % sigma), delimiter=",", skiprows=1)
    ax[0].plot(d[:, 0], d[:, 3], ".", ms=2, color=c, label=r"$\sigma=%.2f$" % sigma)
ax[0].set_xlabel("t")
ax[0].set_ylabel(r"$|\zeta_{Z1}/\zeta-1|$")
ax[0].set_title(r"$\zeta_{Z1}$ from %d supplied ordinates and the clock tail, versus $\zeta$" % N, fontsize=9)
ax[0].legend()
ax[1].plot(tc, Zhardy, "k-", lw=2.5, label=r"$Z(t)$ (mpmath)")
ax[1].plot(tc, Zl[0.0], "r--", lw=1.0, label=r"$Z_{Z1}(t)$ (product)")
ax[1].set_xlabel("t")
ax[1].legend()
ax[1].set_title(r"Critical line: $\lambda=0$ endpoint and Hardy $Z$")
fig.tight_layout()
savefig(fig, "fig02_zetaZ1_vs_zeta.png")

sio.savemat(os.path.join(OUT, "s01_results.mat"), {
    "N": N, "clock_err": clock_err, "eta_blocks": rows, "zetaZ1_cmp": cmp_rows,
    "t_critical": tc, "Z_hardy": Zhardy, "Z_lambda": np.array([Zl[l] for l in lams]), "lambdas": np.array(lams
    )})
say.close()

# ---- zoom: how the smooth clock places the midpoint ordinates next to the zeros
say2 = Tee("s01b_clock_zoom")
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
T = np.linspace(9.0, 52.0, 4000)
ax[0].step(T, np.searchsorted(gam, T, side="right"), where="post", color="k", lw=1.5, label=r"$N_\zeta(T)$")
ax[0].plot(T, clock(T), "r-", lw=1.5, label=r"$\mathcal{N}_{\rm RVM}(T)$")
for i in range(9):
    ax[0].plot([gam[i], gam[i]], [0, i + 1], "k:", lw=0.8)
ax[0].plot(tau[:9], np.arange(9) + 0.5, "ro", ms=5, label=r"$\tau_k$: $\mathcal{N}_{\rm RVM}(\tau_k)=k-\frac
    {1}{2}$")
ax[0].plot(gam[:9], np.arange(9) + 0.5, "ks", ms=4, mfc="none", label=r"$(\gamma_k,\,k-\frac{1}{2})$")
ax[0].set_xlabel("T")
ax[0].set_title("Clock and staircase at the first zeros")
ax[0].legend(loc="upper left", fontsize=8)
kz = np.arange(N - 12, N)
T_OFF = 10.0 * np.floor(gam[kz[0]] / 10.0)          # offsets computed from the file, not hard-coded
N_OFF = 1000 * ((N - 12) // 1000)
T3 = np.linspace(gam[kz[0]] - 0.3, gam[kz[-1]] + 0.3, 6000)
ax[1].step(T3 - T_OFF, np.searchsorted(gam, T3, side="right") - N_OFF, where="post", color="k", lw=1.5, label=
    r"$N_\zeta(T)$")
ax[1].plot(T3 - T_OFF, clock(T3) - N_OFF, "r-", lw=1.5, label=r"$\mathcal{N}_{\rm RVM}(T)$")
ax[1].plot(tau[kz] - T_OFF, kz + 0.5 - N_OFF, "ro", ms=5, label=r"$\tau_k$")
ax[1].plot(gam[kz] - T_OFF, kz + 0.5 - N_OFF, "ks", ms=4, mfc="none", label=r"$(\gamma_k,\,k-\frac{1}{2})$")
ax[1].set_xlabel(r"$T-%d$" % T_OFF)
ax[1].set_ylabel(r"count $-\,%d$" % N_OFF)
```

```
ax[1].set_title(r"Zoom at the last 12 zeros of the file (mean spacing %.3f)" % (2 * np.pi / np.log(gam[-1] /
    (2 * np.pi))))
ax[1].legend(loc="upper left", fontsize=8)
fig.tight_layout()
savefig(fig, "fig00_clock_zoom.png")
fig, ax = plt.subplots(1, 1, figsize=(12, 3))
T2 = np.linspace(9.0, 200.0, 20000)
ax.plot(T2, np.searchsorted(gam, T2, side="right") - clock(T2), "k-", lw=0.8)
ax.axhline(0, color="r", lw=0.8)
ax.set_xlabel("T")
ax.set_ylabel(r"$N_{\rm file}(T)-\mathcal{N}_{\rm RVM}(T)$")
ax.set_title(r"$N_{\rm file}(T)-\mathcal{N}_{\rm RVM}(T)$ from the supplied ordinates (jump $+1$ at each $\
    gamma_k$); "
             r"the file is complete here, so this is $S(T)+O(1/T)$", fontsize=9)
fig.tight_layout()
savefig(fig, "fig00b_remainder.png")
say2("zoom table near the top of the file:")
say2("%9s %18s %18s %10s %10s", "k", "gamma_k", "tau_k", "eta_k", "N*-N_RVM")
for i in kz:
    say2("%9d %18.6f %18.6f %10.6f %10.6f", i + 1, gam[i], tau[i], tau[i] - gam[i],
         np.searchsorted(gam, gam[i], side="right") - 0.5 - clock(gam[i]))
say2.close()
```

B.3. **s02_transition.py.**

```
"""Experiment 2: the transition theorem zeta_RVM = varpi (1 + w)(1 + O(1/t)),
the constant varpi_RVM, the vertical limit of 1/zeta_RVM, and zeta_RVM versus zeta."""
import os
import numpy as np
import mpmath as mp
import scipy.io as sio
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import (Tee, load_ordinates, Product, zeta_from_product, chi, w_transition,
                        write_csv, savefig, OUT)

say = Tee("s02_transition")
gam, tau = load_ordinates()
PR = Product(tau)

# ---- varpi_RVM by Richardson extrapolation on Re s = 2
# The model v(T) = c0 + c1/T + c2/T^2 is an empirical extrapolation: the transition theorem
# gives an error O(1/T), not a full expansion in powers of 1/T.  Its stability is checked
# below with other heights and with the linear model; c0 is kept complex until its
# imaginary part has been checked.


# The extrapolation uses a more accurate evaluation of the product than the default one:
# K1 = 10^6 factors summed exactly and 8 power sums.  With the default evaluation
# (K1 = 10^5, 6 power sums) the relative error of zeta_RVM(2 + iT) is 7e-10 at T = 10^4
# and 1.2e-5 at T = 2*10^4, because |q(s)|/o_{K1}^2 is no longer small there.
PRX = Product(tau, K1=10 ** 6, nterms=8)


def v_at(T):
    return zeta_from_product(PRX, 2 + 1j * T)[0] / (1 + w_transition(2 + 1j * T))


def richardson(Tl, order=2):
    Tl = np.asarray(Tl, dtype=float)
    A = np.column_stack([Tl ** (-j) for j in range(order + 1)]).astype(complex)
    return np.linalg.solve(A, np.array([v_at(T) for T in Tl]))
```

```
Ts = 1.0e4 * np.array([0.25, 0.5, 1.0])
c = richardson(Ts, 2)
if abs(c[0].imag) > 1e-8 * abs(c[0]):
    raise RuntimeError("extrapolated varpi has a large imaginary part: %r" % c[0])
varpi = c[0].real
L = 1.0 / varpi
say("varpi_RVM (Richardson, sigma = 2, T = 2500, 5000, 10000, accurate product) = %.10f %+.2e i", c[0].real, c
    [0].imag)
say("L = 1/varpi_RVM = %.10f", L)
c2 = richardson(2.0 * Ts, 2)
c3 = richardson([4000.0, 7000.0, 10000.0], 2)
c1 = richardson(Ts[1:], 1)
say("stability of the extrapolation (empirical, no error bound):")
say("  quadratic, T = 5000, 10000, 20000: %.10f %+.2e i  (difference %.1e)", c2[0].real, c2[0].imag, abs(c2[0]
    - c[0]))
say("  quadratic, T = 4000, 7000, 10000:  %.10f %+.2e i  (difference %.1e)", c3[0].real, c3[0].imag, abs(c3[0]
    - c[0]))
say("  linear,    T = 5000, 10000:        %.10f %+.2e i  (difference %.1e)", c1[0].real, c1[0].imag, abs(c1[0]
    - c[0]))
for T in (1.0e4, 2.0e4):
    sT = 2 + 1j * T
    say("  evaluation check at T = %.0f: |zeta_RVM(default) / zeta_RVM(accurate) - 1| = %.1e", T,
        abs(zeta_from_product(PR, sT)[0] / zeta_from_product(PRX, sT)[0] - 1))

# ---- relative error of the transition formula
tt = np.exp(np.linspace(np.log(10.0), np.log(1.0e4), 160))
rows = [tt]
hdr = ["t"]
fits = []
err_by_sp = {}
for sp in (0.25, 0.5, 1.5):
    s = 0.5 + sp + 1j * tt
    zr = zeta_from_product(PR, s)
    # computed residual of the transition formula: it contains the theoretical O(1/t) term
    # and the numerical errors (truncated product, tail approximation, extrapolated varpi, rounding)
    err_w = np.abs(zr / (varpi * (1 + w_transition(s))) - 1)
    err_by_sp[sp] = err_w
    rows.append(err_w)
    hdr.append("err_w_sigmap%.2f" % sp)
    sel = tt > 100
    slope = np.polyfit(np.log(tt[sel]), np.log(err_w[sel]), 1)[0]
    fits.append([sp, slope, np.max(err_w[sel] * tt[sel])])
    say("sigma' = %.2f: |zeta_RVM/(varpi(1+w)) - 1|: log-log slope on t>100 = %.3f, max t*err = %.4f",
        sp, slope, np.max(err_w[sel] * tt[sel]))
    if sp < 0.5:
        err_c = np.abs(zr / (varpi * (1 + chi(s))) - 1)
        rows.append(err_c)
        hdr.append("err_chi_sigmap%.2f" % sp)
        say("sigma' = %.2f: |zeta_RVM/(varpi(1+chi)) - 1|: max t*err (t>100) = %.4f",
            sp, np.max(err_c[sel] * tt[sel]))
    wmod = np.abs(w_transition(s))
    rows.append(wmod)
    hdr.append("abs_w_sigmap%.2f" % sp)
    Fv = 1 / zr
    rows.append(np.abs(Fv - L))
    hdr.append("abs_F_minus_L_sigmap%.2f" % sp)
    say("          |w| * t^sigma' in [%.4f, %.4f];  |1/zeta_RVM - L| at t = 1e4: %.3e",
        np.min(wmod * tt ** sp), np.max(wmod * tt ** sp), np.abs(Fv[-1] - L))
write_csv("s02_transition_errors.csv", hdr, rows)

# ---- zeta_RVM versus zeta on two vertical lines and on the critical line
mp.mp.dps = 20
tl = np.linspace(1.0, 120.0, 900)
cmp = {}
for sigma in (2.0, 0.75):
```

```
    s = sigma + 1j * tl
    zr = zeta_from_product(PR, s)
    ze = np.array([complex(mp.zeta(complex(x))) for x in s])
    cmp[sigma] = (zr, ze)
    write_csv("s02_zetaRVM_vs_zeta_sigma%.2f.csv" % sigma, ["t", "abs_zeta", "abs_zetaRVM", "
     abs_varpi_1_plus_w"],
              [tl, np.abs(ze), np.abs(zr), np.abs(varpi * (1 + w_transition(s)))])

fig, ax = plt.subplots(2, 2, figsize=(12, 8))
cols = {0.25: "tab:red", 0.5: "tab:green", 1.5: "tab:blue"}
for sp in (0.25, 0.5, 1.5):
    ax[0, 0].plot(tt, tt * err_by_sp[sp], "-", color=cols[sp], label=r"$\sigma'=%.2f$" % sp)
ax[0, 0].set_xlabel("t")
ax[0, 0].set_ylabel(r"$t\,|\zeta_{\rm RVM}/(\varpi_{\rm RVM}(1+w))-1|$")
ax[0, 0].set_title(r"Computed residual of the transition formula times $t$, $\varpi_{\rm RVM}=%.10f$" % varpi,
      fontsize=9)
ax[0, 0].legend()
for sp in (0.25, 0.5, 1.5):
    s = 0.5 + sp + 1j * tt
    ax[0, 1].plot(tt, np.abs(1 / zeta_from_product(PR, s) - L) * tt ** min(sp, 1.0), "-", color=cols[sp],
     label=r"$\sigma'=%.2f$" % sp)
ax[0, 1].set_xlabel("t")
ax[0, 1].set_ylabel(r"$t^{\min(\sigma',1)}\,|1/\zeta_{\rm RVM}(\sigma+it)-L|$")
ax[0, 1].set_title(r"Vertical limit $L=1/\varpi_{\rm RVM}$, rescaled error", fontsize=9)
ax[0, 1].legend()
for a, sigma in zip((ax[1, 0], ax[1, 1]), (2.0, 0.75)):
    zr, ze = cmp[sigma]
    a.plot(tl, np.abs(ze), "k-", lw=1.2, label=r"$|\zeta(\sigma+it)|$")
    a.plot(tl, np.abs(zr), "r-", lw=1.2, label=r"$|\zeta_{\rm RVM}(\sigma+it)|$")
    a.set_xlabel("t")
    a.set_title(r"$\sigma=%.2f$" % sigma)
    a.legend()
fig.tight_layout()
savefig(fig, "fig03_transition.png")

sio.savemat(os.path.join(OUT, "s02_results.mat"), {"varpi": varpi, "L": L, "richardson": c,
                                                   "richardson_2T": c2, "richardson_alt": c3, "linear": c1,
                                                   "slopes": np.array(fits), "t": tt})
with open(os.path.join(OUT, "s02_varpi.txt"), "w") as fh:     # read by s05, s07, s15
    fh.write("%.15f\n" % varpi)
say.close()
```

B.4. **s03_exponential_sums.py.**

```
"""Experiment 3: exponential sums S(N;u) = sum_{k<=N} exp(i u o_k) over the midpoint
ordinates tau_k and over the ordinates gamma_k of zeta, and the height transforms M_H(x)."""
import os
import numpy as np
import scipy.io as sio
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import Tee, load_ordinates, write_csv, savefig, OUT

say = Tee("s03_exponential_sums")
gam, tau = load_ordinates()
N = gam.size
step = 2000
Nk = np.arange(step, N + 1, step)
xs = [2.0, 3.0, 4.0, 6.0, 10.0]
Lam = {2.0: np.log(2), 3.0: np.log(3), 4.0: np.log(2), 6.0: 0.0, 10.0: 0.0}
res = {}
# max|S_RVM| is taken over ALL N <= N_max (full cumulative sum); the CSV files and the
# figure keep every 2000th N only.  Landau's formula (Landau 1912): for fixed x > 1,
# sum_{0<gamma<=T} x^(i gamma) = -(T/2pi) Lambda(x)/sqrt(x) + O_x(log T); it is used here
```

```
# only as an illustrative comparison.  |M_H^RVM(x)| = |S_RVM(N)|/N with n_RVM(H) = N, the
# normalization of eq. (MH-RVM) of the paper.
say("%6s %14s %14s %14s %14s %16s %14s" % ("x", "max_N|S_RVM|", "max sampled", "max|S|/log N",
                                          "|S_zeta(N)|", "Landau -T/2pi*L/sqrt(x)", "|M_H^RVM(x)|"))
k_all = np.arange(1, N + 1)
for x in xs:
    u = np.log(x)
    SRall = np.cumsum(np.exp(1j * u * tau))
    SR = SRall[Nk - 1]
    SZ = np.cumsum(np.exp(1j * u * gam))[Nk - 1]
    res[x] = (SR, SZ)
    landau = -gam[-1] / (2 * np.pi) * Lam[x] / np.sqrt(x)
    sel = k_all >= 3
    say("%6.1f %14.4f %14.4f %14.4f %14.2f %16.2f %14.3e" % (x, np.max(np.abs(SRall)), np.max(np.abs(SR)),
                                                             np.max(np.abs(SRall[sel]) / np.log(k_all[sel])),
                                                             np.abs(SZ[-1]), landau, np.abs(SR[-1]) / N))
    write_csv("s03_expsum_x%g.csv" % x, ["N", "re_S_RVM", "im_S_RVM", "re_S_zeta", "im_S_zeta"],
              [Nk, SR.real, SR.imag, SZ.real, SZ.imag])

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
cols = {2.0: "tab:blue", 3.0: "tab:green", 4.0: "tab:purple", 6.0: "tab:orange", 10.0: "tab:red"}
for x in xs:
    ax[0].plot(Nk, np.abs(res[x][0]), "-", color=cols[x], lw=0.8, label=r"$x=%g$" % x)
ax[0].plot(Nk, np.log(Nk), "k--", lw=1, label=r"$\log N$")
ax[0].set_xlabel("N")
ax[0].set_ylabel(r"$|S_{\rm RVM}(N;\log x)|$")
ax[0].set_title(r"Midpoint ordinates: observed $|\sum_{k\leq N}x^{i\tau_k}|$ compared with $\log N$", fontsize
    =10)
ax[0].legend(fontsize=8, ncol=2)
for x in xs:
    ax[1].plot(Nk, res[x][1].real, "-", color=cols[x], lw=0.8, label=r"$x=%g$" % x)
T = np.interp(Nk, np.arange(1, N + 1), gam)
for x in (2.0, 3.0, 4.0):
    ax[1].plot(Nk, -T / (2 * np.pi) * Lam[x] / np.sqrt(x), ":", color=cols[x], lw=1.5)
ax[1].set_xlabel("N")
ax[1].set_ylabel(r"$\mathrm{Re}\sum_{k\leq N}x^{i\gamma_k}$")
ax[1].set_title(r"Zeros of $\zeta$: Landau term $-\frac{T}{2\pi}\frac{\Lambda(x)}{\sqrt{x}}$ (dotted),
    illustrative", fontsize=10)
ax[1].legend(fontsize=8, ncol=2)
fig.tight_layout()
savefig(fig, "fig04_exponential_sums.png")
sio.savemat(os.path.join(OUT, "s03_results.mat"), {"N": Nk, "x": np.array(xs),
                                                   "S_RVM": np.array([res[x][0] for x in xs]),
                                                   "S_zeta": np.array([res[x][1] for x in xs])})
say.close()
```

B.5. **s04_primes.py.**

```
"""Experiment 4: the prime-power count J_RVM, the synthetic prime count pi_RVM, the cell
masses c_n^RVM and the real RVM prime levels p_j^RVM, compared with J, pi(x) and the primes.

J_o(x) = li(x) - 2 Re sum_k Ei((1/2 + i o_k) log x) - log 2 + int_x^inf dt/(t(t^2-1) log t)
for an ordinate list o_k (o = tau: J_RVM; o = gamma: Riemann's explicit formula for J),
pi_o(x) = sum_{m <= log2 x} mu(m)/m J_o(x^(1/m)).
The zero sums are evaluated on a uniform grid in u = log x by the recurrence
exp(i o_k (u + du)) = exp(i o_k u) exp(i o_k du), recomputed exactly every 200 steps (the
discrepancy between the recurrence and the exact value is measured and printed); Ei is
exact (scipy exp1) for k <= KE = 300 and uses Ei(w) = e^w (1/w + 1/w^2 + 2/w^3 + 6/w^4) for
k > KE.  For w = (1/2 + i o_k) u, -w has |ph(-w)| = pi/2 + arctan(1/(2 o_k)), and the remainder
of the four-term expansion is bounded by csc|ph(-w)| times the first neglected term,
24 |e^w| / |w|^5 (DLMF 6.12(i)); the program prints min |w| and the resulting bound for the
whole tail sum, and compares the expansion with the exact Ei on a sample."""
import os
import numpy as np
import scipy.io as sio
```

```
from scipy import special, integrate
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import (Tee, load_ordinates, ei_upper, mobius_sieve, primes_upto, write_csv,
                        savefig, clock_inverse, OUT)

say = Tee("s04_primes")
gam, tau = load_ordinates()
KE = 300


def zero_sum_ei(ords, u, K=None, taper=None):
    """sum_k Ei((1/2 + i o_k) u) on a uniform grid u, k <= K, optional weights."""
    o = ords if K is None else ords[:K]
    wts = np.ones(o.size) if taper is None else taper
    rho = 0.5 + 1j * o
    out = np.zeros(u.size, dtype=complex)
    ke = min(KE, o.size)                # safe for lists shorter than KE
    # exact part
    for i in range(ke):
        out += wts[i] * ei_upper(rho[i] * u)
    if ke == o.size:
        return out
    # asymptotic part by recurrence
    r = rho[ke:]
    wmin = np.min(np.abs(r)) * u[0]
    phase = np.pi / 2 + np.arctan(1.0 / (2.0 * o[ke:]))
    csc = 1.0 / np.sin(phase)
    bound = np.max(np.exp(u / 2)) * np.sum(np.abs(wts[ke:]) * csc * 24.0 / (np.abs(r) * u[0]) ** 5)
    samp = np.arange(ke, min(ke + 50, o.size))
    rel = 0.0
    for uu in (u[0], u[-1]):
        w = rho[samp] * uu
        asym = np.exp(w) * (1 / w + 1 / w ** 2 + 2 / w ** 3 + 6 / w ** 4)
        rel = max(rel, np.max(np.abs(asym - ei_upper(w)) / np.abs(np.exp(w) / w)))
    say("  Ei expansion for k > %d: min |w| = %.1f, bound for the total remainder %.2e; "
        "max |expansion - Ei| / |e^w/w| on a sample of %d ordinates = %.2e", ke, wmin, bound, samp.size, rel)
    b = [wts[ke:] / r ** m for m in (1, 2, 3, 4)]
    du = u[1] - u[0]
    stepf = np.exp(1j * o[ke:] * du)
    drift = 0.0
    for j in range(u.size):
        if j % 200 == 0:
            Ex = np.exp(1j * o[ke:] * u[j])
            if j > 0:
                drift = max(drift, np.max(np.abs(E - Ex)))
            E = Ex
        A = [E @ bm for bm in b]
        uj = u[j]
        out[j] += np.exp(uj / 2) * (A[0] / uj + A[1] / uj ** 2 + 2 * A[2] / uj ** 3 + 6 * A[3] / uj ** 4)
        E = E * stepf
    say("  recurrence check: max |recurrent - exact| of exp(i o_k u) before each recomputation = %.2e", drift)
    return out


def tail_integral(x):
    return integrate.quad(lambda t: 1.0 / (t * (t * t - 1.0) * np.log(t)), x, np.inf, limit=200)[0]


def J_from(ords, u, K=None, taper=None):
    x = np.exp(u)
    S = zero_sum_ei(ords, u, K, taper)
    ti = np.array([tail_integral(v) for v in x])
    return special.expi(u) - 2 * S.real - np.log(2.0) + ti
```

```
def pi_from_J(u, J):
    # J(x^(1/m)) = J(e^(u/m)) is read by linear interpolation on the u-grid; this adds an
    # interpolation error of order (du)^2 max|J''|, measured below by a run on the double step
    mu = mobius_sieve(int(np.floor(u.max() / np.log(2.0))) + 2)
    out = np.zeros(u.size)
    for i, ui in enumerate(u):
        m = 1
        while ui / m >= np.log(2.0) - 1e-12:
            if mu[m] != 0:
                out[i] += mu[m] / m * np.interp(ui / m, u, J)
            m += 1
    return out


def exact_J_pi(x):
    P = primes_upto(int(x.max()) + 1)
    pi = np.searchsorted(P, x, side="right").astype(float)
    J = np.zeros(x.size)
    m = 1
    while 2.0 ** m <= x.max():
        J += np.searchsorted(P, x ** (1.0 / m) + 1e-12, side="right") / m
        m += 1
    return J, pi


def run(xmax, npts, K, label):
    u = np.linspace(np.log(2.0), np.log(xmax), npts)
    if K is None:
        tg = tr = None
        Kg = Kr = None
    else:
        Kg = Kr = K
        v = np.arange(K) / K
        tg = tr = np.where(v < 0.6, 1.0, np.cos(np.pi / 2 * (v - 0.6) / 0.4) ** 2)
    JR = J_from(tau, u, Kr, tr)
    JG = J_from(gam, u, Kg, tg)
    PR = pi_from_J(u, JR)
    PG = pi_from_J(u, JG)
    x = np.exp(u)
    Je, Pe = exact_J_pi(x)
    write_csv("s04_prime_counts_%s.csv" % label, ["x", "J_exact", "J_from_zeta_zeros", "J_RVM",
                                                  "pi_exact", "pi_from_zeta_zeros", "pi_RVM", "li"],
              [x, Je, JG, JR, Pe, PG, PR, special.expi(u)])
    return x, Je, JG, JR, Pe, PG, PR


# x <= 100 with all zeros; x <= 1000 with a smooth cut at K zeros
x1, Je1, JG1, JR1, Pe1, PG1, PR1 = run(100.0, 3000, None, "x100_allzeros")
say("x <= 100, all %d zeros of each list, %d grid points:" % (gam.size, x1.size))
say("  max |pi_from_zeta_zeros - pi| away from primes (distance > 0.2): %.4f",
    np.max(np.abs(PG1 - Pe1)[np.min(np.abs(x1[:, None] - primes_upto(101)[None, :]), axis=1) > 0.2]))
say("  pi_RVM(100) = %.4f, pi(100) = %d, li(100) = %.4f, J_RVM(100) = %.4f, J(100) = %.4f",
    PR1[-1], Pe1[-1], special.expi(np.log(100.0)), JR1[-1], Je1[-1])
dPR = np.diff(PR1)
say("  pi_RVM increasing on the sampled grid of [2,100]: %s (min increment on the grid %.3e)", bool(np.all(dPR
      > 0)), dPR.min())

# Cut height: the harmonic m = 1 of the clock resonates at tau ~ 2 pi x (stationary point of
# exp(i tau u) against the clock), so tau_K = 2.5 * 2 pi x_max keeps every ordinate up to
# 2.5 times the resonant height of the largest x, and the cos^2 taper acts only above
# 0.6 K, that is, above 1.5 times that height.  The factor is a choice, not optimized.
tauK = 2 * np.pi * 1000.0 * 2.5
K = int(np.searchsorted(tau, tauK))
```

```
x2, Je2, JG2, JR2, Pe2, PG2, PR2 = run(1000.0, 40000, K, "x1000_K%d" % K)
say("x <= 1000, smooth cut at K = %d zeros (tau_K = %.1f), %d grid points:" % (K, tau[K - 1], x2.size))
say("  pi_RVM(1000) = %.4f, pi(1000) = %d, li(1000) = %.4f", PR2[-1], Pe2[-1], special.expi(np.log(1000.0)))
# interpolation check: the Moebius inversion reads J(x^(1/m)) by linear interpolation in u;
# repeating it on the grid of double step (every second node, same values of J) changes pi_RVM
# by about 3 times the interpolation error of the step used (error O(du^2))
u2 = np.log(x2)
PR2c = pi_from_J(u2[::2], JR2[::2])
dI = np.max(np.abs(PR2c - PR2[::2]))
say("  interpolation check: max |pi_RVM(step du) - pi_RVM(step 2 du)| at the common nodes = %.2e "
    "(estimated interpolation error of the step du: %.1e)", dI, dI / 3)

# cell masses and real prime levels from the fine x<=1000 run
# convention: pi_RVM(1) = 0, so c_2 = pi_RVM(2) and c_n = pi_RVM(n) - pi_RVM(n-1) for n >= 3
n = np.arange(2, 1001)
pin = np.interp(n, x2, PR2)
c = np.diff(np.concatenate([[0.0], pin]))
P = primes_upto(1000)
isp = np.isin(n, P).astype(float)
say("cell masses c_n = pi_RVM(n) - pi_RVM(n-1):")
say("%5s %10s %10s %8s" % ("n", "c_n", "1/log n", "prime"))
for i in range(0, 40):
    say("%5d %10.5f %10.5f %8d" % (n[i], c[i], 1 / np.log(n[i]), isp[i]))
say("sum_{n<=1000} c_n = pi_RVM(1000) = %.4f (telescoping, with pi_RVM(1) = 0);  pi(1000) = %d", c.sum(), len(
    P))
levels = np.arange(1, int(np.floor(PR2.max())) + 1)


def first_crossing(xg, F, j):
    """First grid point with F >= j, refined by linear interpolation from the previous point;
    a discrete version of inf{x : F(x) >= j}.  Raises if the level is not reached."""
    hit = np.nonzero(F >= j)[0]
    if hit.size == 0:
        raise RuntimeError("level %d not reached on the grid" % j)
    i = hit[0]
    if i == 0:
        return xg[0]
    return xg[i - 1] + (j - F[i - 1]) / (F[i] - F[i - 1]) * (xg[i] - xg[i - 1])


pj = np.array([first_crossing(x2, PR2, j) for j in levels])
say("real RVM prime levels p_j^RVM (first grid crossing of level j, refined linearly) versus p_j:")
say("%5s %12s %8s %10s" % ("j", "p_j^RVM", "p_j", "diff"))
for j in range(min(30, levels.size)):
    say("%5d %12.4f %8d %10.4f" % (levels[j], pj[j], P[j], pj[j] - P[j]))
write_csv("s04_cell_masses.csv", ["n", "c_n_RVM", "inv_log_n", "is_prime"], [n, c, 1 / np.log(n), isp])
m = min(levels.size, P.size)
write_csv("s04_rvm_prime_levels.csv", ["j", "p_j_RVM", "p_j"], [levels[:m], pj[:m], P[:m]])

# ---- figures (linear axes)
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
ax[0].step(x1, Pe1, where="post", color="k", lw=1.6, label=r"$\pi(x)$")
ax[0].plot(x1, PG1, "b-", lw=0.6, label=r"$\pi(x)$ from %d zeros of $\zeta$" % gam.size)
ax[0].plot(x1, PR1, "r-", lw=1.4, label=r"$\pi_{\rm RVM}(x)$ from %d $\tau_k$" % tau.size)
ax[0].plot(x1, special.expi(np.log(x1)) - special.expi(np.log(2.0)), "g--", lw=1, label=r"$\mathrm{li}(x)-\
    mathrm{li}(2)$")
ax[0].set_xlabel("x")
ax[0].set_title(r"Prime counts on $[2,100]$")
ax[0].legend(fontsize=7)
ax[1].step(x2, Pe2, where="post", color="k", lw=1.2, label=r"$\pi(x)$")
ax[1].plot(x2, PR2, "r-", lw=1.0, label=r"$\pi_{\rm RVM}(x)$")
ax[1].set_xlabel("x")
ax[1].set_title(r"Prime counts on $[2,1000]$ ($K=%d$ ordinates, smooth cut)" % K)
ax[1].legend(fontsize=8)
fig.tight_layout()
```

```
savefig(fig, "fig05_prime_counts.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
nn = n[:98]
ax[0].bar(nn, isp[:98], width=0.8, color="0.75", label="prime indicator")
ax[0].plot(nn, c[:98], "r.-", lw=0.8, ms=4, label=r"$c_n^{\rm RVM}$")
ax[0].plot(nn, 1 / np.log(nn), "g--", lw=1, label=r"$1/\log n$")
ax[0].set_xlabel("n")
ax[0].set_title(r"Cell masses $c_n^{\rm RVM}=\pi_{\rm RVM}(n)-\pi_{\rm RVM}(n-1)$")
ax[0].legend(fontsize=8)
ax[1].plot(P[:m], pj[:m], "r.", ms=4, label=r"$(p_j,\,p_j^{\rm RVM})$")
ax[1].plot([0, 1000], [0, 1000], "k-", lw=0.8, label="diagonal")
ax[1].set_xlabel(r"$p_j$")
ax[1].set_ylabel(r"$p_j^{\rm RVM}$")
ax[1].set_title(r"First grid crossings $p_j^{\rm RVM}$ of $\pi_{\rm RVM}\geq j$ versus primes")
ax[1].legend(fontsize=8)
fig.tight_layout()
savefig(fig, "fig06_rvm_primes.png")

sio.savemat(os.path.join(OUT, "s04_results.mat"), {"x100": x1, "pi100": Pe1, "piZeta100": PG1, "piRVM100": PR1,

                                                   "x1000": x2, "pi1000": Pe2, "piRVM1000": PR2,
                                                   "c_n": c, "p_RVM": pj, "K": K})
say.close()
```

B.6. **s05_mertens.py.**

```
"""Experiment 5: the Mertens function M, the synthetic Mertens function M_RVM^crit and the
ternary walk M_h with values in {-1,0,+1} on the squarefree support.

Explicit formulas (simple zeros, residues a = 1/(rho zeta'(rho))):
  M_zeta(x) = -2 + 2 Re sum_k a_k x^(rho_k) + trivial terms        (zeros of zeta)
  M_RVM(x)  = -2 + 2 Re sum_k b_k x^(1/2 + i tau_k) + trivial terms (b_k = 1/(rho_k zeta_RVM'(rho_k)))
Ternary walk: mu_h(n) = mu(n)^2 * sign, sign = +1 if M_h(n-1) < target(n) else -1.

The residues b_k are computed from the product for k <= K0 = 1000 and MODELLED for k > K0 by
zeta_RVM'(rho_k) ~ varpi log(tau_k/2pi)(1 + c/tau_k), with c fitted on K0/2 < k <= K0; the
target for M_RVM^crit at the integers is the explicit formula with a smooth x-dependent cut,
evaluated on an adaptive grid and interpolated (PCHIP): the "tapered-and-interpolated target",
not the exact function M_RVM^crit.  varpi_RVM is read from output/s02_varpi.txt (program s02)."""
import os
import math
import numpy as np
import mpmath as mp
import scipy.io as sio
from scipy.interpolate import PchipInterpolator
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import (Tee, load_ordinates, Product, logG, clock_inverse, tail_coefficients,
                        theta_rs, mobius_sieve, write_csv, savefig, zeta_from_product, OUT)

say = Tee("s05_mertens")
gam, tau = load_ordinates()
Nz = gam.size
XMAX = 10 ** 6
K0 = 1000

# ------------------------------------------------------------------ RVM residues
c1, c2, c3 = tail_coefficients(clock_inverse(float(Nz)))
t2 = tau ** 2
zpR = np.zeros(K0, dtype=complex)
# P(s) = prod_j ((s-1/2)^2 + tau_j^2)/(1/4 + tau_j^2), so at rho_k = 1/2 + i tau_k
# P'(rho_k) = 2 i tau_k/(1/4 + tau_k^2) * prod_{j != k} (tau_j^2 - tau_k^2)/(tau_j^2 + 1/4).
# Exactly the factors j < k are negative; with the 0-based index k there are k of them,
```

```
# which gives the sign (-1)**k.  The factors beyond the file use the clock tail.
for k in range(K0):
    tk = tau[k]
    fac = (t2 - tk ** 2) / (t2 + 0.25)
    fac[k] = 1.0
    q = -0.25 - tk ** 2
    lmod = np.sum(np.log(np.abs(fac))) + c1 * q - c2 * q ** 2 + c3 * q ** 3
    rho = 0.5 + 1j * tk
    lP = np.log(2j * tk / (0.25 + tk ** 2)) + lmod
    zpR[k] = (-1) ** k * np.exp(lP - logG(rho))
ratio = zpR / np.log(tau[:K0] / (2 * np.pi))
PR = Product(tau)
varpi = float(open(os.path.join(OUT, "s02_varpi.txt")).read())     # common result of s02
sel = np.arange(K0 // 2, K0)
cfit = np.mean((ratio[sel] / varpi - 1) * tau[sel])
say("zeta_RVM'(rho_k) exact for k <= %d; zeta_RVM'(rho_k)/log(tau_k/2pi) at k = %d: %.10f %+.2e i",
    K0, K0, ratio[-1].real, ratio[-1].imag)
say("varpi_RVM = %.10f; fit zeta_RVM'(rho_k) = varpi log(tau_k/2pi)(1 + c/tau_k), c = %.4f %+.4f i",
    varpi, cfit.real, cfit.imag)
zR = varpi * np.log(tau / (2 * np.pi)) * (1 + cfit / tau)
zR[:K0] = zpR
bR = 1.0 / ((0.5 + 1j * tau) * zR)
# unit test of the product formula and of the sign (-1)**k: central difference of zeta_RVM = P/G
# (default evaluation of the product) across rho_k, k = 1..6
hh = 1e-6
for k in range(6):
    rho = 0.5 + 1j * tau[k]
    num = (zeta_from_product(PR, rho + hh)[0] - zeta_from_product(PR, rho - hh)[0]) / (2 * hh)
    say("  check k = %d: zeta_RVM'(rho_k) product formula %+.8f %+.8fi, central difference %+.8f %+.8fi, "
        "relative difference %.1e", k + 1, zpR[k].real, zpR[k].imag, num.real, num.imag, abs(num / zpR[k] - 1)
     )

# ------------------------------------------------------------------ residues of zeta
def rs_Z(t):
    a = np.sqrt(t / (2 * np.pi))
    Nn = np.floor(a).astype(int)
    p = a - Nn
    nmax = Nn.max()
    n = np.arange(1, nmax + 1)
    ph = theta_rs(t)[:, None] - t[:, None] * np.log(n)[None, :]
    mask = n[None, :] <= Nn[:, None]
    Z = 2 * np.sum(np.cos(ph) * mask / np.sqrt(n)[None, :], axis=1)
    cc = np.cos(2 * np.pi * p)
    C0 = np.cos(2 * np.pi * (p ** 2 - p - 1.0 / 16)) / cc
    return Z + (-1.0) ** (Nn - 1) * (t / (2 * np.pi)) ** (-0.25) * C0


mp.mp.dps = 20
KM = 1000
zG = np.zeros(Nz, dtype=complex)
for k in range(KM + 50):
    zG[k] = complex(mp.zeta(mp.mpc(0.5, gam[k]), 1, 1))
h = 1e-5
blk_checks = []
for ib, i0 in enumerate(range(KM, Nz, 4000)):
    i1 = min(Nz, i0 + 4000)
    t = gam[i0:i1]
    Zp = (rs_Z(t + h) - rs_Z(t - h)) / (2 * h)
    val = -1j * np.exp(-1j * theta_rs(t)) * Zp
    if i0 == KM:
        chk = np.max(np.abs(val[:50] / zG[KM:KM + 50] - 1))
        say("Riemann-Siegel zeta'(rho) versus mpmath at k = %d..%d: max relative difference %.2e",
            KM + 1, KM + 50, chk)
        Zp2 = (rs_Z(t[:50] + 2 * h) - rs_Z(t[:50] - 2 * h)) / (4 * h)
        say("  numerical differentiation: |Z'(h) - Z'(2h)|/|Z'(h)| on the same points: max %.2e",
```

```
            np.max(np.abs(Zp2 - Zp[:50]) / np.abs(Zp[:50])))
        zG[KM + 50:i1] = val[50:]
    else:
        zG[i0:i1] = val
    if ib % 50 == 25:                    # spot check against mpmath inside later blocks
        ref = complex(mp.zeta(mp.mpc(0.5, t[0]), 1, 1))
        blk_checks.append((i0 + 1, abs(val[0] / ref - 1)))
for kk_, d_ in blk_checks:
    say("  spot check at k = %d (start of a block): |RS / mpmath - 1| = %.2e", kk_, d_)
aG = 1.0 / ((0.5 + 1j * gam) * zG)
S2G = 2 * np.sum(np.abs(aG) ** 2)
S2R = 2 * np.sum(np.abs(bR) ** 2)
say("median |zeta'(rho)| = %.4f;  truncated spectral sums over the supplied ordinates 2 sum_{k<=N}|a_k|^2: "
    "zeta %.6f, RVM %.6f", np.median(np.abs(zG)), S2G, S2R)


def trivial_zeta(x, n=6):
    v = np.zeros_like(x)
    for k in range(1, n + 1):
        v += (-1) ** (k - 1) * (2 * np.pi / x) ** (2 * k) / (math.factorial(2 * k) * k * float(mp.zeta(2 * k +
      1)))
    return v


def trivial_rvm(x, n=6):
    v = np.zeros_like(x)
    for k in range(1, n + 1):
        s = -2.0 * k
        lP = np.real(PR.logP(s)[0])
        v -= x ** (-2 * k) * (2 * k + 1) * np.pi ** k * 2 * (-1) ** k / (math.factorial(k) * np.exp(lP))
    return v


def explicit_uniform(ords, wts, u):
    """2 Re sqrt(x) sum_k w_k exp(i o_k u) on a uniform u grid (recurrence)."""
    out = np.zeros(u.size)
    st = np.exp(1j * ords * (u[1] - u[0]))
    for j in range(u.size):
        if j % 200 == 0:
            E = np.exp(1j * ords * u[j])
        out[j] = 2 * np.exp(u[j] / 2) * np.real(E @ wts)
        E = E * st
    return out


# ------------------------------------------------------------------ exact M by sieve
mu = mobius_sieve(XMAX)
M = np.cumsum(mu.astype(np.int64))
sqf = mu != 0
say("M(10^6) = %d;  squarefree density Q(10^6)/10^6 = %.6f (6/pi^2 = %.6f)", M[XMAX], sqf[1:].mean(), 6 / np.
     pi ** 2)

# ------------------------------------------------------------------ explicit formulas on [2, 2000]
u = np.linspace(np.log(2.0), np.log(2000.0), 2500)
xg = np.exp(u)
Mz = -2 + explicit_uniform(gam, aG, u) + trivial_zeta(xg)
Mc = -2 + explicit_uniform(tau, bR, u) + trivial_rvm(xg)
Mfl = M[np.floor(xg).astype(int)]
for thr in (0.02, 0.05, 0.10):
    off = np.abs(xg - np.round(xg)) > thr
    say("explicit formula with %d zeros of zeta on [2,2000]: rms(M_zeta - M) = %.4f off integers (|x-n|>%.2f)",

        Nz, np.sqrt(np.mean((Mz - Mfl)[off] ** 2)), thr)
for nt in (5, 7):
```

```
    say("trivial terms: |trivial_zeta(2; %d) - trivial_zeta(2; 6)| = %.2e, |trivial_rvm(2; %d) - trivial_rvm
     (2; 6)| = %.2e",
        nt, abs(trivial_zeta(np.array([2.0]), nt)[0] - trivial_zeta(np.array([2.0]), 6)[0]),
        nt, abs(trivial_rvm(np.array([2.0]), nt)[0] - trivial_rvm(np.array([2.0]), 6)[0]))
write_csv("s05_explicit_formulas.csv", ["x", "M_floor", "M_zeta_explicit", "M_RVM_explicit"], [xg, Mfl, Mz, Mc
    ])

# ------------------------------------------------------------- integer target for M_RVM^crit
def target_grid(X, div=24):
    x = [1.0]
    while x[-1] < X:
        x.append(min(X, x[-1] + max(1.0, np.floor(np.sqrt(x[-1]) / div))))
    return np.array(x)


def tapered(x, ords, a, C1=2.0, C2=4.0):
    S = np.zeros(x.size, dtype=complex)
    lx = np.log(x)
    i0 = 0
    while i0 < x.size:
        i1 = min(x.size, i0 + 200)
        K = max(1, int(np.searchsorted(ords, C2 * 2 * np.pi * np.sqrt(x[i1 - 1]))))
        uu = ords[None, :K] / (2 * np.pi * np.sqrt(x[i0:i1, None]))
        w = np.where(uu <= C1, 1.0, np.where(uu >= C2, 0.0, np.cos(np.pi / 2 * (uu - C1) / (C2 - C1)) ** 2))
        E = np.exp(1j * lx[i0:i1, None] * ords[None, :K])
        S[i0:i1] = np.sqrt(x[i0:i1]) * np.sum(E * w * a[None, :K], axis=1)
        i0 = i1
    return S


xT = target_grid(float(XMAX))
TT = -2 + 2 * np.real(tapered(xT, tau, bR)) + trivial_rvm(xT)
nn = np.arange(1, XMAX + 1)
T = np.concatenate([[0.0], PchipInterpolator(xT, TT)(nn)])
for nchk in (np.arange(1000, 100001, 997), np.arange(900000, 1000001, 3001)):
    full = -2 + 2 * np.real(np.array([np.sqrt(n) * np.sum(np.exp(1j * np.log(n) * tau) * bR) for n in nchk]))
     + trivial_rvm(nchk.astype(float))
    say("tapered-and-interpolated target on %d grid points; at %d integers in [%d, %d]: "
        "max |M_RVM(n) (all zeros, untapered) - target(n)| = %.4f",
        xT.size, nchk.size, nchk[0], nchk[-1], np.max(np.abs(full - T[nchk])))
# sensitivity of the target to the model of the residues for k > K0: fit range and model form
alt = {}
sel2 = np.arange(K0 // 4, K0)
c_alt = np.mean((ratio[sel2] / varpi - 1) * tau[sel2])
alt["c fitted on 250 < k <= 1000"] = varpi * np.log(tau / (2 * np.pi)) * (1 + c_alt / tau)
Amat = np.column_stack([1 / tau[sel], 1 / tau[sel] ** 2]).astype(complex)
cd = np.linalg.lstsq(Amat, ratio[sel] / varpi - 1, rcond=None)[0]
alt["model c/tau + d/tau^2"] = varpi * np.log(tau / (2 * np.pi)) * (1 + cd[0] / tau + cd[1] / tau ** 2)
nsens = np.r_[np.arange(1000, 100001, 997), np.arange(900000, 1000001, 3001)].astype(float)
base_s = 2 * np.real(tapered(nsens, tau, bR))
for name, zA in alt.items():
    zA[:K0] = zpR
    bA = 1.0 / ((0.5 + 1j * tau) * zA)
    say("  residue model %-28s: max |change of the target| at the %d check integers = %.2e",
        name, nsens.size, np.max(np.abs(2 * np.real(tapered(nsens, tau, bA)) - base_s)))

# ------------------------------------------------------------- ternary walk
mu_h = np.zeros(XMAX + 1, dtype=np.int8)
Mh = np.zeros(XMAX + 1)
sgn = np.zeros(XMAX + 1, dtype=np.int8)
cur = 0.0
for n in range(1, XMAX + 1):
    if sqf[n]:
        sgn[n] = 1 if cur < T[n] else -1
        cur += sgn[n]
```

```
        mu_h[n] = sgn[n]
    Mh[n] = cur
err = Mh[1:] - T[1:]
say("ternary walk: observed max |M_h - target| on [1,1e6] = %.4f, on [1e4,1e6] = %.4f", np.max(np.abs(err)),
     np.max(np.abs(err[9999:])))
dens = lambda a: [np.mean(a[1:] == -1), np.mean(a[1:] == 0), np.mean(a[1:] == 1)]
say("densities of (-1, 0, +1):  mu   %.6f %.6f %.6f", *dens(mu))
say("                           mu_h %.6f %.6f %.6f   (3/pi^2 = %.6f, 1 - 6/pi^2 = %.6f)", *dens(mu_h), 3 / np.
     pi ** 2, 1 - 6 / np.pi ** 2)
say("sign agreement mu_h = mu on squarefrees: %.6f;  sum mu mu_h / Q = %.6f",
    np.mean(mu_h[1:][sqf[1:]] == mu[1:][sqf[1:]]), np.sum(mu_h[1:].astype(int) * mu[1:].astype(int)) / sqf[1:].
     sum())
say("max |M(n)|/sqrt(n) on [10,1e6] = %.4f;  max |target(n)|/sqrt(n) = %.4f;  max |M_h(n)|/sqrt(n) = %.4f",
    np.max(np.abs(M[10:]) / np.sqrt(nn[9:])), np.max(np.abs(T[10:]) / np.sqrt(nn[9:])), np.max(np.abs(Mh[10:])
      / np.sqrt(nn[9:])))
say("\nNUMERICAL EXAMPLE OF THE TERNARY DIVISION (first 40 integers)")
say("%4s %4s %6s %10s %8s %6s %6s %6s %6s" % ("n", "mu", "mu^2", "target", "M_h(n-1)", "sign", "mu_h", "M_h",
     "M"))
rows = []
for n in range(1, 41):
    say("%4d %4d %6d %10.4f %8d %6s %6d %6d %6d" % (n, mu[n], abs(mu[n]), T[n], Mh[n - 1],
                                                    ("+1" if sgn[n] > 0 else ("-1" if sgn[n] < 0 else "-")),
                                                    mu_h[n], Mh[n], M[n]))
    rows.append([n, mu[n], abs(mu[n]), T[n], Mh[n - 1], sgn[n], mu_h[n], Mh[n], M[n]])
rows = np.array(rows, dtype=float)
write_csv("s05_ternary_example.csv", ["n", "mu", "mu2", "target", "Mh_prev", "sign", "mu_h", "Mh", "M"],
          [rows[:, i] for i in range(rows.shape[1])])

Ns = 10 ** np.arange(3, 7)
say("\nENERGY E_N = (1/N) sum_{n<=N} W(n)^2")
say("%10s %14s %14s %14s" % ("N", "E_N(M)", "E_N(M_RVM)", "E_N(M_h)"))
EN = []
for Nn in Ns:
    EN.append([Nn, np.mean(M[1:Nn + 1].astype(float) ** 2), np.mean(T[1:Nn + 1] ** 2), np.mean(Mh[1:Nn + 1] **
      2)])
    say("%10d %14.4f %14.4f %14.4f" % tuple(EN[-1]))
EN = np.array(EN)
write_csv("s05_energy.csv", ["N", "E_M", "E_MRVM", "E_Mh"], [EN[:, i] for i in range(4)])
nall = np.arange(1, XMAX + 1)
write_csv("s05_walks.csv", ["n", "M", "M_RVM_target", "M_h"], [nall, M[nall], T[nall], Mh[nall]])
mu_h[1:].astype(np.int8).tofile(os.path.join(OUT, "mu_h_int8_1e6.bin"))

# ---------------------------------------------------------------- figures (linear axes)
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
w = np.arange(1, 2001)
ax[0].step(w, M[w], where="post", color="k", lw=1.0, label=r"$M(n)$")
ax[0].plot(w, T[w], "r-", lw=1.2, label=r"target for $M_{\rm RVM}^{\rm crit}(n)$ (tapered, interpolated)")
ax[0].step(w, Mh[w], where="post", color="tab:green", lw=0.8, label=r"$M_h(n)$")
ax[0].set_xlabel("n")
ax[0].set_title(r"Mertens functions on $[1,2000]$, every integer")
ax[0].legend(fontsize=8)
nb = 2000
bw = XMAX // nb
xb = (np.arange(nb) + 0.5) * bw
Mb = M[1:].reshape(nb, bw)
Tb = T[1:].reshape(nb, bw)
ax[1].fill_between(xb, Mb.min(1), Mb.max(1), color="0.6", label=r"$M(n)$ (range per block)")
ax[1].plot(xb, Tb.mean(1), "r-", lw=1.0, label=r"target for $M_{\rm RVM}^{\rm crit}(n)$ (block mean)")
ax[1].plot(xb, np.sqrt(xb), "k--", lw=0.8, label=r"$\pm\sqrt{n}$")
ax[1].plot(xb, -np.sqrt(xb), "k--", lw=0.8)
ax[1].set_xlabel("n")
ax[1].set_title(r"$[1,10^6]$, blocks of %d integers" % bw)
ax[1].legend(fontsize=8)
fig.tight_layout()
savefig(fig, "fig07_mertens.png")
```

```
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
w = np.arange(1, 61)
ax[0].vlines(w - 0.15, 0, mu[w], color="k", lw=2, label=r"$\mu(n)$")
ax[0].vlines(w + 0.15, 0, mu_h[w], color="tab:green", lw=2, label=r"$\mu_h(n)$")
ax[0].axhline(0, color="k", lw=0.5)
ax[0].set_xlabel("n")
ax[0].set_yticks([-1, 0, 1])
ax[0].set_title(r"Ternary values: same zero set $\{\mu(n)=0\}$, signs chosen by tracking")
ax[0].legend(fontsize=8)
w = np.arange(1, 301)
ax[1].step(w, M[w], where="post", color="k", lw=1.0, label=r"$M(n)$")
ax[1].plot(w, T[w], "r.-", lw=0.8, ms=3, label=r"tapered-and-interpolated target for $M_{\rm RVM}^{\rm crit}(n
    )$")
ax[1].step(w, Mh[w], where="post", color="tab:green", lw=1.2, label=r"$M_h(n)$")
ax[1].set_xlabel("n")
ax[1].set_title(r"The walk $M_h$ and the target (observed max deviation %.2f on $[1,10^6]$)" % np.max(np.abs(
    err)))
ax[1].legend(fontsize=8)
fig.tight_layout()
savefig(fig, "fig08_ternary_walk.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
sel = xg <= 300
ax[0].step(np.arange(1, 301), M[1:301], where="post", color="k", lw=1.0, label=r"$M(x)$")
ax[0].plot(xg[sel], Mz[sel], "b-", lw=0.7, label=r"explicit formula, %d zeros of $\zeta$" % Nz)
ax[0].plot(xg[sel], Mc[sel], "r-", lw=1.0, label=r"$M_{\rm RVM}^{\rm crit}(x)$, %d $\tau_k$" % Nz)
ax[0].set_xlabel("x")
ax[0].set_title("Explicit formulas on [2,300]")
ax[0].legend(fontsize=7)
kk = np.arange(5000)
ax[1].plot(gam[kk], np.abs(zG[kk]) / np.log(gam[kk] / (2 * np.pi)), "b.", ms=1.5, label=r"$|\zeta'(\rho_k)|/\
    log(\gamma_k/2\pi)$")
ax[1].plot(tau[:K0], np.abs(zR[:K0]) / np.log(tau[:K0] / (2 * np.pi)), "r.", ms=1.5,
           label=r"$|\zeta_{\rm RVM}'(\rho^{\rm RVM}_k)|/\log(\tau_k/2\pi)$, $k\leq1000$ (product)")
ax[1].plot(tau[K0:5000], np.abs(zR[K0:5000]) / np.log(tau[K0:5000] / (2 * np.pi)), ".", color="tab:orange", ms
    =1.0,
           label=r"same, $k>1000$ (fitted model)")
ax[1].axhline(varpi, color="k", lw=0.8, ls="--", label=r"$\varpi_{\rm RVM}$")
ax[1].set_xlabel("ordinate")
ax[1].set_ylim(0, 4)
ax[1].set_title(r"Derivatives at the zeros, $k\leq5000$")
ax[1].legend(fontsize=7)
fig.tight_layout()
savefig(fig, "fig09_explicit_weights.png")

sio.savemat(os.path.join(OUT, "s05_results.mat"), {"varpi": varpi, "cfit": cfit, "S2G": S2G, "S2R": S2R,
                                                   "energy": EN, "ternary_example": rows,
                                                   "x_explicit": xg, "M_zeta_explicit": Mz, "M_RVM_explicit":
    Mc})
say.close()
```

B.7. **s06_prime_ratios.py.**

```
"""Experiment 6: the three ratios li/pi_RVM, pi/li, pi/pi_RVM and the square-root estimate
pi_RVM(x) - li(x) = O(sqrt(x) (log log x)^2 / log x), from the prime counts written by s04_primes.py.

Input: output/s04_prime_counts_x1000_K17061.csv (columns x, pi_exact, pi_RVM, li, ...).
B(x) = sqrt(x) (log log x)^2 / log x is the scale of the square-root estimate."""
import os
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import Tee, savefig, write_csv, OUT
```

```
say = Tee("s06_prime_ratios")
D = np.genfromtxt(os.path.join(OUT, "s04_prime_counts_x1000_K17061.csv"), delimiter=",", names=True)
x, pe, pr, li = D["x"], D["pi_exact"], D["pi_RVM"], D["li"]
say("grid: %d points on [%.0f, %.0f]; pi_RVM from the first 17061 ordinates tau_k (smooth cut)"
    % (x.size, x[0], x[-1]))

sel = x >= 10
r1, r2, r3 = li[sel] / pr[sel], pe[sel] / li[sel], pe[sel] / pr[sel]
xs = x[sel]
say("")
say("ratios at selected x")
say("%8s %12s %12s %12s %12s %12s %12s" % ("x", "pi", "pi_RVM", "li", "li/pi_RVM", "pi/li", "pi/pi_RVM"))
for X in [10, 20, 50, 100, 200, 500, 1000]:
    i = int(np.argmin(np.abs(x - X)))
    say("%8.1f %12.4f %12.4f %12.4f %12.6f %12.6f %12.6f"
        % (x[i], pe[i], pr[i], li[i], li[i] / pr[i], pe[i] / li[i], pe[i] / pr[i]))

s8 = x >= 8
B = np.sqrt(x[s8]) * np.log(np.log(x[s8])) ** 2 / np.log(x[s8])
e_rvm = (pr[s8] - li[s8]) / B
e_pi = (pe[s8] - li[s8]) / B
say("")
say("square-root scale B(x) = sqrt(x) (log log x)^2 / log x on [8, 1000]")
say("max |pi_RVM - li| / B = %.4f   (at x = %.2f)" % (np.max(np.abs(e_rvm)), x[s8][np.argmax(np.abs(e_rvm))]))
say("max |pi - li| / B     = %.4f   (at x = %.2f)" % (np.max(np.abs(e_pi)), x[s8][np.argmax(np.abs(e_pi))]))
for X in [100, 1000]:
    m = x[s8] >= X / 2
    say("max |pi_RVM - li| / B on [%d, 1000] = %.4f" % (X // 2, np.max(np.abs(e_rvm[m]))))

# CSV files: full grid
write_csv("s06_prime_ratios.csv", ["x", "li_over_pi_RVM", "pi_over_li", "pi_over_pi_RVM"],
          [xs, r1, r2, r3])
write_csv("s06_square_root_scale.csv", ["x", "pi_RVM_minus_li_over_B", "pi_minus_li_over_B"],
          [x[s8], e_rvm, e_pi])

# ---- figure (linear axes)
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
ax[0].plot(xs, r3, "k-", lw=0.8, label=r"$\pi(x)/\pi_{\rm RVM}(x)$")
ax[0].plot(xs, r2, "b-", lw=0.8, label=r"$\pi(x)/\mathrm{li}(x)$")
ax[0].plot(xs, r1, "r-", lw=1.4, label=r"$\mathrm{li}(x)/\pi_{\rm RVM}(x)$")
ax[0].axhline(1.0, color="0.5", lw=0.8, ls="--")
ax[0].set_xlabel("x")
ax[0].set_title(r"The three ratios on $[10,1000]$")
ax[0].legend(fontsize=8)
ax[1].plot(x[s8], e_pi, "b-", lw=0.6, label=r"$(\pi-\mathrm{li})/B$")
ax[1].plot(x[s8], e_rvm, "r-", lw=1.4, label=r"$(\pi_{\rm RVM}-\mathrm{li})/B$")
ax[1].axhline(0.0, color="0.5", lw=0.8, ls="--")
ax[1].set_xlabel("x")
ax[1].set_title(r"Square-root scale $B(x)=\sqrt{x}\,(\log\log x)^2/\log x$")
ax[1].legend(fontsize=8)
fig.tight_layout()
savefig(fig, "fig10_prime_ratios.png")
say.close()
```

B.8. **s07_product_identity.py.**

```
"""Experiment 7: the value at infinity of the endpoint.

varpi_Z1 = lim_{sigma -> +inf} zeta_Z1(sigma) = varpi_RVM * prod_k (1/4 + tau_k^2) / (1/4 + gamma_k^2),
so zeta_Z1 = zeta holds exactly when the ordered partial products
P_K = prod_{k <= K} (1/4 + gamma_k^2) / (1/4 + tau_k^2)
converge to varpi_RVM.  The partial products are computed from data/zeross.bin (gamma_k) and
data/zLamb.bin (tau_k) and compared with varpi_RVM read from output/s02_varpi.txt (program
s02_transition.py).  The agreement of P_K with varpi_RVM at the end of the file is an internal
```

```
check of two computations built on the same clock and related data, not a certification of
the infinite product."""
import os
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import Tee, load_ordinates, savefig, write_csv, OUT

say = Tee("s07_product_identity")
gam, tau = load_ordinates()
VARPI = float(open(os.path.join(OUT, "s02_varpi.txt")).read())   # common result of s02_transition.py
N = gam.size
logP = np.cumsum(np.log1p(0.25 / gam ** 2) + 2 * np.log(gam) - np.log1p(0.25 / tau ** 2) - 2 * np.log(tau))
P = np.exp(logP)
say("N = %d ordinates; varpi_RVM = %.10f" % (N, VARPI))
say("%10s %16s %14s" % ("K", "P_K", "P_K - varpi"))
for K in [K for K in [10, 100, 1000, 10 ** 4, 10 ** 5, 10 ** 6] if K < N] + [N]:
    say("%10d %16.10f %14.3e" % (K, P[K - 1], P[K - 1] - VARPI))
for K0 in [K0 for K0 in [10 ** 4, 10 ** 5, 10 ** 6] if K0 <= N]:
    seg = P[K0 - 1:]
    say("K >= %7d: min P_K = %.10f, max P_K = %.10f, mean = %.10f" % (K0, seg.min(), seg.max(), seg.mean()))
# the flow along the family: Upsilon(lam) = sum_k 2 gamma_k(lam) eta_k / (1/4 + gamma_k(lam)^2),
# eta_k = tau_k - gamma_k, gamma_k(lam) = (1 - lam) gamma_k + lam tau_k; d/dlam log varpi_lam = -Upsilon(lam)
eta = tau - gam
say("")
say("flow along the lambda-family")
for lam in [0.0, 0.25, 0.5, 0.75, 1.0]:
    gl = (1 - lam) * gam + lam * tau
    say("Upsilon(%.2f) = %.10f" % (lam, np.sum(2 * gl * eta / (0.25 + gl ** 2))))
say("sum_k 2 eta_k^2 / gamma_k^2 = %.10f   (auxiliary size estimate, not an identity of the flow)" % np.sum(2
    * eta ** 2 / gam ** 2))
total = np.sum(np.log((0.25 + tau ** 2) / (0.25 + gam ** 2)))
say("integral_0^1 Upsilon, truncated at k <= N: sum_{k<=N} log((1/4+tau_k^2)/(1/4+gamma_k^2)) = %.10f" % total
    )
say("log L = log(1/varpi_RVM)                                = %.10f" % np.log(1 / VARPI))
step = 50
idx = np.arange(step - 1, N, step)
write_csv("s07_partial_products.csv", ["K", "P_K"], [idx + 1, P[idx]])

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
k1 = np.arange(1, 10 ** 4 + 1)
ax[0].plot(k1, P[:10 ** 4], "r-", lw=0.6, label=r"$P_K$")
ax[0].axhline(VARPI, color="k", lw=0.8, ls="--", label=r"$\varpi_{\rm RVM}$")
ax[0].set_xlabel("K")
ax[0].set_title(r"Partial products $P_K$, $K\leq 10^4$")
ax[0].legend(fontsize=8)
sel = idx >= 10 ** 5 - 1
ax[1].plot(idx[sel] + 1, (P[idx[sel]] - VARPI) * 1e6, "r-", lw=0.6)
ax[1].axhline(0.0, color="k", lw=0.8, ls="--")
ax[1].set_xlabel("K")
ax[1].set_ylabel(r"$(P_K-\varpi_{\rm RVM})\cdot 10^{6}$")
ax[1].set_title(r"$10^5\leq K\leq %d$" % N)
fig.tight_layout()
savefig(fig, "fig11_product_identity.png")
say.close()
```

B.9. **s08_transport_moments.py.**

```
#!/usr/bin/env python3
"""s08_transport_moments.py
Transport tau -> gamma on the first rung (support of psi inside (0, log 2)).

Input (not recomputed): data/zeross.bin (gamma_k), data/zLamb.bin (tau_k), little-endian float64.
```

```
For a fixed normalized smooth bump psi supported in [A,B] subset (0, log 2):
  c_psi^{(j)}(t) = int psi(u) u^j cos(t u + j pi/2) du
  W_psi    = (1/2) int psi(u) ( e^{u/2} - e^{-u/2}/(e^{2u}-1) ) du       (lambda-independent part)
  R_psi(1) = W_psi - sum_k c_psi(tau_k)                                   (clock defect; clock side only)
  S_psi(lambda) = sum_k [ c_psi(gamma_k + lambda*eta_k) - c_psi(tau_k) ]  (transport), eta_k = tau_k - gamma_k
  m_j(psi) = sum_k eta_k^j c_psi^{(j)}(tau_k)                             (clock-sampled moments)
and the partial sums of  S_psi(0) = sum_{j>=1} (-1)^j m_j / j!.
Outputs: output/s08_transport_console.txt, s08_moments.csv, s08_transport_lambda.csv, s08_first_moment_terms.
    csv
"""
import os, math
import numpy as np

K = 20000
A, B = 0.10, 0.62            # support of psi, inside (0, log 2 = 0.6931)
NU = 4001                    # quadrature nodes
JMAX = 12
HERE = os.path.dirname(os.path.abspath(__file__))
OUT = os.path.join(HERE, '..', 'output')
DATA = os.path.join(HERE, '..', 'data')
os.makedirs(OUT, exist_ok=True)

u = np.linspace(A, B, NU)
v = (u - 0.5 * (A + B)) / (0.5 * (B - A))
psi = np.zeros_like(u)
ins = np.abs(v) < 1
psi[ins] = np.exp(-1.0 / (1.0 - v[ins] ** 2))
du = u[1] - u[0]
psi /= psi.sum() * du                      # int psi = 1

def c(t, j=0, chunk=2000):
    t = np.atleast_1d(np.asarray(t, dtype=float))
    w = psi * u ** j * du
    out = np.empty_like(t)
    for a in range(0, len(t), chunk):
        out[a:a + chunk] = np.cos(np.outer(t[a:a + chunk], u) + j * np.pi / 2) @ w
    return out

def main():
    lines = []
    def P(s=''):
        print(s); lines.append(s)
    g_all = np.fromfile(os.path.join(DATA, 'zeross.bin'), dtype='<f8')
    t_all = np.fromfile(os.path.join(DATA, 'zLamb.bin'), dtype='<f8')
    if len(g_all) != len(t_all) or len(g_all) < K:
        raise ValueError('the ordinate files must have equal lengths >= K = %d' % K)
    g = g_all[:K]; tau = t_all[:K]; eta = tau - g
    W = 0.5 * (psi * (np.exp(u / 2) - np.exp(-u / 2) / (np.exp(2 * u) - 1))).sum() * du
    c_tau = c(tau); c_gam = c(g)
    R1 = W - c_tau.sum()
    S0 = (c_gam - c_tau).sum()
    P('s08: transport tau -> gamma on the first rung')
    P(f'entries in zeross.bin / zLamb.bin: {len(g_all)} / {len(t_all)};  used K = {K}')
    P(f'gamma_1 = {g[0]:.9f}, tau_1 = {tau[0]:.9f};  gamma_K = {g[-1]:.6f}, tau_K = {tau[-1]:.6f}')
    P(f'psi: normalized bump on [{A}, {B}] (log 2 = {math.log(2):.6f}), {NU} nodes')
    P(f'max|eta_k| = {np.abs(eta).max():.6f} at k = {np.abs(eta).argmax()+1};  rms eta = {np.sqrt((eta**2).
    mean()):.6f}')
    for k0 in (100, 300, 1000, 3000, 10000, K - 1):      # k is 1-based: c_tau[k0-1:] is k >= k0
        P(f'  max_(k>={k0:5d}) |c_psi(tau_k)| = {np.abs(c_tau[k0 - 1:]).max():.3e}')
    P('  (values near 1e-13 and below are at the floating-point floor of the quadrature, not a measured decay)
    ')
    P()
    P(f'W_psi                          = {W:+.12f}')
    P(f'sum_k c_psi(tau_k)             = {c_tau.sum():+.12f}')
    P(f'sum_k c_psi(gamma_k)           = {c_gam.sum():+.12f}')
```

```
    P(f'R_psi(1) = W - sum c(tau)      = {R1:+.12f}   (clock defect)')
    P(f'S_psi(0) = sum[c(gam)-c(tau)]  = {S0:+.12f}   (transport)')
    P(f'R_psi(0) = R_psi(1) - S_psi(0) = {R1 - S0:+.3e}   (both terms use the same K ordinates and quadrature)
     ')
    # quadrature check: the integrand is smooth with compact support in [A,B], so the rule used
    # (all nodes, psi = 0 at the ends) is the trapezoidal rule; compare with half the nodes
    u2 = u[::2]; du2 = u2[1] - u2[0]; psi2 = psi[::2] / (psi[::2].sum() * du2)
    def c2(t):
        t = np.atleast_1d(np.asarray(t, dtype=float)); out = np.empty_like(t)
        for a in range(0, len(t), 2000):
            out[a:a + 2000] = np.cos(np.outer(t[a:a + 2000], u2)) @ (psi2 * du2)
        return out
    W2 = 0.5 * (psi2 * (np.exp(u2 / 2) - np.exp(-u2 / 2) / (np.exp(2 * u2) - 1))).sum() * du2
    R1h = W2 - c2(tau).sum(); S0h = (c2(g) - c2(tau)).sum()
    P(f'quadrature check with {len(u2)} nodes: change of R_psi(1) {R1h - R1:+.1e}, of S_psi(0) {S0h - S0:+.1e}
     ')
    P()
    P(' j        m_j(psi)        (-1)^j m_j/j!      partial sum      partial - R_psi(1)')
    rows = []; part = 0.0
    for j in range(1, JMAX + 1):
        mj = (eta ** j * c(tau, j)).sum()
        term = (-1) ** j * mj / math.factorial(j)
        part += term
        rows.append((j, mj, term, part, part - R1))
        P(f'{j:2d}  {mj:+.10e}  {term:+.10e}  {part:+.12f}  {part - R1:+.3e}')
    np.savetxt(os.path.join(OUT, 's08_moments.csv'), np.array(rows), delimiter=',',
               header='j,m_j,term,partial_sum,partial_minus_R1', comments='', fmt=['%d'] + ['%.15e'] * 4)
    # which k carry the first two moments
    t1 = -eta * c(tau, 1); t2 = 0.5 * eta ** 2 * c(tau, 2)
    kk = np.arange(1, K + 1)
    np.savetxt(os.path.join(OUT, 's08_first_moment_terms.csv'), np.c_[kk, tau, eta, c_tau, c_gam - c_tau, t1,
     t2][:400], delimiter=',',
               header='k,tau_k,eta_k,c_psi(tau_k),c_psi(gamma_k)-c_psi(tau_k),-eta_k*c1(tau_k),0.5*eta_k^2*c2(
    tau_k)', comments='',
               fmt=['%d'] + ['%.12e'] * 6)
    P()
    P(' cumulative over k (first K0 ordinates):  K0   sum[c(gam)-c(tau)]   W - sum c(tau)   -m_1 part   +m_2/2
      part')
    for K0 in (1, 2, 3, 5, 10, 20, 50, 100, 200, 400):
        P(f'   {K0:4d}   {(c_gam-c_tau)[:K0].sum():+.10f}   {W - c_tau[:K0].sum():+.10f}   {t1[:K0].sum():+.10
    f}   {t2[:K0].sum():+.10f}')
    P()
    P(' lambda     S_psi(lambda)     R_psi(lambda) = R_psi(1) - S_psi(lambda)')
    P(' (lambda outside [0,1] is a numerical prolongation of gamma_k + lambda eta_k, not part of the family)')
    for L in (-1.0, -0.5, 1.5, 2.0):
        gl = g + L * eta
        P(f'  at lambda = {L:+.1f} the list is increasing: {bool(np.all(np.diff(gl) > 0))}, positive: {bool(np.
    all(gl > 0))}')
    lam = np.linspace(-1.0, 2.0, 61); rows = []
    for L in lam:
        S = (c(g + L * eta) - c_tau).sum()
        rows.append((L, S, R1 - S))
    for L, S, R in rows[::5]:
        P(f'{L:+.2f}   {S:+.12f}   {R:+.12f}')
    np.savetxt(os.path.join(OUT, 's08_transport_lambda.csv'), np.array(rows), delimiter=',',
               header='lambda,S_psi,R_psi', comments='', fmt='%.15e')
    with open(os.path.join(OUT, 's08_transport_console.txt'), 'w') as f:
        f.write('\n'.join(lines) + '\n')

if __name__ == '__main__':
    main()
```

B.10. **s09_clock_defect_split.py.**

```
#!/usr/bin/env python3
```

```
"""s09_clock_defect_split.py
Splits the clock defect R_psi(1) of s08 (same psi) by Poisson summation on the clock:
  sum_{k>=1} c_psi(tau_k) = I_0 + 2 sum_{m>=1} (-1)^m I_m,
  I_m = int_{t_0}^infty c_psi(t) cos(2 pi m N_RVM(t)) N_RVM'(t) dt,   N_RVM(t_0) = 0,
  I_0 = int psi(u) K_0(u) du,  K_0(u) = -(1/(2 pi u)) [ sin(u t_0) log(t_0/(2 pi)) + pi/2 - Si(u t_0) ].
Reports W_psi, I_0, the smooth mismatch W_psi - I_0 and the oscillating part  -2 sum_{m>=1}(-1)^m I_m
(obtained as R_psi(1) - (W_psi - I_0), with sum_k c_psi(tau_k) read from data/zLamb.bin).
Output: output/s09_clock_defect_split_console.txt
"""
import os, math
import numpy as np
HERE = os.path.dirname(os.path.abspath(__file__))
OUT = os.path.join(HERE, '..', 'output'); DATA = os.path.join(HERE, '..', 'data')
A, B, NU, K = 0.10, 0.62, 4001, 20000
u = np.linspace(A, B, NU); v = (u - 0.5*(A+B)) / (0.5*(B-A))
psi = np.zeros_like(u); ins = np.abs(v) < 1; psi[ins] = np.exp(-1.0/(1.0 - v[ins]**2))
du = u[1]-u[0]; psi /= psi.sum()*du

def NR(t): return t/(2*np.pi)*np.log(t/(2*np.pi*np.e)) + 7.0/8.0
t0 = 9.7
for _ in range(60): t0 -= NR(t0) / (np.log(t0/(2*np.pi))/(2*np.pi))

def Si(x, n=20001):
    s = np.linspace(0.0, 1.0, n)[None, :] * np.atleast_1d(x)[:, None]
    f = np.ones_like(s); nz = s != 0; f[nz] = np.sin(s[nz]) / s[nz]
    h = np.atleast_1d(x) / (n - 1)
    return (f.sum(axis=1) - 0.5*(f[:, 0] + f[:, -1])) * h

K0 = -(1.0/(2*np.pi*u)) * (np.sin(u*t0)*np.log(t0/(2*np.pi)) + np.pi/2 - Si(u*t0))
I0 = (psi*K0).sum()*du
W = 0.5*(psi*(np.exp(u/2) - np.exp(-u/2)/(np.exp(2*u)-1))).sum()*du
tau = np.fromfile(os.path.join(DATA, 'zLamb.bin'), dtype='<f8')
if tau.size < K or not np.all(np.isfinite(tau[:K])) or not np.all(np.diff(tau[:K]) > 0):
    raise ValueError('zLamb.bin must contain at least %d finite, strictly increasing ordinates' % K)
tau = tau[:K]
w = psi*du; ctau = sum((np.cos(np.outer(tau[a:a+2000], u)) @ w).sum() for a in range(0, K, 2000))
R1 = W - ctau
L = [f's09: Poisson split of the clock defect (psi of s08 on [{A},{B}])',
     f't_0 (N_RVM(t_0)=0)                = {t0:.12f}   (2 pi = {2*np.pi:.6f}, tau_1 = {tau[0]:.6f})',
     f'W_psi                             = {W:+.12f}',
     f'I_0  (m = 0, smooth clock density) = {I0:+.12f}',
     f'sum_k c_psi(tau_k)                = {ctau:+.12f}',
     f'R_psi(1)                          = {R1:+.12f}',
     f'smooth mismatch  W_psi - I_0      = {W - I0:+.12f}',
     f'oscillating part -2 sum_(m>=1)(-1)^m I_m = {R1 - (W - I0):+.12f}',
     '  (obtained as R_psi(1) - (W_psi - I_0); the I_m, m >= 1, are not summed, so the convergence of the',
     '   oscillatory Poisson series is not verified independently by this program)',
     f'stationary heights 2 pi e^u, u in supp psi: [{2*np.pi*np.exp(A):.4f}, {2*np.pi*np.exp(B):.4f}];  part
     above t_0: u > {np.log(t0/(2*np.pi)):.6f}']
print('\n'.join(L)); open(os.path.join(OUT, 's09_clock_defect_split_console.txt'), 'w').write('\n'.join(L)+'\n
     ')
```

B.11. **s10_clock_to_zeros.py.**

```
#!/usr/bin/env python3
"""s10_clock_to_zeros.py
Solves, starting from the clock (eta = 0), the system of rung equations

   sum_k c_j(gamma_k) + sum_n a_n psi_j(log n) = W_j ,   j = 1..J,

for the unknown ordinates gamma_k (below a height) and the unknown masses a_n at the log-integers.
  psi_j(u) = w(u) cos(omega_j u), w(u) sin(omega_j u);  w = smooth bump on (0, U), U = log(N + 1/2)
  c_j(t)   = int psi_j(u) cos(t u) du
  W_j      = int psi_j(u) (1/2)( e^{u/2} - e^{-u/2}/(e^{2u}-1) ) du      (lambda-independent part)
Ordinates above the free range are kept on the clock (tau_k from data/zLamb.bin).
```

```
The masses enter linearly and are eliminated exactly at every step (ridge RIDGE).
Continuation in height: the band omega <= Omega is enlarged step by step (OMEGAS); at each stage the
ordinates with tau_k <= Omega + MARGIN are free.  Start: gamma_k = tau_k.  Method: Levenberg-Marquardt,
with the damped linear least-squares problems solved by QR/SVD (scipy.linalg.lstsq on the stacked
matrix), not by normal equations; a step is accepted only if it decreases the residual, and the
iteration of a band stops (keeping the last accepted point) if no damping up to 1e12 does.
data/zeross.bin is used only AFTER the iteration, to compare.
Outputs: output/s10_clock_to_zeros_console.txt, s10_ordinates.csv, s10_masses.csv
"""
import os, math
import numpy as np
from scipy import linalg as sla
HERE = os.path.dirname(os.path.abspath(__file__))
OUT = os.path.join(HERE, '..', 'output'); DATA = os.path.join(HERE, '..', 'data')
N = 13; U = math.log(N + 0.5)
KT = 6000; DOM = 0.1; NU = 8001; ITER = 400
OMEGAS = [18, 22, 26, 30, 34, 38, 42, 46, 50]; MARGIN = 12.0
WMIN = 5e-2; STEP = 0.2; RIDGE = 1e-10

u = np.linspace(0.0, U, NU)[1:-1]; du = U / (NU - 1)
v = (u - U/2) / (U/2); w = np.exp(-1.0/(1.0 - v**2)); w /= w.sum()*du
Wfun = 0.5*(np.exp(u/2) - np.exp(-u/2)/(np.exp(2*u) - 1))
ns = np.array([n for n in range(2, N + 1) if np.interp(math.log(n), u, w)/w.max() >= WMIN])
tau = np.fromfile(os.path.join(DATA, 'zLamb.bin'), dtype='<f8')[:KT]
if tau.size < KT:
    raise ValueError('zLamb.bin has fewer than KT = %d entries' % KT)
COS_TAU = np.zeros_like(u)                        # sum over all clock ordinates of cos(tau_k u)
for a in range(0, KT, 500): COS_TAU += np.cos(np.outer(u, tau[a:a+500])).sum(axis=1)

lines = []
def P(s=''):
    print(s); lines.append(s)
P('s10: from the clock to the ordinates through the rung equations (continuation in height)')
P(f'N = {N}, U = log({N}+1/2) = {U:.6f}; free masses n = {ns[0]}..{ns[-1]}; start gamma_k = tau_k')
g_all = tau.copy()
for OM in OMEGAS:
    om = np.arange(0.0, OM + 1e-9, DOM)
    Psi = np.vstack([w*np.cos(o*u) for o in om] + [w*np.sin(o*u) for o in om[1:]]) * du
    Wj = Psi @ Wfun
    Pn = np.vstack([np.interp(np.log(ns), u, Psi[j]/du) for j in range(Psi.shape[0])])
    # Q = (Pn^T Pn + RIDGE I)^(-1) Pn^T, computed by least squares on the stacked matrix
    nt = Pn.shape[0]; nm = len(ns)
    Sm = np.vstack([Pn, math.sqrt(RIDGE) * np.eye(nm)])
    Q = sla.lstsq(Sm, np.vstack([np.eye(nt), np.zeros((nm, nt))]))[0]
    K = int((tau <= OM + MARGIN).sum())
    g = g_all[:K].copy()
    fixed = Psi @ (COS_TAU - np.cos(np.outer(u, tau[:K])).sum(axis=1)) - Wj
    def r0(g): return Psi @ np.cos(np.outer(u, g)).sum(axis=1) + fixed
    def res(g):
        x = r0(g); a = -Q @ x; return x + Pn @ a, a
    r, a = res(g); n0 = np.linalg.norm(r); damp = 1e-3; it = 0; stalled = False
    for it in range(1, ITER + 1):
        Jg = Psi @ (-u[:, None]*np.sin(np.outer(u, g))); Jg = Jg - Pn @ (Q @ Jg)
        Dg = np.sqrt((Jg**2).sum(axis=0) + 1e-14)          # Marquardt scaling sqrt(diag(J^T J))
        accepted = False
        while damp <= 1e12:
            d = sla.lstsq(np.vstack([Jg, math.sqrt(damp) * np.diag(Dg)]),
                          np.concatenate([-r, np.zeros(len(g))]))[0]
            d *= min(1.0, STEP/max(np.abs(d).max(), 1e-300))
            r2, a2 = res(g + d)
            if np.linalg.norm(r2) < np.linalg.norm(r):
                accepted = True; break
            damp *= 4
        if not accepted:
            stalled = True; break                              # keep the last accepted point
```

```
        g = g + d; r, a = r2, a2; damp = max(damp/3, 1e-14)
        if np.abs(d).max() < 1e-11: break
    g_all[:K] = g
    P(f'band omega <= {OM:2d}: {K:2d} free ordinates, {Psi.shape[0]} tests, |residual| {n0:.3e} -> {np.linalg.
     norm(r):.3e} in {it} iterations'
      + ('  (stopped: no decrease up to damping 1e12)' if stalled else ''))
P()
gam = np.fromfile(os.path.join(DATA, 'zeross.bin'), dtype='<f8')[:K]
P('  k      tau_k (start)    solved ordinate    gamma_k (zeross.bin)   solved - gamma_k    tau_k - gamma_k')
for k in range(K):
    P(f'{k+1:3d}   {tau[k]:14.9f}   {g_all[k]:14.9f}   {gam[k]:14.9f}   {g_all[k]-gam[k]:+.3e}   {tau[k]-gam[k
     ]:+.6f}')
P()
def Lam(n):
    """von Mangoldt function by trial division up to sqrt(n)."""
    if n < 2:
        return 0.0
    p = 2
    while p * p <= n:
        if n % p == 0:
            m = n
            while m % p == 0: m //= p
            return math.log(p) if m == 1 else 0.0
        p += 1
    return math.log(n)                                  # n is prime
P('  n     solved mass a_n     Lambda(n)/(2 sqrt n)     difference     w(log n)/max w')
for i, n in enumerate(ns):
    ex = Lam(int(n))/(2*math.sqrt(n))
    P(f'{n:3d}   {a[i]:+.9f}   {ex:+.9f}   {a[i]-ex:+.3e}   {np.interp(math.log(n), u, w)/w.max():.3e}')
np.savetxt(os.path.join(OUT, 's10_ordinates.csv'), np.c_[np.arange(1, K+1), tau[:K], g_all[:K], gam],
     delimiter=',',
           header='k,tau_k,solved,gamma_k', comments='', fmt=['%d'] + ['%.12f']*3)
np.savetxt(os.path.join(OUT, 's10_masses.csv'), np.c_[ns, a, [Lam(int(n))/(2*math.sqrt(n)) for n in ns]],
     delimiter=',',
           header='n,solved_mass,Lambda(n)/(2sqrt(n))', comments='', fmt=['%d'] + ['%.12f']*2)
open(os.path.join(OUT, 's10_clock_to_zeros_console.txt'), 'w').write('\n'.join(lines) + '\n')
```

B.12. **`s11_rvm_z1_difference_analysis.py`.** This supplemental diagnostic reproduces the RVM–$Z1$ difference, shell and convergence calculations. It uses the genuine ordinates only for post-construction diagnostics and is not used in any proof.

```
#!/usr/bin/env python3
"""s11_rvm_z1_difference_analysis.py

Python reproduction of the focused MATLAB program RVM_Z1_DIFFERENCE_ANALYSIS.m.

The script reads the two separately distributed binary ordinate files and
computes, for K = 1e3, 1e4, 1e5, 5e5, Kmax on 2 <= x <= 1e5:
  * the Perron-based Mertens difference M_RVM,K^crit - M_Z1,K^crit;
  * the explicit-formula prime-counting difference pi_RVM,K - pi_Z1,K;
  * K-convergence errors, spectral-shell contributions, normalizations,
    running suprema, and band statistics.

The numerical conventions and default parameters match the MATLAB analysis
used for the figures discussed in the paper.  The raw binaries are not
included in the archive; place zeross.bin and zLamb.bin in data/ before use.
"""

import os
import math
import time
import numpy as np
_trapz = getattr(np, "trapezoid", None) or np.trapz   # np.trapz was removed in NumPy 2.4
from scipy.special import expi, loggamma
```

```
from scipy.integrate import quad
import matplotlib.pyplot as plt

HERE = os.path.dirname(os.path.abspath(__file__))
ROOT = os.path.abspath(os.path.join(HERE, ".."))
DATA = os.path.join(ROOT, "data")
OUT = os.path.join(ROOT, "output")
FIG = os.path.join(ROOT, "figures")
os.makedirs(OUT, exist_ok=True)
os.makedirs(FIG, exist_ok=True)

XMAX = 100000.0
NX = 500
CP = 0.75
TPERRON = 80.0
NTP = 401
KHEAD = 30000
RPROD = 5
SBLOCK = 40
KEI = 5000
REI = 5
ZERO_BLOCK = 100000
ARG_BLOCK = 16


def say(lines, text=""):
    print(text)
    lines.append(text)


def load_ordinates():
    tau_path = os.path.join(DATA, "zLamb.bin")
    gam_path = os.path.join(DATA, "zeross.bin")
    if not os.path.exists(tau_path) or not os.path.exists(gam_path):
        raise FileNotFoundError(
            "Place data/zLamb.bin and data/zeross.bin in the archive tree before running s11."
        )
    tau = np.fromfile(tau_path, dtype="<f8")
    gam = np.fromfile(gam_path, dtype="<f8")
    kmax = min(len(tau), len(gam))
    return tau[:kmax], gam[:kmax]


def zeta_snapshots(s, ords, klist):
    """Canonical-product snapshots using the same head+moment expansion as MATLAB."""
    s = np.asarray(s, dtype=np.complex128).ravel()
    ords = np.asarray(ords, dtype=float).ravel()
    klist = np.asarray(klist, dtype=np.int64).ravel()
    ns = len(s)
    nk = len(klist)
    kmax = int(klist.max())
    aord = 0.25 + ords[:kmax] ** 2

    moments = np.zeros((RPROD, nk), dtype=float)
    for ik, K in enumerate(klist):
        if K > KHEAD:
            tail = aord[KHEAD:K]
            for r in range(1, RPROD + 1):
                moments[r - 1, ik] = np.sum(tail ** (-r))

    zout = np.empty((ns, nk), dtype=np.complex128)
    for i0 in range(0, ns, SBLOCK):
        i1 = min(ns, i0 + SBLOCK)
        ss = s[i0:i1]
        q = ss * (ss - 1.0)
```

```
        logG = np.log(ss * (ss - 1.0)) + loggamma(ss / 2.0) - (ss / 2.0) * math.log(math.pi)

        for ik, K in enumerate(klist):
            kh = min(int(K), KHEAD)
            head = aord[:kh]
            logP = np.sum(np.log(1.0 + q[:, None] / head[None, :]), axis=1)
            if K > KHEAD:
                tail = np.zeros_like(q)
                qpow = np.ones_like(q)
                for r in range(1, RPROD + 1):
                    qpow *= q
                    tail += ((-1) ** (r + 1)) * qpow * moments[r - 1, ik] / r
                logP += tail
            zout[i0:i1, ik] = np.exp(logP - logG)
    return zout


def tail_integral(y):
    val, _ = quad(
        lambda q: 1.0 / (q * (q * q - 1.0) * math.log(q)),
        float(y), np.inf, epsrel=2e-9, epsabs=2e-11, limit=200,
    )
    return val


def j_snapshots(y, ords, klist, label, lines):
    y = np.asarray(y, dtype=float).ravel()
    ords = np.asarray(ords, dtype=float).ravel()
    klist = np.asarray(klist, dtype=np.int64).ravel()
    ny = len(y)
    nk = len(klist)
    kmax = int(klist.max())
    jout = np.zeros((ny, nk), dtype=float)
    nblocks = (ny + ARG_BLOCK - 1) // ARG_BLOCK

    for ib in range(nblocks):
        a0 = ib * ARG_BLOCK
        a1 = min(ny, a0 + ARG_BLOCK)
        yy = y[a0:a1]
        u = np.log(yy)
        B = len(u)
        sum_at_k = np.zeros((B, nk), dtype=np.complex128)

        kbase = min(KEI, kmax)
        rho = 0.5 + 1j * ords[:kbase]
        ehead = expi(rho[:, None] * u[None, :])
        cumulative = np.sum(ehead, axis=0)

        for ik, K in enumerate(klist):
            if K <= KEI:
                sum_at_k[:, ik] = np.sum(expi((0.5 + 1j * ords[:K])[:, None] * u[None, :]), axis=0)
            elif K == kbase:
                sum_at_k[:, ik] = cumulative

        kstart = kbase
        while kstart < kmax:
            higher = klist[klist > kstart]
            next_target = int(higher.min()) if len(higher) else kmax
            kend = min(kmax, kstart + ZERO_BLOCK, next_target)
            ok = ords[kstart:kend]
            rk = 0.5 + 1j * ok
            E = np.exp(1j * ok[:, None] * u[None, :])
            local = np.zeros(B, dtype=np.complex128)
            rpow = 1.0 / rk
            for qidx in range(REI):
```

```
                Sq = np.sum(rpow[:, None] * E, axis=0)
                coeff = math.factorial(qidx) * np.exp(u / 2.0) / (u ** (qidx + 1))
                local += coeff * Sq
                rpow /= rk
            cumulative += local
            hits = np.where(klist == kend)[0]
            for ik in hits:
                sum_at_k[:, ik] = cumulative
            kstart = kend

        lix = expi(u).real
        corr = np.array([tail_integral(v) for v in yy])
        jout[a0:a1, :] = lix[:, None] - 2.0 * sum_at_k.real - math.log(2.0) + corr[:, None]
        say(lines, f"  {label}: argument block {ib+1}/{nblocks}")
    return jout


def mobius_vector(N):
    mu = np.ones(N + 1, dtype=np.int8)
    is_prime = np.ones(N + 1, dtype=bool)
    is_prime[:2] = False
    for p in range(2, N + 1):
        if not is_prime[p]:
            continue
        mu[p::p] *= -1
        p2 = p * p
        if p2 <= N:
            mu[p2::p2] = 0
        if p * p <= N:
            is_prime[p * p::p] = False
    return mu


def save_lines_plot(x, data, labels, ylabel, title, filename, zero=True):
    fig, ax = plt.subplots(figsize=(7.4, 4.8))
    for j, lab in enumerate(labels):
        ax.plot(x, data[:, j], linewidth=1.0, label=lab)
    if zero:
        ax.axhline(0.0, linestyle="--", linewidth=0.8)
    ax.set_xlabel("x")
    ax.set_ylabel(ylabel)
    ax.set_title(title)
    ax.grid(True, alpha=0.25)
    ax.legend(fontsize=8)
    fig.tight_layout()
    fig.savefig(os.path.join(FIG, filename), dpi=180)
    plt.close(fig)


def main():
    t0 = time.time()
    lines = []
    tau, gam = load_ordinates()
    kmax = min(len(tau), len(gam))
    klist = np.unique(np.array([1000, 10000, 100000, 500000, kmax], dtype=np.int64))
    klist = klist[klist <= kmax]
    nk = len(klist)
    x = np.linspace(2.0, XMAX, NX)

    say(lines, "s11: Python reproduction of the MATLAB RVM-Z1 difference analysis")
    say(lines, f"Kmax = {kmax}")
    say(lines, "Klist = " + " ".join(str(int(v)) for v in klist))

    tp = np.linspace(0.0, TPERRON, NTP)
    sp = CP + 1j * tp
```

```
    say(lines, "Computing Perron-line zeta snapshots...")
    zRP = zeta_snapshots(sp, tau, klist)
    zGP = zeta_snapshots(sp, gam, klist)

    MR = np.zeros((NX, nk))
    MG = np.zeros((NX, nk))
    for ik in range(nk):
        baseR = (1.0 / zRP[:, ik]) / sp
        baseG = (1.0 / zGP[:, ik]) / sp
        for ix, xx in enumerate(x):
            ph = np.exp(1j * tp * math.log(xx))
            MR[ix, ik] = xx ** CP / math.pi * np.real(_trapz(baseR * ph, tp))
            MG[ix, ik] = xx ** CP / math.pi * np.real(_trapz(baseG * ph, tp))
    dM = MR - MG

    say(lines, "Building Mobius-inversion arguments...")
    mu = mobius_vector(int(math.floor(XMAX)))
    all_y = []
    all_coeff = []
    maps = []
    for xx in x:
        start = len(all_y)
        mmax = int(math.floor(math.log(xx) / math.log(2.0)))
        for m in range(1, mmax + 1):
            if mu[m] == 0:
                continue
            y = xx ** (1.0 / m)
            if y >= 2.0:
                all_y.append(y)
                all_coeff.append(float(mu[m]) / m)
        maps.append((start, len(all_y) - start))
    all_y = np.asarray(all_y)
    all_coeff = np.asarray(all_coeff)
    y_unique, inverse = np.unique(all_y, return_inverse=True)

    say(lines, "Computing J snapshots...")
    JRu = j_snapshots(y_unique, tau, klist, "RVM", lines)
    JGu = j_snapshots(y_unique, gam, klist, "Z1", lines)
    JRa = JRu[inverse]
    JGa = JGu[inverse]

    piR = np.zeros((NX, nk))
    piG = np.zeros((NX, nk))
    for ix, (start, count) in enumerate(maps):
        if count:
            sl = slice(start, start + count)
            piR[ix, :] = all_coeff[sl] @ JRa[sl, :]
            piG[ix, :] = all_coeff[sl] @ JGa[sl, :]
    dPi = piR - piG

    sqrtx = np.sqrt(x)
    logx = np.log(x)
    nPi = dPi / (sqrtx[:, None] / logx[:, None])
    nM_sqrt = dM / sqrtx[:, None]
    nM_sqrtlog = dM / (sqrtx[:, None] * logx[:, None])

    errM = dM - dM[:, [-1]]
    errPi = dPi - dPi[:, [-1]]
    shellM = np.column_stack([dM[:, 0], np.diff(dM, axis=1)])
    shellPi = np.column_stack([dPi[:, 0], np.diff(dPi, axis=1)])
    runSupPi = np.maximum.accumulate(np.abs(nPi), axis=0)
    runSupMsqrt = np.maximum.accumulate(np.abs(nM_sqrt), axis=0)
    runSupMsqrtlog = np.maximum.accumulate(np.abs(nM_sqrtlog), axis=0)

    rmsErrM = np.sqrt(np.mean(errM ** 2, axis=0))
```

```
    maxErrM = np.max(np.abs(errM), axis=0)
    rmsErrPi = np.sqrt(np.mean(errPi ** 2, axis=0))
    maxErrPi = np.max(np.abs(errPi), axis=0)
    shellRMSM = np.sqrt(np.mean(shellM ** 2, axis=0))
    shellMaxM = np.max(np.abs(shellM), axis=0)
    shellRMSPi = np.sqrt(np.mean(shellPi ** 2, axis=0))
    shellMaxPi = np.max(np.abs(shellPi), axis=0)

    summary = np.column_stack([
        klist, rmsErrM, maxErrM, rmsErrPi, maxErrPi,
        shellRMSM, shellMaxM, shellRMSPi, shellMaxPi,
        runSupPi[-1], runSupMsqrt[-1], runSupMsqrtlog[-1],
    ])
    np.savetxt(
        os.path.join(OUT, "s11_difference_summary.csv"), summary, delimiter=",",
        header=("K,rms_err_M,max_err_M,rms_err_pi,max_err_pi,rms_shell_M,max_shell_M,"
                "rms_shell_pi,max_shell_pi,final_run_sup_pi,final_run_sup_M_sqrt,"
                "final_run_sup_M_sqrtlog"), comments="", fmt="%.15e"
    )
    np.savez_compressed(
        os.path.join(OUT, "s11_difference_analysis.npz"),
        x=x, Klist=klist, MR=MR, MG=MG, dM=dM, piR=piR, piG=piG, dPi=dPi,
        nPi=nPi, nM_sqrt=nM_sqrt, nM_sqrtlog=nM_sqrtlog,
        errM=errM, errPi=errPi, shellM=shellM, shellPi=shellPi,
        runSupPi=runSupPi, runSupMsqrt=runSupMsqrt, runSupMsqrtlog=runSupMsqrtlog,
    )

    labels = [f"K={int(v)}" + (" (max)" if v == kmax else "") for v in klist]
    shell_labels = [f"1...{int(klist[0])}"] + [f"{int(klist[j-1])+1}...{int(klist[j])}" for j in range(1, nk)]
    save_lines_plot(x, dM, labels, "M_RVM,K^crit - M_Z1,K^crit",
                    "Mertens difference: RVM - Z1", "s11_difference_mertens.png")
    save_lines_plot(x, dPi, labels, "pi_RVM,K - pi_Z1,K",
                    "Prime-counting difference: RVM - Z1", "s11_difference_pi.png")
    save_lines_plot(x, shellM, shell_labels, "Mertens shell contribution",
                    "Spectral shell contributions to the Mertens difference", "s11_shells_mertens.png")
    save_lines_plot(x, shellPi, shell_labels, "Prime-counting shell contribution",
                    "Spectral shell contributions to the prime-counting difference", "s11_shells_pi.png")
    save_lines_plot(x, runSupMsqrt, labels, "max_{x<=X} |dM_K(x)| / sqrt(x)",
                    "Running normalized supremum: Mertens / sqrt(x)", "s11_runsup_mertens.png", zero=False)
    save_lines_plot(x, runSupPi, labels, "max_{x<=X} |dpi_K(x)| / (sqrt(x)/log(x))",
                    "Running normalized supremum: prime-counting difference", "s11_runsup_pi.png", zero=False)

    say(lines, "")
    say(lines, "K-convergence and shell summary")
    for j, K in enumerate(klist):
        say(lines, f"K={int(K):8d}  rmsErrM={rmsErrM[j]:.8g}  maxErrM={maxErrM[j]:.8g}  "
                   f"rmsErrPi={rmsErrPi[j]:.8g}  maxErrPi={maxErrPi[j]:.8g}")
        say(lines, f"            rmsShellM={shellRMSM[j]:.8g} maxShellM={shellMaxM[j]:.8g}  "
                   f"rmsShellPi={shellRMSPi[j]:.8g} maxShellPi={shellMaxPi[j]:.8g}")
    say(lines, f"Total elapsed time: {(time.time()-t0)/60.0:.2f} min")
    with open(os.path.join(OUT, "s11_difference_analysis_console.txt"), "w", encoding="utf-8") as f:
        f.write("\n".join(lines) + "\n")


if __name__ == "__main__":
    main()
```

B.13. **s12_multiscale_jacobian.py.** This program produces Experiment 19.17 and Figures 22 and its companion fixed-window diagnostic. It generates the clock from the Lambert formula and does not read the genuine ordinates.

```
#!/usr/bin/env python3
"""s12_multiscale_jacobian.py
```

```
Clock-only diagnostics for the multiscale Jacobian analysis.

No genuine zeta ordinates and no displacement data are used.  The midpoint
ordinates are generated directly from the Lambert-W formula

    tau_k = 2*pi*(k - 11/8) / W_0((k - 11/8)/e).

The program computes:
  1. the minimum clock gap up to several heights and the rigorous lower bound
     2*pi/log(T/(2*pi));
  2. the associated Ingham window threshold 2*pi/min_gap and the natural
     scale log(T/(2*pi));
  3. fixed-window adjacent-pair upper bounds showing that no uniform lower
     frame bound can hold on a fixed u-window as k -> infinity.

Outputs:
  output/s12_multiscale_summary.csv
  output/s12_fixed_window_pairs.csv
  output/s12_multiscale_jacobian_console.txt
  figures/fig14_multiscale_gap.png
  figures/fig15_fixed_window_degeneracy.png
"""

import os
import math
import numpy as np
from scipy.special import lambertw
import matplotlib.pyplot as plt

HERE = os.path.dirname(os.path.abspath(__file__))
ROOT = os.path.abspath(os.path.join(HERE, ".."))
OUT = os.path.join(ROOT, "output")
FIG = os.path.join(ROOT, "figures")
os.makedirs(OUT, exist_ok=True)
os.makedirs(FIG, exist_ok=True)

KMAX = 2_001_052
K_VALUES = np.array([100, 1_000, 10_000, 100_000, 500_000, KMAX], dtype=np.int64)
PAIR_K = np.unique(np.geomspace(10, KMAX - 1, 240).astype(np.int64))
FIXED_WINDOWS = (0.5, 1.0, 2.0, 4.0)


def midpoint_ordinates(k):
    """Return midpoint RVM ordinates for a NumPy array of positive indices."""
    k = np.asarray(k, dtype=np.float64)
    a = k - 11.0 / 8.0
    w = lambertw(a / math.e, k=0).real
    return 2.0 * math.pi * a / w


def main():
    lines = []

    def say(text=""):
        print(text)
        lines.append(text)

    say("s12: multiscale Jacobian diagnostics from the midpoint clock")
    say("No zeross.bin, zLamb.bin, or displacement sequence is used.")
    say(f"KMAX = {KMAX}")
    say()

    indices = np.arange(1, KMAX + 1, dtype=np.float64)
    tau = midpoint_ordinates(indices)
    gaps = np.diff(tau)
```

```
prefix_min_gap = np.minimum.accumulate(gaps)

rows = []
say("       K              T        min gap      gap lower     min/lower      Ucrit       log(T/2pi)
 Ucrit/log")
for K in K_VALUES:
    T = tau[K - 1]
    min_gap = prefix_min_gap[K - 2] if K >= 2 else np.nan
    gap_lower = 2.0 * math.pi / math.log(T / (2.0 * math.pi))
    ucrit = 2.0 * math.pi / min_gap
    unatural = math.log(T / (2.0 * math.pi))
    rows.append((K, T, min_gap, gap_lower, min_gap / gap_lower,
                 ucrit, unatural, ucrit / unatural))
    say(f"{K:9d} {T:14.6f} {min_gap:13.8f} {gap_lower:13.8f} "
        f"{min_gap/gap_lower:13.8f} {ucrit:11.6f} {unatural:14.8f} {ucrit/unatural:11.8f}")

np.savetxt(
    os.path.join(OUT, "s12_multiscale_summary.csv"),
    np.asarray(rows, dtype=float),
    delimiter=",",
    header="K,T,min_gap,gap_lower_bound,min_gap_over_lower,Ucrit,log_T_over_2pi,Ucrit_over_log",
    comments="",
    fmt="%.15e",
)

# Fixed-window adjacent-pair upper bounds.
# For h_k = 1, h_{k+1} = -1, ||h||_2 = sqrt(2), and
# |sin(tau_{k+1}u)-sin(tau_ku)| <= gap_k * u, hence
# ||u[sin(tau_{k+1}u)-sin(tau_ku)]||_L2(0,U) / ||h||_2
# <= gap_k * U^(5/2) / sqrt(10).
pair_rows = []
for K in PAIR_K:
    gap = gaps[K - 1]
    row = [K, tau[K - 1], gap]
    for U in FIXED_WINDOWS:
        row.append(gap * U ** 2.5 / math.sqrt(10.0))
    pair_rows.append(row)
pair_rows = np.asarray(pair_rows, dtype=float)

header = "k,tau_k,gap_k," + ",".join(
    f"adjacent_pair_upper_U_{U:g}" for U in FIXED_WINDOWS
)
np.savetxt(
    os.path.join(OUT, "s12_fixed_window_pairs.csv"),
    pair_rows,
    delimiter=",",
    header=header,
    comments="",
    fmt="%.15e",
)

say()
say("Fixed-window adjacent-pair upper bounds at the largest sampled k:")
K = int(PAIR_K[-1])
gap = gaps[K - 1]
say(f"k = {K}, tau_k = {tau[K-1]:.6f}, gap = {gap:.9f}")
for U in FIXED_WINDOWS:
    bound = gap * U ** 2.5 / math.sqrt(10.0)
    say(f"  U = {U:3.1f}: normalized upper bound <= {bound:.9e}")

# Figure 14: minimum gap and rigorous lower bound.
fig, ax = plt.subplots(figsize=(7.2, 4.6))
arr = np.asarray(rows, dtype=float)
ax.loglog(arr[:, 1], arr[:, 2], marker="o", label="minimum observed clock gap")
ax.loglog(arr[:, 1], arr[:, 3], marker="s", label=r"$2\pi/\log(T/2\pi)$ lower bound")
```

```
    ax.set_xlabel("clock height T")
    ax.set_ylabel("gap")
    ax.set_title("Midpoint-clock gap and the multiscale window threshold")
    ax.grid(True, which="both", alpha=0.25)
    ax.legend()
    fig.tight_layout()
    fig.savefig(os.path.join(FIG, "fig14_multiscale_gap.png"), dpi=180)
    plt.close(fig)

    # Figure 15: failure of a fixed-window lower bound.
    fig, ax = plt.subplots(figsize=(7.2, 4.6))
    for j, U in enumerate(FIXED_WINDOWS, start=3):
        ax.loglog(pair_rows[:, 0], pair_rows[:, j], label=f"U = {U:g}")
    ax.set_xlabel("clock index k")
    ax.set_ylabel("adjacent-pair normalized upper bound")
    ax.set_title("Fixed-window near-degeneracy of adjacent clock frequencies")
    ax.grid(True, which="both", alpha=0.25)
    ax.legend()
    fig.tight_layout()
    fig.savefig(os.path.join(FIG, "fig15_fixed_window_degeneracy.png"), dpi=180)
    plt.close(fig)

    say()
    say("Saved numerical tables and figures.")
    say("The decreasing fixed-window bounds are a diagnostic of the rigorous fact")
    say("that no T-independent lower frame bound can hold on a fixed u-window.")

    with open(os.path.join(OUT, "s12_multiscale_jacobian_console.txt"), "w", encoding="utf-8") as f:
        f.write("\n".join(lines) + "\n")


if __name__ == "__main__":
    main()
```

B.14. **`s13_nested_bounded_endpoint.py`.** This program produces the bounded and nested endpoint diagnostics of Appendix F. The solver uses the midpoint clock; genuine ordinates and von Mangoldt masses, when displayed, are read only after the solve as diagnostics.

```
"""Bounded finite-section experiments for the endpoint rung system.

This program is diagnostic only.  It solves finite sections using the midpoint RVM clock as
input; the ordinates of zeta, if present, are read only after the solves, for diagnostics.
The ordinate shifts are bounded by |kappa_k - tau_k| <= 1.2 and the free masses by
|a_n| <= 0.5 (numerical regularizations, not hypotheses of a theorem).

Part 1 (single sections N = 20, 34, 50): each section is solved alone, started at the clock;
  outputs s13_bounded_mass_summary.csv, _ordinates.csv, _masses.csv, and a cross-validation on
  test bumps not used in the solve (other supports, frequencies on the shifted grid
  omega = (j + 1/2) * 0.25 up to 0.8 Omega): s13_cross_validation.csv.
Part 2 (nested sections N = 13, 20, 34, 50, 70, 100): every earlier test block stays active in
  later sections, each section warm-started from the previous one; output
  s13_nested_metric_summary.csv and a repeatability check of the N = 50 section from three
  perturbed starting points.
Figures: fig16_nested_bounded_endpoint.png (Part 2), fig17_bounded_mass_stability.png (Part 1).
Test functions: a C_c^infinity bump of unit integral, exp(-1/(1-v^2)) on its support, multiplied
by cos(omega u) and sin(omega u).  Part 1: support [0, log(N+1/2)], omega = 0, 0.25, ..., Omega,
one block of equations.  Part 2: support [0.08, log(N+1/2)] for the block introduced by section N,
omega = 0, 1, ..., Omega_N; the rows of each block are divided by sqrt(number of rows), so every
block has the same weight.
Solver: scipy.optimize.least_squares, trust-region reflective with bounds, analytic Jacobian,
x_scale='jac', double precision; Part 1: ftol = xtol = gtol = 1e-11, at most 1200 evaluations;
Part 2: ftol = xtol = gtol = 2e-10, at most MAX_NFEV evaluations.  For every solve the program
prints the termination status, nfev, the numerical rank of the (weighted) Jacobian at the solution
(singular values above 1e-12 of the largest), its condition number on that subspace, the number of
```

```
masses at the edge of the test supports whose columns are below 1e-12 (not determined by the tests),
and the number of shifts and masses at their bounds (active box constraints).
"""

from pathlib import Path
import math
import numpy as np
from scipy.optimize import least_squares
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt

HERE = Path(__file__).resolve().parent
ROOT = HERE.parent
DATA = ROOT / "data"
OUT = ROOT / "output"
FIG = ROOT / "figures"
OUT.mkdir(exist_ok=True)
FIG.mkdir(exist_ok=True)

TAU_FILE = DATA / "zLamb.bin"
GAMMA_FILE = DATA / "zeross.bin"

SHIFT_BOUND = 1.2
MASS_BOUND = 0.5
U_MIN = 0.08
OMEGA_STEP = 1.0
MARGIN = 14.0
LEVELS = [(13, 50.0), (20, 70.0), (34, 110.0), (50, 150.0), (70, 210.0), (100, 300.0)]
MAX_NFEV = 3000            # function evaluations per nested section


LINES = []


def say(msg):
    print(msg, flush=True)
    LINES.append(msg)


def load_data():
    tau = np.fromfile(TAU_FILE, dtype="<f8")
    if tau.size == 0:
        raise RuntimeError("zLamb.bin is required in data/")
    if not np.all(np.isfinite(tau)) or not np.all(np.diff(tau) > 0):
        raise ValueError("zLamb.bin must be finite and strictly increasing")
    gamma = None
    if GAMMA_FILE.exists():
        gamma = np.fromfile(GAMMA_FILE, dtype="<f8")
        if gamma.size != tau.size:
            raise ValueError("the ordinate files have different lengths")
    return tau, gamma


def svd_report(J, mass_cols):
    """Numerical rank of J (singular values above 1e-12 sigma_max), the condition number on that
    subspace, and the number of mass columns whose largest entry is below 1e-12 of the largest
    entry of J (masses at the edge of the test supports, which the tests do not determine)."""
    sv = np.linalg.svd(J, compute_uv=False)
    rank = int(np.sum(sv > 1e-12 * sv[0]))
    weak = int(np.sum(np.max(np.abs(J[:, mass_cols]), axis=0) < 1e-12 * np.max(np.abs(J))))
    return rank, float(sv[0] / sv[rank - 1]), weak


def von_mangoldt(n):
```

```
    """Lambda(n) by trial division (no zeta data)."""
    p = 2
    while p * p <= n:
        if n % p == 0:
            m = n
            while m % p == 0:
                m //= p
            return math.log(p) if m == 1 else 0.0
        p += 1
    return math.log(n) if n >= 2 else 0.0


def choose_grid_size(U, omega):
    """Keep the Nyquist frequency safely above the largest tested frequency."""
    target_nyquist = 2.5 * (omega + MARGIN)
    n = int(math.ceil(target_nyquist * (U - U_MIN) / math.pi)) + 1
    return max(701, n)


def build_block(tau, n_max, omega_max, active_k, mass_nodes):
    U = math.log(n_max + 0.5)
    n_grid = choose_grid_size(U, omega_max)
    grid = np.linspace(U_MIN, U, n_grid)
    u = grid[1:-1]
    du = (U - U_MIN) / (n_grid - 1)

    mid = 0.5 * (U_MIN + U)
    half = 0.5 * (U - U_MIN)
    v = (u - mid) / half
    raw = np.exp(-1.0 / (1.0 - v * v))
    norm = raw.sum() * du
    w = raw / norm

    omega = np.arange(0.0, omega_max + 1e-12, OMEGA_STEP)
    psi = np.vstack([
        np.cos(np.outer(omega, u)) * w[None, :] * du,
        np.sin(np.outer(omega[1:], u)) * w[None, :] * du,
    ])

    smooth = 0.5 * (np.exp(u / 2.0) - np.exp(-u / 2.0) / (np.exp(2.0 * u) - 1.0))
    target = psi @ smooth

    clock = np.zeros_like(u)
    for start in range(0, tau.size, 500):
        clock += np.cos(np.outer(u, tau[start:start + 500])).sum(axis=1)
    low_clock = psi @ np.cos(np.outer(u, tau[:active_k])).sum(axis=1)
    fixed = psi @ clock - low_clock - target

    logs = np.log(mass_nodes.astype(float))
    env = np.zeros_like(logs)
    mask = (logs > U_MIN) & (logs < U)
    if np.any(mask):
        vv = (logs[mask] - mid) / half
        env[mask] = np.exp(-1.0 / (1.0 - vv * vv)) / norm

    mass_matrix = np.vstack([
        np.cos(np.outer(omega, logs)) * env[None, :],
        np.sin(np.outer(omega[1:], logs)) * env[None, :],
    ])

    return {
        "n_max": n_max,
        "omega_max": omega_max,
        "u": u,
        "psi": psi,
```

```
        "fixed": fixed,
        "mass_matrix": mass_matrix,
        "n_grid": n_grid,
    }


def solve_nested_level(tau, gamma, level_index, previous=None, start=None):
    n_max, omega_max = LEVELS[level_index]
    mass_nodes = np.arange(2, n_max + 1, dtype=int)
    active_k = int(np.sum(tau <= omega_max + MARGIN))

    blocks = []
    for j in range(level_index + 1):
        nb, ob = LEVELS[j]
        blocks.append(build_block(tau, nb, ob, active_k, mass_nodes))

    x0 = np.r_[tau[:active_k], np.zeros(mass_nodes.size)]
    if previous is not None:
        k_copy = min(previous["active_k"], active_k)
        n_copy = min(previous["masses"].size, mass_nodes.size)
        x0[:k_copy] = previous["ordinates"][:k_copy]
        x0[active_k:active_k + n_copy] = previous["masses"][:n_copy]

    lower = np.r_[tau[:active_k] - SHIFT_BOUND, -MASS_BOUND * np.ones(mass_nodes.size)]
    upper = np.r_[tau[:active_k] + SHIFT_BOUND, MASS_BOUND * np.ones(mass_nodes.size)]
    if start is not None:
        x0 = start
    x0 = np.clip(x0, lower + 1e-10, upper - 1e-10)

    def residual(z):
        ordinates = z[:active_k]
        masses = z[active_k:]
        pieces = []
        for block in blocks:
            spectral = block["psi"] @ np.cos(np.outer(block["u"], ordinates)).sum(axis=1)
            r = spectral + block["fixed"] + block["mass_matrix"] @ masses
            pieces.append(r / math.sqrt(r.size))
        return np.concatenate(pieces)

    def jacobian(z):
        ordinates = z[:active_k]
        pieces = []
        for block in blocks:
            jk = block["psi"] @ (-block["u"][:, None] * np.sin(np.outer(block["u"], ordinates)))
            j = np.hstack([jk, block["mass_matrix"]])
            pieces.append(j / math.sqrt(j.shape[0]))
        return np.vstack(pieces)

    sol = least_squares(
        residual,
        x0,
        jac=jacobian,
        method="trf",
        tr_solver="lsmr",
        bounds=(lower, upper),
        max_nfev=MAX_NFEV,
        ftol=2e-10,
        xtol=2e-10,
        gtol=2e-10,
        x_scale="jac",
    )

    ordinates = sol.x[:active_k]
    masses = sol.x[active_k:]
    # conditioning and active box constraints at the solution
```

```
    rank, cond, weak = svd_report(jacobian(sol.x), slice(active_k, None))
    n_rows = int(sum(b["psi"].shape[0] for b in blocks))
    act_shift = int(np.sum(np.abs(ordinates - tau[:active_k]) >= SHIFT_BOUND - 1e-6))
    act_mass = int(np.sum(np.abs(masses) >= MASS_BOUND - 1e-6))

    block_rows = []
    for block in blocks:
        spectral = block["psi"] @ np.cos(np.outer(block["u"], ordinates)).sum(axis=1)
        r = spectral + block["fixed"] + block["mass_matrix"] @ masses
        block_rows.append((block["n_max"], float(np.sqrt(np.mean(r * r))), float(np.max(np.abs(r))), block["
     n_grid"]))

    result = {
        "n_max": n_max,
        "omega_max": omega_max,
        "active_k": active_k,
        "mass_nodes": mass_nodes,
        "ordinates": ordinates,
        "masses": masses,
        "blocks": block_rows,
        "nfev": sol.nfev,
        "status": sol.status,
        "rank": rank,
        "cond": cond,
        "weak_masses": weak,
        "rows": n_rows,
        "n_blocks": len(blocks),
        "active_shift": act_shift,
        "active_mass": act_mass,
    }

    if gamma is not None and gamma.size >= min(10, active_k):
        q = min(10, active_k)
        result["rms_first10_diagnostic"] = float(np.sqrt(np.mean((ordinates[:q] - gamma[:q]) ** 2)))
    else:
        result["rms_first10_diagnostic"] = float("nan")
    return result


def single_section(tau, gamma, n_max, omega_max, dom=0.25, nu=2201, kt=14000):
    """One bounded section with free masses, solved alone from the clock (Part 1)."""
    U = math.log(n_max + 0.5)
    grid = np.linspace(0.0, U, nu)
    u = grid[1:-1]
    du = U / (nu - 1)
    v = (u - U / 2) / (U / 2)
    w = np.exp(-1.0 / (1.0 - v * v))
    w /= w.sum() * du
    smooth = 0.5 * (np.exp(u / 2) - np.exp(-u / 2) / (np.exp(2 * u) - 1))
    ns = np.arange(2, n_max + 1)
    om = np.arange(0.0, omega_max + 1e-12, dom)
    Psi = np.vstack([w * np.cos(o * u) for o in om] + [w * np.sin(o * u) for o in om[1:]]) * du
    Pn = np.vstack([np.interp(np.log(ns), u, Psi[j] / du) for j in range(Psi.shape[0])])
    clock = np.zeros_like(u)
    for a in range(0, kt, 400):
        clock += np.cos(np.outer(u, tau[a:min(kt, a + 400)])).sum(axis=1)
    K = int(np.sum(tau <= omega_max + MARGIN))
    tail = clock - np.cos(np.outer(u, tau[:K])).sum(axis=1)

    def fun(z):
        return Psi @ (np.cos(np.outer(u, z[:K])).sum(axis=1) + tail - smooth) + Pn @ z[K:]

    def jac(z):
        return np.hstack([Psi @ (-u[:, None] * np.sin(np.outer(u, z[:K]))), Pn])
```

```
    z0 = np.r_[tau[:K], np.zeros(ns.size)]
    lo = np.r_[tau[:K] - SHIFT_BOUND, -MASS_BOUND * np.ones(ns.size)]
    hi = np.r_[tau[:K] + SHIFT_BOUND, MASS_BOUND * np.ones(ns.size)]
    sol = least_squares(fun, z0, jac=jac, method="trf", bounds=(lo, hi), max_nfev=1200,
                        ftol=1e-11, xtol=1e-11, gtol=1e-11, x_scale="jac")
    x, a = sol.x[:K], sol.x[K:]
    nrms = np.linalg.norm(sol.fun) / math.sqrt(Psi.shape[0])
    rank, cond, weak = svd_report(jac(sol.x), slice(K, None))
    act_shift = int(np.sum(np.abs(x - tau[:K]) >= SHIFT_BOUND - 1e-6))
    act_mass = int(np.sum(np.abs(a) >= MASS_BOUND - 1e-6))
    rms10 = float(np.sqrt(np.mean((x[:10] - gamma[:10]) ** 2))) if gamma is not None else float("nan")
    # cross-validation on tests not used in the solve: other supports, shifted frequency grid
    cv = []
    for name, af, bf in (("full", 0.0, 1.0), ("inner80", 0.1, 0.9), ("left", 0.05, 0.55),
                         ("right", 0.45, 0.95), ("middle", 0.25, 0.75)):
        A, B = af * U, bf * U
        vv = (u - (A + B) / 2) / ((B - A) / 2)
        wc = np.zeros_like(u)
        ins = np.abs(vv) < 1
        wc[ins] = np.exp(-1.0 / (1.0 - vv[ins] ** 2))
        wc /= wc.sum() * du
        omc = (np.arange(int(0.8 * omega_max / dom)) + 0.5) * dom
        Pc = np.vstack([wc * np.cos(o * u) for o in omc] + [wc * np.sin(o * u) for o in omc]) * du
        Pnc = np.vstack([np.interp(np.log(ns), u, Pc[j] / du) for j in range(Pc.shape[0])])
        r = Pc @ (np.cos(np.outer(u, x)).sum(axis=1) + tail - smooth) + Pnc @ a
        cv.append((n_max, name, af, bf, 0.8 * omega_max, float(np.sqrt(np.mean(r * r))), float(np.max(np.abs(r
 )))))
    return dict(N=n_max, Omega=omega_max, K=K, x=x, a=a, ns=ns, nrms=nrms, rms10=rms10, status=sol.status,
                nfev=sol.nfev, rows=Psi.shape[0], cv=cv, rank=rank, cond=cond, weak=weak, act_shift=act_shift,
                act_mass=act_mass)


def main():
    tau, gamma = load_data()
    # ---------------- Part 1: single bounded sections
    say("Part 1: single bounded sections, solved alone from the clock (|shift| <= %.1f, |mass| <= %.1f)"
        % (SHIFT_BOUND, MASS_BOUND))
    singles = [single_section(tau, gamma, N, Om) for N, Om in ((20, 70.0), (34, 110.0), (50, 150.0))]
    summ, ordrows, massrows, cvrows = [], [], [], []
    for r in singles:
        say("N=%3d Omega=%.0f K=%3d masses=%d rows=%d: normalized residual %.3e (status %d, nfev %d); "
            "rms(first 10 - gamma) = %.3e; mass energy %.4f, max |mass| %.4f"
            % (r["N"], r["Omega"], r["K"], r["ns"].size, r["rows"], r["nrms"], r["status"], r["nfev"], r["
 rms10"],
               np.sum(r["a"] ** 2), np.max(np.abs(r["a"]))))
        say("    Jacobian at the solution: numerical rank %d of %d columns, condition number %.2e on that
 subspace, "
            "%d edge masses with weight < 1e-12; active bounds: %d of %d shifts at |shift| = %.1f, %d of %d
 masses at |mass| = %.1f"
            % (r["rank"], r["K"] + r["ns"].size, r["cond"], r["weak"], r["act_shift"], r["K"], SHIFT_BOUND,
               r["act_mass"], r["ns"].size, MASS_BOUND))
        summ.append((r["N"], r["Omega"], r["K"], r["nrms"], r["rms10"], np.sum(r["a"] ** 2), np.max(np.abs(r["
 a"]))))
        for k in range(r["K"]):
            g = gamma[k] if gamma is not None else float("nan")
            ordrows.append((r["N"], k + 1, tau[k], r["x"][k], g, r["x"][k] - g, r["x"][k] - tau[k]))
        for n, a in zip(r["ns"], r["a"]):
            e = von_mangoldt(int(n)) / (2 * math.sqrt(n))
            massrows.append((r["N"], n, a, e, a - e))
        for c in r["cv"]:
            say("    cross-validation %-8s support [%.2f, %.2f] U, omega <= %.0f: rms %.3e, max %.3e" % c[1:])
            cvrows.append(c)
    np.savetxt(OUT / "s13_bounded_mass_summary.csv", np.array(summ, dtype=float), delimiter=",",
               header="N,Omega,K,nrms,rms10_gamma_diagnostic,mass_energy,mass_max", comments="")
    np.savetxt(OUT / "s13_bounded_mass_ordinates.csv", np.array(ordrows, dtype=float), delimiter=",",
```

```
            header="N,k,tau,solved,gamma_diagnostic,solved_minus_gamma,solved_minus_tau", comments="")
np.savetxt(OUT / "s13_bounded_mass_masses.csv", np.array(massrows, dtype=float), delimiter=",",
            header="N,n,solved_mass,von_mangoldt_diagnostic,difference", comments="")
with open(OUT / "s13_cross_validation.csv", "w") as f:
    f.write("N,support,a_frac,b_frac,omega_max,rms,max_abs\n")
    for c in cvrows:
        f.write("%d,%s,%.2f,%.2f,%.1f,%.6e,%.6e\n" % c)

# ---------------- Part 2: nested sections with permanent blocks
say("")
say("Part 2: nested sections; every earlier test block stays active (max_nfev = %d per section)" %
 MAX_NFEV)
rows = []
previous = None
nested = {}
for level_index in range(len(LEVELS)):
    result = solve_nested_level(tau, gamma, level_index, previous)
    previous = result
    nested[result["n_max"]] = result
    say(
        f"N={result['n_max']:3d} K={result['active_k']:3d} "
        f"max_shift={np.max(np.abs(result['ordinates'] - tau[:result['active_k']])):.6f} "
        f"max_mass={np.max(np.abs(result['masses'])):.6f} "
        f"diagnostic_rms10={result['rms_first10_diagnostic']:.3e} status={result['status']} nfev={result['
 nfev']}"
    )
    say(f"  equations={result['rows']} in {result['n_blocks']} blocks, unknowns={result['active_k']}+{
 result['masses'].size}, "
        f"weighted Jacobian: numerical rank {result['rank']} of {result['active_k'] + result['masses'].
 size}, "
        f"condition number {result['cond']:.2e} on that subspace, {result['weak_masses']} edge masses with
  weight < 1e-12; active bounds: "
        f"{result['active_shift']} shifts, {result['active_mass']} masses")
    for block_n, rms, max_abs, n_grid in result["blocks"]:
        say(f"  block N={block_n:3d}: RMS={rms:.3e}, max={max_abs:.3e}, grid={n_grid}")
        rows.append((result["n_max"], block_n, rms, max_abs, n_grid))

np.savetxt(
    OUT / "s13_nested_metric_summary.csv",
    np.asarray(rows, dtype=float),
    delimiter=",",
    header="section_N,block_N,rms,max_abs,n_grid",
    comments="",
)

# repeatability of the N = 50 nested section from perturbed starts
i50 = [L[0] for L in LEVELS].index(50)
base = nested[50]
rng = np.random.default_rng(2025)
say("repeatability of the N = 50 section (same blocks) from three perturbed starts:")
for trial in range(3):
    Kb = base["active_k"]
    x0 = np.r_[tau[:Kb] + rng.uniform(-0.5, 0.5, Kb), rng.uniform(-0.2, 0.2, base["masses"].size)]
    r = solve_nested_level(tau, gamma, i50, None, start=x0)
    say("  start %d: max block RMS %.3e (reference %.3e), max |ordinate difference| %.3e, "
        "max |mass difference| %.3e, status %d, nfev %d"
        % (trial + 1, max(b[1] for b in r["blocks"]), max(b[1] for b in base["blocks"]),
           np.max(np.abs(r["ordinates"] - base["ordinates"])), np.max(np.abs(r["masses"] - base["masses"])
 ),
           r["status"], r["nfev"]))

# ---------------- figures
fig, ax = plt.subplots(1, 1, figsize=(9.6, 6.0))
for N in (50, 70, 100):
    bl = nested[N]["blocks"]
```

```
        r_ = nested[N]
        ax.semilogy([b[0] for b in bl], [b[1] for b in bl], "o-",
                    label="section N=%d: %d equations in %d blocks, %d ordinates + %d masses (%d at the box)"
                    % (N, r_["rows"], r_["n_blocks"], r_["active_k"], r_["masses"].size, r_["active_mass"]))
    ax.set_xlabel("test block (N of the section that introduced it)")
    ax.set_ylabel("RMS residual of the block")
    ax.set_title("Nested bounded endpoint sections: residual on every permanent block")
    ax.legend(fontsize=8)
    fig.tight_layout()
    fig.savefig(FIG / "fig16_nested_bounded_endpoint.png", dpi=150)
    fig, ax = plt.subplots(1, 1, figsize=(9.6, 6.0))
    n15 = np.arange(2, 16)
    ax.plot(n15, [von_mangoldt(int(n)) / (2 * math.sqrt(n)) for n in n15], "ks", ms=8, mfc="none",
            label=r"$\Lambda(n)/(2\sqrt{n})$ (diagnostic only)")
    for r, mk in zip(singles, ("o", "^", "v")):
        ax.plot(n15, r["a"][:n15.size], mk + "-", ms=5, lw=0.8,
                label="free masses, section N=%d (%d ordinates, %d masses, %d equations)"
                % (r["N"], r["K"], r["ns"].size, r["rows"]))
        atb = np.abs(r["a"][:n15.size]) >= MASS_BOUND - 1e-6
        ax.plot(n15[atb], r["a"][:n15.size][atb], mk, ms=9, mfc="none", mec="r", mew=1.5)
    at_box = ["n=" + ",".join(str(int(n)) for n, a in zip(r["ns"], r["a"]) if abs(a) >= MASS_BOUND - 1e-6)
              + " (N=%d)" % r["N"] for r in singles]
    n_in = sum(int(np.sum(np.abs(r["a"][:n15.size]) >= MASS_BOUND - 1e-6)) for r in singles)
    ax.plot([], [], "o", ms=9, mfc="none", mec="r", mew=1.5,
            label="mass at the box $|a_n|=0.5$ (active bound): %d with $n\\leq15$; all at the box: %s"
            % (n_in, "; ".join(at_box)))
    ax.axhline(MASS_BOUND, color="0.6", lw=0.6, ls="--")
    ax.axhline(-MASS_BOUND, color="0.6", lw=0.6, ls="--")
    ax.set_xlabel("n")
    ax.set_title(r"Recovered free masses, $n\leq15$ (all masses of this range shown), single bounded sections;
      box $|a_n|\leq0.5$ dashed",
                 fontsize=10)
    ax.legend(fontsize=8)
    fig.tight_layout()
    fig.savefig(FIG / "fig17_bounded_mass_stability.png", dpi=150)
    with open(OUT / "s13_nested_bounded_endpoint_console.txt", "w") as f:
        f.write("\n".join(LINES) + "\n")


if __name__ == "__main__":
    main()
```

B.15. **s14_varpi_bound.py.** This program verifies the explicit inequalities in the proof of Theorem 9.11 in interval arithmetic (Part A) and computes high-precision floating-point values of ν_4, $P_{\mathrm{RVM}}^{\mathrm{crit}}(\frac{1}{2})$ and ϖ_{RVM} as a cross-check (Part B, not part of the proof). It uses only the Lambert formula for the clock and reads no data file. Run time about 30 seconds.

```
"""Theorem (label thm:varpi-less-one): the transition constant satisfies varpi_RVM < 1.

varpi_RVM = P_RVM^crit(1/2) * exp(nu_3 + nu_4)                    (eq:varpi-def)
  P_RVM^crit(1/2) = prod_k tau_k^2 / (1/4 + tau_k^2)  in (0, 1)      (no computation needed)
  nu_3 = 8 - (33/4) log 2 - 2 log pi                                 (closed form, proof of the theorem,
     step (b))
  nu_4 = -2 int_{tau_0}^inf psi(N_RVM(tau) + 1/2) dtau / tau         (eq:nu4-limit)
       <= -2 M(tau_1)                                                (eq:nu4-bound),
  M(t) = I_0(t) - W_1(t)/12 - (sqrt3/216) |W_1'(t)|,  with  l = log(t/2pi),
  I_0(t) = (t l - 2t - tau_0 log 2 + 2 tau_0)/(2 pi) + (7/8) log(t/tau_0),
  W_1(t) = W(1) = 2 pi/(t l),   |W_1'(t)| = |W'(1)| = 4 pi^2 (l + 1)/(t^2 l^3),   tau_0 = 4 pi,
  W(u) = 2 pi/(T log(T/2pi)), T = inverse clock at u - 1/2, the weight dtau/tau = W(u) du (eq:nu4-weight),
  tau_1 = first midpoint ordinate, N_RVM(tau_1) = 1/2.

PART A is the proof: every inequality is checked in interval arithmetic (mpmath.iv, outward rounding).
PART B is a floating-point cross-check (not part of the proof): high-precision values of nu_4,
```

```
P_RVM^crit(1/2) and varpi_RVM, compared with the Richardson value of s02_transition.py.
The program does not read data/zeross.bin or data/zLamb.bin: the clock is the Lambert formula.
"""
import mpmath as mp
import numpy as np
from scipy.special import lambertw
import os
from rvm_common import Tee, OUT

say = Tee("s14_varpi_bound")
iv = mp.iv
iv.dps = 40


def hi(x):
    """Upper endpoint of an interval, as an exact mpf."""
    return mp.make_mpf(x._mpi_[1])


def lo(x):
    """Lower endpoint of an interval, as an exact mpf."""
    return mp.make_mpf(x._mpi_[0])


# ------------------------------------------------------------------------- PART A: proof
say("PART A. Interval-arithmetic verification of the inequalities in the proof of the theorem varpi_RVM < 1")
say("(mpmath.iv, %d digits, outward rounding)" % iv.dps)
PI = iv.pi
LOG2 = iv.log(2)
TAU0 = 4 * PI

# (a) nu_3 = 8 - 33/4 log 2 - 2 log pi < 0
nu3 = 8 - iv.mpf(33) / 4 * LOG2 - 2 * iv.log(PI)
say("(a) nu_3 = 8 - (33/4) log 2 - 2 log pi in [%s, %s]" % (mp.nstr(lo(nu3), 15), mp.nstr(hi(nu3), 15)))
assert hi(nu3) < 0
say("    sup nu_3 < 0: verified")


def clock(t):
    return t / (2 * PI) * (iv.log(t / (2 * PI)) - 1) + iv.mpf(7) / 8


# (b) N_RVM(tau_0) in (0, 1/2): on [tau_0, tau_1) the sawtooth psi(N + 1/2) equals N
NT0 = clock(TAU0)
say("(b) N_RVM(tau_0) = 2 log(2/e) + 7/8 in [%s, %s]" % (mp.nstr(lo(NT0), 15), mp.nstr(hi(NT0), 15)))
assert lo(NT0) > 0 and hi(NT0) < iv.mpf(1) / 2
say("    0 < N_RVM(tau_0) < 1/2: verified")

# (c) enclosure of tau_1: N_RVM is increasing on [2 pi, inf)
A, B = "14.52134", "14.52135"
NA, NB = clock(iv.mpf(A)), clock(iv.mpf(B))
say("(c) N_RVM(%s) in [%s, %s]" % (A, mp.nstr(lo(NA), 15), mp.nstr(hi(NA), 15)))
say("    N_RVM(%s) in [%s, %s]" % (B, mp.nstr(lo(NB), 15), mp.nstr(hi(NB), 15)))
assert hi(NA) < iv.mpf(1) / 2 < lo(NB)
say("    N_RVM(%s) < 1/2 < N_RVM(%s): tau_1 in [%s, %s] verified" % (A, B, A, B))
T1 = iv.mpf([A, B])

# (d) the lower bound m(tau_1) > 0, evaluated on the whole interval containing tau_1
L1 = iv.log(T1 / (2 * PI))
I0 = (T1 * L1 - 2 * T1 - TAU0 * LOG2 + 2 * TAU0) / (2 * PI) + iv.mpf(7) / 8 * iv.log(T1 / TAU0)
W1 = 2 * PI / (T1 * L1)
DW1 = 4 * PI ** 2 * (L1 + 1) / (T1 ** 2 * L1 ** 3)
C3 = iv.sqrt(3) / 216
M = I0 - W1 / 12 - C3 * DW1
```

```
say("(d) l_1 = log(tau_1/2pi)      in [%s, %s]   (> 0, so W is decreasing and convex on u >= 1)" % (mp.nstr(lo
    (L1), 12), mp.nstr(hi(L1), 12)))
assert lo(L1) > 0
say("    I_0(tau_1)                in [%s, %s]" % (mp.nstr(lo(I0), 12), mp.nstr(hi(I0), 12)))
say("    W(1)/12                   in [%s, %s]" % (mp.nstr(lo(W1 / 12), 12), mp.nstr(hi(W1 / 12), 12)))
say("    (sqrt3/216)|W'(1)|        in [%s, %s]" % (mp.nstr(lo(C3 * DW1), 12), mp.nstr(hi(C3 * DW1), 12)))
say("    M(tau_1)                  in [%s, %s]" % (mp.nstr(lo(M), 12), mp.nstr(hi(M), 12)))
assert lo(M) > 0
say("    inf M(tau_1) > 0: verified")

# (e) conclusion
nu4_up = -2 * lo(M)                       # exact: negation and doubling of a binary float
sum_iv = iv.mpf(hi(nu3)) + iv.mpf(nu4_up)   # outward-rounded sum
sum_up = hi(sum_iv)
varpi_up = hi(iv.exp(sum_iv))
say("(e) nu_4 <= -2 inf M(tau_1) = %s" % mp.nstr(nu4_up, 12))
say("    nu_3 + nu_4 <= %s" % mp.nstr(sum_up, 12))
say("    varpi_RVM = P(1/2) exp(nu_3 + nu_4) < exp(nu_3 + nu_4) <= %s < 1" % mp.nstr(varpi_up, 12))
assert varpi_up < 1
say("    ALL INEQUALITIES OF THE PROOF VERIFIED: varpi_RVM < %s" % mp.nstr(varpi_up, 6))

# ------------------------------------------------------------------------------ PART B: cross-check
say("")
say("PART B. Floating-point cross-check (not part of the proof)")
mp.mp.dps = 25
pi = mp.pi
tau0 = 4 * pi


def N(t):
    return t / (2 * pi) * mp.log(t / (2 * pi * mp.e)) + mp.mpf(7) / 8


def tau_of(u):
    """Inverse clock: N_RVM(tau) = u - 1/2 (Lambert formula)."""
    x = (u - mp.mpf(1) / 2 - mp.mpf(7) / 8) / mp.e
    return 2 * pi * mp.e * mp.exp(mp.lambertw(x))


def w(u):
    t = tau_of(u)
    return 2 * pi / (t * mp.log(t / (2 * pi)))


tau1 = tau_of(1)
I0f = mp.quad(lambda t: N(t) / t, [tau0, tau1])
MMAX = 2000
S = mp.mpf(0)
for n in range(1, MMAX + 1):
    S += mp.quad(lambda u: (u - n - mp.mpf(1) / 2) * w(u), [n, n + 1])
tail = -w(MMAX + 1) / 12
SI = S + tail
nu3f = 8 - mp.mpf(33) / 4 * mp.log(2) - 2 * mp.log(pi)
nu4f = -2 * (I0f + SI)
say("tau_1 = %s (Lambert formula)" % mp.nstr(tau1, 20))
say("I_0 = int_{tau_0}^{tau_1} N_RVM(t) dt/t = %s (quadrature)" % mp.nstr(I0f, 15))
say("sum_{n<=%d} I_n = %s; Euler-Maclaurin tail -W(%d)/12 = %s" % (MMAX, mp.nstr(S, 15), MMAX + 1, mp.nstr(
    tail, 6)))
em = SI + w(1) / 12
say("sum_n I_n + W(1)/12 = %s, Euler-Maclaurin bound (sqrt3/216)|W'(1)| = %s" % (mp.nstr(em, 8), mp.nstr(mp.
    sqrt(3) / 216 * abs(mp.diff(w, 1)), 8)))
say("nu_3 = %s" % mp.nstr(nu3f, 15))
say("nu_4 = %s   (rigorous upper bound of Part A: %s)" % (mp.nstr(nu4f, 15), mp.nstr(nu4_up, 12)))
say("exp(nu_3 + nu_4) = %s" % mp.nstr(mp.e ** (nu3f + nu4f), 15))
```

```
K = 2_000_000
k = np.arange(1, K + 1, dtype=np.float64)
tau = (2 * np.pi * np.e) * np.exp(lambertw((k - 0.5 - 0.875) / np.e).real)
logP_K = -np.sum(np.log1p(0.25 / tau ** 2))
TK = tau[-1]
lK = np.log(TK / (2 * np.pi))
logP_tail = -(lK + 1) / (8 * np.pi * TK)
logP = logP_K + logP_tail
say("tau_1, tau_2, tau_3 (numpy) = %.10f, %.10f, %.10f" % tuple(tau[:3]))
say("log P_RVM^crit(1/2): k <= %d: %.15f; tail (integral): %.3e; total %.15f" % (K, logP_K, logP_tail, logP))
say("P_RVM^crit(1/2) = %.12f" % np.exp(logP))
varpi = mp.e ** (mp.mpf(logP) + nu3f + nu4f)
say("varpi_RVM = P(1/2) exp(nu_3 + nu_4) = %s" % mp.nstr(varpi, 12))
fv = os.path.join(OUT, "s02_varpi.txt")
if os.path.exists(fv):
    vr = open(fv).read().strip()
    say("Richardson value of s02_transition.py (sigma = 2, T = 2500, 5000, 10000), read from output/s02_varpi.
     txt: %s" % vr)
    say("difference = %.2e" % float(varpi - mp.mpf(vr)))
else:
    say("output/s02_varpi.txt not found: run s02_transition.py for the comparison with the Richardson value")
say.close()
```

B.16. **`s15_tables_1000.py`.** This program produces Experiment 19.13, Figures 18–21 and the Tables of Appendix A. It reads `data/zeross.bin` and `data/zLamb.bin`, and needs the package `finufft` (nonuniform fast Fourier transform) in addition to the packages of the other programs. Run time about two minutes. A MATLAB version, `matlab/s15_tables_1000.m`, is distributed with the repository.

```
"""Experiment 15: the first 1000 values of the four comparisons of the paper, side by side.

  (a) the midpoint ordinates tau_k against the ordinates gamma_k of zeta, k <= 1000;
  (b) the real RVM primes p_j^RVM (first grid crossing of the level j by pi_RVM, refined by
      linear interpolation; a discrete version of inf{x : pi_RVM(x) >= j}) against p_j, j <= 1000;
  (c) pi_RVM(n) against pi(n) at the integers n <= 1000;
  (d) the critical RVM Mertens function M_RVM^crit(n) against M(n), n <= 1000.

Data: data/zeross.bin (gamma_k) and data/zLamb.bin (tau_k), 2 001 052 ordinates each.

Method for (b),(c): J_RVM(x) = li(x) - 2 Re sum_k Ei((1/2 + i tau_k) log x) - log 2
+ int_x^inf dt/(t(t^2-1) log t) on a uniform grid in u = log x on [log 2, log XMAX], with the
first K ordinates and the smooth cut of s04_primes.py (tau_K = 2 pi * 2.5 * XMAX); the Ei of
the first KE = 300 ordinates are exact (scipy exp1) on a coarser grid and interpolated by cubic
splines, the remaining ones use Ei(w) = e^w (1/w + 1/w^2 + 2/w^3 + 6/w^4) with the four
exponential sums evaluated by a type-3 nonuniform FFT (finufft).  pi_RVM is the Moebius
inversion (eq. pi-RVM of the paper).
Method for (d): the integer target of s05_mertens.py (tapered explicit formula with the
residues 1/(rho_k zeta_RVM'(rho_k)), exact for k <= 1000), which is the function tabulated in
Table tab:ternary of the paper; the ternary walk M_h is recomputed on n <= 1000.
"""
import os
import math
import numpy as np
import scipy.io as sio
from scipy import special, integrate
from scipy.interpolate import CubicSpline
import finufft
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import (Tee, load_ordinates, ei_upper, mobius_sieve, primes_upto, write_csv,
                        savefig, clock_inverse, tail_coefficients, logG, Product,
                        zeta_from_product, w_transition, OUT)
```

```
say = Tee("s15_tables_1000")
gam, tau = load_ordinates()
Nz = gam.size
NT = 1000
k = np.arange(1, NT + 1)

# ================================================================ (a) clock against zeros
eta = tau[:NT] - gam[:NT]
say("(a) tau_k against gamma_k, k <= %d: max|eta_k| = %.6f at k = %d; mean eta = %.3e; rms = %.6f",
    NT, np.abs(eta).max(), np.argmax(np.abs(eta)) + 1, eta.mean(), np.sqrt(np.mean(eta ** 2)))
say("    gamma_1000 = %.9f, tau_1000 = %.9f; mean spacing at k=1000: %.5f", gam[NT - 1], tau[NT - 1],
    2 * np.pi / np.log(tau[NT - 1] / (2 * np.pi)))
write_csv("s15_clock_1000.csv", ["k", "gamma_k", "tau_k", "eta_k"], [k, gam[:NT], tau[:NT], eta])
# automatic checks of the table data before rounding: indices 1..NT exactly once, eta = tau - gamma
assert np.array_equal(k, np.arange(1, NT + 1)) and np.unique(k).size == NT
assert np.max(np.abs((tau[:NT] - gam[:NT]) - eta)) == 0.0
say("    table checks: indices 1..%d once each; eta_k = tau_k - gamma_k before rounding (displayed values are
     rounded independently)", NT)

# ================================================================ (b),(c) pi_RVM up to XMAX
XMAX = 8200.0
KE = 300
tauK = 2 * np.pi * XMAX * 2.5
K = int(np.searchsorted(tau, tauK))
v = np.arange(K) / K
taper = np.where(v < 0.6, 1.0, np.cos(np.pi / 2 * (v - 0.6) / 0.4) ** 2)
o = tau[:K]
rho = 0.5 + 1j * o
u0, u1 = np.log(2.0), np.log(XMAX)
NF = 1_400_000                          # fine grid: >= 1500 points per period of exp(i tau_K u)
uf = np.linspace(u0, u1, NF)
NC = 140_000                            # coarse grid for the exact Ei of the first KE ordinates
uc = np.linspace(u0, u1, NC)
say("(b),(c) pi_RVM on [2, %g]: K = %d ordinates (tau_K = %.1f), smooth cut; %d fine / %d coarse points",
    XMAX, K, o[-1], NF, NC)

# exact part on the coarse grid, cubic interpolation to the fine grid
Sc = np.zeros(NC, dtype=complex)
for i in range(KE):
    Sc += taper[i] * ei_upper(rho[i] * uc)
S = CubicSpline(uc, Sc.real)(uf) + 1j * CubicSpline(uc, Sc.imag)(uf)
# asymptotic part: A_m(u) = sum_{k>KE} w_k exp(i o_k u) / rho_k^m by a type-3 NUFFT
A = []
for m in (1, 2, 3, 4):
    c = (taper[KE:] / rho[KE:] ** m).astype(np.complex128)
    A.append(finufft.nufft1d3(o[KE:].astype(np.float64), c, uf, eps=1e-13, isign=1))
S += np.exp(uf / 2) * (A[0] / uf + A[1] / uf ** 2 + 2 * A[2] / uf ** 3 + 6 * A[3] / uf ** 4)
# tail integral on a coarse grid, interpolated
ut = np.linspace(u0, u1, 3000)
tfun = lambda t: 1.0 / (t * (t * t - 1.0) * np.log(t))
# the tail beyond 10^7 is below 1/(2 t^2 log t) at t = 10^7, i.e. < 4e-16, and is dropped
ti = np.array([integrate.quad(tfun, np.exp(w), 100 * np.exp(w), limit=200)[0]
               + integrate.quad(tfun, 100 * np.exp(w), 1.0e7, limit=200)[0] for w in ut])
J = special.expi(uf) - 2 * S.real - np.log(2.0) + CubicSpline(ut, ti)(uf)
mu64 = mobius_sieve(64)
PR = np.zeros(NF)
m = 1
while u1 / m >= u0:
    if mu64[m] != 0:
        PR += mu64[m] / m * np.where(uf / m >= u0 - 1e-12, np.interp(uf / m, uf, J), 0.0)
    m += 1
xf = np.exp(uf)
P = primes_upto(int(XMAX) + 1)
# (c) integers n <= 1000
```

```
n = np.arange(1, NT + 1)
pin = np.zeros(NT)
pin[1:] = np.interp(np.log(n[1:].astype(float)), uf, PR)
pi_exact = np.searchsorted(P, n, side="right").astype(float)
li = np.zeros(NT)
li[1:] = special.expi(np.log(n[1:].astype(float)))
say("    pi_RVM(100) = %.4f, pi(100) = %d;  pi_RVM(1000) = %.4f, pi(1000) = %d;  pi_RVM(%g) = %.4f, pi = %d",
    pin[99], int(pi_exact[99]), pin[-1], int(pi_exact[-1]), XMAX, PR[-1], np.searchsorted(P, XMAX, side="right
    "))
d = pin - pi_exact
say("    pi_RVM(n) - pi(n), n <= 1000: max %.4f at n = %d, min %.4f at n = %d, mean %.4f, rms %.4f",
    d.max(), n[np.argmax(d)], d.min(), n[np.argmin(d)], d.mean(), np.sqrt(np.mean(d ** 2)))
say("    pi_RVM(n) - li(n), n <= 1000: mean %.4f;  pi(n) - li(n): mean %.4f", (pin - li)[1:].mean(), (pi_exact
    - li)[1:].mean())
write_csv("s15_pi_1000.csv", ["n", "pi_n", "pi_RVM_n", "pi_RVM_minus_pi", "li_n"], [n, pi_exact, pin, d, li])
# (b) real RVM primes: first crossing of the level j
levels = np.arange(1, NT + 1)
pj = np.zeros(NT)
for j in levels:
    i = int(np.argmax(PR >= j))
    if i == 0:
        raise RuntimeError("level %d not reached below XMAX" % j)
    # linear interpolation of the crossing between the grid points i-1 and i
    x0, x1v = xf[i - 1], xf[i]
    f0, f1 = PR[i - 1], PR[i]
    pj[j - 1] = x0 + (j - f0) / (f1 - f0) * (x1v - x0)
dp = pj - P[:NT]
say("    real RVM primes p_j^RVM, j <= 1000: p_1000^RVM = %.4f, p_1000 = %d; max(p_j^RVM - p_j) = %.4f at j =
    %d, "
    "min = %.4f at j = %d, mean %.4f, rms %.4f", pj[-1], P[NT - 1], dp.max(), np.argmax(dp) + 1, dp.min(),
    np.argmin(dp) + 1, dp.mean(), np.sqrt(np.mean(dp ** 2)))
say("    |p_j^RVM - p_j| / p_j^(1/2): max %.4f;  |p_j^RVM - p_j| / log p_j: max %.4f",
    np.max(np.abs(dp) / np.sqrt(P[:NT])), np.max(np.abs(dp) / np.log(P[:NT])))
say("    first 12 levels: " + ", ".join("%.4f" % x for x in pj[:12]))
write_csv("s15_rvm_primes_1000.csv", ["j", "p_j", "p_j_RVM", "p_j_RVM_minus_p_j"], [levels, P[:NT], pj, dp])
# sensitivity of the levels to the truncation and the grid: s04_primes.py computes the same levels
# with K = 17061 ordinates on [2, 1000] (40000 grid points); here K = %d on [2, 8200]
f04 = os.path.join(OUT, "s04_rvm_prime_levels.csv")
if os.path.exists(f04):
    a04 = np.loadtxt(f04, delimiter=",", skiprows=1, ndmin=2)
    m04 = a04.shape[0]
    d04 = a04[:, 1] - pj[:m04]
    big = np.argsort(-np.abs(d04))[:3]
    say("    sensitivity: levels j <= %d of s04 (K = 17061, x <= 1000) minus these: median |diff| %.1e, "
        "max |diff| %.3f; largest at j = %s (%s); %d levels differ by more than 0.01",
        m04, np.median(np.abs(d04)), np.max(np.abs(d04)), ", ".join(str(i + 1) for i in big),
        ", ".join("%+.3f" % d04[i] for i in big), int(np.sum(np.abs(d04) > 0.01)))
else:
    say("    sensitivity check skipped: run s04_primes.py first (output/s04_rvm_prime_levels.csv)")

# ============================================================== (d) M_RVM^crit(n) against M(n)
K0 = 1000
c1, c2, c3 = tail_coefficients(clock_inverse(float(Nz)))
t2 = tau ** 2
zpR = np.zeros(K0, dtype=complex)
for kk in range(K0):
    tk = tau[kk]
    fac = (t2 - tk ** 2) / (t2 + 0.25)
    fac[kk] = 1.0
    q = -0.25 - tk ** 2
    lmod = np.sum(np.log(np.abs(fac))) + c1 * q - c2 * q ** 2 + c3 * q ** 3
    rr = 0.5 + 1j * tk
    lP = np.log(2j * tk / (0.25 + tk ** 2)) + lmod
    zpR[kk] = (-1) ** kk * np.exp(lP - logG(rr))
ratio = zpR / np.log(tau[:K0] / (2 * np.pi))
```

```
PRD = Product(tau)
varpi = float(open(os.path.join(OUT, "s02_varpi.txt")).read())     # common result of s02_transition.py
sel = np.arange(K0 // 2, K0)
cfit = np.mean((ratio[sel] / varpi - 1) * tau[sel])
zR = varpi * np.log(tau / (2 * np.pi)) * (1 + cfit / tau)
zR[:K0] = zpR
bR = 1.0 / ((0.5 + 1j * tau) * zR)
say("(d) residues: varpi_RVM = %.10f (read from output/s02_varpi.txt), c = %.4f %+.4f i (as in s05_mertens)",
    varpi, cfit.real, cfit.imag)


def trivial_rvm(x, nn=6):
    val = np.zeros_like(x)
    for j in range(1, nn + 1):
        s = -2.0 * j
        lP = np.real(PRD.logP(s)[0])
        val -= x ** (-2 * j) * (2 * j + 1) * np.pi ** j * 2 * (-1) ** j / (math.factorial(j) * np.exp(lP))
    return val


def tapered(x, ords, a, C1=2.0, C2=4.0):
    Sx = np.zeros(x.size, dtype=complex)
    lx = np.log(x)
    Kx = max(1, int(np.searchsorted(ords, C2 * 2 * np.pi * np.sqrt(x[-1]))))
    uu = ords[None, :Kx] / (2 * np.pi * np.sqrt(x[:, None]))
    wt = np.where(uu <= C1, 1.0, np.where(uu >= C2, 0.0, np.cos(np.pi / 2 * (uu - C1) / (C2 - C1)) ** 2))
    E = np.exp(1j * lx[:, None] * ords[None, :Kx])
    Sx = np.sqrt(x) * np.sum(E * wt * a[None, :Kx], axis=1)
    return Sx


xn = n.astype(float)
MR = -2 + 2 * np.real(tapered(xn, tau, bR)) + trivial_rvm(xn)
MR_all = -2 + 2 * np.real(np.sqrt(xn) * np.array([np.sum(np.exp(1j * np.log(x) * tau) * bR) for x in xn])) +
    trivial_rvm(xn)
mu = mobius_sieve(NT)
M = np.cumsum(mu.astype(np.int64))[1:]
Mh = np.zeros(NT)
cur = 0.0
muh = np.zeros(NT, dtype=int)
for i in range(NT):
    if mu[i + 1] != 0:
        sg = 1 if cur < MR[i] else -1
        cur += sg
        muh[i] = sg
    Mh[i] = cur
dM = MR - M
say("    M_RVM^crit(n) (tapered target, Table tab:ternary) against the ordered sum over all %d ordinates: max
    |diff| = %.4f", Nz, np.max(np.abs(MR - MR_all)))
say("    M(1000) = %d, M_RVM^crit(1000) = %.4f, M_h(1000) = %d", M[-1], MR[-1], int(Mh[-1]))
say("    M_RVM^crit(n) - M(n), n <= 1000: max %.4f at n = %d, min %.4f at n = %d, mean %.4f, rms %.4f",
    dM.max(), n[np.argmax(dM)], dM.min(), n[np.argmin(dM)], dM.mean(), np.sqrt(np.mean(dM ** 2)))
say("    max |M(n)|/sqrt n = %.4f, max |M_RVM^crit(n)|/sqrt n = %.4f on [10,1000]; sign agreement mu_h = mu on
     squarefrees: %.4f",
    np.max(np.abs(M[9:]) / np.sqrt(xn[9:])), np.max(np.abs(MR[9:]) / np.sqrt(xn[9:])),
    np.mean(muh[mu[1:] != 0] == mu[1:][mu[1:] != 0]))
write_csv("s15_mertens_1000.csv", ["n", "mu_n", "M_n", "M_RVM_crit_n", "M_RVM_minus_M", "M_h_n", "
    M_RVM_allzeros_n"],
          [n, mu[1:], M, MR, dM, Mh, MR_all])

# ============================================================== LaTeX table bodies
def tex_rows(cols, fmts, groups, fname):
    """cols: list of arrays (same length L); write L/groups rows with `groups` column groups."""
    L = cols[0].size
    R = int(math.ceil(L / groups))
```

```
    lines = []
    for r in range(R):
        cells = []
        for g in range(groups):
            i = r + g * R
            if i < L:
                cells.append(" & ".join(f % c[i] for f, c in zip(fmts, cols)))
            else:
                cells.append(" & ".join("" for _ in cols))
        lines.append(" & ".join(cells) + r" \\")
    with open(os.path.join(OUT, fname), "w") as f:
        f.write("\n".join(lines) + "\n")


tex_rows([k, gam[:NT], tau[:NT], eta], ["%d", "%.6f", "%.6f", "$%+.4f$"], 2, "s15_table_clock_1000.tex")
tex_rows([levels, P[:NT], pj, dp], ["%d", "%d", "%.4f", "$%+.4f$"], 3, "s15_table_primes_1000.tex")
tex_rows([n, pi_exact, pin, d], ["%d", "%d", "%.4f", "$%+.4f$"], 3, "s15_table_pi_1000.tex")
tex_rows([n, mu[1:], M, MR, dM], ["%d", "$%d$", "$%d$", "$%.4f$", "$%+.4f$"], 3, "s15_table_mertens_1000.tex")


def check_tex_table(fname, ncols, ia, ib, idiff, rnd):
    """Read a written table body back and check it: every index 1..NT exactly once (no missing or
    duplicate rows), all cells numeric, and |displayed difference - (displayed a - displayed b)| <= rnd,
    the rounding allowance of the displayed decimals.  The difference columns themselves are
    computed from the unrounded values, so they need not equal the difference of displayed values."""
    cells = []
    for line in open(os.path.join(OUT, fname)).read().split("\\\\"):
        parts = [c.strip().strip("$") for c in line.split("&")]
        for g in range(0, len(parts) - ncols + 1, ncols):
            grp = parts[g:g + ncols]
            if grp[0]:
                cells.append([float(c) for c in grp])
    A = np.array(cells)
    idx = A[:, 0].astype(int)
    ok_idx = np.array_equal(np.sort(idx), np.arange(1, NT + 1))
    dev = np.max(np.abs(A[:, idiff] - (A[:, ia] - A[:, ib])))
    if not ok_idx or dev > rnd:
        raise RuntimeError("table check failed for %s: indices ok %s, deviation %.2e" % (fname, ok_idx, dev))
    say("    table check %s: %d rows, indices 1..%d once each; |displayed difference - difference of displayed
      "
        "values| <= %.1e (allowance %.1e)", fname, A.shape[0], NT, dev, rnd)


check_tex_table("s15_table_clock_1000.tex", 4, 2, 1, 3, 0.5e-4 + 1e-6 + 1e-12)
check_tex_table("s15_table_primes_1000.tex", 4, 2, 1, 3, 1e-4 + 1e-12)
check_tex_table("s15_table_pi_1000.tex", 4, 2, 1, 3, 1e-4 + 1e-12)
check_tex_table("s15_table_mertens_1000.tex", 5, 3, 2, 4, 1e-4 + 1e-12)

# ================================================================ figures (linear axes, PNG)
fig, ax = plt.subplots(2, 2, figsize=(12, 7.5))
ax[0, 0].plot(k, gam[:NT], "k-", lw=0.8, label=r"$\gamma_k$")
ax[0, 0].plot(k, tau[:NT], "r--", lw=0.8, label=r"$\tau_k$")
ax[0, 0].set_xlabel("k"); ax[0, 0].set_title(r"Ordinates $\gamma_k$ and midpoint ordinates $\tau_k$, $k\
    leq1000$")
ax[0, 0].legend(fontsize=8)
ax[0, 1].plot(k, eta, "b.", ms=2.5)
ax[0, 1].axhline(0, color="k", lw=0.6)
ax[0, 1].set_xlabel("k"); ax[0, 1].set_title(r"$\eta_k=\tau_k-\gamma_k$")
kk = np.arange(1, 61)
ax[1, 0].vlines(gam[:60], 0, 1, color="k", lw=0.8, label=r"$\gamma_k$")
ax[1, 0].vlines(tau[:60], 0, -1, color="r", lw=0.8, label=r"$\tau_k$")
ax[1, 0].set_ylim(-1.3, 1.3); ax[1, 0].set_yticks([])
ax[1, 0].set_xlabel("t"); ax[1, 0].set_title(r"The two spectra for $k\leq60$: $\gamma_k$ up, $\tau_k$ down")
ax[1, 0].legend(fontsize=8, loc="upper right")
ax[1, 1].hist(eta, bins=40, color="0.7", edgecolor="k")
```

```
ax[1, 1].set_xlabel(r"$\eta_k$"); ax[1, 1].set_title(r"Empirical distribution of $\eta_k$, $k\leq1000$")
fig.tight_layout()
savefig(fig, "fig20_clock_1000.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4.2))
ax[0].plot(levels, P[:NT], "k-", lw=1.0, label=r"$p_j$")
ax[0].plot(levels, pj, "r--", lw=1.0, label=r"$p_j^{\rm RVM}$")
ax[0].set_xlabel("j"); ax[0].set_title(r"Primes and real RVM primes (first grid crossings), $j\leq1000$"); ax
    [0].legend(fontsize=8)
ax[1].plot(levels, dp, "b.", ms=2.5)
ax[1].axhline(0, color="k", lw=0.6)
ax[1].set_xlabel("j"); ax[1].set_title(r"$p_j^{\rm RVM}-p_j$")
fig.tight_layout()
savefig(fig, "fig21_rvm_primes_1000.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4.2))
ax[0].step(n, pi_exact, where="post", color="k", lw=1.0, label=r"$\pi(n)$")
ax[0].plot(n, pin, "r-", lw=0.9, label=r"$\pi_{\rm RVM}(n)$")
ax[0].plot(n[1:], li[1:] - special.expi(np.log(2.0)), "g--", lw=0.8, label=r"$\mathrm{li}(n)-\mathrm{li}(2)$")
ax[0].set_xlabel("n"); ax[0].set_title(r"$\pi(n)$ and $\pi_{\rm RVM}(n)$, $n\leq1000$"); ax[0].legend(fontsize
    =8)
ax[1].plot(n, d, "b-", lw=0.7)
ax[1].axhline(0, color="k", lw=0.6)
ax[1].set_xlabel("n"); ax[1].set_title(r"$\pi_{\rm RVM}(n)-\pi(n)$")
fig.tight_layout()
savefig(fig, "fig22_pi_1000.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4.2))
ax[0].step(n, M, where="post", color="k", lw=1.0, label=r"$M(n)$")
ax[0].plot(n, MR, "r-", lw=0.9, label=r"$M_{\rm RVM}^{\rm crit}(n)$")
ax[0].plot(n, np.sqrt(xn), "k--", lw=0.6, label=r"$\pm\sqrt{n}$")
ax[0].plot(n, -np.sqrt(xn), "k--", lw=0.6)
ax[0].set_xlabel("n"); ax[0].set_title(r"$M(n)$ and $M_{\rm RVM}^{\rm crit}(n)$ (tapered explicit formula), $n
    \leq1000$"); ax[0].legend(fontsize=8)
ax[1].plot(n, dM, "b-", lw=0.7)
ax[1].axhline(0, color="k", lw=0.6)
ax[1].set_xlabel("n"); ax[1].set_title(r"$M_{\rm RVM}^{\rm crit}(n)-M(n)$")
fig.tight_layout()
savefig(fig, "fig23_mertens_1000.png")

sio.savemat(os.path.join(OUT, "s15_results.mat"), {"k": k, "gamma": gam[:NT], "tau": tau[:NT], "eta": eta,
                                                   "p_j": P[:NT], "p_j_RVM": pj, "pi_n": pi_exact, "pi_RVM_n":
     pin,
                                                   "M_n": M, "M_RVM_crit_n": MR, "M_h_n": Mh, "varpi": varpi,
                                                   "cfit": cfit, "K": K, "XMAX": XMAX})
say.close()
```

B.17. **`s16_walks_3d_1000.py`.** This program produces Experiment 19.8 and Figures 10 and 11. A MATLAB version, `matlab/s16_walks_3d_1000.m`, is distributed with the repository.

```
"""Experiment 16: the three exponential walks of the first 1000 ordinates, drawn in 3D with k
on the vertical axis, and their triangular centroids.

  S_k(x) = sum_{j <= k} exp(i x_j),  x = gamma (zeros of zeta), tau (midpoint clock),
  theta = (gamma + tau)/2 (the midpoint of each zero and its clock ordinate);
  C_N(x) = (1/N) sum_{k <= N} S_k(x) = sum_{k <= N} (1 - (k-1)/N) exp(i x_k).

Data: data/zeross.bin (gamma_k) and data/zLamb.bin (tau_k)."""
import os
import numpy as np
import scipy.io as sio
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
```

```
from matplotlib.ticker import MaxNLocator
from rvm_common import Tee, load_ordinates, write_csv, savefig, OUT

say = Tee("s16_walks_3d_1000")
gam, tau = load_ordinates()
N = 1000
k = np.arange(1, N + 1)
g, t = gam[:N], tau[:N]
th = (g + t) / 2
w = 1 - (k - 1) / N
names = ["gamma", "tau", "theta"]
X = {"gamma": g, "tau": t, "theta": th}
S = {nm: np.cumsum(np.exp(1j * X[nm])) for nm in names}
C = {nm: np.sum(w * np.exp(1j * X[nm])) for nm in names}
for nm in names:
    say("C_%-6s = %+.10f %+.10f i   (|S_N| = %.4f, max_k |S_k| = %.4f, check mean(S_k) - C = %.1e)",
        nm, C[nm].real, C[nm].imag, abs(S[nm][-1]), np.abs(S[nm]).max(), abs(S[nm].mean() - C[nm]))
ch = (C["gamma"] + C["tau"]) / 2
say("(C_gamma + C_tau)/2 = %+.10f %+.10f i;  C_theta - chord midpoint = %+.6f %+.6f i, modulus %.6f",
    ch.real, ch.imag, (C["theta"] - ch).real, (C["theta"] - ch).imag, abs(C["theta"] - ch))
eps = g - t
say("gamma_k - tau_k, k <= %d: mean %+.6f, std %.6f, max |.| %.6f", N, eps.mean(), eps.std(), np.abs(eps).max
    ())
say("reference values with all %d ordinates: C_gamma = %+.10f %+.10f i, C_tau = %+.10f %+.10f i", gam.size,
    *[f(np.sum((1 - (np.arange(gam.size)) / gam.size) * np.exp(1j * o))) for o in (gam, tau) for f in (np.real,
     np.imag)])
write_csv("s16_walks_1000.csv", ["k", "gamma", "tau", "theta", "ReS_gamma", "ImS_gamma", "ReS_tau", "ImS_tau",
                                 "ReS_theta", "ImS_theta"],
          [k, g, t, th, S["gamma"].real, S["gamma"].imag, S["tau"].real, S["tau"].imag, S["theta"].real, S["
    theta"].imag])

cols = {"gamma": "#1f5fa8", "tau": "#d0582a", "theta": "#3d9a3d"}
lab = {"gamma": r"$\gamma_k$ (zeros of $\zeta$)", "tau": r"$\tau_k$ (midpoint clock)", "theta": r"$\theta_k=(\
    gamma_k+\tau_k)/2$"}
mk = {"gamma": "o", "tau": "s", "theta": "D"}
zc = (N + 1) / 2

# one 3D panel per walk, k on the vertical axis
fig = plt.figure(figsize=(16, 6))
for i, nm in enumerate(names):
    ax = fig.add_subplot(1, 3, i + 1, projection="3d")
    ax.plot(S[nm].real, S[nm].imag, k, lw=0.6, color=cols[nm])
    ax.scatter([C[nm].real], [C[nm].imag], [zc], s=60, marker=mk[nm], color=cols[nm], edgecolor="k",
     depthshade=False)
    ax.set_xlabel(r"Re $S_k$"); ax.set_ylabel(r"Im $S_k$"); ax.set_zlabel("$k$")
    ax.set_title(lab[nm] + "\n$C$ = %.4f %+.4f i" % (C[nm].real, C[nm].imag), fontsize=10)
    ax.view_init(22, -55)
    ax.xaxis.set_major_locator(MaxNLocator(5)); ax.yaxis.set_major_locator(MaxNLocator(5))
fig.tight_layout()
savefig(fig, "fig24_walks_3d_1000.png")

# the three walks together, and the projection on the complex plane
fig = plt.figure(figsize=(15, 6.5))
ax = fig.add_subplot(1, 2, 1, projection="3d")
for nm in names:
    ax.plot(S[nm].real, S[nm].imag, k, lw=0.6, color=cols[nm], label=lab[nm])
    ax.scatter([C[nm].real], [C[nm].imag], [zc], s=60, marker=mk[nm], color=cols[nm], edgecolor="k",
     depthshade=False)
ax.set_xlabel(r"Re $S_k$"); ax.set_ylabel(r"Im $S_k$"); ax.set_zlabel("$k$")
ax.view_init(22, -55); ax.set_title("The three walks, $N=%d$ (centroids at height $(N+1)/2$)" % N, fontsize
    =10)
ax.legend(loc="upper left", fontsize=8)
ax = fig.add_subplot(1, 2, 2)
for nm in names:
    ax.plot(S[nm].real, S[nm].imag, lw=0.4, color=cols[nm], label=lab[nm])
```

```
for nm in names:
    ax.plot(C[nm].real, C[nm].imag, mk[nm], ms=10, color=cols[nm], mec="k", mew=1.2, zorder=10,
            label="$C$ = %.4f %+.4f i" % (C[nm].real, C[nm].imag))
ax.plot(ch.real, ch.imag, "kx", ms=12, mew=2, zorder=11, label=r"$(C_\gamma+C_\tau)/2$")
ax.set_aspect("equal"); ax.grid(alpha=0.3)
ax.set_xlabel(r"Re $S_k$"); ax.set_ylabel(r"Im $S_k$"); ax.set_title("Projection on the complex plane",
     fontsize=10)
ax.legend(fontsize=7, loc="upper left")
fig.tight_layout()
savefig(fig, "fig25_walks_projection_1000.png")

sio.savemat(os.path.join(OUT, "s16_results.mat"), {"k": k, "gamma": g, "tau": t, "theta": th,
                                                   "S_gamma": S["gamma"], "S_tau": S["tau"], "S_theta": S["
    theta"],
                                                   "C_gamma": C["gamma"], "C_tau": C["tau"], "C_theta": C["
    theta"]})
say.close()
```

B.18. **`s17_cramer_centroids.py`.** This program evaluates Cramér's function by the zero-free representation of Proposition 5.3 and the centroids of Experiment 19.7. Run time about one minute.

```
"""Experiment 17: Cramer's function V_+(z) = sum_{Im rho > 0} exp(rho z) from its zero-free
representation (Proposition prop:cramer-zero-free of the paper), and the limits of the triangular
centroids of the two spectra.

Zero-free representation (a > 0, Re z > 0, z not of the form log n; contour of the rectangle
-a < Re s < 1 + a, Im s > 0, functional equation xi'/xi(s) = -xi'/xi(1-s)):

  2 pi i V_+(z) = V_b(z) + V_R(z) + V_L(z) + V_p(z),
  V_b(z) = int_{-a}^{1+a} exp(s z) xi'/xi(s) ds                        (real segment),
  V_R(z) = i exp((1+a) z) int_0^{inf e^{i theta}} exp(i t z) g(1 + a + i t) dt,
  V_L(z) = i exp(-a z)    int_0^{inf e^{i theta}} exp(i t z) g(1 + a - i t) dt,
  V_p(z) = sum_n Lambda(n) [ exp((1+a)(z - log n)) / (z - log n) + n^(-1-a) exp(-a z) / (z + log n) ],
  g(s)   = 1/s + 1/(s-1) - (1/2) log pi + (1/2) psi(s/2)   (so that xi'/xi = g + zeta'/zeta),

with the rays rotated by an angle 0 < theta < pi/2.  The value must not depend on a; it is computed
for a = 1 and a = 2.  Checks: 2 Re V_+(u) = e^u - 1/(e^{2u} - 1) for log 2 < u < log 3 (explicit
formula), and the comparison with the triangular centroid of the 2 001 052 ordinates.

Clock centroid: C_RVM^path(N;u) -> C_infty(u) (Theorem thm:clock-centroid), with
S_n - b_n E_{n+1} = C_infty(u) + O(log n / n), b_n = 1/(exp(i u (tau_{n+1}-tau_n)) - 1)."""
import os
import numpy as np
import mpmath as mp
from rvm_common import Tee, load_ordinates, write_csv, primes_upto

say = Tee("s17_cramer_centroids")
mp.mp.dps = 30
PMAX = 2_000_000
P = primes_upto(PMAX)


def lambda_table():
    """von Mangoldt pairs (n, Lambda(n)) for prime powers n <= PMAX."""
    out = []
    for p in P:
        q = int(p)
        lp = mp.log(q)
        while q <= PMAX:
            out.append((q, lp))
            q *= int(p)
    out.sort()
    return out
```

```
LAM = lambda_table()


def xi_logderiv(s):
    s = mp.mpf(s)
    if abs(s - 1) < mp.mpf("1e-20") or abs(s) < mp.mpf("1e-20"):
        s += mp.mpf("1e-18")
    return 1 / s + 1 / (s - 1) - mp.log(mp.pi) / 2 + mp.digamma(s / 2) / 2 + mp.zeta(s, 1, 1) / mp.zeta(s)


def g_arch(s):
    return 1 / s + 1 / (s - 1) - mp.log(mp.pi) / 2 + mp.digamma(s / 2) / 2


def V_plus(z, a, theta=mp.pi / 3):
    z = mp.mpc(z)
    a = mp.mpf(a)
    # real segment, split at 1/2 and use xi'/xi(s) = -xi'/xi(1-s) on the left half
    Vb = mp.quad(lambda s: (mp.exp(s * z) - mp.exp((1 - s) * z)) * xi_logderiv(s), [mp.mpf(1) / 2, 1, 1 + a])
    ray = mp.exp(1j * theta)
    VR = 1j * mp.exp((1 + a) * z) * mp.quad(lambda r: mp.exp(1j * r * ray * z) * g_arch(1 + a + 1j * r * ray)
     * ray, [0, 1, 10, mp.inf])
    VL = 1j * mp.exp(-a * z) * mp.quad(lambda r: mp.exp(1j * r * ray * z) * g_arch(1 + a - 1j * r * ray) * ray,
      [0, 1, 10, mp.inf])
    Vp = mp.mpc(0)
    for n, L in LAM:
        ln = mp.log(n)
        Vp += L * (mp.exp((1 + a) * (z - ln)) / (z - ln) + mp.power(n, -1 - a) * mp.exp(-a * z) / (z + ln))
    return (Vb + VR + VL + Vp) / (2j * mp.pi)


gam, tau = load_ordinates()
N = gam.size
rows = []
for a in (1, 2):
    V1 = V_plus(1, a)
    Z1 = mp.exp(-mp.mpf(1) / 2) * V1
    say("a = %d:  e^{-1/2} V_+(1) = %s %+s i", a, mp.nstr(Z1.real, 16), mp.nstr(Z1.imag, 16))
    rows.append([a, float(Z1.real), float(Z1.imag)])
re_exact = (mp.e - 1 / (mp.e ** 2 - 1)) / (2 * mp.sqrt(mp.e))
say("explicit formula: Re e^{-1/2} V_+(1) = (e - 1/(e^2-1))/(2 sqrt e) = %s", mp.nstr(re_exact, 16))
for u in (0.75, 0.9, 1.05):
    Vu = V_plus(u, 2)
    say("u = %.2f:  2 Re V_+(u) = %s,  e^u - 1/(e^{2u}-1) = %s", u, mp.nstr(2 * Vu.real, 14),
        mp.nstr(mp.e ** u - 1 / (mp.e ** (2 * u) - 1), 14))

# triangular centroids of the ordinates (u = 1)
w = 1 - np.arange(N) / N
Cg = np.sum(w * np.exp(1j * gam))
Ct = np.sum(w * np.exp(1j * tau))
say("triangular centroids with all N = %d ordinates: C_zeta = %.10f %+.10f i,  C_RVM = %.10f %+.10f i",
    N, Cg.real, Cg.imag, Ct.real, Ct.imag)
for K in (10 ** 4, 10 ** 5, 10 ** 6):
    wK = 1 - np.arange(K) / K
    c1 = np.sum(wK * np.exp(1j * gam[:K]))
    c2 = np.sum(wK * np.exp(1j * tau[:K]))
    say("  N = %8d: C_zeta = %.8f %+.8f i,  C_RVM = %.8f %+.8f i", K, c1.real, c1.imag, c2.real, c2.imag)

# clock centroid limit from S_n - b_n E_{n+1}
E = np.exp(1j * tau)
S = np.cumsum(E)
for n in (10 ** 3, 10 ** 4, 10 ** 5, 10 ** 6, N - 1):
    b = 1 / (np.exp(1j * (tau[n] - tau[n - 1])) - 1)
    cinf = S[n - 1] - b * E[n]
```

```
    say("  n = %8d: S_n - b_n E_{n+1} = %.10f %+.10f i", n, cinf.real, cinf.imag)
write_csv("s17_cramer_values.csv", ["a", "Re_e^-1/2V+(1)", "Im_e^-1/2V+(1)"], [np.array(r) for r in zip(*rows)
    ])
say.close()
```

Appendix C. Console output of the programs

The complete console output of each program, as written to `output/*_console.txt`.

The numerical programs that produce the multiscale and bounded-endpoint figures (files `fig14`–`fig17`) and the bounded/nested endpoint tables are also distributed with their machine-readable outputs in the repository. In particular, `s12_multiscale_jacobian.py` writes `output/s12_multiscale_jacobian_console.txt`, `output/s12_multiscale_summary.csv` and `output/s12_fixed_window_pairs.csv`; `s13_nested_bounded_endpoint.py` writes `output/s13_bounded_mass_summary.csv`, `output/s13_bounded_mass_ordinates.csv`, `output/s13_bounded_mass_masses.csv`, `output/s13_cross_validation.csv`, `output/s13_nested_metric_summary.csv` and the console output `output/s13_nested_bounded_endpoint_console.txt`. Here "normalized residual" means the Euclidean residual divided by the square root of the number of test equations. The bump and permanent-block conventions are specified in the corresponding code and in Appendix F.

C.1. `s01_clock_and_family.`

```
zeros loaded: N = 2001052, gamma_1 = 14.134725142, gamma_N = 1132490.658714, tau_1 = 14.521346953, tau_N =
    1132490.722588
max |N_RVM(tau_k) - (k - 1/2)| = 6.985e-10
eta_k = tau_k - gamma_k:  max |eta| = 0.993711 (k = 34),  mean |eta| = 0.151209,  rms = 0.188066,  mean eta =
    1.731e-07
      k from         k to     max|eta|      rms eta
           1           10     0.840022     0.545768
          11          100     0.993711     0.453166
         101         1000     0.915450     0.345177
        1001        10000     0.833881     0.274177
       10001       100000     0.785835     0.226352
      100001      1000000     0.762816     0.192411
     1000001      2001052     0.717359     0.178900
first ordinates:
   k          gamma_k            tau_k        eta_k
   1     14.134725142     14.521346953     0.386622
   2     21.022039639     20.655740356    -0.366299
   3     25.010857580     25.492675432     0.481818
   4     30.424876126     29.739411632    -0.685464
   5     32.935061588     33.624531889     0.689470
   6     37.586178159     37.257370087    -0.328808
   7     40.918719012     40.700551604    -0.218167
   8     43.327073281     43.994013953     0.666941
   9     48.005150881     47.165128427    -0.840022
  10     49.773832478     50.233653391     0.459821
  11     52.970321478     53.214422874     0.244101
  12     56.446247697     56.118921948    -0.327326
sigma = 2.00, t in [1,2000] (499 points): max |zeta_Z1/zeta - 1| = 8.706e-10, median = 2.685e-10; max |zeta_Z1
      - zeta|/max(1,|zeta_Z1|,|zeta|) = 7.544e-10
sigma = 1.25, t in [1,2000] (499 points): max |zeta_Z1/zeta - 1| = 1.128e-09, median = 3.693e-10; max |zeta_Z1
      - zeta|/max(1,|zeta_Z1|,|zeta|) = 1.070e-09
sigma = 0.75, t in [1,2000] (499 points): max |zeta_Z1/zeta - 1| = 2.629e-09, median = 5.179e-10; max |zeta_Z1
      - zeta|/max(1,|zeta_Z1|,|zeta|) = 1.903e-09
lambda = 0.00: max |Im Z_lambda| / max(1,|Z_lambda|) on [0.5,60] = 2.966e-14
lambda = 0.25: max |Im Z_lambda| / max(1,|Z_lambda|) on [0.5,60] = 2.964e-14
lambda = 0.50: max |Im Z_lambda| / max(1,|Z_lambda|) on [0.5,60] = 2.685e-14
lambda = 0.75: max |Im Z_lambda| / max(1,|Z_lambda|) on [0.5,60] = 2.624e-14
lambda = 1.00: max |Im Z_lambda| / max(1,|Z_lambda|) on [0.5,60] = 2.623e-14
critical line, t in [0.5,60]: max |Z_Z1(t) - Z(t)| / max(1,|Z(t)|) = 8.370e-10
total run time 25.8 s
```

C.2. **s01b_clock_zoom.**

```
zoom table near the top of the file:
        k            gamma_k              tau_k      eta_k   N*-N_RVM
  2001041     1132485.303923     1132485.011569  -0.292355  -0.563104
  2001042     1132485.662638     1132485.530752  -0.131886  -0.254025
  2001043     1132486.244063     1132486.049936  -0.194127  -0.373908
  2001044     1132486.392256     1132486.569120   0.176863   0.340656
  2001045     1132486.874881     1132487.088303   0.213422   0.411072
  2001046     1132487.571868     1132487.607487   0.035619   0.068605
  2001047     1132487.950694     1132488.126670   0.175977   0.338949
  2001048     1132488.905328     1132488.645854  -0.259474  -0.499774
  2001049     1132489.162780     1132489.165037   0.002257   0.004348
  2001050     1132489.585215     1132489.684221   0.099006   0.190696
  2001051     1132490.165305     1132490.203404   0.038099   0.073383
  2001052     1132490.658714     1132490.722588   0.063873   0.123026
total run time 0.6 s
```

C.3. **s02_transition.**

```
varpi_RVM (Richardson, sigma = 2, T = 2500, 5000, 10000, accurate product) = 0.9619415789 -1.27e-10 i
L = 1/varpi_RVM = 1.0395641710
stability of the extrapolation (empirical, no error bound):
  quadratic, T = 5000, 10000, 20000: 0.9619415807 +6.21e-11 i  (difference 1.8e-09)
  quadratic, T = 4000, 7000, 10000:  0.9619415789 -3.78e-11 i  (difference 9.0e-11)
  linear,    T = 5000, 10000:        0.9619416885 +9.87e-11 i  (difference 1.1e-07)
  evaluation check at T = 10000: |zeta_RVM(default) / zeta_RVM(accurate) - 1| = 7.1e-10
  evaluation check at T = 20000: |zeta_RVM(default) / zeta_RVM(accurate) - 1| = 1.2e-05
sigma' = 0.25: |zeta_RVM/(varpi(1+w)) - 1|: log-log slope on t>100 = -1.181, max t*err = 0.0617
sigma' = 0.25: |zeta_RVM/(varpi(1+chi)) - 1|: max t*err (t>100) = 0.0790
          |w| * t^sigma' in [1.5832, 1.5832];  |1/zeta_RVM - L| at t = 1e4: 1.425e-01
sigma' = 0.50: |zeta_RVM/(varpi(1+w)) - 1|: log-log slope on t>100 = -1.181, max t*err = 0.0618
          |w| * t^sigma' in [2.5061, 2.5066];  |1/zeta_RVM - L| at t = 1e4: 2.543e-02
sigma' = 1.50: |zeta_RVM/(varpi(1+w)) - 1|: log-log slope on t>100 = -1.182, max t*err = 0.0620
          |w| * t^sigma' in [15.6619, 15.7496];  |1/zeta_RVM - L| at t = 1e4: 1.599e-05
total run time 10.2 s
```

C.4. **s03_exponential_sums.**

```
     x   max_N|S_RVM|    max sampled   max|S|/log N    |S_zeta(N)| Landau -T/2pi*L/sqrt(x)   |M_H^RVM(x)|
   2.0         3.5772         3.5745         0.7334       88300.77        -88341.59      1.722e-06
   3.0         3.9768         3.9750         2.5208      114267.92       -114324.31      1.175e-06
   4.0         3.9578         3.9523         1.9621       62435.29        -62466.94      8.004e-07
   6.0         4.5265         4.5264         1.9236           0.26            -0.00      2.262e-06
  10.0        10.3388         9.6989         3.3959           2.42            -0.00      4.031e-06
total run time 1.5 s
```

C.5. **s04_primes.**

```
  Ei expansion for k > 300: min |w| = 377.0, bound for the total remainder 3.23e-09; max |expansion - Ei| / |e
     ^w/w| on a sample of 50 ordinates = 1.19e-09
  recurrence check: max |recurrent - exact| of exp(i o_k u) before each recomputation = 5.94e-09
  Ei expansion for k > 300: min |w| = 377.3, bound for the total remainder 3.23e-09; max |expansion - Ei| / |e
     ^w/w| on a sample of 50 ordinates = 1.18e-09
  recurrence check: max |recurrent - exact| of exp(i o_k u) before each recomputation = 5.94e-09
x <= 100, all 2001052 zeros of each list, 3000 grid points:
  max |pi_from_zeta_zeros - pi| away from primes (distance > 0.2): 0.4035
  pi_RVM(100) = 26.0782, pi(100) = 25, li(100) = 30.1261, J_RVM(100) = 29.5889, J(100) = 28.5333
  pi_RVM increasing on the sampled grid of [2,100]: False (min increment on the grid -2.528e-01)
  Ei expansion for k > 300: min |w| = 377.0, bound for the total remainder 1.02e-08; max |expansion - Ei| / |e
     ^w/w| on a sample of 50 ordinates = 1.19e-09
  recurrence check: max |recurrent - exact| of exp(i o_k u) before each recomputation = 1.80e-10
  Ei expansion for k > 300: min |w| = 377.3, bound for the total remainder 1.02e-08; max |expansion - Ei| / |e
     ^w/w| on a sample of 50 ordinates = 1.18e-09
  recurrence check: max |recurrent - exact| of exp(i o_k u) before each recomputation = 1.79e-10
x <= 1000, smooth cut at K = 17061 zeros (tau_K = 15707.6), 40000 grid points:
  pi_RVM(1000) = 169.9750, pi(1000) = 168, li(1000) = 177.6097
```

```
  interpolation check: max |pi_RVM(step du) - pi_RVM(step 2 du)| at the common nodes = 5.24e-05 (estimated
     interpolation error of the step du: 1.7e-05)
cell masses c_n = pi_RVM(n) - pi_RVM(n-1):
    n        c_n    1/log n    prime
    2    0.52944    1.44270        1
    3    1.04188    0.91024        1
    4    0.53593    0.72135        0
    5    0.46739    0.62133        1
    6    0.36089    0.55811        0
    7    0.63254    0.51390        1
    8    0.62393    0.48090        0
    9   -0.37669    0.45512        0
   10    0.17151    0.43429        0
   11    0.69779    0.41703        1
   12    0.32703    0.40243        0
   13    0.36227    0.38987        1
   14    0.57662    0.37892        0
   15    0.19671    0.36927        0
   16   -0.05284    0.36067        0
   17    0.39958    0.35296        1
   18    0.76865    0.34598        0
   19    0.43500    0.33962        1
   20   -0.03342    0.33381        0
   21    0.03009    0.32846        0
   22    0.37466    0.32352        0
   23    0.40277    0.31893        1
   24    0.11768    0.31466        0
   25    0.01006    0.31067        0
   26    0.27350    0.30693        0
   27    0.55508    0.30341        0
   28    0.47627    0.30010        0
   29    0.12577    0.29697        1
   30   -0.08367    0.29401        0
   31    0.07329    0.29121        1
   32    0.38040    0.28854        0
   33    0.43927    0.28600        0
   34    0.34619    0.28358        0
   35    0.06419    0.28127        0
   36   -0.02475    0.27906        0
   37    0.15448    0.27694        1
   38    0.40992    0.27491        0
   39    0.49628    0.27296        0
   40    0.35226    0.27109        0
   41    0.14030    0.26928        1
sum_{n<=1000} c_n = pi_RVM(1000) = 169.9750 (telescoping, with pi_RVM(1) = 0);  pi(1000) = 168
real RVM prime levels p_j^RVM (first grid crossing of level j, refined linearly) versus p_j:
    j      p_j^RVM      p_j       diff
    1       2.2138        2     0.2138
    2       3.1546        3     0.1546
    3       5.1880        5     0.1880
    4       7.1588        7     0.1588
    5      11.1391       11     0.1391
    6      14.0201       13     1.0201
    7      17.2551       17     0.2551
    8      19.1767       19     0.1767
    9      26.0524       23     3.0524
   10      28.0419       29    -0.9581
   11      33.0665       31     2.0665
   12      38.0878       37     1.0878
   13      40.2253       41    -0.7747
   14      44.1857       43     1.1857
   15      46.1157       47    -0.8843
   16      57.1722       53     4.1722
   17      59.1154       59     0.1154
   18      61.0717       61     0.0717
```

```
   19      67.0344       67     0.0344
   20      74.0901       71     3.0901
   21      76.1129       73     3.1129
   22      83.0778       79     4.0778
   23      86.0388       83     3.0388
   24      92.0776       89     3.0776
   25      94.0746       97    -2.9254
   26      96.1263      101    -4.8737
   27     103.2186      103     0.2186
   28     112.1163      107     5.1163
   29     115.0245      109     6.0245
   30     118.1254      113     5.1254
total run time 457.3 s
```

C.6. **s05_mertens.**

```
zeta_RVM'(rho_k) exact for k <= 1000; zeta_RVM'(rho_k)/log(tau_k/2pi) at k = 1000: 0.9619388429 -1.41e-05 i
varpi_RVM = 0.9619415789; fit zeta_RVM'(rho_k) = varpi log(tau_k/2pi)(1 + c/tau_k), c = -0.0052 -0.0208 i
  check k = 1: zeta_RVM'(rho_k) product formula +0.76632813 -0.00109973i, central difference +0.76632813
     -0.00109973i, relative difference 2.9e-11
  check k = 2: zeta_RVM'(rho_k) product formula +1.12397259 -0.00113379i, central difference +1.12397259
     -0.00113379i, relative difference 8.1e-12
  check k = 3: zeta_RVM'(rho_k) product formula +1.33270292 -0.00108922i, central difference +1.33270292
     -0.00108922i, relative difference 2.6e-11
  check k = 4: zeta_RVM'(rho_k) product formula +1.48421151 -0.00103980i, central difference +1.48421151
     -0.00103980i, relative difference 3.0e-11
  check k = 5: zeta_RVM'(rho_k) product formula +1.60437102 -0.00099410i, central difference +1.60437102
     -0.00099410i, relative difference 7.2e-11
  check k = 6: zeta_RVM'(rho_k) product formula +1.70447231 -0.00095314i, central difference +1.70447231
     -0.00095314i, relative difference 1.8e-11
Riemann-Siegel zeta'(rho) versus mpmath at k = 1001..1050: max relative difference 1.19e-05
  numerical differentiation: |Z'(h) - Z'(2h)|/|Z'(h)| on the same points: max 5.60e-08
  spot check at k = 101001 (start of a block): |RS / mpmath - 1| = 6.99e-08
  spot check at k = 301001 (start of a block): |RS / mpmath - 1| = 6.73e-06
  spot check at k = 501001 (start of a block): |RS / mpmath - 1| = 5.92e-06
  spot check at k = 701001 (start of a block): |RS / mpmath - 1| = 2.34e-05
  spot check at k = 901001 (start of a block): |RS / mpmath - 1| = 7.08e-06
  spot check at k = 1101001 (start of a block): |RS / mpmath - 1| = 6.41e-06
  spot check at k = 1301001 (start of a block): |RS / mpmath - 1| = 3.46e-06
  spot check at k = 1501001 (start of a block): |RS / mpmath - 1| = 1.13e-05
  spot check at k = 1701001 (start of a block): |RS / mpmath - 1| = 1.80e-05
  spot check at k = 1901001 (start of a block): |RS / mpmath - 1| = 1.12e-05
median |zeta'(rho)| = 7.8899;  truncated spectral sums over the supplied ordinates 2 sum_{k<=N}|a_k|^2: zeta
     0.029034, RVM 0.027203
M(10^6) = 212;  squarefree density Q(10^6)/10^6 = 0.607926 (6/pi^2 = 0.607927)
explicit formula with 2001052 zeros of zeta on [2,2000]: rms(M_zeta - M) = 0.0008 off integers (|x-n|>0.02)
explicit formula with 2001052 zeros of zeta on [2,2000]: rms(M_zeta - M) = 0.0005 off integers (|x-n|>0.05)
explicit formula with 2001052 zeros of zeta on [2,2000]: rms(M_zeta - M) = 0.0004 off integers (|x-n|>0.10)
trivial terms: |trivial_zeta(2; 5) - trivial_zeta(2; 6)| = 3.22e-04, |trivial_rvm(2; 5) - trivial_rvm(2; 6)| =
      3.28e-04
trivial terms: |trivial_zeta(2; 7) - trivial_zeta(2; 6)| = 1.49e-05, |trivial_rvm(2; 7) - trivial_rvm(2; 6)| =
      1.53e-05
tapered-and-interpolated target on 49896 grid points; at 100 integers in [1000, 99703]: max |M_RVM(n) (all
     zeros, untapered) - target(n)| = 0.0191
tapered-and-interpolated target on 49896 grid points; at 34 integers in [900000, 999033]: max |M_RVM(n) (all
     zeros, untapered) - target(n)| = 0.0178
  residue model c fitted on 250 < k <= 1000 : max |change of the target| at the 134 check integers = 3.48e-06
  residue model model c/tau + d/tau^2       : max |change of the target| at the 134 check integers = 5.24e-06
ternary walk: observed max |M_h - target| on [1,1e6] = 2.6155, on [1e4,1e6] = 1.6539
densities of (-1, 0, +1):  mu   0.303857 0.392074 0.304069
                           mu_h 0.304003 0.392074 0.303923   (3/pi^2 = 0.303964, 1 - 6/pi^2 = 0.392073)
sign agreement mu_h = mu on squarefrees: 0.500344;  sum mu mu_h / Q = 0.000688
max |M(n)|/sqrt(n) on [10,1e6] = 0.8321;  max |target(n)|/sqrt(n) = 0.7541;  max |M_h(n)|/sqrt(n) = 0.6882

NUMERICAL EXAMPLE OF THE TERNARY DIVISION (first 40 integers)
```

```
   n   mu   mu^2     target M_h(n-1)   sign   mu_h    M_h      M
   1    1      1     0.2925        0     +1      1      1      1
   2   -1      1     0.4563        1     -1     -1      0      0
   3   -1      1    -0.6523        0     -1     -1     -1     -1
   4    0      0    -0.8720       -1      -      0     -1     -1
   5   -1      1    -1.7255       -1     -1     -1     -2     -2
   6    1      1    -1.3063       -2     +1      1     -1     -1
   7   -1      1    -1.5931       -1     -1     -1     -2     -2
   8    0      0    -2.4029       -2      -      0     -2     -2
   9    0      0    -1.3292       -2      -      0     -2     -2
  10    1      1    -1.3280       -2     +1      1     -1     -1
  11   -1      1    -2.3272       -1     -1     -1     -2     -2
  12    0      0    -2.2067       -2      -      0     -2     -2
  13   -1      1    -1.8468       -2     +1      1     -1     -3
  14    1      1    -1.9324       -1     -1     -1     -2     -2
  15    1      1    -1.5435       -2     +1      1     -1     -1
  16    0      0    -1.0414       -1      -      0     -1     -1
  17   -1      1    -1.6970       -1     -1     -1     -2     -2
  18    0      0    -3.0095       -2      -      0     -2     -2
  19   -1      1    -3.2869       -2     -1     -1     -3     -3
  20    0      0    -2.2604       -3      -      0     -3     -3
  21    1      1    -1.2515       -3     +1      1     -2     -2
  22    1      1    -1.1120       -2     +1      1     -1     -1
  23   -1      1    -1.3607       -1     -1     -1     -2     -2
  24    0      0    -1.3669       -2      -      0     -2     -2
  25    0      0    -1.3810       -2      -      0     -2     -2
  26    1      1    -1.9805       -2     +1      1     -1     -1
  27    0      0    -3.0141       -1      -      0     -1     -1
  28    0      0    -3.6155       -1      -      0     -1     -1
  29   -1      1    -3.2066       -1     -1     -1     -2     -2
  30   -1      1    -2.1599       -2     -1     -1     -3     -3
  31   -1      1    -1.3649       -3     +1      1     -2     -4
  32    0      0    -1.2973       -2      -      0     -2     -4
  33    1      1    -1.6435       -2     +1      1     -1     -3
  34    1      1    -1.7783       -1     -1     -1     -2     -2
  35    1      1    -1.4711       -2     +1      1     -1     -1
  36    0      0    -1.0723       -1      -      0     -1     -1
  37   -1      1    -1.0774       -1     -1     -1     -2     -2
  38    1      1    -1.5897       -2     +1      1     -1     -1
  39    1      1    -2.2227       -1     -1     -1     -2      0
  40    0      0    -2.4881       -2      -      0     -2      0

ENERGY E_N = (1/N) sum_{n<=N} W(n)^2
        N         E_N(M)     E_N(M_RVM)       E_N(M_h)
     1000        14.6400        18.6280        18.5960
    10000       163.6747       195.6965       196.0683
   100000      1622.3189      1308.3041      1308.8186
  1000000     14851.0857     12537.9008     12538.1174
total run time 134.0 s
```

C.7. **s06_prime_ratios.**

```
grid: 40000 points on [2, 1000]; pi_RVM from the first 17061 ordinates tau_k (smooth cut)

ratios at selected x
       x           pi       pi_RVM           li    li/pi_RVM       pi/li    pi/pi_RVM
    10.0       4.0000       3.9876       6.1657     1.546218     0.648747     1.003105
    20.0       8.0000       7.6636       9.9052     1.292496     0.807657     1.043893
    50.0      15.0000      14.7686      18.4695     1.250592     0.812151     1.015670
   100.0      25.0000      26.0813      30.1265     1.155098     0.829835     0.958541
   200.0      46.0000      43.3656      50.1910     1.157393     0.916499     1.060749
   500.0      95.0000      93.7489     101.7975     1.085854     0.933225     1.013346
  1000.0     168.0000     169.9750     177.6097     1.044916     0.945895     0.988381

square-root scale B(x) = sqrt(x) (log log x)^2 / log x on [8, 1000]
```

```
max |pi_RVM - li| / B = 2.7213   (at x = 9.73)
max |pi - li| / B     = 2.4490   (at x = 11.00)
max |pi_RVM - li| / B on [50, 1000] = 1.3455
max |pi_RVM - li| / B on [500, 1000] = 0.7216
total run time 0.6 s
```

C.8. **s07_product_identity.**

```
N = 2001052 ordinates; varpi_RVM = 0.9619415789
         K              P_K    P_K - varpi
        10     0.9625640008      6.224e-04
       100     0.9624787401      5.372e-04
      1000     0.9622891707      3.476e-04
     10000     0.9619513787      9.800e-06
    100000     0.9619301865     -1.139e-05
   1000000     0.9619418189      2.400e-07
   2001052     0.9619415390     -3.985e-08
K >=   10000: min P_K = 0.9618023968, max P_K = 0.9620641927, mean = 0.9619418623
K >=  100000: min P_K = 0.9619210208, max P_K = 0.9619605076, mean = 0.9619417591
K >= 1000000: min P_K = 0.9619389488, max P_K = 0.9619439825, mean = 0.9619416633

flow along the lambda-family
Upsilon(0.00) = 0.0436949578
Upsilon(0.25) = 0.0412445375
Upsilon(0.50) = 0.0387987180
Upsilon(0.75) = 0.0363572218
Upsilon(1.00) = 0.0339197739
sum_k 2 eta_k^2 / gamma_k^2 = 0.0098210688   (auxiliary size estimate, not an identity of the flow)
integral_0^1 Upsilon, truncated at k <= N: sum_{k<=N} log((1/4+tau_k^2)/(1/4+gamma_k^2)) = 0.0388016004
log L = log(1/varpi_RVM)                               = 0.0388015590
total run time 0.8 s
```

C.9. **s08_transport_moments.**

```
s08: transport tau -> gamma on the first rung
entries in zeross.bin / zLamb.bin: 2001052 / 2001052;  used K = 20000
gamma_1 = 14.134725142, tau_1 = 14.521346953;  gamma_K = 18046.464296, tau_K = 18046.544665
psi: normalized bump on [0.1, 0.62] (log 2 = 0.693147), 4001 nodes
max|eta_k| = 0.993711 at k = 34;  rms eta = 0.266395
  max_(k>=  100) |c_psi(tau_k)| = 6.880e-05
  max_(k>=  300) |c_psi(tau_k)| = 7.263e-07
  max_(k>= 1000) |c_psi(tau_k)| = 2.527e-10
  max_(k>= 3000) |c_psi(tau_k)| = 2.074e-13
  max_(k>=10000) |c_psi(tau_k)| = 2.074e-13
  max_(k>=19999) |c_psi(tau_k)| = 3.996e-15
  (values near 1e-13 and below are at the floating-point floor of the quadrature, not a measured decay)

W_psi                          = +0.133643530916
sum_k c_psi(tau_k)             = +0.176106813076
sum_k c_psi(gamma_k)           = +0.133643530929
R_psi(1) = W - sum c(tau)      = -0.042463282161   (clock defect)
S_psi(0) = sum[c(gam)-c(tau)]  = -0.042463282147   (transport)
R_psi(0) = R_psi(1) - S_psi(0) = -1.376e-11   (both terms use the same K ordinates and quadrature)
quadrature check with 2001 nodes: change of R_psi(1) -4.0e-13, of S_psi(0) -5.6e-13

 j        m_j(psi)        (-1)^j m_j/j!      partial sum      partial - R_psi(1)
 1  +4.0395231784e-02  -4.0395231784e-02  -0.040395231784  +2.068e-03
 2  -4.8097910277e-03  -2.4048955139e-03  -0.042800127298  -3.368e-04
 3  -2.0431952768e-03  +3.4053254613e-04  -0.042459594751  +3.687e-06
 4  -5.5184329118e-05  -2.2993470466e-06  -0.042461894098  +1.388e-06
 5  +1.6734348340e-04  -1.3945290283e-06  -0.042463288628  -6.467e-09
 6  +5.5720624242e-06  +7.7389755891e-09  -0.042463280889  +1.272e-09
 7  +7.2607702259e-06  -1.4406290131e-09  -0.042463282329  -1.686e-10
 8  +6.4039357525e-06  +1.5882777164e-10  -0.042463282170  -9.773e-12
 9  -8.6516463634e-06  +2.3841618065e-11  -0.042463282147  +1.407e-11
10  -2.6791658297e-06  -7.3830628022e-13  -0.042463282147  +1.333e-11
```

```
11  +2.8798049874e-06  -7.2145186673e-14  -0.042463282147  +1.326e-11
12  +8.0540801176e-07  +1.6814307338e-15  -0.042463282147  +1.326e-11

 cumulative over k (first K0 ordinates):  K0   sum[c(gam)-c(tau)]   W - sum c(tau)   -m_1 part   +m_2/2 part
      1   -0.0208609181   +0.0155048033   -0.0168182977   -0.0039851021
      2   -0.0179065615   +0.0336283174   -0.0153170893   -0.0024894395
      3   -0.0230428015   -0.0613123463   -0.0185197872   -0.0044492860
      5   -0.0438112092   -0.0723587090   -0.0422769516   -0.0020012704
     10   -0.0464863677   -0.0403092255   -0.0440974925   -0.0029001855
     20   -0.0406060545   -0.0470689184   -0.0387914922   -0.0021299305
     50   -0.0422141654   -0.0432549118   -0.0401314609   -0.0024151466
    100   -0.0424897738   -0.0426540294   -0.0404216192   -0.0024058875
    200   -0.0424647471   -0.0424770847   -0.0403966024   -0.0024050814
    400   -0.0424633643   -0.0424636133   -0.0403953106   -0.0024049002

 lambda     S_psi(lambda)     R_psi(lambda) = R_psi(1) - S_psi(lambda)
 (lambda outside [0,1] is a numerical prolongation of gamma_k + lambda eta_k, not part of the family)
  at lambda = -1.0 the list is increasing: False, positive: True
  at lambda = -0.5 the list is increasing: False, positive: True
  at lambda = +1.5 the list is increasing: True, positive: True
  at lambda = +2.0 the list is increasing: False, positive: True
-1.00   -0.087766836871   +0.045303554710
-0.75   -0.076275893121   +0.033812610960
-0.50   -0.064876726912   +0.022413444752
-0.25   -0.053596431977   +0.011133149816
+0.00   -0.042463282147   -0.000000000014
+0.25   -0.031506572650   -0.010956709511
+0.50   -0.020756460380   -0.021706821780
+0.75   -0.010243803437   -0.032219478724
+1.00   +0.000000000000   -0.042463282161
+1.25   +0.009943173538   -0.052406455698
+1.50   +0.019553725448   -0.062017007609
+1.75   +0.028799612928   -0.071262895088
+2.00   +0.037648908219   -0.080112190380
```

C.10. **s09_clock_defect_split.**

```
s09: Poisson split of the clock defect (psi of s08 on [0.1,0.62])
t_0 (N_RVM(t_0)=0)                = 9.676906787166   (2 pi = 6.283185, tau_1 = 14.521347)
W_psi                             = +0.133643530916
I_0  (m = 0, smooth clock density) = +0.063533498702
sum_k c_psi(tau_k)                = +0.176106813076
R_psi(1)                          = -0.042463282161
smooth mismatch  W_psi - I_0      = +0.070110032214
oscillating part -2 sum_(m>=1)(-1)^m I_m = -0.112573314375
  (obtained as R_psi(1) - (W_psi - I_0); the I_m, m >= 1, are not summed, so the convergence of the
   oscillatory Poisson series is not verified independently by this program)
stationary heights 2 pi e^u, u in supp psi: [6.9440, 11.6800];  part above t_0: u > 0.431865
```

C.11. **s10_clock_to_zeros.**

```
s10: from the clock to the ordinates through the rung equations (continuation in height)
N = 13, U = log(13+1/2) = 2.602690; free masses n = 2..11; start gamma_k = tau_k
band omega <= 18:  4 free ordinates, 361 tests, |residual| 8.136e-01 -> 3.892e-04 in 37 iterations
band omega <= 22:  5 free ordinates, 441 tests, |residual| 1.824e-02 -> 1.520e-03 in 24 iterations
band omega <= 26:  6 free ordinates, 521 tests, |residual| 1.985e-02 -> 8.002e-04 in 26 iterations
band omega <= 30:  7 free ordinates, 601 tests, |residual| 2.363e-02 -> 2.506e-03 in 21 iterations
band omega <= 34:  8 free ordinates, 681 tests, |residual| 4.635e-02 -> 2.827e-03 in 52 iterations  (stopped:
     no decrease up to damping 1e12)
band omega <= 38:  9 free ordinates, 761 tests, |residual| 3.868e-02 -> 1.683e-03 in 11 iterations  (stopped:
     no decrease up to damping 1e12)
band omega <= 42: 11 free ordinates, 841 tests, |residual| 1.881e-01 -> 1.595e-03 in 11 iterations
band omega <= 46: 12 free ordinates, 921 tests, |residual| 1.878e-02 -> 4.591e-03 in 12 iterations
band omega <= 50: 14 free ordinates, 1001 tests, |residual| 5.189e-02 -> 2.885e-03 in 23 iterations

  k      tau_k (start)    solved ordinate    gamma_k (zeross.bin)   solved - gamma_k    tau_k - gamma_k
```

```
  1     14.521346953     14.134660056     14.134725142   -6.509e-05   +0.386622
  2     20.655740356     21.022091631     21.022039639   +5.199e-05   -0.366299
  3     25.492675432     25.010827129     25.010857580   -3.045e-05   +0.481818
  4     29.739411632     30.424917668     30.424876126   +4.154e-05   -0.685464
  5     33.624531889     32.935072953     32.935061588   +1.137e-05   +0.689470
  6     37.257370087     37.586146548     37.586178159   -3.161e-05   -0.328808
  7     40.700551604     40.918919600     40.918719012   +2.006e-04   -0.218167
  8     43.994013953     43.327322117     43.327073281   +2.488e-04   +0.666941
  9     47.165128427     48.004716753     48.005150881   -4.341e-04   -0.840022
 10     50.233653391     49.773657726     49.773832478   -1.748e-04   +0.459821
 11     53.214422874     52.972535790     52.970321478   +2.214e-03   +0.244101
 12     56.118921948     56.548052093     56.446247697   +1.018e-01   -0.327326
 13     58.956266008     58.576523603     59.347044003   -7.705e-01   -0.390778
 14     61.733838843     62.561880182     60.831778525   +1.730e+00   +0.902060

  n     solved mass a_n     Lambda(n)/(2 sqrt n)     difference     w(log n)/max w
  2   +0.245076213   +0.245064536   +1.168e-05   7.562e-01
  3   +0.317145240   +0.317142050   +3.190e-06   9.754e-01
  4   +0.173268937   +0.173286795   -1.786e-05   9.957e-01
  5   +0.359884977   +0.359881258   +3.719e-06   9.423e-01
  6   -0.000004422   +0.000000000   -4.422e-06   8.474e-01
  7   +0.367785301   +0.367742452   +4.285e-05   7.225e-01
  8   +0.122615622   +0.122532268   +8.335e-05   5.733e-01
  9   +0.183143344   +0.183102048   +4.130e-05   4.062e-01
 10   -0.000286111   +0.000000000   -2.861e-04   2.344e-01
 11   +0.362276961   +0.361496314   +7.806e-04   8.642e-02
```

C.12. **s12_multiscale_jacobian.**

```
s12: multiscale Jacobian diagnostics from the midpoint clock
No zeross.bin, zLamb.bin, or displacement sequence is used.
KMAX = 2001052

      K             T        min gap      gap lower     min/lower      Ucrit       log(T/2pi)   Ucrit/log
    100     235.987267    1.73462445    1.73286190    1.00101714    3.622216     3.62590079  0.99898390
   1000    1419.517765    1.15930485    1.15921748    1.00007536    5.419787     5.42019542  0.99992464
  10000    9877.629616    0.85368118    0.85367617    1.00000587    7.360108     7.36015078  0.99999413
 100000   74920.891033    0.66939912    0.66939880    1.00000048    9.386307     9.38631098  0.99999952
 500000  319387.322632    0.57982852    0.57982848    1.00000008   10.836282    10.83628276  0.99999992
2001052 1132490.722588    0.51918344    0.51918343    1.00000002   12.102053    12.10205288  0.99999998

Fixed-window adjacent-pair upper bounds at the largest sampled k:
k = 2001051, tau_k = 1132490.203404, gap = 0.519183439
  U = 0.5: normalized upper bound <= 2.902323655e-02
  U = 1.0: normalized upper bound <= 1.641802190e-01
  U = 2.0: normalized upper bound <= 9.287435694e-01
  U = 4.0: normalized upper bound <= 5.253767008e+00

Saved numerical tables and figures.
The decreasing fixed-window bounds are a diagnostic of the rigorous fact
that no T-independent lower frame bound can hold on a fixed u-window.
```

C.13. **s13_nested_bounded_endpoint.**

```
Part 1: single bounded sections, solved alone from the clock (|shift| <= 1.2, |mass| <= 0.5)
N= 20 Omega=70 K= 22 masses=19 rows=561: normalized residual 6.834e-05 (status 1, nfev 56); rms(first 10 -
     gamma) = 7.457e-05; mass energy 1.3679, max |mass| 0.5000
    Jacobian at the solution: numerical rank 40 of 41 columns, condition number 1.18e+04 on that subspace, 1
     edge masses with weight < 1e-12; active bounds: 0 of 22 shifts at |shift| = 1.2, 1 of 19 masses at |mass|
      = 0.5
    cross-validation full     support [0.00, 1.00] U, omega <= 56: rms 5.196e-05, max 2.258e-04
    cross-validation inner80  support [0.10, 0.90] U, omega <= 56: rms 1.871e-04, max 1.150e-03
    cross-validation left     support [0.05, 0.55] U, omega <= 56: rms 6.937e-04, max 2.814e-03
    cross-validation right    support [0.45, 0.95] U, omega <= 56: rms 7.868e-04, max 4.356e-03
    cross-validation middle   support [0.25, 0.75] U, omega <= 56: rms 4.726e-04, max 2.050e-03
```

```
N= 34 Omega=110 K= 40 masses=33 rows=881: normalized residual 1.390e-04 (status 1, nfev 44); rms(first 10 -
     gamma) = 2.081e-04; mass energy 1.8339, max |mass| 0.5000
    Jacobian at the solution: numerical rank 72 of 73 columns, condition number 2.27e+08 on that subspace, 1
     edge masses with weight < 1e-12; active bounds: 0 of 40 shifts at |shift| = 1.2, 2 of 33 masses at |mass|
      = 0.5
    cross-validation full     support [0.00, 1.00] U, omega <= 88: rms 1.228e-04, max 6.894e-04
    cross-validation inner80  support [0.10, 0.90] U, omega <= 88: rms 1.194e-04, max 6.869e-04
    cross-validation left     support [0.05, 0.55] U, omega <= 88: rms 2.298e-04, max 9.074e-04
    cross-validation right    support [0.45, 0.95] U, omega <= 88: rms 2.509e-04, max 1.044e-03
    cross-validation middle   support [0.25, 0.75] U, omega <= 88: rms 1.660e-04, max 8.267e-04
N= 50 Omega=150 K= 60 masses=49 rows=1201: normalized residual 3.874e-04 (status 2, nfev 127); rms(first 10 -
     gamma) = 4.536e-04; mass energy 2.5110, max |mass| 0.5000
    Jacobian at the solution: numerical rank 107 of 109 columns, condition number 1.24e+08 on that subspace, 2
      edge masses with weight < 1e-12; active bounds: 0 of 60 shifts at |shift| = 1.2, 1 of 49 masses at |mass
     | = 0.5
    cross-validation full     support [0.00, 1.00] U, omega <= 120: rms 3.370e-04, max 2.611e-03
    cross-validation inner80  support [0.10, 0.90] U, omega <= 120: rms 3.564e-04, max 2.802e-03
    cross-validation left     support [0.05, 0.55] U, omega <= 120: rms 4.153e-04, max 1.678e-03
    cross-validation right    support [0.45, 0.95] U, omega <= 120: rms 6.148e-04, max 3.775e-03
    cross-validation middle   support [0.25, 0.75] U, omega <= 120: rms 4.115e-04, max 2.543e-03

Part 2: nested sections; every earlier test block stays active (max_nfev = 3000 per section)
N= 13 K= 14 max_shift=0.839806 max_mass=0.499998 diagnostic_rms10=1.615e-04 status=1 nfev=19
  equations=101 in 1 blocks, unknowns=14+12, weighted Jacobian: numerical rank 26 of 26, condition number 9.68
     e+06 on that subspace, 0 edge masses with weight < 1e-12; active bounds: 0 shifts, 0 masses
  block N= 13: RMS=9.050e-05, max=2.228e-04, grid=701
N= 20 K= 22 max_shift=0.964847 max_mass=0.500000 diagnostic_rms10=3.478e-05 status=1 nfev=39
  equations=242 in 2 blocks, unknowns=22+19, weighted Jacobian: numerical rank 40 of 41, condition number 1.29
     e+04 on that subspace, 1 edge masses with weight < 1e-12; active bounds: 0 shifts, 1 masses
  block N= 13: RMS=4.858e-05, max=2.062e-04, grid=701
  block N= 20: RMS=7.786e-05, max=5.146e-04, grid=701
N= 34 K= 40 max_shift=1.199840 max_mass=0.500000 diagnostic_rms10=3.017e-06 status=1 nfev=38
  equations=463 in 3 blocks, unknowns=40+33, weighted Jacobian: numerical rank 72 of 73, condition number 3.08
     e+08 on that subspace, 1 edge masses with weight < 1e-12; active bounds: 0 shifts, 2 masses
  block N= 13: RMS=4.731e-06, max=1.592e-05, grid=701
  block N= 20: RMS=5.940e-06, max=2.581e-05, grid=701
  block N= 34: RMS=2.351e-05, max=1.399e-04, grid=701
N= 50 K= 60 max_shift=1.189449 max_mass=0.499395 diagnostic_rms10=1.243e-06 status=1 nfev=34
  equations=764 in 4 blocks, unknowns=60+49, weighted Jacobian: numerical rank 107 of 109, condition number
     2.18e+08 on that subspace, 2 edge masses with weight < 1e-12; active bounds: 0 shifts, 0 masses
  block N= 13: RMS=4.356e-07, max=1.067e-06, grid=701
  block N= 20: RMS=6.842e-07, max=2.715e-06, grid=701
  block N= 34: RMS=1.652e-06, max=8.999e-06, grid=701
  block N= 50: RMS=1.688e-05, max=1.261e-04, grid=701
N= 70 K= 93 max_shift=0.993713 max_mass=0.497942 diagnostic_rms10=2.523e-06 status=1 nfev=46
  equations=1185 in 5 blocks, unknowns=93+69, weighted Jacobian: numerical rank 159 of 162, condition number
     1.18e+09 on that subspace, 3 edge masses with weight < 1e-12; active bounds: 0 shifts, 0 masses
  block N= 13: RMS=8.679e-07, max=1.743e-06, grid=701
  block N= 20: RMS=9.065e-07, max=2.435e-06, grid=701
  block N= 34: RMS=1.337e-06, max=3.633e-06, grid=701
  block N= 50: RMS=2.418e-06, max=7.512e-06, grid=701
  block N= 70: RMS=6.243e-05, max=7.705e-04, grid=746
N=100 K=146 max_shift=0.993186 max_mass=0.500000 diagnostic_rms10=3.749e-03 status=2 nfev=402
  equations=1786 in 6 blocks, unknowns=146+99, weighted Jacobian: numerical rank 241 of 245, condition number
     8.67e+10 on that subspace, 4 edge masses with weight < 1e-12; active bounds: 0 shifts, 2 masses
  block N= 13: RMS=1.086e-03, max=2.845e-03, grid=701
  block N= 20: RMS=1.121e-03, max=3.015e-03, grid=701
  block N= 34: RMS=1.344e-03, max=5.010e-03, grid=701
  block N= 50: RMS=1.562e-03, max=5.626e-03, grid=701
  block N= 70: RMS=2.365e-03, max=8.741e-03, grid=746
  block N=100: RMS=2.157e-02, max=2.634e-01, grid=1133
repeatability of the N = 50 section (same blocks) from three perturbed starts:
  start 1: max block RMS 2.707e-04 (reference 1.688e-05), max |ordinate difference| 1.404e+00, max |mass
     difference| 9.200e-01, status 1, nfev 41
```

```
  start 2: max block RMS 9.546e-06 (reference 1.688e-05), max |ordinate difference| 9.932e-01, max |mass
     difference| 6.157e-01, status 1, nfev 33
  start 3: max block RMS 7.567e-06 (reference 1.688e-05), max |ordinate difference| 6.156e-01, max |mass
     difference| 1.068e-01, status 1, nfev 81
```

C.14. **s14_varpi_bound.** The file `output/s14_varpi_bound_console.txt`; Part A is the interval verification used in the proof of Theorem 9.11.

```
PART A. Interval-arithmetic verification of the inequalities in the proof of the theorem varpi_RVM < 1
(mpmath.iv, 40 digits, outward rounding)
(a) nu_3 = 8 - (33/4) log 2 - 2 log pi in [-0.00792401131834915, -0.00792401131834915]
    sup nu_3 < 0: verified
(b) N_RVM(tau_0) = 2 log(2/e) + 7/8 in [0.261294361119891, 0.261294361119891]
    0 < N_RVM(tau_0) < 1/2: verified
(c) N_RVM(14.52134) in [0.49999907294182, 0.49999907294182]
    N_RVM(14.52135) in [0.50000040625053, 0.50000040625053]
    N_RVM(14.52134) < 1/2 < N_RVM(14.52135): tau_1 in [14.52134, 14.52135] verified
(d) l_1 = log(tau_1/2pi)      in [0.837742225231, 0.837742913872]   (> 0, so W is decreasing and convex on u
     >= 1)
    I_0(tau_1)                in [0.0540791514918, 0.05408586201]
    W(1)/12                   in [0.0430408526477, 0.043040917668]
    (sqrt3/216)|W'(1)|        in [0.00469250702868, 0.00469252682202]
    M(tau_1)                  in [0.00634570700179, 0.00635250233362]
    inf M(tau_1) > 0: verified
(e) nu_4 <= -2 inf M(tau_1) = -0.0126914140036
    nu_3 + nu_4 <= -0.0206154253219
    varpi_RVM = P(1/2) exp(nu_3 + nu_4) < exp(nu_3 + nu_4) <= 0.979595619809 < 1
    ALL INEQUALITIES OF THE PROOF VERIFIED: varpi_RVM < 0.979596

PART B. Floating-point cross-check (not part of the proof)
tau_1 = 14.521346953065628168 (Lambert formula)
I_0 = int_{tau_0}^{tau_1} N_RVM(t) dt/t = 0.0540825739989015 (quadrature)
sum_{n<=2000} I_n = -0.0414693995156753; Euler-Maclaurin tail -W(2001)/12 = -3.47203e-5
sum_n I_n + W(1)/12 = 0.0015367526, Euler-Maclaurin bound (sqrt3/216)|W'(1)| = 0.0046925137
nu_3 = -0.00792401131834915
nu_4 = -0.0251569083469753   (rigorous upper bound of Part A: -0.0126914140036)
exp(nu_3 + nu_4) = 0.967460269860234
tau_1, tau_2, tau_3 (numpy) = 14.5213469531, 20.6557403557, 25.4926754323
log P_RVM^crit(1/2): k <= 2000000: -0.005720178750401; tail (integral): -4.605e-07; total -0.005720639280853
P_RVM^crit(1/2) = 0.994295692419
varpi_RVM = P(1/2) exp(nu_3 + nu_4) = 0.961941578908
Richardson value of s02_transition.py (sigma = 2, T = 2500, 5000, 10000), read from output/s02_varpi.txt:
     0.961941578887410
difference = 2.08e-11
total run time 33.0 s
```

C.15. **s15_tables_1000.**

```
(a) tau_k against gamma_k, k <= 1000: max|eta_k| = 0.993711 at k = 34; mean eta = 2.212e-05; rms = 0.358739
    gamma_1000 = 1419.422480946, tau_1000 = 1419.517764572; mean spacing at k=1000: 1.15922
    table checks: indices 1..1000 once each; eta_k = tau_k - gamma_k before rounding (displayed values are
     rounded independently)
(b),(c) pi_RVM on [2, 8200]: K = 183029 ordinates (tau_K = 128805.3), smooth cut; 1400000 fine / 140000 coarse
      points
    pi_RVM(100) = 26.0782, pi(100) = 25;  pi_RVM(1000) = 169.9750, pi(1000) = 168;  pi_RVM(8200) = 1029.6878,
     pi = 1028
    pi_RVM(n) - pi(n), n <= 1000: max 3.7790 at n = 568, min -2.9572 at n = 467, mean 0.0549, rms 1.1871
    pi_RVM(n) - li(n), n <= 1000: mean -6.9833;  pi(n) - li(n): mean -7.0382
    real RVM primes p_j^RVM, j <= 1000: p_1000^RVM = 7928.1950, p_1000 = 7919; max(p_j^RVM - p_j) = 65.2450 at
      j = 775, min = -52.8004 at j = 482, mean 2.4091, rms 19.2600
    |p_j^RVM - p_j| / p_j^(1/2): max 1.0685;  |p_j^RVM - p_j| / log p_j: max 7.5171
    first 12 levels: 2.2138, 3.1546, 5.1880, 7.1588, 11.1391, 14.0201, 17.2551, 19.1767, 26.0524, 28.0419,
     33.0665, 38.0878
    sensitivity: levels j <= 168 of s04 (K = 17061, x <= 1000) minus these: median |diff| 1.7e-03, max |diff|
     0.854; largest at j = 148, 118, 106 (+0.854, +0.851, +0.793); 29 levels differ by more than 0.01
```

```
(d) residues: varpi_RVM = 0.9619415789 (read from output/s02_varpi.txt), c = -0.0052 -0.0208 i (as in
     s05_mertens)
    M_RVM^crit(n) (tapered target, Table tab:ternary) against the ordered sum over all 2001052 ordinates: max
     |diff| = 0.0284
    M(1000) = 2, M_RVM^crit(1000) = -8.6463, M_h(1000) = -8
    M_RVM^crit(n) - M(n), n <= 1000: max 13.8806 at n = 683, min -17.0831 at n = 568, mean -0.8963, rms 6.4436
    max |M(n)|/sqrt n = 0.8321, max |M_RVM^crit(n)|/sqrt n = 0.7541 on [10,1000]; sign agreement mu_h = mu on
     squarefrees: 0.5148
    table check s15_table_clock_1000.tex: 1000 rows, indices 1..1000 once each; |displayed difference -
     difference of displayed values| <= 5.0e-05 (allowance 5.1e-05)
    table check s15_table_primes_1000.tex: 1000 rows, indices 1..1000 once each; |displayed difference -
     difference of displayed values| <= 4.5e-13 (allowance 1.0e-04)
    table check s15_table_pi_1000.tex: 1000 rows, indices 1..1000 once each; |displayed difference -
     difference of displayed values| <= 1.4e-14 (allowance 1.0e-04)
    table check s15_table_mertens_1000.tex: 1000 rows, indices 1..1000 once each; |displayed difference -
     difference of displayed values| <= 1.8e-15 (allowance 1.0e-04)
total run time 99.3 s
```

C.16. **s16_walks_3d_1000.**

```
C_gamma  = +0.7720742579 +0.7626754110 i   (|S_N| = 0.3013, max_k |S_k| = 3.4182, check mean(S_k) - C = 2.4e
     -15)
C_tau    = +0.0401443298 +1.7565288414 i   (|S_N| = 0.8589, max_k |S_k| = 2.6677, check mean(S_k) - C = 3.4e
     -15)
C_theta  = +0.6133423097 +1.3939939459 i   (|S_N| = 0.8084, max_k |S_k| = 3.3102, check mean(S_k) - C = 2.6e
     -15)
(C_gamma + C_tau)/2 = +0.4061092939 +1.2596021262 i;  C_theta - chord midpoint = +0.207233 +0.134392 i,
     modulus 0.246995
gamma_k - tau_k, k <= 1000: mean -0.000022, std 0.358739, max |.| 0.993711
reference values with all 2001052 ordinates: C_gamma = +0.7769071196 +0.7678445908 i, C_tau = +0.0415702193
     +1.7575417496 i
total run time 1.4 s
```

C.17. **s17_cramer_centroids.**

```
a = 1:  e^{-1/2} V_+(1) = 0.7768942607932536 0.7678556452589175 i
a = 2:  e^{-1/2} V_+(1) = 0.7768942607932536 0.7678556688644353 i
explicit formula: Re e^{-1/2} V_+(1) = (e - 1/(e^2-1))/(2 sqrt e) = 0.7768942607932536
u = 0.75:  2 Re V_+(u) = 1.8297830998238,  e^u - 1/(e^{2u}-1) = 1.8297830998238
u = 0.90:  2 Re V_+(u) = 2.2615694846419,  e^u - 1/(e^{2u}-1) = 2.2615694846419
u = 1.05:  2 Re V_+(u) = 2.7181065618775,  e^u - 1/(e^{2u}-1) = 2.7181065618775
triangular centroids with all N = 2001052 ordinates: C_zeta = 0.7769071196 +0.7678445908 i,  C_RVM =
     0.0415702193 +1.7575417496 i
  N =    10000: C_zeta = 0.77776961 +0.76786173 i,  C_RVM = 0.04145468 +1.75735095 i
  N =   100000: C_zeta = 0.77702246 +0.76802715 i,  C_RVM = 0.04154606 +1.75751864 i
  N =  1000000: C_zeta = 0.77689988 +0.76784866 i,  C_RVM = 0.04156483 +1.75754129 i
  n =     1000: S_n - b_n E_{n+1} = 0.0417011468 +1.7575539237 i
  n =    10000: S_n - b_n E_{n+1} = 0.0415800729 +1.7575547937 i
  n =   100000: S_n - b_n E_{n+1} = 0.0415708914 +1.7575422071 i
  n =  1000000: S_n - b_n E_{n+1} = 0.0415691734 +1.7575409824 i
  n =  2001051: S_n - b_n E_{n+1} = 0.0415688005 +1.7575410722 i
total run time 52.6 s
```

References

[1] L. V. Ahlfors, *Complex Analysis*, 3rd ed., McGraw–Hill, New York, 1979. (Cited on p. 91.)

[2] J. Arias de Reyna, Explicit formula and quasicrystal definition, preprint, arXiv:2402.10604 (2024, v2 2025), doi:10.48550/arXiv.2402.10604. (Cited on pp. 28 and 117.)

[3] S. V. Hruščev, N. K. Nikol'skii and B. S. Pavlov, Unconditional bases of exponentials and of reproducing kernels, in: *Complex Analysis and Spectral Theory (Leningrad, 1979/80)*, Lecture Notes in Math. **864**, Springer, Berlin, 1981, 214–335, doi:10.1007/BFb0097000. (Cited on p. 134.)

[4] G. Kozma and N. Lev, Exponential Riesz bases, discrepancy of irrational rotations and BMO, *J. Fourier Anal. Appl.* **17** (2011), no. 5, 879–898, doi:10.1007/s00041-011-9178-1; arXiv:1009.2188. (Cited on p. 134.)

[5] A. Bondarenko, D. Radchenko and K. Seip, Fourier interpolation with zeros of zeta and L-functions, *Constr. Approx.* **57** (2023), 405–461, doi:10.1007/s00365-022-09599-w; arXiv:2005.02996. (Cited on p. 139.)

[6] F. Johansson, Arb: efficient arbitrary-precision midpoint-radius interval arithmetic, *IEEE Trans. Comput.* **66** (2017), no. 8, 1281–1292, doi:10.1109/TC.2017.2690633; arXiv:1611.02831. (Cited on p. 142.)
[7] F. Johansson, Numerical integration in arbitrary-precision ball arithmetic, in: *Mathematical Software – ICMS 2018*, Lecture Notes in Comput. Sci. **10931**, Springer, Cham, 2018, 255–263, doi:10.1007/978-3-319-96418-8_30; arXiv:1802.07942. (Cited on p. 143.)
[8] S. Bernstein, Sur les fonctions absolument monotones, *Acta Math.* **52** (1929), 1–66, doi:10.1007/BF02547400. (Cited on pp. 124 and 146.)
[9] D. V. Widder, *The Laplace Transform*, Princeton Mathematical Series 6, Princeton University Press, Princeton, 1941; reprint doi:10.1515/9781400876457. (Cited on p. 124.)
[10] J. B. Conrey, More than two fifths of the zeros of the Riemann zeta function are on the critical line, *J. Reine Angew. Math.* **399** (1989), 1–26, doi:10.1515/crll.1989.399.1. (Cited on p. 126.)
[11] S. A. Avdonin, On the question of Riesz bases of exponential functions in L^2, *Vestnik Leningrad. Univ. Mat. Mekh. Astronom.* (1974), no. 13, 5–12. (No DOI.) (Cited on pp. 133 and 134.)
[12] S. A. Avdonin and S. A. Ivanov, Exponential Riesz bases of subspaces and divided differences, *St. Petersburg Math. J.* **13** (2002), no. 3, 339–351, preprint version arXiv:math/0103160, doi:10.48550/arXiv.math/0103160. (Cited on pp. 133, 134, 138, and 146.)
[13] P. T. Bateman and H. G. Diamond, Asymptotic distribution of Beurling's generalized prime numbers, in: *Studies in Number Theory*, MAA Studies in Mathematics 6, Mathematical Association of America, 1969. (Cited on p. 12.)
[14] J. J. Benedetto, A. M. Powell and Ö. Yılmaz, Sigma-delta ($\Sigma\Delta$) quantization and finite frames, *IEEE Trans. Inform. Theory* **52** (2006), no. 5, 1990–2005, doi:10.1109/TIT.2006.872849. (Cited on p. 14.)
[15] A. Fujii, On the zeros of Dirichlet L-functions. I, *Trans. Amer. Math. Soc.* **196** (1974), 225–235, doi:10.1090/S0002-9947-1974-0349603-2. (Cited on pp. 138 and 146.)
[16] A. Fujii, On the distribution of the zeros of the Riemann zeta function in short intervals, *Bull. Amer. Math. Soc.* **81** (1975), 139–142, doi:10.1090/S0002-9904-1975-13674-3. (Cited on p. 138.)
[17] A. Fujii, On the difference between r consecutive ordinates of the zeros of the Riemann zeta function, *Proc. Japan Acad.* **51** (1975), no. 10, 741–743, doi:10.3792/pja/1195518466. (Cited on p. 138.)
[18] J. B. Garnett, *Bounded Analytic Functions*, revised first ed., Graduate Texts in Mathematics 236, Springer, New York, 2007, doi:10.1007/0-387-49763-3. (Cited on pp. 132, 133, 137, and 146.)
[19] S. M. Gonek, Finite Euler products and the Riemann hypothesis, *Trans. Amer. Math. Soc.* **364** (2012), 2157–2191, doi:10.1090/S0002-9947-2011-05546-7; arXiv:0704.3448. (Cited on pp. 121 and 146.)
[20] A. P. Guinand, A summation formula in the theory of prime numbers, *Proc. London Math. Soc. (2)* **50** (1948), 107–119, doi:10.1112/plms/s2-50.2.107. (Cited on pp. 11, 27, 28, and 117.)
[21] L. Hörmander, *The Analysis of Linear Partial Differential Operators I*, 2nd ed., Grundlehren der mathematischen Wissenschaften 256, Springer, Berlin, 1990, doi:10.1007/978-3-642-61497-2. (Cited on pp. 101, 104, 105, 116, and 287.)
[22] A. E. Ingham, Some trigonometrical inequalities with applications to the theory of series, *Math. Z.* **41** (1936), 367–379. (Cited on p. 96.)
[23] A. Beurling, Analyse de la loi asymptotique de la distribution des nombres premiers généralisés. I, *Acta Math.* **68** (1937), 255–291, doi:10.1007/BF02546666. (Cited on pp. 12, 13, and 61.)
[24] H. Bohr, Zur Theorie der fastperiodischen Funktionen. III. Dirichletentwicklung analytischer Funktionen, *Acta Math.* **47** (1926), 237–281, doi:10.1007/BF02543846. (Cited on p. 108.)
[25] S. Bochner and K. Chandrasekharan, On Riemann's functional equation, *Ann. of Math. (2)* **63** (1956), 336–360, doi:10.2307/1969614. (Cited on pp. 13, 71, and 103.)
[26] F. Broucke, G. Debruyne and J. Vindas, Beurling integers with RH and large oscillation, *Adv. Math.* **370** (2020), 107240, doi:10.1016/j.aim.2020.107240. (Cited on p. 12.)
[27] F. Broucke and J. Vindas, A new generalized prime random approximation procedure and some of its applications, *Math. Z.* **307** (2024), Paper No. 62, doi:10.1007/s00209-024-03526-4. (Cited on pp. 13, 61, and 63.)
[28] H. Cohn and N. Elkies, New upper bounds on sphere packings I, *Ann. of Math. (2)* **157** (2003), no. 2, 689–714, doi:10.4007/annals.2003.157.689. (Cited on p. 82.)
[29] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey and D. E. Knuth, On the Lambert W function, *Adv. Comput. Math.* **5** (1996), 329–359, doi:10.1007/BF02124750. (Cited on pp. 12 and 21.)
[30] I. Daubechies and R. DeVore, Approximating a bandlimited function using very coarsely quantized data: a family of stable sigma-delta modulators of arbitrary order, *Ann. of Math.* **158** (2003), 679–710, doi:10.4007/annals.2003.158.679. (Cited on p. 14.)
[31] H. Davenport, *Multiplicative Number Theory*, 2nd ed., revised by H. L. Montgomery, Graduate Texts in Mathematics 74, Springer, New York, 1980, doi:10.1007/978-1-4757-5927-3. (Cited on pp. 26, 36, 48, 49, and 55.)
[32] C.-J. de la Vallée Poussin, Recherches analytiques sur la théorie des nombres premiers, *Ann. Soc. Sci. Bruxelles* **20** (1896), 183–256. (Cited on p. 36.)

[33] H. G. Diamond, Asymptotic distribution of Beurling's generalized integers, *Illinois J. Math.* **14** (1970), 12–28, doi:10.1215/ijm/1256053295. (Cited on p. 12.)

[34] H. G. Diamond, H. L. Montgomery and U. M. A. Vorhauer, Beurling primes with large oscillation, *Math. Ann.* **334** (2006), 1–36, doi:10.1007/s00208-005-0638-2. (Cited on pp. 12, 13, 61, and 63.)

[35] H. G. Diamond and W.-B. Zhang, *Beurling Generalized Numbers*, Mathematical Surveys and Monographs 213, American Mathematical Society, Providence, RI, 2016, doi:10.1090/surv/213. (Cited on pp. 13, 61, and 63.)

[36] NIST Digital Library of Mathematical Functions, F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier et al. (eds.), Release 1.2.8 of 2026-09-15, §4.13 (Lambert W-function) and §5.11 (Stirling series), https://dlmf.nist.gov/. (Cited on pp. 12, 21, and 42.)

[37] H. M. Edwards, *Riemann's Zeta Function*, Academic Press, New York, 1974. (Cited on pp. 11, 13, 27, 34, and 70.)

[38] G. França and A. LeClair, Statistical and other properties of Riemann zeros based on an explicit equation for the n-th zero on the critical line, preprint, arXiv:1307.8395 (2013), doi:10.48550/arXiv.1307.8395. (Cited on pp. 11, 12, 17, 21, and 163.)

[39] E. Grosswald and F. J. Schnitzer, A class of modified ζ and L-functions, *Pacific J. Math.* **74** (1978), no. 2, 357–364, doi:10.2140/pjm.1978.74.357. (Cited on pp. 13 and 39.)

[40] C. S. Güntürk and N. T. Thao, Ergodic dynamics in sigma-delta quantization: tiling invariant sets and spectral analysis of error, preprint, arXiv:math/0308223 (2003), doi:10.48550/arXiv.math/0308223. (Cited on p. 14.)

[41] J. Hadamard, Sur la distribution des zéros de la fonction $\zeta(s)$ et ses conséquences arithmétiques, *Bull. Soc. Math. France* **24** (1896), 199–220, doi:10.24033/bsmf.545. (Cited on pp. 11, 36, and 49.)

[42] H. Hamburger, Über die Riemannsche Funktionalgleichung der ζ-Funktion, *Math. Z.* **10** (1921), 240–254, doi:10.1007/BF01211612. (Cited on pp. 11, 13, 14, 53, 69, and 71.)

[43] J. Kaczorowski and A. Perelli, On the structure of the Selberg class, I: $0 \le d \le 1$, *Acta Math.* **182** (1999), 207–241, doi:10.1007/BF02392574. (Cited on pp. 13 and 71.)

[44] M. I. Kadec, The exact value of the Paley–Wiener constant, *Dokl. Akad. Nauk SSSR* **155** (1964), 1253–1254; English transl. Soviet Math. Dokl. **5** (1964), 559–561. (No DOI.) (Cited on p. 133.)

[45] M. I. Knopp, On Dirichlet series satisfying Riemann's functional equation, *Invent. Math.* **117** (1994), 361–372, doi:10.1007/BF01232248. (Cited on pp. 13, 71, and 139.)

[46] P. Kurasov and P. Sarnak, Stable polynomials and crystalline measures, *J. Math. Phys.* **61** (2020), no. 8, 083501, doi:10.1063/5.0012286. (Cited on pp. 29, 106, and 117.)

[47] M. L. Lapidus and M. van Frankenhuysen, *Fractal Geometry and Number Theory: Complex Dimensions of Fractal Strings and Zeros of Zeta Functions*, Birkhäuser, Boston, 2000, doi:10.1007/978-1-4612-5314-3. (Cited on pp. 13 and 63.)

[48] N. Lev and A. Olevskii, Quasicrystals and Poisson's summation formula, *Invent. Math.* **200** (2015), no. 2, 585–606, doi:10.1007/s00222-014-0542-z. (Cited on pp. 29, 82, and 168.)

[49] Yu. I. Lyubarskii and K. Seip, Complete interpolating sequences for Paley–Wiener spaces and Muckenhoupt's (A_p) condition, *Rev. Mat. Iberoam.* **13** (1997), no. 2, 361–376, doi:10.4171/RMI/224. (Cited on p. 134.)

[50] H. L. Montgomery and R. C. Vaughan, Hilbert's inequality, *J. London Math. Soc. (2)* **8** (1974), 73–82, doi:10.1112/jlms/s2-8.1.73. (Cited on p. 138.)

[51] A. Olevskii and A. Ulanovskii, Fourier quasicrystals with unit masses, *C. R. Math. Acad. Sci. Paris* **358** (2020), no. 11–12, 1207–1211, doi:10.5802/crmath.142. (Cited on pp. 106 and 117.)

[52] A. M. Odlyzko and H. J. J. te Riele, Disproof of the Mertens conjecture, *J. reine angew. Math.* **357** (1985), 138–160, doi:10.1515/crll.1985.357.138. (Cited on pp. 11, 14, and 157.)

[53] T. Nakamura, Dirichlet series with periodic coefficients, Riemann's functional equation and real zeros of Dirichlet L-functions, preprint, arXiv:2008.02570 (2020), doi:10.48550/arXiv.2008.02570. (Cited on p. 71.)

[54] A. Perelli, Converse theorems: from the Riemann zeta function to the Selberg class, preprint, arXiv:1605.02354 (2016), doi:10.48550/arXiv.1605.02354. (Cited on p. 14.)

[55] D. J. Platt and T. S. Trudgian, The Riemann hypothesis is true up to $3 \cdot 10^{12}$, *Bull. London Math. Soc.* **53** (2021), 792–797, doi:10.1112/blms.12460; arXiv:2004.09765. (Cited on pp. 11, 63, 118, 146, 158, and 164.)

[56] B. Riemann, Über die Anzahl der Primzahlen unter einer gegebenen Grösse, *Monatsberichte der Königlich Preussischen Akademie der Wissenschaften zu Berlin* (1859), 671–680. (Cited on pp. 11 and 13.)

[57] W. Rudin, *Real and Complex Analysis*, 3rd ed., McGraw–Hill, New York, 1987. (No DOI.) (Cited on pp. 27, 60, 129, and 289.)

[58] A. Selberg, Contributions to the theory of the Riemann zeta-function, *Arch. Math. Naturvid.* **48** (1946), no. 5, 89–155. (No DOI.) (Cited on pp. 11, 33, 138, and 146.)

[59] C. L. Siegel, Bemerkung zu einem Satz von Hamburger über die Funktionalgleichung der Riemannschen Zetafunktion, *Math. Ann.* **86** (1922), 276–279, doi:10.1007/BF01457989. (Cited on pp. 13 and 71.)

[60] E. C. Titchmarsh, *The Theory of Functions*, 2nd ed., Oxford University Press, 1939. (Cited on p. 54.)

[61] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Clarendon Press, Oxford, 1986. (Cited on pp. 11, 13, 14, 21, 25, 35, 38, 41, 49, 50, 53, 54, 56, 57, 62, 71, 76, 82, 129, 135, 137, 138, 146, and 289.)

[62] H. von Mangoldt, Zu Riemanns Abhandlung "Über die Anzahl der Primzahlen unter einer gegebenen Grösse", *J. reine angew. Math.* **114** (1895), 255–305, doi:10.1515/crll.1895.114.255. (Cited on pp. 11 and 21.)

[63] A. Weil, Sur les "formules explicites" de la théorie des nombres premiers, *Comm. Sém. Math. Univ. Lund [Medd. Lunds Univ. Mat. Sem.]* (1952), Tome Supplémentaire, 252–265. (No DOI.) (Cited on pp. 11, 27, 28, and 117.)

[64] E. T. Whittaker and G. N. Watson, *A Course of Modern Analysis*, 4th ed., Cambridge University Press, 1927; reissued in the Cambridge Mathematical Library, 1996, doi:10.1017/CBO9780511608759. (Cited on pp. 115, 122, 128, 129, 130, and 288.)

[65] A. H. Zemanian, *Distribution Theory and Transform Analysis*, McGraw-Hill, New York, 1965; reprinted by Dover, New York, 1987. (No DOI.) (Cited on p. 114.)

[66] H. Cramér, Studien über die Nullstellen der Riemannschen Zetafunktion, *Math. Z.* **4** (1919), 104–130, doi:10.1007/BF01203390. (Cited on pp. 11, 25, 27, 28, and 33.)

[67] E. Landau, Über die Nullstellen der Zetafunktion, *Math. Ann.* **71** (1912), 548–564, doi:10.1007/BF01456808. (Cited on pp. 11 and 32.)

[68] S. M. Gonek, An explicit formula of Landau and its applications to the theory of the zeta-function, in: *A Tribute to Emil Grosswald: Number Theory and Related Analysis*, Contemp. Math. **143**, Amer. Math. Soc., Providence, RI, 1993, 395–413, doi:10.1090/conm/143/1210528. (Cited on pp. 11 and 32.)

[69] G. Debruyne and J. Vindas, Complex Tauberian theorems for Laplace transforms with local pseudofunction boundary behavior, *J. Anal. Math.* **138** (2019), 799–833, doi:10.1007/s11854-019-0045-3; arXiv:1604.05069. (Cited on p. 32.)

[70] J. Korevaar, *Tauberian Theory: A Century of Developments*, Grundlehren der mathematischen Wissenschaften 329, Springer, Berlin, 2004, doi:10.1007/978-3-662-10225-1. (Cited on p. 32.)

[71] G. França and A. LeClair, Transcendental equations satisfied by the individual zeros of Riemann ζ, Dirichlet and modular L-functions, *Commun. Number Theory Phys.* **9** (2015), no. 1, 1–50, doi:10.4310/CNTP.2015.v9.n1.a1; arXiv:1502.06003. (Cited on p. 12.)

[72] E. Hasanalizade, Q. Shen and P.-J. Wong, Counting zeros of the Riemann zeta function, *J. Number Theory* **235** (2022), 219–241, doi:10.1016/j.jnt.2021.06.032; arXiv:2107.06506. (Cited on pp. 11, 118, 146, and 164.)

[73] C. Bellotti and P.-J. Wong, Improved estimates for the argument and zero-counting function of the Riemann zeta-function, *Math. Comp.* (2025, published online), doi:10.1090/mcom/4133; arXiv:2412.15470. (Cited on p. 11.)

[74] T. S. Trudgian, An improved upper bound for the argument of the Riemann zeta-function on the critical line II, *J. Number Theory* **134** (2014), 280–292, doi:10.1016/j.jnt.2013.07.017; arXiv:1208.5846. (Cited on pp. 11 and 63.)

[75] J. E. Littlewood, On the zeros of the Riemann zeta-function, *Proc. Cambridge Philos. Soc.* **22** (1924), 295–318, doi:10.1017/S0305004100014225. (Cited on p. 11.)

[76] J. I. Hutchinson, On the roots of the Riemann zeta function, *Trans. Amer. Math. Soc.* **27** (1925), 49–60, doi:10.1090/S0002-9947-1925-1501297-5. (Cited on p. 11.)

[77] T. S. Trudgian, On the success and failure of Gram's Law and the Rosser Rule, *Acta Arith.* **148** (2011), 225–256, doi:10.4064/aa148-3-2; arXiv:0811.0883. (Cited on p. 11.)

[78] R. P. Brent, On the zeros of the Riemann zeta function in the critical strip, *Math. Comp.* **33** (1979), 1361–1372, doi:10.1090/S0025-5718-1979-0537983-2. (Cited on p. 11.)

[79] D. J. Platt, Isolating some non-trivial zeros of zeta, *Math. Comp.* **86** (2017), 2449–2467, doi:10.1090/mcom/3198. (Cited on p. 11.)

[80] E. Hlawka, Über die Gleichverteilung gewisser Folgen, welche mit den Nullstellen der Zetafunktion zusammenhängen, *Sitzungsber. Österr. Akad. Wiss. Math.-Naturwiss. Kl. Abt. II* **184** (1975), 459–471. (No DOI.) (Cited on pp. 11 and 31.)

[81] K. Ford and A. Zaharescu, On the distribution of imaginary parts of zeros of the Riemann zeta function, *J. reine angew. Math.* **579** (2005), 145–158, doi:10.1515/crll.2005.2005.579.145; arXiv:math/0405459. (Cited on p. 11.)

[82] J. Guillera, Some sums over the non-trivial zeros of the Riemann zeta function, preprint, arXiv:1307.5723 (2013), doi:10.48550/arXiv.1307.5723. (Cited on pp. 28 and 33.)

[83] G. Hurst, Computations of the Mertens function and improved bounds on the Mertens conjecture, *Math. Comp.* **87** (2018), 1013–1028, doi:10.1090/mcom/3275; arXiv:1610.08551. (Cited on p. 11.)

[84] T. Kotnik and J. van de Lune, On the order of the Mertens function, *Experiment. Math.* **13** (2004), 473–481, doi:10.1080/10586458.2004.10504556. (Cited on p. 11.)

[85] N. Ng, The distribution of the summatory function of the Möbius function, *Proc. London Math. Soc. (3)* **89** (2004), 361–389, doi:10.1112/S0024611504014741; arXiv:math/0310381. (Cited on p. 11.)

[86] W.-B. Zhang, Normalization of Beurling generalized primes with Riemann hypothesis, *Ann. Univ. Sci. Budapest. Sect. Comput.* **39** (2013), 459–469, doi:10.71352/ac.39.459. (Cited on p. 12.)

[87] F. Broucke, G. Debruyne and Sz. Gy. Révész, Some examples of well-behaved Beurling number systems, *Trans. Amer. Math. Soc.* **378** (2025), 477–501, doi:10.1090/tran/9274; arXiv:2309.01567. (Cited on p. 12.)

[88] T. W. Hilberdink and M. L. Lapidus, Beurling zeta functions, generalised primes, and fractal membranes, *Acta Appl. Math.* **94** (2006), 21–48, doi:10.1007/s10440-006-9063-0; arXiv:math/0410270. (Cited on p. 12.)

[89] M. Balazard, É. Saias and M. Yor, Notes sur la fonction ζ de Riemann, 2, *Adv. Math.* **143** (1999), 284–287, doi:10.1006/aima.1998.1797. (Cited on p. 31.)

[90] V. V. Volchkov, On an equality equivalent to the Riemann hypothesis, *Ukrainian Math. J.* **47** (1995), 491–493, doi:10.1007/BF01056314. (Cited on p. 31.)

[91] S. K. Sekatskii, S. Beltraminelli and D. Merlini, On equalities involving integrals of the logarithm of the Riemann ζ-function and equivalent to the Riemann hypothesis, *Ukrainian Math. J.* **64** (2012), 247–261, doi:10.1007/s11253-012-0642-0. (Cited on p. 31.)

[92] H. Davenport and H. Heilbronn, On the zeros of certain Dirichlet series (I), *J. London Math. Soc.* **11** (1936), 181–185, doi:10.1112/jlms/s1-11.3.181. (Cited on p. 28.)

[93] H. Davenport and H. Heilbronn, On the zeros of certain Dirichlet series (II), *J. London Math. Soc.* **11** (1936), 307–312, doi:10.1112/jlms/s1-11.4.307. (Cited on p. 28.)

Appendix D. Python programs of Section 18

The programs of `code/routeB/` produce the numbers of Section 18.8 and Figures 1 and 2. They use only the clock, the primes and the Gonek lists; `zeros.py` computes the zeta ordinates used in the diagnostic columns with `mpmath` (at 20 digits, stored as double-precision numbers). They are run from `output/routeB/` in the order `zeros.py`, `gonek_fast.py`, `horizonG.py`, `cert7.py`, `diag_cert7_gamma.py`, `entry.py`, `entry2.py`, `certV.py`, `laplace_check.py`, `figs.py` and `results_summary.py`; the last one writes `RESULTS.txt` from the JSON outputs, with the SHA-256 of the programs and of the files read. All computations are in double precision except `certV.py`, which works in ball arithmetic (python-flint 0.9.0, Arb, 128 bits) and proves Theorem 18.78; a stage marked `FLOAT_PASS` by the other programs satisfies the Banach inequalities in floating point, which is a candidate certificate and not a proof. `entry2.py` repeats the entry test of `entry.py` in ℓ^∞ and in ℓ^2. The program `cert7.py` does not read the zeta ordinates; their distance to the polished points is computed separately by `diag_cert7_gamma.py`. The exploratory scripts (`staged.py`, `lsq*.py`, `basin*.py`, `coupling.py`, `horizon.py`, `certG.py`, `cert2-4.py`) are distributed in the same directory with their logs and are described in `RESULTS.txt`; `cert5.py` and `cert6.py` are variants of `cert7.py` (`cert5.py` evaluates the test at the receding-horizon iterate without polishing, `cert6.py` after the same polishing but without the sampled margin), with the same floating-point labels and defect term, and with the zeta ordinates used only in a final diagnostic block.

D.1. `core.py`.

```
import numpy as np
from scipy.special import lambertw
from scipy.integrate import quad
SQ2PI=np.sqrt(2*np.pi)
def clock(K):
    k=np.arange(1,K+1); a=(k-11/8)/np.e
    return np.real(2*np.pi*(k-11/8)/lambertw(a))
def vonmangoldt(Nmax):
    L=np.zeros(Nmax+1)
    for p in range(2,Nmax+1):
        if all(p%q for q in range(2,int(p**0.5)+1)):
            q=p
            while q<=Nmax: L[q]=np.log(p); q*=p
    return L
LAM=vonmangoldt(200000)
def kfun(u): return np.cosh(u/2)-np.exp(-u/2)/(2*(1-np.exp(-2*u)))
def gauss(c,s,u): return np.exp(-(u-c)**2/(2*s*s))
def Fphi(c,s,kap):  # int_R gauss cos(kap u) du
    return s*SQ2PI*np.exp(-0.5*(s*kap)**2)*np.cos(c*kap)
def dFphi(c,s,kap):
    return s*SQ2PI*np.exp(-0.5*(s*kap)**2)*(-(s*s)*kap*np.cos(c*kap)-c*np.sin(c*kap))
_kcache={}
def kpair(c,s):
    key=(round(c,12),round(s,12))
    if key not in _kcache:
        a=max(c-9*s,1e-9); b=c+9*s
        _kcache[key]=quad(lambda u:kfun(u)*gauss(c,s,u),a,b,limit=400,epsabs=1e-13,epsrel=1e-12)[0]
    return _kcache[key]
def masspair(c,s):
    nmax=int(np.exp(c+9*s))+1
    n=np.arange(2,nmax+1); a=LAM[n]/(2*np.sqrt(n))
    return np.sum(a*gauss(c,s,np.log(n)))
def residual(tests,kap):
    """tests: list of (c,s); kap: full list (must extend far enough)"""
    out=[]
    for c,s in tests:
        out.append(np.sum(Fphi(c,s,kap))+masspair(c,s)-kpair(c,s))
    return np.array(out)
def jac(tests,kap,K):
```

```
    return np.array([dFphi(c,s,kap[:K]) for c,s in tests])
```

D.2. zeros.py.

```
# The first N ordinates of zeta from mpmath.zetazero (20 digits), stored as doubles in gamma.npy.
# The conversion to double precision (relative accuracy 1.1e-16, i.e. better than 1e-13 at t <= 1000)
# is intentional: every Route B computation is done in double precision, and gamma is used only for
# diagnostics (distances of computed lists to the zeros), never in the constructions.
import mpmath as mp, numpy as np, sys
mp.mp.dps=20
N=int(sys.argv[1])
g=[float(mp.zetazero(n).imag) for n in range(1,N+1)]
np.save('gamma.npy',np.array(g)); print(len(g), g[-1])
```

D.3. gonek_fast.py.

```
import numpy as np
from scipy.optimize import brentq
from core import LAM
def theta(t): t=np.asarray(t,float); return t/2*np.log(t/(2*np.pi))-t/2-np.pi/8+1/(48*t)+7/(5760*t**3)
def make_alpha(X):
    n=np.arange(2,int(X*X)+1); L=LAM[n]; w=np.where(n<=X,1.0,2-np.log(n)/np.log(X))
    w=np.maximum(w,0.0)                      # weight log(X^2/n)/log X on (X, X^2]; guard against rounding at
     n = X^2
    c=L*w/(np.sqrt(n)*np.log(n)); ln=np.log(n)
    return lambda t: -(np.sin(np.multiply.outer(np.atleast_1d(t),ln))@c)
def gonek_list(X,tmax,t0=10.0,h=0.005):
    # Crossings of the levels (Theta_X - pi/2)/pi in Z are located between consecutive grid points
    # where the floor changes.  A cell in which the floor changes by more than one level is detected
    # (the number of such cells is printed) and resolved by solving for every intermediate level in
    # the cell.  The largest change of the level function over one cell is printed: when it is far
    # below 1, an excursion that crosses a level and returns inside one cell would need a change of
    # sign of Theta_X' and an excursion of that small size inside the cell.
    al=make_alpha(X); ph=lambda t: theta(t)+al(t)
    ts=np.arange(t0,tmax,h); v=(ph(ts)-np.pi/2)/np.pi; j=np.floor(v); dj=np.diff(j); idx=np.where(dj!=0)[0];
     roots=[]
    nbig=int(np.sum(np.abs(dj)>1))
    for i in idx:
        a,b=ts[i],ts[i+1]
        levs=range(int(min(j[i],j[i+1]))+1,int(max(j[i],j[i+1]))+1)
        for lev in levs:
            f=lambda t,lev=lev:(ph(np.array([t]))[0]-np.pi/2)/np.pi-lev
            if f(a)*f(b)<0: roots.append(brentq(f,a,b,xtol=1e-13))
    dv=np.max(np.abs(np.diff(v)))
    print(f'X={X}: grid h={h}, max |Delta level| per grid cell = {dv:.3f}, cells with a jump of more than one
     level: {nbig}')
    return np.sort(np.array(roots))
if __name__=='__main__':
    import sys
    X=float(sys.argv[1]); r=gonek_list(X,float(sys.argv[2])); np.save(f'gonekF_X{int(X)}.npy',r); print(len(r),
     r[:3],r[-1])
```

D.4. horizonG.py.

```
import numpy as np, json, sys
from scipy.optimize import least_squares
from core import *
g=np.load('gamma.npy'); tau=clock(60000)
q,s0,sp,M,INT,LAG=0.85,0.08,1.0,int(sys.argv[1]),float(sys.argv[2]),int(sys.argv[3])
r=np.load('gonekF_X8.npy'); kap=tau.copy(); kap[:len(r)]=r; tests=[]; level=[]; log=[]
for m in range(1,M+1):
    s=s0*q**(m-1); T=4.0/s; U=np.log(max(T,7)/(2*np.pi))+2.0
    new=[(float(c),s) for c in np.arange(9*s,U+1e-9,sp*s)]; tests+=new; level+=[m]*len(new)
    cs=np.array([t[0] for t in tests]); ss=np.array([t[1] for t in tests])
    ki=int(np.sum(tau<=INT/s))
    const=np.array([masspair(c,s_)-kpair(c,s_) for c,s_ in tests])
    # Frozen tail: the non-interior ordinates are kept at their current values during the solve of the
```

```
    # stage, so every residual printed below is that of a frozen-tail finite model, not of the infinite system.

    tail=np.array([np.sum(Fphi(c,s_,kap[ki:])) for c,s_ in tests])  # edges & beyond: current values (clock)
    F=lambda y: np.sum(Fphi(cs[:,None],ss[:,None],y[None,:]),axis=1)+tail+const
    Jf=lambda y: dFphi(cs[:,None],ss[:,None],y[None,:])
    y0=kap[:ki].copy()
    # The tolerances 1e-15 are at the double-precision resolution of the unknowns; the solver's
    # termination status (3 = step-size test xtol, 1 = gtol, 2 = ftol, 0 = max_nfev) and nfev are
    # printed, and a termination is not read as a certification of convergence.
    sol=least_squares(F,y0,jac=Jf,method='trf',bounds=(tau[:ki]-3,tau[:ki]+3),xtol=1e-15,ftol=1e-15,gtol=1e-15,
     max_nfev=3000)
    # The residual is symmetric in the interior ordinates, and the nondecreasing rearrangement
    # keeps |kappa_k - tau_k| <= 3 (monotone matching, Lemma rb-matching of the paper), so
    # sorting changes neither the residual nor the box; the number of inversions is reported.
    inv=int(np.sum(np.diff(sol.x)<0))
    kap=kap.copy(); kap[:ki]=np.sort(sol.x)
    Fall=F(kap[:ki]); lv=np.array(level)
    lag=np.max(np.abs(Fall[lv<=m-LAG])) if np.any(lv<=m-LAG) else float('nan')
    nk=min(ki,len(g)); dev=np.abs(kap[:nk]-g[:nk])
    front=lambda tol: (int(np.where(dev>tol)[0][0]) if np.any(dev>tol) else nk)
    rec=dict(m=m,s=s,T=T,interior=ki,tests=len(tests),res_all=float(np.max(np.abs(Fall))),res_lag=float(lag),
             dev5=float(dev[:5].max()),dev10=float(dev[:10].max()),dev20=float(dev[:min(20,nk)].max()),
             f6=front(1e-6),f4=front(1e-4),f2=front(1e-2),dev_tau=float(np.max(np.abs(kap[:ki]-tau[:ki]))),
             status=int(sol.status),nfev=int(sol.nfev),inversions=inv)
    log.append(rec)
    print(f"m={m:2d} T={T:5.0f} interior={ki:3d} tests={len(tests):4d} res_all={rec['res_all']:.1e} res_lag{
     LAG}={rec['res_lag']:.1e} |k-g| k<=5:{rec['dev5']:.1e} k<=10:{rec['dev10']:.1e} k<=20:{rec['dev20']:.1e}
     front(1e-6/1e-4/1e-2)={rec['f6']}/{rec['f4']}/{rec['f2']} |k-tau|max={rec['dev_tau']:.2f} status={rec['
     status']} nfev={rec['nfev']} inversions={inv}",flush=True)
json.dump(log,open(f'horizonG_M{M}_I{INT}_L{LAG}.json','w'),indent=1); np.save(f'horizonG_kap_M{M}_I{INT}.npy',
     kap)
```

D.5. cert7.py.

```
# Floating-point candidate certificate for finite square sections (gamma-free construction), at the
# receding-horizon iterate kappa^H (horizonG, Gonek X=8 initialization, M=16).
# Everything is computed in double precision (QR, Newton, the approximate inverse, norms, the
# radius search); no interval arithmetic and no rounding bounds are used.  "FLOAT_PASS" means that
# the inequalities below hold in floating point, NOT that existence has been proved.
# Stage m: tests = levels <= m (spacing 1.0 s), unknowns = interior ordinates (tau_k <= 2.4/s_m),
# other ordinates frozen at kappa^H. Square subset of tests chosen by pivoted QR on J^T.
# F_j separable in kappa_k  =>  J(x)-J(x0) has columns f_j''(x0_k) delta_k + R_jk, |R_jk|<=A3_jk delta_k^2/2.
# x0 = kappa^H polished by 8 Newton steps on the square system (gamma-free).
# Q is an approximate inverse of J_sel(x0), obtained by solving J_sel Q = I (LU); its defect
# d0 = ||I - Q J_sel(x0)||_inf enters the contraction constant.
# Banach for Psi(x)=x-Q F_sel(x) on {||x-x0||_inf<=r}:  k(r)=d0+||QB||r + 0.5*|| |Q|A3(r) || r^2 < 1,
# eta<=(1-k(r)) r.  The margin is the maximum of (1-k(r)) r over 60 sampled radii (a sampled maximum).
# The ordinates of zeta are NOT used here; the distances of the polished points to them are computed
# separately by diag_cert7_gamma.py from cert7_points.npz.
import numpy as np, json
from scipy.linalg import qr, solve
from core import *
tau=clock(60000); kH=np.load('horizonG_kap_M16_I2.4.npy')
q,s0,sp,INT=0.85,0.08,1.0,2.4
def d2(c,s,k):
    a=s*s; e=s*SQ2PI*np.exp(-a*k*k/2)
    return e*(((a*k)**2-a-c*c)*np.cos(c*k)+2*a*k*c*np.sin(c*k))
def A3b(c,s,k,r):
    a=s*s; kl=np.maximum(np.abs(k)-r,0); ku=np.abs(k)+r; w=a*ku+c
    return s*SQ2PI*np.exp(-a*kl*kl/2)*(w**3+3*a*w)
tests=[]; out=[]; points={}
for m in range(1,17):
    s=s0*q**(m-1); T=4.0/s; U=np.log(max(T,7)/(2*np.pi))+2.0
    tests+=[(float(c),s) for c in np.arange(9*s,U+1e-9,sp*s)]
    n=int(np.sum(tau<=INT/s))
```

```
    cs=np.array([t[0] for t in tests]); ss=np.array([t[1] for t in tests])
    x0=kH[:n]
    J=dFphi(cs[:,None],ss[:,None],x0[None,:])
    _,_,piv=qr(J.T,pivoting=True); sel=np.sort(piv[:n])
    kk=kH.copy(); tsel=[tests[i] for i in sel]
    try:
        for it in range(8):                      # gamma-free Newton polishing on the square system
            Jc=dFphi(cs[sel][:,None],ss[sel][:,None],kk[None,:n]); kk[:n]=kk[:n]-np.linalg.solve(Jc,residual(
     tsel,kk))
    except np.linalg.LinAlgError:
        print(f"m={m:2d} Newton polishing failed (singular square Jacobian)"); break
    x0=kk[:n]; J=dFphi(cs[:,None],ss[:,None],x0[None,:])
    Jsel=J[sel]; Q=solve(Jsel,np.eye(n)); F=residual(tsel,kk)
    d0=float(np.max(np.sum(np.abs(np.eye(n)-Q@Jsel),axis=1)))
    eta=np.max(np.abs(Q@F))
    QB=np.max(np.sum(np.abs(Q@d2(cs[sel][:,None],ss[sel][:,None],x0[None,:])),axis=1))
    C3=lambda r: np.max(np.sum(np.abs(Q)@A3b(cs[sel][:,None],ss[sel][:,None],x0[None,:],r),axis=1))
    best=None
    for r in np.geomspace(max(eta,1e-15)*1.0001,1.0,400):
        kr=d0+QB*r+0.5*C3(r)*r*r
        if kr<1 and eta<=(1-kr)*r: best=(r,kr); break
    margin=max((1-(d0+QB*r+0.5*C3(r)*r*r))*r for r in np.geomspace(1e-12,1e-1,60))
    rho_all=float(np.max(np.abs(residual(tests,kk))))
    Linf=np.max(np.sum(np.abs(Q),axis=1))
    devt=float(np.max(np.abs(x0-tau[:n])))
    points[f'm{m}']=x0.copy()
    rec=dict(m=m,T=T,n=n,Linf=float(Linf),eta=float(eta),QB=float(QB),defect=d0,float_pass=best is not None,
     margin=float(margin),
             rho_all_levels=rho_all,r=(float(best[0]) if best else None),k_r=(float(best[1]) if best else None
     ),dev_tau=devt)
    out.append(rec)
    print(f"m={m:2d} T={T:5.0f} n={n:3d} Linf={Linf:.1e} eta={eta:.1e} ||QB||={QB:.1e} ||I-QJ||={d0:.1e} "
          + (f"FLOAT_PASS r={best[0]:.2e} k(r)={best[1]:.2e}" if best else "no float pass")
          + f" sampled margin={margin:.1e} rho(all tests<=m)={rho_all:.1e} | max|x0-tau|={devt:.2f}",flush=
     True)
json.dump(out,open('cert7.json','w'),indent=1)
np.savez('cert7_points.npz',**points)
```

D.6. **entry.py.**

```
# Block-entry test: from the polished stage-m solution x^(m) (square section polished as in
# cert7), put the NEW band (tau_k in (2.4/s_m, 2.4/s_{m+1}]) and everything beyond at Gonek X=8
# values, and ask whether the floating-point Banach inequalities hold for
# (a) the band-only square system (new band unknowns, tests from level m+1),
# (b) the full square system of stage m+1, started there (no polishing).
# Double precision throughout; "FLOAT_PASS" means the inequalities hold in floating point, not that
# existence is proved.  Q solves J_sel Q = I and its defect ||I - Q J_sel||_inf enters k(r); margins
# are maxima over sampled radii.  gamma.npy is read only for the diagnostic column "Gonek band err".
import numpy as np, json
from scipy.linalg import qr, solve
from core import *
g=np.load('gamma.npy'); tau=clock(60000)
q,s0,sp,INT=0.85,0.08,1.0,2.4
def d2(c,s,k):
    a=s*s; e=s*SQ2PI*np.exp(-a*k*k/2)
    return e*(((a*k)**2-a-c*c)*np.cos(c*k)+2*a*k*c*np.sin(c*k))
def A3b(c,s,k,r):
    a=s*s; kl=np.maximum(np.abs(k)-r,0); ku=np.abs(k)+r; w=a*ku+c
    return s*SQ2PI*np.exp(-a*kl*kl/2)*(w**3+3*a*w)
def inv_defect(Jm):
    Q=solve(Jm,np.eye(Jm.shape[0])); return Q,float(np.max(np.sum(np.abs(np.eye(Jm.shape[0])-Q@Jm),axis=1)))
rG=np.load('gonekF_X8.npy'); kG=tau.copy(); kG[:len(rG)]=rG
def certify(Q,F,QBn,A3fun,d0):
    eta=np.max(np.abs(Q@F))
    for r in np.geomspace(max(eta,1e-15)*1.0001,1.0,300):
```

```
        kr=d0+QBn*r+0.5*np.max(np.sum(np.abs(Q)@A3fun(r),axis=1))*r*r
        if kr<1 and eta<=(1-kr)*r: return eta,r
    return eta,None
def margin(QBn,Q,A3fun,d0): return max((1-(d0+QBn*r+0.5*np.max(np.sum(np.abs(Q)@A3fun(r),axis=1))*r*r))*r for
     r in np.geomspace(1e-9,1e-1,50))
tests=[]; lev=[]; out=[]; prev=None
for m in range(1,16):
    s=s0*q**(m-1); T=4.0/s; U=np.log(max(T,7)/(2*np.pi))+2.0
    new=[(float(c),s) for c in np.arange(9*s,U+1e-9,sp*s)]; tests+=new; lev+=[m]*len(new)
    n=int(np.sum(tau<=INT/s)); cs=np.array([t[0] for t in tests]); ss=np.array([t[1] for t in tests]); lv=np.
     array(lev)
    if prev is not None:
        x0=kG.copy(); x0[:prev[1]]=prev[0][:prev[1]]          # old interior: polished stage-(m-1) solution;
     rest Gonek
        band=np.arange(prev[1],n); idx=np.where(lv==m)[0]
        Jb=dFphi(cs[idx][:,None],ss[idx][:,None],x0[None,band]); _,_,pv=qr(Jb.T,pivoting=True); sb=idx[np.sort
     (pv[:len(band)])]
        Qb,db=inv_defect(dFphi(cs[sb][:,None],ss[sb][:,None],x0[None,band])); Fb=residual([tests[i] for i in
     sb],x0)
        QBb=np.max(np.sum(np.abs(Qb@d2(cs[sb][:,None],ss[sb][:,None],x0[None,band])),axis=1))
        A3f=lambda r: A3b(cs[sb][:,None],ss[sb][:,None],x0[None,band],r)
        eb,rb=certify(Qb,Fb,QBb,A3f,db); mb=margin(QBb,Qb,A3f,db)
        J=dFphi(cs[:,None],ss[:,None],x0[None,:n]); _,_,pv=qr(J.T,pivoting=True); sf=np.sort(pv[:n])
        Qf,df=inv_defect(J[sf]); Ff=residual([tests[i] for i in sf],x0)
        QBf=np.max(np.sum(np.abs(Qf@d2(cs[sf][:,None],ss[sf][:,None],x0[None,:n])),axis=1))
        A3g=lambda r: A3b(cs[sf][:,None],ss[sf][:,None],x0[None,:n],r)
        ef,rf=certify(Qf,Ff,QBf,A3g,df); mf=margin(QBf,Qf,A3g,df)
        bandG=float(np.max(np.abs(x0[band]-g[band]))) if band[-1]<len(g) else float('nan')
        rec=dict(m=m,T=T,band=len(band),eta_band=float(eb),margin_band=float(mb),pass_band=rb is not None,
                 eta_full=float(ef),margin_full=float(mf),pass_full=rf is not None,gonek_band_err=bandG)
        out.append(rec)
        print(f"m={m:2d} T={T:5.0f} band={len(band):2d} | band-only: eta={eb:.1e} sampled margin={mb:.1e} {'
     FLOAT_PASS' if rb else 'no pass'} "
              f"| full: eta={ef:.1e} sampled margin={mf:.1e} {'FLOAT_PASS' if rf else 'no pass'} | Gonek band
     err={bandG:.1e}",flush=True)
    kk=kG.copy() if prev is None else x0.copy(); tsel=None
    J=dFphi(cs[:,None],ss[:,None],kk[None,:n]); _,_,pv=qr(J.T,pivoting=True); sel=np.sort(pv[:n]); tsel=[tests
     [i] for i in sel]
    try:
        for it in range(10):
            Jc=dFphi(cs[sel][:,None],ss[sel][:,None],kk[None,:n]); kk[:n]-=np.linalg.solve(Jc,residual(tsel,kk
     ))
    except np.linalg.LinAlgError: print("singular at",m); break
    prev=(kk.copy(),n)
json.dump(out,open('entry.json','w'),indent=1)
```

D.7. **entry2.py.**

```
# Entry step of the staged construction measured in l^2 as well as in l^infinity.
# Same sections, tests (Gaussians), starting points and square selections as entry.py; for each of the
# band-only and full square systems the Banach test for T(x) = x - Q F(x) is evaluated in both norms:
#   l(r) = d0 + ||Q B|| r + (1/2) || |Q| A3(r) || r^2 < 1  and  ||Q F(x0)|| <= (1 - l(r)) r,
# with d0 = ||I - Q J||, B the diagonal-structured matrix of second derivatives and A3(r) an entrywise
# bound for the third derivatives on the ball.  In l^2 the second-order term is bounded by
# ||Q B diag(e)||_2 <= ||Q B||_2 ||e||_inf <= ||Q B||_2 ||e||_2, so the same inequality with 2-norms
# certifies a zero in the l^2 ball, hence in the l^infinity ball of the same radius.
# Double precision throughout (floating-point evidence, not a proof).  The zeta ordinates are not used.
import numpy as np, json
from scipy.linalg import qr, solve
from core import *
tau=clock(60000)
q,s0,sp,INT=0.85,0.08,1.0,2.4
def d2(c,s,k):
    a=s*s; e=s*SQ2PI*np.exp(-a*k*k/2)
    return e*(((a*k)**2-a-c*c)*np.cos(c*k)+2*a*k*c*np.sin(c*k))
```

```
def A3b(c,s,k,r):
    a=s*s; kl=np.maximum(np.abs(k)-r,0); ku=np.abs(k)+r; w=a*ku+c
    return s*SQ2PI*np.exp(-a*kl*kl/2)*(w**3+3*a*w)
def nrm(M,p): return float(np.max(np.sum(np.abs(M),axis=1))) if p=='inf' else float(np.linalg.norm(M,2))
def vn(v,p): return float(np.max(np.abs(v))) if p=='inf' else float(np.linalg.norm(v))
def test(Q,F,B,A3fun,J,p):
    n=Q.shape[0]; d0=nrm(np.eye(n)-Q@J,p); QB=nrm(Q@B,p); eta=vn(Q@F,p)
    radii=np.geomspace(1e-9,1e-1,80)
    ls=[d0+QB*r+0.5*nrm(np.abs(Q)@A3fun(r),p)*r*r for r in radii]
    marg=max((1-l)*r for l,r in zip(ls,radii))
    ok=any(l<1 and eta<=(1-l)*r for l,r in zip(ls,radii))
    return eta,marg,ok,nrm(Q,p),QB
rG=np.load('gonekF_X8.npy'); kG=tau.copy(); kG[:len(rG)]=rG
tests=[]; lev=[]; out=[]; prev=None
for m in range(1,16):
    s=s0*q**(m-1); T=4.0/s; U=np.log(max(T,7)/(2*np.pi))+2.0
    new=[(float(c),s) for c in np.arange(9*s,U+1e-9,sp*s)]; tests+=new; lev+=[m]*len(new)
    n=int(np.sum(tau<=INT/s)); cs=np.array([t[0] for t in tests]); ss=np.array([t[1] for t in tests]); lv=np.
     array(lev)
    if prev is not None:
        x0=kG.copy(); x0[:prev[1]]=prev[0][:prev[1]]
        band=np.arange(prev[1],n); idx=np.where(lv==m)[0]
        Jb=dFphi(cs[idx][:,None],ss[idx][:,None],x0[None,band]); _,_,pv=qr(Jb.T,pivoting=True); sb=idx[np.sort
     (pv[:len(band)])]
        Jbs=dFphi(cs[sb][:,None],ss[sb][:,None],x0[None,band]); Qb=solve(Jbs,np.eye(len(band))); Fb=residual([
     tests[i] for i in sb],x0)
        Bb=d2(cs[sb][:,None],ss[sb][:,None],x0[None,band]); A3f=lambda r: A3b(cs[sb][:,None],ss[sb][:,None],x0
     [None,band],r)
        J=dFphi(cs[:,None],ss[:,None],x0[None,:n]); _,_,pv=qr(J.T,pivoting=True); sf=np.sort(pv[:n])
        Jf=J[sf]; Qf=solve(Jf,np.eye(n)); Ff=residual([tests[i] for i in sf],x0)
        Bf=d2(cs[sf][:,None],ss[sf][:,None],x0[None,:n]); A3g=lambda r: A3b(cs[sf][:,None],ss[sf][:,None],x0[
     None,:n],r)
        rec=dict(m=m,T=T,band=len(band),n=n)
        line=f"m={m:2d} T={T:5.0f} band={len(band):2d} n={n:3d}"
        for name,(Q,F,B,A,Jm) in (("band",(Qb,Fb,Bb,A3f,Jbs)),("full",(Qf,Ff,Bf,A3g,Jf))):
            for p in ('inf','2'):
                eta,marg,ok,LQ,QB=test(Q,F,B,A,Jm,p)
                rec[f"{name}_{p}"]=dict(eta=eta,margin=marg,float_pass=bool(ok),normQ=LQ,QB=QB)
                line+=f" | {name} l{p}: |QF|={eta:.1e} margin={marg:.1e} |Q|={LQ:.1e} {'PASS' if ok else 'no'}
     "
        out.append(rec); print(line,flush=True)
    kk=kG.copy() if prev is None else x0.copy()
    J=dFphi(cs[:,None],ss[:,None],kk[None,:n]); _,_,pv=qr(J.T,pivoting=True); sel=np.sort(pv[:n]); tsel=[tests
     [i] for i in sel]
    try:
        for it in range(10):
            Jc=dFphi(cs[sel][:,None],ss[sel][:,None],kk[None,:n]); kk[:n]-=np.linalg.solve(Jc,residual(tsel,kk
     ))
    except np.linalg.LinAlgError: print("singular at",m); break
    prev=(kk.copy(),n)
json.dump(out,open('entry2.json','w'),indent=1)
```

D.8. **certV.py.**

```
# Verified certificates for square sections of the rung system with prescribed von Mangoldt masses,
# with compactly supported B-spline tests and ball arithmetic (python-flint / Arb).
#
# Tests.  phi_{c,h}(u) = N_R((u-c)/h + R/2), with N_R the cardinal B-spline of order R = 8 on [0,R]
# (piecewise polynomial of degree R-1, class C^{R-2} = C^6, support [c-Rh/2, c+Rh/2] inside (0,inf)).
# Its cosine transform is c_phi(t) = h cos(ct) (sin(ht/2)/(ht/2))^R, and |c_phi(t)| <= h (2/(ht))^R.
# Stage m: s_m = 0.08*0.85^(m-1), h_m = s_m*sqrt(12/R) (same variance as the Gaussian of width s_m used
# before), centres c in [9 s_j, log(max(T_j,7)/2pi)+2] in steps of s_j for every level j <= m, T_j = 4/s_j.
# Unknowns: the interior ordinates kappa_k with tau_k <= 2.4/s_m (n of them).  Frozen list: the receding-
# horizon iterate kappa^H (double precision numbers, taken as exact) for n < k <= 60000, the midpoint
# clock tau_k beyond.  Square system: n tests chosen by pivoted QR on J^T at kappa^H.
```

```
#
# Float phase (double precision): Newton polishing of the square system from kappa^H.
# Ball phase (Arb, 128 bits): every quantity below is an enclosure.
#   F_j(x) = sum_{k<=n} c_j(x_k) + sum_{n<k<=60000} c_j(kappa^H_k) + [tail over k>60000, bounded]
#            + sum_n Lambda(n)/(2 sqrt n) phi_j(log n) - <k, phi_j>,
#   <k,phi> by acb.integral on the R polynomial pieces of phi; tail by
#   sum_{k>K} |c_j(tau_k)| <= h (pi a h)^(-R) (log K)^R K^(1-R) / ((R-1)(1 - R/((R-1) log K))),
#   from tau_k >= 2 pi a k / log k (W(x) <= log x for x >= e), a = 1 - 11/(8K), and the integral test.
#   Two Newton refinements x <- x - Q mid F(x) in Arb (Q = float inverse of the square Jacobian, exact as a
#   matrix of doubles).  Krawczyk/Banach test on the l^infinity ball B = B(x1, rho):
#     l(rho) >= sup_{x in B} ||I - Q J(x)||  (J evaluated on the ball, interval matrix product),
#     eta >= ||Q F(x1)||;  if l(rho) < 1 and eta <= (1 - l(rho)) rho, then x -> x - Q F(x) maps B into
#     itself and contracts, Q is invertible, and F has exactly one zero in B.
# Diagnostics: ordering of the solution (gaps > 2 rho), tube constant max|x_k - tau_k| + rho (tau_k by
# Arb's Lambert W), and the distance to the zeta ordinates gamma_k computed in Arb (acb.zeta_zero,
# rigorous enclosures); the zeros are used only for this diagnostic, after the certificate.
# Bound eps of Proposition rb-nonfinitary for every test of the stage (r = R-1 = 7, H0 = 3e12): the residual
# of the zeta list with the von Mangoldt masses on these tests is at most eps whether or not RH holds.
# A sampled lower bound for the verified margin: max over rho in a grid of (1 - l(rho)) rho, each value
# rigorous.  Output: out_certV.txt (console), certV.json, certV_points.npz (centres as doubles),
# certV_centres.json (exact binary centres and radii of the verified balls).
import numpy as np, json, time
from math import comb, factorial
from scipy.linalg import qr
from scipy.special import lambertw
from flint import arb, acb, arb_mat, ctx
ctx.prec = 128
R = 8
K = 60000
kH = np.load('horizonG_kap_M16_I2.4.npy')
assert len(kH) == K and np.all(np.diff(kH) > 0)
kk = np.arange(1, K + 2); tauF = np.real(2 * np.pi * (kk - 11 / 8) / lambertw((kk - 11 / 8) / np.e))
assert kH[-1] < tauF[K] - 0.5            # the frozen list followed by the clock is increasing
q, s0, sp, INT = 0.85, 0.08, 1.0, 2.4
import sys
MSTAGES = int(sys.argv[1]) if len(sys.argv) > 1 else 16

# ---------------- float part ----------------
def sinc(x): return np.sin(x) / x
def dsinc(x): return (x * np.cos(x) - np.sin(x)) / (x * x)
def cphi(c, h, t):
    x = h * t / 2
    return h * np.cos(c * t) * sinc(x) ** R
def dcphi(c, h, t):
    x = h * t / 2; S = sinc(x)
    return h * (-c * np.sin(c * t) * S ** R + np.cos(c * t) * R * S ** (R - 1) * dsinc(x) * h / 2)
BIN = np.array([comb(R, j) for j in range(R + 1)], float); FAC = float(factorial(R - 1))
def NR_float(x):
    x = np.asarray(x, float); out = np.zeros_like(x)
    for j in range(R + 1):
        y = x - j; out += (-1) ** j * BIN[j] * np.where(y > 0, y, 0.0) ** (R - 1)
    out /= FAC; out[(x <= 0) | (x >= R)] = 0.0
    return out
def kfun(u): return np.cosh(u / 2) - np.exp(-u / 2) / (2 * (1 - np.exp(-2 * u)))
GLx, GLw = np.polynomial.legendre.leggauss(40)
def kpair_float(c, h):
    tot = 0.0
    for i in range(R):
        x = i + (GLx + 1) / 2; u = c + h * (x - R / 2)
        tot += np.sum(GLw / 2 * kfun(u) * NR_float(x))
    return h * tot
def vonmangoldt(N):
    L = np.zeros(N + 1)
    sieve = np.ones(N + 1, bool); sieve[:2] = False
    for p in range(2, N + 1):
```

```
        if sieve[p]:
            sieve[p * p::p] = False
            pk = p
            while pk <= N: L[pk] = np.log(p); pk *= p
    return L
LAM = vonmangoldt(20000)
def nrange(c, h):
    lo = max(2, int(np.floor(np.exp(c - R * h / 2)))); hi = int(np.ceil(np.exp(c + R * h / 2)))
    return np.arange(lo, hi + 1)
def masspair_float(c, h):
    n = nrange(c, h); a = LAM[n] / (2 * np.sqrt(n))
    return np.sum(a * NR_float((np.log(n) - c) / h + R / 2))
_fcache = {}
def frozen_float(c, h, n0):
    key = (c, h)
    if key not in _fcache: _fcache[key] = np.cumsum(cphi(c, h, kH)[::-1])[::-1]   # suffix sums
    S = _fcache[key]
    return S[n0] if n0 < K else 0.0
_kc = {}
def const_float(c, h):
    if (c, h) not in _kc: _kc[(c, h)] = masspair_float(c, h) - kpair_float(c, h)
    return _kc[(c, h)]
def F_float(tests, x, n):
    return np.array([np.sum(cphi(c, h, x[:n])) + frozen_float(c, h, n) + const_float(c, h) for c, h in tests])
def J_float(tests, x, n):
    cs = np.array([t[0] for t in tests])[:, None]; hs = np.array([t[1] for t in tests])[:, None]
    return dcphi(cs, hs, x[None, :n])

# ---------------- ball part ----------------
PI = arb.pi()
def a_(v): return arb(v)
def cphi_b(c, h, t):
    x = h * t / 2
    return h * (c * t).cos() * (x.sin() / x) ** R
def dcphi_b(c, h, t):
    x = h * t / 2; sx, cx = x.sin(), x.cos(); S = sx / x; dS = (x * cx - sx) / (x * x)
    return h * (-c * (c * t).sin() * S ** R + (c * t).cos() * R * S ** (R - 1) * dS * h / 2)
BINb = [arb(comb(R, j)) for j in range(R + 1)]; FACb = arb(factorial(R - 1))
def tpow_b(y):            # (y)_+^(R-1) for a ball y
    if y > 0: return y ** (R - 1)
    if y < 0: return arb(0)
    up = y.upper()
    up = up if up > 0 else arb(0)
    return arb(0).union(up ** (R - 1))
def NR_b(x):
    tot = arb(0)
    for j in range(R + 1): tot += (-1) ** j * BINb[j] * tpow_b(x - j)
    return tot / FACb
def piece_poly(i):
    # polynomial of N_R on [i, i+1]: (1/(R-1)!) sum_{j<=i} (-1)^j C(R,j) (x-j)^(R-1)
    def P(x):
        tot = acb(0)
        for j in range(i + 1): tot += (-1) ** j * comb(R, j) * (x - j) ** (R - 1)
        return tot / factorial(R - 1)
    return P
PIECES = [piece_poly(i) for i in range(R)]
def kpair_b(c, h):
    cb, hb = acb(c), acb(h); tot = acb(0)
    for i in range(R):
        P = PIECES[i]
        def f(x, analytic, P=P):
            u = cb + hb * (x - R / 2)
            return ((u / 2).cosh() - (-u / 2).exp() / (2 * (1 - (-2 * u).exp()))) * P(x)
        tot += acb.integral(f, i, i + 1)
    v = tot * hb
```

```
    assert abs(v.imag) < 1e-30
    return v.real
def logp_b(n):
    for p in range(2, n + 1):
        if n % p == 0:
            m = n
            while m % p == 0: m //= p
            return arb(p).log() if m == 1 else None
def masspair_b(c, h):
    cb, hb = arb(c), arb(h); tot = arb(0)
    for n in nrange(c, h):
        n = int(n); lp = logp_b(n)
        if lp is None: continue
        x = ((arb(n).log() - cb) / hb) + R / 2
        if x < 0 or x > R: continue
        tot += lp / (2 * arb(n).sqrt()) * NR_b(x)
    return tot
aK = 1 - arb(11) / (8 * K); LK = arb(K).log()
TAILI = LK ** R * arb(K) ** (1 - R) / ((R - 1) * (1 - arb(R) / ((R - 1) * LK)))
def tail_b(h):
    hb = arb(h); return hb * (PI * aK * hb) ** (-R) * TAILI
_bfroz = {}
def frozen_b(c, h, n0, N1=1000):
    # sum_{n0<k<=K} c(kH_k) + ball for the clock tail k>K; the part k>N1 is cached per test
    key = (c, h); cb, hb = arb(c), arb(h)
    if key not in _bfroz:
        s = arb(0)
        for t in kH[N1:]: s += cphi_b(cb, hb, arb(float(t)))
        _bfroz[key] = s
    s = _bfroz[key]
    for t in kH[n0:N1]: s += cphi_b(cb, hb, arb(float(t)))
    tb = tail_b(h)
    return s + arb(0, tb.upper())
def tau_b(k):
    a = arb(k) - arb(11) / 8
    return 2 * PI * a / (a / arb(1).exp()).lambertw()
H0 = arb(3) * arb(10) ** 12; RF = R - 1
def eps_bound(c, h):
    # Proposition rb-nonfinitary with r = R-1: |<R(gamma, a^zeta), phi>| <= r/(4(r-1)) W_r(phi) H0^(1-r) log
     H0,
    # W_r(phi) <= (1+u*^2) cosh(u*/2) h (1+2/h)^r for phi = N_R((u-c)/h + R/2), u* = c + R h/2
    hb = arb(h); us = arb(c) + R * hb / 2
    W = (1 + us * us) * (us / 2).cosh() * hb * (1 + 2 / hb) ** RF
    return float((arb(RF) / (4 * (RF - 1)) * W * H0 ** (1 - RF) * H0.log()).upper())
def norm_inf_upper(M, n, m):
    best = 0.0
    for i in range(n):
        r = arb(0)
        for j in range(m): r += M[i, j].abs_upper()
        best = max(best, float(r.upper()))
    return best

# ---------------- the frozen list, rigorously ----------------
# kappa^H is strictly increasing (exact comparison of doubles), its last entry lies below tau_{K+1}, and
# |kappa^H_k - tau_k| <= FROZEN_TUBE for all k <= K, with tau_k enclosed by Arb's Lambert W.
t0 = time.time()
FROZEN_TUBE = 0.0
for k in range(1, K + 1):
    d = float(abs(arb(float(kH[k - 1])) - tau_b(k)).upper())
    if d > FROZEN_TUBE: FROZEN_TUBE = d
assert arb(float(kH[-1])) < tau_b(K + 1)
print(f"frozen list: increasing, max_k<={K} |kappa^H_k - tau_k| <= {FROZEN_TUBE:.6f}, kappa^H_K < tau_(K+1)
     ({time.time()-t0:.0f}s)", flush=True)

# ---------------- stages ----------------
```

```
tests = []; out = []; points = {}; centres = {}   # centres: exact binary centres x_1 = man * 2^exp of the verified balls
t_start = time.time()
for m in range(1, MSTAGES + 1):
    ts = time.time()
    s = s0 * q ** (m - 1); h = s * np.sqrt(12.0 / R); T = 4.0 / s; U = np.log(max(T, 7) / (2 * np.pi)) + 2.0
    tests += [(float(c), float(h)) for c in np.arange(9 * s, U + 1e-9, sp * s)]
    n = int(np.sum(tauF[:K] <= INT / s))
    J = J_float(tests, kH, n); _, _, piv = qr(J.T, pivoting=True); sel = np.sort(piv[:n]); tsel = [tests[i] for i in sel]
    x = kH.copy(); ok = True
    try:
        for it in range(12):
            Fx = F_float(tsel, x, n); x[:n] -= np.linalg.solve(J_float(tsel, x, n), Fx)
    except np.linalg.LinAlgError:
        ok = False
    Fx = F_float(tsel, x, n); Jx = J_float(tsel, x, n)
    condJ = np.linalg.cond(Jx) if ok else np.inf
    if not ok or not np.all(np.isfinite(x[:n])) or np.max(np.abs(Fx)) > 1e-8 or condJ > 1e14:
        print(f"m={m:2d} T={T:5.0f} n={n:3d} float Newton polishing failed (max|F|={np.max(np.abs(Fx)):.1e}, cond={condJ:.1e})", flush=True)
        out.append(dict(m=m, T=T, n=n, verified=False, reason="float polishing failed")); break
    Q = np.linalg.solve(Jx, np.eye(n))
    x0 = x[:n].copy()
    # ---- ball phase ----
    Qb = arb_mat(n, n, [float(v) for v in Q.ravel()])
    const = []
    for c, hh in tsel:
        const.append(frozen_b(c, hh, n) + masspair_b(c, hh) - kpair_b(c, hh))
    xb = [arb(float(v)) for v in x0]
    def Fint(xb):
        return [sum((cphi_b(arb(c), arb(hh), xk) for xk in xb), arb(0)) + const[j] for j, (c, hh) in enumerate(tsel)]
    F0 = Fint(xb)
    for ref in range(2):                          # Newton refinements in ball arithmetic (midpoints kept)
        Fv = arb_mat(n, 1, [f.mid() for f in F0]); dx = Qb * Fv
        xb = [arb((xb[k] - dx[k, 0]).mid()) for k in range(n)]
        F0 = Fint(xb)
    Fv = arb_mat(n, 1, F0); QF = Qb * Fv
    eta = max(float(QF[k, 0].abs_upper().upper()) for k in range(n))
    Frad = max(float(f.rad()) for f in F0)
    def lfun(rho):
        Jb = arb_mat(n, n, [dcphi_b(arb(c), arb(hh), xb[k] + arb(0, rho)) for (c, hh) in tsel for k in range(n)])
        M = arb_mat(n, n, [1 if i == j else 0 for i in range(n) for j in range(n)]) - Qb * Jb
        return norm_inf_upper(M, n, n)
    rho = max(4 * eta, 1e-34); l = lfun(rho)
    verified = (l < 1) and (eta <= (1 - l) * rho)
    d0 = lfun(0.0)
    grid = list(np.geomspace(1e-8, 1e-1, 36))
    margins = []
    for r_ in grid:
        lr = lfun(r_); margins.append((r_, lr, (1 - lr) * r_ if lr < 1 else 0.0))
    vmargin = max(mg[2] for mg in margins)
    xm = np.array([float(v.mid()) for v in xb])
    gaps = np.diff(np.concatenate([xm, [kH[n]]]))
    order_ok = bool(np.min(gaps) > 2 * rho + 1e-300)
    taub = [tau_b(k) for k in range(1, n + 1)]
    tube = max(float(abs(xb[k] - taub[k]).upper()) for k in range(n)) + rho
    gam = [acb.zeta_zero(k).imag for k in range(1, n + 1)]
    dg = [float(abs(xb[k] - gam[k]).upper()) + rho for k in range(n)]
    Linf = float(np.max(np.sum(np.abs(Q), axis=1)))
    allres = F_float(tests, x, n); rho_all = float(np.max(np.abs(allres)))
    eps_sel = max(eps_bound(c, hh) for c, hh in tsel); eps_all = max(eps_bound(c, hh) for c, hh in tests)
    points[f'm{m}'] = xm
```

```
    centres[f'm{m}'] = dict(rho=rho, centre_man_exp=[[int(a), int(b)] for a, b in (v.mid().man_exp() for v in
     xb)])
    rec = dict(m=m, T=T, n=n, R=R, h=h, n_tests_level_le_m=len(tests), verified=bool(verified), rho=rho, l_rho
     =l, eta=eta,
               defect_upper=d0, Frad=Frad, Linf=Linf, verified_margin_lower=vmargin,
               margins=[dict(r=a, l=b, margin=c_) for a, b, c_ in margins], order_ok=order_ok,
               min_gap=float(np.min(gaps)), tube=tube, max_dist_gamma=max(dg), max_dist_gamma_k_le_20=max(dg
     [:20]),
               rho_all_float=rho_all, eps_zeta_selected=eps_sel, eps_zeta_all_tests=eps_all, seconds=time.time
     () - ts)
    out.append(rec)
    print(f"m={m:2d} T={T:5.0f} n={n:3d} h={h:.4f} | {'VERIFIED' if verified else 'not verified'} rho={rho:.2e
     } l(rho)={l:.2e} "
          f"eta={eta:.2e} ||I-QJ||<={d0:.1e} ||Q||={Linf:.1e} margin>={vmargin:.1e} | order={'ok' if order_ok
     else 'NO'} "
          f"mingap={np.min(gaps):.3f} tube<={tube:.3f} | max|x-gamma|<={max(dg):.1e} (k<=20: {max(dg[:20]):.1e
     }) | "
          f"float residual all tests<=m {rho_all:.1e} | zeta-list bound eps<={eps_all:.1e} | {time.time()-ts
     :.0f}s", flush=True)
    json.dump(out, open('certV.json', 'w'), indent=1); np.savez('certV_points.npz', **points)
    json.dump(centres, open('certV_centres.json', 'w'))
json.dump(out, open('certV.json', 'w'), indent=1); np.savez('certV_points.npz', **points)
print(f"total {time.time()-t_start:.0f}s")
```

D.9. **laplace_check.py.**

```
# Numerical check of the identity  (1/2) G'/G(s) = L[K](s)  with
# K = (1/2)e^{u/2} 1_+ - (1/2) g 1_+ - (1/4) fp(1_+/u) + c0 delta_0,
# g(u)=e^{-u/2}/(e^{2u}-1)-1/(2u),  c0=-(1/4)log(2 pi e^{gamma_E}),  z=s-1/2,
# L[fp(1_+/u)](z) = -log z - gamma_E.   Also checks k(u)=cosh(u/2)-e^{-u/2}/(2(1-e^{-2u})) = K|_(0,inf).
from mpmath import mp, mpf, exp, expm1, log, pi, euler, quad, digamma, cosh, inf, mpc
mp.dps=50
def g(u):
    if u<mpf('1e-12'): return -mpf(3)/4       # g(0+)=-3/4; avoids cancellation
    return exp(-u/2)/expm1(2*u)-1/(2*u)
def k_ref(u): return (exp(u/2)-exp(-u/2)/(exp(2*u)-1))/2      # k(u) as written in eq. (transport-defs)
def k_mine(u): return cosh(u/2)-exp(-u/2)/(2*(1-exp(-2*u)))
def K_restr(u): return exp(u/2)/2-g(u)/2-1/(4*u)
for u in [mpf('0.1'),mpf(1),mpf('2.5')]:
    print('u=%s  k_ref=%s  k_mine=%s  K|=%s'%(u,k_ref(u),k_mine(u),K_restr(u)))
def halfGG(s): return (1/s+1/(s-1)-log(pi)/2+digamma(s/2)/2)/2
def LK(s):
    z=s-mpf(1)/2
    c0=-(euler+log(2)+log(pi))/4
    I1=1/(2*(z-mpf(1)/2))                      # L[(1/2)e^{u/2}]
    I2=-quad(lambda u: exp(-z*u)*g(u),[0,1,inf])/2
    I3=-(-log(z)-euler)/4
    return I1+I2+I3+c0
for s in [mpf(2),mpf('1.3'),mpc(2,5),mpc('1.5',-3)]:
    print('s=%s  (1/2)G\'/G=%s  L[K]=%s'%(s,halfGG(s),LK(s)))
```

D.10. **diag_cert7_gamma.py.**

```
# Diagnostic only: distance of the polished points of cert7.py to the ordinates of zeta
# (gamma.npy, from zeros.py).  Kept separate so that cert7.py does not read the zeros.
import numpy as np, json
g=np.load('gamma.npy'); pts=np.load('cert7_points.npz'); rec=json.load(open('cert7.json'))
for r in rec:
    x0=pts[f"m{r['m']}"]; nk=min(len(x0),len(g))
    print(f"m={r['m']:2d} T={r['T']:5.0f} n={r['n']:3d}  max|x0-gamma| over k<={nk}: {np.max(np.abs(x0[:nk]-g[:
     nk])):.1e}")
```

D.11. **figs.py.**

```
# Figures of the Route B numerics: (1) constants of the floating-point candidate certificates, (2) distance to
      gamma of clock and Gonek lists.
import numpy as np, json, matplotlib
matplotlib.use('Agg'); import matplotlib.pyplot as plt
from core import clock
g=np.load('gamma.npy'); K=400; tau=clock(2000)
def align(r,K):
    best=None
    for off in range(0,6):
        a=r[off:off+K]
        if len(a)<K: continue
        d=np.abs(a-g[:K]).mean()
        if best is None or d<best[0]: best=(d,off,a)
    return best[2]
sp=2*np.pi/np.log(g[:K]/(2*np.pi))
fig,ax=plt.subplots(figsize=(7.2,4.2))
ax.plot(np.arange(1,K+1),np.abs(tau[:K]-g[:K])/sp,'.',ms=3,color='0.3',label='midpoint clock $\\tau_k$')
rF=np.load('gonekF_X8.npy'); a=align(rF,K)
ax.plot(np.arange(1,K+1),np.abs(a-g[:K])/sp,'.',ms=3,color='C3',label='Gonek list, $X=8$')
K2=160; sp2=sp[:K2]
for X,c in [(3,'C0'),(20,'C2')]:
    r=np.load(f'gonek_X{X}.npy'); a=align(r,K2)
    ax.plot(np.arange(1,K2+1),np.abs(a-g[:K2])/sp2,'.',ms=3,color=c,label=f'Gonek list, $X={X}$')
ax.axhline(0.25,ls='--',color='k',lw=0.8); ax.text(5,0.26,'$0.25$ spacings',fontsize=8)
ax.set_xlabel('$k$'); ax.set_ylabel('$|\\kappa_k-\\gamma_k|/\\mathrm{spacing}$'); ax.set_ylim(0,0.75)
ax.legend(fontsize=8,loc='upper right'); ax.set_title('Distance to the zeta ordinates, in local mean spacings
     $2\\pi/\\log(\\gamma_k/2\\pi)$',fontsize=9)
fig.tight_layout(); fig.savefig('fig_gonek_vs_clock.png',dpi=160)
d=json.load(open('cert7.json'))
T=np.array([x['T'] for x in d]); L=np.array([x['Linf'] for x in d]); QB=np.array([x['QB'] for x in d]); mg=np.
     array([x['margin'] for x in d])
fig,ax=plt.subplots(figsize=(7.2,4.2))
for y,lab,c in [(L,'$L_m=\\|Q_m\\|_{\\infty\\to\\infty}$','C0'),(QB,'$\\|Q_mB_m\\|_{\\infty\\to\\infty}$','C1'
     ),(1/mg,'1 / (Banach margin)','C3')]:
    p=np.polyfit(np.log(T[3:]),np.log(y[3:]),1); ax.loglog(T,y,'o-',ms=4,color=c,label=lab+f'  (slope {p[0]:.2
     f})')
e=json.load(open('entry.json')); Te=np.array([x['T'] for x in e]); eb=np.array([x['eta_band'] for x in e]); mb
     =np.array([x['margin_band'] for x in e])
ax.loglog(Te,1/mb,'s--',ms=4,color='C2',label='1 / (band-only margin)')
ax.loglog(Te,1/eb,'^--',ms=4,color='C4',label='1 / (Gonek entry residual $\\eta$, band)')
ax.set_xlabel('height $T_m$'); ax.legend(fontsize=7.5); ax.set_title('Constants of the floating-point
     candidate certificates; slopes: log-log fits on stages $m\\geq4$',fontsize=9)
fig.tight_layout(); fig.savefig('fig_cert_scaling.png',dpi=160)
print('ok')
```

D.12. **results_summary.py.**

```
# Writes RESULTS.txt in the current directory (output/routeB) from the JSON outputs of the Route B
# programs, with the SHA-256 of the programs and input files used, so that the summary is tied to
# the code and data rather than to a date.  Run from output/routeB after the other programs.
import hashlib, json, os
import numpy as np

HERE = os.path.dirname(os.path.abspath(__file__))


def sha(path):
    h = hashlib.sha256()
    with open(path, 'rb') as f:
        h.update(f.read())
    return h.hexdigest()


progs = ['core.py', 'zeros.py', 'gonek_fast.py', 'horizon.py', 'horizonG.py', 'cert7.py', 'diag_cert7_gamma.py
     ',
```

```
        'entry.py', 'entry2.py', 'certV.py', 'laplace_check.py', 'figs.py', 'results_summary.py']
inputs = ['gamma.npy', 'gonekF_X8.npy', 'horizonG_kap_M16_I2.4.npy', 'horizon_M16_I2.4_L4.json',
          'horizonG_M16_I2.4_L4.json', 'cert7.json', 'entry.json', 'entry2.json', 'certV.json', 'certV_points.
    npz', 'certV_centres.json']
L = []
P = L.append
P('RVM rung system - staged floating-point candidate certificates (Route B numerics)')
P('Tests: Gaussians phi(u)=exp(-(u-c)^2/(2 s^2)), c>=9s; stage m: s_m=0.08*0.85^(m-1), T_m=4/s_m,')
P('centres spaced 1.0*s_m in [9 s_m, log(max(T_m,7)/2pi)+2]. Interior unknowns: tau_k <= 2.4/s_m.')
P('Masses a_n = Lambda(n)/(2 sqrt n); k(u)=cosh(u/2)-e^{-u/2}/(2(1-e^{-2u})).')
P('Gonek list X=8: roots of (theta(t)+alpha_X(t)-pi/2)/pi in Z (gonek_fast.py).')
P('gamma (400 zeros, mpmath, stored as doubles) is used ONLY for diagnostics, never in the constructions.')
P('Double precision throughout, except certV.py (ball arithmetic, python-flint/Arb, 128 bits), which proves')
P('the theorem on verified square sections with B-spline tests.  FLOAT_PASS = the Banach inequalities hold in'
    )
P('floating point (a candidate certificate), not a proof of existence.')
P('')
P('SHA-256 of the programs (code/routeB) and of the inputs/outputs read here (output/routeB):')
for p in progs:
    P('  %-28s %s' % (p, sha(os.path.join(HERE, p))))
for p in inputs:
    P('  %-28s %s' % (p, sha(p)))
P('')
hz = json.load(open('horizon_M16_I2.4_L4.json'))
hg = json.load(open('horizonG_M16_I2.4_L4.json'))
c7 = json.load(open('cert7.json'))
en = json.load(open('entry.json'))
e2 = json.load(open('entry2.json'))
cv = json.load(open('certV.json'))
P('STATUS CRITERIA: a receding horizon "reaches the threshold" if every one of its 16 stages ends with')
P('res_all < 1e-2 inside the box |kappa_k - tau_k| <= 3; it "does not reach" it if res_all >= 1e-2 at')
P('some stage.  A certificate test "passes" if its floating-point inequalities hold.')
P('')
P('FILES')
P(' core.py          residual, Jacobian, clock, k(u), mass pairing')
P(' validate.py      R(gamma)~1e-16 normalisation check            -> out_validate.txt')
P(' gonek*.py        Gonek lists, distance to gamma                 -> out_gonek.txt, out_gonek_fast.txt,
    gonek*.npy')
P(' horizon.py       receding horizon from the clock: does not reach (res_all in [%.1e, %.1e])  ->
    horizon_M16_I2.4_L4.json'
  % (min(x['res_all'] for x in hz), max(x['res_all'] for x in hz)))
P(' horizonG.py      receding horizon from Gonek X=8: reaches (res_all in [%.1e, %.1e])  -> horizonG_M16_I2.4
    _L4.json, out_horizonG.txt'
  % (min(x['res_all'] for x in hg), max(x['res_all'] for x in hg)))
P(' certG.py         l-inf Kantorovich test at the Gonek list (no stage passes)  -> out_certG.txt')
P(' cert2/3/4.py     l2 / Gauss-Newton / diagonal l-inf constants at Gonek list -> out_cert2/3/4.txt')
P(' cert5/6/7.py     floating-point Banach tests of square finite sections (cert7 = the one reported; cert5,
    cert6 = variants); '
  'cert7: %d stages pass -> out_cert7.txt, cert7.json, cert7_points.npz' % sum(1 for x in c7 if x['float_pass'
    ]))
P(' diag_cert7_gamma.py  distance of the cert7 points to gamma (diagnostic) -> out_cert7_gamma.txt')
P(' entry.py         block-entry test: band-only passes at %d of %d stages, full system at %d  -> out_entry.
    txt, entry.json'
  % (sum(1 for x in en if x['pass_band']), len(en), sum(1 for x in en if x['pass_full'])))
P(' entry2.py        the same entry test in l-inf and l2: band passes at %d (l-inf) and %d (l2) of %d stages
    -> out_entry2.txt, entry2.json'
  % (sum(1 for x in e2 if x['band_inf']['float_pass']), sum(1 for x in e2 if x['band_2']['float_pass']), len(
    e2)))
P(' certV.py         VERIFIED (ball arithmetic) square sections with B-spline tests: %d stages verified, T up
    to %.0f  -> out_certV.txt, certV.json, certV_points.npz, certV_centres.json'
  % (sum(1 for x in cv if x.get('verified')), max(x['T'] for x in cv if x.get('verified'))))
P(' laplace_check.py the identity (1/2)G\'/G = L[K]                  -> out_laplace_check.txt')
P(' staged.py, lsq*.py, floor.py, basin*.py, reg.py, coupling.py: exploratory runs, not used in the paper (
    logs *.json)')
P('')
```

```
for name, d in (('horizon_M16_I2.4_L4.json', hz), ('horizonG_M16_I2.4_L4.json', hg)):
    P(name)
    for x in d:
        P(' m=%2d T=%5.0f n=%3d res_all=%.1e res_lag4=%.1e max|k-g|(k<=20)=%.1e front(1e-2)=%d max|k-tau|=%.2f
    '
          % (x['m'], x['T'], x['interior'], x['res_all'], x['res_lag'], x['dev20'], x['f2'], x['dev_tau']))
T = np.array([x['T'] for x in c7]); fit = {}; err = {}
for key in ('Linf', 'QB', 'margin'):
    y = np.array([x[key] for x in c7])
    coef, cov = np.polyfit(np.log(T[3:]), np.log(y[3:]), 1, cov=True)
    fit[key], err[key] = coef[0], float(np.sqrt(cov[0, 0]))
P('')
P('Least-squares log-log fits over the computed stages m>=4 of cert7 (%d stages; regressions on the computed'
  % (T.size - 3))
P('range, not asymptotic laws; +- one standard error of the slope from the fit residuals):')
P('  Linf ~ T^(%.2f +- %.2f), ||QB|| ~ T^(%.2f +- %.2f), sampled Banach margin ~ T^(%.2f +- %.2f)'
  % (fit['Linf'], err['Linf'], fit['QB'], err['QB'], fit['margin'], err['margin']))
P('')
P('certV.json (verified stages; rho = radius of the verified l-inf ball, l = contraction bound, margin =
    rigorous')
P('lower bound sampled over 36 radii, dist = rigorous bound of max_k<=n |x_k - gamma_k|, eps = bound for the
    zeta list):')
for x in cv:
    if x.get('verified'):
        P(' m=%2d T=%5.0f n=%3d rho=%.2e l=%.1e ||Q||=%.1e margin>=%.1e dist<=%.1e eps<=%.1e'
          % (x['m'], x['T'], x['n'], x['rho'], x['l_rho'], x['Linf'], x['verified_margin_lower'], x['
    max_dist_gamma'],
             x['eps_zeta_all_tests']))
    else:
        P(' m=%2d T=%5.0f n=%3d %s' % (x['m'], x['T'], x['n'], x.get('reason', 'not verified')))
open('RESULTS.txt', 'w').write('\n'.join(L) + '\n')
print('\n'.join(L))
```

Appendix E. Console output of the programs of Section 18

E.1. **RESULTS.txt.**

```
RVM rung system - staged floating-point candidate certificates (Route B numerics)
Tests: Gaussians phi(u)=exp(-(u-c)^2/(2 s^2)), c>=9s; stage m: s_m=0.08*0.85^(m-1), T_m=4/s_m,
centres spaced 1.0*s_m in [9 s_m, log(max(T_m,7)/2pi)+2]. Interior unknowns: tau_k <= 2.4/s_m.
Masses a_n = Lambda(n)/(2 sqrt n); k(u)=cosh(u/2)-e^{-u/2}/(2(1-e^{-2u})).
Gonek list X=8: roots of (theta(t)+alpha_X(t)-pi/2)/pi in Z (gonek_fast.py).
gamma (400 zeros, mpmath, stored as doubles) is used ONLY for diagnostics, never in the constructions.
Double precision throughout, except certV.py (ball arithmetic, python-flint/Arb, 128 bits), which proves
the theorem on verified square sections with B-spline tests.  FLOAT_PASS = the Banach inequalities hold in
floating point (a candidate certificate), not a proof of existence.

SHA-256 of the programs (code/routeB) and of the inputs/outputs read here (output/routeB):
  core.py                      710b7111ec38b524dad838aa319358d0e2866ba6be03a572aa086bf8c7118c5b
  zeros.py                     b5a6ec8967b10fd267752dc0b05515bc97be2d811e4beea550300be6091b5f40
  gonek_fast.py                9206bb86ed4b0b7a1bb48a1dcca222ed41f724b8409aa368609a7b94295ac798
  horizon.py                   990bf9e315c632ba3315db8d7843e26e9f08f2bed0d117ff7423f36b0e391d9c
  horizonG.py                  c36d771c5d8bb0cc2a248127c92c3d1a5430ca9e4518cebdf133ee35d82df73e
  cert7.py                     56dea8c58fc405bda3d3670f8b1c5af2fdcaaa4dcdfa27080cc20085cd933210
  diag_cert7_gamma.py          d105edbd2168a8f9ccbd8cc1e7b94af0eac7302100292c2bf79a4351b60ffe36
  entry.py                     3a30f98f9f86c327ca3f1928fdaf2c1b845dd16148e7d4d83316f71678d3e6f8
  entry2.py                    5f24c380b94d101325a69a5282de30b824ee5676fa3881b4666bcc25e6765b30
  certV.py                     49e6d0f5cc572805d6571f16dece4c39b7b39681517e7392e03ba15004700075
  laplace_check.py             c2a13393958890c067cb3b63fb500aca0abb8f4bb8c709f4220f0c1c3b12a12a
  figs.py                      850a6a441c5ddcfa5500c0a13d3ef9a4a70735bfbb0527d39dec1c4bfddf8306
  results_summary.py           6f09b2f10ea02c20d4255897a7c7ce1e002d257ee9fd9e94daa53a84d5de45ee
  gamma.npy                    2e0b391d143c3ca530e5ef11a6e3e562aa88040a3298175e12e65b2242be7d8f
  gonekF_X8.npy                ae86e2d378cf193a3b61172a5cbd8bee24dbef026eb76fbcaf832d90acbaa176
  horizonG_kap_M16_I2.4.npy    ca4cc55a2359988f8504a3782c373c5989e8091aacda6c4a1bd32fb26a128d68
  horizon_M16_I2.4_L4.json     0e3147c7e2b56af90c96c49f4c8474240b33563bb70f9c68707b8dc9a27ffe1a
```

```
  horizonG_M16_I2.4_L4.json    2b6c55d8f92ccd43814d1612684d53e3a0fbbf0d6f46025a2ca679b08d6d22ad
  cert7.json                   db27be4ce3a62d2597055326d04a3f2280c05f7d8c55a3d01d18c3178555d977
  entry.json                   b559dd7074f39cc39ba8df74b860795b7869b42be13a8b68132583bb8ee3d495
  entry2.json                  a3329ee5f1a309493b538b45060220d4c7268515b1f038a57e344971d12360b2
  certV.json                   1b9796b946a467d324b03a7e1bb59db9552c0b593a12a14b6549390a7b8266eb
  certV_points.npz             4957a5c3409286bf4f8850809954aa6c720075a60a38d0145688cd5426e55ec5
  certV_centres.json           5d19fad20949e2fcec96967c2b54074242d5bb7b678a1849262de4de6be0c64b

STATUS CRITERIA: a receding horizon "reaches the threshold" if every one of its 16 stages ends with
res_all < 1e-2 inside the box |kappa_k - tau_k| <= 3; it "does not reach" it if res_all >= 1e-2 at
some stage.  A certificate test "passes" if its floating-point inequalities hold.

FILES
 core.py          residual, Jacobian, clock, k(u), mass pairing
 validate.py      R(gamma)~1e-16 normalisation check            -> out_validate.txt
 gonek*.py        Gonek lists, distance to gamma                 -> out_gonek.txt, out_gonek_fast.txt, gonek*.
    npy
 horizon.py       receding horizon from the clock: does not reach (res_all in [1.2e-02, 8.6e-02])  ->
    horizon_M16_I2.4_L4.json
 horizonG.py      receding horizon from Gonek X=8: reaches (res_all in [1.1e-04, 2.5e-03])  -> horizonG_M16_I2
    .4_L4.json, out_horizonG.txt
 certG.py         l-inf Kantorovich test at the Gonek list (no stage passes)  -> out_certG.txt
 cert2/3/4.py     l2 / Gauss-Newton / diagonal l-inf constants at Gonek list -> out_cert2/3/4.txt
 cert5/6/7.py     floating-point Banach tests of square finite sections (cert7 = the one reported; cert5,
    cert6 = variants); cert7: 15 stages pass -> out_cert7.txt, cert7.json, cert7_points.npz
 diag_cert7_gamma.py  distance of the cert7 points to gamma (diagnostic) -> out_cert7_gamma.txt
 entry.py         block-entry test: band-only passes at 4 of 10 stages, full system at 1  -> out_entry.txt,
    entry.json
 entry2.py        the same entry test in l-inf and l2: band passes at 4 (l-inf) and 4 (l2) of 10 stages  ->
    out_entry2.txt, entry2.json
 certV.py         VERIFIED (ball arithmetic) square sections with B-spline tests: 15 stages verified, T up to
    487  -> out_certV.txt, certV.json, certV_points.npz, certV_centres.json
 laplace_check.py the identity (1/2)G'/G = L[K]                  -> out_laplace_check.txt
 staged.py, lsq*.py, floor.py, basin*.py, reg.py, coupling.py: exploratory runs, not used in the paper (logs *.
    json)

horizon_M16_I2.4_L4.json
 m= 1 T=   50 n=  4 res_all=1.4e-02 res_lag4=nan max|k-g|(k<=20)=8.1e-02 front(1e-2)=3 max|k-tau|=0.77
 m= 2 T=   59 n=  5 res_all=1.4e-02 res_lag4=nan max|k-g|(k<=20)=4.6e-03 front(1e-2)=5 max|k-tau|=0.69
 m= 3 T=   69 n=  7 res_all=2.1e-02 res_lag4=nan max|k-g|(k<=20)=1.9e-02 front(1e-2)=5 max|k-tau|=0.69
 m= 4 T=   81 n=  9 res_all=1.9e-02 res_lag4=nan max|k-g|(k<=20)=1.2e-02 front(1e-2)=7 max|k-tau|=0.83
 m= 5 T=   96 n= 12 res_all=1.9e-02 res_lag4=2.9e-05 max|k-g|(k<=20)=3.2e-02 front(1e-2)=10 max|k-tau|=0.84
 m= 6 T=  113 n= 16 res_all=1.3e-02 res_lag4=1.7e-04 max|k-g|(k<=20)=5.0e-02 front(1e-2)=11 max|k-tau|=0.88
 m= 7 T=  133 n= 20 res_all=1.7e-02 res_lag4=6.9e-05 max|k-g|(k<=20)=1.1e-01 front(1e-2)=17 max|k-tau|=0.90
 m= 8 T=  156 n= 26 res_all=1.3e-02 res_lag4=8.6e-05 max|k-g|(k<=20)=3.5e-03 front(1e-2)=23 max|k-tau|=0.90
 m= 9 T=  183 n= 34 res_all=1.2e-02 res_lag4=1.8e-05 max|k-g|(k<=20)=3.6e-04 front(1e-2)=31 max|k-tau|=0.90
 m=10 T=  216 n= 43 res_all=2.6e-02 res_lag4=1.5e-04 max|k-g|(k<=20)=9.1e-04 front(1e-2)=30 max|k-tau|=0.90
 m=11 T=  254 n= 54 res_all=3.3e-02 res_lag4=7.2e-04 max|k-g|(k<=20)=1.4e-03 front(1e-2)=31 max|k-tau|=0.90
 m=12 T=  299 n= 68 res_all=4.2e-02 res_lag4=3.1e-03 max|k-g|(k<=20)=2.0e-03 front(1e-2)=30 max|k-tau|=0.90
 m=13 T=  352 n= 85 res_all=5.6e-02 res_lag4=9.0e-03 max|k-g|(k<=20)=1.2e-03 front(1e-2)=33 max|k-tau|=0.90
 m=14 T=  414 n=107 res_all=6.3e-02 res_lag4=1.7e-02 max|k-g|(k<=20)=1.3e-03 front(1e-2)=31 max|k-tau|=0.90
 m=15 T=  487 n=133 res_all=8.0e-02 res_lag4=3.0e-02 max|k-g|(k<=20)=1.4e-03 front(1e-2)=32 max|k-tau|=0.90
 m=16 T=  572 n=165 res_all=8.6e-02 res_lag4=3.7e-02 max|k-g|(k<=20)=1.3e-03 front(1e-2)=32 max|k-tau|=0.90
horizonG_M16_I2.4_L4.json
 m= 1 T=   50 n=  4 res_all=1.1e-04 res_lag4=nan max|k-g|(k<=20)=4.4e-04 front(1e-2)=4 max|k-tau|=0.69
 m= 2 T=   59 n=  5 res_all=1.7e-04 res_lag4=nan max|k-g|(k<=20)=3.1e-04 front(1e-2)=5 max|k-tau|=0.69
 m= 3 T=   69 n=  7 res_all=2.7e-04 res_lag4=nan max|k-g|(k<=20)=1.6e-03 front(1e-2)=7 max|k-tau|=0.69
 m= 4 T=   81 n=  9 res_all=3.8e-04 res_lag4=nan max|k-g|(k<=20)=3.1e-03 front(1e-2)=9 max|k-tau|=0.84
 m= 5 T=   96 n= 12 res_all=7.4e-04 res_lag4=3.0e-06 max|k-g|(k<=20)=2.1e-03 front(1e-2)=12 max|k-tau|=0.84
 m= 6 T=  113 n= 16 res_all=5.7e-04 res_lag4=1.2e-06 max|k-g|(k<=20)=5.9e-04 front(1e-2)=16 max|k-tau|=0.90
 m= 7 T=  133 n= 20 res_all=1.1e-03 res_lag4=2.8e-06 max|k-g|(k<=20)=1.8e-03 front(1e-2)=20 max|k-tau|=0.90
 m= 8 T=  156 n= 26 res_all=1.8e-03 res_lag4=2.4e-06 max|k-g|(k<=20)=2.5e-04 front(1e-2)=26 max|k-tau|=0.90
 m= 9 T=  183 n= 34 res_all=1.1e-03 res_lag4=4.2e-06 max|k-g|(k<=20)=1.4e-04 front(1e-2)=33 max|k-tau|=0.96
 m=10 T=  216 n= 43 res_all=2.0e-03 res_lag4=3.9e-06 max|k-g|(k<=20)=3.0e-05 front(1e-2)=43 max|k-tau|=0.99
 m=11 T=  254 n= 54 res_all=1.5e-03 res_lag4=4.5e-06 max|k-g|(k<=20)=2.3e-05 front(1e-2)=54 max|k-tau|=0.99
```

```
 m=12 T=  299 n= 68 res_all=2.3e-03 res_lag4=6.4e-06 max|k-g|(k<=20)=1.7e-05 front(1e-2)=68 max|k-tau|=0.99
 m=13 T=  352 n= 85 res_all=1.9e-03 res_lag4=3.4e-06 max|k-g|(k<=20)=6.4e-06 front(1e-2)=85 max|k-tau|=0.99
 m=14 T=  414 n=107 res_all=2.5e-03 res_lag4=2.1e-06 max|k-g|(k<=20)=2.8e-06 front(1e-2)=107 max|k-tau|=0.99
 m=15 T=  487 n=133 res_all=2.1e-03 res_lag4=3.8e-06 max|k-g|(k<=20)=4.1e-06 front(1e-2)=133 max|k-tau|=0.99
 m=16 T=  572 n=165 res_all=1.9e-03 res_lag4=8.7e-06 max|k-g|(k<=20)=4.7e-06 front(1e-2)=165 max|k-tau|=0.99

Least-squares log-log fits over the computed stages m>=4 of cert7 (12 stages; regressions on the computed
range, not asymptotic laws; +- one standard error of the slope from the fit residuals):
  Linf ~ T^(1.81 +- 0.10), ||QB|| ~ T^(2.10 +- 0.06), sampled Banach margin ~ T^(-2.08 +- 0.06)

certV.json (verified stages; rho = radius of the verified l-inf ball, l = contraction bound, margin = rigorous
lower bound sampled over 36 radii, dist = rigorous bound of max_k<=n |x_k - gamma_k|, eps = bound for the zeta
     list):
 m= 1 T=   50 n=  4 rho=6.41e-21 l=1.2e-11 ||Q||=1.1e+02 margin>=2.6e-03 dist<=7.7e-07 eps<=2.1e-64
 m= 2 T=   59 n=  5 rho=1.28e-20 l=2.4e-12 ||Q||=6.9e+01 margin>=4.4e-03 dist<=9.0e-08 eps<=6.3e-64
 m= 3 T=   69 n=  7 rho=6.01e-20 l=9.6e-12 ||Q||=1.2e+02 margin>=1.3e-03 dist<=1.1e-06 eps<=1.8e-63
 m= 4 T=   81 n=  9 rho=2.81e-19 l=1.3e-11 ||Q||=1.7e+02 margin>=8.5e-04 dist<=1.2e-06 eps<=4.9e-63
 m= 5 T=   96 n= 12 rho=9.72e-19 l=1.5e-11 ||Q||=1.7e+02 margin>=5.3e-04 dist<=4.8e-07 eps<=1.5e-62
 m= 6 T=  113 n= 16 rho=3.80e-18 l=2.4e-11 ||Q||=2.2e+02 margin>=3.9e-04 dist<=2.5e-06 eps<=4.2e-62
 m= 7 T=  133 n= 20 rho=2.44e-17 l=4.8e-11 ||Q||=4.6e+02 margin>=1.4e-04 dist<=1.7e-05 eps<=1.2e-61
 m= 8 T=  156 n= 26 rho=5.92e-17 l=3.0e-11 ||Q||=4.2e+02 margin>=1.7e-04 dist<=3.8e-06 eps<=3.6e-61
 m= 9 T=  183 n= 34 rho=3.08e-16 l=1.6e-10 ||Q||=6.6e+02 margin>=8.3e-05 dist<=1.7e-05 eps<=1.1e-60
 m=10 T=  216 n= 43 rho=1.25e-15 l=1.3e-10 ||Q||=9.1e+02 margin>=5.1e-05 dist<=2.1e-05 eps<=3.1e-60
 m=11 T=  254 n= 54 rho=5.48e-15 l=1.2e-10 ||Q||=1.2e+03 margin>=3.6e-05 dist<=3.1e-05 eps<=9.4e-60
 m=12 T=  299 n= 68 rho=1.70e-14 l=3.3e-10 ||Q||=1.4e+03 margin>=2.7e-05 dist<=5.3e-05 eps<=2.8e-59
 m=13 T=  352 n= 85 rho=9.32e-14 l=1.8e-09 ||Q||=2.1e+03 margin>=1.5e-05 dist<=3.0e-04 eps<=8.4e-59
 m=14 T=  414 n=107 rho=3.28e-13 l=7.5e-09 ||Q||=2.5e+03 margin>=1.2e-05 dist<=3.8e-03 eps<=2.5e-58
 m=15 T=  487 n=133 rho=1.05e-12 l=3.0e-08 ||Q||=2.8e+03 margin>=8.9e-06 dist<=3.6e-02 eps<=7.5e-58
 m=16 T=  572 n=165 float polishing failed
```

E.2. out_horizonG.txt.

```
m= 1 T=   50 interior=  4 tests=  42 res_all=1.1e-04 res_lag4=nan |k-g| k<=5:4.4e-04 k<=10:4.4e-04 k<=20:4.4e
     -04 front(1e-6/1e-4/1e-2)=1/3/4 |k-tau|max=0.69 status=3 nfev=13 inversions=0
m= 2 T=   59 interior=  5 tests=  96 res_all=1.7e-04 res_lag4=nan |k-g| k<=5:3.1e-04 k<=10:3.1e-04 k<=20:3.1e
     -04 front(1e-6/1e-4/1e-2)=1/4/5 |k-tau|max=0.69 status=3 nfev=7 inversions=0
m= 3 T=   69 interior=  7 tests= 164 res_all=2.7e-04 res_lag4=nan |k-g| k<=5:9.4e-05 k<=10:1.6e-03 k<=20:1.6e
     -03 front(1e-6/1e-4/1e-2)=0/5/7 |k-tau|max=0.69 status=3 nfev=11 inversions=0
m= 4 T=   81 interior=  9 tests= 248 res_all=3.8e-04 res_lag4=nan |k-g| k<=5:5.0e-05 k<=10:3.1e-03 k<=20:3.1e
     -03 front(1e-6/1e-4/1e-2)=0/5/9 |k-tau|max=0.84 status=3 nfev=12 inversions=0
m= 5 T=   96 interior= 12 tests= 353 res_all=7.4e-04 res_lag4=3.0e-06 |k-g| k<=5:5.0e-05 k<=10:1.8e-04 k
     <=20:2.1e-03 front(1e-6/1e-4/1e-2)=0/6/12 |k-tau|max=0.84 status=3 nfev=7 inversions=0
m= 6 T=  113 interior= 16 tests= 482 res_all=5.7e-04 res_lag4=1.2e-06 |k-g| k<=5:3.2e-06 k<=10:3.7e-05 k
     <=20:5.9e-04 front(1e-6/1e-4/1e-2)=1/14/16 |k-tau|max=0.90 status=3 nfev=11 inversions=0
m= 7 T=  133 interior= 20 tests= 641 res_all=1.1e-03 res_lag4=2.8e-06 |k-g| k<=5:5.5e-06 k<=10:3.3e-05 k
     <=20:1.8e-03 front(1e-6/1e-4/1e-2)=0/14/20 |k-tau|max=0.90 status=3 nfev=11 inversions=0
m= 8 T=  156 interior= 26 tests= 836 res_all=1.8e-03 res_lag4=2.4e-06 |k-g| k<=5:2.9e-06 k<=10:1.6e-05 k
     <=20:2.5e-04 front(1e-6/1e-4/1e-2)=0/18/26 |k-tau|max=0.90 status=3 nfev=16 inversions=0
m= 9 T=  183 interior= 34 tests=1074 res_all=1.1e-03 res_lag4=4.2e-06 |k-g| k<=5:9.9e-06 k<=10:1.6e-05 k
     <=20:1.4e-04 front(1e-6/1e-4/1e-2)=3/18/33 |k-tau|max=0.96 status=3 nfev=16 inversions=0
m=10 T=  216 interior= 43 tests=1364 res_all=2.0e-03 res_lag4=3.9e-06 |k-g| k<=5:3.6e-06 k<=10:8.4e-06 k
     <=20:3.0e-05 front(1e-6/1e-4/1e-2)=2/26/43 |k-tau|max=0.99 status=3 nfev=18 inversions=0
m=11 T=  254 interior= 54 tests=1717 res_all=1.5e-03 res_lag4=4.5e-06 |k-g| k<=5:3.6e-06 k<=10:7.2e-06 k
     <=20:2.3e-05 front(1e-6/1e-4/1e-2)=0/34/54 |k-tau|max=0.99 status=3 nfev=13 inversions=0
m=12 T=  299 interior= 68 tests=2146 res_all=2.3e-03 res_lag4=6.4e-06 |k-g| k<=5:4.0e-06 k<=10:5.6e-06 k
     <=20:1.7e-05 front(1e-6/1e-4/1e-2)=0/37/68 |k-tau|max=0.99 status=3 nfev=24 inversions=0
m=13 T=  352 interior= 85 tests=2667 res_all=1.9e-03 res_lag4=3.4e-06 |k-g| k<=5:1.3e-06 k<=10:2.3e-06 k
     <=20:6.4e-06 front(1e-6/1e-4/1e-2)=3/53/85 |k-tau|max=0.99 status=3 nfev=15 inversions=0
m=14 T=  414 interior=107 tests=3298 res_all=2.5e-03 res_lag4=2.1e-06 |k-g| k<=5:6.0e-07 k<=10:1.5e-06 k
     <=20:2.8e-06 front(1e-6/1e-4/1e-2)=6/70/107 |k-tau|max=0.99 status=3 nfev=13 inversions=0
m=15 T=  487 interior=133 tests=4062 res_all=2.1e-03 res_lag4=3.8e-06 |k-g| k<=5:1.4e-06 k<=10:1.5e-06 k
     <=20:4.1e-06 front(1e-6/1e-4/1e-2)=1/90/133 |k-tau|max=0.99 status=3 nfev=16 inversions=0
m=16 T=  572 interior=165 tests=4985 res_all=1.9e-03 res_lag4=8.7e-06 |k-g| k<=5:1.5e-06 k<=10:3.7e-06 k
     <=20:4.7e-06 front(1e-6/1e-4/1e-2)=1/90/165 |k-tau|max=0.99 status=3 nfev=15 inversions=0
```

E.3. **out_cert7.txt.**

```
m= 1 T=   50 n=  4 Linf=4.1e+01 eta=1.7e-15 ||QB||=3.1e+01 ||I-QJ||=4.0e-16 FLOAT_PASS r=1.72e-15 k(r)=5.45e
     -14 sampled margin=7.0e-03 rho(all tests<=m)=7.7e-08 | max|x0-tau|=0.69
m= 2 T=   59 n=  5 Linf=2.8e+01 eta=1.3e-15 ||QB||=2.2e+01 ||I-QJ||=7.0e-16 FLOAT_PASS r=1.31e-15 k(r)=2.89e
     -14 sampled margin=1.0e-02 rho(all tests<=m)=1.1e-07 | max|x0-tau|=0.69
m= 3 T=   69 n=  7 Linf=3.8e+01 eta=1.8e-15 ||QB||=2.9e+01 ||I-QJ||=1.4e-15 FLOAT_PASS r=1.76e-15 k(r)=5.22e
     -14 sampled margin=7.1e-03 rho(all tests<=m)=1.6e-07 | max|x0-tau|=0.69
m= 4 T=   81 n=  9 Linf=5.0e+01 eta=4.3e-15 ||QB||=7.3e+01 ||I-QJ||=2.3e-15 FLOAT_PASS r=4.30e-15 k(r)=3.17e
     -13 sampled margin=3.1e-03 rho(all tests<=m)=2.6e-07 | max|x0-tau|=0.84
m= 5 T=   96 n= 12 Linf=6.7e+01 eta=2.0e-14 ||QB||=1.4e+02 ||I-QJ||=4.0e-15 FLOAT_PASS r=1.99e-14 k(r)=2.76e
     -12 sampled margin=1.7e-03 rho(all tests<=m)=9.4e-08 | max|x0-tau|=0.84
m= 6 T=  113 n= 16 Linf=1.1e+02 eta=1.0e-14 ||QB||=1.4e+02 ||I-QJ||=1.4e-14 FLOAT_PASS r=1.03e-14 k(r)=1.42e
     -12 sampled margin=1.7e-03 rho(all tests<=m)=3.5e-07 | max|x0-tau|=0.90
m= 7 T=  133 n= 20 Linf=1.8e+02 eta=3.4e-14 ||QB||=2.0e+02 ||I-QJ||=3.2e-14 FLOAT_PASS r=3.36e-14 k(r)=6.67e
     -12 sampled margin=1.2e-03 rho(all tests<=m)=1.0e-06 | max|x0-tau|=0.90
m= 8 T=  156 n= 26 Linf=1.5e+02 eta=2.2e-14 ||QB||=3.2e+02 ||I-QJ||=2.1e-14 FLOAT_PASS r=2.19e-14 k(r)=7.09e
     -12 sampled margin=7.5e-04 rho(all tests<=m)=9.6e-07 | max|x0-tau|=0.90
m= 9 T=  183 n= 34 Linf=1.6e+02 eta=2.5e-14 ||QB||=3.7e+02 ||I-QJ||=3.1e-14 FLOAT_PASS r=2.53e-14 k(r)=9.44e
     -12 sampled margin=6.6e-04 rho(all tests<=m)=1.1e-06 | max|x0-tau|=0.99
m=10 T=  216 n= 43 Linf=3.4e+02 eta=4.0e-14 ||QB||=5.5e+02 ||I-QJ||=4.2e-14 FLOAT_PASS r=4.02e-14 k(r)=2.20e
     -11 sampled margin=4.5e-04 rho(all tests<=m)=2.3e-06 | max|x0-tau|=0.99
m=11 T=  254 n= 54 Linf=4.9e+02 eta=6.8e-14 ||QB||=9.2e+02 ||I-QJ||=8.3e-14 FLOAT_PASS r=6.75e-14 k(r)=6.25e
     -11 sampled margin=2.6e-04 rho(all tests<=m)=3.2e-06 | max|x0-tau|=0.99
m=12 T=  299 n= 68 Linf=4.6e+02 eta=1.8e-13 ||QB||=1.1e+03 ||I-QJ||=1.2e-13 FLOAT_PASS r=1.84e-13 k(r)=2.05e
     -10 sampled margin=2.2e-04 rho(all tests<=m)=5.5e-06 | max|x0-tau|=0.99
m=13 T=  352 n= 85 Linf=6.8e+02 eta=2.2e-13 ||QB||=1.5e+03 ||I-QJ||=1.9e-13 FLOAT_PASS r=2.21e-13 k(r)=3.32e
     -10 sampled margin=1.6e-04 rho(all tests<=m)=3.6e-05 | max|x0-tau|=0.99
m=14 T=  414 n=107 Linf=1.0e+03 eta=2.2e-13 ||QB||=2.4e+03 ||I-QJ||=3.3e-13 FLOAT_PASS r=2.21e-13 k(r)=5.29e
     -10 sampled margin=1.0e-04 rho(all tests<=m)=3.7e-04 | max|x0-tau|=0.99
m=15 T=  487 n=133 Linf=1.6e+03 eta=5.3e-13 ||QB||=3.6e+03 ||I-QJ||=6.3e-13 FLOAT_PASS r=5.35e-13 k(r)=1.92e
     -09 sampled margin=6.7e-05 rho(all tests<=m)=3.3e-03 | max|x0-tau|=0.99
m=16 Newton polishing failed (singular square Jacobian)
```

E.4. **out_cert7_gamma.txt.**

```
m= 1 T=   50 n=  4  max|x0-gamma| over k<=4: 7.4e-07
m= 2 T=   59 n=  5  max|x0-gamma| over k<=5: 2.2e-07
m= 3 T=   69 n=  7  max|x0-gamma| over k<=7: 1.0e-06
m= 4 T=   81 n=  9  max|x0-gamma| over k<=9: 1.1e-06
m= 5 T=   96 n= 12  max|x0-gamma| over k<=12: 7.5e-07
m= 6 T=  113 n= 16  max|x0-gamma| over k<=16: 2.6e-06
m= 7 T=  133 n= 20  max|x0-gamma| over k<=20: 2.0e-05
m= 8 T=  156 n= 26  max|x0-gamma| over k<=26: 1.3e-05
m= 9 T=  183 n= 34  max|x0-gamma| over k<=34: 8.7e-06
m=10 T=  216 n= 43  max|x0-gamma| over k<=43: 1.8e-05
m=11 T=  254 n= 54  max|x0-gamma| over k<=54: 4.5e-05
m=12 T=  299 n= 68  max|x0-gamma| over k<=68: 4.5e-05
m=13 T=  352 n= 85  max|x0-gamma| over k<=85: 5.0e-04
m=14 T=  414 n=107  max|x0-gamma| over k<=107: 5.4e-03
m=15 T=  487 n=133  max|x0-gamma| over k<=133: 4.3e-02
```

E.5. **out_entry.txt.**

```
m= 2 T=   59 band= 1 | band-only: eta=1.7e-03 sampled margin=8.8e-02 FLOAT_PASS | full: eta=5.5e-03 sampled
     margin=1.0e-02 FLOAT_PASS | Gonek band err=5.2e-03
m= 3 T=   69 band= 2 | band-only: eta=1.5e-02 sampled margin=3.6e-02 FLOAT_PASS | full: eta=1.1e-02 sampled
     margin=7.1e-03 no pass | Gonek band err=5.7e-03
m= 4 T=   81 band= 2 | band-only: eta=1.1e-02 sampled margin=4.4e-02 FLOAT_PASS | full: eta=1.5e-02 sampled
     margin=3.3e-03 no pass | Gonek band err=1.7e-02
m= 5 T=   96 band= 3 | band-only: eta=1.1e-02 sampled margin=3.1e-02 FLOAT_PASS | full: eta=1.2e-02 sampled
     margin=1.8e-03 no pass | Gonek band err=1.0e-02
m= 6 T=  113 band= 4 | band-only: eta=3.1e-02 sampled margin=2.7e-02 no pass | full: eta=1.8e-02 sampled
     margin=1.3e-03 no pass | Gonek band err=2.4e-02
m= 7 T=  133 band= 4 | band-only: eta=3.2e-02 sampled margin=2.0e-02 no pass | full: eta=3.3e-02 sampled
     margin=1.2e-03 no pass | Gonek band err=4.4e-02
```

```
m= 8 T=  156 band= 6 | band-only: eta=7.0e-02 sampled margin=1.9e-02 no pass | full: eta=4.7e-02 sampled
     margin=8.8e-04 no pass | Gonek band err=4.0e-02
m= 9 T=  183 band= 8 | band-only: eta=1.5e-01 sampled margin=1.7e-02 no pass | full: eta=9.3e-02 sampled
     margin=5.5e-04 no pass | Gonek band err=8.5e-02
m=10 T=  216 band= 9 | band-only: eta=2.6e-01 sampled margin=8.0e-03 no pass | full: eta=1.1e-01 sampled
     margin=3.8e-04 no pass | Gonek band err=5.1e-02
m=11 T=  254 band=11 | band-only: eta=1.6e-01 sampled margin=3.2e-03 no pass | full: eta=2.9e-01 sampled
     margin=2.9e-04 no pass | Gonek band err=6.7e-02
singular at 11
```

E.6. `out_entry2.txt`.

```
m= 2 T=   59 band= 1 n=  5 | band linf: |QF|=1.7e-03 margin=8.8e-02 |Q|=1.7e+01 PASS | band l2: |QF|=1.7e-03
     margin=8.8e-02 |Q|=1.7e+01 PASS | full linf: |QF|=5.5e-03 margin=1.0e-02 |Q|=2.8e+01 PASS | full l2: |QF
     |=6.3e-03 margin=1.3e-02 |Q|=1.7e+01 PASS
m= 3 T=   69 band= 2 n=  7 | band linf: |QF|=1.5e-02 margin=3.6e-02 |Q|=3.4e+01 PASS | band l2: |QF|=1.5e-02
     margin=4.5e-02 |Q|=2.4e+01 PASS | full linf: |QF|=1.1e-02 margin=7.2e-03 |Q|=3.8e+01 no | full l2: |QF
     |=1.1e-02 margin=9.8e-03 |Q|=1.7e+01 no
m= 4 T=   81 band= 2 n=  9 | band linf: |QF|=1.1e-02 margin=4.4e-02 |Q|=3.4e+01 PASS | band l2: |QF|=1.1e-02
     margin=4.8e-02 |Q|=2.4e+01 PASS | full linf: |QF|=1.5e-02 margin=3.3e-03 |Q|=5.0e+01 no | full l2: |QF
     |=1.6e-02 margin=6.9e-03 |Q|=2.1e+01 no
m= 5 T=   96 band= 3 n= 12 | band linf: |QF|=1.1e-02 margin=3.2e-02 |Q|=4.4e+01 PASS | band l2: |QF|=1.3e-02
     margin=3.8e-02 |Q|=2.8e+01 PASS | full linf: |QF|=1.2e-02 margin=1.7e-03 |Q|=6.7e+01 no | full l2: |QF
     |=1.7e-02 margin=3.1e-03 |Q|=2.5e+01 no
m= 6 T=  113 band= 4 n= 16 | band linf: |QF|=3.1e-02 margin=2.7e-02 |Q|=5.3e+01 no | band l2: |QF|=4.9e-02
     margin=3.6e-02 |Q|=3.0e+01 no | full linf: |QF|=1.8e-02 margin=1.3e-03 |Q|=1.2e+02 no | full l2: |QF|=3.1
     e-02 margin=2.6e-03 |Q|=4.7e+01 no
m= 7 T=  133 band= 4 n= 20 | band linf: |QF|=3.2e-02 margin=2.0e-02 |Q|=6.4e+01 no | band l2: |QF|=4.8e-02
     margin=2.9e-02 |Q|=4.3e+01 no | full linf: |QF|=3.3e-02 margin=1.2e-03 |Q|=1.4e+02 no | full l2: |QF|=4.7
     e-02 margin=2.4e-03 |Q|=5.1e+01 no
m= 8 T=  156 band= 6 n= 26 | band linf: |QF|=7.0e-02 margin=1.9e-02 |Q|=7.9e+01 no | band l2: |QF|=1.1e-01
     margin=2.3e-02 |Q|=4.1e+01 no | full linf: |QF|=4.7e-02 margin=8.8e-04 |Q|=1.3e+02 no | full l2: |QF|=7.5
     e-02 margin=1.8e-03 |Q|=4.7e+01 no
m= 9 T=  183 band= 8 n= 34 | band linf: |QF|=1.5e-01 margin=1.7e-02 |Q|=8.7e+01 no | band l2: |QF|=2.1e-01
     margin=2.1e-02 |Q|=3.7e+01 no | full linf: |QF|=9.3e-02 margin=5.4e-04 |Q|=2.2e+02 no | full l2: |QF|=1.5
     e-01 margin=1.1e-03 |Q|=7.8e+01 no
m=10 T=  216 band= 9 n= 43 | band linf: |QF|=2.6e-01 margin=8.0e-03 |Q|=1.5e+02 no | band l2: |QF|=3.3e-01
     margin=1.6e-02 |Q|=7.2e+01 no | full linf: |QF|=1.1e-01 margin=3.8e-04 |Q|=3.0e+02 no | full l2: |QF|=1.8
     e-01 margin=9.0e-04 |Q|=7.5e+01 no
m=11 T=  254 band=11 n= 54 | band linf: |QF|=1.6e-01 margin=3.2e-03 |Q|=2.7e+02 no | band l2: |QF|=2.9e-01
     margin=6.6e-03 |Q|=1.4e+02 no | full linf: |QF|=2.9e-01 margin=2.9e-04 |Q|=4.2e+02 no | full l2: |QF|=5.5
     e-01 margin=8.2e-04 |Q|=1.1e+02 no
singular at 11
```

E.7. `out_certV.txt`.

```
frozen list: increasing, max_k<=60000 |kappa^H_k - tau_k| <= 0.993707, kappa^H_K < tau_(K+1)  (0s)
m= 1 T=   50 n=  4 h=0.0980 | VERIFIED rho=6.41e-21 l(rho)=1.25e-11 eta=1.60e-21 ||I-QJ||<=1.2e-11 ||Q||=1.1e
     +02 margin>=2.6e-03 | order=ok mingap=2.510 tube<=0.685 | max|x-gamma|<=7.7e-07 (k<=20: 7.7e-07) | float
     residual all tests<=m 2.4e-08 | zeta-list bound eps<=2.1e-64 | 1s
m= 2 T=   59 n=  5 h=0.0833 | VERIFIED rho=1.28e-20 l(rho)=2.38e-12 eta=3.19e-21 ||I-QJ||<=2.4e-12 ||Q||=6.9e
     +01 margin>=4.4e-03 | order=ok mingap=2.510 tube<=0.689 | max|x-gamma|<=9.0e-08 (k<=20: 9.0e-08) | float
     residual all tests<=m 3.3e-08 | zeta-list bound eps<=6.3e-64 | 1s
m= 3 T=   69 n=  7 h=0.0708 | VERIFIED rho=6.01e-20 l(rho)=9.63e-12 eta=1.50e-20 ||I-QJ||<=9.6e-12 ||Q||=1.2e
     +02 margin>=1.3e-03 | order=ok mingap=2.408 tube<=0.689 | max|x-gamma|<=1.1e-06 (k<=20: 1.1e-06) | float
     residual all tests<=m 5.4e-08 | zeta-list bound eps<=1.8e-63 | 2s
m= 4 T=   81 n=  9 h=0.0602 | VERIFIED rho=2.81e-19 l(rho)=1.28e-11 eta=7.02e-20 ||I-QJ||<=1.3e-11 ||Q||=1.7e
     +02 margin>=8.5e-04 | order=ok mingap=1.769 tube<=0.840 | max|x-gamma|<=1.2e-06 (k<=20: 1.2e-06) | float
     residual all tests<=m 7.9e-08 | zeta-list bound eps<=4.9e-63 | 2s
m= 5 T=   96 n= 12 h=0.0511 | VERIFIED rho=9.72e-19 l(rho)=1.46e-11 eta=2.43e-19 ||I-QJ||<=1.5e-11 ||Q||=1.7e
     +02 margin>=5.3e-04 | order=ok mingap=1.769 tube<=0.840 | max|x-gamma|<=4.8e-07 (k<=20: 4.8e-07) | float
     residual all tests<=m 3.0e-08 | zeta-list bound eps<=1.5e-62 | 4s
m= 6 T=  113 n= 16 h=0.0435 | VERIFIED rho=3.80e-18 l(rho)=2.43e-11 eta=9.51e-19 ||I-QJ||<=2.4e-11 ||Q||=2.2e
     +02 margin>=3.9e-04 | order=ok mingap=1.485 tube<=0.902 | max|x-gamma|<=2.5e-06 (k<=20: 2.5e-06) | float
     residual all tests<=m 1.0e-07 | zeta-list bound eps<=4.2e-62 | 4s
```

```
m= 7 T=  133 n= 20 h=0.0370 | VERIFIED rho=2.44e-17 l(rho)=4.84e-11 eta=6.09e-18 ||I-QJ||<=4.8e-11 ||Q||=4.6e
     +02 margin>=1.4e-04 | order=ok mingap=1.440 tube<=0.902 | max|x-gamma|<=1.7e-05 (k<=20: 1.7e-05) | float
     residual all tests<=m 3.6e-07 | zeta-list bound eps<=1.2e-61 | 5s
m= 8 T=  156 n= 26 h=0.0314 | VERIFIED rho=5.92e-17 l(rho)=2.96e-11 eta=1.48e-17 ||I-QJ||<=2.9e-11 ||Q||=4.2e
     +02 margin>=1.7e-04 | order=ok mingap=1.384 tube<=0.902 | max|x-gamma|<=3.8e-06 (k<=20: 1.4e-06) | float
     residual all tests<=m 2.0e-07 | zeta-list bound eps<=3.6e-61 | 6s
m= 9 T=  183 n= 34 h=0.0267 | VERIFIED rho=3.08e-16 l(rho)=1.63e-10 eta=7.69e-17 ||I-QJ||<=1.6e-10 ||Q||=6.6e
     +02 margin>=8.3e-05 | order=ok mingap=0.845 tube<=0.994 | max|x-gamma|<=1.7e-05 (k<=20: 2.4e-06) | float
     residual all tests<=m 7.6e-07 | zeta-list bound eps<=1.1e-60 | 9s
m=10 T=  216 n= 43 h=0.0227 | VERIFIED rho=1.25e-15 l(rho)=1.29e-10 eta=3.12e-16 ||I-QJ||<=1.2e-10 ||Q||=9.1e
     +02 margin>=5.1e-05 | order=ok mingap=0.845 tube<=0.994 | max|x-gamma|<=2.1e-05 (k<=20: 2.9e-06) | float
     residual all tests<=m 7.7e-07 | zeta-list bound eps<=3.1e-60 | 10s
m=11 T=  254 n= 54 h=0.0193 | VERIFIED rho=5.48e-15 l(rho)=1.24e-10 eta=1.37e-15 ||I-QJ||<=8.7e-11 ||Q||=1.2e
     +03 margin>=3.6e-05 | order=ok mingap=0.845 tube<=0.994 | max|x-gamma|<=3.1e-05 (k<=20: 1.8e-06) | float
     residual all tests<=m 1.2e-06 | zeta-list bound eps<=9.4e-60 | 13s
m=12 T=  299 n= 68 h=0.0164 | VERIFIED rho=1.70e-14 l(rho)=3.34e-10 eta=4.25e-15 ||I-QJ||<=1.8e-10 ||Q||=1.4e
     +03 margin>=2.7e-05 | order=ok mingap=0.817 tube<=0.994 | max|x-gamma|<=5.3e-05 (k<=20: 5.1e-06) | float
     residual all tests<=m 1.8e-06 | zeta-list bound eps<=2.8e-59 | 17s
m=13 T=  352 n= 85 h=0.0139 | VERIFIED rho=9.32e-14 l(rho)=1.76e-09 eta=2.33e-14 ||I-QJ||<=2.6e-10 ||Q||=2.1e
     +03 margin>=1.5e-05 | order=ok mingap=0.724 tube<=0.994 | max|x-gamma|<=3.0e-04 (k<=20: 3.6e-06) | float
     residual all tests<=m 4.1e-06 | zeta-list bound eps<=8.4e-59 | 22s
m=14 T=  414 n=107 h=0.0118 | VERIFIED rho=3.28e-13 l(rho)=7.51e-09 eta=8.20e-14 ||I-QJ||<=3.7e-10 ||Q||=2.5e
     +03 margin>=1.2e-05 | order=ok mingap=0.716 tube<=0.994 | max|x-gamma|<=3.8e-03 (k<=20: 5.7e-05) | float
     residual all tests<=m 5.3e-05 | zeta-list bound eps<=2.5e-58 | 27s
m=15 T=  487 n=133 h=0.0101 | VERIFIED rho=1.05e-12 l(rho)=2.99e-08 eta=2.62e-13 ||I-QJ||<=7.1e-10 ||Q||=2.8e
     +03 margin>=8.9e-06 | order=ok mingap=0.710 tube<=0.994 | max|x-gamma|<=3.6e-02 (k<=20: 7.6e-04) | float
     residual all tests<=m 7.2e-04 | zeta-list bound eps<=7.5e-58 | 37s
m=16 T=  572 n=165 float Newton polishing failed (max|F|=1.5e-01, cond=inf)
total 162s
```

E.8. `out_certG.txt`.

```
m= 1 T=   50 interior=  4 L=4.16e+01 H=1.23e+01 eta_sel=1.31e-03 eta_all=1.31e-03 A=2L^2H eta=5.61e+01  2L*eta
     =1.09e-01
m= 2 T=   59 interior=  5 L=2.83e+01 H=1.42e+01 eta_sel=1.99e-03 eta_all=2.12e-03 A=2L^2H eta=4.52e+01  2L*eta
     =1.13e-01
m= 3 T=   69 interior=  7 L=4.01e+01 H=1.96e+01 eta_sel=2.48e-03 eta_all=2.53e-03 A=2L^2H eta=1.56e+02  2L*eta
     =1.99e-01
m= 4 T=   81 interior=  9 L=5.63e+01 H=2.26e+01 eta_sel=2.81e-03 eta_all=3.19e-03 A=2L^2H eta=4.03e+02  2L*eta
     =3.16e-01
m= 5 T=   96 interior= 12 L=6.77e+01 H=2.80e+01 eta_sel=4.47e-03 eta_all=4.47e-03 A=2L^2H eta=1.15e+03  2L*eta
     =6.05e-01
m= 6 T=  113 interior= 16 L=9.78e+01 H=3.32e+01 eta_sel=4.61e-03 eta_all=5.00e-03 A=2L^2H eta=2.93e+03  2L*eta
     =9.02e-01
m= 7 T=  133 interior= 20 L=1.41e+02 H=3.97e+01 eta_sel=4.56e-03 eta_all=6.93e-03 A=2L^2H eta=7.17e+03  2L*eta
     =1.28e+00
m= 8 T=  156 interior= 26 L=1.84e+02 H=5.16e+01 eta_sel=8.89e-03 eta_all=8.89e-03 A=2L^2H eta=3.09e+04  2L*eta
     =3.26e+00
m= 9 T=  183 interior= 34 L=2.60e+02 H=6.45e+01 eta_sel=9.13e-03 eta_all=1.01e-02 A=2L^2H eta=7.97e+04  2L*eta
     =4.75e+00
m=10 T=  216 interior= 43 L=3.59e+02 H=8.53e+01 eta_sel=1.45e-02 eta_all=1.45e-02 A=2L^2H eta=3.17e+05  2L*eta
     =1.04e+01
m=11 T=  254 interior= 54 L=4.69e+02 H=1.02e+02 eta_sel=1.51e-02 eta_all=1.83e-02 A=2L^2H eta=6.80e+05  2L*eta
     =1.41e+01
m=12 T=  299 interior= 68 L=7.53e+02 H=1.29e+02 eta_sel=1.66e-02 eta_all=2.11e-02 A=2L^2H eta=2.43e+06  2L*eta
     =2.50e+01
m=13 T=  352 interior= 85 L=6.45e+02 H=1.61e+02 eta_sel=2.36e-02 eta_all=2.61e-02 A=2L^2H eta=3.16e+06  2L*eta
     =3.04e+01
m=14 T=  414 interior=107 L=1.02e+03 H=2.03e+02 eta_sel=2.83e-02 eta_all=2.83e-02 A=2L^2H eta=1.20e+07  2L*eta
     =5.79e+01
```

E.9. `out_cert3.txt`.

```
m= 1 T=   50 n=  4 tests=  42 beta=  8.38 K=6.36e+00 delta=8.85e-03 h=4.72e-01 |kG-g|=8.75e-03 after1GN=4.77e
     -04
```

```
m= 2 T=   59 n=  5 tests=  96 beta=  5.19 K=9.92e+00 delta=1.00e-02 h=5.16e-01 |kG-g|=1.02e-02 after1GN=3.30e
    -04
m= 3 T=   69 n=  7 tests= 164 beta=  6.83 K=1.31e+01 delta=1.27e-02 h=1.13e+00 |kG-g|=1.18e-02 after1GN=1.65e
    -03
m= 4 T=   81 n=  9 tests= 248 beta=  6.99 K=1.59e+01 delta=2.03e-02 h=2.25e+00 |kG-g|=2.26e-02 after1GN=2.98e
    -03
m= 5 T=   96 n= 12 tests= 353 beta=  7.16 K=1.87e+01 delta=2.47e-02 h=3.31e+00 |kG-g|=2.50e-02 after1GN=2.10e
    -03
m= 6 T=  113 n= 16 tests= 482 beta=  8.00 K=2.12e+01 delta=4.45e-02 h=7.55e+00 |kG-g|=4.51e-02 after1GN=7.97e
    -04
m= 7 T=  133 n= 20 tests= 641 beta=  8.18 K=2.36e+01 delta=7.09e-02 h=1.37e+01 |kG-g|=7.14e-02 after1GN=9.39e
    -04
m= 8 T=  156 n= 26 tests= 836 beta=  8.06 K=2.59e+01 delta=9.47e-02 h=1.98e+01 |kG-g|=9.58e-02 after1GN=5.21e
    -03
m= 9 T=  183 n= 34 tests=1074 beta=  9.83 K=2.80e+01 delta=1.60e-01 h=4.42e+01 |kG-g|=1.53e-01 after1GN=4.14e
    -02
m=10 T=  216 n= 43 tests=1364 beta= 10.26 K=3.00e+01 delta=1.72e-01 h=5.28e+01 |kG-g|=1.80e-01 after1GN=1.60e
    -02
m=11 T=  254 n= 54 tests=1717 beta= 17.15 K=3.18e+01 delta=2.33e-01 h=1.27e+02 |kG-g|=2.43e-01 after1GN=2.26e
    -02
m=12 T=  299 n= 68 tests=2146 beta= 15.39 K=3.36e+01 delta=2.66e-01 h=1.37e+02 |kG-g|=2.79e-01 after1GN=2.68e
    -02
m=13 T=  352 n= 85 tests=2667 beta= 10.99 K=3.52e+01 delta=3.38e-01 h=1.31e+02 |kG-g|=3.55e-01 after1GN=3.56e
    -02
m=14 T=  414 n=107 tests=3298 beta= 15.67 K=3.67e+01 delta=4.05e-01 h=2.33e+02 |kG-g|=4.27e-01 after1GN=5.19e
    -02
```

E.10. **out_gonek.txt and out_gonek_fast.txt.**

```
clock : mean|.-g|/sp=0.187 max=0.506  #>0.25sp=50
Gonek X= 3 (offset -3): mean|.-g|/sp=0.048 max=0.184 #>0.25sp=0  k=34: 110.923 (g=111.030)
Gonek X= 5 (offset -3): mean|.-g|/sp=0.029 max=0.081 #>0.25sp=0  k=34: 111.034 (g=111.030)
Gonek X= 8 (offset -3): mean|.-g|/sp=0.019 max=0.053 #>0.25sp=0  k=34: 111.071 (g=111.030)
Gonek X=12 (offset -3): mean|.-g|/sp=0.014 max=0.047 #>0.25sp=0  k=34: 111.092 (g=111.030)
Gonek X=20 (offset -3): mean|.-g|/sp=0.008 max=0.035 #>0.25sp=0  k=34: 111.075 (g=111.030)
```

```
X=8.0: grid h=0.005, max |Delta level| per grid cell = 0.010, cells with a jump of more than one level: 0
983 [14.13395523 21.02631924 25.00355926] 1399.7912834339318
```

E.11. **out_laplace_check.txt.**

```
u=0.1  k_ref=-1.6225522892294519278916331593609088459137814857449  k_mine
    =-1.6225522892294519278916331593609088459137814857449  K
    |=-1.622552289229451927891633159360908845913781485744 9
u=1.0  k_ref=0.7768942607932535817073375768388701722826884980679 7  k_mine
    =0.7768942607932535817073375768388701722826884980679 7  K
    |=0.776894260793253581707337576838870172282688498067 97
u=2.5  k_ref=1.7441997038954705252460103768783663870122500505333  k_mine
    =1.7441997038954705252460103768783663870122500505333  K
    |=1.744199703895470525246010376878366387012250050533 3
s=2.0  (1/2)G'/G=0.319513612312266741312515139641134714327636462786 19  L[K
    ]=0.319513612312266741312515269107123356130616284856 09
s=1.3  (1/2)G'/G=1.422512214600398573948309601670381704130033469369 7  L[K
    ]=1.422512214600398573948309731136370345995730864934 8
s=(2.0 + 5.0j)  (1/2)G'/G=(-0.000004483925696515754762541200984984498851920143985085 5 +
    0.160338457354031730177741946245978073485991510733 49j)  L[K
    ]=(-0.000004483925696515754762411734996342695870171439857178 +
    0.160338457354031730177741946245978073038010423445 5j)
s=(1.5 - 3.0j)  (1/2)G'/G=(-0.092240558736457654753515338039303026644761224786081 -
    0.054136617330789788303900306539987383185752403110009j)  L[K
    ]=(-0.092240558736457654753515208573314384796983337328152 -
    0.054136617330789788303900306539987382916962760406523j)
```

Appendix F. Endpoint-only closure, interpolation tests, and autonomous residual questions

This appendix records the interpolation and finite-section tests. It does not alter any theorem above. Its purpose is to separate the statements that are exact but equivalent to the final step from the one clock-only subproblem that remains autonomous.

F.1. **Three interpolations and their status.** There are three natural ways to interpolate between the midpoint clock and the Riemann zeta function.

The ordinate interpolation. The family ζ_λ of Sections 11–15 moves the ordinates while keeping every nontrivial zero on $\Re s = \frac{1}{2}$. It is an exact analytic family, but for every $0 < \lambda \le 1$ its continuous prime measure is not in the ordinary Dirichlet category. Thus the property that is needed at the endpoint is not preserved along the path; this is exactly the role of condition (D).

The exact-atomic interpolation. One may instead ask for real ordinates κ_k and masses a_n supported at $u = \log n$ throughout an interpolation. Finite sections of this problem were tested numerically. The endpoints can be resolved, whereas the linear interpolation of the target distribution develops a persistent interior residual in the tested sections. This is evidence against that particular interpolation, not a proof that no infinite exact-atomic path exists. In any case the existence of such a path is not supplied by the results of this paper.

The additive interpolation of functions. There is an exact elementary interpolation

$$\mathcal{G}_\varepsilon(s) := (1-\varepsilon)\zeta_{\mathrm{RVM}}(s) + \varepsilon\zeta(s), \qquad 0 \le \varepsilon \le 1. \tag{F.1}$$

Both summands satisfy the same functional equation and reality symmetry, so the same is true of $\mathcal{G}_\varepsilon$. This family exists for every ε, but it does not create a new control quantity: it is a level-set representation of the already defined quotient $\mathcal{Q}_{\mathrm{RVM}} = \zeta/\zeta_{\mathrm{RVM}}$.

Proposition F.1 (The additive family is a level-set reformulation)**.** *Away from the zeros of* ζ_{RVM}*,*

$$\mathcal{G}_\varepsilon(s) = \zeta_{\mathrm{RVM}}(s)\big((1-\varepsilon) + \varepsilon\mathcal{Q}_{\mathrm{RVM}}(s)\big). \tag{F.2}$$

Hence for $0 < \varepsilon < 1$ *every zero of* $\mathcal{G}_\varepsilon$ *which is not a zero of* ζ_{RVM} *lies on the level set*

$$\mathcal{Q}_{\mathrm{RVM}}(s) = -\frac{1-\varepsilon}{\varepsilon}. \tag{F.3}$$

On the critical line $e^{i\vartheta(t)}\mathcal{G}_\varepsilon(\frac{1}{2}+it)$ *is real. At* $\varepsilon = 1$ *the statement that all its nontrivial zeros lie on the critical line is exactly RH.*

Proof. Equation (F.2) is the definition of $\mathcal{Q}_{\mathrm{RVM}}$. The functional equation and the reality statement follow by linearity from those of the two summands. The last assertion is $\mathcal{G}_1 = \zeta$. □

Thus the additive family is useful for visualizing the zero motion, but without an additional monotonicity or flow law it is circular as a route to RH: the terminal level is precisely the original zero problem.

F.2. **The clock as the first transition term.** The transition theorem gives the following $1+\chi$ form on every closed subband of $\frac{1}{2} < \Re s < 1$:

$$\zeta_{\mathrm{RVM}}(s) = \varpi_{\mathrm{RVM}}(1+\chi(s))(1+O(|t|^{-1})). \tag{F.4}$$

Independently, on $s = \frac{1}{2} + it$, where $\chi(s) = e^{-2i\vartheta(t)}$, the exact trigonometric identity

$$e^{i\vartheta(t)}(1+\chi(\tfrac{1}{2}+it)) = 2\cos\vartheta(t). \tag{F.5}$$

Consequently the midpoint clock is asymptotically the zero set selected by the $n = 1$ term of the Hardy–Littlewood/Riemann–Siegel approximate functional equation. This is a structural interpretation of the transition theorem, not an exact identification: the $O(1/t)$ term is visible already in the first ordinates.

For a finite Dirichlet polynomial

$$\mathsf{P}_{\varepsilon,M}(s) := 1 + \varepsilon \sum_{2\le n\le M} n^{-s}, \qquad \mathsf{R}_{\varepsilon,M}(s) := \mathsf{P}_{\varepsilon,M}(s) + \chi(s)\mathsf{P}_{\varepsilon,M}(1-s), \tag{F.6}$$

the functional equation is exact. On the critical line,

$$e^{i\vartheta(t)}\mathsf{R}_{\varepsilon,M}(\tfrac{1}{2}+it) = 2\Re\Big(e^{i\vartheta(t)}\mathsf{P}_{\varepsilon,M}(\tfrac{1}{2}+it)\Big). \tag{F.7}$$

At Gram points the $n=1$ term fixes the sign whenever

$$\varepsilon \sum_{2\le n\le M} n^{-1/2} < 1, \tag{F.8}$$

so in every interval where ϑ is monotone there is a critical-line zero between consecutive Gram points. This dominance statement is only sufficient; it does not count all zeros of $\mathsf{R}_{\varepsilon,M}$ and is not used elsewhere in the paper.

F.3. **What remains genuinely live.** After these tests the status is narrower than the programme suggested by the interpolation pictures.

(L1) *The endpoint problem remains the endpoint problem.* Proving that the cell defect vanishes at $\lambda = 0$, equivalently proving (D), remains the only step that closes the main route. The additive family and the exact-atomic interpolation do not simplify this statement.

(L2) *The clock-only injectivity problem is autonomous.* The one subproblem that does not contain the genuine ordinates is the injectivity of the linearized rung system at the midpoint clock. In the meromorphic form of Remark 16.8, the target statement is: if a bounded real sequence $(\dot\kappa_k)$ and coefficients $\dot{\mathsf{a}}_n = O_\varepsilon(n^{-1/2+\varepsilon})$ give a Dirichlet series H satisfying

$$\mathsf{H}(z) = \nu_8 + \sum_{k\ge1} \dot\kappa_k \left(\frac{\tau_k}{z^2+\tau_k^2} - \frac{1}{\tau_k}\right) = \sum_{n\ge2} \frac{\dot{\mathsf{a}}_n}{\log n}\, n^{-z} \tag{F.9}$$

through a common right half-plane, under the growth hypotheses of the linearization, then $\mathsf{H} \equiv 0$. Theorem 16.7 proves the analogous conclusion at an actual solution, while Remark 16.8 explains why it is not automatic for a general list. Equation (F.9) is therefore a closed clock-only rigidity problem, naturally related to Fourier crystalline uniqueness, but by itself it does not prove (D).

(L3) *Bounded finite sections are a numerical existence programme, not yet an analytic route.* Direct finite sections of the endpoint rung system can be solved while constraining both the ordinate displacement and the free masses. If one could prove, for an exhausting nested family of tests, a uniform boundedness and residual-to-zero statement and then pass to a distributional limit, Theorem 16.6 would identify the limit. Such a theorem would already be an existence theorem for the final system and is therefore not a shortcut around (D).

In particular, the individual displacements $\gamma_k - \tau_k$ are not autonomous data. They are used below only for post-solution numerical diagnostics. None of the live statement (F.9), nor the definition of the bounded finite-section problem, requires them as input.

F.4. **A priori bounded displacement for exact rung solutions.** The following statement is independent of the existence of an interior exact-atomic interpolation. It says only that, if an exact real solution exists, has the stated local density and masses of the stated mean square, then its count follows the midpoint clock up to a linear drift governed by one constant b; its ordinates stay within a bounded distance of a drift-corrected clock, and within a bounded distance of the midpoint clock itself exactly when $\mathrm{b} = 0$. Throughout, the Laplace transform of a distribution $\mathscr{T}$ supported in $[0,\infty)$ is $\mathcal{L}\mathscr{T}(s) = \langle \mathscr{T}, \chi e^{-(s-\frac12)u}\rangle$, as in the proof of Lemma 18.14; thus $\mathcal{L}\delta_{\log n}(s) = n^{\frac12 - s}$ and $\mathcal{L}\delta_0'(s) = s - \frac12$.

Proposition F.2 (A priori displacement bound for an exact rung solution)**.** *Fix* $\mathsf{t} \in [0,1]$*. Let* $\boldsymbol{\kappa} = (\kappa_k)_{k\ge1}$ *satisfy* $0 < \kappa_1 \le \kappa_2 \le \cdots$ *and* $\kappa_k \to \infty$*, and let* $(\mathsf{a}_n)_{n\ge2}$ *be real masses satisfying, in* $\mathcal{D}'(0,\infty)$*,*

$$\sum_{k\ge1} \cos(\kappa_k u) - (1-\mathsf{t}) \sum_{k\ge1} \cos(\tau_k u) - \mathsf{t}\,\mathsf{k}(u) = -\sum_{n\ge2} \mathsf{a}_n\, \delta(u - \log n). \tag{F.10}$$

Assume that for every $\epsilon_0 > 0$ *the local count and the masses satisfy*

$$N_{\boldsymbol{\kappa}}(T+1) - N_{\boldsymbol{\kappa}}(T) \ll_{\epsilon_0} (1+T)^{\epsilon_0} \qquad (T \ge 0), \tag{F.11}$$

$$\sum_{X < n \le 2X} |\mathsf{a}_n|^2 \ll_{\epsilon_0} X^{1+\epsilon_0} \qquad (X \ge 1). \tag{F.12}$$

Then the limit

$$\mathrm{b} := \lim_{\sigma\to+\infty} \Big[\sum_{k\ge1} \frac{\sigma - \frac12}{(\sigma-\frac12)^2 + \kappa_k^2} - (1-\mathsf{t}) \sum_{k\ge1} \frac{\sigma-\frac12}{(\sigma-\frac12)^2+\tau_k^2} - \mathsf{t}\,\frac{G'}{2G}(\sigma) \Big] \tag{F.13}$$

exists and is real. Fix $\epsilon_0 \in (0, \frac14)$ *and a class of such solutions, possibly with different* $\mathsf{t} \in [0,1]$*, on which the following four quantities are bounded: the implied constants in* (F.11) *and* (F.12) *for this* ϵ_0*, the number* $\sum_k \kappa_k^{-2}$ *and the number* $|\mathrm{b}|$*. There are* C^* *and* k_0*, depending only on these four bounds and not on* t*, such that every solution of the class satisfies*

$$\Big| N_{\boldsymbol{\kappa}}(T) - n_{\mathrm{RVM}}(T) - \frac{2\mathrm{b}}{\pi} T \Big| \le C^* \log(2+T) \qquad (T \ge 0) \tag{F.14}$$

and, for every $k \ge k_0$*, with some* s_k *between* κ_k *and* τ_k*,*

$$\Big| \kappa_k - \tau_k + \frac{4\mathrm{b}\,\kappa_k}{\log(\mathrm{s}_k/2\pi)} \Big| \le C^*, \qquad |\kappa_k - \tau_k^{(\mathrm{b})}| \le C^*, \tag{F.15}$$

where $\tau_k^{(\mathrm{b})}$ *is the drift-corrected clock point, the unique solution* $x \ge x_{\mathrm{b}} := 2\pi e^{8|\mathrm{b}|}$ *of* $\mathcal{N}_{\mathrm{RVM}}(x) + \frac{2\mathrm{b}}{\pi} x = k - \frac12$*. In particular* $\kappa_k - \tau_k = -(4\mathrm{b} + o(1))\kappa_k / \log\kappa_k$ *as* $k \to \infty$*, so the displacement* $\kappa_k - \tau_k$ *is bounded if and only if* $\mathrm{b} = 0$*; and if* $\mathrm{b} = 0$*, then* $|\kappa_k - \tau_k| \le C^*$ *for all* $k \ge 1$*. The constant* b *is the mass at* $u = 0$ *of the distribution on* $[0,\infty)$ *that extends the left side of* (F.10) *with the extensions of the proof, and the identity on the open half-line does not determine it: a list whose count exceeds the clock by* $2\mathrm{b}/\pi$ *per unit height satisfies* (F.13) *with that very* b*.*

Proof. Order of $P_{\boldsymbol{\kappa}}$*.* By (F.11), $N_{\boldsymbol{\kappa}}(T) \le \sum_{0\le j\le T} \big(N_{\boldsymbol{\kappa}}(j+1) - N_{\boldsymbol{\kappa}}(j)\big) \ll_{\epsilon_0} (1+T)^{1+\epsilon_0}$ for every $\epsilon_0 > 0$, and $N_{\boldsymbol{\kappa}}(t) = 0$ for $t < \kappa_1$. Hence $\sum_k \kappa_k^{-2} = \int t^{-2} dN_{\boldsymbol{\kappa}}(t) < \infty$; this number is not controlled by the density constant alone, because κ_1 may be small, and it is therefore listed separately. The symmetric canonical product $P_{\boldsymbol{\kappa}}(s) = \prod_k (1 + q(s)/(\frac14 + \kappa_k^2))$ converges locally uniformly, is entire and real on $\mathbb{R}$, is positive for real $s > 1$ (every factor exceeds 1 there, since $q(s) > 0$), and has order at most 1. For $|s| \ge 1$, $|q(s)| \le 2|s|^2 =: R^2$; since $\frac14 + \kappa_k^2 \ge \kappa_k^2$,

$$\log|P_{\boldsymbol{\kappa}}(s)| \le \sum_k \log\Big(1 + \frac{R^2}{\kappa_k^2}\Big) = \int_0^\infty \log\Big(1 + \frac{R^2}{t^2}\Big) dN_{\boldsymbol{\kappa}}(t) = \int_{\kappa_1}^\infty N_{\boldsymbol{\kappa}}(t) \frac{2R^2}{t(t^2+R^2)}\, dt,$$

the boundary terms vanishing because $N_{\boldsymbol{\kappa}}(t) = 0$ for $t < \kappa_1$ and $N_{\boldsymbol{\kappa}}(t)\log(1+R^2/t^2) \ll t^{1+\epsilon_0} R^2 t^{-2} \to 0$ as $t \to \infty$. On $[\kappa_1, 1]$ (if $\kappa_1 < 1$) the integrand is at most $2N_{\boldsymbol{\kappa}}(1)/t$, which contributes $O(1)$; on $[1,\infty)$, with $N_{\boldsymbol{\kappa}}(t) \ll t^{1+\epsilon_0}$ and $t = Rv$, the integral is $\ll R^{1+\epsilon_0} \int_0^\infty \frac{2v^{\epsilon_0}}{1+v^2} dv \ll |s|^{1+\epsilon_0}$. Since $\epsilon_0 > 0$ is arbitrary, $\log|P_{\boldsymbol{\kappa}}(s)| = O_\epsilon(|s|^{1+\epsilon})$ for every $\epsilon > 0$, with constants that may depend on the solution; only this qualitative growth is used below.

From $\mathcal{D}'(0,\infty)$ *to the Laplace transform.* The identity (F.10) holds on the open half-line and says nothing about the point $u = 0$; the exponential $e^{-(s-1/2)u}$ is not a test function of $\mathcal{D}(0,\infty)$, so it cannot be paired with (F.10) directly. We first extend each term to a distribution on $\mathbb{R}$ supported in $[0,\infty)$, as in Lemma 18.14. Put $\mathfrak{H}_{\boldsymbol{\kappa}}(u) := \sum_k \kappa_k^{-2} \cos(\kappa_k u)$, continuous and bounded,

and $\mathscr{S}_{\boldsymbol{\kappa}} := -\partial_u^2(\mathfrak{H}_{\boldsymbol{\kappa}}\mathbf{1}_+) + \mathfrak{H}_{\boldsymbol{\kappa}}(0)\delta_0'$: a distribution of order at most 2, supported in $[0,\infty)$, whose restriction to $(0,\infty)$ is $\sum_k \cos(\kappa_k u)$ (termwise differentiation of a uniformly convergent series in $\mathcal{D}'(0,\infty)$). Its Laplace transform is computed as in (18.17), which uses only $\sum_k \kappa_k^{-2} < \infty$: with $z = s - \frac{1}{2}$, $\mathcal{L}(\partial_u \mathscr{T}) = z\,\mathcal{L}\mathscr{T}$ for $\mathscr{T}$ supported in $[0,\infty)$, $\mathcal{L}\delta_0' = z$, and $\mathcal{L}(\mathfrak{H}_{\boldsymbol{\kappa}}\mathbf{1}_+) = \sum_k \kappa_k^{-2} z/(z^2+\kappa_k^2)$ by termwise integration, so

$$\mathcal{L}\mathscr{S}_{\boldsymbol{\kappa}}(s) = -z^2 \sum_k \frac{z}{\kappa_k^2(z^2+\kappa_k^2)} + z\sum_k \frac{1}{\kappa_k^2} = \sum_{k\ge 1} \frac{z}{z^2+\kappa_k^2} = \tfrac{1}{2}\,\frac{P_{\boldsymbol{\kappa}}'}{P_{\boldsymbol{\kappa}}}(s) \qquad (\Re s > \tfrac{1}{2}): \tag{F.16}$$

the sign of the correction $+\mathfrak{H}_{\boldsymbol{\kappa}}(0)\delta_0'$ is the one that removes the term $-z\mathfrak{H}_{\boldsymbol{\kappa}}(0)$, and no multiple of δ_0 arises. The same extension $\mathscr{S}_{\boldsymbol{\tau}}$ serves the clock, with $P_{\mathrm{RVM}}^{\mathrm{crit}}$ in place of $P_{\boldsymbol{\kappa}}$. The term k is extended by the Archimedean distribution $\mathscr{K}$ of Lemma 18.13, with $\mathcal{L}\mathscr{K} = \frac{1}{2}G'/G$. The mass term $\mathscr{A} := \sum_{n\ge 2} \mathsf{a}_n \delta_{\log n}$ is a locally finite measure on $\mathbb{R}$ supported in $[\log 2,\infty)$, with no mass at 0. For $\sigma_0 > \frac{3}{2}$ choose $\epsilon_0 := \sigma_0 - \frac{3}{2}$, which satisfies $\epsilon_0 < 2\sigma_0 - 3$; by (F.12) for this ϵ_0 and Cauchy–Schwarz on dyadic blocks,

$$\sum_{X<n\le 2X} |\mathsf{a}_n| n^{1/2-\sigma_0} \le \Big(\sum_{X<n\le 2X} |\mathsf{a}_n|^2\Big)^{1/2}\Big(\sum_{X<n\le 2X} n^{1-2\sigma_0}\Big)^{1/2} \ll_{\epsilon_0} X^{3/2-\sigma_0+\epsilon_0/2},$$

a negative power of X; so $\sum_n |\mathsf{a}_n| n^{1/2-\sigma} < \infty$ for every $\sigma > \frac{3}{2}$, and $\mathcal{L}\mathscr{A}(s) = \sum_{n\ge 2} \mathsf{a}_n n^{\frac{1}{2}-s}$ converges absolutely and uniformly on every half-plane $\Re s \ge \sigma_0 > \frac{3}{2}$. The distribution

$$\mathscr{E} := \mathscr{S}_{\boldsymbol{\kappa}} - (1-\mathsf{t})\mathscr{S}_{\boldsymbol{\tau}} - \mathsf{t}\,\mathscr{K} + \mathscr{A} \tag{F.17}$$

is supported in $[0,\infty)$, and by (F.10) its restriction to $(0,\infty)$ vanishes. Hence $\mathscr{E}$ is supported at $\{0\}$; its restriction to a bounded neighbourhood of 0 has finite order (each term has finite order on bounded intervals), so $\mathscr{E} = \sum_{j\le J} c_j \delta_0^{(j)}$ with real c_j [21, Thm. 2.3.4], whose Laplace transform is the real polynomial $p(z) = \sum_{j\le J} c_j z^j$, $z = s - \frac{1}{2}$. Taking Laplace transforms of (F.17) on $\Re s > \frac{3}{2}$,

$$\tfrac{1}{2}\frac{P_{\boldsymbol{\kappa}}'}{P_{\boldsymbol{\kappa}}}(s) - (1-\mathsf{t})\,\tfrac{1}{2}\frac{(P_{\mathrm{RVM}}^{\mathrm{crit}})'}{P_{\mathrm{RVM}}^{\mathrm{crit}}}(s) - \mathsf{t}\,\tfrac{1}{2}\frac{G'}{G}(s) + \sum_{n\ge 2} \mathsf{a}_n n^{\frac{1}{2}-s} = p\big(s-\tfrac{1}{2}\big) \qquad (\Re s > \tfrac{3}{2}). \tag{F.18}$$

The polynomial p is the price of the passage from the open half-line to the closed one (Remark 18.15).

The polynomial is a constant. On the real axis, with $z = \sigma - \frac{1}{2} \ge 1$, each term on the left of (F.18) is $O(z^{\epsilon_0})$ as $z \to +\infty$, with the fixed $\epsilon_0 < \frac{1}{4}$. Integrating by parts against the count (the boundary term $N_{\boldsymbol{\kappa}}(t)\,z/(z^2+t^2)$ vanishes at $t=0$ and, since $N_{\boldsymbol{\kappa}}(t) \ll (1+t)^{1+\epsilon_0}$, at $t\to\infty$),

$$\sum_k \frac{z}{z^2+\kappa_k^2} = \int_0^\infty N_{\boldsymbol{\kappa}}(t)\frac{2zt}{(z^2+t^2)^2}\,dt \ll \int_0^\infty (1+t)^{1+\epsilon_0}\frac{2zt\,dt}{(z^2+t^2)^2} \le z^{\epsilon_0}\int_0^\infty \frac{2(1+v)^{1+\epsilon_0} v\,dv}{(1+v^2)^2} \ll z^{\epsilon_0}$$

(substitute $t = zv$ and use $1 + zv \le z(1+v)$), with the implied constant that of (F.11); the clock term is $O(\log z)$ by the same computation with n_{RVM}; $\frac{1}{2}G'/G(\sigma) = O(\log\sigma)$ by Stirling's formula; and the mass sum tends to 0 by absolute convergence. A polynomial in z that is $O(z^{\epsilon_0})$ as $z\to+\infty$ with $\epsilon_0 < 1$ has degree zero, so $p \equiv \mathrm{b}$ with $\mathrm{b}\in\mathbb{R}$, and (F.18) read at real $\sigma\to+\infty$ is (F.13): the limit exists and equals the constant b.

Integration. All terms of (F.18) are real on $(\frac{3}{2},\infty)$, where $P_{\boldsymbol{\kappa}}$, $P_{\mathrm{RVM}}^{\mathrm{crit}}$ and $G(\sigma) = \sigma(\sigma-1)\Gamma(\frac{\sigma}{2})\pi^{-\sigma/2}$ are positive. Since $\frac{d}{ds}\big(\frac{2\mathsf{a}_n}{\log n} n^{1/2-s}\big) = -2\mathsf{a}_n n^{1/2-s}$, the function

$$Y_{\mathsf{t}}(s) = \exp\left(\sum_{n\ge 2} \frac{2\mathsf{a}_n}{\log n}\, n^{1/2-s}\right) \tag{F.19}$$

has logarithmic derivative $-2\sum_n \mathsf{a}_n n^{1/2-s}$, so twice (F.18) says $(\log P_{\boldsymbol{\kappa}})' = (1-\mathsf{t})(\log P_{\mathrm{RVM}}^{\mathrm{crit}})' + \mathsf{t}(\log G)' + (\log Y_{\mathsf{t}})' + 2\mathrm{b}$. Integrating from 2 to s along a path in $\Re s > \frac{3}{2}$ and exponentiating,

$$P_{\boldsymbol{\kappa}}(s) = C_{\mathsf{t}} \big(P_{\mathrm{RVM}}^{\mathrm{crit}}(s)\big)^{1-\mathsf{t}}\, G(s)^{\mathsf{t}}\, Y_{\mathsf{t}}(s)\, e^{2\mathrm{b}(s-2)} = C_{\mathsf{t}}\, G(s)\, \zeta_{\mathrm{RVM}}(s)^{1-\mathsf{t}}\, Y_{\mathsf{t}}(s)\, e^{2\mathrm{b}(s-2)} \qquad (\Re s > \tfrac{3}{2}), \tag{F.20}$$

with $C_{\mathsf{t}} := P_{\boldsymbol{\kappa}}(2)/\big(G(2)\zeta_{\mathrm{RVM}}(2)^{1-\mathsf{t}} Y_{\mathsf{t}}(2)\big) > 0$, the value that makes both sides agree at $s = 2$. The fractional powers are defined globally, not along paths: the half-plane $\Re s > \frac{3}{2}$ is simply connected, $P_{\mathrm{RVM}}^{\mathrm{crit}}$ has no zeros there (its zeros lie on $\Re s = \frac{1}{2}$), G has no zeros there (its only zeros are $s = 0$ and $s = 1$), and both are positive on $(\frac{3}{2}, \infty)$; so each has a unique holomorphic logarithm on $\Re s > \frac{3}{2}$ that is real on the real axis, and $(P_{\mathrm{RVM}}^{\mathrm{crit}})^{1-\mathsf{t}} := \exp((1-\mathsf{t}) \log P_{\mathrm{RVM}}^{\mathrm{crit}})$, $G^{\mathsf{t}} := \exp(\mathsf{t} \log G)$ with these logarithms. The integration from 2 to s is then path-independent, and the zero count below is performed on the entire function $P_{\boldsymbol{\kappa}}$ itself, so no branch of a fractional factor enters it. By the dyadic bound above with $\sigma_0 = 2$ and the fixed $\epsilon_0 < \frac{1}{4}$ (any $\epsilon_0 < 1$ would do here), the series in (F.19) converges absolutely and uniformly on $\Re s \geq 2$ and

$$\sup_{\Re s \geq 2} |\log Y_{\mathsf{t}}(s)| \leq B_2 := \sum_{n \geq 2} \frac{2|\mathsf{a}_n|}{\log n} n^{-3/2} < \infty, \tag{F.21}$$

with B_2 bounded in terms of the implied constant in (F.12) for the fixed ϵ_0. Since $1 \leq P_{\boldsymbol{\kappa}}(2) = \prod_k (1 + 2/(\frac{1}{4} + \kappa_k^2)) \leq e^{2\sum_k \kappa_k^{-2}}$, $G(2) = 2/\pi$ and $|\log \zeta_{\mathrm{RVM}}(2)| \leq \mathrm{c}_{\mathrm{RVM}}$,

$$|\log C_{\mathsf{t}}| \leq 2 \sum_k \kappa_k^{-2} + \log \tfrac{\pi}{2} + \mathrm{c}_{\mathrm{RVM}} + B_2,$$

a bound in terms of $\sum_k \kappa_k^{-2}$ and B_2 only. No displacement bound is used here.

The right half-strip. Put $\mathbb{W}_{\boldsymbol{\kappa}} := \log|P_{\boldsymbol{\kappa}}| - \log|G|$. On the right half-strip $2 \leq \Re s \leq 4$, (F.20), Theorem 9.10 ($|\log \zeta_{\mathrm{RVM}}| \leq \mathrm{c}_{\mathrm{RVM}}$ on $\Re s \geq 2$) and (F.21) give

$$|\mathbb{W}_{\boldsymbol{\kappa}}(s)| \leq \mathbb{K}_{\mathrm{E}} := |\log C_{\mathsf{t}}| + \mathrm{c}_{\mathrm{RVM}} + B_2 + 4|\mathrm{b}| \qquad (2 \leq \Re s \leq 4), \tag{F.22}$$

uniformly in t and $\Im s$, because $\log|e^{2\mathrm{b}(s-2)}| = 2\mathrm{b}(\Re s - 2)$ lies in $[-4|\mathrm{b}|, 4|\mathrm{b}|]$ there; on the line $\Re s = 2$ itself this term vanishes. The disc used below lies in $0 \leq \Re s \leq 4$; its part in $2 \leq \Re s \leq 4$ is covered by (F.22).

The half-strip to the left. Let $\mathbb{S} := \{-1 \leq \Re s \leq 2,\ \Im s \geq 1\}$. $\mathbb{W}_{\boldsymbol{\kappa}}$ is subharmonic in the interior of $\mathbb{S}$ and upper semicontinuous on $\mathbb{S}$, since $\log|G|$ is harmonic there (G has no zeros or poles with $\Im s \geq 1$). The majorant $\mathrm{h}_0(s) := \frac{3}{2}\log|s+2|$ is harmonic and nonnegative on $\mathbb{S}$. We check the three sides. *Right side:* $\mathbb{W}_{\boldsymbol{\kappa}}(2+it) \leq \mathbb{K}_{\mathrm{E}}$ by (F.22). *Left side:* the symmetry $P_{\boldsymbol{\kappa}}(1-s) = P_{\boldsymbol{\kappa}}(s)$ and the reality of $P_{\boldsymbol{\kappa}}$ give $P_{\boldsymbol{\kappa}}(-1+it) = P_{\boldsymbol{\kappa}}(2-it) = \overline{P_{\boldsymbol{\kappa}}(2+it)}$, so $\log|P_{\boldsymbol{\kappa}}(-1+it)| = \log|P_{\boldsymbol{\kappa}}(2+it)|$. The factor G has no such symmetry, and (18.48) gives, for $t \geq 1$,

$$\mathbb{W}_{\boldsymbol{\kappa}}(-1+it) = \mathbb{W}_{\boldsymbol{\kappa}}(2+it) + \log\Big|\frac{G(2+it)}{G(-1+it)}\Big| \leq \mathbb{K}_{\mathrm{E}} + \tfrac{3}{2}\log(1+t) \leq \mathbb{K}_{\mathrm{E}} + \tfrac{3}{4}\log 2 + \mathrm{h}_0(-1+it),$$

because $1 + t \leq \sqrt{2}|1+it|$ and $|s+2| = |1+it|$ at $s = -1+it$. *Bottom side:* for $s = \sigma + i$ with $-1 \leq \sigma \leq 2$, $|q(s)| = |s|\,|s-1| \leq 5$, so $\log|P_{\boldsymbol{\kappa}}(s)| \leq \sum_k \log(1 + 5/\kappa_k^2) \leq 5\sum_k \kappa_k^{-2}$, and $-\log|G(\sigma+i)| \leq \mathrm{c}_7$, an absolute constant ($G$ is continuous and zero-free on that segment). *Growth:* $\log|P_{\boldsymbol{\kappa}}(s)| = O(|s|^{1+\epsilon})$ by the first step, and $-\log|G(s)| \leq \frac{\pi}{4}|\Im s| + O(\log|s|)$ on $\mathbb{S}$ by Stirling's formula [64, §12.33]; hence $\mathbb{W}_{\boldsymbol{\kappa}} - \mathrm{h}_0 \leq \mathrm{A}_0 e^{c\Im s}$ on $\mathbb{S}$ for every $c > 0$, with A_0 depending on the solution. Lemma 18.45(a), with width 3 and $c < \pi/3$, applied to $\mathbb{W}_{\boldsymbol{\kappa}} - \mathrm{h}_0$, gives

$$\mathbb{W}_{\boldsymbol{\kappa}}(s) \leq \mathbb{K}''_{\mathrm{E}} + \tfrac{3}{2}\log|s+2| \qquad (s \in \mathbb{S}), \qquad \mathbb{K}''_{\mathrm{E}} := \max\Big(\mathbb{K}_{\mathrm{E}} + \tfrac{3}{4}\log 2,\ 5\sum_k \kappa_k^{-2} + \mathrm{c}_7\Big). \tag{F.23}$$

The constant A_0 enters only the applicability of the lemma, not its conclusion, so (F.23) is uniform over the class; the compact part of the strip needs no separate bound, because the bottom side $\Im s = 1$ is part of $\partial\mathbb{S}$.

Horizontal variation of the argument. Fix $T \geq T_0$ with $T \notin \{\kappa_k\}$, where $T_0 \geq 3$ is the absolute constant of Lemma 18.49. As in that lemma, choose $\mathrm{a}_T \in \mathbb{R}$ with $e^{i\mathrm{a}_T} P_{\boldsymbol{\kappa}}(2+iT) > 0$ and put $\mathfrak{F}_T(z) := \frac{1}{2}\big(e^{i\mathrm{a}_T} P_{\boldsymbol{\kappa}}(z+iT) + e^{-i\mathrm{a}_T} P_{\boldsymbol{\kappa}}(z-iT)\big)$, an entire function with $\mathfrak{F}_T(x) = \Re(e^{i\mathrm{a}_T} P_{\boldsymbol{\kappa}}(x+iT))$ for real x and $\mathfrak{F}_T(2) = |P_{\boldsymbol{\kappa}}(2+iT)| > 0$. Let n_T be the number of zeros of $\mathfrak{F}_T$ on $[\frac{1}{2}, 2]$, with multiplicity; the segment splits into at most $n_T + 1$ pieces on which $\Re(e^{i\mathrm{a}_T} P_{\boldsymbol{\kappa}})$ has constant sign, so the variation of $\arg P_{\boldsymbol{\kappa}}$ along the segment from $2 + iT$ to $\frac{1}{2} + iT$ is at most $\pi(n_T + 1)$ [61, §9.4]. Jensen's formula for $\mathfrak{F}_T$ on the disc $|z - 2| \leq 2$, valid also if $\mathfrak{F}_T$ has zeros on the circle (they contribute $\log 1 = 0$) [57, Thm. 15.18], in which the zeros on $[\frac{1}{2}, 2] \subset \{|z-2| \leq \frac{3}{2}\}$ contribute at least $\log \frac{4}{3}$ each and the others contribute nonnegatively, together with the bound of the mean over the circle by the maximum, gives

$$n_T \log \tfrac{4}{3} \leq \max_{|z-2|=2} \log|\mathfrak{F}_T(z)| - \log|\mathfrak{F}_T(2)| \leq \max_{|s-(2+iT)|\leq 2} \log|P_{\boldsymbol{\kappa}}(s)| - \log|P_{\boldsymbol{\kappa}}(2+iT)|,$$

because $|P_{\boldsymbol{\kappa}}(z - iT)| = |P_{\boldsymbol{\kappa}}(\bar{z} + iT)|$. On the disc $\mathbb{D}_T := \{|s - (2+iT)| \leq 2\}$ write $\log|P_{\boldsymbol{\kappa}}| = \mathbb{W}_{\boldsymbol{\kappa}} + \log|G|$. The part of $\mathbb{D}_T$ with $\Re s \leq 2$ lies in $\mathbb{S}$ (there $\Im s \geq T - 2 \geq 1$) and has $|s+2| \leq T + 6$, so (F.23) gives $\mathbb{W}_{\boldsymbol{\kappa}} \leq \mathbb{K}_{\mathrm{E}}'' + \frac{3}{2}\log(T+6)$; the part with $\Re s \geq 2$ has $\mathbb{W}_{\boldsymbol{\kappa}} \leq \mathbb{K}_{\mathrm{E}}$ by (F.22); and at the centre $\mathbb{W}_{\boldsymbol{\kappa}}(2+iT) \geq -\mathbb{K}_{\mathrm{E}}$. The oscillation of $\log|G|$ on $\mathbb{D}_T$ is at most $\log(T+8) + 2\mathrm{c}_5$ (proof of Lemma 18.49). Hence

$$n_T \leq \frac{\mathbb{K}_{\mathrm{E}} + \mathbb{K}_{\mathrm{E}}'' + \frac{5}{2}\log(T+8) + 2\mathrm{c}_5}{\log(4/3)}, \tag{F.24}$$

the numerical constants 6 and 8 coming from deliberately coarse bounds for $|s+2|$ and for Stirling's estimate on $\mathbb{D}_T$. The variation of $\arg G$ along the same segment is at most $\mathrm{c}_6 = \frac{3}{2}$ for $T \geq T_0$ (proof of Theorem 18.51).

Vertical variation of the argument. On the segment from 2 to $2 + iT$, by (F.20),

$$\arg \zeta_{\boldsymbol{\kappa}}(2+it) = \arg \frac{P_{\boldsymbol{\kappa}}}{G}(2+it) = (1-\mathtt{t})\Im \log \zeta_{\mathrm{RVM}}(2+it) + \Im \log Y_{\mathtt{t}}(2+it) + 2\mathrm{b}\, t,$$

with the branches chosen continuous along the segment and real at $t = 0$ (the branch of $\log \zeta_{\mathrm{RVM}}$ is the one of Theorem 9.10, continuous on $\Re s \geq 2$, and $\log Y_{\mathtt{t}}$ is the series (F.19) itself). What the argument principle uses is the *net* variation of the argument along the segment, that is, the difference of the values of the continuous branch at its two ends, not its total variation; the first two terms are bounded by $\mathrm{c}_{\mathrm{RVM}}$ and B_2 in modulus at both ends, so the net variation along the segment is $2\mathrm{b}T + O(2\mathrm{c}_{\mathrm{RVM}} + 2B_2)$. This is where the factor $e^{2\mathrm{b}(s-2)}$ enters the count: its argument is not bounded, it grows linearly.

Zero count. Lemma 18.50 applies verbatim to $P_{\boldsymbol{\kappa}}$: it uses only that $P_{\boldsymbol{\kappa}}$ is real on $\mathbb{R}$, symmetric under $s \mapsto 1 - s$, with zeros exactly at $\frac{1}{2} \pm i\kappa_k$ and none on the contour, the last because $T \notin \{\kappa_k\}$. So $\pi N_{\boldsymbol{\kappa}}(T) = \pi + \vartheta(T) + \Delta_{\mathbb{G}_T} \arg \zeta_{\boldsymbol{\kappa}}$ along the path $2 \to 2 + iT \to \frac{1}{2} + iT$, with $\zeta_{\boldsymbol{\kappa}} := P_{\boldsymbol{\kappa}}/G$. Combining the two variations, and using $|1 + \vartheta(T)/\pi - \mathcal{N}_{\mathrm{RVM}}(T)| \leq 1/T$ for $T \geq T_0$ (Stirling; enlarge T_0 if necessary) and $|n_{\mathrm{RVM}}(T) - \mathcal{N}_{\mathrm{RVM}}(T)| \leq \frac{1}{2}$ for $T \geq 2\pi$, since $n_{\mathrm{RVM}}(T) = \lfloor \mathcal{N}_{\mathrm{RVM}}(T) + \frac{1}{2} \rfloor$,

$$\Big| N_{\boldsymbol{\kappa}}(T) - n_{\mathrm{RVM}}(T) - \frac{2\mathrm{b}}{\pi} T \Big| \leq n_T + 1 + \frac{2\mathrm{c}_{\mathrm{RVM}} + 2B_2 + \mathrm{c}_6}{\pi} + \frac{1}{T} + \frac{1}{2} \qquad (T \geq T_0,\ T \notin \{\kappa_k\}), \tag{F.25}$$

which with (F.24) is (F.14) for these T, with C^* depending only on $\mathbb{K}_{\mathrm{E}}$ and $\mathbb{K}_{\mathrm{E}}''$, hence only on the four bounds of the statement. It extends to $T \in \{\kappa_k\}$, $T \geq T_0$, by right continuity of $N_{\boldsymbol{\kappa}}$ and n_{RVM}, letting T decrease to κ_k through values outside the discrete set $\{\kappa_k\}$; and to $0 \leq T < T_0$ because there $N_{\boldsymbol{\kappa}}(T) \leq N_{\boldsymbol{\kappa}}(T_0) \leq n_{\mathrm{RVM}}(T_0) + 2|\mathrm{b}|T_0/\pi + C^* \log(2 + T_0)$, while $n_{\mathrm{RVM}}(T)$ and $2|\mathrm{b}|T/\pi$ are bounded by constants of the same kind; enlarging C^* gives (F.14) for all $T \geq 0$. The argument uses $P_{\boldsymbol{\kappa}}$, which is entire, so no branch of a fractional factor enters the zero count.

From the count to the displacement. Put $\mathcal{F}_{\mathrm{b}}(x) := \mathcal{N}_{\mathrm{RVM}}(x) + \frac{2\mathrm{b}}{\pi}x$. For $x \geq x_{\mathrm{b}} = 2\pi e^{8|\mathrm{b}|}$, $\mathcal{F}'_{\mathrm{b}}(x) = \frac{1}{2\pi}\log\frac{x}{2\pi} + \frac{2\mathrm{b}}{\pi} \geq \frac{1}{4\pi}\log\frac{x}{2\pi}$, so $\mathcal{F}_{\mathrm{b}}$ is strictly increasing on $[x_{\mathrm{b}}, \infty)$ and tends to ∞. Since $N_{\boldsymbol{\kappa}}(\kappa_k) \geq k$ and $N_{\boldsymbol{\kappa}}(\kappa_k - 0) \leq k - 1$ (multiplicities included), (F.14) at $T = \kappa_k$ and its left limit, together with $|n_{\mathrm{RVM}}(T \pm 0) - \mathcal{N}_{\mathrm{RVM}}(T)| \leq 1$ for all $T > 0$, give

$$\bigl|\mathcal{F}_{\mathrm{b}}(\kappa_k) - k\bigr| \leq C^* \log(2 + \kappa_k) + 2 \qquad (k \geq 1). \tag{F.26}$$

The count bound also places the ordinates uniformly: $\kappa_k > T$ whenever $k > n_{\mathrm{RVM}}(T) + 2|\mathrm{b}|T/\pi + C^*\log(2+T)$, and $\kappa_k \leq T$ whenever $n_{\mathrm{RVM}}(T) - 2|\mathrm{b}|T/\pi - C^*\log(2+T) \geq k$; so for every X there is k_0, and for every k there is a bound for κ_k, depending only on X, k, C^* and $|\mathrm{b}|$. Fix $X \geq x_{\mathrm{b}}$, depending only on C^* and $|\mathrm{b}|$, so large that for $x \geq X$

$$\frac{x}{4\pi}\log\frac{x}{2\pi} \leq \mathcal{F}_{\mathrm{b}}(x) \leq \frac{x}{\pi}\log\frac{x}{2\pi}, \qquad C^*\log(2+x) + \tfrac{5}{2} \leq \frac{x}{8\pi}\log\frac{x}{2\pi},$$

and let k_0 be such that $\kappa_k \geq X$, $\tau_k \geq X$ and $k - \frac{1}{2} \geq \mathcal{F}_{\mathrm{b}}(X)$ for $k \geq k_0$. For $k \geq k_0$, (F.26) and the two-sided bound for $\mathcal{F}_{\mathrm{b}}$ give $\kappa_k \log\kappa_k \asymp k \asymp \tau_k \log\tau_k$, hence $\kappa_k \asymp \tau_k$ and $\kappa_k \asymp \tau_k^{(\mathrm{b})}$, with absolute implied constants; enlarging X, every point s between two of $\kappa_k, \tau_k, \tau_k^{(\mathrm{b})}$ satisfies $\log(\mathrm{s}/2\pi) \geq \frac{1}{2}\log(\kappa_k/2\pi) \geq 1$. The mean-value theorem for $\mathcal{F}_{\mathrm{b}}$ between κ_k and $\tau_k^{(\mathrm{b})}$ gives

$$|\kappa_k - \tau_k^{(\mathrm{b})}| \leq \frac{|\mathcal{F}_{\mathrm{b}}(\kappa_k) - (k - \frac{1}{2})|}{\min \mathcal{F}'_{\mathrm{b}}} \leq \frac{4\pi\bigl(C^*\log(2+\kappa_k) + \frac{5}{2}\bigr)}{\frac{1}{2}\log(\kappa_k/2\pi)},$$

which is bounded; this is the second inequality in (F.15), after enlarging C^*. For the first, write $\mathcal{N}_{\mathrm{RVM}}(\kappa_k) - \mathcal{N}_{\mathrm{RVM}}(\tau_k) = \mathcal{F}_{\mathrm{b}}(\kappa_k) - (k - \frac{1}{2}) - \frac{2\mathrm{b}}{\pi}\kappa_k = -\frac{2\mathrm{b}}{\pi}\kappa_k + E_k$ with $|E_k| \leq C^*\log(2+\kappa_k) + \frac{5}{2}$; the mean-value theorem for $\mathcal{N}_{\mathrm{RVM}}$ gives

$$(\kappa_k - \tau_k)\,\frac{1}{2\pi}\log\frac{\mathrm{s}_k}{2\pi} = -\frac{2\mathrm{b}}{\pi}\kappa_k + E_k, \qquad \mathrm{s}_k \text{ between } \kappa_k \text{ and } \tau_k,$$

so $\kappa_k - \tau_k + 4\mathrm{b}\kappa_k/\log(\mathrm{s}_k/2\pi) = 2\pi E_k/\log(\mathrm{s}_k/2\pi)$, of modulus at most $4\pi(C^*\log(2+\kappa_k) + \frac{5}{2})/\log(\kappa_k/2\pi)$, which is bounded; this is the first inequality in (F.15), after enlarging C^*. The intermediate point s_k is kept: $\log(\mathrm{s}_k/2\pi) = \log(\kappa_k/2\pi) + O(1)$ only gives $\kappa_k - \tau_k = -(4\mathrm{b} + o(1))\kappa_k/\log\kappa_k$, and replacing s_k by κ_k in the denominator would change the main term by $O(\mathrm{b}^2\kappa_k/\log^3\kappa_k)$, which is not bounded when $\mathrm{b} \neq 0$. If $\mathrm{b} = 0$, the first inequality reads $|\kappa_k - \tau_k| \leq C^*$ for $k \geq k_0$, and for $k < k_0$, $|\kappa_k - \tau_k| \leq \kappa_{k_0} + \tau_{k_0}$, which is bounded by the uniform placement of the ordinates in terms of k_0 and C^*; enlarging C^* gives the bound for all k. All constants depend only on the implied constants of (F.11) and (F.12) for the fixed ϵ_0, on $\sum_k \kappa_k^{-2}$ and on $|\mathrm{b}|$, and not on $\mathtt{t}$. □

Corollary F.3 (Exact solutions without mass at the origin)**.** *Under the hypotheses of Proposition F.2, if the extension of the left side of* (F.10) *constructed in the proof has no mass at $u = 0$, that is,* $\mathrm{b} = 0$*, then $|N_{\boldsymbol{\kappa}}(T) - n_{\mathrm{RVM}}(T)| \leq C^*\log(2+T)$ and $|\kappa_k - \tau_k| \leq C^*$ for all $T \geq 0$ and $k \geq 1$, with C^* depending only on the implied constants of* (F.11)–(F.12) *for the fixed ϵ_0 and on $\sum_k \kappa_k^{-2}$.*

The mechanism behind the drift is elementary: a mass $\mathrm{b}\delta_0$ in the extended identity adds the constant b to the Laplace residual, hence the factor $e^{2\mathrm{b}(s-2)}$ to $P_{\boldsymbol{\kappa}}$, whose argument grows linearly along vertical lines; the argument principle converts this into an excess of $2\mathrm{b}T/\pi$ zeros up to height T, and dividing by the clock density $\mathcal{N}'_{\mathrm{RVM}}(\mathrm{s}_k) = \frac{1}{2\pi}\log\frac{\mathrm{s}_k}{2\pi}$ turns the excess into a displacement $-4\mathrm{b}\kappa_k/\log(\mathrm{s}_k/2\pi) = -(4\mathrm{b} + o(1))\kappa_k/\log\kappa_k$, up to a bounded error. The mean-square hypothesis (F.12) on the masses is independent of the local-density hypothesis and is not implied by it for a general list; both are assumptions on the solution, and the proposition is an a priori statement about hypothetical solutions, not an existence theorem.

Remark F.4. Proposition F.2 is an a priori theorem, not an existence theorem. The exact-atomic interpolation has no established interior solution for $0 < \mathtt{t} < 1$; the proposition only says that any exact solution satisfying its hypotheses lies in a uniform drift-corrected class, with ordinates within

C^* of the drift-corrected clock points $\tau_k^{(\mathrm{b})}$; it has uniformly bounded displacement from the midpoint clock only when $\mathrm{b} = 0$. The mean-square bound (F.12) on the masses is a hypothesis: for the lists $(\gamma_k(\lambda))$ it is Theorem 14.11, whose proof uses the tube through Lemma 14.10 and the term (14.23); for a general list satisfying only (F.11) it is not derived here, and the proposition assumes it. The constant b of (F.13) is the mass at $u = 0$ that the identity on the open half-line leaves undetermined; the argument of $e^{2\mathrm{b}(s-2)}$ along $\Re s = 2$ grows linearly and produces the drift $-4\mathrm{b}\kappa_k/\log(\mathrm{s}_k/2\pi)$ in (F.15), so the proposition bounds the displacement from the midpoint clock only for solutions with $\mathrm{b} = 0$, that is, for solutions of the identity extended to $[0, \infty)$ without a mass at the origin. The zero count follows the mechanism of Lemmas 18.49 and 18.50: Backlund–Jensen on a horizontal segment, the argument principle on the symmetric rectangle, and Stirling for ϑ; a disc of fixed radius about $2 + iT$ controls only the horizontal piece of the argument variation, and the count up to T comes from the vertical piece and the contour.

F.5. **Additional finite-section diagnostics.** The calculations of this subsection solve bounded finite sections directly from the midpoint clock with program `s13` (Appendix B.14; console output in Appendix C.13). The unknowns are the ordinates κ_k with $\tau_k \le \Omega + 14$ and the free masses a_n, $2 \le n \le N$; the ordinates beyond are kept on the midpoint clock. Both are bounded,

$$|\kappa_k - \tau_k| \le 1.2, \qquad |a_n| \le 0.5, \tag{F.27}$$

as numerical regularizations, not hypotheses of any theorem. A test is a C_c^∞ bump of unit integral, proportional to $\exp(-1/(1-v^2))$ in the affine coordinate $v \in (-1, 1)$ of its support, multiplied by $\cos(\omega u)$ or $\sin(\omega u)$; the equation is the pairing of the rung system with the test. For a single section of size N the support is $[0, \log(N + \frac{1}{2})]$ and $\omega = 0, 0.25, \dots, \Omega$. In the nested family, the block of tests introduced by the section of size N has support $[0.08, \log(N + \frac{1}{2})]$ and $\omega = 0, 1, \dots, \Omega_N$, and the rows of each block are divided by the square root of their number, so that every block carries the same weight. The normalized residual is the Euclidean residual divided by the square root of the number of equations. The solver is `scipy.optimize.least_squares` in double precision (trust-region reflective method with box constraints, analytic Jacobian, scaling by the Jacobian), with tolerances $\texttt{ftol} = \texttt{xtol} = \texttt{gtol} = 10^{-11}$ and at most 1200 evaluations for the single sections, and $2 \cdot 10^{-10}$ and at most 3000 evaluations for the nested ones. For every solve the program prints the termination status, the number of evaluations, the numerical rank of the Jacobian at the solution (singular values above 10^{-12} times the largest) with the condition number on that subspace, and the number of variables at their bounds. The genuine ordinates and the von Mangoldt masses are read only after the solve, for diagnostics.

Single sections. The sections $N = 20, 34, 50$ ($\Omega = 70, 110, 150$; $22, 40, 60$ ordinates; $19, 33, 49$ masses; $561, 881, 1201$ equations), each solved alone from the clock, end with normalized residuals $6.8 \cdot 10^{-5}$, $1.4 \cdot 10^{-4}$, $3.9 \cdot 10^{-4}$, the last one by the `ftol` test after 127 evaluations. The numerical rank of the Jacobian is one or two below the number of unknowns; the missing directions are the masses a_n with $\log n$ at the edge of the test support, which the tests do not see. On the remaining subspace the condition number is $1.2 \cdot 10^4$, $2.3 \cdot 10^8$, $1.2 \cdot 10^8$. No ordinate reaches its bound; one or two masses reach $|a_n| = 0.5$. The first ten ordinates are within $7.5 \cdot 10^{-5}$, $2.1 \cdot 10^{-4}$, $4.5 \cdot 10^{-4}$ (root mean square) of γ_k, and the masses with $n \le 15$ follow $\Lambda(n)/(2\sqrt{n})$ (Figure 25). Cross-validation with tests not used in the solve (five supports inside $[0, \log(N + \frac{1}{2})]$, frequencies on the shifted grid $(j + \frac{1}{2}) \cdot 0.25 \le 0.8\,\Omega$) gives root mean square residuals between $5 \cdot 10^{-5}$ and $8 \cdot 10^{-4}$.

These endpoint calculations are distinct from the linearized clock problem: in the latter, it is the ordinate-direction coefficients h_k that become large while a residual of roughly 15–18% persists. Imposing bounds there does not remove an artifact; it exposes the lack of an exact solution to that finite tested linear system.

Nested sections. In the nested family every test block, once introduced, stays in all later solves, and each section is warm-started from the previous one ($N = 13, 20, 34, 50, 70, 100$; Figure 24). Up

to $N = 70$ the old blocks are fitted better as the section grows. At $N = 50$ their root mean square residuals are $4.4 \cdot 10^{-7}$ to $1.7 \cdot 10^{-6}$, and the newest block has $1.7 \cdot 10^{-5}$. At $N = 70$ (93 ordinates, 69 masses, 1185 equations in five blocks) they are $8.7 \cdot 10^{-7}$ to $2.4 \cdot 10^{-6}$, with $6.2 \cdot 10^{-5}$ on the newest block. The section $N = 100$ (146 ordinates, 99 masses, 1786 equations in six blocks) does not continue this. The solver stops by the `ftol` test after 402 evaluations, with residuals $1.1 \cdot 10^{-3}$ to $2.4 \cdot 10^{-3}$ on the old blocks and $2.2 \cdot 10^{-2}$ on the newest one, and the first ten ordinates move to $3.7 \cdot 10^{-3}$ (root mean square) from γ_k. The rank deficit grows with the section (one to four edge masses), and the condition number on the determined subspace grows from 10^7 at $N = 13$ to $9 \cdot 10^{10}$ at $N = 100$. In these runs the nested continuation is observed up to $N = 70$ and fails at $N = 100$.

Repeatability. The section $N = 50$ with the same four blocks was solved from three starting points perturbed from the clock (ordinates by up to ± 0.5, masses by up to ± 0.2). The solves end with maximal block residuals $2.7 \cdot 10^{-4}$, $9.5 \cdot 10^{-6}$ and $7.6 \cdot 10^{-6}$, against $1.7 \cdot 10^{-5}$ for the warm-started solution. Their ordinates differ from those of that solution by up to 1.40, 0.99, 0.62, and their masses by up to 0.92, 0.62, 0.11. Bounded finite sections therefore have several near-solutions of comparable residual: the residual alone does not determine the computed ordinates and masses. The closeness to γ_k and to $\Lambda(n)/(2\sqrt{n})$ is a property of the continuation from the clock, not of every near-solution.

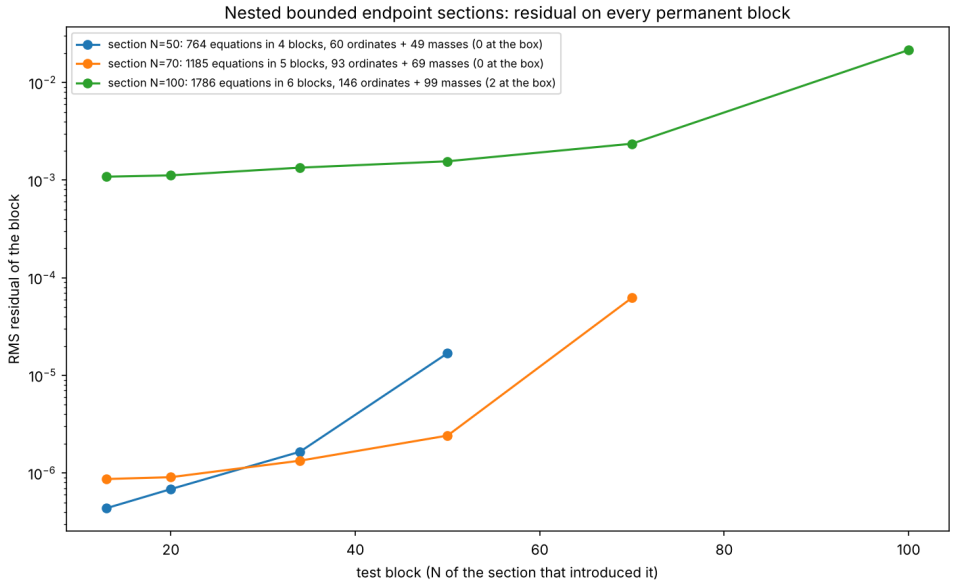


FIGURE 24. Nested bounded endpoint sections $N = 50, 70, 100$: root mean square residual of each section on every permanent test block introduced so far; the horizontal axis is the size N of the section that introduced the block. The legend gives, for each section, the number of equations and blocks, the numbers of ordinates and masses, and how many masses are at the box $|a_n| = 0.5$. Up to $N = 70$ the old blocks are fitted to about 10^{-6}; at $N = 100$ the solver stops with residuals of order 10^{-3} on the old blocks and $2 \cdot 10^{-2}$ on the newest one. A floating-point computation, not an existence proof for the infinite rung system.

Status.

Proved. Proposition F.2: for any exact rung solution satisfying its hypotheses, the count follows the clock with a linear drift governed by one constant b; the ordinates lie within a bounded distance of the drift-corrected clock, and within a bounded distance of the midpoint clock exactly when

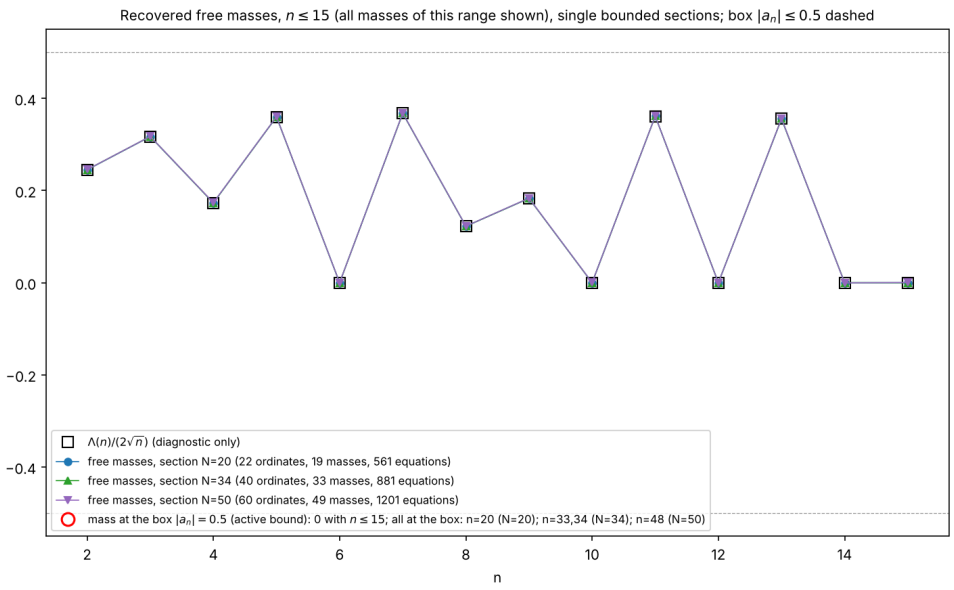


FIGURE 25. Recovered free masses a_n, $n \le 15$, of the single bounded sections $N = 20, 34, 50$, each solved alone from the clock. All masses with $n \le 15$ are shown, and none of them is at the box $|a_n| = 0.5$ (dashed lines; a mass at the box would be circled in red). The masses at the box are edge masses, listed in the legend: $n = 20$ for $N = 20$, $n = 33, 34$ for $N = 34$, $n = 48$ for $N = 50$ (where also $a_{47} = 0.499994$). At $\log n$ the test weights are below $3 \cdot 10^{-6}$ of their peak, so these masses are determined weakly or not at all by the tests; they lie outside the plotted range. The values $\Lambda(n)/(2\sqrt{n})$ (squares) are a post-solution diagnostic and are not supplied to the solver.

$\mathrm{b} = 0$ (Corollary F.3). Theorem 15.47 gives the multiscale linear information at the clock. The additive interpolation is exact, and is a level-set reformulation of $\mathcal{Q}_{\mathrm{RVM}}$.

Numerical evidence. The bounded finite sections of this subsection, in floating point: single sections with normalized residuals of order 10^{-4}; a nested continuation with permanent blocks up to $N = 70$ that fails at $N = 100$; near-solutions that are not unique.

Open. An interior path for the exact-atomic interpolation; the midpoint-clock injectivity problem (F.9), a closed autonomous rigidity question; an infinite solution of the rung system; the endpoint identity (D). None of the statements above supplies existence.